



UTSW

CUH4 Microbiology Lab Reno Phase 2

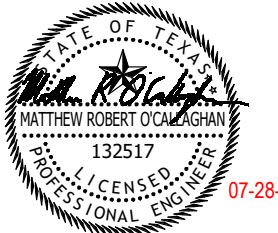
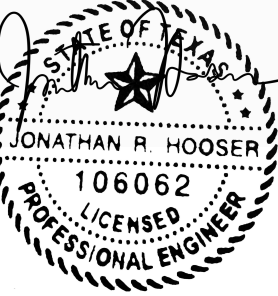
**Construction Documents
Project Manual**

July 28, 2026

HDR Project No. 10385548



SECTION 00 01 07
SEALS PAGE

I hereby certify that the portion of this technical submission described below was prepared by me or under my direct supervision and that I am a duly registered architect under the laws of the State of Texas. <i>Brendon Roger Bangert</i> (BRENDON ROGER BANGERT) Date 07/28/2026 My license renewal date is November 30, 2026 Pages or sheets covered by this seal: Division 01 (Excluding 01 45 29), Division 02, 05 05 10, 05 50 00, Division 06, Division 07, Division 08, Division 09, Division 10, Division 11, Division 12.	
I hereby certify that the portion of this technical submission described below was prepared by me or under my direct supervision and that I am a duly registered engineer under the laws of the State of Texas. <i>Matthew Robert O'Callaghan</i> (MATTHEW ROBERT O'CALLAGHAN) Date 07/28/2026 My license renewal date is September 30, 2026. Pages or sheets covered by this seal: 01 45 29, 03 10 00, 03 20 00, 03 30 00, 05 12 00.	
I hereby certify that the portion of this technical submission described below was prepared by me or under my direct supervision and that I am a duly registered engineer under the laws of the State of Texas. <i>Kyle Mitchell Hansard</i> (Kyle Mitchell Hansard) Date 07/28/2026 My license renewal date is September 30, 2026 Pages or sheets covered by this seal: Division 21, Division 22, Division 23	
I hereby certify that the portion of this technical submission described below was prepared by me or under my direct supervision and that I am a duly registered engineer under the laws of the State of Texas. <i>Jonathan Ross Hooser</i> (JONATHAN ROSS HOOSER) Date 03/35/2026 My license renewal date is March 31, 2026 Pages or sheets covered by this seal: Division 26, 28 31 00	

I hereby certify that the portion of this technical submission described below was prepared by me or under my direct supervision and that I am a duly registered engineer under the laws of the State of Texas.

Josh Kelly

(JOSH KELLY) Date 07/28/2026

My license renewal date is December 31, 2027

Pages or sheets covered by this seal:
Division 27, 28 05 00, 28 13 00.



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SECTION 01 21 00

ALLOWANCES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes administrative and procedural requirements governing allowances.
 - 1. Certain items are specified in the Contract Documents by allowances. Allowances have been established in lieu of additional requirements and to defer selection of actual materials and equipment to a later date when direction will be provided to the Contractor. If necessary, additional requirements will be issued by Change Order.
- B. Types of allowances include the following:
 - 1. Lump-sum allowances.
 - 2. Unit-cost allowances.
 - 3. Quantity allowances.
 - 4. Contingency allowances.
 - 5. Testing and inspecting allowances.

1.3 SELECTION AND PURCHASE

- A. At the earliest practical date after award of the Contract, advise Architect of the date when final selection and purchase of each product or system described by an allowance must be completed to avoid delaying the Work.
- B. At Architect's request, obtain proposals for each allowance for use in making final selections. Include recommendations that are relevant to performing the Work.
- C. Purchase products and systems selected by Architect from the designated supplier.

1.4 SUBMITTALS

- A. Submit proposals for purchase of products or systems included in allowances, in the form specified for Change Orders.
- B. Submit invoices or delivery slips to show actual quantities of materials delivered to the site for use in fulfillment of each allowance.
- C. Submit time sheets and other documentation to show labor time and cost for installation of allowance items that include installation as part of the allowance.
- D. Coordinate and process submittals for allowance items in same manner as for other portions of the Work.

1.5 COORDINATION

- A. Coordinate allowance items with other portions of the Work. Furnish templates as required to coordinate installation.

1.6 LUMP-SUM UNIT-COST AND QUANTITY ALLOWANCES

- A. Allowance shall include cost to Contractor of specific products and materials ordered by Owner or selected by Architect under allowance and shall include [taxes,] freight, and delivery to Project site.
- B. Unless otherwise indicated, Contractor's costs for receiving and handling at Project site, labor, installation, overhead and profit, and similar costs related to products and materials [ordered by

Owner] [selected by Architect] under allowance shall be included as part of the Contract Sum and not part of the allowance.

- C. Unused Materials: Return unused materials purchased under an allowance to manufacturer or supplier for credit to Owner, after installation has been completed and accepted.
 - 1. If requested by Architect, retain and prepare unused material for storage by Owner. Deliver unused material to Owner's storage space as directed.

1.7 ADJUSTMENT OF ALLOWANCES

- A. Allowance Adjustment: To adjust allowance amounts, prepare a Change Order proposal based on the difference between purchase amount and the allowance, multiplied by final measurement of work-in-place
- B. Where applicable. If applicable, include reasonable allowances for cutting losses, tolerances, mixing wastes, normal product imperfections, and similar margins.
 - 1. Include installation costs in purchase amount only where indicated as part of the allowance.
 - 2. If requested, prepare explanation and documentation to substantiate distribution of overhead costs and other margins claimed.
 - 3. Submit substantiation of a change in scope of work, if any, claimed in Change Orders related to unit-cost allowances.
 - 4. Owner reserves the right to establish the quantity of work-in-place by independent quantity survey, measure, or count.
- C. Submit claims for increased costs because of a change in scope or nature of the allowance described in the Contract Documents, whether for the purchase order amount or Contractor's handling, labor, installation, overhead, and profit.
 - 1. Do not include Contractor's or subcontractor's indirect expense in the Change Order cost amount unless it is clearly shown that the nature or extent of work has changed from what could have been foreseen from information in the Contract Documents.
 - 2. No change to Contractor's indirect expense is permitted for selection of higher- or lower-priced materials or systems of the same scope and nature as originally indicated.

PART 2 - - PRODUCTS (NOT USED)

PART 3 - - EXECUTION

3.1 EXAMINATION

- A. Examine products covered by an allowance promptly on delivery for damage or defects. Return damaged or defective products to manufacturer for replacement.

3.2 PREPARATION

- A. Coordinate materials and their installation for each allowance with related materials and installations to ensure that each allowance item is completely integrated and interfaced with related work.

3.3 SCHEDULE OF ALLOWANCES

- A. Allowance No. [Insert number]: Include the sum of [Insert dollar amount]
 - 1. This allowance includes material and installation cost for

END OF SECTION

SECTION 01 23 00

ALTERNATES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes administrative and procedural requirements for alternates.

1.3 DEFINITIONS

- A. Alternate: An amount proposed by bidders and stated on the Bid Form for certain work defined in the Bidding Requirements that may be added to or deducted from the base bid amount if Owner decides to accept a corresponding change either in the amount of construction to be completed or in the products, materials, equipment, systems, or installation methods described in the Contract Documents.
 - 1. Alternates described in this Section are part of the Work only if enumerated in the Agreement.
 - 2. The cost or credit for each alternate is the net addition to or deduction from the Contract Sum to incorporate alternate into the Work. No other adjustments are made to the Contract Sum.

1.4 PROCEDURES

- A. Include as part of each alternate, miscellaneous devices, accessory objects, and similar items incidental to or required for a complete installation whether or not indicated as part of alternate.
- B. Notification: Immediately following award of the Contract, notify each party involved, in writing, of the status of each alternate. Indicate if alternates have been accepted, rejected, or deferred for later consideration. Include a complete description of negotiated modifications to alternates.
- C. Execute accepted alternates under the same conditions as other work of the Contract.
- D. Schedule: A schedule of alternates is included at the end of this Section. Specification Sections referenced in schedule contain requirements for materials necessary to achieve the work described under each alternate.

1.5 ALTERNATES

- A. Alternate 1: Removable Wall Panel System at Existing storefront Wall Chase
- B. Alternate 3: Cleanroom / Hygiene Room Window at AFB/BSL-3 Lab.
- C. Alternate 4: AquaTap RO Water System

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 25 00

SUBSTITUTION PROCEDURES

PART 1 - GENERAL

1.1 QUALITY ASSURANCE

- A. Contract is based upon products and standards established in Contract Documents without consideration of proposed substitutions.
- B. Products specified define standard of quality, type, function, dimension, appearance and performance required.
- C. Substitution Proposals: Permitted for specified products, except where specified otherwise.
- D. Do not substitute products unless substitution has been accepted and approved in writing by Owner.
- E. Substitution procedures for materials and products on LEED projects must comply with sustainable characteristics of originally specified product and be documented in accordance with Sections 016000 and 018113.

1.2 TIME OF SUBSTITUTION REQUESTS

- A. Within 60 days after award of Contract, Architect will consider formal substitution requests from Contractor only. Use attached Substitution Request Form. No requests for substitutions will be accepted after that time without Owner's authorization and with following stipulation.
- B. Architect will record time required for evaluating substitutions proposed by Contractor after receipt of bids, and for making changes in the Contract Documents. Whether or not Architect accepts Contractor proposed substitution, Contractor shall reimburse Owner for charges of Architect and Architect's consultants for evaluating each proposed substitution.
- C. After receipt of bid, three lowest bidders will be requested to submit supporting data for each proposed substitution. Submit supporting data to Architect within seven days of receipt of request; comply with specified substitution procedures and use attached Substitution Request Form.
- D. Acceptance or rejection of proposed substitutions will be determined prior to award of Contract. Accepted substitutions will be identified in Contract.
- E. No additional substitutions will be considered after this initial process unless a substitution is required due to specified product being removed from or unavailable in market place.

1.3 SUBSTITUTION PROCEDURES (SUPPLEMENT AS INDICATED IN OWNER'S SECTION 01 31 00)

- A. Limit each request to one proposed substitution.
- B. Submit substitution requests on attached form complete with attachments necessary to fully document proposed substitution. Submit in number of copies required for Contractor's use and distribution, plus one copy to be retained by Architect and one copy to be retained by Architect.
 - 1. Copy of required Substitution Request Form is bound after last page of this Section. Make additional copies or request original copy from Architect.
- C. Submit substitution requests on attached form complete with sustainability assessment and attachments necessary to fully document proposed substitution. Submit in number of copies required for Contractor's use and distribution, plus one copy to be retained by Architect ~[and one copy to be retained by Architect.

1. Copy of required Substitution Request and Sustainability Assessment Form is bound after last page of this Section. Make additional copies or request original copy from Architect.
- D. Document each request with supporting data substantiating compliance of proposed substitution with Contract Documents, including:
1. Manufacturer's name and address, product, trade name, model, or catalog number, performance and test data, and reference standards.
 2. Itemized point-by-point comparison of proposed substitution with specified product, listing variations in quality, performance and other pertinent characteristics.
 3. Reference to article and paragraph numbers in Specification section.
 4. Cost data comparing proposed substitution with specified product and amount of net change to Contract Sum.
 5. Changes required in other Work.
 6. Availability of maintenance service and source of replacement parts, as applicable.
 7. Certified test data to show compliance with performance characteristics specified.
 8. Samples, when applicable or requested.
 9. Other information as necessary to assist Architect's evaluation.
- E. A request for substitution constitutes representation that Contractor:
1. Has investigated proposed product and determined that it is equal or superior in all respects to specified product.
 2. Will provide identical warranty as required for specified product.
 3. Will coordinate installation and make changes to other Work which may be required.
 4. Waives claims for additional costs or time extension which may subsequently become apparent.
 5. Certifies that proposed product will not affect or delay Construction Progress Schedule.
 6. Will pay for changes to building design, including architectural or Architecting design, detailing, and construction costs caused by the requested substitution. F. Substitutions will not be considered when:
 7. Indicated or implied on Shop Drawings or Product Data submittals without formal request submitted in accord with this Section. Substitution Request Forms may be submitted at the same time as Product Data and Shop Drawing submittals, but time established for Architect review of submittals will not commence until review of substitution has been completed. Indicate on Submittal Transmittal Form as part of Work of Section 01 33 00.
 8. Submittal for substitution request has not been reviewed and approved by Contractor.
 9. Acceptance will require substantial revision of Contract Documents or other items of the Work.
 10. Submittal for substitution request does not include point-by-point comparison of proposed substitution with specified product.

1.4 ARCHITECT'S AND OWNER'S REVIEW

- A. Architect will review requests for proposed substitutions and make recommendations to Owner on Substitution Request Form or Substitution Request and Sustainability Assessment Form with reasonable promptness.
- B. Considerations for acceptance will be based on conformance with Contract Documents, including following as applicable:
1. Physical dimension and clearance requirements to satisfy space limitations.
 2. Static and dynamic weight limitations; structural properties.
 3. Audible noise levels.
 4. Vibration generation.

5. Interchangeability of parts or components.
 6. Accessibility for maintenance to allow possible removal or replacement.
 7. Design.
 8. Colors, textures, patterns, and finishes.
 9. Compatibility with other materials, products, assemblies, and components.
- C. Owner's decision to approve or reject requested substitution will be indicated on Substitution Request Form or ~Substitution Request and Sustainability Assessment Form. Approval of substitution not valid without Owner's signature.
- D. Rejection of proposed substitution by Owner requires use of specified product.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

ATTACHMENT:

SUBSTITUTION REQUEST FORM, 2 PAGES.

SECTION 01 26 00

CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Related Sections:

1. Section 01 21 00 - Allowances.
2. Section 01 25 00 – Substitution Procedures.
3. Section 01 33 00 - Submittal Procedures.
4. Section 01 60 00 - Product Requirements.
5. Section 01 77 00 - Closeout Procedures.

1.2 SUBMITTALS

A. General:

1. Submit names of individuals authorized to receive Contract modification documents.
2. Submit names of individuals responsible for informing Contractor's employees and affected subcontractors of Contract clarifications and modifications.

1.3 CHANGE PROCEDURES (SUPPLEMENT AS INDICATED IN OWNER'S SECTION 01 31 00)

A. General: Change procedures are written instructions issued after execution of Contract Agreement.

B. Minor Changes in the Work:

1. Form: RTKL Form 01 26 00A, Architect's Supplemental Instructions.
2. Description: Written instructions, clarifications, or interpretations of Contract Documents not involving adjustment to Contract Sum or Contract Time. Instructions or interpretations are binding to Owner and Contractor.
3. Procedure: Document is prepared and signed by Architect and distributed to Owner and Contractor. Architect's Supplemental Instructions are effective upon receipt.

C. Proposal Request:

1. Form: RTKL Form 01 26 00C, Proposal Request.
 - a. Description: Written proposed change of Work within Contract scope consisting of additions, deletions, and other revisions. Proposal Request is for information only and does not authorize changes in Contract Sum or Contract Time. Contractor evaluates proposal for pricing and scheduling impact.
2. Procedure:
 - a. Document: Prepared and signed by Architect. Copies are sent to Owner and Contractor.
 - b. Contractor: Review Proposal Request and submit Change Order Request and Proposal Worksheet Detail and Summary Forms with proposed changes in Contract Sum and Contract Time.
 - c. Prepare and submit Change Order Request and Proposal Worksheet Detail and Summary Forms Architect within 20 days of Proposal Request receipt. Proposed Contract Sum and Contract Time changes quoted by Contractor shall remain valid for 30 days from receipt by Architect.

D. Change Order Request:

1. Attached Forms:
 - a. Change Order Request: Describes and summarizes Contractor's proposed changes. Indicates changes in Contract Sum and Contract Time.
 - b. Proposal Worksheet Summary: Summarizes labor, materials, overhead and profit, bonds, insurance, and tax of proposed Contract additions and deductions.
 - c. Proposal Worksheet Detail: Summarizes labor and material costs of each subcontractor involved in proposed change.
 2. Description: Written proposed change of Work consisting of additions, deletions, and other revisions. Submit Change Order Request to Architect for conditions which require Contract Document modifications. Include proposed changes in Contract Sum and Contract Time.
 3. Procedure:
 - a. Proposed changes are documented by Contractor on Change Order Request, Proposal Worksheet Summary, and Proposal Worksheet Detail forms. Documents include description of proposed changes and summary of changes in Contract Sum and Contract Time are prepared and signed by Contractor. Send copies to Owner and Architect.
 - b. Comply with Section 01 60 00 - Product Requirements for proposed changes in Work that includes products or systems not contained in Contract Documents.
 - c. Architect and Owner will review Change Order Request and evaluate proposed changes. Architect and Owner may accept or reject Change Order Request. Upon acceptance Architect will prepare Change Order to document Contract change.
- E. Change Order:
1. Form: As approved by Owner.
 2. Description: Written change of Work within Contract scope consisting of additions, deletions, changes caused by approved substitutions, and other revisions, including proposed basis for adjustment to Contract Sum and Contract Time. Change Orders: Signed by Owner, Contractor, and Architect. Owner's signature authorizes change.
 3. Procedure: Document prepared and signed by Architect; sent to Contractor for acceptance and signature; approved and signed by Owner; distributed to Architect and Contractor. Contractor: Perform changes upon receipt.
- F. Construction Change Directive:
1. Form: RTKL Form 01 26 00G, Construction Change Directive.
 2. Description: Written change of Work within Contract scope consisting of additions, deletions, and other revisions, including a proposed basis for adjustment to Contract Sum and Contract Time. Document: Used in absence of agreement on terms of Change Orders.
 3. Procedure:
 - a. Document: Prepared and signed by Architect and Owner. Contractor shall perform changes upon receipt.
 - b. Adjustments to Contract Sum Should be One of Following:
 - 1) Lump sum.
 - 2) Unit price.
 - 3) Mutually accepted method.
 - 4) As provided in AIA Document A201 Subparagraph 7.3.6; maintain detailed records on time and material basis of Construction Change Directive Work.
 - c. Architect will determine proposed method, time, and amount of Contract adjustment based on reasonable expenditures, and allowance for overhead, profit, and time.

- d. Contractor's signing of Construction Change Directive acknowledges agreement with proposed method for adjusting Contract Sum and Contract Time and is recorded as Change Order.
- e. Contractor disagreement or no response to proposed method for adjusting Contract Sum or Contract Time does not relieve Contractor from responsibility to perform Work.
- f. Payment for Construction Change Directives will be made in accordance with AIA Document A201 Subparagraph 9.3.1.1. subcontractor involved in proposed change.
 - 1) Mutually accepted fixed or percentage fee.

PART 2 - PRODUCTS – (NOT USED)

PART 3 - EXECUTION – (NOT USED)

END OF SECTION

CHANGE ORDER REQUEST (PROPOSAL)

Project: _____

COR Number:

From (Contractor):

To: _____

Date:

A/E Project Number:

RE: _____

Contract For:

This Change Order Request (COR) contains an itemized quotation for changes in the Contract Sum and/or Time in respond to proposed modifications to the Contract Documents base on Proposal Request Number

—

Description of Proposed Change:

Attachment Reason For Change:

Does Proposed Change involve a change in Contract Sum or Contract Time? ☐ Yes ☐ No

If Yes: Proposed Change in Contract Sum

_____ Proposed Change in Contract Time

Attached Pages: _____
Worksheet
Summary
Proposed
Worksheet Detail(s)

Signed by: _____

☐ Attached is supporting information from ☐ Subcontractor ☐ Supplier ☐ _____ ☐ _____

Copies: ☐ Owner ☐ Contractor ☐ Consultants ☐ Field ☐ _____ ☐ _____

PROPOSAL WORKSHEET SUMMARY

Project: _____

COR Number:

From (Contractor):

To: _____

Date:

Proposal Request Number: _____

A/E Project Number:

UT Southwestern Medical Center, New University Hospital

[SectionTitle]

UT Southwestern Medical Center, New University Hospital [SectionTitle]

RTKL Proj 10-09063.00/ OFPC Proj 303-366

[Section#] -

[Page#]

Final Design Development Package -

September 02, 2010

Complete and attach Proposal Worksheet Detail for each Subcontractor. Enter Work Sheet Information below.

SUBCONTRACTORS' ADDITIONS:

	Sheet	Description	Material	Labor	Sub-Total
1					
2					
3					
4					
5					
Sub-Total					

SUBCONTRACTORS' DEDUCTS:

	Sheet	Description	Material	Labor	Sub-Total
1					
2					

3					
4					
5					
Sub-Total					

Subcontractors' Net:

Subcontractors'

OH&P:

Subcontractors' Bond:

Subcontractors' Total:

General Contractor OH&P:

General Contractor Bond:

Insurance:

Tax:

WORKSHEET TOTAL:

PROPOSAL WORKSHEET DETAIL

Project Name: _____ COR Number: _____
 Project Number: _____ Detail Sheet Number: _____
 Subcontractor Name: _____ Date: _____
 Phone Number/Contact Person: _____ DO NOT MARK IN SHADED AREAS.

ADDITIONS

			UNIT PRICES		SUB-TOTALS		
Ref. No.	Item Description	Quantity	Materials	Labor	Materials	Labor	TOTAL
1							
2							
3							
4							
Sub-Total (Enter this number on Work Sheet Summary.)							

DEDUCTIONS

			UNIT PRICES		SUB-TOTALS		
Ref. No.	Item Description	Quantity	Materials	Labor	Materials	Labor	TOTAL
1							
2							
3							
4							
Sub-Total (Enter this number on Work Sheet Summary.)							

SECTION 01 31 00

PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
 - 1. General project coordination procedures.
 - 2. Coordination Drawings.
 - 3. Administrative and supervisory personnel.
 - 4. Conservation.

1.3 COORDINATION

- A. Coordination: Coordinate construction operations included in various Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, that depend on each other for proper installation, connection, and operation.
 - 1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
 - 2. Coordinate installation of different components with other contractors to ensure maximum accessibility for required maintenance, service, and repair.
 - 3. Make adequate provisions to accommodate items scheduled for later installation.
- B. If necessary, prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
 - 1. Prepare similar memoranda for Owner and separate contractors if coordination of their Work is required.
- C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:
 - 1. Preparation of Contractor's Construction Schedule.
 - 2. Preparation of the Schedule of Values.
 - 3. Installation and removal of temporary facilities and controls.
 - 4. Delivery and processing of submittals.
 - 5. Progress meetings.
 - 6. Preinstallation conferences.
 - 7. Project closeout activities.
- D. Conservation: Coordinate construction activities to ensure that operations are carried out with consideration given to conservation of energy, water, and materials.
 - 1. Salvage materials and equipment involved in performance of, but not actually incorporated into, the Work.

1.4 SUBMITTALS

A. Coordination Drawings:

1. Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.
 - a. Indicate relationship of components shown on separate Shop Drawings.
 - b. Indicate required installation sequences.
 - c. Refer to Division 15 and Division 16 for specific Coordination Drawing requirements for mechanical and electrical installations.

1.5 ADMINISTRATIVE AND SUPERVISORY PERSONNEL

- A. General: In addition to Project superintendent, provide other administrative and supervisory personnel as required for proper performance of the Work.

1.6 PROJECT MEETINGS

- A. General: Schedule and conduct meetings and conferences at Project site, unless otherwise indicated.

1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date and time of each meeting. Notify Owner and Architect of scheduled meeting dates and times.
2. Agenda: Prepare the meeting agenda. Distribute the agenda to all invited attendees.
3. Minutes: Record significant discussions and agreements achieved. Distribute the meeting minutes to everyone concerned, including Owner and Architect, within 3 days of the meeting.

- B. Preconstruction Conference: Schedule a preconstruction conference before starting construction, at a time convenient to Owner and Architect, but no later 15 days after execution of the Agreement. Hold the conference at Project site or another convenient location. Conduct the meeting to review responsibilities and personnel assignments.

1. Attendees: Authorized representatives of Owner, Architect, and their consultants; Contractor and its superintendent; major subcontractors; manufacturers; suppliers; and other concerned parties shall attend the conference. All participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Discuss items of significance that could affect progress, including the following:
 - a. Tentative construction schedule.
 - b. Phasing.
 - c. Critical work sequencing.
 - d. Designation of responsible personnel.
 - e. Procedures for processing field decisions and Change Orders.
 - f. Procedures for processing Applications for Payment.
 - g. Distribution of the Contract Documents.
 - h. Submittal procedures.
 - i. Preparation of Record Documents.
 - j. Use of the premises.
 - k. Responsibility for temporary facilities and controls.
 - l. Parking availability.
 - m. Office, work, and storage areas.
 - n. Equipment deliveries and priorities.
 - o. First aid.
 - p. Security.

- q. Progress cleaning.
 - r. Working hours.
- C. Preinstallation Conferences: Conduct a preinstallation conference at Project site before each construction activity that requires coordination with other construction.
- 1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise Architect of scheduled meeting dates.
 - 2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:
 - a. Contract Documents.
 - b. Options.
 - c. Related Change Orders.
 - d. Submittals.
 - e. Review of mockups.
 - f. Possible conflicts.
 - g. Compatibility problems.
 - h. Time schedules.
 - i. Weather limitations.
 - j. Manufacturer's written recommendations.
 - k. Warranty requirements.
 - l. Compatibility of materials.
 - m. Acceptability of substrates.
 - n. Temporary facilities and controls.
 - o. Space and access limitations.
 - p. Regulations of authorities having jurisdiction.
 - q. Testing and inspecting requirements.
 - r. Required performance results.
 - s. Protection of construction and personnel.
 - 3. Record significant conference discussions, agreements, and disagreements.
 - 4. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.
- D. Progress Meetings: Conduct progress meetings at weekly intervals. Coordinate dates of meetings with preparation of payment requests.
- 1. Attendees: In addition to representatives of Owner and Architect, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
 - a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Contractor's Construction Schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.

- b. Review present and future needs of each entity present, including the following:
 - 1) Interface requirements.
 - 2) Sequence of operations.
 - 3) Status of submittals.
 - 4) Deliveries.
 - 5) Off-site fabrication.
 - 6) Access.
 - 7) Site utilization.
 - 8) Temporary facilities and controls.
 - 9) Work hours.
 - 10) Progress cleaning.
 - 11) Change Orders.
 - 12) Documentation of information for payment requests.
- 3. Reporting: Distribute minutes of the meeting to each party present and to parties who should have been present. Include a brief summary, in narrative form, of progress since the previous meeting and report.
 - a. Schedule Updating: Revise Contractor's Construction Schedule after each progress meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with the report of each meeting.

PART 2 - PART 2 - PRODUCTS (NOT USED)

PART 3 - PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 31 13

PROJECT COORDINATION

PART 1 - GENERAL

1.1 PROJECT MANAGEMENT AND COORDINATION – ENVIRONMENTAL

- A. Contractors Environmental Manager: Designate an on-site party responsible for overseeing the environmental goals for the project and implementing procedures for the environmental protection.
 - 1. Qualification: Minimum 5 years construction experience on projects of similar size and scope; minimum 2 years experience with Environmental Management Systems (EMSs) such as ISO 14001; familiarity with environmental regulations applicable to construction operations.
 - 2. Responsibilities: Responsibilities shall include.
 - a. Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.
 - b. Implementation of the Waste Management Plan.
 - c. Implementation of the IAQ Management Plan.
 - d. Implementation of the Environmental Protection Plan.
 - e. Training for Contractor personnel in accordance with their position requirements.
 - f. Monitoring and documentation of environmental procedures.
- B. Perform project quality control in accordance with requirements specified in Related Sections, including:
 - 1. Section 01 74 19 – Construction Waste Management and Disposal.
 - 2. Section 01 81 13 – Sustainable Design Requirements.
 - 3. Section 01 81 19 – Construction Indoor Air Quality (IAQ) Management.
- C. Contractor's Environmental Training program: Contractor shall provide environmental training for workers performing work on the project site. Training shall include the following:
 - 1. Overview of environmental issues related to the building industry.
 - 2. Overview of environmental issues related to the Project.
 - 3. Review of site specific procedures and management plans:
 - a. Section 01 74 19 – Construction Waste Management and Disposal.
 - b. Section 01 81 13 – Sustainable Design Requirements.
 - c. Section 01 81 19 – Construction Indoor Air Quality (IAQ) Management.

1.2 GENERAL COORDINATION PROVISIONS

- A. Carefully study and compare Contract Documents before proceeding with fabrication and installation of Work. Promptly advise Architect of any error, inconsistency, omission, or apparent discrepancy discovered.
- B. Requests for Interpretation (RFI): Allot time in construction scheduling for liaison with Architect; establish procedures for handling queries and clarifications. Use "Request for Interpretation" form for requesting information. Copy of form is bound after last page of this Section. Remove form for making additional copies or request an original copy from Architect. Limit each RFI to a single issue or a group of related issues.
 - 1. If Architect is able to respond to RFI by making specific reference to Drawing sheet or Specification Section, Contractor shall reimburse Owner for charges of Architect and Architect's Consultants for performing review services for Contractor.

2. If Contractor disagrees with Architect's response to Contractor's RFI, Contractor shall notify Architect within seven days of receipt of response. Notification shall acknowledge that Contractor has generated coordination drawings regarding specific conditions and has not been able to resolve issue. Lack of such notification and certification shall be understood to mean that Contractor agrees with response.
 3. Allow minimum 7-working days' response time from receipt of each RFI. RFI's shall include a workable no-cost proposed solution from Contractor. If more than 10 RFIs are received within 4 working day period, Architect's response time will be extended as necessary for professional response; order of responses will be based on priority established by Architect after consultation with Owner and Contractor.
 4. See Section 01 33 00 for additional information regarding review and response to RFI's.
- C. Coordination Meetings: In addition to meetings specified in Section 01 31 19, hold coordination meetings and conferences with personnel and subcontractors to ensure coordination of Work.
- D. Coordination: Coordinate scheduling, submittals, and Work of various Specification sections to avoid conflicts and ensure efficient and orderly sequence of installation of interdependent construction elements.
1. Coordinate Work of various Specification sections having interdependent responsibilities for installation, connection, and operation.
 2. Verify that characteristics of operating equipment are compatible with building utilities and services.
 3. Prepare coordination drawings as required. See requirements elsewhere in this Section.
 4. Conduct pre-installation conferences as required. See Section 01 31 19.
- E. Except as otherwise indicated, conceal pipes, ducts, conduit and wiring in construction. Coordinate locations of fixtures and outlets with finish elements.
- F. Make provision to accommodate items scheduled for later installation.

1.3 COORDINATION DRAWINGS AND LAYOUTS

- A. Coordinate with BIM Execution Plan. Refer to attachment to this section.
- B. Coordination Drawings: Clearly coordinate portions of Work requiring overlap of trades and highly concentrated or congested areas. Include discussion of coordination drawings at preinstallation conferences. See Section 01 33 00.
1. Coordination Drawings: Show relationship and integration of different construction elements that require coordination during fabrication or installation to fit in space provided or to function as intended.
 2. Prior to fabricating or installing work, prepare composite coordination drawings at appropriate scale; detail major elements, components, and systems of architectural, structural, mechanical, and electrical such as equipment, components, and materials in relationship with each other, installations, and building components. Include dimensions.
 3. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are important to efficient flow of Work.
 4. Indicate scheduling, sequencing, movement, and positioning of large equipment into building during construction.
 5. Assembly Penetrations: Prepare drawings as required to indicate penetrations in floors, walls, and ceilings and their relationship to assembly construction, other penetrations and installations. Identify where additional bracing and offsets are required to comply with Contract Documents.
 6. Ceilings: Prepare reflected ceiling plans and other drawings as required to coordinate and integrate installations, air outlets and inlets, light fixtures, communications systems components, sprinklers, other ceiling-mounted devices, components located above suspended ceilings, and suspended ceiling support components.
 7. Show interrelationship of components indicated on separate Shop Drawings.

8. Indicate required installation sequences to minimize cutting and patching.
 9. Make coordination documents available in field office for review by Architect and Owner during entire period of construction.
 - a. Do not submit RFI's which can be solved through prior preparation of coordination drawings.
 10. Prepare coordination drawings required below, in individual Specification sections, and as determined by Contractor to suit project conditions.
- C. Structural Systems: Include, but do not necessarily limit to following:
1. Structural frame showing interface with exterior cladding.
 2. Location of openings in relation to structure.
 3. Show attachments to decking, structural elements, and other systems.
- D. Mechanical Systems: Include, but do not necessarily limit to following:
1. Proposed locations of piping, ductwork, equipment, and materials.
 2. Proposed locations for access panels and doors.
 3. Clearances for installing and maintaining insulation.
 4. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance. Show access locations.
 5. Equipment connections and support details.
 6. Exterior wall and foundation penetrations.
 7. Fire-rated wall and floor penetrations.
 8. Sizes and location of required concrete pads and bases.
 9. Valve stem movement.
- E. Electrical and Communications Systems: Include, but do not necessarily limit to following:
1. Proposed locations of major raceway systems, equipment, and materials.
 2. Clearances for servicing equipment, including space for equipment disassembly required for periodic maintenance. Show access locations.
 3. Exterior wall and foundation penetrations.
 4. Fire-rated wall and floor penetrations.
 5. Equipment connections and support details.
 6. Sizes and location of required concrete pads and bases.
- F. Coordination: Coordinate in field with affected trades for proper relationship to Work based on Project conditions.
1. Notify Architect of conflicts and other coordination issues requiring resolution prior to commencing construction in each affected area.
- G. Submittal: Coordination drawings are Informational Submittals. See Section 01 33 00. Submit coordination drawings to Architect for information only to document proper coordination of all portions of Work and that coordination issues have been identified and resolved prior to commencing construction in each affected area.

1.4 COORDINATION OF SPACE

- A. Coordinate with BIM Execution Plan. Refer to attachment to this section.
- B. General: Coordinate use of Project space and sequence of installation of plumbing, fire protection, mechanical and electrical Work. Follow routings shown for pipes, ducts, and conduits as closely as practicable, with due allowance for available physical space; make runs parallel with building lines. Utilize space efficiency to maximize accessibility for other installations, for maintenance, and for repairs.

1. Layout of plumbing, fire protection, mechanical, electrical, and communications systems, equipment, fixtures, piping, ductwork, conduit, specialty items, and accessories indicated on Drawings are diagrammatic. Variations in alignment, elevation, and details required to avoid interferences and satisfy architectural and structural limitations are not necessarily indicated.
 - a. Prior to installation of material and equipment, review and coordinate Work of all Drawings to establish exact space conditions.
 - b. Prepare coordination drawings where required to coordinate Work. See requirements elsewhere in this Section.
 - c. Conduct pre-installation conferences where required to coordinate Work. See Section 01 31 19.
 - d. Where available space is inadequate or where reasonable modifications are not possible, request information from Architect before proceeding.
 2. Coordinate installation to prevent conflicts and cooperate in making, without extra charge, reasonable modifications in layout as needed.
 3. Provide clear access to control points, valves, strainers, control devices, and specialty items of every nature related to such systems and equipment to obtain maximum head room. Provide adequate clearances as necessary for operation and maintenance.
- C. Order of Priority for Construction Space:
1. Drainlines.
 2. Ductwork.
 3. Fire protection piping.
 4. Other piping.
 5. Conduit.

1.5 COORDINATION OF CONTRACT CLOSEOUT

- A. Coordinate completion and clean up of Work in preparation for Substantial Completion and for portions of Work designated for partial occupancy as indicated in Section 01 10 00 - Summary.
- B. To minimize disruption of Owner's activities after occupancy of premises, coordinate access to site by various trades for correction of defective Work and for correction of Work not in accordance with Contract Documents.

PART 2 - PRODUCTS – (NOT USED)

PART 3 - EXECUTION – (NOT USED)

END OF SECTION

REQUEST FOR INTERPRETATION

RFI Number: Project:

Contractor:

Signed: Date:

Specification Section	Paragraph	Drawing	Detail
-----------------------	-----------	---------	--------

RE:

Contractor's Proposed Solution:

Architect/Engineer:

Signed: Date:

Response:

SECTION 01 31 19

PROJECT MEETINGS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

- A. Schedule meetings and conferences throughout progress of Work; each session scheduled, administered, and presided by entity indicated. Requirements for meetings and conferences include:
 - 1. Prepare agenda for each conference and meeting.
 - 2. Distribute written notice to participants seven days in advance of scheduled date.
 - 3. Make physical arrangements.
 - 4. Record minutes and attendees; include significant proceedings and decisions.
 - 5. Reproduce and distribute copies of minutes within five days after each meeting.
 - 6. Distribute one copy of minutes to each participant and to entities affected by decisions made at meeting.
 - 7. Distribute minutes to Architect, Owner, and all others present.
 - 8. Maintain in field office one copy of agenda and minutes for each conference and meeting.
- B. Representatives attending meetings shall be qualified and authorized to act on behalf of entity each represents.
 - 1. Architect and professional consultants may attend meetings to ascertain that Work is consistent with Contract Documents.
 - 2. Owner may be present at meetings and may propose agenda items.

1.2 PRE-CONSTRUCTION CONFERENCES

- A. Schedule pre-construction conferences no later than seven days prior to commencement of Work. Convene at Project site.
- B. Attendees:
 - 1. Architect and professional consultants; Architect presides over meeting and is responsible for minutes.
 - 2. Owner.
 - 3. Contractor.
 - 4. Major subcontractors.
 - 5. Others as appropriate.
- C. Minimum Agenda:
 - 1. Administrative and Procedural Issues:
 - a. Designation of key personnel.
 - b. Review and clarify responsibilities of parties to contract.
 - c. Communications procedures.
 - d. Review of proposed subcontractors, materials, equipment, and products.
 - e. Application for payment procedures; schedule of values, proposal requests, change orders.
 - f. Critical work sequencing; long lead time items.
 - g. Submittal and construction progress schedules.
 - h. Submittal requirements; complete, correct, and timely submittals; scheduled dates.
 - i. Procedures for submitting product data, shop drawings, samples, and other submittals.

- j. Product options and substitutions procedures.
- k. Procedures for requests for interpretations (RFI), minor changes, field decisions, construction change directives, proposal requests, change orders, and filing claims.
- l. Procedures for testing and inspection, including timely notification.
- m. Responsibilities and limitations of authority of geotechnical engineer and testing laboratories; distribution of reports.
- n. Procedures for maintaining Project Record Documents.
- o. Schedule for progress meetings.
- 2. Site mobilization and utilization:
 - a. Use of premises; office and storage areas; Owner's requirements.
 - b. Temporary utilities and services.
 - c. Environmental requirements and procedures.
 - d. Solid Waste Management Plan.
 - e. IAQ Management Plan.
 - f. Procedures for noise and acoustics management.
 - g. Environmental Management Plan.
 - h. Environmental Regulatory Requirements.

1.3 PROGRESS MEETINGS (SUPPLEMENT AS INDICATED IN OWNER'S SECTION 01 31 00)

- A. Schedule periodic meetings as necessary by progress of Work; day, location, and time to be determined. Convene at Project site.
- B. Attendees:
 - 1. Contractor; presides over meeting and is responsible for minutes.
 - 2. Subcontractors as appropriate.
 - 3. Owner, Architect, and professional consultants may attend as appropriate.
 - 4. Others as appropriate to agenda.
- C. Minimum Agenda:
 - 1. Approval of minutes of previous meeting.
 - 2. Work Progress Since Previous Meeting:
 - a. Current activities.
 - b. Critical activities.
 - c. Deviations from schedule.
 - 3. Field observations, problems, conflicts, and decisions.
 - 4. Deficiencies:
 - a. Identification of items.
 - b. Status of correction.
 - 5. Requests for Interpretations (RFIs):
 - a. Status of clarification.
 - b. Status of proposal requests.
 - 6. Changes and Modifications:
 - a. Status of change orders.
 - b. Pending changes.
 - c. Pending claims and disputes.
 - d. Clarification decisions of Architect or Owner.
 - 7. Problems and conflicts which impede planned progress.
 - 8. Review environmental requirements and procedures. Review status of:

- a. Solid Waste Management.
 - b. IAQ Management Plan.
 - c. Environmental Management Plan.
 - d. Commissioning.
9. Construction Progress and Submittal Schedules:
 - a. Off-site fabrication and delivery schedules.
 - b. Effect of proposed changes on construction progress schedule and coordination.
 - c. Submittal schedules, status of submittals, and effect on construction progress schedule.
 - d. Corrective measures to regain projected schedule.
 10. Planned progress during succeeding Work period.
 11. Adequacy of work forces.
 12. Coordination between elements of Work including coordination drawings and pre-construction conferences.
 13. Waste management procedures.
 14. Indoor air quality procedures.
 15. Environmental Coordination:
 - a. Coordinate activities included in various Sections to assure efficient and orderly installation of each component. Coordinate operations included under different Sections that are dependent on each other for proper installation and operation.
 - b. Interior finishes: Schedule construction operations with consideration for indoor air quality in accordance with Section 01 57 20.
 - c. Commissioning: The project will have selected building systems commissioned as specified in Section 01 91 00 – Commissioning. Coordinate prefunctional test and start-up testing with commissioning.
 16. Maintenance of Project Record Documents.
 17. Other business relating to progress of Work.
- D. Meeting Minutes:
1. Include column to indicate who is required to take action and date action is to be completed. Each of these items requiring action will be carried in subsequent minutes of meeting as "old business" until noted as "resolved."
 2. As minimum, separate into following categories:
 - a. Old business.
 - b. New business.
 - c. Work progress.
 - d. Deficiencies.
 - e. RFIs.
 - f. Proposed changes.
 - g. Schedules.
 - h. Submittals.
 - i. Other business, including events to be accomplished by next meeting.

1.4 PRE-INSTALLATION CONFERENCES

- A. Conduct pre-installation conferences as required to coordinate all portions of Work. Prepare complimentary coordination drawings as required. See Section 01 31 13.
 1. Conduct pre-installation conferences required in individual Specification sections.
 2. Schedule pre-installation conferences at Project site no later than seven days prior to commencement of work to be coordinated.
- B. Attendees:

1. Project superintendent; presides over meeting and is responsible for minutes.
2. Subcontractor (installer, applicator, or erector).
3. Material or equipment supplier.
4. Manufacturers' representative.
5. Others directly affecting, or affected by the work.
6. Testing agency (if necessary).
7. Subcontractors as appropriate.
8. Owner, Architect, and professional consultants may attend as appropriate.
9. Others as appropriate to agenda.

C. Minimum Agenda:

1. Access to work and conditions of proper installation.
2. Conditions of installation, such as substrates, existing and surrounding conditions, and environmental conditions.
3. Conditions detrimental to installation.
4. Preparation procedures, including protection of adjacent work.
5. Verify installers' receipt and understanding of installation instructions.
6. Review submittals, installation procedures, and sequence.
7. Indoor air quality plan and procedures.
8. Review coordination with other work including coordination drawings. See Section 01 31 13.
9. Evaluate delivery schedule and Construction Progress Schedule.
10. Observe sample installation.
11. Required protection procedures.

1.5 CLOSEOUT CONFERENCE

A. Schedule Project Closeout conference with sufficient time to prepare for requesting Substantial Completion.

B. Attendees:

1. Contractor; presides over meeting and is responsible for minutes.
2. Major subcontractors.
3. Owner, Architect, and professional consultants may attend as appropriate.
4. Others as appropriate to agenda.

C. Minimum Agenda:

1. Start-up of facilities and systems.
2. Operations and maintenance manuals.
3. Testing, adjusting, and balancing.
4. System demonstration and observation.
5. Operation and maintenance instructions for the owner's personnel.
6. Indoor air quality plan and procedures.
7. Contractor's inspection of work.
8. Contractor's preparation of an initial "punch list."
9. Procedure to request Architect inspection to determine date of substantial completion.
10. Completion time for correcting deficiencies.
11. Inspections by authorities having jurisdiction.
12. Certificate of occupancy and transfer of insurance responsibilities.
13. Partial release of retainage.
14. Preparation for final inspection.

15. Closeout submittals:
 - a. Project Record Documents.
 - b. Operating and maintenance documents.
 - c. Operating and maintenance materials.
 - d. Warranties and bonds.
 - e. Affidavits.
16. Final application for payment.
17. Final cleaning.
18. Contractor's demobilization of site.
19. Maintenance.

PART 2 - PRODUCTS - NOT USED

PART 3 - EXECUTION - NOT USED

END OF SECTION

SECTION 01 31 26
ELECTRONIC COMMUNICATION PROTOCOLS

PART 1 - GENERAL

1.1 SUMMARY

- A. Use Newforma Info Exchange (Newforma) on this Project for collaboration and certain administrative functions.

1.

1.2 PROJECT INFORMATION MANAGEMENT PLATFORM

- A. Newforma Info Exchange is a web-based application furnished by Architect at no cost to Owner and Contractor.
- B. Contractor will have access to projects and modules for which they have permissions.
- C. Architect will manage and administer Newforma Info Exchange.
- D. Enables project team to receive submission of items, review status of documents and supports management of project information.
- E. Allows tracking, corresponding, sharing, viewing and access to shared project information.
- F. Certain documents will be distributed by Architect by means of Newforma.
- G. Newforma Help Guide is available via <http://help.newforma.com>.

1.3 HARDWARE REQUIREMENTS

- A. Device with high-speed internet connection.

1.4 SOFTWARE REQUIREMENTS

- A. Web Browser: All are supported.
- B. Web Browser: Any web browser supported by Newforma.
- C. Additional applications: MS Word, MS Excel, imaging software to open TIFF and JPEG attachments.
- D. Email application and service.

1.5 ACCESSING HDR NEWFORMA INFO EXCHANGE

- A. Once access is granted by HDR, external users are given permission to join the project team. Electronic hosting will be provided by Newforma.
 - 1. Where an individual works on multiple HDR projects, these credentials will apply to all projects similarly.
- B. External user logging in to Newforma Info Exchange:
 - 1. Access Newforma from a web browser at <https://Newforma.hdrinc.com/userweb>.
 - 2. Newforma will prompt to login with Username and Password.
 - 3. Each individual team member will establish and maintain their own credentials.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 MODULES / FUNCTIONS

- A. Verify with HDR's project Construction Contract Administrator which Newforma modules will be made available to Contractor and the document nomenclature which will be used.

B. Contractor will utilize following Newforma modules:

1. Submittals:
 - a. Comply with requirements of Section 01 33 00 – Submittal Procedures.
 - b. Submit physical samples and finish materials.
 - 1) Digital images of physical samples are not acceptable.
 - c. Submit progress photographs in format described in Section [01 32 26][01 32 33].
 - d. Submit other documents electronically:
 - 1) Submit as PDF documents.
 - 2) Name PDF file as indicated in section 01 33 00 – Submittal Procedures.
 - 3) Include Sender ID where applicable to allow for coordination between Architect's and Contractor's systems.
 - 4) Do not submit file in Adobe PDF/A mode.
 - 5) Transmitting submittal files with Newforma Info Exchange:
 - a) Sign into Info Exchange.
 - b) Select assigned project.
 - c) Select Send Submittal.
 - d) Enter information in fields regarding submittal.
 - e) Attach supporting files.
 - f) No further action is required by the contractor until they receive notice of a response. Access to view the status of each submittal is available via Info Exchange.
 - e. When submittals have been reviewed and posted, they will appear in the Response column of the submittal log.
 - f. Contractor will have access to related files in Newforma Info Exchange.
2. Request for Information (RFI):
 - a. Contractor shall create new RFI using Newforma Info Exchange to request information from Architect.
 - b. Architect will access RFI and any attachments, then review and respond in Newforma.
 - c. Contractor will access Newforma Info Exchange to view status, response, and any attachments.
 - d. See Section 01 26 13 - Requests for Information for additional information.
3. Proposal Requests (PR):
 - a. Architect will notify Contractor by email when document has been issued and available in Newforma Info Exchange.
 - b. Contractor will download electronic documents for further processing directly from the email notification or by accessing Newforma Info Exchange.
 - c. See Section 01 26 00 – Contract Modification Procedures for additional information on processing changes.
4. Architect's Supplemental Instructions /Clarification-Interpretation (ASI/C-I):
 - a. Contractor will be notified by email when document(s) are available in Newforma Info Exchange.
 - b. Contractor may download electronic documents for further processing directly from the email notification or by accessing Newforma Info Exchange.
 - c. See Section 01 26 00 – Contract Modification Procedures for additional information on processing changes.
5. Construction Change Directives (CCD):
 - a. Contractor will be notified by email when document(s) are available in Newforma Info Exchange.

- b. See Section 01 26 00 – Contract Modification Procedures for additional information on processing changes.
- 6. Change Proposal Requests (CPRs):
 - a. Contractor will be notified by email when document(s) are available in Newforma Info Exchange.
 - b. Contractor will download electronic documents for further processing directly from the email notification or by accessing Newforma Info Exchange.
- 7. Deliverables:
 - a. Architect will post a reference copy of Contract Documents on Newforma Info Exchange.
 - b. Contractor may access documents posted to Newforma Info Exchange.

END OF SECTION

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes requirements for the submittal schedule and administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.3 DEFINITIONS

- A. Action Submittals: Written and graphic information and physical samples that require
- B. Architect's responsive action. Action submittals are those submittals indicated in individual Specification Sections as action submittals.
- C. Informational Submittals: Written and graphic information and physical samples that do not require Architect's responsive action. Submittals may be rejected for not complying with requirements. Informational submittals are those submittals indicated in individual Specification Sections as informational submittals.
- D. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.
- E. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.4 ACTION SUBMITTALS

- A. Submittal Schedule: Submit a schedule of submittals, arranged in chronological order by dates required by construction schedule. Include time required for review, ordering, manufacturing, fabrication, and delivery when establishing dates. Include additional time required for making corrections or modifications to submittals noted by the Architect and additional time for handling and reviewing submittals required by those corrections.
 - 1. Coordinate submittal schedule with list of subcontracts, the schedule of values, and Contractor's construction schedule.
 - 2. Initial Submittal: Submit concurrently with start-up construction schedule. Include submittals required during the first 60 days of construction. List those submittals required to maintain orderly progress of the Work and those required early because of long lead time for manufacture or fabrication.
 - 3. Final Submittal: Submit concurrently with the first complete submittal of Contractor's construction schedule.
 - a. Submit revised submittal schedule to reflect changes in current status and timing for submittals.
 - 4. Format: Arrange the following information in a tabular format:
 - a. Scheduled date for first submittal.
 - b. Specification Section number and title.
 - c. Submittal category: Action, informational.
 - d. Name of subcontractor.

- e. Description of the Work covered.
- f. Scheduled date for Architect's final release or approval.
- g. Scheduled dates for purchasing.
- h. Scheduled dates for installation.
- i. Activity or event number.

1.5 SUBMITTAL ADMINISTRATIVE REQUIREMENTS

- A. Architect's Digital Data Files: Electronic copies of CAD Drawings of the Contract Drawings will not be provided by Architect for Contractor's use in preparing submittals.
 - 1. Architect will furnish Contractor one set of digital data drawing files of the Contract Drawings for use in preparing Shop Drawings and Project record drawings.
 - a. Architect makes no representations as to the accuracy or completeness of digital data drawing files as they relate to the Contract Drawings.
 - b. Digital Drawing Software Program: The Contract Drawings are available in AutoCAD 2012.
 - c. Contractor shall execute a data licensing agreement in the form of AIA Document C106, Digital Data Licensing Agreement.
 - d. The following plot files will be furnished for each appropriate discipline:
 - 1) Floor plans.
 - 2) Reflected ceiling plans.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 - 1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Submit all submittal items required for each Specification Section concurrently unless partial submittals for portions of the Work are indicated on approved submittal schedule.
 - 3. Submit action submittals and informational submittals required by the same Specification Section as separate packages under separate transmittals.
 - 4. Coordinate transmittal of different types of submittals for related parts of the Work so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- C. Processing Time: Allow time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Architect's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
 - 1. Initial Review: Allow 15 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. Architect will advise Contractor when a submittal being processed must be delayed for coordination.
 - 2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
 - 3. Resubmittal Review: Allow 15 days for review of each resubmittal.
 - 4. Sequential Review: Where sequential review of submittals by Architect's consultants, Owner, or other parties is indicated, allow 21 days for initial review of each submittal.
 - a. **[Insert list of Specification Sections requiring sequential review].**
 - 5. Concurrent Consultant Review: Where the Contract Documents indicate that submittals may be transmitted simultaneously to Architect and to Architect's consultants, allow 15 days for review of each submittal. Submittal will be returned to Architect before being returned to Contractor.

- D. Identification and Information: Place a permanent label or title block on each paper copy submittal item for identification.
1. Indicate name of firm or entity that prepared each submittal on label or title block.
 2. Provide a space approximately 6 by 8 inches on label or beside title block to record Contractor's review and approval markings and action taken by Architect.
 3. Include the following information for processing and recording action taken:
 - a. Project name.
 - b. Date.
 - c. Name of Architect.
 - d. Name of Contractor.
 - e. Name of subcontractor.
 - f. Name of supplier.
 - g. Name of manufacturer.
Submittal number or other unique identifier, including revision identifier. Submittal number shall use Specification Section number followed by a decimal point and then a sequential number (e.g., 061000.01). Resubmittals shall include an alphabetic suffix after another decimal point (e.g., 061000.01.A).
 - h. Number and title of appropriate Specification Section.
 - i. Drawing number and detail references, as appropriate.
 - j. Location(s) where product is to be installed, as appropriate.
 - k. Other necessary identification.
- E. Identification and Information: Identify and incorporate information in each electronic submittal file as follows:
1. Assemble complete submittal package into a single indexed file with links enabling navigation to each item.
 2. Name file with submittal number or other unique identifier, including revision identifier.
 - a. File name shall use project identifier and Specification Section number followed by a decimal point and then a sequential number (e.g., LNHS-061000.01). Resubmittals shall include an alphabetic suffix after another decimal point (e.g., LNHS-061000.01.A).
 3. Provide means for insertion to permanently record Contractor's review and approval markings and action taken by Architect.
 4. Include the following information on an inserted cover sheet:
 - a. Project name.
 - b. Date.
 - c. Name and address of Architect.
 - d. Name of Contractor.
 - e. Name of firm or entity that prepared submittal.
 - f. Name of subcontractor.
 - g. Name of supplier.
 - h. Name of manufacturer.
 - i. Number and title of appropriate Specification Section.
 - j. Drawing number and detail references, as appropriate.
 - k. Location(s) where product is to be installed, as appropriate.
 - l. Related physical samples submitted directly.
 - m. Other necessary identification.
 5. Include the following information as keywords in the electronic file metadata:
 - a. Project name.
 - b. Number and title of appropriate Specification Section.

- c. Manufacturer name.
- d. Product name.
- e. [Insert additional required information].
- F. Options: Identify options requiring selection by the Architect.
- G. Deviations: Identify deviations from the Contract Documents on submittals.
- H. Additional Paper Copies: Unless additional copies are required for final submittal, and unless Architect observes noncompliance with provisions in the Contract Documents, initial submittal may serve as final submittal.
 - 1. Submit one copy of submittal to concurrent reviewer in addition to specified number of copies to Architect.
- I. Transmittal: Assemble each submittal individually and appropriately for transmittal and handling. Transmit each submittal using a transmittal form. Architect will discard submittals received from sources other than Contractor.
 - 1. Transmittal Form: Use AIA Document G810.
 - 2. On an attached separate sheet, prepared on Contractor's letterhead, record relevant information, requests for data, revisions other than those requested by Architect on previous submittals, and deviations from requirements in the Contract Documents, including minor variations and limitations. Include same identification information as related submittal.
- J. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.
 - 1. Note date and content of previous submittal.
 - 2. Note date and content of revision in label or title block and clearly indicate extent of revision.
 - 3. Resubmit submittals until they are marked with approval notation from Architect's action stamp.
- K. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.
- L. Use for Construction: Use only final submittals that are marked with approval notation from Architect's action stamp.

PART 2 - PRODUCTS

2.1 SUBMITTAL PROCEDURES

- A. General Submittal Procedure Requirements: Prepare and submit submittals required by individual Specification Sections. Types of submittals are indicated in individual Specification Sections.
 - 1. Post electronic submittals as PDF electronic files directly to Project Web site specifically established for Project.
 - a. Architect will return annotated file. Annotate and retain one copy of file as an electronic Project record document file.
 - 2. Action Submittals: After Architect's return of approved submittal, create 6 paper copies of each submittal and transmit to Owner for their review and archive purposes.
 - 3. Informational Submittals: Submit two paper copies of each submittal, unless otherwise indicated. Architect will not return copies.
 - 4. Closeout Submittals and Maintenance Material Submittals: Comply with requirements specified in Division 01 Section "Closeout Procedures."
 - 5. Certificates and Certifications Submittals: Provide a statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.

- a. Provide a digital signature with digital certificate on electronically-submitted certificates and certifications where indicated.
 - b. Provide a notarized statement on original paper copy certificates and certifications where indicated.
6. Test and Inspection Reports Submittals: Comply with requirements specified in Division 01 Section "Quality Requirements."
- B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
 1. If information must be specially prepared for submittal because standard published data are not suitable for use, submit as Shop Drawings, not as Product Data.
 2. Mark each copy of each submittal to show which products and options are applicable.
 3. Include the following information, as applicable:
 - a. Manufacturer's catalog cuts.
 - b. Manufacturer's product specifications.
 - c. Standard color charts.
 - d. Statement of compliance with specified referenced standards.
 - e. Testing by recognized testing agency.
 - f. Application of testing agency labels and seals.
 - g. Notation of coordination requirements.
 - h. Availability and delivery time information.
 4. For equipment, include the following in addition to the above, as applicable:
 - a. Wiring diagrams showing factory-installed wiring.
 - b. Printed performance curves.
 - c. Operational range diagrams.
 - d. Clearances required to other construction, if not indicated on accompanying Shop Drawings.
 5. Submit Product Data before or concurrent with Samples.
 6. Submit Product Data in the following format:
 - a. PDF electronic file.
 - b. After Architect's return of approved electronic submittal, create 6 paper copies of each submittal and transmit to Owner for their review and archive purposes.
- C. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data[, unless submittal based upon Architect's digital data drawing files is otherwise permitted].
 1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
 - a. Identification of products.
 - b. Schedules.
 - c. Compliance with specified standards.
 - d. Notation of coordination requirements.
 - e. Notation of dimensions established by field measurement.
 - f. Relationship and attachment to adjoining construction clearly indicated.
 - g. Shop drawings requiring signature and stamped seal of professional engineer: Ensure that seal and signature appear on every drawing submitted, not just on cover sheet. Place Contractor review on every drawing indicating review for compliance.
 2. Sheet Size: Except for templates, patterns, and similar full-size drawings, submit Shop Drawings on sheets at least 8-1/2 by 11 inches but no larger than 30 by 42 inches.
 3. Submit Shop Drawings in the following format:

- a. PDF electronic file.
 - b. After Architect's return of approved electronic submittal, create 6 paper copies of each submittal and transmit to Owner for their review and archive purposes.
- D. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
 1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 2. Identification: Attach label on unexposed side of Samples that includes the following:
 - a. Generic description of Sample.
 - b. Product name and name of manufacturer.
 - c. Sample source.
 - d. Number and title of applicable Specification Section.
 3. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - a. Samples that may be incorporated into the Work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
 - b. Samples not incorporated into the Work, or otherwise designated as Owner's property, are the property of Contractor.
 4. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
 - a. Number of Samples: Submit one full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Architect will return submittal with options selected.
 5. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit sets of Samples. Architect will retain two Sample sets; remainder will be returned. Mark up and retain one returned Sample set as a Project record sample.
 - 1) Submit a single Sample where assembly details, workmanship, fabrication techniques, connections, operation, and other similar characteristics are to be demonstrated.
 - 2) If variation in color, pattern, texture, or other characteristic is inherent in material or product represented by a Sample, submit at least three sets of paired units that show approximate limits of variations.
- E. Product Schedule: As required in individual Specification Sections, prepare a written summary indicating types of products required for the Work and their intended location. Include the following information in tabular form:
 1. Type of product. Include unique identifier for each product indicated in the Contract Documents.
 2. Manufacturer and product name, and model number if applicable.
 3. Number and name of room or space.
 4. Location within room or space.
 5. Submit product schedule in the following format:
 - a. PDF electronic file.

- b. Three paper copies of product schedule or list, unless otherwise indicated. Architect will return two copies.
- F. Apply Contractor's stamp, sign or initial and date certifying that review, verification of products, field dimensions, adjacent construction Work, and coordination of information, is in accordance with requirements of Work and Contract Documents.
- G. Qualifications of Engineer hired by Contractor: Licensed to practice engineering in jurisdiction where Project is located, with minimum of five years experience in design of systems, products, or assemblies specified, when submittals require signature and seal of registered professional engineer.
- H. Subcontract List: Prepare a written summary identifying individuals or firms proposed for each portion of the Work, including those who are to furnish products or equipment fabricated to a special design. Use CSI Form 1.5A. Include the following information in tabular form:
 - 1. Name, address, and telephone number of entity performing subcontract or supplying products.
 - 2. Number and title of related Specification Section(s) covered by subcontract.
 - 3. Drawing number and detail references, as appropriate, covered by subcontract.
 - 4. Submit subcontract list in the following format:
 - a. PDF electronic file.
 - b. Number of Copies: Three paper copies of subcontractor list, unless otherwise indicated. Architect will return two copies.
- I. Coordination Drawings: Comply with requirements specified in Division 01 Section "Project Management and Coordination."
- J. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, contact information of architects and owners, and other information specified.
- K. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification and Procedure Qualification Record on American Welding Society (AWS) forms. Include names of firms and personnel certified.
- L. Installer Certificates: Submit written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- M. Manufacturer Certificates: Submit written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
- N. Product Certificates: Submit written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
- O. Material Certificates: Submit written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.
- P. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
- Q. Product Test Reports: Submit written reports indicating current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- R. Research Reports: Submit written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project. Include the following information:

1. Name of evaluation organization.
 2. Date of evaluation.
 3. Time period when report is in effect.
 4. Product and manufacturers' names.
 5. Description of product.
 6. Test procedures and results.
 7. Limitations of use.
- S. Schedule of Tests and Inspections: Comply with requirements specified in Division 01 Section "Quality Requirements."
- T. Preconstruction Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
- U. Compatibility Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- V. Field Test Reports: Submit reports indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
- W. Design Data: Prepare and submit written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- X. Changed Condition drawings:
1. General: Submit in pdf format.
 2. When specified in individual Sections, submit changed condition drawings in same quantities as for Shop Drawings.
 - a. Where conditions differ from Contract Documents or Shop Drawings and remedial work is necessary, submit drawing showing changes.
 - b. Submit each drawing bearing seal and signature of professional engineer responsible for design.
 - c. Indicate differing condition and required work caused by differing condition.
- Y. Support Reaction Design data:
1. General: Submit in pdf format.
 2. When specified in individual Sections, submit support reaction design data.
 - a. Furnish separate submittal indicating complete description of loads, forces, and moments transferred to "base building" structure or supplemental framing at each point of contact.
 - b. Include secondary forces resulting from connections used.
 - c. Do not submit engineering calculations for support reactions.
 - d. Design Data: Bear seal and signature of professional engineer responsible for design.
- Z. Calculation:
1. General: Submit in pdf format.
 2. When specified in individual Sections, submit calculations.
 - a. Submit engineering calculations for component sizes, deflections, and connections.
 - b. Submit calculations bearing seal and signature of registered professional engineer responsible for design.

3. Where existing conditions deviate from Contract Documents or Shop Drawings, submit calculations for existing condition, including calculations for anticipated corrective action required, and changes to loads transferred to "base building" structure.
4. Calculations: Provided by properly licensed design professional, whose registration seal shall appear on all design drawings, calculations, specifications, certifications, Shop Drawings and other submittals prepared by such design professional.
 - a. Owner and Architect: Entitled to rely on adequacy, accuracy and completeness of design services, calculations, certifications, and approvals performed by design professional.
 - b. Architect's Review of Such Submittals (including Calculations): Only for limited purpose of checking for conformance of design with such criteria stated by such design professional in submittals and with performance criteria and design intent expressed in Contract Documents.

2.2 DELEGATED-DESIGN SERVICES

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
 1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to Architect.
- B. Delegated-Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally-signed PDF electronic file and three paper copies of certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.
 1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

- A. Review for compliance with Contract Documents and approve submittals prior to transmittal to Architect. Contractor: Responsible for:
 1. Submission to and follow up with building code officials.
 2. Determination and verification of materials including manufacturer's catalog numbers.
 3. Determination and verification of field measurements and field construction criteria.
 4. Checking and coordinating information in submittal with requirements of Work and of Contract Documents.
 5. Determination of accuracy and completeness of dimensions and quantities.
 6. Substantiating instructions for installation and performance of equipment or systems.
 7. Confirmation and coordination of dimensions and field conditions at site.
 8. Construction means, techniques, sequences, and procedures.
 9. Safety precautions.
 10. Fabrication processes.
 11. Errors or omissions on submittals.
 12. Coordination and performance of work of all trades.
- B. Action and Informational Submittals: Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to Architect.
- C. Do not fabricate products or begin Work which requires submittals until approved submittals have been received from Architect.

- D. Project Closeout and Maintenance/Material Submittals: Refer to requirements in Division 01 Section "Closeout Procedures."
- E. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.

3.2 ARCHITECT'S ACTION

- A. General: Architect will not review submittals that do not bear Contractor's approval stamp and will return them without action.
- B. Action Submittals: Architect will review each submittal, make marks to indicate corrections or modifications required, and return it. Architect will stamp each submittal with an action stamp and will mark stamp appropriately to indicate action.
- C. Informational Submittals: Architect will review each submittal and will not return it, or will return it if it does not comply with requirements. Architect will forward each submittal to appropriate party.
- D. Partial submittals prepared for a portion of the Work will be reviewed when use of partial submittals has received prior approval from Architect.
- E. Incomplete submittals are not acceptable, will be considered non-responsive, and will be returned without review.
- F. Submittals not required by the Contract Documents may not be reviewed and may be discarded.

END OF SECTION

SECTION 01 35 13

HOSPITAL PROJECT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes environmental control specific to occupied medical care facilities.

1.3 PUBLISHED REGULATIONS

- A. Abide by published hospital regulations and amendments, including those that may be issued during the Contract. Particular attention is called to regulations pertaining to circulation, noise, sanitation, safety, security, and behavior.

1.4 OWNER'S REPRESENTATIVE

- A. Abide by directions of Owner's Representative in matters affecting operation, safety, and security of Hospital, patients, and visitors.
- B. Abide by directions of Hospital Fire Marshal in matters pertaining to fire safety and preventative measures.
 - 1. Immediately implement oral instructions by local Fire Marshal. Confirmation or explanation of oral instructions will be made by Owner by written notice or at next scheduled progress meeting.
- C. Instruct personnel employed by Contractor, including sub-contractors and their employees, to abide by published regulations, and directives of Owner's Representative and Fire Marshal.

1.5 ENVIRONMENTAL CONTROLS

- A. Noise Control: Execute Work as quietly as practicable to avoid unnecessary disturbance of patients.
 - 1. Complaints duly registered by Owner of unacceptable noise levels is cause for implementation of special precautions and methods of operation, as directed by Owner, to reduce noises to acceptable levels.
 - 2. Owner is sole judge of noise level tolerability.
- B. Temporary Partitions: Provide 1 hour fire rated temporary partitions and door openings.

1.6 PERSONNEL IDENTIFICATION

- A. Employees of Contractor and sub-contractors are required to wear Owner approved numbered identification badges while on Hospital premises. Conspicuously fix identification badges to outer garments above elbow level.

1.7 CONSTRUCTION PERSONNEL PARKING

- A. Parking is allowed only in areas designated by Owner. Receive permit or instructions prior to parking on site.

1.8 NORMAL LIMIT OF OPERATIONS

- A. Normal limit of operations are confined within Limits of Work Area as designated on Drawings.
 - 1. Owner will prohibit Hospital employees, patients, and visitors from using these areas.
- B. Allow access to Owner, Architect, and other personnel performing work within limits of operation.

1.9 PERIODIC OPERATIONS OUTSIDE LIMITS OF OPERATION

- A. Use of certain loading berths, passageways, elevators, and other areas, outside of defined limits of operation will be granted on intermittent basis as required and requested in advance. Owner will judge, and approve in advance, proper time, and extent of such use.
- B. Limit requests for use of berths, corridors, elevators, and other spaces to hour-by-hour basis and approved in advance by Owner.
- C. Comply with designated travel paths, staging areas, dumpster locations, and other restricted items indicated on Drawings.

1.10 SCHEDULING

- A. Schedule arrangements for work which will involve interference with normal Hospital functioning, particularly in occupied patient areas, or adjacent areas, five working days in advance with Owner to provide for minimum of disruption and inconvenience.
 - 1. Owner will schedule and approve interferences to minimize disruption to normal Hospital functioning.

1.11 WORKING HOURS AND OVERTIME NOTIFICATION

- A. Notify Owner and receive permission for work outside of normal daytime working hours either within or outside limits of operation.
- B. If in Owner's judgment, any item of work is being unreasonably delayed and interfering with progress of other dependent work through no fault of Owner, overtime work may be ordered at Contractor's expense.
- C. Owner may direct Contractor to perform overtime work not originally required by Contract. When overtime work is requested for convenience, additional payment to Contractor will be made for premium time only.

1.12 TORCH-CUTTING AND WELDING PERMITS

- A. Welding: Neither welding nor arc-welding will be permitted without appropriate permit from Hospital Fire Marshal.
 - 1. Hospital has right to stop work at any time if it is determined that unsafe conditions exist.
 - 2. Correct unsafe conditions as directed by local Fire Marshal and obtain approval of such corrections prior to commencing further work.
 - 3. Provide electrical power directly from panel box for use by arc-welding equipment. Ascertain that electrical panel has capacity to support welding operations without negative impact on Hospital electrical loads. Do not use Hospital outlets for power sources.
 - 4. Use of single-phase transformer welding equipment is not permitted.
 - 5. No welding will be permitted when critical procedures are taking place in adjacent areas; coordinate with Owner.
- B. Fire Protection: Keep required exit corridors and passageways clear and unobstructed.
 - 1. Remove flammable materials to location not closer than 35 feet from welding operations.
 - 2. Instruct workmen to location of nearest fire alarm.
 - 3. Completely cover fixed flammable items with fire-resistant blankets.
 - 4. Existing smoke detectors and fire sprinkler systems shall remain operational throughout Project. If possible false alarms are anticipated during specific construction operations, detectors may be covered during that specific operation with proper notifications and concurrence of local jurisdiction, but must be cleared and operable at end of each working day.

1.13 BUILDING PERMIT

- A. Obtain building permit and be responsible for securing any necessary inspection fees. Include cost in Contract Sum.

- B. Conspicuously display permit at Project site.

PART 2 - PRODUCTS – (NOT USED)

PART 3 - EXECUTION

3.1 CLEAN-UP

- A. Refer to Section 01 73 00 for additional requirements.
 - 1. Keep premises free from accumulation of waste materials or rubbish. Remove trash daily.
 - 2. Leave existing toilet rooms in clean and sanitary conditions daily.

END OF SECTION

SECTION 01 35 16

ALTERATION PROJECT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes:
 - 1. Products and installation for patching and extending Work within construction areas of existing facilities.
 - 2. Providing transition and adjustments.
 - 3. Repair of damaged surfaces and finishes.
- B. Related Sections include the following:
 - 1. Division 01 Section "Temporary Facilities and Controls" for construction of temporary fire-rated partitions to separate existing occupied areas from construction areas.

1.2 OCCUPANCY, ACCESS, AND PROTECTION

- A. Entire existing facility will be occupied during progress of construction for conduct of normal operations.
- B. Cooperate with Owner in scheduling operations to minimize conflict and to permit continuous usage. Perform work not to interfere with operations of occupied areas.
- C. Existing facilities will remain in full operation during execution of this Work. Exercise every precaution to ensure safety and protection for existing facilities, occupants, merchandise, pedestrians, and vehicles.
 - 1. Maintain safe access and egress at all times for occupants, pedestrians, and vehicles.
 - 2. Provide protection to prevent damage to facilities, merchandise, and vehicles from dust, water, weather, and other similar harmful elements.
 - 3. Maintain exiting from facilities to provide safe passage complying with applicable codes.

1.3 SCHEDULING OF WORK

- A. Make arrangements with Owner and schedule Work to avoid interference with normal operations of occupied areas. Submit schedule and summary of applicable Work within occupied areas and obtain Owner approval not less than two days prior to commencement of such Work.
 - 1. Requests for use of certain existing loading docks, passage ways, and other similar spaces within areas outside limits of construction operations will be limited to day-by-day basis and must be approved in advance by Owner.
- B. Coordinate access and scheduling of Work within tenant areas with Owner.

1.4 TORCH-CUTTING AND WELDING PROCEDURES

- A. Notify Owner in advance of torch-cutting and welding operations performed within occupied areas; obtain approval prior to proceeding with such operations.
 - 1. Neither open-flame torch-cutting, welding nor arc-welding are allowed without having secured appropriate permit from Fire Marshal or authority having jurisdiction.
 - 2. Keep portable fire extinguisher of appropriate class within reach during welding or torch-cutting operations.
 - 3. Screen arc-welding from vision of passersby.
- B. Maintain a "Fire Watch" for minimum of 60 minutes after completion of each torch-cutting and welding operation.

1.5 UTILITY SERVICE OUTAGES

- A. Keep utility and service outages to minimum and perform only after written approval of Owner is received.
 - 1. Requests for outages will not be considered unless they include an identification of areas which will be affected by proposed outage.
 - 2. Schedule outages for times other than normal business hours.
 - 3. Make requests for outages minimum of five calendar days in advance of proposed outage.
- B. Contractor: Responsible for investigating utility and service lines to determine effect of outage upon building operations outside of limit of operations. Obtain approval in advance from Owner to execute investigations.

1.6 KEYS

- A. When necessary to perform Work, Owner will issue keys to existing mechanical/electrical equipment spaces.
- B. Return keys at end of each work day; request keys on succeeding days, if necessary.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Type and Quality of Existing Products: Use products or types of construction that exist in structure, as needed to patch, extend, or match existing Work.
- B. Generally, Contract Documents do not define products or standards of workmanship present in existing construction.
- C. Determine by inspecting and testing products where necessary, referring to existing work as quality standard.
- D. New Materials: Comply with Specifications for each product involved.
- E. Match existing products and work for patching existing work.
- F. Materials for Temporary Fire-Rated Partitions: Comply with provisions of Division 01 Section "Temporary Facilities and Controls."
- G. Salvaged Materials: Salvage sufficient quantities of cut or removed material to replace damaged Work of existing construction, when material is not readily obtainable on current market.
- H. Store salvaged items in dry, secure place on site.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that areas are ready for alteration and remodeling.
- B. Discrepancies: Verify dimensions and elevations indicated in layout of existing work.
 - 1. Prior to commencing work, carefully compare and check Contract Documents for discrepancies in locations or elevations of work to be executed.
 - 2. Refer discrepancies among Drawings and existing conditions to Architect for adjustment before work affected is performed.

3.2 PREPARATION

- A. Construct temporary fire-rated partitions to separate existing occupied areas from construction and alteration areas. Comply with provisions of Division 01 Section "Temporary Facilities and Controls."
- B. Cut, move, or remove items as necessary for access to alteration and renovation Work.

1. Remove unsuitable material not marked for salvage, such as rotted wood, corroded metals, deteriorated masonry and concrete, and other deteriorated materials. Replace materials as specified for finished Work.
 2. Remove debris and abandoned items from area and from concealed spaces.
- C. Cutting and Removal: Perform cutting and removal work to remove minimum necessary, and in manner to avoid damage to adjacent work. Cut finish surfaces such as masonry, tile, plaster, or metals by methods to terminate surfaces in straight line at natural point of division.
- D. Prepare surface and remove surface finishes as necessary to provide for proper installation of new materials and finishes.
- E. Close openings in exterior surfaces to protect existing Work from weather and extremes of temperature and humidity. Insulate ductwork and piping to prevent condensation in exposed areas.
- F. Provide temporary barriers and closures to control operations to prevent spread of dust to occupied portions of building.

3.3 INSTALLATION

- A. Coordinate Work of alterations and renovations to expedite completion and to accommodate Owner occupancy.
- B. Remove, cut, and patch Work in manner to minimize damage and to provide means of restoring products and finishes to specified condition.
1. Refinish visible existing surfaces to remain in renovated rooms and spaces, to specified condition for each material, with neat transition to adjacent finishes.
- C. Install products as specified in individual Specification sections.
- D. Where new Work abuts or aligns with existing, perform smooth and even transition to match existing adjacent surface in texture and appearance.
1. When finished surfaces are cut so that smooth transition with new Work is not possible, terminate existing surface along straight line at natural line of division and request instructions from Architect as to method of making transition.

3.4 ADJUSTMENTS

- A. Where removal of partitions or walls results in adjacent spaces becoming one, rework floors, walls, and ceilings to provide smooth plane without breaks, steps, or soffits.
- B. Trim existing doors as necessary to clear new floor finish. Refinish trim as required.
- C. Fit Work tightly at penetrations of surfaces.
- D. Patch or replace portions of existing surfaces which are damaged, lifted, discolored, or showing other imperfections. Repair substrate prior to application of finishes.

3.5 FINISHES

- A. Finish new surfaces as specified in individual Specification sections.
- B. Finish patches to produce uniform finish and texture over entire area. When finish cannot be matched, refinish entire surface to nearest intersections.

3.6 CLEANING

- A. Thoroughly clean areas and spaces affected by Work. Completely remove paint, mortar, oils, putty and items of similar nature.
- B. Clean Owner-occupied areas daily. Clean spillage, overspray, and heavy collection of dust in Owner- occupied areas immediately.

END OF SECTION

SECTION 01 41 00

REGULATORY REQUIREMENTS

PART 1 - GENERAL

1.1 REGULATORY REQUIREMENTS

- A. General: Reference to codes, standards or regulatory requirements made on Drawings or in Specifications is considered an integral part of Contract Documents as minimum requirements. Nothing in Contract Documents should be understood to conflict with laws, by-laws, or regulations of municipal, State, Federal, and other authorities having jurisdiction.
- B. Should Contractor knowingly perform any work that does not conform with requirements of applicable codes, ordinances, regulations, or standards, Contractor shall assume full responsibility for such work and shall correct non-conforming work at no additional cost to Owner.
- C. Code Requirements:
 - 1. NFPA 101 – 2012.
 - 2. NFPA Codes as applicable, with emphasis on the following:
 - a. NFPA 12A (2009), 20 (2013), 22 (2008), 92 (2012), 204 (2012).
 - b. Additional NFPA For EH&S: 13 (2010), 70 (2011), 72 (2012), 99 (2012), 45 (2011), 30 (2011), 10 (2010).
 - 3. (2011), 10 (2010).
 - 4. International Building Code 2012 Edition.
 - 5. International Mechanical Code 2012 Edition.
 - 6. International Plumbing Code 2012 Edition.
 - 7. International Fuel Gas Code 2012 Edition.
 - 8. National Electrical Code 2011 Edition.
 - 9. Energy Conservation Design Standard for New State Buildings (including major renovation projects), current edition, State Comptroller's Office, Government, Code sec. 447.004 and 34 TAC
 - 10. § 19.32
 - 11. ASHRAE / IESNA 90.1 2010 Edition.
 - 12. International Energy Conservation Code (IECC) 2003 Edition (Residential)
 - 13. American Concrete Institute (ACI – 318 current edition)
 - 14. American institute of Steel Construction (AISC current edition)
 - 15. Texas Department of Insurance Windstorm Inspection Program
 - 16. FEMA 100 year flood plain
 - 17. 2012 Texas Accessibility Standards.
 - 18. Elevators and Escalators, Health & Safety Code chapter 754 and 16TAC § 74 (see 16TAC § 74.100 for effective dates of ASME standards) (see 754.014(k) for date of installation definition)
 - 19. Boilers, Health & Safety Code chapter 755 and 16TAC § 65.
 - 20. Guidelines for Design and Construction of Healthcare Facilities, 2010 Edition, The Facility Guidelines Institute.
 - 21. 2010 Center for Medicare and Medicaid Services
 - 22. Texas Administrative Code For Health Services, Title 25.

PART 2 - PART 2 - PRODUCTS – (NOT USED)

PART 3 - PART 3 - EXECUTION – (NOT USED)

END OF SECTION

SECTION 01 42 00

REFERENCES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 DEFINITIONS

- A. General: Basic Contract definitions are included in the Conditions of the Contract.
- B. "Approved": When used to convey Architect's action on Contractor's submittals, applications, and requests, "approved" is limited to Architect's duties and responsibilities as stated in the Conditions of the Contract.
- C. "Directed": A command or instruction by Architect. Other terms including "requested," "authorized," "selected," "required," and "permitted" have the same meaning as "directed."
- D. "Indicated": Requirements expressed by graphic representations or in written form on Drawings, in Specifications, and in other Contract Documents. Other terms including "shown," "noted," "scheduled," and "specified" have the same meaning as "indicated."
- E. "Regulations": Laws, ordinances, statutes, and lawful orders issued by authorities having jurisdiction, and rules, conventions, and agreements within the construction industry that control performance of the Work.
- F. "Furnish": Supply and deliver to Project site, ready for unloading, unpacking, assembly, installation, and similar operations.
- G. "Install": Operations at Project site including unloading, temporarily storing, unpacking, assembling, erecting, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning, and similar operations.
- H. "Provide": Furnish and install, complete and ready for the intended use.
- I. "Project Site": Space available for performing construction activities. The extent of Project site is shown on Drawings and may or may not be identical with the description of the land on which Project is to be built.

1.3 INDUSTRY STANDARDS

- A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into the Contract Documents to the extent referenced. Such standards are made a part of the Contract Documents by reference.
- B. Publication Dates: Comply with standards in effect as of date of the Contract Documents unless otherwise indicated.
- C. Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with the Contract Documents.
 - 1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

1.4 ABBREVIATIONS AND ACRONYMS

A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities indicated in Gale's "Encyclopedia of Associations" or in Columbia Books' "National Trade & Professional Associations of the United States." Contract Documents, they shall mean the recognized name of the entities in the following list. Names, telephone numbers, and Web sites are subject to change and are believed to be accurate and up-to-date as of the date of the Contract Documents.

1. A/E: Architect/Engineer
2. AHJ: Authority Having Jurisdiction
3. ANSI American National Standards Institute
4. ASTM ASTM International
5. (American Society for Testing and Materials International)
6. BOR: Board of Regents
7. CCD: Construction Change Directive
8. CCL: Construction Cost Limitation
9. CMAR: Construction Manager at Risk
10. CSP: Competitive Sealed Proposal
11. DD: Design Development
12. EPA Environmental Protection Agency
13. FCC Federal Communications Commission
14. FDA Food and Drug Administration
15. FPE: Fire Protection Engineer
16. GA Gypsum Association
17. GCs: General Conditions
18. GMP: Guaranteed Maximum Price
19. GSA General Services Administration
20. GSF: Gross Square Feet
21. HSP: HUB Subcontractor Plan
22. HUB: Historically Underutilized Business
23. HUD Department of Housing and Urban Development LA: Landscape Architect
24. LEED: Leadership in Energy and Environmental Design LDs: Liquidated Damages
25. NASF: Net Assignable Square Feet
26. NFPA (National Fire Protection Association)
27. NTP: Notice to Proceed
28. OAC: Owner/Architect/Contractor
29. OCM: Owner's Construction Manager
30. ODR: Owner's Designated Representative
31. PAR: Progress Assessment Report
32. PE: Professional Engineer
33. PM: Project Manager
34. RID: Registered Interior Designer
35. R&R: Repair and Rehabilitation
36. SD: Schematic Design
37. SDs: Schematic Design Drawings

- 38. SMACNA Sheet Metal and Air Conditioning Contractors' National Association UGC/SGC:Uniform General Conditions/Supplemental General Conditions
- 39. UL Underwriters Laboratories Inc.
- 40. USGBC U.S. Green Building Council

1.5 APPLICABLE PUBLICATIONS

- A. Unless a specific release or publication date is provided in reference to a published specification, code, standard, or other requirement in these Specifications, it shall be understood that the latest published version shall apply.

1.6 REFERENCE SPECIFICATIONS, CODES, AND STANDARDS

- A. Subcontractor shall maintain copies of the codes and reference standards with the Contract Documents at the jobsite at all times.
- B. All conflicts between codes, reference standards, drawings, and the other Contract Documents shall be brought to the attention of UTSW for clarification prior to ordering or providing any materials or furnishing labor.
- C. References to "OSHA Regulations for Construction" shall mean Title 29, Part 1926, Construction Safety and Health Regulations, Code of Federal Regulations (OSHA), including all changes and amendments thereto.
- D. References to "OSHA Standards" shall mean Title 29, Part 1910, Occupational Safety and Health Standards, Code of Federal Regulations (OSHA), including all changes and amendments thereto.

1.7 CODES AND STANDARDS

- A. All work shall comply with the codes and standards promulgated by the following agencies and organizations and those identified in all divisions of these specifications. Bring any conflicts between specifications, drawings, and the referenced documents to the attention of UTSW, in writing, for resolution before taking any related action.
- B. Where differences exist between codes and standards, the one with the most stringent requirement, as determined by the OSBC (OFFICE OF SAFETY AND BUSINESS CONTINUITY Office shall apply.
- C. Codes and References
 - 1. NFPA 101 LIFE SAFETY
 - a. 2015 EDITION LIFE SAFETY CODES (LSC), EFFECTIVE MAY 1, 2016
 - 2. NFPA CODES AS APPLICABLE, WITH EMPHASIS ON THE FOLLOWING: a. NFPA 12A (2009), 92 (2012), 204 (2012)
 - 3. ADDITIONAL NFPA FOR OFFICE OF SAFETY AND BUSINESS CONTINUITY (OSBC):
 - a. 10 (2010), 13 (2010), 20 (2010),
 - b. 30 (2011), 45 (2011),
 - c. 72 (2012), 99 (2012),
 - 4. INTERNATIONAL BUILDING CODE - 2015 EDITION (IBC)
 - 5. ARCHITECTURAL DESIGN
 - a. 2012 TEXAS ACCESSIBILITY STANDARDS
 - b. ELEVATORS AND ESCALATORS, HEALTH & SAFETY CODE CHAPTER 754 AND 16TAC § 74
 - 1) (SEE 16TAC § 74.100 FOR EFFECTIVE DATES OF ASME STANDARDS)
 - 2) (SEE 754.014(K) FOR DATE OF INSTALLATION DEFINITION)

- c. BOILERS, HEALTH & SAFETY CODE CHAPTER 755 AND 16TAC § 65
- d. GUIDELINES FOR DESIGN AND CONSTRUCTION OF HEALTHCARE FACILITIES, 2010 EDITION, THE FACILITY GUIDELINES INSTITUTE
- e. 2010 CENTER FOR MEDICARE AND MEDICAID SERVICES
- f. TEXAS ADMINISTRATIVE CODE FOR HEALTH SERVICES, TITLE 25
- 6. MECHANICAL AND PLUMBING DESIGN
 - a. INTERNATIONAL MECHANICAL CODE 2015 EDITION
 - b. INTERNATIONAL PLUMBING CODE 2015 EDITION
 - c. INTERNATIONAL FUEL GAS CODE 2015 EDITION
- 7. ELECTRICAL DESIGN
 - a. NATIONAL ELECTRICAL CODE 2014 EDITION
- 8. ENERGY CONSERVATION DESIGN
 - a. ENERGY CONSERVATION DESIGN STANDARD FOR NEW STATE BUILDINGS (INCLUDING MAJOR RENOVATION PROJECTS), CURRENT EDITION, STATE COMPTROLLER'S OFFICE, GOVERNMENT, CODE SEC. 447.004 AND 34 TAC § 19.32
 - b. ASHRAE / IESNA 90.1 2013 EDITION (EFFECTIVE JUNE 1, 2016)
 - c. INTERNATIONAL ENERGY CONSERVATION CODE (IECC) 2015 EDITION (RESIDENTIAL) EFFECTIVE JUNE 1, 2016
- 9. AMERICAN CONCRETE INSTITUTE (ACI - 318 CURRENT EDITION)
- 10. AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC CURRENT EDITION)
- 11. TEXAS DEPARTMENT OF INSURANCE WINDSTORM INSPECTION PROGRAM
- 12. FEMA 100 YEAR FLOOD PLAIN

PART 2 - PART 2 - PRODUCTS (NOT USED)

PART 3 - PART 3 - EXECUTION (NOT USED)

END OF SECTION

CODES AND STANDARDS

A. GENERAL REQUIREMENTS

1. The Project Architect/Engineer shall design UT System construction projects to comply with the current Office of Facilities Planning and Construction (OFPC) approved editions of the following codes and standards and advise the Owner of code revisions having impact on the project design.
2. The State Fire Marshal is the code authority having jurisdiction (AHJ) for all issues pertaining to NFPA 101 Life Safety Codes. The UT System Office of Facilities Planning and Construction is the code authority having jurisdiction (AHJ) for UT System construction projects for all codes other than NFPA 101 Life Safety Codes. OFPC is responsible for facilitating resolution of conflicts and interpretations for these non-NFPA 101 codes after a thorough and joint discussion with the Institution.
3. The Project Architect/Engineer shall prepare a written codes and standards analysis, "Building Code Analysis," for each project for review by OFPC. This analysis shall provide a side-by-side comparison of the requirements of the below listed codes and standards for each code issue and an indication of which code requirement is being applied to the project (see sample analysis Exhibit 1). In the absence of a careful and thorough discussion by the design team of a specific conflict between the codes, the default is to design to the more restrictive or more protective code. These code discussions are project-specific and on a point-by-point basis within the codes. The final approved Building Code Analysis shall be placed in the project construction document drawings for future reference by the Owner. See Appendix "L" for submittal requirements.
4. If deemed necessary for local authority to review any aspect of the project, such review shall be arranged to allow an OFPC or institutional representative to attend with the Project Architect/Engineer.
5. In the event of the need for interpretation among the codes and standards, the Project Architect/Engineer shall inform OFPC of the need for an interpretation and OFPC will establish the requirements for compliance.
6. OFPC also requires the Project Architect/Engineer to comply with certain provisions of the local fire department that provides fire protection services for the Institution. These provisions may include locations and dimensions

for fire fighting access, including fire lanes; locations and specifications for stand pipes, fire hose cabinets, fire control room, and fire hose connections; elevator requirements; and other similar matters.

7. The Project Architect/Engineer shall be required to provide an affirmation statement that the project is designed in compliance with applicable codes and standards. The following statement shall be located on the drawing index page or adjacent the project building code summary:

“Life Safety Code Compliance: The Project Architect/Engineer acknowledges that construction projects for the University of Texas System must, at a minimum, be designed in accordance with the requirements of National Fire Protection Association (NFPA) 101, Life Safety Code, as currently adopted by the State Fire Marshal, Texas Government Code sec. 417.008(e). Therefore, the Project Architect/Engineer affirms that, to the best of his/her professional judgment, knowledge, and belief, the design of this project satisfies the requirements of NFPA 101, Life Safety Code, as well as any other codes or standards made applicable to the project by the professional services agreement.”

B. CODE REQUIREMENTS

8. Refer to Section 01 41 00.

C. CODE COMPLIANCE CONFIRMATION REVIEWS

1. An independent project design “Code Compliance Confirmation Review” will be performed, documented and submitted to the Owner at DD and 75% CD submission to ensure compliance with the following codes as they apply to a specific project. The Owner’s Project Manager will direct the A/E to obtain this code confirmation review or will direct the Owner’s Code Consultant to perform this code confirmation review. (non-inclusive code review list, editions as listed in sections above):

- International Building Code (IBC)
- International Mechanical Code (IMC)
- International Plumbing Code (IPC)
- National Electrical Code (NEC)
- NFPA 101 Life Safety Code
- NFPA Codes as applicable, with emphasis on the following:
 - NFPA 101 Referenced Required Codes
 - NFPA 12A
 - NFPA 20
 - NFPA 22
 - NFPA 92A
 - NFPA 92B
 - NFPA 203

- NFPA 204
- NFPA 2001

- FEMA 100 year flood plain verification
- Texas Department of Insurance (TDI) First Tier Coastal Counties wind load criteria

2. This Code Compliance Confirmation Review does not relieve the A/E firm from complying with the approved codes and standards for the project. See Exhibit 2 for sample code review template.

D. ACOUSTICAL DESIGN - BACKGROUND NOISE DESIGN CRITERIA FOR TYPICAL OCCUPANCIES

3. Design in accordance with good practice to achieve conventional ambient noise levels qualified in Noise Criteria (NC) defined in current ASHRAE Applications Volume, Chapter 42 and ANSI S1.8 Reference Quantities for Acoustical Levels – ASA 84.
4. The ambient sound level of an occupied space is not to exceed the following NC listed for its respective typical occupancy unless specifically directed otherwise by the involved component Institution's statement of project program requirements. Spatial forms, materials, assemblies, systems and equipment selections are to be designed as required to achieve a standard quality of specified level of maximum background noise.

a. <u>Typical Occupancy</u>		<u>Max. Noise Criteria</u>
		<u>NC</u>
(2)	Offices:	
	(a) Executive	30
	(b) Conference rooms	30
	(c) Private	35
	(d) Open-plan areas	40
	(e) Computer/Business machine areas	45
	(f) Public circulation	45
(3)	Research, Hospital, and Clinics:	
	(a) Private rooms	30
	(b) Wards	35
	(c) Operating rooms	25
	(d) Laboratories:	
	Research & General	35

- | | | |
|-----|-------------------------------|----|
| | Teaching | 30 |
| | At Hoods: 4' AFF, 3' in front | |
| | 0-50% sash position | 55 |
| | (e) Corridors | 35 |
| | (f) Public areas | 40 |
| (4) | Schools: | |
| | (a) Lecture and classrooms | 30 |
| | (b) Open-plan classrooms | 35 |
| | (b) Lecture theaters | 30 |
5. These conventional standards of the level of ambient noise in a space are independent of and prior to the installation of any Owner-furnished equipment, furniture and furnishings unless specified otherwise. Other resource material describing conventional ambient noise criteria is available in the current edition of Ramsey/Sleeper Architectural Graphic Standards.

EXHIBIT 1

BUILDING CODE ANALYSIS [TEMPLATE]

Project Name:

Institution:

Project No.

Code/Standards Analysis Date:

Project Phase:

Applicable Codes

1. NFPA 101 Life Safety – 2009 Edition
2. International Building Code – 2009 Edition
3. Texas Accessibility Standard
4. etc.

Note: The code requirements selected as the basis for design are bolded.

<u>Code Issue</u>	<u>NFPA 101</u>	<u>IBC</u>
Occupancy Classification		
1. Offices and college classrooms With less than 50 occupants	Business 6.1.2.2	Group B 304.
Construction Classification		
1. Main Building	Not addressed	Type IIA 403.3.1
Stair Pressurization	Not Required	1005.3.2.5
Distance between exits	250 ft. if sprinkled	250 ft. if sprinkled
Etc.		

EXHIBIT 2

CODE COMPLIANCE CONFIRMATION REVIEW [TEMPLATE]

[Date]

[Project Manager]

The University of Texas System

Office of Facilities Planning and Construction 702
Colorado

Suite 4.100

Austin, TX 78701

Reference: Review Comments on [100%DD or 75% CD]
[Project name]

[Institution]

OFPC Project No. XXX-XXX Dear

[Project Manager]:

[Code Consulting Firm (CCF)] has complete its Code Compliance Confirmation review and has prepared review comments on the documents for the [DD or 75% Construction Documents] package for the referenced project.

In performing this current service, [CCF] reviewed the following documents, furnished by A/E.

- DD or 75% Construction Documents Drawings dated [month dd, yyyy].
- DD or 75% Construction Documents Project Manual, Architectural Volume I Divisions 1-14, dated [month dd, yyyy].
- DD or 75% Construction Documents Project Manual, M.E.P. Volume II Divisions 21 - 33, dated [month dd, yyyy].

The principal codes used in this review are as follows:

- International Building Code, 2009 Edition (IBC).
- NFPA 101, *Life Safety Code*, 2009 Edition (LSC).

Other applicable codes, standards, and regulations are listed in the Project Data shown on the Building Code Analysis Drawing 1.1 and in the Project Information Manual. Additionally, FEMA 100 year flood plain verification and TDI First Tier Coastal County wind load criteria were reviewed where applicable.

Description of Project

The [Project] consists of ...

Note: The follow major headings in this Code Compliance Confirmation Review Template are for reference only to demonstrate process.

Building Code Issues

Comments:

[Drawing 5.1, Drawing 10.21, Drawing 11.41 indicates there are accessible dwelling units. There is no table that indicates the discrete Apartment ID and Building ID for each accessible dwelling unit so that a user of the plans can see in one place the summary of accessible units.]

Requirements for Hazardous Materials and Laboratories Insert comments as necessary.

Means of Exit Access

Insert comments as necessary.

Emergency and Standby Power

Insert comments as necessary.

Fire Water Supply

Insert comments as necessary.

FEMA 100 Year Flood Plain

Insert comments as necessary. TDI

Windstorm Inspection Program

Insert comments as necessary.

Other Major Code Headings as Necessary

Insert comments as necessary.

Summary

Compliance with the comments stated in this letter does not relieve the A/E from complying with the Owners Design Guidelines, Owner's insurance/underwriting requirements, applicable NFPA Standards and State requirements.

Sincerely,

Project Manager

Texas License No. xxxxx

cc: UT System Office of Risk Management
OFPC Associate Director for Engineering

Revision Log
Appendix C

Rev. Date	Remarks
3/1/06	Adopt use of Master Format for specs. (change Div 15 & 16 to Div 1-33)
10/1/06	Adopt 2006 NFPA 101 LSC and 2006 IBC
1/1/09	Add Revision Log
	Adopt ASHRAE 90.1 – 2007 edition effective 1-1-09 per SECO
	Add effective date language to TDLR Elevator requirements
	Add language to use current edition of ACI 318 & AISC
10/12/09	Adopt 2009 NFPA 101 LSC and 2009 IBC effective 10-15-09
10/1/10	Changed more stringent to more restrictive or more protective, added 2010 ADA Standard effective date
3/1/11	Updated International Energy Conservation Code Edition to 2009 effective 4/1/11
9/1/11	Adopt International Energy Conservation Code 2009 Edition effective 6/1/11, adopt ASHRAE 90.1 – 2010 edition effective 9-1-11, adopt Water Efficiency Standard for State Buildings and Institutions of Higher Edu. effective 9-1-11

STATE OF TEXAS COMPTROLLER OF PUBLIC
ACCOUNTS STATE ENERGY CONSERVATION OFFICE

Section 19.31 Requirement to Use Design Standards

Pursuant to Government Code, sec. 447.004, state agencies and institutions of higher education shall use the energy and water conservation design standards that the State Energy Conservation Office (SECO) has adopted under this chapter, when constructing new state buildings or conducting major renovations of existing state buildings.

Section 19.32 Energy and Water Conservation Design Standards

- (a) The State Energy Conservation Office (SECO) adopts by reference the following standards for new construction or major renovation projects:
 - (1) for any new construction or major renovation project, except low-rise residential buildings, with a design assignment made prior to September 1, 2011, the energy conservation design standard of the American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) / Illuminating Engineering Society of North America (IESNA), Energy Standard for Buildings, ASHRAE/IESNA Standard 90.1-1999;
 - (2) for any new construction or major renovation project of a public low rise residential buildings with a design assignment made prior to June 1, 2011, the energy conservation design standard of the International Code Council as published in the International Energy Conservation Code for 2000;
 - (3) for any new construction or major renovation project, except low-rise residential buildings, with a design assignment made on or after September 1, 2011, the energy conservation design standard of the American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) / Illuminating Engineering Society of North America (IESNA), Energy Standard for Buildings, ASHRAE/IESNA Standard 90.1-2010, provided however the following buildings or structures are exempt from compliance with this section:
 - (A) a building or structure that is listed in the State or National Register of Historic Places;
 - (B) a building or structure that is designated as a historic property under local or state designation law or survey;
 - (C) a building or structure that is certified as a contributing resource with a National Register listed or locally designated historic district; or

- (D) with an opinion or certification by the State Historic Preservation Officer or Keeper of the National Register of Historic Places, a building or structure that is eligible to be listed on the National or State Registers of Historic Places either individually or as a contributing building to a historic district;
- (1) for any new construction or major renovation project of a low-rise residential building with a design assignment made on or after June 1, 2011, the residential chapter of the International Code Council as published in the International Energy Conservation Code for 2009.
- (b) Effective September 1, 2011, SECO adopts by reference the “Water Efficiency Standards for State Buildings and Institutions of Higher Education Facilities” prepared by the Office of the Comptroller, State Energy Conservation Office dated January 2011 as the water conservation design standards for new state buildings and major renovation projects.
- (c) Copies of the standards are published by the comptroller and are available at the offices of SECO, 111 E. 17th Street, LBJ State Office Building, Suite 1114, Austin, Texas 78774, where they may be viewed during normal office hours as well as on the comptroller's website www.txbuildingenergycode.com.

Section 19.34 Submission of Certification and Compliance Documentation

Before beginning construction of a new state building or a major renovation project including a new building or major renovation project of a state-supported institution of higher education, a state agency or an institution of higher education shall submit to the State Energy Conservation Office (SECO) a copy of the certification by the design architect or engineer that verifies to the agency or institution that the construction or renovation complies with the standards that are established under this chapter, including engineering documentation.

Effective Date: September 28, 2011.

Filed with Secretary of State: September 8, 2011.

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Comptroller of Public Accounts

SECTION 01 45 29
STRUCTURAL TESTING AND INSPECTIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes requirements for quality assurance and quality control to be completed by the Testing Laboratory, the Special Inspector, Contractor, and/or the Geotechnical Engineer for the following structural items:

1. Concrete Forming and Accessories.
2. Concrete Reinforcing.
3. Cast-in-Place Concrete.
4. Structural Steel.
5. Steel Decking.
6. Earthwork.

- B. Related Requirements:

1. Specification 014000 "Regulatory Requirements" for other independent testing agency procedures and administrative requirements.

1.3 PRICE AND PAYMENT PROCEDURES

- A. Unit Prices:

1. Cost Proposal: The Testing Laboratory's proposal to the Owner shall contain unit price stipulations for specified tests and inspections and on an hourly basis for personnel. A total estimated price shall also be submitted.

- B. Measurement and Payment

1. Payment of the Testing Laboratory: The Owner will pay for the initial Laboratory services for inspection and testing of materials for compliance with the requirements of the Contract Documents.
2. Payment for Substitution Testing: The Contractor shall arrange for and pay for any additional samples and tests above those required by the Contract Documents as requested by the Contractor for his convenience in performing the work.
3. Payment for Retesting: When initial tests indicate work does not comply with the requirements of the Contract Documents, the Contractor shall be liable to the Owner for the cost for any additional inspections, sampling, testing, and retesting done by the Testing Laboratory.
4. Payment by Contractor: The Contractor shall furnish and pay for the following items if required:
 - a. Soil survey of the location of borrow soil materials, samples of existing soil materials, and delivery to the Contractor's Testing Laboratory.
 - b. Samples of concrete aggregates and delivery to the Contractor's Testing Laboratory.
 - c. Concrete mix designs as prepared by his concrete supplier.
 - d. Site-situated storage boxes for concrete cylinders
 - e. Concrete coring, tests of below strength concrete, and load tests, if ordered by the Owner, Architect, or Engineer.
 - f. Certification of reinforcing steel and prestressing steel mill order.
 - g. Certification of structural steel mill order.
 - h. Certification of portland cement, lime, fly ash.
 - i. Certification of welders and preparation of Welding Procedure Specifications.
 - j. Tests, samples, and mock-ups of substitute material where the substitution is requested by the Contractor and the tests are necessary in the opinion of the Owner, Architect or Engineer to establish equality with specified items.
 - k. The making and testing of concrete cylinders for the purpose of evaluating strength at time of form stripping or for post-tensioning or the time spent evaluating the in situ strength of concrete using the Maturity Method.
 - l. Any other tests when such costs are required by the Contract Documents to be paid by the Contractor.

5. Payment for Tests of Suspected Deficient Work: If, in the opinion of the Building Official, Owner, Architect, or Engineer, any of the work of the Contractor is not satisfactory, the Contractor shall furnish and pay for all tests that the Owner, Architect, or Engineer deem advisable to determine its proper construction. The Owner shall pay all costs if the tests prove the questioned work to be satisfactory.

1.4 OWNER RESPONSIBILITIES

- A. The Owner shall engage a Geotechnical Engineer to provide inspection services for the soil preparation as outlined below in Article 3.7.
- B. The Owner shall provide a copy of the project plans and specifications to the Testing Laboratory prior to the start of construction and prior to any preinstallation meetings.

1.5 CONTRACTOR RESPONSIBILITIES

- A. The Contractor shall not engage the same Testing Laboratory for construction services as the Owner has for Structural Testing Laboratory Services as defined herein unless agreed to by the Owner.
- B. Furnishing Samples and Certificates: The Contractor shall provide to the laboratory certificates and representative samples of materials proposed for use in the work in quantities sufficient for accurate testing as specified.
- C. Furnishing Casual Labor, Equipment and Facilities: The Contractor shall furnish casual labor, equipment, and facilities as required for sampling and testing by the laboratory and otherwise facilitate the required inspections and tests.

1.6 TESTING LABORATORY RESPONSIBILITIES

- A. The Testing Laboratory shall sample and test materials as they are being installed for compliance with specified acceptance criteria. The Testing Laboratory will report and interpret the test results. The Laboratory shall monitor and report on the installation of construction work and shall perform tests on the completed construction as required to indicate Contractor's compliance with the various material specifications governing this work.
- B. The Testing Laboratory shall provide inspections on the following items:
 1. Reinforcing steel placement.
 2. Concrete work.
 3. Welding of reinforcing steel.
 4. Bolts to be installed in concrete.
 5. Bolts, anchors, and reinforcing bars installed in hardened concrete (post-installed anchors).
 6. Inspection of structural steel, bolting, and welding material.
 7. Welding of structural steel.
 8. High-strength bolting.
 9. Soil improvements as recommended by the Geotechnical Engineer.
- C. Inspections Required by Government Agencies: The Testing Laboratory shall perform inspections and submit reports and certifications as required by government agencies having jurisdiction over the aspects of the project covered by this specification.
- D. Notification of Deficiencies in the Work: The Testing Laboratory shall notify the Architect, Engineer, and Contractor within 24 hours of discovery of observed irregularities and deficiencies of the Work and other conditions not in compliance with the requirements of the Contract Documents. Notification shall be by telephone or e-mail and then in writing.
- E. Accounting: The Testing Laboratory shall be responsible for separating and billing costs attributed to the Owner and costs attributed to the Contractor.
- F. Monitoring Product and Material Certifications: The Testing Laboratory shall be responsible for monitoring the submittals of product and material certifications from manufacturers and suppliers as specified in the Specifications and shall report to the Owner, Architect, and Engineer when those submittals are not made in a timely manner.
- G. Limitations of Authority: The Testing Laboratory is not authorized to revoke, alter, relax, enlarge upon, or release any requirements of the Specifications or to approve or accept any portion of the work or to perform any duties of the General Contractor and his Subcontractors.

1.7 ADMINISTRATIVE REQUIREMENTS

- A. Coordination:
 1. The Testing Laboratory shall cooperate with the Architect, Engineer, and Contractor and provide qualified personnel promptly on notice.
 2. The Contractor shall cooperate with Testing Laboratory personnel and provide access to the work and to manufacturers' operations.

3. Notification of Source Change: The Contractor shall be responsible for notifying the Owner, Architect, Engineer, and Testing Laboratory when the source of any material is changed after the original tests or inspections have been made.
- B. Preinstallation Meetings: The Testing Laboratory shall attend preinstallation meetings with the Architect, Engineer, Contractor, and material suppliers as required to coordinate materials inspection and testing requirements with the planned construction schedule and shall participate in such meetings throughout the course of the project.
- C. Scheduling:
 1. Advance Notice: The Contractor shall be responsible for notifying the Testing Laboratory sufficiently in advance of operations to allow for assignment of personnel and scheduling of tests. Failure to sufficiently notify may result in additional costs incurred by the Testing Laboratory that may be back-charged to the Contractor by the Owner.

1.8 SUBMITTALS

- A. Quality Control Reports:
 1. Information on Reports: The Testing Laboratory shall submit copies of reports of inspections and tests promptly. The reports shall contain at least the following information:
 - a. Project name.
 - b. Date report issued.
 - c. Testing Laboratory name and address.
 - d. Name and signature of inspector/technician.
 - e. Date of inspection and/or sampling.
 - f. Date of test.
 - g. Identification of product and Specification section.
 - h. Location in the project.
 - i. Identification of inspection or test.
 - j. Record of weather conditions and temperature (if applicable).
 - k. Results of test regarding compliance with Contract Documents.
 2. Copies: The Laboratory shall send signed copies of test and inspection reports to the following parties:
 - a. Owner or his/her representative.
 - b. General Contractor.
 - c. Architect.
 - d. Engineer of Record.
- B. Discrepancy Log: The Testing Laboratory shall create and maintain a log of all discrepancies throughout the duration of the project.
 1. Information on Log: This log shall include, but is not limited to:
 - a. Discrepancy date.
 - b. Description of discrepancy.
 - c. Drawing and/or detail reference.
 - d. Description of as-built condition.
 - e. Description of any remedial work performed.
 - f. Status of discrepancy.
 2. Submission Schedule: This log shall be submitted to the Architect/Engineer on a periodic basis for review and comment. Upon completion of the Project, this log shall be submitted in its entirety as an attachment to the final signed report described below under Certifications.
- C. Certification: Upon completion of the job, the Laboratory shall furnish to the Owner, Architect, and Engineer of Record, a statement signed by a licensed professional engineer that, to the best of their knowledge, required tests and inspections were made in accordance with the requirements of the Contract Documents.

1.9 QUALITY ASSURANCE

- A. Qualifications of Special Inspector: The special inspector shall be a qualified person who shall demonstrate competence, to the satisfaction of the Building Official, for inspection of the particular type of construction or operation being inspected. The Special Inspector shall meet the legal qualifications of the building code having jurisdiction.
- B. Qualifications of Testing Laboratory:
 1. The Testing Laboratory shall meet the basic requirements of ASTM E 329 and shall submit to the Owner, Architect, and Engineer evidence of current accreditation from the American Association for Laboratory Accreditation, the AASHTO Accreditation Program or the "NIST" National Voluntary Laboratory Accreditation Program.

2. The Testing Laboratory shall be an Approved Agency by the Building Official to perform Special Inspections and other tests and inspections as outlined in the applicable building code.
 3. Tests and inspections shall be conducted in accordance with specified requirements, and if not specified, in accordance with the applicable standards of the American Society for Testing and Materials or other recognized and accepted authorities in the field.
 4. Qualifications of Welding Inspectors
 - a. Inspectors performing visual weld inspection shall meet the requirements of AWS D1.1 Section 6.1.4. Inspectors shall have current certification as an AWS Certified Welding Inspector (CWI). Assistant inspectors, if any, shall be supervised by an Inspector and shall be qualified by training and experience to perform the specific functions to which they are assigned.
 - b. Inspectors performing nondestructive examinations of welds other than visual inspection (MT, PT, UT, and RT) shall meet the requirements of AWS D1.1, Section 6.14.6.
 5. Qualifications for Post-Tensioning Inspector - The technician for the Testing Laboratory performing the field inspections required for post-tensioned concrete shall possess a currently valid Level 2 Post-Tensioning Inspector Certification issued by the Post-Tensioning Institute. A copy of such certification for each such technician shall be submitted for Engineer review and approval.
- C. The Contractor shall not engage the same testing laboratory for construction services as the Owner has for quality assurance testing, unless agreed to by the Owner.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 SCOPE OF WORK

- A. The work to be performed by the Testing Laboratory shall be as specified in this Section of the Specification and as determined in meetings with the Owner, Architect, and Engineer.

3.2 CONCRETE FORMING AND ACCESSORIES

- A. Field Inspection:
 1. Slabs-on-Grade:
 - a. Verify formwork at turndowns and slab edges is plumb and straight, braced against movement and lubricated for removal.
 2. Suspended Floors Infills (General):
 - a. Verify that formwork conforms to signed and sealed shop drawings.
 - b. Verify that shoring layout conforms to signed and sealed shop drawings.
 - c. Verify that reshores at all levels conform to signed and sealed shop drawings.
 - d. Verify that forms are plumb and straight, braced against movement, and lubricated for removal.
 - e. Verify that the forms used for exposed finish surfaces are of the type specified and provide a joint system as shown on the Architect's drawings.
 - f. Verify the proper dimensions of girders, beams, and joists.
 - g. Verify that the slab thickness and top of slab elevation is correct.
 - h. Verify the top of columns are 1/2 inch below the deck soffit.
 3. In-Situ Concrete Strength Verification Prior to Form Stripping: The Testing Laboratory shall verify that the concrete has reached the required minimum strength before form removal by evaluating the specified tests. Refer to Paragraph 3.4C.2.a for additional information regarding the tests.

3.3 CONCRETE REINFORCING

- A. Quality Assurance:
 1. Review the Welding Procedure Specification (WPS) submitted by the contractor for any reinforcing steel other than ASTM A 706 that is proposed to be welded for consistency with acceptable welding practices and AWS.
 2. Review welder qualifications by certification or verify by retesting. Obtain welder certificates.
- B. Field Inspection: The scope of the work to be performed by the inspector on the jobsite shall be as follows:
 1. Reinforcing Steel: The Testing Laboratory or designated Special Inspector shall inspect 100% of reinforcement before each concrete pour to verify the information noted below. Inspection reports shall be prepared and distributed in accordance with the local building code and as specified in this specification.

- a. Primary and secondary longitudinal reinforcement has correct size and number in proper layers.
- b. Longitudinal reinforcement has correct length and lap.
- c. Ties and stirrups are of correct size, spacing, and number and have the proper termination hook geometry.
- d. Unscheduled face reinforcement in beams are provided and are of correct size, number and spacing and have the proper end terminations.
- e. Proper hooks are provided at bar ends as detailed.
- f. Reinforcement is properly supported and braced to formwork to prevent movement during concrete placement.
- g. Reinforcement has proper cover.
- h. Sufficient spacing between reinforcement for concrete placement.
- i. Dowel reinforcement is of proper size, at proper spacing, and has proper lap length and embedment length.
- j. Welded wire reinforcement is composed of flat sheets, has proper wire gage and spacing, is properly supported, and is properly lapped.
- k. Proper construction/control/expansion joint spacing and reinforcement.
- l. Reinforcement around embedded items is placed according to details.
- m. Welded reinforcement has been done according to AWS requirements.
- n. Proper installation of flat slab shear head reinforcement.
- o. Welded Reinforcing: 100% visual inspection of the welding of reinforcing bars to ensure compliance with the requirements of AWS shall be done for the following items:
 - 1) Reinforcing steel resisting flexural and axial forces.
 - 2) Boundary elements of reinforced concrete walls.
 - 3) Shear reinforcement.

3.4 CAST-IN-PLACE CONCRETE

- A. Quality Assurance:
 - 1. Concrete Mix Designs: The Testing Laboratory shall review the submitted mix designs for conformance to the specifications and for suitability for use in the project.
 - 2. Preinstallation Meetings: The Testing Laboratory shall attend the preinstallation meetings as noted in Specification 033000 "Cast-in-Place Concrete."
- B. Source Inspection:
 - 1. Concrete Batch Plant Inspection: An initial batch plant inspection shall be made by the Testing Laboratory prior to the start of concrete work. The scope of batch plant inspection shall include the following:
 - a. Inspection of Batch Plant Facilities: The Testing Laboratory shall inspect batch plant facilities proposed for use in the work and report in writing inspection results to the Architect, Engineer, and Owner for approval. The inspection shall confirm the batch plant conforms to the standards set forth in ASTM C 94 and can show proof of certification by the National Concrete Ready Mix Association. Inspection shall include:
 - 1) Batch Plant operations and equipment.
 - 2) Truck mixers.
 - 3) Scales.
 - 4) Stockpile placement.
 - 5) Material storage.
 - 6) Admixture dispensers.
 - b. Multiple Batch Plants: The Contractor shall reimburse the Owner for the costs accrued to the Testing Laboratory for visits to more than one batch plant.
- C. Field Testing: The following tests shall be completed by the Testing Laboratory:
 - 1. During Concrete Placement:
 - a. Record the amount of water added and note if it exceeds the amount allowed to be added shown in the approved mix design.
 - b. Mold concrete test cylinders as specified below in Paragraph 3.a.
 - c. Perform tests to determine slump, concrete temperature, unit weight, and air entrainment as specified below.
 - d. Record information for concrete test reports as specified below.
 - e. Pick up and transport to Laboratory cylinders cast the previous day.
 - 2. After Concrete Placement:
 - a. In-situ Concrete Strength Verification for Form Stripping: The Testing Laboratory shall perform the tests necessary to determine the concrete strength prior to form stripping:

- 1) If concrete strength for form stripping is to be determined using field-cured cylinders, the cylinder shall be broken at the time of form removal as directed by the Contractor.
- 2) If concrete strength for form stripping is to be determined using the Maturity Method, the Testing Laboratory shall verify that the requirements of ASTM C 1074 are being followed and that the proper criteria for determining concrete strength by this method has been established and is being followed.
- b. Investigation of Low Strength Concrete Test Results:
 - 1) Cost of Investigations for Low Strength Concrete: The Contractor shall reimburse the Owner for the costs of investigations of low strength concrete, as defined in Part I above.
 - 2) Scope of Investigations: See Specification Section **033000** "Cast-In-Place Concrete" for the investigations that may be required by the Engineer. The Testing Laboratory will conduct these investigations if required.
- c. Post-Installed Anchors in Concrete:
 - 1) Verify maximum anchor tightening torque for all applicable post-installed anchors.
 - 2) Provide pull tests on individual anchors as specified in the ICC Evaluation Services Report, on the drawings, or as directed by the Engineer-of-Record.
- d. Floor Flatness and Levelness Measuring: Perform tests as defined below.
- e. Testing of Concrete Floor Slabs for Acceptability to Receive an Adhesive-Applied, Low-Permeable Floor Covering: Perform tests as defined below.
- f. Testing of Non-Shrink Grout for Base Plates, Bearing Plates, and Precast Wall Panels:
 - 1) Compressive Strength Tests: Compressive strength of grout shall be determined by testing grout cubes according to the requirements of ASTM C 109 - Modified. Test one set of three cubes at one day, and one set of three cubes at 28 days.
 - 2) Frequency of Testing: One set of cubes (6 cubes) shall be made for every ten base plates and bearing plates or fraction thereof but not less than one set for each day's operation. One set of cubes shall be made for each day's operation of grouting wall panels.
3. Standards for Concrete Tests:
 - a. Concrete Test Cylinders: Mold and test concrete cylinders as described below:
 - 1) Cylinder Molding and Testing: Cylinders for strength tests shall be molded and Laboratory cured in accordance with ASTM C 31 and tested in accordance with ASTM C 39. Cylinders may be either 6" in diameter by 12" or 4" in diameter by 8", however, the diameter of the cylinder shall be at least three times the nominal maximum size of the coarse aggregate in the mix tested. All of the cylinders for each class of concrete shall be of the same dimension for all sets of that class.
 - 2) Field Samples: Field samples for strength tests shall be taken in accordance with ASTM C 172 **at the point of placement**.
 - 3) Quantity of Cylinders: Each set of test cylinders shall consist of a minimum of four standard test cylinders. If concrete strength for form stripping is to be determined using field-cured cylinders, one additional cylinder per set will be required for formed slab and pan-formed beam floors for the purpose of evaluating the concrete strength at the time of form stripping. This cylinder shall be stored on the floor where form removal is to occur under the same exposure conditions as the floor concrete. The cylinder shall be cured under field conditions in accordance with ASTM C 31. Field-cured test cylinders shall be molded at the same time and from the same samples as laboratory-cured test specimens. The Contractor shall reimburse the Owner for the cost of making and testing these cylinders.
 - 4) Frequency of Testing: A set of test cylinders shall be made according to the following minimum frequency guidelines:
 - a) One set for each class of concrete taken not less than once a day.
 - b) Piers, Piles, Underreamed Footings: One set for each 50 cubic yards or fraction thereof.
 - c) Spread Footings: One set for each 50 cubic yards or fraction thereof.
 - d) Pile Caps: One set for each 50 cubic yards or fraction thereof.
 - e) Basement Walls: One set for each 150 cubic yards.
 - f) Floors: One set for each 150 cubic yards or fraction thereof but not less than one set for each 5,000 square foot of floor area.
 - g) Columns: One set for each 50 cubic yards or fraction thereof with a minimum of two sets per floor.

- h) All Other Concrete: A minimum of one set for each 150 cubic yards or fraction thereof but not less than one set for each 5,000 square foot of area for walls.
 - i) No more than one set of cylinders at a time shall be made from any single truck.
 - j) If the total volume of concrete is such that the frequency of testing as specified above would provide less than five strength tests for a given class of concrete, tests shall be made from at least five randomly selected batches or from each batch if fewer than five batches are used.
 - k) The above frequencies assume that one batch plant will be used for each pour. If more than one batch plant is used, the frequencies cited above shall apply for each plant used.
- 5) The cylinders shall be numbered, dated, and the point of concrete placement in the building recorded.
 - 6) For concrete specified on the drawings to reach the required strength at 28 days, break one cylinder of the set at seven days, two 6" by 12" cylinders or three 4" by 8" cylinders at 28 days, and keep one in reserve for testing at the Engineer's direction.
 - 7) For concrete specified on the drawings to reach the required strength at 56 days, break one cylinder of the set at seven days, one cylinder at 28 days, two 6" by 12" cylinders or three 4" by 8" cylinders at 56 days, and one kept in reserve for testing at the Engineer's direction.
 - 8) Cylinder Storage Box: The Contractor shall be responsible for providing a protected concrete cylinder wooden storage box at a point on the job site mutually agreeable with the Testing Laboratory for the purpose of storing concrete cylinders until they are transported to the Laboratory. The box shall be constructed and equipped to maintain the environment specified for initial curing in ASTM C 31.
 - 9) Transporting Cylinders: The Testing Laboratory shall be responsible for transporting the cylinders to the Laboratory in a protected environment such that no damage or ill effect will occur to the concrete cylinders including loss of moisture, freezing temperatures or jarring.
 - 10) Information on Concrete Test Reports: The Testing Laboratory shall make and distribute concrete test reports after each job cylinder is broken. Such reports shall contain the following information:
 - a) Truck number and ticket number.
 - b) Concrete Batch Plant.
 - c) Mix design number.
 - d) Accurate location of pour in the structure.
 - e) Strength requirement.
 - f) Date cylinders made and broken.
 - g) Technician making cylinders.
 - h) Concrete temperature at placing.
 - i) Air temperature at point of placement in the structure.
 - j) Amount of water added to the truck at the batch plant and at the site and whether or not it exceeds the amount allowed by the mix design.
 - k) Slump.
 - l) Unit weight.
 - m) Air content.
 - n) Cylinder compressive strengths with type of failure if concrete does not meet Specification requirements. Seven day breaks are to be flagged if they are less than 60% of the required 28 day strength. 28 day breaks are to be brought to the attention of the Architect and Engineer in writing if either cylinder fails to meet specification requirements.
 - b. Slump Tests: Slump Tests (ASTM C 143) shall be completed at the beginning of concrete placement for each batch plant and for each set of test cylinders made. The slump test shall be made from concrete taken from the end of the concrete truck chute. The concrete shall be considered acceptable if the slump is within the slump tolerance noted on the mix design submittal form for that class of concrete.
 - c. Air Entrainment: Air entrainment tests (ASTM C 231 or C 173, C 173 only for lightweight concrete) shall be made at the same time slump tests are made as cited above. Samples for air entrainment tests shall be taken **at the point of placement**.
 - d. Concrete Temperature: Concrete temperature at placement shall be measured (ASTM C 1064) at the same time slump tests are made as cited above.

- e. Floor Flatness and Levelness Measuring:
 - 1) The Testing Laboratory shall measure the floor for flatness and levelness according to ASTM E 1155.
 - 2) Measurement of the finished concrete surface profile for any test section shall be made when requested by the Representative at his option. Notwithstanding, measurements shall be made within 24 hours after completion of finishing operations. For structural elevated floors measurement shall also be made prior to removal of forms and shores. The Contractor shall be notified immediately after the measurements of any section are complete and a written report of the floor measurement results shall be submitted within 72 hours after finishing operations are complete.
 - 3) The concrete surface profile shall be measured using equipment manufactured for the purpose such as a Dipstick Floor Profiler as manufactured by the Edward W. Face Company in Norfolk, Virginia, F-Meters manufactured by Allen Face & Company in Norfolk, Virginia, optical, or laser means or other method specified in ASTM E 1155.
 - 4) Each floor test section and the overall floor area shall conform to the two-tiered measurement standard as specified herein.
 - a) Minimum Local Value (MLV). The minimum local F_F/F_L values represent the absolute minimum surface profile that will be acceptable in any one floor test section.
 - b) Specified Overall Value (SOV). The specified overall F_F/F_L values represent the minimum values acceptable for all combined floor test sections representing the overall floor.
 - 5) For purposes of this specification a floor test section is defined as the smaller of the following areas:
 - a) The area bounded by column and/or wall lines.
 - b) The area bounded by construction and/or control joint lines.
 - c) Any combination of column lines and/or control joint lines.
 - d) Test sample measurement lines within each test section shall be multidirectional along two orthogonal lines as defined by ASTM E 1155.
 - e) The precise layout of each test section shall be determined by the Testing Laboratory and shall be submitted for Architect/Engineer review and approval.
- f. Testing of Concrete Floor Slabs for Acceptability to Receive an Adhesive-Applied, Low-Permeable Floor Covering:
 - 1) The following tests shall be performed by the Testing Laboratory as a part of quality assurance testing to insure that the proper moisture condition and alkalinity of the substrate has been achieved prior to installing adhesive-applied, low-permeability floor coverings such as vinyl composition tile (VCT), linoleum, sheet vinyl, vinyl-backed carpet, rubber, athletic flooring, synthetic turf, wood, acrylic terrazzo, thin-set tile, epoxy overlays and adhesives, waterproofing, et.al.
 - 2) Moisture Vapor Emission Rate: Perform testing according to ASTM F 1869 to determine if the moisture emission rate from the floor is below the flooring manufacturer's maximum recommended value but not greater than five pounds per 1,000 square feet per 24 hours.
 - 3) Relative Humidity Determination Test: As an alternate to the Moisture Vapor Emission Rate Test, and if agreed to by the Contractor, Architect and Owner, perform testing according to ASTM F 2170 to determine if the relative humidity of the concrete slab is below the flooring manufacturer's maximum recommended value but not greater than 75%.
 - 4) Alkalinity Testing: Perform testing in accordance with ASTM F 710, Paragraph 5.3, to determine if the pH level of the concrete slab surface is below the flooring manufacturer's maximum recommended value but not greater than 10. Perform one test per 1,000 square feet with a minimum of three tests within the total area being tested.
- 4. Evaluation and Acceptance of Concrete:
 - a. Strength Test: A strength test shall be defined as the average strength of two six inch cylinder breaks or three four inch cylinder breaks from each set of cylinders tested at the time indicated above.

- b. Quality Control Charts and Logs: The Testing Laboratory shall keep the following quality control logs and charts for each class of concrete containing more than 2,000 cubic yards. The records shall be kept for each batch plant and submitted on a weekly basis with cylinder test reports:
 - 1) Number of strength tests made to date.
 - 2) Strength test results containing the average of all strength tests to date, the high test result, the low test result, the standard deviation, and the coefficient of variation.
 - 3) Number of tests under specified strength.
 - 4) A histogram plotting the number of strength test cylinders versus compressive strength.
 - 5) Quality control chart plotting compressive strength test results for each test.
 - 6) Quality control chart plotting moving average for strength where each point plotted is the average strength of three previous test results.
 - 7) Quality control chart plotting moving average for range where each point plotted is the average of 10 previous ranges.
 - c. Acceptance Criteria: The strength level of an individual class of concrete shall be considered satisfactory if both of the following requirements are met:
 - 1) The average of all sets of three consecutive strength tests equal or exceed the required f'_c .
 - 2) No individual strength test falls below the required f'_c by more than the greater of 10% of f'_c or 500 PSI.
 - d. If either of the above Acceptance Criteria requirements is not met, the Testing Laboratory shall immediately notify the Engineer by telephone. Steps shall immediately be taken to increase the average of subsequent strength tests.
- D. Field Inspection: The scope of the work to be performed by the inspector on the jobsite shall be as follows:
 - 1. Before Concrete Placement:
 - a. Inspect concrete formwork per Article 3.2.
 - b. Inspect concrete reinforcing per Article 3.3.
 - c. Inspect bolts and rods to be embedded in concrete for proper grade, size, length, and embedment.
 - d. For slabs-on-grade, verify that the moisture retarder is provided, is lapped properly, and is not torn or punctured.
 - e. Verify that there is no standing water in pour area and that all debris has been removed from the area and from the formwork.
 - f. Verify that openings and sleeves in slabs or walls are correct size and location. Verify that the openings are shown on the structural drawings and notify the Engineer immediately of any openings in the field that are not shown on the drawings.
 - g. Verify that horizontal and vertical sleeves through girders, beams, or joists have been approved by the Engineer and that approved reinforcement is provided.
 - h. Verify the tops of previously poured columns and/or walls are 1/2 inch below the deck soffit.
 - 2. During Concrete Placement: Provide continuous monitoring to:
 - a. Upon arrival of concrete, inspect the concrete to verify that the proper concrete mix number, type of concrete, concrete strength is being placed at the proper location. Verify that the mix meets the project specifications and is not over 90 minutes old at the time of placement. Report concrete not meeting the specified requirements and immediately notify the Contractor, Batch Plant Inspector, Architect, Engineer, and Owner.
 - b. Inspect plastic concrete upon arrival at the jobsite to verify proper batching. Observe mix consistency and adding of water as required to achieve target slumps in mix designs. The responsibility for adding water to trucks at the job site shall rest only with the Contractor's designated representative. The Contractor is responsible that all concrete placed in the field is in conformance to the Contract Documents.
 - c. Verify that the Contractor is following appropriate Hot Weather or Cold Weather concreting practices consistent with any extreme environmental conditions at the point of placement in the structure.
 - d. Verify that concrete deposited is uniform and that vertical drop does not exceed six feet and is not permitted to drop freely over reinforcement causing segregation.
 - e. Verify that the formwork has remained stable during the concreting operation.
 - f. Verify that there are no cold joints.
 - g. Verify that the concrete is properly vibrated.
 - h. Inspect bolts embedded in concrete during concrete placement for verification that they have been properly installed to the specified embedment.

- i. Verify that the finishing of the concrete surface is done according to specifications. The Testing Laboratory shall report any irregularities that occur in the concrete at the job site or test results to the Contractor, Architect, Owner, and Engineer.
3. After Concrete Placement:
 - a. Verify that the curing process is according to Specifications and that any curing compound used is applied in accordance with the manufacturer's recommendations.
 - b. Verify that sawcut control joints in slab-on-grades are cut within 12 hours of placement.
 - c. Post-Installed Anchors in Concrete: Provide inspection of post-installed anchor installations at the frequency noted in the specifications and in accordance with the published, currently valid, Evaluation Service Report (ESR) for each anchor product. Post-installed anchors include anchors and reinforcing steel. Inspection of post-installed anchors shall include but not be limited to the following:
 - 1) Periodic Inspection: Verify initial installation of post-installed anchors in concrete for each individual installer with each individual anchor product in accordance with the requirements stated below for each type of anchor. Periodically inspect anchor installation after the initial verification.
 - 2) Continuous Inspection: Verify each installation of post-installed anchors in concrete in accordance with the requirements stated below for each type of anchor.
 - 3) All Post-Installed Anchors: Verify that the anchor is installed in accordance with manufacturer's printed installation instructions as well as the following design requirements.
 - a) Concrete type, concrete strength and concrete thickness are in accordance with design drawings.
 - b) Anchor manufacturer and product, including material, is in accordance with design drawings or approved substitution.
 - c) Anchor diameter, length and installed embedment depth.
 - d) Drill bit type and diameter.
 - e) Anchor edge distance and spacing.
 - f) Hole diameter and depth.
 - g) Hole cleaning procedure and cleanliness.
 - h) Anchor maximum tightening torque.
 - 4) Adhesive Anchors: In addition to the requirements for All Post-Installed Anchors, verify adhesive identification and expiration date.
 - a) The installation of all adhesive anchors shall be continuously inspected when anchors are subject to sustained tension loads, such as anchors for shelf angles, or when anchors are installed in an upwardly inclined condition.
- E. Causes for Rejection of Concrete: The Contractor shall reject concrete delivered to the site for any of the following reasons:
 1. Wrong class of concrete (incorrect mix design number).
 2. Environmental Conditions: Environmental condition limits shall be as follows unless appropriate provisions in concreting practices have been made for cold or hot weather:
 - a. Cold Weather: Air temperature must be 40°F and rising or the average daily temperature cannot have been lower than 40°F for 3 consecutive days unless the temperature rose above 50°F for at least one-half of any of those 24 hour periods.
 - b. Hot Weather: Environmental conditions must be such that cause an evaporation rate from the concrete surface of 0.2 lb./sq. ft./hr. or less as determined by Figure 2.1.5 in ACI 305R-91.

Concrete may be placed at other environmental condition ranges only with approval of the job inspector for the Testing Laboratory or other duly appointed representative.
 3. Concrete with temperatures exceeding 95°F shall not be placed in the structure.
 4. Air contents outside the limits specified in the mix designs.
 5. Slumps outside the limits specified.
 6. Excessive Age: Concrete shall be discharged within 90 minutes of plant departure or before it begins to set if sooner than 90 minutes unless approved by the Laboratory job inspector or other duly appointed representative.
- F. Concrete Batch Trip Tickets: Concrete batch trip tickets shall be collected and retained by the Contractor. Compressive strength, slump, air, and temperature tests shall be identified by reference to a particular trip ticket. Tickets shall contain the information specified in ASTM C 94. Each ticket shall also show the amount of water that may be added in the field for the entire batch that will not exceed the specified water cement ratio for the design mix. The Contractor and Testing Laboratory shall immediately notify the Architect/Engineer and each other of tickets not meeting the criteria specified.

3.5 STRUCTURAL STEEL

A. Scope of Work:

1. Contract Obligations:

- a. Owner Responsibility: The Owner shall pay for initial shop and field inspections and tests as required during the fabrication and erection of the structural steel.
- b. Testing Laboratory Responsibility: The inspection by the Testing Laboratory of the Fabricator's work shall be in sequence, timely, and performed in such a manner so that corrections can be made without delaying the progress of the work. Inspections shall be performed by qualified technicians with a minimum of two years of experience in structural steel testing and inspection. Refer to Paragraph 1.9B.4 for special requirements for welding inspectors. The Testing Laboratory shall provide test reports of inspections. All test reports shall indicate types and locations of defects found during inspection, the measures required and performed to correct such defects, statements of final approval of welding and bolting of shop and field connections, and other fabrication and erection data pertinent to the safe and proper welding and bolting of shop and field connections. Weld inspection reports shall be signed by an inspector with current certification as an AWS Certified Welding Inspector (CWI). In addition to the parties listed in this Specification the Fabricator and Erector shall receive copies of the test reports.
- c. Rejection of Material or Workmanship: The Owner, Architect, Engineer, and Testing Laboratory reserve the right to reject any material or workmanship not in conformance with the Contract Documents at any time during the progress of the work. However, this provision does not allow waiving the obligation for timely, in sequence inspections.

B. Quality Assurance:

1. Verify the fabrication shop's certification from AISC.
2. Review field welder qualifications by certification or verify by retesting. Obtain welder certificates.

C. Source Testing: The Testing Laboratory shall provide the following tests at the designated fabrication shops:

1. Test welds completed in the shop according to Paragraph G "Weld Testing" below.
2. Test bolted connections completed in the shop according to Paragraph I "High-Strength Bolt Testing."

D. Field Testing: The Testing Laboratory shall provide the following tests in the field:

1. Test welds completed in the field according to Paragraph G "Weld Testing:" below.
2. Test bolted connections completed in the field according to Paragraph I "High-Strength Bolt Testing."
3. Perform bend tests on completed shear connectors attached to beams as required according to procedures outlined in AWS D1.1. In addition, perform field bend tests on an additional 2% of completed shear connectors on each beam but not less than one connector per beam.

E. Field Inspection: The Testing Laboratory shall provide the following inspections in the field:

1. Inspect galvanized HSS and other cold-worked structural steel members for cracking or other damage resulting from galvanizing process. Endeavor to complete inspections prior to erection of these members. Immediately notify Contractor and Architect/Engineer of any irregularities discovered.
2. Provide continuous or periodic monitoring of field welding as described below in Paragraph F "Weld Inspection and Process Monitoring."
3. Provide continuous or periodic monitoring of field bolting as described below in Paragraph H "High-Strength Bolt Inspection and Process Monitoring" of high-strength bolt installation in pre-tensioned or slip-critical joints using turn-of-the-nut without matchmarking or calibrated wrench method of bolt installation.
4. Inspect welded or bolted connections that were completed, but not inspected, in the shop. Perform inspections according to Paragraph F "Weld Inspection and Process Monitoring" and/or Paragraph H "High-Strength Bolt Inspection and Process Monitoring" as appropriate.
5. Obtain the planned erection procedure, and review with the Erector's supervisory personnel.
6. Check the installation of base plates for proper leveling, grout type, and grout application.
7. Check structural steel as received in the field for possible shipping damage, workmanship, and identification marking to conform to AISC 360 for structural steel and specified ASTM standards for other steel.
8. Verify that surveys are occurring as specified to check plumbness and frame alignment as erection progresses. Review the submitted survey report.
9. Periodically inspect the steel frame for such items as bracing and stiffening details, member locations, and joint details at each connection for compliance with approved construction documents.

10. Inspect 100% of the column compression and base joints for verification that gaps in contact bearing do not exceed 1/16 inch. Gaps greater than 1/16 inch but less than 1/4 inch shall be reported to the Owner and Engineer for assessment. All gaps greater than 1/4 inch shall be shimmed according to Specification **051200** "Structural Steel Framing."
 11. Endeavor to guard the Owner against the Contractor cutting, grinding, reaming, or making any other field modification to structural steel without the prior approval of the Engineer. Report any noted unauthorized modifications to the Owner and Engineer.
- F. Weld Inspection and Process Monitoring: The Testing Laboratory shall make the following inspections of the welds and welding processes. Welds performed in the fabricating shop may be inspected in the field unless continuous monitoring of the welding process is herein specified or if access in the field due to other work or shop finishes makes field inspection impractical:
1. Approve Welding Procedure Specifications submitted by the Contractor. Approve any changes submitted by the Contractor to any WPS that has already been approved. Obtain the Welding Procedure Qualification Record (WPQR) for each successful WPS qualification.
 2. Periodically verify welding electrodes to be used and other welding consumables as the job progresses.
 3. Periodically observe joint preparation, assembly practice, welding techniques including preheating and sequence, and the performance of welders with sufficient frequency to assure compliance with code and contract document requirements. Check preheating to assure conformance with AWS D1.1, Section 5.6. Verify procedure for control of distortion and shrinkage stresses.
 4. Continuously observe joint preparation and fit up, backing strips, and runout plates for welded moment connections and column splices.
 5. Periodically provide visual inspection of the root pass of partial and complete joint penetration welds.
 6. Visually inspect 100 % of welds for proper size, length, location, and weld quality in accordance with AWS D1.1 requirements. Unless specifically noted otherwise, all welding shall be considered statically loaded nontubular connections.
 7. Visually inspect 100% of completed shear connectors on each beam.
 8. Visually inspect 100% of the welds of anchors to embedded plates that are to be cast into concrete elements.
 9. In addition to the inspections above, perform the following:
 - a. Continuously monitor and observe joint preparation, assembly practice, welding techniques including preheating and sequence, and the performance of welders for 100% of complete and partial joint penetration welds, plug and slot welds, multiple-pass fillet welds, and single-pass fillet welds greater than 5/16 inch. Check preheating to assure conformance with AWS D1.1, Section 5.6. Verify procedure for control of distortion and shrinkage stresses.
 - b. Periodically monitor welding of single-pass fillet welds that are less than or equal to 5/16 inch.
 - c. Periodically monitor the welding of headed studs to floor beams.
- G. Weld Testing:
1. Perform nondestructive examination services using a qualified technician with the necessary equipment to perform the following:
 - a. Nondestructive examination conducted in accordance with the specific requirements for the item being examined including radiographic (RT), ultrasonic (UT), magnetic particle (MT), or dye-penetrant inspection (PT). Nondestructive inspection procedures shall conform to AWS D1.1.
 - b. Interpret, record, and report results of the nondestructive tests.
 - c. Mark for repair, any area not meeting Specification requirements. Correction of rejected welds shall be made in accordance with AWS D1.1.
 - d. Re-examine repair areas and interpret, record, and report the results of examinations of repair welds.
 - e. Verify that quality of welds meet the requirements of AWS D1.1.
 2. Fillet Welds: Provide the following:
 - a. MT test a minimum of 10% of the length of each fillet weld exceeding 5/16".
 - b. Periodic MT testing of representative fillet welds 5/16" and less but need not exceed 10% of all such welds, except as required for high rejection rates as indicated in the following paragraph.
 - c. Increase MT testing rate for welders having a high rejection rate as required to ensure acceptable welds.
 3. Partial Joint Penetration (PJP) Welds, including Flare-Bevel Groove Welds: Provide the following:
 - a. MT test a minimum of 25% of the length of each PJP weld exceeding 5/16" effective throat.

- b. Periodic MT testing of representative PJP welds 5/16" and less but need not exceed 10% of all such welds, except as required for high rejection rates as indicated in the following paragraph.
 - c. Increase MT testing rate for welders having a high rejection rate as required to ensure acceptable welds.
 - 4. Complete Joint Penetration (CJP) Welds: Provide the following:
 - a. All CJP welds exceeding 5/16" thickness shall be 100% UT tested per AWS D1.1 Clause 6 Part F. The Testing Laboratory shall review the CJP joints to determine where geometry or accessibility precludes the use of standard scanning patterns per AWS D1.1 Clause 6 Part F. At these locations the testing laboratory shall develop and submit for approval a written testing procedure in accordance with AWS D1.1 Annex S.
 - b. Periodic MT testing of representative CJP welds 5/16" and less not to exceed 10% of all such welds, except as required for high rejection rates as indicated in the following paragraph.
 - c. Increase MT testing rate for welders having a high rejection rate as required to ensure acceptable welds.
 - 5. Acceptance Criteria:
 - a. Visual, MT, PT shall be per AWS D1.1 Table 6.1.
 - b. UT testing shall be per AWS D1.1 6.13.1 and Table 6.2.
 - 6. Welds of Anchors to Embedded Plates:
 - a. Headed Studs: Perform field bend tests according to AWS D1.1 on 2% of the studs welded to plates, but not less than one stud per plate.
 - b. Deformed Bar Anchors: Perform MT testing on 10% of deformed bar anchors larger than #5 bar.
 - 7. The costs of repairing defective welds and the costs of retesting by the Testing Laboratory providing services for the Owner shall be borne by the Contractor. If removal of a backing strip is required by the Testing Laboratory to investigate a suspected weld defect, such cost shall be borne by the Contractor.
 - H. High-Strength Bolt Inspection and Process Monitoring: The Testing Laboratory shall perform the following inspections for connections joined with high-strength bolts. Bolting performed in the shop may be inspected in the field unless continuous monitoring of the bolting operation is specified herein:
 - 1. Observe preinstallation verification testing of the pretensioning method to be used in accordance with the requirements of the "Specification for Structural Joints Using High-Strength Bolts".
 - 2. Check daily the calibration of impact wrenches used in field bolted connections.
 - 3. Inspect bolt installation for 100% of high strength bolted connections according to inspection procedures outlined in the "Specification for Structural Joints Using High-Strength Bolts".
 - 4. Monitoring of Bolting Installation:
 - a. Periodic Monitoring: All other joint types and bolt installation methods shall be monitored on a periodic basis.
 - I. High-Strength Bolt Testing: The Testing Laboratory shall perform the following tests for connections joined with high-strength bolts:
 - 1. Perform Arbitration Testing according to procedures outlined in the "Specification for Structural Joints using High-Strength Bolts" when a disagreement exists between the Testing Laboratory and the Fabricator as to the minimum tension of installed bolts that have been inspected according to paragraph below.
- 3.6 STEEL DECKING
 - A. Field Inspection:
 - 1. Check steel deck as received in the field for possible shipping damage, workmanship, and identification marking to conform to specified ASTM standards for steel deck.
 - 2. Periodically monitor the method of attaching the steel floor and roof decking to the structural frame.
 - 3. Visually inspect 100% of the welding or other attachment method of steel deck to the structure and at sidelaps.
- 3.7 EARTHWORK
 - A. Field Testing: Testing of the remediated soils shall be in accordance with the Geotechnical Engineer.

END OF SECTION

SECTION 01 50 00

TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

- A. Specified Provisions Provided by Contractor:
 - 1. Temporary barriers, barricades, covered walkways, fencing, exterior closures, and interior closures.
 - 2. Temporary field offices.
 - 3. Cleaning during construction.
 - 4. Access roads and approaches.
 - 5. Temporary elevator.
 - 6. Temporary sanitary facilities.
 - 7. Temporary heating and ventilating after enclosure.
 - 8. Temporary electrical service and distribution system for power and lighting.
 - 9. Temporary telephone service.
- B. Each Contractor: Coordinate provisions with General Contractor and provide following items as necessary for execution of their Work, including associated costs:
 - 1. Construction aids.
 - 2. Temporary fire protection, dust control, erosion and sediment control, water control, noise control, and other necessary temporary controls.
 - 3. Temporary barriers, barricades, and similar devices as necessary for safety and protection of construction personnel and public.
 - 4. Upon General Contractor's approval, may provide temporary field office, including electrical service and temporary telephone.
 - 5. Temporary tree and plant protection.
 - 6. Temporary heating prior to building enclosure.
 - 7. Electrical service required in addition to temporary service and distribution provided by General Contractor.

1.2 DEFINITIONS

- A. MOLD GROWTH PRODUCTS: See Section 01 42 00 - References.

1.3 QUALITY ASSURANCE

- A. Environmental Protection: Provide environmental protection as required by authorities having jurisdiction and as indicated in the Contract Documents. Coordinate with requirements of the following:
 - 1. Section 01 74 19 – Construction Waste Management.
 - 2. Section 01 81 19 – Construction Indoor Air Quality (IAQ) Management.
- B. Comply with applicable laws and regulations of authorities having jurisdiction.
 - 1. Arrange for authorities having jurisdiction to inspect and test each temporary utility before use. Obtain required certifications and permits.

1.4 SCHEDULING

- A. Prepare schedule indicating dates for implementation and termination of each temporary utility.
- B. At earliest feasible time, when acceptable to Owner, change over from use of temporary utility service to use of permanent utility service.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Materials may be new or used suitable for intended purpose, but of adequate capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.

- 1. Water: Provide potable water approved by local health officials.

2.2 CONSTRUCTION AIDS

- A. Provide scaffolds, staging, ladders, stairs, ramps, runways, platforms, railings, chutes, and other devices and equipment necessary to facilitate execution and construction contract administration of Work.
- B. Provide cranes, hoists, rigging, material lifts, and other conveyances or apparatus as necessary to facilitate execution of Work.
- C. Provide storage areas and sheds sized to storage requirements for products of individual Sections, allowing for access and orderly provision for maintenance and inspection of products to requirements of Section 01 60 00.

2.3 TRAFFIC REGULATION SIGNS, SIGNALS, AND DEVICES

- A. Provide signs, signals, traffic cones and drums, flares and lights, and flag equipment of types required by authority having jurisdiction.

2.4 TEMPORARY FIELD OFFICES

- A. Field Office: Structurally sound, weathertight, equipped with heating, air conditioning, ventilation system, electric power outlets, lighting, and telephone.
 - 1. Temporary Building: Portable or mobile buildings may be used. Floor raised above ground.
 - 2. When permanent facilities are enclosed with operable utilities, relocate office into building upon written approval of Owner, and remove temporary building.
 - 3. Size: As required for Contractor's operations.
 - a. Provide space in field office for project meetings, with table and chairs to accommodate six people.
 - 4. Equip with telephone modem and facsimile telecopier.
- B. Field Office for Owner and Architect:
 - 1. Provide one separate self-contained space for use by Owner and Architect of same basic construction as field office.
 - 2. Equip with one standard size desk with lockable drawers, swivel chair, one lockable four drawer metal file, one drafting table and stool, and one plan rack.s

2.5 TEMPORARY ENCLOSURES AND PARTITIONS

- A. Partition Performance Requirements:
 - 1. Maximum Flame Spread Index: 75, ASTM E84.
 - 2. Minimum Fire Rating: 2 hours.
 - 3. Insulate to R-Value of 11 at 75 F mean temperature in accordance with ASTM C518.
 - 4. Sound Transmission Class: 35, ASTM E90. B. Partition Framing Options:
- B. Lumber: Stud grade; fire-retardant treated, AWPAC20.
- C. Steel Stud Framing, ASTM C645.
- D. Gypsum Board: ASTM C1396; Type X for fire-rated assemblies.
- E. Plywood: PS-1; fire-retardant treated in accordance with AWPAC27.
- F. Polyethylene: Fire-rated, reinforced, polyethylene sheet.

G. Doors:

1. Non-fire Rated Partitions: Wood or steel doors and frames.
2. Fire Rated Partitions: Steel doors and frames; 1-1/2 hour rating.
3. Hardware: Provide each door with hinges, lockset, closer, and dust-tight gasketing.
 - a. Provide fire-rated hardware on doors in fire-rated partitions.
 - b. Construction masterkey locksets.

H. Temporary Fence:

1. Contractor Options:
 - a. Solid Wood Fencing:
 - 1) Plywood: Exterior type PS-1, C-D Plugged, thickness as appropriate for framing requirements.
 - 2) Framing: 4 by 4 inch treated posts and 2 by 4 inch rails, spaced to support plywood.
 - 3) Paint: Exterior type in colors directed by Architect. Required at surfaces exposed to public view.
 - 4) Provide FSC certified lumber.
 - b. Open Mesh Fencing:
 - 1) Fabric: Hot dip galvanized steel wire woven into 2 inch mesh.
 - 2) Framing: Galvanized steel posts or roll-formed sections spaced to support fabric. Equip with top rail and bottom tension wire.
2. Equip with gates and locks.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install, maintain, and operate temporary utilities and services to ensure continuous operation. Modify and extend systems as Work progresses.
- B. Install temporary facilities and controls in manner to produce reasonable uniform appearance, structurally adequate for required purposes, and properly maintained.
 1. Modify and relocate temporary facilities and controls as necessary to accommodate progress of Work.

3.2 TEMPORARY FIELD OFFICE

- A. Field Office: Provide temporary field offices at time of project mobilization. Maintain during progress of Work.

3.3 TEMPORARY CONTROLS

- A. Fire Protection: Comply with local fire protection code and governing authorities.
 1. Provide and maintain fire protection including, without limitation, fire extinguishers and other appropriate fire-fighting equipment ready for immediate use.
 2. Distribute equipment around site, particularly in immediate vicinity of performance of welding or similar hazardous Work.
 3. Store gasoline and other flammable liquids in Underwriter's Laboratories listed safety containers in conformance with recommendations of National Board of Fire Underwriters. Do not store gasoline or other flammable liquid within building.
 4. Coordination with permanent fire protection systems:
 - a. At earliest feasible date in each area of Project, complete installation of permanent fire protection system, including connected services, and place into operation and use.
 - b. Instruct key construction personnel on use of systems.
- B. Dust Control: Execute Work by methods to minimize raising dust from construction operations.

1. Provide positive means to prevent air-borne dust from dispersing into atmosphere.
- C. Erosion and Sediment Control: Plan and execute construction by methods to control surface drainage from cuts and fills, from borrow, and from waste disposal areas. Prevent erosion and sedimentation.
 1. Minimize amount of bare soil exposed at one time.
 2. Provide temporary measures such as berms, dikes, silt fences, drains, and other soil and erosion control devices required by authorities having jurisdiction .
 3. Construct fill and waste areas by selective placement to avoid erosive surface silts or clays.
 4. Periodically inspect earthwork to detect evidence of erosion and sedimentation; promptly apply corrective measures.
 5. Comply with sediment and erosion control plan shown on Drawings.
- D. Water Control: Provide methods to control surface water to prevent damage to site or adjoining properties.
 1. Control fill, grading, and ditching to direct surface drainage away from excavations, pits, tunnels, and other construction areas; and to direct drainage to proper runoff.
 2. Protect site from puddling or running water.
 3. Maintain excavations free of water. Provide, operate, and maintain pumping equipment.
- E. Security: Provide security and facilities to protect Work from unauthorized entry, vandalism, and theft.
 1. Conduct operations in manner to avoid risk of loss, theft, or damage by vandalism.
 2. Comply with security program established by Owner.
- F. Pest and Rodent Control: Provide methods, means, and facilities to prevent pests, insects, and rodents from accessing or invading premises so Project will be relatively free of them at Substantial Completion.
- G. Pollution Control: Provide methods, means, and facilities to prevent contamination of soil, water, and atmosphere from discharge of noxious, toxic substances, and pollutants produced by construction operations.

3.4 TRAFFIC REGULATION

- A. Construction Parking Control:
 1. Temporary Parking Areas: Provide temporary parking areas to accommodate construction personnel.
 2. Existing on-site streets and parking facilities may not be used by construction personnel.
 3. When site space is not adequate, provide additional remote off-site parking.
 4. Control vehicular parking to prevent interference with public traffic and parking, access by emergency vehicles, and Owner's operations.
 5. Monitor parking of construction personnel's vehicles. Maintain vehicular access to and through parking areas.
 6. Prevent parking on or adjacent to access roads, in pedestrian paths or sidewalks, or nondesignated areas.
- B. Flagpersons: Provide trained and equipped flagpersons to regulate traffic when construction operations or traffic encroaches on public traffic lanes.
- C. Flares and Lights: Use flares and lights during hours of low visibility to delineate traffic lanes and guide traffic.
- D. Access Roads and Approaches:
 1. Construct and maintain temporary roads and approaches accessing public thoroughfares to serve construction area.

2. Extend and relocate as Work progress requires. Provide detours necessary for unimpeded traffic flow.
 3. Provide and maintain access to fire hydrants free of obstructions.
 4. Track-equipped vehicles not allowed on paved areas.
 5. Provide means of removing mud from vehicle wheels before entering streets.
 6. Designated existing on-site roads may be used for construction traffic.
 7. Keep streets, drives, and walks adjacent to site and haul routes clean and free of dirt, debris, and litter caused by construction operations.
- E. Temporary Haul Routes: Consult with authority having jurisdiction, establish public thoroughfares to be used for haul routes and site access. 1.
1. Confine construction traffic to designated haul routes.
 2. Provide traffic control at critical areas of haul routes to regulate traffic, to minimize interference with public traffic.
- F. Traffic Signs and Signals: Comply with requirements of authorities having jurisdiction.

3.5 TEMPORARY BARRIERS AND ENCLOSURES

- A. Temporary Barrier: Provide barriers to prevent unauthorized entry to construction areas, to allow for Owner's use of site, and to protect existing facilities and adjacent properties from damage from construction operations.
1. Barricades And Covered Walkways: Provide as required by governing authorities for public rights-of-way and sidewalks.
- B. Protect non-owned vehicular traffic, stored materials, site, and structures from damage.
- C. Temporary Enclosures:
1. Temporary Exterior Enclosures: Provide temporary weathertight enclosure at exterior openings to accommodate acceptable working conditions and protection for products, to allow for temporary heating and maintenance of required ambient temperatures identified in specification Sections, and to prevent entry of unauthorized persons.
 - a. Mold Prevention: Provide temporary weathertight exterior enclosures as required to keep MOLD GROWTH PRODUCTS dry during construction operations.
 - b. Provide access doors with operating hardware and locks.
 - c. Temporary Fence: Provide minimum 6 foot high fence around construction site.
 - d. Tree and Plant Protection: Preserve and protect existing trees and plants designated to remain.
 - 1) Provide 6 foot high barriers around drip line, with access for maintenance.
 - 2) Consult with Architect, remove agreed-on roots and branches which interfere with construction. Employ qualified tree surgeon to remove roots and branches, and to treat cuts.
 - 3) Protect areas within drip lines from traffic, parking, storage, dumping, chemically injurious materials and liquids, ponding, and continuous running water.
- D. Replace trees and plants damaged by construction operations.

3.6 TEMPORARY ELEVATORS

- A. Building elevators will not be available for use.
- B. One new elevator will be made available for use during construction contingent upon compliance with following:
1. Supply and pay for attendants to operate temporary elevator.
 2. Arrange for, obtain, and pay for necessary approvals, manufacturer's acceptance form, inspections, permits and other provisions as necessary for temporary use.

3. Obtain and pay for elevator maintenance service; coordinate provisions and requirements with elevator manufacturer.
4. Furnish, install, and maintain temporary protection of existing components, including cab and entrances.
5. Remove temporary protection devices after temporary use is no longer required. Restore elevator system to original condition. Replace damaged and worn components.

3.7 TEMPORARY WATER SERVICE

- A. Provide, maintain, and pay for water service required for construction operations.
- B. Extend branch piping with outlets located so water is available by hoses with threaded connections. Provide temporary pipe insulation to prevent freezing.
- C. Wash Facilities: Supply with potable water for personnel to wash-up for sanitary condition. Dispose of drainage properly. Provide cleaning compounds appropriate for each condition.

3.8 TEMPORARY SANITARY FACILITIES

- A. Provide and maintain sanitary facilities and enclosures. Provide temporary facilities until such time that designated permanent facilities become available.
- B. Restore existing facilities used during construction to same or better condition than originally found. Replace damaged fixtures, partitions, and accessories.

3.9 TEMPORARY HEATING AND VENTILATING

- A. Provide heating and ventilation as necessary to protect materials, products, and finishes from damage due to temperature or humidity.
 1. Ventilate enclosed areas to assist cure of materials, to dissipate humidity, and to prevent accumulation of dust, fumes, vapors, or gases.
 2. Temporary Heating and Ventilating System: Comply with codes and regulations.
 3. Comply with Section 01 81 19 for Construction Indoor Air Quality (IAQ) requirements.
- B. Except where indicated otherwise in individual Specification sections, maintain minimum ambient temperature of 50 F in enclosed areas where construction is in progress.
 1. Provide heating system as necessary to maintain specified conditions during construction.
- C. Mold Prevention: Provide heating and ventilation as necessary to keep MOLD GROWTH PRODUCTS dry during construction operations until Substantial Completion.
 1. Heat and ventilate as required to dissipate excessive humidity.
 2. Heat and ventilate as required to properly cure and dry materials.
 3. Heat and ventilate as required to dry wet areas and materials before installation of materials susceptible to moisture damage.
 - a. Exception: Exterior skin of exterior enclosure assemblies.
 - b. Use moisture meter to confirm that materials are sufficiently dry.
 - c. Permanent heating plant may be utilized for source of temporary heat, extended and supplemented with temporary heating devices as required.
 - d. Provide and pay for costs of supervision, operation, maintenance, fuel, and energy consumed.
 - e. Use of permanent heating/ventilating and associated distribution systems will be permitted only upon meeting following requirements:
 - 1) Verify that installation is approved for operation, equipment is lubricated, and filters are in place.
 - 2) Equipment installed complete with accessories, started-up, maintained, serviced, and operated in strict accordance with manufacturer's instructions.
 - 3) Provide and pay for regular replacement of filters and worn or consumed parts.

- 4) Operation of permanent systems or any portion thereof to provide temporary heat/ventilation does not constitute acceptance of system or portion of system.
- 5) Immediately before Substantial Completion, completely clean each permanent unit used, install new filters, and perform service functions required for placing units in use and qualifying for specified warranties.

3.10 TEMPORARY ELECTRIC POWER AND LIGHTING

- A. Temporary Electric Service: Provide as required for power and lighting, arrange provisions with utility company, and pay costs for service and energy consumed. Equip service with meter, main disconnect, and over current protection.
 1. Complement existing power service capacity and characteristics as required.
 2. Provide electrical service sized to provide adequate temporary power and lighting for construction operations.
- B. Temporary Electric Power: Provide branch distribution system from temporary power source with distribution boxes and outlets located so that power and lighting is available throughout active work areas.
 1. Permanent receptacles may not be utilized during construction.
 2. Replace receptacle plates and wiring devices damaged during construction.
- C. Temporary Lighting: Provide lighting to conduct construction operations.
 1. Permanent lighting system may not be utilized during construction.
 2. Restore permanent lighting systems used during construction to original condition. Maintain lighting and provide routine repairs.
 3. Replace burned out lamps with new.

3.11 TEMPORARY TELEPHONE SERVICE

- A. Provide telephone service to temporary field office and field office of Owner and Architect at time of Project mobilization.

3.12 TEMPORARY USE OF PERMANENT SYSTEMS

- A. When allowed by Specifications, certain items of new permanent systems (equipment) may be used prior to Substantial Completion.
- B. Prior to operating permanent equipment, notify Architect in writing of intended usage. Verify equipment is approved for operation and equipment is lubricated and ready for operation. Arrange for, obtain, and pay for necessary approvals, manufacturer's acceptance, inspections, permits, and other provisions necessary for temporary use.
- C. Provide and pay for operation, maintenance, and regular replacement of filters, and worn or consumed parts. Use of permanent equipment shall not affect the warranty which begins at Substantial Completion of Project.

3.13 REMOVAL, CLEANING, AND RESTORATION

- A. Removal:
 1. Remove temporary above grade or buried utilities, equipment, facilities, controls, and materials prior to request for Substantial Completion.
 2. Remove temporary paving that is not intended for or acceptable for integration into permanent paving.
 3. In areas intended for landscaping, remove soil and aggregate fill that does not comply with requirements for fill or subsoil in landscaped areas.
 4. Remove materials contaminated with road oil, asphalt, or other compounds harmful to plant growth.
 5. Remove underground installations to a minimum depth of 2 feet. Grade site as indicated.
 6. Waste Management Procedures: Comply with Section 01 74 19.

- B. Clean and repair evidence or indication of installation or use of temporary Work.
1. Repair or replace street paving, curbs, and sidewalks at temporary entrances as required by authorities having jurisdiction.
 2. Restore existing facilities and equipment used during construction to original condition.
Restore permanent facilities and equipment used during construction to specified condition.

END OF SECTION

SECTION 01 57 19

AIRBORNE CONTAMINANTS CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes:
 - 1. Hospital airborne contaminants control policy and procedures.
 - 2. Ceiling access control of airborne contaminants.

1.3 DEFINITIONS

- A. Containment Areas: (Negative Pressurization) As determined by Architect and as shown on the Drawings as Project Areas. Includes area of construction, adjacent staging and storage areas, and passage areas for contractors, supplies, and waste; includes ceiling spaces above and adjacent to construction.
- B. Protection Areas: (Positive Pressurization) As determined by Architect and as shown on the Drawings as Protection Areas. Includes hospital areas adjacent to Containment Area, either occupied or used for passage, as well as areas connected to construction area by mechanical system air intake, exhaust and ductwork.
- C. "Minor" ceiling access is defined as visual observation or minor adjustments or other activity that does not disturb dust. Acoustical panels shall be replaced or access panel shall be closed immediately when the contractor leaves the work site.
- D. "Major" ceiling access describes any other access not defined as "minor."
- E. "Thorough" cleaning of surfaces which become exposed to dust shall be accomplished by the use of either a HEPA-filtered vacuum cleaner or Water Vac with HEPA Filtration on outlet. Note: Wet mop shall not be used since moisture will activate dormant fungus and mold spores in dust.
- F. Negative Air Machine: Portable mechanical units to provide a negative air pressure in the Containment Areas, as specified in this Section.

1.4 SYSTEM DESCRIPTION

- A. Design Requirements: Owner will establish acceptable, baseline levels of airborne contaminants based on air monitoring of existing conditions prior to start of Work.
 - 1. Aspergillosis and related nosocomial fungal infections are caused through inhalation by immuno-compromised patients of aspergillus spores, or other related spores which can be present in the construction environment. The spores are known to be prolifically present in construction dust, debris, and earthwork excavation dust. Control of construction dust, debris, and excavation dust, as required in this Section, is imperative to help prevent outbreaks of aspergillosis or related nosocomial fungal infections in immuno-compromised patients.
 - a. Inhalation of aspergillus spores or other fungal spores by immuno-compromised patients can lead to serious complications and death.
 - b. Aspergillus and other related spores are present in the natural environment and thus are not a risk to healthy construction workers.
 - c. Construction workers are required to attend an orientation session.
 - 2. Airborne contaminants control is critical in hospital areas. Limit dissemination of airborne contaminants produced by construction-related activities in order to provide protection of

immuno- compromised and other patients, staff, diagnostic operations, or sensitive procedures or equipment, from possible undesirable effects of exposure to such contaminants.

- a. Dust in ceilings and construction debris contains fungus spores. Conduct construction activities causing disturbance of existing dust, or creating new dust, or other airborne contaminants in tight enclosures cutting off flow of particles into patient areas.
 - b. Secure ceilings and walls in Protection Areas and other indicated areas. Follow specified procedures if access into ceiling in occupied areas is required.
3. Design system, including containment and disposal procedures, equipment, and related work, to meet specified requirements.

1.5 SUBMITTALS

- A. Progress Schedules: Submit work areas and procedure schedules for containment of airborne contaminants.
- B. Work Plan: Drawings and details of construction of necessary temporary barriers and description of procedures to be used to achieve and maintain control of construction-related airborne contaminants.
- C. Product Data: Submit manufacturer's literature on:
 1. Vacuum cleaning machines, air compressors, pressure washers, and cleaning related equipment accessories.
 2. Biocide sanitizing fluid, coil cleaning solutions, degreasers, encapsulators.
 3. Duct access doors, access panels.
 4. Labels.
- D. Record Drawings: Upon completion of the project, submit one set of red-marked duct layout drawings showing the location of new access holes and openings installed in the duct systems to accommodate the cleaning process.
- E. Test Reports: Written results of testing specified in Part 3.

1.6 QUALITY ASSURANCE

- A. Air Samples: Owner will provide baseline particle counts and conduct periodic air sampling of areas during construction to monitor effectiveness of containment procedures.
- B. Contractor Qualifications: Company specializing in performing Work of this extent and nature with minimum five years experience.
- C. Regulatory Requirements: Ensure flammable components comply with applicable portions of local, state, and federal codes, laws, and ordinances for flame spread and smoke developed indices.

1.7 OWNER'S USE OF SPACE

- A. Accomplish work with a minimum of interruptions to Owner's operation within the building. Coordinate work with the Owner's Representative.
- B. In the event HVAC systems provide space heating the duration of system shut downs must be limited or the contractor shall provide temporary heating systems to ensure room temperatures of at least 50 degrees F. are maintained.

1.8 PROTECTION

- A. If work is being done above a lay-in ceiling and if work must be performed while the space below is occupied, provide temporary work surfaces to provide a safe working platform and protect the ceiling and the spaces below from falling objects and materials. Take necessary precautions to protect the people and spaces below from injury due to the contractor's operations.

- B. Exercise caution when handling fluids, particularly heating water, in the interstitial space. When working with fluids provide a water-tight barrier beneath the work area to catch and retain spillage before it reaches the ceiling below.
- C. Notify the Architect at least 48 hours prior to commencing work in ceiling or interstitial spaces above occupied areas to allow at-risk patients to be relocated or protected.

1.9 COORDINATION OF WORK

- A. Submit a cleaning work schedule for each HVAC unit/duct system a minimum of 30 days prior to beginning work. Indicate dates, times and activities for each phase or portion of the work. In addition, describe in detail what systems, fans, and HVAC equipment will be affected (no air flow) and what rooms, spaces or areas will require access. Include procedures proposed for the project.
- B. Coordinate work activities with other affected trades and Subcontractors.
- C. Do not begin cleaning activities until other construction activities are complete except for Testing, Adjusting, and Balancing (TAB) activities in which case TAB activities shall be performed after acceptance of cleaning activities.
- D. Prearrange and preschedule with the Architect and Owner's Representative the switching off of any HVAC unit to commence a cleaning operation.

1.10 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 00.
- B. Before Work begins, conduct an orientation session including presentation by Owner's Representative for training and instructing construction and related personnel on precautions to be taken. Do not permit workers to access work areas until successfully completing orientation session.

PART 2 - PRODUCTS

2.1 EQUIPMENT

- A. Negative Air Machines: Include prefilters, final filters, HEPA filters and filter static pressure gauges.
 - 1. HEPA filters: 99.97 percent efficient at 0.3 micron particle size.
 - 2. Acceptable product and manufacturer:
 - a. Micro Trap MT-C Negative Air Filtration Units by Micro-Trap, Inc., 38 North Pine Avenue, Maple Shade, NJ 08052.
 - b. CRSI 2000 by Control Resource System Incorporated, 670 Marine Drive, Michigan City, IN 46360, 1-800-418-1264.
- B. Air Pressure Monitor: Differential switch/gauge to monitor differential pressure between construction Containment Area and Protection Area.
 - 1. Install switch/gauge in NEMA rated enclosure.
 - 2. Provide power wiring, transformers, and relays to operate the system.
 - 3. Provide audio-visual alarm that will activate upon sensing pressure differences beyond the range set points.
 - 4. Provide a switch that will enable activation of either audio, visual, or both alarms.
 - 5. Provide a manual reset switch to reset gauge after an alarm condition.
 - 6. Acceptable product and manufacturer, Switch/gauge:
 - a. Dwyer Model 3000-0 with range of 0 to 0.5 inches of water gauge, and high-low adjustable set points.
- C. Vacuum Collection Machine:
 - 1. Portable vacuum cleaning machine designed for duct cleaning mounted on wheels.

2. Filter section with filter bags and final HEPA exhaust filtration.
 3. Anti-spark construction, made of non-ferrous material.
 4. Fan shall have a minimum of 4,000 CFM 1t 1.5" static pressure.
 5. Unit shall maintain a minimum 1" S.P. on duct at all times.
- D. Air Compressor:
1. Portable air compressor unit consisting of compressors, tanks, and controls ASME rated.
 2. Minimum Capacity: 17 CFM free air at 175 psig.
 3. Accessories: Hoses and velocity nozzles.

2.2 MATERIALS

- A. Carpet or Mats: Provide carpets or mats at enclosure entrances, vacuumed or changed as often as necessary to prevent accumulation of dust. At Owner's option, provide adhesive faced contamination control mats with disposable sheets in lieu of vacuumed mats. Vacuuming of areas not under negative pressure shall be with a certified HEPA-filtered vacuum.
1. Acceptable product and manufacturer – Adhesive faced contamination control mat:
 - a. Tacky Mat by Liberty Industries, 133 Commerce Street, East Berlin, CT 06023, 1-800-828- 5656.
- B. Dust Caps: Block off existing ventilation ducts within construction area. Capping method shall be dust tight and withstand air flow.
- C. Portable Enclosures: Sufficient to seal off area tight to ceiling.
- D. Temporary Prefabricated Partition for work in Sterile Corridors:
1. Acceptable product and manufacturer:
 - a. Kontrol Kube by Fiberlock Technologies, Inc., P.O. Box 432, Cambridge, MA, 617-876-8020, including the following:
 - 1) Adjustable Aluminum Frame #6440.
 - 2) Vinyl Enclosure #6442.
 - 3) Wheel Base Platform #6443.
 - 4) Inspection window and pressure differential porthole.
 - 5) Nilfisk 87 cfm vacuum device and manometer.
- E. Polyethylene: ASTM D210, Minimum thickness: 6 mil, FR treated.
- F. Biocide: Copper-8-quinolinolate.
1. Acceptable product and manufacturer: MAG Chemical, Vero Beach, FL.
- G. Spray Adhesive:
1. Acceptable product and manufacturer: Ram-Tack Adhesive from Aramsco, 906 Gray Street, Elgin, IL 60120.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Before any demolition or construction begins in occupied areas, a complete field review of all airborne contaminant control policies will be conducted. A checklist will be filled out and signed by the Airborne Contaminant Control Nurse and the Contractor, confirming that the area is ready for work to begin.
- B. Provide temporary barriers and other controls to control airborne contaminants. Extend barriers above ceilings as required to seal off and contain airborne contaminants.

3.2 GENERAL CEILING ACCESS DIRECTIVES

- A. Perform Work in accordance with Section 01 73 00, applicable local standards, and approved shop drawings and work plans.
- B. Report to Owner and fill out a ceiling access form. Attach approved work tag to the ceiling access enclosure before work will be allowed to proceed.
- C. Work Tag:
 - 1. Attach a completed, approved work tag on the ceiling access enclosure before work can proceed.
 - 2. Remove work tag only after work is done and cleanup completed.
 - 3. Tags issued from Owner's representative shall be returned the same day to the office from which it was issued, after work and cleanup for the day has been completed.
- D. Spray top of ceiling panels to be removed, and surrounding affected panels, with fine water mist to settle dust prior to removal.
- E. Inform Head Nurse or department manager so that patient room doors near ceiling work will be kept closed while work is in progress.
- F. Contact Owner's Representative regarding ceiling access problems.
- G. Special Techniques:
 - 1. Provide special control of sources of contaminants and waste as determined by Owner's Representative. Contain waste materials during removal, bagging, wrapping, and chute use.
 - 2. Use wet cleaning methods and HEPA-filtered vacuum cleaners to minimize release of airborne contaminants. Disinfect contaminant and protection areas to effect final cleaning.
 - 3. Perform cleaning of heating, ventilating and air conditioning (HVAC) systems and ductwork.
 - 4. Sealing of Openings: Use duct tape or other impenetrable sealant to seal barrier wall seams, cracks around window and door frames, exhaust system ductwork, pipes, joints, and ducts. Seal penetration of dustproof enclosures on all sides and 360 degrees around penetrating objects.
 - 5. Dust Control: Take appropriate steps throughout the term of the Project to prevent airborne dust due to work under this contract. Apply water wherever practical to settle and hold dust to a minimum, particularly during demolition and moving of materials. Prevent accumulation of standing water or saturation of any materials. Use of chemical palliatives is not permitted without permission of Owner's Representative.
 - a. Spray surfaces with water during dust-producing interior demolition activities. Hard surface floors in work area, adjacent hallways and passage areas require vacuuming with HEPA- filtered vacuum cleaners during demolition and construction. Protect adjacent carpeted areas with plastic and plywood, and vacuum with HEPA-filtered vacuum cleaners.
 - b. Vacuum walk-off mats at least once daily.
 - c. Execute work by methods to minimize raising dust from construction operations. Provide positive means to prevent airborne dust from dispersing into atmosphere.
- H. Airborne contaminant enclosures or infection control enclosures shall be dust-tight. Immediately remove dust tracked outside of enclosure. Clean areas outside enclosure with HEPA-filtered vacuum or other approved method.
- I. Implement the following procedure when construction personnel are required to pass through a Protected Area to enter the Containment Area.
 - 1. Provide air lock entry vestibules in dustproof enclosures when shown on Drawings or required by Owner's Representative.

2. Personnel shall wear protective clothing while passing through the Protective Area. Protective clothing shall be removed in the air lock vestibule prior to entering the Containment Area.
3. When exiting the Containment Area, personnel shall put protective clothing on before reentering the Protected Area.
- J. Contractor Personnel: Instruct personnel to refrain from tracking dust into adjacent hospital areas or opening windows or doors that may allow airborne contaminants into adjacent hospital areas.
- K. Exterior Work: Direct exhaust from equipment away from building air intakes. Ensure that filters on building air intakes are operational and protected from excessive amounts of airborne contaminants.
- L. Ceiling access panels opened for investigation beyond sealed areas shall be replaced immediately when unattended.
 1. Whenever access panels are opened in occupied areas, for work above ceiling, provide portable enclosure enclosing ladder and seal off opening, fitted tight to ceiling.
- M. Provide thorough cleaning of existing, exposed surfaces before start of Owner's room occupancy.
- N. Removal of construction barriers and ceiling protection shall be done carefully, possibly outside of normal work hours. Vacuum and clean adjacent surfaces after removal.
- O. Perform vacuuming of areas not under negative pressure with a certified, Owner approved, HEPA-filtered vacuum.

3.3 AIRBORNE CONTAMINANT CONTROL ENCLOSURES AND BARRIERS

- A. Install dustproof enclosures for work as indicated and when required to protect areas occupied by Owner from dust, debris, and damage.
- B. It is the Contractor's responsibility to determine when a dustproof enclosure is required to protect any adjoining area; however, Contractor shall provide a dustproof enclosure where indicated and whenever requested by Owner.
- C. Airborne Contaminant Control General Requirements: Floor to structure, air-tight enclosures, and gypsum board barriers, using tape and foam padding.
 1. Keep traffic between Containment Areas and open areas to a minimum. Keep door to areas closed at all times. Transport materials and refuse into an area from an external site without violating patient care areas by transporting in covered containers.
 2. Provide negative pressure in construction areas as specified herein.
 - a. Provide adequate forced ventilation of enclosed areas to cure installed materials, to prevent excessive humidity, and to prevent hazardous accumulations of dust fumes, vapors, or gases.
- D. Dustproof Enclosures: Full height, noncombustible construction, with minimum 1/2 inch gypsum board both sides with 3-1/2 inch R-11 insulation batts to reduce noise. Use 3-inch wide duct tape to tightly seal top, bottom, and seams, to prevent spread of dust to occupied areas, including above ceiling. Secure tape with Ram-Tack spray adhesive.
 1. Enclosure Doors: Four foot minimum width, unless shown otherwise, solid core wood with metal frame and hardware, including closer, tightly weather-stripped to prevent flow of dust. Locate as directed and swing into construction area. Keep enclosures locked outside of working hours. Provide Owner with keys for emergency access.
 2. Install floor mats on both sides of construction entrance prior to starting demolition or construction. Keep inside mat damp to help remove dust and minimize tracking into adjacent clean areas, vacuum mats daily. As an alternative, provide tacky-mats and remove old surface daily.

3. Obtain Owner's approval of exact location and details of enclosure construction. Precut materials for enclosure in unoccupied areas. No explosive or pneumatic driven fasteners allowed. Provide entrance vestibules as detailed. Provide carpets inside vestibule and inside enclosures at door to vestibule, and keep vacuumed daily.
- E. Enclosure outside work area (including spaces above furred ceilings): Whenever work is necessary outside of construction enclosures, the space where work is being performed, including ladders, shall be contained within a full height portable enclosure or with use of pre-fabricated units as specified herein.
 1. Work performed outside construction enclosure, including work in corridors and lobbies, shall be performed outside of normal working hours and shall be schedule in advance with Owner except where specified otherwise.
 2. Storage of construction equipment or material outside the construction enclosure is prohibited.
 3. Immediately clean up dust tracked outside the construction area. Contractor shall provide necessary manpower and equipment (HEPA filtered vacuum, dust mops, brooms, buckets and clean wiping rags) to keep adjacent occupied areas clean at all times.
- F. Power and Lighting: Provide sufficient temporary lighting and power ventilating equipment to ensure proper workmanship and safety everywhere.
- G. Access Provisions: Provide ramps, stairs, ladders, and similar temporary access elements as reasonably required to perform the work and facilitate its inspection during installation.
- H. Where work occurs in occupied areas, provide access opening through existing plaster, or gypsum board walls, and acoustical ceilings, and to restore walls and ceilings to original condition after work is complete and to ensure dust control within access areas.
 1. Provide temporary plywood panels anchored to existing steel ceiling support grid for support of workers crawling above ceiling. Panel thickness shall suit spans between existing steel support.
- I. Coordinate and phase remodeling work in certain rooms which serve other rooms with the phasing of the remodeled rooms if required, so that at no time are both rooms simultaneously inoperative. Any downtime
 1. necessitated by the remodeling work is to be fully discussed and coordinated with the Owner's Representative in advance of the shutdown.
 2. Dust: Generation of significant quantities of airborne dust will not be tolerated. Clean the work area prior to starting work as necessary to minimize existing dust which may become airborne during construction. Provide drop cloths and dust partitions as necessary to contain dust and debris generated by the work.
 3. Remove demolition material, dust, and dirt in tightly sealed, covered, rubber tired plastic dump carts. Containers shall be fitted with clean polyethylene covers, completely sealed at perimeter by wire tying or taping. Before leaving area, all containers shall be wiped clean to prevent tracking of dust. Place rugs inside barrier entrance, keep them clean or changed daily. Provide debris chutes if allowed.
 4. Hot Processes: Welding and flame cutting which generate smoke shall be scheduled with the Owner's fire detection system deactivated.
 5. Use portable vinyl tunnel or a polyethylene enclosure for single ceiling accesses. Enclosure opening shall have a 3-foot overlap of plastic to decrease risk of airborne dust.
- J. Portable vinyl tunnel or polyethylene enclosures, if used, shall remain in place until the ceiling is secured (all accesses closed).
 1. If access is larger than vinyl tunnel, use a portable polyethylene enclosure also enclosing the ladder. Enclosures opening shall have a 2-foot overlap of plastic to decrease risk of airborne dust.
 2. Secure polyethylene enclosures/barriers in place to walls and floor with use of tape. Reinforce seam on ceiling with frame and flat head screws.

3. When accessing pipes, ducts, or other building infrastructure to investigate a condition, use additional procedures including masks, disposable white coveralls, and disposable shoe covers, before going into the access. Afterwards, strip off the additional procedures carefully, turning the coveralls "inside-out," and depositing the mask, coverall, and shoe covers into a plastic trash bag inside the enclosure. Secure (tie-off) plastic trash bag and discard as directed by Owner's Representative. Do not discard within "patient care areas."
4. When Contractor leaves the work site, the access, especially at ceiling, shall either be completely closed or protected by an appropriate barrier.
5. In patient care areas, dismantle the apparatus (tunnel or enclosure) and replace access (ceiling tiles); or complete work of access at the end of each day.
6. Thoroughly clean surfaces which become exposed to dust before leaving the area of construction. Accomplish cleaning by use of either an HEPA-filtered vacuum cleaner.

3.4 ENFORCEMENT AND FINES

- A. Owner will monitor aspergillus count in vicinity of project in Protection Areas. Such areas will be located as indicated on Drawings. Whenever safe levels are exceeded, Contractor will be notified to correct conditions immediately to avoid fine and work stoppage described.
 1. All work shall be stopped on the project whenever a hazardous airborne contaminant control deficiency exists in occupied areas.
 2. Contractor shall take immediate action to correct all deficiencies.
- B. Process: Failure to maintain containment areas will result in issuance of written warning; if situation is not corrected within 8 hours of receipt of warning, Owner will have cause to stop the work.
 1. Failure of Contractor to correct deficiencies in containment will result in corrective action taken by the Owner and deducting all costs from the Contractor.
 2. Ceiling Access: Each breach of ceiling access policy will cost violators \$500.00 for each occurrence.
- C. The following will be performed by Owner's Representative:
 1. Periodic Rounds: A photograph will be taken to document each violation.
 2. Contractor and Department information will be extracted from the ceiling work tag.
 3. A record of all violations of required ceiling access procedures will be maintained, whether in occupied area or not.

END OF SECTION

SECTION 01 60 00

PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and comparable products.

1.3 DEFINITIONS

- A. Products: Items obtained for incorporating into the Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.
 - 1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature, that is current as of date of the Contract Documents.
 - 2. New Products: Items that have not previously been incorporated into another project or facility. Products salvaged or recycled from other projects are not considered new products.
 - 3. Comparable Product: Product that is demonstrated and approved through submittal process to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Basis-of-Design Product Specification: A specification in which a specific manufacturer's product is named and accompanied by the words "basis-of-design product," including make or model number or other designation, to establish the significant qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics for purposes of evaluating comparable products of additional manufacturers named in the specification.

1.4 ACTION SUBMITTALS

- A. Comparable Product Requests: Submit request for consideration of each comparable product. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.
 - 1. Include data to indicate compliance with the requirements specified in "Comparable Products" Article.
 - 2. Architect's Action: If necessary, Architect will request additional information or documentation for evaluation within one week of receipt of a comparable product request. Architect will notify Contractor of approval or rejection of proposed comparable product request within 15 days of receipt of request, or seven days of receipt of additional information or documentation, whichever is later.
 - a. Form of Approval: As specified in Division 01 Section "Submittal Procedures."
 - b. Use product specified if Architect does not issue a decision on use of a comparable product request within time allocated.
- B. Basis-of-Design Product Specification Submittal: Comply with requirements in Division 01 Section "Submittal Procedures." Show compliance with requirements.

1.5 QUALITY ASSURANCE

- A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, select product compatible with products previously selected, even if previously selected products were also options.

1.6 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft and vandalism. Comply with manufacturer's written instructions.
- B. Delivery and Handling:
 - 1. Schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.
 - 2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
 - 3. Deliver products to Project site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
 - 4. Inspect products on delivery to determine compliance with the Contract Documents and to determine that products are undamaged and properly protected.
- C. Storage:
 - 1. Store products to allow for inspection and measurement of quantity or counting of units.
 - 2. Store materials in a manner that will not endanger Project structure.
 - 3. Store products that are subject to damage by the elements, under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation.
 - 4. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
 - 5. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
 - 6. Protect stored products from damage and liquids from freezing.
 - 7. Provide a secure location and enclosure at Project site for storage of materials and equipment by Owner's construction forces. Coordinate location with Owner.

1.7 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of the Contract Documents.
 - 1. Manufacturer's Warranty: Written warranty furnished by individual manufacturer for a particular product and specifically endorsed by manufacturer to Owner.
 - 2. Special Warranty: Written warranty required by the Contract Documents to provide specific rights for Owner.
- B. Special Warranties: Prepare a written document that contains appropriate terms and identification, ready for execution.
 - 1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
 - 2. Specified Form: When specified forms are included with the Specifications, prepare a written document using indicated form properly executed.
 - 3. Refer to Divisions 02 through 49. Sections for specific content requirements and particular requirements for submitting special warranties.
- C. Submittal Time: Comply with requirements in Division 01 Section "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

- A. General Product Requirements: Provide products that comply with the Contract Documents, are undamaged and, unless otherwise indicated, are new at time of installation.
 - 1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
 - 2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
 - 3. Owner reserves the right to limit selection to products with warranties not in conflict with requirements of the Contract Documents.
 - 4. Where products are accompanied by the term "as selected," Architect will make selection.
 - 5. Descriptive, performance, and reference standard requirements in the Specifications establish salient characteristics of products.
 - 6. Or Equal: For products specified by name and accompanied by the term "or equal," or "or approved equal," or "or approved," comply with requirements in "Comparable Products" Article to obtain approval for use of an unnamed product.
- B. Product Selection Procedures:
 - 1. Product: Where Specifications name a single manufacturer and product, provide the named product that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
 - 2. Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
 - 3. Products:
 - a. Restricted List: Where Specifications include a list of names of both manufacturers and products, provide one of the products listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will be considered, unless otherwise indicated.
 - b. Non-restricted List: Where Specifications include a list of names of both available manufacturers and products, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product.
 - 4. Manufacturers:
 - a. Restricted List: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will be considered, unless otherwise indicated.
 - b. Non-restricted List: Where Specifications include a list of available manufacturers, provide a product by one of the manufacturers listed, or a product by an unnamed manufacturer, that complies with requirements. Comply with requirements in "Comparable Products" Article for consideration of an unnamed manufacturer's product.
 - 5. Basis-of-Design Product: Where Specifications name a product, or refer to a product indicated on Drawings, and include a list of manufacturers, provide the specified or indicated product or a comparable product by one of the other named manufacturers. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product by one of the other named manufacturers.
- C. Visual Matching Specification: Where Specifications require "match Architect's sample", provide a product that complies with requirements and matches Architect's sample. Architect's decision will be final on whether a proposed product matches.

1. If no product available within specified category matches and complies with other specified requirements, comply with requirements in Division 01 Section "Substitution Procedures" for proposal of product.
- D. Visual Selection Specification: Where Specifications include the phrase "as selected by Architect from manufacturer's full range" or similar phrase, select a product that complies with requirements. Architect will select color, gloss, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 COMPARABLE PRODUCTS

- A. Conditions for Consideration: Architect will consider Contractor's request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Architect may return requests without action, except to record noncompliance with these requirements:
 1. Evidence that the proposed product does not require revisions to the Contract Documents, that it is consistent with the Contract Documents and will produce the indicated results, and that it is compatible with other portions of the Work.
 2. Detailed comparison of significant qualities of proposed product with those named in the Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 3. Evidence that proposed product provides specified warranty.
 4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
 5. Samples, if requested.

PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 71 23

FIELD ENGINEERING

PART 1 - GENERAL

1.1 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Name, address, and telephone number of Surveyor before starting survey work.
 - 2. On request, documentation verifying accuracy of survey work.
- C. Closeout Submittals: Submit following in accordance with Section 01 77 00.
 - 1. Project Record Documents:
 - a. Copy of surveyor's log.
 - b. Six copies of final property survey for Owner's records.

1.2 QUALITY ASSURANCE

- A. Employ Land Surveyor registered to perform surveying in State where project is located, acceptable to Owner and Architect.

PART 2 - PRODUCTS - NOT USED

PART 3 - EXECUTION

3.1 SURVEY REFERENCE POINTS

- A. Control datum is indicated on Drawings.
 - 1. Survey Control and Reference Points: Verify locations prior to starting Work. Promptly notify Architect of discovered discrepancies.
 - 2. Bench Marks: Establish and maintain minimum of two permanent bench marks on site, referenced to established control points. Record locations with horizontal and vertical data on Project Record Documents.
- B. Benchmarks and Control Points: Protect survey control points prior to starting site work; preserve permanent reference points during construction.
 - 1. Do not change or relocate benchmarks or control points without prior written approval of Architect.
 - 2. Promptly report to Architect the loss or destruction of any reference point or relocation required because of changes in grades or other reasons.
- C. Replace dislocated survey control points based on original survey control. Make no changes without prior written notice to Architect. Base replacements on original survey control points.
- D. Existing Utilities and Equipment: Existence and location of indicated existing underground utilities and construction are not guaranteed.
 - 1. Before Beginning Sitework:
 - a. Investigate and verify existence, location, and elevations of underground utilities and other construction.
 - b. Verify location and invert elevation at points of connection for sanitary sewer, storm sewer, and water-service piping.
 - 2. Furnish information necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other appurtenances located in or affected by construction.
 - 3. Coordinate with local authorities having jurisdiction.

3.2 SURVEY REQUIREMENTS

- A. Establishing building location and layout on site and first ground floor elevation is responsibility of Contractor. Each Contractor will be responsible for layout, lines, and elevation of their Work based on established datum.
- B. Survey: Work from lines and levels established by property survey.
 - 1. Establish benchmarks and markers to set lines and levels at each story of construction and elsewhere as needed to locate each element of Project.
 - 2. Calculate and measure required dimensions within indicated or recognized tolerances.
 - 3. Do not scale Drawings to determine dimensions.
 - 4. Advise entities engaged in construction activities of marked lines and levels provided for their use.
- C. Establish elevations, lines, and levels. Locate, lay out, and periodically verify layouts, by instrumentation and similar appropriate means:
 - 1. Site improvements including pavements; stakes for grading, fill and topsoil placement; and utility locations, slopes, and invert elevations;
 - 2. Grid or axis for structures.
 - 3. Building foundation, column locations, and ground floor elevations.
 - 4. As construction proceeds, check every major element for line, level, and plumb.
- D. Surveyor's Log: Maintain and make available for references, surveyor's log of control and other survey work as work progresses. Make this log available for reference.
 - 1. Record deviations from required lines and levels, and advise Architect of deviations exceeding indicated or recognized tolerances.
 - 2. Record deviations on Project Record Drawings that are accepted and not corrected.
 - 3. Prepare certified survey showing dimensions, locations, angles, and elevations of construction and sitework on completion of foundation walls, major site improvements, and other work requiring surveying.

3.3 FINAL PROPERTY SURVEY

- A. Prior to Substantial Completion, prepare final property survey illustrating locations, dimensions, angles, and elevations of buildings and site work that have resulted from construction indicating their relationship to permanent bench marks and property lines.
 - 1. Show significant features (real property) for Project.
 - 2. Include certification on survey, signed by surveyor, that principal metes, bounds, lines, levels, and elevations of Project are accurately shown.
- B. Recording: At Substantial Completion, have final property survey recorded by or with local governing authorities as official "property survey."

3.4 SURVEYS FOR MEASUREMENT AND PAYMENT

- A. Perform surveys to determine quantities of unit price work, including control surveys to establish measurement reference lines. Notify Architect prior to starting work.
 - 1. Contractor's Engineer Responsibilities: Sign surveyor's field notes or keep duplicate field, and calculate and certify quantities for payment purposes.

END OF SECTION

SECTION 01 73 00

EXECUTION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes general administrative and procedural requirements governing execution of the Work including, but not limited to, the following:
 - 1. Construction layout.
 - 2. Installation of the Work.
 - 3. Cutting and patching.
 - 4. Coordination of Owner-installed products.
 - 5. Progress cleaning.
 - 6. Starting and adjusting.
 - 7. Protection of installed construction.
 - 8. Correction of the Work.

1.3 DEFINITIONS

- A. Cutting: Removal of in-place construction necessary to permit installation or performance of other work.
- B. Patching: Fitting and repair work required to restore construction to original conditions after installation of other work.

1.4 QUALITY ASSURANCE

- A. Cutting and Patching: Comply with requirements for and limitations on cutting and patching of construction elements.
 - 1. Structural Elements: When cutting and patching structural elements, notify Architect of locations and details of cutting and await directions from the Architect before proceeding. Shore, brace, and support structural element during cutting and patching. Do not cut and patch structural elements in a manner that could change their load-carrying capacity or increase deflection
 - 2. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety. Operational elements include the following:
 - a. Primary operational systems and equipment.
 - b. Fire separation assemblies.
 - c. Air or smoke barriers.
 - d. Fire-suppression systems.
 - e. Mechanical systems piping and ducts.
 - f. Control systems.
 - g. Communication systems.
 - h. Conveying systems.
 - i. Electrical wiring systems.
 - j. Operating systems of special construction.

3. Other Construction Elements: Do not cut and patch other construction elements or components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety. Other construction elements include but are not limited to the following:
 - a. Water, moisture, or vapor barriers.
 - b. Membranes and flashings.
 - c. Exterior curtain-wall construction.
 - d. Equipment supports.
 - e. Piping, ductwork, vessels, and equipment.
 - f. Noise- and vibration-control elements and systems.
 4. Visual Elements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch exposed construction in a manner that would, in Architect's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.
- B. Cutting and Patching Conference: Before proceeding, meet at Project site with parties involved in cutting and patching, including mechanical and electrical trades. Review areas of potential interference and conflict. Coordinate procedures and resolve potential conflicts before proceeding.
- C. Manufacturer's Installation Instructions: Obtain and maintain on-site manufacturer's written recommendations and instructions for installation of products and equipment.

1.5 WARRANTY

- A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during installation or cutting and patching operations, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General: Comply with requirements specified in other Sections.
1. For projects requiring compliance with sustainable design and construction practices and procedures, utilize products for patching that comply with requirements of Division 01 Section "Sustainable Design Requirements."
- B. In-Place Materials: Use materials for patching identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.
1. If identical materials are unavailable or cannot be used, use materials that, when installed, will provide a match acceptable to the Architect for the visual and functional performance of in-place materials.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Existing Conditions: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed.
- B. Examination and Acceptance of Conditions: Before proceeding with each component of the Work, examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
1. Written Report: Where a written report listing conditions detrimental to performance of the Work is required by other Sections, include the following:

- a. Description of the Work.
 - b. List of detrimental conditions, including substrates.
 - c. List of unacceptable installation tolerances.
 - d. Recommended corrections.
2. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
 3. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
 4. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
 5. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Existing Utility Information: Furnish information to Owner that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate with authorities having jurisdiction.
- B. Field Measurements: Take field measurements as required to fit the Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.
- C. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.
- D. Review of Contract Documents and Field Conditions: Immediately on discovery of the need for clarification of the Contract Documents caused by differing field conditions outside the control of the Contractor, submit a request for information to Architect according to requirements in Division 01 Section "Project Management and Coordination."

3.3 INSTALLATION

- A. General: Locate the Work and components of the Work accurately, in correct alignment and elevation, as indicated.
 1. Make vertical work plumb and make horizontal work level.
 2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
 3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
 4. Maintain minimum headroom clearance of 96 inches in occupied spaces and 90 inches in unoccupied spaces.
- B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.
- C. Install products at the time and under conditions that will ensure the best possible results. Maintain conditions required for product performance until Substantial Completion.
- D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.
- E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.
- F. Templates: Obtain and distribute to the parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing products to comply with indicated requirements.

- G. Attachment: Provide blocking and attachment plates and anchors and fasteners of adequate size and number to securely anchor each component in place, accurately located and aligned with other portions of the Work. Where size and type of attachments are not indicated, verify size and type required for load conditions.
 - 1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Architect.
 - 2. Allow for building movement, including thermal expansion and contraction.
 - 3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.
- H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for the best visual effect. Fit exposed connections together to form hairline joints.
- I. Hazardous Materials: Use products, cleaners, and installation materials that are not considered hazardous.

3.4 CUTTING AND PATCHING

- A. Cutting and Patching, General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.
 - 1. Cut in-place construction to provide for installation of other components or performance of other construction, and subsequently patch as required to restore surfaces to their original condition.
- B. Temporary Support: Provide temporary support of work to be cut.
- C. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.
- D. Adjacent Occupied Areas: Where interference with use of adjoining areas or interruption of free passage to adjoining areas is unavoidable, coordinate cutting and patching in accordance with requirements of Division 01 Section "Summary."
- E. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to prevent interruption to occupied areas.
- F. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer; comply with original Installer's written recommendations.
 - 1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots neatly to minimum size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
 - 2. Finished Surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
 - 3. Concrete and Masonry: Cut using a cutting machine, such as an abrasive saw or a diamond-core drill.
 - 4. Excavating and Backfilling: Comply with requirements in applicable Division 31 Sections where required by cutting and patching operations.
 - 5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. Cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
 - 6. Proceed with patching after construction operations requiring cutting are complete.
- G. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other work. Patch with durable seams that are as invisible as

practicable. Provide materials and comply with installation requirements specified in other Sections, where applicable.

1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate physical integrity of installation.
 2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will minimize evidence of patching and refinishing.
 - a. Clean piping, conduit, and similar features before applying paint or other finishing materials.
 - b. Restore damaged pipe covering to its original condition.
 3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another, patch and repair floor and wall surfaces in the new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
 - a. Where patching occurs in a painted surface, prepare substrate and apply primer and intermediate paint coats appropriate for substrate over the patch, and apply final paint coat over entire unbroken surface containing the patch. Provide additional coats until patch blends with adjacent surfaces.
 4. Ceilings: Patch, repair, or rehang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
 5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weathertight condition.
- H. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials from adjacent finished surfaces.

3.5 OWNER-INSTALLED PRODUCTS

- A. Site Access: Provide access to Project site for Owner's construction personnel.
- B. Coordination: Coordinate construction and operations of the Work with work performed by Owner's construction personnel.
1. Construction Schedule: Inform Owner of Contractor's preferred construction schedule for Owner's portion of the Work. Adjust construction schedule based on a mutually agreeable timetable. Notify Owner if changes to schedule are required due to differences in actual construction progress.
 2. Preinstallation Conferences: Include Owner's construction personnel at preinstallation conferences covering portions of the Work that are to receive Owner's work. Attend preinstallation conferences conducted by Owner's construction personnel if portions of the Work depend on Owner's construction.

3.6 PROGRESS CLEANING

- A. General: Clean Project site and work areas daily, including common areas. Enforce requirements strictly. Dispose of materials lawfully.
1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.
 2. Do not hold waste materials more than seven days during normal weather or three days if the temperature is expected to rise above 80 deg F.
 3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
 - a. Utilize containers intended for holding waste materials of type to be stored.
 4. Coordinate progress cleaning for joint-use areas where more than one installer has worked.
- B. Site: Maintain Project site free of waste materials and debris.

- C. Work Areas: Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the Work.
 - 1. Remove liquid spills promptly.
 - 2. Where dust would impair proper execution of the Work, broom-clean or vacuum the entire work area, as appropriate.
- D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and that will not damage exposed surfaces.
- E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.
- F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at time of Substantial Completion.
- G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways. Comply with waste disposal requirements in Division 01 Section "Temporary Facilities and Controls." Division 01 Section "Construction Waste Management and Disposal."
- H. During handling and installation, clean and protect construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.
- I. Clean and provide maintenance on completed construction as frequently as necessary through the remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.
- J. Limiting Exposures: Supervise construction operations to assure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period.

3.7 STARTING AND ADJUSTING

- A. Coordinate startup and adjusting of equipment and operating components with requirements in Division 01 Section "General Commissioning Requirements."
- B. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.
- C. Adjust equipment for proper operation. Adjust operating components for proper operation without binding.
- D. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- E. Manufacturer's Field Service: Comply with qualification requirements in Division 01 Section "Quality Requirements."

3.8 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.
- B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.9 CORRECTION OF THE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes.
 - 1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.
- B. Restore permanent facilities used during construction to their specified condition.

- C. Remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence of repair.
- D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
- E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION

SECTION 01 73 29

CUTTING AND PATCHING

PART 1 - GENERAL

1.1 SUMMARY

- A. Related Documents: Conditions of the Contract, this Section and other sections of Division 1 - General Requirements, and Drawings apply to entire Work of the Contract.
- B. Section includes incidental cutting, fitting, and patching within new construction required to complete work or to make its several parts fit together.
- C. Related Sections:
 - 1. Section 07 84 00 - Firestopping.

1.2 SUBMITTALS

- A. Submit written request to perform cutting and patching two weeks in advance of cutting or alteration which affects:
 - 1. Structural value or integrity of any element of Project.
 - 2. Integrity or effectiveness of weather exposed or moisture resistant elements or systems.
 - 3. Efficiency, operation, maintenance, or safety of operational equipment.
 - 4. Visual qualities of elements exposed to view.
 - 5. Work of Owner, Tenant, or separate contractor.
- B. Include in request:
 - 1. Identification of Project.
 - 2. Location and description of affected Work.
 - 3. Description of proposed Work:
 - a. Extent of cutting, fitting, patching, or alteration.
 - b. Listing of applicable trades.
 - c. Proposed products and materials.
 - d. Extent of refinishing.
 - 4. Necessity for cutting or alteration.
 - 5. Alternatives to cutting and patching.
 - 6. Effect on structural integrity of Work.
 - 7. Effect on weatherproof integrity of Work.
 - 8. Effect on the building's appearance and significant visual elements.
 - 9. Effect on utilities:
 - a. List utilities affected by cutting and patching.
 - b. List utilities that will be relocated.
 - c. List utilities that will be temporarily out-of-service. Indicate time period of service outage.
 - 10. Effect on work of Owner, Tenant, or separate contractor.
 - 11. Date and time of execution.
- C. Should conditions or schedule require change of products or methods different than original installation, submit written recommendation to Architect explaining conditions necessitating change and requirements of alternative materials or methods.
- D. Approval by Architect to proceed with cutting and patching does not waive Architect's right to later require complete removal and replacement of unsatisfactory work.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Primary Products and Materials: Those required for original installation; comply with Specifications for each specific product involved.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Comply with Section 01 73 00.
 - 1. After uncovering existing Work, examine conditions affecting installation of products and performance of Work.

3.2 PREPARATION

- A. Provide temporary supports to ensure structural integrity of affected portions of Work.
 - 1. Provide devices and methods to protect other portions of Project from damage.
 - 2. Provide protection from elements for areas which may be exposed by uncovering work; maintain excavations free of water.
- B. Provide materials and control operations to prevent spread of dust in surrounding area. Provide drop cloths or other suitable barriers.
- C. Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.
- D. Avoid cutting in service pipes, ducts, or conduit until provisions have been made to bypass them.

3.3 PERFORMANCE

- A. Cutting and Patching: Cut into construction to provide for installation of other Work and subsequent fitting and patching required to restore surface to original condition.
- B. Cut, fit and patch, including excavation and backfill, to complete Work and to:
 - 1. Fit several parts together, to integrate with other work.
 - 2. Uncover portions of work to provide for installation of ill-timed work.
 - 3. Remove and replace defective work.
 - 4. Remove and replace work not conforming to requirements of Contract Documents.
 - 5. Remove samples of installed work as necessary for testing.
 - 6. Provide openings in elements of work for penetrations of plumbing, mechanical, and electrical work.
 - 7. Uncover work to allow for Architect's observation of covered work which has been covered up prior to required observation by Architect.
- C. Cutting and Patching: Execute in manner which does not void required or existing warranties.
 - 1. Execute by methods which will prevent damage to other Work and which will produce appropriate surfaces to receive installation of new Work:
 - a. Use hand or small power tools designed for sawing or grinding, not hammering or chopping.
 - b. Cut holes and slots as small as possible, neatly to size required, with minimum disturbance of adjacent surfaces.
 - c. Temporarily cover openings when not in use.
 - d. To avoid marring existing finished surfaces, cut or drill from exposed or finished side into concealed surfaces.
 - e. Cut through concrete and masonry using cutting machine, such as Carborundum saw or diamond-core drill.

2. Execute excavating and backfilling by methods in accordance with applicable Sections of Division 31 which will prevent settlement or damage to Project.
 3. Execute fitting and adjustment to produce finished installation complying with specified products, functions, tolerances, and finishes.
- D. Restore surfaces which have been cut, removed, or damaged, to match existing conditions.
- E. Install products and materials to complete Work in accordance with requirements of Contract Documents.
1. Weather Exposed and Moisture Resistant Elements, and Sight Exposed Surfaces: Use installer of the materials to cut and patch the materials.
- F. Cutting and Patching:
1. Cut rigid materials using masonry saw or core drill. Pneumatic tools not allowed without prior approval of Architect.
 2. Do not cut and patch structural elements in manner that would result in reduction of load carrying capacity or of load deflection ratio.
 3. Do not cut and patch operational elements or safety related components in manner that would result in reduction of their capacity to perform in manner intended, including energy performance, that would result in increased maintenance, decreased operational life, or decreased safety.
- G. Utilities and Penetrations: Fit work tight to pipes, sleeves, ducts, conduit, and other penetrations through surfaces.
1. At penetrations of fire-rated assemblies, completely seal with firestops in accordance with Section 07 84 00.
 2. Where utilities are to be removed, relocated, or abandoned, by-pass before cutting. Cut-off pipe or conduit in walls or partitions to be removed. Cap, valve, or plug and seal the remaining portion of pipe, duct, or conduit to prevent entrance of moisture or matter after by-passing and cutting.
- H. Restoration: Except where indicated otherwise, restore exposed finishes of patched areas to match existing and where necessary extend finish restoration into retained adjoining surfaces in manner which will eliminate evidence of patching and refinishing. Thoroughly clean surfaces prior to application of paint and other finishes.
1. Where patching occurs in previously painted surface, provide appropriate prime coat followed by first finish coat of paint. Provide final finish coat over entire area containing patch; for continuous surface extend to nearest vertical break or intersection, for an assembly refinish entire unit. Except where indicated otherwise, finish in sheen and color to match existing.

3.4 CLEANING

- A. Comply with Section 01 50 00. Thoroughly clean areas and spaces affected by Work. Completely remove paint, mortar, oils, putty, and items of similar nature.
- B. Restore damaged surfaces to their original condition.

END OF SECTION

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Administrative and procedural requirements:
 - 1. Salvaging nonhazardous ~ construction waste.
 - 2. Recycling nonhazardous construction waste.
 - 3. Disposing of nonhazardous construction waste.
- B. Related Sections include the following:
 - 1. Section 01 10 00 - Summary for coordination of responsibilities for waste management.
 - 2. Section 01 81 13 - Sustainable Design Requirements for additional LEED requirements.
 - 3. Section 01 50 00 - Temporary Facilities and Controls for environmental-protection measures during construction, and location of waste containers at Project site.
 - 4. Section 31 10 00 - Site Clearing for disposition of waste resulting from site clearing and removal of above- and below-grade improvements.
 - 5. Section 04 20 00 - Unit Masonry for disposal requirements for masonry waste.

1.2 DEFINITIONS

- A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- C. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.
- D. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.
- E. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.
- F. Salvage and Reuse: Recovery of demolition or construction waste and subsequent incorporation into the Work.

1.3 PERFORMANCE GOALS

- A. General: Develop waste management plan that results in end-of-Project rates for salvage/recycling of 75 percent by weight of total waste generated by the Work.
- B. Salvage/Recycle Goals: Owner's goal is to salvage and recycle as much nonhazardous construction waste as possible including the following materials:
 - 1. Construction Waste:
 - a. Site-clearing waste.
 - b. Masonry and CMU.
 - c. Lumber.

- d. Wood sheet materials.
- e. Wood trim.
- f. Metals.
- g. Roofing.
- h. Insulation.
- i. Carpet and pad.
- j. Gypsum board.
- k. Piping.
- l. Electrical conduit.
- m. Packaging: Regardless of salvage/recycle goal indicated above, salvage or recycle 100 percent of the following uncontaminated packaging materials:
 - 1) Paper.
 - 2) Cardboard.
 - 3) Boxes.
 - 4) Plastic sheet and film.
 - 5) Polystyrene packaging.
 - 6) Wood crates.
 - 7) Plastic pails.

1.4 SUBMITTALS

- A. Waste Management Plan: Submit three copies of plan within seven days of date established for commencement of the Work .
 - 1. Waste Reduction Progress Reports: Concurrent with each Application for Payment, submit three copies of report. Include the following information.
 - 2. Material category.
 - 3. Generation point of waste.
 - 4. Total quantity of waste in tons.
 - 5. Quantity of waste salvaged, both estimated and actual in tons.
 - 6. Quantity of waste recycled, both estimated and actual in tons.
 - 7. Total quantity of waste recovered (salvaged plus recycled) in tons.
 - 8. Total quantity of waste recovered (salvaged plus recycled) as a percentage of total waste.
- B. Waste Reduction Calculations: Before request for Substantial Completion, submit three copies of calculated end-of-Project rates for salvage, recycling, and disposal as a percentage of total waste generated by the Work.
- C. Records of Donations: Indicate receipt and acceptance of salvageable waste donated to individuals and organizations. Indicate whether organization is tax exempt.
- D. Records of Sales: Indicate receipt and acceptance of salvageable waste sold to individuals and organizations. Indicate whether organization is tax exempt.
- E. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- F. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

- G. LEED Submittal: LEED letter template for Credit MR 2.1, signed by Contractor, tabulating total waste material, quantities diverted and means by which it is diverted, and statement that requirements for the credit have been met.
- H. Qualification Data: For Waste Management Coordinator .
- I. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating that all refrigerant that was present was recovered and that recovery was performed according to EPA regulations. Include name and address of technician and date refrigerant was recovered.

1.5 QUALITY ASSURANCE

- A. Refrigerant Recovery Technician Qualifications: Certified by EPA-approved certification program.
- B. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.
- C. Waste Management Conference: Conduct conference at Project site to comply with requirements in Section 01 31 13 - Project Coordination. Review methods and procedures related to waste management including, but not limited to, the following:
 - 1. Review and discuss waste management plan including responsibilities of Waste Management Coordinator.
 - 2. Review requirements for documenting quantities of each type of waste and its disposition.
 - 3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
 - 4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
 - 5. Review waste management requirements for each trade.

1.6 WASTE MANAGEMENT PLAN

- A. General: Develop plan consisting of waste identification, waste reduction work plan, and cost/revenue analysis. Indicate quantities by weight or volume, but use same units of measure throughout waste management plan.
- B. Waste Identification: Indicate anticipated types and quantities of site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.
- C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.
 - 1. Salvaged Materials for Reuse: For materials that will be salvaged and reused in this Project, describe methods for preparing salvaged materials before incorporation into the Work.
 - 2. Salvaged Materials for Sale: For materials that will be sold to individuals and organizations, include list of their names, addresses, and telephone numbers.
 - 3. Salvaged Materials for Donation: For materials that will be donated to individuals and organizations, include list of their names, addresses, and telephone numbers.
 - 4. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
 - 5. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.

6. Handling and Transportation Procedures: Include method that will be used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.
- D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include the following:
1. Total quantity of waste.
 2. Estimated cost of disposal (cost per unit). Include hauling and tipping fees and cost of collection containers for each type of waste.
 3. Total cost of disposal (with no waste management).
 4. Revenue from salvaged materials.
 5. Revenue from recycled materials.
 6. Savings in hauling and tipping fees by donating materials.
 7. Savings in hauling and tipping fees that are avoided.
 8. Handling and transportation costs. Include cost of collection containers for each type of waste.
 9. Net additional cost or net savings from waste management plan.
- E. Forms: Prepare waste management plan on forms included at end of Part 3.

PART 2 - PRODUCTS – (NOT USED)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

- A. General: Implement waste management plan as approved by Owner. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during the entire duration of the Contract.
1. Comply with Section 01 50 00 - Temporary Facilities and Controls for operation, termination, and removal requirements.
- B. Waste Management Coordinator: Engage a waste management coordinator to be responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.
- C. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.
1. Distribute waste management plan to everyone concerned within three days of submittal return.
 2. Distribute waste management plan to entities when they first begin work on-site.
Review plan procedures and locations established for salvage, recycling, and disposal.
- D. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
1. Designate and label specific areas on Project site necessary for separating materials that are to be salvaged, recycled, reused, donated, and sold.
 2. Comply with Section 01 50 00 - Temporary Facilities and Controls for controlling dust and dirt, environmental protection, and noise control.

3.2 SALVAGING DEMOLITION WASTE

- A. Salvaged Items for Reuse in the Work:

1. Clean salvaged items.
 2. Pack or crate items after cleaning. Identify contents of containers.
 3. Store items in a secure area until installation.
 4. Protect items from damage during transport and storage.
 5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.
- B. Salvaged Items for Sale and Donation: Not permitted on Project site.
- C. Salvaged Items for Owner's Use:
1. Clean salvaged items.
 2. Pack or crate items after cleaning. Identify contents of containers.
 3. Store items in a secure area until delivery to Owner.
 4. Transport items to Owner's storage area on-site designated by Owner.
 5. Protect items from damage during transport and storage.
- D. Doors and Hardware: Brace open end of door frames. Except for removing door closers, leave door hardware attached to doors.

3.3 RECYCLING CONSTRUCTION WASTE, GENERAL

- A. General: Recycle paper and beverage containers used by on-site workers.
- B. Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall become the Contractor's property.
- C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to the maximum extent practical.
1. Provide appropriately marked containers or bins for controlling recyclable waste until they are removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
 - a. Inspect containers and bins for contamination and remove contaminated materials if found.
 2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
 3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
 4. Store components off the ground and protect from the weather.
 5. Remove recyclable waste off Owner's property and transport to recycling receiver or processor.
 6. Separate suspension system, trim, and other metals from panels and tile and sort with other metals.
- D. Carpet: Roll large pieces tightly after removing debris, trash, adhesive, and tack strips.
1. Store clean, dry carpet in a closed container or trailer provided by Carpet Reclamation Agency or carpet recycler.
- E. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.
- F. Plumbing Fixtures: Separate by type and size.
- G. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, valves, sprinklers, and other components by type and size.

- H. Lighting Fixtures: Separate lamps by type and protect from breakage.
- I. Electrical Devices: Separate switches, receptacles, switchgear, transformers, meters, panelboards, circuit breakers, and other devices by type.
- J. Conduit: Reduce conduit to straight lengths and store by type and size.

3.4 RECYCLING CONSTRUCTION WASTE

- A. Packaging:
 - 1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in a dry location.
 - 2. Polystyrene Packaging: Separate and bag materials.
 - 3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
 - 4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.
- B. Site-Clearing Wastes: Chip brush, branches, and trees on-site.
 - 1. Comply with requirements in Section 32 93 00 - Plants for use of chipped organic waste as organic mulch.
- C. Wood Materials:
 - 1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
 - 2. Clean Sawdust: Bag sawdust that does not contain painted or treated wood.
 - a. Comply with requirements in Section 32 93 00 - Plants for use of clean sawdust as organic mulch.
- D. Gypsum Board: Stack large clean pieces on wood pallets and store in a dry location.
 - 1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.
 - a. Comply with requirements in Section 32 93 00 - Plants for use of clean ground gypsum board as inorganic soil amendment.

3.5 DISPOSAL OF WASTE

- A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose of them in a landfill or incinerator acceptable to authorities having jurisdiction.
 - 1. Except as otherwise specified, do not allow waste materials that are to be disposed of accumulate on- site.
 - 2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
- B. Burning: Do not burn waste materials.
- C. Burning: Burning of waste materials is permitted only at designated areas on Owner's property, provided required permits are obtained. Provide full-time monitoring for burning materials until fires are extinguished.
- D. Disposal: Transport waste materials and dispose of at designated spoil areas on Owner's property.
- E. Disposal: Transport waste materials off Owner's property and legally dispose of them.

SUMMARY OF SOLID WASTE DISPOSAL AND DIVERSION

Project Name: _____ Project Number: _____
Contractor Name: _____ License Number: _____

Contractor Address: _____

Solid Waste Material	Date Material Disposed/ Diverted	Amount Disposed Diverted (ton or cu. Yd)	Municipal Solid Waste Facility (name, address, & phone number)	Recycling/Reuse Facility (name, address, & Phone number)	Comments (if disposed, state why not diverted)
Land Clearing Debris					
Asphalt					
Concrete					
Metal					
Wood					
Debris					
Glass					
Clay brick					
Paper/ Cardboard					
Plastic					
Gypsum					
Paint					
Carpet					
Other					

Signature: _____ Date: _____

END OF SECTION

SECTION 01 77 01
CLOSEOUT PROCEDURES (CM)

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes administrative and procedural requirements for contract closeout.
- B. Provisions followed by an asterisk (*) include some or all of the provisions as obtained from AIA Document A201 - General Conditions of the Contract for Construction.

1.2 SUBMITTALS

- A. Contract closeout information:
 - 1. For substantial completion:
 - a. Comprehensive list of all items to be completed or corrected.
 - b. Contractor's Notice of Substantial Completion.
 - c. Certificates of governing authorities.
 - d. Submittals required by other Sections.
 - 2. For final completion:
 - a. Contractor's Certificate of Completion.
 - b. Evidence of payments and release or waiver of liens in triplicate.
 - 1) Contractor's Affidavit of Payments of Debts and Claims: AIA Document G706.
 - 2) Contractor's Affidavit of Release of Liens: AIA Document G706A.
 - 3) Contractor's release or waiver of liens.
 - 4) Separate releases or waivers of liens for subcontractors, suppliers, and others with lien rights against Owner, together with list of all such parties.
 - 5) If required by Owner, other data establishing payment or satisfaction of obligations arising out of Contract.
 - c. Consent of Surety to Final Payment: AIA Document G707.
 - d. Certificates evidencing that insurance to remain enforce.
 - e. Final application for payment.
 - f. Initialed list(s) of items to be completed or corrected verifying completion of each items.
 - g. List of Contractors, Subcontractors and equipment suppliers. Include:
 - 1) Name.
 - 2) Address.
 - 3) Telephone number.
 - 4) Representative.
 - h. Letter of site conformance.
 - i. Closeout submittals required by other Sections.

1.3 SUBSTANTIAL COMPLETION

- A. Substantial Completion is the stage in the progress of Work when the Work or designated portion thereof is sufficiently complete in general accordance with Contract Documents so Owner can occupy or utilize Work for its intended use. *
- 1. Work will not be considered for Substantial Completion until all systems and equipment are operational; all designated or required governing agencies inspections and certifications have been made and posted, instruction of designated Owner's personnel in operation of systems and equipment has been completed, operation and maintenance data has been satisfactorily turned over to Owner, and finishes are in place. In general, the only remaining Work shall be minor in nature, such that Owner may occupy or utilize Work or designated

- portion thereof, and completion or correction of Work by Contractor would not materially interfere or hamper Owner's intended business use or operation.
2. Construction Manager and Contractors shall certify that all remaining Work will be completed within 30 consecutive calendar days following date of Substantial Completion, or as agreed to in writing, and failure to do so shall automatically reinstate provision for damages due Owner as contained elsewhere in Contract Documents or as provided by law for such period of time as may be required by Construction Manager Contractors to fully complete Work whether Owner has occupied Work or not.
- B. Obtain evidence of compliance with requirements of governing authorities:
1. Certificates of inspection of:
 - a. Mechanical.
 - b. Electrical.
 - c. Plumbing.
 - d. Fire protection and life safety systems.
 - e. Elevators.
 2. Health Department and other governing authorities as required.
 3. Certificate of Occupancy.
- C. When Construction Manager and Contractor considers that Work, or a portion thereof which Owner agrees to accept separately, is substantially complete, Contractors shall thoroughly inspect Work, and prepare and submit to Construction Manager comprehensive lists of items to be corrected or completed, and Contractor's Notice of Substantial Completion(s) (utilize form at end of this section). *
- D. Construction Manager certify that:
1. Work performed under this Contract has been thoroughly inspected and considered to be sufficiently complete, in accordance with the Contract Documents, so Owner can occupy or utilize Work for its intended use.
- E. Failure of Construction Manager or Contractor to include an item on such list(s) does not alter responsibility of Contractor to complete all Work in accordance with Contract Documents. *
- F. Contractor shall proceed promptly to complete and correct the items on the list.
- G. After receipt of Contractor's comprehensive list(s) of items to be completed or corrected, and Contractor's Notice of Substantial Completion, Construction Manager [, Inspector of Record,] and Owner will inspect Work and review list(s) of items to be completed or corrected, to determine if Work can be considered to be Substantially Complete.
- H. If review or inspection discloses any item, whether or not included on such lists, which is not sufficiently complete in general accordance with Contract Documents, so Owner may occupy or utilize Work or designated portion thereof for its intended use:
1. Construction Manager shall notify Contractors stating reasons.
 2. Contractors shall substantially complete or correct Work.
 3. Contractor shall thoroughly re-inspect Work and revise list.
 4. Contractor shall submit another Contractor's Notice of Substantial Completion, revised list of items to be completed or corrected, and a request for another inspection.
 5. Construction Manager [, Inspector of Record,] and Owner will again review Work and list of items to be completed or corrected.
- I. If Construction Manager [, Inspector of Record,] and Owner agree Work for all Work Categories can be considered substantially complete:
1. Construction Manager [and Inspector of Record] will thoroughly inspect Work and prepare a comprehensive list of items to be completed or corrected and provide copies to Contractors for their required action.

2. Construction Manager will prepare and submit to Architect, Contractor's Notice of Substantial Completion (utilize form at end of this Section), along with comprehensive list(s) of items to be completed or corrected.
- J. After receipt of list(s) of items to be completed or corrected, and Construction Manager's Notice of Substantial Completion, Architect will, within reasonable period after notification, review list(s) of items to be completed or corrected, or Work to determine whether Work is Substantially Complete. *
- K. If Architect's review discloses any item, whether or not included on list(s), which is not sufficiently complete in general accordance with Contract Documents so Owner can occupy or utilize Work or designated portion thereof for its intended use: *
 1. Construction Manager will be notified stating reasons.
 2. Contractor(s) will substantially complete or correct Work.
 3. Construction Manager will submit another Construction Manager's Notice of Substantial Completion, including revised list(s) of items to be completed or corrected.
 4. Architect will again review list of items to be completed or corrected, or Work.
- L. If Construction Manager prematurely submits a Contractor's Notice of Substantial Completion or requests Architect's review of Work, and Architect determines that Project or designated portion thereof is not Substantially Complete, Architect may invoice Owner as a change in services for such cost involved in evaluating and reviewing Work, and associated travel costs. Construction Manager shall reimburse Owner for such costs.
- M. Architect will not perform more reviews of sub-projects or phases than number indicated in Contract Documents or Owner - Architect Agreement, unless otherwise mutually agreed to by Architect, Owner and Construction Manager.
- N. When Work or designated portion thereof is considered Substantially Complete, Architect will prepare a Certificate of Substantial Completion.
 1. The Certificate of Substantial Completion shall establish date of Substantial Completion, shall establish responsibilities of Owner and Contractor for security, maintenance, heat, utilities, damage to Work and insurance, and shall fix time within which Contractor shall complete and correct Work.
 2. Warranties and guarantees required by Contract Documents shall commence on date of Substantial Completion of Work or designated portion thereof unless otherwise provided in Certificate of Substantial Completion.
 3. The Certificate of Substantial Completion shall be submitted to Owner and Contractor for their written acceptance of responsibilities assigned to them in such Certificate. *
- O. Owner may occupy Project, or designated portion thereof, under provisions agreed to in Certificate of Substantial Completion, and if required, a certificate of occupancy has been issued by governing authorities.
 1. If Owner is going to occupy Project, or designated portion thereof, Contractor shall perform final cleaning immediately.
 2. If Owner, Construction Manager or Architect discovers any Work which is not complete and/or is not in conformance with Contract Documents, during or after occupying or utilizing Work, whether included on a list or not, Owner or Construction Manager shall notify Contractor to complete or correct item(s) identified.
- P. Contractors shall proceed expeditiously with adequate forces to complete or correct Work and to complete all Project closeout requirements within designated time.
- Q. Upon completion of Work, Construction Manager [and Inspector of Record] employ Licensed Surveyor to make survey of site to assure conformance of elevations, grade, and site work to contours shown.

1.4 FOR FINAL COMPLETION

- A. After Construction Manager [, Inspector of Record,] and Contractors have thoroughly inspected Project to determine that Work is sufficiently complete, is in general accordance with Contract Documents and Contract is fully performed. *
- B. Construction Manager shall submit to Architect:
 - 1. Contractor's Certificate of Completion (utilize form at end of this Section).
 - 2. List(s) of items which were to be completed and corrected, with each item initialed to indicate Construction Manager [and Inspector of Record] has verified completion or correction of each.
- C. Construction Manager certifies that:
 - 1. Work has been thoroughly inspected by Construction Manager and Contractors for compliance with Contract Documents.
 - 2. Work has been completed in accordance with Contract Documents.
 - 3. Equipment and systems have been tested and are operating satisfactorily.
 - 4. Contract closeout requirements have been completed satisfactorily and submitted.
 - 5. Contractor knows of no reason that insurance will not be renewable to cover period required by Contract Documents.
 - 6. Work is ready for final inspection and acceptance.
- D. Contractors submit final closeout submittals required by this and other Sections to Construction Manager for processing and forwarding to Architect and/or Owner.
- E. Owner, Architect [, Inspector of Record,] and Construction Manager will make final walk through of Project within a reasonable time after receipt of Contractor's Certificate of Completion and final Application for Payment. *
 - 1. If Construction Manager prematurely submits a Contractor's Notice of Final Completion or requests Architect's final review of Project, and Architect determines that the Project is not satisfactorily complete, Architect may invoice Owner as a change in services for such cost involved in evaluating and reviewing Work, and associated travel costs. Construction Manager shall reimburse Owner for such costs.
- F. Contractor(s) shall remedy any remaining deficiencies or incomplete Work at Contractor's expense.
- G. When Architect, [Inspector of Record,] Owner, and Construction Manager find Work acceptable under Contract Documents and Contract fully performed, Architect will promptly issue a final Certificate for Payment. *
- H. Neither final payment nor any remaining retained percentage shall become due until Construction Manager and Contractor submits to Architect;
 - 1. affidavit that payrolls, bills for materials and equipment, and other indebtedness connected with Work for which Owner or Owner's property might be responsible or encumbered (less amounts withheld by Owner) have been paid or otherwise satisfied (AIA Documents G706 and G706A),
 - 2. certificate evidencing that insurance required by Contract Documents to remain in force after final payment is currently in effect and will not be canceled or allowed to expire until at least 30 days' prior written notice has been given to Owner,
 - 3. written statement that Contractor knows of no substantial reason that insurance will not be renewable to cover period required by Contract Documents,
 - 4. consent of surety, if any, to final payment (AIA Document G707),
 - 5. Contractor's and Subcontractor's final release or waiver of liens,
 - 6. if required by Owner, other data establishing payment or satisfaction of obligations, such as receipts, releases and waivers of liens, claims, security interests or encumbrances arising

out of Contract, to extent and in such form as may be designated by Owner, for Owner's review, and

7. if Construction Manager or any Contractor or a Subcontractor refuses to furnish a final release or waiver required by Owner, they may furnish a bond satisfactory to Owner to indemnify Owner against such lien. If such lien remains unsatisfied after payments are made, the Contractor shall refund to Owner all money that Owner may be compelled to pay in discharging such lien, including all costs and reasonable attorneys' fees. *
- I. If Substantial Completion or Final Completion is delayed through no fault of Owner [, Inspector of Record,] or Architect, Architect may invoice Owner as a change in services for such costs, and associated travel costs. Contractor shall reimburse Owner for such cost.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

CONTRACTOR'S NOTICE OF SUBSTANTIAL COMPLETION

PROJECT: _____

ARCH PROJ. NO.: _____ CONTRACT DATE: _____

CONTRACT FOR: _____

WORK OR DESIGNATED PORTION SHALL INCLUDE: _____

Work performed under this Contract has been thoroughly inspected and is considered to be sufficiently complete, in accordance with Contract Documents, so Owner can occupy or utilize Work or designated portion thereof for its intended use.

- ☐ Certificates of inspections indicating compliance with requirements of governing authorities, are attached hereto.
- ☐ Certificate of Occupancy have been obtained from governing authorities, are attached hereto.
- ☐ A comprehensive list of items to be completed or corrected, prepared by C.M./Contractor is attached, hereto. Failure to include any items on such list does not alter responsibility of C.M./Contractor to complete all Work in accordance with Contract Documents.

Construction Manager/Contractor will complete or correct Work by: _____

CONSTRUCTION MANAGER /
CONTRACTOR:

BY: _____ DATE: _____

OWNER (agrees) (does not agree) to accept portion designated above separately from rest of Project.

Owner intends to utilize, occupy or take use on: _____

OWNER: _____

BY: _____ DATE: _____

The Work designated above, has been determined to be:

- ☐ Substantially Complete and a Certificate of Substantial Completion will be issued.
- ☐ Not substantially complete for following reasons: _____

ARCHITECT: HDR Architecture, Inc.

BY: _____ DATE: _____

DISTRIBUTION:

☐ OWNER ☐ ARCHITECT ☐ CONSTRUCTION MANAGER ☐ CONTRACTOR

END OF CONTRACTOR'S NOTICE OF SUBSTANTIAL COMPLETION

CONTRACTOR'S CERTIFICATE OF COMPLETION

PROJECT: _____

ARCH PROJ. NO.: _____ CONTRACT DATE: _____

CONTRACT FOR: _____

This is to certify that I am an authorized official of, and have been properly authorized by said firm or corporation to certify following:

I know of my own personal knowledge, and do hereby certify on behalf of C.M./Contractor that Work has been reviewed and thoroughly inspected for compliance with Contract Documents, that Work has been completed in accordance with Contract Documents and Contract fully performed, that all equipment and systems have been tested and are operating satisfactorily, that all Contract closeout requirements have been completed satisfactorily and submitted, know of no substantial reason that insurance will not be renewable to cover period required by Contract Documents, and work is ready for final inspection and acceptance.

Attached are three (3) copies of following documents, which are required prior to final payment:

- ☐ Final Application for Payment.
- ☐ Contractor's Affidavit of Payments of Debts and Claims: AIA Document G706.
- ☐ Contractor's Affidavit of Release of Liens: AIA Document G706A.
- ☐ Contractor's Final Release or Waiver of Liens.
- ☐ Consent of Surety to Final Payment: AIA Document G707.
- ☐ Certificate evidencing that insurance required by the Contract Documents to remain in force after final payment is currently in effect, and will not be canceled or allowed to expire until at least thirty (30) days prior written notice has been given to Owner.
- ☐ The list(s) of if items which were to be completed and corrected, with each item initialed to indicate Construction Manager has verified completion or correction of each.
- ☐ List of subcontractors and equipment suppliers.
- ☐ Certified list of all sales and service taxes paid.
- ☐ Letter of site conformance by licensed surveyor.
- ☐ If required by Owner, other data establishing payment or satisfaction of obligations arising out of Contract.
- ☐ Bond(s) satisfactory to Owner to indemnify Owner against liens from Contractors or Subcontractors
- ☐ Transmittal indicating Owner has received Project Record Documents.

I understand that acceptance of final payment by Construction Manager, Contractor, a Subcontractor or material supplier shall constitute a waiver of claims by that payee except those previously made in writing and identified by that payee as unsettled at time of final Application for Payment.

C.M./CONTRACTOR: _____ BY: _____

TITLE: _____ DATE: _____

Subscribed and sworn to me this _____ day of _____

NOTARY PUBLIC: _____

My commission expires: _____

DISTRIBUTION: ☐ OWNER ☐ ARCHITECT

END OF CONTRACTOR'S CERTIFICATE OF COMPLETION

SECTION 01 78 36

WARRANTIES

PART 1 - GENERAL

1.1 DESCRIPTION

- A. Warranties specified in Divisions 02 through 48 Sections shall be in addition to and run concurrent with other warranties required by Contract Documents.
 - 1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for a particular product and specifically endorsed by manufacturer to the Owner.
 - 2. Special Warranty: Written warranty required by or incorporated into the Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for the Owner.
- B. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.
- C. Manufacturer's Warranties:
 - 1. Provide for products, equipment, systems, and installations required by Divisions 02 through 48 Sections of Contract Documents for duration indicated.
 - 2. Where manufacturer's standard warranties or guarantees or both expire before duration required by other Sections of Contract Documents, obtain and pay for extensions as part of Contract Price.
- D. Special Warranties:
 - 1. Refer to Divisions 02 through 48 Sections for specific content requirements and particular requirements for submitting special warranties.
 - 2. Provide written Special Warranties for products, equipment, systems, installations, and joint responsibilities as noted and required by Divisions 02 through 48 Sections of Contract Documents for duration indicated.
 - 3. Prepare a written document that contains appropriate terms and identification, ready for execution.
 - a. Modified and properly executed Manufacturer's standard form to include project specific information.
 - b. Submit draft for approval before final execution.
 - 1) See Section 01 33 00.
- E. Provide Warranties. Special Warranties and Guarantees prior to final payment.
 - 1. Provide in electronic data format.
 - a. Coordinate format with Owner.
- F. Warranties. Special Warranties and Guarantees required by Contract Documents shall commence on date of Substantial Completion of Work unless otherwise indicated in Certificate of Substantial Completion.

1.2 SUBMITTALS

- A. Contract Closeout Information:
 - 1. Transmittal letter indicating Owner's receipt of electronic data format containing product equipment and system warranties or guarantees or both required by other Sections of Contract Documents.

1.3 JOB CONDITIONS

- A. If for any reason, Contractor cannot warrant or guarantee or both any portion of Work using products or construction methods indicated or required by other Sections of Contract

Documents, notify Architect in writing during bid period, and before contracts are awarded, indicating reasons and names of products and data on substitutions that can be warranted or guaranteed or both.

1. Should Contractor fail to notify Architect, Contractor will be considered as having agreed to warrant or guarantee the Work indicated.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 PRODUCT, EQUIPMENT AND SYSTEM WARRANTIES AND GUARANTEES

- A. Compile approved warranties and guarantees or both required by other Sections of Contract Documents.
 1. Index by Section, with each warranty, guarantee, or both clearly labeled.
 - a. Identify each volume with project name and contents.
 2. Identify each warranty or guarantee or both in manner consistent with names and identification numbers used in Contract Documents.
 3. Provide transmittal letter containing:
 - a. Date
 - b. Project title
 - c. Contractor's name and address
 - d. Title and number of warranties, guarantees, or both.
 - e. Indication of Owner's receipt
 4. Deliver to Owner prior to final payment with copy of transmittal letter indicating Owner's receipt.

END OF SECTION

SECTION 01 81 19

CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes:
 - 1. Special requirements for Indoor Air Quality (IAQ) management during construction operations.
 - 2. Procedures for testing baseline IAQ. Baseline IAQ requirements specify maximum indoor pollutant concentrations for acceptance of the facility.
 - 3. Requirements for Independent Materials Testing of specific materials anticipated to have measurable impact on IAQ.
- B. Related Sections:
 - 1. Section 01 31 19 – Project Meetings.
 - 2. Section 01 74 00 – Cleaning.
 - 3. Section 01 78 00 – Closeout Submittals: Final Submittals.

1.2 DEFINITIONS

- A. Definitions pertaining to sustainable development: As defined in ASTM E2114.
- B. Adequate ventilation: Ventilation, including air circulation and air changes, required to cure materials, dissipate humidity, and prevent accumulation of dust fumes, vapors, or gasses.
- C. Environmental pollution and damage: The presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade the utility of the environment for aesthetic, cultural, or historical purpose.
- D. Hazardous Materials: Any material that regulated as a hazardous material in accordance with 49 CFR 173, requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have the potential to meet the definition of hazardous Waste in accordance with 40 CFR 261. Throughout this specification, hazardous material includes hazardous chemicals.
 - 1. Hazardous materials include: pesticides, biocides, and carcinogens as listed by recognized authorities, such as the Environmental Protection Agency (EPA) and the international Agency for Research on Cancer (IARC).
 - 2. Do NOT submit MSDS sheets to Architect as construction submittals.
- E. Indoor Air Quality (IAQ): The composition and characteristics of the air in an enclosed space that affect the occupants of that space. The indoor air quality of the space refers to the relative quality of air in a building with respect to contaminants and hazards and is determined by the level of indoor air pollution and other characteristics of the air, including those that impact thermal comfort such as air temperature, relative humidity and air speed.
- F. Interior final finishes: materials and products will be exposed at interior, occupied spaces: including flooring, wallcovering, finish carpentry, and ceilings.
- G. Packaged dry products: materials and products that are installed in dry form and are delivered to the site in manufacturer's packaging; including carpets, resilient flooring, ceiling tiles, and insulation.
- H. Wet products: Materials and products installed in wet form, including paints, sealants, adhesives, and special coatings.

1.3 QUALITY ASSURANCE

- A. Inspection and Testing Lab Qualifications: Minimum of 5 years experience in performing the types of testing specified herein.

1.4 PRECONSTRUCTION MEETING

- A. After award of Contract and prior to the commencement of the Work, schedule and conduct meeting with Owner and Architect to discuss the proposed IAQ Management Plan and to develop mutual understanding relative to details of environmental protection.

PART 2 -

2.1 SUBMITTALS

- A. Indoor Air Quality (IAQ) Management Plan: Not less than 10 days before the Preconstruction meeting, prepare and submit an IAQ Management Plan including, but not limited to, the following:
 - 1. Schedule for application of interior finishes.
 - 2. Revise and resubmit Plan as required by Owner.
 - a. Approval of Contractor's Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations.
 - b. Baseline Indoor Air Quality (IAQ) Test Reports.
 - c. Independent Materials Testing Reports. Submit for the following products:
 - 1) Emissions:
 - a) Fireproofing ;material on appropriate substrate.
 - b) Ceiling tile.
 - c) Resilient flooring.
 - d) Carpet including adhesive and concrete flooring.
 - e) Interior paint on appropriate substrate, including any primer coat.
 - f) Wall covering.
 - 2) Lethal Toxic Potency.
 - a) Ceiling tile.
 - b) Resilient flooring.
 - c) Carpet including adhesive and concrete flooring
 - d) Wall covering.
 - e) Office equipment.
 - 3) Microbial Growth:
 - a) Fireproofing material on appropriate substrate.
 - b) Ceiling tile.
 - c) Wall covering.
 - d) Product Data: Submit product data for filtration media used during construction and during operation. Include Minimum Efficiency Reporting Value (MERV).
- B. Material Safety Data Sheets: Submit MSDS when required as support for a material meeting a particular LEED requirement. However, submit the MSDS data as a LEED submittal and not as a construction submittal. Provide MSDS for the following products.
 - 1. Adhesives.
 - 2. Floor and wall patching/leveling materials.
 - 3. Caulking and sealants.
 - 4. Insulating materials.
 - 5. Fireproofing and firestopping.
 - 6. Carpet.

7. Paint.
8. Clear finish for wood surfaces.
9. Lubricants.
10. Cleaning products.

PART 3 - PRODUCTS – NOT USED

PART 4 - EXECUTION

4.1 INDOOR AIR QUALITY (IAQ) MANAGEMENT

- A. During construction, comply with SMACNA IAQ Guidelines for Occupied Buildings under construction.
- B. HVAC Protection: To the greatest extent possible, isolate and/or shut down the return side of the HVAC system during construction. When ventilation system must be operational during construction activities, provide temporary filters.
- C. Source Control: Provide low and zero VOC materials as specified.
- D. Pathway Interruption: Isolate areas of work as necessary to prevent contamination of clean or occupied spaces. Provide pressure differentials and/or physical barriers to protect clean or occupied spaces.
- E. Housekeeping: During construction, maintain project and building during the products systems to prevent contamination of building spaces.
 1. Protect stored on-site and installed absorptive materials from moisture damage.
 2. Provide minimum 48 hour pre-ventilation of packaged dry products prior to installation. Remove from packaging and ventilate in a secure, dry, well-ventilated space free from strong contaminant sources and residues. Provide a temperature range of 60 degrees F minimum to 90 degree F maximum continuously during the ventilation period. Do not ventilate within limits of Work unless otherwise approved by Architect.
 3. Provide adequate ventilation during and after installation of interior wet products and interior final finishes.
 4. Provide filtration media with a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2 during construction and during Owner occupancy. Coordinate with work of Division 23.
- F. Scheduling: Schedule construction operations involving wet products prior to packaged dry products to the greatest extent possible.

4.2 INDOOR AIR QUALITY (IAQ) BASELINE

- A. Coordinate with commissioning as specified in Section 01 91 00. Upon verification of HVAC system operation, perform baseline IAQ testing.
 1. Perform testing for minimum 3 locations in each air handling zone. Perform in the breathing zone; between 3 feet and 6 feet from the floor.
 2. Collect air samples on three consecutive days during normal business hours (between the hours of 8:00 am and 5:00 pm) with building operating at normal HVAC rates. Average the result of each three-day test cycle to determine compliance or non-compliance of indoor air quality for each air handling zone tested.
 3. Sample and record outside air levels of formaldehyde and TVOC contaminants air outside air intake of each respective air handling unit simultaneously with indoor tests to establish basis of comparison for these contaminant levels.
- B. Baseline IAQ shall conform to the following standards and limits:
 1. Carbon Monoxide: Not to exceed 9 ppm.
 2. Carbon Dioxide: Set points not to exceed 530 ppm higher than outdoor ambient levels. Assess indoor Carbon Dioxide concentrations in accordance with ASTM D6245.

3. Airborne Mold and Mildew: Simultaneous indoor and outdoor reading.
- C. VOCs and particulates: Monitor VOCs (volatile organic compounds) in indoor air rates, and qualities of the listed contaminants shall not exceed the following limits. The levels do not account for contributions from office furniture, occupants, and occupant activities.
 1. Carbon Monoxide: Not more than 9 parts per million and not more than 2 parts per million higher than outdoor air.
 2. Airborne Mold and Mildew: Measure in relation to outside air; not higher than outside air.
 3. Formaldehyde: Not more than 50 parts per billion.
 4. Formaldehyde: Measure in micrograms per cubic meter, in relation to outside air; not more than 20 micrograms per cubic meter higher than outside air.
 5. Total Volatile Organic Compounds (TVOC): Not more than 500 micrograms per cubic meter.
 6. Total Volatile Organic Compounds (TVOC): Measure in micrograms per cubic meter, in relation to outside air; not more than 200 micrograms per cubic meter higher than outside air.
 7. Particulates (PM10): Not more than 50 micrograms per cubic meter.
 8. Total Particulates (PM): Measure in micrograms per cubic meter, in relation to outside air; not more than 20 micrograms per cubic meter higher than outside air.
- D. Test Reports: Prepare test reports showing the results and location of each test, a summary of the HVAC operating conditions, a listing of any discrepancies and recommendations for corrective actions, if required.
 1. Include certification of test equipment calibration with each test report.
 2. If an test fails the standard, the Contractor is responsible to ventilate the building with 100 percent outside air until the building passes both air quality test and duct inspections. Retesting shall be performed at no additional expense to the Owner

4.3 INDEPENDENT MATERIALS TESTING

- A. Emissions: Indicate type and rate of emissions in a 24 hour period at 35 degrees Centigrade and 50 percent relative humidity per unit of product. Indicate type and rate of emissions under fire condition.
 1. Small Scale Chamber: Test and report emissions from products and materials indicated in accordance with ASTM D5116.
 2. Full Scale Chamber: Test and report emissions from products and materials indicated in accordance with ASTM D6670.
- B. Lethal Toxic Potency: Test for lethal toxic potency of smoke produced from the materials and products indicated under fire conditions in accordance with ASTM E1678.
 1. Report results in accordance with Section 13 of ASTM E1678.
- C. Support of Microbial Growth; Test and report in accordance with ASTM D6329. Indicate susceptibility of product or material to colonization and amplification of microorganisms. Identify microorganisms and conditions of testing.
 1. Normal conditions: Perform testing at 35 degrees Centigrade and 50 percent relative humidity.
 2. Extreme conditions: Perform worst case scenarios screening tests by providing an atmosphere where environmental conditions may be favorable for microbial growth.

END OF SECTION

SECTION 02 41 00

DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Demolition, as indicated, in accordance with provisions of Contract Documents.
- B. Completely coordinate with work of other trades.

1.2 QUALITY ASSURANCE

- A. Conduct work in accordance with OSHA and EPA requirements.
- B. National Fire Protection Association (NFPA):
 - 1. NFPA 241 Standards for Safeguarding Construction, Alteration, and Demolition Operations.
- C. Design, engineering, and construction of shoring, bracing and supports are responsibility of Contractor.
 - 1. Employ a Registered Engineer, licensed to practice structural engineering in state in which project is located.
 - 2. Design to support dead, live, and lateral, wind or seismic loads required by code or as otherwise indicated, along with construction loads during demolition until permanent construction is in place.

1.3 DESCRIPTION

- A. Work Includes:
 - 1. Demolition of designated structures.
 - 2. Removal of demolition debris.
 - 3. Protection of construction to remain, including:
 - a. Utilities.
 - b. Other items indicated.
- B. Condition of Existing Structures to be demolished:
 - 1. Owner assumes no responsibility for actual condition of structures to be demolished.
 - 2. Owner will maintain building conditions existing at time of inspection for bidding purposes insofar as practicable.
- C. Standpipes:
 - 1. Maintain in operable condition and available for use by fire department.
 - 2. Do not demolish standpipe more than one floor below floor being demolished.

1.4 JOB CONDITIONS

- A. Perform preliminary investigations to ascertain extent of work.
 - 1. Conditions apparent by investigation shall not be allowed as claim for extra cost.
- B. Obtain and pay for permits required by authorities having jurisdiction and notify interested utilities companies prior to commencement of activities.
- C. Obtain approval of authorities having jurisdiction for work affecting existing means of egress.
 - 1. Review with and obtain approval of authorities for temporary construction which affects such areas.
 - 2. Obtain approval of fire authorities.

- D. Separate, store and dispose of hazardous materials and toxic wastes in accordance with local and EPA regulations and criteria listed below:
 - 1. Disposal of fluorescent light tubes in open containers is not permitted.
 - 2. Disposal of ballasts and other building elements containing PCBs in open containers is not permitted.
 - 3. Disposal of building elements containing mercury in open containers is not permitted.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 GENERAL DEMOLITION PROCEDURES

- A. Items Designated for Re-use:
 - 1. Remove, tag, protect from damage, store and deliver to locations designated.
 - 2. Brace motors attached to flexible mountings until reinstallation.
- B. Structural Demolition:
 - 1. Demolish concrete and masonry in small sections.
 - 2. Perform removal to avoid excessive loads on supporting walls, floors or framing.
- C. Existing Utilities to Remain.
 - 1. Keep in service and protect against damage during demolition.
 - 2. Do not interrupt existing utilities serving occupied or facilities in use, except as authorized by Owner.
 - 3. Provide temporary services during interruptions to existing utilities, as acceptable to Owner.
- D. Conduct operations to ensure minimum interference with roads, walks, entrances, exits, and adjacent facilities.
 - 1. Do not close or obstruct private drives, walks or other facilities unless approved in writing.
 - 2. Do not close or obstruct exits from existing facilities or obstruct public thoroughfares and walks without approval of authorities having jurisdiction.
 - 3. Provide alternate routes around closed or obstructed traffic ways.
- E. Provide covered passageways to ensure safe passage of persons in or near areas of work.
- F. Provide barricades and safety lights as required.

3.2 PROTECTION OF EXISTING BUILDING TO REMAIN

- A. Protect Property to Remain:
 - 1. Conduct operations to prevent damage by falling debris or other cause to adjacent buildings, structures, and other facilities as well as persons.
 - 2. Provide interior and exterior shoring, bracing, or support to prevent movement, settlement or collapse of structures.
- B. Protect occupants from injury and discomfort.
- C. Provide temporary dustproof partitions between demolition areas and occupied areas.
 - 1. In public areas use clean, painted 1/2 inches thick plywood.
 - 2. Utilize fire rated construction where required by Authorities Having Jurisdiction.
- D. Provide temporary weather protection and insulation to prevent damage to existing facilities and discomfort to persons in occupied areas.
 - 1. Insulation value: R 19.

3.3 ITEMS SALVAGED FOR OWNER

- A. Remove salvage items at appropriate stage of demolition, but early enough to prevent damage to them by demolition operations:
 - 1. Coordinate with Owner items Owner wishes to save.
- B. Remove salvage items as a unit:
 - 1. Clean, list, and tag for storage.
 - 2. Protect from damage.
 - 3. Salvage each item with auxiliary or associated equipment required for operation.
 - 4. Store in an area designated within building.

3.4 ITEMS SALVAGED FOR CONTRACTOR

- A. Items of salvage value to Contractor may be removed from structure as work progresses.
- B. Transport salvaged items from site as they are removed.
- C. Storage or sale of removed items not permitted on site.

3.5 CLEAN-UP AND DISPOSAL OF DEMOLITION MATERIALS

- A. Remove debris, rubbish, and materials resulting from demolition operations.
 - 1. Remove and legally dispose of off-site.
 - 2. Do not burn materials on site.
- B. Dispose of items and materials not designated for Owner salvage or reuse.
 - 1. Promptly remove from site.
 - 2. Do not store or sell Contractor salvaged items or materials on site.
- C. Clean adjacent structures and improvements of dust, dirt, and debris caused by demolition operations.
- D. Return adjacent areas to condition existing prior to start of work.

END OF SECTION

SECTION 031000

CONCRETE FORMING AND ACCESSORIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification sections, apply to work of this section.

1.2 SUMMARY

- A. Section includes formwork, shoring, reshoring, backshoring, falsework, bracing, and other temporary supports required to form and support all cast-in-place concrete shown on the drawings.
- B. Related Requirements:
 - 1. Specification **014000 "Quality Requirements"** for requirements of material testing and inspection.
 - 2. Specification **014529 "Structural Testing and Inspections"** for inspection requirements associated with forming and accessories.
 - 3. Specification **018113 "Sustainable Design Requirements"** for sustainable design requirements.
 - 4. Specification **032000 "Concrete Reinforcing"** for reinforcement associated with cast-in-place concrete.
 - 5. Specification **033000 "Cast-in-Place Concrete"** for cast-in-place concrete and related products.

1.3 REFERENCES

- A. Definitions:
 - 1. Backshores: Shores placed snugly under a stripped concrete structural member after the original formwork and shores have been removed from a small area without allowing the structural member to deflect or support its own weight or superimposed construction loads. It is assumed that backshores carry the same load as that carried by the original shores they replace.
 - 2. Formwork: The total system of support for freshly placed concrete, including the mold or sheathing that contacts the concrete and all supporting members, hardware, and necessary bracing.
 - 3. Professional Engineer: A professional engineer who is licensed to practice engineering in the state where the project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for projects with concrete formwork that are similar to that indicated for this Project in material.
 - 4. Reshores: Shores placed snugly under a stripped concrete structural member after the original forms and shores have been removed from the member, thus requiring the member to carry its own weight and superimposed construction loads at the time of installation. Reshores are assumed to carry no load at the time of installation. After the installation of reshores, superimposed construction loads are assumed to be distributed among all members connected by reshores.
 - 5. Shores: Vertical or inclined support members designed to carry the weight of formwork, concrete, and construction loads above.
- B. Reference Standards:

1. Comply with the provision of the following codes, specifications, and standards except where more stringent requirements are shown or specified:
 - a. ACI 117 - Standard Specifications for Tolerances for Concrete Construction and Materials.
 - b. ACI 301 - Specifications for Structural Concrete.
 - c. ACI 318 - Building Code Requirements for Structural Concrete and Commentary.
 - d. ACI 347R - Guide to Formwork for Concrete; 2014.
 - e. ASME A17.1 - Safety Code for Elevators and Escalators; 2013.
 - f. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2015.
 - g. ASTM B221 - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes; 2014.
 - h. ASTM B221M - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes [Metric]; 2013.
 - i. ASTM C177 - Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus; 2013.
 - j. ASTM C578 - Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation; 2015a.
 - k. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2015a.
 - l. PS 1 - Structural Plywood; 2009.
 - m. CRSI, "Manual of Standard Practice."

1.4 RESPONSIBILITY

- A. The design, construction, and safety of all formwork shall be the responsibility of the Contractor. All forms, shores, reshores, backshores, falsework, bracing, and other temporary supports shall be engineered to support all loads imposed including the wet weight of concrete, construction equipment, live loads, lateral loads due to wind and wet concrete imbalance. The Contractor shall also be responsible for determining when temporary supports, shores, reshores, backshores, and other bracing may be safely removed.

1.5 SUBMITTALS

- A. Product Data: Submit technical data and brochures for carton forms.
- B. Shop Drawings:
 1. Formwork Drawings: Formwork drawings, prepared under the supervision and sealed by the formwork design engineer, shall be submitted for Owner's record and shall be reviewed by the Engineer for conformance to structural layout only. Such shop drawings shall indicate all dimensions and types of materials, sizes, lengths, connection details, design allowance for construction loads, anchors, form ties, shores, braces, construction joints, reveals, camber, openings, formwork coatings, and all other pertinent information.
 2. Pan Form Shop Drawings: The Contractor shall submit pan shop drawings for Engineer's review and approval. Approval will be for conformance to structural layout only.
 3. Shoring Plan: Submit drawings to indicate the number of levels of shoring, proposed time and sequence of formwork and shore removal, minimum concrete strength for stripping of forms and shore removal, assumed construction loads, amount and layout of shores (specify whether backshores or reshores), and length of time shores are to be left in place. This plan shall be strictly followed by the Contractor. Shoring plans are to be submitted for Owner's record only and will not be reviewed or returned.
- C. Temporary Structure Design Submittals: Submit the following items for the Owner's records:
 1. Design Calculations: Submit, for record purposes, calculations of all concrete formwork sealed by the formwork design engineer.

1.6 QUALITY ASSURANCE

A. Qualifications:

1. Licensed Professionals: The formwork design engineer retained by the Contractor shall be a professional engineer registered in the state where the project is located and shall be experienced in the design of concrete formwork.

PART 2 - PRODUCTS

2.1 PAN FORMS

A. Specification: Unless specified otherwise, concrete joist and pan-formed beam construction shall conform to Manual of Standard Practice, Chapter 10, as published by CRSI.

B. Material and Pan Type:

1. Material: Pans shall be fabricated either of steel that is free of dents, irregularities, sag and rust or of glass-fiber reinforced plastic that is molded under pressure with matched dies. Pan forms allowing warped surfaces, leakage of concrete at joints, and uneven surfaces beyond tolerance levels will not be acceptable.
2. Subject to pan tolerance and the surface finish required by the Surface Finish Class SF-1.1 as shown on the drawings, pan forms may be either new pans or reconditioned pans at Contractor's option. Forms may be "long forms", "flange forms", "long flange forms", or "adjustable forms" at Contractor's option. Pan splices may be lapped, reinforced butt jointed, or semi-butt jointed (using end caps welded back-to-back with 2" maximum distance between pan ends).
3. Subject to pan tolerance and the surface finish required by the Surface Finish Class SF-1.0 as shown on the drawings, pan forms may be either new pans or reconditioned pans at Contractor's option. Forms may be "long forms", "flange forms", "long flange forms", or "adjustable forms" at Contractor's option. Pan splices may be lapped, reinforced butt jointed, or semi-butt jointed (using end caps welded back-to-back with 2" maximum distance between pan ends).
4. New Pans: Subject to pan tolerance and the surface finish required by the Surface Finish Class SF-1.1 as shown on the drawings, pan forms shall be new pans only. Forms may be "long forms", "flange forms", or "long flange forms" at Contractor's option. Pan splices may be reinforced butt jointed, or semi-butt jointed (using end caps welded back-to-back with 2" maximum distance between pan ends).
5. New Pans: All pan forms used in areas designated to have Surface Finish-2.3 shall be new pans either one piece continuous from beam to beam or beam to header ("long forms", "long flange forms", or "adjustable forms") without splices or with reinforced butt joint spliced. "Flange forms" are not acceptable, nor will forms be permitted that are lapped spliced or semi-butt joint spliced (using end caps welded back-to-back). Pans shall meet tolerances and the surface finish required for Surface Finish Class 2.3.
6. New Pans. All pan forms used in areas designated to have Surface Finish-3.3 shall be new pans either one piece continuous from beam to beam or beam to header ("long forms", "long flange forms", or "adjustable forms") without splices or reinforced butt joint spliced. "Flange forms" are not acceptable, nor will forms be permitted that are lapped spliced or semi-butt joint spliced (using end caps welded back-to-back). Pans shall meet tolerances and the surface finish required for Surface Finish Class 3.3.
The pan form surfaces specified herein are intended to be architecturally exposed to view and require a high quality surface finish.

2.2 FORM-FACING MATERIALS

- A. Smooth-Formed Finished Concrete: Unless otherwise specified, formwork for exposed concrete surfaces as defined by the Surface Finish Class noted on the drawings, shall consist of plywood, metal, metal framed plywood, or other acceptable surface. Formwork shall provide a continuous straight and smooth surface conforming to the joint system as specified on the Architect's drawings. Form material shall have sufficient thickness to withstand pressure of concrete without bow or deflection. Plywood shall be exterior grade plywood panels, suitable for concrete forms, complying with U.S. Product Standard PS-1, each piece bearing a legible inspection trademark, and as follows:
1. **[Phenolic Surface Film Overlay over Hardwood Face, Class 1 or better.]**
 2. **[High Density Overlay (100/30 min. rating) on Hardwood Face, Class 1 or better.]**
 3. **[High Density Overlay (100/30 min. rating) on Softwood Face, Class 1 or better.]**
 4. **[Medium Density Overlay on Hardwood Face, Class 1 or better, mill-release agent treated and edge sealed.]**
 5. **[Medium Density Overlay on Softwood Face, Class 1 or better, mill-release agent treated and edge sealed.]**
 6. **[Structural 1, B-B, or better, mill oiled and edged sealed.]**
 7. **["B-B (Concrete Form) Plywood", Class 1, or better, mill-oiled and edge sealed.]**
- B. Non-specific formed concrete: Unless otherwise specified, the default finish for formed surfaces shall be rough-form finish constructed with plywood, lumber, metal, or other acceptable material. Lumber shall be dressed on at least two edges and one side for tight fit. The minimum grade shall be B-C, exterior grade.
- C. Textured-form finished concrete: For exposed surfaces as noted on the drawings provide units of form face design, size, arrangement and configuration that matches Architect's control sample. Provide solid backing and form supports to ensure stability of textured form liners. See Architect's drawings, specifications and control sample for special form textured finish concrete.

2.3 CYLINDRICAL COLUMNS AND SUPPORTS

- A. Round section members shall be formed with metal, fiberglass, reinforced plastic, paper, or fiber tubes, unless otherwise specified. Paper or fiber tubes shall be constructed of laminated plies using water-resistant adhesive with wax impregnated exterior for weather and moisture protection. Units shall have sufficient wall thickness to resist loads imposed by wet concrete without detrimental deformation.

2.4 CARTON FORMS

- A. Carton forms shall be manufactured using corrugated paper material with a moisture resistant exterior surface and specifically designed for foundation support. Carton forms shall be designed to support the wet weight of the concrete that is shown by the details to be poured on top of the form, but the weight assumed shall not be less than 600 pounds per square foot. Refer to the General Notes for the restriction on horizontal construction joints. The forms shall be designed in such a way that the bottom of the form will collapse when acted upon by upward movement of the soil.
1. Form Configuration: Carton forms shall be of a vertical cellular configuration only, except as permitted by Paragraph 2 below and shall be rectangular as shown on the details. The depth of the carton forms is shown on the details. Forms shall be manufactured to fit snugly against round piers and shall be baffled in such a way as to prevent concrete from flowing back into the form during the concrete pour.
 2. Other types of forms using different types of paper and different configurations will be accepted if it can be shown by independent tests that the form will properly function and will deteriorate due to moisture in an appropriate period.
 3. Products: Subject to requirements, acceptable manufacturers include, but are not limited to, the following:
 - a. SureVoid Products, Inc.

2.5 PRECAST CONCRETE RETAINERS

- A. Retainers shall be used with voids or carton forms where shown on the drawings.
- B. Retainers shall have a minimum compressive strength of 2,500 PSI and be reinforced with 6x6xW1.4xW1.4 welded wire reinforcement and minimum thickness of 1 5/8".

2.6 SOIL RETAINERS

- A. Soil retainers shall be provided where specified and shown on the drawings to prevent migration of backfill under suspended foundation elements.
- B. Retainers shall be composed of high density polyethylene materials that are not adversely affected by moisture. They must be flexible, impact resistant and have sufficient strength to resist lateral loads applied by soil.
- C. Soil retainers shall extend 3 inches above the void forms and a minimum of 3 inches below the void forms. To allow for construction tolerances, the height of the soil retainer material ordered shall be at least 10 inches greater than the specified void depth. Soil retainers shall not be attached to concrete. Waler ties shall be twisted off to allow soil retainers to rise the specified void depth.
- D. Use 3/8" thickness for void spaces 8" and less. Use 1/2" thickness for void spaces greater than 8" but less than 12". Use 3/4" thickness for void spaces equal to 12". Soil retainers shall be equal to the following:
 - 1. BackFill Retainer by SureVoid
- E. For void spaces greater than 12" but less than or equal to 16", retainers shall be ribbed and made from high density polyethylene. Soil retainers shall be equal to the following:
 - 1. Motzblock by M&M Construction specialties
 - 2. SureRetainer by VoidForm Products

2.7 FORMWORK COATINGS

- A. Formwork coatings shall be of a commercial formulation that will not bond with, stain, nor adversely affect concrete surfaces or impair subsequent treatment of concrete surfaces requiring bond or adhesion, nor impede curing with water or curing compounds. Provide a product that has a maximum VOC (Volatile Organic Compounds) of 50 g/l but not greater than that permitted by the local government agency having jurisdiction in the area where the project is located.
- B. Products: Subject to compliance with requirements, provide one of the following:
 - 1. Dayton Superior; Clean Strip J1EF.
 - 2. Dayton Superior; Farm Fresh XL.
 - 3. W.R. Meadows; Duogard II.

2.8 NAILS AND FASTENERS

- A. Use only galvanized nails and fasteners for securing formwork in structures exposed to weather or unconditioned spaces such as garages, canopies, and porte-cochères.

2.9 FORM TIES

- A. Factory-fabricated, removable or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to minimize spalling of concrete on removal.

1. Exposed Surfaces: For surfaces designated with Surface Finish Class SF-2.x or SF-3.x, furnish units that will leave no portion of the tie closer than 3/4 inch to the plane of the concrete surface and that will leave holes not larger than one inch in diameter in concrete surface when the ends or end-fasteners have been removed.
2. Dampproofed Surfaces: Furnish ties with integral water-barrier plates to walls indicated to receive dampproofing or waterproofing.
3. Exposed to Weather or Unconditioned Space: Provide removable, glass-fiber-reinforced plastic, stainless steel, or galvanized form ties that will leave no corrodible metal closer than 1 1/2 inches in surfaces that will be exposed to weather or in an unconditioned space in the final structure. The ties shall leave holes no larger than one inch in diameter in concrete surfaces when the ends or end-fasteners are removed.

2.10 CHAMFER STRIPS

- A. Provide wood, metal, PVC, or rubber strips, 3/4 by 3/4 inch, minimum.

PART 3 - EXECUTION

3.1 FABRICATION AND CONSTRUCTION

- A. Design, erect, support, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic construction loads that might be applied until the concrete structure can support such loads.
 1. The formwork design engineer shall design the concrete formwork, formwork removal, shoring, reshoring, and backshoring.
- B. Construct forms to sizes, shapes, lines, and dimensions shown, and to obtain accurate alignment, location, grades, level and plumb work in finished structures. Provide for openings, offsets, sinkages, keyways, recesses, moldings, rustications, reglets, chamfers, blocking, screeds, bulkheads, anchorages, inserts, and other features required in work. Use selected materials to obtain required finishes. Solidly butt joints and provide back-up at joints to prevent leakage of concrete mortar.
- C. Carton Forms:
 1. Carton forms shall be kept dry and protected until concrete is poured. Wet, compressed, or deteriorated carton forms shall not be used. Do not wrap or cover carton forms with polyethylene sheets or permanent waterproof cover as that will prevent proper deterioration of the forms.
 2. The Contractor shall use expandable foam to fill all gaps and holes between carton forms and at intersections with foundations.
 3. For slab conditions, cover carton forms with a 1/4 inch masonite protection cover board to prevent puncture and other damage during construction.
- D. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush plates or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical. Kerf wood inserts for forming keyways, reglets, recesses, and the like, to prevent swelling and for easy removal.
- E. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.
- F. Provide temporary openings where interior area of formwork is inaccessible for cleanout, for inspection before concrete placement, and for placement of concrete. Securely brace temporary openings and patch forms to prevent loss of concrete mortar. Locate temporary openings on forms at inconspicuous locations.

- G. Chamfer exposed corners and edges as indicated, using specified chamfer strips fabricated to produce uniform smooth lines and tight edge joints.
- H. Provisions for Other Trades: Provide openings in concrete formwork to accommodate work of other trades. Determine size and location of openings, recesses and chases from trades providing such items. Accurately place and securely support items built into forms.
- I. Pan Form Fabrication and Construction:
 - 1. Factory fabricate pan form units to specified sizes and shapes as indicated on the drawings. Units shall be designed for easy removal without damaging placed concrete. Units shall be shored properly and adjoining pan units shall be blocked if required to prevent lateral or vertical deflection of formwork during concrete placement.
 - 2. Load Distribution Ribs: Provide load distribution ribs at least 5" wide for all pans 30" wide and narrower and elsewhere where indicated on the drawings. Minimum rib spacing shall be:
 - a. None in spans less than 20 feet.
 - b. One near the center of spans 20 to 30 feet.
 - c. Two near the third points of spans over 30 feet.

Discontinue ribs between two adjacent joists or beams that have differences in span larger than 33%, between a joist and an adjacent parallel wall, and between a joist and an adjacent parallel beam that is 1 ½ or more times wider than the joist.

3.2 CLEANING AND TIGHTENING

- A. Thoroughly clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and all other debris just prior to concrete placement. Retighten forms and bracing prior to concrete placement as required to prevent concrete mortar leaks and maintain proper alignment.

3.3 CLEANING AND RE-USE OF FORMS

- A. Forms reused in the work shall be repaired and cleaned. Split, frayed, delaminated, or otherwise damaged facing material will not be acceptable for exposed surfaces. Forms intended for successive concrete placement shall have surfaces cleaned, fins and laitance removed, and joints tightened to avoid surface offsets. New form coating compound shall be applied to reused forms. Thin form-coating compounds only with thinning agent of type, and in amount, and under conditions of form-coating compound manufacturer's directions. Do not allow excess form-coating material to accumulate in forms or to come into contact with in-place concrete surfaces against which fresh concrete will be placed. Apply in compliance with manufacturer's instructions. Coat steel forms with a non-staining, rust-preventative form oil or otherwise protect against rusting. Rust-stained steel formwork is not acceptable.

3.4 TOLERANCES

- A. Unless specified otherwise, all tolerances for concrete formwork shall conform to ACI Standard 117, "Standard Tolerances for Concrete Construction and Materials". Before concrete placement the Contractor shall check lines and levels of erected formwork and make any corrections and adjustments as required to ensure proper size and location of concrete members and stability of forming systems. During concrete placement the Contractor shall check formwork and supports to ensure that forms have not displaced and that completed work will be within specified tolerances.
- B. Construct forms so as to limit the offset between adjacent pieces of formwork facing material in accordance with the surface tolerance class as defined in ACI 117 corresponding to the Surface Finish Class noted on the drawings. The offset limits shall apply to both abrupt and gradual variations in the surface.

- C. Pan Construction Surface Tolerance: Discontinuities in the concrete surface formed by pan construction shall be limited to the values noted in the surface tolerance classes of ACI 117 corresponding to the Surface Finish Classes noted on the drawings.
- D. Prior to each concrete pour, the Contractor shall engage a qualified surveyor to verify that work is within specified tolerances. The surveyor shall report in writing to the Architect, Engineer and Contractor certifying that the work is acceptable or indicating any deviations from allowable tolerances.
- E. The Owner shall hire an independent qualified surveyor to verify the proper form, line, position, and elevation of the finished concrete work. The results of each survey shall be sent to the Owner, Architect/Engineer, and Contractor and shall identify any deviation from specified tolerances. All work not in conformance with specified tolerances shall be removed at the Contractor's sole expense if so specified by the Owner.

3.5 SHORES AND SUPPORTS

- A. Comply with requirements of ACI 301 for shoring, reshoring and backshoring in concrete construction and as herein specified where more stringent.
- B. Design: Shores and reshores or backshores must be designed to carry all loads transmitted to them. A rational analysis should be used to determine the number of floors to be shored, reshored, or backshored, subject to the minimums stated in the following paragraph, and to determine the loads transmitted to the floors, shores and reshores or backshores as a result of the construction sequence. The analysis should consider, but should not necessarily be limited to, the following:
 - 1. Structural design load of the slab or member including live load, partition loads, and other loads for which the engineer designed the slab. The live load reduction method for the design of certain members is shown on the structural drawings. The reduced live load and an allowance for construction loads shall be taken into consideration when performing the analysis.
 - 2. Dead load weight of the concrete and formwork.
 - 3. Construction live loads, such as placing crews and equipment or stored materials.
 - 4. Design strength of concrete specified.
 - 5. Cycle time between placement of successive floors.
 - 6. Strength of concrete at time it is required to support shoring loads from above.
 - 7. The distribution of loads between floors, shores, and reshores or backshores at the time of placing concrete, stripping formwork, and removal of reshoring or backshoring.
 - 8. Span of slab or structural member between permanent supports.
 - 9. Type of formwork systems, i.e., span of horizontal formwork components, individual shore loads, etc.
 - 10. Minimum age of concrete where appropriate.
 - 11. Alignment of shores: Where possible, shores for any floor shall be placed directly above previously placed shores so that load will be transferred directly to such shores. Where shores are not vertically aligned, calculations shall include verification that the structure can support the reaction.

3.6 REMOVAL OF FORMS AND SUPPORTS

- A. Determination by Contractor's Registered Engineer: The Contractor's registered professional engineer shall determine and submit for Owner's record the time and sequence of formwork and shore removal subject to the criteria as specified below. The submittal shall clearly distinguish between reshoring and backshoring procedures.
- B. Determining in situ Strength of Concrete: The General Contractor shall be responsible for making and curing concrete cylinders, cured under field conditions, for the purpose of determining concrete strength at time of form and shore removal. Such cylinders shall be made by the Contractor and tested by his testing laboratory. Alternatively, the in situ strength of concrete may be determined by the Maturity Method following the requirements of ASTM C 1074. An acceptable system for this method is the "intelliRock" system manufactured and supplied by EngiUS Constructive Intelligence of Stillwater, OK.
- C. Records of Weather Conditions: The Contractor shall be responsible for keeping records of weather conditions to be used in the decision on when to remove forms.

- D. Formwork Not Supporting Concrete: Formwork not supporting concrete, such as sides of beams, walls, columns and similar parts of the structure, may be removed after cumulatively (not necessarily consecutively) curing at not less than 50°F for 12 hours after placing concrete, provided the concrete is sufficiently hard so as not to be damaged by form removal operations and provided curing and protection operations are maintained. If ambient air temperatures remain below 50°F, if retarding agents are used, or if Type II and Type V Portland cement is used, then this specified minimum period shall be increased as required to safely remove the forms without damage to the concrete. Where such forms also support formwork for slab or beam soffits, the removal times of the latter shall govern.
- E. Formwork Supporting Weight of Concrete: Formwork supporting weight of concrete such as beam soffits, joists, slabs and other structural elements shall not be removed until concrete has attained at least the following percentages of the design compressive strength:
1. Joists, Beam Bottoms: 70%, but not less than 2,800 psi.
 2. Slabs: 70%, but not less than 2,800 psi.
- F. Placing Reshores and Backshores:
1. All shoring operations shall be carried out in accordance with a planned sequence as determined by the Contractor's shoring engineer.
 2. Shoring operations shall be performed so that at no time will areas of new construction be required to support combined dead and construction loads in excess of the available strength as determined by the design loads (as specified in the General Notes) and the developed concrete strength (as determined by field cured cylinders or maturity method) at the time of stripping and reshoring or backshoring.
 3. Shores (backshores or reshores) shall not be removed until the structural member supported has sufficient strength to support all applied loads.
 4. For backshoring operations, the forms shall be removed in such a manner that individual structural members are not allowed to deflect and carry load.
 5. Reshoring operations require that the structural members be strong enough to safely support their own weight before stripping of formwork.
 6. For reshoring operations, no structural member shall be overstressed under its own dead weight plus the weight of the floors above and construction loads assigned to the structural member by a rational analysis that accounts for the relative stiffness of each floor with due consideration of concrete age and strength. While reshoring is underway, no construction loads shall be permitted on the new construction unless it can safely support the construction loads.
 7. Where possible, shores shall be located in the same position on each floor so that they will be continuous in their support from floor to floor.
- G. Post-Tensioned Construction: Formwork supporting post-tensioned floor construction, including shores, reshores and backshores shall be designed to support any additional loads produced by the stressing operation. Design of the formwork shall take into account the possibility of the slab, beam, or girder lifting off the formwork during tensioning, thereby transferring the entire load to the support areas. Construct the formwork to permit movement of the member without damage during application of the post-tensioning force. Removal of formwork supporting post-tensioned construction shall be coordinated with the post-tensioning operations and formwork shall not be removed until the concrete has been stressed fully.

3.7 FIELD QUALITY CONTROL

- A. Field Inspection: Refer to Specification **014529 "Structural Testing and Inspections"** for inspection requirements associated with forming and accessories.

END OF SECTION 031000

SECTION 032000

CONCRETE REINFORCING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification sections, apply to work of this section.

1.2 SUMMARY

- A. Section includes labor, materials, hardware, equipment, transportation and services required to fabricate and place all reinforcement for cast-in-place concrete including bars, welded wire reinforcement, ties and supports shown on the drawings and as specified. Prestressing reinforcement is specified in Post-Tensioned Concrete and/or Precast Concrete sections of the specifications.
- B. Related Requirements:
 - 1. Specification **014000 "Quality Requirements"** for requirements of material testing and inspection.
 - 2. Specification **014529 "Structural Testing and Inspections"** for testing and inspection requirements associated with concrete reinforcing.
 - 3. Specification **018113 "Sustainable Design Requirements"** for sustainable design requirements.
 - 4. Specification **031000 "Concrete Forming and Accessories"** for forming associated with cast-in-place concrete.
 - 5. Specification **033000 "Cast-in-Place Concrete"** for cast-in-place concrete and related products.

1.3 REFERENCES

- A. Reference Standards:
 - 1. Comply with all provisions of the following codes, specifications, and standards except where more stringent requirements are shown or specified:
 - a. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."
 - b. ACI 301, "Specifications for Structural Concrete for Buildings."
 - c. ANSI/AWS D1.4, "Structural Welding Code – Reinforcing Steel."
 - d. CRSI, "Manual of Standard Practice."

1.4 ADMINISTRATIVE REQUIREMENTS

- A. Coordination:
 - 1. Quality Control: The Contractor is responsible for quality control, including workmanship and materials furnished by subcontractors and suppliers.
 - 2. Document Conflict and Precedence: In case of conflict among documents, including architectural and structural drawings and specifications, notify the Architect/Engineer prior to submitting proposal. In case of conflict between and/or among the structural drawings and specifications, the strictest interpretation shall govern, unless specified otherwise in writing by the Architect/Engineer.

- B. Preinstallation Meetings: The Reinforcing-Placing subcontractor shall attend the Pre-Concrete Conference conducted by the Concrete Contractor as described in Specification **033000** "Cast-in-Place Concrete."

1.5 SUBMITTALS

- A. Product Data: Submit manufacturer's product data with application and installation instructions for proprietary materials and items including mechanical splices, hooked anchorage systems, large-headed stud punching shear reinforcement, dowel bar replacement systems, and dowel bar sleeves. For fiber reinforcement, submit manufacturer's product data, including application rate and mixing instructions.
- B. Shop Drawings:
 - 1. Submit shop drawings for all reinforcing steel and related accessories for the Engineer's approval. Shop drawings shall show arrangement and layout, bending and assembly diagrams, bar schedules, stirrup spacing, splicing and laps of bars and shall be prepared in accordance with CRSI Standards.
 - 2. Submit shop drawings indicating which members will use fusion welding process for assembly. Shop drawings shall show complete structural details indicating the size of stirrups, the size of holding wires, and welding requirements.
- C. Certificates:
 - 1. Submit, for record, mill certificates and/or test results signed by Producer, for all reinforcement.
 - 2. Provide certification from fiber reinforcement manufacturer that fiber reinforcement complies with specified requirements.
- D. Test and Evaluation Reports:
 - 1. Submit International Code Council (ICC) Evaluation Service Reports indicating approval from ICC Evaluation Service, Inc. for mechanical splices, hooked anchorage systems, large-headed stud punching shear reinforcement and dowel bar replacement systems.
 - 2. Submit test results for deformed bar material not identifiable as outlined in Part 2 below.
- E. Special Procedure Submittals: Submit shop welding program for fusion welding including the type of the specific fusion welding machine and the quality control/inspection protocol for the shop welding.
- F. Qualification Statements: Submit welding certificates.

1.6 QUALITY ASSURANCE

- A. Testing Laboratory Requirements: The Owner's Testing Laboratory shall:
 - 1. Review the Welding Procedure Specification (WPS) submitted by the Contractor for any reinforcing steel other than ASTM A 706 that is proposed to be welded for consistency with acceptable welding practices and AWS.
 - 2. Review the welder qualifications by certification or verify by retesting and shall obtain the welder certificates.
- B. Welder Qualifications: Qualify procedures and personnel according to ANSI/AWS D1.4.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Steel Reinforcement:

1. Reinforcing materials shall be delivered from the mill in bundles that are identified as to heat number and manufacturer and accompanied with mill and analysis test reports and an affidavit from the supplier stating that the material conforms to the requirements of the governing ASTM specification listed herein.
2. Use reinforcing steel made from 90% recycled material, two-thirds of which shall be post-consumer material. A minimum of 50% of the material in the reinforcement must have been extracted, harvested, or recovered as well as manufactured, within 500 miles of the project site.
3. Deformed bar material that is not identifiable according to the criteria listed above shall be tested for tensile strength and bend tests according to ASTM A 615 on a sample of two bars for each ten tons or fraction thereof of unidentified material for each bar size. The bars shall be a minimum of 24 inches long. Bend tests are not required for #14 and #18 bars. Submit the results of such tests for record.
4. Reinforcing Bars: Reinforcing bars shall conform to ASTM A 615, Grade 60 or Grade 80 as noted on the drawings.
5. Reinforcing Bars: Reinforcing bars used as longitudinal reinforcing in Seismic Special Moment Frames, Special Structural Walls, Coupling Beams, and as noted on the drawings shall conform to ASTM A 706, Grade 60. As an alternate, ASTM A 615 Grade 60 reinforcement is permitted to be used such that it complies with the additional following requirements:
 - a. The actual yield strength based on mill tests shall not exceed the nominal yield strength F_y by more than 18,000 PSI.
 - b. The ratio of the actual tensile strength to the actual yield strength is not less than 1.25.
6. Reinforcing Steel: Reinforcing steel used as transverse reinforcing or as spiral reinforcing as noted on the drawings shall conform to ASTM A 1035.
7. Weldable Reinforcing Bars: All reinforcing bars noted on the drawings as being required to be welded shall conform to ASTM A 706, Grade 60.
8. Deformed Bar Anchors: 3/8" to 5/8" diameter AWS Type C studs manufactured in conformance with ASTM A 1064 with a minimum yield strength of 70,000 psi and a tensile strength of 80,000 PSI. 3/4" or larger diameter, ASTM A 706 bars of equal size with welds to steel substrate that develop the full strength of the anchor. ASTM A 615 reinforcing bars may not be substituted for deformed bar anchors. Reinforcement shall be approved by the ICC-Evaluation Service, Inc and shall have the Evaluation Service Report submitted for Engineer review. The following are acceptable products, provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. Nelson Stud Welding, Inc.; Nelson D2L Deformed Bar Anchor Studs (ESR-2907).
 - b. Tru-Weld Division, TFP Corporation; Deformed Bar Anchors (ESR-2823).
9. Galvanized Reinforcing Steel: Provide galvanized reinforcing bars at the locations indicated on the drawings. Galvanized reinforcing bars shall conform to ASTM A 767 Class II (2.0 ounces zinc per square foot), hot dipped galvanized after fabrication and bending. Bars that are to be galvanized shall conform to the type of steel required for the given situation as noted on the drawings.
10. Epoxy-Coated Reinforcing Steel: Provide epoxy coated reinforcing bars at the locations indicated on the drawings. Epoxy coated reinforcing bars shall conform to ASTM A 775. Bars that are to be epoxy coated shall conform to the type of steel required for the given situation as noted on the drawings.
11. Epoxy-Coated Fabricated Reinforcing Steel: Provide reinforcing bars that are epoxy-coated after fabrication at the locations indicated on the drawings. Reinforcing bars that are epoxy-coated after fabrication shall conform to ASTM A 934. Bars that are to be epoxy-coated shall conform to the type of steel required for the given situation as noted on the drawings.
12. Plain Steel Welded Wire Reinforcement: ASTM A 1064 with a yield strength of 65,000 PSI. Provide in flat sheets only.
13. Deformed-Steel Welded Wire Reinforcement: ASTM A 1064 with a yield strength of 70,000 PSI. Provide in flat sheets only.
14. Galvanized Plain-Steel Welded Wire Reinforcement: ASTM A 1064, fabricated from galvanized steel wire into flat sheets.
15. Epoxy Coated Plain-Steel Welded Wire Reinforcement: ASTM A 884, Class A, plain steel.
16. Epoxy Coated Deformed-Steel Welded Wire Reinforcement: ASTM A 884, Class A, deformed steel.
17. Strands: Uncoated seven wire, one half inch diameter, stress relieved 270 KSI strand low relaxation type, ASTM A 416 and "Specification for Unbonded Single Strand Tendons" as published by the Post-Tensioning Institute.

18. Prestressing Bars: All prestressing bars shall be deformed threadbars conforming to ASTM A 722 with a minimum ultimate tensile strength of 150 KSI and other properties as specified on page 15-31 of the PCI Design Handbook, Seventh Edition. Threadbars, plate anchorages and couplings shall be furnished by Dywidag Systems International or Williams unless approved otherwise in writing by the Engineer.
19. Wire: Smooth wire for spiral reinforcement shall conform to ASTM A 82 with a minimum yield strength of 70,000 PSI.
20. Epoxy-Coated Plain-Steel Wire: ASTM A 884, Class A, plain-steel wire.
21. Headed Steel Stud Punching Shear Reinforcement: Punching shear reinforcement using headed studs welded to flat bars shall be manufactured in conformance with ASTM A 1044 **<and approved by the ICC Evaluation Service, Inc. as expressed in an ICC Evaluation Report for use as punching shear reinforcement for slabs and footings designed in accordance with ACI 421.1>**. The following are acceptable products for the studs and for the assemblies:
 - a. Studs:
 - 1) Nelson Stud Welding Inc.; Nelson Punching Shear Resistor Studs (ESR-1170).
 - 2) Tru-Weld Division, TFP Corporation; Tru-Weld Punching Shear Resistor Studs (ESR-2822).
 - b. Assemblies:
 - 1) Dayton Superior Corporation; Dayton Shear Resistance System (ESR-2696).
 - 2) Decon, Inc.; Decon Studrails (ESR-2494).
 - 3) Dywidag Systems International USA, Inc.; Dywidag Shear System (ESR-2534).
 - 4) Jobsite Stud Welding, Inc.; Jobsite Stud Welding Punching Shear Resistor Shear Rail Assemblies (ESR-3264).
 - 5) SRL Industries LTD.; SRL Punching Shear Resistor Shear Rails (ESR-2938).
 - 6) Suncoast Post-Tension, LTD.; Suncoast Stud Reinforcement System (ESR-2708).
 - 7) Tendon Systems, LLC; Shearails.

B. Fiber Reinforcement:

1. Synthetic Micro Fiber Reinforcement: Collated, fibrillated, or monofilament polypropylene, cellulose, or multi-filament nylon fibers conforming to ASTM C 1116, Type III or Type IV. Reinforcement shall be approved by the ICC-Evaluation Service, Inc and shall have the Evaluation Service Report submitted for Engineer review. The following are acceptable products, provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. The Euclid Chemical Company; Fiberstrand Series.
 - b. Forta Corporation; Econo-Mono or Econo-Net (ESR-2720).
 - c. Propex Concrete Systems Corp.; Fibermesh 300 (ESR-1165).
 - d. W.R. Grace & Company; Grace Microfibers (ESR-1506).
 - e. Nycon, Inc.; Nycon RC.
 - f. Buckeye Technologies; UltraFiber 500 (ESR-1032).
 - g. BASF Construction Chemicals; MasterFiber M or F series.
 - h. Maccaferri; Fibromac.
2. Synthetic Macro Fiber Reinforcement: Monofilament polypropylene/polyethylene fibers conforming to ASTM C 1116, Type III having an aspect ratio between 50 and 90 and a minimum tensile strength of 90 KSI. The fiber lengths shall be between 1.5 and 2 inches long. Reinforcement shall be approved by the ICC-Evaluation Service, Inc. and shall have the Evaluation Service Report submitted for Engineer review. The following are acceptable products, provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. The Euclid Chemical Company; Tuf-Strand SF.
 - b. Forta Corp.; Forta-Ferro.
 - c. W.R. Grace; Strux 90/40.
 - d. Propex Concrete Systems, Corp.; Fibermesh 650.
 - e. Bekaert Corp.; Synmix.
 - f. BASF Construction Chemicals; MasterFiber MAC.

3. Steel Fibers: Provide deformed cold-drawn wire or modified cold-drawn steel fibers meeting the requirements of ASTM A 820, Types I or V, and that are listed as an acceptable product for use in the D900 series of UL Fire Rating Assemblies. The fibers shall have a minimum tensile strength of 145,000 PSI when tested in accordance with ASTM A 370. The fibers shall have a minimum aspect ratio of 50 and maximum aspect ratio of 100. Acceptable products include:
 - a. Bekaert Corp; Dramix 3D 65/60 BG (Type 1), or Dramix 3D 80/60 BG (Type 1).
 - b. Propex Concrete Systems, Corp.; Novocon 1050 (Type I), Novocon 1050 HE (Type I), or Novomesh 850 (Type I).
 - c. BASF Construction Chemicals; MasterFiber FF or FS series.
 - d. Maccaferri; Wirand series.

2.2 SPLICES

- A. End Bearing Compression Splices: Members with end bearing compression splices shall have vertical bars saw cut or otherwise finished for true bearing. Bar ends shall terminate in flat surfaces within 1 1/2 degrees of a right angle to the axis of the bars and shall be fitted within three degrees of full bearing after assembly. Splice bars shall be held in concentric contact by a suitable device. The following are acceptable end bearing compression devices:
 1. nVent Electric, plc.; Speed Sleeve.
 2. Other Engineer-approved product.
- B. Mechanical Tension Splices:
 1. Mechanical splices shall conform to Type 2 splices.
 - a. Type 1 splice shall develop, in tension and compression, 1.25 times the specified yield strength of the splice bar.
 - b. Type 2 splice shall meet the requirements of Type 1 splice and, in addition, develop the full tensile strength of the splice bar.
 2. Splices shall be approved by the ICC-Evaluation Service, Inc and shall have the Evaluation Service Report submitted for Engineer review.
 3. The bar ends that are to attach to the splice shall be prepared and installed in accordance with the manufacturer's requirements.
 4. The following are acceptable mechanical tension splices (splices qualified for use with grade 80 bars are parenthetically noted), provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. BarSplice Products, Inc.; BPI-Grip XL System (ESR-2299). (Type 1 or Type 2)
 - b. BarSplice Products, Inc.; Taper Threaded Grip-Twist System (ESR-2299). (Type 1 or Type 2)
 - c. BarSplice Products, Inc.; Position Taper Threaded Grip-Twist System (ESR-2299). (Type 1 or Type 2)
 - d. Headed Reinforcement Corporation; HRC 500/510 Xtender Mechanical Coupler System (ESR-2764). (Type 1 or Type 2)
 - e. Dayton Superior Corporation; DBDI Reinforcing Bar Mechanical Splice System (ESR-2649). (Type 1 or Type 2).
 - f. Dayton Superior Corporation; Bar-Lock Coupler Systems for Splicing Reinforcement Bars, S-Series (ESR-2495). (Type 1)
 - g. Dayton Superior Corporation; Bar-Lock Coupler Systems for Splicing Reinforcement Bars, L-Series (ESR-2495). (Type 1 or Type 2)
 - h. Dayton Superior Corporation; Taperlok Reinforcing Bar Mechanical Splice Couplers (ESR-2481). (Type 1 or Type 2)
 - i. Dextra Manufacturing Co., Ltd.; Bartec Mechanical Splice System for Steel Reinforcing Bars in Concrete (ESR-1705). (Type 1 or Type 2)
 - a. nVent Electric, plc.; Lenton Standard Coupler (IAPMO-UES 0129). (Type 1 or Type 2)

- b. nVent Electric, plc.; Lenton Ultimate Standard Coupler (IAPMO-UES 0129). (Type 1 or Type 2)
 - c. nVent Electric, plc.; Lenton Ultimate Position Coupler (IAPMO-UES 0129). (Type 1 or Type 2)
 - d. BarSplice Products, Inc.; ZAP Screwlok (qualified for use with grade 80 bars) (ER-5461). (Type 1 and Type 2)
 - e. Splice Sleeve North America; NMB Splice-Sleeve (ER-5645). (Type 1 or Type 2).
- C. Dowel Bar Replacement: All grade 60 reinforcing steel dowel bars shown on the drawings crossing concrete construction joint surfaces with inserts cast flush against the form and having reinforcing bars connected to the insert in a subsequent concrete pour shall conform to the following:
 - 1. Splice connection to the insert shall develop the 1.25 times the specified yield strength and the full tensile strength of the spliced bar.
 - 2. Splices shall be approved by the ICC Evaluation Service, Inc. as expressed in an ICC Evaluation Service Report which shall be submitted for review.
 - 3. The following are acceptable products (for use only with grade 60 bars), provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. Dextra Manufacturing Co., Ltd.; Bartec Mechanical Splice System for Steel Reinforcing Bars in Concrete (ESR-1705).
 - b. nVent Electric, plc.; Lenton Form Saver SA (IAPMO-UES 0129).
- D. Hooked Anchorage Replacement: Reinforcing bar terminations shall be manufactured out of ASTM A 576, ASTM A 615, or A 706 material and shall develop the full tensile strength of the bar when installed at the manufacturer's recommended depth.
 - 1. The anchorage shall be approved by the ICC Evaluation Service Inc. as expressed in an ICC Evaluation Service Report which shall be submitted for review.
 - 2. The following are acceptable products (for use only with grade 60 bars), provided that their Evaluation Service Reports are still valid at the time of intended use on the project:
 - a. Dextra Manufacturing Co., Ltd; Bartec Mechanical Anchorages for Steel Reinforcing Bars in Concrete (ESR-2166).
 - b. Headed Reinforcement Corporation; HRC 555 Headed Reinforcing Bars (ESR-2935).
 - c. Headed Reinforcement Corporation; HRC 670 HeadLock (IAPMO-UES 0177).
 - d. nVent Electric plc.; Lenton Terminator (IAPMO-UES 0188).
 - e. nVent Electric plc.; Lenton Ultimate (IAPMO-UES 0188)

2.3 REINFORCEMENT ACCESSORIES

- A. Joint Dowel Bars: Smooth bars used to dowel across slab-on-grade construction joints shall conform to ASTM A 615, Grade 40 or ASTM A 36, plain-steel bars. Cut bars true to length with ends square and free of burrs.
- B. Epoxy-Coated Joint Dowel Bars: Smooth epoxy-coated bars used to dowel across slab-on-grade construction joints shall conform to ASTM A 775 with ASTM A 615, Grade 40 or ASTM A 36 plain-steel bars. Cut bars true to length with ends square and free of burrs.
- C. Dowel Bar Sleeves: Plastic or gage metal (26 gauge minimum) sleeves with an inside diameter of 1/16 inch greater than the dowel bar that it encases, that have the strength, durability, and design to provide free movement of the dowel relative to the concrete slab and that are specifically manufactured for this purpose.
- D. Alternate Slab-on-Grade Joint Load Transfer Systems: A system that consists of flat, ASTM A 36 plate that is saw cut into a square or rectangular shape and is embedded into or encased by a plastic sleeve that allows movement in both lateral directions but not in the vertical direction. Acceptable systems are manufactured by PNA Construction Technologies with products known by the names "Diamond Dowel System" and "PD³ Basket" and Greenstreak Group Inc. with products known as "Speed Plate" and "Double-Tapered Basket".

- E. Tie Wire: Tie wire shall be annealed steel tie wire, minimum 16 gauge.
- a. Tie wire in architecturally exposed concrete shall be plastic coated or stainless steel.
 - b. Tie wire for epoxy-coated reinforcement shall be epoxy-coated.
 - c. Tie wire for galvanized reinforcement shall be galvanized.
- F. Holding Wire: Holding wire shall conform to ASTM A 82 or ASTM A 1064.
- G. Coating Repair Materials: Repair damaged areas of epoxy-coated or galvanized reinforcement using the following products.
1. Epoxy Repair Coating: Liquid, two-part, epoxy repair coating compatible with epoxy coating on reinforcement and complying with ASTM A 775.
 2. Zinc Repair Material: ASTM A 780, zinc-based solder, paint containing zinc dust, or sprayed zinc shall be used to repair damaged areas of galvanized reinforcement.
- H. Supports for Reinforcement: Provide supports for reinforcement including bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Use wire bar type supports complying with CRSI recommendations.
1. Slabs-on-Grade: Use precast concrete bar supports (dobies) or supports with sand plates or horizontal runners designed for use on ground.
 2. Spread Footing Bottom Reinforcement: Use precast concrete bar supports (dobies) or chairs designed for soil-supported slabs.
 3. Mat Foundation: Use precast concrete bar supports (dobies), chairs designed for soil-supported slabs, or cast-in-place concrete curbs.
 4. Exposed to View Concrete: Provide supports with legs which are plastic protected (CRSI, Class 1) or stainless steel protected (CRSI, Class 2).
 5. Support of Epoxy-Coated Reinforcement: Provide epoxy-coated or other dielectric-polymer-coated wire bar supports to support epoxy-coated reinforcement.
 6. Support of Galvanized Reinforcement: Provide galvanized wire bar supports to support galvanized reinforcement.

2.4 ALTERNATES:

1. Products Requiring International Code Council (ICC) Evaluation Service Reports:
 - a. For those products listed in Part 2 as requiring Evaluation Service Reports (ESRs), alternate products that do not have ESRs will be considered by the Engineer only if valid research reports or test data from an independent and approved agency is provided and use of the product receives prior approval from the Building Official.

PART 3 - EXECUTION

3.1 FABRICATION AND DELIVERY

- A. Bending and Forming: Fabricate bars of indicated sizes and accurately form to shapes and lengths indicated and required, by methods not injurious to materials. Do not heat reinforcement for bending. Bars shall be free from injurious defects, have a workman-like finish with no excessive rust and/or pitting, and have no unusual kinks or bends.
- B. Marking and Shipping: Bundle reinforcement and tag in accordance with Section 7.4.5 of the CRSI "Manual of Standard Practice." Transport and store at site so as not to damage material. Keep sufficient supply of tested, approved, and proper reinforcement at the site to avoid delays. Maintain reinforcing bars free of mud, dirt, grease, or other coating.

- C. Repair of Epoxy-Coated Reinforcing: Repair cut and damaged epoxy coatings on fabricated reinforcing before delivery with epoxy repair coating according to ASTM D 3963

3.2 PLACING REINFORCEMENT

- A. Comply with CRSI recommended practice for "Placing Reinforcing Bars", for details and methods of reinforcement placement and supports and as herein specified.
- B. Before placing reinforcement and again before concrete is placed, clean reinforcement of loose rust and mill scale, earth, ice, and other materials which reduce or destroy bond with concrete.
- C. Accurately position, support, and secure reinforcement against displacement by formwork, construction, or concrete placement operations. Locate and support reinforcing by chairs, runners, bolsters, spacers, and hangers as required. Exercise particular care to maintain proper distance and clearance between parallel bars and between bars and forms. Provide spreaders and spacers to hold steel in position. Support steel at proper height upon approved chairs.
- D. Place reinforcement to obtain at least minimum coverages for concrete protection. Arrange, space, and securely tie bars and bar supports to hold reinforcement in position during concrete placement operations. Set tie wires so ends are directed into concrete, not toward exposed concrete surfaces.
- E. Support of Column and Wall Reinforcing Steel:
 - 1. It is the responsibility of the contractor to ensure that all reinforcing assemblies have adequate strength and stability to resist loads imposed during construction.
- F. Support of Spread Footing Reinforcing Steel:
 - 1. Bottom Steel: Support bottom reinforcing mat to provide the specified clearance to the bars. Spacing between supports shall not exceed 4'-0" centers each way.
 - 2. Top Steel: Support top reinforcing on steel angle frames braced in both directions or on special standee support bars. Spacing between supports shall not exceed 4'-0" centers each way. The depth of the supports shall provide the specified clearance from the bars to the top of the concrete. The design of the support steel shall be the responsibility of the Contractor.
- G. Support of Mat Foundation Reinforcing Steel:
 - 1. Bottom Steel: Support bottom reinforcing mat to provide the specified clearance to the bars. Spacing between supports shall not exceed 4'-0" centers each way.
 - 2. Top Steel: Support top reinforcing on steel angle frames braced in both directions or on special standee support bars. Spacing between supports shall not exceed 4'-0" centers each way. The depth of the supports shall provide the specified clearance from the bars to the top of the concrete. The design of the support steel shall be the responsibility of the Contractor.
- H. Install welded wire reinforcement in as long lengths as practicable. Provide lap splice for wires of adjoining pieces per ACI 318 Chapter 25.5.3 or 25.5.4 and lace splices with wire. Offset end laps in adjacent widths to prevent continuous laps in either direction.
- I. Coordinate with other trades and expedite materials and labor to avoid omissions and delay.
- J. Install waterproof membrane or vapor retarder as specified prior to placing steel for concrete slabs-on-grade.
- K. Extend reinforcement continuous through construction joints unless otherwise shown on the drawings.
- L. Slab-on-Grade Joint Dowel Bars: Support slab-on-grade joint dowel bars independently of support for slab reinforcement on soil supported slab bolsters or specially manufactured cradles such that dowel bar remains parallel to slab surface and at right angles to joint during concreting operations. Lightly coat the exposed

end of the dowel with a paraffin-base lubricant, asphalt emulsion, form oil, or grease or use a dowel bar sleeve.

- M. Alternate Slab-on-Grade Joint Load Transfer Systems: Install the alternate load transfer system in accordance with the manufacturer's instructions such that the largest plane of the flat plate is parallel to the plane of the subgrade on which the slab is bearing.
- N. Provide and place additional reinforcing steel at all sleeves and openings in beams, slabs, and walls as specified on the drawings. Where sleeves or openings not shown on the drawings interrupt the reinforcement, consult with Engineer for instructions for placing and splicing of bars. Provide required additional reinforcing steel at no additional cost to the Owner.
- O. Epoxy-Coated Reinforcement: Use epoxy-coated steel tie wires to fasten epoxy-coated reinforcement. Repair cut and damaged epoxy coatings with epoxy repair coating according to ASTM D 3963.
- P. Galvanized Reinforcement: Use galvanized steel tie wires to fasten galvanized reinforcement. Repair cut and damaged zinc coatings with zinc repair material.
- Q. Do not bend reinforcement that is embedded partially in concrete except in locations noted on the drawings or approved by the Engineer.

3.3 SPLICING REINFORCING STEEL

- A. Provide splices as indicated on the drawings. Splice reinforcing bars only at locations shown on the structural drawings and approved shop drawings. Unauthorized or unscheduled splices not approved by the Engineer in writing will not be accepted.
- B. All lap splices in reinforcing steel shall be contact lap splices unless detailed otherwise on the drawings.
- C. Maintain proper cover and spacing between reinforcing bars at splices.
- D. Lap unscheduled reinforcing bars not otherwise specified with a Class B lap splice. Lap welded wire reinforcement per ACI 318 Chapter 25.5.3 or 25.5.4.
- E. Reinforcing Steel Placement in Mat Foundations:
 - 1. Size, length, spacing, and location of all mat reinforcing steel is shown on the mat plans and details. See details on the drawings for required stagger pattern of top and bottom bar splices and for sequence of placing mat reinforcing steel layers.
 - 2. The number of splices shall be minimized by using bar runs of 60'-0" as much as possible. Unless noted otherwise, continuous top reinforcing bars shall be spliced along column centerlines. Continuous bottom reinforcing bars shall be spliced mid-way between columns.
 - 3. Provide Class B tension lap splices for all bars #11 and smaller. Stagger splices as shown in the typical details.
 - 4. Avoid splices of #14 and #18 bars where possible. Where required, a mechanical tension splice as specified shall be provided. No more than 50% of such bars shall be spliced in any 5'-0" width of mat cross-section. Spliced bars shall be staggered with un-spliced bars.
- F. Manufacturer of mechanical tension splice shall be present for first day's installation.

3.4 WELDING REINFORCING STEEL

- A. Welding reinforcing steel is permitted only where specifically shown on the drawings. All welding shall conform to AWS D1.4. Only weldable reinforcing steel conforming to ASTM A 706 or deformed bar anchors conforming to ASTM A 1064 shall be permitted. ASTM A 615 bars may not be welded for structural use.
- B. Tack welding of reinforcement shall not be permitted.

- C. Fusion welding of preassembled cages shall be permitted only under the following conditions:
1. Written approval is received from the Building Official and the Engineer.
 2. Fusion welding of holding wires to ties, stirrups, and hoops in beams, columns and grade beams to preassemble reinforcing cages is permissible. The holding wire area shall not exceed five percent of the beam, column, or grade beam cross sectional longitudinal steel area. Fusion welding is not allowed to longitudinal reinforcing steel in any beam, column, or grade beam.
 3. Fusion welding of holding wires to the ends of the reinforcing steel placed in spread footings or mats is permitted if the fusion weld occurs within six bar diameters of the free end of the bar. Fusion welding is not allowed at the end of coupled, T-headed, or weld spliced bars.
 4. Fusion welding of holding wires shall not occur on a bent portion of a reinforcing bar. Bars shall not be bent where a fusion weld occurs.

3.5 SHRINKAGE AND TEMPERATURE REINFORCEMENT

- A. Provide shrinkage and temperature reinforcement as indicated on the drawings at right angles to main top and bottom bars for all structural slabs unless detailed otherwise on the drawings.

3.6 PLACEMENT OF WELDED WIRE REINFORCEMENT

- A. Wherever welded wire reinforcement is specified as reinforcement in pan-formed beams or slabs, it shall be continuous and properly lapped per ACI 318 Chapter 25.5.3 or 25.5.4 across the entire concrete surface and not interrupted by beam or girders.

3.7 REINFORCEMENT IN JOIST DISTRIBUTION RIBS

- A. Provide reinforcement in ribs, minimum one - #5 continuous top and bottom unless indicated otherwise on the drawings.

3.8 REINFORCEMENT IN COMPOSITE STEEL DECK SLAB

- A. Composite steel deck slabs shall be reinforced as indicated on the drawings.
- B. Extra Reinforcement over Girders: Provide additional reinforcing steel over interior girders as shown on the drawings.
- C. Placement of Slab Reinforcement: Provide bolsters, high chairs, and/or additional reinforcing as shown in details on the drawings to support the reinforcing with the clear cover shown on the drawings.
- D. Fiber-Reinforced Concrete in Composite Steel Deck Slabs:
1. Provide fibers of the type and at the dosage rate shown on the drawings in concrete cast over composite steel deck.
 2. The fiber-reinforced concrete shall be produced in accordance with ASTM C 1116 and have a residual strength of 80 PSI when tested in accordance with ASTM C 1399.

3.9 REINFORCEMENT AROUND OPENINGS IN COMPOSITE STEEL DECK SLABS

- A. For all openings in steel deck not framed with structural steel and greater than 10" in width in either direction, provide additional reinforcing steel as shown in details on the drawings.

3.10 REINFORCEMENT IN PAN-FORMED BEAM SLABS

- A. Reinforcement: Provide reinforcing in pan-formed beam slabs as shown on the drawings.

- B. Placement of Slab Reinforcement: Provide required bar supports and additional reinforcing as shown in details on the drawings to support slab reinforcing with the clear cover shown on the drawings.

3.11 REINFORCEMENT IN GRADE BEAMS

- A. Provide reinforcing in grade beams as shown on the drawings.
- B. Bar Support for Grade Beam Cages: Grade beam bottom steel shall be supported at 5'-0" maximum centers using beam bolsters that provide bottom cover to the reinforcing steel as noted on the drawings. Beam bolsters used shall be designed and manufactured for support on soil.

3.12 REINFORCEMENT IN TOPPING SLABS

- A. Provide welded smooth wire reinforcement minimum 6 x 6 W1.4 x W1.4 in all topping slabs unless specified otherwise on the drawings.
- B. Provide reinforcing in topping slabs as shown on the drawings.

3.13 REINFORCEMENT IN HOUSEKEEPING PADS

- A. Provide welded smooth wire reinforcement 6 x 6 W2.9 x W2.9 minimum in all housekeeping pads supporting mechanical equipment unless detailed otherwise on the drawings.
- B. Provide reinforcing in housekeeping pads as shown on the drawings.

3.14 MECHANICAL AND PLUMBING REQUIREMENTS

- A. Refer to Mechanical and Plumbing Drawings for concrete requiring reinforcing steel. Such reinforcement shall be furnished as part of the work of this section.

3.15 FIELD QUALITY CONTROL

- A. Field Testing and Inspection: Refer to Specification **014529 "Structural Testing and Inspections"** for testing and inspection requirements associated with concrete reinforcing.p

END OF SECTION 032000

SECTION 03 30 00
CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division **01** Specification sections, apply to work of this section.

1.2 SUMMARY

- A. Section includes all labor, materials, services, equipment, and hardware required in conjunction with or related to the forming, delivery, and pouring of all cast-in-place concrete work. Concrete paving and walks are specified in Division **32**. Architectural Concrete, Precast Concrete and Post-Tensioned Concrete are specified in other Division **03** sections.
- B. Related Requirements:
1. Specification **014000** "Regulatory Requirements" for requirements of material testing and inspection.
 2. Specification **014529** "Structural Testing and Inspections" for inspection requirements associated with cast-in-place concrete.
 3. Specification **031000** "Concrete Forming and Accessories" for forming associated with cast-in-place concrete.
 4. Specification **032000** "Concrete Reinforcing" for reinforcement for cast-in-place concrete.

REFERENCES

- C. Reference Standards:

1. Codes and Standards: Comply with provisions of following codes, specifications and standards, except where more stringent requirements are shown or specified:
 - a. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."
 - b. ACI 301, "Specifications for Structural Concrete for Buildings."
 - c. ACI 305.1, "Specification for Hot Weather Concreting."
 - d. ACI 318, "Building Code Requirements for Structural Concrete."
 - e. CRSI, "Manual of Standard Practice."

1.3 ADMINISTRATIVE REQUIREMENTS

- A. Coordination:

1. Quality Control: The Contractor is responsible for quality control, including workmanship and materials furnished by subcontractors and suppliers.
2. Document Conflict and Precedence: In case of conflict among documents, including architectural and structural drawings and specifications, notify the Architect/Engineer prior to submitting proposal. In case of conflict between and/or among the structural drawings and specifications, the strictest interpretation shall govern, unless specified otherwise in writing by the Architect/Engineer.
3. Materials and installed work may require testing and retesting, as directed by the governing building code or the Architect/Engineer, at any time during progress of work.
 - a. The Contractor shall provide adequate notification to the Owner's Testing Agency of construction operations including the project schedule to allow the Testing Agency to schedule inspections. Failure to notify sufficiently may result in additional costs incurred by the Testing Laboratory that may be back-charged to the Contractor by the Owner.
 - b. The Contractor shall cooperate with laboratory personnel, provide access to the work, and provide access to manufacturer's operations.
 - c. The Contractor shall make adequate arrangement with the Owner's Testing Agency for inspection of material stockpiles and facilities.
 - d. The Contractor shall provide to the laboratory certificates and representative samples of materials proposed for use in the work in quantities sufficient for accurate testing as specified.
 - e. The Contractor shall furnish casual labor, equipment, and facilities as required for sampling and testing by the laboratory and otherwise facilitate the required inspections and tests.
 - f. Inspection or testing by the Owner does not relieve the Contractor of his responsibility to perform the Work in accordance with the Contract Documents. Tests not specifically indicated to be done at the Owner's expense, including retesting of rejected materials and installed work, shall be done at the Contractor's expense. See **Structural Testing and Inspections** section of the Specifications.

4. Responsibility for Selection and Use of Concrete Admixtures and Chemical Treatments: The Contractor shall be responsible for selecting admixtures and surface treatments that are compatible with the intended use of the concrete including all final surface treatments called for within this or other specifications or on the structural or architectural drawings. The Contractor is responsible for following the manufacturer's instructions for the use of their product including abiding by any limitations placed by the manufacturer on the use of any of its products.
- B. Preinstallation Meetings:
1. Design Mixture Conference: At least 30 days prior to submittal of design concrete mixtures, the Contractor shall hold a meeting or telephone conference to review the detailed requirements for preparing the design concrete mixtures. Participants shall include representatives from the Contractor, Owner's Testing Laboratory, Concrete Supplier, Architect, and Engineer.
 2. Pre-Concrete Conference:
 - a. At least seven days prior to beginning concrete work, the Contractor shall conduct a meeting to review the proposed design mixtures and to discuss required methods and procedures to produce concrete construction of the required quality. Also, review requirements for submittals, status of coordinating work and availability of materials. Establish work progress schedule and procedures for materials inspection, testing, and certifications. The contractor shall send a pre-concrete conference agenda to all attendees seven days prior to the scheduled date of the conference.
 - b. The Contractor shall require responsible representatives of every party who is concerned with the concrete work to attend the conference, including but not limited to the following:
 - 1) Contractor's Superintendent.
 - 2) Laboratory responsible for the concrete design mix.
 - 3) Laboratory responsible for field quality control.
 - 4) Concrete Subcontractor.
 - 5) Ready-Mix Concrete Producer.
 - 6) Admixture Supplier.
 - 7) Concrete Pumping Contractor.
 - 8) Fiber Reinforcement Representative.
 - 9) Owner's and Architect's/Engineer's Representative.
 - c. Minutes of the meeting shall be recorded, typed and printed by the Contractor and distributed by him to all parties concerned within five days of the meeting. One copy of the minutes shall be transmitted to the following for information purposes:
 - 1) Owner's Representative.
 - 2) Architect.
 - 3) Engineer-of-Record.
 - d. The Engineer shall be present at the conference. The Contractor shall notify the Engineer at least seven days prior to the scheduled date of the conference.
- C. Sequencing:
1. Provide for installation of inserts, hangers, metal ties, anchors, bolts, angle guards, dowels, thimbles, slots, nailing strips, blocking, grounds, and other fastening devices required for attachment of work. Properly locate in cooperation with other trades and secure in position before concrete is poured. Do not install sleeves in any concrete slabs, beams, or columns except where shown on the drawings or upon written approval of the Architect/Engineer.
- 1.4 SUBMITTALS
- A. Product Data: Submit manufacturer's product data with application and installation instructions for proprietary materials and items, including admixtures, patching compounds, epoxies, grouts, waterstops, joint systems, curing compounds, dry-shake finish materials, hardeners, sealers, joint fillers, and others as requested by Architect/Engineer.
- B. Shop Drawings:
1. Construction Joints: Submit drawings of proposed construction joint locations in concrete for slab-on-grade, mat foundations, structural floors, roofs and walls. Submit any additional or changed reinforcing that is required at construction joints that differs from that shown on the drawings.
 2. Openings, Sleeves, and Cores: Submit drawings of all openings to be formed, sleeved, cored, or sawcut in cast-in-place elements. Drawings shall indicate size and location of openings, sleeves, or cores.
 3. Penetrations: Submit drawings locating all horizontal and vertical penetrations in beams and joists. Drawings shall indicate location, size, orientation, and type of penetrations.
 4. Embedded Items: Submit drawings showing all items to be embedded in concrete elements, including plates, angles, bolts, and any non-structural items, such as conduit. Drawings shall indicate location, size, orientation, and type of embedded item.

- C. Samples: Submit samples of materials specified if requested by Architect/Engineer, including names, sources, and descriptions.
- D. Certificates:
 - 1. Material and Mill Certificates:
 - a. Provide material and mill certificates as specified herein and in the Testing Laboratory section of the Specifications. The Manufacturer and Contractor shall sign the material and mill certificates certifying that each material item complies with specified requirements.
 - b. Provide certification from admixture manufacturers that chloride ion content complies with specified requirements.
- E. Design Mixtures: Submit for each concrete mixture as specified herein.
- F. Field Quality Control Submittals:
 - 1. Surveys: Submit report certifying that all anchor rods and reinforcing dowels into columns above are in their proper location prior to placing of concrete.
- G. Qualification Statements: Submit certifications for adhesive anchor installers.
- H. Minutes of Preinstallation Meetings: Submit for review.

1.5 QUALITY ASSURANCE

- A. Qualifications:
 - 1. Concrete Supplier: The concrete supplier shall have a minimum of five years of experience in manufacturing ready-mixed concrete products complying with ASTM C 94 requirements for production facilities and equipment. The supplier must be certified according to the National Ready Mixed Concrete Association's Certification of Ready Mixed Concrete Production Facilities.
 - 2. Concrete Contractor: The concrete contractor shall have a minimum of five years of experience with installation of concrete similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful service performance.
 - 3. Adhesive Anchor Installers: The individuals performing the installation of adhesive anchors that are horizontally or upwardly inclined shall be certified in accordance with the ACI/CRSI Adhesive Anchor Installer Certification program.
- B. Survey for Anchor Rods and Reinforcing Steel Dowels: The Contractor shall use a qualified professional engineer or land surveyor to lay out the proper location of all embedded anchor rods and reinforcing steel dowels for columns above before they are encased in concrete. The surveyed locations of such elements shall be submitted to the Architect/Engineer for record, if requested.
- C. Manufacturer Representative Presence:
 - 1. Post-installed anchors: The manufacturer's representative for each post-installed anchor product (adhesive, expansion, undercut, screw, or insert anchor) shall be present during the first day's installation of the product to provide instruction for the correct installation of each type of any to be installed in accordance with the manufacturer's recommendation and the current ICC-ES Evaluation Report.
- D. Mockups: Provide mock-ups as required.

PART 2 - PRODUCTS

2.1 CONCRETE MATERIALS

- A. Refer to the drawings for classes and strengths of concrete required.
- B. Hydraulic Cement:
 - 1. Use ASTM C 150, Type I or Type III, or ASTM C 1157, Type GU or HE unless otherwise specified. Do not use Type III cement in slabs-on-grade unless approved in advance by the Engineer.
 - 2. Concrete Exposed to Sulfates in Soil or Water:
 - a. Exposure Class S1: For concrete designated on the drawings as Exposure Class S1, use ASTM C 150, Type II or ASTM C 1157, Type MS.
 - b. Exposure Class S2: For concrete designated on the drawings as Exposure Class S2, use ASTM C 150, Type V or ASTM C 1157, Type HS.
 - c. Alternate Cement Types for Exposure Classes S1 and S2: ASTM C 150, Type I or III cement may be used for concrete designated as Exposure Class S1 or S2 if the tricalcium aluminate (C₃A) content is less than eight percent for Exposure Class S1 or five percent for Exposure Class S2. ASTM C 150, Type I or III cement may be used for exposure to seawater if the tricalcium aluminate content does not exceed 10 percent and the water/cementitious material ratio of the concrete mix does not exceed 0.40.
 - d. Exposure Class S3: For concrete designated on the drawings as Exposure Class S3, use ASTM C 150, Type V plus pozzolan or slag or ASTM C 1157, Type HS plus pozzolan or slag or ASTM C 595, Type IP (HS) or Type IS (HS). The amount of pozzolan or slag added or in

a blended mix shall be such that has been determined by service record to improve sulfate resistance when used with Type V cement or the amount that when tested according to ASTM C 1012 meets the criteria of Table 4.5.1 in ACI 318-08.

3. Use one brand of cement, for each class of concrete, throughout the project, unless approved otherwise by the Architect/Engineer and the Owner's Testing Laboratory. Submit mill certificates certifying conformance to this specification for each brand and type of cement.
4. Testing of cement in lieu of mill certificate submittal will be required if:
 - a. The cement has been in storage at the mixing site for over 30 days.
 - b. It is suspected by the Owner, Architect, Engineer, or Owner's Testing Laboratory that the cement has been damaged in storage or in transit or is in any way defective.
- C. Low-alkali cement: Cement that has the additional requirement that equivalent alkalis ($\text{Na}_2\text{O} + 0.658\text{K}_2\text{O}$) do not exceed 0.60% according to ASTM C 150-00, Table 2.
- D. Fly Ash: ASTM C 618, Class C or F.
- E. Silica Fume: ASTM C 1240, Amorphous Silica.
- F. Slag Cement: ASTM C 989, Grade 100 or 120 or ASTM C 595, Type IS or Type S.
- G. Normalweight Aggregates: ASTM C 33, and as herein specified. Submit material certificates from aggregate supplier or test results from an independent testing agency certifying conformance to this specification for each source of aggregate.
 1. For concrete identified on the drawings as exposed to Exposure Classes C1 and C2, submit certification that aggregate does not contain any deleterious materials that react with alkalis in the concrete mix to cause excessive expansion of the concrete for concrete that is exposed to wetting, has extended exposure to humid atmosphere, or is in contact with moist ground unless low-alkali cement is used.
- H. Lightweight Aggregates: ASTM C 330. Submit material certificates from aggregate supplier or test results from an independent testing agency certifying conformance to this specification for each source of aggregate.
- I. Water: Comply with the requirements of ASTM C 1602.

2.2 ADMIXTURES

- A. Air-Entraining Admixture: ASTM C 260.
 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. W.R. Grace & Co.; Darex or Daravair series.
 - b. BASF Admixtures, Inc.; MB-VR, MB-AE90, or Micro-Air.
 - c. Sika Corporation; Sika AER.
 - d. The Euclid Chemical Company; Air Mix or AEA-92.
 - e. The Euclid Chemical Company; Eucon Air 30 or Eucon Air 40.
 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
- B. Water-Reducing Admixture: ASTM C 494, Type A. See maximum permissible chloride ion content in concrete specified below.
 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. BASF Construction Chemicals; Pozzolith series or Glenium 7000.
 - b. Sika Chemical Corp.; Plastocrete 161.
 - c. The Euclid Chemical Company; Eucon WR-75 or WR-91.
 - d. W.R. Grace & Co.; WRDA series.
 - e. The Euclid Chemical Company; Eucon NW or Eucon LW.
 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
- C. Mid-Range Water-Reducing Admixture: ASTM C 494, Type A and Type F. See maximum permissible chloride ion content in concrete specified below.
 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. BASF Construction Chemicals; Polyheed series or Glenium 7000.
 - b. The Euclid Chemical Company; Eucon MR.
 - c. Sika Chemical Corp.; Sikament HP.
 - d. W.R. Grace & Co.; Daracem or Mira series.
 - e. The Euclid Chemical Company; Eucon X-15 or Eucon X-20.
 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
- D. High-Range Water-Reducing Admixture (Superplasticizer): ASTM C 494, Type F or Type G. See maximum permissible chloride ion content in concrete specified below.
 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. W.R. Grace & Co.; ADVA or Daracem Series.

- b. BASF Construction Chemicals; Rheobuild 1000 or Glenium series.
 - c. Sika Chemical Corp.; Sikament.
 - d. The Euclid Chemical Company; Eucon 37/1037 or Plastol series.
 - e. The Euclid Chemical Company; Eucon SP or Eucon RD2.
 - 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
 - E. Water-Reducing, Accelerator Admixture (Non-Corrosive, Non-Chloride): ASTM C 494, Type C or E. See maximum permissible chloride ion content in concrete specified below.
 - 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. W.R. Grace & Co.; Polarset, Gilco, Lubricon NCA, Daraset 400, or DCI.
 - b. BASF Admixtures, Inc.; Pozzutec 20+ or Pozzoloth NC 534.
 - c. The Euclid Chemical Company; Accelguard 80/90, NCA, or AcN.
 - d. Sika Chemical Co.; Plastocrete 161FL.
 - e. The Euclid Chemical Company; Eucon AcN.
 - 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
 - F. Water-Reducing, Retarding Admixture: ASTM C 494, Type D. See maximum permissible chloride ion content in concrete specified below.
 - 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. W.R. Grace & Co.; Daratard series.
 - b. BASF Construction Chemicals; Pozzoloth series or DELVO series.
 - c. Sika Chemical Co.; Plastiment.
 - d. The Euclid Chemical Company; Eucon Retarder series.
 - 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all other admixtures to be used.
 - G. Viscosity Modifying Admixture: Used to enhance plastic concrete properties such as workability, pumpability, and stability for "Self-Consolidating Concrete".
 - 1. BASF Building Systems; Rheomac VMA series.
 - 2. The Euclid Chemical Company; Eucon SL or Viscrol.
 - 3. Sika Chemical Co.; VisoCrete series.
 - 4. W.R. Grace & Co.; VMAR series.
 - H. Shrinkage Reducing Admixture.
 - 1. Subject to compliance with requirements, provide one of the following products and manufacturers:
 - a. For Air-Entrained Concrete:
 - 1) Grace Construction Products; Eclipse 4500.
 - 2) The Euclid Chemical Company; Eucon SRA.
 - b. For Non Air-Entrained Concrete
 - 1) Grace Construction Products; Eclipse Floor 200.
 - 2) BASF Construction Chemicals; Tetraguard AS20.
 - I. Calcium Chloride and Chloride Ion Content: Calcium chloride or admixtures containing more than 0.5% chloride ions by weight of the admixture are not permitted. For shrinkage compensating concrete, industrial slabs, and concrete designated as Exposure Class S2 or S3 as noted on the drawings, admixtures must be completely free of chloride ions.
 - J. Certification: Written conformance to all the above-mentioned requirements and the chloride ion content of the admixture as tested by an accredited laboratory will be required from the admixture manufacturer at the time of design mixture review by the Engineer.
- 2.3 WATERSTOPS
- A. Provide waterstops at all construction joints and other joints in all foundation walls below grade and where shown on the drawings. Size to suit joints.
 - 1. Products:
 - a. Swell Hydrophilic Waterstops:
 - 1) Manufacturers: W.R. Grace & Co.; ADCOR ES.
 - b. Polyvinyl Chloride (PVC) Waterstops: Comply with Corps of Engineers CRD-C 572. Provide flat, dumbbell type or centerbulb type as noted on the drawings.
 - c. Rubber Waterstops: Comply with Corps of Engineers CRD-C 513. Provide flat, dumbbell type or centerbulb type as noted on the drawings.
 - d. Preformed Plastic Waterstops: Comply with Federal Specifications SS-S-210A "Sealing Compound for Expansion Joints".
 - 1) Manufacturers: Synko-Flex Products, Inc.
 - 2. American Volclay Products; Bentonite Waterstop RX.

2.4 VAPOR RETARDERS

- A. Provide vapor retarder cover chosen from products specified below over prepared base material where indicated Reference 07 26 00 "Under-slab Vapor Barrier".
1. Plastic Vapor Retarder: Provide a flexible, preformed sheet membrane conforming to ASTM E 1745 with the following properties:
 - a. Class A material.
 - b. Minimum of 15 mils thick.
 - c. Maximum water vapor permeance rating of 0.01 perms after mandatory conditioning as tested by ASTM E 96.
 - d. Manufacturer's recommended tape and mastic.
 - e. Acceptable products include the following:
 - 1) Stego Industries, LLC; Stego Wrap Vapor Barrier (15 mil).
 - 2) Epro; Ecoshield-E (15 mil).
 - 3) Raven Industries; VAPORBLOCK 15 (15 mil).
 2. Bituminous Vapor Retarders: Provide a pre-molded membrane consisting of reinforced core and carrier sheet with fortified bitumen layers, protective weather coating, and plastic anti-stick sheet conforming to ASTM E 1993 with the following properties:
 - a. Maximum water vapor permeance rating of 0.002 perms after mandatory conditioning as tested by ASTM E 96.
 - b. Manufacturer's recommended tape and mastic.
 - c. Acceptable products include the following:
 - 1) W.R. Meadows; Premoulded Membrane Vapor Seal with Plasmatic Core.
 3. Tape for Plastic Vapor Retarders: High-density polyethylene tape with pressure sensitive adhesive having a minimum width of 3.75" having a maximum water vapor transmission rate of 0.3 perms in accordance with Manufacturer's recommendations.

2.5 FLOOR AND SLAB TREATMENTS

- A. Slip-Resistive Emery Aggregate or Aluminum Granule Finish: Provide fused aluminum-oxide granules, or crushed emery, as abrasive aggregate for slip-resistive finish. The emery aggregate shall contain not less than 50% aluminum oxide and not less than 20% ferric oxide. The aluminum aggregate material shall contain not less than 95% fused aluminum-oxide granules. Use material that is factory-graded, packaged, rustproof, and non-glazing, and is unaffected by freezing, moisture, and cleaning materials.
1. Subject to compliance with requirements, provide one of the following:
 - a. Dayton-Superior; Emery Tuff Non-Slip.
 - b. L&M Construction Chemicals, Inc.; Grip-It or Grip-It AO.
 - c. Sonneborn-ChemRex; Frictex NS.
- B. Colored, Mineral Aggregate, Dry Shake Surface Hardener: Packaged, dry, combination of materials, consisting of portland cement, graded quartz aggregate, coloring pigments (if required) and plasticizing admixtures. Use coloring pigments that are finely ground, non-fading mineral oxides, interground with cement. Color, as selected by Architect, unless otherwise indicated.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. The Euclid Chemical Company; Surfex.
 - b. L&M Construction Chemicals, Inc.; Quartz Plate.
 - c. LM Scofield Construction Chemical Co.; Lithochrome.
 - d. BASF Building Systems; Mastercron.
 - e. Dayton-Superior; Quartz-Tuff.
 - f. US Mix Co.; US Spec Dense Top.
 - g. SpecChem; Quartz Floor Hardener.
 2. Submit manufacturer's certification that product conforms to the requirements specified.
- C. Metallic Aggregate Hardener Finish: Packaged dry, combination of materials consisting of Portland Cement, specially processed and graded iron aggregate, coloring pigments (if required) and plasticizing admixtures. The hardener shall be formulated, processed, and packaged under stringent quality control. Use coloring pigments that are finely ground, non-fading mineral oxides inter-ground with cement. Color as selected by Architect unless otherwise indicated.
1. The Euclid Chemical Company; Euco-Plate HD.
 2. BASF Building Systems; Masterplate 200.
 3. Dayton-Superior; Ferro Tuff.
 4. SpecChem; SPECPLATE Floor Topping.
- D. Non-Oxidizing Metallic Floor Hardener: Packaged dry, combination of materials consisting of portland cement, non-rusting aggregate and plasticizing admixtures.
1. The Euclid Chemical Company; Diamond Plate.
 2. BASF Building Systems; Lumiplate.

2.6 CURING MATERIALS

A. Liquid Membrane-Forming Curing and Curing and Sealing Compounds:

1. Water-Based Dissipating Resin Type Curing Compound: Curing Compound shall be a dissipating resin type, which chemically breaks down after approximately four weeks. Membrane forming compound shall meet ASTM C 309, Types 1 or 1D, Class B with a VOC content less than 350 grams per liter.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) The Euclid Chemical Company; Kurez DR VOX.
 - 2) L&M Construction Chemicals; L&M Cure R.
 - 3) Unitex; Hydro Cure 309.
 - 4) W.R. Meadows; Sealtight 1100-Clear.
 - 5) US Mix Co.; US Spec Maxcure Resin Clear.
 - 6) SpecChem; SpecRez.
 - b. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with any covering or surface treatments to be applied. Submit any instructions that must be followed prior to any subsequent surface treatments and floor coverings.
2. High Solids, Water-Based Acrylic Curing and Sealing Compound with Moderate Yellowing Characteristics: Water-Based membrane-forming curing and sealing compound conforming to ASTM C 1315, Type 1, Class B, classified as low odor with a VOC content less than 350 grams per liter. Product shall provide a maximum moisture loss of 0.030 kilograms per square meter in 72 hours when applied at a coverage rate of 300 square feet per gallon. Do not apply to surfaces that are to receive subsequent cementitious toppings, sealers, hardeners, ceramic tile, resilient flooring, vinyl-backed carpet, wood, terrazzo, epoxy overlays or adhesives, or other coating or finishing products.
 - a. Products: Subject to compliance with above requirements, provide one of the following products or equivalent products:
 - 1) Dayton-Superior; Safe Cure and Seal (J-19).
 - 2) The Euclid Chemical Company; Super Aqua-Cure VOX.
 - 3) L&M Construction Chemicals; Dress & Seal, 30 WB.
 - 4) BASF Building Systems; Masterkure 200W.
 - 5) SpecChem; Cure & Seal WB 25.
 - b. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with any covering or surface treatments to be applied. Submit any instructions that must be followed prior to any subsequent surface treatments.
3. High Solids, Water-Based, Non-Yellowing Curing and Sealing Compound: Water based membrane-forming curing and sealing compound, acrylic type, complying with ASTM C 1315, Type 1, Class A classified as low odor with a VOC content less than 350 grams per liter. Do not apply to surfaces that are to receive subsequent cementitious toppings, sealers, hardeners, ceramic tile resilient flooring, vinyl-backed carpet, wood, terrazzo, epoxy overlays or adhesives, or other coating or finishing products.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) The Euclid Chemical Company; Super Diamond Clear VOX.
 - 2) L&M Construction Chemicals; Lumiseal 30 WB.
 - 3) BASF Building Systems; Kure 1315.
 - 4) Unitex; Hydro Seal 30.
 - 5) W.R. Meadows; Vocomp 30.
 - 6) US Mix Co.; US Spec Radiance UV-25.
 - 7) SpecChem; Cure & Seal WB 30.
 - b. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with any covering or surface treatments to be applied. Submit any instructions that must be followed prior to any subsequent surface treatments.

B. Evaporation Control: Monomolecular film forming compound applied to exposed concrete slab surfaces for temporary protection from rapid moisture loss in hot, dry, or windy weather conditions.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. The Euclid Chemical Company; Eucobar.
 - b. L&M Construction Chemicals; E-Con.
 - c. BASF Building Systems; Confilm.
 - d. Dayton-Superior; Sure Film (J-74).
 - e. Sika Chemical Co.; SikaFilm.
 - f. Unitex; Pro-Film.
 - g. W.R. Meadows; Sealtight Evapre.
 - h. US Mix Co.; US Spec Monofilm ER.
 - i. SpecChem; SpecFilm RTU.

2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all coverings and surface treatments to be applied. Submit any instructions that must be followed prior to any subsequent surface treatments.
- C. Absorptive Cover: Burlap cloth made from jute or kenaf, weighing approximately nine ounces per square yard, complying with AASHTO M 182, Class 2.
- D. Moisture-Retaining Cover: One of the following, complying with ANSI/ASTM C 171:
 1. Waterproof paper.
 2. Polyethylene film.
 3. Polyethylene-coated burlap.
 4. Polyethylene-coated natural cellulose fabric such as Greenstreak Group, Inc.; Aquacure.
 5. Cover for Industrial Slab: Provide a low permeance moisture-retaining cover that allows a moisture loss of no more than one pound per square yard in 72 hours when tested in accordance with ASTM C 156 for industrial slabs. The material shall be non-staining and meet with requirements of ASTM C 171.

2.7 LIQUID FLOOR TREATMENTS

- A. Chemical Hardener: Colorless aqueous solution containing a blend of magnesium fluosilicate and zinc fluosilicate combined with a wetting agent, containing not less than two pounds of fluosilicates per gallon.
 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. The Euclid Chemical Company; Surfhard.
 - b. BASF Building Systems; Lapidolith.
 - c. Dayton-Superior; Day-Chem Hardener (J-15).
 - d. L&M Construction Chemicals; Fluohard.
 - e. W.R. Meadows; Penalith.
 - f. SpecChem; Spec-O-Lith.
 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all coverings or surface treatments to be received. Submit any instructions that must be followed prior to any subsequent surface treatments.
- B. Chemical Curing/Floor Hardener Compound: Sodium silicate based compound that reacts with concrete constituents to harden the surface, resulting in a surface having a maximum abrasion coefficient of 0.25 cubic centimeters per square centimeter when tested in accordance with ASTM C 418.
 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. The Euclid Chemical Company; Eucosil.
 - b. BASF Building Systems; Sonosil.
 - c. Dayton-Superior; Day-Chem S.1-Cure (J-13).
 - d. L&M Construction Chemicals; Chem Hard.
 - e. Unitex; Uni Cure HD.
 - f. W.R. Meadows; Med-Cure.
 - g. US Mix Co.; US Spec Permasil.
 2. Submit manufacturer's certification that product conforms to the requirements specified and is compatible with all coverings and surface treatments to be applied. Submit any instructions that must be followed prior to any subsequent surface treatments.
- C. Liquid Sealer/Densifier: High performance, deeply penetrating concrete densifier that is an odorless, colorless, VOC-compliant, non-yellowing siliconate-based solution containing a minimum solids content of 20%, 50% of which is siliconate.
 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. The Euclid Chemical Company; Euco Diamond Hard.
 - b. L&M Construction Chemicals; Seal Hard.
 - c. W.R. Meadows; Luqui-Hard.
- D. Water and Chloride Ion Repelling Penetrating Sealer: Clear, solvent based Silane or Siloxane penetrating sealer which reacts chemically with the concrete surface to function as a Chloride Ion screen with a minimum 90% factor when tested in accordance with NCHRP #244, Series II, 100% solids, and applied in accordance with the manufacturer's recommendation.
 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Building Systems; Hydrozo 100.
 - b. Lyntal International, Inc.; Iso-flex 618-100 CRS.
 - c. Evonik Industries; Protectosil Chem-Trete BSM-400.
 - d. SpecChem; SpecSilane 100.
- E. Water and Chloride Ion Repelling Penetrating Sealer: Clear, solvent free, Silane penetrating sealer which reacts chemically with the concrete surface to function as a Chloride Ion screen with a minimum 83% factor when tested in accordance with NCHRP #244, Series II and applied in accordance with the manufacturer's recommendation.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. 40% solids:
 - 1) BASF Building Systems; Enviroseal 40.
 - 2) Lyntal International, Inc.; Iso-flex 618-40 WB.
 - 3) SpecChem; SpecSilane WB 40.
 - b. 100% solids:
 - 1) Evonik Industries; Protectosil BH-N.
- F. Moisture Vapor Reduction Sealer: ASTM C 1315 Type 1 Class A, ASTM C 309 Type 1 Class A, penetrating product to have no less than 34% solids content, leaving no sheen. Acceptable products include:
 1. Creteseal; CS2000 Spray Apply System.

2.8 RELATED MATERIALS

- A. Post-Installed Anchors:
 1. ICC Approval: Only anchors evaluated by the ICC Evaluations Service, Inc. (ICC-ES) with a published Evaluation Report, valid at the time of installation, showing it as having passed Acceptance Criteria 193 and approval for use in cracked concrete and resisting wind and seismic loads shall be approved for use.
 2. Alternate Anchor Approval: Install only anchors identified on the drawings by manufacturer and product. Substitutions using products approved by this Specification may be permitted provided complete design calculations, as required by and in accordance with the proposed product's current and valid ICC Evaluations Service Reports (ESR) and ACI 318 Appendix D, are signed and sealed by a registered professional engineer licensed in the state where the project is located and furnished to the Engineer for review and approval prior to commencement of work. The Contractor shall request design criteria for all conditions where a product substitution is considered. Failure to obtain approval for an anchor substitution may result in the request by the Engineer to remove installed anchors and replace with the product specified on the drawings at the Contractor's expense.
 3. Installation: All installation of post-installed anchors shall be in accordance with the Manufacturer's Printed Installation Instructions (MPII).
 4. Interior Use: All anchors for use in interior conditioned environments free of potential moisture shall be manufactured from carbon steel zinc plated in accordance with Federal Specification QQ-Z-325C, Type II, Class 3.
 5. Exterior or Exposed Use: All anchors for use in exposed or potentially wet environments or for attachment of exterior cladding materials shall be galvanized or stainless steel. Galvanized anchors shall conform to ASTM A 153. Stainless steel anchors shall be manufactured from 300 series stainless steel.
 6. Nuts and Washers: Nuts and washers shall be furnished from the manufacturer and used with the anchors.
 7. Anchor Types:
 - a. Expansion and Undercut Anchors in Concrete:
 - 1) Type: All expansion and undercut anchors in concrete shall be wedge type expansion, sleeve type expansion, or undercut type anchors.
 - 2) Acceptable Products and Manufacturers – Normalweight and Sand-Lightweight Concrete Not on Corrugated Steel Deck:
 - 3) Acceptable Products and Manufacturers – Normalweight and Sand-Lightweight Concrete on Corrugated Steel Deck:
 - a) Hilti, Inc.; Kwik Bolt TZ (ESR-1917).
 - b) Simpson Strong-Tie Co., Inc.; Strong-Bolt Wedge-Anchor (ESR-1771).
 - c) Powers Fasteners, Inc.; Power Stud+ SD1 Expansion Anchor (ESR-2818).
 - d) Powers Fasteners, Inc.; Power Stud+ SD2 Anchor (ESR-2502).
 - b. Screw and Insert Anchors in Concrete
 - 1) Acceptable Products and Manufacturers:
 - a) Hilti, Inc.; Kwik HUS-EZ Anchor (ESR-3027).
 - b) Simpson Strong-Tie Co., Inc.; Titen HD (ESR-2713).
 - c) Powers Fasteners, Inc.; Snake+ Anchor (ESR-2272).
 - c. Adhesive Anchoring Systems in Normalweight Concrete: Product that conforms to ASTM C 881-02, Type IV, Grade 3, Class A, B, & C except gel times, and that is dispensed from a two-component cartridge system through a mixing nozzle that thoroughly mixes the two components as it is injected into the hole.
 - 1) Chemical anchoring of anchors, rods, or reinforcing steel is not allowed for fire-rated assemblies, unless specified provided for in the drawings.
 - 2) Consult with the manufacturer for the minimum temperature of the concrete substrate allowed.

- 3) Only personnel trained to install adhesive anchors and certified in accordance with the ACI/CRSI Adhesive Anchor Installer Certification Program shall install adhesive anchors, including reinforcing steel.
 - 4) All anchors installed horizontally or upwardly inclined require continuous inspection.
 - 5) All adhesive anchors shall be installed in concrete having a minimum age of 21 days at the time of anchor installation.
 - 6) Normalweight Concrete:
 - a) Hilti, Inc.; HIT-HY 150 MAX-SD (ESR-3013).
 - b) Hilti, Inc.; HIT-HY 200 (ESR-3187).
 - c) Hilti, Inc.; HIT-RE 500-SD (ESR-2322).
 - d) ITW Red Head; EPCON G5 (ESR-1137).
 - e) ITW Red Head; EPCON S7 (ESR-2308).
 - f) Powers Fasteners, Inc.; PE 1000+ (ESR-2583).
 - g) Simpson Strong-Tie; SET-XP Adhesive (ESR-2508).
 - 7) Lightweight Concrete:
 - a) No approved products.
 - 8) These products may not be used in concrete cast over corrugated deck.
 - 9) Threaded Rods Chemically Anchored in Concrete:
 - a) Type: Threaded rods installed in holes using a chemical anchoring process shall have a 45° chiseled end on the embedded end.
 - b) Interior Application: Meet the requirements of ASTM A 307, A 36 or A 193, grade B7.
 - c) Exterior Application: Meet the requirements of ASTM A 153 galvanized steel, or F 593, Group 1 or 2, condition CW stainless steel.
- B. Bonding Compound: Polyvinyl acetate or acrylic base, for use in cosmetic and/or nonstructural repairs.
1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Acrylic or Styrene Butadiene:
 - 1) Dayton-Superior; Day-Chem Ad Bond (J-40).
 - 2) The Euclid Chemical Company; SBR Latex.
 - 3) W.R. Grace; Daraweld C.
 - 4) BASF Building Systems; Acrylic Additive.
 - 5) Sika Chemical Co.; SikaLatex.
 - 6) W.R. Meadows; Intralok.
 - 7) US Mix Co.; US Spec Acrylcoat.
 - 8) The Euclid Chemical Company; Akkro-7T.
 - 9) SpecChem; Acrylic Bonder.
 - b. Polyvinyl Acetate (Interior Use Only):
 - 1) The Euclid Chemical Company; Tammself.
 - 2) L&M Construction Chemicals; Everweld.
 - 3) Dayton-Superior; Superior Concrete Bonder (J-41).
 - 4) US Mix Co.; US Spec Bondcoat.
 - 5) SpecChem; SpecWeld.
- C. Epoxy Products: Two-component material suitable for use on dry or damp surface, complying with ASTM C 881.
1. Products for Crack Repair:
 - a. Sika Chemical Co.; Sikadur 35 Hi Mod LV – injection type.
 - b. Sika Chemical Co.; Sikadur 52 – injection type.
 - c. Sika Chemical Co.; Sikadur 55 SLV – gravity feed.
 - d. The Euclid Chemical Company; Eucopoxy Injection Resin.
 - e. Dayton-Superior; Sure-Inject (J-56).
 - f. BASF Building Systems; Epofil SLV.
 - g. Simpson Strong-Tie Co., Inc.; ETI-LV or ETI-GV – injection type.
 - h. Unitex; Pro-Poxy 100 LV or Pro-Poxy 50-1.
 - i. Adhesives Technology; Crackbond.
 - j. W.R. Meadows; Rezi-Weld LV.
 - k. US Mix Co.; US Spec Maxibond – injection or gravity feed.
 - l. US Mix Co.; US Spec Eposeal LVS – gravity feed.
 - m. The Euclid Chemical Company; Duralcrete LV.
 - n. SpecChem; SpecPoxxy 1000.
 2. Products for Epoxy Mortar Patches:
 - a. Sika Chemical Co.; Sikadur Lo-Mod LV.
 - b. The Euclid Chemical Company; Duracrete Series.

- c. Dayton-Superior; Sure Level Epoxy (J-57).
 - d. BASF Building Systems; Epofil.
 - e. Unitex; Pro-Poxy 2500.
 - f. W.R. Meadows; Rezi-Weld 1000.
 - g. US Mix Co.; US Spec EPM 3000.
 - h. The Euclid Chemical Company; Duralcrete LV.
 - i. SpecChem; SpecPoxy Binder.
3. Products for Epoxying Steel Plates to Concrete: Conform to ASTM C 881-90, Type IV, Grade 3, Class A, B, & C except gel times.
- a. Sika Chemical Co.; Sikadur 31 Hi-Mod Gel.
 - b. Dayton-Superior; Sure Anchor I (J-S1).
 - c. BASF Building Systems; Epo Gel or Rapid Gel.
 - d. Unitex; Pro-Poxy 300.
 - e. US Mix Co.; US Spec Gelbond NS.
 - f. The Euclid Chemical Company; Duralcrete Gel.
 - g. SpecChem; SpecPoxy 3000.
- D. Reglets: Where resilient or elastomeric sheet flashing or bituminous membrane is terminated in reglets, provide reglets of not less than 26 gauge galvanized sheet steel. Fill reglet or cover face opening to prevent intrusion of concrete or debris.
- E. Contraction and Construction Joint-Filler Material for Slabs-on-Grade: Provide a two-component semi-rigid, 100% solids epoxy having a minimum Shore A Hardness of 80 when tested in accordance with ASTM D 2240. Subject to compliance with requirements, provide one of the following:
- 1. The Euclid Chemical Company; Euco 700.
 - 2. Conspec Marketing and Manufacturing Co.; Inc.; Spec-Joint CJ.
 - 3. BASF Building Systems; Masterfill 300 I.
 - 4. Metzger/McGuire Co.; MM-80.
 - 5. W.R. Meadows; Rezi-Weld Flex.
 - 6. US Mix Co.; US Spec SR-50 EJF.
 - 7. SpecChem; SpecPoxy CJ.
- F. Bondbreaker for Construction Joints in Slabs-on-Grade: A dissipating bondbreaking compound containing no silicones, resins, or waxes, and that conforms to ASTM C 309. Subject to compliance with requirements, acceptable manufacturers include the following:
- 1. Dayton-Superior Corporation, Inc.; Sure-Lift.
 - 2. Conspec Marketing and Manufacturing Co.; Tilt-Eez.
 - 3. SpecChem; SpecTilt 100.
- G. Joint-Filler Strips for Isolation Joints in Slabs-on-Grade: ASTM D 1751, asphalt-saturated cellulosic fiber, or ASTM D 1752, cork or self-expanding cork.
- H. Rigid-Cellular-Polystyrene Boards use as Fill under Topping Slabs or Slabs-on-Grade: Provide rigid, expanded (EPS) or extruded (XPS) cellular polystyrene boards that conform to ASTM D 6817 or ASTM C 578 with a minimum density of 19 kilograms per cubic meter. Subject to compliance with requirements, acceptable manufacturers include the following:
- 1. Dow Chemical Company; STYROFOAM Brand.
 - 2. R-Control Buiding Systems; R-Control EPS Geofoam – All grades.
 - 3. Carpenter Co.; EPS Geofoam.
 - 4. Knauf Polystyrene; Knauf Geofoam.
 - 5. Premier Industries; Insulfill.

2.9 REPAIR MATERIALS

- A. Self-Leveling Mortars, Underlayment Compound: Freeflowing, self-leveling, pumpable cementitious base compound. Follow manufacturer's instruction regarding the use of a bonding agent.
1. Products: Unless specified otherwise, provide one of the following:
 - a. BASF Building Systems; Sonoflow.
 - b. Sika Chemical Co.; Sikatop 111.
 - c. The Euclid Chemical Company; Flo-Top or Super Flo-Top.
 - d. Dayton-Superior; Levelayer I.
 - e. US Mix Co.; US Spec Self-Leveling Underlayment.
 - f. The Euclid Chemical Company; Level Magic.
 - g. SpecChem; SpecFlow.
- B. Polymer Patching Mortar: Polymer and microsilica modified cementitious-based compounds.
1. Products:
 - a. Horizontal Application:
 - 1) The Euclid Chemical Company; Thin Top Supreme, Concrete Top Supreme.
 - 2) Sika Chemical Co.; Sikatop 121 or 122.
 - 3) BASF Building Systems; Emaco R310 CI.
 - 4) BASF Building Systems; Sonopatch 100 or 200.
 - 5) US Mix Co.; US Spec H2 or NuTop.
 - 6) The Euclid Chemical Company; Speed Crete PM.
 - 7) SpecChem; RepCon H.
 - b. Upwardly Inclined Application:
 - 1) The Euclid Chemical Company; Verticoat/Verticoat Supreme.
 - 2) Sika Chemical Co.; Sikatop 123.
 - 3) BASF Building Systems; Emaco R350 CI.
 - 4) BASF Building Systems; Sonopatch 200.
 - 5) US Mix Co.; US Spec V/O Patch.
 - 6) The Euclid Chemical Company; Speed Crete PM.
 - 7) SpecChem; RepCon V/O.
- C. High Strength Flowing Repair Mortar: For forming and pouring structural members, or large horizontal repairs, provide flowable one-part, high strength microsilica polymer modified repair mortar with 3/8" aggregate. The product shall achieve 9,000 PSI at 28-days at a nine inch slump.
1. Products:
 - a. BASF Building Systems; Road Patch.
 - b. US Mix Co.; US Spec STR Mortar.
 - c. The Euclid Chemical Company; Eucocrete.
 - d. The Euclid Chemical Company; Tamms Form and Pour.
 - e. SpecChem; RepCon 928.
- D. Anti-Corrosive Epoxy/Cementitious Adhesive: Water-based epoxy/cementitious compound for adhesion and corrosion protection or reinforcing members (20 hour maximum open time).
1. Products:
 - a. The Euclid Chemical Company; Duralprep A.C.
 - b. Sika Chemical Co.; Armatec 110.
 - c. BASF Building Systems; Sonoprep Plus.

2.10 PROPORTIONING AND DESIGN OF CONCRETE MIXTURES

- A. The Contractor shall submit design concrete mixtures for each class of concrete indicated on the structural drawings and in the Specifications for approval by the Engineer and Owner's Testing Laboratory at least 15

working days prior to the start of construction. If required, the Contractor shall engage the services of an independent Testing Laboratory to assist in preparing the design mixtures. The Contractor shall not begin work with a particular mixture until that design mixture has been approved.

- B. The Contractor, acting in conjunction with his Concrete Supplier and his Testing Laboratory, shall submit in writing, with his design mixtures, the method used to select mixture proportions. Either of the following methods, as outlined in ACI 301, may be used:
 - 1. Field Experience Method.
 - 2. Laboratory Trial Mixture Method.
- C. Required types of concrete and compressive strengths shall be as indicated on the Structural Drawings.
- D. All design mixtures shall state the following information:
 - 1. Design mixture number or code designation by which the Contractor shall order the concrete from the Supplier.
 - 2. Identify design mixture usage (i.e., columns, shear walls, footings, slab-on-grade, etc.).
 - 3. Wet and dry unit weights.
 - 4. Compressive strength and associated age (28-day, 56-day, etc.).
 - 5. Aggregate type, source, size, gradation, fineness modulus.
 - 6. Cement type and brand.
 - 7. Fly ash or other pozzolan type and brand (if any).
 - 8. Admixtures including air entrainment, water reducers, high-range water reducers, accelerators, and retarders.
 - 9. Design slump or slump/flow.
 - 10. Proportions of each material used.
 - 11. Water/cementitious ratio and maximum allowable water content.
 - 12. Method by which the concrete is intended to be placed (bucket, chute, or pump).
 - 13. Required average strength qualification calculations per ACI 301 4.2.3.3a and 4.2.3.3b. Submit separate qualification calculations for each production facility that will supply concrete to the project.
 - 14. Documentation of Average Strength (Trial Mixture Data or Field Test Data) per ACI 301: When field test data is used to qualify average strength, submit separate documentation for each production facility that will supply concrete to the project.
 - 15. Field test data submitted for qualification of average strength under ACI 301 shall include copies of the Concrete Testing Agency's reports from which the data was compiled.
- E. Low Alkali Concrete: For concrete identified on the drawings as Exposure Classes C1 and C2, the total alkali contribution from cementitious materials in the concrete mix shall not exceed 4.0 pounds per cubic yard of concrete unless the aggregate used is certified to contain no deleterious materials that react with alkalis in the concrete mix as defined in ASTM C 33. This requirement may be met by the use of low-alkali cement.
- F. Supplementary Cementitious Materials: Fly ash and/or ground granulated blast-furnace slag replacement of Portland cement shall be within percentage replacement levels listed on the drawings unless noted otherwise. Every effort should be made to reduce the amount of cement to the minimum practical amount, and still achieve performance requirements contained in the Contract Documents.
 - 1. Cement replacement shall not exceed a percentage level that has been shown by experience on other projects to exhibit satisfactory performance using materials from identical sources as proposed for this project. As an alternate, trial concrete batches can be performed to identify design mixtures that maximize cement replacement while meeting strength requirements per ACI 318 Section 5.3 and finishability criteria.
 - 2. The use of fly ash or slag in architecturally exposed structural concrete shall be coordinated with the Architect, Engineer, and Contractor.
 - 3. If fly ash is used, it must be at a minimum replacement percentage of 15%.
 - 4. Overall replacement percentages with combined fly ash and slag shall not exceed the maximum identified with slag or be less than the minimum identified with fly ash for each type of element. In addition, the replacement percentage of fly ash within the combined mixture shall not exceed the maximum identified with fly ash alone.
 - 5. Replacement percentages exceeding the maximum may be permitted at the discretion of the Architect, Engineer of Record, and Contractor.
 - 6. For concrete identified on the drawings as being subject to Exposure Class F3, the maximum amount of supplementary cementitious materials shall not exceed the limits noted in Table 4.2.2.7.b.2 "Maximum cementitious materials requirements for concrete exposed to deicing chemicals" of ACI 301.
 - 7. Except for Mass Concrete, the Contractor may submit for approval a revised design mixture with lower supplementary cementitious material percentages than herein specified should finishability or other issues arise due to changing weather conditions.
- G. Aggregate: Comply with the following special requirements:

1. For exposed concrete, provide aggregates from a single source.
 2. For exposed surfaces subject to Exposure Class C1 or C2, do not use aggregates containing spalling-causing deleterious substances.
 3. For slabs and other designated concrete, combined aggregate gradation shall be 8% - 18% for large top size aggregates (1 1/2 inches) or 8% - 22% for smaller top size aggregates (1 inch or 3/4 inch) retained on each sieve below the top size and above the No. 100. Deviations from this gradation may be allowed upon the approval of the Engineer subject to the following limitations:
 - a. The percent retained on two adjacent sieves shall be not less than 5%.
 - b. The percent retained on three adjacent sieves shall be not less than 8%.
 - c. If the percent retained on two adjacent sieves is less than 8%, the total percent retained on either of those sieves and the adjacent outside sieve shall be not less than 13%.
- H. Admixtures:
1. Admixtures to be used in concrete shall be subject to the approval of the Engineer and Owner's Testing Laboratory and shall be used for the purpose intended by the manufacturer to produce concrete to meet the specified requirements.
 2. Quantities of admixtures to be used shall be in strict accordance with the manufacturer's instructions.
 3. Air Content Requirements: For concrete subject to Exposure Class F1, F2 or F3 as noted on the drawings, use air-entrainment admixtures to provide concrete such that the air content at the point of placement shall conform to the requirements of ACI 301 Table 4.2.2.7.b "For Exposure Category F: Freezing and thawing exposures" within plus or minus 1.5%. Required air content levels may be reduced by 1.0 percent for concrete strengths above 5,000 PSI.
 - a. Interior steel troweled surfaces subjected to vehicular traffic shall not have more than 3% total air content.
 - b. Surfaces scheduled to receive hardeners shall not have more than 3% total air content.
 - c. Air-entraining admixtures are not permitted in industrial slabs.
- I. Lightweight Structural Concrete:
1. Comply with the requirements of ACI 301, Section 7.
 2. Provide concrete with a dry unit weight of not more than 116 pounds per cubic foot and not less than 110 pounds per cubic foot. Design mixture to produce strengths as indicated on the drawings with a split cylinder strength factor ($f_{ct}/(f_c)^{0.5}$) of not less than 5.7.
- J. Adjustments of Concrete Mixtures: Design mixture adjustments may be requested by the Contractor when characteristics of materials, job conditions, weather, test results, or other circumstances warrant. Such adjustments shall be provided at no additional cost to the Owner. Any adjustments in approved design mixtures including changes in admixtures shall be submitted in writing to the Engineer and Owner's Testing Laboratory for approval prior to field use.
- K. Chloride Ion Content:
1. Unless noted otherwise, the maximum water soluble chloride ion concentration in hardened concrete measured at ages from 28 to 42 days contributed from all ingredients including water, aggregates, cementitious materials, and admixtures shall not exceed the limits specified in ACI 301 Table 4.2.2.7.a.1 "Requirements for establishing suitability of cementitious materials combinations exposed to water-soluble sulfate" depending on to which Corrosion Exposure Class (C0, C1 or C2) the concrete is subject as noted on the drawings. Water-soluble chloride ion tests shall conform to ASTM C 1218. One test shall be run for each class of concrete before the design mixture submittal and each time a change is made to the design mixture (such as change in aggregate type or source).
 2. The chloride ion content in all concrete used for prestressed or post-tensioned concrete shall not exceed 0.06 percent by weight of cement.
 3. The Concrete Supplier shall certify that the chloride ion content in all concrete design mixtures used on the project does not exceed the limits stated above.
- 2.11 CONCRETE MIXING
- A. Ready-Mix Concrete: Comply with requirements of ANSI/ASTM C 94 and the Structural Testing and Inspections section of the specifications.
- 2.12 SOURCE QUALITY CONTROL
- A. Source Inspection: Refer to Specification **014529 "Structural Testing and Inspections"** for inspection requirements associated with cast-in-place concrete.
- B. Source Inspection:

PART 3 - EXECUTION

3.1 SLUMP LIMIT

- A. The slump, as measured in the field where concrete cylinders are taken, shall be within plus or minus one inch of the design slump noted in the approved Design Mixture submittal. Self-Consolidating Concrete shall have a slump/flow of plus or minus two inches of the design slump/flow noted on the approved Design Mixture submittal. Water may be added to the concrete in the field only to the extent that the prescribed water/cementitious ratio noted in the approved Design Mixture submittal is not exceeded. The responsibility for adding water to trucks at the job site shall rest only with the Contractor's designated representative. The Contractor is responsible that all concrete placed in the field is in conformance with the Contract Documents.

3.2 VAPOR RETARDER INSTALLATION

- A. **Reference 07 26 00 "Under-slab Vapor Barrier"**. Install and repair damaged vapor retarder in accordance with ASTM E 1643 and manufacturer's instructions.
- B. Lap all seams per manufacturer's instruction (6" minimum lap) and seal all joints in the field with the specified pressure sensitive tape. Heat-welded joints done in a shop prior to delivery is an acceptable method to minimize the number of field joints.
- C. Seal all pipe penetrations through the vapor retarder with a boot made from the vapor retarder material and tape or mastic.

3.3 JOINTS IN CONCRETE

- A. Construction Joints: Locate and install construction joints as indicated on the drawings or if not shown on drawings, located so as not to impair strength and appearance of the structure, as acceptable to Architect/Engineer.
 - 1. Keyways: Provide keyways with a depth of one tenth of the member thickness (1 1/2" minimum or as shown on the drawings) in construction joints only where shown on the drawings.
 - 2. Joint Construction: Place construction joints in the center one third of suspended spans and grade beams and as shown on the drawings for slabs-on-grade and walls unless shown otherwise. Offset joints in girders a minimum distance of twice the beam width from a beam-girder intersection. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints unless otherwise shown on the drawings. Dowels that cross construction joints shall be supported during concreting operations so as to remain parallel with the slab or wall surface and at right angles to the joint. Submit all construction joint locations as a shop drawing submittal.
 - 3. Waterstops: Provide waterstops in construction joints as indicated on the Architectural and Structural Drawings. Install waterstops to form continuous diaphragm in each joint. Make provisions to support and protect exposed waterstops during progress of work. Fabricate field joints in waterstops in accordance with manufacturer's printed instructions.
 - 4. Isolation Joints in Slabs-on-Grade: Construct isolation joints (without dowels) in slabs-on-grade at points of contact between slabs-on-grade and vertical surfaces only where specifically detailed on the drawings. Install joint-filler strips at joints where indicated. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface, unless otherwise indicated on the drawings. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together. Provide construction joints with dowels at all locations unless isolation joints are detailed.
- B. Contraction Joints in Slabs-on-Grade and Unbonded Topping Slabs: Install contraction joints at locations and spacings as indicated on the drawings or if not shown on drawings, located so as not to impair strength and appearance of the structure, as acceptable to Architect/Engineer. Maximum joint spacing shall be per the drawings and be perpendicular to the slab surface. Use one of the two following methods (sawed or formed) to create the joints. Do not use the formed joint in areas subject to vehicular traffic or in industrial slabs.
 - 1. Sawed Joints:
 - a. Primary Method: Early-Entry, dry-cut method, using Soff-Cut saws. Finisher must have documented successful experience in the use of this method prior to this project. Install cuts within one to four hours, depending on air temperature, after final finish as soon as the concrete surface is firm enough to not be torn or damaged by the blade at each saw cut location. Use 1/8 inch thick blade, cutting to a depth of one quarter of the slab thickness but not less than one inch. Cut to a depth of one third of the slab thickness for slabs reinforced with steel fibers or synthetic fibers.
 - b. Optional Method (where Soff-Cut System method equipment is not available, subject to limitations): This method may not be used when there is no dowel passing through the contraction joint. Use a conventional saw to cut joints within four to 12 hours after finishing

as soon as the concrete has hardened sufficiently to prevent aggregates from being dislodged by the saw. Complete cutting before shrinkage stresses become sufficient to produce cracking. Use 1/8 inch thick blade, cutting to a depth of one quarter of the slab thickness but not less than one inch. Cut to a depth of one third of the slab thickness for slabs reinforced with steel fibers.

2. Formed Joints: Form contraction joints by inserting premolded plastic hardboard or fiberboard strip into fresh concrete until top surface of strip is flush with slab surface. The depth is to be one quarter of the slab thickness, but not less than one inch. Tool slab edges round on each side of insert. After concrete has cured, remove inserts and clean groove of loose debris.
3. Joint Filler: Provide in both contraction and saw-cut construction joints when specified.
 - a. Remove dirt and debris from the joint by vacuuming immediately prior to filling joint. Clean the joint of curing compounds and sealers.
 - b. Filler material shall be applied to the joints when the building is under permanent temperature control, but no less than 90 days after slab construction.
 - c. Follow the manufacturer's recommended procedure for installing filler material. The joint filler must be flush with the adjacent concrete. A concave profile on the top of the joint filler is unacceptable and will be grounds for removal and replacement.
4. The Contractor shall protect the joints from damage caused by wheeled traffic or other sources during construction until a joint-filler material (if specified) has been installed.

3.4 INSTALLATION OF EMBEDDED ITEMS

- A. General: Set and build into work anchorage devices and other embedded items required for other work that is attached to, or supported by, cast-in-place concrete. Use setting drawings, diagrams, instructions and directions provided by suppliers of items to be attached thereto unless directed otherwise by these specifications. Install reglets to receive top edge of foundation sheet waterproofing where specified by the Architect, and to receive thru-wall flashings in outer face of concrete frame at exterior walls, where flashing is shown at lintels, relieving angles and other conditions.
- B. Edge Forms and Screed Strips for Slabs: Set edge forms or bulkheads and intermediate screed strips for slabs to obtain required elevations and contours in finished slab surface. Provide and secure units sufficiently strong to support types of screed strips by use of strike-off templates or accepted compacting type screeds.
- C. Do not install sleeves in any concrete member except where shown on the structural drawings or approved by the Architect and Engineer.
- D. Securely fasten embedded plates, angles, anchor rods and other items to be built into the concrete to the formwork or hold in place with templates. Insertion of these items into concrete after concrete placement is prohibited.

3.5 CONCRETE PLACEMENT

Pre-placement Inspection: Before placing concrete, inspect and complete formwork installation, reinforcing steel, and items to be embedded or cast-in. Notify other crafts to permit installation of their work; cooperate with other trades in setting such work. Moisten wood forms immediately before placing concrete where form coatings are not used.

- A. Concrete Batch Trip Tickets: The Contractor shall collect and retain concrete batch trip tickets. Compressive strength, slump, air content, and temperature tests shall be identified by reference to a particular trip ticket. Tickets shall contain the information specified in ASTM C 94. Each ticket shall also show the amount of water that may be added in the field for the entire batch that will not exceed the specified water cement ratio for the design mixture. The Contractor and Testing Laboratory shall immediately notify the Architect/Engineer and each other of tickets not meeting the criteria specified.
- B. Causes for Rejection of Concrete: The Contractor shall reject concrete delivered to the site for any of the following reasons:
 1. Wrong class of concrete (incorrect design mixture number).
 2. Environmental condition limits shall be as follows unless appropriate provisions in concrete practices have been made for cold or hot weather:
 - a. Cold Weather: Air temperature must be 40°F and rising or the average daily temperature cannot have been lower than 40°F for three consecutive days unless the temperature rose about 50°F for at least one-half of any of those 24 hour periods.
 - b. Hot Weather: Environmental conditions must be such that cause an evaporation rate from the concrete surface of 0.2 pounds per square foot per hour or less as determined by the figure "NRMCA Nomograph for Estimating Evaporation Rate on the Basis of Menzel Formula" in Appendix A of ACI 305.1.
 - c. Concrete may be placed at other environmental condition ranges only with the approval of the job inspector for the Testing Laboratory or other duly appointed representative.

3. Concrete with temperatures exceeding 95°F.
 4. Air contents outside the limits specified in the design mixtures.
 5. Slumps outside the limits specified.
 6. Water added to the mix that exceeds the maximum allowed water-to-cementitious material ratio.
 7. Excessive Age: Concrete shall be discharged within 90 minutes of plant departure or before it begins to set if sooner than 90 minutes and it shall be discharged before the drum has revolved 300 revolutions, unless approved by the Testing Laboratory job inspector or other duly appointed representative.
- C. Coordinate the installation of joint materials and vapor retarders with placement of forms and reinforcing steel.
- D. Comply with ACI 301 and as herein specified:
1. Concrete Temperature: The maximum acceptable concrete temperature at the truck discharge point shall be 95°F.
 2. Deposit concrete continuously or in layers of such thickness that no concrete will be placed on concrete that has hardened sufficiently to cause the formation of seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as herein specified. Deposit concrete as nearly as practicable to its final location to avoid segregation. Spread concrete using short-handled, square-ended shovels, or come-alongs.
 3. Placing Concrete in Forms: Deposit concrete in forms in horizontal layers not deeper than 24" and in a manner to avoid inclined construction joints. Where placement consists of several layers, place each layer while preceding layer is still plastic to avoid cold joints.
 4. Consolidate placed concrete by mechanical vibrating equipment supplemented by hand-spading, rodding or tamping. Use internal vibrators of the largest size and power that can properly be used in the work.
 5. Do not vibrate Self-Consolidating Concrete.
 6. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations not farther than visible effectiveness of machine. Place vibrators to penetrate rapidly placed layer and at least 6" into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to set. At each insertion limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing segregation of mix.
 7. Placing Concrete Slabs: Deposit and consolidate concrete slabs in a continuous operation, within limits of construction joints, until the placing of a panel or section is completed. Place concrete for beams, girders, brackets, column capitals, haunches, and drop panels at the same time as concrete for slabs. Do not place concrete over columns and walls until concrete in columns and walls is no longer plastic and has been in place at least one hour.
 8. Consolidate concrete during placing operations so that concrete is thoroughly worked around reinforcement and other embedded items and into corners of forms, eliminating air and stone pockets that may cause honeycombing, pitting, or planes of weakness.
 9. Bring slab surfaces to correct level with straightedge and strikeoff. Use highway straightedges, bull floats, or darbies to smooth surface free of humps or hollows before excess moisture or bleedwater appears on the surface. Do not disturb slab surfaces prior to beginning finishing operations.
 10. Maintain reinforcing in proper position during concrete placement operations.
 11. Protect adjacent finish materials against damage and spatter during concrete placement.
 12. Placing Concrete by Pump: If concrete is placed by using a pump, the grout used for pump priming must not become a part of the completed structure unless an engineered grout design mix and grout location are approved in advance by the Engineer.
- 3.6 FINISH OF FORMED SURFACES
- A. General: Formed surfaces shall have the finishes as described below and as shown on the drawings after formwork is removed and repairs made.
- B. Matching Mockup Finish: In all areas where a special finish is required or a mock-up is required below, Contractor shall prepare a 100 square foot mock-up to match the required finish. The mock-up should match the finish on a sample panel furnished to the Contractor. If a sample is not furnished, provide finish to match SF2.0 or any other finish specified for the project. Protect mock-up from damage for the duration of project. Approval of mock-up by Architect is required before proceeding with application of finish in project.
- C. Classifications and Finish Requirements:
1. Surface Finish 1.0 (SF-1.0):
 - a. No formwork facing material is specified.
 - b. Patch voids larger than 1-1/2 inch wide or 1/2 inch deep.
 - c. Remove projections larger than 1.0 inch.
 - d. Provide surface tolerance Class D as specified in ACI 117.

- e. Tie holes need not be patched.
- 2. Surface Finish 1.1 (SF-1.1):
 - a. No formwork facing material is specified.
 - b. Patch voids larger than 1 inch wide or 1/2 inch deep.
 - c. Remove projections larger than 1/2 inch.
 - d. Provide surface tolerance Class C as specified in ACI 117.
 - e. Tie holes need not be patched.
- 3. Surface Finish 2.0 (SF-2.0):
 - a. Provide specified formwork-facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/4 inch.
 - e. Provide surface tolerance Class B as specified in ACI 117.
 - f. Provide mock-up of concrete surface appearance.
- 4. Surface Finish 2.1 (SF-2.1):
 - a. Provide specified formwork-facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/4 inch.
 - e. Provide surface tolerance Class B as specified in ACI 117.
 - f. Provide specified rubbed finish after formwork removal.
 - g. Provide mock-up of concrete surface appearance.
- 5. Surface Finish 2.2 (SF-2.2):
 - a. Provide specified formwork-facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/4 inch.
 - e. Provide surface tolerance Class B as specified in ACI 117.
- 6. Surface Finish 2.3 (SF-2.3):
 - a. No formwork-facing material is specified.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/4 inch.
 - e. Provide surface tolerance Class B as specified in ACI 117.
- 7. Surface Finish 3.0 (SF-3.0):
 - a. Provide specified formwork facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Remove projections larger than 1/8 inch.
 - d. Patch tie holes.
 - e. Provide surface tolerance Class A as specified in ACI 117.
 - f. Provide mock-up of concrete surface appearance.
- 8. Surface Finish 3.1 (SF-3.1):
 - a. Provide specified formwork-facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/8 inch.
 - e. Provide surface tolerance Class A as specified in ACI 117.
 - f. Provide specified rubbed finish after formwork removal.
 - g. Provide mock-up of concrete surface appearance.
- 9. Surface Finish 3.2 (SF-3.2):
 - a. Provide specified formwork-facing material.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/8 inch.
 - e. Provide surface tolerance Class A as specified in ACI 117.
- 10. Surface Finish 3.3 (SF-3.3):
 - a. No formwork-facing material is specified.
 - b. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
 - c. Patch tie holes.
 - d. Remove projections larger than 1/8 inch.
 - e. Provide surface tolerance Class A as specified in ACI 117.

- D. Standard Finish: Provide SF-1.0 on all concrete surfaces not exposed to view in the final condition unless otherwise specified.
- E. Exposed Finishes: Provide SF-2.0 on all concrete surfaces exposed to view in final condition unless otherwise specified.
- F. Rubbed Finishes: Remove forms as early as permitted by these specifications and perform any necessary repairs and patches.
 - 1. Smooth Rubbed Finish: Provide smooth rubbed finish to scheduled or specified concrete surfaces where indicated, not later than one day after form removal. Moisten concrete surfaces and rub with carborundum brick or other abrasive until a uniform color and texture is produced. Do not apply cement grout other than that created by the rubbing process.
- G. Related Unformed Surfaces: At tops of walls, horizontal offsets and similar unformed surfaces occurring adjacent to formed surfaces, strike-off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces, unless otherwise indicated.

3.7 MONOLITHIC SLAB FINISHES

- A. Place, consolidate, strike off, and level concrete, eliminating high spots and low spots, before proceeding with any other finish operation. Do not add water to the surface of the concrete during finishing operation.
- B. Scratch Finish: Apply scratch finish to monolithic slab surfaces that are to receive concrete floor topping or mortar setting beds for tile, portland cement terrazzo and other bonded applied cementitious finish flooring material, and as otherwise indicated. After placing slabs, plane surface to tolerance specified below. Slope surfaces uniformly to drains where required. After leveling, roughen surface before final set, with stiff brushes, brooms, or rakes.
- C. Float Finish: Apply float finish to monolithic slab surfaces to receive trowel finish and other finishes as hereinafter specified, and slab surfaces which are to be covered with membrane or elastic waterproofing, membrane or elastic roofing, or sand-bed terrazzo, and as otherwise indicated. After screeding, consolidating and leveling concrete slabs, do not work surface until ready for floating. Begin floating, using a hand float, a bladed power float equipped with float shoes, or a powered disk float, when the bleed water sheen has disappeared and the concrete surface has stiffened sufficiently to permit the operation. Check and level surface plane to a tolerance as specified below. Cut down high spots and fill low spots. Uniformly slope surfaces to drains. Immediately after leveling, refloat surface to a uniform, smooth, granular texture.
- D. Trowel Finish: Apply trowel finish to monolithic slab surfaces to be exposed-to-view, and slab surfaces to be covered with resilient flooring, carpet, ceramic or quarry tile, paint or other thin film finish coating system. After floating, begin first trowel finish operation by hand or power-driven trowel. Begin final troweling when surface produces a ringing sound as trowel is moved over surface. Consolidate concrete surface by final hand-troweling operation, free of trowel marks, uniform in texture and appearance, and with a level surface to a tolerance as specified below. Grind smooth surface defects that would telegraph through applied floor covering system.
- E. Trowel and Fine Broom Finish: Where ceramic or quarry tile is to be installed with thin-set mortar, apply initial trowel finish as specified above, then immediately follow with slightly scarifying surface by fine brooming.
- F. Slip-Resistive Broom Finish: Apply slip-resistive broom finish to garage floors and ramps less than 6% slope, exterior concrete platforms, steps, and ramps and elsewhere as indicated. Immediately after float finishing, slightly roughen concrete surface by brooming with fiber bristle broom perpendicular to main traffic route. Coordinate required final finish with Architect before application. For concrete containing fibers, broom once only in one direction.
- G. Roller-Bug Finish: Provide a roller-bug finish with minimum 1/4" amplitude to all ramps exceeding a 6% slope. Extend the finish at least 12 feet beyond the beginning and ending of the greater-than-6% ramp. The finish shall be imprinted on the concrete by the use of a roller-bug tamper.
- H. Chemical-Hardener Finish: Apply chemical-hardener finish to interior concrete floors where indicated. Apply liquid chemical-hardener after complete curing and drying of the concrete surface. Apply proprietary chemical hardeners, in strict accordance with manufacturer's printed instructions. After final coat of chemical-hardener solution is applied and dried, remove surplus hardener by scrubbing and mopping with water.
- I. Liquid Sealer/Densifier Finish: Apply liquid sealer/densifier finish to exposed interior concrete floors where indicated. Apply after complete curing and drying of the concrete surface and in strict accordance with manufacturer's printed instructions.
- J. Penetrating Sealer Finish: Apply a chloride-and-water-repelling-penetrating-sealer finish to surfaces as described below and where indicated on the drawings. Apply liquid penetrating sealer after complete curing and drying of the concrete surface. Apply proprietary sealers in strict accordance with manufacturer's printed instructions. The Contractor shall verify the compatibility of the sealer product with the paint used to stripe

parking decks and coordinate the sequencing of the sealing and striping operations. Apply to the following surfaces:

1. Sloping and horizontal surfaces of parking garages.
 2. Top surfaces of exposed exterior balconies.
- K. Slip-Resistive Aggregate Finish: Apply slip-resistive aggregate finish to concrete stair treads, platforms, ramps, and elsewhere as indicated on the Architect's or Structural Drawings.
1. After completion of float finishing, and before starting trowel finish, uniformly spread 25 pounds of dampened slip-resistive aggregate per 100 square feet of surface. Tamp aggregate flush with surface using a steel trowel, but do not force below surface. After broadcasting and tamping, apply trowel finishing as herein specified.
 2. After curing, lightly work surface with a steel wire brush, or an abrasive stone, and water to expose slip-resistive aggregate.
- L. Colored, Mineral Aggregate Surface Hardener: Provide colored, mineral aggregate surface hardener to monolithic slab surface indicated.
1. Apply dry shake materials for colored wear-resistant finish at rate of not less than 100 pounds per 100 square feet, unless greater amount is recommended by material manufacturer.
 2. Cast a trial slab approximately 20 feet by 20 feet to determine actual application rate, color, and finish as acceptable to Architect/Engineer.
 3. Immediately following first floating operation using wood floats, uniformly distribute approximately two thirds of required weight of dry shake material over concrete surface, and embed by means of power floating using float shoes or pan floats. Follow floating operation with second shake application, uniformly distributing remainder of dry shake material at right angles to first application, and embed by power floating.
 4. After completion of broadcasting and floating, apply trowel finish as herein specified. Cure slab surface with curing compound recommended by dry shake hardener manufacturer. Apply curing compound immediately after final finishing.
- M. Non-Oxidizing Metallic Floor Hardener: Slabs in areas noted on the drawings shall receive an application of the non-oxidizing, metallic floor hardener applied at the rate of 150 pounds per 100 square feet. Immediately following the first floating operation using wood floats, uniformly distribute approximately two thirds of the required weight of the hardener over the concrete surface by mechanical spreader and embedded by means of power floating using float shoes or pan floats. The hardener shall be floated in and the second application made. The surface shall be floated again to bond properly the hardener to the base concrete slab. The surface shall then be troweled at least twice to a smooth dense finish.

3.8 CONCRETE FINISH MEASUREMENT AND TOLERANCES

- A. Testing Procedure: ASTM E 1155.
- B. Tolerance on Floor Elevations: Construction tolerance on absolute floor elevation from the specified elevation as shown on the drawings shall be as specified below, taken from ACI 117:
1. Slab-on-Grade Construction: $\pm 3/4"$.
 2. Top Surfaces of Formed Slabs Measured Prior to Removal of Supporting Shores: $\pm 3/4"$.
 3. Top Surfaces of All Other Slabs: $\pm 3/4"$.
- C. Random Traffic Floor Finish Tolerances:
1. Specified overall values for flatness (SO_F) and levelness (SO_L) shall conform to the values listed below for the floor surface classification noted for each slab category noted.
 - a. Conventional:
 - 1) SO_F : 20.
 - 2) SO_L : 15.
 - b. Moderately Flat:
 - 1) SO_F : 25.
 - 2) SO_L : 20.
 - c. Flat:
 - 1) SO_F : 35.
 - 2) SO_L : 25.
 - d. Very Flat:
 - 1) SO_F : 45.
 - 2) SO_L : 35.
 2. Minimum local values for flatness (ML_F) and levelness (ML_L) shall equal $3/5$ of the SO_F and SO_L values, respectively, unless noted otherwise. The ML_F and ML_L values shall apply to the minimum areas bounded by the column lines and half-column lines, or the minimum areas bounded by the construction and contraction joints, whichever are the smaller areas.

3. The $SOFL$ and MFL tolerance values shall apply only to level slabs-on-ground or to level, uncambered suspended slabs that are shored such that it cannot deflect from the time the floor is placed to the time it is measured.
4. Slabs specified to slope shall have a tolerance from the specified slope of 3/8" in 10 feet at any point.
- D. Construction Requirements to Achieve Specified Floor Finish Tolerances:
 1. Forms shall be properly leveled, in good condition, and securely anchored including special attention to ends and transitions.
 2. Bearing surfaces for straightedges such as form edges or previously poured slabs shall be kept clean of laitance, sand, gravel, or other foreign elements.
 3. Screeds shall be maintained in good condition with true round rolling wheels and level cutting edges. The use of optical sighting equipment such as lasers is recommended for checking levelness and straightness. The Contractor shall promptly adjust or replace equipment when test results indicate substandard work.
 4. Highway straightedges are recommended for use in lieu of bullfloats for all slab placement and finishing operations. If mineral, non-oxidizing metallic, or metallic floor hardeners are used, the slab shall be wood bullfloated immediately after the straightedge.
- E. Contractor Responsibility for Concrete Floor Finish Requirements: Floor finish requirements shown below (flatness and levelness tolerances) are minimum requirements that apply unless stricter requirements are contained in instructions for installation of applied floor products in which case the Contractor is responsible for attaining the values prescribed by the manufacturer of such products.
- F. Concrete Floor Finish Tolerance for Slab-on-Grade Construction:
 1. Concrete Placement: Concrete shall be placed and screeded to predetermined marks set to elevations prescribed on the drawings.
 2. Finish Tolerances of Random Traffic Floor Surfaces:
 - a. Slabs in nonpublic areas, mechanical rooms, surfaces to receive raised computer flooring, surfaces to have thick-set tile or a topping, and parking structures: Conventional.
 - b. Carpeted Areas: Moderately Flat.
 - c. Exposed Slabs in Public Spaces, Slabs to Receive Thin-Set Flooring: Flat.
- G. Concrete Floor Finish Tolerance for Shored, Cast-in-Place Suspended Slab Construction:
 1. Concrete Placement: Formwork shall be set and securely braced so that soffits are positioned to allow scheduled concrete member sizes and thicknesses within tolerances specified in ACI 117. Concrete shall be placed and screeded to predetermined marks on the form surface conforming to elevations prescribed on the drawings.
 2. Camber: Formwork camber, as indicated on the drawings, shall be set to provide a uniform, smooth soffit profile in each direction. Minimum slab thickness, as specified on the drawings, shall be maintained throughout the slab surface to a tolerance as specified in ACI 117. Tolerance on camber shall be $\pm 1/4"$. Levelness F-Number tolerances specified below do not apply to areas of the floor where camber or intentional slope is shown.
 3. Finish Tolerances of Random Traffic Floor Surfaces:
 - a. Slabs in Nonpublic Areas, Mechanical Rooms, Surfaces to Receive Raised Computer Flooring, Surfaces to Have Thick-Set Tile or a Topping, and Parking Structures: Conventional.
 - b. Carpeted Areas: Moderately Flat.
 - c. Exposed Slabs in Public Spaces, Slabs to Receive Thin-Set Flooring: Flat.
 - d. Movie or Television studios: Super Flat.
 4. Extra Concrete: The contractor shall include in his bid any additional concrete required to achieve the specified slab surface finish tolerance.
 5. Concrete Placement at Column Bays Supported by Unshored Transfer Girders: Concrete in floor areas supported by unshored transfer girders shall be placed and screeded to predetermined marks placed over the slab conforming to elevations as specified on the drawings. At least the minimum slab thickness, as specified on the drawings, shall be maintained throughout the slab surface. The Contractor shall conform to the F_F values specified above.
- H. Concrete Floor Finish Tolerance – Unshored Steel Deck on Shored or Unshored Steel Beam or Steel Joist Floor Construction:
 1. Concrete Placement: Concrete over steel deck shall be placed and screeded level and flat to the tolerance specified below, maintaining at least the minimum slab thickness at all locations as specified on the drawings. The Contractor shall increase the slab thickness as required to compensate for steel deck deflection, and in unshored beam construction, residual beam camber and beam deflection in order to achieve a level and flat floor within specified tolerances.
 2. Finish Tolerance of Random Traffic Floor Surfaces:
 - a. Slabs in Nonpublic Areas, Mechanical Rooms, Surfaces to Receive Raised Computer Flooring, Surfaces to Have Thick-Set Tile or a Topping, and Parking Structures: Conventional.

- b. Carpeted Areas: Moderately Flat.
 - c. Exposed slabs in public spaces, slabs to receive thin-set flooring: Flat.
 - d. Eighty percent (80%) of the final floor surface shall fall within an envelope of 0.75" centered about the mean elevation of all the readings. The mean elevation of all readings shall not deviate from the specified design grade by more than ± 0.375 ".
- 3. Extra Concrete: The Contractor shall include in his bid any additional concrete required to achieve the specified slab surface finish tolerance and to compensate for steel deck deflection, beam camber and beam deflection.
- I. Remedial Measures for Slab Finish Construction Not Meeting Specified Tolerances:
 - 1. Application of Remedial Measures. Remedial measures specified herein are required whenever either or both of the following occur:
 - a. The composite overall values of F_F or F_L of the entire floor installation measure less than specified values.
 - b. Any individual test section measures less than the specified absolute minimum F_F or F_L value.
 - 2. Modification of Existing Surface:
 - a. If, in the opinion of the Architect/Engineer or Owner's Representative, all or any portion of the substandard work can be repaired without sacrifice to the appearance or serviceability of the area, then the Contractor shall immediately undertake the approved repair method.
 - b. The Contractor shall submit for review and approval a detailed work plan of the proposed repair showing areas to be repaired, method of repair, and time to affect the repair.
 - c. Repair method(s), at the sole discretion of the Architect/Engineer or Owner's Representative, may include grinding (floor stoning), planing, retopping with self-leveling underlayment compound or repair topping, or any combination of the above.
 - d. The Architect/Engineer or Owner's Representative maintains the right to require a test repair section using the approved method of repair for review and approval to demonstrate a satisfactory end product. If, in the opinion of the Architect/Engineer or Owner's Representative, the repair is not satisfactory an alternate method of repair shall be submitted or the defective area shall be replaced.
 - e. The judgment of the Architect/Engineer or Owner's Representative on the appropriateness of a repair method and its ability to achieve the desired end product shall be final.
 - f. All repair work shall be performed at no additional cost to the Owner and with no extension to the construction schedule.
 - 3. Removal and Replacement:
 - a. If, in the opinion of the Architect/Engineer or Owner's Representative, all or any portion of the substandard work cannot be satisfactorily repaired without sacrifice to the appearance or serviceability of the area, then the Contractor shall immediately commence to remove and replace the defective work.
 - b. Replacement section boundaries shall be made to coincide with the test section boundaries as previously defined.
 - c. Sections requiring replacement shall be removed by sawcutting along the section boundary lines to provide a neat clean joint between new replacement floor and existing floor.
 - d. The new section shall be reinforced the same as the removed section and doweled into the existing floor as required by the Engineer. No existing removed reinforcing steel may be used. All reinforcing steel shall be new steel.
 - e. Replacement sections may be retested for compliance at the discretion of the Architect/Engineer or Owner's Representative.
 - f. The judgment of the Architect/Engineer or Owner's Representative on the need for replacement shall be final.
 - g. All replacement work shall be performed at no additional cost to the Owner and with no extension to the construction schedule.

3.9 CONCRETE CURING AND PROTECTION

A. General:

- 1. Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Maintain concrete with minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of concrete. Limit moisture loss to a maximum of 0.05 pounds per square foot per hour for concrete containing silica fume and 0.2 pounds per square foot per hour for all other concrete before and during finishing operations. If using an evaporation retarder, apply in accordance with manufacturer's instructions after screeding and bull floating, but before power floating and troweling.

2. Curing shall commence as soon as free water has disappeared from the concrete surface after placing and finishing. The curing period shall be seven days for all concrete except high early strength concrete that shall be cured for three days minimum.
 3. Alternatively, curing times may be reduced if either of the following provisions is complied with:
 - a. If tests are made of cylinders kept adjacent to the structure and cured by the same methods, curing measures may be terminated when the average compressive strength has reached 70% of the specified compressive strength.
 - b. If the temperature of the concrete is maintained at a minimum of 50°F for the same length of time required for laboratory cured cylinders of the same concrete to reach 85% of the specified compressive strength, then curing may be terminated thereafter.
 4. Curing shall be in accordance with ACI 301 procedures. Avoid rapid drying at the end of the curing period.
- B. Curing Formed Surfaces: Where wooden forms are used, cure formed concrete surfaces, including undersides of beams, supported slabs and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. When forms are removed, continue curing by one or a combination of the methods specified below, as applicable:
1. Columns and Shear Walls Not Exposed to View: Moist cure in forms or by one or a combination of Methods 1, 2, or 3 specified below. Use a high-solids, liquid membrane-forming curing and sealing compound conforming to ASTM C 1315, Type I, Class A or B for Method 3.
 2. Columns and Shear Walls Exposed to View: Moist cure in forms or by one or a combination of Methods 1, 2, or 3 specified below. Use a high-solids, non-yellowing, liquid membrane-forming curing and sealing compound conforming to ASTM C 1315, Type 1, class A for Method 3.
 3. Sides and Soffits of Beams and Pan-Joist Ribs, Soffits of Slabs: Moist cure in forms or by one or a combination of Methods 1, 2, or 3 specified below. Use a liquid membrane-forming dissipating resin curing compound conforming to ASTM C 309, Type 1, class A or B for Method 3.
 4. Basement Walls, Sides of Exterior Retaining Walls: Moist cure in forms or by one or a combination of Methods 1, 2, or 3 specified below. Use a liquid membrane-forming dissipating resin curing compound conforming to ASTM C 309, Type 1, class A or B for Method 3.
- C. Curing Unformed Surfaces: Cure unformed surfaces, such as slabs, floor topping, and other flat surfaces by one or a combination of the methods specified below, as applicable. The Contractor shall choose a curing method that is compatible with the requirements for subsequent material usage on the concrete surface.
1. Ramps and Horizontal Surfaces of Parking Areas, Exposed Exterior Balconies: Cure using only Methods 1 or 2 as specified below.
 2. Floors in Non-Public Spaces that are Left Exposed to View and Not Receiving Sealers or Hardeners, Floors Involved in Under-Floor Air Distribution Systems: Apply one coat of a high-solids, water-based, non-yellowing, liquid membrane-forming curing and sealing compound conforming to ASTM C 1315, Type 1, Class A or B in accordance with Method 3 as specified below.
 3. Floors that are to Receive Subsequent Cementitious Toppings, Sealers, Hardeners, Ceramic Tile, Acrylic Terrazzo, Vinyl Composition Tile, Sheet Vinyl, Linoleum, Vinyl-Backed Carpet, Rubber, Athletic Flooring, Synthetic Turf, Wood, Epoxy Overlay or Adhesive, or Other Coating or Finishing Products: Cure using Methods 2 or 3 as specified below. Use a water-based dissipating resin type curing compound conforming to ASTM C 309, Type 1, class A or B for Method 3.
 4. Industrial Slabs: Cure using Methods 1 or 2 as specified below for seven days. The temperature of applied water shall be within 10°F of concrete surface temperature.
 5. All Other Surfaces: Cure using Methods 1, 2, or 3 as specified below. Use a water-based dissipating resin type curing compound conforming to ASTM C 309, Type 1, class A or B for Method 3.
- D. Curing Methods:
1. Method 1 – Moisture Curing: Provide moisture curing by one of the following methods:
 - a. Keep concrete surface continuously wet by covering with water.
 - b. Continuous water-fog spray.
 - c. Covering concrete surface with specified absorptive cover, thoroughly saturating cover with water, and keeping continuously wet. Place absorptive cover to provide coverage of concrete surfaces and edges, with 4" lap over adjacent absorptive covers.
 2. Method 2 – Moisture-Retaining Cover Curing: Provide moisture-retaining cover curing as follows:
 - a. Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width with sides and ends lapped at least 3" and sealed by waterproof tape or adhesive. Immediately repair any holes or tears during curing period using cover material and waterproof tape. Water may be added to concrete surface to prevent drying before the cover is installed, but the surface shall not be flooded with water if a non-absorptive cover is used.

3. Method 3 – Curing or Curing and Sealing Compound: Provide curing, liquid membrane-forming curing, or curing and sealing compound as follows:
 - a. Apply specified compound to concrete slabs as soon as final finishing operations are complete (within two hours and after surface water sheen has disappeared). Apply uniformly in continuous operation by power-spray or roller in accordance with manufacturer's directions. Do not allow to puddle. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period. Apply second coat for sealing two to three hours after the first coat was applied.
 - b. Do not use membrane-forming curing and sealing compounds on surfaces which are to be covered with coating material applied directly to concrete, liquid floor hardener, waterproofing, dampproofing, membrane roofing, flooring (such as ceramic or quarry tile, glued-down carpet, vinyl composition tile, linoleum, sheet vinyl, rubber, athletic flooring, synthetic turf, or wood), paint, or other coatings and finish materials. Dissipating resin type cures are acceptable in these locations.

3.10 HOT WEATHER CONCRETING

A. Definition:

1. Conditions warranting hot weather concreting practices are defined as any combination of high air temperature, low relative humidity, and wind velocity tending to impair the quality of fresh or hardened concrete or otherwise result in abnormal properties. If conditions cause an evaporation rate of 0.2 pounds per square foot per hour or greater as calculated by the figure "NRMCA Nomograph for Estimating Evaporation Rate on the Basis of Menzel Formula" in Appendix A of ACI 305.1, then precautions shall be taken to prevent plastic shrinkage cracks from occurring.
- B. Specification: Follow hot weather concreting practices specified below when required to limit the concrete temperature at the truck discharge point to the stated maximum acceptable temperature.
- C. Records: Under hot weather conditions, the Contractor shall keep records of outside air temperature, concrete temperature at truck discharge and general weather conditions.
- D. Hot Weather Concreting Requirements: The following items, all or in part as required, shall be followed to limit the concrete temperature to the stated maximum acceptable temperature and to minimize the possibility of plastic shrinkage cracks from developing.
 1. Design the concrete mixtures specifically for hot weather conditions replacing some cement with fly ash or other pozzolan and using a water reducing retarding admixture (ASTM C 494 Type D).
 2. Use the largest size and amount of coarse aggregate compatible with the job.
 3. Use sunshades and/or windbreaks.
 4. Delay construction of indoor slabs-on-grade until the walls and roof are constructed.
 5. Cool and shade aggregate stockpiles.
 6. Use ice as part of the mixing water or cool the water with liquid nitrogen. Do not place concrete that contains unmelted ice.
 7. Limit the number of revolutions at mixing speed to 125 maximum.
 8. Reduce time between mixing and placing as much as possible.
 9. Do not add water to ready-mixed concrete at the job site unless it is part of the amount required initially for the specified water-cement ratio and the specified slump.
 10. Schedule concrete placement for early morning, late afternoon, or night.
 11. Have all forms, equipment, and workers ready to receive and handle concrete.
 12. Maintain one standby vibrator for every three vibrators used.
 13. Keep all equipment and material cool by spraying with water including exteriors of forms, reinforcing steel, subgrade, chutes, conveyors, pump lines, tremies, and buggies.
 14. Protect slab concrete at all stages against undue evaporation by applying a fog spray or mist above the surface or applying a monomolecular film. Where high temperatures and/or placing conditions dictate, use water-reducing retarding admixture (Type D) in lieu of the water-reducing admixture (Type A) as directed by the Owner's Testing Laboratory.
 15. Provide continuous curing, preferably with water, during the first 24 hours using wet burlap, cotton mats, continuous spray mist, or by applying a curing compound meeting ASTM C 1315. Continue curing for three days minimum.
 16. Cover reinforcing steel with water soaked burlap so that steel temperature will not exceed ambient air temperature immediately before placement of concrete.
 17. As soon as possible, loosen forms and run water down the inside. When forms are removed, provide a wet cover to newly exposed surfaces.

3.11 COLD WEATHER CONCRETING

A. Definition:

1. Concrete shall not be placed when the outside air temperature is 40°F or less unless cold weather concreting practices are followed as specified below.
 2. Cold weather concreting practices should also be followed whenever the average daily air temperature is expected to be less than 40°F for more than three successive days. The average daily air temperature is the average of the highest and lowest temperature occurring during the period from midnight to midnight. The requirement for adhering to these cold-weather concreting practices may be terminated when the air temperature is above 50° F for more than half of any 24 hour duration.
 3. Cold-weather concreting practices invoked shall keep the temperature of the concrete immediately after placing within the following temperature ranges:
 - a. 55° to 75° F for sections less than 12 inches in the least dimension.
 - b. 50° to 70° F for sections 12 to 36 inches in the least dimension.
 - c. 45° to 65° F for sections 36 to 72 inches in the least dimension.
 - d. 40° to 60° F for sections greater than 72 inches in the least dimension.
 4. Concrete Protection: Protect the concrete immediately after placing and during the defined protection period such that the concrete does not freeze nor fall below the temperature levels stated in the above paragraph. For concrete not loaded during construction, the protection period shall be for a minimum of three days if cold-weather conditions persist. The time may be reduced to a minimum of two days if Type III cement or an accelerating admixture is used or if an additional 100 pounds of cement per cubic yard is added to the concrete mix. Concrete fully loaded during construction shall be protected during cold weather conditions for whatever time is required to obtain the required strength as determined by nondestructive strength tests (Windsor probe, Swiss Hammer Test) on the in-place concrete. Protect concrete surfaces from freezing for the first 24 hours even if cold-weather conditions do not officially exist due to high volatility in ambient temperatures.
 5. Protection Deficiency: If the temperature requirements during any portion of the protection period are not met but the concrete surface did not freeze, the protection period shall be extended until twice the deficiency expressed in degree-hours is made up. Deficiency degree-hours are defined as the average deficiency in temperature below the required value times the number of hours the deficiency persisted. Make-up degree hours are the average increase in temperature above the minimum value times the hours required to make up twice the deficiency degree-hours. Contact the Architect/Engineer if the concrete surface was allowed to freeze during the protection period.
 6. Protection Removal: As the protection is being removed the decrease in temperature measured at the surface of the concrete in a 24 hour period shall not exceed the following:
 - a. 50° F for sections less than 12 inches in the least dimension.
 - b. 40° F for sections 12 to 36 inches in the least dimension.
 - c. 30° F for sections 36 to 72 inches in the least dimension.
 - d. 20° F for sections greater than 72 inches in the least dimension.
 7. The maximum concrete temperature heated by artificial means at point of placement shall not exceed 90°F.
- B. Records: Under cold weather conditions, the Contractor shall keep records of outside air temperature, concrete temperature as placed and general weather conditions. The temperature record shall be taken no less than two times per 24 hour duration.
- C. Cold Weather Concreting Requirements: The following items, all or in part as required, should be followed to assure acceptable concrete in cold weather conditions:
1. Design the concrete mixture to obtain high early strength by using higher cement content, a high early strength cement (Type III), or a specified non-chloride accelerator (ASTM C 494 Type C or E).
 2. Protect the concrete during curing period using insulating blankets, insulated forms, enclosures, and/or heaters.
 3. Concrete cured in heated enclosures shall have heaters vented to prevent exposure of concrete and workmen to noxious gases.
 4. Frozen subgrade shall be thawed prior to concrete placement and snow and ice shall be removed from forms.
 5. Temperature of embedments in concrete must be heated to above 32°F prior to placing concrete
 6. Heat the mixing water and then blend hot and cold water to obtain concrete no more than 10°F above the required temperature.
 7. Heat the aggregates by circulating steam in pipes placed in the storage bins for air temperatures consistently below 32°F. When either water or aggregate is heated to over 140°F, combine them in the mixer first to obtain a maximum temperature of the mixture not to exceed 140°F in order to prevent flash set of the concrete.
 8. Uniformly thaw aggregates far in advance of batching to prevent moisture variations in the stockpile.
 9. Cover warmed stockpiles with tarps to retain heat.
 10. Place air entraining admixture in the batch after the water temperature has been reduced by mixing with cooler solid materials.

11. Use wind screens to protect concrete from rapid cooling.
12. Place vertical pump lines inside the building, if possible, for concrete being pumped.
13. Maintain artificial heat as low as possible to reduce temperature stresses during cooling.
14. Avoid water curing of concrete except for parking garage structures. Apply the required curing compound to unformed surfaces as soon as possible to prevent drying of concrete from heated enclosures.
15. Delay form stripping as long as possible to help prevent drying from heated enclosures and to reduce damage to formed surfaces caused by premature stripping.
16. Provide triple thickness of insulating materials at corners and edges vulnerable to freezing.
17. Wrap protruding reinforcing bars with insulation to avoid heat drain from the warm concrete.
18. Gradually reduce the heat at the end of the heating period to reduce likelihood of thermal shock.

3.12 MISCELLANEOUS CONCRETE ITEMS

- A. Filling-In: Fill-in holes and openings left in concrete structures for passage of work by other trades, unless otherwise shown or directed, after work of other trades is in place. Mix, place, and cure concrete as herein specified, to blend with in-place construction. Provide other miscellaneous concrete filling shown or required to complete work.
- B. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.
- C. Equipment Bases and Foundations: Provide machine and equipment bases and foundations, as shown on drawings. Set anchor rods for machines and equipment to template at correct elevations, complying with certified diagrams or templates of manufacturer furnishing machines and equipment.
- D. Steel Pan Stairs: Provide concrete fill for steel pan stair treads and landings and associated items. Cast-in safety inserts and accessories as shown on drawings. Screed, tamp and finish concrete surfaces as scheduled.

3.13 INVESTIGATION OF LOW CONCRETE STRENGTH TEST RESULTS

- A. Contractor Responsibility for Low Strength Concrete:
 1. If the average of any three consecutive strength tests falls below the required f'_c for a class of concrete but no individual strength test is more than 500 PSI below the required f'_c , the Contractor shall immediately notify the Engineer by telephone or email and take immediate steps to increase the average of subsequent strength tests.
 2. If any individual strength test falls more than 500 PSI below the required f'_c , the Contractor shall immediately notify the Engineer by telephone or e-mail and take immediate steps to assure that the load-carrying capacity of the structure is not jeopardized.
- B. Additional Field Tests to Confirm Low Concrete Strengths:
 1. The cost of all investigations of low-strength concrete, as defined by any individual strength test being more than 500 PSI below the required f'_c , shall be borne by the Contractor.
 2. Code-Prescribed Acceptance: The only accepted field-test methods of determining actual in-situ concrete strength is by the way of core tests as prescribed by ACI 318.
 3. Non-Destructive Tests: If any individual strength test falls more than 500 PSI below the required f'_c , the Engineer may request that non-destructive field tests be performed on the concrete in question using Swiss Hammer, Windsor Probe, or other appropriate methods as approved by the Engineer. Report the comparative test results of the suspect concrete under consideration with identical tests done on concrete of known strength and of the same class. The Engineer considers these test results as only approximate indicators of strength and may not necessarily, by themselves, resolve the low concrete strength issue. These test results will be considered as additional information by which to make an informed judgment. The Engineer reserves the right to accept the concrete based on the results of these approximate tests or order that core tests be taken as prescribed below. At the Contractor's option, the approximate non-destructive field-tests may be waived and core tests immediately initiated.
 4. Core Tests: If, in the opinion of the Engineer, the likelihood of low-strength concrete is confirmed and it has been determined that the load-carrying capacity of the structure is significantly reduced as a result, the Engineer may request that core tests be taken from the area in question as directed by the Engineer. There shall be a minimum of three cores taken for each strength test more than 500 PSI below the required f'_c in accordance with ASTM C 42. If concrete in the structure will be dry under service conditions, cores shall be air dried (temperature 60° to 80°F, relative humidity less than 60 percent) for seven days before test and shall be tested dry. If concrete in the structure will be more than superficially wet under service conditions, cores shall be immersed in water for at least 40 hours and tested wet. The Contractor shall fill all holes made by drilling cores with an approved drypack concrete.

5. Acceptance Criteria for Core Test: Concrete in an area represented by core tests shall be considered adequate if the average of three cores is equal to at least 85% of the required f'_c and no single core is less than 75% of the required f'_c . If approved by the Engineer, locations of erratic core strengths may be retested to check testing accuracy.
6. Load Test: If the concrete strength is not considered adequate based on core tests and the structural adequacy remains in doubt, the Engineer may order a load test as specified in ACI 318 be conducted for the questionable portion of the structure.
7. Strengthening or Demolition of the Structure: If the structural adequacy of the affected portion of the structure remains in doubt following the load test, the Engineer may order the structure to be strengthened by an appropriate means or demolished and rebuilt at the Contractor's expense.

3.14 CONCRETE SURFACE REPAIRS

A. Defective Areas:

1. Formed Surfaces: Concrete surfaces requiring repairs shall include all cracks in excess of 0.01" and any other defects that affect the durability or structural integrity of the concrete. Voids, including honeycombing and rock pockets, and tie holes shall be repaired as required by the specified Surface Finish.
2. Unformed Surfaces: Concrete surfaces requiring repair shall include all surface defects such as crazing, cracks in excess of 0.01" wide or cracks that penetrate to reinforcement or through the member, popouts, spalling, and honeycombs.

B. Classification:

1. Structural Concrete Repair: Major defective areas in concrete members that are load carrying (such as shear walls, beams, joists and slabs), are highly stressed, and are vital to the structural integrity of the structure shall require structural repairs. Structural concrete repairs shall be made using a two-part epoxy bonder, epoxy mortar, or specified polymer repair mortar. The Engineer shall determine the locations of required structural concrete repairs.
2. Cosmetic Concrete Repair: Defective areas in concrete members that are non-load carrying and minor defective areas in load carrying concrete members shall require cosmetic concrete repair when exposed to view and not covered up by architectural finishes. Cosmetic concrete repairs may be made using a polymer repair mortar and compatible bonding agent. The Architect/Engineer shall determine the locations of required cosmetic concrete repairs. Stains and other discolorations that cannot be removed by cleaning and are exposed to view will require cosmetic repair. Cosmetic concrete repair in exposed-to-view surfaces will require Architect's approval prior to patching operation.
3. Slab Repairs: High and low areas in concrete slabs shall be repaired by removing and replacing defective slab areas unless an alternate method, such as grinding and/or filling with self-leveling underlayment compound or repair mortar is approved by the Architect/Engineer. Repair of slab spalls and other surface defects shall be made using epoxy products as specified above and as determined by the Engineer. The high strength flowing repair mortar may be used for areas greater than one inch in depth.

3.15 FIELD QUALITY CONTROL

- #### A. Field Testing and Inspection:
- Refer to Specification **014529 "Structural Testing and Inspections"** for testing and inspection requirements associated with cast-in-place concrete.

END OF SECTION

SECTION 05 05 10

METAL FINISHES

PART 1 - GENERAL

1.1 SUMMARY

A. Sections Includes:

1. Anodizing.
2. Baked enamel coating.
3. Fluorourethane coating
4. PVDF coating.
5. Powder coated finish.
6. Stainless steel finish.

B. Related Sections:

1. Sections with Metals: Galvanizing.
2. Sections with Metals: Factory or shop applied primers for field painting or coating.
3. Section 08 71 00 – Door Hardware: Hardware finishes.
4. Division 9 – Finishes: Field-applied paints and coatings.

1.2 REFERENCES

- A. Reference Standards: See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following:
1. NAAMM Metal Finishes Manual for finish designations and application recommendations.

1.3 SUBMITTALS

A. General: Submit in accordance with Section 01 33 00.

1. Submit submittals of this section simultaneously with submittals of sections with components with finishes specified in this section.

B. Product Data: Submit following:

1. Product data for each coating.
2. Color charts for finish indicating manufacturer's colors available for selection.
3. Include sample of warranty customized for this Project.

C. Closeout Submittals: Submit following in accordance with Section 01 78 00.

PART 2 - PRODUCTS

1.4 PREPARATION

A. Sheet Steel to be Coated:

1. Mechanical Finishes: Complete mechanical finishes of flat sheet metal surfaces before fabrication. After fabrication, finish joints, bends, abrasions, and other surface blemishes to match sheet finish. Protect mechanical finishes on exposed surfaces from damage by application of adhesive paper or other temporary protective covering, prior to shipment.
2. Surface Preparation: Solvent-clean surfaces in compliance with SSPC-SP1 to remove dirt, oil, grease and other contaminants that could impair paint bond. Remove mill scale and rust, if present, from uncoated steel in compliance with SSPC-SP5 (White Metal Blast Cleaning) or SSPC-SP8 (Pickling).

3. Chemical Pretreatment: Apply hot phosphate surface treatment to uncoated steel sheet to comply with SSPC-PT4.
 4. Comply with SSPC-PA1 "Paint Application Specification No. 1" for shop painting.
- B. Protective Coatings: Do not use coatings containing lead.
1. Primed Carbon Steel: Touch-up with primer.
 2. Galvanized Steel: Touch-up with primer.
- C. Metal Patching Compound:
1. Two-part epoxy and metal filler, putty grade, composed of metal alloy blended with high molecular weight polymer.
 2. Match base metal of putty to item being repaired.
 3. Use steel based alloy for repairing cast iron.
 4. Machinable after curing.
 5. Paste consistency: Negligible slump at 1 inch thickness or less.
 6. Shrinkage: Show no evidence of measurable shrinkage from plastic state to hardened state.
 7. Acceptable Manufacturers and Products:
 - a. Super Metal 1111, Belzona; Miami, FL. Use only on steel or cast iron items.
 - b. Bondo, Bondo/Mar-Hide; Medina, OH.
 - c. Accepted Substitute in accordance with Section 01 25 00.

1.5 SHOP FINISHING

- A. General: Apply finishes in factory after products are assembled.
- B. Protect finish with factory applied protective covering prior to shipment.
1. Remove scratches and blemishes from exposed surfaces which will be visible after completing finishing process.
 2. Finish accessories such as trim, flashing, screens, blank-off panels, and fasteners to match assembly.

1.6 ANODIZING

- A. Anodizing: Comply with AAMA 611.
- B. Clear Anodized: AA-M12C22A41, Architectural Class I, nonspecular as fabricated mechanical finish, etched medium matte, clear anodic coating, minimum 0.7 mil thick.
1. Interior Locations: May be architectural Class II anodic coating, AA-M12C22A31.

1.7 BAKED ENAMEL COATING

- A. Baked Enamel Finish System: Includes dipped, electrostatic, powder coat, and other forms of baked enamel shop finishing.
1. Galvanized Steel Surfaces: Cleaned and phosphate conversion coated prior to application of 0.2 mil dry film thickness rust-inhibitive prime coat.
 2. Aluminum Surfaces: Cleaned, etched and given chromate conversion pre-treatment prior to application 0.2 mil dry film thickness of prime coat.
 3. Finish Coat: Manufacturer's standard thermo-cured acrylic, polyester or alkyd enamel, 1.0 mil minimum dry film thickness.
 4. Total Coating Dry Film Thickness: 1.5 mils.
 5. Acceptable Product: Duracron 900 by PPG.
 6. Color: As selected by Architect from manufacturer's full range of standard colors.

1.8 FLUOROPOLYMER (PVDF) COATING

- A. Acceptable Manufacturers:

1. Akzo Nobel Coatings, Inc., Columbus, OH.
 2. BASF Corporation, Decatur, AL.
 3. PPG Industries Inc., Delaware, OH and Springdale, PA.
 4. Valspar Corporation, Garland, TX.
 5. Accepted Substitute in accordance with Section 01 25 00.
- B. Fluoropolymer (PVDF) Coating: AAMA 2605.
1. Resin: 70 percent polyvinylidene fluoride (PVDF).
 2. Substrate, Aluminum: Cleaned and chrome phosphate pre-treated.
 3. Primer: Manufacturer's standard epoxy or acrylic coating.
 - a. Dry Film Thickness: Minimum 0.20 mil.
 - 1) Topcoat: PVDF, Dry Film Thickness:
 - a) Coil: 0.80 mil.
 - b) Extrusion: 1.0 mil.
 - b. Fluoropolymer (PVDF) Coating with Clear Coat: AAMA 2605.
 - 1) Resin: 70 percent polyvinylidene fluoride (PVDF).
 - 2) Substrate, Aluminum: Cleaned and chrome phosphate pre-treated.
 - 3) Primer: Manufacturer's standard epoxy or acrylic coating.
 - a) Dry Film Thickness: Minimum 0.20 0.80 mil.
 - 4) Topcoat: PVDF, Dry Film Thickness:
 - a) Coil: 0.80 mil.
 - b) Extrusion: 1.0 mil.
 - 5) Clear Coat: PVDF, Dry Film Thickness:
 - a) Coil: 0.50 mil.
 - b) Extrusion: 0.50 mil.

1.9 POWDER COATED FINISH

- A. Shop-Applied Powder Coated Finish: Fabricator's standard .
1. Clean and phosphatize surfaces prior to application of coating.
 2. Apply chrome pretreatment to aluminum surfaces to receive PVDF finish.
 3. Apply powder coat finish system for sheet steel immediately following surface preparation and chemical pretreatment.
 4. Comply with paint manufacturer's recommendations for application and baking to achieve minimum recommended dry film thickness.

1.10 STAINLESS STEEL FINISH

- A. Stainless Steel Finishes: ASTM A480.
1. Protect finishes with factory applied adhesive backed paper covering.
 2. Unless otherwise Scheduled or Indicated: No. 4 - General Purpose Polished, vertical grain.

PART 2 - EXECUTION – NOT USED

END OF SECTION

SECTION 05 12 00
STRUCTURAL STEEL FRAMING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification sections, apply to work of this section.

1.2 SUMMARY

- A. Section includes labor, materials, services, equipment, and appliances required in conjunction with or related to the furnishing, fabrication, delivery, and erection of all structural steel, as defined below. Include all supplementary parts, members, and connections necessary to complete the structural steel work, regardless of whether all such items specifically are shown or specified on the drawings. Miscellaneous metal fabrications, architecturally exposed structural steel, metal stairs and ladders, cold-formed metal framing, and steel deck are specified in other Division 05 sections.
- B. Related Requirements:
1. Specification 014529 "Structural Testing and Inspections for testing and inspection requirements associated with structural steel.

1.3 PRICE AND PAYMENT PROCEDURES

- A. Alternates: Substitutions for the member sizes, type(s) of steel connection details, or any other modifications proposed will be considered by the Architect/Engineer only under the following conditions:
1. The request has been made and accepted prior to the submission of shop drawings. All substitutions shall be marked clearly and indicated on the shop drawings as a substitute.
 2. There is a substantial cost advantage or time advantage to the Owner or that the proposed revision is necessary to obtain the required materials or methods at the proper times to accomplish the work in the time scheduled.
 3. Sufficient sketches, engineering calculations, and other data have been submitted to facilitate checking by the Architect/Engineer, including cost reductions or savings in time to complete the work.
 4. In no case shall such substitutions result in additional cost to the Owner.

1.4 REFERENCES

- A. Definitions:
1. Erection Drawings: Field installation or member-placement drawings that are prepared by the Fabricator to show the location and attachment of the individual shipping pieces.
 2. Erection-Bracing Drawings: Drawings that are prepared by the Erector to illustrate the sequence of erection, any requirements for temporary supports, and the requirements for raising, bolting, and or/welding. These drawings are in addition to and separate from the Erection Drawings.
 3. Professional Engineer: A professional engineer who is licensed to practice engineering in the state where the project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for projects with structural steel framing that are similar to that indicated for this Project in material.
 4. Shop Drawings: Drawings of the individual structural steel shipping pieces that are to be produced in the fabrication shop.
 5. Structural Steel: Structural steel shall be defined as that work prescribed in Section 2.1 of AISC 303 and all steel support for elevator guide rails and catwalks (including support members and attached structural steel shapes and plates such as hangers, toe plates, and the grating walking surface).
- B. Reference Standards:
1. Codes and Standards: Comply with provisions of following codes, specifications, and standards, except where more stringent requirements are shown or specified.
 - a. All federal (OSHA), state, and local laws that govern safety requirements for steel erection and other requirements if more stringent than the codes and standards enumerated below. OSHA requirements include regulation 29 CFR 1926, Part R, "Safety Standard for Steel Erection".
 - b. AISC, "Steel Construction Manual."
 - c. AISC 303, "Code of Standard Practice for Steel Buildings and Bridges," except as noted herein.

- 1) Certain sections in this specification contain requirements that are more restrictive and/or different than contained in this standard. In such cases, the requirements of this specification shall control.
- d. ANSI/AISC 360, "Specification for Structural Steel Buildings."
- e. ANSI/AWS D1.1, "Structural Welding Code – Steel."
- f. ANSI/AWS D1.3, "Structural Welding Code – Sheet Steel."
- g. ANSI/AWS D1.4, "Structural Welding Code – Reinforcing Steel."
- h. Research Council on Structural Connections (RCSC), "Specification for Structural Joints using High-Strength Bolts."
- i. The Society of Protective Coatings, "SSPC Painting Manual", Volumes 1 and 2.

1.5 ADMINISTRATIVE REQUIREMENTS

A. Coordination:

1. Quality Control:

- a. The Contractor is responsible for quality control, including workmanship and materials furnished by his subcontractors and suppliers.
- b. The Contractor shall coordinate the fabrication and erection of all structural steel work with the work of other trades.
- c. The Fabricator alone shall be responsible for all errors of detailing, fabrication, and for the correct fitting of the structural members.
- d. The Fabricator shall coordinate connection details, joint fit-up procedures, and field adjustment requirements with Erector. The Contractor shall coordinate provision of all erection bolts, lifting lugs, or other devices required for erection with the Fabricator and the Erector and for interference with architectural finishes and constraints.

2. Document Conflict and Precedence:

- a. In case of conflict among documents, including architectural and structural drawings and specifications, notify Architect/Engineer prior to submitting proposal. In case of conflict between and/or among the structural drawings and specifications, the strictest interpretation shall govern, unless specified otherwise in writing by the Architect/Engineer.
- b. Questions about Contract Documents: The Contractor shall notify promptly the Architect/Engineer whenever design of members and connections for any portion of the structure are not indicated clearly or when other questions exist about the Contract Documents. Such questions shall be resolved prior to the submission of shop drawings.

3. Materials and installed work may require testing and retesting, as directed by the governing building code or the Architect/Engineer, at any time during progress of work.

- a. The Contractor shall provide adequate notification to the Owner's Testing Agency of construction operation including the project schedule to allow the Testing Agency to schedule inspections. Failure to notify sufficiently may result in additional costs incurred by the Testing Laboratory that may be back-charge to the Contractor by the Owner.
- b. The Contractor shall cooperate with laboratory personnel, provide access to the work, and provide access to manufacturer's operations.
- c. The Contractor shall cooperate with the Owner's Testing Laboratory when Arbitration Testing and Inspection is called for due to a disagreement regarding the tension in installed bolts that have been inspected according to the Structural Testing and Inspections specification.
- d. The Contractor shall make adequate arrangement with the Owner's Testing Agency for inspection of material stockpiles and facilities.
- e. The Contractor shall provide to the laboratory certificates and representative samples of materials proposed for use in the work in quantities sufficient for accurate testing as specified.
- f. The Contractor shall furnish labor, equipment, and facilities as required for sampling and testing by the laboratory and other facilitate the required inspections and test.
- g. Inspection or testing by the Owner does not relieve the Contractor of his responsibility to perform the Work in accordance with the Contract Documents. Test not specifically indicated to be done at the Owner's expense, including retesting of rejected materials and installed work, shall be done at the Contractor's expense. See the Structural Testing and Inspections Specification.

B. Preinstallation Meetings:

1. At least 14 days prior to beginning structural steel erection, the Contractor shall hold a meeting to review the detailed quality control and construction requirements and to determine the procedures for producing proper structural steel construction. Also, review requirements for submittals, status of coordinated work, and availability of materials. Establish work progress schedule and procedures for materials inspection, testing, and certification.

2. The Contractor shall require responsible representatives of every party who is concerned with the structural steel work to attend the conference, including, but not limited to, the following:
 - a. Contractor's Superintendent.
 - b. Laboratory responsible for field quality control.
 - c. Special Inspector or Laboratory responsible for shop inspection or testing.
 - d. Structural steel detailer.
 - e. Structural steel fabricator.
 - f. Structural steel erector.
 - g. Owner's and Architect's/Engineer's Representative.
 - h. Engineer.
3. Minutes of the meeting shall be record, typed, and printed by the contractor and distributed to all parties concerned within five days of the meeting. One copy of the minutes shall be transmitted to the following for information purposes:
 - a. Owner's Representative.
 - b. Architect.
 - c. Engineer.
4. The Engineer shall be present at the conference. The Contractor shall notify the Engineer at least seven days prior to the scheduled date of the conference.

1.6 SUBMITTALS

- A. Product Data: Submit producer's or manufacturer's specifications and installation instructions for following products to show compliance with specifications, including the specified standards):
 1. Shrinkage-Resistant Grout.
 2. Welding Electrodes.
 3. Structural Steel Primer Paint.
 4. Inorganic or Other Protective Paint.
 5. Shear Studs.
 6. Direct Tension Indicators.
- B. Shop Drawings:
 1. Preliminary Connection Review: Submit preliminary details of proposed connections not less than 14 days in advance of the start of preparation of detailed shop drawings. Proposed variations from the details shown on the drawings will be considered and such variations must have preliminary approval from the Engineer prior to the preparation of detailed shop drawings. Failure to adhere to the requirements of this paragraph obligates the Contractor to take responsibility for any and all resulting delays in the detailing and fabrication of structural steel.
 2. Detailed Shop Drawings: Submit drawings showing complete details and schedules for fabrication and assembly of structural steel members. Drawings shall include the following minimum information:
 - a. Details of cuts, connections, camber, holes, and other pertinent data.
 - b. Indication of welds by standard AWS symbols, and show size, length, and type of each weld.
 - c. Indication of type, size, and length of bolts, distinguishing between shop and field bolts. Identify the type of high-strength bolted connection (slip-critical, direct-tension, or bearing connections). Indicate locations of pretensioned bolts.
 - d. Connection material specification and sizes.
 - e. Joints or groups of joints in which a specific assembly order, welding sequence, welding technique, or other special precautions are required.
 - f. Holes, flange cuts, slots, and openings shall be made as required by the structural drawings, all of which shall be properly located by means of templates.
 - g. Setting drawings, templates, and directions for installation of anchor rods and other anchorages to be installed by others.
 - h. Non-Destructive Testing (NDT) to be performed by the Fabricator, if any.
 - i. A letter sealed by the Fabricator's Professional Engineer responsible for the design of any of the connections shown on the shop drawings attesting that the engineer has reviewed the shop drawings and that the connections detailed and shown on the shop drawings conform to the engineer's design.
 3. Erection Drawings: Submit complete erection drawings showing field installation and member-placing instructions for locating and attaching the individual shipping pieces.
 4. Erection-Bracing Drawings: Submit, for record purposes only, complete erection-bracing drawings.
 5. Shear Connector Placement Drawings: Provide drawings showing proper placement (longitudinal and transverse spacing) of shear connectors on each composite beam requiring such connectors. The drawings shall show the proper relationship of the shear connectors to the flutes in the steel deck and the arrangement of shear connectors along the span of the composite beam. Show the

- method of attachment of shear connectors and the proposed brand and model of equipment to be used.
6. All drawings submitted for review shall have blank space for a 2" high and 3.5" wide shop drawing stamp of the Engineer as part of the title block
- C. Certificates:
1. Structural Steel: Submit for each type.
 2. High-Strength Bolts: Submit for each type, including nuts and washers.
 3. Unfinished Bolts and Nuts.
- D. Delegated Design Submittals:
1. Preliminary Connection Design Review: In conjunction with the Preliminary Connection Review submittal, the Fabricator's licensed professional engineer shall submit example design calculations for each connection type not less than 14 days in advance of the start of preparation of detailed shop drawings.
 2. Connection Design Submittals: The Fabricator's licensed professional engineer shall submit complete design calculations show all information as specified in the "Connections" section under Part 2. The Engineer reserves the right to reject all shop drawings submitted without complete design calculations.
 3. Connection Design Validation Letter: The Fabricator's licensed professional engineer responsible for the design of any of the connections shown on the shop drawings shall submit a letter that is sealed attesting that the connection design engineer has reviewed the shop drawings and that the connections detailed and shown on the shop drawings conform to the engineer's design.
- E. Test and Evaluations Reports: Submit certified reports of tests required by this Specification. Include data on type(s) of tests conducted and test results.
- F. Field Quality Control Submittals:
1. Surveys: Submit for each survey required.
- G. Sustainable Design Submittals:
1. Recycled Content – Credit MR 4.1/MR 4.2: Provide documentation indicating percentage of post-consumer and pre-consumer recycled content by weight per unit of product or assembly containing the product. Indicate the percentage of the dollar value of the recycled content compared to the total dollar value of the product or assembly containing the product.
 2. Material Proximity – Credit MR 5.1/MR 5.2: When the distance to the project site is 500 miles or less, indicate location and distance to project site of extraction, harvesting, recovery and manufacturing of all materials. Indicate the dollar value of the material cost of the product containing local/regional materials. Where product components are sourced or manufactured in separate locations, provide location and percentage by weight of each component per unit of product.
- H. Qualification Statements:
1. Submit qualification data, including required certifications, for firms and persons specified in the "Qualifications" section under Part 1, to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
 2. Submit a resume from the structural steel detailer showing a minimum of two years of experience selecting or completing structural steel connection details using information found in tables in the AISC "Steel Construction Manual".
 3. Submit Welding Procedure Specifications (WPS) in accordance with ANSI/AWS D1.1 for all welded joints. Submit test reports showing successful passage of qualification tests for all non-prequalified WPSs.
 4. Provide certification that welders to be employed in work have satisfactorily passed AWS qualification tests as specified in the "Qualifications" section under Part 1. If recertification of welders is required, retesting will be at Contractor's responsibility.
 5. A fabricator that is registered with the local building official and is approved to perform fabrication without special inspection shall submit a certificate of compliance stating that the work was performed in accordance with the approved construction documents.
- I. Record Documentation:
- J. Minutes of Preinstallation Meeting: Submit for review.
- 1.7 QUALITY ASSURANCE
- A. Qualifications:
1. Fabricator:
 - a. The structural steel fabricator shall have not less than five years of experience in the successful fabrication of structural steel similar to this project.
 2. Detailer:

- a. The structural steel detailer shall have not less than two years of experience in the successful detailing of structural steel similar to this project including experience in selecting or completing structural steel connection details using information found in tables in the AISI "Steel Construction Manual".
 3. Erector:
 - a. The structural steel erector shall have not less than two years of successful experience in the erection of structural steel of a similar nature to this project.
 4. Welding Qualifications: Qualify welding processes and welding operators in accordance with AWS "Structural Welding Code – Steel". Retain the section below if the fabricator is required to design the connections or when a professional engineer is required to be involved in designing the temporary shoring and bracing of the structure.
 5. Professional Engineer:
 - a. The Professional Engineer employed by the Fabricator for connection design shall be experienced in the specific area of structural steel connection design with demonstrated experience of not less than three projects of similar scope and complexity.
 - b. The Professional Engineer employed by the Erector for preparation of Erection Bracing Drawings shall be experienced in the specific area of structural frame bracing during erection design with demonstrated experience of not less than three projects of similar scope and complexity.
 6. Independent Testing Laboratory:
 - a. Any testing laboratory retained to perform tests that are required by this specification shall meet the basic requirements of ASTM E 329 and shall submit to the Owner, Architect, and Engineer evidence of current accreditation from the American Association for Laboratory Accreditation, the AASTHO Accreditation Program or the "NIST" National Voluntary Laboratory Accreditation Program.
 - b. The Testing Laboratory shall be an Approved Agency by the Building Official to perform Special Inspections and other tests and inspections as outlined in the applicable building code.
 - c. Tests and inspections shall be conducted in accordance with specified requirements, and if not specified, in accordance with the applicable standards of the American Society for Testing and Materials or other recognized and accepted authorities in the field.
 - d. Qualification of Welding Inspectors:
 - 1) Inspectors performing visual weld inspection shall meet the requirements of AWS D1.1 Section 6.1.4. Inspectors shall have current certification as an AWS Certified Welding Inspector (CWI). Assistant inspectors, if any, shall be supervised by an Inspector and shall be qualified by training and experience to perform the specific functions to which they are assigned.
 - 2) Inspectors performing nondestructive examinations of welds other than visual inspection (MT, PT, UT, and RT) shall meet the requirements of AWS D1.1, Section 6.14.6.
- 1.8 DELIVERY, STORAGE, AND HANDLING
- A. Deliver materials to site at such intervals to ensure uninterrupted progress of work.
 - B. Deliver anchor rods and anchorage devices, which are to be embedded in cast-in-place concrete or masonry, in ample time so as not to delay work.
 - C. Store materials to permit easy access for inspection and identification. Keep steel members off ground, using pallets, platforms, or other supports. Protect steel members and packaged materials from corrosion and deterioration. Do not store materials on structure in a manner that might exceed allowable loads on or cause distortion or damage to members or supporting structures. Repair or replace damaged materials or structures as directed by Architect/Engineer.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Structural Steel:
 1. All hot rolled steel plates, shapes, sheet piling, and bars shall be new steel conforming to ASTM A 6.
 2. Comply with the provisions of the following ASTM Specifications as appropriate for the grades and types, and at the locations as specified on the drawings:
 - a. Structural Steel Wide Flange and WT Shapes: High Strength Steel, ASTM A 572, Grade 60.
 - b. Angle Shapes: Carbon Steel, ASTM A 36.
 - c. Structural Steel Plates and Bars: High Strength Steel, ASTM A 572, Grade 60.

3. Connection Material: Unless noted otherwise on the drawings, column stiffener plates and doubler plates at moment connections shall be the same grade of steel as the beam connecting the column (highest grade if more than one grade is used). All other connection material except as noted otherwise on the drawings including bearing plates, gusset plates, stiffener plates, filler plates, angles, etc. shall be A36 steel unless a higher or matching grade of steel with the members connected is required by strength or stiffness calculations and provided the resulting sizes are compatible with the members connected.
 4. Structural Steel Surfaces: For fabrication of work which will be exposed to view in the completed structure, use only materials which are smooth and free of surface blemishes including pitting, seam marks, roller marks, rolled trade names and roughness. Remove such blemishes by grinding, or by welding and grinding, prior to cleaning, treating and application of surface finishes.
 5. Potential Non-conforming Material: For structural steel for which evidence exists that the steel may not conform to ASTM requirements, the Contractor, where permitted by the Engineer, shall engage the services of an independent testing laboratory to test the material according to ASTM A 6 and submit certified test reports that verify conformity to ASTM standards. Tests shall be made for each 10 tons of affected material unless otherwise directed by the Engineer.
- B. Structural Bolts and Threaded Fasteners: Structural bolts and threaded fasteners shall comply with the following ASTM Specifications as appropriate for the types and at the locations as specified on the drawings:
1. ASTM A 325 Type 1.
 2. ASTM A 490 Type 1.
 3. Twist-Off-Type Tension-Control Bolt Assemblies:
 - a. Bolts that are manufactured to conform to ASTM A 325 shall additionally conform to ASTM F 1852.
 - b. Bolts that are manufactured to conform to ASTM A 490 shall additionally conform to ASTM F 2280.
 - c. Subject to conformance with specified requirements, acceptable manufacturers include but are not limited to:
 - 1) Nucor Fastener, A Division of Nucor Corporation.
 - 2) Lake Erie Screw Corp.
 - 3) Vermont Fasteners Manufacturing.
 - 4) Lohr Structural Fasteners.
 4. Threaded Round Stock:
 - a. ASTM A 572 Grade 50 (to 2 inches in diameter).
 5. Bolts and Nuts, High Strength Bolts: Bolts and nuts for all high strength bolts shall be heavy hex head conforming to ANSI Standards B18.2.1 and B18.2.2 respectively. Nuts shall conform to ASTM A 563.
 6. Washers: All washers shall be circular, flat and smooth and shall conform to the requirements of Type A washers in ANSI Standard B23.1. Washers for high strength bolts shall be hardened and conform to ASTM F 436. Beveled washers for American Standard Beams and channels shall be square or rectangular, shall taper in thickness (16 2/3% slope) with an average thickness of 5/16". When an outer face of a bolted part has a slope greater than 1:20 with respect to a plane normal to the bolt axis, a beveled washer shall be used. Washers to be used with A490 bolts larger than 1 inch in diameter and installed over oversized or short-slotted holes and other similar situations shall conform to ASTM F 436 except with 5/16 inch minimum thickness.
 7. Zinc-Coated Bolts: ASTM A 325 bolts, with their nuts and washers, that are used to connect steel called for on the drawings or in the specifications as hot-dip galvanized after fabrication shall be zinc-coated either by the hot-dip process in accordance with ASTM A 153, Class C or by the mechanical deposition process in accordance with ASTM B 695, Class 50, Type 1. The bolts, nuts, and washers shall all be zinc-coated using the same process and they shall be considered together as an assembly and shall be tested and shipped together as such. Comply with all the requirements of ASTM A 325 and ASTM A 563 as they relate to zinc-coated materials. ASTM F 1852 bolts with their nuts, and washers shall be zinc-coated only by the mechanical deposition process in accordance with ASTM B 695, Class 50, Type 1. Do not zinc-coat ASTM A 490 bolts.
 8. Atmospheric Corrosion Resistant Bolts: High strength bolts, nuts and washers connecting steel specified as ASTM A 588 or A242 weathering steel shall be weather resistant Type 3 bolts and similarly treated nuts and washers.
 9. Direct Tension Indicators: Compressible washer-type direct-tension indicators conforming to ASTM F 959.
 - a. Subject to conformance with specified requirements, acceptable manufacturers include but are not limited to:
 - 1) Applied Bolting Technology.
 - 2) Turnasure, LLC.

10. Bolt Lubrication: All bolts shall be well lubricated at time of installation. Dry, rusty bolts will not be allowed.
11. New Bolts: All bolts shall be new and shall not be reused.
- C. Electrodes for Welding:
 1. Provide electrodes that comply with AWS D1.1, "Structural Welding Code - Steel" and that can produce welds that have a minimum Charpy V-notch toughness of 20 ft-lbs at 40° F, unless noted otherwise in these specifications or on the drawings.
 2. Electrodes for various welding processes shall be as specified below:
 - a. SMAW:
 - 1) E70XX low hydrogen.
 - b. SAW:
 - 1) F7X-EXXX.
 - c. GMAW:
 - 1) ER70S-X.
 - d. FCAW:
 - 1) E7XT-X.
 3. Electrodes shall be compatible with parent metal joined.
- D. Headed Studs used as Anchors for Structural Steel Plates and Members connecting to Concrete: AWS Type A studs manufactured in conformance with ASTM A 29 with a minimum tensile strength of 61,000 psi of sizes as specified on the drawings.
- E. Headed Studs used as Composite Member Shear Connectors: AWS type B studs manufactured in conformance with ASTM A 29 with a minimum tensile strength of 65,000 psi of sizes as specified on the drawings.
- F. Deformed Bar Anchors: AWS Type C studs manufactured in conformance with ASTM A 496 with a minimum tensile strength of 80,000 PSI. ASTM A 615 reinforcing bars may not be substituted for deformed bar anchors.
- G. Anchor Rods:
 1. All anchor rods shall conform to ASTM F 1554, unless noted otherwise on the drawings and shall be of the yield strength as specified below as appropriate for the types and at the locations as specified on the drawings:
 - a. Grade 55 (1/4 inch to 4 inches in diameter), complying with Supplementary Requirement S1 of ASTM F 1554.
 2. Anchor rods used with galvanized baseplates shall be galvanized.
 3. Nuts: All nuts with anchor rods shall be heavy hex head conforming to ASTM A 563.
 4. Washers: Unless indicated otherwise, washers for all base plates shall be in accordance with the AISC "Steel Construction Manual", Table 14-2 with holes 1/16" larger than the anchor rod diameter. Washers shall conform to ASTM A 36 steel.
- H. Structural Steel Primer Paint:
 1. Primer paint shall produce a Class B coating on all painted faying surfaces that are a part of a slip-critical connection as noted on the drawings; surface prepared according to SSPC-SP-6 (Commercial Blast Cleaning) and shall be of the following types.
 - a. Organic zinc-rich primer utilizing either an epoxy or urethane binder with a minimum volume solids ratio of 50 percent with a minimum zinc content of 80 percent by weight in the dry film. Apply primer at a rate to achieve a dry film thickness of 3.0 to 4.0 mils. The primer shall comply with the AISC Class B slip critical requirement. (SSPC-SP6 Commercial Blast Cleaning).
 - b. Ethyl Silicate Inorganic zinc-rich primer with a minimum volume solids ratio of 60 percent and with a minimum zinc content of 75 percent by weight in the dry film. The primer shall comply with the AISC Class B slip critical requirement. (SSPC-SP6 Commercial Blast Cleaning).
 - c. Polyamide Epoxy with a minimum volume solids ratio of 65 percent. The primer shall comply with the AISC Class A slip critical requirement. (SSPC-SP6 Commercial Blast Cleaning).
 2. Unless noted otherwise, primer paint shall be one of the following types with the indicated surface preparation:
 - a. SSPC-Paint 25.1, Type II; zinc oxide, raw linseed oil and alkyd primer, surface prepared according to SSPC-SP-2 (Hand Tool Cleaning) unless noted otherwise in this specification.
 - b. Hydrophobic Acrylic Polymer with dry film thickness of not less than 2.0 mils: Tnemec Series 30, Spra-Saf EN or Sherwin Williams, Spraylastic Exterior Waterborn Dryfall (SSPC-SP6 Commercial Blast Cleaning).
 3. Refer to Architect's drawings and specifications for final paint finish requirements of structural steel. Primer paint shall be compatible with final paint requirements.
- I. Grating: Welded steel bar grating of the type, depth, and finish noted on the drawings capable of carrying not less than the stated live load and deflecting not more than span/360 under that load.

- J. Hot-Dip Galvanizing:
1. Scope: All structural steel items and their connections permanently exposed to exterior conditions or that are within areas of unconditioned airspace, whether specified on the drawings or not, shall be hot-dip galvanized after fabrication unless indicated on the drawings or in Specification 099100 to receive a primer and/or finish coat. Such items include, but are not limited to:
 - a. Base plates and anchor rods supporting galvanized members.
 - b. Shelf angles.
 - c. Parapet wall supporting members.
 - d. Screen wall supporting members.
 - e. Window washing support members.
 - f. Exterior covered walkways.
 - g. Embedded plates in concrete exposed to unconditioned airspace.
 - h. Garage guardrail steel and connections.
 - i. Cooling tower support steel.
 - j. Building skin support steel exposed to moisture outside the exterior waterproofing surface.
 - k. Examine the architectural and structural drawings for other items required to be hot-dip galvanized.
 2. Zinc-coat all ASTM A 325 bolts nuts, and washers used in the connection of such steel. Field welded connections shall have welds protected and the exposed portions of ASTM A 490 bolts, nuts, and washers shall be protected with galvanizing repair paint.
 3. Surface Preparation: All steel to be hot-dip galvanized shall undergo the following surface preparation as specified by the Society for Protective Coatings (SSPC), Volume 2.
 - a. Remove all grease, oil, grime and foreign contaminants by thorough cleaning with an alkaline or organic solvent followed by thorough rinsing in cold water.
 - b. Remove scale by pickling in diluted sulfuric or hydrochloric acid. Pickling shall be followed by a rinse in warm water and a second rinse in cold water. As an alternative to pickling, the steel may be white metal blast cleaned according to SSPC-SP-5.
 - c. Dip in a flux solution of zinc ammonia chloride followed by drying at room temperature.
 4. Zinc Coating: The zinc coating for steel shapes and plates shall conform to ASTM A 123. Weight of zinc coating per square foot of surface for 1/8 inch and 3/16 inch thick steels shall average not less than 3.0 mils with no individual thickness less than 2.6 mils. The coating weight shall average not less than 3.9 mils for 1/4" thick and heavier steel with no individual thickness less than 3.3 mils.
- K. Galvanizing Repair Paint: Galvanizing repair paint shall be "ZRC Cold Galvanizing Compound" as manufactured by ZRC Chemical Products or a paint complying with SSPC-Paint 20, Level 1.

2.2 FABRICATION

- A. Structural steel members for which shop drawings have not been reviewed shall not be fabricated. Any steel detailed or fabricated prior to the Initial Survey from Part 3 below is at contractor's risk.
- B. All fabricated material and connections shall fit within architectural constraints.
- C. The omission from the shop drawings of any materials required by the Contract Documents shall not relieve the Contractor of the responsibility of furnishing and installing such materials, even though the shop drawings may have been reviewed.
- D. Shop Fabrication and Assembly:
1. Fabricate and assemble structural assemblies in shop to greatest extent possible. Fabricate items of structural steel in accordance with AISC Specification and as indicated on approved final shop drawings. Provide camber in structural members where indicated.
 2. Properly mark and match-mark materials for field assembly. Fabricate for delivery sequence which will expedite erection and minimize field handling of materials.
 3. Milled surfaces of built-up sections shall be completely assembled or welded before milling.
 4. Fitted stiffeners shall be fabricated neatly between flanges, and the ends of stiffeners shall be milled or ground to secure an even bearing against abutting surfaces. All milled or ground joints shall bear throughout their contact length.
- E. Dimensional Tolerances: Dimensional tolerances of fabricated structural steel shall conform to Section 6.4 of the AISC Code of Standard Practice.
- F. Camber:
1. Camber of structural steel members is indicated on the drawings. Camber shall be measured in the Fabricator's shop in the unstressed condition, prior to erection. The Fabricator shall provide camber measurements of all beams and a report to the Testing Laboratory confirming this has been done.
 2. Where possible, camber of beams shall be applied by a cold bend process.

3. The local application of heat may be used to introduce or correct camber, curvature, or straightness provided the temperature of the heated area, as measured by temperature crayons or other approved means, does not exceed 1,200°F.
 4. Where indicated on the drawings in a camber diagram, cantilever or double cantilever beams shall be cambered for the main span and cantilever end separately, either by a staged cold bending process or by the application of heat.
 5. Beams <and trusses> detailed without specified camber shall be fabricated so that after erection any natural camber due to rolling or shop fabrication is upward.
 6. Truss Camber Tolerances: Unless noted otherwise, the tolerance for any specified camber point shall be plus or minus 1/1600 (three-quarter inch per 100 feet) of the distance from the nearest specified support point.
- G. Splices in Structural Steel: Splicing of structural steel members in the shop or the field is prohibited without prior approval of the Engineer. Any member having a splice not shown and detailed on approved shop drawings will be rejected.
- H. Compression Joints: Ends of columns, except as otherwise noted, and other compression joints at splices and other connections as noted on the drawings which depend on contact bearing as part of the splice strength shall be finished to bear in accordance with AISC Specification M2.6 so as to provide complete true bearing in accordance with AISC Specification M4.4.
- I. Cutting: Manual oxygen cutting shall be done only with a mechanically guided torch. An unguided torch may be used provided the cut is not within 1/8 inch of the finished dimension and final removal is completed by means such as chipping or grinding to produce a smooth surface quality free of notches or jagged edges. All corners shall be smooth and rounded to a minimum 1/2" radius.
- J. Anchor Rods: Furnish anchor rods and other connectors required for securing structural steel to foundations and other in-place work. Furnish 1/8" minimum steel templates for presetting bolts and other anchors to accurate locations.
- K. Large Plates to be Embedded in Concrete: For steel plates that are larger than 24"x24" and are to be embedded horizontally in and at the top surface of concrete, provide one-inch diameter holes to prevent trapped air underneath plates and to achieve full consolidation. The location of holes shall be shown on the shop drawings and shall not impair the strength of the plate.
- L. Holes for Other Work: Provide holes required for securing other work to structural steel framing, and for passage of other work through steel framing members as shown on the contract documents, and/or the final shop drawings.
1. Provide specialty items as indicated to receive other work.
 2. Cut, drill, or punch holes perpendicular to metal surfaces. Do not flame cut holes or enlarge holes by burning. Drill holes in bearing plates.
- M. Lifting and Erection Devices: The Fabricator shall be responsible for designing, detailing, and furnishing all lifting devices and erection aids required for erection. Such devices shall be removed after erection if they interfere with architectural finish requirements.
1. Edge Preparations: Comply with the requirements of AISC Specification Section J1.6 and M2.2 for beam copes and weld access holes. Use of weld access hole geometry meeting the requirements of AWS D1.1 is acceptable. <Unless noted otherwise weld access holes in beam to column connections of the Seismic Load Resisting System (SLRS) as designated shall conform to the requirements of AWS D1.8, Section 6.9.>
- N. Drainage Holes: Provide 1 inch diameter drainage (weep) holes in all members (trusses, girders, beams, etc.) exposed to weather where rain water could collect (at low points and/or behind dams caused by connections, stiffener plates, etc.). Show all holes on shop drawings for review by the Engineer.
- 2.3 WELDING
- A. Code: All shop and field welding shall conform to all requirements in the "Structural Welding Code – Steel", ANSI/AWS D1.1, as published by the American Welding Society (AWS). Welder certifications never expire unless the welder has not welded for more than 6 months or there is a specific reason to question a welder's ability. For complicated welding jobs, consider requiring requalification if certification is over two years old. Requalification testing should be at contractor's expense.
- B. Welder Certification: All shop and field welders shall be certified according to all the applicable AWS procedures for the welding process and welding position used. Each welder shall be assigned an identifying symbol or mark and all shop and field welded connections containing complete or partial joint penetration welds, multi-pass fillet welds, and fillet welds greater than 5/16" shall be identified by the symbol or mark of the welder responsible for the connection.
- C. Minimum Size and Strength:
1. Fillet Welds: Minimum size of fillet welds shall be as specified in Table J2.4 in AISC Specification, Chapter J.

2. Partial-Penetration Groove Welds: The minimum effective throat thickness of partial-penetration groove welds shall be as specified in Table J2.1 in AISC Specification, Chapter J.
 3. Minimum Strength of Welded Connections: Except as specified below in "Connections" or noted otherwise on the drawings, all shop and field welds shall develop the full tensile strength of the member or element joined. All members with moment connections as indicated on the drawings shall be welded to develop the full flexural capacity of the member, unless noted otherwise on the drawings.
- D. Filler Metal Requirements: Weld metal shall be as specified in Table J2.5 in AISC Specification, Chapter J and other requirements of this specification.
- E. Welding Procedure Specification:
1. All welding shall be performed in accordance with a Welding Procedure Specification (WPS) as required in AWS D1.1 and reviewed by the Owner's Testing Laboratory and by the Architect/Engineer. The WPS variables shall be within the parameters established by the filler-metal manufacturer. Engage the services of an independent Testing Laboratory to provide the qualification testing required by AWS D 1.1, Chapter 4, part B to qualify any non-prequalified WPS needed for the project. The independent Testing Laboratory shall prepare Welding Procedure Qualification Records (WPQR) documenting the successful qualification of each Welding Procedure Specification.
- F. Welding Procedures:
1. All welding processes shall comply with the requirements of ANSI/AWS D1.1 unless noted otherwise.
 2. Built-up sections assembled by welding shall be free of warpage and all axes shall have true alignment.
 3. Welds not specified shall, if possible, be continuous fillet welds developing the minimum strength, as specified above, using not less than the minimum fillet welds as specified by AISC.
 4. The toughness and notch sensitivity of the steel shall be considered in the formation of all welding procedures to prevent brittle and premature fracture during fabrication and erection.
 5. The Welding Procedure Specification shall be followed without deviation unless specific approval for change is obtained from the Owner's Testing Laboratory and the Architect/Engineer.
 6. Before welding, particular attention shall be paid to surface preparation, fit up, and cleanliness of surfaces to be welded.
 7. Minimum preheat and interpass temperatures for structural steel welding shall be as specified in ANSI/AWS D1.1, except that no welding shall be performed when the ambient temperature is lower than 0 degrees F. The temperature shall be measured from the side opposite that upon which the preheat is applied.
 8. The heat, input, length of weld, and sequence of weld shall be controlled to prevent distortions. The surfaces to be welded and the filler metals to be used shall be subject to inspection before any welding is performed.
 9. Welds shall be sound throughout. There shall be no crack in any weld or weld pass. Welds shall be considered sound if they conform to AWS requirements, as confirmed by non-destructive testing.
 10. Welds shall be free from overlap.
 11. Craters shall be filled to the full cross section of the welds.
 12. For high-strength low-alloy steels, follow welding procedures as recommended by steel producer for exposed and concealed connections.
 13. Fabricator and Erector shall coordinate welding responsibility at all welded joints.
- G. Stress Relieving: All welding sequences shall be such as to reduce the residual stresses due to welding to a minimum value. If high residual stresses are present, stress relieving of joints shall be required. Welded connections shall be detailed and designed to minimize the accumulation and concentration of through-thickness strains due to weld shrinkage.
- 2.4 BOLTING
- A. Bolt Diameter: Minimum bolt diameter shall be 3/4 inch. The difference in diameter between bolts of differing sizes used on the project shall be not less than 1/4".
- B. Connection Type: Unless noted otherwise on the drawings, all bolted connections shall be snug-tightened using high-strength bolts in standard holes (hole diameter nominally 1/16 inch greater than the nominal bolt diameter) with threads included in the shear planes. Notwithstanding, the contractor shall be responsible to adhere to provisions of ANSI/AISC 360 Section J1.10, which lists circumstances under which certain connections require pretensioned high strength bolts.
- C. Oversize, Short-Slotted and Long-Slotted Holes: The dimensions and washer requirements of oversize, short-slotted, and long-slotted holes shall conform to ANSI/AISC 360 Table J3.3.
- D. Fastener Tension:
1. High strength bolts in snug-tightened joints shall be tightened to a snug tight condition only. Do not pretension bolts in snug-tightened joints the same as if they were in slip-critical joints. The snug-tightened condition is defined as the tightness that exists when all plies are in firm contact. This may

usually be attained by a few impacts of an impact wrench or the full effort of an ironworker using an ordinary spud wrench. If Twist-Off-Type Tension-Control Bolt Assemblies are used in snug-tightened joints, do not fully tension bolts and leave the splines intact.

2. High-strength Bolts in Slip-critical and Pretensioned Joints:
 - a. High-strength bolts in slip-critical and pretensioned joints shall be tightened to achieve the minimum bolt tension as specified in the RCSC's "Specification for Structural Joints using High-Strength Bolts" when all the fasteners of a joint are tight.
 - b. Any of the four methods to tighten bolts specified in the RCSC Specification may be used to achieve the minimum bolt tension. The tightening procedure that uses direct tension indicator washers shall conform to the requirements of ASTM F 959. Conform to the requirements of ASTM F 1852 for a Twist-Off-Type Tension-Control bolt pretensioning.
- E. Washers: Washers under the bolt head and/or nut shall be used as required by the RCSC Specification.
- F. Bolt Lubrication: All bolts shall be well lubricated at time of installation. Dry, rusty bolts are not be allowed.
- G. Impact Wrenches: Properly sized and lubricated air impact wrenches with adequate air pressure shall be utilized for all bolt installation.
- H. New Bolts: All bolts shall be new and shall not be reused.

2.5 CONNECTIONS

- A. Conceptual connection details with the required member design forces are shown on the drawings for bidding purposes and are applicable to all connections not designed and completely detailed on the drawings. The conceptual details are provided only to indicate the connection type required and may not fully represent the complexity of the connection as required by the final connection design for the forces they must resist. Except as noted below, the Fabricator is responsible for engaging the services of a professional engineer to prepare a final connection design for submission that meets the requirements of the conceptual connection details and resists the indicated design forces. Refer to the drawings and specifications for complete requirements.
- B. Typical connection details are indicated on the drawings.
- C. Design and Detailing Procedure:
 1. Unless noted otherwise or specifically detailed on the drawings, end connections of beams, girders, and trusses shall be designed as flexible and the connection shall accommodate end rotations of the unrestrained beams. Restrained end connections, as indicated on the drawings, shall be designed for the combined effect of bending moment and shears induced by the rigidity of the connection. Forces to be used in the design are described below.
 2. The Fabricator's licensed professional engineer shall design and submit sealed calculations documenting the design and showing details of the assembled joint with the bolts and welds required for the conditions noted below:
 - a. For each connection not otherwise completely detailed on the drawings.
 - b. Where connections are encountered on the project that do not match those assumed in the AISC Manual.
 3. Where connections are of the type than can be selected or completed using information found in tables in the AISC "Steel Construction Manual", sealed calculations need not be submitted provided the project design conditions precisely match those assumed in the referenced publications. For conditions encountered on the project that do not conform to the AISC Manual, a complete design shall be prepared and submitted for Engineer's review.
 4. The Fabricator's licensed professional engineer shall seal all design calculations.
 5. The Engineer reserves the right to reject all shop drawings submitted without complete design calculations if required. Failure to adhere to the requirements of this section obligates the Contractor to take responsibility for any and all resulting delays in the detailing and fabrication of structural steel.
 6. The Fabricator, his detailer, and professional engineer shall coordinate all connection requirements with the Erector. The Fabricator is responsible to detail connections that contain the adjustability and all other requirements that allow the Erector to erect the structural steel in conformance to all specified tolerances. The Fabricator shall be responsible for providing adjustability in all connections between exterior-cladding systems, skylights, and other architectural features and the supporting structural steel as required in achieving the specified tolerances for the architectural feature as specified in the contract documents or per the manufacturer's requirements.
- D. Design Intent: It is the intention of the plans and specifications that shop connections be welded or bolted and that field connections be bolted, unless detailed otherwise on the drawings.
- E. Preliminary Connection Review: Proposed variations from the details shown on the drawings will be considered and such variations must obtain preliminary approval from the Engineer prior to preparation of detailed shop drawings.
- F. Flexible (Simple) Beam Connections:

1. All typical beam simple connections shall conform to requirements of the AISC specifications. Refer to the drawings for typical connection types.
 - G. Restrained (Moment) Connections:
 1. Refer to the drawings for moment connection details.
 2. Design Reactions for Moment Connected Beams: Shear connections for moment-connected beams shall be designed for the factored reactions shown on the drawings.
 3. Design and Furnishing of Reinforcement in Moment Connected Joints: As part of the design responsibility outlined above, the fabricator shall design and furnish all additional reinforcement in moment connected joints to resist the specified design forces unless otherwise specifically detailed on the drawings. Column sections shall be investigated for web shear, web yielding, web buckling, and tension. Stiffeners and/or doubler plates shall be furnished as required by the AISC Specification Section J10.
 - H. Tightening of Bolts in Welded Moment Connections: At moment connections where beams are complete-joint penetration welded directly to columns or girders in the field, welds shall be made after installation of erection bolts to draw the pieces together and before the final shear connection bolts are tightened. Where loose moment plates are used, such plates shall be groove welded to columns prior to connecting these plates to the beams.
 - I. Column Splices:
 1. Compression Splice: Unless indicated otherwise on the drawings, all column splices shall be either a bolted compression splice using high strength snug-tightened bolts or a welded compression splice. Splice and filler plate sizes, thicknesses, and number of fasteners or weld information shall be as shown in Table 14-3 of AISC "Steel Construction Manual". It shall be the Fabricator's responsibility to examine the architectural drawings to verify that splice plates and fasteners do not violate architectural finish requirements.
 2. Bearing and Fit-Up of Column Compression Joints: All column splices, except those that are direct welded with complete-joint penetration welds, shall be considered as a compression joint as defined herein unless noted otherwise on the drawings.
 - J. Base Plates and Bearing Plates:
 1. Finish: All baseplates and bearing plates shall be finished in accordance with AISC Specification M2.8.
 2. A complete-joint penetration weld that joins a column that is a part of the Seismic Load Resisting System to a base is a Demand Critical Weld.
 3. Anchor Rod Holes in Baseplates: Hole sizes in baseplates for anchor rods shall be per the AISC "Steel Construction Manual", Table 14-2.
 - K. Hangers and Braces:
 1. Connections for all hangers and braces shall have connections designed to develop the factored axial force shown on the drawings. Contact the Engineer if no force for a member is shown on the drawings.
 2. Compression members composed of two or more rolled shapes separated from one another by intermittent fillers shall be connected to one another at such fillers at intervals (not to exceed 48") so that the slenderness ratio l/r of either shape, between the fasteners, does not exceed 75% of the governing slenderness ratio of the built-up member. The least radius of gyration, r , shall be used in computing the slenderness ratio of each component part.
 - L. Stiffeners: Provide stiffeners finished to bear under load concentrations where shown on the drawings.
 - M. Steel Shelf Angles: Shelf angles supporting veneer shown on the drawings to be continuous shall be furnished to a maximum length of 20'-0". Provide a 1/4" gap at each joint. The gap shall not be welded. The distance from the joint to the first supporting bolt shall not exceed 40% of the bolt spacing (12" maximum). Shelf angles shall be continuous around corners with corner joint complete-joint penetration welded. The distance to the first supporting bolt from the corner shall not exceed 12".
 - N. Limitations on Use of A307 Bolts: ASTM A 307 bolts shall not be used in any permanent steel-to-steel or concrete-to-steel connection.
 - O. Bolts in Combination with Welds: Bolts shall not be considered as sharing the load in combination with welds, except as allowed in AISC Specification Section J1.8.
- 2.6 SURFACE PREPARATION AND SHOP PRIME PAINTING
- A. Specification: Surface preparation, paint, and painting practices shall conform to the "SSPC Painting Manual", Volumes 1 and 2.
 - B. Scope: All steel shall receive a primer coat unless shown to be fire-proofed or as noted below:
 1. Shop paint surfaces that are to remain exposed to view in the final construction.
 2. Shop paint any steel other than weathering steel that, in the final construction, will not be in a controlled environment and is therefore subject to moisture or high humidity infiltration and that has not been specified to be galvanized.

3. Shop paint any steel that is shown on the drawings to receive a finished paint system as defined in Specification 099100.
 4. Coordinate all shop painting of structural steel with Architect's painting requirements as specified on the architectural drawings and in the specifications. The Fabricator shall be responsible for determining all painting requirements (which surfaces are to be painted or left unpainted) on the project prior to fabrication.
- C. Additional Painting Requirements:
1. Extend shop paint to 2" from location of welds on surfaces that are to be field welded.
 2. All unpainted mating surfaces of all elements that are welded together into an assembly that is permanently exposed to the exterior shall be seal welded in addition to structural welding requirements.
 3. If individual elements (including the mating surfaces) of an assembly that is required to be painted are painted prior to welding into an assembly, then all painted surfaces affected by welding shall be touched-up and repaired (according to manufacturer's instructions, if any) to prevent corrosion bleeding.
 4. The fabricator shall be responsible to ensure that all elements of all assemblies that are to be painted are fabricated so that no exposed surface shall be subject to stains due to corrosion bleeding during the warranty period of the paint.
 5. Structural steel elements that are bolted with slip-critical joints and are required on the drawings to be painted shall have all faying surfaces (including all surfaces of filler plates, member end supplement plates, and welds) painted to comply with the specified slip-critical coating requirement.
- D. Surface Preparation – Unpainted Steel: All structural steel that is not specified to receive a shop coat of primer paint shall be prepared in accordance with Society for Protective Coatings specifications as follows:
1. SSPC-SP 2, "Hand Tool Cleaning" or SSPC-SP 3, "Power Tool Cleaning" unless otherwise specified.
 2. SSPC-SP 6, "Commercial Blast Cleaning" shall be applied to the faying surfaces of connections that are noted on the drawings as slip-critical connections requiring a Class B surface. Apply this surface preparation to the area surrounding all bolt holes including the area up to 2" outside the outer-most holes.
- E. Surface Preparation and Primer Paint – Shop Painted Steel:
1. Surface Preparation: Prepare the surface of all structural steel specified to be shop painted as required by the paint manufacturer or the Society for Protective Coatings specifications, but not less than the following:
 - a. SSPC-SP 2, "Hand Tool Cleaning" or SSPC-SP 3, "Power Tool Cleaning" unless otherwise specified.
 - b. SSPC-SP 6, "Commercial Blast Cleaning" shall be applied to the faying surfaces (including filler and member-end supplement plates, if any) of connections that are noted on the drawings as requiring a slip-critical coating. At a minimum, apply this surface preparation to the area between and surrounding all bolt holes including the area up to 2" outside the outer-most holes.
 2. Priming: Immediately after surface preparation, apply primer to all structural steel specified to be shop primed in strict accordance with manufacturer's instructions and the Society for Protective Coatings specifications. Apply paint at a rate to conform to the manufacturer's written instructions and to provide a dry film thickness of not less than 1.5 mils. Use priming methods that result in full coverage of joints, corners, edges, welds, and all exposed surfaces. Apply two coats to surfaces that are inaccessible after assembly or erection. Change the color of the second coat to distinguish it from the first coat.
 3. Finish Coat: Coordinate shop primer paint requirements with architectural drawings and specifications. The primer selected must be compatible with any specified finish coat.
- F. Shop Touch-Up Painting: The Fabricator shall provide for cleaning and touch-up painting of welds, bolted connections (including nuts, bolts, washers, filler plates, member end supplement plates and welds, if any), and abraded areas. Prior to shipment, apply paint to exposed areas using same materials and surface preparation as used for shop painting. Paint shall be applied by brush or spray with minimum dry film thickness of 1.5 mils.
- 2.7 SOURCE QUALITY CONTROL
- A. Source Testing and Inspection: Refer to Specification 014529 "Structural Testing and Inspections" for testing and inspection requirements associated with structural steel.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Inspection Prior to Erection: Erector shall examine areas and conditions under which structural steel work is to be installed and notify the Contractor and the Architect/Engineer in writing of conditions detrimental to proper and timely completion of the work.

3.2 PREPARATION

- A. Temporary Shoring and Bracing:
 - 1. The lateral-load resisting or stability-providing system and connecting diaphragms are identified on the drawings. Comply with the provisions of the Code of Standard Practice regarding stability of the structure during the erection process, except where stricter requirements are noted herein.
 - 2. Where architectural or MEP requirements do not allow for any temporary supports, members, erection devices, or connections to be left in place permanently or where such items affect the final structural behavior, they shall be removed by the Erector. All costs associated therewith shall be included in the bid price.

3.3 ERECTION

- A. The erection work shall comply with the requirements of AISC Specification Section M4.
- B. Surveys: The following surveys shall be performed by a qualified land surveyor:
 - 1. Initial Survey: Check elevations of concrete and masonry bearing surfaces, anchor bolt locations, embedded connection plates, and all dimensions of existing structures to which new connections are to be made prior to erection and submit any discrepancies to the Engineer prior to the start of erection. Corrections or compensating adjustments to the structural steel shall be made and approved prior to the start of erection.
 - 2. Final Survey: Upon completion of erection of the steel frame, and before the start of work by other trades that may be supported, attached, or applied to the frame, a final survey shall be made and a report submitted certifying compliance with specified tolerances.
- C. Erection Tolerances: Erection tolerances of anchor rods, embedded items, and all structural steel shall conform to the AISC Code of Standard Practice, Section 7, unless stricter tolerances are specified elsewhere in the contract documents.
- D. Wherever the erection equipment is supported by the structure, the contractor shall be responsible for the retention of a licensed professional engineer to determine the adequacy of the member supporting the erection equipment in relation to the loads imposed thereon. The Contractor shall submit to the Architect/Engineer, for review, the loads that will be imposed by the erection equipment on the building structure. Where the imposed load exceeds the allowable strength, the Contractor shall be responsible for any additional materials, supports, bracing, connections and similar measures required to support the imposed load of the equipment while in use, subject to review by the Architect/Engineer.
- E. Anchor Rods: Tighten anchor rods after supported members have been positioned and plumbed. Do not remove wedges or shims, but if protruding, cut off flush with edge of base or bearing plate prior to packing with grout. Use only steel wedges or shims.
- F. Base Plates and Bearing Plates: Remove loose latent material from bearing surfaces and base and bearing plates. Set plates to the elevation indicated on the drawings and level using steel shims (plastic shims will not be allowed) or by three leveling screws with weldments at the plate edges. After all protruding plates have been trimmed, grout plates solidly between bearing surfaces using the specified grout, ensuring no voids are present. Finish exposed surfaces, protect installed materials, and allow to wet cure. For proprietary grout materials, comply with manufacturer's instructions. Tighten anchor bolts after supported members have been positioned and plumbed.
- G. Splices: Splices will be permitted only where indicated on the contract drawings and approved shop drawings. Fastenings of splices of compression members shall be done after the abutting surfaces have been brought completely into contact within AISC tolerances. Bearing surfaces and surfaces that will be in permanent contact are to be cleaned before the members are assembled.
- H. Cellular and/or Castellated Beams: The handling and erection of cellular and/or castellated beams shall be in accordance with the manufacturer's recommendations and the applicable rules of OSHA.
- I. Field Assembly of Structural Steel:
 - 1. As erection of the steel progresses, the work shall be fastened securely to safely carry all dead load, wind, and erection forces. Particular care shall be exercised to ensure straightness and tautness of bracing immediately upon raising a steel column.
 - 2. Provide temporary planking and working platforms as necessary to effectively complete work.
 - 3. Set structural frames accurately to lines and elevations indicated. Align and adjust various members forming part of complete frame or structure before permanently fastening. Clean bearing surfaces and other surfaces which will be in permanent contact before assembly. Perform necessary

adjustments to compensate for discrepancies in elevations and alignment. Level and plumb individual members of structure within specified AISC tolerances. The Contractor shall coordinate with Erector and Fabricator regarding possible discrepancies in member lengths between temperature at time of fabrication and temperatures during erection, and shall make necessary adjustments to ensure plumbness within AISC tolerances at 70°F. Compensate for cumulative welding draw, construction loadings, sequential applications of dead loads, or any other predictable conditions that could cause distortions to exceed tolerance limitations.

4. On welded construction exposed to view or weather, remove erection bolts, fill holes with plug welds or filler and grind smooth at exposed surfaces.
 5. Comply with AISC Specifications for bearing, adequacy of temporary connections, alignment, and removal of paint on surfaces receiving field welds.
 6. Comply with all bolting and welding requirements of Part 2 of this specification.
- J. Field Modifications to Structural Steel: Errors in shop fabrication or deformation resulting from handling and transportation that prevent the proper assembly and structural fitting of parts shall be reported immediately to the Architect/Engineer, and approval of the method of correction shall be obtained. Approved corrections shall be made at no additional cost to the Owner. Do not use cutting torches, reamers, or other devices in the field for unauthorized correction of fabrication errors.
- K. Miscellaneous Framing: Provide supplemental structural steel support framing for steel deck where columns, or other framing members or floor openings interrupt normal deck bearing whether shown or not on the architectural, mechanical, or structural drawings.
- L. Removal of Erection Aids and Devices: The Erector shall remove all erection aids and devices that interfere with architectural finish or MEP requirements.
- M. Field Touch-Up Painting:
1. Clean field welds, unpainted areas of bolted connections (including all exposed areas of nuts, bolts, washers, filler plates, member end supplement plates, and welds), and any shop painted areas that are abraded. Apply paint to all exposed areas using same material and surface preparation as used for shop painting. Apply by brush or spray to provide minimum dry film thickness of 1.5 mils.
 2. Clean field welds, ungalvanized areas of bolted connections (including all exposed areas of nuts, bolts, washers, filler plates, member end supplement plates, and welds), and any galvanized areas that are abraded. Prepare surfaces and apply two coats of the specified galvanizing repair paint in accordance with ASTM A 780.
 3. The Contractor shall ensure that, at the substantial completion of the project, all structural steel, bolted and/or welded, required to be painted shall have all necessary steel surfaces painted (including touch-up painting as required) to prevent corrosion bleeding.
- N. Shear Connector Installation:
1. Composite Beams:
 - a. Studs shall be welded in the field (not the shop) using automatically timed stud welding equipment.
 - b. The top flange of the beams must be unpainted and free of heavy rust, mill scale, dirt, sand or other foreign material which will interfere with the welding operation.
 - c. The steel deck must be free of dirt, sand, oil, or other foreign material and must be dry and free of moisture. Steel deck must rest tightly on the beam flange. Welding must take place through only one thickness of deck.
 - d. Stud Spacing: Studs shall be spaced on beams and girders as shown on the drawings.
- O. Headed Stud Anchor Installation:
1. Steel Plates Embedded in Concrete:
 - a. Studs shall be welded using automatically timed stud welding equipment.
 - b. Plates must be unpainted and free of heavy rust, mill scale, dirt, sand or other foreign material that will interfere with the welding operation.
- P. Clean Up: Clean up all debris caused by the Work of this Section, keeping the premises neat and clean at all times.
- 3.4 FIELD QUALITY CONTROL
- A. Field Testing and Inspection: Refer to Specification 014529 "Structural Testing and Inspections" for testing and inspection requirements associated with structural steel.

END OF SECTION

SECTION 05 50 00

METAL FABRICATIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Steel framing and supports for ceiling-hung toilet compartments.
 - 2. Steel framing and supports for operable partitions.
 - 3. Steel framing and supports for overhead doors and grilles.
 - 4. Steel framing and supports for countertops.
 - 5. Steel framing and supports for mechanical and electrical equipment.
 - 6. Steel framing and supports for applications where framing and supports are not specified in other Sections.
 - 7. Steel framing and supports (outriggers) for window-washing equipment.
 - 8. Elevator machine beams, hoist beams, and divider beams.
 - 9. Steel shapes for supporting elevator door sills.
 - 10. Shelf angles.
 - 11. Metal ladders.
- B. Products furnished, but not installed, under this Section:
 - 1. Loose steel lintels.
 - 2. Anchor bolts, steel pipe sleeves, slotted-channel inserts, and wedge-type inserts indicated to be cast into concrete or built into unit masonry.

1.3 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Design ladders alternating tread devices, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

1.4 SUBMITTALS

- A. Product Data: For the following:
 - 1. Nonslip aggregates and nonslip-aggregate surface finishes.
 - 2. Paint products.
 - 3. Grout.
- B. Sustainable Submittals:
 - 1. Product Data indicating percentages by weight of postconsumer and preconsumer recycled content for products having recycled content. Include statement indicating costs for each product having recycled content.
- C. Shop Drawings: Show fabrication and installation details for metal fabrications.
 - 1. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items.

1.5 QUALITY ASSURANCE

- A. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- B. Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
 - 2. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.6 PROJECT CONDITIONS

- A. Field Measurements: Verify actual locations of walls and other construction contiguous with metal fabrications by field measurements before fabrication.

1.7 COORDINATION

- A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written recommendations to ensure that shop primers and topcoats are compatible with one another.
- B. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

PART 2 - PRODUCTS

2.1 METALS, GENERAL

- A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.

2.2 FERROUS METALS

- A. Recycled Content of Steel Products: Provide products with average recycled content of steel products so postconsumer recycled content plus one-half of preconsumer recycled content is not less than 25 percent.
- B. Steel Plates, Shapes, and Bars: ASTM A 36.
- C. Steel Pipe: ASTM A 53, standard weight (Schedule 40) unless otherwise indicated.
- D. Slotted Channel Framing: Cold-formed metal box channels (struts) complying with MFMA-4.
 - 1. Size of Channels: 1-5/8 by 1-5/8 inches.
 - 2. Material: Galvanized steel, ASTM A 653, commercial steel, Type B, with G90 Z275 coating; 0.079-inch nominal thickness.

2.3 NONFERROUS METALS

- A. Aluminum Castings: ASTM B 26, Alloy 443.0-F.

2.4 FASTENERS

- A. General: Unless otherwise indicated, provide Type 304 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633 or ASTM F 1941, Class Fe/Zn 5, at exterior walls. Select fasteners for type, grade, and class required.
 - 1. Provide stainless-steel fasteners for fastening aluminum.
- B. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A 307, Grade AASTM F 568M, Property Class 4.6; with hex nuts, ASTM A 563; and, where indicated, flat washers.
- C. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A 325, Type 3; with hex nuts, ASTM A 563, Grade C3ASTM A 563M, Class 8S3; and, where indicated, flat washers.

- D. Stainless-Steel Bolts and Nuts: Regular hexagon-head annealed stainless-steel bolts, ASTM F 593; with hex nuts, ASTM F 594/ASTM F 836M; and, where indicated, flat washers; Alloy Group 1A1.
- E. Anchor Bolts: ASTM F 1554, Grade 36, of dimensions indicated; with nuts, ASTM A 563; and, where indicated, flat washers.
 - 1. Hot-dip galvanize or provide mechanically deposited, zinc coating where item being fastened is indicated to be galvanized.
 - a. Eyebolts: ASTM A 489.
 - b. Machine Screws: ASME B18.6.
 - c. Lag Screws: ASME B18.2.
 - d. Wood Screws: Flat head, ASME B18.6.1.
 - e. Plain Washers: Round, ASME B18.22..
 - f. Lock Washers: Helical, spring type, ASME B18.21.
 - g. Anchors, General: Anchors capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency.
 - h. Cast-in-Place Anchors in Concrete: Either threaded type or wedge type unless otherwise indicated; galvanized ferrous castings, either ASTM A 47 malleable iron or ASTM A 27 cast steel. Provide bolts, washers, and shims as needed, all hot-dip galvanized per ASTM F 2329.
 - i. Post-Installed Anchors: chemical anchors.
 - 1) Material for Exterior Locations and Where Stainless Steel is Indicated: Alloy Group 1A1 stainless-steel bolts, ASTM F 593, and nuts, ASTM F 594.

2.5 MISCELLANEOUS MATERIALS

- A. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.
- B. Universal Shop Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with MPI#79 and compatible with topcoat.
- C. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.
- D. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187.
- E. Nonshrink, Metallic Grout: Factory-packaged, ferrous-aggregate grout complying with ASTM C 1107, specifically recommended by manufacturer for heavy-duty loading applications.
- F. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by manufacturer for interior and exterior applications.

2.6 FABRICATION, GENERAL

- A. Shop Assembly: Preassemble items in the shop to greatest extent possible. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.
- B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- C. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.
- D. Form exposed work with accurate angles and surfaces and straight edges.
- E. Weld corners and seams continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 2. Obtain fusion without undercut or overlap.
 3. Remove welding flux immediately.
 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- F. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) fasteners unless otherwise indicated. Locate joints where least conspicuous.
- G. Fabricate seams and other connections that will be exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.
- a. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.
 - b. Provide for anchorage of type indicated; coordinate with supporting structure. Space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.
 - 1) Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors, 1/8 by 1-1/2 inches, with a minimum 6-inch embedment and 2-inch hook, not less than 8 inches from ends and corners of units and 24 inches o.c., unless otherwise indicated.

2.7 MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.
- B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.
1. Fabricate units from slotted channel framing where indicated.
- C. Fabricate supports for operable partitions from continuous steel beams of sizes indicated with attached bearing plates, anchors, and braces as indicated. Drill or punch bottom flanges of beams to receive partition track hanger rods; locate holes where indicated on operable partition Shop Drawings.

2.8 SHELF ANGLES

- A. Fabricate shelf angles from steel angles of sizes indicated and for attachment to concrete framing. Provide horizontally slotted holes to receive 3/4-inch bolts, spaced not more than 6 inches from ends and 24 inches o.c., unless otherwise indicated.
1. Provide mitered and welded units at corners.
 2. Provide open joints in shelf angles at expansion and control joints. Make open joint approximately 2 inches larger than expansion or control joint.
- B. For cavity walls, provide vertical channel brackets to support angles from backup masonry and concrete.
- C. Galvanize shelf angles located in exterior walls.
- D. Furnish wedge-type concrete inserts, complete with fasteners, to attach shelf angles to cast-in-place concrete.

2.9 ABRASIVE METAL

- A. Cast-Metal Units: Cast aluminum, with an integral-abrasive, as-cast finish consisting of aluminum oxide, silicon carbide, or a combination of both. Fabricate units in lengths necessary to accurately fit openings or conditions.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. American Safety Tread Co., Inc.
 - b. Balco Inc.
 - c. Wooster Products Inc.
 2. Nosings: Cross-hatched units, 4 inches wide with [1/4-inch] [1-inch] lip, for casting into concrete steps.
 3. Treads: Cross-hatched units, full depth of tread with 3/4-by-3/4-inch nosing, for application over bent plate treads or existing stairs.
- B. Provide anchors for embedding units in concrete, either integral or applied to units, as standard with manufacturer.
- C. Drill for mechanical anchors and countersink. Locate holes not more than 4 inches from ends and not more than 12 inches, evenly spaced between ends, unless otherwise indicated. Provide closer spacing if recommended by manufacturer.
1. Provide two rows of holes for units more than 5 inches wide, with two holes aligned at ends and intermediate holes staggered.
- D. Apply bituminous paint to concealed surfaces of cast-metal units.

2.10 LOOSE BEARING AND LEVELING PLATES

- A. Provide loose bearing and leveling plates for steel items bearing on masonry or concrete construction. Drill plates to receive anchor bolts and for grouting.
- B. Galvanize plates.
- C. Prime plates with zinc-rich primer.

2.11 FINISHES, GENERAL

- A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.
- B. Finish metal fabrications after assembly.
- C. Finish exposed surfaces to remove tool and die marks and stretch lines, and to blend into surrounding surface.

2.12 STEEL AND IRON FINISHES

- A. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A 153 for steel and iron hardware and with ASTM A 123 for other steel and iron products.
 1. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.
- B. Shop prime iron and steel items not indicated to be galvanized unless they are to be embedded in concrete, sprayed-on fireproofing, or masonry, or unless otherwise indicated.
 1. Shop prime with universal shop primer unless zinc-rich primer is indicated.

2.13 ALUMINUM FINISHES

- A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.
- B. Provide aluminum finish in class 1 clear anodized.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges

and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.

- B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.
- C. Field Welding: Comply with the following requirements:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.
- E. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.
- F. Corrosion Protection: Coat concealed surfaces of aluminum that will come into contact with grout, concrete, masonry, wood, or dissimilar metals with the following:
 - 1. Cast Aluminum: Heavy coat of bituminous paint.
 - 2. Extruded Aluminum: Two coats of clear lacquer.

3.2 INSTALLING MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings.
- B. Anchor supports for operable partitions securely to and rigidly brace from building structure.

3.3 INSTALLING NOSINGS, TREADS, AND THRESHOLDS

- A. Center nosings on tread widths unless otherwise indicated.
- B. For nosings embedded in concrete steps or curbs, align nosings flush with riser faces and level with tread surfaces.

3.4 INSTALLING BEARING AND LEVELING PLATES

- A. Clean concrete and masonry bearing surfaces of bond-reducing materials, and roughen to improve bond to surfaces. Clean bottom surface of plates.
- B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with grout.
 - 1. Use non-shrink grout, either metallic or non-metallic, in concealed locations where not exposed to moisture; use non-shrink, non-metallic grout in exposed locations unless otherwise indicated.
 - 2. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.5 ADJUSTING AND CLEANING

- A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

1. Apply by brush or spray to provide a minimum 2.0-mil dry film thickness.
- B. Touchup Painting: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint are specified in Division 09 painting Sections.
- C. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

END OF SECTION

SECTION 06 05 73

WOOD TREATMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Sections Includes:

1. Preservative pressure treatment.
2. Fire-retardant pressure treatment.
3. Applied preservative treatment.

1.2 RELATED SECTIONS:

1. Section 06 10 00 – Rough Carpentry.
2. Section 06 11 00 – Wood Framing.
3. Section 06 16 00 – Sheathing, Subflooring, and Underlayment.
4. Section 06 20 23 – Interior Finish Carpentry.
5. Section 06 40 00 – Architectural Woodwork.

1.3 DEFINITIONS

A. Pressure Treatment: Pressure impregnation with preservative and fire-retardant chemicals.

1.4 SUBMITTALS

A. General: Submit in accordance with Section 01 33 00.

1. Submit Sections 06 10 00, 06 20 23, and 064116 submittals simultaneously with submittals for this section.

B. Product Data: Submit following:

1. Product data for:
 - a. Wood preservative treatment, including type of preservative used, net amount of preservative retained, and chemical treatment manufacturer's written instructions for handling, storing, installing, and finishing treated material.
 - b. Fire-retardant treatment, including physical properties of treated materials, both before and after exposure to elevated temperatures, ASTM D5516 and ASTM D5664.
 - c. Waterborne treatment, including statement identifying required moisture content of treated materials before shipment to Project site.

C. Informational Submittals: Submit following:

1. Test/Evaluation Reports showing Code compliance as specified in Quality Assurance Article.
2. Certifications specified in Quality Assurance Article.
3. Manufacturer's instructions.

1.5 QUALITY ASSURANCE

A. Grade Marks:

1. Pressure Treatment: Include quality mark of grading agency which maintains continued supervision, testing, inspection, and re-examination service over product quality as described in AWP standards.
2. Fire-Retardant Treated Wood: Imprint each piece with mark attesting to FR-S rating.

B. Manufacturer Qualifications: Company specializing in manufacturing Products specified in this Section with minimum five years' experience.

- C. Applicator Qualifications: Acceptable to manufacturer with experience on at least five projects of similar nature in past five years.
- D. Regulatory Requirements: Submit research/evaluation reports showing Code compliance for following:
 - 1. Preservative-treated wood products.
 - 2. Fire-retardant-treated wood products.
- E. Certifications: Manufacturer's certification that products furnished for Project meet or exceed specified requirements.
 - 1. Certify that fire-retardant treated plywood used as roof sheathing or decking will not result in the following:
 - a. Delamination of layers
 - b. Corrosion of fasteners
 - c. Loss of loading or spanning capabilities
 - d. Loss of screw holding capacity.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. General: Comply with Section 01 60 00.
- B. Stack lumber, plywood, and other panels; place spacers between each bundle to provide air circulation. Provide for air circulation around stacks and under coverings.
 - 1. Fire-Retardant Treated Wood: Keep materials dry during delivery and storage.
 - a. Protect against exposure to weather.

PART 2 - PRODUCTS

2.1 WOOD TREATMENTS

- A. Kiln dry pressure-treated products after treatment to following maximum moisture contents:
 - 1. Lumber: 19-percent.
 - 2. Plywood: 15-percent.
- B. Dual treatment using separate wood preservative and fire-retardant pressure treatments: Not permitted.
- C. Wood Preservative Pressure Treatment:
 - 1. Solid Sawn Lumber, Decking, and Timber: AWPAC2.
 - a. Waterborne Treatments: AWPAC Standard P5. Do not use for glue laminated members after gluing.
 - 2. Structural Glued Laminated Timber: AITC 109.
 - 3. Plywood: AWPAC9.
 - 4. Use of chromated copper arsenate (CCA) not allowed.
 - 5. Provide preservative treatment of quaternary ammonia, copper, and ammonia or amine.
 - 6. Acceptable Above Grade Products and Manufacturers:
 - a. ACQ Preserve, Chicago Flameproof, Chicago, IL.
 - b. Wolmanized Outdoor Wood, Arch Wood Protection, Inc., Smyrna, GA.
 - c. Dura-Guard, Hoover Treated Wood Products, Inc., Thomson, GA.
 - d. Accepted Substitute in accordance with Section 01 25 00.
 - 7. For applications listed, provide following retention rates:
 - a. Above ground and applications with ground contact or fresh water contact: 0.40 pound/cubic foot.
 - 8. Do not incise surfaces of lumber at exposed to view locations.

9. Treat following items:
 - a. Surfaces in contact with concrete or ground.
 - b. Floor screeds.
 - c. Sills, plates, soles, and furring in contact with exterior concrete or masonry walls or in direct contact with soils or located within 8 inches of soils.
 - d. Blocking and sleepers in contact with exterior concrete or masonry walls or in direct contact with soils.
 - e. Wood used with roofing, flashing, and waterproofing such as
 - 1) Curbs.
 - 2) Nailers.
- D. Preservative Cut Surface Treatment Applied at Site: AWP4 M4.
 1. Water repellent containing 2-percent copper naphthenate solution.
 2. Compatible with preservative pressure treatment.
 3. Pigment: Colored.
 4. Acceptable Products and Manufacturers:
 - a. Number 10 Green Preservative, Cuprinol Group, Cleveland, OH.
 - b. Green EndCoat, Osmose Wood Preserving, Inc., Griffen, GA.
 - c. Accepted Substitute in accordance with Section 01 25 00.
- E. Wood Fire Retardant Pressure Treatment:
 1. AWP4 C20 for lumber.
 2. AWP4 C27 for plywood.
 3. Chemically treat and pressure impregnate wood products.
 4. Capable of providing maximum flame spread/smoke development rating of 25/25 (FR-S Rating).
 5. Not required to have brush treatment of cuts made in field.
 6. Equilibrium moisture content of not more than 28 percent, ASTM D3201 procedures at 92 percent relative humidity.
 7. Not detrimental to structural properties of plywood when exposed to elevated temperatures and high humidity, ASTM D5516.
 8. Not capable of bleeding through or adversely affecting type of finish indicated.
 9. Acceptable Products and Manufacturers, Interior:
 - a. D-Blaze FRT, Chicago Flameproof, Chicago, IL.
 - b. Dricon, Arch Wood Protection, Inc., Smyrna, GA.
 - c. Pyroguard, Hoover Treated Wood Products, Thomson, GA.
 - d. Accepted Substitute in accordance with Section 01 25 00.
 10. Acceptable Products and Manufacturers, Exterior:
 - a. Exterior Fire-X, Hoover Treated Wood Products, Thomson, GA.
 - b. FRX Outdoor Wood, Arch Wood Protection, Inc., Smyrna, GA.
 - c. Exterior Fire Retardant Wood, Chicago Flameproof, Chicago, IL.
 - d. Accepted Substitute in accordance with Section 01 25 00.
 11. Treat following wood items with Interior Type A treatment:
 - a. Blocking above ceilings.
 - b. Blocking within return air plenums.
 - c. Blocking within walls.
 - d. Plywood panel boards.
 12. Treat following wood items with Exterior treatment:

- a. Roofing Items:
 - 1) Curbs.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. General: Examine conditions and proceed with work in accordance with Section 01 73 00.

3.2 APPLICATION

- A. Wood Treatment Applied to Cut Surfaces at Site: Comply with AWPA M4 and with Section 01 73 00.
 - 1. Apply preservative treatment in accordance with manufacturer's instructions to:
 - a. Preservative pressure treated wood site-sawn ends.
 - b. Holes cut through preservative pressure treated wood.
 - 2. Allow preservative to cure prior to erecting members.
- B. Wood Fire Retardant Pressure Treated Wood: Do not rip cut.
 - 1. Do not mill.
 - 2. Only end cuts and bored holes are permitted.

END OF SECTION

SECTION 06 10 00

ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

- A. Products Specified But Not Installed Under This Section:
 - 1. Section 03 10 00 - Concrete Forms and Accessories: Attachment of anchor plates to formwork.
 - 2. Section 03 30 00 - Cast-In-Place Concrete: Placement of anchor bolts.

1.2 DEFINITIONS

- A. Blocking: Wood used for plates, furring, shimming, stripping, sleepers, grounds, nailers, curbing, cants, bracing, and filling in between framing members.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.

1.4 QUALITY ASSURANCE

- A. Grade Marks: Identify lumber and plywood by official grade mark.
 - 1. Lumber: Include symbol of grading agency, mill name, grade, species, grading rules, and condition of seasoning at time of manufacturer.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Storage and Protection: Store products above ground, on platforms or skids, and covered with waterproof coverings.
 - a. Store products with ventilation, drainage, and protection against damp or wet locations.
 - b. Support products to prevent warping and distortion.
- B. Mold Prevention: Comply with Section 01 60 00.

PART 2 - PRODUCTS

2.1 SOLID SAWN LUMBER

- A. General: Provide from an FSC certified source.
- B. Lumber: PS-20
 - 1. Lumber Grading Agency: Certified by American Lumber Standards Committee or NLGA.
 - 2. Grading Rules: NLGA, SPIB, WCLIB, and WWPA.
 - 3. Dimensions: Lumber dimensions are nominal; actual dimensions conform to PS-20 and applicable rules writing agencies.
 - 4. Maximum Moisture Content at Time Of Dressing: 19 percent.
 - 5. Surfacing: Surface four sides (S4S), unless noted otherwise.
 - 6. Finger-jointed lumber not allowed.
 - 7. Size Classification: 2 inches to 4 inches thick, 2 inches to 4 inches wide.
- C. General Non-Structural Framing: Stud grade, any species.
- D. Wood Blocking: Utility grade, any species.
- E. Plywood Equipment Panel Boards: APA Rated Sheathing.
 - 1. Exposure 1.

- F. Grade: C-D.
- G. Thickness: 23/32 inches minimum.
- H. Backer Board (Cementitious): Cementitious composition with glass fiber reinforcement, ANSI A108.11 and A118.9, UL fire resistant rated.
 - 1. Product specifically manufactured as substrate material for application of ceramic tile
 - 2. Thickness: 5/8 inches.
 - 3. Acceptable Products and Manufacturers:
 - I. WonderBoard, BackerBoard, Custom Building Products, Seal Beach, CA.
 - J. PermaBase Cement Board, National Gypsum Company, Charlotte, NC.
 - K. Durock Cement Board, United States Gypsum Company, Chicago, IL.
 - L. Accepted Substitute in accordance with Section 01 25 00.
- M. Wall Protection:
 - 1. Exposure 1.
 - 2. Grade: A-D.
 - 3. Thickness: 5/8 inches minimum.
 - 4. Chamfer all edges.
 - 5. Fire retardant treat to meet code.

2.2 ACCESSORIES

- A. Adhesive: APA AFG-01, waterproof, air cured type, cartridge dispensed.

2.3 FASTENERS

- A. Provide fasteners in sizes, spacings, and locations to suit applications.
- B. At preservative treated wood, provide corrosion resistant fasteners:
 - 1. Provide stainless steel Type 304 or 316.
 - 2. At typical exterior conditions: ASTM A153/A153M for fasteners and ASTM A653/A653M for connectors; minimum 1.85 ounces of zinc coating per square foot of surface area.
 - 3. Ensure that fasteners and connectors used together are of the same type.
 - 4. Carbon steel and aluminum products: Not permitted.
- C. Anchors:
 - 1. Toggle bolt type for anchorage to hollow masonry.
 - 2. Expansion shield and lag bolt type for anchorage to solid masonry or concrete.
 - 3. Bolts or ballistic fasteners for anchorage to steel.
- D. Bolts: ASTM A307 with nuts and washers.
- E. Anchor Bolts: ASTM A307 with nuts and washers.
- F. Lag Screws and Lag Bolts: ANSI B18.2.1 with washers.
- G. Wood Screws: ANSI B18.6.1.
- H. Nails, Staples, and Spikes: FS FF-N-105.
- I. Gypsum Board Screws: ASTM C1002
 - 1. Length: Three times sheathing thickness.
 - 2. Finish: Manufacturer's standard rust-inhibitive paint.

2.4 FRAMING ACCESSORIES

- A. Adhesives & Sealants: Only use adhesives and sealants in the interior of the building that meet or do not exceed the VOC limits of the CURRENT requirements of South Coast Air Quality Management District (SCAQMD) Rule No. 1168 on the interior of the building.
 - 1. Current requirement refers to the date on which the materials are installed in the building.
 - 2. A copy of SCAQMD Rule #1168 is referenced in Section 01 81 13 that was current as of the date of this specification. Refer to <http://www.aqmd.gov/rules> for the actual current version of the rule that will be applicable at the date of installation during construction.
 - 3. Interior refers to all building construction that is inside of the exterior weatherproofing material.

2.5 WOOD TREATMENTS

- A. Kiln dry pressure-treated products after treatment to following maximum moisture contents:
 - 1. Lumber: 19-percent.
- B. Preservative Treated: Comply with Section 06 05 73.
- C. Fire Retardant Treated: Comply with Section 06 05 73.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 - 1. Verify end supports are ready to receive framing.
 - 2. Before installation, check members for damage, and proper dimensions.

3.2 PREPARATION

- A. Wood Treatment Applied to Cut Surfaces at Site: Treatment. Comply with Section 06 05 73 – Wood

3.3 CONSTRUCTION

- A. General: Construct plumb, level, true to line, square, and free from warp or twist while maintaining dimensional tolerances and alignment with surrounding construction.
 - 1. Comply with AFPA NDS for attaching to substrates by anchoring and fastening to draw and securely hold members rigidly in place.
 - 2. Install fasteners at spacings required by more stringent requirements of AFPA NDS or applicable building code.
 - 3. Place horizontal and sloped members with crown edge up.
 - 4. Place vertical members with crown edge facing in same direction.
 - 5. Discard Material:
 - a. With defects which might impair quality of work.
 - b. Which are too small to fabricate work with minimum joints or optimum joint arrangement.
 - 6. Scribe, cope, and construct members accurately cut and fitted.
 - 7. Make tight connections between members to develop full member strength.
 - 8. Locate members as indicated. Do not change size, spacing, or spans without Architect's specific approval. Take care to place species and grades of members where indicated.
 - 9. Do not splice framing members between support points.
 - 10. Cut, notch, or bore members for passage of pipes and conduits in accordance with AFPA WCD. Reinforce members by use of formed sheet metal accessories.
 - 11. Fasteners: Use washers under bolt heads and nuts.

- a. Select fasteners of size that will not penetrate members where opposite side will be exposed to view or will receive finish materials.
 - b. Install fasteners without splitting wood; predrill as necessary.
- 12. Shimming:
 - a. Concrete and masonry bearing: Use steel or slate shims.
 - b. Metal and wood bearing: Do not use shims.
- 13. Wood Fire Retardant Pressure Treatment: Do not rip cut.
 - a. Do not mill.
 - b. Only end cuts and bored holes are permitted.
- B. Wood Blocking: Construct using maximum practical lengths.
 - 1. Provide and locate continuous blocking to facilitate installation and attachment of wall mounted equipment, cabinets, millwork, finishing materials, fixtures, specialty items, and trim.
 - 2. Cut and form to shapes for true line and level of work to be attached.
 - 3. Coordinate location with other work involved.
 - 4. Secure to masonry with metal plugs, toggle bolts, or expansion bolts set in masonry.
 - 5. Attach to substrates to support applied loading.
 - 6. Countersink bolts and nuts flush with wood surfaces.
 - 7. Where possible, anchor grounds to concrete forms before concrete placement.
 - 8. Provide permanent grounds of dressed, preservative pressure-treated, key beveled lumber not less than 1-1/2 inches wide, and of thickness required to bring face of ground to exact thickness of finish material involved.
 - 9. Remove temporary grounds when no longer required.
 - 10. Size as necessary unless specific size is indicated.
- C. Plywood Equipment Panel Boards: Install for equipment where indicated.
 - 1. Oversize panel by 12 inches beyond equipment perimeter.
- D. Wood Nailers: Provide at roof openings, terminations, and other locations as indicated to match thickness of roof insulation.
- E. Wood Curbs: Provide at roof openings except where prefabricated curbs are provided. Form corners by lapping side members alternately.

3.4 TOLERANCES

- A. Lumber Framing Members: 1/4 inch maximum from true position.

3.5 PROTECTION

- A. Protect finished work in accordance with Section 01 73 00.
 - 1. Protect products from moisture absorption and subsequent warping or deterioration until subsequent construction can proceed.
- B. Mold Prevention: Comply with Sections 01 50 00 and 01 60 00.

END OF SECTION

SECTION 06 16 90

SHEATHING ON METAL FRAMING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Sheathing board.
 - 2. Sheathing barrier.
 - 3. Joint treatments.
- B. Related Sections:
 - 1. Section 01 83 00 – Facility Shell Performance Requirements.
 - 2. Section 05 41 00 – Structural Metal Stud Framing.
 - 3. Section 07 21 00 – Thermal Insulation.
 - 4. Section 07 27 26 – Fluid-Applied Membrane Air/Moisture Barrier.

1.2 SYSTEM DESCRIPTION

- A. Fire Resistance Ratings: Where assemblies with fire ratings are indicated, provide materials and installations which are identical to assemblies tested in accordance with ASTM E119 by testing laboratories acceptable to authorities having jurisdiction.
 - 1. Construct assemblies identical to those indicated by reference in GA 600, UL Fire Resistance Directory, listing of FM, ITS Directory of Listed Products (ETL)(Warnock Hersey), or other agencies acceptable to authorities having jurisdiction.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
 - 1. Submit for components of each wall assembly containing components specified in this section simultaneously.
- B. Product Data: Submit for:
 - 1. Sheathing.
 - 2. Sheathing barrier.
 - 3. Joint treatments.
 - 4. Fasteners.
- C. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Certifications specified in Quality Assurance article.
 - 2. Manufacturer's Instructions: Include applicable temperature and humidity ranges, special procedures, and perimeter conditions requiring special attention.

1.4 QUALITY ASSURANCE

- A. Single Source Responsibility: Except where specified otherwise, obtain gypsum board products, joint treatment, and accessories from single manufacturer or from manufacturers recommended by prime manufacturer of gypsum board products.
- B. Certifications:
 - 1. Air barrier manufacturer's certification that primer is specifically formulated for adhesion to glass fiber faced gypsum sheathing and joint treatment materials.

1.5 FIELD SAMPLES

- A. General: Comply with Section 01 40 00.
- B. Sample Installation: Sheathing field sample may be combined with field samples of other exterior skin products, such as masonry, EIFS, metal panel, or siding, when used as part of the total wall assembly including acceptable size of joints between sheathing panels and between sheathing and dissimilar materials.
 - 1. Show finishing of sheathing joints for watertight installation.
 - 2. Show stepped sample installation of primer, sheathing barrier, and seams and termination of sheathing barrier, including conditions at openings, such as doors and windows, and penetrations such as anchors.
 - 3. Include typical attachment clip for metal panel system, and penetration.
 - a. Field sample area showing sheathing barrier conditions: 100 SQ FT.
 - 4. Install as directed by Architect.
 - 5. Sheathing with applied sheathing barrier shall be left exposed to weather for not less than three weeks before review and acceptance or rejection by Architect. Failure of barrier to remain fully adhered to sheathing shall be grounds for rejection. Do not apply or install other exterior skin items over sheathing and barrier until review by Architect for adhesion of barrier to sheathing.
 - 6. Accepted Field Sample: May remain part of completed Work.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Store in dry ventilated space off ground.
 - 2. Protect materials from surface contamination, soiling, corrosion, construction traffic, and damage.
 - 3. Support on level platform and fully protect from weather and direct sunlight exposure.
 - 4. Store and support gypsum board in flat stacks to prevent sagging.
 - 5. Protect materials to keep them dry. Remove wet gypsum board from Project site except for gypsum board wetted for application to curved surfaces.
 - 6. Protect gypsum board panels to prevent damage to edges, ends, and surfaces.
 - 7. Do not bend or damage metal trim.
- B. Mold Prevention: Comply with Section 01 60 00.

1.7 PROJECT CONDITIONS

- A. Environmental Requirements: Comply with manufacturer's written requirements under which products can be installed.
 - 1. Maintain minimum 50 F surface and ambient temperature for 48 hours before and continuously until joint treatment and bonding adhesives are thoroughly dry.
 - 2. Do not apply when ambient temperature to exceed 95 F.
 - 3. Avoid drafts during dry, hot weather to prevent too rapid drying.

1.8 SEQUENCING

- A. Mold Prevention: Comply with Sections 015000 and 01 60 00.
 - 1. Schedule and sequence construction to protect paper faced products from moisture exposure.

1.9 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 78 00.
 - 1. Warrant installed product exposed to weather to be without failure, when installed in accordance with manufacturer's requirements, for period of not less than 6 months.

PART 2 - PRODUCTS

2.1 SHEATHING

- A. Water-Resistant Sheathing:
 - 1. Glass fiber faced gypsum sheathing, ASTM C1177 or fiber-reinforced gypsum panel sheathing, ASTM C1278. ASTM C79 not permitted.
 - 2. Thickness: 5/8 inch.
 - 3. Type X for fire-rated and typical assemblies.
 - 4. Flame Spread: ASTM E84: 5 maximum.
 - 5. Smoke Developed: ASTM E84: 0.
- B. Acceptable Products and Manufacturers:
 - 1. G-P Dens-Glass Gold Exterior Sheathing, Georgia-Pacific, Atlanta, GA.
 - 2. GlasRoc Sheathing, Certainteed, Tampa, FL.
 - 3. e2XP Extended Exposure Sheathing, National Gypsum, Charlotte, NC.
- C. Where used as substrate for roofing applications, refer to Section 07 52 00.

2.2 SHEATHING JOINT TREATMENT

- A. Joint Sealant: For use at Water-Resistant Sheathing. Silicone—General Purpose (Designation S-GP): ASTM C920, Type S, Grade NS:
 - 1. Class: 25. Joint movement range without cohesive/adhesive failure: Plus 50 percent to minus 50 percent of joint width.
 - 2. Uses: NT, M, G, A, O
 - 3. Low modulus, single component, neutral curing, non-staining, non-bleeding silicone sealant.
 - 4. Color: Selected by Architect from manufacturer's full color range.
 - 5. Acceptable Products:
 - a. 795, Dow Corning.
 - b. Silpruf, General Electric, Waterford, NY.
 - c. Rhodorsil 5C, Rhone-Poulenc, Monmouth Junction, NJ.
 - d. Accepted Substitute in accordance with Section 01 25 00.

2.3 SHEATHING BARRIERS

- A. Seamless Air Barriers: Fluid-applied, self-adhering membrane specified in Section 07 27 00.

2.4 FASTENERS

- A. Provide fasteners in sizes, spacings, and locations to suit applications.
 - 1. Gypsum Board Fasteners: Self-drilling, self-tapping, bugle head screws, ASTM C954.
 - a. Type S screws for 16 to 26 mil thick metal framing and furring.
 - b. Type S-12 screws for 33 to 112 mil thick metal framing and furring.
 - c. Type G screws for gypsum board to gypsum board.
 - d. Length: Three times sheathing thickness.
 - e. Finish: Manufacturer's standard rust-inhibitive coating.

2. Fasteners at Fiber Reinforced Gypsum Panel Sheathing: As required by sheathing manufacturer to suit warranty requirements.
 - a. Acceptable Products and Manufacturers:
 - 1) USG Exterior Screws, sized to suit application, USG, Chicago, IL.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 1. Verify blocking is in proper location.
 2. Before erection, check members for damage, proper dimensions, proper camber, and accuracy of anchorage connections.

3.2 PREPARATION

- A. Items Which Require Backer Plates or Blocking: Coordinate sizes and locations.
 1. Install additional studs for attachment of backer plates and blocking in required locations to receive surface mounted accessories as indicated or as required by accessory manufacturer.
 2. Elimination of backer plates and blocking not allowed.
 3. Direct attachment of items to studs not allowed.

3.3 SHEATHING

- A. Water-Resistant Sheathing: Erect horizontally, with edge butted tight and ends occurring over framing member. Space framing members at not more than 16 inches OC.
 1. Maximum allowable gap between panels: As identified by sheathing manufacturer.
 2. Secure to steel framing to each support in accordance with manufacturer's recommendations but with fasteners spaced at not more than 8 inches OC vertically.
 3. Sheathing Joint Treatment, Joint Sealant:
 - a. Apply continuous bead of silicone sealant at joints between panels.
 - b. Apply continuous bead of silicone sealant at board perimeter, terminations, around penetrations, and where in contact with dissimilar materials.
 - c. Trowel sealant flat to establish watertight condition. Limit sealant application to width of joint. Do not extend troweled sealant over face of sheathing.
- B. Mold Prevention: Comply with Sections 015000 and 01 60 00.

3.4 PROTECTION

- A. Mold Prevention: Comply with Sections 015000 and 01 60 00.
 1. Protect paper-faced products from moisture exposure.
- B. Provide protection of installed work at joints and walls where if water was to penetrate behind the wall surface could result in deterioration of the installed work.

END OF SECTION

SECTION 07 13 52

MODIFIED BITUMINOUS SHEET WATERPROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Modified bituminous sheet waterproofing membrane at vertical surfaces with subsurface drainage mat/protection board. Indicated on the drawings as WP-1.
2. The waterproofing per 07 13 52 and 071354 compromise the foundation waterproofing system. The systems specified shall be provided by a single-source manufacturer and shall be installed by a single- source installer
3. The foundation waterproofing systems shall be placed in concert with expansion assemblies where applicable and shall form a continuous, watertight seal with those expansion assemblies.
4. The system shall be approved by the manufacturer for use in an application that recognizes the contents of the Geotechnical Report May 19, 2017, UTSW OFPC 11 Tech 32 – Expansion Project. The bidder shall review in full with the manufacturer, all reports, specifications, drawings and all other aspects of the project before submitting a bid. The bidder shall include all materials and means and methods necessary to provide a system as recommended by the manufacturer in regards to anticipated site conditions and information per the Geotechnical report.
5. Provide waterstops in concert with requirements of 03 30 00 and as part of the manufacturer's system warranty.

1.2 REFERENCES

- A. Reference Standards: See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following for design, materials, fabrication, and installation of component parts:
1. NRCA Roofing and Waterproofing Manual, National Roofing Contractors Association - NRCA Waterproofing and Dampproofing Manual.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Submit for components of each wall assembly containing components specified in this section simultaneously and as may be required to provide a complete water proofing system where indicated on the drawings as WP-
1. This is a performance based specification. The Contractor shall submit a complete waterproofing system in accordance with the selected manufacturers recommendations and requirements consisting of compatible products applicable for each project condition. The Contractor shall provide shop drawings and details for project conditions outlined below that will produce a complete water proof finished product. The contractor shall submit correspondence from a technical representative of the manufacturer that states that the manufacturer has reviewed the Geotechnical report and that the submitted system will provide the performance as noted herein in concert with project specific geotechnical conditions and structural design.
- C. Shop Drawings: Submit project specific shop drawings that describe all typical and non-typical waterproofing conditions including but not limited to the following:
1. Details of surface preparation of substrate and crack preparation.
 2. Details at penetration, corners and surface undulations.
 3. Details at transition between different types of substrates.

4. Details at transition between different types of waterproofing membranes.
 5. Details at product terminations, laps and sealant applications at site specific conditions.
 6. Details at product terminations at expansion joints
 7. Details for proper application of protection board.
 8. Details for proper application of protection board at transition, uneven surfaces and terminations.
 9. Detail for waterproofing application at sheet metal "slip flashing"
 10. Detail for waterproofing at precast coping
- D. Product Data: Submit for materials and accessories.
1. Include sample of warranty customized for this project.
- E. Informational Submittals: Submit following packaged separately from other submittals:
1. Certifications specified in Quality Assurance article.
 2. Manufacturer's instructions; include applicable temperature ranges.
 3. Manufacturer's field reports.
- F. Sample warranty and list of required inspections in order to achieve requested warranty.
- G. Closeout Sample warranty and list of required inspections in order to achieve requested warranty.
1. Closeout Submittals: Submit specified warranty in accordance with Section 01 77 00.
 2. Provide letter from manufacturer, indicating substrates are acceptable, including any additives and surface curing agents.

1.4 QUALITY ASSURANCE

- A. Manufacturer:
1. Qualifications: Not less than five years experience in manufacturing of specified waterproofing.
 2. Primary Waterproofing Materials: From single manufacturer.
 - a. Manufacturer's Name: Appear on containers.
 - b. Secondary Materials: As required by manufacturer of primary materials.
- B. Applicator Qualifications: Acceptable to membrane manufacturer prior to execution of this Contract.
1. Company specializing in application of specified waterproofing with documented successful project experience of similar size and scope using specified product.
 2. Minimum ten years experience with submitted product.
- C. Owner reserves right to hire independent waterproofing consultant to review submittals, procedures, and installation.
- D. Certifications: Manufacturer's Certification on manufacturer's letterhead:
1. Submit manufacturer's approval of installer, dated within one year of date of submission.
 2. Certify materials ordered and supplied meet membrane manufacturer's published performance standards and requirements of this Specification.
 3. Certify materials ordered and supplied are compatible with each other, suited for locale and purpose intended and shipped in sufficient quantity to ensure proper timely installation.
 4. Certify waterproofing materials have express warranty of fitness for particular purposes of this Project.
 5. Certify submittal details meet manufacturer's warranty requirements for this specific Project.
 6. Certify manufacturer has reviewed Drawings, Specifications, submittals, and project conditions and will issue warranty upon successful completion of installation.

- E. Mock-up: Provide mock-up of both typical and non-typical conditions for all applications, including sheet metal “slip flashing” for review by Architect and manufacturer’s representative.

1.5 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 19.
- B. Agenda:
 - 1. Review Project Specifications and Drawings.
 - 2. Establish installation schedules and sequence.
 - 3. Coordinate work with in-place and subsequent construction.
 - 4. Review weather and working conditions.
 - 5. Review installation procedures, including:
 - a. Substrate requirements for Project acceptance (curing of concrete surface, form release agents, temperature).
 - b. Waterproofing installation.
 - c. Phasing and sequencing requirements.
 - d. Termination, flashing, expansion joint, and penetration requirements.
 - e. Review inspection, testing, and quality control procedures.
 - f. Review protection requirements for construction period beyond waterproofing installation.
 - g. Include concrete installer at meeting.
 - h. Waterproofing Manufacturer
 - i. Waterproofing Consultant
- C. Conduct tour of areas to be waterproofed and report on surface acceptance, possible problem areas, and recommended remedies, and coordination with surrounding construction.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Do not double stack membrane pallets.
 - 2. Keep primer, mastics and adhesives in dry area away from flames, sparks and excessive heat.
 - 3. Store material in dry area out of direct sunlight.
 - 4. Cover materials and allow for adequate ventilation.

1.7 PROJECT CONDITIONS

- A. Environmental Requirements: Do not apply products when surface or ambient temperature is below 40 F unless special low temperature products are used.
 - 1. Do not apply products in any instances where surface temperature is lower than minus 25 F.
 - 2. Do not apply to damp or frozen surfaces or during inclement weather.
- B. Do not work or walk on exposed waterproofing membrane. Install permanent drainage mat/ protection board immediately to protect membrane during subsequent work operations.

1.8 SEQUENCING

- A. Coordinate and sequence work to ensure that construction materials placed against or over waterproofing and protection system will occur within seven days of membrane installation. Do not expose membrane to ultraviolet rays beyond period of time recommended by system manufacturer.
 - 1. Install drainage mat/protection board within 24 hours of membrane installation.
- B. Mold Prevention: Comply with Sections 015000 and 01 60 00.

1. Schedule and sequence work to protect mold growth products from exposure to moisture.

1.9 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 77 00.
1. Warranty with attachments for full replacement value of completed installation signed by manufacturer, applicator and Contractor warranting against water infiltration and defects of materials and workmanship for period of 10 years.
 2. If manufacturer will not allow applicator and Contractor to sign manufacturer's warranty, append applicator and Contractor's warranty to manufacturer's warranty to create warranty that covers labor and workmanship, including labor for access to waterproofing, for watertight warranty.
 3. Warrant penetrations, terminations, sealants, expansion joints, membrane, and deck drainage mat/protection board.
 4. Warranty: Include removing and reinstalling superimposed work covering waterproofing.

PART 2 - PRODUCTS

2.1 PRODUCTS AND MANUFACTURERS

- A. Acceptable Products and Manufacturers, Vertical Applications:
1. Bituthene 3000 Waterproofing System, WR Grace & Co., Cambridge, MA.
 2. CCW MiraDRI 860/861, MiraDRI, Wylie, TX.
 3. WR Meadows Mel-Rol Hampshire, IL
 4. Cetco Envirosheet, Hoffman Estates
- B. Submit for components of each wall assembly containing components specified in this section simultaneously and as may be required to provide a complete water proofing system where indicated on the drawings as WP-
1. This is a performance based specification. The Contractor shall submit a complete waterproofing system in accordance with the selected manufacturers recommendations and requirements consisting of compatible products applicable for each project condition. The Contractor shall provide shop drawings and details for project conditions outlined below that will produce a complete water proof finished product. The contractor shall submit correspondence from a technical representative of the manufacturer that states that the manufacturer has reviewed the Geotechnical report and that the submitted system will provide the performance as noted herein in concert with project specific geotechnical conditions and structural design.

2.2 MATERIALS

- A. SHEET WATERPROOFING Membrane, Vertical: Self-adhering membrane of rubberized asphalt laminated to polyethylene sheet.
1. Low temperature grade when dictated by temperature at time of application.
 2. Compatible with water based primer.
 3. Thickness: 60 mils min.
 4. Color: Black or gray.
 5. Tensile Strength (Membrane): 250 PSI minimum, ASTM D412.
 6. Ultimate Elongation: 300 percent minimum, ASTM D412.
 7. Pliability Temperature: Minus 25 F, ASTM D146.
 8. Hydrostatic Head Resistance at Lap: 150 feet minimum.
 9. Water Absorption: 0.5 percent maximum weight gain after 48 hour immersion at 70F, ASTM D570.
- B. Primer/Conditioner: Water-based VOC compliant type required by membrane manufacturer. No solvent or 1,1,1 - Trichloroethane based primers will be accepted.

1. If green concrete application is required: solvent based primer will be required, similar to GCP B2 LVC. Confirm with consultant before application.
- C. Mastic: Rubberized asphaltic type required by membrane manufacturer.
 1. Liquid Membrane: Two-component elastomeric, mastic grade as furnished by membrane manufacturer.
 2. Cement Mortar: Epoxy or latex modified cementitious composition acceptable to membrane manufacturer.
 3. Concrete Patching Compound: Fast setting, non-shrinking patching compound, of type acceptable to membrane manufacturer.

2.3 ACCESSORY MATERIALS FOR VERTICAL APPLICATIONS

- A. Subsurface Drainage Mat/Protection Board: Vertical subsurface drainage mat at typical backfill conditions, fabricated from compression-resistant, three-dimensional, non-biodegradable, single-sided, polymeric core sheet, with construction designed to effectively conduct water to foundation drainage system.
 1. Laminate geotextile fabric to drainage face of drainage mat and plastic film to membrane side.
 2. Overall Thickness: 7/16 inch minimum.
 3. Compressive Strength, Mat: 15,000 LBF/SQ FT minimum, ASTM D1621.
 4. Flow Capacity: 16 GPM per foot of width; ASTM D4716.
 5. Geotextile Fabric:
 - a. Material: Polypropylene
 - b. Weight: 4 oz/yd² minimum.
 - c. Puncture Strength: ASTM D4833, 65 pound.
 - d. Mullen Burst Strength: ASTM D3786, 215 PSI.
 - e. Elongation at Break: ASTM D4632, 60 percent.
 - f. Permeability: ASTM D4491, 0.39 cm/sec
 - g. Flow Rate: ASTM D4491, 150 GPM/SQ FT.
 6. Accessories: As necessary or recommended by drainage mat manufacturer for complete installation, including stainless steel termination bars, primers, stainless steel flashings, and sealants.
 - a. Adhesive: Manufacturer's standard adhesive to prevent displacement of product during installation and during subsequent operations.
 - b. When plastic film not available at membrane side of mat, provide protection board over membrane. Comply with membrane manufacturer's recommendations.
 7. Acceptable Products:
 - a. Hydroduct 220 Drainage Composite, Grace Construction Products, Cambridge, MA.
 - b. MiraDRAIN 6200, TC MiraDRI, Duluth, GA.
 - c. MelDrain, WR Meadows
 - d. Accepted Substitute in accordance with Section 01 60 00.
- B. Metal Termination Bars: Aluminum bars, approximately 1 by 1/8 inch thick, predrilled at 9-inch centers.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 1. Ensure surfaces are reasonably smooth and free of holes, cracks or projections which might be detrimental to successful installation.

2. Verify that curing methods used for concrete are compatible with membrane system.
3. Verify that horizontal surfaces have smooth wood float finish, free from defects. Broom finish not allowed.
4. Verify that items penetrating waterproofing system are securely installed.
5. Verify that concrete surfaces have cured for period of time acceptable to membrane manufacturer.
6. Verify that masonry joints are struck flush with face of unit.
7. Verify that any curing agents employed are compatible with the WP product or properly removed or properly treated with primer or other products recommended by the WP membrane manufacturer.
8. Verify and form release agents used during concrete placement are compatible with adhesive for sheet waterproofing.
9. Provide written waterproofing applicators acceptance of the substrate prior to placement of membrane.

3.2 PREPARATION

- A. Protection:
 1. Take precautions to prevent odors from waterproofing operations from entering adjacent inhabited enclosed spaces.
 2. Protect adjacent surfaces not designated to receive waterproofing.
- B. General: Clean and prepare surfaces in accordance with manufacturer's instructions.
 1. Seal cracks and joints in accordance with manufacturer's instructions. Use proper depth-width ratio required by sealant manufacturer.
 2. Remove sharp projections, fins, and loose material. Remove form ties to 3/4 inch minimum behind face of wall. Fill holes, voids, and honeycomb areas flush with concrete patching compound or cement mortar.
 3. Mastic: Seal penetrations with mastic.
 4. Provide fillet or cant at junction of vertical and horizontal surfaces using liquid membrane in configuration acceptable to membrane manufacturer.

3.3 INSTALLATION

- A. Waterproofing System: Install in accordance with Section 01 73 00 and, approved Shop Drawings.
 1. Apply primer/conditioner at rate required by manufacturer. Install only as much primer/conditioner as can be covered in same day.
 2. Prior to placing full membrane, provide membrane strips at inside corners, outside corners, and working joints. Center strips along axis of corner and joint.
- B. Sheet Waterproofing Membrane: Install membrane in shingle fashion with edges and ends overlapped at dimensions required by manufacturer.
 1. Remove release paper layer. Roll out laps and surface with mechanical roller to encourage full contact bond.
 2. Completely bond membrane to substrate, except those areas directly over or within 3 inches of working cracks or expansion joints.
 3. Mastic: Place uniform bead of mastic to joint edges at locations required by manufacturer.
 4. Seal perimeter ends and edges to adjoining surfaces.
 5. Liquid Membrane: Seal items penetrating membrane with flashing membrane material and liquid membrane. Ensuring positive seal with membrane and penetrating member.
 6. Vertical Application: Extend membrane and down face 6 inches minimum, except terminate at point 12 inches below floor slab of protected space where footings are at greater depth.

7. Integrate waterproofing with adjacent waterproofing systems per manufacturer's recommendations, vertical to horizontal
- C. Subsurface Drainage Mat: At typical, vertical, positive side waterproofing and backfill conditions, install mats in accordance with manufacturer's written instructions with fabric covered surface exposed to backfill and flat side of core placed against waterproofing.
 1. Lap ends of core and filters fabric 3 inches and secure in place with manufacturer's approved system.
 2. Install mat over waterproofing with seams overlapped in direction of waterflow.
 3. Prior to placement of backfill, inspect drainage mat for defects, voids or displacement. Repair or replace as necessary to provide complete coverage.
 4. Extend and wrap fabric around foundation drainage pipe in accordance with mat manufacturer's recommendations.
 5. Protect in-place matting during backfill operations in accordance with matting manufacturer's instructions.
 6. Adhere drainage mat in a vertical manner to sheet waterproofing with light spray adhesive per manufacturer's recommendations.

3.4 FIELD QUALITY CONTROL

- A. Comply with requirements of Section 01 40 00.
- B. Manufacturer's Field Services: Comply with Section 01 40 00.
 1. Notify manufacturer in timely manner to arrange for manufacturer's technical representative's site visits to ensure proper installation, verify work is in accordance with manufacturer's requirements, and warranty requirements have been met.
 2. Manufacturer's Qualified Technical Representative: Monitor activities and advise applicator of proper installation procedures and precautions.
 3. Minimum Site Visits:
 - a. Pre-construction conference.
 - b. First day of waterproofing on site including acceptance of substrate conditions.
 - c. Periodic Visits: Weekly during installation and immediately prior to installation of subsurface drainage mat. Notify the Owner and Design team of the date and time of for site visits.
 4. Submit reports; include site observations, instructions, and monitoring activities.
- C. Coordinate installation with commissioning work

3.5 PROTECTION AND CLEANING

- A. Protect finished work in accordance with Section 01 73 00.
 1. Protect adjacent surfaces from damages and stains. Clean materials from surfaces where inadvertently applied.

END OF SECTION

SECTION 07 21 00
THERMAL INSULATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Batt (blanket) insulation.
 - 2. Curtain wall (semi-rigid mineral fiber) insulation.
- B. Related Sections:
 - 1. Section 01 83 00 – Building Envelope Design Requirements and Testing.
 - 2. Section 07 26 00 – Vapor Retarders.
 - 3. Section 07 84 00 - Firestopping: Fire Safing.
 - 4. Section 08 44 13 - Glazed Aluminum Curtain Wall.

1.2 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
 - 1. Submit for components of each wall assembly containing components specified in this section simultaneously. See Schedule of Wall Assemblies at end of Section 01 83 00.
- B. Product Data: Submit product data for each product, including attachment devices.
- C. Samples: Submit two samples, 12 by 12 inches in size of each type of insulation.
- D. Informational Submittals: Submit manufacturer's instructions packaged separately from other submittals.

1.3 QUALITY ASSURANCE

- A. Single Source Responsibility: Furnish each insulation type from one manufacturer for entire Project, unless otherwise acceptable to Architect.
- B. Provide mock-up of each wall conditioning showing proper means of installation of insulation

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Identify products with appropriate markings of applicable testing and inspecting organization.
 - 2. Storage and Protection: Store materials raised off floor or ground and under cover to keep dry. Protect from weather, direct sun light, contamination, sources of ignition, and damage from construction operations.

1.5 PROJECT CONDITIONS

- A. Mold Prevention: Comply with Section 01 50 00. Provide temporary heating and ventilation as required to keep insulation products dry during construction operations. Material that does get wet will not be used.

1.6 SEQUENCING

- A. Sequence work under provisions of Section 01 10 00.
 - 1. Do not begin work until adjacent substrate work is complete and work of other trades which will be concealed by work of this section has been approved.
 - 2. Any gutter or water drainage system in the curtain wall system shall be cleaned of any foreign debris prior installation of the curtain wall insulation. This cleaning operation shall be

part of the quality control installation procedures and shall be document as being completed before installation of the insulation. Insulation contractor shall take proper measures it not create debris from the insulation installation that will deposit materials in the drainage system that will block the drainage of eh curtain wall system.

B. Mold Prevention: Comply with Sections 01 50 00 and 01 60 00.

1. Schedule and sequence work to protect mold growth products from exposure to moisture.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Glass Fiber Insulation Manufacturers:

1. CertainTeed Corp., Valley Forge, PA.
2. Johns Manville, Denver, CO.
3. Owens-Corning, Toledo, OH.
4. Knauf Fiber Glass, Shelbyville, IN.
5. Accepted Substitute in accordance with Section 01 60 00.

2.2 INSULATION MATERIALS

A. Insulation Materials: Use of chlorofluorocarbons (CFCs) not allowed.

B. Batt (Blanket) Insulation:

1. Maximize use of recycled material with minimum of 20 percent recycled glass cullet [plus 20 percent post-consumer].
2. Acceptable Formaldehyde Levels: Formaldehyde free.
3. Batt Insulation - Unfaced: Glass fiber composition, friction fit type, unfaced.
 - a. Thermal Resistance (R-Values): As indicated.
 - b. Classification: ASTM C665, Type I.
 - c. Density: 1 PCF minimum.
 - d. Thermal Resistance: ASTM C518, R Value of 3.2 per inch of thickness at 75 F mean temperature.
 - e. Fire Rating: ASTM E84, Flame spread 25 or less and smoke development 50 or less.
 - f. Acceptable Products:
 - 1) Commercial Blanket Insulation, CertainTeed Corp.
 - 2) Comfort-Therm Commercial Insulation, Johns Manville.
 - 3) Unfaced Light Density Thermal Insulation, Owens Corning.

C. Existing Glazing Furr-out Curtain Wall Insulation: Rockwool based mineral fiber semi-rigid board insulation for thermal insulation purposes only. See Section 07 84 00 for fire safing systems at rated deck to non-rated curtain wall interface.

1. Maximize use of recycled material with minimum of 75 percent.
2. Thermal Resistance (R-Values) as scheduled.
3. Factory faced with fire retardant reinforced foil laminate on one face.
4. Dark color insulation at glass spandrel areas.
5. Classification: ASTM C612, Class 1 or 2.
6. Thermal Resistance: ASTM C518, R value of 4.3 per inch of thickness, at 75 F mean temperature.
7. Thermal Conductivity: ASTM C518, K Value of 0.24 minimum, at 75 F mean temperature.
8. Permeance: 0.05 perms maximum.
9. Fire Rating: ASTM E84, flame spread 25 or less, smoke development 0.
10. Acceptable Products:

- a. FBX Curtainwall Insulation, Fibrex Co.
 - b. MW -CW-6, Owens-Corning.
 - c. ThermafiberCW-60, Thermafiber, Inc.
- D. Ceiling Insulation: Encapsulated Unfaced glass fiber batt (blanket) insulation.
- 1. Thermal Resistance (R-Values): As indicated.
 - 2. Classification: ASTM C665, Type II, Class 1, Category 1.
 - 3. Density: 1 PCF minimum.
 - 4. Formadehyde free.
 - a. Thermal Resistance: ASTM C518, R Value of 3.2 per inch of thickness at 75 F mean temperature.
 - b. Fire Rating: ASTM E84, Flame spread 25 or less and smoke development 50 or less.
 - c. Vapor Permeability: ASTM E96.
 - d. Facer: 0.5 perms
 - 1) Perforated Backer Film: 25 perms.
 - e. Acceptable Products:
 - 1) Comfort-Therm Commercial Insulation, Johns Manville.
 - 2) Sonobatts Insulation, Owens Corning.

2.3 ACCESSORIES

- A. Insulation Fasteners: Impale clip type with retaining disc or plate, galvanized steel, adhered or mechanically fastened to surface to receive insulation, length to suit insulation thickness, capable of securely fastening insulation in place. Do not adhere any accessories to any glass surface
- B. Tape: Self-adhering pressure sensitive, compatible with insulation, foil type recommended by manufacturer of insulation.
- C. Adhesives & Sealants: Only use adhesives and sealants in the interior of the building that meet or do not exceed the VOC limits of the CURRENT requirements of South Coast Air Quality Management District (SCAQMD) Rule No. 1168 on the interior of the building.
 - 1. Current requirement refers to the date on which the materials are installed in the building.
 - 2. SCAQMD Rule #1168 is referenced in section 01 81 13 that was current as of the date of this specification. Refer to <http://www.aqmd.gov/rules> for the actual current version of the rule that will be applicable at the date of installation during construction.
 - 3. Interior refers to all building construction that is inside of the exterior weatherproofing material.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 40 00.
 - 1. Verify that work of other trades which will be covered by insulation is complete, approved, and tested.
- B. Mold Prevention: Comply with Sections 01 50 00 and 01 60 00. Do not install wet insulation, insulation adjacent to wet materials, or insulation in wet conditions, or insulation that can get wet prior to the building being dried in.

3.2 BATT (BLANKET) INSULATION AND CURTAIN WALL INSULATION

- A. General: Comply with Section 01 60 00.
- B. Batt (Blanket) Installation: Install insulation after plumbing, mechanical, and electrical services have been installed.

1. Provide mechanical fasteners, wire mesh, or other accessories to ensure insulation remains in specified position.
2. Fit insulation tight within spaces and tight to exterior side of plumbing, mechanical, and electric services within plane of insulation leaving no gaps or voids.
3. Butt insulation tightly.
4. Cut and fit tightly around items penetrating insulation.
5. Stagger butt joints.
6. Use batts free of damage.
7. Install insulation within metal framing systems full height and width. Do not allow voids or openings to occur. Insulation is required for full width between studs, including cavity of each stud.
8. Cut and trim insulation neatly, to fit spaces.
9. Cut insulation oversize to ensure tight butt joints when installed. Cut insulation to fit around protrusions and irregularly shaped projections.

3.3 CEILING INSULATION

- A. Ceilings: Loose lay encapsulated batt (blanket) insulation on topside of suspended ceiling. Tightly fit to prevent gaps or voids..voids.

3.4 EXISTING GLAZING FURR-OUT CURTAIN WALL INSULATION

- A. General: Comply with Section 01 60 00.
- B. Existing Glazing Furr-out Wall Curtain Wall Insulation: Install insulation after window film applications have been curtain wall system has been installed, glazed and made watertight.
 1. Provide mechanical fasteners, wire mesh, or other accessories to ensure insulation remains in specified position and a minimum of 1" from the interior surface of the glass.
 2. Fit insulation tight within spaces and tight to the side of the curtain wall framing within plane of insulation leaving no gaps or voids.
 3. Butt insulation tightly.
 4. Cut and fit tightly around items penetrating insulation or adjacent to the insulation such as anchors used to secure curtain wall to structure.
 5. Stagger butt joints.
 6. Use semi rigid insulation free of damage.
 7. Install insulation within metal framing systems full height and width. Do not allow voids or openings to occur. Insulation is required for full width between vertical framing elements or provide supplemental framing to support the semi-rigid board insulation if spans exceed tohe available sizes of insulation.
 8. Cut and trim insulation neatly, to tightly fit spaces.
 9. Cut insulation oversize to ensure tight butt joints when installed. Cut insulation to fit around protrusions and irregularly shaped projections.
 10. Install insulation over interior surface of the vertical mullions and lap onto adjacent insulation to comply with fire tested systems.
 11. If vapor retarder is used, seal all joints in insulation field and at perimeter with kraft reinforced foil scrim tape. Tape any punctures in the foil facing. If there is no vapor retarder required, then leave joints un-taped.

3.5 PROTECTION

- A. Protect finished work in accordance with Section 01 50 00.
 1. Protect insulation from moisture until final completion.
- B. Mold Prevention: Comply with Sections 01 50 00 and 01 60 00.

PART 4 -

4.1 SCHEDULE

- A. Provide thermal resistance (R Values) for insulation in locations as follows:
1. Exterior Walls: Thermal resistance R Value of 19.
 2. Soffits and Fascias: Thermal resistance R Value of 19.
 3. Curtain Wall Spandrel Panels: Thermal resistance R Value of 11.5

END OF SECTION

SECTION 07 26 00

VAPOR RETARDERS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Building slab vapor retarder.

1.2 DEFINITIONS

- A. Perm: 1 grain/h•ft²•in-Hg.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit for each product, including tape.
- C. Shop Drawings: Showing termination transitions for below grade waterproofing and penetrations

1.4 SEQUENCING

- A. Begin installation only after substrate work is complete and penetrations are securely anchored.

1.5 QUALITY CONTROL

- A. Manufacturer's representative to perform inspection prior to concrete placement and provide a written report

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Building Slab Vapor Retarder : ASTM E1745, Class A, with the following limitations:
 - 1. Use: Typical conditions.
 - 2. Material: Polyethylene or polyolefin membrane, non-reinforced.
 - 3. Thickness: 15 mils.
 - 4. Permeance: ASTM E1745, 0.10 perms maximum.
 - 5. Puncture Resistance: ASTM E1745, 1970 grams.
 - 6. Tensile Strength: ASTM E1745, 45 lbf/in.
 - 7. Acceptable Products and Manufacturers:
 - a. Stego Wrap 15 mil, Stego Industries, LLC, San Juan Capistrano, CA.
 - b. VaporBlock VB15, Raven Industries, Sioux Falls, SD.
 - c. GCP FlorPrufe 120
 - d. WR Meadows Perminator 15 mil
 - e. Griffolyn 15 mil Green, Reef Industries, Houston, TX.
 - f. Accepted Substitute in accordance with Section 01 25 00.
- B. Joint Tape: Manufacturer's recommended, pressure sensitive type, self adhering, mesh reinforced, and of perm rating not less than vapor retarder.
- C. Gusset Tape: Provide manufacturer's recommended 6 inch gusset tape.
- D. Adhesive: Type recommended by manufacturer of sheet products, non-sagging grade, compatible with sheet and substrate.
- E. Seam Tape: High Density Polyethylene (HDPE) tape with pressure sensitive adhesive; minimum 4 inches wide.

PART 3 - EXECUTION

3.1 EXAMINATION AND PREPARATION

- A. Examination: Examine conditions and proceed with Work in accordance with Section 01 73 00.
 - 1. Verify that substrate work is complete, clean and dry before beginning installation of sheet products.

3.2 PREPARATION:

- A. See Schedule of Slab-On-Grade Systems at end of Section 03 30 00.
- B. Coordinate installation with Section 31 00 00.

3.3 INSTALLATION

- A. Building Slab Vapor Retarder: ASTM E1643.
 - 1. Lay-out sheets to minimize quantity of joints. Lap edge and end joints 12 inches minimum and continuously seal with joint tape.
 - 2. Turn up sheets at perimeter; at footers and vertical walls, and against penetrations. Seal joints with joint tape. Install pipe boots at penetrations and seal.

3.4 QUALITY CONTROL

- A. Manufacturer's representative to perform inspection prior to concrete placement and provide a written report

3.5 PROTECTION

- A. Protect sheets from puncture during installation. Patch and seal punctures before proceeding with subsequent construction and just before placing concrete.
- B. Install runway planks in construction traffic lanes until slabs are placed.

END OF SECTION

SECTION 07 52 00

MODIFIED BITUMINOUS MEMBRANE ROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Related Sections:

1. Section 06 10 00 - Rough Carpentry: Wood nailers, wood curbs,.
2. Section 07 71 00 - Roof Specialties: Hatches, Curbs, Equipment Supports Rooftop Expansion Joints.
3. Division 22 – Plumbing: Roof drains and accessories.

1.2 REFERENCES

A. Reference Standards: See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following for design, materials, fabrication, and installation of component parts:

1. NRCA Roofing and Waterproofing Manual, National Roofing Contractors Association.
 - a. NRCA Specification MBS(2)-I-(M)-M insulated deck, SBS modified bitumen Torch-down roof system with modified bitumen base layer and mineral surfaced or white reflective cap sheet.
 - b. NRCA Modified Bitumen Roofing Details (MB Series).

1.3 DEFINITIONS

A. Roof Expansion Joint: A non-structural gap or discontinuation of materials extending through decking and roofing and flashing systems and components occurring at conditions identified by NRCA Roofing and Waterproofing Manual.

1.4 SYSTEM DESCRIPTION

A. Performance Requirements:

1. Prevent water migration from entering building through roof membrane and related flashings.
2. Provide insulation board attachment requirements which results in FM wind uplift rating of 1- 90 (and higher as identified on structural drawings) and Fire Hazard Classification Class 1, in accordance with FM Loss Prevention Data Sheet 1-28.
3. Roofing System Design: Provide a membrane roofing system that is identical to systems that have been successfully tested by a qualified testing and inspection agency to resist uplift pressures based on ASCE-7 and according to SPRI's "WD-1 2008" and conforming to requirements contained in Section 1609 of the International Building Code.

B. Fire Resistance Requirements:

1. Maintain UL Class A rating per ASTM E108 for roof.

C. Interface with Adjacent Systems:

1. Provide for watertight interface with related flashings, counterflashings and copings.

D. Low Roofs (All roofs below elevation 175'-0")

1. Prime concrete deck
2. Torch-down ASTM D6164 Temp Base Sheet per spec.
 - a. This shall be used for the construction dry in – patch as required to maintain weather proof condition.
 - b. This is the "Additional base layer – fully adhered shown on the documents".

3. Patch and clean
 4. Mop-down insulation and cover board.
 5. Torch-down ASTM D6163 Base Sheet Membrane per spec.
 6. Torch-down Cap Sheet per spec.
- E. Patch and clean
- F. Install insulation and cover board using adhesive per specification.
- G. Torch-down ASTM D6163 Base Sheet Membrane per spec.
- H. Torch-down Cap Sheet per spec.
- I. Low Roofs (to be used for future floors)
1. Install insulation and cover board using adhesive per specification.
 2. Torch-down ASTM D6164 Temp Base Sheet per spec.
 - a. This shall be used for the construction dry in – patch as required to maintain weather proof condition.
 - b. This is the “Additional base layer – fully adhered shown on the documents”.
 3. Patch and clean
 4. Torch-down ASTM D6163 Base Sheet Membrane per spec.
 5. Torch-down Cap Sheet per spec.

1.5 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
1. Submit Section 07 62 00 submittals simultaneously with submittals for this section.
- B. Product Data: For primary and secondary roofing materials.
1. List of materials and data sheets describing physical characteristics and performance criteria for materials proposed for use, including membrane, flashings, vapor retarder, and insulation.
 2. List of projects of similar assembly by both manufacturer and applicator which have been installed during last five years.
 3. Include sample of warranty customized for this project.
- C. Shop Drawings:
1. Description: Provide total system assembly description including materials and installation methods for:
 - a. Insulation, including installation methods, insulating value, and establishment of slope.
 - b. Membrane
 - c. Flashing
 - d. Accessories
 - e. Identify type of deck if more than one deck or substrate is shown.
 - f. Termination and flashing at precast panels.
 2. Details: Project specific details for flashings; expansion joints; penetrations; control joints; drains; terminations; transitions; detailing at substrate cracks; construction of night seals; window washing equipment and anchors; and other details not shown on Contract Documents.
 3. Manufacturer: Sign Description and Shop Drawings stating that information as submitted meets manufacturer’s warranty requirements for this specific Project.
- D. Samples:
1. 12 by 12 inch samples of granular surfaced membrane and walkway surfacing.
- E. Informational Submittals: Submit following; may be packaged separately from other submittals:

1. Certifications specified in Quality Assurance article.
2. Qualification Data: Manufacturer's qualification data.
3. Qualification Data: Applicator's qualification data.
4. Manufacturer's installation instructions.
 - a. Manufacturer's field reports.

F. Closeout Submittals: Submit specified warranty in accordance with Section 01 77 00.

1.6 QUALITY ASSURANCE

- A. Manufacturer's Qualifications: Not less than five years experience in manufacturing of modified bituminous sheet roofing.
 1. Obtain primary roofing materials from single manufacturer. Manufacturer's name shall appear on containers.
 2. Provide secondary materials as required by manufacturer of primary materials.
- B. Applicator Qualifications: Approved by manufacturer prior to execution of this Contract, with experience on at least five projects using submitted modified bituminous sheet roofing membrane.
 1. Foreman of Field Crew: Minimum five years experience with modified bituminous sheet roofing.
- C. Owner reserves right to hire independent roofing consultant to review submittals, procedures, installation, and perform nondestructive testing to ensure dry roof and components.
- D. Certifications: Manufacturer's certifications on manufacturer's letterhead:
 1. Roof system design; penetration, transition, and perimeter details; and system specifications are appropriate and satisfactory for this particular project, including equipment mounts, lightning rod supports, and fall protection davits are suitable for use with intended roofing system.
 2. Products proposed for use comply with referenced standards; with UL and FM attached.
 3. Materials ordered and supplied are compatible with each other, suited for locale and purpose intended and shipped in sufficient quantity to ensure proper timely installation.
 4. Roofing materials have express warranty of merchantability and fitness for particular purposes of this Project.
 5. Manufacturer has reviewed Project and will issue warranty upon successful completion of installation.
 6. Materials shipped to site meet membrane manufacturer's published performance standards and requirements of this Specification.
 7. Membrane manufacturer's approval of insulation type and method of installation.
 8. Manufacturer's approval of installer.

1.7 PREINSTALLATION CONFERENCE

- A. Conduct preinstallation conference in accordance with of Section 01 31 19. Discuss sequence and scheduling of work and interface with other trades.
- B. Minimum Agenda:
 1. Review requirements for Work and conditions which could possibly interfere with successful performance of Work.
 2. Review Project Specifications and Drawings.
 3. Establish installation schedules and sequence.
 4. Coordinate work with in-place and subsequent construction.
 5. Review weather and working conditions.
 6. Review installation procedures, including:

- a. Substrate requirements.
 - b. Insulation installation.
 - c. Membrane installation.
 - d. Surfacing requirements, including sequencing, and walkway protection.
 - e. Roof terminations, night seals, flashings, and roof drain requirements.
 - f. Mechanical equipment placement, supports, and height requirements.
 - g. Inspection, testing, and quality control procedures.
 - h. Protection requirements for construction period beyond roofing installation.
 - i. Procedures for making roof penetrations after membrane installation.
 - j. Window washing procedures.
 - k. Warranty requirements, Owner expectations and exclusions.
- C. Conduct tour of roof deck and review substrate surfaces to receive roofing and flashings. Report on substrate acceptability, possible problem areas, and recommended remedies.

1.8 DELIVERY, STORAGE AND HANDLING

- A. Comply with Section 01 60 00.
- 1. Store materials in weather protected environment, clear of ground and moisture.
 - 2. Protect light sensitive insulation from direct sunlight exposure.
 - 3. Store materials (except membrane) between 60 F and 80 F. If exposed to lower temperatures, restore to proper temperature before using.
 - 4. Stand roll materials on end.
 - 5. Do not store materials on roof in such concentrations as to cause deck or structural membranes to be overloaded.

1.9 PROJECT CONDITIONS

- A. Environmental Requirements: Proceed with roofing work only when existing and forecasted weather conditions permit work to be performed in accordance with manufacturer's requirements.
- 1. Verify roof surface is not frozen and is free of ponded water, dew, frost, ice, and snow.
 - 2. Maintain membrane above 40 F prior to installation.

1.10 SEQUENCING, SCHEDULING AND COORDINATION

- A. Do not install greater amount of insulation than can be covered by membrane in same day. Complete with night seals and appropriate tieoffs.
- 1. Sequence work to avoid traversing over completed areas in order to continue roofing operations.
- B. Manufacturer: Responsible for details and dimensions not shown on Drawings or controlled by Project conditions. Furnish Contractor necessary details and field measurements to ensure roofing is covered by warranty. Show required field measurements beyond manufacturer's control on Shop Drawings. Coordinate with responsible trades to establish, verify and maintain field dimension and Project conditions.
- C. Mold Prevention: Comply with Sections 015000 and 01 60 00.
- 1. Schedule and sequence work to protect mold growth products from exposure to moisture.

1.11 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 78 00.
- 1. Warranty for full replacement value, non-prorated, NDL (no dollar limit) of completed installation signed by manufacturer, applicator and Contractor warranting against water infiltration and defects of materials and workmanship for period of 20 years. If manufacturer will not allow installer and Contractor to sign manufacturer's warranty, append installer and

Contractor's warranty to manufacturer's warranty to create warranty that is complete in covering labor and workmanship, including labor for access to roof membrane and replacement, for watertight warranty.

- a. Segmented warranties or warranties requiring renewal to achieve specified period not permitted.
 - b. Warranties requiring periodic inspections and non-covered repairs by manufacturer not permitted.
 - c. Warranty: Cover responsibility of parties to repair membrane, supply materials, and also include procedures for repair and payment to Owner. If more than one warranty is required, Contractor: Define and identify responsibility of each warrantor so Owner receives complete warranty for labor, materials, repair of defects, and other items necessary to obtain watertight roofing system for full specified time period. Contractor: Responsible for compliance with terms and limitations of manufacturer's warranty.
 - d. Warrant penetrations, terminations, flashings, sealants, expansion joints, pitch pockets, membrane, pavers, vapor retarders insulation, and fasteners. Warrant full height of flashing extending up back of parapet wall. Defects include (but are not limited to) blisters, fishmouths, wrinkles, aggregate bare spots, ridges, splits, punctures, and loose flashings.
 - e. Refer to end of section for sample warranty requirements.
2. Inspection of roof be made when the roof system has been in place for an 18 month period. This re-Inspection is to be made by representatives of both the Installer and the material manufacturer. Written notification to the Owner of the date and time of this re-Inspection is to be made. Any issues affecting the extended system warranty must be recorded and corrected prior to the expiration of the Installer's 2 year warranty. –

PART 2 - PRODUCTS

2.1 MANUFACTURERS AND PRODUCTS

- A. Acceptable Products and Manufacturers, Torch-down System:
1. Paradiene 20 TG/30 FR CR Series, Siplast, Irving, TX.
 2. Comparable products by:
 - a. Soprema with Sprastar cap sheet, Wadsworth, OH.
 - b. Firestone Building Products with Ultra White Granular cap sheet, Carmel, IN. (preferred product)
 - c. Johns Manville
 - d. Tremco.
 3. Accepted Substitute in accordance with Section 01 25 00.

2.2 MATERIALS

- A. General: Provide products required by manufacturers to be fully compatible with each other and with indicated substrates, or provide separation materials as required to eliminate contact between incompatible materials.
- B. Modified Bituminous Membrane:
1. Base Sheet Membrane: SBS modified bituminous membrane
 - a. Temp Base Sheet, ASTM D6164 Grade S, Type I, as modified below.
 - b. Base Sheet, ASTM D6163 Grade S, Type I, as modified below
 - c. Thickness: 0.090 inch minimum.
 - d. Size: Manufacturer's standard sheet width in longest practical rolls.
 - e. Reinforcing: Manufacturer's standard reinforcing.
 - f. Surface: Smooth.

2. Cap Sheet Membrane: SBS modified bituminous membrane, fire retardant treated, ASTM D6164 Grade S, Type I, as modified below.
 - a. Thickness: 0.160 inch minimum.
 - b. Size: Manufacturer's standard sheet width in longest practical rolls.
 - c. Reinforcing: Manufacturer's standard reinforcing.
 - d. Surface: Integral ceramic granules.
 - e. Surface Color:
 - 1) White reflective surface, meeting USGBC SRI for reflectance of .78 or higher.
 - 2) Meet Current City of Dallas requirements
 - 3) Meet Owner Project Requirements (OPR) of SRI rating of .65 at installation and .50 after 3 years.
- C. Base Flashing: Manufacturer's two-ply polyester reinforced, SBS modified bitumen system for torch application.
 1. Manufacturer's heavy duty system including sanded base sheet and fire-rated cap sheet color to match membrane cap sheet.
 2. Cold-adhesive and peel-and-stick/self-adhering flashing not permitted.
- D. Flexible Flashing: Factory-fabricated, elastomeric, vulcanized, continuous EPDM sheet; fleece-faced at flanges for full adhesion; sized to accommodate anticipated movement:
 1. Elongation: ASTM D412, Minimum plus/minus 200 percent.
 2. Elongation to Break: ASTM D412, Minimum 500 percent.
 3. Seam Tensile Strength: ASTM D816, Equivalent to field of membrane.
 4. Acceptable Product and Manufacturer:
 - a. RedLINE, Situra, Inc., North Tonawanda, NY.
- E. Termination Bar: Manufacturer's standard metal termination bar, prepunched to permit proper spacing of fasteners. Continuous cleat may be used in lieu of termination bar when fabricated and installed in accordance with requirements specified in Section 07 62 00.
- F. Asphalt Bitumen: ASTM D312, Type III at insulation installation; Type IV for slopes over ¼ Inch in 1 foot.
- G. Asphalt Primer: ASTM D41.
- H. Roof Insulation Materials:
 1. Insulation Materials: Use of chlorofluorocarbons (CFCs) not allowed.
 2. Base Insulation: Polyisocyanurate, ASTM C1289, Type II.
 - a. Thermal Resistance: ASTM C518, R Value of 5.64 per inch of thickness at 75 F mean temperature; no exceptions.
 - b. Maximum Allowable Thickness per Insulation Layer:
 - 1) 2.0 inches.
 - c. Facers: Both faces finished with asphalt-saturated fiberglass face.
 - d. Manufacturer: Approved by membrane manufacturer.
 3. Tapered Roof Insulation:
 - a. Type: Polyisocyanurate as specified above.
 - b. Minimum Slope: 1/4 inch per foot.
 - 1) ½ inch per foot slope at valleys, crickets and saddles for improved drainage
 4. Compressible Insulation: Glass fiber insulation.
 - a. Classification: ASTM C665, Type I.
 - b. Density: 1 PCF minimum.
 5. Average LTTR Insulation Value: R-30, Minimum R-20

- I. Cover Board - Glass Fiber Faced Gypsum:
 - 1. 4'-0" by 8'-0" by 1/2 inch roof board.
 - 2. ASTM C1177; Type X; FM approved; silicone treated core with glass mat facing.
 - 3. Non-asphaltic coating to enhance bonding of hot-mop modified bitumen roofing systems, including for cold mastic and torch applied modified bitumen
 - 4. Flame Spread: 0.
 - 5. Acceptable Product: Dens-Deck Prime Gypsum Roof Board, G-P Gypsum Corporation, Atlanta, GA 30303. At back of parapets, provide 1/2 inch Dense Deck Duraguard.
 - 6. The overlay board shall be one recommended for use by the roofing membrane manufacturer and shall be part of a tested assembly.
- J. Wood Nailers: Solid, pressure treated softwood as specified in Section 06 10 00.
- K. Cants Strips: Wood fiberboard, preformed to 45 degree angle or configuration detailed.
- L. Surfacing Granules: Loose granules matching cap sheet for use in hot bitumen bleedout at laps and seams.
- M. Insulation and Cover Board Adhesive for High Roof Applications: 3M Polyurethane Foam Insulation Adhesive CR-20 or approved equal. Total assembly shall meet uplift requirements of the specifications.
- N. Flashing at irregular roof penetrations: Where roof detailing conditions are such that conventional bituminous and/or sheet metal flashings must be used, use a manufacturer's approved, fabric reinforced 3 part elastomeric (2 parts product and 1 part color) to form field flashings. Sprinkle additional mineral granules on horizontal surfaces adjacent to all such flashing installations. Identify all proposed locations for such materials on the shop drawings.
- O. No pitch pans

2.3 ACCESSORIES

- A. Roof Walkway Protection: Manufacturer's standard walk plank, not less than 1/2 inch thickness.
 - 1. Granular surface color: Selected by Architect
- B. Roof Hatches, Premanufactured Curbs, and other Accessories: See Section 07 71 00.
- C. Roof Expansion Joints: Flame Line Torch-Grade by Situra, Soprajoint by Soprema, or as required by manufacturer for warranty inclusion.
 - 1. Acceptable Alternate: Johns Manville Expando-Flash

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 - 1. Verify deck is clean and smooth, free of depressions, waves or projections, properly sloped to points of drains. Verify flutes of steel deck are clean and dry.
 - 2. Verify roof openings and penetrating elements through roof are solidly set, and cant strips, wood blocking, nailing strips, and reglets are in place. Verify deck is properly supported and secured.
 - 3. Do not apply roofing materials to damp, frozen, dirty, dusty, or other deck surface conditions which are unacceptable to manufacturer and applicator.
 - 4. Verify deck surfaces are dry and free of snow or ice.
 - 5. Concrete Substrate: Verify concrete substrate has cured minimum 28 days.
 - a. Concrete with greater than 12 percent moisture content not acceptable. Test with Wagner moisture meter or in accordance with manufacturer's other requirements for warranty issuance. Allow to fully cure and dry to specified requirements prior to applying membrane.

6. Primer: Apply to substrate surfaces at rates as required by membrane manufacturer.

3.2 PREPARATION

- A. Protection: Take precautions to prevent odors from roofing operations from entering adjacent inhabited enclosed spaces.
- B. Wood Nailers and Curbs: Install securely to substrate in accordance with Section 06 10 00 and roofing manufacturer's recommendations.
 1. Nailers: Match thickness of roof insulation system unless otherwise indicated.

3.3 APPLICATION

- A. Modified Bituminous Membrane Roofing System: Install in accordance with manufacturer's printed instruction and NRCA Manual.
- B. Roof Insulation: Install in accordance with manufacturer's printed instruction.
 1. Verify surfaces are clean and dry and that concrete is fully cured and sufficiently dry to receive work of this section.
 2. Base Insulation: Place minimum two layers of base and tapered insulation in accordance with insulation manufacturer's instructions. Install second and subsequent layers of insulation with joints in both directions staggered from previous layers.
 3. Metal Deck: Mechanical Fasteners: Mechanically fasten first layer of insulation directly to metal deck with fasteners, penetration depths, and patterns complying with FM requirements. Install insulation parallel to flute direction. Ensure edges are supported by decking. Install subsequent layers in mopping of hot bitumen at EVT.
 4. Concrete Deck: Embed insulation in mopping of hot bitumen at EVT temperature ensuring complete adhesion to concrete substrate. Install second and subsequent layers in mopping of hot bitumen with joints staggered from previous layer.
 5. Lay adjacent insulation boards to make contact without forcing joints. Cut insulation to fit neatly to perimeter blocking and protrusions through roof. Do not install cracked or broken boards.
 6. Tapered Roof Insulation: Lay tapered boards to provide drainage slope and to form crickets and edge strips to provide minimum 1/4 inch slope per foot drainage at areas where structure has not provided slope to points of roof drainage. Stagger joints in both directions from previous layer.
 - a. Drains: Lay tapered boards for distance of 24 inches back from roof drains for positive drainage.
 7. Minimum Total Base and Tapered Roof Insulation Thickness:
 - a. At Drain Sumps: Maintain minimum R-20.
 8. Cover Board: Install as continuous top layer with joints in both directions staggered from previous layer.
 9. Cant Strips: Install cant strips as recommended by manufacturer.
 10. Install insulation in hot asphalt mopping bitumen in accordance with bitumen manufacturers heating and mopping requirements.
 - a. Apply with hot-mopped asphalt at rates required by manufacturer with asphalt target temperature of 425 F or 20 F above EVT, whichever is higher to ensure proper asphalt temperature at point of application.
 - b. Maintain mopping bitumen within recommended temperatures at point of insulation embedment.
 - c. Ensure bitumen is applied in quantity to provide full mopping at each insulation board. Point or ribbon mopping not permitted.
 - d. Do not heat mopping bitumen above flash point.
 - e. Hot asphalt bitumen not permitted at membrane installation.
- C. Modified Bituminous Membrane:

1. General: Install two plies of membrane in accordance with manufacturer's requirements with minimum 4 inch side laps and 6 inch end laps.
 2. Unroll membrane over prepared substrate and install using hot-mopped torch-applied method.
 - a. Stagger laps between plies.
 3. Overlap edges and ends of membrane distance as required by manufacturer, and seal as required by manufacturer.
 - a. Stagger end laps 3 feet minimum.
 4. Shingle joints on sloped substrate in direction of drainage. Verify proper direction with manufacturer for high slope areas.
 - a. Torch apply each ply of membrane, applying heat evenly to fully embed membrane; prevent burning membrane and reinforcing. Use hand held torches only. Gang burners and "dragon wagons" not permitted.
 5. Install membrane smooth, free of tears, wrinkles and bubbles. Cut membrane to fit neatly around projections and penetrations.
 6. Surfacing Granules: Apply additional granules in same color as cap sheet granules into hot bitumen bleedout at laps and edges.
 7. Seal membrane to adjoining surfaces.
 8. Continue membrane up vertical surfaces minimum 8 inches above cant top or edge strip unless otherwise noted. Adhere and fasten top of membrane; fasten at frequency as required by manufacturer.
 9. Apply membrane and expansion joint materials to isolate roof into areas as required. Seal roofing membrane sheet to joint flange, apply sealant to edge or seam.
 10. Night Seals: Provide manufacturer's required night seal at end of each day to protect installed roofing from water penetration. Provide detail requirements to Shop Drawings
 11. Do not expose materials vulnerable to water or sun damage in quantities greater than can be weatherproofed during same day.
- D. Membrane Base Flashing: Torch apply two ply flashings at vertical terminations and penetrations in accordance with manufacturer's instructions.
1. Apply base flashing over cants. Avoid application of flashing at 90 degree conditions.
 2. Ensure flashing extends at least 9 inches above termination of roof membrane.
 3. Use longest pieces practicable.
 4. Take measures to ensure flashing is not bridging or ridging where there is change of direction.
 5. Fully adhere flashing cap sheet to flashing base sheet. Ensure 100 percent bond with no voids, gaps, or holidays.
 6. Fasten top of flashing under metal counterflashing using termination bar, fastened at manufacturer's required spacing. Seal counterflashing into, or tight to, wall.
 7. Flash penetrations passing through membrane and counterflash in accordance with manufacturer's instructions.
 8. Install additional ply of base sheet under cap sheet flashing at inside and at outside corners of base flashing, extending minimum 12 inches in both directions.
- E. Flexible Flashing: Set flange of flexible flashing with fleece facing in hot bitumen. Mop fleece - faced flange with second coat of bitumen, and install two-ply base flashing system. Leave area of flexible flashing exposed where not faced with fleece to permit elongation and movement.
- F. Termination Bar: Mechanically fasten termination bar at maximum 8 inches on center to establish compression seal at top of membrane flashing for typical conditions.
- G. Roof Walkway Protection: Lay out walkway in locations as shown.
1. In event locations are not shown, provide walkway in direct route from point of roof access to, and surrounding, roof equipment normally needing servicing.

2. Apply protection surfacing in accordance with manufacturer's requirements.

3.4 FIELD QUALITY CONTROL

- A. Request site attendance of roofing membrane manufacturer technical representative during various stages of installation of roof assembly.
- B. Manufacturer's Field Services: Comply with Section 01 40 00.
 1. Notify manufacturer in timely manner to arrange for manufacturer's technical representative's site visits to ensure proper installation, verify work is in accordance with manufacturer's requirements, and warranty requirements have been met.
 2. Manufacturer's Qualified Technical Representative: Monitor activities and advise applicator of proper installation procedures and precautions.
 3. Minimum Site Visits:
 - a. Pre-construction conference.
 - b. First day of roofing work on site including acceptance of substrate conditions.
 - c. Periodic Visits: Daily during installation.
 - d. Inspection of completed roofing.
 4. Submit reports; include site observations, instructions, and monitoring activities.
- C. Inspection and Testing:
 1. Visually inspect surfaces for voids or ruptures and make necessary repairs.
- D. Testing: Utilize non-destructive electronic leak detection equipment as follows:
 1. Confirm the integrity of membrane and flashing seams at all completed roof areas. Utilize IR Roof scan (or equal) electrical capacitance deck scanner and scan all membrane seams.
 2. Verify the presence of a seam defect using a hand-held Wagner hand-held electrical resistance meter and field mark defects for correction.
 3. At vertical flashing seams, scan all using a hand-held IR Roof scan (or equal) hand held electrical capacitance leak seeker. Field mark any defects encountered.
- E. Correct identified defects or irregularities.

3.5 CLEANING

- A. General: Comply with Section 01 74 00. Periodically remove debris from roof as installation progresses.
 1. Replace work or materials damaged during handling and installation of roofing materials.
 2. Remove foreign matter from finished membrane surface.

3.6 ROOFING INSTALLER'S WARRANTY

- A. WHEREAS [Insert name] of [Insert address], herein called the "Roofing Installer," has performed roofing and associated work ("work") on the following project:
 1. Owner: [Insert name of Owner].
 2. Address: [Insert address].
 3. Building Name/Type: [Insert information].
 4. Address: [Insert address].
 5. Area of Work: [Insert information].
 6. Acceptance Date: [Insert date].
 7. Warranty Period: [Insert time].
 8. Expiration Date: [Insert date].
- B. AND WHEREAS Roofing Installer has contracted (either directly with Owner or indirectly as a subcontractor) to warrant said work against leaks and faulty or defective materials and workmanship for designated Warranty Period,

- C. NOW THEREFORE Roofing Installer hereby warrants, subject to terms and conditions herein set forth, that during Warranty Period he will, at his own cost and expense, make or cause to be made such repairs to or replacements of said work as are necessary to correct faulty and defective work and as are necessary to maintain said work in a watertight condition.
- D. This Warranty is made subject to the following terms and conditions:
1. Specifically excluded from this Warranty are damages to work and other parts of the building, and to building contents, caused by:
 - a. lightning;
 - b. peak gust wind speed exceeding 90 mph;
 - c. fire;
 - d. failure of roofing system substrate, including cracking, settlement, excessive deflection, deterioration, and decomposition;
 - e. faulty construction of parapet walls, copings, chimneys, skylights, vents, equipment supports, and other edge conditions and penetrations of the work;
 - f. vapor condensation on bottom of roofing; and
 - g. activity on roofing by others, including construction contractors, maintenance personnel, other persons, and animals, whether authorized or unauthorized by Owner.
 2. When work has been damaged by any of foregoing causes, Warranty shall be null and void until such damage has been repaired by Roofing Installer and until cost and expense thereof have been paid by Owner or by another responsible party so designated.
 3. Roofing Installer is responsible for damage to work covered by this Warranty but is not liable for consequential damages to building or building contents resulting from leaks or faults or defects of work.
 4. During Warranty Period, if Owner allows alteration of work by anyone other than Roofing Installer, including cutting, patching, and maintenance in connection with penetrations, attachment of other work, and positioning of anything on roof, this Warranty shall become null and void on date of said alterations, but only to the extent said alterations affect work covered by this Warranty. If Owner engages Roofing Installer to perform said alterations, Warranty shall not become null and void unless Roofing Installer, before starting said work, shall have notified Owner in writing, showing reasonable cause for claim, that said alterations would likely damage or deteriorate work, thereby reasonably justifying a limitation or termination of this Warranty.
 5. During Warranty Period, if original use of roof is changed and it becomes used for, but was not originally specified for, a promenade, work deck, spray-cooled surface, flooded basin, or other use or service more severe than originally specified, this Warranty shall become null and void on date of said change, but only to the extent said change affects work covered by this Warranty.
 6. Owner shall promptly notify Roofing Installer of observed, known, or suspected leaks, defects, or deterioration and shall afford reasonable opportunity for Roofing Installer to inspect work and to examine evidence of such leaks, defects, or deterioration.
 7. This Warranty is recognized to be the only warranty of Roofing Installer on said work and shall not operate to restrict or cut off Owner from other remedies and resources lawfully available to Owner in cases of roofing failure. Specifically, this Warranty shall not operate to relieve Roofing Installer of responsibility for performance of original work according to requirements of the Contract Documents, regardless of whether Contract was a contract directly with Owner or a subcontract with Owner's General Contractor.
- E. IN WITNESS THEREOF, this instrument has been duly executed this [Insert day] day of [Insert month], [Insert year].
1. Authorized Signature: [Insert signature].
 2. Name: [Insert name].
 3. Title: [Insert title].

END OF SECTION

SECTION 07 62 10

FLEXIBLE FLASHING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Formed Products: Concealed flashing within wall assemblies to protect and shed incidental water to the exterior.

1.3 PERFORMANCE REQUIREMENTS

- A. General: Flashing and trim assemblies as indicated shall withstand structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.
- B. Thermal Movements: Provide sheet metal flashing and trim that allows for thermal movements from ambient and surface temperature changes.
 - 1. Temperature Change (Range): 120 deg F, ambient; 180 deg F, material surfaces.

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each manufactured product and accessory.
- B. Shop Drawings: Show fabrication and installation layouts of sheet metal flashing and trim, including plans, elevations, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled work.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Do not store flashing materials in contact with other materials that might cause staining, denting, or other surface damage. Store flashing materials away from uncured concrete and masonry.
- B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to the extent necessary for the period of sheet metal flashing and trim installation.

PART 2 - PRODUCTS

2.1 FLEXIBLE FLASHING

- A. Self-Adhesive flexible flashing product consisting of a pliable, adhesive rubberized-asphalt compound, bonded to a high-density, cross-laminated polyethylene film to produce an overall thickness of not less than 40 mils.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Advanced Building Products Inc.; Strip-N-Flash.
 - b. Carlisle Coatings & Waterproofing; CCW-705 Air & Vapor Barrier Strips.
 - c. Grace Construction Products; Perm-A-Barrier Detail Membrane.
 - d. Henry; Blueskin SA.

2.2 THRU-WALL FLASHING

- A. Self-Adhesive thru-wall flashing product consisting of a pliable, adhesive rubberized-asphalt compound, bonded to a high-density, cross-laminated polyethylene film to produce an overall thickness of not less than 40 mils.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Advanced Building Products Inc.; Strip-N-Flash.
 - b. Carlisle Coatings & Waterproofing; CCW-705-TWF Thru-Wall Flashing.
 - c. Grace Construction Products; Perm-A-Barrier Wall Flashing.
 - d. Henry; Blueskin TWF Thru-Wall Flashing.

2.3 HIGH TEMPERATURE FLASHING

- A. Self-Adhering, High-Temperature Sheet: Minimum 30 to 40 mils thick, consisting of slip-resisting polyethylene-film top surface laminated to layer of butyl or SBS-modified asphalt adhesive, with release-paper backing; cold applied. Provide primer when recommended by manufacturer.
 - 1. Thermal Stability: ASTM D 1970; stable after testing at 240 deg F.
 - 2. Low-Temperature Flexibility: ASTM D 1970; passes after testing at minus 20 deg F.
 - 3. Products: Subject to compliance with requirements, provide one of the following:
 - a. Carlisle Coatings & Waterproofing Inc.; CCW WIP 300HT.
 - b. Grace Construction Products, a unit of W. R. Grace & Co.; Ultra.
 - c. Henry Company; Blueskin PE200 HT.
 - d. Owens Corning; WeatherLock Metal High Temperature Underlayment.

2.4 COPPER FABRIC FLASHING:

- A. 5-oz./sq. ft. (1.5-kg/sq. m) copper sheet laminated on both sides with a treated glass fabric, bonded together with asphalt.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Advanced Building Products Inc.; Copper Fabric Flashing.
 - b. Hohmann & Barnard, Inc.; H & B C-Fab Flashing.
 - c. Sandell Manufacturing Co., Inc.; Copper Fabric Flashing.
 - d. York Manufacturing, Inc.; Multi-Flash 500.

2.5 MISCELLANEOUS MATERIALS

- A. General: Provide materials and types of fasteners, separators, sealants, and other miscellaneous items as required for complete metal flashing installation and recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.
- B. Elastomeric Sealant: ASTM C 920, elastomeric silicone polymer sealant; low modulus; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, to verify actual locations, dimensions and other conditions affecting performance of the Work.
 - 1. Verify compliance with requirements for installation tolerances of substrates.
 - 2. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
- B. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 FLASHING INSTALLATION

A. General: Install as indicated on Drawings and per Manufacturer's recommendations.

B. Self-Adhering Sheet Flashing: Install self-adhering sheet flashing wrinkle free. Apply primer if required by flashing manufacturer. Comply with temperature restrictions of flashing manufacturer for installation. Apply in shingle fashion to shed water, with end laps of not less than 6 inches staggered 24 inches between courses. Overlap side edges not less than 3-1/2 inches. Roll laps with roller. Cover flashing with subsequent construction within 14 days.

C. Location:

1. Flexible Flashing: As indicated on drawings, or at all exterior windows, doors or other penetrations where high temperature flashing is not required.
2. Thru-Wall Flashing: As indicated on drawings, or at all masonry cavity wall conditions requiring flashing (head of windows, doors, openings, shelf angles, base of wall).
3. High Temperature Flashing: As indicated on drawings, or at all locations where flashing will be in contact with metal coping or metal panels where high temperatures exist.
4. Copper Laminated Flashing: As indicated on drawings, or at masonry cavity wall base of wall conditions (sill and lintel).

END OF SECTION

SECTION 07 71 00

ROOF SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Preformed roof trim.
 - 2. Roof curbs and supports.
 - 3. Pipe portals.
 - 4. Roof hatches.
- B. Related Requirements: Comply with following:
 - 1. Metal Finishes: Section 05 05 10.

1.2 SYSTEM DESCRIPTION

- A. Curbs, Equipment Supports, Pipe Roller Supports, and Pipe Portals:
 - 1. Design Requirements: Manufacturer is responsible for designing units, including anchorage to structural system and necessary modifications to meet specified requirements.
 - 2. Structural Requirements: Design to support load of curb-mounted equipment.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit for each product.
 - 1. Indicate profiles, anchorages, jointing details, flashings, and accessories.
 - 2. Include color charts for finish indicating manufacturer's standard colors available for selection.
 - 3. Include sample of warranty customized for this project.
- C. Shop Drawings: Indicate typical layout including dimensions, configuration, locations, interface with adjacent systems, clearances, tolerances, frequency of attachment, and fabrication details.
 - 1. Submit detail drawings of transitions, intersections, and connections.
 - 2. Submit detail drawings of accessory components not included in manufacturer's product data.
- D. Certifications:
 - 1. Certification Letter: ANSI SPRI ES1 for all parapet caps and roof termination metal.
- E. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Manufacturer's instructions.
- F. Closeout Submittals: Submit following in accordance with Section 01 78 00.
 - 1. Warranty: Specified warranty.

1.4 QUALITY ASSURANCE

- A. Single Source Responsibility: Furnish each product from one manufacturer, unless otherwise acceptable to Architect.
- B. Certifications: Submit following:
 - 1. Manufacturer's certification that products furnished for Project meet or exceed specified requirements.

2. Contractor's and installer's certification that products are installed in accordance with Contract Documents.
- C. IBC 2009/1504.5: Use the current edition of ANSI/SPRI ES-1 Wind Design Standard for Edge Systems Used with Low Slope Roofing Systems. All configurations, gauges, attachments and other items shall meet or exceed testing and design criteria related to ES-1 and IBC 1504.5.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
 1. Stack materials to prevent twisting, bending and abrasions, and to provide ventilation.

1.6 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 78 00.
 1. Warrant installed roof curbs, roof expansion joint covers, fascia, and copings to be free from defects in material and workmanship for time period to match roof system specified in Section 07 52 00 .

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Preservative Pressure Treated Wood: Softwood lumber treated in accordance with AWPAC2 for above grade use.
- B. Bituminous Paint: SSPC Paint 12
- C. Roofing Cement: ASTM D4586, Type I.

2.2 PREFORMED ROOF TRIM

- A. Preformed Fascia:
 1. Acceptable Manufacturers:
 - a. WP Hickman Company, Asheville, NC 28803.
 - b. Metal Era, Inc., Waukesha, WI 53188.
 - c. Peterson Aluminum Corporation, Elk Grove Village, IL 60007.
 - d. Accepted Substitute in accordance with Section 01 25 00.
 2. Description: Metal water dam with fascia.
 - a. Provide with continuous clamp mechanically fastened to be secured and engaged against roofing membrane.
 - b. SPRI ES-1 wind uplift rating.
 - c. Provide with flow over scupper.
 - d. Dam: 24 gage commercial quality galvanized steel.
 - e. Fascia: 0.063 inch] thick aluminum.
 - f. Provide with splice plates to conceal and weatherseal joints between sections of dam and fascia.
- B. Preformed Coping:
 1. Acceptable Products and Manufacturers:
 - a. Permasnap Coping, WP Hickman Company, Asheville, NC 28803.
 - b. Anchor-Tite Coping, Metal Era, Inc., Waukesha, WI 53188.
 - c. Tite-Loc Coping, Peterson Aluminum Corporation, Elk Grove Village, IL 60007.
 - d. Accepted Substitute in accordance with Section 01 25 00.
 2. Description:
 - a. Anchor PlateAnchor Plate/Cleat: 20 gage thick commercial quality galvanized steel or manufacturer's standard aluminum.

- b. Coping: 0.063 inch thick 0.063-inch-thick aluminum, smooth surface.
 - c. Gutter/Splice Plate: Anchor cleat with integral drainable gutter or manufacturer's standard closed cell composition compressible material gasket between anchor plate and coping finish top match coping.
 - d. SPRI ES-1 wind uplift rating.
 - e. Slope coping towards roof.
 - f. Provide gutter/splice plates at joints between sections of coping.
- C. Splice Plate: Anchor scupper to conductor head for water-tight fabrication.
- D. SPRI ES-1 wind uplift rating.
- E. Sump scuppers; coordinate with roof system, including tapered insulation.
- F. Fabrication: Form sections true to shape, accurate in size, square, free from distortion and defects, from distortion and defects to profiles indicated.
- 1. Shop fabricate fabricates intersections, inside corners, and outside corners with miters welded in factory prior to finishing.
- G. Accessories: Provide fasteners, clamping devices, splice strips adhesives, and sealants required by system for proper performance and as recommended by manufacturer.
- H. Finish:
- 1. Fluoropolymer (PVDF) Coating: In accordance with Section 05 05 10.
 - 2. Clear Anodized Finish: In accordance with Section 05 05 10.

2.3 ROOF CURBS AND SUPPORTS

- A. Acceptable Insulated Roof Curb Products and Manufacturers:
- 1. Style PC-5 by The Pate Company, Lombard, IL.
 - 2. Model RC-4A by Roof Products & Systems Corporation, Bensenville, IL.
 - 3. Model TC-3 by ThyCurb Division of Thybar Corporation, Addison, IL.
 - 4. Accepted Substitute in accordance with Section 01 25 00.
- B. Acceptable Insulated Expansion Curb Products and Manufacturers:
- 1. Style EJ-5 by The Pate Company, Lombard, IL.
- C. Model EC-4A by Roof Products Systems Corporation, Bensenville, IL.
- D. Model TXJ-1 by ThyCurb Division of Thybar Corporation, Addison, IL.
- E. Accepted Substitute in accordance with Section 01 25 00.
- F. Acceptable Equipment Support Rail Products and Manufacturers:
- 1. Style ES-5, The Pate Company, Lombard, IL.
 - 2. Model ER-4A, Roof Products Systems Corporation, Bensenville, IL.
 - 3. Model TEMS-1, ThyCurb Division of Thybar Corporation, Addison, IL.
 - 4. Accepted Substitute in accordance with Section 01 25 00.
- G. Acceptable Pipe Roller Support Products and Manufacturers:
- 1. Style PRS-5, The Pate Company, Lombard, IL.
 - 2. Model DMP, Roof Products Systems Corporation, Bensenville, IL.
 - 3. Model TEMS-1 with pipe roller assembly, ThyCurb Division of Thybar Corporation, Addison, IL.
 - 4. Accepted Substitute in accordance with Section 01 25 00.
- H. General:

1. Sheet Metal: Structural quality galvanized steel, thickness to suit spans and imposed loads with corner seams mitered and welded.
2. Insulation: 3 PCF density, rigid glass fiber board, minimum 1-1/2 inches thick.
3. Wood Nailers and Grounds: Preservative pressure treated wood, minimum 2 inch cross sectional dimensions.
4. T12 inches minimum above elevation of finished roofing, except where indicated otherwise.
5. Built-in Cant: 3 inch wide; 45 angleangles.
6. Fabricate with bottom edge of built-in cant raised above roof deck surface to accommodate roof deck insulation.
7. Roof Curbs: Accommodate curb-mounted equipment ~ and pipe portals; coordinate requirements prior to fabrication.
8. Pipe Rollers Supports: Provide with single channel atop curb to support threaded rods holding pipe rollers.
9. Size pipe rollers to accommodate pipe sizes.

2.4 PIPE PORTALS

- A. Acceptable Products and Manufacturers:
 1. PCC Cap, The Pate Company, Broadview, IL.
 2. Pipe Portal System, Roof Products Systems Corporation, Bensenville, IL.
 3. TP-2 Piping Cover, ThyCurb Division of Thybar Corporation., Addison, IL.
 4. Accepted Substitute in accordance with Section 01 25 00.
- B. Description: Manufacturer's standard ABS and EPDM rubber boots to accommodate 3/8 through 6 inch6-inch diameter pipe.
 1. Furnish complete with stainless steel hose clamps.
 2. Accommodate quantity and size of piping to pass through portal caps.
 3. Fabricate for mounting atop manufacturer's curb.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 1. Verify that deck, curbs, blocking, cants, roof membrane, and base flashing are in place and positioned correctly.
- B. Preformed Fascia, Gravel Guard, and Coping: Verify that coverage onto vertical finish materials is sufficient to result to watertight installation.
 1. Verify membrane terminations and base flashings are in place, sealed, and secure.

3.2 INSTALLATION

- A. Roof Specialties and Accessories: Install units plumb, level, square, and free from warp or twist while maintaining dimensional tolerances and alignment with surrounding construction.
 1. Apply bituminous paint on metal surfaces of units in contact with cementitious materials and dissimilar metals.
 2. Securely anchor roof accessories to supporting substrates with appropriate type fasteners.
 3. Coordinate with installation of roofing system, tapered insulation, roof slope, and related flashings.
- B. Expansion Joint Covers: Install as indicated and in accordance with NRCA MB-9A
 1. Where installed over asphaltic roof membrane and base flashings set metal flanges in full bed of roofing cement.
 2. Provide in continuous pieces without splices where ever possible.

- a. Where required, splice sections in accordance with manufacturer's instructions producing watertight installation.
- 3. Utilize factory fabricated intersections and transitions wherever possible. Field fabricate/fabricates where pre-manufactured sections are not available.
- 4. Maintain uniform profile; do not stretch or compress bellows.
- C. Preformed Fascia/Preformed Fascia and Coping: Secure in place with concealed fasteners using methods as recommended by manufacturer to comply with FM rating.
 - 1. Seal joints watertight.
- D. Preformed Through-Wall Scuppers, Conductor Heads, and Downspouts: Install watertight in accordance with manufacturer's requirements. Allow for expansion and contraction, including at roof-to-wall interface. Accommodate movement at outside face of wall of wall, not between wall and roof. Extend parapet cap membrane under through-wall scupper for continuous, watertight substrate.
- E. Parapet Cap Membrane:
 - 1. Clean substrate of dirt, dust, and materials which may impair adhesion.
 - 2. Apply primer, when required, in accordance with manufacturer's requirements.
 - 3. Apply to top of parapet wall under coping and gravel stops.
 - 4. Turn membrane down exterior wall face and parapet wall face 2 inches.
 - 5. Install without fishmouths and wrinkles.
 - 6. Press tape into firm contact with substrate.
 - 7. Lap tape ends minimum of 2 inches.
- F. Roof Curbs: Integrate curbs with adjacent roofing systems, base flashings, and counter flashings to create watertight conditions.
- G. Roof Hatches: Secure flanges to deck by bolting or welding.
 - 1. Access Roof Hatch for Ladders: Orient hatch with opening side and operating hardware location close to top of ladder.
 - 2. Signage: Provide permanent signage with minimum 1 inch high letters at top surface and underside surface of hatch cover to read: KEEP HATCH CLOSED WHEN NOT IN USE. Comply with CABO/ANSI A117.1 requirements.

3.3 ADJUSTING AND CLEANING

- A. Adjusting: Adjust roof hatch covers and hardware for smooth, uniform operation.
- B. Cleaning: Comply with Section 01 74 00. Clean as recommended by manufacturer. Do not use materials or methods which may damage finish or surrounding construction.
 - 1. Clean primer, adhesive, flashing cements, and other products from surfaces, exposed sheet metal and bellows.

END OF SECTION

SECTION 07 84 00

FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Firestop devices and systems tested in accordance with ASTM E814 (ANSI/UL 1479) and listed in UL Fire Resistance Directory.
2. Fire resistant construction joints.
3. Duct and damper firestops.
4. Dynamic partition head details.
5. Penetrations through fire-rated floors, walls, and shafts.

B. Related Sections:

1. Section 01 73 29 - Cutting and Patching; Repair of openings with original materials.
2. Section 07 21 00 – Thermal Insulation: Curtain wall insulation.
3. Section 09 29 00 - Gypsum Board
4. Division 21 – Fire Suppression.
5. Division 22 – Plumbing.
6. Division 23 – Heating, Ventilating, and Air Conditioning.
7. Division 26 - Electrical.

1.2 DEFINITIONS

- A. See Section 01 42 00 for typical definitions.
- B. Annulus or Annular Space: Opening around penetrant.
- C. Assembly: Condition requiring firestop system, fire containment, smoke barrier, fire safing system, or fire-resistive joint system.
- D. Compartmentalization or Fire Area: The aggregate area or floor area enclosed and bounded by building separation walls, fire barrier walls, exterior walls, or fire-resistance rated horizontal assemblies of a building.
- E. Combustible: Penetrant which burns or melts when exposed to flame.
- F. DWV: Drain, waste, or vent pipes.
- G. Engineering Judgments: To meet actual field conditions, manufacturers make recommendations based on available testing that approximate the condition encountered. Testing laboratories verify judgments by manufacturers for validity.
- H. Fill Material: Firestopping material which seals opening, often used with Forming Material.
- I. Fire Safing System: Perimeter fire containment systems for terminating fire-rated horizontal decks into non-fire-rated exterior curtain wall or spandrel conditions. Specific construction consisting of floor assembly with hourly fire endurance rating, exterior curtain wall or spandrel with no hourly fire endurance rating, and fill material, such as Fire Safing with flexible coating installed between the floor and the curtain wall or spandrel to prevent vertical spread of fire and smoke. Hourly ratings only apply to complete systems.
- J. Firestop System: Tested system of maintaining flame spread rating of fire-rated walls and floors by filling penetrations, joints, and gaps with firestopping and forming materials.
- K. Forming Material: Damming material used prior to application of fill material to substantially fill penetration or gap.

- L. Fire-resistive Joint System: An assemblage of specific materials or products that are designed, tested and fire rated in accordance with ASTM E 119 to resist, for a prescribed period of time, the spread of fire through joints made in or between fire rated assemblies.
- M. Gap: Linear opening in wall or in ceiling assemblies created by abutting two surfaces together. May be vertical or horizontal. Dynamic gaps occur at moving construction joints. Static gaps occur at joints without movement.
- N. Joint: The linear opening in or between adjacent fire-resistance rated assemblies that is designed to allow independent movement of the building or building portion, in any plane, caused by thermal, seismic, wind, or other loading.
- O. Passive Fire Protection: Act of preventing fire from spreading from one area to another using containment methods such as Firestopping Systems. Typically used in conjunction with Preventative Fire Protection methods, such as use of non-combustible or fire-retardant-treated materials, and Active Fire Protection such as extinguishing systems which suppress fires.
- P. Penetrant: Item penetrating wall or floor assembly, such as pipes, tubes, conduits, cabling, ducts, cable trays, receptacles, switches.
- Q. Penetrations: Assembly requiring firestopping. Types of Penetrations include:
 - 1. Through Penetration: Penetrants run completely through rated wall or floor.
 - 2. Membrane Penetration: Penetrants run through one side of rated wall or floor.

1.3 SYSTEM DESCRIPTION

- A. General: Make firestop and smoke seal assembly selections that comply with ETL, UL Fire Resistance Directory, or other nationally recognized testing agency or laboratory, AHJ, and applicable codes for:
 - 1. Materials, fabrication, and installation of Firestop Systems.
 - 2. Penetrations.
 - 3. Gaps.
- B. Firestop Penetrations, Joints, and Gaps as follows:
 - 1. At Floors: Firestop from side where flame spread is anticipated, such as Top-Down applications and Bottom-Up applications.
 - 2. At Walls: Firestop both sides of wall as potential for flame spread is equal on both sides.
 - 3. Openings between floor slab edges and exterior wall spandrel conditions.
 - 4. Gaps such as openings and cracks at abutting fire rated assemblies and components, such as wall-to-wall and wall-to-floor conditions, and overhead floor and roof decks.
 - 5. Penetrations into or through fire rated floors and walls.
 - 6. Other locations indicated or scheduled.
- C. Design Requirements:
 - 1. Firestop materials used to fill floor openings in which smallest dimension is 4 inches shall support same loads that floor was designed to support. If equal floor loading capacity cannot be obtained with firestop material, provide fire rated permanent covering to support loads and traffic, capable of being removed to allow access.
 - 2. Design firestopping at dynamic gaps to accommodate cyclical movement in accordance with ASTM E1399.
 - 3. Design grease duct firestopping in accordance with UL 1978.
 - 4. Insulated Piping and Duct Penetrations: Install firestop systems tested for use with type of insulation on penetrating item.
 - a. If compatible firestop system is unavailable, remove insulation at contact area with firestop material.
 - b. Coordinate with trades who installed insulation to ensure proper re-sealing of cut edges of insulation.

5. Encapsulate friable materials, such as ceramic fiber, in reinforced aluminum facing material.
6. Provide Products that Do Not Deteriorate when Exposed to Following Conditions:
 - a. Plumbing and Wet-Pipe Sprinkler Systems: Moisture-resistant through-penetration firestop.
 - b. Fire Safing Systems: Moisture-resistant system designed to withstand dynamic movement. Air erosion.
 - c. Ultraviolet light.
- D. Exposed to View: Flame-spread value of less than 25 and smoke-developed value of less than 450, ASTM E84.
 1. Compatible with applied finishes.
- E. F and T Rating Requirements: Conform to F and T ratings, ASTM E814 (ANSI/UL 1479).
 1. Comply with applicable codes and authority having jurisdiction.
 2. F Ratings: Prevent passage of flame; Equal to fire resistance rating of assembly being penetrated but not less than one hour.
 3. T Ratings: Temperature rise equal to fire resistance rating of assembly being penetrated but not less than one hour, except as otherwise required by Code or by authority having jurisdiction.
- F. Testing Requirements: Utilize systems and materials tested and approved by UL or other nationally recognized independent testing agency acceptable to authorities having jurisdiction.
 1. Determine fire ratings in accordance with ASTM E814 (ANSI/UL 1479) for through penetration firestops, ASTM E119 (UL263) for fire rated assemblies, and as required by applicable codes and authority having jurisdiction.
 2. Provide typical dynamic assemblies complying with ASTM E1399 and UL 2079 for fire rated assemblies exposed to movement such as: head of wall joints; floor to floor joints; floor to wall joints; wall to wall joints; undersides of metal decks; tops of walls; undersides of composite decks; and fire-rated control, construction, and expansion joints.
 3. At Fire Safing Systems, provide assemblies tested and evaluated using assimilation of NFPA 285 and UL 2079. Engineering Judgments for systems based solely on UL 2079 not allowed if tested assembly from other manufacturers available.
- G. Large openings and annular spaces may be closed with same type construction as adjacent floor, roof, and wall assembly.
- H. Sealing around penetrations in fire rated assemblies without approved firestop system not allowed. Methods and materials not allowed include but are not limited to:
 1. Joint compound at gypsum board assemblies.
 2. Mortar at masonry and concrete assemblies.
 3. Non-fire-rated joint sealants.
- I. Whenever finished firestop materials are scheduled to receive finish paint or other coatings, test compatibility of firestop materials with coatings to be applied. Ensure that paint or coating will not fail due to joint movement.

1.4 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Provide manufacturer's certification stating:
 1. Each penetration of fire rated walls and floor, partition heads, and edge of slabs will be firestopped with a firestopping system tested by UL or other recognized testing agency for substrate and penetrating item.
 2. Authorities having jurisdiction have approved firestopping systems for this project.
 3. Products and Classifications Schedule:

- a. Provide tabular form schedule for firestops, fire containment, and fire resistant construction joints.
- b. Schedule to identify:
 - 1) Construction penetrated including fire resistance rating.
 - 2) Penetrating item.
 - 3) Products and manufacturers included in each system.
 - 4) Form material used.
 - 5) Firestop classification and description from UL or other nationally recognized independent testing agency acceptable to authority having jurisdiction.
 - 6) Fire containment and fire resistant construction joint description.
- C. F and T ratings.
 - 1. Update schedule periodically to include addition and changes.
- D. Informational Submittals: Submit following, packaged separately from other submittals:
 - 1. Certifications specified in Quality Assurance article.
 - 2. Qualification Data: Manufacturer's and installer's qualification data.
 - 3. Manufacturer's instructions.
 - 4. Inspection Reports: ASTM E2174 as modified in this Section.
 - 5. Product Data: Submit for each product.
 - 6. Manufacturer's field reports.
- E. Closeout Submittals: Submit following in accordance with Section 01 77 00.
 - 1. Project Record Documents: Submit final completed Products and Classifications Schedule to reflect installed firestopping in accordance with Section 01 78 39.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing Products specified in this Section with minimum five years' experience.
 - 1. Manufacturer: Provide on-site training of installers and field services. Manufacturers who do not provide on-site training and field services not allowed.
- B. Installer Qualifications: Single company for work of entire Project specializing in installation of Firestopping systems specified, licensed, trained and approved by manufacturers of firestop systems and materials submitted, with documented experience on at least five projects of similar nature in past five years.
- C. Installer Qualifications: Single company for work of entire project; FM Certified Firestop Installer in accordance with FM Approval Standard 4991.
 - 1. Certify that Installer, including superintendent and workers, has successfully completed training course by manufacturer of systems and products.
 - a. Course Includes:
 - 1) Training for selecting proper firestopping materials and systems for project specific needs.
 - 2) Code requirements for applicable conditions.
 - 3) Hands-on application and installation techniques for each product and system. This requirement may be conducted in conjunction with field sample work.
 - 4) Completion of course: Not later than 3 months preceding commencement of Work.
- D. Installer Responsibility: Select firestop, fire containment, fire safing system, and fire resistant construction joint products based on tested assemblies for each penetration, gap, and joint and schedule at end of section.
 - 1. Consult with manufacturer on situations requiring engineering judgements.

2. Obtain approval of authorities having jurisdiction for selected methods where required by Code.
 3. Submit Schedule, approved by independent inspection agency providing Field Quality Control.
- E. Regulatory Requirements: Ensure firestop, fire containment, and construction joint components comply with applicable portions of local, state, and federal codes, laws, and ordinances for flame spread and smoke developed indices.
- F. Engineering Judgments: Comply with IFC Guidelines for Evaluating Firestop Systems Engineering Judgments. Do not submit Engineering Judgments when tested assemblies are available from other manufacturers.
- G. Testing and Inspection Agency: Section 01 45 29. Hired by Contractor, to provide services specified.
- H. Certifications:
1. Installer's documentation showing compliance with certification, training, approval, and licensing requirements.
 2. Contractor's and installer's certification that products are installed in accordance with Contract Documents, based on inspection and testing specified as part of Field Quality Control.
 3. Certificates of compliance from authority having jurisdiction indicating approval of firestops, fire containments, and construction joints.
 4. Certificate of inspection and acceptance by authority having jurisdiction of firestops, fire containments, and construction joints.

1.6 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 19.
- B. Agenda: Include discussion and agreement on following:
1. Product and Classification Schedule.
 2. Current certified training of Installer by firestopping manufacturer. Test firestop materials to confirm compatibility with adjacent materials and chemicals and solvents with which they may come into contact during construction.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
1. Packing, Shipping, Handling, and Unloading: Deliver materials in manufacturer's unopened containers with manufacturers name, product identification, lot number, UL labels or labels of other nationally recognized independent testing agency, and mixing and installation instructions.
 2. Storage and Protection: Store materials to prevent deterioration and damage due to moisture, temperature change, and contamination.

1.8 PROJECT CONDITIONS

- A. Environmental Requirements:
1. Comply with manufacturer's temperature and humidity limitations before, during, and after installation.
 2. Comply with ventilation requirements specified in Section 01 50 00.

1.9 SEQUENCING

- A. Sequence Work under provisions of Section 01 10 00.
1. Identify assemblies, penetrations, and openings requiring firestops, fire containments, and construction joints.

2. Schedule installation of firestopping after completion of work involving penetrating items, but prior to covering, concealing, and eliminating access to penetrations.
 3. Coordinate with work of other trades
- B. Inspection: Request inspection of firestops by authority having jurisdiction and testing consultant before concealment.
1. Sequence work to permit installation to be inspected and approved prior to being concealed.
 2. Ensure that subsequent openings and penetrations are reported, properly firestopped, and inspected.

PART 2 - PRODUCTS

2.1 FIRESTOPPING SYSTEMS

- A. Firestopping Devices and Systems: Products of one manufacturer. Products from multiple manufacturers not allowed without specific approval.
1. Engineering Judgements: Not accepted at assemblies where tested systems from other manufacturers are available.
- B. Acceptable Manufacturers.
1. 3M Fire Protection Products, St. Paul, MN.
 2. Hilti, Tulsa, OK.
 3. The Rectorseal Corp., Houston, TX.
 4. Specified Technologies Inc., Sommerville, NJ.
 5. Accepted Substitute in accordance with Section 01 25 00.
- C. Acceptable Edge of Slab Firestop Manufacturers:
1. Specified Technologies Inc., Sommerville, NJ.
 2. 3M Fire Protection Products, St. Paul, MN.
 3. Hilti, Tulsa, OK.
 4. Accepted Substitute in accordance with Section 01 25 00.
- D. Materials: Provide materials and products as required by manufacturer, UL or other testing agency and classification to provide firestopping systems for each specific application.
- E. Accessories: Provide accessories as required by manufacturer, UL or other testing agency and classification to provide firestopping systems for each specific application, including:
1. Retaining Collars: Manufacturer's standard.
 2. Steel wire, wire mesh, clips, sleeves, anchoring devices, primers, and other materials.
 3. Metal Sheets and Shapes: Size and thickness as required by fire resistant system.
 4. Fibrous Fire Safing Adhesive: As instructed by manufacturer.
 5. Fibrous Fire Safing Clips/Fasteners: As instructed by manufacturer.
 6. Sealant Primers: As instructed by manufacturer.
 7. Sealant Damming Materials:
 - a. Non-combustible.
 - b. Chemically compatible with sealant.
 - c. Mineral fiberboard, mineral fiber matting, or fibrous fire safing.
 8. Cleaning Solvents: As instructed by manufacturer.
 9. Labels:
 - a. Provide label for each firestop condition.
 - b. Type information in non-fading ink on 20 pound (minimum) paper.
 - c. Include following information on each label:

- 1) Manufacturer's name.
- 2) Product name.
- 3) Product type (sealant, putty, mortar, or other generic material description).
- 4) F-Rating.
- 5) T-Rating. State when not required for condition.
- 6) Testing and listing agency filing number, such as UL System number.

F. Description:

1. Sealant, putty, or mortar material.
 2. Non-corrosive and compatible with synthetic cable jackets.
 3. Flame spread less than 25 when tested according to ASTM E84.
 4. Mixes: If mixing is required, mix components as instructed by manufacturer.
 5. Top of partition assemblies: Combination of safing insulation and flexible fire rated smoke seal tested and approved for dynamic movement complying with ANSI/UL2079 Test for Fire Resistance of Building Joint Systems (cyclic test).
 6. Provide typical dynamic assemblies complying with ASTM E1399 and UL 2079 for fire rated assemblies exposed to movement such as: head of wall joints; floor to floor joints; floor to wall joints; wall to wall joints; undersides of metal decks; tops of walls; undersides of composite decks; and fire-rated control, construction, and expansion joints.
- G. Adhesives & Sealants: Only use adhesives and sealants in the interior of the building that meet or do not exceed the VOC limits of the CURRENT requirements of South Coast Air Quality Management District (SCAQMD) Rule No. 1168 on the interior of the building.
1. Current requirement refers to the date on which the materials are installed in the building.
 - a. SCAQMD Rule #1168 is referenced in section 01 81 13 that was current as of the date of this specification. Refer to <http://www.aqmd.gov/rules> for the actual current version of the rule that will be applicable at the date of installation during construction.
 - b. Interior refers to all building construction that is inside of the exterior weatherproofing material.

2.2 FIRE SAFING SYSTEMS

- A. Fire Safing Systems: Combination of products listed in submitted system testing design number.
- B. Materials: Provide forming materials and containment products as required by UL or other testing agency and classification to provide perimeter fire containment systems for each specific application.
1. Seal at Fire Safing System: Water-tight and capable of withstanding movement; tested as part of fire-rated assembly.
- C. Accessories: Provide clips, netting, mesh, and other accessories as required by assembly design, material manufacturers, UL or other testing agency, and classification to provide perimeter fire containment systems for each specific application.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
1. Verify that permanent penetration items have been installed and that temporary penetrating items have been removed.
 2. Verify that supports have been installed on both sides of penetrated construction as required by UL or other testing agency classifications.
 3. Inspect and verify that surfaces and condition of openings have no defects that could interfere with installation and performance of firestop materials.
 4. Verify sleeves installed under Divisions 21, 22, 23, and 26 are properly installed.

3.2 PREPARATION

- A. General: Clean surfaces of opening substrates free of dirt, oil, grease, loose and harmful materials which may adversely affect bond of materials to surfaces in accordance with manufacturers recommendations.
 - 1. Test surfaces which have been previously painted, sealed, and treated with other coatings and compounds to ensure compatibility with materials and proper bond capability.
 - 2. Remove incompatible coatings and materials which may affect firestop bond with surrounding surfaces.
 - 3. Mask and protect adjacent surfaces from damage.
 - 4. Prime surfaces as instructed by manufacturer.

3.3 FIRESTOPPING INSTALLATION

- A. General: Install in accordance with Section 01 73 00, manufacturer's details, applicable codes, approved Schedule, and UL or other testing agency classification requirements.
 - 1. Fire resistant systems without UL or other testing agency classification requirements shall be approved by authorities having jurisdiction before installation.
 - 2. Install firestopping material in manner required to achieve F rating and T rating required by UL classification, applicable codes, and authorities having jurisdiction.
 - 3. Install firestopping material with sufficient pressure to ensure uniform density and texture, and to ensure proper filling and sealing of openings to create smoke seal.
 - 4. Install forms and supports to arrest liquid and flowable material leakage and retain materials in openings.
 - 5. Remove form materials after firestopping material has cured unless materials used are permitted or required to remain according to test classifications.
- B. Through Penetration Firestopping Systems: Comply with classification design requirements. Separate cables not in conduit and maintain required separation of penetrating items from edges of openings and from each other.
 - 1. Tool and trowel exposed surfaces to smooth finish, flush with surrounding surfaces unless otherwise required by test classification.
 - 2. Remove excess firestop material promptly as work progresses.
- C. Through Penetration Firestopping: Securely attach device frames to supporting construction.
 - 1. Assembly component parts to ensure proper contact and sealing of gaps and openings around penetrating items.
- D. Construction Joint Firestopping: Provide fire resistant systems to match fire rating of adjacent construction. Comply with ASTM E1399 and UL 2079 for joints requiring movement and deflection capabilities.
 - 1. Provide fire resistant systems at following locations:
 - a. Voids and gaps in fire rated construction, including control joints and gap at top of fire-rated CMU walls.
 - b. Fire rated partition and metal deck flutes.
 - c. Changes in partition material.
 - d. Floor joints not requiring expansion joint.
 - e. Other locations indicated and required by applicable codes.
- E. Curtain Wall Fire Containment, Foil Faced Safing Insulation:
 - 1. Fill voids between curtain wall and edge of slabs at floors and roofs in accordance with manufacturer's instructions. Do not leave voids in safing.
 - 2. Tape and seal tears and cuts in facing.
 - 3. Seal joints with manufacturer's recommended sealant.

4. Provide flexible fire rated smoke seal tested and approved for dynamic movement.

3.4 FIRE SAFING SYSTEMS INSTALLATION

- A. Fire Safing System: Fill voids between curtain wall and edge of slabs at floors and roofs in accordance with tested assembly requirements and with manufacturer's instructions. Do not leave voids.
 1. Forming Material: Extend minimum 36 inches vertically at spandrel.
 2. Secure in place with tested accessories.
 3. Tape and seal tears and cuts in forming material facing.
 4. Seal joints with material required by submitted testing assembly.

3.5 FIELD QUALITY CONTROL

- A. Manufacturer's Field Services: Comply with Section 01 40 00.
 1. Notify manufacturer in timely manner to arrange for manufacturer's technical representative's site visits to ensure proper installation, verify work is in accordance with manufacturer's requirements, and warranty requirements have been met.
 2. Manufacturer's Qualified Technical Representative: Monitor activities and advise applicator of proper installation procedures and precautions.
- B. Minimum Site Visits:
 1. Pre-construction conference.
 2. On-site training of each installer.
 3. Preparation of field samples.
 4. First day of fire stopping work on site.
 5. Periodic visits as required to properly monitor fire stopping work.
 6. Times of Independent Testing Consultant inspections.
 7. Inspections of completed work before concealment.
- C. Submit reports; include site observations, instructions, and monitoring activities.
- D. Site Tests and Inspections: Comply with Section 01 45 29.
- E. Inspection: Contractor will engage and pay for services of independent testing consultant to perform quality control testing and inspection.
 1. Field inspections: Provide testing and certification of firestopping, fire containments, and fire resistant construction joints.
 2. Do not conceal firestopping, fire containments, and fire resistant construction joints prior to required testing and inspection.
 3. Notify Architect, authority having jurisdiction, and designated inspectors of work released for inspection.
- F. Inspection and Testing Requirements: ASTM E2174.
 1. Review: Review Schedule prepared by Firestopping Installer. Certify that approved Schedule addresses applicable, project-specific conditions and provides acceptable systems and products for each condition encountered.
 2. Inspection: Visually examine firestopping, fire containments, and fire resistant construction joints to verify compliance with Contract Documents.
 - a. Examine assembly for proper installation, adhesion, and curing appropriate for each material.
 3. Testing Requirements: Test 10 percent of firestops, fire containments, and fire resistant construction joints but not less than one of each type.
 - a. Measure installed thickness of fire retardant material.
 4. Submit written inspection and testing report including following information:

- a. Location of assembly such as Room Number, Floor Level, and Column Grid Coordinates, or other designator approved by Architect. When using Room Number as designator, identify direction and whether assembly is located in wall, ceiling, or floor.
 - b. Type of construction penetrated including fire resistance rating.
 - c. Penetrating item.
 - d. List of generic descriptions and product names and manufacturers included in each system including form material, containment system, such as those used at gang assemblies or to control size of annular space, and sealer, topcoat, or intumescent materials.
 - e. Listing or Design Number for assembly classification, including description, from UL or other nationally recognized independent testing agency acceptable to authority having jurisdiction.
 - f. F and T ratings.
 - g. State whether assembly is or is not in full compliance with testing agency classification, description and manufacturer's requirements. If variations occur, confirm acceptance of variation by manufacturer and authority having jurisdiction.
- G. Re-examine firestopping, fire containments, and fire resistant construction joints immediately prior to concealment by other construction to ensure no damage has occurred since initial inspection.
- H. After correction of unacceptable firestopping, fire containments, and fire resistant construction joints, provide additional inspection and testing, to verify compliance with this Section, at no additional cost to Owner.

3.6 REPAIRS AND MODIFICATIONS

- A. Repair or remove and install new materials at inspected, unacceptable fire stopping, fire containments, and fire resistant joints. Request additional inspection upon completion of corrective work.
- B. Identify damaged and re-entered seals requiring repair and modification.
 - 1. Remove loose and damaged materials.
 - 2. If penetrating items are to be added, remove enough material to permit penetration by new elements, being careful not to damage balance of seal.
 - 3. Repair holes, cracks, and damage in accordance with manufacturer's instructions to ensure complete smoke seal.
 - 4. Use only materials approved by manufacturer of original seal as suitable for repair.

3.7 CLEANING AND PROTECTION

- A. General: Comply with Section 01 74 00. Clean as instructed by manufacturer. Do not use materials or methods which may damage firestop or surrounding construction.
 - 1. Remove stains and correct damage to adjacent surfaces.
- B. Protect finished work in accordance with Section 01 73 00.
 - 1. Protect material subject to traffic from damage.

3.8 THROUGH-PENETRATION FIRESTOP SYSTEM SCHEDULES

- A. Systems Listed Using Alpha-Alpha-Numeric Identification System in UL's Fire Resistance Directory, Vol. 2

TYPE OF PENETRANT	CONSTRUCTION					
	FLOOR PENETRATION SYSTEMS (First Alpha Component = C or F)			WALL PENETRATION SYSTEMS (First Alpha Component = C or W)		
	Concrete Floors with Minimum Thickness Less Than or Equal to 5 Inches	Concrete Floors with Minimum Thickness Greater Than 5 Inches	Framed Floors	Concrete or Masonry Walls with Minimum Thickness Less Than or Equal to 8 Inches	Concrete or Masonry Walls with Minimum Thickness Greater Than 8 Inches	Framed Walls
NO PENETRATING ITEMS	C-AJ-0001-0999 or F-A-0001-0999	C-BJ-0001-0999		C-AJ-0001-0999, C-BJ-0001-0999, Or W-J-0001-0999		W-L-0001-0999
METALLIC PIPE, CONDUIT, OR TUBING	C-AJ-1001-1999 or F-A-1001-1999	C-BJ-1001-1999, C-BK-1001-1999, or F-B-1001-1999	F-C-1001-1999	C-AJ-1001-1999, C-BJ-1001-1999, Or W-J-1001-1999	C-BK-1001-1999 Or W-K-1001-1999	W-L-1001-1999
NONMETALLIC PIPE, CONDUIT, OR TUBING	C-AJ-2001-2999 or F-A-2001-2999	C-BJ-2001-2999 or F-B-2001-2999	F-C-2001-2999	C-AJ-2001-2999, C-BJ-2001-2999, Or W-J-2001-2999		W-L-2001-2999
ELECTRICAL CABLES	C-AJ-3001-3999 or F-A-3001-3999	C-BJ-3001-3999 Or F-B-3001-3999	F-C-3001-3999	C-AJ-3001-3999, C-BJ-3001-3999, Or W-J-3001-3999		W-L-3001-3999
CABLE TRAYS WITH ELECTRICAL CABLES	C-AJ-4001-4999 or F-A-4001-4999	C-BJ-4001-4999		C-AJ-4001-4999, C-BJ-4001-4999, Or W-J-4001-4999	W-K-4001-4999	W-L-4001-4999
INSULATED PIPES	C-AJ-5001-5999 or F-A-5001-5999	C-BJ-5001-5999	F-C-5001-5999	C-AJ-5001-5999 Or W-J-5001-5999		W-L-5001-5999

MISCELLANEOUS ELECTRICAL PENETRANTS	C-AJ-6001-6999 or F-A-6001-6999			C-AJ-6001-6999		W-L-6001-6999
MISCELLANEOUS MECHANICAL PENETRANTS	C-AJ-7001-7999		F-C-7001-7999	C-AJ-7001-7999 Or W-J-7001-7999		W-L-7001-7999
GROUPINGS OF PENETRATIONS	C-AJ-8001-8999 or F-A-8001-8999	C-BJ-8001-8999	F-C-8001-8999	C-AJ-8001-8999, C-BJ-8001-8999, Or W-J-8001-8999		W-L-8001-8999
For penetrations of fire-resistance-rated floor or wall assemblies, provide UL-listed through-penetration firestop system from applicable UL number range listed above that complies with this section and is suitable for penetration conditions indicated.						

3.1 FIRE RESISTANCE JOINT SYSTEM SCHEDULE

- A. Systems Listed Using Alpha-Alpha-Numeric Identification System in UL's *Fire Resistance Directory*, Vol. 2 Typically, conditions have movement capabilities unless specifically identified otherwise for individual conditions.

JOINT WIDTH		LESS THAN OR EQUAL TO 2 INCHES	GREATER THAN 2 INCHES AND LESS THAN OR EQUAL TO 6 INCHES	GREATER THAN 6 INCHES AND LESS THAN OR EQUAL TO 12 INCHES	GREATER THAN 12 INCHES AND LESS THAN OR EQUAL TO 24 INCHES	GREATER THAN 24 INCHES
TYPE OF JOINT						
Floor-to-Floor (FF)	No Movement (S)	FF-S-0001-0999	FF-S-1001-4999	FF-S-2001-4999	FF-S-3001-3999	FF-S-4001-4999
	Have Movement (D)	FF-D-0001-0999	FF-D-1001-4999	FF-D-2001-4999	FF-D-3001-3999	FF-D-4001-4999
Wall-to-Wall (WW)	No Movement (S)	WW-S-0001-0999	WW-S-1001-4999	WW-S-2001-4999	WW-S-3001-3999	WW-S-4001-4999
	Have Movement (D)	WW-D-0001-0999	WW-D-1001-4999	WW-D-2001-4999	WW-D-3001-3999	WW-D-4001-4999
Floor-to-Wall (FW)	No Movement (S)	FW-S-0001-0999	FW-S-1001-4999	FW-S-2001-4999	FW-S-3001-3999	FW-S-4001-4999
	Have Movement (D)	FW-D-0001-0999	FW-D-1001-4999	FW-D-2001-4999	FW-D-3001-3999	FW-D-4001-4999
Head-of-Wall (HW)	No Movement (S)	HW-S-0001-0999	HW-S-1001-4999	HW-S-2001-4999	HW-S-3001-3999	HW-S-4001-4999
	Have Movement (D)	HW-D-0001-0999	HW-D-1001-4999	HW-D-2001-4999	HW-D-3001-3999	HW-D-4001-4999
Wall-to-Wall Joints Intended for Use as Corner Guards (CG)	No Movement (S)	CG-S-0001-0999	CG-S-1001-4999	CG-S-2001-4999	CG-S-3001-3999	CG-S-4001-4999
	Have Movement (D)	CG-D-0001-0999	CG-D-1001-4999	CG-D-2001-4999	CG-D-3001-3999	CG-D-4001-4999
For joints between fire-rated assemblies, provide UL-listed Joint Systems from applicable UL number range listed above that complies with this section and is suitable for joint conditions indicated.						

1.1 PERIMETER FIRE CONTAINMENT SYSTEM SCHEDULE

- A. Systems Listed Using Alpha-Alpha-Numeric Identification System in UL's *Fire Resistance Directory*, Vol. 2. Typically, conditions have movement capabilities unless specifically identified otherwise for individual conditions.

	NONFIRE-RATED EXTERIOR CURTAIN WALL (First Two Alpha Components = C W)	
MINIMUM CLEARANCE DISTANCE BETWEEN CURTAIN WALL & PERIMETER OF FLOOR	NO MOVEMENT CAPABILITIES S (Third Alpha Component = S)	HAVE MOVEMENT CAPABILITIES (Third Alpha Component = D)
LESS THAN OR EQUAL TO 2 INCHES	CW-S-0001-0999	CW-D-0001-0999
GREATER THAN 2 INCHES AND LESS THAN OR EQUAL TO 6 INCHES	CW-S -1001-1999	CW-D -1001-1999
GREATER THAN 2 INCHES AND LESS THAN OR EQUAL TO 6 INCHES	CW-S -2001-2999	CW-D -2001-2999
For perimeter with nonfire-rated curtain walls, provide UL-listed Perimeter Fire Containment Systems from applicable UL number range listed above that complies with this section and is suitable for perimeter conditions indicated. For perimeter with nonfire-rated spandrel, provide Perimeter Fire Containment System with Engineering Judgment certifying that system provides level of protection suitable for perimeter conditions indicated.		

END OF SECTION

SECTION 07 84 13

PENETRATION FIRESTOPPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Penetrations in fire-resistance-rated walls.
 - 2. Penetrations in horizontal assemblies.
 - 3. Penetrations in smoke barriers.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Product Schedule: For each penetration firestopping system. Include location and design designation of qualified testing and inspecting agency.
 - 1. Where Project conditions require modification to a qualified testing and inspecting agency's illustration for a particular penetration firestopping condition, submit illustration, with modifications marked, approved by penetration firestopping manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, for penetration firestopping.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: A firm that has been approved by FM Global according to FM Global 4991, "Approval of Firestop Contractors," or been evaluated by UL and found to comply with its "Qualified Firestop Contractor Program Requirements."
- B. Installer Qualifications: A firm experienced in installing penetration firestopping similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful performance. Qualifications include having the necessary experience, staff, and training to install manufacturer's products per specified requirements. Manufacturer's willingness to sell its penetration firestopping products to Contractor or to Installer engaged by Contractor does not in itself confer qualification on buyer.
- C. Fire-Test-Response Characteristics: Penetration firestopping shall comply with the following requirements:
 - 1. Penetration firestopping tests are performed by a qualified testing agency acceptable to authorities having jurisdiction.
 - 2. Penetration firestopping is identical to those tested per testing standard referenced in "Penetration Firestopping" Article. Provide rated systems complying with the following requirements:
 - a. Penetration firestopping products bear classification marking of qualified testing and inspecting agency.
 - b. Classification markings on penetration firestopping correspond to designations listed by the following:
 - 1) UL in its "Fire Resistance Directory."

- c. Preinstallation Conference: Conduct conference at Project site.

1.6 PROJECT CONDITIONS

- A. Environmental Limitations: Do not install penetration firestopping when ambient or substrate temperatures are outside limits permitted by penetration firestopping manufacturers or when substrates are wet because of rain, frost, condensation, or other causes.
- B. Install and cure penetration firestopping per manufacturer's written instructions using natural means of ventilations or, where this is inadequate, forced-air circulation.

1.7 COORDINATION

- A. Coordinate construction of openings and penetrating items to ensure that penetration firestopping is installed according to specified requirements.
- B. Coordinate sizing of sleeves, openings, core-drilled holes, or cut openings to accommodate penetration firestopping.
- C. Notify Owner's testing agency at least seven days in advance of penetration firestopping installations; confirm dates and times on day preceding each series of installations.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Grace Construction Products.
 - 2. Hilti, Inc.
 - 3. RectorSeal Corporation.
 - 4. Specified Technologies Inc.
 - 5. 3M Fire Protection Products.
 - 6. Tremco, Inc.; Tremco Fire Protection Systems Group.
 - 7. USG Corporation.

2.2 PENETRATION FIRESTOPPING

- A. Provide penetration firestopping that is produced and installed to resist spread of fire according to requirements indicated, resist passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated. Penetration firestopping systems shall be compatible with one another, with the substrates forming openings, and with penetrating items if any.
- B. Penetrations in Fire-Resistance-Rated Walls: Provide penetration firestopping with ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg.
 - 1. Fire-resistance-rated walls include fire walls fire-barrier walls smoke-barrier walls and fire partitions.
 - 2. F-Rating: Not less than the fire-resistance rating of constructions penetrated.
- C. Penetrations in Horizontal Assemblies: Provide penetration firestopping with ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg.
 - 1. Horizontal assemblies include floors floor/ceiling assemblies and ceiling membranes of roof/ceiling assemblies.
 - 2. F-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated.
 - 3. T-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated except for floor penetrations within the cavity of a wall.

- D. Penetrations in Smoke Barriers: Provide penetration firestopping with ratings determined per UL 1479.
 - 1. L-Rating: Not exceeding 5.0 cfm/sq. ft. of penetration opening at 0.30-inch wg at both ambient and elevated temperatures.
- E. W-Rating: Provide penetration firestopping showing no evidence of water leakage when tested according to UL 1479.
- F. Exposed Penetration Firestopping: Provide products with flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E 84.
- G. Accessories: Provide components for each penetration firestopping system that are needed to install fill materials and to maintain ratings required. Use only those components specified by penetration firestopping manufacturer and approved by qualified testing and inspecting agency for firestopping indicated.
 - 1. Permanent forming/damming/backing materials, including the following:
 - a. Slag-wool-fiber or rock-wool-fiber insulation.
 - b. Sealants used in combination with other forming/damming/backing materials to prevent leakage of fill materials in liquid state.
 - c. Fire-rated form board.
 - d. Fillers for sealants.
 - 2. Temporary forming materials.
 - 3. Substrate primers.
 - 4. Collars.
 - 5. Steel sleeves.

2.3 FILL MATERIALS

- A. Latex Sealants: Single-component latex formulations that do not re-emulsify after cure during exposure to moisture.
- B. Firestop Devices: Factory-assembled collars formed from galvanized steel and lined with intumescent material sized to fit specific diameter of penetrant.
- C. Intumescent Composite Sheets: Rigid panels consisting of aluminum-foil-faced elastomeric sheet bonded to galvanized-steel sheet.
- D. Intumescent Putties: Nonhardening dielectric, water-resistant putties containing no solvents, inorganic fibers, or silicone compounds.
- E. Intumescent Wrap Strips: Single-component intumescent elastomeric sheets with aluminum foil on one side.
- F. Mortars: Prepackaged dry mixes consisting of a blend of inorganic binders, hydraulic cement, fillers, and lightweight aggregate formulated for mixing with water at Project site to form a nonshrinking, homogeneous mortar.
- G. Pillows/Bags: Reusable heat-expanding pillows/bags consisting of glass-fiber cloth cases filled with a combination of mineral-fiber, water-insoluble expansion agents, and fire-retardant additives. Where exposed, cover openings with steel-reinforcing wire mesh to protect pillows/bags from being easily removed.
- H. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.
- I. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below:
 - 1. Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces, and nonsag formulation for openings in vertical and sloped surfaces, unless indicated firestopping limits use of nonsag grade for both opening conditions.

2.4 MIXING

- A. For those products requiring mixing before application, comply with penetration firestopping manufacturer's written instructions for accurate proportioning of materials, water (if required), type of mixing equipment, selection of mixer speeds, mixing containers, mixing time, and other items or procedures needed to produce products of uniform quality with optimum performance characteristics for application indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Cleaning: Clean out openings immediately before installing penetration firestopping to comply with manufacturer's written instructions and with the following requirements:
 - 1. Remove from surfaces of opening substrates and from penetrating items foreign materials that could interfere with adhesion of penetration firestopping.
 - 2. Clean opening substrates and penetrating items to produce clean, sound surfaces capable of developing optimum bond with penetration firestopping. Remove loose particles remaining from cleaning operation.
 - 3. Remove laitance and form-release agents from concrete.
- B. Priming: Prime substrates where recommended in writing by manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.
- C. Masking Tape: Use masking tape to prevent penetration firestopping from contacting adjoining surfaces that will remain exposed on completion of the Work and that would otherwise be permanently stained or damaged by such contact or by cleaning methods used to remove stains. Remove tape as soon as possible without disturbing firestopping's seal with substrates.

3.3 INSTALLATION

- A. General: Install penetration firestopping to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.
- B. Install forming materials and other accessories of types required to support fill materials during their application and in the position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.
 - 1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of firestopping.
- C. Install fill materials for firestopping by proven techniques to produce the following results:
 - 1. Fill voids and cavities formed by openings, forming materials, accessories, and penetrating items as required to achieve fire-resistance ratings indicated.
 - 2. Apply materials so they contact and adhere to substrates formed by openings and penetrating items.
 - 3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.4 IDENTIFICATION

- A. Identify penetration firestopping with preprinted metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches of firestopping edge so labels will be visible to anyone seeking to remove penetrating items or firestopping. Use mechanical

fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:

1. The words "Warning - Penetration Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
2. Contractor's name, address, and phone number.
3. Designation of applicable testing and inspecting agency.
4. Date of installation.
5. Manufacturer's name.
6. Installer's name.

3.5 FIELD QUALITY CONTROL

- A. Owner will engage a qualified testing agency to perform tests and inspections.
- B. Where deficiencies are found or penetration firestopping is damaged or removed because of testing, repair or replace penetration firestopping to comply with requirements.
- C. Proceed with enclosing penetration firestopping with other construction only after inspection reports are issued and installations comply with requirements.

3.6 CLEANING AND PROTECTION

- A. Clean off excess fill materials adjacent to openings as the Work progresses by methods and with cleaning materials that are approved in writing by penetration firestopping manufacturers and that do not damage materials in which openings occur.
- B. Provide final protection and maintain conditions during and after installation that ensure that penetration firestopping is without damage or deterioration at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, immediately cut out and remove damaged or deteriorated penetration firestopping and install new materials to produce systems complying with specified requirements.

3.7 PENETRATION FIRESTOPPING SCHEDULE

- A. Fire Resistance Joint System Schedule –
 1. Greater than 2 inches and less than or equal to 6 inches: 1000-1999
 2. Greater than 6 inches and less than or equal to 12 inches: 2000-2999
- B. Perimeter Fire Containment System Schedule –
 1. The first two alpha components CW do not apply to joint systems, these digits are used to label penetrations in floor and / or walls. Perimeter Fire Barrier joints utilize the Omega Point Numbering System. The first three components should be CEJ (Construction Expansion Joint) follow by the system listed and ending with P for perimeter (curtain wall).

END OF SECTION 07 84 13

SECTION 07 92 00

JOINT SEALANTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Silicone joint sealants.
 - 2. Urethane joint sealants.
 - 3. Latex joint sealants.
 - 4. Solvent-release-curing joint sealants.
 - 5. Acoustical joint sealants.
 - 6. Preformed sealants.

1.3 PRECONSTRUCTION TESTING

- A. Preconstruction Compatibility and Adhesion Testing: Submit to joint-sealant manufacturers, for testing indicated below, samples of materials that will contact or affect joint sealants.
 - 1. Use ASTM C 1087 to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of joint sealants to joint substrates.
 - 2. Submit not fewer than four pieces of each kind of material, including joint substrates, shims, joint-sealant backings, secondary seals, and miscellaneous materials.
 - 3. Schedule sufficient time for testing and analyzing results to prevent delaying the Work.
 - 4. For materials failing tests, obtain joint-sealant manufacturer's written instructions for corrective measures including use of specially formulated primers.
 - 5. Testing will not be required if joint-sealant manufacturers submit joint preparation data that are based on previous testing, not older than 24 months, of sealant products for adhesion to, and compatibility with, joint substrates and other materials matching those submitted.

1.4 SUBMITTALS

- A. Product Data: For each joint-sealant product indicated.
- B. Samples for Initial Selection: Manufacturer's color charts consisting of strips of cured sealants showing the full range of colors available for each product exposed to view.
- C. Joint-Sealant Schedule: Include the following information:
 - 1. Joint-sealant application, joint location, and designation.
 - 2. Joint-sealant manufacturer and product name.
 - 3. Joint-sealant formulation.
 - 4. Joint-sealant color.
- D. Qualification Data: For qualified Installer.
- E. Product Certificates: For each kind of joint sealant and accessory, from manufacturer.
- F. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, indicating that sealants comply with requirements.
- G. Field-Adhesion Test Reports: For each sealant application tested.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.
- B. Source Limitations: Obtain each kind of joint sealant from single source from single manufacturer.
- C. Mockups: Install sealant in mockups of assemblies specified in other Sections that are indicated to receive joint sealants specified in this Section. Use materials and installation methods specified in this Section.

1.6 PROJECT CONDITIONS

- A. Do not proceed with installation of joint sealants under the following conditions:
 - 1. When ambient and substrate temperature conditions are outside limits permitted by joint-sealant manufacturer or are below 40 deg F.
 - 2. When joint substrates are wet.
 - 3. Where joint widths are less than those allowed by joint-sealant manufacturer for applications indicated.
 - 4. Where contaminants capable of interfering with adhesion have not yet been removed from joint substrates.

1.7 WARRANTY

- A. Special Installer's Warranty: Manufacturer's standard form in which Installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.
 - 1. Warranty Period: Two years from date of Substantial Completion.
- B. Special Manufacturer's Warranty: Manufacturer's standard form in which joint-sealant manufacturer agrees to furnish joint sealants to repair or replace those that do not comply with performance and other requirements specified in this Section within specified warranty period.
 - 1. Warranty Period: 20 years from date of Substantial Completion for silicone sealants.
- C. Special warranties specified in this article exclude deterioration or failure of joint sealants from the following:
 - 1. Movement of the structure caused by structural settlement or errors attributable to design or construction resulting in stresses on the sealant exceeding sealant manufacturer's written specifications for sealant elongation and compression.
 - 2. Disintegration of joint substrates from natural causes exceeding design specifications.
 - 3. Mechanical damage caused by individuals, tools, or other outside agents.
 - 4. Changes in sealant appearance caused by accumulation of dirt or other atmospheric contaminants.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

- A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer, based on testing and field experience.
- B. VOC Content of Interior Sealants: Provide sealants and sealant primers for use inside the weatherproofing system that comply with the following limits for VOC content when calculated according to 40 CFR 59, Part 59, Subpart D (EPA Method 24):
 - 1. Architectural Sealants: 250 g/L.
 - 2. Sealant Primers for Nonporous Substrates: 250 g/L.
 - 3. Sealant Primers for Porous Substrates: 775 g/L.

- C. Liquid-Applied Joint Sealants: Comply with ASTM C 920 and other requirements indicated for each liquid-applied joint sealant specified, including those referencing ASTM C 920 classifications for type, grade, class, and uses related to exposure and joint substrates.
 - 1. Suitability for Immersion in Liquids: Where sealants are indicated for Use I for joints that will be continuously immersed in liquids, provide products that have undergone testing according to ASTM C 1247. Liquid used for testing sealants is de-ionized water, unless otherwise indicated.
- D. Stain-Test-Response Characteristics: Where sealants are specified to be non-staining to porous substrates, provide products that have undergone testing according to ASTM C 1248 and have not stained porous joint substrates indicated for Project.
- E. Suitability for Contact with Food: Where sealants are indicated for joints that will come in repeated contact with food, provide products that comply with 21 CFR 177.2600.
- F. Colors of Exposed Joint Sealants: As selected by Architect from manufacturer's full range.

2.2 SILICONE JOINT SEALANTS

- A. Single-Component, Nonsag, Neutral-Curing Silicone Joint Sealant (S:) ASTM C 920, Type S, Grade NS, Class 50, for Use NT.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Building Systems; Omniseal 50.
 - b. Dow Corning Corporation; 795.
 - c. GE Advanced Materials - Silicones; SilGlaze II SCS2800.
 - d. Pecora Corporation; 864.
- B. Mildew-Resistant, Single-Component, Acid-Curing Silicone Joint Sealant(S-S:) ASTM C 920, Type S, Grade NS, Class 25, for Use NT.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. BASF Building Systems; Omniplus.
 - b. Dow Corning Corporation; 786 Mildew Resistant.
 - c. GE Advanced Materials - Silicones; Sanitary SCS1700.
 - d. Tremco Incorporated; Tremsil 200 Sanitary.

2.3 URETHANE JOINT SEALANTS

- A. Multi-component, Non-sag, Urethane Joint Sealant (U-NS): ASTM C 920, Type M, Grade NS, Class 50, for Use NT.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Pecora Corporation; Dynatrol II.
 - b. Polymeric Systems, Inc.; PSI-270.
 - c. Tremco Incorporated
- B. Multi-component, Self-Leveling, Traffic-Grade, Urethane Joint Sealant (U-TB): ASTM C 920, Type M, Grade SL, Class 50, for Use T.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Polymeric Systems, Inc.; PSI-270.
 - b. Tremco Incorporated; Dymeric 240 FC.
 - c. Pecora; Dynatread.

2.4 LATEX JOINT SEALANTS

- A. Latex Joint Sealant (L): Acrylic latex or siliconized acrylic latex, ASTM C 834, Type OP, Grade NF.
 - 1. Products: Subject to compliance with requirements, provide one of the following:

- a. BASF Building Systems; Sonolac.
- b. Bostik, Inc.; Chem-Calk 600.
- c. May National Associates, Inc.
- d. Pecora Corporation; AC-20+.
- e. Tremco Incorporated; Tremflex 834.

2.5 ACOUSTICAL JOINT SEALANTS (AC)

- A. Acoustical Joint Sealant: Manufacturer's standard non-sag, paintable, non-staining latex sealant complying with ASTM C 834. Product effectively reduces airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Pecora Corporation; AC-20 FTR.
 - b. USG Corporation; SHEETROCK Acoustical Sealant.

2.6 PREFORMED JOINT SEALANTS (PF)

- A. Preformed Foam Joint Sealant: Manufacturer's standard preformed, pre-compressed, open-cell foam sealant manufactured from urethane foam with minimum density of 10 lb/cu. ft. and impregnated with a nondrying, water-repellent agent. Factory produce in pre-compressed sizes in roll or stick form to fit joint widths indicated; coated on one side with a pressure-sensitive adhesive and covered with protective wrapping.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. EMSEAL Joint Systems, Ltd.; Emseal 25V.
 - b. Sandell Manufacturing Co., Inc.; Polyseal.
 - c. Willseal USA, LLC; Willseal 150.
 - d. MM Systems, Color Joint Silicone
 - e. BASF, WABO Weather Seal II
- B. PICK RESISTANT ADHESIVE
 - 1. Pick Resistant Rubber Adhesive – Psych Unit
 - a. Surebond SB-190 Everseal

2.7 MISCELLANEOUS MATERIALS

- A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from pre-construction joint-sealant-substrate tests and field tests.
- B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants to joint substrates.
- C. Masking Tape: Non-staining, non-absorbent material compatible with joint sealants and surfaces adjacent to joints.
- D. Backer Rod:
 - 1. General: Provide sealant backings of material and type that are non-staining; are compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing
 - 2. Elastomeric Tubing Sealant Backings: Neoprene, butyl, EPDM, or silicone tubing complying with ASTM D 1056, nonabsorbent to water and gas, and capable of remaining resilient at temperatures down to minus 26 deg F. Provide products with low compression set and of

size and shape to provide a secondary seal, to control sealant depth, and otherwise contribute to optimum sealant performance

3. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting joint-sealant performance.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions and the following requirements:
 1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
 2. Clean porous joint substrate surfaces by brushing, grinding, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining after cleaning operations above by vacuuming or blowing out joints with oil-free compressed air.
 - a. Porous joint substrates include the following:
 - 1) Concrete.
 - 2) Masonry.
 - 3) Unglazed surfaces of ceramic tile.
 - 4) Exterior insulation and finish systems.
 3. Remove laitance and form-release agents from concrete.
 4. Clean nonporous joint substrate surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants. Nonporous joint substrates include the following:
 - a. Metal.
 - b. Glass.
 - c. Porcelain enamel.
 - d. Glazed surfaces of ceramic tile.
- B. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.
- C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 INSTALLATION OF JOINT SEALANTS

- A. General: Comply with joint-sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.

- B. Sealant Installation Standard: Comply with recommendations in ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.
- C. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
 - 1. Do not leave gaps between ends of sealant backings.
 - 2. Do not stretch, twist, puncture, or tear sealant backings.
 - 3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.
- D. Install sealants using proven techniques that comply with the following and at the same time backings are installed:
 - 1. Place sealants so they directly contact and fully wet joint substrates.
 - 2. Completely fill recesses in each joint configuration.
 - 3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.
- E. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified in subparagraphs below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.
 - 1. Remove excess sealant from surfaces adjacent to joints.
 - 2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
 - 3. Provide concave joint profile per Figure 8A in ASTM C 1193, unless otherwise indicated.
 - 4. Provide flush joint profile where indicated per Figure 8B in ASTM C 1193.
 - 5. Provide recessed joint configuration of recess depth and at locations indicated per Figure 8C in ASTM C 1193.
 - a. Use masking tape to protect surfaces adjacent to recessed tooled joints.
- F. Acoustical Sealant Installation: At sound-rated assemblies and elsewhere as indicated, seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Comply with ASTM C 919 and with manufacturer's written recommendations.

3.4 CLEANING

- A. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

3.5 PROTECTION

- A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from original work.

3.6 JOINT-SEALANT SCHEDULE

- A. Sealant Schedule:
 - 1. Interior Joints:
 - a. Wall and ceiling joints subject to movement: Designation U-MC.
 - b. Wall and ceiling joints not subject to movement: Designation AL.
 - c. Interior side of exterior openings: U-MC.

- d. Floor joints: Designation U-TB.
 - e. Wall and ceiling joints between frames and their rough opening: Designation AL.
 - f. Wall and ceiling joints between frames and adjoining surfaces: Designation AL.
 - g. Interior Sanitary Joints; Joints Between Plumbing Fixtures and Adjoining Floor, Wall, and Ceiling Surfaces; Joints Between Shower Door Enclosure Components and Adjacent Finish Surfaces; Joints in Dietary and Food Preparation Areas, Kitchens, Food Storage Areas, and Areas Subject to Frequent Wet Cleaning, including joints between walls and floors, Joints Between Back Splashes and Wall Substrates: Designation S-S.
2. Exterior locations:
- a. Wall joints:
 - 1) Bordered on both sides by porous building material (concrete, stone, masonry, exterior insulation and finish systems): Designation S-GP [PF]
 - 2) Bordered on both sides by non-porous building material (coated and uncoated metals, anodized aluminum, porcelain tile, and glass): Designation S-GP [PF]
 - 3) Bordered on one side by porous building material (concrete, stone, masonry) and other side by non-porous building material (coated and uncoated metals, anodized aluminum, porcelain tile, and glass): Designation S-GP [PF]
 - b. Perimeter of penetrations through walls: Designation S-GP
 - c. Control joints (filling of V-grooves) and perimeter of penetrations in Portland cement plaster walls: Designation S-GP.
 - d. Expansion joints in ceilings, soffits, and overhead surfaces: Designation S-GP
 - e. Control joints and perimeter of penetrations in ceilings, soffits, and overhead surfaces: Designation S-GP
 - f. Wall and ceiling joints between frames and their rough opening: Designation S-GP.
 - g. Wall and ceiling joints between frames and adjoining surfaces: Designation S-GP.
 - h. Joints and perimeter of penetrations in horizontal pedestrian and vehicle traffic surfaces: Designation U-TB.
 - i. Joints in Division 07 Section 07 "Sheet Metal Flashing and Trim:" Designation S- GP.

Guide to Sealant Types - Interior				
Location	Materials	Sealant Type	Base Product	Remarks / Exceptions
Interior (General)	Window Sills and Stools	100 percent silicone	Color-Sil Poly-Sil Pecora 898NST	Do not use in areas subject to abrasion or physical abuse
	Cabinets and Casework to wall			
	Countertops and Backsplashes			
	Sinks in Countertops			
	Fixtures and Casework to floor			
	Interior Alum Doors and Window Frame Perimeters	One-part neutral curing silane-modified polyurethane (STPU)	Pecora Dynatrol I-XL Hybrid	Paintable
	Non-rated wall, floor, and deck penetrations.			
	Interior joints at exterior wall.			
	Hollow Metal Door and Window Frames	Siliconized Acrylic Latex (paintable)	Tremco Tremflex 834 Pecora AC-20+ Silicone	Exception: Where sealant will not be painted and white color will not be visually compatible with adjacent finishes use Dynatrol I-XL Hybrid of matching color.

Guide to Sealant Types - Interior				
Location	Materials	Sealant Type	Base Product	Remarks / Exceptions
	Acoustic Sealant Joints at top and bottom terminations of Interior Non-rated Walls.	Water-based acrylic; paintable	Pecora AC-20 FTR or AIS-919	
		Water-based acrylic spray	Hilti CP 572 Pecora AIS-919	
	Acoustic Sealant Joints at top and bottom terminations of Interior Fire-Rated Walls and Building Perimeter Gaps between slabs and facades.	Acrylic	Pecora AC-20 FTR	Coordinate with 07 84 00 and 07 84 53.
		Silicone	Pecora 864NST	
Interior Flatwork	Control Joints in Concrete Floors in Mechanical Rooms and other unfinished spaces	Multi-part Polyurethane	Pecora DynaTred	DynaTred may be subjected to continual water immersion.
	Stone and Precast Flooring			For resistance to high heel traffic, use Pecora Dynaflex or Dynaflex SC
Interior Wet Areas	Porcelain, Ceramic Tile, Metals, and surfaces with Epoxy Paints	Silicone; Air cure	Pecora 860 (acid cure) or Pecora 898NST	--
Laboratories, biocontainment facilities, and cGMP cleanrooms	Sanitary seal at joints between ceramic and prefinished surfaces	Sanitary Silicone	Tremco Tremsil 200 Pecora 860 (acid cure) or Pecora 898NST	White only
	Sanitary seal at joints between epoxy painted and prefinished surfaces	Sanitary Silicone	Pecora 898NST	White only
	Airtight seal at openings in walls and ceilings	Structural Silicone	Pecora 895NST	Includes electrical conduits
		Heat Shrink Polyolefin Tubing	NP-700md	
Microelectronics & Aeronautics Clean Zone / Cleanrooms	Cleanroom Sealant	Polyurethane	Sikaflex-1A	White only

Guide to Sealant Types - Exterior				
Location	Materials	Sealant Type	Base Product	Remarks / Exceptions
General Exterior	Cast in Place Concrete	Multi-part Polyurethane, chemically curing, epoxidized	Pecora Dynatrol II	Exception: Use Dymonic where used as bedding sealant for frames, sills, thresholds etc.
	Brick and Concrete Masonry			
	Portland Cement Plaster			
	Hollow Metal Door and Window Perimeters	Silane-Modified Polymer (STPU, STPE, Polyurea)	Pecora Dynatrol I-XL Hybrid	--
	Aluminum Composite Panels (ACM) and Metal Column Covers			
	Weatherseal of Aluminum Window Frames (including perimeter joints)			

	Precast Concrete Panels	Silicone or Silyl Terminated Polyether (STPE)	Pecora 890NST or 890FTS or Pecora Dynatrol I-XL Hybrid	--
	EIFS Systems			
	Stone Work	Silicone or Silane-Modified Polymer (STPU, STPE, Polyurea)	Pecora 864NST or Pecora Dynatrol I-XL Hybrid	Exception: Pre-test for staining potential per ASTM C1248, with stain-sensitive stone Note: Silane-modified Polyurethane will not stain.
	General Exterior Glazing	Silicone; Neutral-cure	Pecora 895NST	Exception: Select alternate silicone sealant types as appropriate for specific glazing application.
	Butt Glazing and Structural Silicone Joints	Silicone; 1-part, Neutral-cure	Pecora 895NST	--
	Fabrication of Insulating Glass Units (IGU)	Primary Seal: Polyisobutylene	Select high quality sealants, of basic type listed, as appropriate for specific application.	
		Secondary Seal: Silicone	Dowsil 982	--
	Zone dams, shear blocks and other internal component of Aluminum Window Systems	Silicone	Use product which offers optimal adhesion and performance for application.	
	Sheet Metal Gutters, Downspouts, Scuppers, etc.	Butyl Rubber	Pecora BC-158	--
	Existing joints where Silicone was previously used	Silicone	Use product which offers optimal adhesion and performance for condition, and which offers suitable color choices for matching.	

Notes

1. The above is intended to be an overall guide. Additional conditions and materials may be required. Notify Architect if additional Guidance is required to select unlisted items.
2. Optional sealant products shall offer same number of color choices as the Base Product listed.
3. All conditions and materials listed may not apply to subject project.
4. Not all project conditions may be addressed on above table; Refer also to other specification sections and install sealants where called for by other sections.
5. Materials and Conditions conventionally occurring on Exterior but used on Interior (e.g. Brick Masonry on interior) may not be listed on this Table. Refer to Exterior Guide (Section 07 92 13) for appropriate sealant type.

END OF SECTION

SECTION 08 11 14

HOLLOW METAL DOORS AND FRAMES – INTERIOR

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Standard hollow metal doors frames for interior applications.

1.3 DEFINITIONS

- A. Minimum Thickness: Minimum thickness of base metal without coatings.
- B. Standard Hollow Metal Work: Hollow metal work fabricated according to ANSI/SDI A250.8.

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated. Include construction details, material descriptions, core descriptions, fire-resistance rating, and finishes.
- B. Shop Drawings: Include the following:
 - 1. Elevations of each door design.
 - 2. Details of doors, including vertical and horizontal edge details and metal thicknesses.
 - 3. Frame details for each frame type, including dimensioned profiles and metal thicknesses.
 - 4. Locations of reinforcement and preparations for hardware.
 - 5. Details of each different wall opening condition.
 - 6. Details of anchorages, joints, field splices, and connections.
 - 7. Details of accessories.
 - 8. Details of moldings, removable stops, and glazing.
 - 9. Details of conduit and preparations for power, signal, and control systems. Other Action Submittals:
 - a. Schedule: Provide a schedule of hollow metal work prepared by or under the supervision of supplier, using same reference numbers for details and openings as those on Drawings. Coordinate with door hardware schedule.
- C. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, for each type of hollow metal door and frame assembly.

1.5 QUALITY ASSURANCE

- A. Source Limitations: Obtain hollow metal work from single source from single manufacturer.
- B. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to NFPA 252 and UL10C.
 - 1. Oversize Fire-Rated Door Assemblies: For units exceeding sizes of tested assemblies, provide certification by a qualified testing agency that doors comply with standard construction requirements for tested and labeled fire-rated door assemblies except for size.
 - 2. Temperature-Rise Limit: At vertical exit enclosures and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F above ambient after 30 minutes of standard fire-test exposure.

- C. Fire-Rated, Borrowed-Light Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled, by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on testing according to NFPA 257 or UL 9. Label each individual glazed lite.
- D. Smoke-Control Door Assemblies: Comply with NFPA 105 or UL 1784.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver hollow metal work palletized, wrapped, or crated to provide protection during transit and Project-site storage. Do not use nonvented plastic.
 - 1. Provide additional protection to prevent damage to finish of factory-finished units.
- B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.
- C. Store hollow metal work under cover at Project site. Place in stacks of five units maximum in a vertical position with heads up, spaced by blocking, on minimum 4-inch- high wood blocking. Do not store in a manner that traps excess humidity.
 - 1. Provide minimum 1/4-inch space between each stacked door to permit air circulation.

1.7 PROJECT CONDITIONS

- A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

1.8 COORDINATION

- A. Coordinate installation of anchorages for hollow metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Amweld Building Products, LLC.
 - 2. Benchmark; a division of Therma-Tru Corporation.
 - 3. Ceco Door Products; an Assa Abloy Group company.
 - 4. Curries Company; an Assa Abloy Group company.
 - 5. Steelcraft; an Ingersoll-Rand company.

2.2 MATERIALS

- A. Cold-Rolled Steel Sheet: ASTM A 1008, Commercial Steel (CS), Type B; suitable for exposed applications.
- B. Hot-Rolled Steel Sheet: ASTM A 1011, Commercial Steel (CS), Type B; free of scale, pitting, or surface defects; pickled and oiled.
- C. Frame Anchors: ASTM A 591, Commercial Steel (CS), 40Z coating designation; mill phosphatized.
 - 1. For anchors built into exterior walls, steel sheet complying with ASTM A 1008 or ASTM A 1011, hot-dip galvanized according to ASTM A 153, Class B.
- D. Inserts, Bolts, and Fasteners: Hot-dip galvanized according to ASTM A 153.
- E. Powder-Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hollow metal frames of type indicated.

- F. Mineral-Fiber Insulation: ASTM C 665, Type I (blankets without membrane facing); consisting of fibers manufactured from slag or rock wool with 6- to 12-lb/cu. ft. density; with maximum flame-spread and smoke-development indexes of 25 and 50, respectively; passing ASTM E 136 for combustion characteristics.
- G. Glazing: Comply with requirements in Division 08 Section "Glazing."

2.3 STANDARD HOLLOW METAL DOORS

- A. General: Provide doors of design indicated, not less than thickness indicated; fabricated with smooth surfaces, without visible joints or seams on exposed faces unless otherwise indicated. Comply with ANSI/SDI A250.8.
 - 1. Design: Flush panel.
 - 2. Thickness: 14 Gauge Seamless Construction with internal reinforcement
 - 3. Core Construction: Manufacturer's standard Kraft-paper honeycomb, polystyrene, polyurethane, polyisocyanurate, mineral-board, or vertical steel-stiffener core.
 - a. Fire Door Core: As required to provide fire-protection and temperature-rise ratings indicated.
 - 4. Vertical Edges for Single-Acting Doors: Square edge.
 - a. Beveled Edge: 1/8 inch in 2 inches.
 - 5. Vertical Edges for Double-Acting Doors: Round vertical edges with 2-1/8-inch radius.
 - 6. Top and Bottom Edges: Closed with flush or inverted 0.042-inch- thick, end closures or channels of same material as face sheets.
 - 7. Tolerances: Comply with SDI 117, "Manufacturing Tolerances for Standard Steel Doors and Frames."
- B. Interior Doors: Face sheets fabricated from cold-rolled steel sheet unless metallic-coated sheet is indicated. Provide doors complying with requirements indicated below by referencing ANSI/SDI A250.8 for level and model and ANSI/SDI A250.4 for physical performance level: Level 2 and Physical Performance Level B (Heavy Duty), Model 1 (Full Flush).
- C. Exterior Doors: Provide doors complying with requirements indicated below by referencing ANSI A250.8 for level and model and ANSI A250.4 for physical-endurance level:
 - 1. Level 2 and Physical Performance Level B (Heavy Duty), Model 2 (Seamless)
- D. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.
- E. Fabricate concealed stiffeners and hardware reinforcement from either cold- or hot-rolled steel sheet.

2.4 STANDARD HOLLOW METAL FRAMES

- A. General: Comply with ANSI/SDI A250.8 and with details indicated for type and profile.
- B. Interior Frames: Fabricated from cold-rolled steel sheet unless metallic-coated sheet is indicated.
 - 1. Fabricate frames with mitered or coped corners.
 - 2. Fabricate frames as full profile welded unless otherwise indicated.
 - 3. Frames for Level 2 Steel Doors: 0.053-inch (16 gage) thick steel sheet.
 - 4. Frames for Wood Doors: 0.053-inch (16 gage) thick steel sheet.
 - 5. Frames for Borrowed Lights: 0.053-inch (16 gage) thick steel sheet.
- C. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 with reinforcement plates from same material as frames.

2.5 FRAME ANCHORS

A. Jamb Anchors:

1. Masonry Type: Adjustable strap-and-stirrup or T-shaped anchors to suit frame size, not less than 0.042 inch thick, with corrugated or perforated straps not less than 2 inches wide by 10 inches long; or wire anchors not less than 0.177 inch thick.
2. Stud-Wall Type: Designed to engage stud, welded to back of frames; not less than 0.042 inch thick.
3. Compression Type for Drywall Slip-on Frames: Adjustable compression anchors.
4. Postinstalled Expansion Type for In-Place Concrete or Masonry: Minimum 3/8-inch-diameter bolts with expansion shields or inserts. Provide pipe spacer from frame to wall, with throat reinforcement plate, welded to frame at each anchor location.

B. Floor Anchors: Formed from same material as frames, not less than 0.042 inch thick, and as follows:

1. Monolithic Concrete Slabs: Clip-type anchors, with two holes to receive fasteners.
2. Separate Topping Concrete Slabs: Adjustable-type anchors with extension clips, allowing not less than 2-inch height adjustment. Terminate bottom of frames at finish floor surface.

2.6 STOPS AND MOLDINGS

A. Moldings for Glazed Lites in Doors: Minimum 0.032 inch thick, fabricated from same material as door face sheet in which they are installed.

B. Fixed Frame Moldings: Formed integral with hollow metal frames, a minimum of 5/8 inch high unless otherwise indicated.

C. Loose Stops for Glazed Lites in Frames: Minimum 0.032 inch thick, fabricated from same material as frames in which they are installed.

D. Terminated Stops: Where indicated on interior door frames, terminate stops 6 inches above finish floor with a 45-degree angle cut, and close open end of stop with steel sheet closure. Cover opening in extension of frame with welded-steel filler plate, with welds ground smooth and flush with frame.

1. Provide terminated stops unless otherwise indicated.

2.7 ACCESSORIES

A. Mullions and Transom Bars: Join to adjacent members by welding or rigid mechanical anchors.

2.8 FABRICATION

A. Fabricate hollow metal work to be rigid and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. To ensure proper assembly at Project site, clearly identify work that cannot be permanently factory assembled before shipment.

B. Tolerances: Fabricate hollow metal work to tolerances indicated in SDI 117.

C. Hollow Metal Doors:

1. Glazed Lites: Factory cut openings in doors.
2. Astragals: Provide overlapping astragal on one leaf of pairs of doors where required by NFPA 80 for fire-performance rating or where indicated. Extend minimum 3/4 inch beyond edge of door on which astragal is mounted.

D. Hollow Metal Frames: Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.

- E. Welded Frames: Weld flush face joints continuously; grind, fill, dress, and make smooth, flush, and invisible.
- F. Sidelight Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material as door frame. Fasten members at crossings and to jambs by butt welding.
- G. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated.
- H. Floor Anchors: Weld anchors to bottom of jambs and mullions with at least four spot welds per anchor.
- I. Jamb Anchors: Provide number and spacing of anchors as follows:
 - 1. Masonry Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
 - a. Two anchors per jamb up to 60 inches high.
 - b. Three anchors per jamb from 60 to 90 inches high.
 - c. Four anchors per jamb from 90 to 120 inches high.
 - d. Four anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 120 inches high.
 - 2. Stud-Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
 - a. Three anchors per jamb up to 60 inches high.
 - b. Four anchors per jamb from 60 to 90 inches high.
 - c. Five anchors per jamb from 90 to 96 inches high.
 - d. Five anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 96 inches high.
 - 3. Two anchors per head for frames above 42 inches wide and mounted in metal-stud partitions.
 - a. Compression Type: Not less than two anchors in each jamb.
- J. Post installed Expansion Type: Locate anchors not more than 6 inches from top and bottom of frame. Space anchors not more than 26 inches o.c.
 - 1. Door Silencers: Except on weather-stripped doors, drill stops to receive door silencers as follows. Keep holes clear during construction.
- K. Single-Door Frames: Drill stop in strike jamb to receive three door silencers.
- L. Double-Door Frames: Drill stop in head jamb to receive two door silencers.
- M. Fabricate concealed stiffeners, edge channels, and hardware reinforcement from either cold- or hot-rolled steel sheet.
- N. Hardware Preparation: Factory prepare hollow metal work to receive templated mortised hardware; include cutouts, reinforcement, mortising, drilling, and tapping according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware."
 - 1. Locate hardware as indicated, or if not indicated, according to ANSI/SDI A250.8.
 - 2. Reinforce doors and frames to receive nontemplated, mortised and surface-mounted door hardware.
 - 3. Comply with applicable requirements in ANSI/SDI A250.6 and ANSI/DHI A115 Series specifications for preparation of hollow metal work for hardware.
 - 4. Coordinate locations of conduit and wiring boxes for electrical connections with Division 26 Sections.
- O. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with butted or mitered hairline joints.

1. Single Glazed Lites: Provide fixed stops and moldings welded on secure side of hollow metal work.
 2. Multiple Glazed Lites: Provide fixed and removable stops and moldings so that each glazed lite is capable of being removed independently.
 3. Provide fixed frame moldings on outside of exterior and on secure side of interior doors and frames.
 4. Provide loose stops and moldings on inside of hollow metal work.
 5. Coordinate rabbet width between fixed and removable stops with type of glazing and type of installation indicated.
- P. Lead Lined Hollow Metal Frames: Equip frames with lead lining where scheduled or indicated. Match thickness of lead with adjacent lead lined walls.
1. Equip frames with angle reinforcing welded to each jamb extending full height from floor to structure above complete with provisions for securing to floor and structure above.

2.9 STEEL FINISHES

- A. Prime Finish: Apply manufacturer's standard primer immediately after cleaning and pretreating.
1. Shop Primer: Manufacturer's standard, fast-curing, lead- and chromate-free primer complying with ANSI/SDI A250.10 acceptance criteria; recommended by primer manufacturer for substrate; compatible with substrate and field-applied coatings despite prolonged exposure.
- B. Color and Gloss: As indicated by manufacturer's designations.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine roughing-in for embedded and built-in anchors to verify actual locations before frame installation.
- C. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Remove welded-in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces.
- B. Prior to installation, adjust and securely brace welded hollow metal frames for squareness, alignment, twist, and plumbness to the following tolerances:
 1. Squareness: Plus or minus 1/16 inch, measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
 2. Alignment: Plus or minus 1/16 inch, measured at jambs on a horizontal line parallel to plane of wall.
 3. Twist: Plus or minus 1/16 inch, measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
 4. Plumbness: Plus or minus 1/16 inch, measured at jambs on a perpendicular line from head to floor.
- C. Drill and tap doors and frames to receive nontemplated, mortised, and surface-mounted door hardware.

3.3 INSTALLATION

- A. General: Install hollow metal work plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions.
- B. Hollow Metal Frames: Install hollow metal frames of size and profile indicated. Comply with ANSI/SDI A250.11 HMMA 840.
 - 1. Set frames accurately in position, plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces, leaving surfaces smooth and undamaged.
 - a. At fire-protection-rated openings, install frames according to NFPA 80.
 - b. Where frames are fabricated in sections because of shipping or handling limitations, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces.
 - c. Install frames with removable glazing stops located on secure side of opening.
 - d. Install door silencers in frames before grouting.
 - e. Remove temporary braces necessary for installation only after frames have been properly set and secured.
 - f. Check plumbness, squareness, and twist of frames as walls are constructed. Shim as necessary to comply with installation tolerances.
 - g. Field apply bituminous coating to backs of frames that are filled with grout containing anti-freezing agents.
 - 2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with postinstalled expansion anchors.
- C. Floor anchors may be set with powder-actuated fasteners instead of postinstalled expansion anchors if so indicated and approved on Shop Drawings.
- D. Metal-Stud Partitions: Solidly pack mineral-fiber insulation behind frames.
- E. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with grout.
- F. Concrete Walls: Solidly fill space between frames and concrete with grout. Take precautions, including bracing frames, to ensure that frames are not deformed or damaged by grout forces.
- G. In-Place Concrete or Masonry Construction: Secure frames in place with postinstalled expansion anchors. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
- H. In-Place Gypsum Board Partitions: Secure frames in place with postinstalled expansion anchors through floor anchors at each jamb. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
- I. Installation Tolerances: Adjust hollow metal door frames for squareness, alignment, twist, and plumb to the following tolerances:
 - 1. Squareness: Plus or minus 1/16 inch, measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
 - 2. Alignment: Plus or minus 1/16 inch, measured at jambs on a horizontal line parallel to plane of wall.
 - 3. Twist: Plus or minus 1/16 inch, measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
 - 4. Plumbness: Plus or minus 1/16 inch, measured at jambs at floor.
- J. Hollow Metal Doors: Fit hollow metal doors accurately in frames, within clearances specified below. Shim as necessary.
- K. Non-Fire-Rated Standard Steel Doors:

- a. Jamb and Head: 1/8 inch plus or minus 1/16 inch.
- b. Between Edges of Pairs of Doors: 1/8 inch plus or minus 1/16 inch.
- c. Between Bottom of Door and Top of Threshold: Maximum 3/8 inch.
- d. Between Bottom of Door and Top of Finish Floor (No Threshold): Maximum 3/4 inch.
- 2. Fire-Rated Doors: Install doors with clearances according to NFPA 80.
- 3. Smoke-Control Doors: Install doors according to NFPA 105.
- L. Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.
 - 1. Secure stops with countersunk flat- or oval-head machine screws spaced uniformly not more than 9 inches o.c. and not more than 2 inches o.c. from each corner.

3.4 ADJUSTING AND CLEANING

- A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or otherwise unacceptable.
- B. Remove grout and other bonding material from hollow metal work immediately after installation.
- C. Prime-Coat Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat and apply touchup of compatible air-drying, rust-inhibitive primer.
- D. Metallic-Coated Surfaces: Clean abraded areas and repair with galvanizing repair paint according to manufacturer's written instructions.

END OF SECTION

SECTION 08 14 16

FLUSH WOOD DOORS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Solid-core doors with wood-veneer, plastic-laminate faces.
 - 2. Factory finishing flush wood doors.
 - 3. Factory fitting flush wood doors to frames and factory machining for hardware.

1.3 SUBMITTALS

- A. Product Data: For each type of door indicated. Include details of core and edge construction, louvers, and trim for openings. Include factory-finishing specifications.
- B. Shop Drawings: Indicate location, size, and hand of each door; elevation of each kind of door; construction details not covered in Product Data; location and extent of hardware blocking; and other pertinent data.
 - 1. Indicate dimensions and locations of mortises and holes for hardware.
 - 2. Indicate dimensions and locations of cutouts.
 - 3. Indicate requirements for veneer matching.
 - 4. Indicate doors to be factory finished and finish requirements.
 - 5. Indicate fire-protection ratings for fire-rated doors.
- C. Samples for Initial Selection: For plastic-laminate door faces factory-finished doors.
- D. Warranty: Sample of special warranty.

1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain flush wood doors from single manufacturer.
- B. Quality Standard: In addition to requirements specified, comply with AWI's "Architectural Woodwork Quality Standards Illustrated." WDMA I.S.1-A, "Architectural Wood Flush Doors." WI's "Manual of Millwork."
 - 1. Provide AWI Quality Certification Labels or an AWI letter of licensing for Project indicating that doors comply with requirements of grades specified.
 - 2. Provide WI-Certified Compliance Certificate indicating that doors comply with requirements of grades specified.
 - 3. Provide WI-Certified Compliance Certificate for installation.
- C. Fire-Rated Wood Doors: Doors complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to NFPA 252 or UL 10C.
 - 1. Oversize Fire-Rated Door Assemblies: For units exceeding sizes of tested assemblies, provide certification by a qualified testing agency that doors comply with standard construction requirements for tested and labeled fire-rated door assemblies except for size.
 - 2. Temperature-Rise Limit: At vertical exit enclosures and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F above ambient after 30 minutes of standard fire-test exposure.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Comply with requirements of referenced standard and manufacturer's written instructions.
- B. Package doors individually in plastic bags or cardboard cartons.
- C. Mark each door on top and bottom rail with opening number used on Shop Drawings.

1.6 PROJECT CONDITIONS

- A. Environmental Limitations: Do not deliver or install doors until spaces are enclosed and weathertight, wet work in spaces is complete and dry, and HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.
- B. Environmental Limitations: Do not deliver or install doors until spaces are enclosed and weathertight, wet work in spaces is complete and dry, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 25 and 55 percent during the remainder of the construction period.

1.7 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Warping (bow, cup, or twist) more than 1/4 inch in a 42-by-84-inch section.
 - b. Telegraphing of core construction in face veneers exceeding 0.01 inch in a 3-inch span.
 - 2. Warranty shall also include installation and finishing that may be required due to repair or replacement of defective doors.
 - 3. Warranty Period for Solid-Core Interior Doors: Life of installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
- B. Algoma Hardwoods, Inc.
- C. Ampco, Inc.
- D. Eggers Industries.
- E. Graham; an Assa Abloy Group company.
- F. Marshfield Door Systems, Inc.
- G. Mohawk Flush Doors, Inc.; a Masonite company.
- H. VT Industries Inc.

2.2 DOOR CONSTRUCTION, GENERAL

- A. Low-Emitting Materials: Provide doors made with adhesives and composite wood products that do not contain urea formaldehyde.
- B. WDMA I.S.1-A Performance Grade: Standard Duty.
- C. Particleboard-Core Doors:
 - 1. Particleboard: ANSI A208.1, Grade LD-1.
 - 2. Particleboard: Straw-based particleboard complying with ANSI A208.1, Grade LD-2 or M-2, except for density.
 - 3. Blocking: Provide wood blocking in particleboard-core doors as needed to eliminate through-bolting hardware.

- a. 5-inch top-rail blocking, in doors indicated to have closers.
- b. 5-inch bottom-rail blocking, in exterior doors and doors indicated to have kick, mop, or armor plates.
- c. 5-inch midrail blocking, in doors indicated to have exit devices.
- 4. Provide doors with either glued-wood-stave or structural-composite-lumber cores instead of particleboard cores for doors indicated to receive exit devices.

D. Mineral-Core Doors:

- 1. Core: Noncombustible mineral product complying with requirements of referenced quality standard and testing and inspecting agency for fire-protection rating indicated.
- 2. Blocking: Provide composite blocking with improved screw-holding capability approved for use in doors of fire-protection ratings indicated as needed to eliminate through-bolting hardware.
 - a. 5-inch top-rail blocking.
 - b. 5-inch bottom-rail blocking, in doors indicated to have protection plates.
 - c. 5-inch midrail blocking, in doors indicated to have armor plates.
 - d. 4-1/2-by-10-inch lock blocks, in doors indicated to have exit devices.
- 3. Edge Construction: At hinge stiles, provide laminated-edge construction with improved screw-holding capability and split resistance. Comply with specified requirements for exposed edges.

E. Lead Sheet: FS QQ-L-201.

2.3 VENEERED-FACED DOORS FOR TRANSPARENT FINISH

A. Interior Solid-Core Doors :

- 1. Grade: Custom (Grade A faces)
- 2. Species: Refer to Finish Legend.
- 3. Cut: .Refer to Finish Legend.

2.4 MATCH BETWEEN VENEER LEAVES: BOOK MATCH.

- 1. Assembly of Veneer Leaves on Door Faces: Center-balance match.
- 2. Pair and Set Match: Provide for doors hung in same opening.
- 3. Room Match: Match door faces within each separate room or area of building. Corridor-door faces do not need to match where they are separated by 20 feet or more.
- 4. Room Match: Provide door faces of compatible color and grain within each separate room or area of building.
- 5. Exposed Vertical Edges: Same species as faces or a compatible species.
- 6. Core: Particleboard .
- 7. Construction: Five plies. Stiles and rails are bonded to core, then entire unit abrasive planed before veneering.
- 8. WDMA I.S.1-A Performance Grade: Standard Duty.

2.5 DOORS FOR OPAQUE FINISH

A. Interior Solid-Core Doors:

- 1. Grade: Custom.
- 2. Faces: Any closed-grain hardwood of mill option.
 - a. Apply medium-density overlay to standard-thickness, closed-grain, hardwood face veneers.
 - b. Hardboard Faces: AHA A135.4, Class 1 (tempered) or Class 2 (standard).
 - c. MDF Faces: ANSI A208.2, Grade 150 or 160.
- 3. Exposed Vertical and Top Edges: Any closed-grain hardwood.

4. Core: Particleboard .
5. Construction: Five plies. Stiles and rails are bonded to core, then entire unit abrasive planed before veneering. Faces are bonded to core using a hot press.
6. Construction: plies, either bonded or non-bonded construction.
7. WDMA I.S.1-A Performance Grade: Standard Duty.

2.6 LIGHT FRAMES

- A. Metal Frames for Light Openings in Fire-Rated Doors: Manufacturer's standard frame formed of 0.048-inch- thick, cold-rolled steel sheet; factory primed for paint finish; and approved for use in doors of fire-protection rating indicated.

2.7 FABRICATION

- A. Factory fit doors to suit frame-opening sizes indicated. Comply with clearance requirements of referenced quality standard for fitting unless otherwise indicated.
 1. Comply with requirements in NFPA 80 for fire-rated doors.
- B. Factory machine doors for hardware that is not surface applied. Locate hardware to comply with DHI-WDHS-3. Comply with final hardware schedules, door frame Shop Drawings, DHI A115-W series standards, and hardware templates.
 1. Coordinate with hardware mortises in metal frames to verify dimensions and alignment before factory machining.
 2. Metal Astragals: Factory machine astragals and formed-steel edges for hardware for pairs of fire-rated doors.
- C. Openings: Cut and trim openings through doors in factory.
 1. Light Openings: Trim openings with moldings of material and profile indicated.
 2. Glazing: Factory install glazing in doors indicated to be factory finished. Comply with applicable requirements in Division 08 Section "Glazing."
 3. Louvers: Factory install louvers in prepared openings.
- D. Lead-Lined Doors:
 1. Construction: Five-ply (exclusive of lead).
 2. Core: Formed particleboard, bonded to stiles and rails.
 3. Lead Sheet: Bonded or secured with lead covered bolts to each side of core or centered in divided core.
 - a. Maximum combined lead thickness: Match thickness of lead in adjacent lead lined walls.

2.8 FACTORY FINISHING

- A. General: Comply with referenced quality standard for factory finishing. Complete fabrication, including fitting doors for openings and machining for hardware that is not surface applied, before finishing.
 1. Finish faces, all four edges, edges of cutouts, and mortises. Stains and fillers may be omitted on top and bottom edges, edges of cutouts, and mortises.
- B. Finish doors at factory.
- C. Finish doors at factory that are indicated to receive transparent finish. Field finish doors indicated to receive opaque finish.
- D. Finish doors at factory where indicated in schedules or on Drawings as factory finished.
- E. Transparent Finish:
 1. Grade: Custom.
 2. Finish: AWI system.

3. Staining: As selected by Architect from manufacturer's full range.
 4. Effect: Open-grain finish.
 5. Sheen: Satin.
- F. Opaque Finish:
1. Grade: Custom.
 2. Finish: AWI catalyzed polyurethane system.
 3. Color: As selected by Architect from manufacturer's full range.
 4. Sheen: Satin.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine doors and installed door frames before hanging doors.
1. Verify that frames comply with indicated requirements for type, size, location, and swing characteristics and have been installed with level heads and plumb jambs.
 2. Reject doors with defects.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Hardware: For installation, see Division 08 Section "Door Hardware."
- B. Installation Instructions: Install doors to comply with manufacturer's written instructions and the referenced quality standard, and as indicated.
1. Install fire-rated doors in corresponding fire-rated frames according to NFPA 80.
- C. Factory-Fitted Doors: Align in frames for uniform clearance at each edge.
- D. Factory-Finished Doors: Restore finish before installation if fitting or machining is required at Project site.

3.3 ADJUSTING

- A. Operation: Re-hang or replace doors that do not swing or operate freely.
- B. Finished Doors: Replace doors that are damaged or that do not comply with requirements. Doors may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing.

END OF SECTION

SECTION 08 31 00

ACCESS DOORS AND PANELS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Furnishing and installation of access doors and panels where indicated or noted on Architectural Drawings.
- B. Product Installed But Not Furnished Under This Section:
 - 1. Install access doors and panels furnished by other trades under plumbing, fire protection, mechanical and electrical work where necessary for operation and maintenance of concealed equipment, valves, cleanouts, controls, and other similar devices requiring access.
 - 2. Coordinate requirements with Divisions 21, 22, 23 and 26.
- C. Related Sections:
 - 1. Section 08 71 00 - Door Hardware: Keyed cylinders.
 - 2. Section 09 91 00 - Painting: Field paint finish.

1.2 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit for each type of access door and panel.
- C. Shop Drawings: Include complete schedule indicating types, general locations, sizes, wall and ceiling construction details, latching and locking provisions, and other pertinent data.

1.3 QUALITY ASSURANCE

- A. Fire-Resistance Ratings: Appropriate UL or Warnock Hersey label required on fire rated assemblies.
- B. Size Variations: Obtain Architect's acceptance of manufacturer's standard size units which may vary from sizes indicated.
- C. Coordination: Furnish inserts and anchoring devices which must be built into other work for installation of access panels. Coordinate delivery with other work to avoid delay.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers:
 - 1. Bar-Co., Enterprise, AL.
 - 2. Cesco Products, Minneapolis, MN.
 - 3. Elmdor/Stoneman Division of Acorn Engineering, City of Industry, CA.
 - 4. JL Industries, Bloomington, MN.
 - 5. Karp Associates, Inc., Maspeth, NY.
 - 6. Milcor Incorporated, Lima, OH.
 - 7. Nystrom, Inc., Minneapolis, MN.
 - 8. The Williams Brothers Corporation of America, Front Royal, VA.
 - 9. No Substitutions.

2.2 ACCESS DOORS MATERIALS AND FABRICATION

- A. Steel Assemblies: Fabricate of sheet steel, ASTM A1008.
- B. Galvanized Steel Assemblies (In Wet Areas): ASTM A653/A653M, Quality SS, Grade as required by design, Coating Designation G90.
- C. Frames: Fabricate typical units with metal thickness of not lighter than 16 gage unless otherwise required for fire-rated or security conditions.
 - 1. Fabricate with exposed flange of nominal 1 inch wide around perimeter for assemblies installed in following construction:
 - a. Exposed masonry or concrete.
 - b. Ceramic tile.
 - c. Suspended ceiling, acoustical tile.
 - 2. Fabricate with galvanized casing bead welded to perimeter of frame for assemblies installed in gypsum board surfaces.
- D. Non-Rated Doors: For openings 12 inches by 12 inches or less, fabricate units with metal thickness of not lighter than 16 gage; for openings greater than 12 inches by 12 inches, fabricate units with metal thickness of not lighter than 14 gage sheet materials.
 - 1. Non-Rated Recessed Doors: Fabricate with recessed door panel to allow application of gypsum board finish on door. Equip with concealed center pivot hinge.
 - 2. Typical: Equip with concealed spring hinges or concealed continuous piano hinge designed to open 175 degree minimum.
- E. Fire-Rated Flush Doors: Fabricate of not lighter than 20 gage sheet materials of sandwich type construction with manufacturer's standard core. Equip with continuous piano hinge, self-closing mechanism, and interior latch release mechanism.
- F. Locking Devices: Provide flush, screwdriver-operated cam locks of number required to hold door in flush, smooth plane when closed.

2.3 FINISHES

- A. Steel Assemblies:
 - 1. Phosphatize or chemically treat surfaces before application of prime coat finish.
 - 2. Apply rust inhibitive primer to produce uniform smooth coat at 2.0 mils minimum dry film thickness.
- B. Galvanized Steel Assemblies, Field Finish: See Section 09 91 00.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 40 00.
 - 1. Verify rough openings are correctly located, sized and prepared for installation.

3.2 INSTALLATION

- A. Access Doors: Install plumb, level, square and rigidly secured in accordance with Section 01 73 00.
 - 1. Position for convenient access to concealed work requiring access.

3.3 ADJUSTING

- A. After installation, test and adjust doors and hardware for smooth operation.

END OF SECTION

SECTION 08 34 00

SPECIAL FUNCTION DOORS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings And General Provisions of The Contract, Including General And Supplementary Conditions And Division 01 Specification Sections, Apply To This Section.

1.2 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.3 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Manual Sliding Doors - Surface Mount, as indicated, in accordance with provisions of Contract Documents.
- B. Completely coordinate with work of other trades.
- C. Section includes:
 - 1. Interior Aluminum-Framed Top-Hung Sliding Doors
 - 2. Related Sections:
 - a. Section 08 14 16 – Flush Wood Door
 - b. Section 08 80 00 - Glazing
 - c. Section 08 71 00 – Door Hardware

1.4 REFERENCES

- A. ANSI – American National Standards Institute
- B. ANSI 156.18 Materials and Finishes
- C. ANSI A117.1 Specifications for making buildings and facilities usable by physically handicapped people.
- D. ANSI Z97.1 Standards for Safety Glazing Material Used in Buildings
- E. BHMA – Builders Hardware Manufacturers Association
- F. DHI – Door and Hardware Institute
- G. NFPA – National Fire Protection Association
- H. NFPA 80 – Fire Doors and Windows
- I. NFPA 101 – Life Safety code
- J. NFPA 105 – Smoke and Draft Control Door Assemblies
- K. NFPA 252 – Fire Tests of Doors Assemblies
- L. AWS – Architectural Woodwork Standards
- M. ASTM International (ASTM):
 - 1. ASTM B209 Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate.
 - 2. ASTM B221 Standard Specification for Aluminum and Aluminum Alloy Extruded Bars, Rods, Wire, Profiles and Tubes.
- N. American Architectural Manufacturers Association (AAMA):
 - 1. AAMA 611 Voluntary Specification for Anodized Architectural Aluminum
- O. The Aluminum Association (AA):

1. Aluminum Finishes Manual.

1.5 SUBMITTALS

- A. Comply with Section 01 33 00 – Submittal Procedures
- B. Product Data: Submit manufacturer's product data, including installation instructions.
- C. Shop Drawings: Submit manufacturer's shop drawings, including plans, elevations, sections, and details, indicating dimensions, tolerances, materials, components, hardware, finish, options, and accessories. Shop Drawings to show required blocking by others.
- D. Samples: Submit manufacturer's samples of the following sliding door components:
 1. Door veneer or laminate sample.
 2. Framing Members: 12-inch-long (300-mm-long) section with seals, infill panels, glazing bead, and factory-applied color finish.
 3. Hardware: Full-size units with factory-applied finish.
 4. Manufacturer's Certification: Submit manufacturer's certification that materials comply with specified requirements and are suitable for intended application.
- E. Warranty Documentation: Submit manufacturer's standard warranty for complete system
- F. Contract Closeout Information:
 1. Operation and Maintenance Data.
- G. See Section 01 78 23.

1.6 QUALITY ASSURANCE

- A. Product Options: Drawings indicate size, profiles, and dimensional requirements of interior aluminum frames and doors.
- B. Source: Obtain sliding aluminum framed doors and hardware from single source.
- C. Manufacturer's Qualifications: Manufacturer regularly engaged for past 5 years in manufacture of sliding doors similar to that specified.
- D. Installer Qualifications:
- E. Employed or trained by product manufacturer, with a minimum five (5) years of experience on installation and maintenance of manual sliding doors and equipment.

1.7 PERFORMANCE

- A. Aluminum perimeter frames with integral acoustic seals at all door/frame interfaces
 1. Architect to verify frame thickness suitable for required application
- B. Soft-closing mechanism at both sides of door integrated with top track. Soft Closers tested to a minimum of 150,000 cycles.
- C. Fully Concealed door guide
- D. Manufacturer to provide 3rd party report demonstrating compliance with UL1784, UL10B as required.
- E. Noise Isolation Class (NIC): Rated for not less than 39 NIC according FGI 2018 requirements for Inpatient/Outpatient facilities.
- F. frame option permitted on openings with 42" clear opening width.

1.8 DOOR TYPES

- A. Provide the following types of configurations with options and accessories as scheduled in 087100 and in the door schedule provided. .
 1. SD-01: OfficeSlide flush sliding Barn-style door assembly by AD Systems.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Delivery and Acceptance Requirements: Deliver materials to site in manufacturer's original, unopened containers and packaging, with labels clearly identifying product name and manufacturer.
- B. Notify manufacturer immediately of any shipping damage.
- C. Storage and Handling Requirements:
 - 1. Store and handle materials in accordance with manufacturer's instructions.
 - 2. Keep materials in manufacturer's original, unopened containers and packaging until installation.
 - 3. Store materials in clean, dry area indoors.
 - 4. Protect materials and finish during storage, handling, and installation to prevent damage.

1.10 WARRANTY

- A. Special Warranties: Prepare and submit in accordance with Section 01 78 00.
 - 1. Warrant to repair or replace components of aluminum sliding glass doors that fail in materials or workmanship within three years. Failures include, but are not limited to, following:
 - a. Structural failures, including excessive deflection, water leakage, air infiltration, or condensation.
 - b. Faulty operation of movable panels and hardware.
 - 2. Warranty organic coating finish to be free from fading more than 10 percent, chalking, yellowing, peeling, cracking, pitting, corroding or non-uniformity of color, or gloss deterioration beyond manufacturer's descriptive standards for 10 years.
 - 3. Warrant insulated glass units to be free from condensation, fogging and obstruction of vision due to film on internal surfaces for 10 years.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. AD Systems 2201 100th St. Sw, Everett, WA 98204 | Website: [Http://Specadsystems.Com](http://Specadsystems.Com) | Phone: 425-740-6011 | Adsystems.Estimating@Allegion.com
- B. Basis Of Design: Flush Wood Single Sliding & Flush Aluminum Sliding

2.2 INTERIOR SLIDING ALUMINUM-FRAMED DOORS AND PARTITIONS

- A. Ad systems high performance sliding door system
- B. No substitutions
- C. Specified Wall Thickness: <Insert Wall Thickness>. Refer to Partition Schedule
- D. Frame Profiles: Extruded aluminum frame "wrap" frame with integral vertical jamb (stile pocket). Frames required to complete seal around door leaf. Gasketing required at all frame to door interfacing. Exposed gaskets at jamb to be silicone.
 - 1. Header track:
 - a. Extruded aluminum with closed header and end cap with removable face plate.
 - b. 4 inches deep x 2 1/2 inches high.
 - c. Provide sloped top valance at AFB Lab Only.
 - 2. Jamb:
 - a. Manufacturer's standard extruded aluminum framing.
 - 3. 1-3/4 inches
- E. Finish:

1. Standard: Painted Hardcoat (Kynar) Finish. Meets AAMA 2604 Standard Colors: Light Sequin 789G048
- F. Door Leafs. All Doors to be factory machined for hardware including pilot and function holes. Leading edge of door to be fully finished.
1. 1-3/4" Flush Wood Door: Reference Spec Section 08 20 008 14 16 Wood Doors or other section as applicable. OR OPTION: Specify: [Solid Timber-Strand Core] [Solid Particle Core] <Insert veneer species, cut, stain or plastic laminate>.
 - a. Optional Glazing: [monolithic clear tempered] [safety laminated glass] [sound enhanced laminated glass] [integral blinds] [other glass as specified by architect].
 - b. Standard stile widths are 6" with a 10" bottom rail.
 2. 1-3/4" Aluminum Stile & Rail Door: 3-1/2" stiles plus 1/2" stop. p.
 - a. 10" bottom Rail.
 - b. Aluminum Composite Panel at BSL-3 Only.
 - c. Glazing: [monolithic clear tempered] [safety laminated glass] [sound enhanced laminated glass] [integral blinds] [other glass as specified by architect].
 - d. Other 1-3/4" Doors. Contact AD Systems.
- G. Door Components (Required):
1. Single Top Track: Anodized finish aluminum track
 2. Valances: Extruded aluminum with integral end caps
 3. Standard square valance.
 - a. Sloped top valance at BSL-3 only.
 4. Top Rollers: tandem nylon roller sized to match door weight
 5. Concealed Floor Guide: Integral Jamb floor guide by AD Systems
 6. Soft-Closers: Soft-closing dampeners at both sides of door leaf.
- H. Hardware:
1. Privacy lock:
 - a. Privacy with occupancy indicator.
 - b. 2001SDL by Accurate Lock and Hardware
 - c. ADA compliant thumb turn No. 7200ADA.
 - d. Coin emergency release No. 7200ER-I.
 2. Privacy lock and pull:
 - a. Privacy with occupancy indicator.
 - b. ADA compliant thumb turn.
 - c. Coin emergency release.
 - d. 9100ADAL-5i by Accurate Lock and Hardware
 3. Privacy lock:
 - a. CL400 by CaviLock
 - b. Privacy lock function.
 - c. Barn door strike.
 - d. Mortise manual hookbolt.
 - 1) Latching device to include ADA compliant thumbturn or lever on restroom side with coin style emergency release on outside.
 - e. Mortise self-latching lockbolt.
 - 1) Latching device to include ADA compliant lever handles each side.
 - 2) Include compatible strike and strike box.
 - f. Recessed pull installed on outside and outside of active door leaves.

4. Surface-mounted pull.
 - a. Round Wrought Door Pull.
 - b. Mount on both sides of door.
 - c. 12 inches x 3/4 inches.
 - d. Finish Code:
 - 1) US32D Satin Stainless Steel.
 - e. Comply with quality and standards specified in Section 08 71 00.
 - f. Hager H3L.
5. Weatherstripping:
 - a. Captured.
 - b. Door nose: Spring loaded and adjustable.
 - c. Interlock rail and sliding panel edge: Blade type.
6. Privacy and Office Locksets.
 - a. Refer to specified hardware in Section 08 71 00.
- I. Aluminum Finish:
 1. Anodized:
 - a. 204-R1 Clear: Arch. Class 2 Clear Anodized Coating, AA-MI2C22A31.
- J. Supports:
 1. Provide structural components required to support and brace units.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine wall openings to receive sliding doors for plumb, level, and square. Note: finish door operation will be affected by out of tolerance framing. Examine surfaces to receive top and bottom guide.
- B. Verify dimensions of wall openings.
- C. Examine surfaces to receive top and bottom guide.
- D. Notify architect of conditions that would adversely affect installation or subsequent use of sliding doors.
- E. Do not begin installation until unacceptable conditions are corrected.
- F. Base of door side to be flush or minimal. Rubber base acceptable.
- G. Perform corrective work as required to allow successful installation.
- H. Installation constitutes acceptance of responsibility for performance

3.2 INSTALLATION

- A. Install sliding doors in accordance with manufacturer's instructions at locations indicated on the drawings.
- B. Install sliding doors plumb, level, square, and in proper alignment.
- C. Install sliding doors to close against walls without gaps
- D. Install sliding doors to open and close smoothly.
- E. Anchor sliding doors securely in place to supports. Required: fire treated 2 x 6 blocking required full length of track.

3.3 ADJUSTING

- A. Adjust sliding doors for proper operation in accordance with manufacturer's instructions.

- B. Adjust sliding doors to operate smoothly without binding. Adjust in accordance with manufacturer's instructions to ensure smooth operation.
- C. Repair minor damages to finish in accordance with manufacturer's instructions and as approved by Architect.

3.4 PREPARATION

- A. Prepare substrate in accordance with manufacturer's instructions.
- B. If required by flooring manufacturer, comply with requirements of Section 07 16 04 Concrete Floor Moisture Testing, and Section 07 16 05 Water Vapor Emission Control System.
- C. Fill construction joints and other non-moving joints with elastomeric sealant approved by manufacturer.
- D. Protect adjacent surfaces or both, including fixtures, cabinetwork, equipment, and similar items

3.5 ADJUSTMENT AND CLEANING

- A. Adjust door and hardware for smooth, safe operation.
- B. Remove excess sealants, compounds, dirt, and other substances.
- C. Clean surfaces soiled by door installation.
- D. Repair or replace damaged items as required.

3.6 CLEANING

- A. Cleaning: Comply with Section 01 74 23017400. Clean surfaces in compliance with manufacturer's recommendations; remove excess mastic, mastic smears, foreign materials and other unsightly marks.
 - 1. Clean metal surfaces exercising care to avoid damage.
- B. Clean sliding doors promptly after installation in accordance with manufacturer's instructions.
- C. Do not use harsh cleaning materials or methods that could damage materials or finish.

3.7 PROTECTION

- A. Protect installed sliding doors from damage during construction.

END OF SECTION

SECTION 08 34 23
SPECIALTY CLEANROOM WINDOWS

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Specialty Cleanroom Windows, as indicated, in accordance with provisions of Contract Documents.
- B. Completely coordinate with work of other trades.

1.2 REFERENCES

- A. NAAMM, National Association of Architectural Metal Manufacturers:
 - 1. ANSI/NAAMM HMMA 866-01 Guide Specifications for Stainless Steel Hollow Metal Doors and Frames
 - 2. NAAMM HMMA 802 Manufacturing of Hollow Metal Doors and Frames
- B. SDI, Steel Door Institute:
 - 1. ANSI/SDI A250.4 Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames and Frame Anchors
- C. NFPA, National Fire Protection Association:
 - 1. NFPA 80 Standard for Fire Doors and Other Opening Protectives
 - 2. NFPA 252 Standard Methods of Fire Tests of Door Assemblies
 - 3. NFPA 257 Standard on Fire Test for Window and Glass Block Assemblies
- D. ASTM International:
 - 1. ASTM A480 Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
 - 2. ASTM C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
 - 3. ASTM D256 Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
 - 4. ASTM D638 Standard Test Method for Tensile Properties of Plastics
 - 5. ASTM D790 Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
 - 6. ASTM D792 Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
 - 7. ASTM D1622 Standard Test Method for Apparent Density of Rigid Cellular Plastics
 - 8. ASTM D2583 Standard Test Method for Apparent Density of Rigid Cellular Plastics
 - 9. ASTM E84 Standard Test Method for Surface Burning Characteristics of Building Materials
 - 10. ASTM E119 Standard Test Methods for Fire Tests of Building Construction and Materials
- E. ANSI, American National Standards Institute:
 - 1. ANSI/UL 263 Standard for Safety of Fire Tests of Building Construction Materials

1.3 SUBMITTALS

- A. Product Data:
 - 1. For each type of material and accessory.
- B. Shop Drawings:

1. Indicate door locations and numbers, material thickness, dimensions, elevations, assembly methods, hardware reinforcement locations and preparations, fabrication, anchorage details specific to project and wiring and electrical connections.
- C. Samples:
1. Manufacturer's standard colors on actual door and frame material for selection.
 2. Frame: 12 by 12-inch corner section showing welding of head to jamb. Include hinge mortise, reinforcement, and plaster guard in one rabbet and glazing stop applied as specified in the opposite rabbet. Glazing stop shall be applied to both head and jamb section to show corner joint.
 3. Samples shall be of the production type and shall represent in all respects the minimum quality of work to be furnished by the Manufacturer. Downgrading of quality demonstrated by comparison with the samples may be cause for rejection of the work.
- D. Project Information:
1. Certification of installer qualifications.
 2. Document certifying work meets or exceeds specified physical properties.
 3. Manufacturers installation instructions.
- E. Contract Closeout Information:
1. Operation and Maintenance Data: Refer to Section 01 78 23.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications:
1. Door manufacturer, or manufacturer trained, approved, and licensed door installer.
- B. Manufacturer Qualifications:
1. Manufacturer shall provide evidence of having personnel and plant equipment capable of fabricating door and frame assemblies of the type specified.
 2. Manufacturer shall provide evidence of having a quality control system in place.
 3. Manufacturer shall have been regularly engaged in the manufacture of stainless steel doors and frames for a period of no less than 5 years.
- C. Fabrication methods and product quality shall meet ANSI/NAAMM HMMA 866-01.

1.5 SPECIAL WARRANTY

- A. Warrant frames against failure due to workmanship or materials for 5 years from date of Substantial Completion.
1. Include failure to resist thermal warp beyond that verified by calculation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Specialty Cleanroom Windows:
1. Base:
 - a. DorteK
 2. Optional:
 - a. ASI Doors
- B. Other manufacturers desiring approval comply with Section 01 61 00.

2.2 PERFORMANCE REQUIREMENTS

- A. Performance Testing:

1. Test specimen: 3 by 7 feet nominal size, 1-3/4-inch thick door, representative of the material to be provided.
2. Test specimen shall be tested in accordance with the ANSI A250.4 procedure and shall meet the Acceptance Criteria for the Level A frames.
3. Test reports or certificates of compliance shall include a description of the test specimen, procedures used in testing, and indicate compliance with the acceptance criteria for the test.

B. Fire Rated Window Frames:

1. Frames shall be provided for those openings requiring fire protection ratings as determined and scheduled in the contract documents. Such frames shall be constructed as tested in accordance with ANSI/NFPA 252 and listed and classified for labeling by a recognized testing agency having a factory inspection service.
2. If frames specified in the contract documents to be fire-rated cannot qualify for appropriate labeling because of design, hardware, or other reasons, the Architect shall be so advised before fabricating work on that item is started.
3. Window frames shall be provided for those openings requiring fire protection ratings as determined and scheduled in the contract documents. Such windows shall be constructed as tested in accordance with ANSI/NFPA 257 and listed and classified for labeling by a recognized testing agency having a factory inspection service.

2.3 MATERIALS

A. Window Frames:

1. Manufacturers standard except as noted in this specification.

B. Window Frames:

1. Basis of Design: - Dortek Stainless Steel Observation Windows.
2. Seamless stainless steel, Type 304 with #4 satin finish conforming to ASTM A480.
3. Thickness:
 - a. For door openings 4 feet or less in width and for window frames, stainless steel shall be 0.053 inches minimum thickness.
 - b. For door openings greater than 4 feet in width, stainless steel shall be 0.067 inches minimum thickness.
4. Frames shall be constructed in accordance with NAAMM HMMA 820 with regard to joint design and welding techniques.
5. Corners: Mitered and fully welded with molded pockets and corner reinforcement.
6. Flush face joints at mullions, sills, and transoms shall be continuously welded and smoothly finished.
7. Reinforce for specified finish hardware.
 - a. Frames shall be mortised, reinforced, drilled, and tapped at the factory for templated mortised hardware only, in accordance with the final approved hardware schedule and templates provided by the hardware supplier. Hardware reinforcing shall be Type 304 stainless steel.
 - b. Minimum thickness of hardware reinforcing plates shall be as follows:
 - c. Hinge reinforcements 0.167 by 1.25 by 10-inch .
 - d. Strike, closer, flush bolt, surface applied hardware, hold open arms, surface panic devices reinforcements 0.093 inches .
8. Anchors:
 - a. Metal stud:
 - 1) Frames for installation in gypsum board, metal stud partitions shall be provided with anchors of suitable design, not less than 0.042-inch thickness, securely welded inside each jamb.

- 2) Anchors shall be placed not greater than 18 inches from top and bottom of openings.
- 3) Maximum spacing for intermediate anchors at 32 inches .
- b. Existing openings:
 - 1) Frames for installation in existing masonry or concrete walls shall be prepared for expansion bolt type anchors.
 - 2) The preparation shall consist of a countersunk hole for a 3/8-inch diameter bolt and a spacer from the unexposed surface of the frame to the wall. The spacer shall be welded to the frame and spaced a maximum of 6 inches from the top and bottom of the door openings, with intermediate spacing at a maximum of 26 inches o.c.
- c. Floor anchors:
 - 1) Thickness of floor anchors shall be minimum of 0.067 inches . Floor anchors shall be Type 304 stainless steel.
 - 2) Where applicable, floor anchors shall be provided with two holes for fasteners and shall be fastened inside jamb with at least 4 welds per anchor.
 - 3) Where scheduled, adjustable floor anchors, providing no less than 2-inch height adjustment, shall be fastened in places with at least 4 welds per anchor.
 - 4) For applications that do not permit the use of a floor anchor, an additional jamb anchor at the base of the frame can be substituted.
- C. Window Glazing:
 1. Non-Fire rated Windows:
 - a. Manufacturer's standard Reinforced Glazing. Glazing shall be flush with frame surface on both sides of the window frame.
 - b. Glazing: 1/4 inches tempered glass.
 - 1) 1" Ceramic Border
- D. Finishes: All exposed surfaces of stainless steel frames shall be finished in accordance with NAAMM HMMA 802 Manufacturing of Hollow Metal Doors and Frames "Finishes for Stainless Steel."

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify installation conditions are satisfactory to receive work of this Section.
- B. Verify opening size, dimensions, and tolerances.
- C. Do not install until unsatisfactory conditions are corrected.
- D. Beginning work constitutes acceptance of conditions as satisfactory.

3.2 PREPARATION

- A. Protect surrounding areas and surfaces to prevent damage during work of this section.
- B. Prior to installation, frames shall be checked and corrected for size, swing, squareness, alignment, twist and plumbness. Permissible installation tolerances shall not exceed the following:
 1. Squareness: +/- 1/16 inches measured on a line, 90 degrees from one jamb, at the upper corner of the frame at the other jamb.
 2. Alignment: +/- 1/16 inches measured on the jambs on a horizontal line parallel to the plane of the wall.
 3. Twist: +/- 1/16 inches measured at face corners of jambs on parallel lines perpendicular to the plane of the wall.
 4. Plumbness: +/- 1/16 inches measured on the jamb at the floor.

3.3 INSTALLATION

- A. Install frames plumb, level, square and securely anchored to opening.
- B. Install frames with specified hardware and adjust for smooth non-binding operation.
- C. Install labeled fire rated frames in accordance with the terms of their listings and in accordance with NFPA 80.
- D. Maximum Clearances:
 - 1. Jambs and Head: 3/16 inches.
 - 2. Meeting Stiles: 3/16 inches.
 - 3. Sills:
 - a. 3/4 inches between door bottom and finished floor.
 - b. 1/2 inches between door bottom and surface of floor covering at fire rated openings per ANSI/NFPA 80.

3.4 CLEANING AND REPAIR

- A. Finished and polished surfaces that have been scratched or otherwise marred during installation, including field welding, shall be refinished to match the original polish or finish.
- B. Stainless steel surfaces shall be kept free of grout, tar and other bonding materials or sealers. Grout, tar, and other bonding materials or sealers shall be promptly cleaned off of frames and/or doors to avoid permanent staining.

END OF SECTION

SECTION 08 41 13

ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Aluminum doors complete with hardware.
 - 2. Accessories necessary to complete work.
- B. Products Furnished But Not Installed Under This Section:
 - 1. Anchoring devices which are built into masonry.
 - 2. Anchoring devices which are cast in concrete.
 - 3. Glazing accessories which are part of aluminum framing system.
- C. Related Requirements: Comply with following:
 - 1. Metal Finishes: Section 05 05 10.
 - 2. Perimeter Joint Sealants: Section 07 92 00 - Joint Sealants.
 - 3. Special Function Doors: Section 08 34 00
 - 4. Glass and Glazing: Section 08 80 00 - Glazing.
- D. Related Sections:
 - 1. Section 07 92 00 - Joint Sealants: Perimeter sealants and backup materials.
 - 2. Section 08 71 00 - Door Hardware: Keyed cylinders.
 - 3. Section 08 80 00 - Glazing.

1.2 SYSTEM REQUIREMENTS

- A. General: In addition to requirements shown or specified, comply with design requirements of Section 01 83 00.
- B. Design Requirements: Manufacturer: Responsible for designing system, including anchorage to structural system and necessary modifications to meet specified requirements and maintain visual design concepts.
 - 1. Employ a registered professional engineer, licensed to practice structural engineering in jurisdiction where the Project is located, to engineer each component of storefront system.
 - 2. Drawings: Diagrammatic and do not purport to identify nor solve problems of thermal or structural movement, glazing, anchorage, or moisture disposal.
 - 3. System: Metal stick-framed system, steel reinforced, with exposed interior and exterior metal framing members.
 - 4. Requirements Shown by Details: Establish basic dimension of units, sight lines and profiles of members.
 - 5. Provide entrance and storefront systems, including necessary modifications, to meet specified requirements and maintaining visual design concepts.
 - 6. Assemblies: Free from rattles, wind whistles and noise due to thermal and structural movement and wind pressure.
 - a. Provide for expansion and contraction due to structural movement without detriment to appearance or performance.
 - b. Attachments: Account for site peculiarities and expansion and contraction movements so there is no possibility of loosening, weakening and fracturing connection between units and building structure or between units themselves.

- c. Anchors, Fasteners and Braces: Structurally stressed not more than 50 percent of allowable stress when maximum loads are applied.
 - d. Provide concealed fastening wherever possible.
- 7. Glazing Requirements: Comply with Section 08 80 00.
- 8. Anticipated Live Load Interior Beam Deflections: 1/360 of beam length.
- C. Thermal Requirements: Framing Systems: Accommodate expansion and contraction movement due to surface temperature differentials of 180 F without causing buckling, stress on glass, failure of joint seals, excessive stress on structural elements, reduction of performance, or other detrimental effects.
 - 1. Ensure doors function normally within limits of specified temperature range.
 - 2. Deflection Under Uniform Loading: When tested in accordance with ASTM E330 at design pressure, maximum deflection of exterior member shall not exceed L/175 or 3/4 inch maximum for spans under 13 feet – 6 inches and L/240 plus 1/4 inch for spans 13 feet – 6 inches or greater. Maximum deflections of members supporting plaster surfaces shall not exceed 1/360 of its span.
- D. Sealant Performance and Requirements: Section 07 92 00.
- E. Testing Requirements: Provide components that have been previously tested by independent testing laboratory.
- F. Interface With Adjacent Systems: Integrate design and connections with adjacent construction.
 - 1. Accommodate allowable tolerances and deflections for structural members in installation.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: For the storefront system.
 - 1. Include information for factory finishes, hardware, accessories and other required components.
 - 2. Include sample of warranty customized for this Project.
- C. Shop Drawings: Indicate fabrication, installation and finish of specified systems.
 - 1. Stamp with seal and signature of professional engineer responsible for design.
 - 2. Include Following:
 - a. Fully dimensioned plans and elevations with detail coordination keys.
 - b. Locations of exposed fasteners and joints.
 - 3. Provide Detailed Drawings of:
 - a. Composite members.
 - b. Joint connections for framing systems and for entrance doors.
 - c. Anchorage.
 - d. System reinforcements.
 - e. Hardware, including locations, mounting heights, reinforcements and special installation provisions.
 - f. Glazing methods and accessories.
 - g. Internal sealant requirements and recommended types.
 - 4. Schedule of finishes.
- D. Changed Condition Drawings: Submit in accordance with Section 01 33 00.
- E. Samples: Indicate quality of finish on alloys used for work, 12 inches long for extrusions and 6 inches square for sheet materials.
 - 1. Where normal texture or color variations are expected, include additional samples illustrating range of variation.

2. Each type of glass, 12 by 12 inch size.
3. Joint sealants for color selection.
4. Informational Submittals: Submit following packaged separately from other submittals:
5. Test Reports for Standard Systems: Certified copies of previous test reports substantiating performance of system in lieu of retesting. Include other supportive data as necessary.
6. Support Reactions Design Data: Submit in accordance with Section 01 33 00.
7. Certifications specified in Quality Assurance article.
8. Manufacturer's instructions.

F. Closeout Submittals: Submit specified warranty in accordance with Section 01 78 00.

1.4 QUALITY ASSURANCE

- A. Single Source Responsibility: To ensure quality of appearance and performance, obtain materials for systems from either single manufacturer or from manufacturer approved by systems manufacturer.
 1. Provide glass, glazing, flashing, and perimeter joint sealants for entrance and storefront systems as work of this Section.
- B. Engineer Qualifications: Registered professional engineer licensed to practice structural engineering in jurisdiction where Project is located, with minimum of five years experience in design of storefront systems.
- C. Installer Qualifications: Certified in writing by system manufacturer as qualified for installation of specified systems.
- D. Certifications:
 1. Manufacturer's approval of installer.
 2. Engineering certifications.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
 1. Protect finished surfaces as necessary to prevent damage.
 2. Do not use adhesive papers or sprayed coatings which become firmly bonded when exposed to sun.
 3. Do not leave coating residue on any surfaces.
 4. Replace damaged units.
 5. Glass: Comply with Section 08 80 00.

1.6 WARRANTY

- A. Special Warranties: Prepare and submit following in accordance with Section 01 78 00.
 1. Warranty: Cover Following:
 - a. Glazing sealants and gaskets will remain free from abnormal deterioration or dislocation due to sunlight, weathering or oxidation.
 2. Warranty stating anodized finish will be free from fading more than 10 percent, chalking, yellowing, pitting, corroding or non-uniformity of color beyond manufacturer's descriptive standards for 10 years and agreeing to promptly correct or replace defects.
 3. Sealant manufacturer's warranty stating joint sealant will remain watertight for 20 years.

PART 2 - PRODUCTS

2.1 MANUFACTURERS AND PRODUCTS

- A. Acceptable Entrance and Storefront Manufacturers:
 1. Kawneer Company, Inc., Norcross, GA.

2. US Aluminum Co., Houston, TX
 3. Vistawall Architectural Products, Terrell, TX.
 4. YKK AP America, Inc., Atlanta, GA.
 5. Accura Systems Inc., Sunnyvale, TX
 6. Accepted Substitute in accordance with Section 01 25 00.
- B. Acceptable Entrance and Storefront Products:
1. Storefront and Entrance Framing Systems: Series Trifab 451 by Kawneer.
 2. Equivalent system by listed acceptable manufacturers is acceptable.

2.2 FRAMING MATERIALS AND ACCESSORIES

- A. Aluminum: ASTM B221, alloy 6063-T5 for extrusions; ASTM B209, alloy 5005-H16 for sheets; or other alloys and temper recommended by manufacturer appropriate for specified finish.
1. Minimum thickness of 0.125 inch for framing members and rails, 0.090 inch minimum for sheet trim, flashing, closures, and other sheet goods, and 0.050 inch for glazing stops and similar components.
 2. Provide aluminum with 20% minimum post-consumer recycled content.
- B. Internal Reinforcement: ASTM A36 for carbon steel; or ASTM B308 for structural aluminum.
1. Shapes and sizes to suit installation.
 2. Steel components factory coated with alkyd-type zinc chromate primer, FS TT-P-645, applied after fabrication.
- C. Anchorage Devices: Manufacturer's standard formed or fabricated steel or aluminum assemblies of shapes, plates, bars or tubes.
1. Hot-dip galvanize steel assemblies after fabrication, ASTM A123, 2.0 ounce minimum coating.
- D. Fasteners: Aluminum, non-magnetic stainless steel or other non-corrosive materials compatible with items being fastened.
1. Provide concealed fasteners wherever possible.
 2. Exposed Locations: Phillips flathead screws with finish matching item fastened.
 3. Concealed Locations: Manufacturer's standard fasteners.
 4. At locations where securing dissimilar metals, provide dual alloy fasteners.
- E. Expansion Anchor Devices: Drilled-in, expansion bolt anchors.
- F. Protective Coatings and Barriers:
1. Cold-applied asphalt mastic, SSPC-Paint 12, alkyd-type, bituminous mastic capable of preventing galvanic action between dissimilar metals, free of lead, chromates, or sulfur and containing no asbestos fibers, compounded for 30 mil dry film thickness per coat.
 2. Shims: Molded or extruded plastic, establishing sufficient space between dissimilar metals to prevent galvanic action or arcing of current.
- G. Touch-Up Primer for Galvanized Components: Zinc oxide, FS TT-P-641.
- H. Internal Sealants and Baffles: Types recommended by systems manufacturers.

2.3 FABRICATION

- A. Coordination of Fabrication: Check actual frame or door openings required in construction work by accurate field measurements before fabrication.
1. Coordinate Storefront side lite systems to align with AG specialty sliding doors.
- B. General: Provide each unit of framework continuous.
1. Disassemble only to extent necessary for shipment and installation.

2. Conceal fasteners wherever possible.
 3. Reinforce work as necessary for performance requirements, and for support to structure.
 4. Form end dams at flashings to prevent water infiltration.
 5. Separate dissimilar metals and aluminum in contact with concrete utilizing protective coating or preformed separators which will prevent contact and corrosion.
 6. Glass and Glazing: Comply with Section 08 80 00.
- C. Aluminum Framing: Provide members of size, shape and profile indicated, designed to provide for glazing from exterior.
1. Components: Fabricate to allow for accurate and rigid fit of joints and corners.
 - a. Match components carefully ensuring continuity of line and design.
 - b. Fabricate frame assemblies with mitered or coped joints.
 - c. Joints and Connections: Flushsealed tight.
 - d. Slip Joints: Full, tight contact and sealed tight.
 - e. Maintain accurate relation of planes and angles, with hairline fit of contacting members.
 2. Reinforcement: Reinforce to develop full strength and maximum rigidity in framework. Reinforce components at anchorage and support points, at joints, and at attachment points for interfacing work.
 - a. Internal Reinforcement: Provide within framing members where required to maintain rigidity and accommodate design loads.
 3. Make provisions in framing for minimum edge clearance, nominal edge cover and nominal pocket width for thickness and type of glazing or infill used in accordance with recommendations of manufacturer and GANA Glazing Manual.
 4. Allow for adequate clearance around perimeter of system to enable proper installation and for thermal movement within system.
 5. Separate dissimilar metals with Protective Coating or preformed separators to prevent contact and corrosion.
- D. Hardware:
1. Receive hardware supplied in accordance with Section 08 71 00 and install in accordance with requirements of this Section.
 2. Cut, reinforce, drill and tap frames and doors as required to receive hardware.
 3. Comply with hardware manufacturer's templates and instructions.
 4. Use concealed fasteners wherever possible.
- E. Welding: Comply with recommendations of American Welding Society (AWS).
1. Use recommended electrodes and methods to avoid distortion and discoloration.
 2. Grind exposed welds smooth and flush with adjacent surfaces; restore mechanical finish.

2.4 FINISHES

- A. Clear Anodized Finish: In accordance with Section 05 05 10.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 73 00.
- B. Verify dimensions, tolerances, and method of attachment with other Work.

3.2 INSTALLATION

- A. Erection Tolerances:
1. Limit Variations from Plumb and Level:
 - a. 1/8 inch in 10 feet vertically.

- b. 1/8 inch in 20 feet horizontally.
- 2. Limit Variations from Theoretical Locations: 1/4 inch for any member at any location.
- 3. Limit Offsets in Theoretical End-To-End and Edge-To-Edge Alignment: 1/16 inch from flush surfaces not more than 2 inches apart or out-of-flush by more than 1/4 inch.
- B. Aluminum Framing and Entrance Doors: Install in accordance with manufacturer's recommendations and approved Shop Drawings.
 - 1. Set units plumb, level and true to line, without warp or rack of frame.
 - 2. Anchor securely in place, allowing for required movement, including expansion and contraction.
 - 3. Protective Coating: Separate dissimilar materials at contact points, including metal in contact with masonry or concrete surfaces, with protective coating or preformed separators to prevent contact and corrosion.
 - 4. Set sill members in bed of sealant.
- C. Thresholds: Set in full, continuous bed of self-levelling sealant, specified in Section 07 92 00. Mechanically attach to substrate in accordance with manufacturer's standard requirements.
- D. Joint Sealant: Install perimeter joint sealant and backing materials between assemblies and adjacent construction in accordance with Section 07 92 00.
- E. Glazing: Install glass in preformed glazing gaskets (Dry Method) (Method 71) in accordance with Section 08 80 00.

3.3 ADJUSTING AND CLEANING

- A. Adjusting: Test door operating functions. Adjust closing and latching speeds and other hardware in accordance with manufacturer's instructions to ensure smooth operation.
- B. Cleaning: Comply with Section 01 74 00. Clean surfaces in compliance with manufacturer's recommendations; remove excess mastic, mastic smears, foreign materials and other unsightly marks.
 - 1. Clean metal surfaces exercising care to avoid damage.
 - 2. Cleaning: Comply with Sections 07 92 00 and 08 80 00.

END OF SECTION

SECTION 08710

DOOR HARDWARE

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Provide all items of finish hardware required to adequately trim, hang, and operate all doors, as is hereinafter specified and listed in the Hardware Schedule.
 - 1. Provide hardware for doors and frames of unusual profile or shape or other special conditions.
 - 2. Provide all necessary standard and special fasteners, screws, bolts, expansion shields or anchors to properly secure hardware to its intended door, frame, or other surface.
- B. Related Sections include the following:
 - 1. Hollow Metal Doors and Frames: Section 08 11 00
 - 2. Impact Resistant Doors: Section 08 14 23
 - 3. Flush Laminate Clad Wood Doors: Section 08 15 13
 - 4. Aluminum Storefront: Section 08 41 13
 - 5. Security Control and Monitoring System: Division 27 and 28.

1.2 REFERENCES

- A. The following reference standards and model code documents shall be used in estimating and detailing door hardware, and shall be considered as a standard of quality, function, and performance, as applicable:
 - 1. I.B.C. International Building Code, (current year adopted).
 - 2. NFPA-80 Fire Doors & Windows (current year adopted).
 - 3. NFPA-101 Life Safety Code (current year adopted).
 - 4. NFPA-105 Smoke Control Door Assembly. (current year adopted)
 - 5. ANSI-117.1 1992 Edition Providing Accessibility and Usability for Physically Handicapped People.
 - 6. A.D.A.A.G Americans with Disabilities Act Accessibility Guidelines.
 - 7. T.A.S. Texas Accessibility Standards.

1.3 SUBMITTALS

- A. General: Submit the following in accordance with Section 01 33 00.
- B. Product Data: Provide a catalog cut sheet, clearly marked and identified, illustrating and describing each product included in the Hardware Schedule.
 - 1. Include construction and installation details, material descriptions, dimensions of individual components and profiles, and finishes.
 - 2. Formulate catalog cut sheets into sets and include a set with each copy of the Hardware Schedule submitted.
- C. Door Hardware Schedule: Prepared by or under the supervision of Architectural Hardware Consultant, detailing fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
 - 1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."

- a. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Content: Include the following information: Type, style, function, size, label, hand, and finish of each door hardware item.
 - b. Complete designations of every item required for each door or opening including name and manufacturer.
 - c. Fastenings and other pertinent information.
 - d. Location of each door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule. Use same scheduling sequence and format and use same door numbers and hardware set numbers as in the Contract Documents.
 - e. Explanation of abbreviations, symbols, and codes contained in schedule.
 - f. Mounting locations for door hardware.
 - g. Door and frame sizes and materials.
 - h. Description of each electrified door hardware function, including location, sequence of operation, and interface with other building control systems.
2. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other Work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.
- D. Wiring Diagrams: For electrified hardware items specified for this Project, Provide complete wiring diagrams along with riser drawings and elevations, showing locations where such material is to be installed. Wiring Diagrams shall be submitted with Hardware Schedule. Verify and coordinate with the electrical systems installer. Integration shall take effect into central system as specified by Owner.
1. Operation Narrative: Describe the operation of doors controlled by electrified door hardware.
 2. Sequence of Operation: Include description of component functions that occur in the following situations:
 - a. authorized person wants to enter;
 - b. authorized person wants to exit;
 - c. unauthorized person wants to enter;
 - d. unauthorized person wants to exit.
- E. Samples for Verification: If so requested by the Architect, provide a sample of any product or item requested, properly marked and tagged, for the opening for which it is intended.
- F. Keying: Provide a keying schedule, listing the levels of keying, (GGMK, GKD, MKD or KA) as well as an explanation of the key system's function, the key symbols used and the numbers of the doors controlled. Provide in conjunction with the Door Index/Keying Schedule (which lists the door number, schedule heading, lock type and individual key symbol and remarks or special instructions) mentioned in above. Project shall be Masterkeyed and/or Grand Masterkeyed and provide two (2) keys per lockset or cylinder.
- G. Operation and Maintenance Data: For each type of door hardware to include in maintenance manuals. Provide latest, revised and updated schedule of finish hardware, complete with catalog cuts and keying schedule. In addition, furnish one (1) copy of maintenance and parts manuals for those items for which they are readily available and normally provided.
- H. Submit in accordance with provisions of Section 01 77 00.
- I. LEED Submittals:
1. Credit MR 4: For products having recycled content, documentation indicating percentages by weight of postconsumer and preconsumer recycled content. Include statement indicating costs for each product having recycled content.

2. Credit MR 5: List of proposed regional materials. Identify each regional material along with the location of its manufacture, processing and raw material source, and cost.

1.4 QUALITY ASSURANCE

- A. Substitutions: Request for substitutions for alternative hardware items will not be accepted on this Project unless specifically indicated. Specification indicates one (1) specified product, listed hereinafter in the Hardware Schedule, and two (2) acceptable alternative manufacturers for that product. If any specified product is listed as a "No Substitution" product, only that specified product shall be provided as indicated.
- B. Installer Qualifications: An experienced installer who has completed door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.
- C. Supplier Qualifications: Door hardware supplier with warehousing facilities in Project's vicinity and who is or employs a qualified Architectural Hardware Consultant, available during the course of the Work to consult with Contractor, Architect, and Owner about door hardware and keying.
 1. The hardware supplier shall be engaged regularly in the furnishing, delivery and servicing of contract builder's hardware and must be experienced and knowledgeable in all phases of estimating, detailing, scheduling, masterkeying, shipping and installation practices.
 2. When electro-mechanical or electronic hardware is supplied, a qualified individual with a minimum five- (5) years experience shall be available for assistance.
- D. Architectural Hardware Consultant Qualifications: A person who is currently certified by the Door and Hardware Institute as an Architectural Hardware Consultant and who is experienced in providing consulting services for door hardware installations that are comparable in material, design, and extent to that indicated for this Project.
- E. Source Limitations: Obtain each type and variety of door hardware from a single manufacturer, unless otherwise indicated.
- F. Regulatory Requirements: Comply with provisions of the following:
 1. Provide hardware that complies with Americans with Disabilities Act (ADA), "Accessibility Guidelines for Buildings and Facilities (ADAAG)," and ANSI A117.1.
- G. Fire-Rated Door Assemblies: Provide door hardware for assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to NFPA 252.
- H. Electrified Door Hardware: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- I. Keying Conference: Conduct conference at Project site to comply with requirements in Division
- J. 1 Section "Project Meetings." Incorporate keying conference decisions into final keying schedule after reviewing door hardware keying system including, but not limited to, the following:
 1. Function of building, flow of traffic, purpose of each area, degree of security required, and plans for future expansion.
 2. Preliminary key system schematic diagram.
 3. Requirements for key control system.
 4. Address for delivery of keys.
 5. Location of Key Cabinet.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Marking and Packaging: All items of hardware shall be delivered to the site in manufacturer's original cartons or boxes. Each item of hardware shall be marked with the abbreviation set

forth on the Shop Drawings to ensure that the product reaches its installation destination without needing specific hardware product number knowledge.

- B. Inventory door hardware on receipt and provide secure lock-up for door hardware delivered to
- A. Project site.
- B. Tag each item or package separately with identification related to the final Door Hardware Schedule, and include basic installation instructions with each item or package.

1.6 COORDINATION

- A. Templates: Obtain and distribute to the parties involved templates for doors, frames, and other work specified to be factory prepared for installing door hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing door hardware to comply with indicated requirements.
- B. Electrical System Roughing-in: Coordinate layout and installation of electrified door hardware with connections to power supplies, fire alarm system and detection devices, access control system, security system, and building control system, as applicable.

1.7 MAINTENANCE

- A. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of door hardware.
- B. Maintenance Service: If there are any products listed hereinafter that normally require a maintenance or service contract, provide the Owner and Architect with details and costs of standard maintenance or service contract.

PART 2 - PRODUCTS

2.1 ENVIRONMENTALLY PREFERABLE PRODUCTS

- A. Provide environmentally preferable products (EPP) to the greatest extent possible. Provide products and materials that promote stewardship of the earth's resources, promote good indoor environmental quality, and promote efficiencies in operational performance. EPP's include products that have low VOC content, high recycled content, and are manufactured, fabricated, or extracted within 500 miles of the construction site.
- B. Provide an aggregate recycled content of not less than 20% for all hardware devices.

2.2 MANUFACTURERS

- A. Designations: Requirements for design, grade, function, finish, size, and other distinctive qualities of each type of door hardware are indicated in Part 3 "Hardware Schedule" Article. Products are identified by using door hardware designations, as follows:
 - 1. Named Manufacturers' Products: Manufacturer and product designation are listed for each door hardware type required for the purpose of establishing minimum requirements. Manufacturers' names are abbreviated in Part 3 "Hardware Schedule" Article.
- B. Product manufacturers listed with an asterisk (*) denote the specified manufacturers listed in the Hardware Schedule. The remaining two (2) listed manufacturers will be acceptable substitutions. If only one manufacturer is listed this shall be considered a "No Substitution" specification as set forth in "Quality Assurance" Article, for that particular item.

2.3 MATERIALS

- A. Screws and Fasteners: Provide all screws and fasteners of the proper size and type to properly anchor or attach the item of hardware scheduled. Provide all fasteners with Phillips heads, unless security type screws (spanner-head or torx-head) are hereinafter specified.
- B. Hinges: Provide as follows:

1. On doors to exterior openings and main corridor doors, and other doors of high frequency use, provide a continuous, gear type hinge of appropriate weight.
 2. Where regular ball bearing hinges are listed for other doors, provide one hinge for each 30-inch of door height.
 3. The width of the hinges shall be sufficient to clear all trim that is mounted to the doorframe.
 4. Hinges shall be guaranteed for life of opening if installed per manufacturer's recommendations.
 5. Acceptable Manufacturers:
 - a. Ives*
 - b. Stanley.
 - c. Hager.
- C. Continuous Hinges: Continuous hinges shall consist of three (3)-interlocking extrusions in a pinless assembly applied to the full height of the door. All continuous geared hinges shall be manufactured to template screw locations and be non-handed. All mortise hinges and half mortise hinges shall cover and wrap the door edge completely. Doorframe heads shall be extended for clearance on full or half mortise hinges versus downsizing doors for ease of repair and replacement. All frames shall be properly reinforced per manufacturer's standards.
1. Standard warranty shall be for the life of opening.
 2. Acceptable Manufacturers:
 - a. Ives*
 - b. Select.
 - c. Pemko.
 3. THE DOOR SWITCH: Where noted in the hardware sets – continuous hinge to be provided with THE DOOR SWITCH system.
- D. Locks: All locks shall incorporate a seven pin removable core patented tumbler system and be keyed to a GRANDMASTER SYSTEM as not to breach security of system in place. Keying system must be guaranteed of no duplication of existing change keys, master keys or grandmaster keys located in this Project. All keying shall be coordinated with Owner. Locks shall be Grade 1 mortise as hereinafter listed in the Hardware Schedule.
1. Acceptable Manufacturers:
 - a. Schlage* (Everest)
 - b. No Substitutions.
- E. Lock Trim: Mortise locks are to be furnished with lever handle trim, with levers having a return to within 1/2 inch of the door face, as is hereinafter listed in the Hardware Schedule.
- F. Electronic Locksets: Heavy-duty lever handled, Grade 1, mortise lockset for commercial, industrial and institutional applications. Lockset shall combine key-in-lever design with motorized, programmable, stand-alone electronics for high security access control, which require no external wiring. Entry shall be by keypad.
1. Acceptable Manufacturers:
 - a. Schlage Commercial Electronics*
 - b. No Substitutions.
- G. Flush Bolts: Manual flush bolts to have 12-inch rods for doors 7'-6". Doors over 7'-6" high shall have bolts with top rods of 18 inch or 24 inch to allow ease of access to bolt lever. Furnish dust proof strikes for all bottom bolts.
1. Acceptable Manufacturers:
 - a. Ives*
 - b. Trimco.
 - c. Rockwood.

- H. Power Supply: Power supply shall integrate with selected switching for maintained switching with an emergency interface relay wired into the fire alarm system to insure fail secure application. Battery backup shall be included to produce backup power at full load during power failure.
1. Acceptable Manufacturers:
 - a. Von Duprin*
 - b. No Substitutions.
- I. Exit Devices: Exit Devices shall be rim, mortise or vertical rod type as called for in the Hardware Schedule. Devices shall be of the touch-pad type as is hereinafter specified in the Hardware Schedule. Exit devices shall be constructed to allow cylinder to be removed and rekeyed without removing the device from the door either by removable core cylinders or construction of exit device. Exit devices shall be constructed to allow the conversion from one function to another simply within lock stile case and selecting proper outside trim as specified hereinafter in the Hardware Schedule. Devices shall be furnished with outside trim lever handles matching locks.
1. Acceptable Manufacturers:
 - a. Von Duprin*
 - b. No Substitutions.
- J. Exit Device (QEL): Q u i t e Electric latch retraction, exit devices shall provide remote unlocking ability. A control switch or wiring schematic as specified shall allow an "exit" only or latched door to push-pull operation by a continuous duty solenoid retracting the latch bolt.
1. Acceptable Manufacturers:
 - a. Von Duprin*
 - b. No Substitutions.
- K. Card Reader/Controller: Access credential reader shall be capable of reading keypad codes to insure flexibility of control and management.
1. Acceptable Manufacturers: As specified in Section 13713.
- L. Door Closers: Door closers shall be of cast iron and rectangular design, furnished with a full cover. Provide complete with backcheck, delayed action and hold-open as indicated. Closers shall be mounted out of the line of sight wherever possible (i.e., room side of corridor doors, etc.) with parallel arm mounting on out-swinging doors. Mount closers to jamb or on brackets and/or drop plates, where special conditions require.
1. Acceptable Manufacturers:
 - a. LCN*
 - b. No Substitutions.
- M. Push Plates: Push plates are to be .050 brass, bronze or stainless steel with four (4) beveled edges, drilled and countersunk for screws, as is hereinafter specified in the Hardware Schedule.
1. Acceptable Manufacturers:
 - a. Ives*
 - b. Trimco.
 - c. Rockwood.
- N. Door Pulls: Door pulls shall be ADA compliant with a 2 1/2 inch projection from back of pull to face of door. All door pulls shall be thru-bolted or back-to-back mounted.
1. Acceptable Manufacturers:
 - a. Ives*
 - b. Trimco.
 - c. Rockwood.

- O. Protective Plates: Protective plates shall be mop (6"), kick (10") or armor (34") and shall be minimum .050 thick brass, bronze, or stainless steel, with three (3) beveled edges, drilled and countersunk for screws. Plates shall be mounted to avoid louvers and/or glass kits.
1. Acceptable Manufacturers:
 - a. Ives*
 - b. Trimco.
 - c. Rockwood.
- P. Door Stops and Holders: Where a door strikes a wall at approximately 90 degrees, a suitable door stop shall be provided, either a wall bumper or floor stop. Where doors are undercut, provide floor stops with adequate height to properly stop the door. If door would not otherwise strike a wall, an overhead stop shall be provided. In-wall blocking for wall bumpers at stud walls shall be provided in accordance with Section 06105. Provide reinforcing in frame and door for overhead stops.
1. Acceptable Manufacturers:
 - a. Ives*
 - b. Trimco.
 - c. Rockwood.
 - d. Glynn-Johnson*.
- Q. Thresholds and Weatherstrip: Weatherstripping to have aluminum housing, specified insert, and elongated mounting holes. Door sweeps shall be surface mounted, of aluminum/stainless steel housing with specified insert. Overhead drip caps to be of aluminum, have a 2 1/2-inch projection and be 4 inches wider than the door opening. Thresholds shall be of saddle type with no more than 1/2 inch rise. Weatherstripping and smoke seals shall be surface-mounted on doorstop and have 1/4" adjustment slots.
1. Acceptable Manufacturers:
 - a. Zero*
 - b. NGP.
 - c. Pemko.
- R. Smoke Gasket: Smoke gasket shall comply with door and frame manufacturers for positive pressure tests for fire and smoke. (UBC 7-2, Parts 1 & 2/UL10C).
1. Acceptable Manufacturers:
 - a. Zero*
 - b. NGP.
 - c. Pemko.
- S. DOOR ALARM – Provide THE DOOR SWITCH assembly for over-the-door pressure sensitive alarm on patient room doors as noted in the hardware sets.

2.4 FINISHES

- A. Hardware finishes shall match and be maintained to BHMA symbols, as indicated in the Hardware Schedule. Strict adherence to base metals and finish is required.
- B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

2.5 KEYING

- A. Keying of locks and cylinders throughout project shall be scheduled through a key meeting with Architect, Owner, and hardware supplier. Key schedule shall be prepared and submitted to the Owner for approval. Copies of final key schedule with the biting instructions shall be submitted as part of the Project Record Documents.

2.6 KEY CONTROL

- A. Provide key cabinet(s) manufactured by of sufficient capacity to handle all keys, plus 50 percent expansion. Provide key control cross-reference chart and accountability (sign-out) tags.
 - 1. Acceptable Manufacturers:
 - a. Telkee*
 - b. Lund.
 - c. Key Control Systems.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.
- B. Examine roughing-in for electrical power systems to verify actual locations of wiring connections before electrified door hardware installation.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Steel Doors and Frames: Comply with DHI A115 series.
 - 1. Surface-Applied Door Hardware: Drill and tap doors and frames according to SDI 107 or ANSI A250.6, whichever is more stringent.
- B. Wood Doors: Comply with DHI A115-W series.

3.3 INSTALLATION

- A. Installation shall be by a qualified installer with a minimum five (5) year's experience in the installation of commercial grade hardware. Manufacturer's instructions shall dictate templating and installation.
- B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:
 - 1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
 - 2. Wood Doors: DHI WDHS.3, "Recommended Locations for Architectural Hardware for Wood Flush Doors."
- C. Install each door hardware item to comply with manufacturer's written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work specified in Division 9 Sections. Do not install surface-mounted items until finishes have been completed on substrates involved.
 - 1. Set units level, plumb, and true to line and location. Adjust and reinforce attachment substrates as necessary for proper installation and operation.
 - 2. Drill and countersink units that are not factory prepared for anchorage fasteners. Space fasteners and anchors according to industry standards.
- D. Key Control System: Place keys on markers and hooks in key control system cabinet, as determined by final keying schedule.
- E. Boxed Power Supplies: Locate power supplies as indicated or, if not indicated, above accessible ceilings. Verify location with Architect prior to installation.
- F. Thresholds: Set thresholds for exterior and acoustical doors in full bed of sealant complying with requirements specified in Division 7 Section "Joint Sealants."

3.4 THE DOOR SWITCH:

- A. Reference Division 28 for installation and coordination.

3.5 FIELD QUALITY CONTROL

- A. Perform final inspection with hardware installer and hardware supplier present to ensure correct installation and operation, and check for any damaged or defective items. Observe and inspect that all hardware has been installed to its correct destination in proper working order.
- B. Independent Architectural Hardware Consultant: Owner reserves the right to engage a qualified independent Architectural Hardware Consultant to perform a separate independent inspection and to prepare an inspection report.

3.6 ADJUSTING

- A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended.
 - 1. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.
 - 2. Electric Strikes: Adjust horizontal and vertical alignment of keeper to properly engage lock bolt.
 - 3. Door Closers: Adjust sweep period so that, from an open position of 70 degrees, the door will take at least 3 seconds to move to a point 3 inches (75 mm) from the latch, measured to the leading edge of the door.
- B. At completion of the installation and prior to Substantial Completion, make final adjustments to door closures and other items of hardware. Leave all hardware clean and fully operable. Should any item be found to be defective, it shall be repaired or replaced as directed.
- C. Occupancy Adjustment: Approximately three months after date of Substantial Completion, Installer's Architectural Hardware Consultant shall examine and readjust, including adjusting operating forces, each item of door hardware as necessary to ensure function of doors, door hardware, and electrified door hardware.

3.7 CLEANING AND PROTECTION

- A. Clean adjacent surfaces soiled by door hardware installation.
- B. Clean operating items as necessary to restore proper function and finish.
- C. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of Substantial Completion.

3.8 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain door hardware and door hardware finishes. Refer to Division 01 Section "Project Closeout."

3.9 HARDWARE SCHEDULE

HARDWARE GROUP NO. 005

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
1	EA	SLIDING DOOR SYSTEM	EXAMSLIDE SYSTEM - SECTION 08 34 00		ADS
1	EA	DEADBOLT PRIVACY LOCK NOTE	AD5540-WP - SECTION 08 34 00 (VERIFY LEVER TYPE) BALANCE OF HARDWARE BY DOOR MFR	626	ADS

-COMPLETE OPENING BY AD SYSTEMS.

-COORDINATE DOOR, FRAME AND HARDWARE OPERATION WITH AD SYSTEMS.

-REFER TO 083400 FOR SPECIAL FUNCTION DOORS.

HARDWARE GROUP NO. 210CS1

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
6	EA	HINGE	5BB1 4.5 X 4.5	652	IVE
1	EA	CONST LATCHING BOLT	FB51T/FB61T AS REQ (USE FB52/FB62 @ RATED DOORS)	630	IVE
1	EA	STOREROOM LOCK	L9080HD 06A	626	SCH
1	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	OH STOP	100S SERIES X SIZE & MOUNTING AS REQ	630	GLY
1	EA	SURFACE CLOSER	4040XP SCUSH TBWMS X MTG BRKT, SPCR & PLATE AS REQ	689	LCN
1	EA	KICK PLATE	8400 10" X 1" LDW B-CS	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	EA	MEETING STILE	8193AA (2 PIECES - 1 SET) HEIGHT AS REQ (OMIT @ NON- RATED OPENINGS)	AA	ZER

HARDWARE GROUP NO. AC710

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
6	EA	HINGE	5BB1HW 4.5 X 4.5	652	IVE
2	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-9927-L-DT-LBR-06-CON (WDC @ WD) LENGTH & HEIGHT AS REQ	626	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-9927-L-NL-LBR-06-CON (WDC @ WD) LENGTH & HEIGHT AS REQ	626	VON
1	EA	SFIC RIM CYLINDER	TYPE AS REQ W/KEYED CONST. CORE	626	SCH
1	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	AUTO OPERATOR	QUAD 4000 SERIES WITH CLOSE ASSIST MODULE (COORDINATE WITH RELATED SECTION)		BEA

1	EA	ACTUATOR, TOUCHLESS	MS31 (COORDINATE WITH RELATED SECTION) (OMIT WHERE NOT DESIRED BY OWNER)		BEA
2	EA	KICK PLATE	8400 10" X 1" LDW B-CS	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	SET	MEETING STILE	8193AA (2 PCS - 1 SET) (OMIT @ NON-RATED DOORS)	AA	ZER
2	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
2	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
1	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28		
2	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
2	EA	MOTION SENSOR	SUPERSCANII (MOUNT ON BOTH SIDES OF DOOR)		BEA
1	EA	POWER SUPPLY	PS902 900-4RL 120/240 VAC (OMIT 4RL BOARD WHERE NOT REQ)		VON

-INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.

-FREE EGRESS BY THE ACTUATOR OR THE PUSH PAD.

-THE ELECTRIFIED LATCH BOLTS WILL BE SEQUENCED WITH THE AUTOMATIC OPENER AND RETRACT PRIOR TO THE AUTOMATIC OPENER ACTIVATING.

-COORDINATE POWER SUPPLY WITH SECURITY CONTRACTOR PRIOR TO SUBMITTALS.

-OMIT POWER SUPPLY WHERE PROVIDED BY SECURITY.

-PROVIDE LOCAL MEANS TO DISCONNECT AUTOMATIC OPERATOR IN HEADER.

HARDWARE GROUP NO. AC711

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 5 X 4.5	652	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-99-L-NL-06-CON LENGTH AS REQ	626	VON
1	EA	SFIC RIM CYLINDER	TYPE AS REQ W/KEYED CONST. CORE	626	SCH
1	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	AUTO OPERATOR	QUAD 4000 SERIES WITH CLOSE ASSIST MODULE (COORDINATE WITH RELATED SECTION)		BEA
1	EA	ACTUATOR, TOUCHLESS	MS31 (COORDINATE WITH RELATED SECTION) (OMIT WHERE NOT DESIRED BY OWNER)		BEA
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH

1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)	SCH
1	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28	
1	EA	DOOR CONTACT	1070 SERIES AS REQ	SEN
2	EA	MOTION SENSOR	SUPERSCANII (MOUNT ON BOTH SIDES OF DOOR)	BEA
1	EA	POWER SUPPLY	PS902 900-4RL 120/240 VAC (OMIT 4RL BOARD WHERE NOT REQ)	VON

-INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.

-FREE EGRESS BY THE ACTUATOR OR THE PUSH PAD.

-THE ELECTRIFIED LATCH BOLT WILL BE SEQUENCED WITH THE AUTOMATIC OPENER AND RETRACT PRIOR TO THE AUTOMATIC OPENER ACTIVATING.

-COORDINATE POWER SUPPLY WITH SECURITY CONTRACTOR PRIOR TO SUBMITTALS.

-OMIT POWER SUPPLY WHERE PROVIDED BY SECURITY.

-PROVIDE LOCAL MEANS TO DISCONNECT AUTOMATIC OPERATOR IN HEADER.

HARDWARE GROUP NO. AC711W

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 5 X 4.5	652	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-99-L-NL-06-CON LENGTH AS REQ	626	VON
1	EA	SFIC RIM CYLINDER	TYPE AS REQ W/KEYED CONST. CORE	626	SCH
1	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	AUTO OPERATOR	QUAD 4000 SERIES WITH CLOSE ASSIST MODULE (COORDINATE WITH RELATED SECTION)		BEA
1	EA	ACTUATOR, TOUCHLESS	MS31 (COORDINATE WITH RELATED SECTION) (OMIT WHERE NOT DESIRED BY OWNER)		BEA
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
1	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28		
1	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
2	EA	MOTION SENSOR	SUPERSCANII (MOUNT ON BOTH SIDES OF DOOR)		BEA
1	EA	POWER SUPPLY	PS902 900-4RL 120/240 VAC (OMIT 4RL BOARD WHERE NOT REQ)		VON

- INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.
- FREE EGRESS BY THE ACTUATOR OR THE PUSH PAD.
- THE ELECTRIFIED LATCH BOLT WILL BE SEQUENCED WITH THE AUTOMATIC OPENER AND RETRACT PRIOR TO THE AUTOMATIC OPENER ACTIVATING.
- COORDINATE POWER SUPPLY WITH SECURITY CONTRACTOR PRIOR TO SUBMITTALS.
- OMIT POWER SUPPLY WHERE PROVIDED BY SECURITY.
- PROVIDE LOCAL MEANS TO DISCONNECT AUTOMATIC OPERATOR IN HEADER.

HARDWARE GROUP NO. ADOF101

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
1	EA	SLIDING DOOR SYSTEM	OFFICESLIDE SYSTEM - SECTION 08 34 00		ADS
1	EA	SELF-LATCHING OFFICE LOCK	AD6450-J-L17 - SECTION 08 34 00 (VERIFY LEVER TYPE)	626	ADS
1	EA	MORTISE CYLINDER	TYPE AS REQ W/KEYED CONST. CORE	626	SCH
1	EA	FSIC PERMANENT CORE	23-030	626	SCH

-COMPLETE OPENING BY AD SYSTEMS.
-COORDINATE DOOR, FRAME AND HARDWARE OPERATION WITH AD SYSTEMS.
-REFER TO 083400 FOR SPECIAL FUNCTION DOORS.

HARDWARE GROUP NO. C201GPW

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 5 X 4.5	652	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	EU MORTISE LOCK	L9092HDEU 06A RX CON (FAIL SECURE)	626	SCH
1	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH TBWMS X MTG BRKT, SPCR & PLATE AS REQ	689	LCN
1	EA	ARMOR PLATE	8400 34" X 2" LDW B-CS	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	EA	DOOR SWEEP	39A LENGTH AS REQ	A	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
1	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28		
1	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
1	EA	POWER SUPPLY	PS902 120/240 VAC (COORDINATE POWER SUPPLIES WITH SECURITY PRIOR TO SUBMITTAL. OMIT WHERE PROVIDED BY SECURITY)	LGR	SCE

- INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.
- FREE EGRESS BY LEVER.

HARDWARE GROUP NO. CPZ291PZ

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
1	EA	CONT. HINGE	112XY EPT HEIGHT AS REQ	628	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	EL MORTISE LOCK	L9095BDC EL 06A CON (FAIL SAFE)	626	SCH
2	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	SURFACE CLOSER	4040XP RW/PA TBWMS X MTG BRKT, SPCR & PLATE AS REQ	689	LCN
1	EA	ARMOR PLATE	8400 34" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	SEALS	488S PSA H & J	BK	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
1	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28		
1	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
1	EA	PUSH BUTTON	PUSH BUTTON BY DIV 28	603	SCE
1	EA	POWER SUPPLY	PS902 FA900 120/240 VAC	LGR	SCE
1	EA	EMERGENCY DOOR RELEASE & ALARM SYSTEM	EMERGENCY DOOR RELEASE & ALARM SYSTEM BY DIV 28		

-INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.

-EGRESS BY PUSH BUTTON OR KEY OVERRIDE.

-EMERGENCY EGRESS BY PULL STATION.

HARDWARE GROUP NO. CZ291GPZ

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
1	EA	CONT. HINGE	112XY EPT HEIGHT AS REQ	628	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	EL MORTISE LOCK	L9095BDC EL 06A CON (FAIL SAFE)	626	SCH
2	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	SURFACE CLOSER	4040XP RW/PA TBWMS X MTG BRKT, SPCR & PLATE AS REQ	689	LCN
1	EA	ARMOR PLATE	8400 34" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	GASKETING	870AA H & J	AA	ZER
1	EA	HEAVY DUTY DOOR BOTTOM	364AA-Z49-PULL OUT TYPE AS REQ	AA	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
2	EA	CREDENTIAL READER	CREDENTIAL READER BY DIV 28		
1	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
1	EA	POWER SUPPLY	PS902 FA900 120/240 VAC	LGR	SCE

-INGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.

-EGRESS BY THE CREDENTIAL READER OR KEY OVERRIDE.

- EMERGENCY EGRESS BY PULL STATION.
- COORDINATE POWER SUPPLY WITH SECURITY CONTRACTOR PRIOR TO SUBMITTALS.
- OMIT POWER SUPPLY WHERE PROVIDED BY SECURITY.

HARDWARE GROUP NO. KPZ291EPZ

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
1	EA	CONT. HINGE	112XY EPT HEIGHT AS REQ	628	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	EL MORTISE LOCK	L9095BDC EL 06A CON (FAIL SAFE)	626	SCH
1	EA	KEY PAD	KEY PAD BY DIV 28		B/O
2	EA	SFIC EVEREST CORE	80-037 CKC	626	SCH
1	EA	SURFACE CLOSER	4040XP RW/PA TBWMS X MTG BRKT, SPCR & PLATE AS REQ	689	LCN
1	EA	ARMOR PLATE	8400 34" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407CCV	630	IVE
1	EA	GASKETING	870AA H & J	AA	ZER
1	EA	HARNESS (IN DOOR)	ALLEGION CONNECT TYPE & LENGTH AS REQ		SCH
1	EA	HARNESS (TO POWER SUPPLY)	CON-6W (CONNECTION LEADS)		SCH
1	EA	DOOR CONTACT	1070 SERIES AS REQ		SEN
1	EA	PUSH BUTTON	PUSH BUTTON BY DIV 28	603	SCE
1	EA	POWER SUPPLY	PS902 FA900 120/240 VAC	LGR	SCE
1	EA	EMERGENCY DOOR RELEASE & ALARM SYSTEM	EMERGENCY DOOR RELEASE & ALARM SYSTEM BY DIV 28		

- INGRESS BY KEYPAD OR KEY OVERRIDE.
- EGRESS BY PUSH BUTTON OR KEY OVERRIDE.
- EMERGENCY EGRESS BY PULL STATION.
- INTERLOCK BY SECURITY CONTRACTOR.
- COORDINATE POWER SUPPLY WITH SECURITY CONTRACTOR PRIOR TO SUBMITTALS.
- OMIT POWER SUPPLY WHERE PROVIDED BY SECURITY.

**END OF
SECTION**

SECTION 08 71 13

AUTOMATIC DOOR OPERATORS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes
 - 1. Exterior and interior Full Energy and Low Energy Swing Door Operators with visible mounting
 - 2. Automatic door operators shall be configured for doors as follows
 - a. Simultaneous pairs, outswing, inswing or double egress
 - b. Single doors out swing or in swing
- B. Related Sections:
 - 1. Section 08 71 00 - Door Hardware.
 - 2. Division 26 - Electrical.

1.2 REFERENCES

- A. Reference Standards: See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following for design, materials, fabrication, and installation of component parts American National Standards Institute (ANSI) / Builders Hardware Manufacturers Association (BHMA).
 - 1. ANSI/BHMA A156.19 Standards for Power Assist and Low Energy Power Operated Doors.
 - 2. ANSI/BHMA A156.10 American National Standard for Power Operated Pedestrian Doors.

1.3 SYSTEM DESCRIPTION

- A. Design Requirements:
 - 1. Manufacturer: Responsible for providing complete operating system.
 - 2. Provide concealed fastening wherever possible.
 - 3. Operate in accordance with manufacturer's published literature including power operation, emergency breakaway, manual use, entrapment protection, and "soft start" features.
 - 4. Size Operator Capacity for:
 - a. Door size, weight, and movement.
 - b. Condition of exposure.
 - c. Long-term, maintenance-free operation under heavy pedestrian traffic load for indicated occupancy.
 - d. Opening to back check position in time period no faster than 3 seconds.
 - e. Requiring no more than 15 pounds of force to stop door movement.
 - 5. Emergency Exit Doors: Comply with applicable code when doors serve as required means of egress.
 - a. Permit automatic release of control to permit manual opening of doors.
 - 6. Provide for manual opening and closing operation of door leaves in event of power failure with opening force complying with code requirement.
 - 7. Products requiring electrical connection: Listed and classified by FM, UL, Warnock Hersey, or other testing agencies acceptable to authorities having jurisdiction as being suitable for specified and indicated purpose.
- B. Structural Requirements: Fabricate system to operate, hold open, and hold close doors under design wind loads as calculated in accordance with applicable building code.

- C. Fire Resistance Requirements: Provide door operators, electro-magnetic latch mechanisms, activating and sensor safety control devices which comply with fire-rated door requirements, as certified by manufacturer for fire door rating and application indicated.
 - 1. Provide fire-rated operators with electrical connection to fire alarm system that releases doors when alarm is activated.
- D. Thermal Requirements: Provide units with operating temperature range of minus 29 and minus 20 and plus 120 F.
- E. Interface With Adjacent Systems: Provide units coordinated with door size, weight, and swing type.
 - 1. Coordinate operators according to fire ratings, relationship with door head, and alarm system.

1.4 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00 and 08 71 00.
 - 1. Submit Sections 081000, 081513, submittals simultaneously with submittals for this section.
- B. Product Data: Submit for door operators.
 - 1. Include operating hardware, accessories and other required components.
 - 2. Include sample of warranty.
 - 3. Include information for factory finishes and manufacturer's color charts showing full range of colors available for selection.
 - 4. Include sample of warranty customized for this project.
- C. Shop Drawings: Submit indicating locations to receive automatic door equipment.
 - 1. Indicate design, fabrication, and installation details including relationship to adjoining construction.
 - 2. Clearly indicate locations of exposed fasteners.
 - 3. Submit detail drawings of special accessory components not included in manufacturer's product data.
- D. Templates and Diagrams: Furnish templates, diagrams, and other data to fabricators and installers of related work, as necessary for coordination of operators with doors, frames, hardware, electrical, and other work.
- E. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Certifications specified in Quality Assurance article.
 - 2. Qualification Data: Manufacturer's and installer's qualification data.
 - 3. Manufacturer's instructions.
- F. Closeout Submittals: Submit following in accordance with Section 01 78 00.
 - 1. Operation and maintenance data.
 - 2. Warranty.

1.5 QUALITY ASSURANCE

- A. Single Source Responsibility: Obtain each type of operator and sensor components specified in this Section from a single source, same manufacturer unless otherwise indicated
- B. Certified Inspector Qualifications: Certified by AAADM.
- C. Power-Operated Pedestrian Door Standard: ANSI/BHMA A156.10 (current version).
- D. Power Assist and Low Energy Power Operated Doors. Standard: ANSI/BHMA A156.19 (current version)

- E. Manufacturer Qualifications: Company specializing in manufacturing Products specified in this Section with minimum five years experience.
- F. Installer Qualifications: Acceptable to manufacturer with experience on at least five projects of similar nature in past five years.
 - 1. Certified in writing by system manufacturing as qualified for installation of specified system.

1.6 DELIVERY STORAGE AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Protect electrical components and finished surfaces as necessary to prevent damage.
 - 2. Replace damaged units.

1.7 WARRANTY

- A. General Warranty: Special warranty specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.
- B. Special Warranty: Written warranty, executed by manufacturer agreeing to repair or replace components of the entrance door system that fail in materials or workmanship within specified warranty period. Failures include, but are not limited to, the following:
 - 1. Deterioration of metals, metal finishes, and other materials beyond normal use.
 - a. Warranty Period: Two years from date of Substantial Completion.

1.8 MAINTENANCE

- A. Maintenance Service: Provide service and maintenance of operating equipment for 12 months from date of Substantial Completion.
- B. Maintenance Materials: Provide wrenches and tools for maintenance of equipment.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers-Basis of Design:
 - 1. Besam Entrance Solutions, Inc (see below for configurations)
- B. Manufacturer: Subject to compliance with requirements, provide products by one of the following:
 - 1. Besam Entrance Solutions, Inc.
 - 2. Norton, Monroe, NC
 - 3. LCN Princeton, IL
 - 4. Accepted Substitute in accordance with Section 01 73 00.
- C. Swing Door Operators
 - 1. Surgical Doors- Besam SW100 Low Energy Swing Door Operator (Basis of Design)
 - a. Comply with ANSI/BHMA A156.19 Standards for Power Assist and Low Energy Power Operated Doors.
 - b. Configuration: Operator to control single swinging doors and pairs of swinging doors as indicated on the drawings and specified below:
 - 1) Traffic Pattern: Two way.
 - 2) Pairs of Doors: Simultaneous swing
 - c. Door Operator: Electro-mechanical, non-handed operator, powered by 24 volt, 1/8 hp motor. Spring shall be adjustable to compensate for different manual push forces required on varying door widths.

- d. Operator Housing: Operator is completely contained in a 5-3/16" (132mm) deep x 4-5/16" (110mm) high extruded aluminum housing.
- e. Header Length: Extended Width, full width of door frame.
 - 1) Connecting Hardware: Steel arm from the operator, secured to the top of the swing door.
- f. UL Listed R-9469 Fire Door Operator with Automatic Closer.
- g. Electrical Characteristics: Nominal current draw 75 watts (.625 amps at 120 VAC), built-in thermal overload protection.
- h. Door Operation:
 - 1) Opening Cycle: The adjustable speed operator shall control the door opening to the back check position, where the opening speed is reduced.
 - 2) Hold Open: The operator shall stop and hold the door open at the selected door opening angle for an adjustable period of time.
 - 3) Closing Cycle: Power closing shall be provided by means of compression spring and motor. The door will slow to low speed at latch check before it reaches the fully closed position.
 - 4) Obstruction Control: The operator will stop and reverse the door movement.
 - 5) Electronic Dampening: Operator to include standard electric dampening system which automatically counteracts additional forces applied to the door during the opening or closing cycle by reducing door speed.
 - 6) Stack Pressure Compensation: Electronic control allows for increases of forces to overcome minor stack pressures while compensating to lower manual push forces when the door is used in manual mode in order to comply with ANSI/BHMA A156.19.
 - a) Lock retry circuit: If attempt to fully close the door is unsuccessful, the operator will automatically reverse open 10 degrees and reclose in an attempt to successfully close the door.
- i. Electronic Controls: Microprocessor controlled unit shall control the operation and switching of the swing power operator. The microprocessor unit provides low voltage power supply for all means of actuation. The controls include time delay (1.5 to 30 seconds) for normal cycle.
- j. Operator Interface:
 - 1) Safety Sensor Integration: To interface operator with door mounted reactivation safety device.
- k. Electric Strike Integration: To interface operator with electrified door hardware.
 - 1) Cross Corridor Applications- Besam Swingmaster 900 Low Energy Swing Door Operator (Basis of Design)
- l. Allow future upgrade to Besam model, based on construction schedule
- m. Comply with ANSI/BHMA A156.19 Standards for Power Assist and Low Energy Power Operated Doors.
- n. Configuration: Operator to control single swinging doors and pairs of swinging doors as indicated on the drawings and specified below:
 - 1) Traffic Pattern: Two way.
 - 2) Pairs of Doors: Simultaneous swing and Dual Egress
 - 3) Emergency Breakaway: Where inswing doors also serve as required exits, provide emergency breakaway feature to allow doors to swing in the direction of egress. Forces to comply with ANSI/BHMA A156.10. Discontinue power to power door operator when door is in emergency breakaway position, and to automatically reset when door is returns to the full closed position.
- o. Door Operator: Electro-mechanical, non-handed operator, powered by 24 volt, 1/4 hp motor. Spring shall be adjustable to compensate for different manual push forces required on varying door widths.

- 1) Side Access Operator Housing: Operator is contained in a 6 inch (152 mm) x 6 inch (152 mm) side access, extruded aluminum housing.
- 2) Surface Mounted Header: Extended Width, full width of door frame.
 - a) Connecting Hardware: Surface mounted operators to have a cast steel arm from the operator, secured to the top face of the swing door.
- p. UL Listed R-9469 Fire Door Operator with Automatic Closer (surface mounted operator).
- q. Operator can be field upgraded to a full energy operator by the addition of the required safety sensors, and guard rails to comply with ANSI/BHMA A156.10 American National Standard for Power Operated Pedestrian Doors.
- r. Electrical Characteristics: Maximum power consumption is 4 amps at 120 VAC, built-in thermal overload protection.
 - 1) Door Operation:
 - 2) Opening Cycle: The adjustable speed operator mechanically powers the drive shaft which maintains constant engagement throughout the opening cycle. Operator shall allow manual door operation with operational forces of 15 lbf maximum to fully open the door applied at 1" (25 mm) from the latch edge of the door.
 - 3) Hold Open: The operator shall stop and hold the door open at the selected door opening angle for an adjustable period of time.
 - 4) Closing Cycle: Closing shall be provided by means of spring. Closing forces regulated by utilizing the motor and gear assembly as a dynamic brake.
 - 5) Obstruction Control: The operator will stop and reverse the door movement.
 - 6) Stack Pressure Compensation: Operator is designed to counteract most wind and/or interior stack pressures without an additional power assist mechanism.
- s. Electronic Controls: Solid state integrated circuit controls the operation and switching of the swing power operator. The electronic control provides low voltage power supply for all means of actuation. The controls include time delay (1 to 60 seconds) for normal cycle.
- t. Operator Interface:
 - 1) Safety Sensor Integration for overhead presence safety device and door mounted reactivation safety sensors.
 - 2) Electric Strike Integration for electrified door hardware.
 - a) Shielded Surgical Doors- Besam Brookfield NB-500 Swing Door Operator (Basis of Design)
- u. Configuration: Operator to control single swinging doors and pairs of swinging doors as indicated on the drawings and specified below:
 - 1) Traffic Pattern: Two way.
 - 2) Pairs of Doors: Simultaneous swing
- v. Door Operator: Electro-mechanical, power open, power close, non-handed operator, powered by a 1/4hp permanent magnet 90 volt DC motor 1750 RPM TENV.
- w. Electrical Characteristics:
 - 1) Current consumption: 5 amperes
 - 2) Supply voltage: 115 VAC +/- 10% 50/60 Hertz single phase. In-line circuit breakers supplied with motor control an PLC
 - 3) Surge protection included.
- x. Manual operation shall be in accordance with UL 325 during a power loss.
- y. Operator Interface:
 - 1) Safety Sensor Integration for overhead presence safety device and door mounted reactivation safety sensors.
 - 2) Electric Strike Integration for electrified door hardware.

- z. Smoke Evacuation- Besam Swingmaster 900 Full Energy Swing Door Operator (Basis of Design)
- aa. Allow future upgrade to Besam model, based on construction schedule
- bb. Configuration: Operator to control single swinging doors and pairs of swinging doors as indicated on the drawings and specified below:
 - 1) Traffic Pattern: One way, no foot traffic.
 - 2) Pairs of Doors: Dual Egress
 - 3) Doors in controlled perimeter, no sensors required
 - 4) Surface applied configuration
 - 5) Activated by fire/smoke alarm

2.2 ACTIVATION AND SAFETY DEVICES

- A. General: Provide the types of activation and safety devices specified in accordance with ANSI/BHMA standards, for the condition of exposure and for long-term, maintenance-free operation under normal traffic load for type of occupancy indicated. Coordinate activation and safety devices with door operation and door operator mechanisms.
- B. Knowing Act Activation:
 - 1. Sensor Plate: Touchless, 2-3/4 inch x 4-1/2 inch activation sensor plates, black polycarbonate with white letters. Microwave technology has an adjustable range of 2 inches to 24 inches.
 - 2. Push Plate: Hard wired, 4-1/2 inch square stainless steel push plate switches engraved with "Push to Open" with a blue handicap logo.
 - 3. Card Reader: Shall be provided by others.
 - 4. Key Pad: Shall be provided by others.
- C. Safety Devices:
 - 1. Double Egress Pairs
 - a. Door Mounted Presence Detector: Shall be the Besam door mounted (mounted at top of each door), infrared presence safety device; adjustable to provide detection field sizes and functions. Unit to provide detection during the travel of the door.
 - 1) The door mounted presence detector shall be mounted on the swing side of the door (1 sensors per leaf), providing detection on the pull side.
 - 2) Upon detection the sensor shall provide a signal to stop or reverse the door action.
 - b. Presence Detector: Shall be the Besam header mounted, overhead infrared presence safety device utilizing infrared technology for detection (2 sensors per pair of doors); adjustable to provide detection field sizes and functions. Unit to provide the following two independently adjustable patterns of detection:
 - 1) The door closed position covering the area on the swing side of the door.
 - 2) The door open position including an area of detection that reaches through the threshold toward the non-swing side of the door.
 - 3) The unit is not active during the door closing cycle.
 - 2. Simultaneous Pairs
 - a. Door Mounted Presence Detector: Shall be the Besam door mounted (mounted at top of each door), infrared presence safety device; adjustable to provide detection field sizes and functions. Unit to provide detection during the travel of the door.
 - 1) The door mounted presence detector shall be mounted on the swing side of the door (1 sensors per leaf), providing detection on the push side.
 - 2) Upon detection the sensor shall provide a signal to stop or reverse the door action.
 - b. Presence Detector: Shall be the Besam header mounted, overhead infrared presence safety device utilizing infrared technology for detection (1 sensor per pair of doors); adjustable to provide detection field sizes and functions. Unit to provide the following two independently adjustable patterns of detection:

- 1) The door closed position covering the area on the swing side of the door.
- 2) The door open position including an area of detection that reaches through the threshold toward the non-swing side of the door.
- 3) The unit is not active during the door closing cycle.

2.3 FINISHES

- A. Clear Anodized Finish: In accordance with Section 05 05 10.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
1. Verify electrical power is available and of correct characteristics.

3.2 INSTALLATION

- A. Low Energy Automatic Door Operator: Install assemblies and operating equipment complete with operators, controls, and related accessories in accordance with Section 01 73 00 and approved Shop Drawings.
1. Install units plumb, level, square, and free from warp or twist while maintaining dimensional tolerances and alignment with surrounding construction.
 2. Coordinate installation of operator with installation of:
 - a. Door frame.
 - b. Door.
 - c. Door hardware.
 - d. Electrical service.
 3. Provide necessary blocking, supports, and miscellaneous items for complete installation.
 4. Do not install surface mounted items until finishes have been completed on substrate.
 5. Install power connections, connection to fire alarm system, and control wiring.
 6. Signage: Install on door.

3.3 ADJUSTING

- A. Adjust parts for smooth, uniform, and optimum operation.
1. Test door operating functions. Adjust opening and closing speeds and breakout forces in accordance with manufacturer's instructions.
 2. After repeated operation of completed installation, equivalent to three days use by normal traffic (100 to 300 cycles), readjust door operators and controls for optimum operating condition and safety and for weather tight closure.
 3. Lubricate hardware, operating equipment, and other moving parts in accordance with manufacturer's requirements.
 4. Adjust motion detector sensors, adjust door operator, and actuating controls to operate correctly.

3.4 CLEANING AND PROTECTION

- A. Cleaning: Clean in accordance with manufacturer requirements. Do not use materials or methods which may damage finish or surrounding construction.
1. Remove protective material from surfaces.
 2. Remove excess mastic, mastic smears, harmful materials, and other unsightly marks.
 3. Avoid damage to finishes during cleaning.
- B. Protection: Protect finished work in accordance with Section 01 73 00.

3.5 DEMONSTRATION

- A. Demonstration and Instruction of Owner's Personnel: Provide in accordance with Section 01 92 00.
 - 1. Demonstrate operating components, door operation, adjustment features, and lubrication requirements.

END OF SECTION

SECTION 08 80 00

GLAZING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Glass for hollow metal, wood, laminate faced and aluminum doors.
 - 2. Glass for interior borrowed light and sidelight frames.
 - 3. Glass for aluminum entrances and storefronts.
 - 4. Glass for aluminum curtain walls.
 - 5. Associated glazing sealants and accessories.
- B. Related Sections:
 - 1. Section 01 35 43 – Environmental Procedures: Indoor air quality procedures.
 - 2. Section 08 41 13 – Aluminum-Framed Entrances and Storefronts.
 - 3. Section 08 44 13 - Glazed Aluminum Curtain Walls.

1.2 REFERENCES

- A. Reference Standards: See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following for design, materials, fabrication, and installation of component parts:
 - 1. GANA Glazing Manual and GANA Sealant Manual.

1.3 SYSTEM REQUIREMENTS

- A. General: In addition to requirements shown or specified, comply with design requirements of Section 01 83 00.
- B. Design Requirements: Provide continuity of building enclosure to maintain continuous air and vapor barrier throughout glazed assembly from glass pane to heel bead of sealant.
 - 1. Employ registered professional engineer, licensed to practice structural engineering who specializes in this type of work and who is licensed to practice in state where project is located, , to engineer each component of glass and glazing system.
- C. Performance Requirements: Section 01 45 29. Provide thickness of glass units to withstand specified wind loads.
 - 1. Limit glass deflection to flexure limit of glass with full recovery of glazing materials, or 1" center deflection whichever is less. At the architects sole discretion, the 1' center deflection may be adjusted upward by the architect based on special conditions as determined by calculations presented by the engineer for the architect's consideration.
- D. Sound Requirements: At exterior openings, provide STC rating at glass of not less than 34.
- E. Glazing Requirements: Comply with CPSC 16 CFR 1201 and ANSI Z97.1 for safety requirements of glazing materials.
 - 1. Glass thickness, where indicated, are minimum requirements and are to be confirmed by glass manufacturer.
 - 2. Provide glass of thickness and heat treatment (annealed, heat strengthened or fully tempered) as necessary to prevent temperature stress breakage.
 - 3. Use 2.5 safety factor for glass to arrive at a statistical probability of failure (8 lites/1000).for glass in a vertical position or up to a 15 degree from vertical slope when it is in its installed position.

4. Maximum Height to Width Ratio: For glass units not dimensioned, determine minimum acceptable width of glass by using ratio of 6:1, height to width.
 5. Obtain safety glazing products permanently marked with certification label of the Safety Glazing Certification Council or other certification agency acceptable to authorities having jurisdiction.
- F. Sealant Performance and Requirements: Section 07 92 00.
- G. Furnish and install glass that meets safety requirements as established by the building code being used for the project.

1.4 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: For each type of glass and glazing material specified, including glazing accessories and glazing sealants.
1. Include sample of warranty customized for this Project.
- C. Shop Drawings: Sections and details of glass and glazing materials installation at framing members including head, mullions, transoms, jambs and sills.
1. Stamp with seal and signature of professional engineer responsible for design.
- D. Samples:
1. 12 inches by 12 inches in size illustrating color , reflective value, laminated construction of glass units.
 2. 12 inches long bead of glazing sealant, in full range of manufacturer's standard colors for selection by Architect.
- E. Informational Submittals: Submit following packaged separately from other submittals:
1. Test Reports:
 - a. Glazing sealant indicating substrate adhesion.
 - b. Glazing sealant compatibility.
 - c. Glazing sealant manufacturer's recommendations.
 - d. Glazing gaskets, identifying materials, compositions, durometers and other physical properties
 2. Certifications specified in Quality Assurance article.
 3. Qualification Data: Installer's qualification data.
 4. Manufacturer's instructions.
- F. Closeout Submittals: Submit specified warranty in accordance with Section 01 78 00.

1.5 QUALITY ASSURANCE

- A. Single Source Responsibility: Glass of each type to be produced by same manufacturer. The float glass used shall be from one manufacture.
- B. Engineer Qualifications: Registered professional engineer licensed to practice structural engineering, with minimum of five years' experience in design of glass and glazing.
- C. Installer Qualifications: Acceptable to manufacturer with experience on at least five projects of similar nature in past five years.
- D. Regulatory Requirements: Fabricate glass to comply with ASTM C1036, ASTM C1048, and ANSI Z97.1. Quality stated in these standards applies to all the various glass thickness used on the project.
- E. Certifications: Submit following:
1. Manufacturer's letter certifying that sealed insulating glass units meet or exceed specification.

2. Engineering certifications.

1.6 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 19.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products in accordance with Section 01 60 00.
 1. Deliver glass units with manufacturer's labels intact on interior side of glass. Ensure labels indicate glass thickness, unit location, glass strength and orientation of units in vertical position.
 2. Protect glass edges and corners against chipping, spalls, oysters or other edge damage and cracking. Do not install glass with any of these defects.

1.8 PROJECT CONDITIONS

- A. Environmental Requirements: Perform glazing when ambient temperature is above 40 F.
 1. Perform glazing on dry surfaces only.
 2. Indoor Air Quality Procedures: Ventilate in accordance with Section 01 35 43.

1.9 WARRANTY

- A. Special Warranties: Prepare and submit following in accordance with Section 01 78 00.
 1. Warrant hermetically sealed insulating glass units to be free from defects in material and workmanship for ten years.
 2. Warrant laminated glass units to be free from defects in material and workmanship for five years.
 3. Warrant polycarbonate clad units to be free from defects in material and workmanship for ten years.
 4. Warrant spandrel glass to be free of defects in materials and workmanship for ten years.

PART 2 - PRODUCTS

2.1 GLASS MATERIALS

- A. Type 01 - Clear Float Glass: Glazing select, float, ASTM C1036, Type I, Class 1, Quality q3.
 1. Type: Annealed.
 2. Thickness: 1/4 inch minimum.
- B. Type T - Clear Tempered Glass: Same Type 01 above except tempered, ASTM C1048, Kind FT fully tempered; 1/4 and 3/8 inch thickness; frosted appearance where indicated.

2.2 GLAZING GASKETS

- A. Dense Compression Gaskets: Molded or extruded gaskets of profile and hardness required to maintain watertight seal, made from the following:
 1. Silicone complying with ASTM C 1115.
- B. Soft Compression Gaskets: Extruded or molded, closed-cell, integral-skinned silicone gaskets complying with ASTM C 509, Type II, black; of profile and hardness required to maintain watertight seal.
 1. Application: Use where soft compression gaskets will be compressed by inserting dense compression gaskets on opposite side of glazing or pressure applied by means of pressure-glazing stops on opposite side of glazing.

2.3 GLAZING ACCESSORIES

- A. Setting Blocks: Preformed silicone (polymer based), compatible with sealant.
 1. Hardness: 80-90 Shore A durometer.

2. Size: 0.10 inch for each square foot of glazing, not less than 4 inch length by width of channel minus 1/16 inch by 1/4 inch high.
 3. Location: Sill quarter points, centered minimum 4 inches from each edge.
 4. Requirement: Resistant to sunlight, weathering oxidation and permanent deformation under load.
- B. Spacer Shims: Preformed silicone compatible EPDM , compatible with sealant.
1. Hardness: 50-60 Shore A durometer.
 2. Size: Minimum 3 inch length by 1/2 height of glazing stop by thickness to suit application.
 3. Requirement: Self-adhesive one face.
- C. Edge Blocks: Preformed silicone compatible EPDM , compatible with sealant.
1. Hardness: 60-70 Shore A durometer.
 2. Size: Minimum 4 inch length by width to support thickness of glass, allow nominal 1/8 inch clearance between edge of glass and edge block..
 3. Location: Place in vertical channel.
 4. Requirement: Resistant to sunlight, weathering, oxidation and permanent deformation under load.
- D. Glazing Tapes:
1. Exterior: Preformed urethane foam with integral spacing device and containing paper release.
 2. Interior: Preformed urethane foam, butyl, or closed cell PVC foam with integral spacing device and containing paper release.
 3. Acceptable Products:
 - a. SGT-900 Urethane Tape, Tremco, Cleveland, OH.
 - b. Polyshim II Pre-Shimmed 440 Butyl Tape, Tremco, Cleveland, OH.
 - c. Norseal V-980, Norton, Granville, NY.
 - d. 330 Glazing Tape, PTI, Dayton, OH.

2.4 GLAZING SEALANTS

- A. Glazing sealants formulated with aromatic solvents (organic solvent with benzene ring in its molecular structure), fibrous talc or asbestos, formaldehyde, halogenated solvents, mercury, lead, cadmium, hexavalent chromium, or their components not allowed.
- B. Glazing sealants used shall be compatible with and adhere to sealants used for frame joint seals and ther perimeter seals around the framing systems, If conflicts between what is specified in this section and those sealants identified in other sections of this specification, contractor shall advise architect of such conflict and resolve conflict at no additional cost to project. Sealant manufacture sall be a single source for all sealants that contact each other.
- C. Glazing Sealants: Silicone, one-part, neutral cure, ASTM C920, Type S, Grade NS, Class 25.
1. Joint Movement: ASTM C719; plus 100, minus 50.
 2. Colors: As selected by Architect from full range of manufacturer's standard colors.
- D. Primer: When required by sealant manufacturer, type as recommended.
- E. Acceptable Products, Structural Glazing (Glass to Glass):
1. 995 or 795 Silicone Building and Glazing Sealant, Dow Corning Corp., Midland, MI.
 2. Ultraglaze , General Electric Co., Waterford, NY.
 3. Spectrem 2, Tremco
- F. Acceptable Products, Weather Seal (Glass to Metal):
1. 795 Silicone Sealant, Dow Corning Corp., Midland, MI.

2. Silpruf, General Electric Co., Waterford, NY.
 3. Spectrem 2, Tremco
- G. Adhesives & Sealants: Only use adhesives and sealants in the interior of the building that meet or do not exceed the VOC limits of the CURRENT requirements of South Coast Air Quality Management District (SCAQMD) Rule No. 1168 on the interior of the building.
1. Current requirement refers to the date on which the materials are installed in the building.
 2. SCAQMD Rule #1168 referenced in section 01 81 13 is current as of the date of this specification. Refer to <http://www.aqmd.gov/rules> for the actual current version of the rule that will be applicable at the date of installation during construction.
 3. Interior refers to all building construction that is inside of the exterior weatherproofing material.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 73 00.
1. Verify that openings for glazing are correctly sized and within tolerances.
 2. Verify that glazing channel surfaces or recesses are clear, free of burrs, obstructions, irregularities, and glass is free of edge damage or imperfections.

3.2 PREPARATION

- A. Clean contact surfaces with solvent and wipe dry.
- B. Seal porous glazing channels or recesses with substrate compatible primer or sealer.
- C. Prime surfaces scheduled to receive sealant, if required by sealant manufacturer.
- D. Verify that materials used for cleaning edges of sealed insulating units are compatible with sealants and components and will not damage or cause deterioration or integrity of sealed insulating unit.

3.3 INSTALLATION

- A. Install glass units in accordance with Section 01 73 00. Ensure weep and drainage holes are not blocked by sealants or setting blocks.
- B. Preformed Glazing Gaskets (Dry Method): Factory cut and seam gasket to proper length.
1. Prefabricate vulcanized joints in shop.
 2. Place setting blocks at quarter points, with edge blocks no closer than 6 inches from corner.
 3. Rest glass on setting blocks and push against stop with sufficient pressure to ensure full contact and adhesion at perimeter.
 4. Install removable stops, avoiding displacement of gasket and exert pressure for full continuous contact.
 5. Remove gaskets which twist or which otherwise become torn or damaged and replace with new gasket.
 6. Install storefront glass in gaskets as specified in Section 08 41 13.
 7. Install curtain wall glass in gaskets as specified in Section 08 44 13.
- C. Exterior Wet/Dry Method (Tape and Sealant): Cut glazing tape to length and set against permanent stops, 3/16 inch below sight line.
1. Weld joints by butting tape and by sealing junctions with sealant. Do not overlap tape.
 2. Apply heel bead of sealant along intersection of permanent stop with frame ensuring full perimeter seal between glass and frame to complete continuity of air and vapor seal.
 3. Place setting blocks at 1/4 points with edge blocks no closer than 6 inches from corners.

4. Rest glass on setting blocks and push against tape and heel bead of sealant with sufficient pressure to ensure full, continuous contact and adhesion at perimeter.
5. Install removable stops, spacer strips inserted between glass and removable stops at 2 foot intervals, 1/4 inch below sight line.
 - a. Fill gap between glass and removable stop with sealant to depth equal to bite of frame on glass but not more than 3/8 inch below sight line.
 - b. Apply cap bead of sealant along void between stop and glass, to uniform and level line, flush with sight line.
 - c. Tool or wipe cap bead surface with solvent for smooth appearance.
 - d. Glaze insulated units with sealant compatible with manufacturer's sealant.
- D. Interior Dry Method (Tape and Tape): Cut glazing tape to length and install against permanent stop, projecting 1/16 inch above sight line.
 1. Weld joints by butting tape and by sealing junctions as directed by tape manufacturer. Do not overlap tape.
 2. Place setting blocks at 1/4 points with edge blocks no more than 6 inches from corners.
 3. Rest glass on setting blocks and push against stop for full contact and adhesion at perimeter.
 4. Place glazing tape on free perimeter of glass in same manner described above.
 5. Install removable stop, avoid displacement of tape, exert pressure on tape for full continuous contact.
 6. Knife trim excess or protruding tape.
- E. Interior Wet/Dry Method (Tape and Sealant): Cut glazing tape to length and set against permanent stops, projecting 1/16 inch above sight line.
 1. Weld joints by butting tape and by sealing junctions with sealant. Do not overlap tape.
 2. Place setting blocks at 1/4 points with edge blocks no more than 6 inches from corners.
 3. Rest glass on setting blocks and push against tape with sufficient pressure to ensure full contact and adhesion at perimeter.
 4. Install removable stops, insert spacer strips between glass and removable stops at 2 foot intervals, 1/4 inch below sight line.
 5. Fill gap between glass and removable stop with sealant to depth equal to bite of frame on glass to uniform and level line.
 6. Neatly trim off excess tape to sight line.
- F. Tempered Glass: Do not cut, seam, nip or abrade tempered glass.
 1. Install in windows and sidelights where required by code.
- G. Butt Glazing: Clean areas to receive silicone glazing with products recommended by sealant manufacturer. Use clean white lint-free cloth.
 1. Prime as recommended by sealant manufacturer.
 2. Mask interior face of joint with tape runs of approximately 3 to 6 feet starting at top down overlapping runs.
 3. Apply sealant using cartridge type caulk gun or bulk type, hand or air pressure activated, apply as full bead starting from bottom of vertical joint and working up.
 4. Push bead ahead of nozzle to ensure complete filling of joint, do not leave air pockets or voids along edge.
 5. Neatly tool joint immediately, do not use soap or detergent solution.
 6. Remove masking tape immediately following sealant tooling.

3.4 PROTECTION AND CLEANING

- A. Protection: Protect finished Work under provisions of Section 01 73 00.
 - 1. After installation, mark glass pane with X by using removable plastic tape or paste. Do not mark heat absorbing or reflective glass units.
- B. Cleaning: Comply with Section 01 74 00.
 - 1. Remove excess glazing materials from finished surfaces.
 - 2. Remove labels after work is completed.
 - 3. Wash and clean both surfaces of vision glass and exterior surface of spandrel glass using methods and materials as recommended by the glass manufactures requirements and guidelines., not more than seven days prior to Owner's acceptance of work.
 - 4. Comply with glass manufacturer's recommendations for final cleaning.

END OF SECTION

SECTION 08 87 00

GLAZING SURFACE FILMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes film products applied to glass surfaces to impart the following characteristics:
 - 1. Decorative applications.

1.2 ACTION SUBMITTALS

- A. Product Data: For each film product indicated.
- B. Samples for Color Selection: Manufacturer's standard sample sets showing the full range of colors available for each type of product indicated.
- C. Samples for Verification: 12-inch square samples of each type of glazing film specified, in color specified.

1.3 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

1.4 CLOSEOUT SUBMITTALS

- A. Warranty: Special warranty specified in this Section.
- B. Maintenance Data and Replacement Instructions: For each type of film overlay to include in maintenance manuals.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Engage a firm experienced in manufacturing systems similar to those indicated for this Project and meeting the standards of the International Standards Organization (ISO), ISO 9001 Quality Assurance in Production and Installation.
- B. Installer Qualifications: Engage an experienced installer certified, licensed, or otherwise qualified by film manufacturer as having the necessary experience, staff, and training to install the manufacturer's products according to specified requirements.
- C. Source Limitations: Obtain each type of film overlay through one source from a single manufacturer to provide products of consistent quality in appearance and physical properties.
- D. Mockups: Apply glazing films in locations as directed to verify selections made under sample Submittals and to demonstrate aesthetic effects and qualities of materials and execution.
 - 1. Obtain approval of field samples before continuing with the remainder of the installation.
 - 2. Maintain field samples during the remainder of installation in an undisturbed condition as a standard for judging the completed Work.
 - 3. Approved field samples may become part of the completed Work.
- E. Preinstallation Conference: Conduct a conference at the project site to comply with requirements in Division 01 Section "Project Management and Coordination."
 - 1. Conduct a pre-installation conference in conjunction with the installation of the mockup.
 - 2. Meet with Owner, Architect, glazing film Installer and glazing film manufacturer's representative.

3. Review methods and procedures related to installation, including the manufacturer's written instructions.
4. Examine substrate conditions for compliance with requirements.
5. Review temporary protection measures required during and after installation.
6. Document proceedings, including corrective measures or actions required, and furnish copy of record to each participant.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery: Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels intact.
- B. Store and protect glazing films according to the manufacturer's written instructions as needed to prevent damage, condensation, temperature changes, direct exposure to sun, or other causes.

1.7 PROJECT CONDITIONS

- A. Environmental Limitations: Do not proceed with film installation when ambient and substrate temperature conditions are outside limits permitted by manufacturer and when glass when substrates glass substrates are wet from frost, condensation, or other causes.

1.8 WARRANTY

- A. Manufacturer's Warranty: Fully executed warranty written in favor of the Owner, agreeing to replace films that deteriorate as defined in "Definitions" Article, within 5 years from date of original installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. 3M.
 2. Substitutions: Submit in accordance with Section 01 25 00.

2.2 GLAZING SURFACE FILMS

- A. Film Overlay: Single-layered applied glazing film products, 2-mil minimum thickness (or as otherwise standard with manufacturer), applied to interior glass surfaces, consisting of the following (from outboard surface to inboard surface), as applicable to each type of film indicated:
 1. Removable release liner.
 2. Pressure-sensitive adhesive with integral ultraviolet absorbers.
 3. Printed pattern layer of polyester film.
- B. Colors and Patterns: Refer to Finish Legend.

2.3 GLAZING FILM ACCESSORIES

- A. General: Provide products complying with the requirements of the glazing film manufacturer for the indicated, and with a proven record of compatibility with surfaces contacted in installation.
- B. Adhesive: Pressure-sensitive acrylic adhesive system.
- C. Cleaners, Primers, and Sealers: Types recommended by the glazing film manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine glass and surrounding adjacent surfaces for conditions affecting installation.

- B. Report conditions that may adversely affect installation. In report include a description of any glass that is broken, chipped, cracked, abraded, or damaged in any way.
- C. Proceed with installation only after unsatisfactory conditions have been corrected

3.2 PREPARATION

- A. Comply with the manufacturer's written instructions for surface preparation.
- B. Immediately before beginning installation of films, clean glass surfaces of substances that could impair glazing film's bond, including mold, mildew, oil, grease, dirt and other foreign materials.
- C. Blade the inside surface of window glass with industrial razors to ensure removal of foreign contaminants.
- D. Protect window frames and surrounding surfaces and materials from damage during installation.

3.3 INSTALLATION

- A. General: Comply with glazing film manufacturers' written instructions and installation instructions applicable to products and applications indicated, except where more stringent requirements are indicated.
- B. Install film continuously, but not necessarily in one continuous length. Install with no gaps or overlaps.
- C. If seams, install with no gaps or overlaps. Install seams vertically and plumb.
- D. No horizontal seams allowed.
- E. Do not remove the release liner from the film until just before each piece of film is cut and ready for installation.
- F. Install film with mounting solution and custom cut to the glass with neat, square corners and edges to within 1/8 inch of the window frame.
- G. Remove air bubbles, wrinkles, blisters, and other defects.
- H. After installation, view film from a distance of 10 feet against a bright, uniform sky or background. Film shall appear uniform in appearance with no visible streaks, banding, thin spots or pinholes.
 - 1. If the installed film does not meet this criterion, remove and replace it with new film.

3.4 CLEANING

- A. Remove excess mounting solution at finished seams, perimeter edges, and adjacent surfaces.
- B. After application of the film, wash the film using the cleaning methods recommended by the glazing film manufacturer. Do not use abrasive-type cleaning agents or bristle brushes.
- C. Replace films that cannot be cleaned.

END OF SECTION

SECTION 08 92 00

LOUVERED EQUIPMENT SCREENS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Exterior metal equipment screen panels with concealed fasteners at the following locations:
- B. Roof top equipment screens.

1.3 PERFORMANCE REQUIREMENTS

- A. General: Provide manufactured wall panel assemblies complying with performance requirements indicated and capable of withstanding structural movement, thermally induced movement, and exposure to weather without failure or infiltration of water into the building interior.
- B. Structural Performance: Provide manufactured wall panel assemblies capable of withstanding design wind loads indicated under in-service conditions with deflection no greater than the following, based on testing manufacturer's standard units according to ASTM E 330 by a qualified independent testing and inspecting agency.
 - 1. Maximum Deflection: 1/180 of the span.

1.4 ACTION SUBMITTALS

- A. Product Data: Include manufacturer's product specifications, standard details, certified product test results, and general recommendations, as applicable to materials and finishes for each component and for total panel assemblies.
- B. Shop Drawings: Show layouts of panels, details of corner conditions, joints, panel profiles, supports, anchorages, trim, flashings, closures, and special details. Distinguish between factory- and field-assembled work.
- C. Samples for Verification: Provide sample panels 12 inches (300 mm) long by actual panel width, in the profile, style, color, and texture indicated. Include clips, caps, battens, fasteners, closures, and other exposed panel accessories.

1.5 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- B. Product Test Reports: Indicate compliance of manufactured wall panel assemblies and materials with performance and other requirements based on comprehensive testing of current products.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer who has completed metal wall panel projects similar in material, design, and extent to that indicated for this Project and with a record of successful in-service performance.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver panels and other components so they will not be damaged or deformed. Package panels for protection against damage during transportation or handling.
- B. Handling: Exercise care in unloading, storing, and erecting wall panels to prevent bending, warping, twisting, and surface damage.
- C. Stack materials on platforms or pallets, covered with tarpaulins or other suitable weathertight and ventilated covering. Store panels to ensure dryness. Do not store panels in contact with other materials that might cause staining, denting, or other surface damage.

1.8 PROJECT CONDITIONS

- A. Field Measurements: Verify location of structural members and openings in substrates by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.9 WARRANTY

- A. General Warranty: Special warranties specified in this Article shall not deprive the Owner of other rights the Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by the Contractor under requirements of the Contract Documents.
- B. Special Finish Warranty: Submit a written warranty, signed by manufacturer, covering failure of the factory-applied exterior finish on metal wall panels within the specified warranty period and agreeing to repair finish or replace wall panels that show evidence of finish deterioration. Deterioration of finish includes, but is not limited to, color fade, chalking, cracking, peeling, and loss of film integrity.
 - 1. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturer and Product:
- B. Louvered Metal Panels: Construction Specialties, Inc. (C/S), Cranford, NJ.
- C. Acceptable Product: Vert-A-Cade model 301.

2.2 METALS AND FINISHES

- A. Aluminum Extrusions: ASTM B211, Alloy 6063-T52.
 - 1. Louvers to be finished with a minimum 1.0 mil (0.025mm) thick full strength 70% resin, 2 coat Fluoropolymer system.
 - a. All aluminum shall be thoroughly cleaned, etched and given a chromated conversion pre-treatment before application of the MICA II coating. The coating shall consist of a primer and a pearlescent pigmented PFV2 topcoat. It shall receive a bake cycle of 17 minutes at 450F. All finishing procedures shall be one continuous operation in the plant of the manufacturer.
 - b. Color to match CS Pewter # 747
 - c. Finish to comply with AAMA 2605
 - d. Louver Manufacturer to furnish an extended 20 limited warranty for the Kynar/Hylar coating. This limited warranty shall begin on the date of material shipment

2.3 LOUVERED WALL PANEL ASSEMBLIES

- A. Panels: Fabricate louvers to the profile or configuration indicated; and of the material, finish, and thickness indicated. Design joints between panels to form weathertight seals.
 - 1. Thickness: 0.081 inch thick minimum.
 - 2. Free Area: 50 percent.

3. No blade joints in sections less than 30 feet in length.
4. Blades: Fixed, horizontal; reinforced.
5. Blade Depth: 4 inches.
6. 3 inch Vertical Aluminum Support Channel with Blade Braces Attached
7. Panel Height: As shown on Drawings.

2.4 MISCELLANEOUS MATERIALS

- A. Fasteners: Concealed type; self-tapping screws, bolts, nuts, self-locking rivets and bolts, end welded studs, and other suitable fasteners designed to withstand design loads.
 1. Use stainless-steel fasteners.
- B. Accessories: Unless otherwise specified, provide components required for a complete assembly including trim, mullions, sills, corner units, clips, seam covers, flashings, louvers, sealants, gaskets, fillers, closure strips, and similar items. Match materials and finishes of panels.
 1. Joint Sealant: One-part elastomeric polyurethane, polysulfide, or silicone-rubber sealant as recommended by panel manufacturer.
- C. Bituminous Coating: Cold-applied asphalt mastic, SSPC-Paint 12, compounded for 15-mil(0.4-mm) dry film thickness per coat, unless otherwise indicated. Provide inert-type non-corrosive compound free of asbestos fibers, sulfur components, and other deleterious impurities.

2.5 FABRICATION

- A. General: Fabricate and finish panels and accessories at the factory to greatest extent possible, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.
- B. Apply bituminous coating or other permanent separation materials on concealed panel surfaces where panels would otherwise be in direct contact with substrate materials that are non-compatible or could result in corrosion or deterioration of either materials or finishes.

2.6 SECONDARY FRAMING

- A. Panel Supports and Anchorage: Provide girts, furring channels, angles, plates, bracing, and other secondary framing members.
 1. Girts: C- or Z-shaped sections fabricated from 0.0598-inch-(1.5-mm-) thick, shop-painted, roll-formed steel.
 2. Flange and Sag Bracing: 1-5/8-by-1-5/8-inch(41-by-41-mm) angles, fabricated from 0.0598-inch-(1.5-mm-) thick, shop-painted, roll-formed steel.
 3. Base or Sill Angles: Fabricate from 0.079-inch-(2.0-mm-) thick, cold-formed, galvanized steel sections.
 4. Secondary structural members, except columns and beams, shall be manufacturer's standard sections fabricated from 0.079-inch-(2.0-mm-) thick, cold-formed galvanized steel.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements indicated for conditions affecting performance of metal panel walls.
 1. Panel Supports and Anchorage: Examine framing to verify that girts, angles, and other secondary structural panel support members and anchorage have been installed to meet requirements of panel manufacturer.
 2. Do not proceed with wall panel installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Coordinate louvered panels with rain drainage work; flashing; trim; and construction of roofing, parapets, walls, and other adjoining work to provide a leakproof, secure, and non-corrosive installation.
- B. Promptly remove protective film, if any, from exposed surfaces of metal panels. Strip with care to avoid damage to finish.
- C. Secondary Structural Supports: Install girts, angles, and other secondary structural panel support members and anchorage according to the Light Gage Structural Institute's "Guide Specifications," Section 07 41 13 and 07 42 00, "Manufactured Roof and Wall Panels."

3.3 PANEL INSTALLATION

- A. General: Comply with panel manufacturer's written instructions and recommendations for installation, as applicable to project conditions and supporting substrates. Anchor panels and, other components of the Work securely in place, with provisions for thermal and structural movement.
 - 1. Field cutting exterior panels by torch is not permitted.
 - 2. Install panels with concealed fasteners.
- B. Accessories: Install components required for a complete panel assembly including trim, mullions, sills, corner units, clips, seam covers, flashings, louvers, sealants, gaskets, fillers, closure strips, and similar items.
- C. Joint Sealers: Install gaskets, joint fillers, and sealants where indicated and where required for weatherproof performance of wall panel assemblies. Provide types of gaskets, fillers, and sealants indicated or, if not otherwise indicated, types recommended by panel manufacturer.
 - 1. Seal panel end laps with a bead of tape or sealant, full width of panel. Seal side joints where recommended by panel manufacturer.
 - 2. Prepare joints and apply sealants to comply with requirements of Division 07 Section "Joint Sealants."
- D. Wall Panels: Apply elastomeric sealant continuously between metal base channel (sill angle) and concrete, and elsewhere as necessary for waterproofing. Handle and apply sealant and back-up according to sealant manufacturer's written instructions.
 - 1. Align bottom of wall panels and fasten with blind rivets, bolts, or self-tapping screws. Fasten flashings and trim around openings and similar elements with self-tapping screws.
 - 2. Install screw fasteners with power tools having controlled torque adjusted to compress neoprene washer tightly without damage to washer, screw threads, or panels. Install screws in predrilled holes.
 - 3. Provide weatherproof escutcheons for pipe and conduit penetrating exterior walls.
- E. Separate dissimilar metals by painting each metal surface in area of contact with a bituminous coating or by other permanent separation as recommended by manufacturers of dissimilar metals.
 - 1. Installation Tolerances: Shim and align panel units within installed tolerance of 1/4 inch in 20 feet(6 mm in 6 m) on level, plumb, and location lines as indicated and within 1/8-inch(3-mm) offset of adjoining faces and of alignment of matching profiles.

3.4 CLEANING AND PROTECTING

- A. Damaged Units: Replace panels and other components of the Work that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.
- B. Cleaning: Remove temporary protective coverings and strippable films, if any, as soon as each panel is installed. On completion of panel installation, clean finished surfaces as recommended by panel manufacturer and maintain in a clean condition during construction.

END OF SECTION

NON-STRUCTURAL METAL FRAMING

SECTION 09 22 16

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal stud framing systems for interior non-load bearing portland cement plaster and gypsum plaster partitions.
2. Metal furring systems for interior non-load bearing portland cement plaster and gypsum plaster assemblies.
3. Metal framing systems for interior suspended portland cement plaster and gypsum plaster ceilings and soffits.
4. Framing accessories.

B. Metal framing at exterior vertical applications: See Section 05 41 00.

1.2 SYSTEM DESCRIPTION

A. Design Requirements: Contractor: Responsible for designing supplemental metal framing used to comply with performance requirements, including anchorage to structural system and necessary modifications to meet specified requirements and maintain visual design concepts.

B. Performance Requirements: Design and install systems to:

1. Interior Suspended Ceilings, Soffits, and Bullheads: Maintain deflection of not more than $L/360$ of distance between supports.
2. Interior Metal Stud/Gypsum Board Assemblies: Withstand lateral loading (air pressure) of 5 PSF with deflection limit not more than $1/240$ of partition height.
3. Requirements at Deflection Tracks, Deflection Clips and Bridge Clips: Accommodate building structure deflections in connections to structure. Maintain positive attachment of units while allowing vertical free movement due to Live Load Deflection of structure, $1/2$ inch minimum. Partially tightened fasteners not permitted.
 - a. Anticipated Live Load Deflection for Internal Beams: The deflection of the structure is a function of the beam lengths or spacing between the four columns forming the corners of the structural bay. The deflection to be accommodated is equal to $1/360$ multiplied by the longest distance between columns forming the structural bay surrounding the studs in question.
4. Anticipated Live Load Interior Beam Deflections: $1/240$ of span.

C. Fire Resistance Ratings: Where assemblies with fire ratings are indicated, provide materials and installations which are identical to assemblies tested in accordance with ASTM E119 by testing laboratories acceptable to authorities having jurisdiction.

1. Construct assemblies identical to those indicated by reference to design designations listed in UL Fire Resistance Directory or other agencies acceptable to authorities having jurisdiction.

D. Acoustical Requirements: Where sound-rated assemblies are indicated, provide materials and installation for sound-rated assembly tested and classified by manufacturer to achieve Sound Transmission Class (STC) rating scheduled or indicated, ASTM E90.

E. Interface With Adjacent Systems: Integrate design and connections with adjacent construction.

1. Accommodate allowable tolerances and deflections for structural members in installation.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Submit Sections , , 092400, and 092900 submittals simultaneously with submittals for this section.
- C. Product Data: Submit for each product.
- D. Shop Drawings: Submit for unusual conditions in connection with metal framing work, including standard partition connections to suspended ceilings and building structure, and ceiling and soffit layout, control joints, expansion joints, access panels and other recessed items.
- E. Changed Condition Drawings: Submit in accordance with Section 01 33 00.
- F. Informational Submittals: Submit following packaged separately from other submittals:
- G. Manufacturer's instructions for each product.
- H. Support Reactions Design Data: Submit in accordance with Section 01 33 00.
- I. Certifications specified in Quality Assurance article.
- J. QUALITY ASSURANCE
- K. Engineer Qualifications: Registered professional engineer licensed to practice structural engineering in jurisdiction where P roject is located, with minimum of five years experience in design of metal framing systems used to support surfaces on building exterior.
- L. Welder Qualifications: AWS certified within past 12 months for each type of weld required.
- M. Certifications:
- N. Certificates verifying AWS qualifications for each welder employed on Project.
- O. Engineering certifications.

1.4 FIELD SAMPLES

- A. General: Comply with Section 01 40 00.
- B. Sample Installation:
- C. Provide metal support systems as required in conjunction with field samples specified in following sections:
- D. Section 09 23 00 - Gypsum Plastering.
- E. Show backer plate construction techniques.
- F. Accepted Field Sample: May not remain part of completed Work.

1.5 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 19.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
 - 1. Deliver to project site promptly without undue exposure to weather.
 - 2. Store in dry ventilated space off ground.
 - 3. Protect materials from soiling, rusting and damage

PART 2 - PRODUCTS

2.1 ACCEPTABLE PRODUCTS AND MANUFACTURERS

- A. Products and Manufacturers, Deflection Clips:

1. Verti-Clip Series, Signature Industries, LLC/The Steel Network, Raleigh, NC.
 2. Accepted Substitute in accordance with Section 01 60 00.
- B. Products and Manufacturers, Bridge Clips:
1. BridgeClip Series, Signature Industries, LLC/The Steel Network, Raleigh, NC.
 2. Accepted Substitute in accordance with Section 01 60 00.

2.2 WALL AND PARTITION FRAMING MATERIALS

- A. Steel Framing: Maximize use of recycled steel content with minimum of 60 percent.
- B. Metal Studs, Runners, and Furring Channels: ASTM C645.
1. Hot dip galvanize in accordance with ASTM A653M, G60, or electrolytic zinc-coat in accordance with ASTM A591M.
 2. Framing, furring, and hanger thickness specified below identifies bare metal thickness prior to coating.
- C. Shop fabricate stud profiles. Field fabricated stud profiles not permitted.
- D. Metal Stud Framing:
1. Metal Stud: Non-load bearing C-shaped design with punched web, roll formed, galvanized steel, 18 mils thickness unless indicated otherwise, and designed for screw attachment of finishes.
 2. Size: As scheduled or indicated.
 3. Height: As scheduled or indicated.
 4. Metal Floor and Ceiling Runner Tracks: Channel type members, roll formed galvanized steel with 1-1/4 inch flanges, designed for screw attachment of metal studs and finishes.
- E. Resilient Furring Channels: Where furring is indicated as resilient, provide 1/2 inch high, 25 gage specially shaped furring designed to reduce sound transmission.
- F. Deflection and Bridge Clips: ASTM A653, Quality SS, 50-KSI yield strength; 65-ksi ultimate strength; Coating Designation G60.

2.3 CEILING AND SOFFIT FRAMING MATERIALS

- A. Steel Framing: Maximize use of recycled steel content with minimum of 60 percent.
- B. Protective Finish for Exterior Work: Hot-dip galvanize exterior suspension components including hangers.
- C. Attachment Devices: Size devices for five times loads imposed by completed system.
1. Hanger Wire Concrete Inserts: No. 6 galvanized wire loop and 25 gage galvanized shell.
 2. Hanger Anchorage Devices: Screws, clips, bolts, inserts or other devices applicable to indicated method of structural anchorage for ceiling hangers and whose suitability for use intended has been proven through standard construction practices or by certified test data.
- D. Hangers: Wire, rod or rigid steel, ASTM C754 for maximum ceiling area and loads to be supported.
- E. Framing - Channel and Cross Furring System:
1. Metal Runner Channels: Cold-rolled or hot-rolled steel, ASTM C754, 1-1/2 inch minimum size.
 2. Metal Furring Channels: Hat-shape, 25 gage, 7/8 inch high.
 3. Resilient Furring Channels: 1/2 inch high, 25 gage specially shaped furring channel designed to reduce sound transmission.
 4. Framing Ties or Clips: 14 gage galvanized wire ties, or wire-type clips.
 5. Exterior Soffit Wind Bracing: Provide cross and vertical bracing with additional framing using anchors and fasteners to comply with performance requirements.

6. Suspended Ceiling Bracing Assembly: Provide Compression Struts and Bracing Wire in accordance with applicable codes and regulations.
- F. Framing - Proprietary Direct Hung Suspension System: At Contractor's option, provide factory fabricated, proprietary system in lieu of channel and cross furring framing system.
 1. Interlocking cold-rolled sheet steel grid, ASTM C635 Heavy Duty structural classification.
 2. Acceptable Product: 640 Drywall Furring System, Chicago Metallic Corp., Chicago, IL.
 3. Equivalent systems by National Rolling Mills, Inc., Paoli, PA and Unimast, Inc., Franklin Park, IL are also acceptable.
- G. Framing - Fire Rated Proprietary Direct Hung Suspension System: Interlocking Cold-Rolled Sheet Steel Grid: ASTM C635, Heavy Duty structural classification.
 1. Acceptable Product: 650 Fire Front Drywall Furring System, Chicago Metallic Corp., Chicago, IL.
 2. Equivalent systems by National Rolling Mills, Inc., Paoli, PA and Unimast, Inc., Franklin Park, IL are also acceptable.

2.4 ACCESSORIES

- A. Backer Plates: 14 gage steel sheet, galvanized in accordance with ASTM A525, G60.
 1. Length: Spanning 3 studs minimum or longer if required.
 2. Height: 6 inch minimum or larger where required to accommodate item being fastened.
 3. Provide backer plate for securing surface mounted fittings and accessories, including, but not limited to handrails, grab bars, toilet partitions, towel bars, wall mounted door stops, and similar items.
 4. Refer to Section 06 10 00 for wood blocking to secure surface mounted equipment, cabinets, and millwork.
- B. Fasteners: Types and sizes recommended by framing manufacturer for substrate and applications indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 40 00.

3.2 PREPARATION

- A. Coordinate sizes and locations of surface mounted accessories which require backer plates. Install additional studs for attachment of backer plates in required locations to receive surface mounted accessories as indicated or as recommended by accessory manufacturer.
- B. Elimination of backer plates or direct attachment of accessories to studs not allowed.
- C. Coordinate installation of concealed blocking and furring required for carpentry and millwork.

3.3 INSTALLATION, GENERAL

- A. Install metal support systems in accordance with ASTM C754, except for more stringent requirements of manufacturer or these Specifications.
- B. Install components true to lines and levels indicated to provide surface flatness of finished assemblies with maximum variation of 1/8 inch in 10 feet in any direction.

3.4 METAL STUD AND RUNNER INSTALLATION

- A. Metal Stud Spacing: 16 inches OC, unless otherwise indicated.
- B. Metal Runner Tracks: Provide continuous tracks sized to match studs.
 1. Align runner tracks accurately to partition layout at both floor and ceiling.

2. Secure runner tracks as recommended by stud manufacturer for floor and ceiling construction involved, except do not exceed 24 inch OC spacing for fasteners.
 3. In addition, provide fasteners at corners and ends of runner tracks.
- C. Where partitions abut ceilings, deck construction, or structural elements, provide slip or cushion type joint between partition and structure to prevent transfer of structural loads or movements to partitions.
- D. Terminate top of partitions at ceiling construction, or as scheduled unless otherwise indicated.
- E. Brace partitions which are terminated at suspended ceilings to structure above as indicated, or if not indicated, in accordance with stud manufacturer's instructions.
- F. Use full length studs vertically positioned between runner tracks. If splicing is required due to partition height, splice studs by nesting with minimum lap of 8 inches and fasten laps with minimum of two screws through each flange.
1. Friction fit studs to runner tracks by positioning and rotating into place. Secure studs to runner tracks at partition corners and intersections, and adjacent to openings using self tapping screws or stud clinching tool on both flanges of stud.
 2. Align and plumb partition framing accurately according to partition layout.
- G. Door Framing: Provide two 18 mil studs at each jamb and one additional stud not more than 6 inches from jamb studs; or provide one 33 mil stud adjacent to each jamb.
1. Provide runner track and typical studs above door openings with studs spaced not more than 24 inches OC.
 2. At welded frames with fixed anchor clips, secure studs to jamb anchors clips with not less than two self tapping screws per clip.
- H. Additional Framing: Provide additional studs and framing to support inside corners, partition intersections, outside corners, termination of partitions, both sides of control joints, at openings and cut-outs and to support built-in anchorage and attachment devices for other work.
1. Locate studs no more than 2 inches from abutting partitions, partition corners and other construction. Start typical partition studs 6 inches either side of stud reinforcing or frames.
 2. Electrical Outlets and Similar Penetrations: Install in locations indicated. Provide additional blocking and straps for proper locations; do not mount on "nearest" stud.

3.5 WALL FURRING INSTALLATION

- A. Metal Furring: Install where indicated and where gypsum board finishes are scheduled or indicated over concrete or masonry substrates.
1. Metal Furring Channels: Install either vertically or horizontally .
- B. Space furring at maximum of 16 inches OC and attach to substrate with fasteners at 24 inches OC staggered through alternate wing flanges.
1. Install furring channels within 4 inches of floor line and ceiling line.
 2. Install corner furring channels in similar manner.

3.6 INSTALLATION OF BACKER PLATES

- A. Provide backer plate for securing surface mounted fittings.
1. Secure with sufficient number of self-tapping sheet metal screws to sustain loads imposed by items attached to backer plates.
 2. Obtain observation of framing with backer plates by Architect before concealing.

3.7 SUSPENDED CEILING AND SOFFIT INSTALLATION

- A. Provide attachments and hangers from structural support above, spaced on grid of 4 feet by 2 feet. Do not attach hangers directly to metal roof deck. Provide supplementary framing if necessary.

1. Furnish hanger anchors and similar devices to other trades for installation well in advance of time needed for coordination with other work.
 2. Hangers: Secure as required to structural support by connecting directly to structure where possible, otherwise connect to inserts, clips or other anchorage devices or fasteners as indicated.
- B. Framing Attachment:
1. Runner Channel and Furring Channel Framing: Attach runner channels to wire hangers by tying. Attach furring channels to runner channels with framing ties or clips.
 2. Proprietary Direct Hung Framing: Tie direct to suspension; interconnect components in accordance with framing system manufacturer's instructions.
- C. Metal Runner Channels: Space at maximum 4 feet OC.
- D. Metal Furring Channels: Space at maximum 16 inches OC, unless otherwise indicated.
- E. Provide additional framing to fulfill structural requirements and for support at recessed fixtures and similar items.
- F. Exterior Soffit Wind Bracing: Install cross and vertical bracing with additional framing using anchors and fasteners sufficient to comply with performance requirements.
- G. Suspended Ceiling Bracing Assembly: Install Compression Struts and Bracing Wire and other measures as required by applicable codes and regulations.

3.8 ADJUSTING AND PROTECTION

- A. Adjusting: Adjust and align metal framing to properly receive final finishes in accordance with required tolerances.
- B. Protection: Protect metal framing so that it will be without any evidence of damage which would be detrimental to finished work.

END OF SECTION

SECTION 09 22 49

METAL FURRING AND LATHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal furring for attachment of metal lath.
2. Metal suspension systems for attachment of metal lath.
3. Metal lath.
4. Metal trim and accessories.

B. Related Sections:

1. Section 07 27 00 – Air Barrier.
2. Section 05 41 00 – Structural Metal Stud Framing
3. Section 09 24 00 - Portland Cement Plastering.

1.2 REFERENCES

- A. Reference Standards:** See Section 01 42 00. In addition to requirements shown or specified, comply with applicable provisions of following for design, materials, fabrication, and installation of component parts:
1. NAAMM ML/SFA by National Association of Architectural Metal Manufacturers.
- B. ASTM 1063-08, Standard Specification for Installation of Metal Lath and Furring to Receive Interior and Exterior Portland Cement Based Plaster.**

1.3 SYSTEM REQUIREMENTS

- A. Design Requirements:** Contractor: Responsible for designing supplemental metal framing used to comply with performance requirements, including anchorage to structural system and necessary modifications to meet specified requirements and maintain visual design concepts.
- B. Performance Requirements:** Design and install systems to:
1. Exterior Soffits: Withstand minimum positive and negative wind pressure of 20 PSF and maintain deflection of not more than L/360 of distance between supports.
 - a. Provide cross and vertical bracing with additional framing using anchors and fasteners as required.
 2. Interior Suspended Ceilings, Soffits, and Bullheads: Maintain deflection of not more than L/360 of distance between supports.
- C. Fire Resistance Ratings:** Where assemblies with fire ratings are indicated, provide materials and installations which are identical to assemblies tested in accordance with ASTM E119 by testing laboratories acceptable to authorities having jurisdiction.
1. Construct assemblies identical to those indicated by reference to design designations listed in UL Fire Resistance Directory or other agencies acceptable to authorities having jurisdiction.

1.4 SUBMITTALS

- A. General:** Submit in accordance with Section 01 33 00.
- B. Product Data:** Submit for each product.
- C. Shop Drawings:** Submit for conditions in connection with furring and lathing work, including control and expansion joints, reveals, special hangers, special runners, deflection channels and details at light fixtures and other recessed items.

D. Informational Submittals: Submit following packaged separately from other submittals:

1. Manufacturer's instructions.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in metal furring and lathing work having minimum of five years successful experience with work comparable to that indicated and specified.

1.6 FIELD SAMPLES

A. General: Comply with Section 01 40 00.

B. Sample Installation:

1. Provide furring and lathing systems in conjunction with field samples specified in following sections:
 - a. Section 09 22 16 – Non-Structural Metal Framing.
 - b. Section 09 23 00 - Gypsum Plastering.
 - c. Section 09 24 00 - Portland Cement Plastering.
2. Accepted Field Sample: May remain part of completed Work.

1.7 DELIVERY, STORAGE, AND HANDLING

A. General: Deliver, store, handle, and protect products in accordance with Section 01 60 00.

1. Deliver to project site promptly without undue exposure to weather.
2. Deliver materials in manufacturer's unopened container or bundles, fully identified with name, brand, type and grade.
3. Store in dry ventilated space off the ground.
4. Protect materials from soiling, rusting and damage.

B. Mold Prevention: Comply with Section 01 60 00.

1. Protect paper faced products from moisture exposure.

1.8 PROJECT CONDITIONS

A. Mold Prevention: Comply with Section 01 50 00. Provide temporary heating and ventilation as required to keep paper faced products dry during construction operations.

PART 2 - PRODUCTS

2.1 PRODUCTS AND MANUFACTURERS

A. Acceptable Manufacturers - Metal Furring and Lath:

1. Alabama Metal Industries Corp. (AMICO), Birmingham, AL.
2. California Expanded Metals Company (CEMCO), City of Industry, CA
3. Marino/Ware, South Plainfield, NJ
4. Accepted Substitute in accordance with Section 01 25 00.

2.2 FURRING MATERIALS

A. Steel Framing: Maximize use of recycled steel content with minimum of 60 percent.

B. General: Size components as indicated and to conform to requirements of following reference standards to fulfill specified structural requirements.

1. Comply with ASTM C1063 for interior and exterior lathing and furring to receive Portland cement plasterwork.

C. Provide metal stud framing in accordance with section 05 41 00.

2.3 LATH MATERIALS

A. Metal Lath: Fabricate from galvanized cold-rolled steel, ASTM C847.

1. Maximize use of recycled steel.
2. Galvanized typical.
3. Acceptable Product: Structalath by Structa Wire Corporation.
4. Lath Attachment Devices: Screw attach with neoprene washers. Provide special care to only attach through sheathing and structural members.
5. Each lath fastener shall be dabbed with a sealant compatible with waterproofing membrane below. All penetrations shall be sealed with same sealer.

2.4 METAL TRIM MATERIALS

- A. General: Galvanized for interior use, zinc alloy for exterior use.
 1. Shapes Used as Grounds: Sized and dimensioned to produce required plaster thickness.
 2. Flanges: Designed to permit complete embedment of accessory in plaster, alignment, and attachment to underlying surface.
- B. Corner Beads: Minimum 26 gage.
- C. Casing Beads: Minimum 24 gage.
- D. Control Joints: Minimum 26 gage, "M" or "VV" type. Provide prefabricated control joint intersections.
- E. Ventilated Soffit Vent: Equal to Fry Reglet model WPM-75-V, width as indicated on Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.

3.2 INSTALLATION, GENERAL

- A. Install furring and lathing in accordance with ML/SFA and with ASTM C1063 for interior and exterior Portland cement plaster work, except for more stringent specified or manufacturer requirements.
 1. Install members true to lines and levels and to maintain surface flatness with maximum variation of 1/8 inch in 10 feet in any direction.
 2. Isolate furring and lathing work from structural movement to prevent transfer of loading into work from building structure.
 3. Do not allow suspension grillage to come in contact with abutting partitions or walls.
- B. Control or Expansion Joints: Do not continue lath and furring across control or expansion joints.
 1. Provide where indicated and at supporting structure expansion or control joints and where panel sizes or dimensions change.
 2. Control Joints for Interior Work: Space maximum 30 feet apart in both directions.

3.3 WALL AND FURRED SPACE FRAMING

- A. General:
 1. Steel Channels (Furring): Provide where indicated and specified and as required to provide secondary support for furring, lath, and base for plasterwork. See Section 09 22 16 for metal support assemblies.
 2. Include steel shapes, clips, wire and other attachments necessary to bring plaster to required lines.
 3. Secure to supporting construction with masonry clips or other approved type anchorage.
- B. Masonry and Concrete Substrates: Install horizontal metal furring channels directly to substrate; space at maximum 4 feet OC, not more than 4 inches from floor or ceiling, shim as necessary.

1. Attach vertical furring to horizontal steel furring channels; space as necessary for adequate support of metal lath, not more than 16 inches OC.
- C. Metal Stud Substrates: Install horizontal metal furring channels directly to metal support assemblies; space as necessary for adequate support of metal lath, not more than 16 inches OC.
 1. Provide channel stiffeners between studs as required to rigid support for furring and lathing.
- D. Structural Steel Column Substrates: Install horizontal steel channels (furring) directly to substrate; space at maximum 4 feet OC, not more than 4 inches from floor or ceiling.
 1. Attach vertical metal furring channels to horizontal steel channels (furring); space as necessary for adequate support of metal lath, not more than 16 inches OC.
- E. Coordinate installation with required placement of control and expansion joints. Discontinue and provide independent support at joints.

3.4 SUSPENDED CEILING AND SOFFIT FRAMING

- A. System Description: See Section 09 22 16 for metal support assemblies. Provide steel channels (furring), suspended by hangers from construction above, with perpendicular metal cross furring channels.
 1. Provide rigid bracing and hangers as required to fulfill structural performance requirements.
 2. Provide component sizes and gages as required to rigidly support respective ceiling and soffit loads.
- B. Hanger Spacing: Space hanger inserts or other anchorage devices as required to fulfill structural performance requirements.
 1. Provide sufficient spacing on each side of light fixtures, diffusers, grilles, access panels and other items penetrating assembly.
 2. Space hangers not more than 3 feet OC parallel with runner channel and 4 feet OC maximum at right angles.
 3. Provide hangers not more than 6 inches from end of runners.
- C. Hanger Installation: Securely anchor hangers to structural support by connecting directly to structural members, by embedment in structural slabs, or by connecting to inserts, clips or other anchorage devices.
 1. Do not attach hangers to metal deck tabs.
- D. Steel Channels: Provide 1-1/2 inch minimum size.
 1. Install at maximum 4 feet OC.
 2. Extend to within 6 inches maximum of wall surfaces.
 3. Lap splice securely.
 4. Securely fix channels to hangers to prevent turning or twisting and to transmit full load to hangers.
- E. Metal Furring Channels: Provide 3/4 inch minimum size.
 1. Install at maximum 13-1/2 inches OC.
 2. Extend to within 2 inches maximum of wall surfaces.
 3. Lap splice securely.
 4. Framing Ties: Rigidly secure metal furring channels to steel channels with framing ties to prevent turning or twisting and to transmit full load to runners.
- F. Control and Expansion Joints: Coordinate installation of hangers and framing with required placement of joints. Discontinue framing and provide independent support at joints.

3.5 METAL LATH

- A. Metal Lath: Apply lath in proper manner to form true surfaces, straight without sags or buckles.

1. Install with long dimension of lath at right angles to direction of supports.
2. Lap at sides not less than 1/2 inch and at ends not less than 1 inch.
3. Stagger end laps at adjoining sheets.
4. Secure to supports by wire ties at intervals not exceeding 6 inches OC.
5. Corner Mesh: Continuously reinforce internal corners with corner mesh, except where metal lath returns 3 inches from corner to form angle reinforcement; fasten at perimeter edges only.
6. Strip Lath: Install strip lath centered over junctions of dissimilar backing materials and provide minimum 24 inch long strip diagonally at corners of doors, windows and other openings in plastered surfaces.

3.6 METAL TRIM

- A. Corner Beads: Provide at external corners and in single lengths where length of corner or jamb does not exceed standard stock lengths.
 1. Miter or cope at corners, and fasten securely with tie wire, hardened galvanized nails, staples or offset head or hookhead lath nails.
 2. Space not more than 8 inches OC staggered at outer edges.
- B. Casing Beads: Provide where indicated and where plastering terminates and is not covered by other finish.
 1. [Provide sheet metal closures at reveals, unless special reveals are indicated or specified.]
- C. Set beads level, plumb and true to line.
- D. Install in lengths as long as practicable and align joints with concealed splice or tie plates.
- E. Secure beads with tie wire.
- F. Space fastenings not more than 8 inches OC.
- G. Control Joints: Terminate metal lath on each side of joint and secure to metal lath with ties.
 1. Space ties not more than 6 inches OC.
 2. Control Joints for Exterior Work :
 - a. Install at 100 sq. ft. maximum with 18 feet longest length and no greater than 2- 1/2 to 1 length to width ratio.
 - b. Provide continuous 6 inch wide strip of foil faced flexible flashing under each reveal and control joint complying with Technical Services Information Bureau (www.tsib.org) Technical Bulletin No. 220.

3.7 CLEANING

- A. General: Comply with Section 01 74 00.

3.8 PROTECTION

- A. Mold Prevention: Comply with Sections 01 50 00 and 01 60 00.
 1. Protect paper-faced products from moisture exposure.

END OF SECTION

SECTION 09 29 00

GYPSUM BOARD

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
- B. SUMMARY
- C. Section Includes:
 - 1. Interior gypsum board.
 - 2. Tile backing panels.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For the following products:
 - 1. Trim Accessories: Full-size Sample in 12-inch- long length for each trim accessory indicated.

1.3 QUALITY ASSURANCE

- A. Mockups: Before beginning gypsum board installation, install mockups of at least 100 sq. ft. in surface area to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Install mockups for the following:
 - a. Each level of gypsum board finish indicated for use in exposed locations.
 - b. Each texture finish indicated.
 - 2. Apply or install final decoration indicated, including painting and wall coverings, on exposed surfaces for review of mockups.
 - 3. Simulate finished lighting conditions for review of mockups.
 - 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store materials inside under cover and keep them dry and protected against weather, condensation, direct sunlight, construction traffic, and other potential causes of damage. Stack panels flat and supported on risers on a flat platform to prevent sagging.

1.5 FIELD CONDITIONS

- A. Environmental Limitations: Comply with ASTM C 840 requirements or gypsum board manufacturer's written recommendations, whichever are more stringent.
- B. Do not install paper-faced gypsum panels until installation areas are enclosed and conditioned.
- C. Do not install panels that are wet, those that are moisture damaged, and those that are mold damaged.
 - 1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 - 2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Fire-Resistance-Rated Assemblies: For fire-resistance-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 119 by an independent testing agency.
- B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.

2.2 GYPSUM BOARD, GENERAL

- A. Size: Provide maximum lengths and widths available that will minimize joints in each area and that correspond with support system indicated.

2.3 INTERIOR GYPSUM BOARD

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. American Gypsum.
 - 2. CertainTeed Corp.
 - 3. Georgia-Pacific Gypsum LLC.
 - 4. National Gypsum Company.
 - 5. Temple-Inland.
 - 6. USG Corporation.
- B. Gypsum Board, Type X: ASTM C 1396.
- C. Thickness: 5/8 inch.
 - 1. Long Edges: Tapered.
- D. Flexible Gypsum Board: ASTM C 1396. Manufactured to bend to fit radii and to be more flexible than standard regular-type gypsum board of same thickness.
 - 1. Thickness: 1/4 inch.
 - 2. Long Edges: Tapered.
- E. Gypsum Ceiling Board: ASTM C 1396.
 - 1. Thickness: 1/2 inch.
 - 2. Long Edges: Tapered.
- F. Moisture- and Mold-Resistant Gypsum Board: ASTM C 1396. With moisture- and mold-resistant core and paper surfaces.
 - 1. Core: 5/8 inch, Type X.
 - 2. Long Edges: Tapered.
 - 3. Mold Resistance: ASTM D 3273, score of 10.
 - 4. Locations: Restrooms, janitor closets, other locations where water exposure is probable.

2.4 SPECIALTY GYPSUM BOARD

- A. Gypsum Board, Type C: ASTM C 1396. Manufactured to have increased fire-resistive capability.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. American Gypsum; Firebloc Type C.
 - b. CertainTeed Corp.; ProRoc Type C.
 - c. Georgia-Pacific Gypsum LLC; Fireguard C.
 - d. National Gypsum Company; Gold Bond Fire-Shield C.

- e. Temple-Inland; Type TG-C.
 - f. USG Corporation; Firecode C Core.
- 2. Thickness: As required by fire-resistance-rated assembly indicated on Drawings.
 - a. Long Edges: Tapered.
- B. Mold Resistant Glass-Mat Interior Gypsum Board: ASTM C 1658. With fiberglass mat laminated to both sides. Specifically designed for interior use.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Georgia-Pacific Gypsum LLC; DensArmour Plus.
 - 2. Core: 5/8 inch, Type X.
 - 3. Long Edges: Tapered.
 - 4. Mold Resistance: ASTM D 3273, score of 10.

2.5 TRIM ACCESSORIES

- A. Interior Trim: ASTM C 1047.
 - 1. Material: Galvanized or aluminum-coated steel sheet or rolled zinc.
 - 2. Shapes:
 - a. Cornerbead.
 - b. Bullnose bead.
 - c. LC-Bead: J-shaped; exposed long flange receives joint compound.
 - d. L-Bead: L-shaped; exposed long flange receives joint compound.
 - e. U-Bead: J-shaped; exposed short flange does not receive joint compound.
 - f. Expansion (control) joint.
 - g. Curved-Edge Cornerbead: With notched or flexible flanges.
- B. Aluminum Trim: Extruded accessories of profiles and dimensions indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Fry Reglet Corp.
 - b. Gordon, Inc.
 - c. Pittcon Industries.
 - 2. Aluminum: Alloy and temper with not less than the strength and durability properties of ASTM B 221, Alloy 6063-T5.
 - 3. Finish: Corrosion-resistant primer compatible with joint compound and finish materials specified.

2.6 JOINT TREATMENT MATERIALS

- A. General: Comply with ASTM C 475/C 475M.
- B. Joint Tape:
 - 1. Interior Gypsum Board: Paper.
 - 2. Exterior Gypsum Soffit Board: Paper.
 - 3. Glass-Mat Gypsum Sheathing Board: 10-by-10 glass mesh.
 - 4. Tile Backing Panels: As recommended by panel manufacturer.
- C. Joint Compound for Interior Gypsum Board: For each coat use formulation that is compatible with other compounds applied on previous or for successive coats.
 - 1. Prefilling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting-type taping compound.
 - 2. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use drying-type, all-purpose compound.

- a. Use setting-type compound for installing paper-faced metal trim accessories.
- 3. Fill Coat: For second coat, use drying-type, all-purpose compound.
 - a. Finish Coat: For third coat, use drying-type, all-purpose compound.
 - b. Skim Coat: For final coat of Level 5 finish, use drying-type, all-purpose compound.

2.7 AUXILIARY MATERIALS

- A. General: Provide auxiliary materials that comply with referenced installation standards and manufacturer's written recommendations.
- B. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.
 - 1. Laminating adhesive shall have a VOC content of 50 Insert value g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- C. Steel Drill Screws: ASTM C 1002, unless otherwise indicated.
 - 1. Use screws complying with ASTM C 954 for fastening panels to steel members from 0.033 to 0.112 inch thick.
 - 2. For fastening cementitious backer units, use screws of type and size recommended by panel manufacturer.
- D. Sound Attenuation Blankets: ASTM C 665, Type I (blankets without membrane facing) produced by combining thermosetting resins with mineral fibers manufactured from glass, slag wool, or rock wool.
 - 1. Fire-Resistance-Rated Assemblies: Comply with mineral-fiber requirements of assembly.
- E. Acoustical Joint Sealant: Manufacturer's standard nonsag, paintable, nonstaining latex sealant complying with ASTM C 834. Product effectively reduces airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Accumetric LLC; BOSS 824 Acoustical Sound Sealant.
 - b. Pecora Corporation; AC-20 FTR AIS-919.
 - c. Specified Technologies, Inc.; Smoke N Sound Acoustical Sealant.
 - d. USG Corporation; SHEETROCK Acoustical Sealant.
 - 2. Acoustical joint sealant shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and substrates including welded hollow-metal frames and framing, with Installer present, for compliance with requirements and other conditions affecting performance.
- B. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 APPLYING AND FINISHING PANELS, GENERAL

- A. Comply with ASTM C 840.
- B. Install ceiling panels across framing to minimize the number of abutting end joints and to avoid abutting end joints in central area of each ceiling. Stagger abutting end joints of adjacent panels not less than one framing member.
- C. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch of open space between panels. Do not force into place.

- D. Locate edge and end joints over supports, except in ceiling applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Do not place tapered edges against cut edges or ends. Stagger vertical joints on opposite sides of partitions. Do not make joints other than control joints at corners of framed openings.
- E. Form control and expansion joints with space between edges of adjoining gypsum panels.
- F. Cover both faces of support framing with gypsum panels in concealed spaces (above ceilings, etc., except in chases braced internally).
 - 1. Unless concealed application is indicated or required for sound, fire, air, or smoke ratings, coverage may be accomplished with scraps of not less than 8 sq. ft. in area.
 - 2. Fit gypsum panels around ducts, pipes, and conduits.
 - 3. Where partitions intersect structural members projecting below underside of floor/roof slabs and decks, cut gypsum panels to fit profile formed by structural members; allow 1/4- to 3/8-inch- wide joints to install sealant.
- G. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments, except floors. Provide 1/4- to 1/2-inch- wide spaces at these locations and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.
- H. Attachment to Steel Framing: Attach panels so leading edge or end of each panel is attached to open (unsupported) edges of stud flanges first.
- I. Wood Framing: Install gypsum panels over wood framing, with floating internal corner construction. Do not attach gypsum panels across the flat grain of wide-dimension lumber, including floor joists and headers. Float gypsum panels over these members or provide control joints to counteract wood shrinkage.
- J. STC-Rated Assemblies: Seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Comply with ASTM C 919 and with manufacturer's written recommendations for locating edge trim and closing off sound-flanking paths around or through assemblies, including sealing partitions above acoustical ceilings.
- K. Install sound attenuation blankets before installing gypsum panels unless blankets are readily installed after panels have been installed on one side.

3.3 APPLYING INTERIOR GYPSUM BOARD

- A. Install interior gypsum board in the following locations:
 - 1. Wallboard Type: Vertical surfaces unless otherwise indicated.
 - 2. Type X: Vertical surfaces unless otherwise indicated.
 - 3. Flexible Type: Apply in double layer at curved assemblies.
 - 4. Ceiling Type: Ceiling surfaces.
 - 5. Abuse-Resistant Type: As indicated on Drawings.
 - 6. Moisture- and Mold-Resistant Type: As indicated on Drawings.
 - 7. Type C: Where required for specific fire-resistance-rated assembly indicated.
 - 8. Glass-Mat Interior Type: Behind all wall tile.
- B. Single-Layer Application:
 - 1. On ceilings, apply gypsum panels before wall/partition board application to greatest extent possible and at right angles to framing unless otherwise indicated.
 - 2. On partitions/walls, apply gypsum panels vertically (parallel to framing) unless otherwise indicated or required by fire-resistance-rated assembly, and minimize end joints.
 - a. Stagger abutting end joints not less than one framing member in alternate courses of panels.

- b. At stairwells and other high walls, install panels horizontally unless otherwise indicated or required by fire-resistance-rated assembly.
- 3. On Z-furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.
- 4. Fastening Methods: Apply gypsum panels to supports with steel drill screws.
- C. Multilayer Application:
 - 1. On ceilings, apply gypsum board indicated for base layers before applying base layers on walls/partitions; apply face layers in same sequence. Apply base layers at right angles to framing members and offset face-layer joints one framing member, 16 inches minimum, from parallel base-layer joints, unless otherwise indicated or required by fire-resistance-rated assembly.
 - 2. On partitions/walls, apply gypsum board indicated for base layers and face layers vertically (parallel to framing) with joints of base layers located over stud or furring member and face-layer joints offset at least one stud or furring member with base-layer joints, unless otherwise indicated or required by fire-resistance-rated assembly. Stagger joints on opposite sides of partitions.
 - 3. On Z-furring members, apply base layer vertically (parallel to framing) and face layer either vertically (parallel to framing) or horizontally (perpendicular to framing) with vertical joints offset at least one furring member. Locate edge joints of base layer over furring members.
 - 4. Fastening Methods: Fasten base layers and face layers separately to supports with screws.
- D. Laminating to Substrate: Where gypsum panels are indicated as directly adhered to a substrate (other than studs, joists, furring members, or base layer of gypsum board), comply with gypsum board manufacturer's written recommendations and temporarily brace or fasten gypsum panels until fastening adhesive has set.
- E. Curved Surfaces:
 - 1. Install panels horizontally (perpendicular to supports) and unbroken, to extent possible, across curved surface plus 12-inch- long straight sections at ends of curves and tangent to them.
 - 2. For double-layer construction, fasten base layer to studs with screws 16 inches o.c. Center gypsum board face layer over joints in base layer, and fasten to studs with screws spaced 12 inches o.c.

3.4 INSTALLING TRIM ACCESSORIES

- A. General: For trim with back flanges intended for fasteners, attach to framing with same fasteners used for panels. Otherwise, attach trim according to manufacturer's written instructions.
- B. Control Joints: Install control joints according to ASTM C 840 and in specific locations approved by Architect for visual effect.
- C. Interior Trim: Install in the following locations:
 - 1. Cornerbead: Use at outside corners unless otherwise indicated.
 - 2. LC-Bead: Use at exposed panel edges.
 - 3. Curved-Edge Cornerbead: Use at curved openings.
- D. Exterior Trim: Install in the following locations:
 - 1. Cornerbead: Use at outside corners.
 - 2. LC-Bead: Use at exposed panel edges.
- E. Aluminum Trim: Install in locations indicated on Drawings.

3.5 FINISHING GYPSUM BOARD

- A. General: Treat gypsum board joints, interior angles, edge trim, control joints, penetrations, fastener heads, surface defects, and elsewhere as required to prepare gypsum board surfaces for decoration. Promptly remove residual joint compound from adjacent surfaces.
- B. Prefill open joints, rounded or beveled edges, and damaged surface areas.
- C. Apply joint tape over gypsum board joints, except for trim products specifically indicated as not intended to receive tape.
- D. Gypsum Board Finish Levels: Finish panels to levels indicated below and according to ASTM C 840:
 - 1. Level 1: Ceiling plenum areas, concealed areas, and where indicated.
 - 2. Level 2: Panels that are substrate for tile Panels that are substrate for acoustical tile Where indicated on Drawings.
 - 3. Level 4: At panel surfaces that will be exposed to view unless otherwise indicated.
 - a. Primer and its application to surfaces are specified in other Division 09 Sections.
 - 4. Level 5: Where indicated on Drawings.
 - a. Primer and its application to surfaces are specified in other Division 09 Sections.
- E. Glass-Mat Gypsum Sheathing Board: Finish according to manufacturer's written instructions for use as exposed soffit board.
- F. Glass-Mat Faced Panels: Finish according to manufacturer's written instructions.
- G. Cementitious Backer Units: Finish according to manufacturer's written instructions.

3.6 PROTECTION

- A. Protect adjacent surfaces from drywall compound and promptly remove from floors and other non-drywall surfaces. Repair surfaces stained, marred, or otherwise damaged during drywall application.
- B. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.
- C. Remove and replace panels that are wet, moisture damaged, and mold damaged.
 - 1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 - 2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION

SECTION 09 51 13

ACOUSTICAL PANEL CEILINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes acoustical panels and exposed suspension systems for ceilings.
- B. Products furnished, but not installed under this Section, include anchors, clips, and other ceiling attachment devices to be cast in concrete at ceilings.

1.3 DEFINITIONS

- A. Articulation Class.
- B. CAC: Ceiling Attenuation Class. LR: Light Reflectance coefficient. NRC: Noise Reduction Coefficient.

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 - 1. Ceiling suspension system members.
 - 2. Method of attaching hangers to building structure.
- C. Furnish layouts for cast-in-place anchors, clips, and other ceiling attachment devices whose installation is specified in other Sections.
 - 1. Ceiling-mounted items including lighting fixtures, diffusers, grilles, speakers, sprinklers, access panels, and special moldings.
 - 2. Minimum Drawing Scale: 1/4 inch = 1 foot
- D. Samples for Initial Selection: For components with factory-applied color finishes.
- E. Maintenance Data: For finishes to include in maintenance manuals.

1.5 QUALITY ASSURANCE

- A. Acoustical Testing Agency Qualifications: An independent testing laboratory, or an NVLAP-accredited laboratory, with the experience and capability to conduct the testing indicated.
- B. NVLAP-accredited laboratories must document accreditation, based on a "Certificate of Accreditation" and a "Scope of Accreditation" listing the test methods specified.
- C. Source Limitations:
 - 1. Acoustical Ceiling Panel: Obtain each type through one source from a single manufacturer.
 - 2. Suspension System: Obtain each type through one source from a single manufacturer.Source Limitations: Obtain each type of acoustical ceiling panel and supporting suspension system through one source from a single manufacturer.
- D. Fire-Test-Response Characteristics: Provide acoustical panel ceilings that comply with the following requirements:
 - 1. Surface-Burning Characteristics: Provide acoustical panels with the following surface-burning characteristics complying with ASTM E 1264 for Class A materials as determined by testing identical products per ASTM E 84:

- a. Smoke-Developed Index: 450 or less.
- E. Mockups: Build mockups to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Approved mockups may become part of the completed Work if undisturbed at time of
- F. Substantial Completion.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver acoustical panels, suspension system components, and accessories to Project site in original, unopened packages and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.
- B. Before installing acoustical panels, permit them to reach room temperature and a stabilized moisture content.
- C. Handle acoustical panels carefully to avoid chipping edges or damaging units in any way.

1.7 PROJECT CONDITIONS

- A. Environmental Limitations: Do not install acoustical panel ceilings until spaces are enclosed and weatherproof, wet work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.
 - 1. Pressurized Plenums: Operate ventilation system for not less than 48 hours before beginning acoustical panel ceiling installation.

1.8 COORDINATION

- A. Coordinate layout and installation of acoustical panels and suspension system with other construction that penetrates ceilings or is supported by them, including light fixtures, HVAC equipment, fire-suppression system, and partition assemblies.

1.9 EXTRA MATERIALS

- A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Acoustical Ceiling Panels: Full-size panels equal to 2.0 percent of quantity installed, but not less than one unopened box of tile for each 50 boxes or fraction thereof installed of each type, pattern, and color
 - 2. Suspension System Components: Quantity of each exposed component equal to 2.0 percent of quantity installed.
 - 3. Hold-Down Clips: Equal to 2.0 percent of quantity installed.

PART 2 - PRODUCTS

2.1 ACOUSTICAL PANELS, GENERAL(ACT1)

- A. Acoustical Panel Standard: Provide manufacturer's standard panels of configuration indicated that comply with ASTM E 1264 classifications as designated by types, patterns, acoustical ratings, and light reflectance's, unless otherwise indicated.
 - 1. Mounting Method for Measuring NRC: Type E-400; plenum mounting in which face of test specimen is 15-3/4 inches away from test surface per ASTM E 795.
- B. Acoustical Panel Colors and Patterns: Match appearance characteristics indicated for each product type.
 - 1. Where appearance characteristics of acoustical panels are indicated by referencing pattern designations in ASTM E 1264 and not manufacturers' proprietary product designations, provide products selected by Architect from each manufacturer's full range that comply with

requirements indicated for type, pattern, color, light reflectance, acoustical performance, edge detail, and size.

Broad Spectrum Anti-microbial Fungicide and Bactericide Treatment: Provide acoustical panels treated with manufacturer's standard anti-microbial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested according to ASTM D 3273 and evaluated according to ASTM D 3274 or ASTM G 21.

2.2 ACOUSTICAL PANELS FOR ACOUSTICAL PANEL CEILING

- A. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
- B. Products: Subject to compliance with requirements, provide one of the following:
 - 1. Armstrong World Industries, Inc.
 - 2. BPB USA.
 - 3. Chicago Metallic Corporation.
 - 4. USG Interiors, Inc.
- C. Basis-of-Design Product: Subject to compliance with requirements, provide the product indicated on the Finish Legend or a comparable product by one of the above.

2.3 METAL SUSPENSION SYSTEMS, GENERAL

- A. Metal Suspension System Standard: Provide manufacturer's standard direct-hung metal suspension systems of types, structural classifications, and finishes indicated that comply with applicable requirements in ASTM C 635.

2.4 METAL SUSPENSION SYSTEM FOR ACOUSTICAL PANEL CEILING

- A. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
- B. Products: Subject to compliance with requirements, provide one of the following:
 - 1. Armstrong World Industries, Inc.
 - 2. BPB USA.
 - 3. Chicago Metallic Corporation.
 - 4. USG Interiors, Inc.
- C. Wide-Face, Double-Web, Steel Suspension System: Main and cross runners roll formed from cold-rolled steel sheet, prepainted, electrolytically zinc coated, or hot-dip galvanized according to ASTM A 653/A 653M, not less than G30 coating designation, with prefinished 15/16-inch-wide metal caps on flanges.
 - 1. Structural Classification: Intermediate -duty system.
 - 2. End Condition of Cross Runners: Override (stepped) or butt-edge type.
 - 3. Face Design: Flat, flush.
- D. Aluminum Grid: For use in MRI rooms.

2.5 METAL EDGE MOLDINGS AND TRIM

- A. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
- B. Products: Subject to compliance with requirements, provide one of the following:
 - 1. Armstrong World Industries, Inc.
 - 2. BPB USA.
 - 3. Chicago Metallic Corporation.
 - 4. Fry Reglet Corporation.
 - 5. Gordon, Inc.

6. USG Interiors, Inc.
- C. Roll-Formed, Sheet-Metal Edge Moldings and Trim: Type and profile indicated or, if not indicated, manufacturer's standard moldings for edges and penetrations that comply with seismic design requirements; formed from sheet metal of same material, finish, and color as that used for exposed flanges of suspension system runners.
1. Provide manufacturer's standard edge moldings that fit acoustical panel edge details and suspension systems indicated and that match width and configuration of exposed runners, unless otherwise indicated.
 2. For lay-in panels with reveal edge details, provide stepped edge molding that forms reveal of same depth and width as that formed between edge of panel and flange at exposed suspension member
 3. For circular penetrations of ceiling, provide edge moldings fabricated to diameter required to fit penetration exactly.
- D. Extruded-Aluminum Edge Moldings and Trim: Where indicated, provide manufacturer's extruded-aluminum edge moldings and trim of profile indicated or referenced by manufacturer's designations, including splice plates, corner pieces, and attachment and other clips, complying with seismic design requirements and the following:
1. Aluminum Alloy: Alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than the strength and durability properties of aluminum extrusions complying with ASTM B 221 for Alloy and Temper 6063-T5.
 2. Finish designations prefixed by AA comply with system established by the Aluminum Association for designating aluminum finishes.
 3. Baked-Enamel Finish: AA-C12C42R1x (Chemical Finish: cleaned with inhibited chemicals; Chemical Finish: acid-chromate-fluoride-phosphate conversion coating; organic coating: as specified below). Apply baked enamel complying with paint manufacturer's written instructions for cleaning, conversion coating, and painting.
- E. Organic Coating: Thermosetting, primer/topcoat system with a minimum dry film thickness of 0.8 to 1.2 mils.

2.6 ACOUSTICAL SEALANT

- A. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
- B. Products: Subject to compliance with requirements, provide one of the following:
1. Acoustical Sealant for Exposed and Concealed Joints:
- C. Pecora Corporation; AC-20 FTR Acoustical and Insulation Sealant.
- D. USG Corporation; SHEETROCK Acoustical Sealant.
- E. Acoustical Sealant for Exposed and Concealed Joints: Manufacturer's standard non-sag, paintable, non-staining latex sealant, with a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24), complying with ASTM C 834 and effective in reducing airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.
- F. Acoustical Sealant for Concealed Joints: Manufacturer's standard nondrying, non-hardening, non-skinning, non-staining, gunnable, synthetic-rubber sealant, with a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24), recommended for sealing interior concealed joints to reduce airborne sound transmission.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, including structural framing to which acoustical panel ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and with requirements for installation tolerances and other conditions affecting performance of acoustical panel ceilings.

- 1. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Measure each ceiling area and establish layout of acoustical panels to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width panels at borders, and comply with layout shown on reflected ceiling plans.

3.3 INSTALLATION

- A. General: Install acoustical panel ceilings to comply with ASTM C 636 and seismic design requirements indicated, per manufacturer's written instructions and CISCA's "Ceiling Systems Handbook."
- B. Suspend ceiling hangers from building's structural members and as follows:
 - 1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
 - 2. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 - 3. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 - 4. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
 - 5. Secure wire hangers to ceiling suspension members and to supports above with a minimum of three tight turns. Connect hangers directly either to structures or to inserts, eye screws, or other devices that are secure and appropriate for substrate and that will not deteriorate or otherwise fail due to age, corrosion, or elevated temperatures.
 - 6. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both structure to which hangers are attached and type of hanger involved. Install hangers in a manner that will not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
 - 7. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to cast-in-place hanger inserts, postinstalled mechanical or adhesive anchors, or power-actuated fasteners that extend through forms into concrete.
 - 8. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
 - 9. Do not attach hangers to steel deck tabs.
 - 10. Do not attach hangers to steel roof deck. Attach hangers to structural members.
 - 11. Space hangers not more than 48 inches o.c. along each member supported directly from hangers, unless otherwise indicated; provide hangers not more than 8 inches from ends of each member.
 - 12. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards and publications.

- a. Secure bracing wires to ceiling suspension members and to supports with a minimum of four tight turns. Suspend bracing from building's structural members as required for hangers, without attaching to permanent metal forms, steel deck, or steel deck tabs. Fasten bracing wires into concrete with cast-in-place or postinstalled anchors.
 - b. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical panels.
- 13. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
- 14. Screw attach moldings to substrate at intervals not more than 16 inches o.c. and not more than 3 inches from ends, leveling with ceiling suspension system to a tolerance of 1/8 inch in 12 feet. Miter corners accurately and connect securely.
- 15. Do not use exposed fasteners, including pop rivets, on moldings and trim.
- C. Install suspension system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.
- D. Install acoustical panels with undamaged edges and fit accurately into suspension system runners and edge moldings. Scribe and cut panels at borders and penetrations to provide a neat, precise fit.
 - 1. Arrange directionally patterned acoustical panels as follows:
 - a. As indicated on reflected ceiling plans.
 - b. Install panels with pattern running in one direction parallel to long axis of space.
 - c. Install panels in a basket-weave pattern.
 - 2. For square-edged panels, install panels with edges fully hidden from view by flanges of suspension system runners and moldings.
 - 3. For reveal-edged panels on suspension system runners, install panels with bottom of reveal in firm contact with top surface of runner flanges.
 - 4. For reveal-edged panels on suspension system members with box-shaped flanges, install panels with reveal surfaces in firm contact with suspension system surfaces and panel faces flush with bottom face of runners.
 - 5. Paint cut edges of panel remaining exposed after installation; match color of exposed panel surfaces using coating recommended in writing for this purpose by acoustical panel manufacturer.
 - 6. Install hold-down clips in areas indicated, in areas required by authorities having jurisdiction, and for fire-resistance ratings; space as recommended by panel manufacturer's written instructions, unless otherwise indicated.
 - 7. Install clean-room gasket system in areas indicated, sealing each panel and fixture as recommended by panel manufacturer's written instructions.
 - 8. Protect lighting fixtures and air ducts to comply with requirements indicated for fire-resistance-rated assembly.

3.4 CLEANING

- A. Clean exposed surfaces of acoustical panel ceilings, including trim, edge moldings, and suspension system members. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage. Remove and replace ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

END OF SECTION

SECTION 09 65 13

RESILIENT BASE AND ACCESSORIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Resilient base.
 - 2. Resilient stair accessories.
 - 3. Resilient molding accessories.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples for Initial Selection: For each type of product indicated.
- C. Product Schedule: For resilient products. Use same designations indicated on Drawings.

1.4 QUALITY ASSURANCE

- A. Mockups: Provide resilient products with mockups specified in other Sections.
- B. Manufacturer's representative to be on site during installation and final inspection.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store resilient products and installation materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than 50 deg F or more than 90 deg F.

1.6 PROJECT CONDITIONS

- A. Maintain ambient temperatures within range recommended by manufacturer, but not less than 70 deg F or more than 95 deg F, in spaces to receive resilient products during the following time periods:
 - 1. 48 hours before installation.
 - 2. During installation.
 - 3. 48 hours after installation.
- B. Until Substantial Completion, maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F or more than 95 deg F.
- C. Install resilient products after other finishing operations, including painting, have been completed.

1.7 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Furnish not less than 10 linear feet for every 500 linear feet or fraction thereof, of each type, color, pattern, and size of resilient product installed.
- B. Furnish 6 inside and outside molded corners matching each base

PART 2 - PRODUCTS

2.1 RESILIENT BASE

A. Resilient Base:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Allstate Rubber Corp.; Stoler Industries.
 - b. Armstrong World Industries, Inc.
 - c. Burke Mercer Flooring Products; Division of Burke Industries, Inc.
 - d. Johnsonite.
 - e. Musson, R. C. Rubber Co.
 - f. Roppe Corporation, USA.
 - g. VPI, LLC; Floor Products Division.

B. Resilient Base Standard: ASTM F 1861.

1. Material Requirement: Type TS (rubber, vulcanized thermoset).
2. Manufacturing Method: Group I (solid, homogeneous).
3. Style: Cove (base with toe) typically. Straight where indicated.

C. Minimum Thickness: 0.125 inch.

D. Height: 4 inches typically; 6 inches where indicated.

E. Lengths: Coils in manufacturer's standard length.

F. Outside Corners: Job formed.

G. Inside Corners: Job formed.

H. Finish: Satin.

I. Colors and Patterns: Refer to Finish Legend.

2.2 RESILIENT MOLDING ACCESSORY

A. Resilient Molding Accessory

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Burke Mercer Flooring Products; Division of Burke Industries, Inc.
 - b. Flexco, Inc.
 - c. Johnsonite.
 - d. R.C.A. Rubber Company (The).
 - e. Roppe Corporation, USA.
 - f. VPI, LLC; Floor Products Division.

B. Description: Carpet edge for glue-down applications Nosing for resilient floor covering; Reducer strip for resilient floor covering; Joiner for tile and carpet; Transition strips.

C. Material: Rubber.

D. Profile and Dimensions: As indicated.

E. Colors and Patterns: Refer to Finish Legend.

2.3 INSTALLATION MATERIALS

- A. Trowelable Leveling and Patching Compounds: Latex-modified, portland cement based or blended hydraulic-cement-based formulation provided or approved by manufacturer for applications indicated.

- B. Adhesives: Water-resistant type recommended by manufacturer to suit resilient products and substrate conditions indicated.
 - 1. Use adhesives that comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 - a. Cove Base Adhesives: Not more than 50 g/L.
 - b. Rubber Floor Adhesives: Not more than 60 g/L.
 - c. Pick Resistant Rubber Adhesive – Psych Unit
 - d. Surebond SB-190 Everseal
- C. Metal Edge Strips: Extruded aluminum with mill finish of width shown, of height required to protect exposed edges of tiles, and in maximum available lengths to minimize running joints.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resilient products.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of resilient products.
- B. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound and remove bumps and ridges to produce a uniform and smooth substrate.
- C. Do not install resilient products until they are same temperature as the space where they are to be installed.
 - 1. Move resilient products and installation materials into spaces where they will be installed at least 48 hours in advance of installation.
- D. Sweep and vacuum clean substrates to be covered by resilient products immediately before installation.

3.3 RESILIENT BASE INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient base.
- B. Apply resilient base to walls, columns, pilasters, casework and cabinets in toe spaces, and other permanent fixtures in rooms and areas where base is required.
- C. Install resilient base in lengths as long as practicable without gaps at seams and with tops of adjacent pieces aligned.
- D. Tightly adhere resilient base to substrate throughout length of each piece, with base in continuous contact with horizontal and vertical substrates.
- E. Do not stretch resilient base during installation.
- F. On masonry surfaces or other similar irregular substrates, fill voids along top edge of resilient base with manufacturer's recommended adhesive filler material.
- G. Job-Formed Corners:
 - 1. Outside Corners: Use straight pieces of maximum lengths possible. Form without producing discoloration (whitening) at bends.
 - 2. Inside Corners: Use straight pieces of maximum lengths possible.

3.4 RESILIENT ACCESSORY INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient accessories.
- B. Resilient Stair Accessories:
 - 1. Use stair-tread-nose filler to fill nosing substrates that do not conform to tread contours.
 - 2. Tightly adhere to substrates throughout length of each piece.
 - 3. For treads installed as separate, equal-length units, install to produce a flush joint between units.
- C. Resilient Molding Accessories: Butt to adjacent materials and tightly adhere to substrates throughout length of each piece. Install reducer strips at edges of carpet and resilient floor covering that would otherwise be exposed.

3.5 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protection of resilient products.
- B. Perform the following operations immediately after completing resilient product installation:
 - 1. Remove adhesive and other blemishes from exposed surfaces.
 - 2. Sweep and vacuum surfaces thoroughly.
 - 3. Damp-mop surfaces to remove marks and soil.
- C. Protect resilient products from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.
- D. Floor Polish: Remove soil, visible adhesive, and surface blemishes from resilient stair treads before applying liquid floor polish.
 - 1. Apply two coat(s).
- E. Cover resilient products until Substantial Completion.

END OF SECTION

SECTION 09 65 16

RESILIENT SHEET FLOORING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Vinyl sheet floor covering.
 - 2. Rubber sheet floor covering.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: For each type of floor covering. Include floor covering layouts, locations of seams, edges, columns, doorways, enclosing partitions, built-in furniture, cabinets, and cutouts.
 - 1. Show details of special patterns.
- C. Samples for Initial Selection: For each type of floor covering indicated.
- D. Product Schedule: For floor coverings. Use same designations indicated on Drawings.
- E. Qualification Data: For qualified Installer.
- F. Maintenance Data: For each type of floor covering to include in maintenance manuals.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: A qualified installer who employs workers for this Project who are competent in techniques required by manufacturer for floor covering installation and seaming method indicated.
 - 1. Engage an installer who employs workers for this Project who are trained or certified by floor covering manufacturer for installation techniques required.
- B. Manufacturer's representative to be on site during installation and final inspection.
- C. Mockups: Build mockups to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Build mockups for floor coverings including resilient base and accessories.
 - a. Size: Minimum 100 sq. ft. for each type, color and pattern in locations directed by Architect.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store floor coverings and installation materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than

1.6 PROJECT CONDITIONS

- A. Maintain ambient temperatures within range recommended by manufacturer, but not less than 70 deg F or more than 85 deg F, in spaces to receive floor coverings during the following time periods:
 - 1. 48 hours before installation.
 - 2. During installation.

3. 48 hours after installation.
- B. Until Substantial Completion, maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F or more than 95 deg F.
- C. Close spaces to traffic during floor covering installation.
- D. Close spaces to traffic for 48 hours after floor covering installation.
- E. Install floor coverings after other finishing operations, including painting, have been completed.

1.7 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 1. Floor Covering: Furnish quantity not less than 10 linear feet for every 500 linear feet or fraction thereof, in roll form and in full roll width for each color, pattern, and type of floor covering installed.

PART 2 - PRODUCTS

2.1 RUBBER SHEET FLOOR COVERING (SR1, SR1B)

- A. Products: Subject to compliance with requirements, provide one of the following:
- B. Flexco
 1. Johnsonite
 2. Mondo Rubber International, Inc.
 3. Nora Rubber Flooring, Freudenberg Building Systems, Inc.
 4. R.C.A. Rubber Company (The)
- C. Acceptable Product: Refer to Finish Legend.
- D. Rubber Sheet Floor Covering with Backing: ASTM F 1860.
 1. Type: Type I, homogeneous rubber sheet with backing.
 2. Wear-Layer Thickness: As standard with manufacturer.
 3. Overall Thickness: As standard with manufacturer.
 4. Interlayer Material: As standard with manufacturer.
- E. Hardness: Manufacturer's standard hardness, measured using Shore, Type A durometer per ASTM D 2240.
- F. Wearing Surface: Smooth.
 1. Molded-Pattern Figure: Raised discs.
- G. Sheet Width: As standard with manufacturer .
 1. Seaming Method: Heat welded.
 2. Colors and Patterns: Refer to Finish Legend.

2.2 INSTALLATION MATERIALS

- A. Trowelable Leveling and Patching Compounds: Latex-modified, portland cement based or blended hydraulic-cement-based formulation provided or approved by manufacturer for applications indicated.
- B. Adhesives: Water-resistant type recommended by manufacturer to suit floor covering and substrate conditions indicated.
 1. Use adhesives that have a VOC content of not more than 50 g/L when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- C. Seamless-Installation Accessories:

1. Heat-Welding Bead: Manufacturer's solid-strand product for heat welding seams.
 - a. Color: Match floor covering.
- D. Integral-Flash-Cove-Base Accessories:
 1. Cove Strip: 1-inch radius provided or approved by manufacturer.
 2. Cap Strip: Square metal, vinyl, or rubber cap provided or approved by manufacturer.
 3. Corners: Metal inside and outside corners and end stops provided or approved by manufacturer.
- E. Floor Polish: Provide protective liquid floor polish products as recommended by manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of floor coverings.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of floor coverings.
- B. Concrete Substrates: Prepare according to ASTM F 710.
 1. Verify that substrates are dry and free of curing compounds, sealers, and hardeners.
 2. Remove substrate coatings and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, using mechanical methods recommended by manufacturer. Do not use solvents.
 3. Alkalinity and Adhesion Testing: Perform tests recommended by manufacturer. Proceed with installation only after substrates pass testing.
 4. Moisture Testing: Perform tests recommended by manufacturer and as follows. Proceed with installation only after substrates pass testing.
 - a. Perform anhydrous calcium chloride test, ASTM F 1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.
 - b. Perform relative humidity test using in situ probes, ASTM F 2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.
- C. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound and remove bumps and ridges to produce a uniform and smooth substrate.
- D. Do not install floor coverings until they are same temperature as space where they are to be installed.
 - a. Move floor coverings and installation materials into spaces where they will be installed at least 48 hours in advance of installation.
- E. Sweep and vacuum clean substrates to be covered by floor coverings immediately before installation.

3.3 FLOOR COVERING INSTALLATION

- A. Comply with manufacturer's written instructions for installing floor coverings.
- B. Unroll floor coverings and allow them to stabilize before cutting and fitting.

- C. Lay out floor coverings as follows:
 - 1. Maintain uniformity of floor covering direction.
 - 2. Minimize number of seams; place seams in inconspicuous and low-traffic areas, at least 6 inches away from parallel joints in floor covering substrates.
 - 3. Match edges of floor coverings for color shading at seams.
 - 4. Avoid cross seams.
- D. Scribe and cut floor coverings to butt neatly and tightly to vertical surfaces, permanent fixtures, and built-in furniture including cabinets, pipes, outlets, and door frames.
- E. Extend floor coverings into toe spaces, door reveals, closets, and similar openings.
- F. Maintain reference markers, holes, or openings that are in place or marked for future cutting by repeating on floor coverings as marked on substrates. Use chalk or other nonpermanent marking device.
- G. Install floor coverings on covers for telephone and electrical ducts and similar items in installation areas. Maintain overall continuity of color and pattern between pieces of floor coverings installed on covers and adjoining floor covering. Tightly adhere floor covering edges to substrates that abut covers and to cover perimeters.
- H. Adhere floor coverings to substrates using a full spread of adhesive applied to substrate to produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.
- I. Seamless Installation:
 - 1. Heat-Welded Seams: Comply with ASTM F 1516. Rout joints and use welding bead to permanently fuse sections into a seamless floor covering. Prepare, weld, and finish seams to produce surfaces flush with adjoining floor covering surfaces.
- J. Integral-Flash-Cove Base: Cove floor coverings 6 inches up vertical surfaces. Support floor coverings at horizontal and vertical junction by cove strip. Butt at top against cap strip.
 - 1. Install metal corners at inside and outside corners.

3.4 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protection of floor coverings.
- B. PROHIBIT TRAFFIC ON FLOOR FINISH FOR A MINIMUM OF 7 DAYS AFTER INSTALLATION.**
- C. Perform the following operations immediately after completing floor covering installation:
 - 1. Remove adhesive and other blemishes from floor covering surfaces.
 - 2. Sweep and vacuum floor coverings thoroughly.
 - 3. Damp-mop floor coverings to remove marks and soil.
- D. Protect floor coverings from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.
- E. Floor Polish: Remove soil, visible adhesive, and surface blemishes from floor covering before applying liquid floor polish.
 - 1. Apply two coat(s).
- F. Cover floor coverings until Substantial Completion.

END OF SECTION

SECTION 09 67 23

RESINOUS FLOORING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Decorative resinous flooring systems in dietary areas laboratories.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include manufacturer's technical data, application instructions, and recommendations for each resinous flooring component required.
- B. Samples for Initial Selection: For each type of exposed finish required.
- C. Product Schedule: For resinous flooring.
- D. Installer Certificates: Signed by manufacturer certifying that installers comply with specified requirements.
- E. Material Certificates: For each resinous flooring component, from manufacturer.
- F. Material Test Reports: For each resinous flooring system.
- G. Maintenance Data: For resinous flooring to include in maintenance manuals.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of flooring systems required for this Project.
 - 1. Engage an installer who is certified in writing by resinous flooring manufacturer as qualified to apply resinous flooring systems indicated.
- B. Source Limitations: Obtain primary resinous flooring materials, including primers, resins, hardening agents, grouting coats, and topcoats, from single source from single manufacturer. Provide secondary materials, including patching and fill material, joint sealant, and repair materials, of type and from source recommended by manufacturer of primary materials.
- C. Mockups: Apply mockups to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Apply full-thickness mockups on 48-inch- square floor area selected by Architect.
 - 2. Simulate finished lighting conditions for Architect's review of mockups.
 - 3. Approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating brand name and directions for storage and mixing with other components.

1.6 PROJECT CONDITIONS

- A. Environmental Limitations: Comply with resinous flooring manufacturer's written instructions for substrate temperature, ambient temperature, moisture, ventilation, and other conditions affecting resinous flooring application.

- B. Lighting: Provide permanent lighting or, if permanent lighting is not in place, simulate permanent lighting conditions during resinous flooring application.
- C. Close spaces to traffic during resinous flooring application and for not less than 24 hours after application unless manufacturer recommends a longer period

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Crossfield Products Corp.; Dex-O-Tex.
 - 2. Key Resin Company.
 - 3. Pacific Polymers, Inc.
 - 4. Sherwin-Williams Company; General Polymers.
 - 5. Stonhard, Inc.

2.2 MATERIALS

- A. VOC Content of Resinous Flooring: Provide resinous flooring systems, for use inside the weatherproofing system, that comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 - 1. Resinous Flooring: 100 g/L.

2.3 DECORATIVE RESINOUS FLOORING

- A. Resinous Flooring: Abrasion-, impact- and chemical-resistant, decorative-aggregate-filled, epoxy-resin-based, monolithic floor surfacing designed to produce a seamless floor.
- B. System Characteristics:
 - 1. Color and Pattern: As selected by Architect from manufacturer's full range.
 - 2. Wearing Surface: Textured for slip resistance.
 - 3. Overall System Thickness: 3/16 inch.
- C. Body Coats:
 - 1. Resin: Epoxy.
 - 2. Formulation Description: 100 percent solids .
 - 3. Application Method: Self-leveling slurry with broadcast aggregates .
 - a. Thickness of Coats: 3/16 inch.
 - b. Number of Coats: Two.
 - 4. Aggregates: Colored quartz (ceramic-coated silica).
- D. Topcoat: Sealing or finish coats.
 - 1. Resin: Urethane.
 - 2. Formulation Description: 100 percent solids.
 - 3. Type: Clear.
 - 4. Finish: Matte.
 - 5. Number of Coats: One.
- E. System Physical Properties: Provide resinous flooring system with the following minimum physical property requirements when tested according to test methods indicated:
 - 1. Compressive Strength: 3500 per ASTM C 579.
 - 2. Impact Resistance: No chipping, cracking, or delamination and not more than 1/16-inch permanent indentation per MIL-D-3134.

3. Resistance to Elevated Temperature: No slip or flow of more than 1/16 inch per MIL-D-3134.
 4. Flammability: Self-extinguishing per ASTM D 635.
 5. Critical Radiant Flux: 0.45 W/sq. cm or greater per NFPA 253.
- F. System Chemical Resistance: Test specimens of cured resinous flooring system are unaffected when tested according to ASTM D 1308 for 50 percent immersion.

2.4 ACCESSORIES

- A. Patching and Fill Material: Resinous product of or approved by resinous flooring manufacturer and recommended by manufacturer for application indicated.

PART 3 - EXECUTION

3.1 PREPARATION

- A. General: Prepare and clean substrates according to resinous flooring manufacturer's written instructions for substrate indicated. Provide clean, dry substrate for resinous flooring application.
- B. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.
1. Roughen concrete substrates as follows:
 - a. Comply with ASTM C 811 requirements unless manufacturer's written instructions are more stringent.
 2. Repair damaged and deteriorated concrete according to resinous flooring manufacturer's written instructions.
 3. Verify that concrete substrates are dry and moisture-vapor emissions are within acceptable levels according to manufacturer's written instructions.
 - a. Perform anhydrous calcium chloride test, ASTM F 1869. Proceed with application of resinous flooring only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. of slab area in 24 hours.
 - b. Perform plastic sheet test, ASTM D 4263. Proceed with application only after testing indicates absence of moisture in substrates.
- C. Perform relative humidity test using in situ probes, ASTM F 2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.
1. Alkalinity and Adhesion Testing: Verify that concrete substrates have pH within acceptable range. Perform tests recommended by manufacturer. Proceed with application only after substrates pass testing.
- D. Resinous Materials: Mix components and prepare materials according to resinous flooring manufacturer's written instructions.
- E. Use patching and fill material to fill holes and depressions in substrates according to manufacturer's written instructions.
- F. Treat control joints and other nonmoving substrate cracks to prevent cracks from reflecting through resinous flooring according to manufacturer's written instructions.

3.2 SLIDING DOOR PREPERATION

- A. Verify that surfaces are smooth and flat with maximum variation of 3/16 inch in 10 feet and are ready to receive Work.
- B. Prepare substrate in accordance with ASTM F710 and flooring manufacturer's recommendations.
- C. Remove sub-floor ridges and bumps. Fill low spots, cracks, joints, holes, and other defects with subfloor filler.

1. Apply, trowel, and float filler to leave a smooth, flat, hard surface. Repair all floor irregularities.
2. Fill or level cracks, holes and depressions 1/8 inch wide or wider, and protrusions more than 1/32 inch, unless more stringent requirements are required by manufacturer's written instructions.

3.3 APPLICATION

- A. General: Apply components of resinous flooring system according to manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness indicated.
 1. Coordinate application of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.
 2. Cure resinous flooring components according to manufacturer's written instructions. Prevent contamination during application and curing processes.
 3. At substrate expansion and isolation joints, comply with resinous flooring manufacturer's written instructions.
- B. Apply self-leveling slurry body coats in thickness indicated for flooring system.
 1. Broadcast aggregates at rate recommended by manufacturer and, after resin is cured, remove excess aggregates to provide surface texture indicated.
- C. Apply troweled or screeded body coats in thickness indicated for flooring system. Hand or power trowel and grout to fill voids. When cured, remove trowel marks and roughness using method recommended by manufacturer.
- D. Apply grout coat, of type recommended by resinous flooring manufacturer, to fill voids in surface of final body coat and to produce wearing surface indicated.
- E. Apply topcoats in number indicated for flooring system and at spreading rates recommended in writing by manufacturer.

3.4 FIELD QUALITY CONTROL

- A. Material Sampling: Owner may at any time and any number of times during resinous flooring application requires material samples for testing for compliance with requirements.
 1. Testing agency will test samples for compliance with requirements, using applicable referenced testing procedures or, if not referenced, using testing procedures listed in manufacturer's product data.
 2. If test results show applied materials do not comply with specified requirements, pay for testing, remove noncomplying materials, prepare surfaces coated with unacceptable materials, and reapply flooring materials to comply with requirements.

3.5 PROTECTION

- A. Protect resinous flooring from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by resinous flooring manufacturer.

END OF SECTION

SECTION 09 91 00

PAINTING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Preparation and priming of surfaces scheduled in PART 2 to receive finish painting.
- B. Finish Painting of Exterior and Interior Items and Surfaces, Including:
 - 1. Exposed interior surfaces.
 - 2. Factory-primed surfaces.
 - 3. Scheduled and otherwise identified exterior surfaces.
 - 4. Exposed items on site.
 - 5. Roof top equipment visible from ground level.
 - 6. Luminous paint to identify exits and exist passageways as required by IBC 1024.
- C. Exterior and Interior Items and Surfaces Not Requiring Painting, Unless Noted Otherwise:
 - 1. Surfaces coated by other specification sections.
 - 2. Items with factory applied finishes other than factory-applied primer only applications.
 - 3. Aluminum, stainless steel, brass, bronze, chromium plate, copper, nickel, plated architectural metals, prefinished metals specified in Section 05 05 10, and "weathering" metals.
 - 4. Brick, stone, ceramic tile, plastic laminate, glass unit masonry, decorative concrete unit masonry, and natural and manufactured or simulated stone..
 - 5. Moving parts of operating units.
 - 6. Code required labels or equipment identification plates.
 - 7. Acoustical ceilings.
 - 8. Architectural concrete, architectural precast concrete, cast stone.
 - 9. Decorative plastic and metal laminates, and synthetic countertops.
 - 10. Elastomeric membranes and flashings, roofing materials, and exterior joint sealants.
 - 11. Acoustic materials.
 - 12. Rubber, vinyl, or plastic seals and bumpers.
- D. Surfaces concealed or inaccessible in finished construction unless specifically required.
 - 1. Other surfaces specifically scheduled or indicated to remain unfinished or unpainted.
- E. Materials and Products Not To Be Painted:
 - 1. Items with integral or factory-applied final finish unless indicated otherwise.
 - 2. Wire fencing and areaway grating.
 - 3. Cast metal stair nosings, trench drain grates, manhole covers, and curb inlets.
 - 4. Wire mesh partitions and gates, metal and storage shelving.
 - 5. Moving parts of operating equipment such as valve and damper operators, linkages, sensing devices, motor and fan shafts.
 - 6. UL, FM or other code-required labels, name plates, identification or performance rating labels.
- F. Sprinkler heads.
 - 1. Mechanical and electrical items within unfinished spaces unless noted otherwise.

- G. Products Requiring Field Applied Final Finish:
 - 1. Coiling and sectional overhead doors.
 - 2. Dock levelers and lifts.
 - 3. Sight exposed surfaces of exterior rooftop mechanical and electrical equipment.
 - 4. Air outlets and inlets.
- H. Field finish coating of shop or factory primed items. Refer to individual Sections for priming requirements. Use intermediate and finish coat products of same manufacturer as primer.
- I. Preparation work and coatings specified in this Section are in addition to shop and factory applied finishes and surface treatment specified in other Sections.
- J. Refer to Divisions 21, 22, 23, and 26 for painting requirements for items in dedicated mechanical and electrical spaces.
- K. Paint items unless specifically shown or specified not to be painted.
 - 1. Refer to Color Schedule on Drawings or at end of PART 2.
- L. Related Sections:
 - 1. Division 21: Fire Suppression Identification.
 - 2. Division 22: Plumbing Systems Identification.
 - 3. Division 23: HVAC Identification.
 - 4. Division 26 - Electrical: Electrical Identification.

1.2 DEFINITIONS

- A. Conform to Painting and Decorating Contractors of America (PDCA) Glossary for interpretation of terms used in this Section except as modified below.
 - 1. EXPOSED SURFACES: Surfaces of products, assemblies, and components visible from any angle after final installation. Includes internal surfaces visible when operable doors, panels or drawers are open, and surfaces visible behind registers, grilles, or louvers.
 - 2. CONCEALED SURFACES: Surfaces permanently hidden from view in finished construction and which are only visible after removal or disassembly of part or all of product or assembly.
 - 3. INACCESSIBLE SPACES: Spaces not intended for human use.
 - 4. Spaces listed below are defined as CONCEALED or INACCESSIBLE:
 - a. Space between suspended ceilings and floor or roof construction above.
 - b. Inside furred spaces.
 - c. Inside of partitions.
 - d. Mechanical and electrical items enclosed within casework or equipment.
 - e. Foundation spaces.
 - f. Crawl spaces.
 - g. Trenches and manholes.
 - h. Mechanical shafts or chases.
 - i. Enclosed elevator shafts unless visible through glass panels.
 - j. Utility tunnels.
 - 5. SHEEN: Degree of luster as measured with specular gloss meter

Flat:	85 degree meter	Below 15
Eggshell:	60 degree meter	5 to 20
Satin:	60 degree meter	15 to 35
Semi-gloss:	60 degree meter	30 to 65

Gloss:	60 degree meter	65 to
80 High Gloss:	60 degree meter	Over
80		

6. ARCHITECTURAL PAINT: Paints, field applied to stationary structures and their appurtenances, mobile homes, pavements, or curbs.
7. INDUSTRIAL MAINTENANCE PRIMERS and TOPCOATS: High performance paints formulated for and applied to substrates in industrial, commercial, or institutional situations for purpose resisting heavy abrasion, immersion, prolonged exposure to temperatures in excess of 250 F, prolonged moisture condensation, chemical corrosion, solvent cleaning, or exterior exposure of metal structures.
8. METALLIC PIGMENTED PAINTS: Paints containing at least 0.4 pounds of metallic pigment per gallon of paint as applied.
9. SYSTEM DFT: Dry film thickness of entire paint system unless otherwise noted.
10. VOC: Volatile organic compounds.

1.3 SYSTEM DESCRIPTION

- A. Application Requirements: Apply scheduled paints to exposed surfaces of items and spaces unless specifically indicated otherwise.
- B. Interface with Adjacent Systems: Review other Sections specifying prime coats to ensure compatibility of total paint system for various substrates.
 1. Upon request from other trades, furnish information on characteristics of finish materials proposed for use to ensure compatibility of various paints.
 2. Test compatibility of existing coatings, including shop applied primers and previously applied coatings, by applying specified special paint to small, inconspicuous area.
 3. If specified paint lifts or blisters existing coating, apply barrier or tie coat as recommended by paint manufacturer.
- C. If no compatible barrier or tie coat exists, remove existing coating completely and apply paint system as specified for new work.

1.4 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
 1. Submit submittals for Division 9 Sections involving exposed interior finishes with color, texture, pattern, etc. selections simultaneously with submittals for this section.
- B. Product Data: Submit product data, including label analysis for each product.
 1. Specifically include percent solids-by-volume, VOC content pound/gallon, and lead content (percent of weight of dried film).
 2. Schedule: List each material proposed for use, and cross-reference to specific paint system and substrate application.
 - a. Identify each material by manufacturer's catalog number, product name, and generic classification.
 - b. Include typewritten list identifying paint systems and colors applied to each room, space, or item.
 3. Include sample of warranty customized for this Project.
- C. Samples: Color and Sheen: Prepare two samples of each opaque finish paint specified in each color and sheen scheduled for appearance verification.
 1. Apply to 12 by 12 by 1/4 inch hardboard. Apply sufficient paint thickness to provide proper hiding and appearance.
 2. Label each sample to indicate material, color, and sheen.

D. Samples: Painting Systems:

1. Prepare two samples of each opaque paint system scheduled on actual substrate materials proposed for use. Apply in most common top coat color scheduled.
2. Step back each coat and process at least one inch to show bare substrate and each coat and process in system build-up.
3. Minimum sample size of 4 by 8 inches.
4. Label each sample to indicate materials, color, sheen, DFT of each coat applied, and total system DFT.

E. Informational Submittals: Submit following packaged separately from other submittals:

1. Certifications specified in Quality Assurance article.
2. Qualification Data: Applicator's qualification data.
3. Manufacturer's instructions.

F. Closeout Submittals: Submit specified warranty in accordance with Section 01 77 00.

1.5 QUALITY ASSURANCE

- A. Single Source Responsibility: Provide products of single manufacturer for use in each paint system. Do not mix products of different manufacturers without approval of Architect and manufacturers involved.
- B. Applicator Qualifications: Company specializing in commercial painting and finishing with three years experience.
- C. Regulatory Requirements:
1. Lead Content: Comply with CPSC 16 CFR 1303 and other applicable federal, state, and local regulations.
 2. Volatile Organic Compound Content: ASTM D3960.
 3. Comply with applicable regulations limiting volatile organic compound (VOC) content of paints to be applied. Conduct and report measurement of volatile organic compounds in paints in accordance with ASTM D3960 or Architect approved method.
- D. Certifications:
1. Certification from manufacturer that materials furnished for use on this Project comply with applicable federal, state, and local requirements regarding lead and VOC content.

1.6 FIELD SAMPLES

- A. General: Comply with Section 01 40 00.
- B. Sample Installation: Duplicate finishes of approved paint system samples on wall surfaces and other interior and exterior components selected by Architect.
1. Provide full-coat finish on at least 100 square feet of surface until required color, sheen, and texture are obtained. Simulate finished lighting conditions for review of in-place work.
 2. Request review by Architect of first finished room, space, or item for each paint system for color, texture, quality, and workmanship before proceeding with rest of Work.
 3. Accepted Field Sample: May remain part of completed Work.

1.7 PRE-INSTALLATION CONFERENCE

- A. Conference attendance by Owner, Architect, Contractor, Installer, and manufacturer's representative authorized to certify Installer as an approved applicator and participant in extended warranty program specified.
- B. Minimum Agenda:
1. Discuss sequence and scheduling, installation procedures and interface with other trades.

2. Review requirements and conditions which could possibly interfere with successful performance of work.
 3. Review Project Specifications and Drawings.
 4. Review basis and coordination to determine acceptability of surfaces to receive paints and coatings.
 5. Review environmental and ventilation requirements.
 6. Indoor Air Quality Procedures: Review requirements of Section 01 81 19.
- C. Where required by warranty requirements, include scheduling requirements for manufacturer representative to confirm acceptability of surfaces. Include surface preparation to receive paints and coating, acceptable application methods and techniques for painting and coating, and acceptable completion of work to permit issuance of warranty.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Comply with Section 01 60 00.
1. Deliver products to site in manufacturer's sealed and labeled containers; inspect to verify compliance with specified requirements.
 2. Label containers to indicate manufacturer's name, product name and type of paint, brand code or stock number, date of manufacture, coverage, surface preparation, drying time, cleanup, color designation and instructions for mixing and reducing.
 3. Store paint materials in tightly covered containers in well ventilated area at ambient temperatures of 45 F minimum and 90 F maximum, unless required otherwise by manufacturer. Maintain containers in clean condition, free of foreign materials and residue with labels in legible condition.
 4. Take precautionary measures to prevent fire hazards and spontaneous combustion.

1.9 PROJECT CONDITIONS

- A. Environmental Conditions: Comply with more restrictive of following or manufacturer's requirements under which systems can be applied.
1. Provide continuous ventilation during application of paints to exhaust hazardous fumes.
 2. Indoor Air Quality Procedures: Ventilate in accordance with Section 01 35 43 and 01 81 19.
 3. Provide heating necessary to maintain surface and ambient temperatures within specified limits.
 4. Maintain temperature and humidity conditions for minimum 24 hours before, during, and 48 hours after application of finishes, unless longer times are required by manufacturer.
 5. Do not permit wide variations in ambient temperatures which might result in condensation on freshly coated surfaces.
 6. Provide illumination of not less than 80 foot candles measured mid-height at substrate surface during application of paints.
 7. Apply water reducible paints only when ambient and surface temperatures are between 50 F and 90 F.
 8. Apply solvent reducible paints only when ambient and surface temperatures are between 45 F and 90 F.
- B. Do not apply paints under any of following conditions:
1. When surfaces are damp or wet.
 2. During snow, rain, fog, or mist.
 3. When relative humidity is less than 20 percent or exceeds 85 percent.
 4. When temperature is less than 5 F above dew point.
 5. When dust may be generated before paints have dried.
 6. In direct sunlight.
 7. When wind velocity is above 20 mph.

- C. Application of paints may continue during inclement weather provided work areas and surfaces to be coated are enclosed and specified environmental conditions are maintained.

1.10 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 78 00.
 - 1. Warrant installation to be free from defects in material and workmanship for five years.
 - 2. Repair or replace defects occurring during warranty period.
 - 3. Defects include, but are not limited to, pinholes, crazing or cracking, loss of adhesion to substrate, deficient thickness, bleed-through, improper materials, defective materials, and defective workmanship.

PART 2 - PRODUCTS

2.1 MANUFACTURERS (PT1) (PTE1) (PTM1)

- A. Acceptable Manufacturers:
 - Sherwin-Williams, Per UTSW, all contractors must use Sherwin Williams store on Mockingbird Ln (2727 W Mockingbird Ln #100, Dallas, TX 75235).
- B. Listing of products by manufacturer's trade name is not intended to exclude equivalent products by other manufacturers identified above. For products not specifically named, submit substitution request in accordance with Section 01 25 00.

2.2 PAINT MATERIALS - GENERAL

- A. All paints to be from one manufacturer.
- B. Paints: Ready-mixed, factory tinted, best professional grade produced by manufacturer.
 - 1. Fully grind pigments to maintain soft paste consistency in vehicle.
 - 2. Capable of being dispersed into uniform, homogeneous mixture.
 - 3. Possess good flowing and brushing properties.
 - 4. Capable of drying or curing free of streaks or sags, and yielding specified finish.
 - 5. Paints formulated with formaldehyde, halogenated solvents, mercury or mercury compounds, or tinted with pigments of lead, cadmium, chromium VI and their oxides not allowed.
 - a. Water Based Paints formulated with aromatic hydrocarbons (organic solvent with benzene ring in its molecular structure) not allowed.
 - b. Solvent Based Paints formulated with more than 10 percent aromatic hydrocarbons by weight not allowed.
 - 6. Comply with local regulatory requirements for VOC content of field applied paints.
- C. Paint and Primer Maximum Product Emissions Limits: Top coat and primer interior paints must meet or not exceed the VOC (Volatile Organic Compounds) limits of the current requirements of Green Seal Standards GS-11 - Paints in the building. GS-11 VOC limits for interior paints are as follows. Interior refers to all building construction that is inside of the exterior weatherproofing material:
 - 1. Interior, Non-flats: 150 grams per liter of product minus water
 - 2. Interior, Flats: 50 grams per liter of product minus water.

2.3 FINISH PAINTING SYSTEMS

- A. General: System DFT specified identifies the minimum of the acceptable range. See Sections 099653 and 099726 for exterior paints and coatings.
- B. Metal Surfaces:
 - 1. Paint System No. PTM1 (Latex Finish):
 - a. Surface: Interior ferrous metals - uncoated.

- b. Sheen: Eggshell.
 - c. Prime Coat: Ethyl Silicate Zinc Primer at 3.0 mils.
 - d. Under Coat: Interior Latex Enamel at 1.5 mils.
 - e. Top Coat: Interior Latex Enamel at 1.5 mils.
 - f. System DFT: 6.0 mils.
- C. Gypsum Board Surfaces:
 - 1. Paint System No. PT1 (Latex Finish):
 - a. Surface: Interior gypsum board walls and ceilings.
 - b. Sheen: Eggshell.
 - c. Prime Coat: Latex Primer at 1.0 mils.
 - d. Under Coat: Interior Latex at 1.4 mils.
 - e. Top Coat: Interior Latex at 1.4 mils.
 - f. System DFT: 3.9 mils.
 - 2. Paint System No. PTE1 (Water Based Epoxy):
 - a. Surface: Interior gypsum board walls and ceilings
 - b. Sheen: Satin
 - c. Prime Coat: Latex Primer at 1.0 mils.
 - d. Under Coat: Water Based Epoxy at 2.5 mils.
 - e. Top Coat: Water Based Epoxy 2.5 mils.
 - f. System DFT: 6 mils.

2.4 PRIMERS

- A. Interior Alkyd Enamel Undercoater:
 - 1. Benjamin Moore: Alkyd Underbody No. 217.
 - 2. ICI Paints: Prep & Prime Wall and Woodwork Penetrating Primer Sealer No.1120.
 - 3. PPG: Speedhide Quick Drying Enamel Undercoater, 6-6.
 - 4. Pratt & Lambert: Vitralite Enamel Undercoater.
 - 5. Sherwin-Williams: Wall and Wood Primer B49WZ2.
- B. Alkyd-Phenolic Primer:
 - 1. ICI/Devoe Paints: Devguard Multi-Purpose Metal Primer No. 4160.
 - 2. PPG: Multiprime Universal Primer No. 97-682.
 - 3. Sherwin-Williams: Ken Kromik Universal B50NZ Series.
- C. Etching Metal Primer:
 - 1. PPG: Polyclutch Wash Primer No. 97-687.
 - 2. Sherwin-Williams: DTM Wash Primer B71Y1.
 - 3. Tnemec: Tneme-Grip No. 32-1210.
- D. Galvanized Primer:
 - 1. ICI/Devoe Paints: Devguard Multi-Purpose Metal Primer No. 4160.
 - 2. PPG: Galvanized Steel Primer No. 6-209.
 - 3. Sherwin-Williams: Galvite HS B50WZ30.
- E. Latex Primer:
 - 1. Benjamin Moore: Latex Quick Dry Prime Seal No.023.
 - 2. ICI Paints: Ultra-Hide No. 1030 for Gypsum Wallboard or 3210 Gripper for Masonry.
 - 3. PPG: Speedhide Latex Wall Sealer No. 6-2.

4. Pratt & Lambert: Pro-Hide Plus PVA Latex Wall Primer.
 5. Sherwin-Williams: PrepRite Primer B28W200.
- F. Low VOC Latex Primer:
1. Benjamin Moore and Company:.
 2. ICI Paints: Prep & Prime Odor-less Interior Water Based Primer-Sealer No. LM9116.
 3. PPG Industries, Inc.:.
 4. Sherwin-Williams: ProGreen 200 Primer B28W600.

2.5 WATER REDUCIBLE PAINTS

- A. Industrial Acrylic:
1. ICI Paints:
 - a. Semi-Gloss: Devflex Acrylic Semi-Gloss Enamel No. 4216HP.
 - b. Gloss: Devflex Acrylic Gloss Enamel No. 4208.
 2. PPG: Water Base Inhibitive Primer No. 6-712.
 3. Sherwin-Williams: DTM Acrylic B66W1.
- B. Low-VOC Interior Latex/Acrylic Based Paint:
1. Scuffmaster Paint:
 - a. Eggshell: ScrubTough.
- C. Water Based Epoxy:
1. Benjamin Moore: Moorcraft Acrylic Epoxy Enamel.
 2. ICI/Devoe Paints:
 - a. Semi-Gloss: Tru-Glaze Waterborne Semi-Gloss Epoxy No. 4406.
- D. Gloss: Tru-Glaze Waterborne Gloss Epoxy No. 4408.
1. PPG: Pitt-Glaze Water Based Acrylic Epoxy Enamel.
 2. Sherwin-Williams: Water Based Catalyzed Epoxy B70-200.
- E. Dry Fall Coating:
1. Sherwin-Williams: Waterborne Acrylic Dry Fall Coatings; B42W00002 Eggshell, B42W00001 Flat, or B42T00001 Flat Tint Base.

2.6 ACCESSORY MATERIALS

- A. Muriatic Acid, Mildewcide, TSP (Tri-Sodium Phosphate), Acidic-Detergent, Zinc Sulfate, Sodium Metasilicate, And Solvent: Commercially available, non-damaging to surface being cleaned; as specified in PDCA Specification Manual; acceptable to paint manufacturer.
- B. Metal Conditioner: Proprietary phosphoric acid based, etching type solution; acceptable to paint manufacturer.
- C. Rust Inhibitor: Water containing 0.32 percent of sodium nitrite and 1.28 percent by weight of secondary ammonium phosphate (dibasic); or water containing 0.2 percent by weight of chromic acid or sodium chromate or sodium dichromate or potassium dichromate.
- D. Spackling Compound, Putty, Plastic Wood Filler, Liquid De-Glosser, Latex Patching Plaster, Latex Base Filler, Thinners, and Other Materials Not Specifically Indicated But Required To Achieve Finishes Specified: Pure, of highest commercial quality, compatible with paints and acceptable to paint manufacturer.
- E. Do not use products of different manufacturers in combination.

2.7 MIXING

- A. Use factory prepared colors matching approved samples. Site Tinting: Not allowed.

1. Thoroughly mix and stir paints before use to ensure homogeneous dispersion of ingredients. Prior to application, blend multiple containers of same material and color by pouring from one container to another several times to ensure uniform consistency, color, and smoothness.
2. Mix only in clean mixing pails of material recommended by manufacturer to avoid contamination.
3. Remove film which may form on surface of material in containers and strain material before using. Stir frequently during use to maintain pigments in suspension. Do not stir film into material.
4. Apply paints of consistency recommended by manufacturer. Thin only within recommended limits using thinners approved by paint manufacturer.

2.8 COLORS AND FINISHES

- A. Colors: Refer to color schedule at end of this Section.
- B. Sheen:
 1. Ceilings: Semi-gloss in patient areas; Flat elsewhere unless noted otherwise.
 2. Walls: Semi-gloss in patient areas; Satin/Eggshell elsewhere unless noted otherwise.
 3. Metal Doors and Frames: Semi-gloss, unless noted otherwise.
 4. Metals with accent colors – Semi-gloss, unless noted otherwise.
 5. Painted Wood: Semi-gloss, unless noted otherwise.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
- B. Measure moisture content of substrates using recently calibrated electronic moisture meter. Do not apply paints if moisture content of surfaces exceeds lesser of percentages listed below or those required by paint manufacturer. If excess moisture content exists and cannot be reduced, obtain written approval of paint manufacturer before application of paints.
 1. Gypsum Board and Gypsum Plaster: 17 percent.
 2. Architectural Woodwork, Trim, Cabinets, and Casework: 10 percent; measure with resistance-type meter in accordance with ASTM D4442.
 3. Common Board and Dimension Lumber: 12 percent; measure with resistance-type meter in accordance with ASTM D4442.
 4. Canvas and Cotton Insulation Coverings: 12 percent max.
- C. Prior to applying alkali and acid sensitive paints, test surface pH with universal pH paper placed against wetted surface. Substrate pH shall not exceed pH of clean wash water.
- D. Beginning of execution constitutes acceptance of existing conditions.

3.2 PREPARATION - GENERAL

- A. Protect completed construction from damage. Furnish drop cloths, shields, and protective methods to prevent spray, splatter, or droppings from disfiguring other surfaces.
- B. Remove surface hardware, mechanical diffusers, escutcheons, registers, electrical plates, light fixture trim, fittings, fastenings and similar items prior to preparing surfaces for finishing. Provide surface-applied protective masking for non-removable items. Carefully store removed items for reinstallation.
 1. Remove mildew by scrubbing with mildewcide. Rinse thoroughly with clean water.
 2. Before beginning application of paints, ensure surfaces are clean, dry, and free of dirt, dust, rust or rust scale, oil, grease, mold, mildew, algae, efflorescence, release agents, or any other foreign material which could adversely affect paint adhesion or finished appearance.

3.3 SURFACE PREPARATION FOR NEW WORK

- A. General: Correct minor defects.
 - 1. Remove temporary labels, wrappings, and protective coverings from surfaces to be coated.
 - 2. Seal stains, marks, and other imperfections which may bleed through surface finishes.
- B. Aluminum: Clean in accordance with SSPC SP1 - Solvent Cleaning.
 - 1. Apply etching type primer.
- C. Cloth Insulation Coverings: Remove dirt, grease, oil, and other foreign substances.
 - 1. Seal with thin coating of drywall compound thinned with latex PVA primer to working consistency.
- D. Concrete: Prior to application of paints, allow surfaces to cure minimum 60 days.
 - 1. Remove dirt, scale, powder, laitance, and bond breakers by light sandblasting to minimum mil profile.
 - 1. Remove oil and grease with solution of TSP; rinse well.
 - 2. Remove stains caused by weathering or corroding metals with solution of sodium metasilicate applied after thoroughly wetting surface with potable water; allow to dry.
 - a. Fill cracks and voids with compatible filler.
- F. Gypsum Board: Comply with Gypsum Association Publication GA 232. Refer to Section 09 29 00 for level of gypsum board finish and for general surface preparation.
 - 1. Fill remaining cracks, depressions, holes and other irregularities with spackling compound.
 - 2. Sand rough or high spots left by joint cement or spackling compound without damaging paper face.
 - 3. Remove dust by wiping with damp cloths or vacuuming.
- G. Masonry: Prior to application of paints, allow surfaces to cure minimum 28 days.
 - 1. Remove dirt, scale, loose mortar, efflorescence, and powder by wire brushing or by other approved methods.
 - 2. Remove oil and grease with solution of TSP, rinse, and allow to dry.
 - 3. Remove stains caused by weathering or corroding metals with solution of sodium metasilicate applied after thoroughly wetting surface with potable water; allow to dry.
 - 4. Wash and neutralize surfaces as recommended by paint manufacturer, rinse, and allow to dry.
- H. Frames: At interface between wall and frame surfaces, apply Sealant Designation AL , tool and prepare to receive paint in accordance with sealant and paint manufacturers requirements. See Section 07 92 00.
- I. Sills: At interface between sills and adjacent wall and window frame surfaces, apply Sealant Designation AL , tool and prepare to receive paint in accordance with sealant and paint manufacturers requirements. See Section 07920.
- J. Steel - Uncoated: Remove weld spatter by chipping or grinding.
 - 1. Clean interior and weather protected steel in accordance with SSPC SP2 - Hand Tool Cleaning and SP3 - Power Tool Cleaning. Clean areas of excessive corrosion or scale in accordance with SSPC SP7 - Brush-Off Blast Cleaning.
 - 2. Clean exterior steel permanently exposed to elements in accordance with SSPC SP6 - Commercial Blast Cleaning .
 - 3. Apply metal conditioner to bare surfaces in accordance with manufacturer's recommendations, paying particular attention to abrasions, welds, bolts, and nuts. Allow to set as recommended by solution manufacturer. Rinse with clean water with rust inhibitor mixed with water or applied immediately following rinse. Allow to dry.

4. Prime coat immediately.
- K. Steel – Factory or Shop Prime Coated: Remove loose primer and rust to feather-edge at adjacent sound primer by cleaning in accordance with SSPC SP2 - Hand Tool Cleaning and SP3 - Power Tool Cleaning.
1. Apply metal conditioner to abrasions, welds, bolts, and nuts in accordance with manufacturer's recommendations. Allow to set as recommended by manufacturer. Rinse with clean water with rust inhibitor mixed with water or applied immediately following rinse. Allow to dry.
 2. Prime entire surface immediately.
- L. Steel - Galvanized: Remove white rust by cleaning in accordance with SSPC SP2 - Hand Tool Cleaning and SP3 - Power Tool Cleaning. Exercise care not to remove galvanizing.
1. Pretreat surfaces to receive solvent reducible coatings immediately.
- M. Wood - Opaque Finish: Remove excess residue from knots, pitch streaks, cracks, open joints, and sappy spots. Remove or seal over grade stamp markings.
1. Sand wood surfaces and edges smooth. Remove dust after each sanding.
 2. Apply compatible stain sealer to knots, pitch and resinous sapwood before applying prime coat. Do not apply shellac to exterior surfaces, or under latex or urethane finishes.
 3. After primer is dry and before second coat, countersink nails and fill nail holes, cracks, open joints and other defects with putty or plastic wood filler.

3.4 SURFACE PREPARATION OF PREVIOUSLY COATED SURFACES

- A. General: Remove cracked and deteriorated joint sealants and caulk.
1. Remove chalk deposits and loose, blistered, peeling, scaling, or crazed finish to bare base material or sound substrate by scraping and sanding.
 2. Wash surfaces with solution of TSP to remove wax, oil, grease, and other foreign material; rinse, and allow to dry. Exercise caution that TSP solution does not soften existing coating.
 3. Abrade glossy surfaces by sanding or wiping with liquid deglosser.
 4. Remove mildew as specified above.
 5. Test compatibility of existing coatings by applying new paint to small, inconspicuous area. If new paints lift or blister existing coatings, request recommendation from Architect.
 6. Apply specified primer to surfaces scheduled to receive paints.
- B. Concrete, Masonry, and Portland Cement Plaster: Fill cracks and voids with latex base filler.
1. Apply masonry conditioner to masonry surfaces in accordance with manufacturer's instructions.
 2. Apply primer over bare surfaces and filler material.
- C. Gypsum Wallboard and Gypsum Plaster: Fill cracks and voids with spackling compound.
1. Apply primer over bare surfaces and newly applied texture coatings.
- D. Metal: Remove rust from surfaces to bare metal in accordance with SSPC SP6 - Commercial Blast Cleaning.
1. Exercise care not to remove galvanizing.
 2. Complete preparation as specified for new work.
- E. Wood: Fill cracks, crevices and nail holes with putty or wood filler.
1. Apply primer over bare surfaces and filler material.

3.5 APPLICATION

- A. General Requirements: Coat all surfaces specified, scheduled, illustrated, and otherwise exposed unless specifically noted otherwise.
 - 1. Apply paints of type, color, and sheen as scheduled.
 - 2. Apply products in accordance with Section 01 73 00. Use application materials, equipment, and techniques as recommended by paint manufacturer and best suited for substrate and type of material being applied.
 - 3. Do not apply finishes to surfaces that are improperly prepared.
 - 4. Number of coats specified are minimum number acceptable.
 - 5. Apply paint systems to total dry film thickness scheduled. Apply material at not less than manufacturer's recommended spreading rate. Do not exceed maximum single coat thickness recommended by paint manufacturer. Do not double-back with spray equipment building up film thickness of two coats in one pass.
 - 6. Ensure that edges, corners, crevices, welds, and exposed fasteners receive dry film thickness equivalent of flat surfaces.
 - 7. Finish edges of paints adjoining other materials or colors sharp and clean, without overlapping.
- B. Prime Coats: Apply initial coat to surfaces as soon as practical after preparation and before subsequent surface deterioration.
 - 1. Prime factory or shop primed metal surfaces prior to application of finish coatings.
- C. Intermediate and Top Coats: Allow previously applied coat to dry before next coat is applied.
 - 1. Sand and dust lightly between coats as recommended by paint manufacturer.
 - 2. Apply each coat to achieve uniform finish, color, appearance, and coverage free of brush and roller marks, runs, misses, visible laps or shadows, hazing, bubbles, pin holes, or other defects.
 - 3. If stains, undercoats, or other conditions show through final topcoat, correct defects and apply additional topcoats until paint film is of uniform finish, color, and appearance.
- D. Finish Matching: Finish closets same as adjoining rooms, unless otherwise specified.
 - 1. Finish tops, bottoms, and edges of doors same as door faces. Apply sanding sealer to cut-outs. When faces are different colors, finish edges of doors to match space from which they are visible when door is in partly open position.
 - 2. Finish other surfaces not specifically mentioned to match adjoining surfaces.
- E. Mechanical and Electrical Items: Refer to Divisions 22/23 and Division 26 - Electrical for schedule of color coding and identification banding of equipment, ductwork, piping, and conduit. Color code equipment, piping, conduit and exposed ductwork in accordance with requirements indicated.
- F. Prior to finishing mechanical and electrical items, remove louvers, grilles, covers, and access panels and finish separately. Replace when dry.
 - 1. Paint interior surfaces of ducts, and heating cabinets that are visible or reflective behind grilles and registers with one coat of flat black paint.
 - 2. Finish dampers visible behind grilles and registers to match surface finish.
 - 3. Paint both sides and edges of plywood equipment backboards before installing equipment.
 - 4. Do not apply paints over name plates, tags, or other equipment identification.
- G. Reinstall trim, fittings, and other items removed for finishing.

3.6 FIELD QUALITY CONTROL

- A. General: Comply with requirements of Section 01 40 00.

1. Periodically test film thickness of each coat with wet film gage to ensure paints are being applied to proper thickness.
2. Request review of each applied coat by Architect before application of successive coats. Only reviewed coats will be considered in determining number of coats applied.
3. Immediately prior to Substantial Completion, perform detailed inspection of painted surfaces and repair or refinish abraded, stained, or otherwise disfigured surfaces.

B. Manufacturer Field Observations:

1. Manufacturer: Periodically visit site to review surface preparation and coating application procedures to ensure proper application.
2. When commencing surface preparation and application of coatings at each type of surface, obtain manufacturer's review of process. Manufacturer shall submit certification that work was observed and shall note whether proper procedures were followed in accordance with manufacturer's printed instructions.

3.7 CLEANING AND PROTECTION

A. Cleaning: Comply with Section 01 74 00. Promptly remove spilled, splashed, or spattered paints. Clean spots, oil, and other soiling from finished surfaces using cleaning agents and methods which will not damage materials.

1. If completed construction is damaged beyond normal cleaning or repair by painting operations, replace damaged items at no additional cost to Owner.
2. Maintain premises and storage areas free of unnecessary accumulation of tools, equipment, surplus materials, and debris.
3. Collect waste, cloths, and material which may constitute fire hazards and place in closed metal containers; remove from site daily along with empty containers.

B. Protection: Protect finished work in accordance with Section 01 73 00.

1. Protect work of other trades against damage from paint activities. Correct damage by cleaning, repairing, replacing, and repaint as acceptable to Architect.
2. Provide "Wet Paint" signs and other methods to protect newly coated surfaces. Remove when directed or when no longer needed.

END OF SECTION

SECTION 09 96 00
HIGH PERFORMANCE COATINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Textured Urethane Flooring in accordance with provisions of Contract Documents.
- B. Completely coordinate with work of other trades.

1.2 REFERENCES

- A. ASTM International (ASTM):
 - 1. ASTM C307 Standard Test Method for Tensile Strength of Chemical-Resistant Mortar, Grouts, and Monolithic Surfacing
 - 2. ASTM C531 Standard Test Method for Linear Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
 - 3. ASTM C579 Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
 - 4. ASTM C580
 - 5. ASTM D2240 Standard Test Method for Rubber Property - Durometer Hardness
 - 6. ASTM D2794 Standard Test Method for Flexural Strength and Modulus of Elasticity of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
 - 7. ASTM D4060 Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser
 - 8. ASTM E648 Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

1.3 SUBMITTALS

- A. Product Data:
 - 1. For each type of material and accessory.
- B. Samples:
 - 1. Three samples, 12 inches square of each material specified in Drawing Finish Schedule.
 - 2. Samples for review of color and texture.
- C. Contract Closeout Information:
 - 1. Maintenance data:
 - a. See Section 01 78 23.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications:
 - 1. Certified by manufacturer of flooring, with a minimum of 7 years of experience and minimum of 10 completed projects of similar size and scope.
- B. Since Source Responsibility: Materials recommended and manufactured by single supplier to ensure compatibility and proper chemical and mechanical bond.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Textured Urethane Flooring:

1. Base:
 - a. Stonhard
- B. Other manufacturers desiring approval comply with Section 01 61 00.

2.2 MATERIALS

- A. Textured Urethane Flooring System:
 1. Self-priming, textured, four-component, polyurethane mortar and broadcast system consisting of a urethane-urea binder, pigments, powders, and quartz aggregates.
 2. Base Product: Stonshield UTS by Stonhard
 3. Color:
 - a. See Drawing Finish Schedule.
- B. Mortar: Four-component urethane-urea slurry.
- C. Aggregate: Quartz, of color selected.
- D. Seal Coat: Two-component, UV-resistant, aliphatic polyaspartic urethane sealer.
- E. System Performance Characteristics:
 1. Tensile Strength: Minimum 1,000 psi.
 2. Hardness: Shore D; 80-84
 3. Abrasion Resistance: Maximum 0.10 g weight loss.
 4. Impact Resistance: 160 inches/LBS; no cracking, chipping, or delamination.
 5. Flammability: Class I.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine surfaces for defects, irregularities and conditions under which flooring is to be installed.
- B. Verify substrates are free of materials that may affect adhesion.
- C. Identify cracks and other surface defects which need repair prior to application of floor system.
- D. Inspect substrate for markers, paint and similar materials used for layout by others and take remedial action to remove and prevent bleed-through.
- E. Verify floors are level or meet indicated slope.
- F. Do not proceed with installation until unsatisfactory conditions have been corrected.
- G. Installation indicates acceptance of substrates and responsibility for performance.

3.2 PREPARATION

- A. Prepare substrate in accordance with manufacturer's instructions.
- B. If required by flooring manufacturer, comply with requirements of Section 07 16 04 Concrete Floor Moisture Testing, and Section 07 16 05 Water Vapor Emission Control System.
- C. Fill construction joints and other non-moving joints with elastomeric sealant approved by manufacturer.
- D. Protect adjacent surfaces or both, including fixtures, cabinetwork, equipment, and similar items.

3.3 INSTALLATION

- A. Apply flooring in accordance with manufacturer's printed instruction, employing qualified and experienced lead mechanic and using equipment specifically designed for this purpose.
- B. Install integral cove base with 1 inch radius cove.

1. Height: 4 inches.
 2. Caulk top edge of base to wall.
- C. Finish work to match approved samples; uniform in thickness, sheen, color, pattern, and texture; and be free from defects detrimental to performance.

3.4 PROTECTION

- A. Provide non-staining protection for finished flooring until building is ready for occupancy.

3.5 CLEANING

- A. Clean floors as recommended by manufacturer just prior to occupancy.

END OF SECTION

SECTION 10 14 00

SIGNAGE

PART 1 - GENERAL

1.1 SUMMARY

- A. Related Documents: Provisions established in Conditions of the Contract, Division 01 - General Requirements, and the Drawings are collectively applicable to this Section.
- B. Section Includes
 - 1. Identifying devices where shown on the Drawings complete and as specified including the following:
 - a. Parking signs indicating accessible spaces.
 - b. Directional and traffic signs.
 - c. Pin mounted building identification signs.
 - d. Interior code required signs.
 - 2. Coordination for installation of signage provided by others.

1.2 SUBMITTALS

- A. Product Data: Include manufacturer's construction details relative to materials, dimensions of individual components, profiles, and finishes for each type of sign required.
- B. Shop Drawings: Provide shop drawings for fabrication and erection of signs. Include plans, elevations, and large-scale sections of typical members and other components. Show anchors, accessories, layout, and installation details.
- C. Samples for Verification:
 - 1. Physical: Submit samples of one competed sign for review and approval. Approved sample may be incorporated into Project.
 - 2. Color: Submit manufacturer's standard color selection chart. Do not proceed until colors have been selected.

1.3 QUALITY ASSURANCE

- A. Single-Source Responsibility: For each separate type of sign required, obtain signs from one source from a single manufacturer.
- B. Manufacturer shall have a minimum of five years experience in the manufacturing of signs specified.
- C. Codes and Standards:
 - 1. Panel signs shall have 1/32-inch raised copy and grade 2 Braille, and shall comply with all existing federal, state, and local accessibility standards.
 - 2. Code and Standards: Comply with American with Disabilities Act of 1990, Title 3 Provisions, Public Accommodations and Commercial Facilities.
 - 3. Comply with the State of Texas Accessibility Act as administered by the Texas Department of Licensing and Regulation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers: Subject to compliance with requirements herein, provide products from one of the following:
 - 1. Best Manufacturing Company, Montrose, Colorado.

2. Mohawk Sign Systems, Schenectady, New York.
3. Nelson-Harkins, Chicago, Illinois.
4. ASI Signs, Dallas, Texas.

B. Substitutions: Under provisions of Section 01 25 00.

2.2 DIRECTIONAL SIGNS

A. Screen Printed Signs:

1. Extruded aluminum panels with anodic finish and white screen printed copy.
2. Size and Configuration: As indicated on Drawings.
3. Copy: As indicated on Drawings.
4. Acceptable Product: Best Post and Plank as detailed on Drawings.

2.3 ROOM SIGNAGE SYSTEMS

A. Acceptable Manufacturers:

1. ASI Sign Systems, 3890 W. Northwest Highway, Suite 102, Dallas, TX 75220; telephone; (214) 352 9741 facsimile; (800) ASI-SPEC 274-7446
2. Substitutions: Submit in accordance with Section 01600.
 - a. (214) 352 9140

B. Acceptable Product: ASI Unframed SP Series Signs with requirements indicated for materials, thickness, finish colors, designs, shapes, sizes and details..

C. Sign Face: Clear acrylic, 0.080 inch thick, matte first surface.

1. Adhesive: Pressure sensitive adhesive film, second surface.

D. Tactile Graphics and Text:

1. Fabrication: Provide tactile copy and grade 2 Braille raised 1/32 inch minimum from plaque first surface by manufacturer's stratification process as follows:
 - a. ASI Intouch™, photo-mechanical method.
2. Provide lettering and graphics precisely formed, uniformly opaque to comply with relevant ADA regulations and requirements indicated for size, style, spacing, content, position, and colors.

E. Non-Tactile Graphics and Text:

1. Fabrication options:
 - a. Series SPE/SPJ: Non-tactile graphic plaque, no back plate.
2. Text or graphic technique:
 - a. Screen process using subsurface method.
3. Provide lettering and graphics precisely formed, uniformly opaque, and consistent in size, style, spacing, content, position, and colors.

F. Overall panel size: Refer to Drawings.

G. Panel colors: As selected by Architect.

H. Text or graphic colors: As selected by Architect.

I. Letter styles, colors, letter sizes and layout position: As selected by Architect.

J. Installation Method: System SA, silicone adhesive

PART 3 - EXECUTION

3.1 DELIVERY AND STORAGE

- A. Deliver and store identifying devices in protective wrappings until ready for installation. Install letters in protective wrappings and remove wrappings just prior to substantial completion.

3.2 INSTALLATION

- A. Install signs plumb, level and square and in proper planes with other work, at heights required by accessibility codes and standards.
- B. Anchor each plastic laminate sign with adhesive.
- C. Install signs with sufficient amount of foam tape for proper installation.
- D. Attach as recommended by sign manufacturer.
- E. Anchor each sign with adhesive.
- F. Coordinate arrival and installation of graphic signs with hardware installation. Graphic signs function as and are coordinated with the hardware as shown on the Drawings.
- G. Room name signs shall be placed on the public side of the door except where noted otherwise.
- H. Single Door Sign: Provide one sign as specified above, mounted to wall adjacent to door on knob side.
- I. Pair of Doors: Provide one sign as specified above, mounted to adjacent wall closest to active leaf of door. Do not install sign where it will be obstructed by door when door is in the 'open' position.
- J. Attachment: Mounting to surfaces shall be done by pressure sensitive frame double-faced tape. Signs shall be delivered to the project site with the tape in place and trimmed on each sign, but with the protective paper layer not removed. Paper layer shall be removed just prior to installation of signs.

3.3 EXTERIOR INSTALLATION - PARKING AND DIRECTIONAL SIGNS

- A. Mount posts in 12 inch round by 2'-6" deep concrete footing.
- B. Handicapped Signs: Mount signs at height to comply with accessibility codes.

3.4 COORDINATION

- A. Coordinate the installation of the identifying devices with the hardware manufacturer for lock set and knob leave outs as detailed and scheduled.

3.5 DAMAGE

- A. Any identifying device which is scratched or defaced will be rejected.

3.6 CLEANING

- A. Remove protective materials and clean all signs. Clean surfaces with plain water or water with soap or household detergent.

END OF SECTION

SECTION 10 26 00

WALL AND DOOR PROTECTION

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

- A. Performance Requirements: Comply with UL 723 Class I characteristics as follows:
 - 1. Flame Spread: 25 maximum.
 - 2. Smoke Developed: 350 - 450.
 - 3. When used as part of fire rated assembly, provide devices capable of maintaining specified or indicated hourly rating when tested in accordance with ASTM E119.

1.2 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit manufacturer's descriptive technical data including test performance data and performance characteristics for each product.
- C. Shop Drawings: Show built-in items, including sizes, types, materials, construction, finishing, anchoring, accessories, and preparation for installation of corner guards.
- D. Samples: Submit actual samples of standard vinyl colors for Architect's selection.
- E. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Certifications specified in Quality Assurance article.
 - 2. Qualification Data: Manufacturer's qualification data.
 - 3. Manufacturer's instructions.

1.3 QUALITY ASSURANCE

- A. Installer Qualifications: An employer of workers trained and approved by manufacturer.
- B. Source Limitations: Obtain impact-resistant wall protection units from single source from single manufacturer.
- C. Product Options: Drawings indicate size, profiles, and dimensional requirements of impact-resistant wall protection units and are based on the specific system indicated. Refer to Division 01 Section "Quality Requirements."
 - 1. Do not modify intended aesthetic effects, as judged solely by Architect, except with Architect's approval. If modifications are proposed, submit comprehensive explanatory data to Architect for review.
- D. Forest Certification: Fabricate wood rails from wood obtained from forests certified by an FSC-accredited certification body to comply with FSC STD-01-001, "FSC Principles and Criteria for Forest Stewardship."
- E. Surface-Burning Characteristics: Provide impact-resistant, plastic wall protection units with surface-burning characteristics as determined by testing identical products per ASTM E 84, NFPA 255, or UL 723 by UL or another qualified testing agency.
- F. Regulatory Requirements: Comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Store impact-resistant wall protection units in original undamaged packages and containers inside well-ventilated area protected from weather, moisture, soiling, extreme temperatures, and humidity.
 - 1. Maintain room temperature within storage area at not less than 70 deg F 21 deg C during the period plastic materials are stored.
 - 2. Keep plastic sheet material out of direct sunlight.
 - 3. Store plastic wall protection components for a minimum of 72 hours, or until plastic material attains a minimum room temperature of 70 deg F 21 deg C.
 - a. Store corner-guard covers in a vertical position.
 - b. Store wall-guard bed-locator handrail covers in a horizontal position.

1.5 FIELD SAMPLES

- A. General: Comply with Section 01 40 00.
- B. Sample Installation: Install one of each type of unit in location as directed.
 - 1. Accepted Field Sample: May not remain part of completed Work.

1.6 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Wall-Guard and Handrail Covers: Full-size plastic covers of maximum length equal to 2 percent of each type, color, and texture of units installed, but no fewer than two, 8-foot-2.4-m- long units.
 - 2. Bed-Locator Covers: Full-size plastic covers equal to 2 percent of each type, color, and texture of units installed, but no fewer than two units.
 - 3. Corner-Guard Covers: Full-size plastic covers of maximum length equal to 2 percent of each type, color, and texture of units installed, but no fewer than two, 4-foot-1.2-m- long units.
- B. Include mounting and accessory components. Replacement materials shall be from same production run as installed units.
- C. Provide necessary tools for installation and removal of extra material.

PART 2 - PRODUCTS

2.1 CORNER GUARDS – (CG1)

- A. Surface Mounted Vinyl With Retainer Type Resilient vinyl cover over continuous extruded aluminum retainer, surface applied.
 - 1. Provide units complete with color matching end caps.
 - 2. Cover Thickness: 0.078 inch minimum.
 - 3. Aluminum Thickness: 0.063 inch minimum.
 - 4. Provide UL classified materials tested in accordance with UL723 achieving Class I flame and smoke rating.
 - 5. Size: 2 by 2 inches by height as indicated on Drawings.
 - 6. Color: Refer to Finish Legend.
 - 7. Corners: Use custom product at other than 90 degrees and 135 degrees angle corners.
 - 8. Acceptable Manufacturers:
 - a. Construction Specialties Inc., Muncy, PA.
 - b. Inpro.

2.2 BUMER GUARD AND WALL PROTECTION (BR-1)(WP1)9(WP-2)

A. Floor-Mounted Crash Rail:

1. Design: Floor mounted crash rail with stainless steel finish, and symmetrical bracket for floor mounting applications. System to be impact resistant that can withstand 550 lbf while producing no visual blemish upon rails surface and no deformations in the stainless steel retainers:
 - a. Stainless Steel 304, No. 4 Satin Finish.
 - b. Height 4 3/8" In height
 - c. Tube 1 1/2" Wide Tube
 - d. Provide UL classified materials, ASTM E-84 Class A
2. Acceptable Manufacturers:
 - a. Basis of Design: Inpro Model 61SS.
 - 1) Construction Specialties Inc., Muncy, PA.
 - 2) Inpro.

B. Bumper Rail Type and End Guard Type :

1. Design: Extended mounted crash rail with resilient free cover, aluminum retainer and continuous resilient bumper cushion and mounting brackets; color coordinate end caps and returns.
 - a. Vinyl Thickness: 0.110 inch minimum.
 - b. Aluminum Thickness: 0.093 inch minimum.
 - c. Height: 5 1/2"
 - d. Provide UL classified materials tested in accordance with UL723 achieving Class I flame and smoke rating.
 - e. Color: Refer to Finish Legend.
2. Acceptable Manufacturers:
 - a. Basis of Design: Inpro Model 800
 - 1) Construction Specialties Inc., Muncy, PA.
 - 2) Inpro.

C. Wall Covering Type :

1. Design: Free surface applied with adhesive.
 - a. Thickness: 0.060 inch minimum.
 - b. Sheet Size: 4 feet by 8 feet.
 - c. Texture: Pebblette.
 - d. Color: Refer to Finish Legend.
2. Provide UL classified materials tested in accordance with UL723 achieving Class II flame and smoke rating for 0.060 inch thick material.
3. Accessories:
 - a. Formed inside and outside corner trim, expansion joint strips, joint dividers, J-molding caps, and edge trim.
 - b. Adhesive: As recommended by manufacturer to maintain fire rating.
4. Acceptable manufacturers:
 - a. Construction Specialties Inc., Muncy, PA.
 - b. Inpro.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 40 00.
- B. Verify that substrate finishes are complete and attachment devices in hollow walls are accurately located.

3.2 INSTALLATION

- A. Install in accordance with Section 01 73 00 and approved Shop Drawings.
 - 1. Locate devices where indicated.
 - 2. Use attachment devices or adhesive as specified and as recommended by manufacturer.
 - a. Corner Guards: Stop at heights indicated. If no heights are indicated, extend from top of base to 48 inch above finished floor.
 - 3. Built-in Corner Guards: Coordinate with wall construction for built-in corner guards.
- B. Fasten corner guards to drywall framing on 8 inch centers.
- C. Anchor stainless steel corner guards with epoxy adhesive and mechanical anchors.
- D. Horizontal Devices: Install at elevations indicated.

3.3 ADJUSTING AND CLEANING

- A. On flush mounted devices, verify that device is flush with adjacent wall surface. Adjust for proper fit and alignment.
- B. Remove protective coverings from devices at final cleaning stage.

END OF SECTION

SECTION 10 28 13

HEALTHCARE ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

- A. Related Sections: Section 13 49 00 - Radiation Protection.

1.2 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit manufacturer's catalog cut sheets, and data sheets.
- C. Shop Drawings: Submit setting drawings, templates, instructions, and directions for installing anchorage devices and cut-out requirements in other work.
- D. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Certifications specified in Quality Assurance article.
 - 2. Manufacturer's instructions.
- E. Closeout Submittals: Submit maintenance data in accordance with Section 01 77 00.

1.3 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with Authorities Having Jurisdiction to accommodate barrier free design.
- B. Certification: Provide verification of grab bar strength and installation.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Comply with requirements of Section 01 60 00.
- B. Pack accessories individually with protective wrappings.

1.5 KEYING

- A. Dispensing units keyed alike, furnish six keys. Key coin boxes separately from dispensing unit; furnish six keys.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers:
 - 1. A&J Washroom Accessories, New Windsor, NY.
 - 2. American Specialties, Inc., Yonkers, NY.
 - 3. Bobrick Washroom Equipment, Inc., Clifton Park, NY.
 - 4. Bradley Corporation, Milwaukee, WI.
 - 5. Accepted Substitute in accordance with Section 01 25 00.
- B. Bobrick accessories are specified, but items of equivalent design, sightlines, construction, size, function and capacity by manufacturers listed in paragraph above are also acceptable.

2.2 MATERIALS

- A. Stainless Steel:
 - 1. Sheet: ASTM A240, UNS S30400, 22 gage, except where specified otherwise.
 - 2. Tubing: ASTM A269, UNS S30400.
- B. Sheet Steel:

1. Cold Rolled: Commercial quality ASTM A1008, 20 gage minimum. Surface preparation and metal pretreatment as required for applied finish.
 2. Galvanized Steel: ASTM A563, G60 zinc coating.
- C. Aluminum Casting: ASTM B85.
- D. Fasteners: Screws, bolts, and other devices of same material and finish as accessory item, or of galvanized steel, ASTM A123/A123M where concealed; theft-proof design at exposed conditions.
- E. Expansion Shields: Type as recommended by accessory manufacturer for component and substrate.
- F. Sanitary Sealant: One part silicone, FS-TT-S-001543, FDA Regulation 21 CFR177.2600, and FDA Food Additive Regulation 121.2514.
1. Color: White.
 2. Acceptable Products:
 - a. No. 786 Silicone Rubber Sealant, Dow Corning.
 - b. SCS1700, General Electric.

2.3 FABRICATION

- A. Weld and grind smooth joints of fabricated components.
1. Form exposed surfaces from single sheet of stock, free of joints.
 2. Form surfaces flat without distortion. Maintain flat surfaces without scratches or dents.
 3. Back paint components where contact is made with building finishes to prevent electrolysis.
 4. Shop assemble components and package complete with fasteners, anchors, and fittings.
 5. Provide anchor plates, adapters, and anchor components necessary for installation.

2.4 FACTORY FINISHING

- A. Galvanizing After Fabrication: ASTM A123/A123M.
- B. Shop Primed Ferrous Metals: Pretreat and clean, spray apply one coat primer and bake.
- C. Stainless Steel: Stainless Steel Finish: In accordance with Section 05 05 10.
- D. Chrome/Nickel Plating: ASTM B456, Type SC2, satin finish.
- E. Aluminum: Clear Anodized Finish: In accordance with Section 05 05 10.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 73 00.
1. Verify that site conditions are ready to receive work and dimensions are as indicated on shop drawings and instructed by manufacturer.
 2. Check openings for plumbness of blocking and frames.
 3. Beginning of installation means acceptance of existing conditions.

3.2 PREPARATION

- A. Deliver inserts and rough-in frames to site at appropriate time for installation.
1. Provide templates and rough-in measurements as required.
 2. Verify exact location of accessories for installation.
 3. Protect adjacent or adjoining finished surfaces and work from damage during installation.
 4. Coordinate work with placement of wall reinforcement and reinforcement of toilet partitions to receive anchor attachments.
 5. Supply rough-in data in sufficient time to be built into other work.

6. Do not install accessories until room finishes are completed.

3.3 INSTALLATION

- A. Accessories: Install in accordance Section 01 73 00 and approved Shop Drawings.
 1. Install plumb, level, and securely anchored to substrate.
 2. Locate accessories in order that they do not interfere with door swings or use of fixtures. Install accessories after wall finishes have been completed.
 3. Anchor accessories with bolts, plates, and approved type fasteners.
 4. Install surface mounted accessories to backup material with toggle bolts, plumb and align.
- B. Grab Bars:
 1. At Gypsum Board Partitions: Extend grab bars through wall surface; anchor to concealed 16 gage steel anchor plates secured to steel studs supporting partition.
 2. At CMU Partitions: Fasten with expansion anchors to grouted CMU cells.
- C. Performance Requirements, Grab Bars: Install to support 500 pound hanging load placed at any point along bar length.
 1. Comply with ADA requirements.
 2. Mount horizontally unless otherwise shown.
- D. Sanitary Sealant: Seal fastener holes at showers with sanitary silicone sealant prior to mounting accessories and grab bars. Set entire plate and perimeter trim in sealant to ensure watertight installation at penetrations.
- E. Lead Shielded Areas: Install accessories to ensure continuous X-ray shielding. Refer to Section 13 49 00.
- F. Adjusting and Cleaning:
 1. Adjust accessories for proper operation and smooth mechanical function.
 2. Clean and polish exposed surfaces after removal of protective coverings.

3.4 SCHEDULE OF HEALTHCARE ACCESSORIES

- A. Refer to schedule on Drawings.

END OF SECTION

SECTION 10 44 16

FIRE EXTINGUISHERS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes portable, hand-carried fire extinguishers and mounting brackets for fire extinguishers.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include rating and classification, material descriptions, dimensions of individual components and profiles, and finishes for fire extinguisher and mounting brackets.
- B. Warranty: Sample of special warranty.
- C. Product Location: The location of the fire extinguishers shall be in compliance with NFPA 10: Installation of Portable Fire Extinguishers for the hazard that they are intended to protect. This information will be provided by the architect; reflected on the architectural or life safety plans.
- D. Contractor shall maintain a valid State of Texas Extinguisher Certificate of Registration (ECR). Installing employees must maintain a valid Type B portable License.

1.4 QUALITY ASSURANCE

- A. NFPA Compliance: Fabricate and label fire extinguishers to comply with NFPA 10, "Portable Fire Extinguishers."
- B. Fire Extinguishers: Listed and labeled for type, rating, and classification by an independent testing agency acceptable to authorities having jurisdiction.
 - 1. Provide fire extinguishers approved, listed, and labeled by UL.

1.5 COORDINATION

- A. Coordinate type and capacity of fire extinguishers with fire protection cabinets to ensure fit and function.

1.6 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace fire extinguishers that fail in materials or workmanship within specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Failure of hydrostatic test according to NFPA 10.
 - b. Faulty operation of valves or release levers.
 - 2. Warranty Period: Six years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PORTABLE, HAND-CARRIED FIRE EXTINGUISHERS

- A. Fire Extinguishers: Type, size, and capacity for each mounting bracket indicated.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Amerex

2. Valves: Manufacturer's standard.
 3. Handles and Levers: Manufacturer's standard.
 4. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B and bar coding for documenting fire extinguisher location, inspections, maintenance, and recharging.
- B. Multipurpose Dry Chemical Type in Steel Container: UL Rated 4-A: 60-B:C 10lb – FEC 1
 - C. Purple K Dry Chemical Type in Aluminum Container (Areas where cooking producing grease laden vapors): UL rated 10-B:C, 2.5lb – FE 1
 - D. Carbon Dioxide Type (OR's, Cath Labs, EP and Interventional Radiology) : UL rated 10-B:C, 10 lb – FEC 2
 - E. Water Extinguisher – UL rated 2-A:2.5 gallon pressurized steel water based extinguisher – FEC 3
 - F. Water Extinguisher (Hospital Operating Rooms) – UL rated 2-A 2.5 gallon pressurized steel extinguisher containing di-ionized water. Shall be identified with a with a label stating di-ionized water – FEC 4
 - G. Combustible Metal Extinguisher (storage locations or laboratories using combustible metals) – UL rated sodium chloride based dry powder extinguishing agent – FEC 5
 - H. Multipurpose Dry-Chemical Type in Steel Container: UL-rated 4-A: 60-B:C, 10-lb nominal capacity, with monoammonium phosphate-based dry chemical in enameled-steel container.
 - I. Purple-K Dry-Chemical Type in Aluminum Container (Areas where cooking producing grease laden vapors): UL-rated 10-B:C, 2.5-lb nominal capacity, with potassium bicarbonate-based dry chemical in enameled-aluminum container.
 - J. Carbon Dioxide Type (at OR's, Cath Labs, EP, and Interventional Radiology): UL-rated 10-B:C, 10-lb nominal capacity, with carbon dioxide in manufacturer's standard enameled-metal container. Extinguishers must be constructed completely of nonferrous metal and have an "MRI SAFE" extinguisher logo
 - K. Water Extinguisher – UL rated 2- A: 2.5 gallon pressurized steel water based extinguisher.
 - L. Water Extinguisher (Hospital Operating Rooms) – UL rated 2-A 2.5 gallon pressurized steel extinguisher containing di-ionized water. Shall be identified with a label stating di-ionized water.
 - M. Combustible Metal Extinguisher (storage locations or laboratories using combustible metals) – UL rated, sodium chloride based dry powder extinguishing agent

2.2 MOUNTING BRACKETS

- A. Mounting Brackets: Manufacturer's standard galvanized steel, designed to secure fire extinguisher to wall or structure, of sizes required for types and capacities of fire extinguishers indicated, with plated or red baked-enamel finish.
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Larsen's Manufacturing Company.
 - b. Identification: Lettering complying with UT Southwestern Fire Safety for letter style, size, spacing, and location. Locate as indicated by Architect.
 2. Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.
 - a. Orientation: Vertical.

2.3 FIRE EXTINGUISHER CABINETS

- A. Fire Extinguisher Cabinet see Fire Extinguisher Cabinets (104413)
 1. Building Interior: Larsen semi- recessed fire extinguisher with laminated safety class, exposed handle, and aluminum finish

2. Building Exterior: Larsen semi-recessed or exposed fire extinguisher cabinet with laminated safety glass, exposed handle, and aluminum finish
3. Cabinets shall be sized appropriately to hold the type and size extinguisher(s) proposed
4. Triangle fire extinguisher signs shall be mounted about all semi-recessed fire extinguisher boxes. Height to be verified on architectural drawings.
5. Fire extinguisher boxes installed in fire rated walls shall be fire rated as required

EXECUTION

2.4 EXAMINATION

- A. Examine fire extinguishers for proper charging and tagging.
 1. Remove and replace damaged, defective, or undercharged fire extinguishers.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

2.5 INSTALLATION

- A. General: Install fire extinguishers and mounting brackets in locations indicated and in compliance with requirements of UT Southwestern Fire Safety.
 1. Mounting Brackets: As required by NFPA 10.
- B. Mounting Brackets: Fasten mounting brackets to surfaces, square and plumb, at locations indicated.

END OF SECTION

SECTION 10 99 00

MISCELLANEOUS SPECIALTIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Miscellaneous specialty items as listed herein.

1.3 ACTION SUBMITTALS

- A. Product Data: Including all pertinent performance characteristics and criteria.
- B. Shop Drawings: Indicate materials, construction, sizes, quantities, finishes, and installation details.

1.4 INFORMATIONAL SUBMITTALS

- A. Manufacturer's Instructions: For installation, maintenance, and repair.

PART 2 - PRODUCTS

2.1 ANALOG WALL CLOCK

- A. Peter Pepper Products (800) 496-0204
 - 1. Model 300 10" Diameter Clock without Acrylic Cover

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that surfaces and conditions are ready to receive work of this Section.
- B. Notify Architect of any existing conditions which will adversely affect execution.
- C. Beginning of execution will constitute acceptance of existing conditions.

3.2 PREPARATION

- A. Prepare substrate surfaces as recommended by manufacturer.

3.3 INSTALLATION

- A. Install using skilled workers in accordance with manufacturer's published instructions and recommendations.

3.4 ADJUSTING

- A. Adjust and fit items to be flush with adjacent construction.
- B. Fasten or adhere for tight connections and joints.

END OF SECTION

SECTION 11 53 43
LABORATORY SERVICE FITTINGS AND FIXTURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Laboratory Service Fittings and Fixtures, as indicated, in accordance with provisions of Contract Documents.
- B. Section includes:
 - 1. Laboratory service fittings, valves, electrical pedestal boxes, and related components.
 - 2. Laboratory emergency plumbing fixtures.
 - 3. Medical gas outlets.
 - 4. Laboratory sinks.
 - 5. CO2 Point of Use Panel.
- C. Completely coordinate with work of other trades.

1.2 DESCRIPTION

- A. Work includes but is not necessarily limited to furnishing to the project site for installation by Division 22, laboratory fittings, emergency plumbing fixtures, and fixtures described herein and shown on the Laboratory Furnishings Drawings.

1.3 SUBMITTALS

- A. Shop Drawings:
 - 1. Submit complete shop fabrication and installation drawings, including plans, elevations, sections, dimensions, details, and schedules.
 - 2. Show relationship to adjoining materials and construction.
- B. Product Data:
 - 1. Submit complete materials list, including catalogue data of materials, equipment, and products for Work in this Section.
- C. Samples:
 - 1. Submit 2 samples of each type of specified finish and color.
- D. Contract Closeout Information:
 - 1. Operation and Maintenance Data:
 - a. Operating and maintenance manuals describing operating and maintenance procedures.
 - b. Maintenance and replacement schedules.
 - c. Component parts list.
 - d. Nearest local factory representative for components and repairs.
 - e. See Section 01 78 23.

1.4 QUALITY ASSURANCE

- A. Unless specified otherwise, because of special coordination requirements, the scope of work described in this Section shall be provided by the supplier of the Section 12 35 53 scope of work.
- B. Work in this Section requires close coordination with Work in Section 12 35 53, Division 22 Plumbing, Division 23 HVAC and Division 26 Electrical.

1. Coordinate Work to assure an orderly progress in the Project, without removal of previously installed Work, and so as to prevent damage to finishes and products.
- C. Review conditions of installation, procedures, and coordination with related Work.
- D. Carefully inspect the installed Work specified in other Sections and verify that Work is complete and ready for the installation of this Work to properly commence.
- E. Install Work in accordance with original design, reviewed submittals, and manufacturer's recommendations.
- F. Manufacturer and Installer Qualifications:
 1. 5 years of documented experience specializing in manufacture of type of equipment specified, with demonstrated ability to produce the specified equipment of required quality and quantity for complete installation in a project of this type and size within required time limits.
 2. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.
- G. Conform to the recommended practices for laboratory service fittings and fixtures published by the Scientific Equipment and Furniture Association (SEFA) 7: Fixtures, current version.
- H. Emergency plumbing fixtures shall comply with requirements of ANSI Standard Z358.1: American National Standard for Emergency Eyewash and Shower Equipment, current edition.
- I. Where identified, service fittings and sinks shall be accessible in compliance with the requirements of the federal Americans with Disabilities Act (ADA), ADA Accessibility Guidelines (ADAAG), and state accessibility regulations.
- J. Emergency plumbing fixtures shall be accessible in compliance with requirements of federal Americans with Disabilities Act (ADA), ADA Accessibility Guidelines (ADAAG), and state accessibility regulations.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver fittings and fixtures to job site in recommended packaging, with each fitting individually packaged, marked, and scheduled for point of use.
- B. Inventory fittings, at job site, verify that type and quantity are correct, and re-package until installed.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Service fittings and emergency plumbing fixtures shall be designed for laboratory use.
- B. Service fittings and emergency plumbing fixtures shall be factory pre-assembled including the assembly of valves to turrets, mounting shanks to turrets, etc., and individually factory tested.
- C. Laboratory service fittings shall be the product of one service fitting manufacturer to assure ease of replacement and maintenance.
- D. Emergency plumbing fixtures shall be the product of one manufacturer to assure ease of replacement and maintenance.
- E. Service valves, fittings, and accessories shall be of cast brass with a minimum copper content of 85 percent, except for items which are to be brass forging or bar stock or are specifically specified to be of another material.
- F. Provide fittings at locations indicated on the Laboratory Furnishings Drawings. Refer to Service Fitting Schedule.
- G. Assembly components and operating parts such as valve stems, renewable units, packing nuts, outlet nozzles and straight serrated hose ends shall be made from solid brass stock.

- H. Replaceable seats, needle cones, valve disc screws and other accessories shall be Monel or stainless steel alloys especially selected for use intended.
- I. Fittings shall be factory tested and shall be supplied with nipples, lock nuts, shanks, etc.
- J. Serrated tip fittings shall have 3/8 inches IPS thread with the hose end being tapered. Diameter of orifice in serrated tip shall be 1/8 inches, except where otherwise specified.
- K. Turrets shall be brass drop forging of design indicated in detail shown elsewhere in the Section and shall be one or two-way, as required, with 3/8 inches IPS female inlet thread for connections.
 - 1. Units shall be furnished with brass shanks, brass locknuts, and washers.
- L. Fittings located on the same plane shall have their handles project the same distance from the plane of reference to present a uniform related appearance, regardless of valve type construction.
- M. Flanges shall be brass forging of approved design with 3/8 inches IPS female inlet and outlet.
- N. Goosenecks shall provide full thread for attachment of anti-splash outlet fittings, serrated tips, and filter pumps.
- O. Hot water/cold water gooseneck mixers and wall-mounted cold water goosenecks shall swivel.
 - 1. Swivel point shall be at turret or at valve level if wall mounted.
 - 2. Swing joints shall have heavy Teflon type packings.
 - 3. O-rings will not be permitted.
 - 4. Cold water goosenecks at cup sinks shall be rigid.
- P. Fittings shall have plastic colored service index buttons.
- Q. Provide vacuum breakers for hot and cold water fittings including water fittings at fume hoods.
- R. Fittings and fixtures designated to be accessible (ADA) with operable parts shall be operable with one hand and shall not require tight grasping, pinching, or twisting of the wrist.
 - 1. Force required to activate operable parts shall be 5 pounds, maximum.

2.2 LABORATORY SERVICE FITTINGS

- A. Laboratory service fittings specified in this section shall be provided by a single manufacturer.
- B. Manufacturers:
 - 1. Base:
 - a. Water Saver Faucet Co.
 - 2. Optional:
 - a. The Chicago Faucet Company.
 - b. T&S Brass and Bronze Works, Inc.
 - 3. Other manufacturers desiring approval comply with Section 01 61 00.
- C. Body Profile:
 - 1. Service fittings shall have cylindrical body profiles.
 - a. Mushroom profile not allowed.
 - b. Turrets shall be provided without deck flanges.
 - 2. Service fittings shall have mushroom body profile with integral flange at work surface.
 - 3. Service fittings shall have cylindrical body profile with integral flange at work surface.
- D. Handles:
 - 1. Locations designated to be accessible (ADA).
 - a. Water faucets: wrist-blade handles with screw on index identification discs.

- b. Laboratory gases, air, and vacuum valves: ball valves fitted with lever-type handles and screw on index identification discs.
 - 2. Other fittings shall be fitted with four arm handles.
- E. Finish: Satin chrome, with clear, acid-resistant (epoxy) coating; refer to Finishes section below for requirements.
- F. Water Valves:
 - 1. General:
 - a. Water fixtures shall be fully assembled and factory tested at 80 psi water pressure.
 - b. Water valves shall include a renewable unit containing working parts which are subject to wear, including stainless steel or monel seat, monel screw and heavy duty seat disk and Teflon packing, and an integral adjustable volume control.
 - c. Unit shall be convertible from compression to self-closing, and vice versa, without disturbing faucet body proper and shall also be convertible from water construction to needle valve or steam valve construction having outside packing gland without disturbing faucet body.
 - d. Unit shall be sealed in valve body with special composition gasket.
 - 1) Metal-to-metal or ground joint type of sealing is not acceptable.
 - e. Exposed metal shall be finished as specified elsewhere in this Section.
 - f. Water service fixtures shall satisfy the requirements of ANSI/ASME A 112.18.1M and be certified by the Canadian Standards Association (CSA) under Standard CAN/CSA B.125.M89.
 - 2. DI1: Bench-mounted hot and cold water faucet.
 - a. Base Product: Water Saver L414VB-8BH.
 - b. Wrist blade handles .
 - c. Renewable water valves and valve body.
 - d. Swing gooseneck with vacuum breaker and 8 inches spread.
 - e. Removable aerator.
 - 1) Aerator should terminate approximately 3 inches above deck.
 - f. Threaded mounting with locknut, washer, and coupling nuts.
 - g. Adjustable volume control.
- G. High Purity Water Valves:
 - 1. General:
 - a. Suitable for purified water.
 - b. Components in contact with purified water shall be polypropylene or PVDF.
 - c. Provide Georg Fischer's (GF) required Inline Flow Diverter (IFD) and 5/8 IN PFA Flexible Tubing for each purified water valve.
 - d. Water fixtures shall be fully assembled and factory tested.
 - e.
 - 2. DI-1A: Panel-mounted purified water fixture (Alternate 004)
 - a. Base Product: Aquatap 175 530 106 (Short)
 - b. Dual head
 - c. Recirculating faucet.
 - d. SYGEF PVDF with 5/8 IN PFA tubing, flare connections.
 - e. Two isolation ball valves for flared tubing.
 - f. Provide mounting flange, all applications:
 - 1) At wall-mounted applications: Flange shall be threaded in a manner to be installed tight to wall surface and allow fitting to be installed plumb.

- g. Mounting shank.
 - h. End of serrated hose end shall be 5-1/2 inches, nominal, above centerline of panel mounting.
3. DI-2: Bench-mounted purified water fixture.
- a. Base Product: Water Saver CT7833MSC.
 - b. Forged brass valve body and 8 inches spread riser with polypropylene interior and lining.
 - c. Self-closing lever that can also be turned to operate in a stay-open mode.
 - 1) Handle to be approved by UTSW prior to finalization.
 - d. Polypropylene serrated hose end.
 - e. Deck mounting flange.
 - f. Mounting shank.
 - g. End of serrated hose end shall be 7-7/8 inches, nominal, above bench top.
4. DI-3: Wall-mounted purified water valve.
- a. Base Product: Water Saver L73100F-PP-WSA.
 - b. All-polypropylene angle pattern valve body, stem, and mounting shank.
 - c. Four-arm polypropylene handle with index button.
 - d. Serrated, polypropylene hose end.
 - e. Length of valve from wall to index button shall be 5-1/8 inches, nominal.
- H. Needle Valves:
1. General:
- a. Fully assembled and factory tested at 225 psi air pressure.
 - b. Gas, air, vacuum, and steam needle valve fittings shall have stainless steel replaceable floating cone that is precision ground and self-centering which shall seat against a stainless steel or monel renewable valve seat.
 - c. Valve shall be for standard control under pressure up to 150 psi and shall have easily replaceable, subject-to-wear parts.
 - d. Provide pressure regulators designed for use with the appropriate service at locations indicated on the Laboratory Furnishing drawings.
 - e. Exposed metal shall be finished as specified elsewhere in this Section.
 - f. Needle valves for natural laboratory gas service shall be certified for use with natural gas by the Canadian Standards Association under ANSI Z21.15-1997/CGA9.1-M97.
 - g. Needle valves in fume hoods shall be mounted on the front panel of the fume hood, with components subject to wear accessible from the exterior face of hood.
2. G5: Wall/panel-mounted, single, angle pattern needle valve fitting.
- a. Base Product: Water Saver L5180F-225WSA.
 - b. Four-arm handle with colored plastic index button.
 - c. Provide mounting flange, all applications:
 - 1) At wall-mounted applications: Flange shall be threaded in a manner to be installed tight to wall surface and allow fitting to be installed plumb.
 - d. Serrated hose end.
 - e. Mounting shank assembly for panel installations, only.
 - f. Length of valve from wall to index button shall be 5-1/2 inches, nominal.
- I. Laboratory Ball Valves:
1. General:
- a. Suitable for laboratory gas, air and vacuum and be supplied fully assembled and factory tested at 125 psi air pressure.

- b. Exposed metal shall be finished as specified elsewhere in this Section.
- c. Ball valves shall be of quarter-turn (closed to fully open) design, be fitted with lever handle, except where indicated otherwise, requiring less than 5 feet/LBS force to operate, and shall have easily replaceable, subject-to-wear parts.
- d. Ball valves for natural laboratory gas service shall be certified for use with natural gas by the Canadian Standards Association under ANSI Z21.15/CGA9.1, current editions. +
- 2. B1: Wall-mounted, double, laboratory valve fitting with pressure regulator
 - a. Base Product: Water Saver L4203-364EL-179FT-WSA
 - b. For use on carbon dioxide gas systems.
 - c. Lever handle with colored plastic index button.
 - d. Forged brass valve body.
 - e. 1/4 IN outside diameter tube compression fitting end.
 - f. Mounted with SI136-02-08R stainless steel mounting shank.
 - g. Fixture has female inlet for connection to supply.
 - h. Floating escutcheon is threaded to conceal any gap between valve and wall.
 - i. Pressure regulator fitting with brass body with BUNA-N diaphragm and seals.
 - 1) Inlet pressure range of 0-300 PSIG.
 - 2) Outlet pressure range of 0-25 PSIG.
- J. Service Fitting Color Index: for colored plastic index buttons:

Service Name	Disc Color	Letters	Letter Color
Lab Air	Orange	AIR	Black
Compressed Air	Orange	AIR60,90,100	White
Gas	Dark Blue	GAS	White
Vacuum	Yellow	VAC	Black
Industrial (Laboratory) Cold Water	Dark Green	ICW	White
Industrial (Laboratory) Hot Water	Red	IHW	White
Cold Water (Potable)	Dark Green	CW	White
Hot Water (Potable)	Red	HW	White
High Purity Water	White	DI	Black
Argon	Violet	AR	White
Nitrogen	Brown	N2	White
Carbon Dioxide	Pink	CO2	Black
Helium	Black	HE	White
Oxygen	Light Green	O2	Black
CWS/R	Green	CWS/CWR	Black
Steam	Black	STM	White
Sea Water	Dark Green	SEAWAT	White
Cylinder Gas	Light Blue	CYL GAS	Black

2.3 LABORATORY SAFETY EQUIPMENT

- A. Manufacturers:
 - 1. Laboratory safety fixtures specified in this section shall be provided by a single manufacturer.
 - 2. Base:
 - a. Water Saver Faucet Co.
 - 3. Optional:
 - a. Haws Corporation.
 - b. Guardian Equipment.

4. Other manufacturers desiring approval comply with Section 01 61 00.
- B. Laboratory safety equipment shall be certified in accordance with ANSI Z358.1, current edition.
- C. General: Recessed cabinet in concrete masonry unit walls shall be fabricated so that piping does not interfere with the block face shell (outer wall).
- D. DH1: Dual-purpose eye wash/drench hose, deck mounted.
 1. Basis for design: Water Saver Model No. EW1022-BP.
 2. Dual-purpose eye wash/drench hose unit with dual gentle spray outlet heads and squeeze handle/valve with locking clip.
 3. Heads shall be equipped with flip top dust cover that automatically releases with water pressure.
 4. Flexible, reinforced PVC hose shall be furnished with swivel fitting at inlet end.
 5. Furnish with in-line backflow preventer at the hose inlet.
 6. Mounting flange and squeeze valve shall be chrome-plated brass with clear epoxy coating.
 7. Mounting flange and squeeze valve shall be color epoxy coated.
 8. Mounting shank.

2.4 FINISHES

- A. Satin Chrome Finish with Clear Epoxy Coating:
 1. Applicable to:
 - a. Laboratory service fittings except fittings inside fume hoods.
 2. Satin Chrome Finish:
 - a. Exposed surfaces shall be polished and electroplated with one layer of nickel.
 - b. Exposed surfaces shall then be further polished to an ANSI No. 6 brushed finish that is fine-grained and uniform.
 - c. Components shall then be electroplated with one layer of chrome.
 3. Clear epoxy coating:
 - a. Following plating, exposed surfaces shall be cleaned and degreased.
 - b. Clear epoxy coating shall be applied to exposed surfaces and then baked to permit curing.
 - c. Surfaces shall have a minimum coating thickness of 3 mils.
- B. Performance requirements for coated finishes:
 1. Chemical resistance:
 - a. Fume Test:
 - 1) Suspend coated samples in a container of at least 6 cubic feet capacity, approximately 12 inches above open beakers, each containing 100ml of 70 percent nitric acid, 94 percent sulfuric acid and 35 percent hydrochloric acid, respectively.
 - 2) After exposure to these fumes for 150 hours, the finish on the samples shall show no discoloration, disintegration, or other effects.
 - b. Direct Application Test:
 - 1) Subject coated samples to the direct action of the following reagents and solvents at a temperature of 25 deg C dropping from a burette at the rate of 60 drops per minute for ten minutes.
 - 2) Finish on samples shall not rupture, though slight discoloration or temporary softening is permissible.

Reagent	Concentration
Acetic Acid	98 percent
Acetone	

Reagent	Concentration
Ammonium Hydroxide	28 percent
Amyl Acetate	
Amyl Alcohol	
Benzene	
Butyl Alcohol	
Calcium Hypochlorite	
Carbon Disulfide	
Carbon Tetrachloride	
Chloroform	
Chromic Trioxide Acid	
Cresol	
Crude Oil	
Dioxane	
Distilled Water	
Ether	
Ethyl Acetate	
Ethyl Alcohol	
Ethyl Ether	
Formaldehyde	37 percent
Formic Acid	90 percent
Gasoline	
Glacial Acetic Acid	99.5 percent
Glycerine	
Hydrochloric Acid	38 percent
Hydrofluoric Acid	48 percent
Hydrogen Peroxide	5 percent
Isopropyl Alcohol	
Lactic Acid	10 percent
Kerosene	
Methanol	
Methyl Alcohol	
Methyl Ethyl Ketone	
Methylene Chloride	
Mineral Oil	
Monochlor Benzene	
N-Hexane	
Naphthalene	
Nitric Acid	70 percent
Perchloric Acid	70 percent
Phenol	
Phosphoric Acid	75 percent
Sea Water	
Silver Nitrate	30 percent
Sodium Bichromate	saturated
Sodium Carbonate	10 percent
Sodium Chloride	20 percent
Sodium Hydroxide	50 percent
Sodium Hypochlorite	
Sodium Sulfide	
Sulfuric Acid	87 percent
Toluene	
Trichlorethylene	
Turpentine	
Urea	saturated
Xylene	
Zinc Chloride	saturated

2. Mar and abrasion resistance:

- a. Coating material shall have a pencil hardness of 2H – 4H with adhesion substantial enough to withstand both direct and reverse impacts of 160 inch-LBS.
 - b. Coating shall have excellent mar resistance and be capable of withstanding scuffing, marring and other ordinary wear.
- 3. Reparability:
 - a. Scratches and other localized surface damage shall be field-repairable.

2.5 LABORATORY SINKS

A. General:

- 1. Inside dimensions of sink bowl are provided on drawings.
- 2. Provide tailpiece compatible with waste piping system for sinks unless otherwise specified.
 - a. Refer to Division 22 for piping requirements.

B. Epoxy Resin:

- 1. Acceptable Manufacturer: Manufacturer shall be the manufacturer of the epoxy resin work surfaces specified in Section 12 35 53.
- 2. Laboratory Sinks:
 - a. Drop-in Type:
 - 1) Drop-in installation by Division 11 in epoxy resin work surface.
 - 2) Color to match work surface.
 - b. Comply with the requirements of Section 12 35 53 for epoxy resin.
 - c. Exposed edges shall be radiused not less than 1/4 inches.
 - d. Drain grooves in top, when indicated on drawings.
 - 1) Sink shall be set 1/8 inches below the lowest drain groove level.
 - e. Tops without drain grooves: Sink shall be set 1/8 inches below the level of the adjacent surface.
 - f. Provide epoxy resin sink outlet with strainer, stopper, and open-end overflow, and install with continuous bead of silicone sealant.
- 3. Cup Sinks:
 - a. Laboratory Work Surface Installations:
 - 1) Raised rim, color to match work surface, sizes as indicated on drawings, with integral threaded tailpiece.
 - 2) Flush with work surface, color to match work surface, sizes as indicated on drawings, with integral outlet and threaded tailpiece.
 - b. Comply with the requirements of Section 12 35 53 for epoxy resin.
 - c. Provide strainer for cup sinks.

C. Stainless steel:

- 1. Laboratory Sinks, integral with stainless steel work surface:
 - a. Refer to Section 12 35 53, Stainless Steel Fabrications.
 - b. Provide stainless steel strainer, outlet, standpipe overflow and stopper for sinks unless otherwise specified.
 - c. Provide tailpieces compatible with waste piping system for sinks unless otherwise specified. Refer to Division 22 for piping requirements.
 - d. Undercoating: Waterborne and non-flammable in liquid state, with no volatile organic compounds (VOCs) providing sound deadening and preventing condensation.
- 2. Cup Sink:
 - a. Laboratory Work Surface Installations:
 - 1) Refer to Section 12 35 53, Stainless Steel Fabrications.
 - b. Provide strainer, outlet, and tailpiece for cup sinks.

- c. Provide mounting bracket for wall-mounted cup sinks.

PART 3 - EXECUTION

3.1 SITE CONDITIONS

A. Inspection:

1. Prior to installation of fixtures, carefully inspect the installed Work specified in other Sections and verify that Work is complete to the point where this installation may properly commence.
2. Verify that Work has been installed in complete accordance with the original design, approved submittals, and the manufacturer's recommendations.

B. Discrepancy:

1. In the event of discrepancy, immediately notify the Architect.

3.2 INSTALLATION

A. General:

1. Rough-in fixture piping connections in accordance with minimum sizes indicated in fixture rough-in schedule for particular fixtures.
2. Install fixtures plumb and level.
3. Provide piping connections to fixtures with valves and escutcheons as specified in Division 22.
4. Verify that fixtures and trim are tight, leak-free and function properly.

B. Set internal volume control on cup sink water fittings so water does not splash out of sink.

C. Set sinks in chemical resistant sealing compound, secure and support, as recommended by the manufacturer.

D. Safety Stations, Emergency Showers, and Emergency Eyewash:

1. Emergency showers must be installed with a minimum of 16 inches of clearance from obstruction to the center of the showerhead.
 - a. Eyewash or eye/face wash fixture is not considered an obstruction.
2. Install emergency showers with showerhead positioned not lower than 82 inches or above 96 inches measured from the operational standing surface.
 - a. Refer to drawings for mounting height within that range.
 - b. If no range is provided, shower head shall be installed 96 inches above operational standing surface.
3. Recessed wall cabinets in fire rated partitions shall be installed in accordance with UL approved design.

E. Hose Reels:

1. Manufacturer shall confirm weight of hose reel assembly when hose is filled with water.
2. Structural Support:
 - a. Framed walls and partitions:
 - 1) Provide slotted channel framing between structural flooring and structural deck above in the wall or partition construction to support hose reel assembly.
 - 2) Framing shall be designed to support live load of hose reel assembly.
 - b. Concrete Masonry Walls:
 - 1) Grout concrete masonry solid to support hose reel assembly.
3. Provide masonry anchors as recommended by manufacturer to support live load of hose reel assembly.

END OF SECTION

SECTION 11 53 63

LABORATORY STERILIZERS

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish labor, materials, tools, equipment, and services for Laboratory Sterilizers, as indicated, in accordance with provisions of Contract Documents.
- B. Section includes:
 - 1. Laboratory Sterilizers (Autoclaves): Small.
 - 2. Owner Furnished and Contractor Installed equipment:

1.2 REFERENCES

- A. ASTM International (ASTM):
 - 1. ASTM A240 Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Application.
 - 2. ASTM C795 Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel.
 - 3. ASTM E84 Standard Test Method for Surface Burning Characteristics of Building Materials; current edition.
- B. Underwriters Laboratory (UL):
 - 1. UL Standard 508.
 - 2. UL Standard 61010-1.
- C. National Electrical Code (NEC) and National Fire Protection Association (NFPA):
 - 1. NFPA 70, current edition, for clearances at electrical equipment,
- D. American Society of Mechanical Engineers (ASME):
 - 1. ASME Code, Section VIII, Division 1, for unfired pressure vessels, current edition.
 - a. ASME Code, Section I, Part PMB, for power boilers, current edition.
- E. Electro-Magnetic Engineering (EMC):
 - 1. EMC Directive 89/336/EEC, 92/31/EC, 93/68/EEC.
- F. Test autoclaves to pass biological monitoring as described in ANSI/AAMI ST79, current edition, Comprehensive Guide to Steam Sterilization and Sterility Assurance in Health Care Facilities.

1.3 SUBMITTALS

- A. Shop Drawings:
 - 1. Submit complete shop fabrication and installation drawings, including plans, elevations, sections, details, finished and materials dimensions, and schedules.
 - 2. Submit manufacturer certified pit drawings within 60 days of Award of Contract from Owner to Contractor.
 - 3. Show relationship to adjoining materials, and construction and requisite service, operating and installation clearances (width, depth, and height).
 - 4. Layout shall include relationship to light fixtures and ventilation registers, grilles, and diffusers.
 - 5. Identify connection points and sizes to building services and systems.
 - 6. Provide clear identification where equipment requirements deviate from the service/utility provisions identified in the Construction Documents.

7. Verify that component parts and assembly of each item will support superimposed loads, without deflection detrimental to function, appearance, and safety.
 8. Provide flow rates, pressure drops, temperature and pressure requirements, voltage, and amperage, etc.
 9. Indicate service requirements, loads, and rough-in locations of piped services and electrical connections.
- B. Product Data:
1. For each type of sterilizer, submit for review a complete materials list, including catalog data and performance data of materials, equipment, and products for work in this Section.
- C. Project Information:
1. Provide piping, wiring, and/or control diagrams, including connection points and sizes to building services and systems.
 2. Structural:
 - a. Submit detailed anchorage and attachment drawings and calculations to show compliance with seismic restraint requirements of applicable building codes.
 - b. Engineering design shall be performed and sealed by registered Engineer, licensed to practice structural engineering in the state of the project location.
 - c. Refer to the Structural Drawings for seismic design criteria.
 3. Statement of installer qualifications.
 4. Notice of factory testing.
 - a. Factory Authorized Testing (FAT) Procedures:
 - 1) Requirements and Documentation.
 - 2) Include NIST calibration documentation.
 - b. Manufacturer's installation, start-up, and adjustment instructions.
 - c. Start-up and field test reports.
 - d. Demonstration and instruction statement from Owner
- D. Contract Closeout Information:
1. See Section 01 78 23.
 2. Operations/Maintenance Manuals:
 - a. Complete operating and maintenance manuals that describe proper operating procedures, maintenance and replacement schedules, PLC, components parts list, including non-proprietary parts with original part manufacturer's parts numbers, and closest factory representative for components and service.
 - b. Manual shall include non-proprietary list of valves and other serviceable components.

1.4 QUALITY ASSURANCE

- A. Laboratory sterilizers specified in this section shall be the provided by a single manufacturer.
- B. Work of this section requires close coordination with work of Division 22 and 26 as well as installation of Owner furnished components and work specified in other Sections.
1. Sequence work to assure an orderly progress in the project without removal of previously installed work and so as to prevent damage to finishes and products.
- C. Refer to General Laboratory Equipment Schedule.
- D. Manufacturer's Qualifications:
1. Manufacturer shall specialize in manufacture of products specified herein with minimum ten years of experience.
 2. Manufacturers for work in this section shall have established organizations and production facilities including tools, equipment, and special machinery necessary for the fabrication

- and installation of the type of equipment specified, with skilled personnel, factory trained workmen and an experienced engineering department.
3. Each shall have the demonstrated knowledge, ability, and the proven capability to produce the specified equipment of the required quality and the proven capacity to complete an installation of this size and type within the required time limits.
 - a. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.
 - b. Upon request, manufacturer shall provide three references of similar installations.
- E. Installer Qualifications:
1. Installer for work in this section shall have an established organization including tools, equipment, and special machinery necessary for specializing in the installation of the type of equipment specified, with skilled personnel and factory-trained workmen.
 - a. Each shall have the demonstrated knowledge, ability, and the proven capability to install the specified equipment of the required quality and the proven capacity to complete an installation of this size and type within the required time limits.
 2. Firm specializing in installation of products specified, with minimum five years of experience, and authorized by manufacturer to install product.
 3. Upon request, installer shall provide three references of similar installations.
- F. Specified products, materials, or systems for Project may include engineering or on file standards required by the Regulatory Agency.
1. Contractor's substitution of products, materials or systems may require additional engineering, testing, reviews, approvals, assurances, or other information for compliance with Regulatory Agency requirements or both.
 2. Contractor shall provide Agency approvals or other additional information required and pay for additional costs required Architect's services made necessary by the substitution at no increase in Contract Sum or schedule time, and as a part of substitution proposal.

1.5 PRODUCT HANDLING

- A. Schedule delivery of steam sterilizers only after wet operations in building are completed.
- B. Provide receiving, distribution, and storage areas of sufficient size and capacity to accommodate crated equipment.
- C. Store steam sterilizers in a ventilated place, protected from the weather, with relative humidity therein of 50 percent or less at 21.1 deg C 70 deg F.
- D. Protect work of this section before, during and after installation including installed work and materials of other trades.
- E. Any damage as a result of this contractors work will be replaced, repaired, and restored to original condition to the approval of the Architect at no additional cost or inconvenience to the Owner.
- F. At no time shall personnel use the steam sterilizer equipment as a workbench, scaffolding, or other uses.
- G. Protect steam sterilizer equipment from debris, paint, damage in the course of construction.

1.6 WARRANTY

- A. Warrant each pressure vessel against failure under normal operating conditions for ten (10) years from date of Substantial Completion.

1.7 MAINTENANCE SERVICE

- A. Furnish routine preventative and emergency maintenance service for 2 years from final acceptance.
 1. Maintenance service shall supplement warranty requirements.

- B. Include periodic examination, adjustment, and maintenance of equipment.
 - 1. Repair or replace worn parts, using only standard parts produced by equipment manufacturer.
- C. Furnish 7-day per week emergency service during maintenance period, providing maintenance personnel on site within 24 hours, should a shutdown or emergency develop between regular examinations.

1.8 MAINTENANCE PARTS AND MATERIALS

- A. Furnish 2-year supply of selected mechanical and electrical components based on normal maintenance schedule and sterilizer operation for each unit.
- B. Provide two additional rolls of printer paper with each sterilizer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Laboratory Sterilizers:
 - 1. Base:
 - a. Steris Corporation.
 - 2. Optional:
 - a. Beta Star Life Science Products of R-V Industries.
 - b. GetingeUSA, Inc.
- B. Other manufacturers desiring approval shall comply with Section 01 61 00.

2.2 DESIGN CRITERIA

- A. Units Serviced with Building Steam:
 - 1. Manufacturer's unit-mounted valves, pressure reducing valves, safety relief valves, strainer, joints, fittings, flanges, unions, and piping shall be in compliance with Section 23 05 19.
 - a. Provide union connections for pipe sizes 1-1/2 inches and smaller, and flange connections for pipe 2 inches and larger pipe sizes.
 - 2. Manufacturer's unit-mounted steam pressure gauges shall be provided with gage cock with siphon loop in compliance with Section 23 05 19.
- B. Utility Requirements:
 - 1. Mechanical and electrical services have been designed for the services and loads as described for individual equipment items herein.
 - 2. If a manufacturer's product requires services in excess of those indicated, either of type, size, quantity, or quality, that cost shall be borne by the Contractor and cannot be the justification for a change order.
- C. Equipment Pits:
 - 1. Design to meet standards of successful bidder.
 - 2. Refer to Submittals.
 - 3. Manufacturer shall provide height adjustments, thresholds, and other items required to finish off machines for pit mounted arrangement.
- D. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

2.3 LABORATORY STERILIZER AUTOCLAVE: SMALL

- A. Description:

1. Vertical sliding door sterilizers using steam under pressure and mechanical air removal or gravity displacement principles; designed for sterilization of hard goods, porous loads, liquid loads in vented containers, and laboratory animal caging systems.
- B. Base Product:
1. Lab Series Small Sterilizer by Steris.
 2. Model 533LS Sterilizer by Getinge.
- C. Chamber Size (nominal):
1. Chamber size: 20 inches wide by 20 inches high by 38 inches long.
- D. Configuration:
1. Steam source: Building steam. Sterilizer shall connect to a single point of connection to building steam.
 2. Doors/installation: Double doors, recessed through one wall, cabinet enclosed.
 3. Process:
 - a. Prevacuum process is for the sterilization of porous and non-porous heat- and moisture-stable goods, sterilization of liquids and media in borosilicate glass containers with vented closures, and decontamination of supplies after laboratory procedures.
 - 1) Prevacuum sterilizers shall be equipped with prevacuum, gravity, liquid, leak test, and daily air removal test cycles.
- E. Pre-programmed Cycle Descriptions:
1. Gravity Cycle:
 - a. Provided on gravity and prevacuum sterilizers for the sterilization of heat- and moisture-stable goods at 212-285 deg F, decontamination of bagged laboratory wastes.
 - b. Gravity cycle utilizes the gravity air-displacement principle.
 2. Liquid Cycle:
 - a. Provided on gravity and prevacuum sterilizers for the sterilization of liquids and media in vented borosilicate glass or metal containers at 212 deg F to 254 deg F. Liquid cycle utilizes the optimal solution cooling feature, during exhaust cooling phase, to control the exhaust rate.
 - b. Provide automatic jacket blowdown to exhaust steam from chamber and jacket to prevent liquid loads from boiling over.
 3. Prevacuum Cycle:
 - a. Provided only on prevacuum sterilizers for the efficient, high-volume sterilization of porous, heat- and moisture-stable materials at 250 deg F to 285 deg F. Prevacuum cycle utilizes a mechanical air-evacuation system.
 4. Leak Test Cycle:
 - a. Provided only on prevacuum sterilizers for verification of chamber integrity.
 - b. Cycle parameters shall be preprogrammed and fixed.
 - c. The acceptable maximum leak rate is 1 mm Hg/min over a 10-minute period following a fixed stabilization time.
 5. DART (Daily Air Removal Test or Bowie-Dick) Test Cycle: Provided on prevacuum sterilizers; is used to conduct a Bowie-Dick test on the sterilizer.
 6. Effluent Decontamination Cycle:
 - a. Provide effluent decontamination cycle designed for applications with risk of contaminating the sterilizer drain line or building drain system with effluents from the chamber.
 - b. The cycle shall sterilize effluents from the unit prior to discharge by injecting steam through the drain.

- c. Air to door seal back-up system shall be provided to ensure door seal is maintained in case of steam pressure loss.
- d. The system shall be provided with following features:
- e. A fixed decontamination pre-conditioning phase shall be added to pre-conditioning menu selection.
 - 1) The pre-conditioning phase shall use one vacuum pulse pulled through a dedicated decontamination filter assembly (steam-heated, 0.2 micron hydrophobic cartridge and housing).
- f. During heating-up and exposure phases the steam shall be introduced into the bottom of the chamber through the drain line to ensure that the entire system is sterilized.
- g. The chamber steam shall be bled through the decontamination filter during exposure to ensure sterilization.
- h. The controlling temperature probe shall be located after the decontamination filter housing to ensure that the load and decontamination filter assembly is sterilized.
- i. The sterilized condensate accumulated in the drain line and chamber shall be held until draining in the post-conditioning drying or equalization phase.
- j. Air evacuated from the chamber shall pass through a 0.22µm sterile membrane filter or other system to sterilize the chamber air prior to release.
 - 1) Filters shall be steam sterilized in subsequent processes.

F. Control system:

- 1. Allen-Bradley PLC microcomputer control system.
- 2. Operator Interface: Provide at operating load, non-sterile end, adjacent to chamber.
 - a. Touch-sensitive color screen with graphic display area.
 - 1) Screen shall display sterilizer functions, alarm conditions, and be used for cycle configuration and initiation.
 - 2) Displayed messages shall be in complete phrases with no cross-referenced codes. Screen shall be a minimum of 5 inches by 4 inches.
 - b. Thermal printer to provide record of all cycle data.
- 3. Non-Operating End Control Panel:
 - a. Touch-sensitive screen with graphic display area, showing same messages as operating end display.
 - b. Provide second thermal printer to provide record of all cycle data located on hazard side of double door chambers with a biological seal.
- 4. Internal Battery:
 - a. Battery shall be used to back-up all cycle memory for up to ten years.
 - b. Controller shall require complete restart of cycle upon interruption of power.

G. Construction:

- 1. Shell assembly:
 - a. Double walled, insulated, Type 316L stainless steel, welded sterilizer vessel.
 - 1) Shielded steam entry inside chamber.
 - b. End Frame: Type 316L stainless steel, welded to door end.
 - c. Vessel: ASME rated at 45 psiG.
 - d. Baffle: Shield steam supply opening inside chamber by a Type 316L stainless steel baffle.
 - e. Hinged front cabinet panel to access sterilizer piping and control board housing.
 - f. Equipment Stand: Support sterilizers on height-adjustable stainless or carbon-steel stand, shop-coated for corrosion protection.
 - g. Door assembly:

- 1) Formed from a single sheet of Type 316L stainless steel, insulated.
 - 2) Provide safety lock to prevent door opening when chamber pressure exceeds 2 psi.
 - 3) Door shall be vertical sliding with foot pedal activation; not required for chambers furnished with biological seal.
 - 4) Door shall have replaceable, one-piece, compressed air-activated polymer gasket. Upon completion of cycle, gasket shall retract under vacuum into a machined groove in the end frame of the sterilizer.
- h. Insulation:
- 1) 1 inch thick, fiberglass insulation sleeve, sealed and held in place with hook-and-loop-type closures.
 - a) Insulation shall be asbestos- and chloride-free, silicone impregnated, oil- and water-resistant.
2. Piping Valves, Switches, and Gauges:
- a. Piping:
- 1) General:
 - a) Brass, including steam strainer, shutoff valve, and brass pressure regulator.
- b. Insulation:
- 1) General: Insulate hot and cold piping.
 - 2) Hot Piping:
 - a) Piping conveying water or steam greater than 100 deg F.
 - b) Provide vinyl wrap.
 - 3) Refer to Insulation specification in Division 23 Mechanical for insulation type and R-value.
- c. Check Valves: Provide check valves on water feed piping.
- d. Valves:
- 1) Valves and major components arranged to be easily accessed and replaceable.
 - 2) All valves shall be non-proprietary.
 - 3) Air operated, solenoid valves with DIN connectors, or piston valves located in manifold.
 - 4) All piping connections shall terminate within the confines of the sterilizer.
 - 5) Utility Shut-Off Valves:
 - a) Provide utility shut-off valves for incoming piped utilities.
 - b) Manual Steam Shutoff:
 - (1) Pressure-rated at 125 psiG for saturated steam.
 - (2) Valve handles shall be low heat conducting-type.
 - 6) Safety Relief Valve:
 - a) Provide ASME-certified and labeled safety relief valve (SRV) to protect vessel, jacket, and heat exchanger (where applicable), from pressure build-up beyond the equipment's maximum rated pressure.
 - b) Coordinate relief valve discharge with Mechanical installer for vent piping to the exterior of the building.
 - c) Manufacturer shall pipe multiple steam safety relief valves on a unit to a single point for field connection.
 - d) Pressure relief shall not release potential biological hazards into non-contained spaces or where they could re-enter building.
 - 7) Pressure Reducing Valves (PRV):

- a) Provide pressure reducing valves if incoming steam, water, and compressed air pressures need to be reduced to manufacturer-recommend operating pressures.
- 8) Decontamination Isolation Valves and Injection Ports:
 - a) Provide inlet and outlet ports with associated valves for the decontamination of sterilizer, filter, and incinerator, if present, requiring service.
 - b) Inlet port shall enter the chamber at the bottom and allow for the decontaminating agent to flow up through the chamber to the outlet port.
 - c) System shall allow for steam, formaldehyde gas, Vaporized hydrogen peroxide, or Chlorine dioxide gas as the decontaminant.
- e. Gauges:
 - 1) Locations:
 - a) Chamber and jacket pressure gauges shall be mounted in service area.
 - b) Locate visible to the operator in fascia panel or in service panel adjacent to door.
 - c) Provide on each end of double door units.
 - 2) Provide diaphragm or electric gauges for units serving BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 applications.
- f. Steam Valve Interlock:
 - 1) Provide steam valve interlock to prevent steam valve from opening when door is open.
- g. Pressure Interlock:
 - 1) Provide pressure interlock to prevent opening of door when the unit is above or below atmospheric pressure.
- h. Chamber Float Switch Alarms:
 - 1) Provide with chamber float switch to indicate alarm that water is in chamber.
- 3. Electrical Connections:
 - a. Provide lighted DIN Connectors.
 - b. All electrical connections shall terminate within the confines of the sterilizer.
- 4. Vacuum System (Prevacuum units only):
 - a. Water ring seal type pump with non-asbestos seal and packing.
 - 1) Install on vibration-isolation mounts.
 - b. Provide cooling water saving package to reduce water consumption.
 - 1) Initial charge of vacuum pump water shall be cooled and recirculated through a heat exchanger to a break tank and cool the chamber effluent with a heat exchanger. Tank, recirculation pump, heat exchanger and associated components shall be factory piped and provided by autoclave manufacturer.
 - 2) Vacuum pump seal water shall be cooled through the use of a heat exchanger and a closed loop chilled water supply, reducing water consumption by approximately 99 percent.
 - 3) Vacuum pump seal water shall be drawn from tank, cooled by heat exchanger to a maximum of 60 deg F and delivered to vacuum pump.
 - 4) Tank overflow shall be cooled not to exceed 140 deg F prior to discharge.
 - 5) Tank shall have inlet for cold water make-up.
 - 6) Chilled water will be supplied at temperature not to exceed 42 deg F and should return from heat exchanger not to exceed 57 deg F (15 deg F temperature rise).
 - 7) System shall accommodate incoming chilled water pressures up to 150 psiG and shall have a pressure drop not to exceed 5 psiG.

H. Features:

1. Resistance Temperature Detectors (RTD):
 - a. Installed in chamber drain line and chamber jacket to sense and control temperature variations within the chamber.
2. Jacket Temperature Control:
 - a. Separate jacket RTD installed in the jacket drain line to allow control of jacket temperature, for optimal cycle performance.
3. Steam Bleed:
 - a. Constant steam flow supplied across chamber RTD to assure even temperature distribution and temperature control.
4. Steam Purge:
 - a. To assist in air removal and to preheat load.
5. RS-485 Interface Port:
 - a. For downloading cycle information to user-furnished data acquisition system.
6. Electronic Water Saving Control:
 - a. To control the amount of water used in condensing exhausted chamber steam.
7. Automatic Utilities Start-up/Shut-down:
 - a. To shut off utility valves, permitting slow cooling of the entire vessel and load.
 - b. Programmed and manual restart options.
8. Control Lockout Switch:
 - a. Limit switch on chamber door to prevent cycle from starting unless door seal is tight against the door.
9. Chamber Drain System:
 - a. To prevent pollutants from entering into the water supply system and sterilizer.
 - b. Automatic stainless steel plate-type condensing system to convert chamber steam to condensate and to dispose it to waste.
 - c. Provide non-clogging chamber drain line.
10. Provide emergency manual exhaust valve.
11. Effluent Temperature Control System:
 - a. System shall not discharge effluent until it has cooled to less than 140 deg F.
 - b. Provide Type 316 stainless steel tank to collapse steam and receive hot effluent for cooling.
 - 1) Tank shall be equipped with temperature probe and overflow to drain at top.
 - 2) Hot effluent shall flow into bottom of tank.
 - 3) Provide cold water for make-up and for cooling if demanded by tank prior to discharging effluent.
 - 4) Temperature probe shall adjust flow of cold water in to effluent discharge to reduce tank water temperature prior to discharge at overflow only when tank water is 140 deg F or higher.
 - 5) System shall only require cold water supply. System shall not require power.
12. Door Interlocks, Double Door units only:
 - a. To allow only one door to be opened at a time, and to prevent non-operating end door from being opened until a satisfactory cycle is complete.
13. Air Differential Seal, Double Door units only:
 - a. Air differential seal shall be provided to maintain room air differential pressure; provide on unload end of sterilizer.
 - b. Seal shall be fabricated from Type 304 stainless steel.
 - c. Air differential seal shall also serve as vermin-proof seal.
14. Biological Seal (Bioseal):

- a. To prevent passage of airborne microorganisms and other hazards from containment to non-containment zones via the space between the vessel body and the structural wall opening.
- b. Provide on load (containment) end of sterilizer.
- c. Infiltration tests shall indicate no cross contamination leakage through the bioseal at a pressure differential 0.022 psiG.
 - 1) Wall Frame and Extension Panels:
 - 2) Four-sided frame consisting of stainless steel angle or channel, minimum 14 GA or as indicated on drawings, shall be secured to wall opening.
 - 3) Provide continuous sealant under width and length of wall frame. Mechanically fasten wall frame to containment barrier wall.
 - 4) Where wall frame is greater than 4 inches from the chamber floor, provide stainless steel, 14 GA extension panels, continuously welded to wall frame and to intersecting extension panels.
 - 5) If required, flange for compression bar attachment shall be continuously welded to extension panel.
 - 6) Weld threaded stainless steel studs (bolts) maximum spacing of 6 inches on center to wall flange or extension panel flange on load containment side of sterilizer.
 - 7) Fully Welded Stainless Steel Chamber Flange: 1/4 inches stainless steel flange shall be continuously welded around the periphery of the chamber. Weld threaded stainless steel studs (bolts) maximum spacing of 6 inches on center to chamber flange on load side of sterilizer.
 - 8) Gasket: 1/4 inches minimum thick BUNA rubber or silicone (or NIH approved equal) designed to provide airtight seal between wall flange extension panels and chamber flange.
 - a) There shall be approximately 2 inches to 4 inches unsupported material between the extension panel flange and the sterilizer chamber flange.
 - b) The gasket shall bridge the distance between flanges or extension panels.
 - c) Gasket shall have uniform thickness and have no seams.
 - d) Gasket flexibility shall allow for vibration and thermal expansion/contraction of sterilizer unit and maintain an airtight seal.
 - e) Gasket material shall be compatible with multiple cycles of decontamination by fumigation.
 - f) Caulking is not allowed.
 - g) Holes in gasket shall be made with appropriate tools to avoid tearing.
 - 9) Compression Bar:
 - a) 1/4 inches thick stainless steel, or mild steel if bolts are on unload non-containment side, to compress gasket material.
 - (1) Clamping bar shall be bolted to flanges or extension panels on welded studs with stainless steel nuts and lock washers.
 - (2) Ensure compression bar fasteners are reasonably accessible from service side of sterilizer to allow periodic inspection and tightening as required for life-cycle maintenance.
 - 10) Compression bar and gasket shall extend a minimum of 1 inch beyond the stud bolt holes.
 - 11) Manufacturer and installer of sterilizer shall verify adequate clearance and access to parts of biological seal and sterilizer that require servicing and inspection.
15. RTD Load Probe and F0 Sterilization:
 - a. To measure load temperature during processing; allows control system to start and terminate cycles based on recorded data.
16. Exterior Enclosure:

- a. Type 304 stainless steel side and rear panels with louvered stainless steel top panel with No. 3 finish to enclose sterilizer body and piping for freestanding units or where extends into finished space.
- 17. Gasket:
 - a. Synthetic neoprene or silicone on sterilizer sub-frame to ensure tight fit between cabinet end panels, and wall partitions, if provided.
- 18. Seismic tie down to conform to building code requirements.
- I. Accessories required for each unit:
 - 1. Thermocouple Feed-Through Assembly: Tri-clamp chamber penetration allowing a minimum 12 thermocouple feed through assembly into the chamber.
 - 2. Intermediate Shelf.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Site Conditions:
 - 1. Prior to installation of steam sterilization equipment, carefully inspect the installed work specified in other Sections and verify that work is complete to the point where this installation may properly commence.
 - 2. In the event of discrepancy, immediately notify the Architect.
- B. Examine surfaces designated to receive work for conditions that would adversely affect the finished work.
- C. Repair or replace surfaces not meeting tolerances or quality requirements governing substrate construction prior to start of work.
- D. Verify that surfaces, prepared openings, or support structures are ready to receive work.
- E. Verify field measurements and opening dimensions are as instructed by manufacturer.
- F. Inspect and verify that the required utilities are available, in proper locations and ready for use, prior to equipment installation.

3.2 PREPARATION

- A. Coordinate with Divisions 22, 23, 25, and 26 for location, size, and type of mechanical, power and communications services required.
- B. Before shipping, equipment shall be cleaned inside and outside, be free of rust, loose scale, and other deposits.
- C. Finished surfaces shall be protected to prevent shipping and/or storage damage. Threaded connections, flanges, and couplings shall be protected.
- D. Delivery and installation shall be coordinated so that equipment can be positioned in-place prior within the sequence of construction:
- E. Contractor shall verify the delivery route through building is adequate for equipment.
- F. Equipment to be securely crated and/or packaged to prevent damage during shipment.
- G. Loose parts shipped inside of the unit shall be secured.
- H. The vendor shall be responsible for delivery of the unit to the job site, setting the equipment in place, unpacking and reassembly.
- I. The vendor shall verify that required utilities are available, in proper locations, and ready for use.
- J. Beginning of installation means acceptance of existing conditions by the vendor.

- K. Upon unpacking of the equipment, the vendor shall remove debris, crating material, and packaging from the location.

3.3 WORK REQUIRED OF OTHER SECTIONS PRIOR TO INSTALLATION

- A. Provide building service lines supplying specified pressures and flow rates.
- B. Install shutoff valves on service lines.
- C. Install fused disconnect switches with lockout in OFF position in electric supply lines near the equipment.
- D. Provide illumination of service area, with provision of convenience outlet for maintenance.

3.4 INSTALLATION

- A. General:
 - 1. Install equipment in accordance with manufacturer's printed instructions and the reviewed submittals.
 - 2. Installation shall be with manufacturer certified, approved, trained, or provided personnel.
 - 3. Properly align and position equipment.
 - 4. Manufacturer's Field Services:
 - a. Provide manufacturer's field services to supervise installation.

3.5 FACTORY ACCEPTANCE TESTING (FAT)

- A. Sterilizers requiring FAT:
 - 1. Small autoclaves.
 - 2. Medium autoclaves.
 - 3. Bulk autoclaves.
 - 4. All autoclaves
- B. Sterilizer shall be provided with service identical to those that will be provided on site, including electrical service, and piped services connections, flows, and temperature, as applicable.
- C. The manufacturer shall:
 - 1. Issue Factory Acceptance Test protocols and notification to the Owner at least 30 days in advance of scheduled testing for Owner comment and approval.
 - 2. Execute tests and measurements.
 - 3. Analyze test results.
 - 4. Describe deviations in test report and make corrections. Retest/repeat procedures, if required.
 - 5. Check that register deviations are complete and documentation is correct.
 - 6. Verify calibration of test equipment.
 - 7. Check the conclusion of the test conformity.
 - 8. Sign test report after completion and submit to Owner.
- D. Test requirements:
 - 1. Performance testing of sterilizer on cages in accordance with the inspection procedure below:
 - a. Manufacturer and Owner will coordinate the provision of test cages and racks supplied either by Owner or from a research facility in local proximity to the equipment supplier's factory.
 - b. Manufacturer and Owner will coordinate number and types of cages/racks to be processed during factory testing.
 - c. During performance testing, caging/racks washed will be visually inspected and must be accepted as being physically clean, free of dirt, unsoiled, and unstained.

- d. Cycle time performance will be compared to target cycle times specified above. Equipment supplier shall work with Owner to make adjustments at the factory to minimize cycle times if these times exceed target performance.
- E. Owner's authorized representative shall have the right to inspect work during and after fabrication completion and shall perform and witness testing.
 - 1. This inspection shall occur at the vendor's manufacturing facility.
 - 2. Facility shall have utilities as required to perform full testing of the sterilizer.
 - 3. The representative's decisions to reject material or workmanship for non-compliance with the Specification, Drawings and purchase order shall be final.
- F. Owner's representative(s) inspection shall include, but not be limited to the following items:
 - 1. Manufacturer's adherence to the terms of Purchase Order, particularly in the areas of schedule and required documents.
 - 2. Manufacturer's adherence to the Contract Documents, including revisions thereof.
 - 3. Manufacturer's adherence to fabrication procedures as mandated by Code, Specifications and Drawings.
 - 4. Fabrication within tolerances as established by Code, Specifications, and Drawings, whichever is most stringent.
 - 5. Full operation of the sterilizer including proper functioning of the following items:
 - a. Microprocessor controls.
 - b. Valves.
 - c. Pumps.
 - d. Sensors.
 - e. Alarms.
 - f. Printers.
 - g. Safety features.
 - h. All operational cycles.
 - i. All other operational components of the sterilizer.
 - 6. Overall workmanship of equipment and accessories.
- G. FATs shall include the following:
 - a. Verification of equipment and scope of delivery.
 - b. Verification of the pipe work.
 - c. Verification of mechanical installation.
 - d. Verification of electrical installation.
 - e. Verification of utilities.
 - f. Verification of documentation.
 - g. Vacuum leak rate test.
 - h. Hydrostatic pressure test.
 - i. Test protocol run analysis.
- H. Manufacturer shall prepare and maintain a Deviation Tracking List identifying deviations or faults that occur during the FAT.
- I. Acceptance by the Owner's representative shall not relieve the manufacturer of the responsibility for workmanship conforming to the Contract Documents and the Purchase Order.

3.6 START-UP AND TESTING

- A. Test, clean, and adjust equipment and apparatus installed to ensure performance meets specified requirements.
- B. Operate each unit and test full range of cycles over a continuous period. Record test data.

- C. Adjust and re-test units not meeting requirements.

3.7 CLEANING AND PROTECTION

- A. Packaging and debris and other waste resulting from installation of equipment shall be removed.
- B. Clean finished equipment, touch up as required and remove and refinish damaged or soiled areas.

3.8 DEMONSTRATION AND INSTRUCTIONS

- A. Refer to Section 01 79 00 for equipment specified in Part 2.
- B. Test equipment prior to demonstration. Ensure equipment, including specified accessories, is operational.
- C. Provide demonstration of equipment operation and instruction of Owner's personnel.
- D. Demonstrate operating capability of equipment and systems.
- E. Include control and safety features, and service and maintenance procedures.
- F. Engage services of qualified instructor to instruct and train Owner's operating and maintenance personnel in operation, service, and maintenance of equipment.

3.9 PROTECTION

- A. Equipment shall be protected before, during and after installation. Damage to material due to improper protection shall be cause for rejection.

END OF SECTION

SECTION 11 70 00

HEALTHCARE EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Related Documents: The Equipment List is a multi-volume book of medical and hospital specialty equipment that contains information on items that are either Owner furnished/Owner installed or Owner furnished/Contractor Installed. Contractor furnished/Contractor installed equipment is specified in the Project Manual.
- B. Sections Includes:
 - 1. Owner furnished, Contractor installed healthcare equipment.
 - 2. Owner furnished, Owner installed healthcare equipment.
 - 3. List of Architecturally Significant Equipment with matrix as to who furnishes and who installs.
- C. Responsibility requirements for furnishing and installing healthcare equipment.

1.2 OWNER FURNISHED PRODUCTS

- A. Owner Responsibilities:
 - 1. Arrange for delivery of product data, shop drawings, samples, manufacturer's installation instructions, certificates, and other pertinent data to Contractor.
 - 2. Deliver supplier's bill of materials to Contractor.
 - 3. Arrange and pay for delivery to site in accordance with Construction Progress Schedule.
 - 4. Inspect deliveries jointly with Contractor.
 - 5. Submit claims for transportation damage.
 - 6. Arrange for replacement of damaged, defective, or missing items.
 - 7. Arrange for manufacturer's field services; arrange for and deliver manufacturer's warranties and bonds to Contractor.
- B. Contractor Responsibilities:
 - 1. Indicate submittal and delivery dates for each product in Construction Progress Schedule.
 - 2. Review product data, shop drawings, samples, and other submittals. Forward submittals to Architect with notification of any observed discrepancies or problems anticipated due to non-conformance with Contract Documents.
 - 3. Receive and unload products at site.
- C. Inspect deliveries jointly with Owner; record shortages, and damaged or defective items.
 - 1. Handle products at site, including uncrating and storage.
 - 2. Protect products from damage and from exposure to elements.
 - 3. Assemble, install, connect, adjust, and finish products as specified.
 - 4. Provide installation inspections required by public authorities.
 - 5. Repair or replace items damaged or lost by Contractor.

1.3 DIVIDED RESPONSIBILITY ON FURNISHING AND INSTALLING CERTAIN PRODUCTS

- A. OF/OI - Owner Furnished and Owner Installed Products:
 - 1. Owner Responsibilities:
 - a. Furnish Contractor with quantity of time required for product installation.

- b. Furnish Contractor with product literature, blocking requirements, and rough-in dimensions in accordance with Construction Progress Schedule.
 - c. Coordinate delivery time and location with Contractor. Arrange and pay for delivery to site.
 - d. Receive and unload products at site.
 - e. Handle products at site, including uncrating and storage. Furnish manufacturer's field services as necessary.
 - f. Protect products from damage.
 - g. Assemble and install as stipulated in manufacturer's written instructions.
 - h. Perform final connection and make adjustments.
 - i. Provide installation inspections required by public authorities.
2. Contractor Responsibilities:
- a. Construction Progress Schedule: Establish deadlines for receipt of Owner's product literature, blocking requirements, and rough-in requirements.
 - b. Include Owner's product installation time periods.
 - c. Establish on-site staging areas for Owner's product receipt and installation.
 - d. Provide necessary blocking and utility rough-ins required by product literature for proper installation.
 - e. Coordinate supplying and energizing utilities.
- B. OF/CI - Owner Furnished and Contractor Installed Products:
1. Owner Responsibilities:
- a. Furnish Contractor with product data, shop drawings, samples, manufacturer's installation instructions and anchorage requirements, certificates, utility rough-in requirements, and other pertinent data in coordination with Construction Progress Schedule.
 - b. Deliver supplier's bill of materials to Contractor.
 - c. Arrange and pay for delivery to site in accordance with Construction Progress Schedule. Inspect deliveries jointly with Contractor.
 - d. Submit claims for transportation damage.
 - e. Arrange for replacement of damaged, defective, or missing products. Arrange for manufacturer's field services.
 - f. Provide testing, calibration, and adjusting of installed products.
2. Contractor Responsibilities:
- a. Construction Progress Schedule: Establish deadlines for receipt of Owner's product literature, blocking requirements, and rough-in requirements.
 - b. Indicate submittal and delivery dates for each product.
 - c. Review product data, shop drawings, samples, and other submittals from Owner. Notify Owner of observed discrepancies or problems anticipated.
 - d. Receive and unload products at site.
 - e. Inspect deliveries jointly with Owner; record shortages, and damaged or defective products. Handle products at site, including uncrating and storage.
 - f. Protect products from damage and from exposure to elements. Provide necessary blocking required for proper installation.
 - g. Assemble, install, and connect products as required in manufacturer's written instructions and respective Specification section.
 - h. Provide installation inspections required by public authorities. Repair or replace products damaged or lost by Contractor.
- C. CF/CI - Contractor Furnished and Contractor Installed Products: Comply with contract documents.

1.4 SUBMITTALS

- A. General: Submit in accordance with Section 01 33 00.
- B. Product Data: Submit for each product.
 - 1. Include data to indicate standard mounting and utility connection details.
 - 2. Include information for factory finishes, hardware, glass, sealants, accessories and other required components.
 - 3. Include color charts for color finished items indicating manufacturer's full range of colors available for selection.
 - 4. Include wiring diagrams and rough-in requirements for items requiring electrical connections.
- C. Shop Drawings: Provide for non-standard custom-fabricated items.
 - 1. Indicate typical layout including dimensions, mounting locations and sizes, service accesses, utility connections, mounting sequences, and division of installation responsibilities.
 - 2. Submit detail drawings of non-standard mounting details and utility connections.
 - 3. Submit detail drawings of special accessory components not included in manufacturer's product data.
- D. Samples: Provide color samples for items which have colored finish.
 - 1. Submit 6 by 6 inch in size illustrating finish colors.
- E. Informational Submittals: Submit following packaged separately from other submittals:
 - 1. Qualification Data for manufacturer's and installer.
 - 2. Manufacturer's Instructions: Manufacturer's printed installation instructions.
- F. Manufacturer's Field Reports: Written results and findings of manufacturer's field services specified as part of Field Quality Control.
 - 1. Provide when required by product description.
- G. Closeout Submittals: Submit following in accordance with Section 01 78 00.
 - 1. Operation and Maintenance Data: Manufacturer's printed, recommended operation and maintenance data.
- H. Provide when specified by product description.
 - 1. Warranty.

1.5 QUALITY ASSURANCE

- A. Single Source Responsibility: Furnish each product type from one manufacturer for entire Project, unless otherwise acceptable to Architect.
 - 1. Provide each product as complete unit, including accessory items necessary for proper operation.
- B. Manufacturer Qualifications: Company specializing in manufacturing Products specified in this Section with minimum five years experience.
- C. Installer Qualifications: Acceptable to manufacturer with experience on at least five projects of similar nature in past five years.
 - 1. Where required by manufacturer, provide factory-trained installers.
- D. Conduct pre-installation conference in accordance with Section 01 31 19.
- E. The Contractor will be responsible for scheduling equipment installations with completed rough-in requirements, coordinating equipment requirements with Owner and various manufacturers
- F. and protection of installed equipment.

- G. Finished installation of a piece of equipment means that that item will function to the satisfaction of the Owner and will function within the manufacturer's stated parameters
- H. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.
 - 1. American National Standards Institute/Association for the Advancement of Medical Instrumentation (ANSI/AAMI) for installation of equipment listed.
 - 2. National Association of Architectural Metal Manufacturers (NAAMM):
 - 3. Metal Finishes Manual for Architectural and Metal Products

1.6 PRE-INSTALLATION CONFERENCE

- A. Conduct pre-installation conference in accordance with Section 01 31 19.
- B. Convene pre-installation conference one week prior to commencing work of this Section when specified under product description.
- C. Attendance Required: Architect, Owner, Contractor, manufacturer's representative, and installer.
- D. Agenda: Discuss and agree upon acceptable substrate and mounting conditions, preparatory work, utility connections, and methods of installation.

1.7 SEQUENCING

- A. General: Sequence work in accordance with Section 01 10 00.
- B. Prior to fabrication of mounting plates, furnish mounting plate templates to trades installing structure to support mounting plates.
- C. Install mounting plates to structural supports prior to covering-up by subsequent construction operations.

1.8 WARRANTY

- A. Special Warranty: Prepare and submit in accordance with Section 01 78 00.
 - 1. Warrant installed products to be free from defects in material and workmanship for time period specified under product descriptions.

1.9 MAINTENANCE

- A. Service: Provide service and maintenance of specific products as specified under product descriptions for specified time period from date of Substantial Completion in accordance with Section 01 92 00.

1.10 EXTRA STOCK MATERIALS

- A. Furnish in accordance with Section 01 78 00.
 - 1. Furnish as specified by product descriptions.

PART 2 - PRODUCTS

2.1 REFER TO EQUIPMENT MATRIX.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with work in accordance with Section 01 73 00.
 - 1. Verify utility connections are installed.
 - 2. Verify mounting brackets, plates, and supports are installed.

3.2 INSTALLATION

- A. Install in accordance with manufacturer's printed instructions.
- B. Install equipment plumb, level, square, and free from warp or twist while maintaining dimensional tolerances and alignment with surrounding construction.
- C. Provide anchorages in conformance with recommendations of the equipment manufacturer and as shown on approved shop drawings and calculations.
- D. Refer to Equipment Item Index for:
 - 1. Mounting heights.
 - 2. Mounting and anchoring details.
- E. Related Requirements:
 - 1. Plumbing Connections which may include treated and/or potable water systems supply and waste and waste water treatment
 - 2. Connections to Compressed Air System
 - 3. Electrical Connections
 - 4. Steam Connections.
 - 5. Natural Gas Connections
 - 6. Monitoring and alarm connections; connections to data networks:.
 - 7. Direct or indirect connections to Fire alarm monitoring or extinguishing systems
 - 8. Monitoring and connection to UPS systems, units or emergency power.

3.3 FIELD QUALITY CONTROL

- A. General: Comply with requirements of Section 01 40 00.
- B. Field test installed equipment after utilities are energized, pressurized or fully activated for proper operation of equipment. Use of temporary utilities will not be acceptable for testing.
 - 1. Operate each unit for one hour through repeated full cycles. During and after testing, there shall be no evidence of leaks, overheating, electrical failure, excessive vibration or other symptom of failure.
 - 2. For units that fail testing, make adjustments and corrections to installation, or inform Owner for replacement of equipment and repeat tests until equipment complies with requirements.

3.4 ADJUSTING, CLEANING, AND PROTECTION

- A. Adjusting: Adjust parts for smooth, uniform operation.
- B. Cleaning: Comply with Section 01 74 00. Remove protective covering from pre-finished items.
 - 1. Clean as recommended by manufacturer. Do not use materials or methods which may damage finish or surrounding construction.
 - 2. Touch-up minor surface coating damaged during installation; replace damaged units as directed by Architect.
- C. Protect finished work in accordance with Section 01 73 00.

3.5 DEMONSTRATION

- A. Demonstration and Instruction of Owner's Personnel: Provide in accordance with Section 01 92 00.
 - 1. Demonstrate equipment operation, maintenance and trouble shooting procedures and techniques.
 - 2. Provide as specified under product descriptions.

END OF SECTION

APPENDIX: RESPONSIBILITY MATRIX

SECTION 12 35 53
LABORATORY CASEWORK AND OTHER FURNISHINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Metal Laboratory Casework, Tables, and Casework Systems:
 - a. Metal Laboratory Casework.
 - b. Adaptable Laboratory Furniture Systems.
 - 2. Stainless Steel Laboratory Casework and Tables.
 - 3. Laboratory Work Surfaces.
 - 4. Balance Tables.
 - 5. Shelving Assemblies.
 - 6. Cylinder Restraint Assembly.
 - 7. Ceiling Service Panels.
 - 8. Drying Rack.
 - 9. Task Light Fixtures.
 - 10. Blackout Curtain and Track Assembly.
 - 11. Stainless Steel Fabrications:
 - a. Work Surfaces.
 - b. Laboratory Sinks and Cup Sinks.
 - c. Canopy Hoods.
 - 12. Sealant.
- B. Completely coordinate with work of other trades.

1.2 REFERENCES

- A. Scientific Equipment and Furniture Association (SEFA):
 - 1. SEFA 2 Installation.
 - 2. SEFA 3 Work Surfaces.
 - 3. SEFA 7 Fixtures.
 - 4. SEFA 8-M Laboratory Grade Metal Casework.
- B. American National Standards Institute (ANSI):
 - 1. ANSI/BHMA A156.9 Cabinet Hardware Standards, Grade 1.

1.3 SUBMITTALS

- A. Product Data:
 - 1. Submit materials list, including catalogue data, of materials, equipment, and products for work in this section.
- B. Shop Drawings:
 - 1. Shop fabrication and installation drawings, including plans, elevations, sections, details, and schedules.
 - 2. Show relationship to adjoining materials and construction.
- C. Project Information:
 - 1. Structural:

- a. Submit detailed anchorage and attachment drawings and calculations to show compliance with seismic restraint requirements.
 - b. Engineering design shall be performed and sealed by registered Engineer, licensed to practice structural engineering in the state of the project location.
 - c. Refer to the Structural Drawings for seismic design criteria.
2. Statement of Installer Qualifications.
3. Load Tests: Provide on request, load test results certified by an independent testing laboratory for cabinet box, drawers, doors, suspensions slides, and unit shelving as identified in SEFA 8.
4. Certificates:
 - a. Certify that factory tests specified for mechanical service fixtures have been performed and that products or systems meet or exceed specified requirements.
 - b. As a condition of acceptance, submit certification stating that equipment is complete and ready for intended function.
- D. Contract Closeout Information:
 1. Operations/Maintenance Manuals: Accompanying certification, submit for Architect's review and Owner's use, including:
 - a. Complete operating and maintenance manuals that describe proper operating procedures.
 - b. Maintenance and replacement schedules.
 - c. Component parts list.
 - d. Nearest factory representative for components and service.

1.4 QUALITY ASSURANCE

- A. Qualifications:
 1. Work in this section shall be performed by a company having a minimum of eight (8) years documented experience manufacturing the respective products specified herein, and an established organization and production facilities including tools, equipment, and special machinery necessary for the fabrication and installation of the type of equipment required, with skilled personnel, factory trained workmen and an experienced engineering department. Each shall have the demonstrated knowledge, ability, and the proven capability to produce the specified equipment of the required quality and the proven capacity to complete an installation of this size and type within the required time limits. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.
 2. Casework manufacturer must have at least one project in past twelve (12) months where the value of the laboratory casework was within 20 percent of the cost of the laboratory casework for this project.
 3. Casework installers shall be approved in writing by the casework manufacturer for the installation of specified products.
 4. Work shall be in accordance with the Grade or Grades specified of the Architectural Woodwork Standards for applicable products.
 5. Fabricators of wood, plastic laminate, and solid phenolic casework shall be accredited through the AWI Quality Certification Program (QCP).
- B. Undivided Responsibility:
 1. Unless specified otherwise, because of special coordination requirements, the supplier of scope of work described in this Section shall also provide scope of work described in following Sections:
 2. Section 11 53 43 Laboratory Service Fittings and Fixtures.
- C. Conform to recommended practices of the Scientific Equipment and Furniture Association (SEFA), current version, except as superseded by this specification:

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Contractor shall schedule the delivery of casework and furnishings when spaces are sufficiently complete so materials can be installed immediately following delivery.
- B. Delivery of laboratory casework shall be made only when the area of operation is enclosed, plaster and concrete work is dry, and the area broom clean.
- C. Product shall be stored in a clean storage area.

1.6 FIELD CONDITIONS

- A. Field verify all dimensions prior to fabrication of casework.
- B. Casework and furnishings shall not be delivered and installed prior to completion of the following items:
 - 1. Windows and doors shall be installed, and the building shall be weather-tight.
 - 2. Finished ceilings, if specified, overhead ductwork, piping, electrical, and lighting work shall be installed.
 - 3. Painting shall be complete.
 - 4. Flooring shall be installed, except when an integral base is specified to be installed over the casework toe kick.
- C. Maintain interior building temperature between 65 deg F and 80 deg F.
- D. Maintain ambient relative humidity between 25 and 55 percent prior to delivery, during installation, and for the remainder of the project.
- E. Frequent and/or excessive changes in temperature and/or humidity levels during casework installation, or once casework is installed, must be avoided to prevent damage to materials.

PART 2 - PRODUCTS

2.1 CASEWORK DESIGN AND PERFORMANCE CRITERIA

- A. Cabinets may be manufacturer's standard height and depth, provided such standard is not more or less than 1/2 inches of the height and depth indicated on the Drawings, unless noted otherwise in this Section.
- B. System Structural Performance:
 - 1. Laboratory casework and support framing system shall withstand the effects of the following loads and stresses without permanent deformation, excessive deflection, or binding of drawers and doors:
 - a. Casework, doors, drawers, work surfaces and shelving shall comply with SEFA requirements for the respective casework material.
 - b. Work Surfaces: In addition to SEFA test requirements, work surface spans without continuous base cabinet support shall support 50 psf; deflection shall be limited to 1/180 of the length of the span, not to exceed 1/4 inches.
 - c. Shelving: In addition to SEFA test requirements:
 - 1) Shelf spans without continuous support shall support 50 psf.
 - a) Polypropylene shelving shall comply with SEFA loading requirements.
 - 2) Deflection shall be limited to 1/180 of the length of the span, not to exceed 1/4 inches.
 - d. Seismic Anchor: Provide seismic anchor for freestanding cabinets and cabinets located below fume hoods designated to be removable for access for the disabled. Seismic anchors may be floor or wall attachments but shall not attach to adjacent casework or work surfaces. Seismic anchors shall be accessible without removal of laboratory casework, furnishings, or equipment. Anchor attachment shall not void UL listing.

2.2 METAL LABORATORY CASEWORK, TABLES AND CASEWORK SYSTEMS

A. Manufacturers:

1. Laboratory Casework:
 - a. Kewaunee Scientific Corporation.
 - b. Lab Crafters, Inc.
 - c. A.T. Villa USA, Inc.
 - d. Bedcolab Ltd.
 - e. Mott Manufacturing Limited.
 - f. Other manufacturers desiring approval comply with Section 01 61 00.
 - g. Substitutions are not permitted.

B. Components:

1. Cold rolled sheet steel:
 - a. Recycled Steel Content: A minimum of 20 percent of the steel used in the fabrication of laboratory cabinets and modular laboratory systems, if specified, shall consist of the sum of post-consumer recycled content plus one-half of the pre-consumer content, based on the cost of the total value of materials.
 - b. Fabricators Scrap: Fabricators shall provide documentation that manufacturing fall-off is recycled to respective steel mills and neither enters the solid waste system nor becomes a product of landfill space.
 - c. Prime grade, roller leveled, and treated at the mill to be free of scale, ragged edges, deep scratches, or other injurious effects. Thickness of metal used in construction of cases shall be 18 GA, except as follows:
 - 1) 20 GA: Solid door interior panels, drawer fronts, scribe strips, filler panels, enclosures, drawer bodies, shelves, security panels, and sloping tops.
 - 2) 16 GA: Top front rails, top rear gussets, intermediate horizontal rails, table legs and frames, leg rails and stretchers.
 - 3) 14 GA: Drawer suspensions, door and case hinge reinforcements, and front corner reinforcements.
 - 4) 12 GA: Table leg corner brackets and gussets for leveling screws.
 - d. Metal Casework Finish Requirements: Refer to Painted Metal Finish Performance Requirements elsewhere in this Section
2. Hardware: As specified elsewhere in this Section.

C. Design Requirements:

1. Door and Drawer Design:
 - a. Full flush overlay:
 - 1) Provide applied panels in areas such as sink cabinets and knee spaces with apron panels to provide a full flush overlay appearance.
 - a) Apron panels shall align with plane of adjacent doors and drawers for full flush overlay appearance.
 - 2) Material:
 - a) Painted Metal.
2. Pulls on doors shall be mounted vertically and on drawers horizontally.

D. Fabrication:

1. General:
 - a. Cabinets shall be constructed and finished to be suitable for use as stand-alone units and to permit future rearrangement without the need for additional parts or finish.
 - b. Units shall have a cleanable smooth interior. Front and rear posts, reinforcing members or channel uprights shall be enclosed full heights on cabinet openings.

- c. Exterior corners: shall be spot and arc welded with gussets at exterior corners. Face joints shall be arc welded and ground smooth to provide a continuous flat plane.
 - d. Units less than 49 inches tall: Provide internal reinforcing and rear posts for end panels and cabinet backs.
 - e. Units 49 inches tall and greater: Provide formed end panels with front and rear reinforcing posts. Back shall be formed steel panel, recessed 3/4 inches for mounting purposes.
 - f. Posts: Front post fully closed with full height reinforcing upright to facilitate cleaning.
 - g. The front edge shall be formed to provide a strike for doors and drawers and shall be pre-drilled for intermediate rails and hinge screws.
 - h. Intermediate Vertical Uprights: shall be furnished to enclose cabinets when used in a unit in combination with a half width bank of drawers. However, to allow storage of large or bulky objects, no upright of any type shall be used at the center of double door cupboard units.
 - i. End Uprights shall be formed into not less than a channel formation at top, bottom, back and front.
 - j. The upright filler shall be perforated with shelf adjustment holes at no more than 2 inches centers.
 - k. The inside front of the upright shall be further reinforced with a full height 14 GA hinge reinforcement angle.
 - l. Exposed fasteners are not allowed without prior approval of the Architect. If allowed, exposed fasteners in BSL-3, ABSL-3, BSL-3Ag, BSL-4 or ABSL-4 shall be Type 304 or 316 stainless steel or protected with epoxy paint or sealant.
2. Drawer Construction:
- a. Drawer bodies shall be made in one-piece construction including the bottom, two sides, back and inner front. Front and back faces should be spot welded to sides or center section.
 - b. Fully coved at interior bottom on all four sides for easy cleaning.
 - c. Full height sides.
 - d. 1/2 inch side clearance to frame opening.
 - e. Drawers shall be a minimum of 18 inches front to back.
 - f. Drawers shall be easily removable in the field without the use of special tools.
 - g. Drawers shall be sized on a modular basis for interchange to satisfy varying storage requirements.
 - h. Drawer Suspension: Refer to Drawer Slides under Hardware section.
 - i. Drawer Heads:
 - 1) Thickness to match doors.
 - 2) Widths of drawer front in aprons shall be lesser of the width of the apron or 24 inches.
 - 3) Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel formation on all four sides welded and ground to eliminate exposure of sharp raw edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.
 - a) Welds shall be ground smooth.
 - b) Painted inside and out prior to assembly.
3. Doors: Doors shall be readily removable and hinges easily replaceable. Hinges shall be applied to the case and door with screws. Welding of hinges to either case or door will not be acceptable.
- a. Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel

formation on all four sides welded and ground to eliminate exposure of sharp raw edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.

- 1) Welds shall be ground smooth.
- 2) Painted inside and out prior to assembly.
- b. Unframed Sliding Glass Doors: Glass with ground edges shall be set in extruded Aluminum shoe with integral pulls, wheel assemblies and top and bottom extruded Aluminum track. Provide rubber bumpers at fully opened and closed door position.
4. Cabinet Bottom:
 - a. Case bottom and bottom rail shall be formed of one piece of metal except in corner units and shall have both sides and back formed up or down and shall be rabbeted in front for drawers and swinging doors.
5. Cabinet Top:
 - a. Wall and Tall Case Top: One-piece, with front edge formed into front rail.
 - b. Provide sloping tops when indicated on the Drawings.
6. Cabinet Back, Unexposed: Cabinet back shall consist of a top and bottom rail, channel formed for maximum strength and welded to back and top flange of end uprights, with space between left open for access to plumbing lines.
 - a. Provide removable back panels for all fixed base cabinets. Removable panels shall not be provided in BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 installations.
 - b. Drawer cabinets shall have split back panel allowing for removal from cabinet.
 - c. Sink base back shall be full half-height construction to allow for plumbing and sink waste connection.
 - d. Back panels shall be removable from inside cabinet without the use of special tools.
7. Toe Bases, Kicks, and Sleepers:
 - a. Toe Space Rail:
 - 1) Provide 3 inches deep and 4 inches high formed steel base with corner gussets. Whenever the base is omitted for units to be set on building bases or separate metal bases, the toe space rail shall extend back 4-1/2 inches.
 - 2) Provide 3/8 inches diameter leveling screw with integral bottom flange of minimum 0.56 inches² area at each corner, accessible through openings in toe space.
8. Shelves:
 - a. Full-depth turned down 1 inch on four sides, and returned under front and back 1 inch, nominal.
 - b. Shelves over 36 inches in length shall be additionally reinforced by a flanged channel shaped member electro-welded to underside of shelf.
 - c. Shelves shall be adjustable.
 - d. Shelf adjustment posts shall be perfectly aligned for level setting, with holes for shelf adjustment at 1/2 inches on center.
9. Suspended casework: Provide as detailed on Laboratory Furnishing drawings. Construction of suspended casework units shall be the same as floor mounted cabinets with the following exceptions:
 - a. Finish back and side panels as exposed surface.
 - b. Omit sub-base and provide casework with finished bottom panels.
 - c. Provide fixings and fastenings necessary for its attachment to the supporting structure.
 - d. Cabinet and suspending frame shall allow locating at any point between supports.
 - e. Top reinforcement to prevent cabinet top deflection.
10. Metal-Framed Laboratory Tables:
 - a. Legs: Not less than 2 inches by 2 inches by 16 GA square tubular steel sections with welded leg brackets.

- 1) Table legs shall be telescoping to allow vertical height adjustment of work surface from 30 inches to 36 inches above finished floor.
 - b. Leg rails and spreader rail: Not less than 1-1/4 inches by 2-1/2 inches by 16 GA steel sections, reinforced as necessary for leg attachment.
 - c. Aprons and Rails: 1-1/2 inches, minimum, by 4-1/2 inches, maximum, 16 GA channel steel sections, reinforced as necessary for leg attachment. Provide 1-1/2 inches by 2 inches by 16 GA channel steel sections, reinforced as necessary for leg attachment, where no drawers are required.
 - d. Work Surface: Refer to Laboratory Furnishing drawings for laboratory work surface materials, described in the Laboratory Work Surfaces section of this specification.
 - e. Electrical receptacles: Where indicated on the Laboratory Electrical drawings provide cutouts for electrical receptacles as work of this Section and coordinate with Division 26.
 - f. Cord and plug shall be provided under the work of Division 26.
 - g. Leveling Glide and Leg Shoe: Each leg, other than those fitted with casters, shall have a recessed leveling screw and a leg shoe.
- E. Self-Supporting Movable Workstation System:
1. Base Product: Kewaunee Enterprise System.
 - a. Alternates:
 - 1) Alternative manufacturers will be accepted as long as basic system requirements are satisfied.
 - 2) Alternate manufacturer's product data and description must be submitted to the Architect for approval during the Bid Period.
 - 3) Systems proposed after the Bid Date will not be considered.
 2. Description: A self-supporting movable laboratory workstation table design system. Modular system shall consist of tubular steel framing components to create support structures for tables and island, wall, or peninsula laboratory benches.
 - a. Frames structures shall support worksurfaces and shelving, and support distribution for piped services, power distribution, lighting, and communications cabling, as indicated on Drawings.
 - b. Structural Capacity:
 - 1) Rear Support Frame:
 - a) Single Sided: Three shelves; 540 pounds total.
 - b) Shared Frame: Three shelves/side; 1080 pounds total
 - 2) Table Frames: 100 pounds/LF; 800 pounds maximum.
 - 3) Shelves: 180 pounds each.
 - c. Compliance: UL 962.
 1. Components:
 - a. Rear Support Frames:
 - 1) Dimensions:
 - a) Width: Refer to Drawings.
 - b) Depth: 2 inches diameter.
 - c) Height: 84 inches.
 - 2) Uprights:
 - a) Provide center upright from frames 60 inches, or greater in width.
 - b) Leveling device.
 - c) Slotted for shelf brackets at 1 inch increments from 55 inches above finished floor to the top of the upright.

- d) Route power circuitry in one upright and piped services and data in other upright.
 - e) Provide one NEMA 5-20 receptacle in upright with power circuitry.
 - 3) Cross Rails:
 - a) Thickness: 11 GA.
 - b) House two-channel electrical raceway.
- b. Table Frames:
 - 1) Dimensions:
 - a) Width: Refer to Drawings.
 - b) Depth: 30 inches, nominal.
 - c) Height: Adjustable.
 - (1) Maximum low position: 30 inches.
 - (2) Minimum high position: 36 inches.
 - (3) Adjustable in 1 inch increments.
 - d) Thickness:
 - (1) Single Sided: 2 inches diameter tubular steel, 11 GA.
 - (2) Shared Frame: 2 inches by 6 inches, 14 GA formed steel with full height, removable cover.
 - 2) Tubular Legs:
 - a) Outer Tube: 2 inches outside diameter, 12 GA.
 - b) Inner Telescoping Tube: 1-3/4 inches, 11 GA.
 - c) Leveling device.
 - 3) Worksurface Frame:
 - a) Thickness: 11 GA brake formed.
 - b) Provide fittings to attach frame to legs.
 - c) Fasten to rear support frame with 3/8 inches bolts.
 - d) Provide hanger rails when suspended casework is specified.
- c. Worksurface: Refer to Drawings and worksurface specifications in this Section.
- d. Adjustable Shelves:
 - 1) Refer to shelving specifications elsewhere in this Section.
 - 2) Adjustable in 1 inch increments.
 - 3) Depth: 12 inches.
 - 4) Material:
 - a) Painted Metal:
- e. Casework:
 - 1) Type:
 - a) Provide mobile base cabinets, as specified elsewhere in this Section.
 - b) Provide suspended base cabinets, as specified elsewhere in this Section.
 - 2) Construction:
 - a) Painted Metal.
- f. Piped Services Distribution:
 - 1) Rear support frame shall house up to three piped services and associated fittings.
 - 2) Piped Service Distribution:
 - a) Internal Tubing:
 - (1) Single Frame: 1/4 inches diameter copper tubing.
 - (2) Shared Frame: 3/8 inches diameter copper tubing.

- (3) Tubing Material:
 - (a) Typical: Copper.
 - (b) Natural and Burning Gases: Stainless steel.
 - (4) Arrange so services cannot be intermixed.
 - (5) Quick connect termination at top of rear support frame.
 - b) Ceiling Panel-Connecting Service Hose:
 - (1) Non-Burning Hose: 3/8 inches coiled polyurethane hose.
 - (2) Non-Burning Hose: 3/8 inches reinforced PVC hose; color-coded.
 - (3) Natural and Burning, Reactive Gases: 3/8 inches braided stainless steel hose.
 - (4) Quick connect fitting at each end of hose.
 - 3) Quick Connect Fitting:
 - a) Provide on each tube at top of rear support frame.
 - b) Size: 3/8 inches.
 - c) Color coded to match service fittings.
 - 4) Laboratory Service Fittings: Mounted in rear support frame as specified in Section 11 53 43 and indicated on Drawings.
- g. Electrical:
- 1) Terminate power and communications cables extending 48 inches above top of rear support frame.
 - 2) Terminate power and communications cables extending 3m 10 feet above top of rear support frame.
 - 3) Power Cord: SJ 12/3 cord satisfying requirements of Division 26.
 - a) Plug: NEMA L5-20P.
 - 4) Power/Data Bar:
 - a) Extruded Aluminum Electrical Raceway:
 - (1) Single Frame, 60 inches wide and less: Provide two NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector.
 - (2) Single Frame, greater than 60 inches wide: Provide three NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector.
 - (3) Shared Frame, 60 inches wide and less: Provide two NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector on each side.
 - (4) Shared Frame, greater than 60 inches wide: Provide three NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector on each side.
 - 5) Electrical Receptacles:
 - a) NEMA 5-20R, duplex receptacle, black.
 - b) Provide GFIC type when located within 1.8m 6 feet of a sink.
 - c) Locate in lower portion of rear support frame where indicated on Drawings.
 - 6) Data Cable: Refer to Division 27.
 - a) Terminate cable with male cable end connector.
 - 7) Task Lighting: Refer to specifications in this Section

F. Adaptable Laboratory Furniture System:

- 1. Support structures
 - a. General:
 - 1) Uprights: 16 GA steel with levelers.

- 2) Frames: 16 GA welded rolled steel frame and tie rail brackets, with 14 GA corner gussets.
 - 3) Tie rails: 16 GA steel
 - 4) Base cover: 18 GA steel.
 - 5) Closure panels: 20 GA steel with dual lock fasteners.
 - 6) Upright and plug caps: Flame resistant ABS plastic, color-matched.
 - 7) Adjustable floor clamps: Two per core or frame; 80-55-06 ductile cast iron.
- b. Island cores:
- 1) Cores shall be capable of having removable uprights and reagent shelf or adding uprights and reagent shelves after installation. When secured to floor, core shall be capable of supporting minimum structural requirements without end riggers.
 - 2) Closure panels shall snap on without the use of tools and shall be removable without the removal of cantilever table frames or suspended cabinetry.
 - 3) Floor clamps shall be supplied with three rawl bolts for concrete deck anchorage.
 - 4) Service chase area: Minimum 5-1/2 inches by 18-1/4 inches between uprights and tie rails.
- c. Wall cores:
- 1) Service chase area: Minimum 5-1/2 inches by 18-1/4 inches between uprights and tie rails.
- d. Island panels:
- 1) Upper and lower panels shall be of the same materials and construction. Lower panel shall be capable of having upper panel removable or added after installation.
 - 2) Suspended components, including cantilevered table frames, wall cabinets, shelving, shall be vertically adjustable on 1 inch increments.
 - 3) Closure panels shall snap on without the use of tools and shall be removable without the removal of cantilever table frames or suspended cabinetry.
- e. Freestanding column:
- 1) Corner uprights and column covers: 18 GA steel.
 - 2) Upright connectors: 20 GA steel.
 - 3) Column shall be capable of connecting to island cores when assembled in L, T, or cross configurations. Column shall increase modularity of island core 12 inches. Column shall provide means of routing services from ceiling to an island core from 84 inches to 120 inches.
 - 4) When used with an island core, suspended components, including cantilevered table frames, wall cabinets, shelving, shall be vertically adjustable on 1 inch increments within the resulting module width.
- f. Service column:
- 1) Uprights: 20 GA steel.
 - 2) Angles: 18 GA steel.
 - 3) Columns shall be capable of enclosing services which enter through the center of an island core and shall be capable of being constructed around existing services.
 - 4) Closure panels shall snap on without tools.

G. Cantilevered Casework Metal Frames

1. Piped services requirements:
 - a. Upright frame shall house a minimum of three plumbing services per side.
 - b. Fittings:
 - 1) Service fittings: Refer to drawings and Section 11 53 43.
 - 2) Quick connect fittings; Refer to Section 11 53 43.
 - a) Hoses shall be arranged so services cannot be intermixed.

- 3) Service fittings and quick connects shall be color-coded and media-keyed so that connects cannot be switched.
- c. Non-flammable gases:
 - 1) 1/2 inches diameter, PVC flexible hose, connected to service fitting and with quick connect fitting at hose end; mechanical breaks in hose not allowed.
- d. Flammable gases:
 - 1) 9.5mm 3/8 inches diameter, No. 304 braided stainless steel hose, connected to service fitting and with quick connect fitting at hose end; mechanical breaks in hose not allowed.
2. Electrical Requirements:
 - a. Components shall be U.L. Listed.
 - b. Provide SJ 12/3 cord with NEMA L5-20P plug to extend to ceiling receptacle or other location as indicated on drawings.
 - c. Provide data and telephone cable if indicated on drawings. Refer to Division 26 for cable specifications.
 - d. Electrical cords and cables shall be continuous.

2.3 STAINLESS STEEL LABORATORY CASEWORK AND TABLES

A. Manufacturers:

1. Acceptable Laboratory Casework:
 - a. Mott Manufacturing Limited
 - b. Bedcolab Ltd.
 - c. Kewaunee Scientific Corporation.
 - d. Other manufacturers desiring approval comply with Section 01 61 00.
 - e. Substitutions are not permitted.
 - f. Other manufacturers desiring approval comply with Section 01 61 00.
 - g. Substitutions are not permitted.

B. Components:

1. Recycled steel content:
 - a. Minimum 20 percent of stainless steel used in fabrication of laboratory cabinets shall consist of sum of post-consumer recycled content plus one-half of pre-consumer content, based on cost of total value of materials.
2. Stainless sheet steel:
 - a. ASTM A240, Type 304 or Type 316.
 - b. Finish:
 - 1) Stainless steel casework shall be completely finished inside and out, whether exposed to view or not, to a No. 4 satin finish.
 - 2) All welds shall be made without discoloration and shall be ground, polished and passivated to match the surrounding surfaces.
3. Minimum sheet metal thickness:
 - a. 18 GA, except as follows:
 - b. 20 GA: Solid door interior panels, drawer fronts, scribe strips, filler panels, enclosures, drawer bodies, shelves, security panels, and sloping tops.
 - c. 16 GA: Top front rails, top rear gussets, intermediate horizontal rails, table legs and frames, leg rails and stretchers.
 - d. 14 GA: Drawer suspensions, door and case hinge reinforcements, and front corner reinforcements.
 - e. 12 GA: Table leg corner brackets and gussets for leveling screws.
4. Hardware: As specified elsewhere in this Section.

C. Design Requirements:

1. Pulls on doors shall be mounted vertically and on drawers horizontally.

D. Fabrication:

1. General:

- a. Cabinets shall be constructed and finished to be suitable for use as stand-alone units and to permit future rearrangement without the need for additional parts or finish.
- b. Units shall have a cleanable smooth interior.
 - 1) Front and rear posts, reinforcing members or channel uprights shall be enclosed full heights on cabinet openings.
- c. Exterior corners:
 - 1) Spot and arc welded with gussets at exterior corners. Face joints shall be arc welded and ground smooth to provide a continuous flat plane.
- d. Units less than 1245mm 49 inches tall:
 - 1) Provide internal reinforcing and rear posts for end panels and cabinet backs.
- e. Units 1245mm 49 inches tall and greater:
 - 1) Provide formed end panels with front and rear reinforcing posts.
 - 2) Back shall be formed steel panel, recessed 3/4 inches for mounting purposes.
- f. Posts:
 - 1) Front post fully closed with full height reinforcing upright to facilitate cleaning.
- g. Form front edge to provide door and drawer strikes.
 - 1) Pre-drill for intermediate rails and hinge screws.
- h. Intermediate Vertical Uprights:
 - 1) Furnish to enclose cabinets when used in a unit in combination with a half width bank of drawers.
 - 2) To allow storage of large or bulky objects, no upright of any type shall be used at the center of double door cupboard units.
- i. End Uprights shall be formed into not less than a channel formation at top, bottom, back and front.
- j. An upright filler shall be screwed in place in wall cabinet units to close the back of the channel at front of the upright and to provide a smooth interior for the wall cabinet to facilitate cleaning.
- k. The upright filler shall be perforated with shelf adjustment holes at no more than 51mm /2 inches centers.
- l. The inside front of the upright shall be further reinforced with a full height 14 GA hinge reinforcement angle.
- m. Exposed fasteners are not allowed without prior approval of the Architect. If allowed, exposed fasteners shall be Type 304 or 316 stainless steel.

2. Drawer Construction:

- a. Drawer bodies shall be made in one-piece construction including the bottom, two sides, back and inner front.
- b. Fully coved at interior bottom on all four sides for easy cleaning.
- c. Full height sides.
- d. 1/2 inches side clearance to frame opening.
- e. Drawers shall be a minimum of 18 inches front to back.
- f. Drawers shall be easily removable in the field without the use of special tools.
- g. Drawers shall be sized on a modular basis for interchange to satisfy varying storage requirements.
- h. Drawer Suspension:

- 1) Refer to Drawer Slides under Hardware section.
- i. Drawer Heads:
 - 1) Thickness to match doors.
 - 2) Widths of drawer front in aprons shall be lesser of the width of the apron or 24 inches; for longer aprons, provide false fronts on either side of drawer front when flush overlay is specified.
 - 3) Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel formation on all four sides welded and ground to eliminate exposure of sharp raw edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.
 - a) Welds shall be ground smooth.
3. Doors shall be readily removable and hinges easily replaceable.
 - a. Hinges shall be applied to the case and door with screws.
 - b. Welding of hinges to either case or door will not be acceptable.
 - c. Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel formation on four sides welded and ground to eliminate exposure of sharp raw edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.
 - 1) Welds shall be ground smooth.
4. Cabinet Bottom:
 - a. Case bottom and bottom rail shall be formed of one piece of metal except in corner units and shall have both sides and back formed up or down and shall be rabbeted in front for drawers and swinging doors.
5. Cabinet Top:
 - a. Wall and Tall Case Top:
 - 1) One-piece, with front edge formed into front rail.
 - b. Provide sloping tops when indicated on the Drawings.
6. Cabinet Back, Concealed:
 - a. Cabinet back shall consist of a top and bottom rail, channel formed for maximum strength and welded to back and top flange of end uprights, with space between left open for access to plumbing lines.
 - b. Provide removable back panels for fixed base cabinets.
 - 1) Removable panels shall not be provided in BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 installations.
 - c. Drawer cabinets shall have split back panel allowing for removal from cabinet.
 - d. Sink base back shall be full half-height construction to allow for plumbing and sink waste connection.
 - e. Back panels shall be removable from inside cabinet without the use of special tools.
7. Toe Bases, Kicks, and Sleepers:
 - a. Toe Space Rail:
 - 1) Provide 3 inches deep and 4 inches high formed steel base with corner gussets.
 - a) Whenever the base is omitted for units to be set on building bases or separate metal bases, the toe space rail shall extend back 4-1/2 inches.
 - 2) Provide 3/8 inches diameter leveling screw with integral bottom flange of minimum 3.6cm² 0.56 inches² area at each corner, accessible through openings in toe space.
8. Shelves:

- a. Full-depth turned down 1 inch on four sides, and returned under front and back 1 inch, nominal.
 - b. Shelves over 36 inches in length shall be additionally reinforced by a flanged channel shaped member electro-welded to underside of shelf.
 - c. Shelves shall be adjustable.
 - d. Shelf adjustment posts shall be perfectly aligned for level setting, with holes for shelf adjustment at 1/2 inches on center.
9. Suspended casework:
- a. Provide as detailed on Laboratory Furnishing drawings. Construction of suspended casework units shall be the same as floor mounted cabinets with the following exceptions:
 - 1) Omit sub-base and provide casework with finished bottom panels.
 - 2) Fixings and fastenings necessary for its attachment to the supporting structure, including any necessary wall rails and work surface and casework supports.
 - 3) Top reinforcement to prevent cabinet top deflection.
 - 4) Cabinet and suspending frame shall allow locating at any point between supports.
10. Stainless Steel-Framed Laboratory Tables:
- a. Legs: Not less than 2 inches by 2 inches by 16 GA square tubular stainless steel sections with welded leg brackets.
 - 1) Table legs shall be telescoping to allow vertical height adjustment of work surface from 30 inches to 36 inches above finished floor.
 - b. Leg rails and spreader rail:
 - 1) Not less than 1-1/4 inches by 2-1/2 inches by 16 GA steel sections, reinforced as necessary for leg attachment.
 - c. Aprons and Rails:
 - 1) 1-1/2 inches, minimum, by 4-1/2 inches, maximum, 16 GA channel stainless steel sections, reinforced as necessary for leg attachment.
 - 2) Provide 1-1/2 inches by 2 inches by 16 GA channel stainless steel sections, reinforced as necessary for leg attachment, where no drawers are required.
 - d. Work Surface:
 - 1) Refer to Laboratory Furnishing drawings for worktop materials, described in the Laboratory Work Surfaces section of this specification.
 - e. Electrical receptacles:
 - 1) Where indicated on the Laboratory Electrical drawings provide cutouts for electrical receptacles as work of this Section and coordinate with Division 26.
 - f. Cord and plug shall be provided under the work of Division 26.
 - g. Leveling Glide and Leg Shoe:
 - 1) Each leg, other than those fitted with casters, shall have a recessed leveling screw and a leg shoe.
 - h. Casters:
 - 1) Where indicated on Laboratory Furnishing drawings or specifications, as specified under Cabinet Hardware.
 - i. Table Drawers:
 - 1) Where indicated on the drawings, provide front and back rails; drawer unit, hardware, and suspension same as specified for casework base unit drawers.
- E. Acceptable Self-Supporting Movable Workstation System:
- 1. Piped Services Distribution:
 - a. Rear support frame shall house up to three piped services and associated fittings.
 - b. Piped Service Distribution:

- 1) Internal Tubing:
 - a) Single Frame: 1/4 inches diameter copper tubing.
 - b) Shared Frame: 3/8 inches diameter copper tubing.
 - c) Tubing Material:
 - (1) Typical: Copper.
 - (2) Natural and Burning Gases: Stainless steel.
 - d) Arrange so services cannot be intermixed.
 - e) Quick connect termination at top of rear support frame.
 - 2) Ceiling Panel-Connecting Service Hose:
 - a) Non-Burning Hose: 3/8 inches coiled polyurethane hose.
 - b) Non-Burning Hose: 3/8 inches reinforced PVC hose; color-coded.
 - c) Natural and Burning, Reactive Gases: 3/8 inches braided stainless steel hose.
 - d) Quick connect fitting at each end of hose.
 - c. Quick Connect Fitting:
 - 1) Provide on each tube at top of rear support frame.
 - 2) Size: 3/8 inches.
 - 3) Color coded to match service fittings.
 - d. Laboratory Service Fittings: Mounted in rear support frame as specified in Section 11 53 43 and indicated on Drawings.
2. Electrical:
- a. Terminate power and communications cables extending 48 inches above top of rear support frame.
 - b. Terminate power and communications cables extending 3m 10 feet above top of rear support frame.
 - c. Power Cord: SJ 12/3 cord satisfying requirements of Division 26.
 - 1) Plug: NEMA L5-20P.
 - d. Power/Data Bar:
 - 1) Extruded Aluminum Electrical Raceway:
 - a) Single Frame, 60 inches wide and less: Provide two NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector.
 - b) Single Frame, greater than 60 inches wide: Provide three NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector.
 - c) Shared Frame, 60 inches wide and less: Provide two NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector on each side.
 - d) Shared Frame, greater than 60 inches wide: Provide three NEMA 5-20R, duplex receptacles, black, and one black RJ45 duplex female data connector on each side.
 - e. Electrical Receptacles:
 - 1) NEMA 5-20R, duplex receptacle, black.
 - 2) Provide GFIC type when located within 1.8m 6 feet of a sink.
 - 3) Locate in lower portion of rear support frame where indicated on Drawings.
 - f. Data Cable: Refer to Division 27.
 - 1) Terminate cable with male cable end connector.
 - g. Task Lighting: Refer to specifications in this Section

2.4 CABINET HARDWARE

A. General:

1. Special metal cabinets, such as corrosives storage, flammable liquid and solvent storage, rock storage, map storage, museum storage, radioisotope storage, and narcotics locker, may be provided with the manufacturer's standard hardware.
 2. Exception: Door and drawer pulls shall match, regardless of type of casework, except for polypropylene casework, if specified.
 - a. Polypropylene casework hardware is specified under Polypropylene Casework.
- B. Hinges:
1. General:
 - a. Satisfy ANSI/BHMA Grade 1 requirements.
 2. Five Knuckle Hinges:
 - a. Characteristics:
 - 1) Institutional grade butt hinge.
 - 2) Hospital tip.
 - 3) Edge mounted.
 - 4) Height: 70mm 2-3/4 inches, nominal.
 - 5) Thickness: 2.4mm 0.095 inches.
 - 6) Material:
 - a) Steel with US26D satin chrome finish with zinc-plated screws.
 3. Semi-Concealed Hinges for Flush Overlay Doors:
 - a. Three-way adjustable clip-on institutional quality hinge with built in catch.
 - 1) With door in closed position, hinge site line shall be limited to the 5mm diameter knuckle; remainder of hinge should be concealed.
 - 2) Back mounted.
 - 3) Roller catch is not required when hinge has built-in catch.
- C. Shelving Pilaster Standards and Clips:
1. Manufacturers:
 - a. Knape & Vogt Manufacturing Co.
 - b. Bainbridge Manufacturing, Inc.
 - c. The Engineered Products Company (EpcO).
 - d. Fixture Hardware Manufacturing.
 - e. Rockford Process Control, Inc.
 - f. Sugatsune America, Inc
 - g. Other manufacturers desiring approval comply with Section 01 61 00.
 2. The following systems are acceptable:
 - a. Surface Mounted Pilaster Standard for painted metal, stainless steel casework, only:
 - 1) 23 GA high strength steel.
 - 2) 21mm 13/16 inches wide by 3/16 inches deep.
 - 3) Pre-drilled or pre-punched mounting holes.
 - 4) Zinc finish.
 - 5) Shelf clip with vertical upper tab and horizontal lower tab to engage two slots in pilaster standard.
 - a) Clip shall be profiled to engage sides of pilaster standard.
 - b) Zinc finish.
 - b. Mortised Mounted Pilaster Standard for wood, plastic laminate, solid phenolic casework, only:
 - 1) 23 GA high strength steel.

- 2) 5/8 inches wide by 3/16 inches deep.
 - 3) Pre-drilled or pre-punched mounting holes.
 - 4) Zinc finish.
 - 5) Shelf clip with vertical upper tab and horizontal lower tab to engage two slots in pilaster standard.
 - a) Clip shall install flush against face of pilaster standard.
 - b) Zinc finish
 - c. Die-formed, zinc-plated steel, designed to engage in shelf adjustment holes in side of cabinet.
 - d. Where casework requires seismic restraint:
 - 1) Wood plastic laminate and/or solid phenolic cabinets: Adjustable plastic shelf support with lockdown clips designed to engage in shelf adjustment holes in side of cabinet.
 - 2) Metal shelves: Adjustable shelf clip support with lockdown clips designed to engage in shelf adjustment holes in side of cabinet or pilaster standards, or shelves may be screwed to shelf clip. Shelves shall be vertically adjustable on 1/2 inches centers.
 - e. Shelf hardware shall support loading in accordance with Part 1.
- D. Ball Bearing Drawer Slides:
1. Manufacturers:
 - a. Base:
 - 1) Accuride.
 - b. Optional:
 - 1) Waterloo Furniture Components Inc.
 - 2) Fulterer USA.
 - 3) Other manufacturers desiring approval comply with Section 01 61 00.
 2. Typical Drawers:
 - a. Satisfy ANSI/BHMA Grade 1HD requirements.
 - b. Base Product:
 - 1) For drawer widths up to 18 inches: Accuride 4033.
 - 2) For drawer widths greater than 18 inches and up to 27 inches: Accuride 4032.
 - 3) For drawer widths greater than 27 inches and up to 42 inches: Accuride 930.
 - c. Full extension.
 - d. Dynamic Load Rating: 100 pounds/pair capacity, minimum:
 3. Pull-out shelf/slide suspension:
 - a. Base Product: Accuride 322.
 - b. Dynamic Load Rating: 100 pounds/pair.
 4. Base Metal:
 - a. Electrocoat white steel.
- E. Roller Catch:
1. Manufacturers:
 - a. The Engineered Products Company (Epc).
 - b. Amerock.
 - c. Ives.
 - d. Other manufacturers desiring approval comply with Section 01 61 00.

2. Description:

- a. Adjustable, spring-loaded dual polyethylene roller with a steel strike plate.
- b. Double doors without locks: Provide a catch on each door.
- c. Tall cabinets: Locate catch on the upper and lower part of each door.
- d. Cabinets equipped with locks: Right hand door shall have a roller catch; left hand door shall have a positive catch.

F. Levelers:

1. Manufacturers:

- a. Blum, Inc
- b. Camar, distributed by Peter Meier Inc.
- c. Häfele America Co.
- d. Hettich America LLP.
- e. Other manufacturers desiring approval comply with Section 01 61 00.

2. Provide levelers consisting of sockets and levelers.

- a. Load Rating: 330 pounds minimum capacity.
- b. Quantity: four per cabinet, minimum.
- c. Service Access: Levelers shall be accessible through a hole in bottom of cabinet. Provide plastic cap for each leveler hole.
- d. Provide clip or other hardware for attachment of toe kick.

G. Casters:

1. Location: Where indicated on Laboratory Furnishing drawings or specifications.

2. Description:

- a. Self-lubricating precision swivel casters with locking brakes.
- b. Wheel Material: Urethane or polyurethane.
- c. Load Rating: 200 pounds minimum, each.
- d. Wheel Diameter and Load Height:
 - 1) Tables: 4 inches diameter. Load Height: 5 inches. Stem mount.
 - 2) Cabinets: 3 inches. Load Height: 4 3/8 inches. Plate mount.
- e. Caster Housing: Heavy gauge cold rolled steel with epoxy powder coating.

H. Pulls:

1. Drawer and hinged doors: Full width hardwood pulls. Hardwood species to match casework exposed hardwood selection.

2. Drawer and hinged doors: Attach with machine screws. Two pulls shall be furnished on drawers wider than 28 inches.

a. Type:

- 1) Pulls shall be round wire.
- 2) Rectangular box style.

b. Material and Finish:

- 1) Extruded Aluminum with US26D satin chrome finish.
- 2) Stainless steel with US32D satin finish.
- 3) Plastic pulls or other types subject to breakage are not acceptable.
- 4) Back mounted.

c. Length: 4 inches center to center of screw holes.

d. Diameter: 1/4 inches.

I. Drawer Stops: Provide integral stops or drawer bumpers on each side of drawer body to prevent drawer head impact with cabinet body.

- J. Cabinet Door Bumper Pads: Non-staining, non-marring, clear polyurethane pads with pressure-sensitive, adhesive backing for sound and vibration dampening, preventing direct contact between door and cabinet. Pad should have raised tip in middle of pad.
 - 1. Location:
 - a. Doors 36 inches and less in height: At top and bottom corners along pull edge.
 - b. Doors greater than 36 inches in height: At top and bottom corners and at an intermediate location along pull edge.
 - c. Drawers: At center of each side.
- K. Glides: Provide non-marring material on movable tables, unless otherwise indicated.
 - 1. Diameter: 1 inch diameter, minimum.
 - 2. Vertical Adjustment: 5/8 inches, minimum.
- L. Leveling devices: Provide leveling bolt with floor clip at each table leg.
 - 1. Bolt Diameter: 3/8 inches, minimum.
 - 2. Nylon Pad: 2 inches diameter.

2.5 LABORATORY WORK SURFACES

- A. Epoxy Resin:
 - 1. Countertop Grade: AWS Premium Grade.
 - a. Modifications: In addition to satisfying the specified grade, products shall incorporate the identified modified requirements, characteristics, and features as specified herein.
 - 2. Manufacturers:
 - a. Durcon Inc.
 - b. Epoxyn Products.
 - c. Kewaunee Scientific Corporation.
 - d. Other manufacturers desiring approval comply with Section 01 61 00.
 - e. Substitutions are not permitted.
 - 3. Thickness:
 - a. Typical work surface: 1 inch.
 - 4. Color:
 - a. Dark gray (Platinum gray as supplied by Durcon Inc. or similar).
 - b. Light gray
 - c. Color sample shall be submitted for approved by Architect.
 - 5. Provide the following:
 - a. Drip Grooves:
 - 1) Provide under work surface exposed edges, unless noted otherwise on the Laboratory Furnishing Drawings.
 - 2) Where the top overhangs 1 inch: 1/2 inches from the edge.
 - 3) Where the top overhangs 1/2 inches: 1/4 inches from the edge.
 - b. Edge profile:
 - 1) All exposed upper edges and corners: 1/4 inches radius, or 1/8 inches bevel.
 - c. Marine edges:
 - 1) Where indicated on the Laboratory Furnishing Drawings.
 - 2) 1 inch wide and 1/4 inches high with chamfered or radiused transition to and an integral part of the work surface.
 - d. Indented areas:
 - 1) Where indicated on the Laboratory Furnishing Drawings.
 - 2) Profile: 1/4 inches deep with chamfered or radiused sides.

- 3) Internal and external corners: 1/4 inches to 1/2 inches radius.
 - 4) Marine edges formed around indented areas shall not be less than 1 inch wide.
 - e. Sink Mounting:
 - 1) Drop-in Sink Cutouts:
 - a) Profiled to provide support for the sink, and to ensure that the rim of the installed sink is 1/8 inches below the surrounding work surface level or bottom of drain grooves if present.
 - b) Top edge of the cutout shall have 1/8 inches bevel.
 - c) No gaps between the installed sink rim and work surface.
 - f. Curbs and Splashes:
 - 1) Curbs and Splashes: 1 inch thick.
 - 2) Curbs and Splashes: 3/4 inches thick.
 - 3) Height: 4 inches, unless noted otherwise on Laboratory Furnishing Drawings.
 - 4) Bonded to the surface of the top to form a square joint.
 - g. Provide holes and cutouts as required for built-in equipment and mechanical and electrical service fixtures. Verify size of opening with actual size of equipment to be used prior to making openings. Form inside corners to a radius of not less than 1/8 inches. After sawing, rout and file cutouts to ensure smooth, crack-free edges. Seal exposed edges after cutting with a waterproofing material recommended by the manufacturer.
6. Physical Properties:
- a. Chemical resistance:
 - 1) Organic solvents test:
 - a) Saturate a cotton ball with the test chemical; place; in a one ounce bottle with a reservoir of liquid above the ball.
 - b) Invert the container on the test material surface for a period of 24 hours.
 - c) Test temperature: 23 deg C $\pm$ 2 deg C.
 - 2) Other test chemicals test:
 - a) Place five drops (1/4 cc) of the test chemical on the test material surface.
 - b) Cover the chemical with a 1 inch diameter watch glass for a period of 24 hours.
 - c) Test temperature: 23 deg C $\pm$ 2 deg C.
 - 3) Evaluation:
 - a) After 24 hours exposure: Wash exposed areas with water, then with a detergent solution, finally with naphtha, then rinse with distilled water, dry with a cloth, and rate as follows:

0	No effect	No detectable change in the material surface.
1	Excellent	Slight detectable change in color or gloss but no change in function or life of the surface.
2	Good	A clearly discernible change in color or gloss but no significant impairment of surface life or function.
3	Fair	Objectionable change in appearance due to discoloration or etch, possibly resulting in deterioration of function over an extended period of time.
4	Failure	Pitting, cratering, or erosion of the surface. Obvious and significant deterioration.

- 4) Test results:

Test chemical	Concentration Percent	Black	Dark gray	Light gray	Beige
Chromic acid	40	3	2	2	2
Hydrochloric acid	10	0	0	0	0
Hydrochloric acid (conc.)	37	0	0	0	0
Nitric acid	40	0	0	0	0
Nitric acid (conc.)	70	0	0	0	0
Sulfuric acid	60	0	0	0	0
Sulfuric acid (conc.)	96	4	4	4	4
Acetic acid	5	0	0	0	0
Acetic acid (glacial)		0	0	0	0
Citric acid	1	0	0	0	0
Oleic acid		0	0	0	0
Phenol solution	5	0	0	0	0
Ammonium hydroxide	10	0	0	0	0
Sodium carbonate sol.	20	0	0	0	0
Sodium hydroxide sol.	60	0	0	0	0
Sodium hypochlorite sol.	4	0	0	0	0
Acetone		1	1	1	1
Benzene		1	1	1	1
Carbon tetrachloride		1	1	0	0
Diethyl ether		0	0	1	1
Dimethyl formamide		0	0	0	0
Ethyl acetate		0	1	1	0
Ethyl alcohol	95	0	0	0	0
Ethylene dichloride		0	0	0	0
Heptane		0	0	1	0
Isooctane		0	0	0	0
Kerosene		0	0	0	0
Methyl alcohol		0	0	0	0
Toluene		0	0	0	0
Aniline		0	0	0	0
Mineral oil		0	0	0	0
Olive oil		0	0	0	0
Soap solution	1	0	0	0	0
Transformer oil		0	0	0	0
Turpentine		0	0	0	0

b. Heat resistance tests:

1) High temperature test:

- a) Heat a porcelain crucible to a dull red color, place on the test material, and allow to cool to ambient temperature.
- b) Result: No observable surface deformation.

2) Flame test:

- a) Overturn a 3/8 inches Bunsen burner, adjusted to a quiet flame, with a 1-1/2 inches inner cone, on the test material, and allow to stay for 5 minutes.
- b) Result: no observable surface deformation.

c. Physical properties:

Physical Properties

Compressive strength	ASTM D695	31,400 psi
Tensile strength	ASTM D638	8,000 psi
Flexural strength	ASTM D790	11,700 psi
Rockwell hardness M	ASTM D785	122
Specific density	ASTM D792	122.4 psf
Water absorption	ASTM D570	0.01 percent
Fire Resistance	ASTM D635	ATB (sec)=0
Heat deflection at 264 psi (1.82 MPa)	ASTM D648	342 deg F

B. Stainless Steel: Refer to Stainless Steel Fabrications section of this specification.

2.6 SHELVING ASSEMBLIES

A. Shelf Material:

1. Metal Shelving:

- Provide 18 GA steel shelves with integral seismic lip and hat-section stiffener with 14 GA, minimum, bookend brackets.
- Rear of bracket shall be profiled to fit into slots of shelf standards as described below.
- Refer to details on Laboratory Furnishings Drawings.
- Provide safety edging at front of shelf.
- Provide safety edging at rear of shelf.

B. Adjustable Wall Shelves:

1. Shelving:

- Metal shelving as specified above.

2. Double Slot Shelf Standards:

a. Manufacturers:

1) Base:

- Knape & Vogt Manufacturing Co.

2) Optional:

- Fixture Hardware Manufacturing.
- Other manufacturers desiring approval comply with Section 01 61 00.
- Substitutions are not permitted.

- Base Product: Knape & Vogt 85 ANO series uprights. Length as indicated on the drawings.

3. Shelf standards shall be slotted channel framing as specified elsewhere in this Section.

4. Shelf Brackets: 16 GA cold formed steel, bookend type, as detailed on drawings.

a. Unless indicated otherwise on Drawings:

- Brackets shall be full depth of shelf,
- Height: 5 inches above top of shelf.
- Radius: 3 inches.
- Rear edge of bracket shall be profiled for attachment to shelf standards.

5. Finish: Factory finish standards and brackets, and metal shelving when specified, with epoxy powder coating.

- Color to be selected by the Architect.

C. Stainless Steel Shelving System:

1. Manufacturers:

a. Base:

- InterMetro Industries Corporation.

b. Optional:

- 1) Eagle Group.
 - 2) Nexel Industries Inc.
 - 3) Other manufacturers desiring approval comply with Section 01 61 00.
 - 4) Substitutions are not permitted.
2. Base Product: Metro Super-Erecta stainless steel shelf system, floor mounted post supported.
 - a. Posts: Floor mounted stainless steel posts with stainless steel leveling glide, grooved at 1 inch increments and numbered at 2 inches increments for shelf adjustment, length as shown on drawings, Metro PS series.
 - 1) Posts shall be 74-5/8 inches high, nominal, unless indicated otherwise on the Drawings.
 - b. Shelves:
 - 1) Open Wire: Stainless steel wire.
 - 2) Provide five shelves per unit, unless indicated otherwise on the Drawings.
 - c. Accessories:
 - 1) Foot Plate: Stainless steel with adjustable leveling bolt, Metro No. 9993S.
 - 2) Post Clamps: Zinc-plated, to join adjacent posts, Metro No. 9994Z.
 - 3) Casters: 5 inches diameter, cart-washable, stainless steel swivel stem caster with brake and polyurethane wheel, Metro No. 5MPBGSA.

2.7 GAS AND LIQUID CYLINDER RESTRAINT ASSEMBLY

- A. Gas and Liquid Cylinder Restraint Assembly:
 1. Slotted Framing Channel, Fittings, Swivel Hangers, and End Caps:
 - a. Slotted channel framing as specified elsewhere in this Section.
 - 1) Color: To be selected by Architect.
 - b. Provide two swivel hangers per gas cylinder or liquid cylinder per wall bracket.
 2. Chain and Associated Hardware:
 - a. All chain and hardware shall be Type 304 or Type 316 stainless steel.
 - b. Chain:
 - 1) 1/4 inches diameter welded chain.
 - 2) Chain shall be long enough to secure 9 inches gas cylinder or 20 inches liquid cylinder, as indicated on drawings.
 - c. Provide one threaded connector chain link at each end, and one harness clip at one end: McMaster-Carr Supply Company, Suncor Stainless, Inc. Other manufacturers desiring approval comply with Section 01 61 00.
 - d. Provide two assemblies per gas or liquid cylinder to anchor at a high and low point.

2.8 DRYING RACK

- A. Epoxy Drying Rack:
 1. Comply with requirements for molded epoxy resin specified under Laboratory Tops in this Section and as described herein.
 2. Drying rack bodies shall be of 1 inch thick white epoxy with a 3/16 inches to 1/4 inches radius on edges and corners. Each rack shall be of the size and with the peg arrangement shown on the Laboratory Furnishing drawings.
 3. Pegs shall be of injection molded white polypropylene. Pegs shall not be bonded into the body but shall be held in position by mechanical design.
 4. Provide a drip trough of Type 304 stainless steel with a 16 GA, Type 304 stainless steel screen of 14 by 14 mesh, 0.02 inches wire. Drip trough shall be continuously welded.
 5. Provide stainless steel fixing screws of appropriate type for attachment to support structure.
 6. Provide clear, tight-fitting hose to drain from drip tray into sink.

B. Clear Acrylic Drying Rack:

1. Drying rack bodies shall be of 3/4 inches thick clear acrylic with eased corners and edges. Bottom edge shall be cut to form 60 degree drip edge with 1/8 inches flat edge at bottom to prevent damage. Each rack shall be of the size and with the peg arrangement shown on the Laboratory Furnishing drawings. Edges shall be polished.
2. Pegs shall be of injection molded white polypropylene, 6 inches long. Pegs shall not be bonded into the body but shall be held in position by mechanical design.
3. Provide 1-1/2 inches diameter by 3/4 inches thick clear acrylic rod spacers and stainless steel fixing screws of appropriate type for attachment to support structure.

C. Stainless Steel Drying Rack with White Polypropylene Pegs:

1. Manufacturers:
 - a. Base:
 - 1) Inter Dyne Systems, Inc.
 - b. Optional:
 - 1) Other manufacturers desiring approval comply with Section 01 61 00 (no known equal).
2. Base Product: Inter Dyne Systems V- Victoria Series, modified as indicated on the drawings.
 - a. Body: One-piece design.
 - 1) Material 20 GA Type 304 stainless steel with a No. 4 finish.
 - 2) Construction:
 - a) The top shall have two 90-degree bends, and sides shall have one 90 degree bend.
 - b) Punch with T-shaped holes to accommodate the peg arrangement shown on the drawings. Holes shall be designed to support pegs for holding single or multiple utensil drip trays, drain shelves, funnel racks or pipette holders.
 - b. Drip Trough: Integral, full-width
 - 1) Material: 20 GA, Type 304 stainless steel.
 - 2) Construction:
 - a) Stainless steel drain tube.
 - b) Continuously welded at corners.
 - c) Screen Insert: Full-length, Type 304 stainless steel wire mesh. Turn down on sides.
 - c. Pegs:
 - 1) Material: Injection molded white polypropylene.
 - 2) Peg base shall have a T-shaped protrusion to allow easy removal and replacement without the need for tools. Pegs shall fit into the punched holes in the rack body.
 - d. Wall Hangers: Provide for each rack, designed to enable the removal and replacement of the entire rack for cleaning without the need for tools.
 - e. Hardware: Provide stainless steel fixing screws of appropriate type for attachment to support structure.
 - f. Drain Tube: Clear, tight-fitting hose to drain from drip trough drain tube into sink.
 - g. Back Panel: Provide finished stainless steel back panel when any portion of the back of drying rack is exposed.

2.9 TASK LIGHT FIXTURES

A. Manufacturers:

1. Base:
 - a. Alkco.

2. Optional:
 - a. Peerless Lighting Corporation.
 - b. Other manufacturers desiring approval comply with Section 01 61 00.
- B. Basis for Design: Alkco Little Inch2 satisfying the identified requirements, characteristics, and features as specified herein.
- C. Description: Fluorescent fixture for undershelf or undercabinet mounting. Provide task lights at locations indicated on the drawings.
- D. Construction:
 1. Dimensions: Fixture shall nominally cover the entire underside of the length of shelf.
 2. Housing: 20 GA cold rolled steel with neatly formed corners and de-burred exposed edges. No exposed fasteners.
 3. Finish: Baked-on polyester resin powder coat finish in color selected by the Architect.
 4. Diffuser/Lens: Prismatic acrylic lens.
 5. Electrical: Pre-wired with a single circuit. Fixture shall be controlled by fixture-mounted rocker switch. Fixture shall be furnished with 36 inches, maximum, grounded cord and NEMA 5-15P or 5-20P plug. Fixture shall be provided with white, ground convenience receptacle to enable task lights to be daisy chained.
 6. Ballast: Electronic, high power factor, thermally protected, Class P.
 7. Lamps : T-5, 3500K.
 8. Labels: Fixture shall be U.L. listed and bear an IBEW label.

2.10 BLACKOUT CURTAIN AND TRACK ASSEMBLY - BC-1

- A. Manufacturers:
 1. Darkrooms USA, Inc
 2. Blackout Curtains.
 3. Cubicle Curtain Factory.
 4. Other manufacturers desiring approval comply with Section 01 61 00.
 5. Substitutions are not permitted.
- B. Curtain Materials and Fabrication:
 1. Fabric:
 - a. 38 percent SEF Modacrylic, 30 percent Saran Flat Monofilament, 26 percent Viscose Rayon and 6 percent Polyester.
 - b. Material shall have a double-vinyl black color laminated back for opacity (samples must be submitted).
 - c. Comply with applicable Federal, State and Municipal codes, laws and regulations regarding flammability and smoke generation of interior finishes.
 - 1) Testing:
 - a) Flame Spread: 0 – 25; ASTM E 84.
 - b) Smoke Developed: 450 or less; ASTM E 84.
 - c) NFPA 701, current edition, Test Method #1 and #2: Pass.
 - d) Fabric must pass the flame resistance requirements specified by the State of California test.
 2. Curtain:
 - a. Finishing: Sewn flat, French-style seams (no raw edges visible).
 - 1) All sewing shall be done in a manner so that light is not allowed to pass through the needle holes.
 - 2) Stitching:

- a) Double-stitch critical seams.
 - b) Lock stitch for durability and to prevent unraveling.
 - c) Seam ends shall be back-stitched.
 - d) All seams shall be pucker-free and lay flat.
 - 3) Fullness: 10 percent additional width.
- b. Grommets:
 - 1) Material: Brass.
 - 2) Spacing: 12 inches centers.
 - 3) Hems:
 - a) Top: Insert a heavy gauge fabric-reinforcing strip to provide addition strength for the insertion of grommets.
 - b) Bottom: Weighted, to maintain a light-tight seal to floor.
- c. Panel and Wall Connections: Hook and loop seal strips creating light-trap overlaps for light-tight attachments to walls or adjacent panels.
- 3. Valance: 11 inches high minimum, front and rear light-trap valances.
 - a. Material: Blackout curtain fabric.
 - b. Finishing:
 - 1) Same as for curtain, except sewn flat (no fullness).
 - 2) Curtain Track Attachment: Hook and loop-seal strips to facilitate light-trap overlaps for easy light-tight attachment to curtain track.
- 4. Double curtain installation: Where two separate curtain panels meet, they shall overlap each other approximately 24 inches.
- C. Track Material and Assembly:
 - 1. Material: Satin anodized extruded Aluminum box-channel.
 - 2. Dimensions: 1-3/8 inches by 3/4 inches.
 - 3. Design:
 - a. Slotted on the underside to receive two wheeled carriers.
 - b. Designed for surface mounting to the underside of the ceiling.
 - c. The track shall also serve as an integral part of the valance assembly; valance shall be permanently adhered or fastened to track at not more than 18 inches on center.
 - 1) Ceiling curves, where required, shall be one-piece, 90 degree track sections.
Nominal Radius: 26 inches.
 - 2) Track corners, where required, shall be one-piece, 12 inches radius 90 degree track sections.
 - d. Hardware: Supply complete with end-caps, snap-outs, and sleeve connectors. Bypass rollers shall have 9 inches extension; double bypass shall have 18 inches overlap.
 - 1) Rollers: Two-wheeled nylon rollers with rustproof wire hook with or without bead chain.
 - 4. Installation: Curtain Supplier/Installer shall provide any necessary bracing at ceiling. Coordinate with ceiling Manufacturer/Installer.

2.11 METAL FABRICATIONS AND FINISH REQUIREMENTS

- A. Applicability:
 - 1. Applies to metal fabrications specified in this Section, including, but not limited to, pipe drop enclosures, radioisotope storage cabinets, shelving support systems, metal-framed laboratory tables, metal-framed balance tables, cylinder racks, and other miscellaneous brake-formed and shop fabricated components and trim, such as required for overhead service carriers.
- B. Materials:

1. Cold rolled sheet steel: Prime grade, roller leveled, and treated at the mill to be free of scale, ragged edges, deep scratches, or other injurious effects.
 - a. Recycled Steel Content: Post-consumer recycled content plus one-half of the pre-consumer content shall be 20 percent, minimum, of the cost of the total value of materials.
- C. Finish Requirements:
1. Description:
 - a. Finish Type: Dry powder coating.
 - b. Performance Requirements: Meet or exceed SEFA 8 M Cabinet Surface Finish requirements.
 - c. Operator Protection:
 - 1) Convenient and easily mastered through robotic application plus manual detailing.
 - 2) Process shall be contained and have no solvent odor and be performed in an air conditioned room.
 - d. Overspray:
 - 1) Overspray shall be captured and re-sprayed.
 - 2) 99 percent efficiency in coating usage, reducing waste generated.
 - 3) A closed collection system shall be used for overspray that is not reused.
 - 4) Powder overspray shall not escape the facility and shall be collected in bulk, eliminating the need for daily replacement/disposal of filter media.
 - e. VOC Emissions:
 - 1) Sprayed and baked with a near zero (0.29 pounds/gallon maximum) VOC (Volatile Organic Compounds) emissions.
 - 2) Comply with the GS (Green Seal Standard) 11 allowable emissions.
 - f. Off gassing:
 - 1) Finish shall be firm and stable after final cure.
 - 2) No further emissions or off gassing/decomposition vapors shall occur at room temperature.
 2. Metal Casework Color: Selected by the Architect from manufacturer's full color line.

2.12 STAINLESS STEEL FABRICATIONS

- A. Applicability:
1. Applies to stainless steel fabrications specified in this Section, including, but not limited to, work surfaces, drying racks, canopy hoods, sinks, low slot exhausts, and other miscellaneous brake-formed and shop fabricated components and trim.
- B. Materials:
1. Stainless Steel Sheet: Type 304, unless otherwise noted, and of gauge indicated on Laboratory Furnishing drawings or this specification.
 - a. Recycled Steel Content:
 - 1) Post-consumer recycled content plus one-half of the pre-consumer content shall be 20 percent, minimum, of the cost of the total value of materials.
 2. Stainless steel nuts, screws, bolts, and rivets, etc.: Same type stainless steel as in the sheet material and shall have a tumbled finish closely resembling that of a No. 4 finish.
 3. Welds:
 - a. Welding material shall be of type similar to the sheet material or a richer quality.
 - b. All welds shall be made without discoloration and shall be ground, polished, and passivated to blend harmoniously with the specified finish.
 - c. Joints in stainless steel tops and work surfaces shall be welded, unless noted otherwise.

- C. Finish: Exposed surfaces shall be ground and polished to a No. 4 finish, unless noted otherwise for one-piece, seamless appearance.
- D. Fabrication:
1. Form to shapes indicated with straight lines, sharp angles, and smooth curves.
 2. Drill or punch holes with smooth edges for temporary field connections and attachment of work by other trades.
 3. Make permanent shop and field connections with continuous fillet type welds.
 4. Grind exposed welds smooth.
 5. Shop-fabricate in as large assemblies as practicable.
 6. Qualify welding processes and welding operators in accordance with American Welding Society.
 7. Supply items required to complete construction and installation.
- E. Work Surfaces:
1. Thickness:
 - a. Typical: 16 GA.
 - b. Tops with Field-Welded Joints: 14 GA.
 2. Fabrication:
 - a. Edges:
 - 1) Turn down flange in the same dimension as the adjacent non-stainless steel top, if any, with 1 inch being a minimum.
 - 2) Return 1/2 inches, minimum, over reinforcing channels.
 - b. Reinforcement:
 - 1) General: 16 GA stainless steel channels or equivalent at formed edges, rear and middle top, below surface, to ensure rigidity and prevent buckling, warping, or oil canning.
 - 2) At bench mounted fittings: Provide reinforcement to allow for rigid, secure mounting of fittings.
 - c. Undercoating: Waterborne and non-flammable in liquid state, with no volatile organic compounds (VOCs), providing sound deadening and to prevent condensation.
 - 1) Undercoating shall not be provided at any BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 locations.
 - d. Stainless steel side and back-splashes:
 - 1) Integrally welded to top, coved, and finished as indicated above.
 - 2) Back side of exposed backsplashes shall be finished to match front and sides.
 - e. Joints: Electrically weld shop joints; grind smooth and polish. Field joints may be bolted with full length supported, resulting in a hairline seam with flat, level surfaces on each side of joint.
 - f. Provide holes and cutouts as required for built-in equipment and mechanical and electrical service fixtures.
 - 1) Verify size of opening with actual size of equipment to be used prior to making openings.
 - 2) Form inside corners to a radius of not less than 1/8 inches.
 - 3) After sawing, rout and file cutouts to ensure smooth, crack-free edges with no burrs.
 - g. Marine edges: Where indicated on the Laboratory Furnishing Drawings:
 - 1) 1 inch wide and 1/4 inches high with chamfered or radiused transition to and be an integral part of the work surface.
 - 2) Seamless die-formed edges.
 3. Tops with Sinks:

- a. Integral, fabricated with a marine edge, pitched to sink bowl for proper drainage.
 - b. Provide sanitary radii where sink bowls meet drainboards.
- 4. Wall-Supported Benchtop:
 - a. Provide stainless steel wall support and bracket angles as indicated on Laboratory Furnishings Drawings.
 - b. Refer to Wall Hung Benchtop requirements elsewhere in this Section.
- F. Laboratory Sink: Integral one piece construction with stainless steel work surface.
 - 1. Thickness: 16 GA, unless otherwise noted.
 - 2. Construction:
 - a. Provide sufficient reinforcement to prevent oil canning.
 - b. Joints:
 - 1) Butt-welded, ground smooth by the heliarc welding process.
 - 2) Inside radii shall be 1 inch, minimum, 1 3/4 inches maximum.
 - 3) Pitch bottom to the drain indent.
 - 4) No soldering will be permitted in connection with sink construction.
 - c. Sink bowl dimensions provided are inside dimensions.
- G. Canopy Hood: Provide stainless steel canopy with hangers and miscellaneous hardware at locations and sizes as indicated on the Laboratory Furnishing drawings.
 - 1. Canopy hood shall be constructed to avoid interference with autoclave pressure gauges if located in fascia panel above autoclave door. Coordinate with autoclave manufacturer.
 - 2. Materials:
 - a. Type 304 stainless steel.
 - 1) Finish, exposed to view components: No. 4.
 - b. Canopy Hood, Baffle, and Fascia Panel Thickness: 18 GA, minimum.
 - c. Hood Support Brackets; 16 GA, 1-1/4 inches wide.
 - d. Baffle Support:
 - 1) Stainless steel slotted channel framing.
 - 2) Brackets: 16 GA, 1-1/4 inches wide.
 - e. Threaded Rod: 3/8 inches minimum.
 - f. Hardware and Fasteners: Stainless steel nuts and washers
 - 3. Construction:
 - a. Provide reinforcing necessary to prevent oil canning or deflection of panel between supports.
 - b. Continuously weld corners and joints, grind smooth, free of defects. Welded joints with visible burn marks will not be accepted.
 - c. Form with 1/2 inches deep condensation gutter with continuous welds to be watertight.
 - d. Provide welded 18 GA exhaust collar with no open seams.
 - 4. Accessories: Provide stainless steel hangers and miscellaneous support hardware as required for a complete installation.
 - 5. Light fixtures and wiring shall be provided under Division 26.
 - a. Holes for electrical work shall be made by canopy hood fabricator.
 - 6. Provide exhaust duct transition pieces for mechanical connection above the ceiling.
 - 7. Refer to Exhaust Schedule for exhaust rate and Mechanical Drawings for duct connection sizes.

2.13 LAB COAT HOOK

- A. Manufacturers: Bobrick

B. Base Product: Bobrick B-6827

1. Description: Surface mounted single hook, satin finish

2.14 AUTOMATION LINE SERVICE COLUMN – ALSC

A. Manufacturers: Legrand

B. Base Product: Vista - VFL662345CMGYF

1. Description: Automation line required services to be oriented and positioned as required by equipment manufacturer.
2. Frame Shape: Rounded End Channels with Large Profiled Frame
3. Height: Height to be coordinated with ceiling height and adjacent work surfaces
4. Location: Finalized location to be coordinated with Automation Line Requirements.

2.15 SLOTTED CHANNEL FRAMING

A. Manufacturers:

1. Unistrut.
2. Power Engineering Co.
3. Kumar Industries.
4. Cooper B-Line Inc. (B-Line).
5. Other manufacturers desiring approval comply with Section 01 61 00.
6. Substitutions are not permitted.

B. Materials: Channel and framing members shall be fabricated from steel conforming to the following requirements:

1. Framing Members:
 - a. Concealed Framing Members and Fittings: ASTM A570 GR 33.
 - b. Exposed Framing Members and Fittings: ASTM A446 GR A with zinc coating conforming to ASTM A525.
 - c. Stainless Steel Framing Members and Fittings: ASTM A240 (Type 304), where indicated.
2. Fittings:
 - a. Concealed Fittings: Fabricate from steel satisfying the requirements of ASTM A570 GR 33, and conform to the following ASTM specifications: A575, A576, A36, or A635. Nuts shall conform to ASTM A576 GR 1015 and screws shall conform to SAE J429 GR 2 and ASTM A307.
 - b. Exposed Fittings: Fabricate from steel satisfying the requirements of ASTM A570 GR 33, and conform to the following ASTM specifications: A575, A576, A36, or A635. Nuts shall conform to ASTM A576 GR 1015 and screws shall conform to SAE J429 GR 2 and ASTM A307. Exposed fittings shall receive zinc coating conforming to ASTM A525.
 - c. Stainless Steel Fittings and Hardware: Sintered Nuts shall be of ASTM B783 (Type 316N2-33) stainless steel and fittings shall be of ASTM A240 (Type 304) stainless steel. Stainless steel fittings and hardware shall be used with stainless steel framing members, or where indicated.
3. Thickness: 12 GA, unless noted otherwise.
4. Size: 1-5/8 inches by 1-5/8 inches cross-section, unless noted otherwise.

C. Components:

1. The following components shall be provided, unless otherwise noted:
 - a. Framing Channel: 1-5/8 inches by 1-5/8 inches by 12 GA: Unistrut P1000, Powerstrut PS 200, Kumar Industries N-200, B-Line Systems, Inc. B22.

- b. Suspended Framing Channel, 3-1/4 inches by 1-5/8 inches by 12 GA: Unistrut P5000, Powerstrut PS 100, Kumar Industries N-150, B-Line Systems, Inc. B11.
 - c. Slotted Hole Framing Channel, 1-5/8 inches by 1-5/8 inches by 12 GA framing channel with 13/32 inches by 3 inches slotted holes, 4 inches on center: Unistrut P1000 SL, Powerstrut P 200 S, Kumar Industries N-200-SL, B-Line Systems, Inc. B22S.
 - d. Slotted Framing Channel for installation in Chemical Fume Hoods, 1-5/8 inches by 13/16 inches by 16 GA Type 316 stainless steel framing channel: Unistrut P4000 SS, Powerstrut PS 560 SS, Kumar Industries, B-Line Systems, Inc.
 - 1) Attach channel to side of fume hood with 2-5/16 inches by 1-7/8 inches by 1/8 inches, 4 hole, stainless steel 90 degree fitting: Unistrut P6325 SS, Powerstrut, Kumar Industries, B-Line Systems, Inc.
 - e. Vertical Posts: 3-1/4 inches by 1-5/8 inches by 12 GA, double channel section: Unistrut P1001, Powerstrut PS 200 2T3, Kumar Industries N-200-A, B-Line Systems, Inc. B22A.
 - f. Horizontal Support Members: 1-5/8 inches by 1-5/8 inches by 12 GA framing channel with 13/32 inches by 3 inches slotted holes, 4 inches on center: Unistrut P1000 SL, Powerstrut P 200 S, Kumar Industries N-200-SL, B-Line Systems, Inc. B22S.
 - g. Diagonal Brace Supports: Framing Channel, 1-5/8 inches by 1-5/8 inches by 12 GA: Unistrut P1000, Powerstrut PS 200, Kumar Industries N-200, B-Line Systems, Inc. B22.
 - h. 90 Degree Angle Fitting: 4-1/8 inches by 3-1/2 inches by 1/4 inches with two holes, each leg: Unistrut P1325, Powerstrut PS 607, Kumar Industries N-1123, B-Line Systems, Inc. B104.
 - i. 135 Degree Angle Fitting: 3 inches by 2-5/16 inches by 1/4 inches with one hole, each leg: Unistrut P1546, Powerstrut PS 633-45 degrees, Kumar Industries N-1425, B-Line Systems, Inc. B154.
 - j. T-Shaped Flat Plate Fitting: 5-3/8 inches by 3-1/2 inches by 1/4 inches plate, T-shaped, with four holes: Unistrut P1031, Powerstrut PS 714, Kumar Industries N-1022, B-Line Systems, Inc. B133.
 - k. Wing Shape Fitting: 9-5/32 inches by 3-7/8 inches ten holes, two holes in each wing section and two holes in each of three channel section sides: Unistrut P2347, Powerstrut PS 913, B-Line Systems, Inc. B273.
 - l. Closure Strip: 1-5/8 inches wide, 0.04 inches thick snap-in cover for framing channel: Unistrut P3184, Powerstrut PS 6152, Kumar Industries N-1920, B-Line Systems, Inc. B217-24. Provide closure strips over exposed vertical post sections.
 - m. End Caps: 0.06 inches thick for framing channel: Unistrut P1280, Powerstrut PS 707, Kumar Industries N-2500, B-Line Systems, Inc. B205. Provide end caps for exposed horizontal framing channels.
 - n. Ceiling Escutcheon: Provide 18 GA steel, finished to match framing members, as indicated on the Laboratory Furnishing drawings, at ceiling penetrations.
 - o. Other components, hardware, and fasteners, as required for a complete assembly and as indicated on the drawings.
2. Service Struts and Leding:
- a. 16 GA, 13/16 inches by 1-5/8 inches cold-formed framing uprights: Unistrut P4000, Powerstrut PS 560, Kumar Industries N-400, B-Line Systems, Inc. B56. Uprights shall be provided at 48 inches, maximum, and fastened top and bottom by two adjustable U-shaped spreaders.
 - b. U-shaped spreaders: 12 GA by 1-1/2 inches wide by length required, galvanized steel.
 - c. Locations:
 - 1) Provide to support tops at pipe service chase space, support drain troughs, under fume hood superstructures, and other abnormal loads.
 - 2) Support struts with U-shaped spreaders shall be provided at 48 inches on center below island and peninsula benches, as indicated on drawings. Support struts shall

be provided along wall 48 inches on center below island and peninsula benches. Struts will be used to support piped and electrical services installed under Divisions 23 and 26. Provide bolts, expansion sleeves, and fastening devices for a complete assembly. Pipe and conduit hangers shall be provided by Division 23 and 26 installers.

3. Adjustable and Heavy Duty Wall Shelving:
 - a. Shelf Standards: Framing channel, spaced equally, 36 inches on center, maximum. Secure to wall. Provide bolts and fastening devices for a complete assembly.
 - b. Brackets: Cold-formed framing channel brackets, as required for maximum cover of shelf depth:
 - 1) Shelves at least than 9 inches and less than 11 inches deep: Unistrut P1769, Powerstrut PS 732-8, B-Line Systems, Inc. B187. Secure to steel uprights and underside of shelf with removable bolt fasteners.
 - 2) Shelves at least than 11 inches and less than 13 inches deep: Unistrut P1771, Powerstrut PS 732-10, B-Line Systems, Inc. B541. Secure to steel uprights and underside of shelf with removable bolt fasteners.
 - 3) Shelves at least 13 inches and less than 15 inches deep: Unistrut P1773, Powerstrut PS 732-12, B-Line Systems, Inc. B289-12. Secure to steel uprights and underside of shelf with removable bolt fasteners.
 - 4) Shelves at least 15 inches and less than 17 inches deep: Unistrut P1775, Powerstrut PS 732-14, B-Line Systems, Inc. B289-14. Secure to steel uprights and underside of shelf with removable bolt fasteners.
 - 5) Shelves at least 17 inches and not exceeding 20 inches deep: Unistrut P1777, Powerstrut PS 732-16, B-Line Systems, Inc. B290. Secure to steel uprights and underside of shelf with removable bolt fasteners.
 4. Suspended Adjustable Shelf:
 - a. Vertical members: Telescoping 12 GA tube or strut, 1-5/8 inches by 1-5/8 inches and 1-7/8 inches by 1-7/8 inches, with 9/16 inches diameter pre-punched holes at 1-7/8 inches on center: Unistrut Telestrut System, Allied Tube & Conduit Square-Fit telescoping channel.
 - b. Provide fittings designed to connect and attach telescoping tubing.
 - c. Gravity pins shall be used to connect telescoping members.
 5. Gas and Liquid Cylinder Restraint:
 - a. Swivel Hanger: 1-3/4 inches long by 3/8 inches diameter link welded to threaded stud; provide two per cylinder: Unistrut M2350, Powerstrut PS205, Kumar N-2911, B-Line 446B.
 6. Laser Shelf:
 - a. Vertical members: Telescoping 12 GA tube or strut, 1-5/8 inches by 1-5/8 inches and 1-7/8 inches by 1-7/8 inches, with 9/16 inches diameter pre-punched holes at 1-7/8 inches on center: Unistrut Telestrut System, Allied Tube & Conduit Square-Fit telescoping channel.
 - b. Provide fittings designed to connect and attach telescoping tubing.
 - c. Gravity pins shall be used to connect telescoping members.
 7. Overhead Support Grid:
 - a. Exposed horizontal support members for user attachment, 3-1/4 inches by 1-5/8 inches by 12 GA framing channel: Unistrut P5000, Powerstrut P 100, Kumar Industries N-150, B-Line Systems, Inc. B11.
- D. Finish:
- a. Provide finish coating for cold-formed framing components, except for stainless steel components.

- b. Concealed Framing Members and Fittings: Rust inhibiting acrylic enamel paint applied by electrostatic deposition, after cleaning and phosphating, and thoroughly baked. Finish shall withstand a minimum of 400 hours salt spray when tested in accordance with ASTM B117. Color: Green.
- c. Exposed Framing Members and Fittings: Factory applied epoxy powder coat. Color: To be selected by the Architect.

2.16 SEALANT

- A. Refer to Section 07 92 16.
- B. Sealant shall be installed by installer of the work of this Section.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Prior to installation of the work of this Section, verify installed work is complete to the point where this installation may properly commence.
- B. Verify that work may be installed in complete accordance with the original design, reviewed submittals, and the manufacturer's recommendations.
- C. Verify adequacy and proper location of any required backing or support framing.
- D. Verify that mechanical, electrical, plumbing, and other building components affecting work in this Section are in place and ready.
- E. Installation of the Work constitutes Contractor's acceptance of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 INSTALLATION

- A. Installation of items specified in this Section shall be performed by installers experienced in the installation of the respective item as determined by the respective manufacturer.
- B. Coordinate work with any Owner furnished and/or installed components indicated on drawings.
- C. Casework:
 - 1. Set casework items square, plumb, and level. Shim as required, using concealed shims and securely anchor.
 - 2. Where required, assemble units into one integral unit with joints flush, tight, and uniform.
 - 3. Provide matching filler pieces where casework abuts walls or columns or should be closed off.
 - a. All wood, plastic laminate, solid phenolic and polypropylene work abutting other building components shall be properly scribed.
 - 4. Mechanical fasteners used at exposed and semi-exposed surfaces, excluding installation attachment screws and those securing cabinets end to end, shall be countersunk.
 - 5. Cutouts for equipment, mechanical, electrical, or plumbing services shall be made by the casework manufacturer or casework installer.
 - 6. Install hardware uniformly and precisely. Set hinges snug and flat in mortises. Adjust doors, drawers, hardware, fixtures and other moving or operating parts to function smoothly and correctly, and so that doors and drawers operate without warp or bind, and contact points meet accurately.
 - 7. Secure base cabinets to service struts and ledging, partition framing, wood blocking, or reinforcements in partitions with fasteners spaced not more than 24 inches on center. Bolt adjacent floor mounted cabinets together with joints flush, tight, and uniform.
 - 8. Floor mounted casework shall receive top set or integral base as specified under Division 09 and on the Finishes drawings.
- D. Suspended Casework, Wall Cabinets, and Shelving:

- a. Fasten suspended and wall cabinets to hanging strips, masonry, partition framing, blocking, or reinforcements in partitions. Fasten each cabinet through the back, near top, at not less than 24 inches on center.
- b. Securely fasten to solid supporting material; not plaster, lath, or wallboard. Anchor, adjust, and align suspended casework, wall cabinets, and shelving supports as specified for base cabinets.
- c. Blocking and backing in cavity wall construction for suspended casework, wall cabinets, and shelving shall be as specified under Division 09, and shall be installed under the scope of work of other Sections. General Contractor shall coordinate the location of in-wall blocking and backing using the shop drawings provided under this Section. Verify that backing and reinforcement necessary to support wall-mounted units is in place, secure, and accurately located.

E. Laboratory Tops:

1. Scribe tops as necessary for close and accurate fit.
2. Field Joints: Factory-prepared and identical to factory joints, locate only where indicated on approved Shop Drawings. Field processing of top and edge surfaces is not acceptable, except as described by manufacturer in approved Submittal Data. Provide full length, one-piece tops and backsplashes wherever possible, and keep field joints to an absolute minimum.
3. Abut top and edge surface in one true plane, with internal supports placed to prevent any deflection. Joints in top units shall be flush and the narrowest for the respective materials of construction.
4. Epoxy Resin: Cement joint in accordance with the manufacturers' specifications.
5. Stainless Steel: Field-weld joints in stainless steel tight, without open seams. Finish material to match adjacent to weld.
6. Plastic Laminate: Seal unfinished edges and cutouts in plastic laminate work surfaces with heavy coat of polyurethane varnish.
7. Polypropylene: Adjacent worksurface sheets shall be beveled along the top edge, continuously welded to fill bevel, and buffed smooth with the worksurface. In addition, joint between adjacent worksurface sheets shall be continuously welded from below.

F. Tolerances: Casework shall not exceed the following tolerances:

1. Variation of Bottoms of Wall Cabinets from Level: 1/8 inches in 10 feet.
2. Variation of Cabinet Faces from a True Plane: 1/8 inches in 10 feet.
3. Variation of Adjacent Surfaces from a True Plane: 1/32 inches.
4. Variation in Alignment of Adjacent Door and Drawer Edges: 1/16 inches.
5. Variation of Work Surfaces from Level: 1/16 inches in 10 feet.

G. Laboratory Sinks:

1. Epoxy Resin, Undermount: Sinks shall be set in work surface with chemical-resisting sealing compound, secured and supported in accordance with manufacturer's instructions. Adjust sink and securely support to prevent movement. Remove excess sealant or adhesive while still wet and finish joint for neat appearance.
2. Epoxy Resin, Drop In: Set sink in adhesive and fill remainder of rabbet with sealant or adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.
3. Epoxy Resin and Polypropylene Cup Sinks, Drop In: Set sink in adhesive and fill remainder of rabbet with sealant or adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.
4. Epoxy Resin and Polypropylene Cup Sinks, Top Mount: Set sink in adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.

5. Stainless Steel, Top Mount: Before setting, apply sink and work surface manufacturers' recommended sealant under rim lip and along work surface. Remove excess sealant while still wet and finish joint for neat appearance.
- H. Miscellaneous Furnishings and Accessories:
1. Install in accordance with manufacturer's instructions.
 2. Securely fasten wall mounted adjustable shelving supports, stainless steel shelves, drying racks, etc. to partition framing, wood blocking, or reinforcements in partitions.
 3. Install shelf standards plumb and at heights to align shelf brackets for level shelves. Install shelving level and straight, closely fitted to other work where indicated.
 4. Tighten screws to seal flat; do not drive.
- I. Sealant:
1. Caulk edges of tops, backsplashes, and side splashes to adjacent wall surface, and around work surface penetrations, with sealant.
 2. Sealant application shall be in accordance with manufacturer's published recommendations.
- J. Repair or remove and replace defective work as approved by the Architect at no additional cost to the Owner.
1. Where repair is approved by Architect, touch-up finishes applied to damaged surfaces shall have a VOC content of no more than 250 g/L in accord with SCAQMD Rule #1168.
- K. Adjustable Laboratory Furniture System:
1. Support system locations shall be established from approved shop drawings so that mechanical and electrical work can be installed without interfering with installation.
 2. Installation of support system shall be coordinated with the trades to maintain the integrity of the installed system.
 3. Support system assemblies, ancillary components and accessories shall be installed with the supervision of the manufacturer's authorized representative and according to manufacturer's recommendations.

3.3 CLEANING

- A. Clean counter tops with diluted dishwashing liquid and water leaving tops free of grease and streaks. Use no wax or oils.
- B. Before completion, adjust moving and operating parts to function smoothly and correctly.
- C. Damaged work shall be replaced, repaired, or restored to original condition to the approval of the Architect at no additional cost to the Owner.
1. Nicks, chips, and scratches in the finish shall be filled and retouched. Damaged items that cannot be repaired shall be replaced.

3.4 PROTECTION

- A. Protect work of this section before, during and after installation including installed work and materials of other trades.
- B. Cover work surfaces with 1/4 inches corrugated cardboard, secured in place, after installation for protection against scratching, soiling, and deterioration during remainder of construction period. Remove protection prior to final cleaning.
- C. Standing or staging work on protected or unprotected work surfaces is not allowed.
- D. Replacement: Damaged work shall be replaced, repaired, and restored to original condition to the approval of the Architect at no additional cost or inconvenience to the Owner.

END OF SECTION

SECTION 21 05 00

COMMON WORK RESULTS FOR FIRE SUPPRESSION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Common work results for requirements specifically applicable to Division 21.
- B. Requirements of Division 01 Specifications, General Provisions of the Contract and General and Supplementary Conditions apply to this Division.

1.2 REGULATORY REQUIREMENTS

- A. Perform work specified in Division 21, in accordance with the codes and standards listed below of the latest applicable edition adopted by the authority having jurisdiction. Where these Specifications are more stringent, they shall take precedence. In case of conflict, obtain a decision from the Architect.
 - 1. NFPA 13: Standard for the Installation of Sprinkler System
 - 2. NFPA 14: Standard for the Installation of Standpipe and Hose Systems.
 - 3. NFPA 20: Standard for the Installation of Stationary Pumps for Fire Protection.
 - 4. NFPA 99: Healthcare Facilities Code.
 - 5. NFPA 101: Life Safety Code.
 - 6. ANSI Handicapped Code-A117.1
 - 7. U.L Fire Resistance Index
 - 8. ASTM E814-08B: Standard Test Method for Fire Tests of Penetration Firestop Systems.
 - 9. Texas Department of State Health Services, Hospital Licensing Regulations
 - 10. Texas Accessibility Standard.
 - 11. IBC: International Building Code, with Mechanical and Plumbing Codes.
 - 12. NFPA 10: Standard for Portable Fire Extinguishers
 - 13. NFPA 17: Standard for Dry Chemical Extinguishing Systems
 - 14. NFPA 17A: Standard for Wet Chemical Extinguishing Systems
 - 15. NFPA 30: Flammable and Combustible Liquids Code
 - 16. NFPA 54: National Fuel and Gas Code
 - 17. NFPA 70: National Electrical Code
 - 18. NFPA 72: National Fire Alarm and Signalling Code
 - 19. NFPA 75: Standard for the Protection of Information Technology Equipment
 - 20. NFPA 96: Standard for Ventilation Control and Fire Protection for Commercial Cooking Operations
 - 21. NFPA 110: Standard for Emergency and Standby Power Systems
 - 22. NFPA 241: Standard for Safeguarding Building Construction, Alteration and Demolition Operations
 - 23. Special regulations, supplement, and amendments of the State and/or local authorities having jurisdiction.

1.3 REFERENCE STANDARDS

- A. ANSI: American National Standards Institute.
- B. ASME: American Society for Mechanical Engineers.
- C. ASTM: American Society for Testing and Materials.
- D. AWWA: American Water Works Association.
- E. FM: Factory Mutual
- F. IRI: Industrial Risk Insurers
- G. MSS: Manufacturer's Standardization Society of the Valve and Fitting Industry.
- H. NEMA: National Electrical Manufacturers' Association.
- I. NFPA: National Fire Protection Association.
- J. U.L.: Underwriters' Laboratories, Inc.
- K. U. L. Fire Resistance Index.

1.4 SUBMITTALS

- A. Submit shop drawings and product data in accordance with requirements of Division 01 including the required number of copies.
- B. Include Products as specified in the individual sections of Division 21.
- C. Submit shop drawing and product data grouped to include complete submittals of related systems, products, and accessories in a single submittal.
- D. Prepare shop drawings completely independent of the Engineer of Record's CADD files. Should the Contractor or Vendor wish to use the Engineer of Record's CADD files or model as the basis for developing their shop drawings, a release form, obtainable from the Engineer or Architect, must be signed.
- E. Submit copies of shop drawings in accordance with Division 01, including:
 - 1. Concrete pads and foundations including anchor bolt and sleeve locations
 - 2. Fire protection systems and hydraulic calculations.
 - 3. Coordinated room layouts shall include:
 - a. Room dimensions
 - b. Support column locations.
 - c. Locations and dimensions of equipment foundations and pads required.
 - d. Locations and dimensions of equipment and apparatus, including electrical control panels and starters, and service and coil pull areas.
 - e. Dimensioned floor drain locations
 - f. Trench locations and sizes.
 - g. Sleeve locations in mechanical and equipment rooms.
 - h. Piping 2" and larger.
 - i. Conduit 2" and larger.
- F. Brochures: Submit manufacturer's product data and brochures including:
 - 1. Complete descriptions.
 - 2. Illustrations and wiring diagrams.
 - 3. Rating data, accessories, dimensional data, and applicable options and features marked for the specific items scheduled on drawings and specified herein.
 - 4. Capacities stated in the terms specified and scheduled.
 - 5. Performance curves for fire pumps and jockey pumps.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing the type of products specified in this Division with a minimum of 5 years of documented experience.
- B. Comply with all codes and standards adopted by the authorities having jurisdiction.
- C. Perform all required tests of systems, record results and deliver as part of the project closing file.

1.6 FIELD CONDITIONS

- A. Layouts indicated on drawings are diagrammatical and intended to show relative positions and arrangement of piping and equipment. Coordinate work with other trades and with measurements obtained at the job site, as applicable, prior to installation. Generally, install work in locations shown on Drawings. Provide necessary rises, drops, and offsets to fit in the available space unless prevented by Project conditions.
- B. If prevented by project conditions, prepare drawings showing proposed rearrangement of The Work, including changes to Work specified in other sections. Obtain permission of the Architect before proceeding.
- C. Place anchors, sleeves, and supports prior to pouring concrete or installation of masonry work.
- D. Cause as little interference or interruption of existing utilities and services as possible. Schedule work which will cause interference or interruption in advance with Owner and all affected trades.
- E. Determine sizes and verify locations of existing utilities on or near site.
- F. Keep roads clear of materials and debris.

- G. Visit site and be informed of conditions under which Work must be performed.
- H. Locate equipment requiring periodic servicing so that it is readily accessible. Provide means of service access, following appropriate manufacturer's recommended service clearance space or, as applicable, means of access using wall or ceiling access doors.
- I. Install piping to leave sufficient space for AHJ inspection of rated wall construction. Coordinate pipe routing with other trades including but not limited to the work of Divisions 22, 23, 26 and 28.

1.7 FEES AND PERMITS

- A. Obtain and pay for all necessary permits and inspection fees required to perform Division 21 work.

1.8 COORDINATION DRAWINGS

- A. Prior to commencement of installation, prepare coordination drawings for work under this Division, as specified in Division 01. Fully cooperate with persons coordinating and performing work under other Divisions.
- B. Drawings shall not be formally submitted but shall be kept on site for reference. Notify the Architect of conflicts that cannot be resolved.
- C. Coordination drawings shall be prepared to include the following:
 - 1. Drawn to a scale of 1/4" = 1' - 0".
 - 2. Room dimensions
 - 3. Sheet size matching contract documents.
 - 4. Ductwork sizes with bottom elevation from finished floor.
 - 5. Show fire protection equipment, structural columns and beams.
 - 6. Concrete pad and foundation layouts including anchor bolt and sleeve locations.
 - 7. Dimensioned floor drain locations.
 - 8. Wall mounted equipment.
 - 9. Piping 2" and larger.
 - 10. Suspended equipment and piping to illustrate sprinkler types and heights.
 - 11. Dry pipe cabinets and valves for pre-action systems.
 - 12. Floor to floor and ceiling heights.

1.9 COMPLETENESS OF WORK

- A. The Contract Documents depict fire suppression systems which are intended to be complete and functioning systems. All products, materials, and labor necessary to render a fully functional system to fulfill the design intent shown on the documents shall be provided by the Contractor.
- B. Catalog numbers referenced throughout the Division 21 Drawings and Specifications are intended to convey a general understanding of the type and quality of the product required. Where written descriptions differ from information conveyed by a catalog number, the written description shall govern. No extra shall be allowed because a catalog number is found to be incomplete or obsolete.

1.10 PRODUCT SUBSTITUTIONS

- A. Comply with provisions of Division 01.

1.11 RECORD DRAWINGS

- A. Provide record drawings that illustrate the work of Division 21 as finally constructed. Deliver record drawings to the Architect in a form suitable for reproduction. Comply with the provisions and requirements of Division 01.
- B. Record drawings shall reflect all changes made to the Contract Documents, whether generated by addenda, change orders, or field conditions. Maintain a daily record of these changes and keep current set of drawings showing these changes.
- C. Deliver record drawings in a form suitable for re-production to the Architect within 30 days of Substantial Completion.

1.12 OWNING AND OPERATING MANUALS

- A. Comply with the requirements of Division 01, but provide a minimum of three sets.

- B. Manuals shall include clear and comprehensive instructions with appropriate graphics and project specific marked data to enable owner to operate and maintain all systems specified in this Division.
- C. Copies of reviewed submittals on furnished equipment shall be included.

PART 2 – PRODUCTS

3.1 EQUIPMENT SUPPORTS

- A. Structural steel for supports: ASTM A36.
 - 1. Use galvanized members installed areas of high humidity or condensation, and outside.
 - 2. Furnish other members with shop coat of red primer.
 - 3. Retouch primer after field welding.

3.2 WALL AND CEILING ACCESS PANELS

- A. Style and type as required for material in which installed.
- B. Size: 24"x24" minimum, as indicated, or as required to allow inspection, service and removal of items served.
- C. 14 gauge minimum sheet metal for doors, 16 gauge frames of cadmium-plated or galvanized construction. Doors shall have expanded plaster rings where located in plaster walls or flanged finish where located in drywall or block construction.
- D. Panels shall have spring hinges with screwdriver locks in non-public areas. Key lock, keyed alike, for panels in public areas.
- E. Prime painted or rust inhibitive paint finish.
- F. UL labeled when in fire-rated construction, 1-1/2 hour rating.
- G. Provide in walls, floors, and ceilings to permit access to all equipment and piping requiring service or adjustment. Examples of such equipment needing access are valves, and equipment needing periodic or replacement maintenance.
- H. Furnish and locate access panels under this Division. Coordinate with trades who are responsible for building system in which panels are to be installed.
- I. Acceptable manufactures: Milcor, Nystrom, Karp, J.L. Industries, or Williams Brothers.
 - 1. For masonry and drywall construction: Milcor Style M
 - 2. For plastered masonry walls and ceiling: Milcor Style K
 - 3. For ceramic tile or glazed structural tile: Use stainless steel panels

3.3 SLEEVES

- A. Materials
 - 1. Concrete floors, concrete and masonry walls: 18-gauge galvanized steel tube with welded longitudinal seam.
 - 2. Drywall partitions: 18 gauge galvanized steel sheet metal.
- B. Sleeves shall be sized such that the annular space between outside surface of pipe or pipe insulation and the inside surface of the sleeve is not less than 1/2". Provide larger annular space if required by firestopping product installation instructions.
- C. Sleeves supporting riser piping 4" and larger shall have three 6" long reinforcing rods welded radially at 120 degree spacing to the sleeve and shall be installed with the rods embedded in the concrete slab as the floor slabs are poured.

3.4 ESCUTCHEON PLATES

- A. Provide B & C No. 10 or equal chrome plated escutcheon plates where pipes penetrate partitions or ceilings in finished spaces or areas.

PART 3 - EXECUTION

4.1 CUTTING AND PATCHING

- A. Repair or replace damage caused by cutting or installation of work specified in Division 21.

- B. Perform repairs with materials which match existing and install in accordance with the appropriate section of these specifications.

4.2 DELIVERY, STORAGE, AND PROTECTION

- A. Insofar as possible, deliver items in manufacturer's original unopened packaging. Where deliver in original packaging is not practical, provide cover and shielding for all items with protective materials to keep them from being damaged. Use care in loading, transporting, unloading, and storing to keep items from being damaged.
- B. Store items in a clean, dry place, and protect from damage. Equipment may not be staged or stored outdoors unless intended for outdoor use.
- C. Protect nameplates on motors, pumps, and similar equipment. Do not paint or insulate over nameplate data.
- D. Keep dirt and debris out of pipes.
- E. Repair, restore, and replace damaged items.

4.3 SLEEVES

- A. Floors: Sleeve all pipe penetrations. Extend sleeve 2" above finished floor, except piping within fire rated pipe chases. Sleeve shall be flush with underside of floor.
- B. Masonry or concrete walls: Sleeve all pipe penetrations. Sleeves shall be flush on both sides of wall.
- C. Drywall partitions: Sleeve all penetrations of piping through fire or smoke rated partitions.
- D. Non-Rated drywall partitions: Sleeves are not required. Seal around penetration with appropriate joint sealing material.
- E. Seal voids between outside surface of sleeve and wall, partition or floor. Seals shall be airtight.
- F. Install piping and sleeves in strict accordance with applicable U.L. floor or partition assembly instructions. Coordinate with Division 07 Firestop and manufacturer's installation instructions.
- G. Penetrations not sleeved or firestopped:
 - 1. Seal voids between pipe and partition. Seals shall be airtight with appropriate joint sealing material.

4.4 ESCUTCHEON PLATES

- A. Provide chromium plated escutcheon plates for exposed uninsulated pipes projecting through floors and walls in finished spaces. Mechanical rooms, storage rooms, electrical closets and housekeeping rooms closets are not considered finished spaces.
- B. Clearance between sleeve and pipe: Minimum of 1/2" or as otherwise dictated by the UL Fire Resistance Directory.

4.5 CLEANING FIRE SUPPRESSION SYSTEMS

- A. General Cleanup:
 - 1. As work proceeds throughout the construction schedule, clean up dirt, debris, old materials, etc., and remove from site, keeping premises in neat and clean condition. See Division 01 of specifications for further requirements.
 - 2. Seepage, discoloration or other damage to parts of the building, its finish, or furnishings due to Contractor's failure to properly clean piping systems shall be repaired without cost to the Owner.
- B. Factory Finishes:
 - 1. Clean items with factory finishes. Touch up bare places, scratches and other minor damage to finishes. Use only factory supplied paint of matching color and formula. If finishes are badly damaged or if there are many damaged, scratched or bare places, refinish the entire item.

4.6 TESTING FIRE SUPPRESSION SYSTEMS

- A. Test all systems and equipment to demonstrate proper operation to the AHJ, Owner and Architect.
- B. Advise the Architect of scheduled systems testing and system demonstration/operation schedules so these tests may be witnessed if desired.
- C. Correct and retest work found to be defective when tested.

- D. Make all repairs to piping systems with new materials. Peening, doping or caulking of joints or holes will not be acceptable.
 - E. Test and Balance fire pumps, flow meters and test headers in accordance with applicable standards. Record results and deliver as part of the project closing file.
 - F. Test, verify and record standpipe system pressures.
 - G. Test, verify and record all floor control and test valves and all fire alarm connections for proper operation.
 - H. Test and verify all supervised fire valves for proper function and annunciation at the fire alarm panel. Record and submit proof of testing for Owner's record.
- 4.7 INFECTION CONTROL REQUIREMENTS
- A. Coordinate with the Owner the exact requirements for the infection control measures to be executed and performed during the course of this Project.
 - B. Prior to execution, present to the Owner for approval a written execution plan for each infection control measure.
 - C. Coordinate infection control measures as needed with all other trades and disciplines.
 - D. Provide documentation of infection control measures to the Owner, as required and specified in the ICRA.

END OF SECTION

SECTION 211313

AUTOMATIC FIRE SPRINKLER SYSTEMS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Wet-pipe sprinkler system.
- B. System design, installation, and certification.

1.2 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.3 REFERENCES

- A. ANSI/ASME B16.1 - Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800.
- B. ANSI/ASME B16.3 - Malleable Iron Threaded Fittings, Class 150 and 300.
- C. ANSI/ASME B16.4 - Cast Iron Threaded Fittings, Class 125 and 250.
- D. ANSI/ASME B16.5 - Pipe Flanges and Flanged Fittings.
- E. ANSI/ASME B16.9 - Factory-made Wrought Steel Buttwelding Fittings.
- F. ANSI/ASME B16.11 - Forged Steel Fittings, Socket-welding and Threaded.
- G. ANSI/ASME B16.18 - Cast Copper Alloy Solder-Joint Pressure Fittings.
- H. ANSI/ASME B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings.
- I. ANSI/ASME B16.25 - Buttwelding Ends.
- J. ANSI/ASME B36.10 - Welded and Seamless Wrought Steel Pipe.
- K. ANSI/ASME Sec 9 - Welding and Brazing Qualifications.
- L. ANSI/ASTM A135 - Electric-Resistance-Welded Steel Pipe.
- M. ANSI/ASTM A47 - Malleable Iron Castings.
- N. ANSI/ASTM B32 - Solder Metal.
- O. ANSI/AWS A5.8 - Brazing Filler Metal.
- P. ANSI/AWWA C110 - Ductile Iron and Gray Iron Fittings.
- Q. ANSI/AWWA C151 - Ductile Iron Pipe, Centrifugally Cast.
- R. ASTM A53 - Pipe, Steel, Black and Hot-Dipped, Zinc-coated Welded and Seamless.
- S. ASTM A120 - Pipe, Steel, Black and Hot-Dipped, Zinc-coated (Galvanized) Welded and Seamless, for Ordinary Uses.
- T. ASTM A234 - Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures.
- U. ASTM A795 - Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Fire Protection Use.
- V. ASTM B75 - Seamless Copper Tube.
- W. ASTM B88 - Seamless Copper Water Tube.
- X. ASTM B251 - General Requirements for Wrought Seamless Copper and Copper-Alloy Tube.
- Y. ASTM D3309 - Polybutylene (PB) Plastic Hot- and Cold-Water Distribution Systems.
- Z. ASTM F438 - Socket-Type Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 40.
- AA. ASTM F439 - Socket-Type Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
- BB. ASTM F442 - Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe (SDR-PR).
- CC. ASTM F493 - Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings.
- DD. AWS D10.9 - Specifications for Qualification of Welding Procedures and Welders for Piping and Tubing.
- EE. NFPA 13 - Installation of Sprinkler Systems.
- FF. NFPA 14 - Standpipe and Hose Systems.
- GG. NFPA 24 - Installation of Private Fire Service Mains and Their Appurtenances
- HH. NFPA 45 - Standard on Fire Protection for Laboratories Using Chemicals

II. NFPA 20 – Standard for the Installation of Stationary Pumps for Fire Protection

1.4 SYSTEM DESCRIPTION

- A. System to provide adequate coverage for the area of renovation..
- B. Design system in accordance with the currently adopted edition of the National Fire Protection Association Standard for the Installation of Sprinkler Systems (NFPA 13), Standpipe and Hose System (NFPA 14), Standard for the Installation of Stationary Pumps for Fire Protection (NFPA 20).

1.5 SUBMITTALS

- A. Separate submittals are required by companies and/or individuals who are licensed by the Texas State Fire Marshal to install the following:
 - 1. Automatic Fire Sprinkler System (Division 21: Section 211313)
 - 2. Refer to Section 4.1 of this document regarding Approval and Permitting.
- B. Preliminary Shop Drawings: Prior to detailed submission, submit preliminary layout of finished ceiling areas indicating only head locations coordinated with ceiling installation.
- C. Shop Drawings:
 - 1. Automatic Fire Sprinkler: At a minimum, all information referenced in NFPA 13 necessary to approve the installation of a fire sprinkler system shall be provided.
- D. Product Data: Provide data on all manufacturers' catalogue information for products used in the building. Submit performance ratings rough-in details, weights, support requirements, and piping connections.
- E. Shop drawings, product data, and hydraulic calculations must be submitted to the University Of Texas Southwestern Medical Center Director Of Fire and Occupational Safety (University Fire Marshal) or his designee for review and approval. Information shall be directed through the assigned UT Southwestern Project Manager.
- F. Manufacturer's Certificate: Certify that system has been tested and meets or exceeds requirements specified, and suggested by listed codes.
- G. Manufacturer's Field Report: Submit under provisions of Section 22 0010.
- H. Manufacturer's Field Report: Indicate time of start-up of treatment systems and include analysis of system water after cleaning and treatment.

1.6 PROJECT RECORD DOCUMENTS

- A. Submit under provisions of Section 220010 (Basic Mechanical Requirements) and Section 213113 (Automatic Fire Sprinkler System).
- B. As-builts shall be provided which indicate any variances or changes to the original design.

1.7 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Section 220010 (Basic Mechanical Requirements) and Section 213113 (Automatic Fire Sprinkler System).
- B. Maintenance Data: Include components of system, servicing requirements, Record Drawings, inspection data, replacement part numbers and availability, and location and numbers of service depot.

1.8 QUALITY ASSURANCE

- A. Sprinkler Systems: Perform work to NFPA 13.
- B. Welding Materials and Procedures: Perform to ASME Code.
- C. Valves: Bear UL label or marking. Provide manufacturer's name and pressure rating marked on valve body.
- D. Maintain one copy of each document on site.

1.9 QUALIFICATIONS

- A. Installer: Company licensed to perform work in accordance with the requirements of the State of Texas Fire Marshal Sprinkler Certificate of Registration (SCR). Additionally, three successful years of working with organizations similar to UT Southwestern Medical Center is required.
- B. Design sprinkler system under direct supervision of a Responsible Managing Employee- General (RME-G) licensed by the Texas State Fire Marshal or a Professional Engineer with experience in design of this work and licensed in the State of Texas. All design submittal documents and shop drawings shall bear the responsible parties signature or Professional Engineer's Seal.

1.10 REGULATORY REQUIREMENTS

- A. Hydraulic Calculations, Product Data, and Shop Drawings: All products must be Underwriters Laboratory Listed (UL).

1.11 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect, and protect products to site under provisions of Section 220010 (Basic Mechanical Requirements) and Section 213113 (Automatic Fire Sprinkler System).
- B. Store products in shipping containers and maintain in place until installation. Provide temporary inlet and outlet caps. Maintain caps in place until installation.

1.12 EXTRA MATERIALS

- A. Furnish under provisions of Section 220010 (Basic Mechanical Requirements) and Section 213113 (Automatic Fire Sprinkler System).
- B. Provide extra sprinkler heads as required (type and size) under provisions of NFPA 13.
- C. Provide suitable wrenches for each head type.
- D. Provide lockable metal sprinkler head storage cabinet in location designated.

PART 2 - PRODUCTS

2.1 GENERAL

- A. General: The Contractor shall provide all components required for the complete installation of automatic sprinkler systems as hereinafter specified and indicated on the Drawings.

2.2 QUALIFICATIONS OF THE INSTALLER

- A. The system shall be installed by a firm regularly engaged in the design and installation of automatic sprinkler systems in accordance with requirements of the National Fire Protection Association and licensed by the Texas State Fire Marshal as a Responsible Managing Employee- General (RME-G) licensed by the Texas State Fire Marshal or a Professional Engineer with experience in fire protection. Underground fire line installations must be

done by a licensed Texas State Fire Marshal Responsible Managing Employee Underground.

2.3 SYSTEM LAYOUT

- A. The fire sprinkler areas, piping, head locations, etc. as indicated is only for Contractor's reference as to areas to be protected and possible piping routes. If header or manifold sizes are given in the drawings, then the sizes given shall be the minimum sizes installed. Actual number, spacing and location of heads, size and routes of piping shall be provided in accordance with the applicable Specifications and acceptable Shop Drawings. All layouts, head spacing, coverage, etc., as may be required by the referenced authorities and/or Architectural and Structural conditions shall be made without increase in cost to the Owner or the Architect/Engineer.
- B. Modifications to head spacing, pipe routes, etc. shall be closely coordinated with the work of all other trades. The Fire Sprinkler Subcontractor shall be responsible for the design and installation of the fire sprinkler system as described herein and on the project drawings.

2.4 SHOP DRAWINGS

- A. Shop Drawings shall be submitted and approved by Director of Fire and Occupational Safety (University Fire Marshal) or his designee prior to fabrication. Shop drawings shall include detail plans of underground fire line, sprinkler systems, or fire pump to include but not limited to all components necessary to properly install the system in accordance with the codes. The Shop Drawings shall become an integral part of these Specifications.

2.5 MATERIALS AND EQUIPMENT

- A. All materials and equipment used in the installation of the sprinkler system shall be listed as approved by the Underwriters' Laboratories, Inc., List of Inspected Fire Protection Equipment and Materials, , Fire Protection Devices and Devices Involving Fire Hazard, and shall be the latest design of the manufacturer. All piping, control valves, drain valves, fittings, etc. shall be as specified under this Section, Fire Protection System, utilizing welded, flanged, and threaded fittings only. Roll-grooved couplings and mechanical fittings are also permitted and shall be manufactured by Victaulic. Roll-grooved couplings and mechanical fittings shall be UL listed and. Cut-grooved couplings and mechanical fittings shall not be permitted. Where valves are not specified by Figure No. they shall be of specified manufacture, U.L. listed for service, and of same quality level as Figure Nos. specified. All steel pipe shall be Schedule 40. 300 pound malleable iron, A120 Schedule 40 black steel pipe and fittings. Note that if galvanized pipe or fittings are installed, the contractor shall be responsible to remove the galvanized pipe or fittings and replace them with specified materials as soon as possible prior to further installation of the system.

2.6 SPRINKLER HEADS

- A. Unless otherwise specified or indicated on the Drawings, sprinkler heads shall be regular automatic closed type spray heads of the upright, pendant, or sidewall ordinary degree temperature rating type except that sprinkler heads to be installed in the vicinity of heating equipment and lights shall be of the temperature rating required for such locations by NFPA 13.
- B. Heads shall be located in a symmetrical pattern related to ceiling features such as grid, beams, light fixtures, diffusers, etc., and where applicable, heads shall be located symmetrically with the ceiling grid, centered in two directions.
- C. The Contractor shall provide spare heads equal to one percent (1%) of the total number of heads installed under the Contract. The heads shall be packed in a suitable wall mounted sprinkler cabinet and shall be representative of, and in proportion to, the number of each type and temperature rating heads installed. In addition to the spare heads, the Contractor shall provide not less than one special sprinkler head wrench for each type of head. The cabinet shall be located where directed by the Director of Fire and Occupational Safety (University Fire Marshal) or his designee.
- C. Sprinkler heads manufactured by Reliable Sprinkler Corp. shall be used.
- D. Sprinkler heads shall be the concealed type in all areas with ceilings. Paint cover plates to match ceilings.

2.7 PIPING

- A. Installation of piping, fittings and valves shall be as specified in the System Components; NFPA 13, except where noted otherwise. Piping shall be concealed in all areas with finished ceilings. Piping shall be cleaned and free of dirt, oil, and other contaminants prior to and following installation. The O.S. & Y. valves shall be provided under this section. Materials shall be as specified. Note that the use of piping bushings for any purpose is explicitly prohibited.
- B. All pipe used for fire protection sprinkler systems shall be Schedule 40 black steel pipe except in imaging areas. This pipe shall, in every detail, conform to Standard Specifications for WELDED AND SEAMLESS STEEL PIPE A.S.T.M. SERIAL DESIGNATION: A120. Sizes 2" and smaller shall conform to requirements of ASTM A120. All steel piping shall be American Made.
- C. All piping above imaging suites shall be non-ferrous piping (CPVC).
 - 1. CPVC Pipe: ASTM F 442/F 442M and UL 1821, SDR 13.5, for 175-psig rated pressure at 150 deg F, with plain ends. Include "LISTED" and "CPVC SPRINKLER PIPE" markings.
 - 2. CPVC Fittings: UL listed or FM approved, for 175-psig rated pressure at 150 deg F, socket type. Include "LISTED" and "CPVC SPRINKLER FITTING" markings.
 - a. NPS 3/4 to NPS 1-1/2: ASTM F 438 and UL 1821, Schedule 40, socket type.
 - b. NPS 2 to NPS 3: ASTM F 439 and UL 1821, Schedule 80, socket type.
 - c. CPVC-to-Metal Transition Fittings: CPVC, one piece, with dimensions equivalent to pipe; one end with threaded brass insert, and one socket end.
 - d. CPVC-to-Metal Transition Unions: CPVC, with dimensions equivalent to pipe; one end with threaded brass insert, and one socket end.
 - e. Flanges: CPVC, one or two pieces.

2.8 PIPE PAINTING

- A. Piping shall be cleaned and free of dirt, oil, and other contaminants prior to and following installation. All exposed piping shall be painted Sherwin-Williams – Real Red #SW6868 or equivalent. Painting shall coordinate with project architect and/or Section 099100 - "Painting" of UTSW specifications regarding method of application to ensure the appropriate finish.
- B. Following painting, protective bags, tape, wrappings etc. shall be removed and the equipment shall be visually examined to ensure there is no paint applied to areas not intended. Any overspray shall be cleaned. The application of paint on items which prevent, damage, or alter the proper operation of the device (gauges, sprinkler heads, fire alarm devices etc.); the device shall be removed and replaced with the brand new device of same specification. Any variations will require the formal approval of the Director of Fire and Occupational Safety (University Fire Marshal).
- C. For the identification of all piping, see Section 2.26 MARKING OF VALVES, SUPPLY MAINS, and PIPING.

2.9 TESTS

- A. Upon completion and prior to acceptance of the installation, the Contractor shall subject the system to the test procedures as described by National Fire Protection Association Standard for the Installation of Fire Sprinklers (NFPA 13), which shall be witnessed by the Director of Fire and Occupational Safety (University Fire Marshal) or his designee. Where dry system or double interlock pre-action, additional air tests shall be required in accordance with NFPA 13.

2.10 FITTINGS

- A. All welding type steel fittings employed in fabricating fire protection system including fire sprinkler systems shall conform to A.S.T.M. Specification A-234 and ANSI Standard B16.9-1964. All threaded fittings shall be 300 pound

malleable iron fittings. **Grooved type fittings will not be accepted for use in standpipe systems.** Pipe size changes shall be performed through the use of reducing tees or reducers designed for that purpose. **The use of bushings is explicitly prohibited.**

- B. Threaded fittings shall be used when shown and shall be used from the point of connection of the pipe to the riser to each fire hose cabinet. Threaded fittings shall be Crane or Grinnell Company's 300 pound malleable iron fittings.
- C. Extra heavy "Thread-o-lets" shall be used at each point of departure from the riser to the fire hose or valve cabinet. A "Thread-o-let" shall be installed below the level of the valve in the cabinet and a minimum of two (2) threaded ells shall be used to provide a swing joint connection from the riser to the valve in the cabinet.
- D. Roll-grooved couplings and mechanical fittings are permitted and shall be manufactured by Victaulic. Roll-grooved couplings and mechanical fittings shall be UL listed. Cut-grooved couplings and mechanical fittings shall not be permitted.

2.11 VALVES

- A. All valves shall be located such that the removal of their bonnets is possible. All flanged valves shown in horizontal lines with the valve stem in a horizontal position. Screw pattern valves placed in horizontal lines shall be installed with their valve stems inclined at an angle of a minimum of 30 degrees above the horizontal position. All valves must be true and straight at the time the system is tested and inspected for final acceptance. Valves shall be installed as nearly as possible to the locations indicated in the Construction Drawings. Any change in valve location must be so indicated on the Record Drawings. All valves must be of threaded or flanged type. No solder connected fitting valves shall be used on this project. All bronze and iron body gate and globe valves shall be the product of one manufacture for each project. Manufacturers of other types may not be mixed on the same project; i.e., all butterfly valves shall be of the same manufacture, all ball valves shall be of the same manufacture, etc.
- B. Class 300 valves shall be constructed of all ASTM B-61 composition. All gate, globe and angle valves shall be union or screw-over-bonnet design. Metal used in the stems of all bronze gate, globe and angle valves shall conform to ASTM B371 Alloy 694, ASTM B99 Alloy 651, or other corrosion resistant equivalents. Written approvals must be secured for the use of alternative materials.
- C. All iron body valves shall have the pressure containing parts constructed of ASTM designated of 126 class B iron. Stem material shall meet ASTM B16 Alloy 360 or ASTM 371 Alloy 876 silicon bronze or its equivalent. Gates and globes shall be bolted bonnet with OS&Y (outside screw and yoke) and rising stem design. A lubrication fitting is preferred on yoke cap for maintenance lubrication of the yoke bushing.
- D. All cast steel body valves shall have the pressure containing parts constructed of ASTM designation A-216-GR-WCB carbon steel. Gate and globe valves shall be bolted bonnet outside and screw and yoke design with pressure-temperature rating conforming to ANSI B16-34-1977. Stems shall meet ASTM designation A-186-F6 chromium stainless steel. Wedge (gate valves) may be solid or flexible type and shall meet ASTM A-182-F6 chromium stainless steel on valves from 2" to 6". Sizes 8" and larger may be A-216-WCB with forged rings or overlay equal to 182-F6. Seat ring shall be hard faced carbon steel or 13% chromium A-182-F6 stainless. Handwheels shall be A47 Grade 35018 malleable iron or Ductile Iron ASTM A536.
- E. All forged steel body valves shall have the pressure containing parts constructed of ASTM 105, Grade 2 forged carbon steel. Seat and wedges shall meet ASTM A-182-F6 chromium stainless steel. Seat rings shall be hard faced. Valves shall conform to ANSI B16-34 pressure-temperature rating.
- F. All valves shall be repackable, under pressure, with the valve in the full open position. All gate valves, globe valves, angle valves and shutoff valves of every character shall have malleable iron hand wheels, except iron body valves 2-1/2" and larger which may have either malleable iron or ASTM A-126 Class B, gray iron hand wheels.
- G. Packing for all valves shall be free of asbestos fibers and selected for the pressure-temperature service of the valve. It is incumbent upon the manufacturer to select the best quality, standard packing for the intended valve service. At the end of one year, period spot checks will be made, and should the packing show signs of hardening

or causing stem corrosion then all valves supplied by the manufacturer shall be repacked by the Contractor, at no expense to the Owner, with a packing material selected by the Owner.

- H. Valves located with stem in horizontal position shall be drilled and tapped in accordance with MSS-SP-45 at Boss G to accommodate a drain valve.
- I. Gate Valves for Fire Protection Service: 2" and smaller, bronze body, outside screw and yoke, rising stem, solid wedge, Underwriters' Laboratories Listed screw pattern. Iron body, 2-1/2" and larger wedge, flanged pattern, OS&Y rising stem. Underwriters Laboratories Listed Approved. Butterfly valves U. L. listed with tapped full lug body and gear operated with malleable mop hand-wheel and position indicator may also be used.
- J. Check Valves Fire Protection System: Iron body, swing-check, bronze disc, seat ring and hinge pin, 175 psi rated working pressure, Underwriters' Laboratories approved. Complete with ball drip assembly.
- K. Standards of Quality for Valves:
Standard of Quality for Valves:

			<u>Milwaukee</u>	<u>Nibco</u>	Stockham Or as <u>Noted</u>
2" & Smaller	Gate Valve	Fire Protection	--	T-104-0	B-133
2-1/2" & larger	Gate Valve	Fire Protection	--	F-607-0	
2" & Smaller	Check Valve	All Water Systems	510	T-433	B-345
2-1/2" & Larger	Check Valve	*Fire Protection	--	F-908-W	G-940
*Check valve requires ball drip assembly. Note: For valves not listed, see Section 22 1116.					

- L. Suction diffusers shall be Paco or approved equal, cast iron body and cover, steel diffuser, and stainless steel strainer, 125 pound ASA (flat face) flange for a working pressure of 175 psi and temperature of 300 ° F.

2.12 UNIONS

- A. Provide and install unions at proper points to permit removal of pipe and various equipment and machinery items without injury to other parts of the system. No unions will be required in welded lines or lines assembled with solder joint fittings except at equipment items, machinery items and other special pieces of apparatus. Unions in 2" and smaller in ferrous lines shall be Class 300 AAR malleable iron unions with iron to brass seats, and 2-1/2" and larger shall be ground flange unions. Unions in copper lines shall be Class 125 ground joint brass unions or Class 150 brass flanges if required by the mating item of equipment. Companion flanges on lines at various items of equipment, machines and pieces of apparatus shall serve as unions to permit removal of the particular items. See particular Specifications for special fittings and pressure.
- B. Unions connecting ferrous pipe to copper or brass pipe shall be dielectric type equal to Epcor.
- C. In all domestic water lines where the material of the pipe is changed from ferrous to copper or brass, a dielectric coupling shall be used at the transition.

2.13 WELDED PIPING

- A. SCOPE: This section applies to all piping systems providing for welded piping, fittings, and other appurtenances. Specific systems requiring welded piping include, but are not limited to: chilled water, steam, steam condensate, and fire protection systems.
- B. MATERIALS: All weld fittings shall be USA factory made wrought carbon steel butt-welding fittings conforming to ASTM A234 and ASME/ANSI B16.9, latest edition, as made by Weld Bend, Tube Turn, Hackney, or Ladish Company. Each fitting shall be stamped as specified by ASME/ANSI B16.9 and, in addition, shall have the laboratory control number metal stenciled on each fitting for ready reference as to physical properties required for any fittings selected at random. Only one manufacturer of weld fittings will be approved for the project. Fittings which have been machined, remarked, printed or otherwise produced domestically from nondomestic forgings or materials will not be acceptable. Each in accordance with MSS SP-25. Markings shall be placed on the fittings at the farthest point from the edge to be welded to prevent disfiguring from the welding process. Submittal data for these fittings shall include a letter signed by an official of the manufacturing firm certifying compliance with these specifications.
- C. EXECUTION:
1. Piping and fittings shall be welded and fabricated in accordance with ASME/ANSI the latest editions of Standards B31.1 for all systems, and B31.3 for Steam and Condensate systems, from the Code for Pressure Piping. Machine beveling in shop is preferred. Field beveling may be done by flame cutting to recognized standards.
 2. Contractor shall ensure complete penetration of deposited metal with base metal. Contractor shall keep inside of fittings free from globules of weld metal. All welded pipe joints shall be made by the fusion welding process, employing a metallic arc or gas welding process. All pipe shall have the ends beveled 37-1/2 degrees and all joints shall be aligned true before welding. Except as specified otherwise, all changes in direction, intersection of lines, reduction in pipe size and the like shall be made with factory-fabricated welding fittings. Mitering of pipe to form elbows, notching of straight runs to form tees, or any similar construction will not be permitted.
 3. Contractor shall align piping and equipment so that no part is offset more than 1/16 inch. Set all fittings and joints square and true, and preserve alignment during welding operation. Use of alignment rods inside pipe is prohibited.
 4. Contractor shall not permit any weld to project within the pipe so as to restrict it. Tack welds, if used, must be of the same material and made by the same procedure as the completed weld. Otherwise, remove tack welds during welding operation.
 5. Contractor shall not split, bend, flatten or otherwise damage piping before, during or after installation.
 6. Contractor shall remove dirt, scale and other foreign matter from the inside of piping, by swabbing or flushing, prior to the connection of other piping sections, fittings, valves or equipment.
 7. In no cases shall Schedule 40 pipe be welded with less than three passes including one stringer/root, one filler and one lacer. Schedule 80 pipe shall be welded with not less than four passes including one stringer/root, two filler and one lacer. In all cases, however, the weld must be filled before the cap weld is added.
- D. TESTING:
1. All welds are subject to inspection, visual and/or Xray, for compliance with specifications. The owner will, at the owner's option, provide employees or employ a testing laboratory for the purposes of performing said inspections and/or Xray testing. Initial visual and Xray inspections will be provided by the owner. The contractor shall be responsible for all labor, material and travel expenses involved in the reinspection and retesting of any welds found to be unacceptable. In addition, the contractor shall be responsible for the costs involved in any and all additional testing required or recommended by ASME/ANSI Standards B31.1 and B31.3 due to the discovery of poor, unacceptable or rejected welds.
 2. Welds lacking penetration, containing excessive porosity or cracks, or are found to be unacceptable for any reason, must be removed and replaced with an original quality weld as specified herein. All qualifying tests, welding and stress relieving procedures shall, moreover, be in accord with Standard Qualification for Welding Procedures, Welders and Welding Operators, Appendix A, Section 6 of the Code, current edition.

- E. All weld fittings shall be USA factory made wrought carbon steel butt welding fittings conforming to ASTM Spec. A234 and ANSI standard B16.9-1964 as made by Grinnell, Tube Turn, Hackney, Taylor Forge, or Ladish Company. Each fitting shall be stamped as specified by ANSI B16.9 and, in addition, shall have the laboratory control number metal stenciled on each fitting for ready reference as to physical properties and chemical composition of the material. Complete test reports may be required for any fittings selected at random. Only one manufacturer of weld fittings will be approved for each project. Fittings which have been machined, remarked, printed or otherwise produced domestically from imported forgings or materials will not be acceptable. Each fitting shall have the manufacturer's trademark permanently identified in accordance with MSS SP-25. Markings shall be placed on the fittings at the farthest point from the edge to be welded to prevent disfiguring from the welding process. Submittal data for these fittings shall include a letter signed by an official of the manufacturing firm certifying compliance with these Specifications.

2.14 FLANGES

- A. All 150 lb. and 300 lb. ANSI flanges shall be weld neck and shall be domestically manufactured, forged carbon steel, conforming to ANSI B16.5 and ASTM A-181 Grade I or II or A-105-71 as made by Tube Turn, Hackney or Ladish Company. Slip on flanges shall not be used. Each fitting shall be stamped as specified by ANSI B16.9 and, in addition, shall have the laboratory control number stenciled on each fitting for ready reference as to physical properties and chemical composition of the material. Complete test reports may be required for any fitting selected at random. Flanges which have been machined, remarked, painted or otherwise produced domestically from imported forges will not be acceptable. Flanges shall have the manufacturer's trademark permanently identified in accordance with MSS SP-25. Contractor shall submit data for firm certifying compliance with these Specifications. Bolts used shall be carbon steel bolts with semi-finished hexagon nuts of American Standard Heavy dimensions. Allthread rods will not be an acceptable for flange bolts. Bolts shall have a tensile strength of 60,000 psi and an elastic limit of 30,000 psi. Flat faced flanges shall be furnished where required to match flanges on pumps, check valves, strainers, etc. Only one manufacturer of weld flanges will be approved for each project.
- B. All flanges shall be gasketed. Contractor shall place gasket between flanges of flanged joints. Gaskets shall fit within the bolt circle on raised face flanges and shall be full face on flat face flanges. Gaskets shall be cut from 1/16" thick, non metallic, non asbestos gasket material suitable for operating temperatures from -150 °F to +750 °F, Klingerseal C-4400, Manville Style 60 service sheet packing, or equal.

2.15 WALL, FLOOR AND CEILING PLATES

- A. Except as otherwise noted, provide C.P. (Chrome plated) brass floor and ceiling plates around all pipes, conduits, etc., passing exposed through walls, floors, or ceilings, in any spaces accept underfloor and attic spaces. Plates shall be sized to fit snugly against the outside of the pipe or against the insulation on lines which are insulated and positively secured to such pipe or insulation. Plates will not be required for piping where pipe sleeves extend 3/4" above finished floor. All equipment rooms are classified as finished areas. Round and rectangular ducts shall have closure plates (NOT chrome plated) made to fit accurately at all floor, wall and ceiling penetrations.

2.16 SLEEVES, INSERTS, AND FASTENINGS

- A. General: All openings through all floors, walls, and roofs, etc., regardless of material for the passage of piping, ductwork, conduit, cable trays, etc., shall be sleeved. All penetrations must pass through sleeves except soil pipe installed under concrete slabs on fill. Sleeves shall be set in new construction before concrete is poured, as cutting holes through any part of the concrete will not be permitted unless acceptable to the Architect/Engineer. If a penetration is cored into an existing concrete, masonry or stone structure, then the installation of a sleeve will not be necessary.
- B. The minimum clearance between horizontal penetrations including insulation where applicable, and sleeve shall be 1/4", except that the minimum clearance shall accommodate a Thunderline Link-seal closure where piping exits the building, or penetrates a wall below ground level. Contractor shall be responsible for the accurate location of penetrations in the slab for his pipe, duct, etc. All penetrations shall be of ample size to accommodate the pipe, duct, etc. plus any specified insulation. Sleeve materials shall be rigid metal of adequate strength. Void between

sleeve and pipe shall be filled with Nelson Flameseal Firestop or approved equal caulk or putty.

- C. Installation of sleeves in walls shall be the same as for floors. Refer to the details on the project drawings. Where the details differ from these specifications, the drawings take precedence.
- D. Sleeves for penetrations passing through walls or floors on or below grade shall be removed, if practical, and after the pipes have been installed, the void space around the pipe shall be caulked with a suitable material to effect a waterproof penetration or fire rated material depending on wall type. Note that the practicality of the removal of the sleeve shall be the decision of the Construction Inspector. The decision of the Inspector shall be final.
- E. Inserts: Suitable concrete inserts for pipe and equipment hangers shall be set and properly located for all pipe and equipment to be suspended from concrete construction. If the inserts are later found not to be in the proper location for the placement of hangers, then drilled anchors shall be installed. Drilled anchors in concrete or masonry shall be submitted for the approval by the Owner.
- F. Fasteners: Fastening of pipes, conduits, etc., in the building shall be as follows: To wood members - by wood screws; to masonry - by threaded metal inserts, metal expansion screws, or toggle bolts, whichever is appropriate for the particular type of masonry; to steel - machine screws or welding (when specifically permitted or directed), or bolts, and to concrete by suitable inserts anchored to reinforcing steel, and poured in place unless other means are indicated on the plans. Power-actuated fasteners (shooting) will not be acceptable under any circumstances. If it is necessary to install a method of fastening a hanger after the structure has been installed, then only clamps or drilled anchors shall be used.

Note: Under no circumstances will the use of plastic anchors or plastic expansion shields be permitted for any purpose whatsoever.

- G. Air Plenums: The space around piping, ductwork, etc., passing through air plenums shall be made airtight in a manner acceptable to the Architect/Engineer.
- H. Fireproofing: Each fire protection, mechanical and electrical contractor shall seal all cable trays, pipe, conduit, duct, etc., penetrations through roof, fire rated walls and floors with a foam or sealant as described below that will form a watertight, vermintight barrier that is capable of containing smoke and fire up to 2000 degrees F for two hours. Sealing of cable trays and conduits that extend through rated walls from ends of cable tray shall be done after conductors have been installed. For wet locations, the foam material shall be a silicone RTV foam or an approved equal. For dry locations, a premixed putty equal to Nelson Flameseal Firestop putty may be used.

2.17 HANGERS, AND SUPPORTS

- A. General: All special foundations and supports, hangers, anchors, and guides required for the proper installation of equipment and pipe shall be provided as hereinafter specified, unless otherwise indicated on the Drawings.
- B. Pipe Supports, Hangers, Anchors, and Guides:
 - 1. Supports, hangers, anchors and guides shall be provided for all horizontal and vertical piping. Shop Drawings shall be provided, indicating locations and details of anchors, guides, expansion loops and joints, hangers, etc. The hanger design shall conform to the ASME Code for Pressure Piping.
 - 2. All auxiliary steel required for pipe supports, anchors, guides, etc. shall be provided by the Mechanical Trades unless specifically indicated to be provided by others.
 - 3. Contractor shall review all Drawings, including Structural Drawings, for details regarding pipe supports, anchors, hangers, and guides.
 - 4. All pipe supports shall be of type and arrangement to prevent excessive deflection, to avoid excessive bending stresses between supports, and to eliminate transmission of vibration.
 - 5. All rod sizes indicated in this Specification are minimum sizes only. This trade shall be responsible for structural integrity of all supports, anchors, guides, etc. All structural hanging materials shall have a safety factor of 5 built in.
 - 6. Anchor points as indicated on Drawings or as required shall be located and constructed to permit the piping system to take up its expansion and contraction freely in opposite directions away from the

anchored points.

7. Guide points shall be located and constructed wherever required or indicated on Drawings and at each side of an expansion joint or loop, to permit free axial movement only.
8. Pipe supports, hangers, anchors, and guides shall be fastened to the structure only at such points where the structure is capable of restraining the forces in the piping system.

C. Pipe Hangers:

1. Hangers supporting and contacting brass or copper lines 3" in size and smaller shall be Grinnell Fig. CT-65 carbon steel clevis type hanger with a copper finish. Hangers supporting and contacting brass or copper lines 4" and larger shall be Grinnell Fig. 260 with high density rubber tape wrapped around the pipe for protection. Isolate all copper or brass lines from ferrous metals with approved dielectric materials.
2. Hangers supporting insulated lines where the outside diameter of the insulation is the equivalent of 8" dia. pipe or smaller in size and supporting all ferrous lines 6" and smaller in size shall be clevis type hangers.
3. Hangers supporting and contacting lines larger than 6" in size and outside of insulation on lines with the outside dia. equivalent to 10" dia. pipe shall be Grinnell Fig. 260, clevis hangers with two nuts on each support rod.
4. Other special type of hangers may be employed where so specified or indicated on the Drawings, or where required by the particular conditions. In any case, all hangers must be acceptable to the Engineer.
5. Each hanger shall be properly sized to fit the supported pipe or fit the outside of the insulation on lines where specified. Hangers for dual or low temperature insulation pipes shall bear on the outside of the insulation, which shall be protected by support shields as specified in Section F, INSULATION. Protect insulation from crushing by means of a section of rigid insulation to be installed at hanger points. Hangers for insulated pipes shall be encased in the insulation unless supported by trapezes in which case shield and rigid insulation shall be provided as specified above for low temperature insulated pipes.
6. Hanger Rods:

<u>Pipe Sizes</u>	<u>Rod Diameter</u>
4" and Smaller	3/8"
5" thru 8"	1/2"
10" and 12"	5/8"
14" and larger	3/4" or as necessary

7. Supports for vertical piping shall be as manufactured by: Grinnell, or other approved equal, with each end having equal bearing on the building structure, and located at each floor. Note that two-bolt riser clamps installed at the floor in exposed stairwells are not acceptable. Where brass or copper lines are supported on trapeze hangers or Kindorf channels the pipes shall be isolated from these supports with plastic tape with insulating qualities, or strut clamps as manufactured by specialty Products Company, Stanton, California, or equal.
8. Hangers shall be attached to the structure in a manner to support the minimum weight requirements of Standard for the Installation of Fire Sprinklers (NFPA 13)

D. Perforated strap iron or wire will not, under any circumstances, be acceptable as hanger material.

- E. Vibration Isolation: Resilient hangers shall be provided on all piping connected to rotating equipment (pumps, etc.). Piping that may vibrate and create an audible noise shall also be isolated. Spring hangers or supports shall be provided where indicated and/or specified.

F. Attachment:

1. The load and spacing on each hanger and/or insert shall not exceed the safe allowable load for any component of the support system, including the concrete which holds the inserts. Reinforcement at inserts shall be provided as required to develop the strength required.
2. Inserts for piping shall be of a type which will not interfere with reinforcing as shown on the structural Drawings and which will not displace excessive amounts of structural concrete.
3. All pipe supports shall be designed and installed to avoid interference with other piping, hangers, ducts, electrical conduit, supports, building structures, equipment, etc. All piping shall be installed with due regard to expansion and contraction and the type of hanger method of support, location of support, etc. shall be

governed in part by this Specification.

4. Pipe hangers shall be attached to the structure as follows:

- a. Poured In Place Concrete: Where pipes and equipment are supported under poured in place concrete construction, each hanger rod shall be fitted with a nut at its upper end, which nut shall be set into an Underwriters' Laboratories, Inc. listed universal concrete insert placed in the form work before concrete is poured. Where inserts are placed in the bottom faces of concrete joists which are too narrow to provide adequate strength of concrete to hold the insert properly or where a larger insert would require displacement of the bottom joist steel, the hanger rod shall be suspended from the center of a horizontal angle iron, channel iron, I-beam, etc. spanning across two adjacent joists. The horizontal support shall be bolted to non-adjustable concrete inserts of the "spot" type, of physical size small enough to avoid the bottom joist steel.
- b. Steel Bar Joists: Where pipes and loads are supported under bar joists, hanger rods may be run through the space between the bottom angles and secured with a washer and two nuts. Where larger lines are supported beneath bar joists, hanger rods shall be secured to angle irons of adequate size; each angle shall span across two or more joists as required to distribute the weight properly and shall be welded to the joists or otherwise permanently fixed thereto.
- c. Steel Beams: Where pipes and loads are supported under steel beams, approved type beam clamps shall be used.
- d. Wood Framing: Where pipes and loads are supported from wood framing, hanger rods shall be attached to framing with side beam brackets or angle clips.
- e. Pre-Cast Tee Structural Concrete: Hanger supports, anchors, etc. required for mechanical systems attached to the precast, double tee, and structural concrete system are to be installed in accord with approved shop Drawings only. Holes required for hanger rods shall be core drilled in the "flange" of the double tee only; impact type tools are not allowed under any circumstances. Core drilling in the "stem" portions of the double tee is not allowed. Holes core drilled through the "flange" for hanger rods shall be no greater than 1/4" larger than the diameter of the hanger rod. Hanger rods shall be supported by means of bearing plates of size and shape acceptable to the Architect, with welded double nuts on the hanger rod above the bearing plate. Cinch anchors, lead shields, expansion bolts, and studs driven by explosion charges are not allowed under any circumstances in the lower 15" of each stem and in the "shadow" of the stem on the top side of the "double tees".

G. Hanger spacing shall be in accordance with Standard for the Installation of Fire Sprinklers (NFPA 13).

H. Trapezes: Where multiple lines are run horizontally at the same elevation and grade, they may be supported on trapezes of Kindorf, Elcen, or approved equal, channel-suspended on rods or pipes. Trapeze members including suspension rods shall each be properly sized for the number, size, and loaded weight of the lines they are to support.

I. Finishes: All hangers on piping including clevis hangers, rods, inserts, clamps, stanchions, and brackets, shall be dipped in Zinc Chromate Primer before installation. Rods shall be galvanized or cadmium plated after threading, in lieu of dipping in zinc chromate primer. Universal concrete inserts shall be cadmium plated.

2.18 SIAMESE CONNECTIONS

A. At the points designated on the accompanying Drawings, install Siamese fittings required for fire protection purposes. From a point on the incoming water supply line, this Contractor shall extend water line for fire protection purposes to Siamese connections.

B. Wall type Siamese shall be equal to W.D. Allen Company's 276 double clapper flush type Siamese connections having threads complying with the requirements of the Fire Department of the City Dallas, Texas. They shall have proper caps with pin type lugs attached to the body of the Siamese connections with substantial chains. A sign in accordance with the State Fire Marshal of Texas specification for the Fire Department Connection (FDC) as referenced by State of Texas Government Code 417.008 implemented in accordance with the 2012 Edition of the State Fire Code – Issued May of 2016 shall be provided at a location indicated by the Director of Fire and Occupational Safety (University Fire Marshal) or designee.

- C. Sidewalk type Siamese shall be equal to W.D. Allen Company's No. 231 cast brass body and escutcheon. All external surfaces shall be chromium plated polished surfaces. A sign in accordance with the State Fire Marshal of Texas specification for the Fire Department Connection (FDC) as referenced by State of Texas Government Code 417.008 implemented in accordance with the 2012 Edition of the State Fire Code – Issued May of 2016 shall be provided at a location indicated by the Director of Fire and Occupational Safety (University Fire Marshal) or designee.

D. GENERAL FINISH REQUIREMENTS

1. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.
2. Protect mechanical finishes on exposed surfaces of fire protection cabinets from damage by applying a strippable, temporary protective covering before shipping.
3. Finish fire protection cabinets after assembly.
4. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

E. ALUMINUM FINISHES

1. Clear Anodic Finish: AAMA 611, AA-M12C22A31, Class II, 0.010 mm or thicker.

E. EXECUTION

1. EXAMINATION

- a. Examine roughing-in for cabinets to verify actual locations of piping connections before cabinet installation.
- b. Examine walls and partitions for suitable framing depth and blocking where semi-recessed recessed and semi-recessed cabinets will be installed. Proceed with installation only after unsatisfactory conditions have been corrected.

2. PREPARATION

- a. Prepare recesses for semi-recessed fire protection cabinets as required by type and size of cabinet and trim style.

3. INSTALLATION

- a. General: Install fire protection cabinets in locations and at mounting heights indicated or, if not indicated, at heights acceptable to authorities having jurisdiction.
- b. Fire Protection Cabinets: 54 inches above finished floor to top of cabinet.
- c. Fire Protection Cabinets: Fasten cabinets to structure, square and plumb.
- d. Unless otherwise indicated, provide recessed fire protection cabinets. If wall thickness is not adequate for recessed cabinets, provide semi-recessed fire protection cabinets.
- e. Provide inside latch and lock for break-glass panels.
- f. Fasten mounting brackets to inside surface of fire protection cabinets, square and plumb.
- g. Identification: Apply vinyl lettering at locations indicated.

4. ADJUSTING AND CLEANING

- a. Remove temporary protective coverings and strippable films, if any, as fire protection cabinets are installed unless otherwise indicated in manufacturers written installation instructions.
- b. Adjust fire protection cabinet doors to operate easily without binding. Verify that integral

- locking devices operate properly.
- c. On completion of fire protection cabinet installation, clean interior and exterior surfaces as recommended by manufacturer.
- d. Touch up marred finishes, or replace fire protection cabinets that cannot be restored to factory-finished appearance. Use only materials and procedures recommended or furnished by fire protection cabinet and mounting bracket manufacturers.
- e. Replace fire protection cabinets that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

2.19 MARKING OF VALVES, SUPPLY MAINS, and PIPING OF CONCEALED PIPING

- A. Supply Piping shall have a red label with 1" white letters a minimum of size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) shall be permanently affixed to all of the incoming supply piping indicating the building designation, "FIRE PROTECTION WATER", and the direction of flow. The label shall be provided within 10' of all fire sprinkler risers and every 30' until the supply line returns back underground. A label shall also be located immediately where the piping comes out of the ground inside the building.
- B. Mains and Branch lines shall have a red label with 1" white letters a minimum size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) permanently affixed along the piping every 20'. The label shall indicate the two letter building designation and "FIRE PROTECTION WATER."
- C. Risers – Hydraulic information as required by NFPA 13 shall be permanently affixed to the piping direction over the alarm valve or clapper valve. A red label with 1" white letters a minimum of size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) shall also be permanently affixed to the piping indicating which building designation, "FIRE PROTECTION WATER", and area being feed by the system.
- D. Floor Control or Isolation Valves – A red label with 1" white letters a minimum of size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) shall be permanently affixed to the piping at the valve indicating the building designation, "FIRE PROTECTION WATER," and area being feed by the system.
- E. Fire Department Connection Piping – A red label with 1" white letters a minimum of size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) shall be permanently affixed to the piping indicating which building designation and FIRE DEPARTMENT CONNECTION Line. The label shall be provided every 20' along the FDC. A 2" Label with black letter on a white background shall be provided at the top of any remote fire department connection where other signage is not provided.
- F. Drains – A red label with 1" white letters a minimum of size of 7" x 1-1/2" (Bold Face Font: Calibri or equivalent) shall be permanently affixed to all drain piping. The label shall indicate building designation, and "DRAIN". The label shall be located every 20' or as necessary to clearly indicate the pipe.
- G. All labels shall be applied in a manner that they can be read from the ground without the assistance of a ladder.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Coordinate work of this Section with other affected work.

3.2 GENERAL INSTALLATION

- A. Install equipment in accordance with manufacturer's instructions.
- B. Locate fire department connection with sufficient clearance from walls, obstructions, or adjacent Siamese connectors to allow full swing of fire department wrench handle.
- C. Place pipe runs to minimize obstructions with other work.
- D. Place piping in concealed spaces above finished ceilings.
- E. Center heads in two directions in ceiling tile and provide piping offsets as required.

- F. Apply paper cover to ensure concealed sprinkler head and cover plates do not receive field paint finish. Install and connect fire pumps in accordance with NFPA 20.
- G. Flush entire piping system of foreign matter.
- H. Hydrostatically test entire system at 150 percent of charge or working pressure, whichever is greater.
- I. Require test be witnessed by the UT Southwestern Medical Center Director of Fire and Occupational Safety (University Fire Marshal) or his designee

3.3 PREPARATION

- A. Ream pipe and tube ends. Remove burrs. Bevel plain end ferrous pipe.
- B. Remove scale and foreign material, from inside and outside, before assembly.
- C. Prepare piping connections to equipment with flanges or unions.

3.4 INSTALLATION

- A. Install piping in accordance with NFPA 13 for sprinkler systems, NFPA 14 for standpipe and hose systems, and NFPA 13 for service mains. Note that the piping sizes indicated in the plans are the minimum acceptable. The Qualified Contractor shall provide proper sizes, materials and installation as required in the appropriate NFPA Standard.
- B. Route piping in orderly manner, plumb and parallel to building structure. Maintain gradient.
- C. Install piping to conserve building space, and not interfere with use of space and other work.
- D. Group piping whenever practical at common elevations.
- E. Install piping to allow for expansion and contraction without stressing pipe, joints, or connected equipment.
- F. Slope piping and arrange systems to drain at low points. Use eccentric reducers to maintain top of pipe level.
- G. Prepare pipe, fittings, supports, and accessories for finish painting. Where pipe support members are welded to structural building framing, scrape, brush clean, and apply one coat of zinc rich primer to welding.
- H. Do not penetrate building structural members unless indicated.
- I. Provide sleeves when penetrating footings, floors, and walls. Seal pipe and sleeve penetrations to achieve fire resistance equivalent to fire separation required. Refer to Section 230529.
- J. Die cut screw joints with full cut standard taper pipe threads with red lead and linseed oil or other non-toxic joint compound applied to male threads only.
- K. Install valves with stems upright or horizontal, not inverted. Remove protective coatings after installation.
- L. Provide ball valves for shut-off or isolating service.
- M. Provide drain valves at main shut-off valves, low points of piping and apparatus.

3.5 GENERAL FABRICATION OF PIPE (except for welded piping):

- A. All the various piping systems shall be made up straight and true and run at proper grades to permit proper flow of the contained material. Lines shall also be graded for proper drainage.
- B. Piping shall follow as closely as possible the routes shown on Drawings which take into consideration conditions to be met at the site.
- C. Should any unforeseen conditions arise, lines shall be changed or rerouted as required after proper approval has been obtained.
- D. All piping shall be installed with due regard to expansion and contraction and so as to prevent excessive strain and stress in the piping, in connections, and in equipment to which the lines are connected.
- E. All piping shall be clean when it is installed. Before installation, it shall be checked, upended, swabbed, if necessary, and all rust or dirt from storage or from lying on the ground shall be removed.
- F. Procedure of Assembling Screw Pipe Fittings: All screw joints shall be made with taper threads, properly cut. Joints shall be made tight with Teflon applied to the pipe threads only and not to fittings. When threads are cut on pipes, the ends shall be carefully reamed to remove any burrs. Before installing pipe that has been cut and threaded, the lengths of pipe shall be upended and hammered to remove all shavings and foreign material.
- G. Procedure for Assembling Other Joints: Procedures for assembling joints in cast iron and copper lines have been set forth elsewhere in these Specifications. For any special materials, consult the manufacturers for the recommended procedures in assembling the joints.

3.6 TESTING AND FLUSHING

- A. After the fabrication of fire protection piping systems has been completed, these systems shall be subjected to a hydrostatic test at a minimum pressure of 200 pounds per square inch without loss for 2 hours and shall be flushed at each valve through a temporary hose until the system is clean. Any leaks found shall be remedied in each instance in a manner approved in advance by the Owner's duly authorized representative. The systems shall be alternately tested and repaired where necessary until they have demonstrated their capability to withstand such a minimum 200 pound per square inch gauge hydraulic pressure for 2 hours without any drop in the test pressure initially applied. The test shall be witnessed by Director of Fire and Occupational Safety (University Fire Marshal) or his designee.

PART 4 – PERMITTING

4.1 PLAN REVIEW & PERMITTING

- A. Plans shall be submitted for review and approval a minimum of 3 business days through the UT Southwestern Project Manager for delivery to The Office of Safety and Business Continuity – Fire Safety.
- B. Plans shall include all appropriate information, calculations, and manufactures information as required in National Fire Protection Association Standard on Installation of Sprinkler Systems (NFPA 13), Standpipe and Hose System (NFPA 14), Standard for the Installation of Stationary Pumps for Fire Protection (NFPA 20). Failure to provide the necessary information will result in a delay in the review of the project until all of the information is provided.
- C. Submittal of plans does not constitute approval. Contractors working on the project without a permit do so at their own risk. The contractor will be responsible to replace, without cost to the owner, any components necessary as required by the approved plans.
- D. Three copies of submittal information shall be provided to the UT Southwestern Medical Center Office of Safety and Business Continuity – Attention Director of Fire and Occupational Safety. A copy of the approved, stamped, plans shall be maintained on the job at all times.

E. A permit shall be provided with the approval of the submitted plans.

F. A separate permit shall be issued for the following:

1. Fire Sprinkler System

4.2 INSPECTION

A. All inspections shall be witnessed and approved by the Director of Fire and Occupational Safety or his designee prior to acceptance and operation.

B. A minimum of three (3) business day's notification is required prior to inspection. Inspections shall be coordinated through the UT Southwestern Medical Center Project Manager.

C. All state paperwork for testing shall be copied at the time of the conclusion for the test and signed by the contractor and UTSW Fire Safety representative. A copy of the test shall be maintained by the contractor and provided immediately to the Fire Safety representative. The test will not be considered passed under the paperwork for the test is complete.

4.3 AS BUILTS

A. As built drawings shall be provided on the day of final approval of the system. Final approval shall not be granted without accurate as-builts provided to the UTSW Fire and Safety Inspector.

END OF SECTION

SECTION 22 00 13

UTSW PLUMBING DESIGN REQUIREMENTS

PART 1 - GENERAL

1.1 PURPOSE

- A. This Specification establishes University of Texas Southwestern Medical Center (UTSW) requirements for plumbing projects on campus. This document address design criterion not specifically covered by the Plumbing Code (UPC), or set requirements that may exceed the minimum requirements of the Code.

1.2 APPLICABILITY

- A. This standard applies to all plumbing projects designed and constructed by physical plant personnel, outside Architect/Engineering (A/E) firms, and all construction contractors. No substitutions to the standard are acceptable without the written authorization of the ENGINEER.

1.3 TERMS AND DEFINITIONS

- A. CDAS – Central Data Acquisition System. CDAS is the campus-wide central monitoring station, which is located in the Physical Plant Office (P Building).
- B. Dry Lab – Laboratories, which are not equipped with multiple utilities, but require a greater degree of electrical power and HVAC than an office area.
- C. ENGINEER – UTSW Engineer
- D. Gas Cylinder Room – Any room that contains carbon dioxide, nitrogen, or argon cylinders in excess of 50 Lbs. capacity.
- E. Wet Lab – Any laboratory equipped with sinks, fume hoods, biological safety cabinets, or other equipment, which requires multiple utilities (such as gas, air or vacuum), and a greater degree of HVAC than a dry lab.

1.4 GENERAL PLUMBING DESIGN DRAWING CRITERIA

- A. Flow diagrams shall be drawn for each piping system including but not limited to steam, heating water, chilled water, hot and cold water, distilled water, fire standpipe, oxygen, compressed air, condensing water, gas, vacuum, and refrigerant systems. Mains and major branches shall show quantities of flow with size. All valve sizes shall be indicated. Architectural room names and numbers shall be used to indicate locations.
- B. HVAC plumbing and air-conditioning systems shall be drawn as separate drawings. A complete roof plan shall be included both for air-conditioning plans and plumbing plans; one plan may serve for both.
- C. Floor plans for mechanical systems shall be drawn to show pipes, ducts, etc. on the floor in which they are installed.
- D. Plumbing riser diagrams must be drawn with one for each riser on the project. The risers must show all piping from the under-floor through the roof.

1.5 PLUMBING DESIGN FOR WET LABS

- A. UTSWMC has standardized the arrangement of horizontal plumbing utilities for laboratory counters and peninsulas. An AutoCAD detail of the horizontal plumbing chase is readily available from the ENGINEER.
- B. Acid waste piping is composed of glass. No acid waste or vent pipe shall be sized smaller than 2 inches diameter. Acid waste exposed to the potential for damage shall be constructed of either high silica cast iron (Duriron) or protected with unistrut guards.
- C. All fixtures serving laboratory Millipore filtration systems shall be equipped with a positive connection, such as a compression fitting. No "tube-and-turret-barb" configurations are acceptable. The standard procedure is to provide a Hayward ½" PVC Needle valve with double female connections, stock # NV10050T. The department provides the connections to the valve.
- D. All connections between dissimilar materials in the piping system shall be made with dielectric unions or couplings.
- E. At every point where piping penetrate a floor slab, except slabs on grade, a cast-in sleeve or other waterproof curbing at least 2" high shall be provided.
- F. Rebar shall be located prior to coring any holes.

1.6 DESIGN OF GAS PIPING IN GAS CYLINDER ROOMS AND LABORATORIES

- A. All piping for vacuum and gases other than natural gas shall be ASTM B88, Type L hard drawn seamless copper pipe and tubing. Use brass mechanical connections where required. All vacuum and oxygen piping shall be dry nitrogen purged while being soldered.
- B. An isolation valve shall be provided for each branch circuit. Gas turrets shall not be relied upon as isolation valves.
- C. Greater than two tank installations require a manifold system and shall be hard piped. Piping for two tank installations is normally the responsibility of the Department.

1.7 LAVATORY AND MECHANICAL ROOM DESIGN CONSIDERATIONS

- A. Waste lines from lavatories shall be limited to two fixtures per arm. The wastes shall discharge into a Cast Iron (C.I.) stack behind the fixture. Back to back lavatories are permitted if connected to sanitary tapped crosses. Straight tapped crosses are not permitted.
- B. Floor drains shall be 4 inches in size serving 80 or more square feet. Areas less than 80 square feet shall be 3-inch diameter. Coordinate with architect to provide minimum 1" in 10 feet slope to room and area drains.
- C. Drains serving as indirect receptors for other drain piping shall be floor sink style.
- D. Clean-outs shall be shown on plans and on riser diagrams.
- E. Vent pipes shall be carried up adjoining soil and waste pipe, and they shall be connected into the main stack at top and bottom. Vents may be one size smaller than the traps they serve, except that no vent shall be less than 2". The size of vent lines accommodating more than one fixture shall be sized in accordance with the International Plumbing Code.
- F. Specified fixtures shall conform to the requirements of the Texas Department of Health water saving performance standards. Photoelectric activated plumbing fixtures are generally not in use at the campus, and as a result of past maintenance problems their introduction is discouraged.

1.8 MISCELLANEOUS

- A. Domestic water lines shall not be run to coffee makers, unless a drain is provided or the coffee maker is immediately adjacent to a sink.
- B. All machine room floor sinks shall be a minimum of 4-inch diameter, with acid resisting interior, and shall be connected with trap primers. Floor drains in areas above grade and over crawlspaces shall have mechanical joints for easy access.

PART 2 – PRODUCTS

NOT USED

PART 3 – EXECUTION

NOT USED

END OF SECTION

SECTION 22 05 23

VALVES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Valves for plumbing systems.

1.2 RELATED REQUIREMENTS

- A. Section 22 11 16 - Domestic Water Piping

1.3 SUBMITTALS

- A. Submit product data for review in accordance with the requirements of Division 01. Valves used or indicated to be used in domestic potable water systems must be lead free in accordance with the Reduction of Lead in Drinking Water Act effective January 4, 2014.

1.4 QUALITY ASSURANCE

- A. Lead Free: All wetted surface of pipe, fittings and fixtures in potable water systems shall have a weighted average lead content equal to or less than 0.25% per the Safe Drinking Water Act (Section 1417) as amended January 4, 2011.
 - 1. NSF Compliance: NSF/ANSI 61 and/or NSF/ANSI 372 for valve materials for potable-water service. Valves for domestic water must be 3rd Party Certified.
- B. Boiler valves: ASME Boiler Code Specifications.
- C. Valve bodies, shells, and seats: Factory tested.
- D. Bronze body valves:
 - 1. Materials for pressure containing parts: ASTM B-62 (less than 200 psi), B-61 (200 psi and above)
 - 2. Design, workmanship, testing: MSS-SP-80
- E. Iron body valves:
 - 1. Materials for pressure containing parts: ASTM A126, Grade B
 - 2. Face-to-face and end-to-end dimensions: ANSI B16.10
 - 3. Design, workmanship, testing: MSS-SP-70, 71
- F. Butterfly valves:
 - 1. Face-to-face and end-to-end dimensions: MSS-SP-67
- G. Valve stems: ASTM B584-78, Class 13C (cast silicon brass), ASTM B-371-79, Alloy A (rolled silicon brass), or other material equally resistant to dezincification.

- H. Pressure castings: Free of impregnating materials.
- I. Valve name or trademark and working pressure stamped or cast into body.
- J. Standard for 200 PSI and 300 PSI valves with metallic seats: ASTM B61-76.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Iron body valves: Nibco, Apollo, Stockham, Crane, Milwaukee, Dezurik, Mueller or Kennedy
- B. Bronze body valves: Nibco, Apollo, Stockham, Milwaukee, Dezurik or Kennedy
- C. Butterfly valves: Nibco, Apollo, Stockham, Crane, Milwaukee, Mueller, Victaulic (grooved only) or Dezurik
- D. Ball valves: Nibco, Apollo, Watts, Milwaukee, Jamesbury or Hammond
- E. Pressure Reducing Valves: Watts, Apollo, Nibco.

2.2 MATERIALS

- A. Nibco Figure numbers are indicated below unless noted otherwise:
- B. Check Valves:
 - 1. Domestic Water:
 - a. 2" and less, Figure T-413-Y-LF, or S-413-Y-LF, threaded or solder, Lead Free silicone bronze body, Class 150, PTFE seat
 - b. 2-1/2" and up: Iron body, bronze alloy disc, bronze alloy or Buna-N seat, stainless steel spring, flanged, Class 125, Lead Free, globe style, F-910-LF; or F-960-LF, Class 250.
- C. Ball Valves
 - 1. Domestic Water
 - a. 2" and less, Figure T585-80-LF or S-585-80-LF, 2-piece, full port, 600 psi WOG, PTFE packing seal and seat ring, lead free silicone bronze alloy body and ball.
 - b. 2-1/2" and up, Watts G4000-FDA-Lead Free-200 psi, 2 piece, full port, cast iron, flanged, heat fused epoxy coating, stainless steel ball and stem, PTFE seat, Class 125 meeting MSS-SP-72-92 or equal.
 - 2. Provide ball valves with locking handles.
 - 3. Provide extended lever for insulated service.
- D. Butterfly Valves - 2-1/2" And Up:
 - 1. Domestic Water: Figure LD-2000, ductile iron body, lug type, 200 psi, Class 125, EPDM liner and seals, and aluminum bronze disc, meeting AWWA C504-88.
 - 2. Butterfly valves rated bubble tight for dead end service at full pressure in both directions without the need for downstream blind flange.

- 3. Provide hand wheel and closed housing worm gear on valves 6 inches and larger. Provide clamp lock hand lever operators on valves less than 6 inches.
- 4. Grooved piping: Victaulic Series 608N, brass body, rated bi-directionally to 300 psi, aluminum bronze disc, fluoroelastomer seat, -20 degrees F to +250 degrees F temperature range.
- E. Valve connections: Two inches and smaller - threaded; 2-1/2 inches and larger - flanged.
- F. Provide chain operators for gate valves, butterfly valves, and plug cocks located in mechanical rooms as required by plumbing plans or where valves are mounted above 7'-0" A.F.F.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Furnish and install valves in each piping connection at each piece of plumbing equipment to allow equipment to be isolated from piping systems.
- B. Furnish and install valves in all piping systems to isolate each floor or main section of the building. Install sufficient number of valves to minimize the portion of the system which must be shut down for service or maintenance purposes.
- C. Furnish and install valves above each group of plumbing fixtures.
- D. Install valves in water piping systems so ordinary maintenance work can be performed on the equipment that the valves isolate, without having to drain the system beyond the valve.
- E. All valves above drywall access panels shall be located within one foot of access panels including valves located above ceilings.
- F. Locate valves so as to be easily accessible by maintenance personnel.
- G. Pressure reducing valves shall be installed on equipment as required per manufacturer's recommendation and on main lines serving any facility when the static pressure exceeds 80 psi as required by the State or Local Plumbing Code.
- H. All plumbing systems including pumps, domestic water piping and valve settings and hot water recirculation systems must be tested and balanced. Coordinate with Section 23 05 93 for test and balance requirements.

END OF SECTION

SECTION 22 05 29

HANGERS FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Hangers for plumbing piping

1.2 RELATED REQUIREMENTS

- A. Section 22 11 16 - Domestic Water Piping
- B. Section 22 13 16 - Storm And Sanitary Waste And Vent Piping

1.3 SUBMITTALS

- A. Submit product data for review in accordance with the requirements of Division 01.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Anvil, Carpenter and Patterson, Fee and Mason, B-Line, Viking, Reliable, and Michigan. Anvil model numbers are used for reference.

2.2 HANGERS

- A. Anvil Figure #260 MSS Type 1, clevis hangers for:
 - 1. Non-insulated steel and galvanized piping 2" through 24" diameter
 - 2. Non-insulated cast iron pipe
 - 3. Non-insulated PVC piping
- B. Anvil Figure #260 clevis hangers with Figure 167, MSS Type 40 galvanized insulation protection shields (sized for supporting insulation having a compressive strength of 4 psi). Support piping on outside of insulation. Size hangers so that pipe insulation passes through them without interruption.
 - 1. Domestic hot water piping above 160 degrees F. 4" diameter and less
 - 2. All other insulated piping
- C. Anvil Figure CT-69, MSS Type 10 with adjustable wrought tubing ring hanger, copper plated for:
 - 1. Non-insulated copper tubing with no longitudinal movement
 - 2. Isolation of copper tubing from dissimilar material shall also be accomplished through the use of PHD Manufacturing, Model Numbers 2501 - 2514 Unistrut clamps with PVC inserts or PHD model number 143 PVC coated swivel ring hangers.

- D. Anvil Figure #171, MSS Type 41 with pipe roller, Anvil Figure #167 protection saddle, MSS Type 40 galvanized insulation protection shields (sized for supporting insulation having a compressive strength of 4 psi, at 8 foot intervals). Support piping on outside of insulation. Size hangers so that pipe insulation passes through them without interruption. Use these for:
 - 1. Domestic hot water above 160 degrees F, 6" diameter and larger.
- E. Anvil Figure #CT-121, MSS Type 8, riser clamps (at floor penetrations) to support:
 - 1. Copper pipe risers
- F. Anvil Figure #261, MSS Type 8, riser clamps (at floor slab penetrations) to support:
 - 1. Steel pipe risers
 - 2. PVC pipe risers
- G. Anvil Powerstrut Trapeze Hangers: Where three or more lines of pipe run parallel, support them with trapeze hangers.
- H. Water piping supports within walls to be by Caddy, Holdrite, Sioux Chief or approved equivalent. Support vertical drops and piping at fixture supplies in wall. Hanger material to be suitable for piping material installed. Piping supports shall be installed per manufacturer's recommendations.

2.3 INSERTS

- A. Concrete Insert: Anvil Figure #281, MSS Type 18, universal concrete inserts, adequately sized and correctly positioned to support full load operating systems.
- B. Concrete Insert, Wedge Type: Anvil Figure #281, 1/4" to 7/8"
- C. Lightweight Concrete Insert: Anvil Figure #285
- D. Continuous Concrete Insert: Anvil Powerstrut Figure #PS-349 pre-galvanized

2.4 EXPANSION ANCHORS

- A. Hilti Kwik-bolt, zinc-plated, metal expansion anchor.
- B. Anchor to meet U.L., ICBO-4627 and FM listings.

2.5 HANGER RODS

- A. Provide mild steel all-thread rods with maximum loads as follows:
 - 1. 3/8" - 300 lbs
 - 2. 1/2" - 600 lbs
 - 3. 5/8" - 1,200 lbs
 - 4. 3/4" - 2,000 lbs
 - 5. 1" - 5,000 lbs

2.6 CLAMPS

- A. C-Clamps: Anvil Figure #92, MSS Type 23.
 - 1. Use these for attaching hangers to steel beams. Do not weld hanger rods to structural steel members.
- B. Malleable Beam Clamps: Anvil Figure #218, MSS Type 30: Use these for attaching hangers to bar joists. Attach clamps to top chord of bar joists only. Confirm with structural engineer for maximum loading and restrictions.

PART 3 - EXECUTION

3.1 PIPE HANGERS

- A. Support pipes on specified hangers so that equipment, pumps, and fittings do not bear weight or stresses from vibration and swaying of pipe. Support pipe risers at regular intervals in pipe shafts at least once at each floor level or a maximum of 12'-0" apart. Do not use perforated metal, strap iron, or band iron. Do not make offsets in hangers.
- B. Maximum allowable spacing of pipe hangers is listed below. Space hangers and brackets at closer intervals where necessary to maintain levels, slopes, and drainage, or to prevent sagging or swaying of pipe.
- C. STEEL PIPE - Water
 - 1. 1/4" to 1-1/2" - 7' 0" O.C.
 - 2. 2" to 2-1/2" - 10' 0" O.C.
 - 3. 3" to 4" - 12' 0" O.C.
 - 4. 5" and above - 14'0" O.C.
- D. STEEL - Vapor
 - 1. 1/4" to 1-1/2" - 8' 0" O.C.
 - 2. 2" to 2-1/2" - 13' 0" O.C.
 - 3. 3" and above - 15' 0" O.C.
- E. COPPER PIPE - Water
 - 1. 1/4" to 1-1/4" - 5'0" O.C.
 - 2. 2" to 2-1/2" - 8'0" O.C.
 - 3. 3" and above - 10'0" O.C.
- F. COPPER PIPE - Vapor
 - 1. 1/4" to 1" - 5'0" O.C.
 - 2. 1-1/4" to 2" - 8'0" O.C.
 - 3. 2-1/2" to 4" - 10' 0" O.C.
 - 4. 5" and above -15' 0" O.C.

G. CAST IRON PIPE

1. Space hangers not to exceed 5 feet on centers. Provide minimum of two hangers per section within 18" of joint on barrel and at change of direction and branch connection. Install hanger and supports per CISPI 301-12.

END OF SECTION

SECTION 22 07 00

PLUMBING INSULATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Insulation of plumbing piping
- B. Insulation of plumbing equipment

1.2 RELATED REQUIREMENTS

- A. Section 23 07 00 - HVAC Insulation

1.3 DEFINITIONS

- A. Exposed - Equipment and piping in areas which will be visible without removing ceilings or opening access panels.
- B. Concealed - Installed above ceiling, in walls or chases.
- C. Outdoors - Exposed to the weather or ambient conditions.
- D. Underground - Buried.

1.4 SUBMITTALS

- A. Submit manufacturer's product data and installation procedures for review in accordance with the requirements of Division 01.

1.5 QUALITY ASSURANCE

- A. Insulation, adhesives, coatings, sealers, jackets and tapes shall have a flame spread rating of 25 or less and smoke development of 50 or less in accordance with ASTM E-84 and UL 723.
- B. Materials shall meet the requirements of NFPA 90-A.
- C. Manufacturer qualifications: ISO 9001-2000 Certified specializing in manufacturing the products specified in this section.
- D. Applicator qualifications: Company specializing in the installation of the specified products and the work required to install the products with not less than 5 years experience.

PART 2 - PRODUCTS

2.1 PIPE AND EQUIPMENT INSULATION

- A. Materials for Pipe and Equipment: Provide factory premolded insulation for pipe, pipe fittings, and valves.

- B. Fitting insulation: Same thickness and material as adjoining pipe insulation.
- C. Cellular Glass (Foamglass):
 - 1. Acceptable manufacturers: Pittsburgh Corning "Foamglass" type; minimum "R" value of 2.63
 - 2. Mastic:
 - a. Inside applications: Foster Vapor Fas 30-65, Childers CP-34 or Vimasco 739. Permeance shall be 0.03 perms or less at 45 mils dry per ASTM E 96."
 - 3. Use on the following services:
 - a. Domestic water piping above grade at cooling tower to 18" below grade, 1-1/2" thick.
- D. Flexible Tubular Elastomeric:
 - 1. Provide fire-retardant closed-cell slip-on flexible type; minimum "R" value of 2.57
 - 2. Acceptable manufacturers: Armacell; Nomaco K-Flex
 - 3. Use on the following services:
 - a. Moisture condensate drains - 1/2" thick
- E. Fiberglass Pipe Insulation:
 - 1. Acceptable manufacturers: Johns-Manville "Micro-Lok 850"; CertainTeed; Knauf; Owens Corning, Foster. Jacket: ASJ fiberglass reinforced kraft paper with aluminum foil; minimum R value of 3.7.
 - 2. Use on the following services:
 - a. Domestic hot water supply piping - 1" thick
 - b. Domestic hot water recirculation piping - 1" thick
 - c. Domestic cold water piping - 1/2" thick
 - d. Horizontal rain water leaders and roof drain bodies - 1" thick
 - e. Drain bodies, traps and horizontal drain lines receiving cold condensate - 1/2" thick
- F. Fiberglass Equipment insulation:
 - 1. Acceptable Manufacturers: Johns-Manville; CertainTeed; Knauf; Owens Corning, Foster.
 - 2. Apply to equipment as directed by the manufacturer.
 - 3. Install rigid or flexible insulation, 2" minimum thickness with 'K' value of 0.27 at 75 degrees F with a maximum service temperature of 850 degrees F. (1200 degrees F for engine exhaust insulation).
 - 4. Provide vapor barrier as required when the liquid temperature is below 65 degrees F.

2.2 MATERIALS FOR FITTINGS, VALVES, AND SPECIAL COVERINGS

- A. For all services, use premolded insulation for pipe fittings, elbows, tees, butterfly valves, and couplings 2-1/2 and larger. Finish shall be as specified under Products above or as specified below. PVC fitting covers with fiberglass inserts may be used on piping fittings elbows and valves 2" and less for the following services:
 - 1. Domestic Cold Water

2. Domestic Hot Water
 3. Domestic Hot Water Recirc.
- B. For piping installed above grade exposed-to-the-weather outside the building, cover straight pipe insulation with 0.016" thick aluminum jacket equivalent to Childers and cover fittings with factory formed covers equivalent to Elljacs.
 - C. For flexible tubular elastomeric pipe and fitting insulation when exposed-to-view inside building or exposed to the weather, finish with two coats of fire retardant self-extinguishing vinyl lacquer type highly flexible coating equivalent to Armacell "Armaflex Finish", custom color blended to match surrounding surfaces.

2.3 JACKETS

- A. Canvas Jacket: UL listed 6oz/sq. yd. plain weave cotton fabric treated with dilute fire retardant lagging adhesive compatible with insulation.
- B. Aluminum Jacket: ASTM B209 formed aluminum sheet of 0.016 inch, smooth finish with longitudinal slip joints and 2" laps, 0.016" thick die shaped fittings with factory attached protective liner. Adhere with 3/8" wide aluminum bands.
- C. PVC Jacket: One piece molded type fitting covers and sheet material, off-white in color, 15 mil thickness, 0.002 perm inch maximum in accordance with ASTM E96. Adhere with pressure sensitive color matching vinyl tape.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Deliver and store insulation materials in manufacturer's containers and keep free from dirt, water, chemical and mechanical damage.
- B. Complete piping pressure testing prior to applying insulation.
- C. Apply insulation in workmanlike manner by experienced and qualified workmen.
- D. Surfaces shall be clean and dry when covering is applied. Covering to be dry when installed and before and during application of any finish, unless such finish requires specifically a wetted surface for application.
- E. Adhesives, cements and mastics shall be compatible with materials applied and shall not attack materials in either wet or dry state.

3.2 FLEXIBLE SHEET ELASTOMERIC INSULATION

- A. Prior to application of flexible sheet elastomeric insulation, thoroughly clean all metal surfaces, making sure that all dirt, scale, loose paint, plaster, and oil has been removed and that surfaces are dry. If surface has been primed, test a two square foot section using adhesive equivalent to Armstrong No. 520 in order to determine whether solvent in adhesive will loosen or lift the primer. If primer is loosened, then remove it. When testing proves acceptable, adhere insulation with smooth side out, using thin but adequate coating of same adhesive. Follow manufacturer's instructions. Coat all butt edges of each sheet. Stagger all joints. Insulate all standing seams or flanges with same thickness of insulation material as that used on main surface.

3.3 INSTALLATION OF PIPE AND EQUIPMENT COVERING

- A. Where glass fiber or flexible tubular elastomeric insulation is used on piping sized 2" and larger, insert a section of foamglass or calcium silicate insulation, at hanger or support points, between pipe and metal shield for full length of shield, to prevent crushing of insulation. Where insulation passes through pipe hangers and across trapeze supports, 12" long metal saddles shall be used. Insulation thickness to be same as adjoining glass fiber insulation. On cold pipe, vapor barrier should be carried through the hanger and sealed. Saddles shall be used where rigid foamglass inserts are not acceptable.
- B. Pipe exposed in mechanical equipment rooms within 8 feet of the finished floor: Finish with canvas jacket ready for finish painting.
 - 1. 15 mil PVC jacket shall be applied to pipes and shall be color coded for the system can be accepted as a contractor option.C. Exterior Applications: Provide vapor barrier jacket. Insulate fittings, joints and valves with insulation of like material and thickness as adjoining pipe and finish with glass mesh reinforced vapor barrier cement. Cover with aluminum jacket with seams located on bottom side of horizontal piping.
- C. Install equipment insulation in accordance with manufacturer's requirements and OSHA requirements.
- D. Apply foamglass insulation as follows:
 - 1. Both the circumferential and longitudinal joints shall be buttered with fire-resistive pliable sealer. Voids and cracks shall be filled with sealer. Apply appropriate mastic as specified under Part 2 - Products. Secure insulation with 3/4" wide x 0.010" thick aluminum bands on 8" centers.
 - 2. The circumferential joints shall be staggered.
 - 3. Fittings, valves, flanges, traps, and air vents shall be insulated with the same thickness of insulation using factory fabricated fitting sections or pre-molded insulated fittings.
 - 4. Block type insulation shall be adhered by stick-clips or bands, in addition to the sealer, as required to provide support for the insulation.
 - 5. Finish above furred ceilings and in chases shall be the bare insulation.
 - 6. Finish in equipment rooms and elsewhere where exposed-to-view shall be White ASJ jacket.

END OF SECTION

SECTION 22 11 16

DOMESTIC WATER PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Piping and pipe fittings for domestic cold, hot and recirculating water piping.

1.2 RELATED REQUIREMENTS

- A. Section 22 05 00 - Common Work Results For Plumbing
- B. Section 22 05 53 - Identification for Plumbing Piping and Equipment
- C. Section 22 11 19 - Domestic Water Piping Specialties

1.3 SUBMITTALS

- A. Submit product data for review on piping and fittings in accordance with the requirements of Division 01. Submittal data shall include:
 - 1. Manufacturer of pipe.
 - 2. Tests or listings by recognized testing laboratory that certifies material composition is in accordance with ANSI/ASTM requirements.
 - 3. Product data for pipe and fittings to be used on each piping system.
 - 4. Solder and brazing product data and installation procedures for copper pipe.
 - 5. ProPress installation methods and instructions for copper pipe.

1.4 QUALITY ASSURANCE

- A. Lead Free: All wetted surface of pipe, fittings and fixtures in potable water systems shall have a weighted average lead content equal to or less than 0.25% per the Safe Drinking Water Act (Section 1417) as amended January 4, 2011.
- B. NSF Compliance: NSF/ANSI 61, NSF 61-G and/or NSF/ANSI 372 for valve materials for potable-water service. Valves for domestic water must be 3rd Party Certified.
- C. Identify pipe with marking including size, ASTM material classification and specification and water pressure rating.
- D. Compliance with ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings.
- E. Compliance with ASME B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings.
- F. Compliance with ASME B16.51 - Copper and Copper Press-Connect Pressure Fittings.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Provide temporary end caps and closures on piping and fittings. Maintain in place until installation.
- B. Protect piping systems from entry of foreign materials by temporary covers, completing sections of the work and isolating parts of the completed system.

PART 2 - PRODUCTS

2.1 COPPER PIPE

- A. Conform to ASTM B-88 specification for wrought seamless copper.
- B. Type K, hard for:
 - 1. Domestic cold water piping 3" and larger
 - 2. Domestic hot water 3" and larger
 - 3. Domestic hot water recirculating piping 3" and larger
- C. Type L, hard for:
 - 1. Domestic cold water piping 2 1/2" and smaller
 - 2. Domestic hot water 2 1/2" and smaller
 - 3. Domestic hot water recirculating piping 2 1/2" and smaller
- D. Use Type K, rolled, soft for: Piping installed under floor slabs.

2.2 COPPER PIPE FITTINGS

- A. Mechanical Couplings: Roll groove rigid type by Victaulic, Gruvlok or Anvil.
- B. Mechanical joints: Domestic cold and hot water piping 3" and larger may be joined with rolled grooved copper fittings (Type K) and valves by Victaulic or Anvil, copper pipe grooved systems. Gaskets shall be rated for not less than 220 degrees. Joints, materials, and gaskets for grooved systems shall be approved for use in potable water systems.
- C. Grooved copper system:
 - 1. Victaulic Style 607 or Style 641 rolled grooved copper fittings for domestic cold water and domestic hot water for sizes 3" and larger. (Type K)
- D. Dielectric Connections:
 - 1. Provide at junction of copper pipe and equipment with steel piping systems.
 - 2. Central, Dielectric insulating unions, and insulating flange unions, as manufactured by Central Plastic Company, or CTS Fabrication USA (1-1/2" thru 8").
 - 3. Brass fittings and valves may not be used for dielectric union locations.
- E. Unions: Brass ground joint, 250 lb. working pressure.
- F. Nipples: Brass.

2.3 ROLL GROOVE JOINT FOR COPPER

- A. Copper tube shall conform to ASTM B-88. Types K with the Anvil Gruvlok, Victaulic , or equivalent roll grooved fittings. Gruvlok model numbers are used below.
- B. Grooved copper fittings shall be Gruvlok Wrot Copper fittings per ASTM B-75 and ANSI B-16.22, alloy C12200 copper. Wrot Copper fittings size 3" - 8" diameter fittings. Copper fittings are 99.9% lead free. Intended for usage with Gruvlok series 7400 Rigidlite couplings, and series 7012 flange adapters. Couplings and Wrot Copper Fittings are NSF, Plumbing Code approved and UL/ULC listed.
- C. Copper tube system couplings shall be Gruvlok Fig. 7400 Rigidlite and Fig. 7012 coupling flange adapters. The Grade "E" EPDM gasket for the Gruvlok Copper Method covers a service temperature range from -40 degrees F to +190 degrees F. Use "Gruvlok Xtreme Lubricant" only. Coupling working pressure is 300 psig maximum.
- D. All Gruvlok coupling gaskets except DRI-SEAL must be lubricated with approved lubricant as provided by Gruvlok. Gruvlok Xtreme Lubricant is strongly recommended below -20 degrees F, and above 180 degrees F and systems subject to continuous cycle temperature changes. Standard and other specific Gruvlok lubricants can be used on other applications as recommended by Gruvlok.

2.4 PRESSED COPPER FITTINGS

- A. Pressed Copper Fittings: (VIEGA) Wrought copper, ASTM B88, conforming to ASME B16.51, NSF 61 and NSF 61-G and/or NSF 372 with dimensions conforming to ANSI B16.22 or B16.18. Pressed copper fittings 1/2" through 2 1/2" shall be suitable for use with ASTM B88 copper tube Type L and 1/2" up to and including 2 1/2" annealed copper tube. Pressed copper fittings shall have an EPDM sealing element conforming to IAMPO PS 117 performance criteria and shall have a Smart Connect feature.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Clean inside of pipe before installation. Keep installed piping clean, and protect ends from foreign matter by capping or plugging them.
- B. Install pipe so that it does not interfere with opening of doors or apparatus, access to equipment, or to electrical equipment.
- C. Run pipes in straight lines and square with building. Install risers plumb. Make offsets only where indicated and where necessary.
- D. Identify piping and systems in accordance with Section 22 05 53.

3.2 PIPE JOINTING

- A. Preparing Pipe Ends:
 - 1. Machine cut pipe ends square
 - 2. Ream pipe ends, after cutting, to full diameter
- B. Grooved Coupling Installation:
 - 1. Pipe ends shall be clean and free from indentations, projections and roll marks in the area from pipe end to groove for proper gasket sealing.

2. The gasket style and elastomeric material (grade) shall be verified as suitable for the intended service as specified.
3. All grooved components (couplings, fittings, valves, gaskets, bolts and nuts) shall be of one manufacturer.

C. Pressed Copper Fittings:

1. Press copper fittings shall be made in accordance with the manufacturer's most current installation instructions. Tube ends shall be cut on a right angle (square) to the tube. Tube ends shall be reamed and chamfered; all grease, oil and dirt removed from the tube end with a clean rag. The tubing shall be fully examined to ensure there is no damage. Fully insert the tube into the fitting and mark the tubing at the shoulder of the fitting. The fitting alignment shall be checked against the mark on the tubing to assure the tubing is fully engaged (inserted) in the fitting prior to pressing the joint. The joints shall be pressed using Ridgid Pressed Copper tools or other tool(s) approved by the manufacturer. Fitting shall have a 5 year warranty that the system shall be free of failure from manufacturing defects.
2. Installers shall attend a Pressed Copper installation training class from the manufacturer prior to any installation. Records of the installers training shall be maintained at the job site.

D. Bracing Joints:

1. Provide braces and bridle rods as required to reinforce joints.
2. If mechanical couplings are used, then prepare pipe ends and make joints in accordance with pipe coupling manufacturer's printed instructions.
3. Where large pipes underground are subject to shock because of sudden changes in liquid flow rate, provide concrete "kicker" blocks at joints, fittings, and changes of pipe direction. Provide "kicker" blocks in accordance with applicable pipe industry trade or research organization recommendations.
 - a. For example, for ductile iron pipe follow recommendations of Ductile Iron Pipe Research Association.

3.3 TESTING

- A. Before piping is concealed or insulated, recheck it for leaks.
- B. Rework or replace defective and leaking joints, and joints which are otherwise unsatisfactory. Peening, caulking, and doping are not permitted.
- C. After Pressed Copper fittings have been installed, a "step test" shall be followed:
 1. Pressurize the system with air or dry nitrogen between 0.5psig and 45psig, or with water between 15psig and 85psig for a minimum of 4 hours.
 2. Check the pressure gauge for pressure loss. If the system does not hold pressure, walk the system and check for un-pressed fittings.
 3. When un-pressed fittings are found, ensure the tube is fully inserted into the fitting and the tube is properly marked prior to pressing the fitting.
 4. Make appropriate repairs and retest the system per local code and these specifications.

END OF SECTION

SECTION 22 11 19

DOMESTIC WATER PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. This section includes requirements for:
 - 1. Backflow preventers.
 - 2. Shock arrestors.

1.2 RELATED REQUIREMENTS

- A. Section 22 05 00 - Common Work Results For Plumbing
- B. Section 22 05 23 - Valves For Plumbing Piping
- C. Section 22 07 00 - Plumbing Insulation

1.3 QUALITY ASSURANCE

- A. Lead Free: All wetted surface of pipe, fittings and fixtures in potable water systems shall have a weighted average lead content equal to or less than 0.25% per the Safe Drinking Water Act (Section 1417) as amended January 4, 2011.
- B. NSF Compliance: NSF/ANSI 61 and/or NSF/ANSI 372 for valve materials for potable-water service. Valves for domestic water must be 3rd Party Certified.

1.4 SUBMITTALS

- A. Submit product data for review in accordance with the requirements of Division 01.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Acceptable manufacturers are indicated in subsequent paragraphs.

2.2 BACKFLOW PREVENTERS

- A. Acceptable manufacturers:
 - 1. Beeco
 - 2. Apollo
 - 3. Febco
 - 4. Cla-Val
 - 5. Ames

6. Watts
 7. Wilkins Regulator Company
- B. Provide completely automatic unit, fitted with tight closing shut-off valves and test cocks at each end.
 - C. Construct such that all parts are replaceable without removing unit from line.
 - D. Total pressure drop through complete backflow preventer does not exceed 12 PSI at rated flow.
 - E. Provide per local requirements and dimensions as detailed on drawings.
 - F. Backflow preventers shall meet the following standards: (Apollo Figure numbers are used for reference.)
 1. Atmospheric vacuum breaker - Apollo 38-200 series, ASSE 1001-2008
 2. Hose-connected vacuum breakers - Apollo 38-414, ASSE 1011-2004
 3. Back siphonage vacuum breakers - Apollo 4A-500-04F, ASSE 1056-2013
 4. Reduced pressure principle backflow preventers - Apollo 4ALF-200 Series, ASSE 1013-2011
 5. Pressure type vacuum breaker - ASSE-1020-2004

2.3 SHOCK ARRESTORS

- A. Acceptable manufacturers:
 1. Josam
 2. Wade
 3. Jay R. Smith
 4. Precision Products
 5. Zurn
 6. Sioux Chief
- B. Arrestor shall be piston type, polycarbonate with two EPDM O-rings, lubricated with FDA-approved Dow Corning #111 silicone compound in Type L or K copper body, suitable for 200 psig minimum pressure at 200 degrees F.
- C. Arrestor shall be ANSI/ASSE 1010 Certified and be maintenance free with no access panel required.

PART 3 - EXECUTION

3.1 INSTALLATION AND TESTING

- A. Backflow Preventers:
 1. Pipe relief through fixed air gap and discharge to sewer.
 2. Install adjacent to wall and/or floor utilizing stand-off brackets, angle frame, and/or concrete piers.
 3. Test unit for leaks and pressure drop. Clean and/or replace soiled strainer media.

4. Provide dual parallel reduced pressure backflow preventers on the main domestic water entrance to the facility.
5. Provide a reduced pressure backflow preventer on the domestic hot and cold water lines feeding the laboratory and morgue.
6. Provide a reduced pressure backflow preventer on the domestic cold water serving carbonators at soda machines.
7. Provide backflow prevention vacuum breaker on any water line feeding any piece of equipment which could cause back siphonage such as mechanical equipment, trap primer lines, etc.

B. Shock Arrestors:

1. Install shock arrestors at each quick closing valve, solenoid type valve, and flush valve. Size shock arrestors in accordance with manufacturer's instructions.
2. Install shock arrestors within five feet of valve, provide wall access panel as required.
3. Test and certify shock arrestors by Plumbing and Drainage Institute in accordance with ANSI/ASSE 1010.

END OF SECTION

SECTION 22 13 16

STORM AND SANITARY WASTE AND VENT PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Sanitary waste and vent piping

1.2 RELATED REQUIREMENTS

- A. Section 22 05 00 - Common Work Results For Plumbing
- B. Section 22 05 29 - Hangers for Plumbing Piping
- C. Section 22 13 19 - Sanitary Waste Piping Specialties
- D. Section 23 21 13 - HVAC Piping

1.3 SUBMITTALS

- A. Submit product data for review on piping and fittings in accordance with the requirements of Division 01. Submittal data shall include:
 - 1. Manufacturer of pipe.
 - 2. Tests or listing by recognized testing laboratory that certifies material composition is in accordance with ANSI/ASTM requirements.
 - 3. Product data for pipe and fittings to be used on each piping system.
 - 4. Identification of where each pipe type will be used.

1.4 QUALITY ASSURANCE

- A. Identify pipe with marking including size, ASTM material classification and ASTM specification.

1.5 DELIVERY, STORAGE AND PROTECTION

- A. Protect piping systems from entry of foreign materials by temporary covers, completing sections of the work and isolating parts of the completed system.

PART 2 - PRODUCTS

2.1 CAST IRON PIPE AND FITTINGS

- A. Conform to ASTM A-74, A-888, and CISPI 301-12 and CISPI 310-12.
- B. Pipe and fittings shall be marked with the collective trademark of Cast Iron Soil Pipe Institute (CISPI) and be listed by NSF International.

- C. Acceptable Manufacturers: Charlotte Pipe and Foundry Company; NewAge Casting; Tyler Pipe, a part of McWane family of companies
- D. Standard weight (pipes larger than 12") and Schedule 40 (12" and smaller) pipe with drainage fittings for:
 - 1. Sanitary waste, vent, and drainage pipe 2" and larger above ground.
 - 2. Building storm drains.
 - 3. Rainwater conductors inside building.
 - 4. Drain lines under buildings, and under exterior concrete or other paving. Extend cast iron piping at least 5 feet outside of building.
- E. Joints in Cast Iron Pipe:
 - 1. Below grade: Bell and spigot with neoprene compression gaskets
 - a. Gaskets: ASTM C 564
 - b. Caulking Materials: ASTM B29, pure lead and oakum or hemp fiber.
 - 2. Above grade: No-Hub using stainless couplings.
 - a. Acceptable Manufacturers: ANACO-Husky; MIFAB, Inc.; NewAge Casting; Tyler Pipe.
 - b. Provide 4-band, heavy duty couplings for piping 2" through 10" and 6-band heavy duty couplings for piping 12" and larger. Couplings shall comply with ASTM C 1540/ FM-1680, ASTM C1277 and CISPI 310, rated no hub bands for all cast iron piping material above slab-on-grade.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Cast Iron Pipe Joints:
 - 1. Install compression gaskets and No-Hub bands in accordance with CISPI installation methods and manufacturer's instructions.
- B. Grading Pipes for Drainage:
 - 1. Uniformly place storm drainage pipes and footing drain pipes at elevations and slopes indicated. If no elevations or slopes are indicated, slope pipes at not less than 1/8" per foot.
 - 2. Uniformly place sanitary sewer pipes at elevations and slopes required by the local codes
- C. Bracing Joints:
 - 1. Provide braces and all-thread bridle rods as required to reinforce sanitary and storm piping joints at each change of direction within building and where otherwise not direct buried below ground.
 - 2. If mechanical lock type couplings are used, then prepare pipe ends and make joints in accordance with pipe coupling manufacturer's printed instructions.

3. Where large pipes underground are subject to shock because of sudden changes in liquid flow rate, provide concrete "kicker" blocks at joints, fittings, and changes of pipe direction. Provide "kicker" blocks in accordance with applicable pipe industry trade or research organization recommendations.
- D. Clean inside of pipe before installation. Keep installed piping clean, and protect ends from foreign matter by capping or plugging them.
- E. Do not install piping above electrical equipment such as starters, variable frequency motor controllers, motor control center's, or disconnects. Maintain code required clearance above, below and to sides of electrical equipment.
- F. Do not install piping above or passing through any IT rooms, IDF rooms, or service entrance rooms.
- G. Run pipes in straight lines and square with building. Install risers plumb. Make offsets only where indicated and where necessary.
- H. Identify all waste and vent piping in accordance with and as specified in Section 22 05 53.

END OF SECTION

SECTION 22 13 19

SANITARY WASTE PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. This section includes requirements for:
 - 1. Cleanouts
 - 2. Trap primers

1.2 RELATED REQUIREMENTS

- A. Section 22 05 00 - Common Work Results For Plumbing
- B. Section 22 11 16 - Domestic Water Piping
- C. Section 22 13 16 - Storm And Sanitary Waste And Vent Piping

1.3 SUBMITTALS

- A. Submit product data for review in accordance with the requirements of Division 01.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Acceptable manufacturers are indicated in subsequent paragraphs.

2.2 CLEANOUTS

- A. Acceptable manufacturers:
 - 1. Zurn (Zurn model numbers are used below)
 - 2. Josam
 - 3. Wade
 - 4. Jay R. Smith
 - 5. Sioux Chief
 - 6. Watts
- B. Finished concrete floor: ZN1400 cast iron body with round adjustable polished nickel bronze top, ABS plug and carpet marker where required.
- C. Ceramic tile: ZN1400 Series, cast iron body, polished nickel bronze top, 1/2" terrazzo recess and closure plug.

- D. Vinyl tile floor: ZN1400-X, cast iron body, round nickel bronze top, 1/8" tile recess and closure plug.
- E. Carpet: ZN1400-CM. Inside caulk round brass scoriated frame and cover and provide carpet marker.
- F. Wall: Z1441, cast iron caulking ferrule with stainless round access cover and screws.
- G. Access covers: Minimum size 12" x 12" located for access to valves, shock absorbers, trap primers, wall cleanouts, etc.
- H. Furnish cleanouts occurring in waterproof floors with clamping devices.

2.3 TRAP PRIMERS

- A. Acceptable manufacturers:
 - 1. Josam
 - 2. Zurn
 - 3. Wade
 - 4. Jay R. Smith
 - 5. Precision Plumbing Products
 - 6. Sioux Chief
- B. Provide trap primer of brass construction, with removable operating parts, and integral vacuum breaker.
- C. See Plumbing fixture section for specifications.

PART 3 - EXECUTION

3.1 INSTALLATION AND TESTING

- A. Cleanouts
 - 1. Provide line size cleanouts up to 4"; 4" cleanout for lines larger than 4".
 - 2. Locate cleanouts at all changes in direction greater than 45 degrees and in straight runs as shown 100 feet outside the building on drawing or spaced not greater than required by applicable Plumbing Code.
 - 3. Extend inaccessible cleanouts up through floor and/or wall to provide easy accessibility.
- B. Trap Primers
 - 1. Install primers in accessible location or as shown on drawings.
 - 2. Trap primers shall be Plumbing and Drainage Institute approved.

OF SECTION

SECTION 22 43 00

HEALTH CARE PLUMBING FIXTURES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Compliance with the provisions of Section 22 05 00.
- B. Plumbing fixtures, trim and related items such as supplies, traps, drains, cleanouts, water closet flanges, bolts, seats and covers, fixture supports and other accessory items.
- C. Coordination of fixture requirements by reviewing architectural, structural, and equipment drawings. Install fixtures in accordance with Contract Drawings and manufacturer's rough-in drawings.
- D. Installation of water tempering devices at all public lavatories and similar hand washing fixtures and elsewhere noted on the drawings.

1.2 RELATED REQUIREMENTS

- A. Section 22 05 23 - Valves For Plumbing Piping
- B. Section 22 11 16 - Domestic Water Piping
- C. Section 22 13 19 - Sanitary Waste Piping Specialties

1.3 REFERENCE STANDARDS

- A. Perform work in accordance with applicable codes and standards enforced by local authorities.
- B. All barrier free fixtures shall be installed in accordance with the Americans with Disabilities Act (ADA) Rules and Regulations.

1.4 SUBMITTALS

- A. Submit manufacturer's product data: fixtures; fittings; accessories and supplies for review in accordance with Division 01 requirements.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing the type of products specified in this section with a minimum of five years of documented experience.
- B. Products requiring electrical connections must be UL or ETL listed and classified suitable for the purpose specified.
- C. All fixtures, faucets, trim and accessories must be protected from damage at all times including after installation to prevent unauthorized use.
- D. All wetted surfaces of faucets, mixing valves, isolation valves, balancing valves and shower valves in potable water systems must be lead free and ASSE 1070 and NSF 61 Certified.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Fixtures: Kohler, American Standard, Zurn
- B. Brass Trim: McGuire Engineered Brass Company, Brasscraft, Zurn.
- C. Faucets: T&S Brass and Bronze Works, Chicago Faucet Company, Sloan, Kohler, Zurn, Symmons.
- D. Stainless Steel Sinks: Elkay, Just, Kohler.
- E. Flow Control Devices: Dole Flow Controls Company.
- F. Provide fixtures and trim as a complete unit as required in the individual "P" numbers listed below.
- G. Pre-fabricated insulation on water lines and P-trap under barrier free lavatories and sinks: Plumberex, Trap Wrap TrueBro, Inc. Handi Lav-Guard, McGuire Pro-Wrap. Products must meet 25/50 flame and smoke spread ratings.
- H. Thermostatic Mixing Valves: Symmons, Holby, Lawler, Powers, Leonard, Acorn, Guardian.
- I. Emergency Equipment. Encon, Haws, Guardian, and Acorn.

2.2 MATERIALS

- A. Wall Hung Lavatories: Furnish complete with floor mounted carriers Zurn Z1231.
- B. Provide mixing valves per ASSE or CSA Standards as required by the local adopted code. Mixing valves used in drinking water shall be lead free. Mixing valves shall be used to supply tempered water to public hand-washing facilities and shall conform to ASSE 1070 or CSA B125.3.
- C. Clamping Device: Provide for drains installed in slabs above grade.
- D. Trap Primer: Provide connections for floor drain as shown on drawings.
- E. Caulking: General Electric silicon sanitary sealant or approved equal. Color to match fixture, clear for SS sinks.
- F. Wrist Blade Operation:
 - 1. Pull wrist blade to operator to turn on.
 - 2. Push toward backsplash to turn off faucet.

2.3 PLUMBING FIXTURES

- 1. Refer to plumbing fixture schedule on sheet P0.00.PART 3 - EXECUTION

3.1 FIXTURE CONNECTIONS

- A. Connect to plumbing fixtures and equipment provided under this and other sections of the specifications, architectural drawings, and manufacturer's shop drawings. Provide rough-in connections as shown on drawings.

- B. Use schedule and details on drawings and/or manufacturer's shop drawings for connection sizes to fixtures.
- C. Provide separate p-trap for each fixture, floor drain, and piece of equipment.
- D. Provide cast iron p-traps under floor drains.
- E. Provide deep seal traps under floor drains in air conditioning unit plenums, walk-in cooler and freezer units, storage rooms, toilet rooms, and elsewhere as indicated on drawings.
- F. Coordinate dimensions required for minimum fixture clearances with other Divisions.
- G. Caulk around joints at fixtures mounted on wall or floor, or backed up to walls.
- H. Mount fixtures rigid to walls as shown on drawings and details.
- I. Gooseneck faucets intended for staff and patient, use shall not contain aerators but shall contain plain end outlet trim ring. Flow control devices shall be installed at the base of the gooseneck.
- J. Projects under the International Plumbing Code shall include a mixing valve below each sink and/or lavatory to limit water temperature per IPC requirements. Mixing valves to be similar to Symmons # 7-210 or Powers LM495 and compliant with ASSE 1070 requirements.

3.2 TESTING AND CLEANING

- A. Inspect and test work to insure that it is installed in accordance with drawings, specifications and manufacturer's requirements and is functioning as designed and required. Use test procedures and pressures as specified and required under this Division.
- B. Correct all deficiencies found and retest.

END OF SECTION

SECTION 22 60 00

MEDICAL GAS PIPING SYSTEMS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Furnish and test the following medical gas systems:
 - 1. Oxygen (O₂).
 - 2. Carbon Dioxide (CO₂)
 - 3. Mixed Gas
- B. Outlets, valve boxes, valves, alarm systems, pressure and vacuum switches, and miscellaneous accessories for complete systems.
- C. Mixed Gas manifold system.
- D. Pressure testing, cross connection testing and final testing, including purging, outlet flow and analyzing.
- E. Electrical wiring for ceiling columns/hose drops and other electrical accessories associated with the system shall be furnished and installed as follows:
 - 1. Line voltage wiring by Division 26
 - 2. Low voltage alarm wiring by Division 22
- F. Anesthesia evacuation connectors

1.2 RELATED REQUIREMENTS

- A. Section 22 05 00 - Common Work Results for Plumbing
- B. Section 22 60 13 - Medical Gas Systems - Startup and Certification
- C. Division 26: Electrical

1.3 SUBMITTALS

- A. In accordance with the requirements of Division 01, submit manufacturer's product data for all components indicating size, dimensions and configuration including:
 - 1. Operating instructions
 - 2. Maintenance, service and repair data
 - 3. Manufacturer's installation instructions
- B. Brazier performance qualification test records for each brazer used on the installation. Brazers shall be qualified per N.I.T.C./ASSE 6010 with a minimum of 37 hours of training prior to installing any medical gas components.
- C. Manufacturer certification to indicate that all tubes, valves, fittings, station outlets, and other piping components have been cleaned for oxygen service in accordance with CGA G-4.1.
- D. Project record drawings and lists recording actual locations of piping, valving and outlets.
- E. Independent "System Verification" testing report and certification including (1) a one-page management certification summary, (2) fully documented test data and analysis and (3) a test instrument quality assurance and calibration summary. "System Verification" report must document that each component in each individual system is complete and is functioning in accordance with NFPA 99.
- F. Submit proof of supplying pre-cleaned copper piping by submitting invoices from suppliers.

1.4 QUALITY ASSURANCE

- A. Install in compliance NFPA 99 as required and enforced by authority having jurisdiction.
- B. Comply with local, state and federal codes applicable in this jurisdiction.
- C. Employ only qualified journeymen for this work. Employ a competent qualified mechanic/piping foreman, who has satisfactorily completed at least five other similar installations, for this work. Qualification of brazers shall meet the installation requirements per NFPA 99.

- D. All individuals engaged in installing medical gas piping systems shall have a medical gas certification card issued by either N.I.T.C. or ASSE 6000 Series. A current certification card shall be carried on their person while installing any medical gas system components during construction.
- E. All individuals engaged in installing medical gas piping systems shall have a medical gas endorsement issued by the Texas State Board of Plumbing Examiners.

1.5 COORDINATION

- A. Coordinate with other trades to assure timely installations and to avoid conflicts and interference.
- B. Work closely with the metal stud partition installer and/or mason to assure that anchors, sleeves and similar items are provided in sufficient time to avoid delays; chases and openings are properly sized and prepared.
- C. Coordinate layout of medical gas systems in all spaces and identify all piping accurately and as specified in Section 22 05 53 - Identification for Plumbing Piping and Equipment.
- D. On renovations projects, prior to receipt of medical gas outlets, contractor shall confirm compatibility of specified outlets with existing project equipment requirements. Notify Architect/Engineer if incompatibility is discovered.

1.6 MEDICAL GAS ALARM SYSTEM

- A. All local, master, and area alarm panels shall have a separate visual indicator for each condition monitored. Each alarm panel shall have a cancelable audible indicator that produces a minimum of 80 dBA measured at 3 feet from panel.
- B. All local, master and area alarm panels shall indicate visually and audible if the wiring to the sensor or switch is disconnected or the monitored condition occurs.
- C. The local, master and area alarm panels shall be powered from the life safety branch of the emergency power system.
- D. All pressure switches and/or sensors shall have a demand check fitting for testing, servicing or replacement.
- E. Wiring from switches to be in parallel and direct to each master alarms. Refer to master alarm panel wiring diagram.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Piping System Components:
 - 1. BeaconMedaes.
 - 2. Allied Health Products, Chemetron Division.
 - 3. Ohio Medical
 - 4. Amico.
 - 5. Powerex medical gas pipeline
 - 6. Patton's Medical
- B. BeaconMedaes manufacturer model numbers are used to establish product quality and scope, not to limit competition.

2.2 QUALIFICATION OF MANUFACTURER

- A. Pipeline System Components:
 - 1. One manufacturer shall supply the medical gas piping system components to include outlets, valves and gauges, valve boxes, alarm boxes, nitrous oxide manifold system, nitrogen manifold system, reserve oxygen manifold system, and nitrogen control cabinet.
 - 2. Component manufacturer shall have a pipeline system engineer or product specialist available to periodically check with the Contractor during installation of pipeline system equipment and provide a service organization to certify the system in accordance with NFPA 99.
- B. Provide ongoing service support to the Owner's Representative after acceptance of system.

2.3 PIPING MATERIALS

- A. Positive Pressure Piping: Seamless Type K or L (ASTM B819) copper tubing, in accordance with NFPA 99. Piping shall be precleaned and plugged by supplier before shipment to jobsite. Piping on-site cleaning

shall not be permitted. Manufacturer pipe material markings shall not be defaced. Pipe markings shall be one or more of the following:

1. OXY, Medical, OXY/MED, OXY/ACR, or ACR/MED as required by NFPA 99.
- B. Medical air system intake, medical vacuum systems, vacuum exhaust, anesthesia evacuation system and anesthesia evacuation exhaust pipe shall be ASTM B 88, B 280, or B 819 Type "K", "L", or "M" hard drawn copper tube with wrought copper fittings. "Medical oxygen service cleanliness" is not required for this pipe.
- C. Fittings: Wrought copper, brass or bronze designed expressly for brazed connection. Fittings shall be preclean and sealed by the manufacturer, and prior to shipping to job site, on site cleaning shall not be permitted. Cast fittings shall not be used for brazed joints. Flared and compression-type connections shall be prohibited throughout the piping system.
- D. Silver brazing material: Stay-Silv-15, Silvaloy-15, Aircosil No. 15, or Phos-Silver-15.
- E. Silver brazing alloy composition: 15% silver, 80% copper, and 5% phosphorus, no cadmium.
 1. Minimum of 1000 degrees F liquid melting point with ASTM rating of "BCuP5".
- F. The use of flux is prohibited for the making of joints between copper to copper pipes and fittings. Appropriate flux similar to "Stay-Silv-Black Flux" or "Stay-Silv-White Flux" is required between dissimilar metals such as copper to brass or bronze material, when parts are heated over a prolonged period. Use flux sparingly. Apply only a thin coat of flux to minimize possibility of flux continuous piping system.
- G. Flux: Do not use for copper-to-copper joints. Use flux for joining copper to brass or bronze. In those cases where flux is used, exercise particular care in applying the flux to avoid leaving any excess inside the completed joints.
- H. Piping Supports:
 1. Isolation of copper tubing from dissimilar metal shall be accomplished through use of PHD Manufacturing, model numbers 2501-2514 Unistrut clamps with PVC inserts or PHD Model No. 143 PVC coated swivel ring hangers.

2.4 HIGH PRESSURE CYLINDER MANIFOLDS

- A. Provide automatic manifold control including self-shifting to secondary reserve bank on exhaustion of primary bank and automatic reset of replaced bank to secondary reserve status.
- B. Incorporate actuating switches in control for the purpose of actuating designated signal to master panel when service bank is exhausted.
- C. Incorporate in control unit a visible means of determining when either bank is exhausted and operation has automatically switched to "reserve in use".
 1. Visible indication shall be by red light over exhausted bank and by gauges showing remainder pressure in each bank.
 2. Continuous visible indication of electrical circuits in effective operation shall be by means of green light.
 3. Mechanical means shall be provided to ensure the connection of cylinders containing the proper gas to the manifold.
 4. Line pressure shall be shown by separate visible gauge. Manifolds shall be installed as shown on drawings. Manifold to include Dome Bias Regulators with identifiable premounted gauge and locking ring to ensure constant bank regulator dome pressure.
 5. Each manifold shall meet the requirements of NFPA 99.
- D. Oxygen manifolds:
 1. BeaconMedaes Medical Products #MNE-HH_x_-O2-AS, to accommodate multiple cylinders.
- E. Nitrous oxide (N2O) manifold:
 1. BeaconMedaes Medical Products #MNE-HH_x_-NO2-AS heater not required to accommodate multiple cylinders divided into 2 equal banks.
- F. Nitrogen (N2) manifold:
 1. BeaconMedaes Medical Products #MNE-HH_x_-N2-CS to accommodate multiple cylinders divided into 2 equal banks.
- G. Carbon dioxide (CO2) manifold:
 1. BeaconMedaes Medical Products #MNE-HH_x_-CO2-AS heater not required to accommodate multiple cylinders to accommodate divided into 2 equal banks.

2.5 MEDICAL GAS VALVING

- A. Main, source, base of riser and service shutoff valves (Valves Not in Boxes)
 - 1. Valves and tubing shall be specially prepared for oxygen service and shall conform in all particulars to NFPA 99.
 - 2. Valves shall be quarter-turn ball-type, with extensions for brazing, Teflon seats and adjustable stem packing gland with Teflon stem seal, through 2". 2-1/2" to 3" valves shall have Teflon seats and double Teflon stem seal. 4-inch valves shall have Buna-N ball seats. Valves shall be three piece type with full ports.
 - 3. Ball valves shall be rated at 400 psig, actuate from full "ON" to full "OFF" by 90 degrees turn of vinyl gripped valve handle. Factory installed copper tubing shall be extended sufficiently to help prevent valve seat damage during brazing. Threaded joints are not acceptable for medical gas valves.
 - 4. Valve manufacturer:
 - a. BeaconMedaes Medical Products #6-21160x-00 (1/2" - 4")
 - 5. Service shutoff valves shall have lockable handles (locks provided by owner) and tagged to prevent accidental opening and closing.
 - 6. Main valve shall be located to permit access by authorized personnel only. The main valve shall be located outside the source room. When the source valve is accessible from within the building a main valve shall not be required.
 - 7. Provide ball valves with gauge ports and plugs.
 - 8. For large vacuum lines, butterfly valves, cleaned for oxygen service, may be used.
- B. Area/Zone Valves (Valves in Boxes)
 - 1. Zone valve boxes shall be constructed of 18" gauge sheet steel with white epoxy finish. The cover frame shall be made of a polymer (ABS) treated with InstaCure Guardian and attached to the box by concealed screws. The finished assembly shall be substantially dust-tight. The frame assembly shall be capable of adjusting for variances in wall thickness up to one inch. The frame assembly shall contain an easily removable opaque cover window with pull ring. The window shall conceal exposed piping and valves within the box and shall be labeled "Caution - Medical Gas Control Valves - Close Only in Emergency". Tinted transparent window shall be provided to display the gas service, the area controlled by the valve, and pressure gauges on units so equipped.
 - 2. Frames for valve boxes shall have uniform width for balanced appearance. Manufacturer shall provide color-coded self adhesive gas service labels for compliance with NFPA 99 labeling requirements. Apply labels to each valve within the assembly for proper gas service identification according to the manufacturer's instructions.
 - 3. Placement of the valve within the zone valve box shall be such that the removable window cannot be replaced when any valve is closed. Factory installed Type K copper pipe extensions shall extend three (3) inches outside the valve box. Design of the valve box shall be such that valves may be removed prior to brazing, without disassembly of the box, to permit field rearrangement of valves if necessary. Valves shall be ball type, cleaned for oxygen service, supplied with capped ends, and shall operate full open to closed position with 90 degree handle rotation. Verify wall depth required with all valves boxes.
 - 4. Valve boxes shall include 1-1/2" pressure gauges reading 0-100 psig for oxygen, nitrous oxide and air; 0-300 psi for nitrogen, and 0-30" Hg for vacuum or evacuation vacuum. The gauge port shall be equipped with removable plug for pressure testing prior to final assembly of gauge.
 - 5. Gauge model zone valve box assemblies shall read pressure downstream of the valve per NFPA 99.
 - 6. All valve assemblies shall have an 1/8" NPT port for sensor installation. Provide Sensor Installation Kit.
 - 7. Valve boxes:
 - a. BeaconMedaes ZVBX-XXX-ENG
 - 8. Provide individual medical gas valve identification as follows:
 - a. Installer shall provide identification for valves in box to identify room name and room numbers per graphic and/or signage package.

2.6 MEDICAL GAS ALARM SYSTEMS (A.A.P.)

- A. Area Alarm (Line Pressure Alarm) -Digital
 - 1. The area alarm as required by the NFPA 99, CSA standard Z 305.1 and shall be U.L. Listed. The alarm shall meet the Federal Electromagnetic Compatibility Standard for Medical Devices 201-0004 to reduce the possibility of magnetic radiation interference with other equipment interference. Each

- area alarm shall include one power supply module, a sensor module for each specific gas, one digital module for each specific gas.
2. The power supply module shall be capable of eight digital modules and converting 120VAC to low voltage DC. This module to contain a fuse to protect the system from voltage and amperage surges and yellow POWER ON indicator. Each power module shall provide an audible signal activated by a fault signal at the Digital Module.
 3. The Digital Module shall provide an audible and visual signal when an advisory or a fault signal is received. An audible alarm silencing switch shall be available, and automatically canceled when the fault is correct. Signal limits shall be factory set, with the ability to be field adjusted without the use of tools. The display shall be programmable to read psig, in. Hg or kPa. Digital modules shall provide three additional audible and visual signals with the ability to field label and function as a master alarm.
 4. The Sensor Module shall contain a transducer capable of providing calibrated signals to the Digital Module. Sensor Modules shall be gas specific and capable of mounting directly in the gas pipeline system, either in the ceiling or directly above the Digital Module in standard wall construction. Pipeline connections shall be 3/8" nominal copper tubing. Connectors shall be provided for attaching field wiring. Sensors shall be capable of being removed without shutting down the medical gas system.
 5. Furnish and install the alarm system and associated wiring under Division 22.
 6. Termination of signal wire at alarm location by manufacturer supplier/contractor.
 7. BeaconMedaes Medical Products Mega 3 or Total Alert Infinity.

PART 3 - EXECUTION

3.1 IDENTIFICATION

- A. Piping shall be labeled as specified in Sections 22 05 00 - Common Work Results For Plumbing and 22 05 53 - Identification for Plumbing Piping and Equipment.
- B. Prior to cutting into existing system, Certifier shall perform a PPM white cloth test to certify that existing system is clean and free of contamination.

3.2 INSTALLATION

- A. Joints in the piping, except those at equipment requiring screwed connections, shall be made with silver brazing alloy or similar high melting point (above 1000 degrees F) brazing metal.
- B. Contractor shall provide a copy of the installers brazing certificate to the engineer for record prior to any installation. Certification and training shall be in accordance with ASME Section 9 or AWS B2.2 and to assure that the pipeline certification report and installation complies with all Chapters of NFPA 99.
- C. During the brazing of pipe connections, the interior of the pipe shall be purged continuously with 5 SCFH of oil free nitrogen (NF) to prevent the formation of copper oxide inside the joint. Contractors shall use oxygen analyzer and purge regulator alarm as required by NFPA 99 requirements. The flow of purge gas shall be maintained until the joint is cool to the touch. Hold pipe full of nitrogen at night during erection of a pipe system. The outside of the tube and fittings shall be cleaned by washing with hot water after brazing the assembly, and prior to inspection of joints and pressure testing.
- D. Threaded joints in piping systems shall be made up with polytetrafluoroethylene (such as Teflon) tape or other thread sealants suitable for oxygen service. Apply sealants to the male threads only.
- E. During erection of the piping systems, the systems shall be free from induction of gaseous hydrocarbons. During brazing, use medical gas equipment shipment plugs or black electrical tape to seal off purged piping. At the most remote point from the pipe being brazed, provide a small opening in the plug or tape to allow purge gas to escape during the brazing process.
 1. The use of duct tape or masking tape for sealing the purged piping systems is strictly prohibited.
- F. Sand cloth/sand paper type abrasive cleaning material shall not be allowed. Joints shall be cleaned with Scotch Brite pads, prior to brazing.
- G. All installation tools shall be cleaned and oil free before installation of piping. Tools shall remain clean during all medical gas system piping installation.
- H. All piping, valves and components shall be inspected prior to erection. If found to be contaminated, do not install the piping, valves or other components in the medical gas piping system.
- I. All pipe ends shall be cut square, and deburred with a sharp clean deburring tool. Care shall be taken to prevent chips from entering the tube or pipe.

- J. Support medical gas piping at distances listed per NFPA-99. Piping shall not be supported from other piping. Unistrut type trapeze hangers shall contain plastic inserts to isolate medical gas piping from dissimilar metals and straps.
1. Isolate copper piping from dissimilar metals. Do not use duct tape as an isolation material. Where medical gas piping is in contact with construction materials. Wrap the exposed medical gas piping with 10 mil thick adhesive Polyken brand protective wrap. Wrap shall extend 6" either side of contact point.
- K. After installation of the piping, but before installation of the outlet valves, blow lines clear by means of oil-free, dry nitrogen (NF).
- L. The qualified brazer/certifier shall inspect the piping system for:
1. Flux or flux residue.
 2. Excessive oxidation of all joints.
 3. Presence of unmelted filler metal.
 4. Failure of the filler metal to be clearly visible all around the joint at interface between the socket and the tube.
 5. Cracks in the tube or component.
 6. Cracks in the braze filler metal.
 7. Failure of the joint to hold the test pressure per NFPA 99 standing pressure test.
- M. Piping protection:
1. Piping shall be protected against freezing, corrosion, and physical damage. Buried piping outside of buildings shall be installed below the level of frost penetration. Piping shall be installed in a continuous PVC Sch-40 encasement when installed below grade. Continuous encasement sleeves shall terminate above slab level at both ends. Encasement shall be sealed water tight on each end.
 2. Buried piping that will be subject to surface loads shall be buried at a sufficient depth (3' minimum) to protect the piping from excessive stresses. Ducts or casings shall be used in road way, drive way, parking lots or other areas subject to surface loading.
 3. The minimum backfilled cover above the top of buried piping outside of buildings shall be 36 in. (91.4 cm), except that the minimum cover shall be permitted to be reduced to 18 in. (45.7 cm) where physical damage to the piping is not likely to occur. Trenches shall be excavated so that the pipe has a firm, substantially continuous bearing on the bottom of the trench. Underground piping shall be installed in a continuous split enclosure, Schedule 40 PVC, to protect the pipe from damage while backfilling. Backfill shall be clean and compacted so as to protect and uniformly support the piping.
 4. A continuous tape or marker, placed immediately above the enclosure, shall clearly identify the pipeline by specific name. In addition, a continuous tape or warning means shall be provided above the pipeline at approximately one-half the depth of bury.
 5. Where underground piping is installed through a wall/floor sleeves, the ends of the sleeve shall be sealed to prevent the entrance of ground water.
 6. Piping exposed to physical damage shall be adequately protected.
 7. Medical gas piping shall be protected from contact with concrete floor slabs and other building materials. For protection from concrete floor slabs, acceptable means shall be PVC encasement sleeves or Polyken brand 10 mil thick adhesive wrap. For dissimilar metal contact, provide 10 mil thick adhesive wrap, extending 6" either side of contact surfaces.
- N. Contractor shall instruct Owner's Representative to obtain certificate from oxygen supplier that the supplier's piping has been installed per brazing and purging requirements per NFPA 99. Oxygen supplier shall clean piping and equipment per pamphlet CGAG-4.1 before connection to certified pipeline system supplied by Contractor. Contractor shall verify cleaning of oxygen supplier piping from bulk system before making final connection to house system.

3.3 INSTALLER PERFORMANCE TESTING

- A. After installation of the piping and valves, but before installation of the service outlets, alarm actuating switches and gauges, and before piping and valves are concealed in walls or ceiling the line shall be blown clear by means of oil-free, dry air or nitrogen.
- B. Next, with station outlets with test caps in place each section of the piping system shall be subjected to a test pressure of one and one-half (1-1/2) times the maximum working pressure, but not less than 150 psig, with oil-free, dry nitrogen (NF). At no time shall any two medical gas pipelines be interconnected for testing such as a piping manifold. Maintain test pressure for at least 24 hours. Then each joint shall be

examined for leakage by means of soapy water or other effective means of leak detection safe for use with oxygen.

- C. Repair leaks and retest the section.
- D. After completing the testing of each individual piping system, all of the medical gas systems shall be subjected to a pressure test at one and one-half (1-1/2) times the maximum working pressure, but not less than 150 psig. The test gas shall be oil-free, dry air or nitrogen. The main line shut-off valve shall be closed during the test.
- E. After completion of the above test procedure the finishing assemblies of alarms, and all components (e.g. pressure switches, gauges, relief valves, etc.) shall be installed and all medical gas piping systems shall be subjected to a 24 hour standing pressure test at 20% above the normal operating line pressure. The main line shut-off valve shall be closed during this test.
- F. Leaks, if any, shall be located, repaired, and the system retested.
- G. Purge testing of each medical gas outlet at a minimum rate of 225 LPM shall be performed with a white cloth loosely held over the adaptor until purge produces no discoloration of the white cloth.
- H. To determine that no cross connection to other pipeline systems exists, reduce all systems to atmospheric pressure. Disconnect all sources to test gas from all of the systems with the exception of the one system to be checked. Pressure this system with oil-free, dry nitrogen (NF) to a pressure of 50 psig. With appropriate adapters matching outlet labels, check each individual station outlet of all systems installed to determine that test gas is being dispensed from only the outlets of this system.

END OF SECTION

SECTION 22 60 13

MEDICAL GAS SYSTEMS - STARTUP AND CERTIFICATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Testing, certification and startup of the medical gas equipment and pipeline systems.
- B. The Certification of the medical gas pipeline system must be performed by a certifier technically competent and experienced in the field of medical gas pipeline testing. Certifiers shall be licensed in accordance with the requirements of ASSE (American Society of Sanitary Engineering) Series #6030 (N.I.T.C. Level M-4 or equivalent). Testing shall be performed by a party other than the installing contractor.
- C. The air compressor and vacuum pump supplier shall be present at site for system startup to verify equipment is installed and operating properly in accordance with the requirements of NFPA 99.
- D. When new piping is "cut-in" to previously installed piping, it is possible that previously installed piping could be contaminated or even cross-connected. In conjunction with and in addition to inspection and certification of the new medical gases system pipelines, it is imperative that previously installed systems be inspected for crossed-connections in the piping systems and purity tested prior to connection of new system.

1.2 RELATED REQUIREMENTS

- A. Section 22 60 00 - Medical Gas Piping Systems
- B. Section 22 60 01 - Medical Gas Piping Systems
- C. Electrical Connections:
 - 1. Under Division 26, provide power wiring for vacuum pumps and air compressors.
 - 2. Under Division 22, provide local and remote wiring from sensors and contacts to alarm panels and annunciation devices in accordance with the requirements of NFPA 99. Provide conductors and raceways in accordance with the requirements of Division 26.

1.3 SUBMITTALS

- A. Submit certifier credentials indicating certification of personnel to perform this work on the medical gas system.
- B. Provide written test reports as to the procedures to be followed regarding the certification.
- C. Provide completed reports and certifications of test results described herein.

1.4 QUALITY ASSURANCE

- A. Provide certificate of compliance from the authority having jurisdiction, indicating approval of the system.
- B. Testing Laboratory: Company specializing in performing the work of this section with a minimum of 5 years of documented experience.

PART 2 - PRODUCTS

2.1 NOT APPLICABLE

PART 3 - EXECUTION

3.1 CERTIFICATION PROCEDURE

- A. Perform mechanical check of all medical gas outlets prior to the certification inspection. Any necessary repairs or rework shall be done prior to system certification inspection. Check to include:
 - 1. Outlets properly supported
 - 2. Installation complete
 - 3. Appropriate adapters fit and securely lock in place
- B. The Owner is responsible for ensuring that bulk tank and/or cylinder supplies are installed, connected, and filled (or partially filled) prior to system certification inspection.
- C. Medical gas system suppliers and/or certification company and Plumbing Contractor sign Medical Gas Pipeline Inspection Agreement describing service to be performed by the medical gas system supplier,

services not performed by the medical gas system supplier, price quotation, and exceptions affecting quoted price.

- D. The Contractor shall provide a qualified certifier to perform pipeline certification inspection and to provide report and certification in accordance with previously signed agreement.
 - 1. Any discrepancies discovered during the inspection shall be noted, corrected, and any and all portions of the system affected by corrective action shall be retested and findings recorded after retest.
- E. Provide representative who shall serve as customer contact person and who shall witness the certification inspection and certify that all outlets have been checked and are in accordance with inspection procedure and findings as witnessed.
- F. Hospital Facilities Department to provide representative who shall witness the certification and certify that all spaces and the outlets have been checked in accordance with inspection procedure.
- G. All air, oxygen, nitrous oxide and carbon dioxide outlets shall be tested to deliver 3.5 SCFM at 50 psi with a maximum pressure drop of 5 psi. All nitrogen outlets shall be tested to deliver 5 SCFM at 160 psi with a maximum pressure drop of 5 psi. Medical surgical vacuum inlets shall draw 3.0 SCFM without reducing the vacuum below 12" Hg at any adjacent station inlet. These tests shall be performed with oil-free, dry nitrogen.
- H. Each medical gas outlet shall be purge tested to verify that a minimum of 35 cubic feet of oil free dry nitrogen shall be filtered through a clean white 0.45 micron filter at a minimum flow of 3.5 SCFM. Filter shall not accrue more than 0.1 mg of matter. This test shall be performed at twenty-five percent of the zones for each gas type. Samples taken within each zone shall be taken at the outlet furthest from the zone valve.
- I. For clinical air, oxygen, nitrogen, nitrous oxide and carbon dioxide systems measure and record the dew point, total hydrocarbons, and halogenated hydrocarbons. The maximum allowable variation for each parameter shall be:
 - 1. Dew point: .5 degrees C @ 50 PSI
 - 2. Total hydrocarbons: 1.0 PPM
 - 3. Halogenated hydrocarbons: 2.0 PPM
 - 4. Tests shall be performed on the outlet most remote from the source and compared with source gas tests. The two test shall not exceed the measurements outlined above. These tests shall be performed with oil-free dry nitrogen.
- J. The valves shall be tested to verify proper operation and rooms or areas of control. Each valve shall be labeled for the rooms or area they control. Verify room names with owner's representative, prior to labeling valves.
- K. The alarm panels wiring shall be tested to ensure that all components are functioning properly. The alarm panels shall be tested per NFPA 99.
- L. The certifier shall furnish copies of Medical Gas Pipeline Inspection Report and Medical Gas Pipeline Certification. Transmit a copy of the report and certification to each of the following parties:
 - 1. Contractor
 - 2. Owner's Construction Manager
 - 3. Architect
 - 4. Consulting Engineer
 - 5. Hospital
- M. The following procedure should be followed in addition to the above on extensions to existing systems:
 - 1. Architect shall, with adequate advance notice, request that certifier or inspection team be on-site when old piping is cut-in for installation of new lines.
 - 2. Architect or Contractor shall arrange to have certifier or inspection team on-site when the existing piping is cut into for installation of new lines.
 - 3. Main line shut-off valves shall be installed in new piping as close as possible to point to cut-in to previously installed piping.
 - 4. After verification of proper labeling, pressure and proper gas distribution of previously installed piping and after cut-in procedure, the aforementioned valves shall be considered the service valve of supply for the new piping.
 - 5. Before connection to the existing system and before system is put into service, the system shall be pressure tested, purged, outlet flow, tested, and gases analyzed per NFPA 99.

6. After re-establishing pressure in each previously installed supply line after "cut-in", the previously installed outlets immediately upstream for each gas and for each cut-in point shall be opened to ensure that the proper gas is being delivered. Gas should be bled to atmosphere prior to analyzation to ensure purging from point of cut-in to test point.
- N. The presence of the desired gas shall be confirmed with the combined use of an analyzer designed to measure the specific gas dispensed, and a pressure gauge attached to an appropriate adaptor with results tabulated below:
1. Oxygen: 99-100% Oxygen, Pressure: 50 psi, +/- 5 psi
 2. Carbon Dioxide: +/- 1% Oxygen; Temporarily reduce pressure at source to 30 psi. Outlet pressure shall be 30 psi, +/- 5 psi
- 3.2 CROSS CONNECTION TESTING AND CERTIFICATION
- A. Cross connection testing and certification of the medical gas system must be performed by party technically competent and experienced in the field of medical gas pipeline testing.
- B. Medical gas system shall be tested in accordance with NFPA 99 and Section 22 60 00 - Medical Gas Piping Systems of these specifications.
- C. In addition to cross connection testing, this specification shall require the technical certifier to test each individual pipeline systems component for performance to design specifications and make any necessary adjustments to ensure a complete and working system.
- D. The system shall be tested for cross connection in one of the following methods:
1. Test one pipeline system at a time at 50 psi while the others are at atmospheric pressure including the vacuum system.
 2. Reduce the pressure in all medical gas systems to atmospheric pressure. Increase the test pressure to the piping to the following pressures:
 - a. Gas Mixtures: 20 psi
 - b. Compressed Air: 60 psi
 3. Any medical-surgical vacuum system shall be in operation at same time that medical gases are tested.
- E. Obtain and present to the Owner a complete bound report of pipeline certification from the equipment manufacturer. Also include a letter of certification indicating:
1. That the system is free of crossed connections.

END OF SECTION

SECTION 22 62 19

MEDICAL GAS SYSTEMS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Medical gas systems.

1.2 REFERENCES

- A. ANSI B16.18 - Cast Copper Alloy Solder-Joint Pressure Fittings.
- B. ANSI B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings.
- C. ANSI B40.1 - Gauges, Pressure and Vacuum, Indicating Dial Type-Elastic Element.
- D. ASME Boiler and Pressure Vessel Code.
- E. ASTM A167 - Stainless and Heat-Resisting Chromium - Nickel Steel Plate.
- F. ASTM A269 - Stainless and Welded Austenitic Stainless Steel Tubing for General Service.
- G. ASTM A403 - Wrought Austenitic Stainless Steel Piping Fittings.
- H. ASTM B32 - Solder Metal
- I. ASTM B88 - Seamless Copper Water Tube.
- J. ASTM B280 - Seamless Copper Tube for Air Conditioning and Refrigeration field Service.
- K. ASTM D1785 - Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
- L. ASTM D2466 - Poly (Vinyl Chloride) (PVC) Plastic Pipe and Fittings.
- M. ASTM D2564 - Solvent Cements for Poly (Vinyl Chloride) (PVC) Plastic Pipe and Fittings.
- N. AWS A5.8 - Brazing Filler Metal.
- O. CGA G-7 - Compressed Air for Human Respiration.
- P. CGA P-2.1 - Medical-Surgical Vacuum Systems in Health Care Facilities.
- Q. CGA V-5 - Diameter Index Safety System Non Interchangeable Low Pressure Connections for Medical Gas Applications.
- R. FM - Factory Mutual System - Approval Guide.
- S. FS TT-P-645 - Primer, Paint, Zinc Chromate, Alkyd Type.

- T. FS W-C-596 - Electrical Power Connector, Plug, Receptacle, and Cable Outlet.
- U. FS WW-V-35 - Valve Ball.
- V. FS WW-V-54 - Valve, Gate, Bronze (125, 150 and 200 Pound, Screwed, Flanged, Solder End, For Land Use).
- W. MIL-R-36557 - Regulator, Pressure, Medical Gas Administration Apparatus.
- X. MIL-V-82026 - Valves, Diaphragm, Stop.
- Y. MSS SP-58 - Pipe Hangers and Supports - Materials, Design and Manufacture.
- Z. MSS SP-69 - Pipe Hangers and Supports - Selection and Application.
- AA. NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum).
- AB. NFPA 50 - Bulk Oxygen Systems at Consumer Sites
- AC. NFPA 99 - Standard for Health Care Facilities.

1.3 SUBMITTALS

- A. Shop Drawings: Indicate general assembly of components, mounting and installation details, and general layout of control and alarm panels. Submit detailed medical wall assembly drawings.
- B. Product Data: Provide manufacturers literature and illustrations for all components indicating size, dimensions and configuration.
- C. Independent Testing Agency Reports: Indicate systems are complete, zone valves installed, alarm systems functional, and pressure and cross connections tests performed. Document tests.
- D. Manufacturer's Installation Instruction: Indicate requirements for equipment and systems.

1.4 PROJECT RECORD DOCUMENTS

- A. Submit 3 copies of as-built drawings, and provide electronic file in AutoCAD format.
- B. Record actual locations of piping, valving, and outlets.

1.5 OPERATION AND MAINTENANCE DATA

- A. Submit 5 copies of operation and maintenance manuals.
- B. Operation Data: Include installation instructions, assembly views, lubrication instructions, and assembly views.
- C. Maintenance Data: Include maintenance and inspection data, replacement part numbers and availability, and service depot location and telephone.

1.6 QUALITY ASSURANCE

- A. Perform Work in accordance with NFPA 99.
- B. Maintain one copy of each document on site.

1.7 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing the products specified in this Section with minimum three years documented experience.
- B. Installer: Company specializing in performing the work of this Section with minimum three years documented experience.
- C. Testing Laboratory: Company specializing in performing the testing of this Section with minimum three years documented experience.

1.8 REGULATORY REQUIREMENTS

- A. Conform with applicable codes for medical gas systems.
- B. Provide certificate of compliance indicating approval of systems.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Accept material on site in factory containers and packing. Inspect for damage.
- B. Protect from damage and contamination by maintaining factory packaging and caps in place until installation.

1.10 SCHEDULING

- A. Schedule Work to ensure equipment is installed and systems tested and certified prior to substantial completion.

1.11 SCOPE

- A. Provide all labor, materials, equipment, tools and services and perform all operations required in connection with or properly incidental to the installation and testing of oxygen, nitrogen, nitrous oxide, compressed air and vacuum piping systems including fittings, valves, medical gas outlets, air compressors, nitrogen and nitrous oxide manifolds, vacuum pumps, alarms, etc. for complete operable systems.
- B. The stock and model numbers of equipment listed hereinafter identify equipment manufactured by NCG Division of Chemetron Corp., Chicago, Ill. Equal equipment by approved manufacturers will be acceptable. PART 2 - PRODUCTS

2.1 PIPING

- A. All piping for vacuum and gases of every character shall be ASTM Specification B-88, Type L, hard drawn, seamless copper tubing with wrought copper solder fittings. No ferrous piping will be permitted in

the system. Where threaded nipples are required these shall be I.P.S. brass. All vacuum piping shall be purged with dry nitrogen while being soldered.

- B. All piping shall be pitched back so as to drain to the point shown on the Drawings. All branch takeoffs shall be made from the top of the mains.
- C. Not Used.
- D. Fittings for copper tube shall be wrought copper fittings and attached with silver solder alloy containing not less than 50% silver. All soldering shall be done with dry nitrogen flowing through the pipe to prevent oxidation and scale information.
- E. Before erection, all pipe, tubing, valves and fittings (except those supplied expressly cleaned for oxygen, nitrogen, nitrous oxide, air-and-vacuum service by manufacturer) shall be thoroughly cleansed of all grease, oil and other combustible materials by washing in a hot solution of sodium carbonate or trisodium phosphate mixed in equal proportions of one pound to three gallons of water. Scrubbing and continuous agitation of the parts shall be employed where necessary to remove all deposits and to insure complete cleansing. After washing, all materials shall be rinsed thoroughly in clean, hot water. After rinsing, great care must be exercised in the storage and handling of all materials and in the condition of tools used in cutting and reaming to prevent oil or grease being introduced into the tubing. Where such contamination is known to have occurred, the materials affected must be rewashed and then rinsed.
- F. Where screwed connections are required at equipment, suitable adapters shall be provided with threaded connections. A thin paste of litharge and glycerin shall be applied to the external threads only.
- G. After erection of pipe and tubing, but prior to installation of the service outlet valves, each system shall be blown clear of moisture and foreign matter by means of dry nitrogen or oil free air.
- H. After installing service outlet valves, each system shall be subjected to a test pressure of 150 psig by means of water-pumped (oil free) nitrogen or air. This test pressure shall be maintained until each joint has been thoroughly examined for leaks by means of soapy water. A soap solution mixed in the following proportions should be used: one ounce of castile or palm oil soap, eight ounces of water, and four ounces of glycerine. Dissolve the soap in the water, add the glycerin and mix thoroughly. Wipe joints clean after test. All leaks shall be properly repaired and the system retested.
- I. A final test shall be 24 hours standing pressure test with water pumped (oil free) air or dry nitrogen at 150 psig to check the completeness of prior joint pressure tests. If water pumped nitrogen is used, particular care must be exercised to assure that it is all flushed out with oxygen before placing the system in service.

2.2 MEDICAL GAS VALVES

- A. Valves not in boxes shall be NCG bronze bodied, double seal, full flow ball type, with Teflon seat seals and O-ring packing designed for working pressures up to 300 psi with a chrome plated brass ball which seals in both directions. The valves shall be so designed that only a quarter turn of the lever type handle is necessary between the open and closed positions. Valves shall be supplied and properly washed for oxygen service. Gas service labels shall be provided for each service as required.
- B. Shutoff valves in recessed boxes shall be installed in boxes with back box constructed of 18 gauge Paintlok steel with a gray baked on semigloss finish and with plaster flanges on all four sides for securing to wall. The valves shall be bronze bodied, double seal, full flow ball type with Teflon seat seals, O-ring packing designed for working pressures up to 300 psi, chrome plated bronze balls which seal in both directions, and adjustable Teflon stem seals and bearings, self compensating to guard against leakage due to wear. Tubing extensions shall be factory soldered to the valve flanges for connection to piping

outside the box. Valves and tubing extensions shall be chrome plated, preassembled, pressure tested and rigidly mounted to the box for ease of installation. The valves shall require only a quarter turn of the handle to completely open or close. A color coded gas label shall be supplied with each valve, which can be marked to indicate the area controlled. The cover shall be of 18 gauge stainless steel with No. 4 brushed finish, and shall incorporate an internal service identification cover and shield, providing shutoff directions. The combination internal and external fascia shall attach to the box assembly without the use of screws, and shall compensate for variations in plastic thickness. Mounted in the finishing frame shall be a clear, rigid, vinyl window for easy access to the valve. Window shall contain a caution label reading: "CAUTION, CLOSE ONLY IN EMERGENCY" and "PULL RING TO BREAK WINDOW".

2.3 VACUUM PUMP AND ACCESSORIES

- A. Provide vacuum pumping units as scheduled on the drawings. Each unit shall be arranged for triplex operation and shall be complete with driving motors, tanks, automatic controls including alternates and accessories as hereinafter specified.
- B. Pumps shall have capacity as scheduled on the Drawings and shall be rotary liquid sealed type, cast iron body, bronze fitted; BeaconMedaes.
- C. Tank shall be welded steel, hot dipped galvanized after fabrication, inside and out; size as scheduled with gauge and necessary taps for pipe connections.
- D. Provide automatic control consisting of tank mounted adjustable vacuum switches, motor starters and disconnect switches, automatically starting and stopping vacuum pumps in duplex operation. Provide controls as required for automatic alternating of each pump.
- E. Provide vacuum relief valve, silencer and solenoid valve for water seal line to each vacuum pump. Wire solenoid valve to open and close with starting and stopping of motor. Install Crane No. 8H-550 backflow preventer in water seal line to vacuum pump.
- F. Pipe air and water discharge from each pump to nearest floor drain with silencer and pipe trap installed as shown on the Drawings.
- G. A vacuum exhaust line shall be installed to exhaust vapors and odors to outside of building to point approved by Architect/Engineer. The exhaust line shall terminate adjacent to pumps in a drip leg of minimum 10" length with cock valve drain, for purpose of trapping condensates. Flexible hose connections and in-line exhaust mufflers shall be installed as shown on Drawings.

2.5 FINAL CHECKING AND OPERATING INSTRUCTIONS

- A. A representative of the equipment manufacturer shall periodically check with the Contractor during initial installation of the pipeline systems equipment. He shall assist the Contractor in final check to make certain that all systems are in perfect operating condition. The equipment manufacturer's representative shall provide 8 hours of instruction to the hospital personnel in the use of the piping systems and the related equipment which is operated from those systems.

2.6 STANDARDS AND CODES

- A. The recommendations of the National Fire Protection Association (NFPA) as set forth in Pamphlet No. 56, 565 and 566, and the "Standard for Medical/Surgical Vacuum Systems in Hospitals" as set forth in Compressed Gas Association (CGA) Pamphlet No. P-2.1, Second Edition 1967, shall apply to this installation and shall be adhered to in all respects.

2.7 ALARM SYSTEMS

- A. The alarm systems for all services shall be consolidated into one integrated panel and shall be installed complete with all necessary sensing, sending and display devices. All signals shall be flush mounted with facility for job site service identification and shall incorporate:
1. Brushed stainless steel fascia.
 2. Visual indication of whether circuit is energized.
 3. Audible buzzer recording of all signal impulses.
 4. Reset permitting silencing of audible buzzer but not allowing cancellation of visual identify until situation has been corrected.
 5. Shall be listed as an assembly by Underwriters' Laboratories, Inc.
 6. Ability to receive and properly record by audible buzzer of any combination of sequential signal.
 7. The signal panel shall be equipped with a "test" button which shall actuate for test, all visual and audible alarms.
 8. The panel shall be of modular construction, facilitating service and maintenance.
 9. Power supply boxes shall remain mounted and connected as required by manufacturer.
 10. Pressure switches shall be mounted in the pipe lines to be sensed and shall be connected as required by manufacturer.
- B. Monitoring Gauges: Monitoring gauges shall be NCG No. 480991-64 and shall be complete with brushed stainless steel fascia with services identified, plastic protective dial free, back box, check valves, and 3/8" sensing line. Provide valves to isolate gauge from pressure line. These gauges shall be installed in the locations as required by NFPA Code and where indicated on the Drawings.

3.1 INSTALLATION: INSTALL IN COMPLETE COMPLIANCE WITH GOVERNING CODES AND MANUFACTURERS INSTRUCTIONS.

3.2 MANUFACTURER: EXCEPT FOR PIPING AND PIPE FITTINGS, ALL COMPONENTS SHALL BE SUPPLIED BY A SINGLE MANUFACTURER AND SHALL BE FULLY COMPATIBLE WITH OWNER'S EXISTING SYSTEM AND SERVICE DEVICES.

3.3 TESTING: TEST IN ACCORDANCE WITH NFPA 99, SECTIONS 4-3 THROUGH 4-10. PROVIDE TO OWNER A NOTARIZED LETTER OF CERTIFICATION FROM EQUIPMENT MANUFACTURER CERTIFYING THE FOLLOWING:

- A. No cross connections exist.
- B. Alarm system is adjusted and performing to manufacturer's design.
- C. All components have been installed, adjusted and are functioning in accordance with manufacturer's recommendations.3.4 COMPATIBILITY: VERIFY COMPATIBILITY OF ALL NEW COMPONENTS WITH EXISTING SYSTEM AND SERVICES.

END OF SECTION

SECTION 22 67 13

PURIFIED WATER SYSTEM AND PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Labor, materials, tools, equipment, and services for purified water equipment and piping system noted below and as indicated on drawings.
- B. Deionized Water systems

1.2 OWNER PROVIDED EQUIPMENT

- A. Coordinate and verify with the Owner equipment provided for contractor hookup.
- B. Install and connect and test all owner provided pure water equipment and systems.

1.3 SUBMITTALS

- A. Submit product data for owner and engineer review for water purification equipment, piping and jointing methods. Submit product data for piping, valves, fittings, and supports for review in accordance with the requirements of Division 01.
- B. Submit test reports including pressure tests, sterilization tests and purity tests to Owner at completion of job.

1.4 QUALITY ASSURANCE

- A. Standard for Disinfection of water mains: AWWA C601.
- B. NSF Standard 61.
- C. Lead Free Drinking Water Act of January 4, 2014.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. As indicated for individual items.

2.2 DEIONIZATION UNITS

- A. Acceptable Manufacturers: Ultrapure; Culligan Water Systems; Aqua Systems; United Industries
- B. Components:
 - 1. Deionization tanks
 - 2. Operating valve assembly
 - 3. Flow Controls
 - 4. Exchange resin
 - 5. Control panels

2.3 VALVES

- A. All valves: Schedule 80 PVDF rated for 125 PSIG working pressure, non-shock.
- B. Polyvinylidene Fluoride (PVDF): Valves will be manufactured from virgin unpigmented PVDF (Polyvinylidene fluoride) to be fully compatible with high purity PVDF system. All valves test to 150 psi @ 73°F.
- C. Polypropylene valves: Valves will be manufactured from virgin unpigmented Type I homopolymer polypropylene to be fully compatible with the pipe material.
- D. Shut-off valves: Ball type, socket or threaded ends, and Teflon seating.
- E. Check valves: Ball type for either vertical or horizontal mounting, with threaded or socket ends.
- F. Provide reduced pressure backflow preventer in piping serving equipment producing distilled or deionized water.
- G. Acceptable manufacturers: IPEX (Enfield), Orion, George Fisher, Nibco

2.4 PIPE AND FITTINGS

- A. Deionized and distilled water piping:

1. Polyvinylidene Fluoride (PVDF) pipe and fittings: Schedule 80, iron pipe dimensions, from virgin, unpigmented PVDF (Polyvinylidene fluoride) resin, meeting ASTM D3222, using no plasticizers or pigments. Meet all dimensional tolerances of ASTM D2447. Fittings: Join by using the socket fusion method, conforming to ASTM 2657.
2. Polypropylene pipe and fittings: Schedule 80 iron pipe dimensions from virgin unpigmented, Type I homopolymer polypropylene, meeting ASTM D4101, using no plasticizers or antioxidants. Meet the dimensional tolerances of ASTM D2447. Manufacture fittings for joining by the socket fusion heat method, conforming to ASTM D2657.
3. Acceptable manufacturers: IPEX (Enfield), Orion, George Fisher.
4. Piping and accessories shall meet flame spread and smoke developed ratings of 25/50.

PART 3 - EXECUTION

3.1 PIPING INSTALLATION

- A. Install and hang piping as required by manufacturer's recommendations.
- B. Install piping to provide every fixture and item of equipment requiring pure and deionized water with suitable supply connection.
- C. Consult manufacturer's data and large scale details of rooms containing plumbing fixtures on architectural drawings before roughing in piping. Plug or cap piping immediately after installation.
- D. Connect equipment furnished by others in accordance with Section 22 05 00 - Common Work Results For Plumbing.
 1. For each water supply piping connection to equipment by others, furnish and install a Tru-union and a gate, ball or angle valve.. Provide a stop valve at each sink water supply. Minimum line size 1/2 inch.
 2. Cap and protect until such time as installation is performed.
- E. Furnish and install wiring and conduit under Division 26.

3.2 INSTALLATION

- A. Install all equipment and systems in accordance with the manufacturer's written instructions and requirements.

3.3 STERILIZATION OF SYSTEMS

- A. Sterilize system as indicated, or in accordance with AWWA C601.
- B. Thoroughly flush system.
- C. After flushing, introduce chlorine or chlorine compound into system with dosage sufficient to give an initial residual chlorine content of 50 ppm.
- D. Collect samples from various taps and fixtures throughout buildings during introduction of chlorine to assure uniform distribution. Open and close valves several times.
- E. After 24-hour contact period, flush all traces of heavily chlorinated water from systems.
- F. After flushing is complete, provide evidence of effectiveness of disinfection by filing with Engineer and owner's representative, laboratory reports of bacteriological tests on samples taken from system at each faucet and piece of equipment.
- G. If satisfactory results are not obtained, repeat above disinfection process until satisfactory results are obtained.
- H. Do not put system into service until satisfactory tests are reviewed by Architect.

3.4 TESTING

- A. Do not insulate, conceal or furred in pipe until it has been tested to satisfaction of Architect.
- B. Upon completion of a section of system, test hydrostatically to pressure not less than 50 percent in excess of maximum pressure to which pipe will ordinarily be subjected, but in no case less than 100 PSI.
- C. If test shows defects, replace such defective work or material and repeat tests.
 1. Make repair to piping with new material.
 2. No caulking of screwed joints or holes will be acceptable.

- D. Test all systems and water for required purity, cleanliness and suitability for the application it is intended. Obtain from the Owner the requirements of each system. Coordinate all work with the equipment vendor as necessary.

END OF SECTION

SECTION 23 05 00

COMMON WORK RESULTS FOR HVAC

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Basic Mechanical Requirements specifically applicable to Division 23 Sections, in addition to Division 1 - General Requirements

1.2 APPLICABILITY

- A. THE UNIFORM GENERAL CONDITIONS, SUPPLEMENTARY GENERAL CONDITIONS, and DIVISION 1 of the Specifications apply to the work specified in this Section.
- B. All work covered by this Section of these Specifications shall be accomplished in accordance with all applicable provisions of the Contract Documents and any addenda or directives which may be issued herewith, or otherwise.

1.3 GENERAL

- A. The Contractor shall execute all work hereinafter specified or indicated on accompanying Drawings. Contractor shall provide all equipment necessary and usually furnished in connection with such work and systems whether or not mentioned specifically herein or on the Drawings.
- B. The Contractor shall be responsible for fitting his material and apparatus into the building and shall carefully lay out his work at the site to conform to the structural conditions, to avoid all obstructions, to conform to the details of the installation and thereby to provide an integrated satisfactory operating installation.
- C. The mechanical, electrical, and plumbing Drawings are necessarily diagrammatic by their nature, and are not intended to show every connection in detail or every pipe or conduit in its exact location. These details are subject to the requirements of standards referenced elsewhere in these specifications, and structural and architectural conditions.
- D. The Contractor shall carefully investigate structural and finish conditions and shall coordinate the separate trades in order to avoid interference between the various phases of work. Work shall be organized and laid out so that it will be concealed in furred chases and suspended ceilings, etc., in finished portions of the building, unless specifically noted otherwise. All exposed work shall be installed parallel or perpendicular to the lines of the building unless otherwise noted.
- E. When the mechanical, electrical, and plumbing Drawings do not give exact details as to the elevation of pipe, conduit and ducts, the Contractor shall physically arrange the systems to fit in the space available at the elevations intended with proper grades for the functioning of the system involved.
- F. New construction projects will be provided with a Hierarchy Drawing or sections and elevations, which clearly show the general elevations that utilities will be routed in N-S and E-W directions. The contractor shall not start work until this drawing has been provided.
- G. Piping, exposed conduit and the duct systems are generally intended to be installed true and square to the building construction, and located as high as possible against the structure in a neat and workmanlike

manner. The drawings do not show all required offsets, control lines, pilot lines and other location details. Work shall be concealed in all finished areas.

1.4 TERMS AND DEFINITIONS: (NOTE: THESE DEFINITIONS ARE INCLUDED HERE TO CLARIFY THE DIRECTION AND INTENTION OF THIS SPECIFICATION. THE LIST GIVEN HERE IS NOT BY ANY MEANS COMPLETE. FOR FURTHER CLARIFICATION AS REQUIRED, CONTRACTOR SHALL CONTACT THE DESIGNATED OWNER'S REPRESENTATIVE.)

- A. ENGINEER – UTSW Engineer, Owner, or Owner's designated representative
- B. CONCEALED / EXPOSED: "Concealed" areas are those areas which cannot be seen by the building occupants. "Exposed" areas are all areas which are exposed to view by the building occupants, including under counters, inside cabinets and closets, plus all mechanical rooms.

1.5 PERMITS, UTILITY CONNECTIONS AND INSPECTIONS

- A. General: Refer to DIVISION 1 for construction phasing and time increments.
- B. Fees and Costs: If, during the course of the construction, a need arises to buy utilities, the Contractor shall pay all fees attendant thereto. If City or privately owned utility piping or electrical cable needs to be extended, relocated, or terminated, the Contractor will pay all permits and construction/inspection fees associated with that particular work.
- C. All work performed on this project is under the authority of the State of Texas, therefore no local construction fees or construction permits will be required except as may be required for new service taps, or new or modified connections to City controlled services. If inspections by City personnel are specifically required by this document, then the Contractor is responsible for any fees or permits in connection to those requirements.
- D. Compliance: The Contractor shall comply in every respect with all requirements of National Fire Protection Association, local Fire Department regulations, and utility company requirements. In no case does this relieve the Contractor of the responsibility of complying with these Specifications and Drawings where specified conditions are of higher quality than the requirements of the above specified authorities. Where requirements of the Specifications and Drawings are more lenient than the requirements of the above authorities having jurisdiction, the Contractor shall make installations in compliance with the requirements of the above authorities with no extra compensation.

1.6 CONTRACT DOCUMENTS

- A. All dimensional information related to new structures shall be taken from the appropriate Drawings. All dimensional information related to existing facilities shall be taken from actual measurements made by the Contractor on the site.
- B. The interrelation of the Specifications, the Drawings, and the schedules are as follows: The Specifications determine the nature and setting of the several materials, the Drawings establish the quantities, dimensions and details, and the schedules give the performance characteristics. If the Contractor requires additional clarification, he shall request it in writing, following the contractually prescribed information flow requirements.
- C. Should the Drawings or Specifications conflict within themselves or with each other, the better quality, or greater size or quantity of work or materials shall be performed or furnished except where directed otherwise in writing by the design professional.

1.7 OWNER FURNISHED PRODUCTS

- A. Products furnished to the site and paid for by Owner will be noted on the drawings.

1.8 FUTURE WORK

- A. Future work will be noted on the Drawings.

1.9 ALTERNATES

- A. Alternates quoted on Bid Forms will be reviewed and accepted or rejected at the Owner's option. Accepted Alternates will be identified in Owner Contractor Agreement.
- B. Coordinate related work and modify surrounding work as required.
- C. Schedule of Alternates: See "Special Conditions" and Bid Form.
- D. Any Alternate Proposals are summarized in Division 1 of the Specifications. The Contractor is directed to refer to all Sections of the Specifications and Drawings for this project to determine the exact extent and scope of the various Alternate Proposals as each pertains to the work of all trades.

1.10 SUBMITTALS

- A. Refer to the enclosed Specifications.
- B. Submit shop drawings and product data grouped to include complete submittals of related systems, products, and accessories in a single submittal.
- C. Mark dimensions and values in units to match those specified.
- D. Submit Fabrication Drawings whenever (1) equipment proposed varies in physical size and arrangement from that indicated on the Drawings, thus causing rearrangement of equipment space, (2) where tight spaces require extreme coordination between ductwork, piping, conduit, and other equipment, (3) where called for elsewhere in these Specifications; or (4) where specifically requested by the Architect/Engineer. Fabrication Drawings shall be made at no additional charge to the Owner or the Architect/Engineer.
- E. All required Fabrication Drawings, except as noted otherwise, shall be prepared at a scale of not less than $1/4" = 1'-0"$. Fabrication Drawings for ductwork, air handling units, and sections in Mechanical Rooms shall be drawn at a minimum scale of $3/8" = 1'-0"$. Submit Fabrication Drawing to the Architect/Engineer for review in the quantity and format as specified in Division 1. Reproduction and submittal of the Construction Documents is not acceptable. The Architect/Engineer will review the drawing and return with comments.

1.11 SUBSTITUTION OF MATERIALS AND EQUIPMENT

- A. Refer to General Conditions for substitution of materials and equipment.
- B. General: Within thirty days after the date of contract award or work order, whichever is later, and before purchasing or starting installation of materials or equipment, the Contractor shall submit for review, a complete list of suppliers, contractors and manufacturers for all materials and equipment which will be submitted for incorporation into the project. The list shall be arranged in accordance with the organization of the Specifications. This initial list shall include the manufacturer's name and type or catalog number as required to identify the quality of material or equipment proposed. This list will be reviewed by the

Engineer and the Owner and will be returned to the Contractor with comments as to which items are acceptable without further submittal data and which items will require detailed submittal data for further review and subsequent approval. The initial list shall be submitted as herein specified. Materials and equipment requiring detailed submittal data shall be submitted with sufficient data to indicate that all requirements of these Specifications have been met and samples shall be furnished when requested. All manufacturers' data used as part of the submittal shall have all inapplicable features crossed out or deleted in a manner that will clearly indicate exactly what is to be furnished.

- C. It is not the intent of the Drawings and/or Specifications to limit products to any particular manufacturer or to discriminate against an "APPROVED EQUAL" product as produced by another manufacturer. Some proprietary products are mentioned to set a definite standard for acceptance and to serve as a reference in comparison with other products. When a manufacturer's name appears in these Specifications, it is not to be construed that the manufacturer is unconditionally acceptable as a provider of equipment for this project. The successful manufacturer or supplier shall meet all of the provisions of the appropriate specification(s).
- D. The specified products have been used in preparing the Drawings and Specifications and thus establish minimum qualities with which substitutes must at least equal to be considered acceptable. The burden of proof of equality rests with the Contractor. The decision of the designer is final.
- E. When requested by the Architect/Engineer, the Contractor shall provide a sample of the proposed substitute item. In some cases, samples of both the specified item and the proposed item shall be provided for comparison purposes.
- F. Timeliness: The burden of timeliness in the complete cycle of submittal data, shop Drawings, and sample processing is on the Contractor. The Contractor shall allow a minimum of six (6) weeks' time frame for review of each submission by the office of the design discipline involved after receipt of such submissions by that design discipline. The Contractor is responsible for allowing sufficient time in the construction schedule to cover the aforementioned cycles of data processing, including time for all resubmittal cycles on unacceptable materials, equipment, etc. covered by the data submitted. Construction delays and/or lack of timeliness in the above regard are the responsibility of the Contractor and will not be considered in any request for scheduled construction time extensions and/or additional costs to the Owner.
- G. All equipment installed on this project shall have local representation; local factory authorized service, and a local stock of repair parts.
- H. Acceptance of materials and equipment will be based on manufacturer's published data and will be tentative subject to the submission of complete shop Drawings indicating compliance with the contract documents and that adequate and acceptable clearances for entry, servicing, and maintenance will exist. Acceptance of materials and equipment under this provision shall not be construed as authorizing any deviations from the Specifications, unless the attention of the Architect/Engineer has been directed in writing to the specific deviations. Data submitted shall not contain unrelated information unless all pertinent information is properly identified.
 - 1. Certification: The Contractor shall carefully examine all data forwarded for approval and shall sign a certificate to the effect that the data has been carefully checked and found to be correct with respect to dimensions and available space and that the equipment complies with all requirements of the Specifications and is suitable for its intended use on this project.
 - 2. Physical Size of Equipment: Space is critical; therefore, equipment of larger sizes than shown, even though of specified manufacturer, will not be acceptable unless it can be demonstrated that ample space exists for proper installation, operation, and maintenance.
 - 3. Materials and Equipment Lists: The list of materials and equipment, the name of manufacturer, trade name, type, and catalog number shall be submitted to the Architect/Engineer in quantity and format as described in Division 1. The lists shall be accompanied by pictorial and descriptive data

derived from the manufacturers' catalogs, sales literature, or incorporated in the Shop Drawings. Should a substitution be accepted, and should the substitute material prove defective, or otherwise unsatisfactory for the service intended within the guarantee period, this material or equipment shall be replaced with the material or equipment specified at no additional cost to the Owner.

1.12 MATERIALS AND WORKMANSHIP

- A. All materials, unless otherwise specified, shall be new, free from all defects, suitable for the intended use, and of the best quality of their respective kinds. Materials and equipment shall be installed in accordance with the manufacturer's recommendations and the best standard practice for the type of work involved. All work shall be executed by mechanics skilled in their respective trades, and the installations shall provide a neat, precise appearance. Materials and/or equipment damaged in shipment or otherwise damaged prior to installation shall not be repaired at the job site but shall be replaced with new materials and/or equipment.
- B. The responsibility for the furnishing of the proper equipment and/or material and seeing that it is installed as intended by the manufacturer rests entirely upon the Contractor who shall request advice and supervisory assistance from the representative of specific manufacturers during the installation.

1.13 FLAME SPREAD PROPERTIES OF MATERIALS

- A. Materials and adhesives incorporated in this project shall conform to NFPA Standard 255 (1984), "Method of Test of Surface Burning Characteristics of Building Materials". The classification shall not exceed a flame spread rating of 25 for all materials, adhesives, finishes, etc., specified for each system, and shall not exceed a smoke developed rating of 50.

1.14 REGULATORY REQUIREMENTS

- A. The "Authority Having Jurisdiction" over the project described by these documents is the Owner, as an Agency of the State of Texas. As such, it is required that the installation shall meet the minimum standards prescribed in the latest editions of the following listed codes and standards in effect as of January 1, 2010 (current edition) unless otherwise noted, which are made a part of these Specifications.
- B. National Fire Protection Association Standards (NFPA):
 - 1. NFPA No. 13, Sprinkler System, Installation
 - 2. NFPA No. 14, Standpipes and Hose Systems
 - 3. NFPA No. 20, Centrifugal Fire Pumps
 - 4. NFPA No. 37, Stationary Combustion Engines & Gas Turbines
 - 5. NFPA No. 45, Fire Protection for Laboratories Using Chemicals
 - 6. NFPA No. 51, Welding & Cutting, Oxygen Fuel Gas Systems
 - 7. NFPA No. 54, Gas Appliances, Piping, National Fuel Gas Code
 - 8. NFPA No. 70, National Electrical Code
 - 9. NFPA No. 72D, Proprietary Signaling Systems
 - 10. NFPA No. 78, Lightning Protection Code
 - 11. NFPA No. 88A, Standard for Parking Structures

12. NFPA No. 90A, Air Conditioning Systems
 13. NFPA No. 91, Blower & Exhaust Systems
 14. NFPA No. 99, Health Care Facilities
 15. NFPA No. 101, Life Safety Code
 16. NFPA No. 200, Series, Building Construction
 17. NFPA No. 211, Chimneys, Fireplaces, Vent Systems
 18. NFPA No. 241, Standard for Safeguarding Construction, Alteration and Demolition Operations
 19. NFPA No. 255, Method of Test of Surface Burning Characteristics of Building Materials
 20. NFPA No. 258, Standard Research Test Method for Determining Smoke Generation of Solid Materials
- C. ASHRAE 90.1, 2010
- D. American National Standards Institute (ANSI):
1. A40.8, National Plumbing Code
 2. B31.1, Power Piping
 3. B9.1, Safety Code for Mechanical Refrigeration
- E. American Gas Association Publications (AGA): Directory of Approved Gas Appliances and Tested Accessories
- F. American Society of Mechanical Engineers (ASME): Boiler and Pressure Vessel Codes
- G. Air Conditioning and Refrigeration Institute Standards (ARI): All standards related to refrigeration and air conditioning equipment and piping furnished under these Specifications.
- H. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA): All current editions of applicable manuals and standards (See Sections 233100 and 233300).
- I. Air Moving and Conditioning Association (AMCA): All current editions of applicable manuals and standards.
- J. American Society of Testing Materials (ASTM): All current editions of applicable manuals and standards.
- K. American Water Works Association (AWWA): All current editions of applicable manuals and standards.
- L. National Electrical Manufacturers' Association (NEMA): All current editions of applicable manuals and standards.
- M. International Codes, current edition or as listed elsewhere in the contract.
- N. Texas Occupational Safety Act: All applicable safety standards.
- O. Occupational Safety and Health Act (OSHA).

- P. TAS, ADA, and ANSI Standards: All work shall be in accord with all regulations and requirements of the Standards and Specifications for Handicapped and Disabled for the Construction of Public Buildings and Facilities in the State of Texas Usable by Physically Handicapped and Disabled persons, ANSI Standards, and the requirements of the American Disabilities Act.
- Q. Refer to Specification Sections hereinafter bound for additional Codes and Standards.
- R. All materials and workmanship shall comply with all applicable state and national codes, Specifications, and industry standards. In all cases where Underwriters' Laboratories, Inc. has established standards for a particular type material, such material shall comply with these standards. Evidence of compliance shall be the UL "label" or "listing" under Reexamination Service.
- S. The Contract Documents are intended to comply with the aforementioned rules and regulations; however, some discrepancies may occur. Where such discrepancies occur, the Contractor shall immediately notify the Architect/Engineer in writing of said discrepancies and apply for an interpretation. Should the discovery and notification occur after the execution of a contract, any additional work required for compliance with said regulations shall be paid for as covered by Division 1 of these Contract Documents, providing no work of fabrication of materials has been accomplished in a manner of noncompliance. Should the Contractor fabricate and/or install materials and/or workmanship in such a manner that does not comply with the applicable codes, rules and regulations, the Contractor who performed such work shall bear all costs arising in correcting these deficiencies to comply with said rules and regulations.

1.15 GENERAL MATERIALS AND EQUIPMENT REQUIREMENTS

- A. Storage at Site: The Contractor shall not receive material or equipment at the job site until there is suitable space provided to properly protect equipment from rust, drip, humidity, and dust damage.
- B. Capacities shall be not less than those indicated but shall be such that no component or system becomes inoperative or is damaged because of startup or other overload conditions.
- C. Conformance with Agency Requirements: Where materials or equipment are specified to be approved, listed, tested, or labeled by the Underwriters' Laboratories, Inc., or constructed and/or tested in accordance with the standards of the American Society of Mechanical Engineers or the Air Moving and Conditioning Association, the Contractor shall submit proof that the items furnished under this Section of the Specifications conform to such requirements. The label of the Underwriters Laboratories, Inc., applied to the item will be acceptable as sufficient evidence that the items conform to such requirements. The ASME stamp or the AMCA label will be acceptable as sufficient evidence that the items conform to the respective requirements.
- D. Nameplates: Each major component of equipment shall have the manufacturer's name, address, and catalog number on a plate securely attached to the item of equipment. All data on nameplates shall be legible at the time of Final Inspection.
- E. Prevention of Rust: Standard factory finish will be acceptable on equipment specified by model number; otherwise, surfaces of ferrous metal shall be given a rust inhibiting coating. The treatment shall withstand 200 hours in salt spray fog test, in accordance with Method 6061 of Federal Standard No. 141. Immediately after completion of the test, the specimen shall show no signs of wrinkling or cracking and no signs of rust creep beyond 1/8" on either side of the scratch mark. Where rust inhibitor coating is specified hereinafter, any treatment that will pass the above test is acceptable unless a specific coating is specified except that coal tar or asphalt type coating will not be acceptable unless so stated for a specific item. Where steel is specified to be hot-dip galvanized, mill-galvanized sheet steel may be used provided all raw edges are painted with a zinc-pigmented paint conforming to Military Specification MIL-P-26915.

- F. Protection from Moving Parts: Belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts shall be fully enclosed or properly guarded for personnel protection. Guards shall be compliant with OSHA requirements.
- G. Verification of Dimensions: The Contractor shall be responsible for the coordination and proper relation of his work to the building structure and to the work of all trades. The Contractor shall visit the premises and become thoroughly familiar with all details of the work and working conditions, to verify all dimensions in the field, and to advise the Architect/Engineer of any discrepancy before performing any work. Adjustments to the work required in order to facilitate a coordinated installation shall be made at no additional cost to the Owner or the Architect/Engineer.

1.16 WALL, FLOOR AND CEILING PLATES

- A. Except as otherwise noted, provide C.P. (Chrome plated) brass floor and ceiling plates around all pipes, ducts, conduits, etc., passing exposed through walls, floors, or ceilings, in any finished spaces except underfloor and attic spaces. Plates shall be sized to fit snugly against the outside of the pipe or against the insulation on lines which are insulated and positively secured to such pipe or insulation. Plates will not be required for piping where pipe sleeves extend 3/4" above finished floor. All equipment rooms are classified as finished areas. Round and rectangular ducts shall have closure plates (NOT chrome plated) made to fit accurately at all floor, wall and ceiling penetrations.

1.17 SLEEVES, INSERTS, AND FASTENINGS

- A. General: All openings through all floors, walls, and roofs, etc., regardless of material for the passage of piping, ductwork, conduit, cable trays, etc., shall be sleeved. All penetrations must pass through sleeves except soil pipe installed under concrete slabs on fill. Sleeves shall be set in new construction before concrete is poured, as cutting holes through any part of the concrete will not be permitted unless acceptable to the Architect/Engineer. If a penetration is cored into an existing solid concrete or stone structure, then the installation of a sleeve will not be necessary. (This does not apply to cases in which a sleeve was not placed in a timely fashion before the concrete was poured. In such cases, direction shall be requested from the Architect/Engineer.) Sleeves set in floors shall extend 4" above finished floor elevation and be sealed water tight to the floor.
- B. The minimum clearance between horizontal penetrations including insulation where applicable, and sleeve shall be 1/4", except that the minimum clearance shall accommodate a Thunderline Link-seal (or approved equal) closure where piping exits the building, or penetrates a wall below ground level. Contractor shall be responsible for the accurate location of penetrations in the slab for his pipe, duct, etc. All penetrations shall be of ample size to accommodate the pipe, duct, etc. plus any specified insulation. Sleeve materials shall be rigid metal of adequate strength. Void between sleeve and pipe shall be filled with Nelson Flameseal Firestop or approved equal caulk or putty.
- C. Installation of sleeves in walls shall be the same as for floors. Refer to the details on the project drawings. Where the details differ from these specifications, the drawings take precedence.
- D. Sleeves for penetrations passing through walls or floors on or below grade shall be removed, if practical, and after the pipes have been installed, the void space around the pipe shall be caulked with a suitable material to effect a waterproof penetration. Note that the practicality of the removal of the sleeve shall be the decision of the Construction Inspector. The decision of the Inspector shall be final.
- E. Inserts: Where the construction schedule allows, suitable concrete inserts for pipe and equipment hangers shall be set and properly located for all pipe and equipment to be suspended from concrete construction. If the inserts are later found not to be in the proper location for the placement of hangers or if the construction schedule does not allow inserts to be installed, then drilled anchors shall be installed. Drilled anchors in concrete or masonry shall be submitted for approval.

1. Fasteners: Fastening of pipes, conduits, etc., in the building shall be as follows: To wood members - by wood screws; to masonry - by threaded metal inserts, metal expansion screws, or toggle bolts, whichever is appropriate for the particular type of masonry; to steel - machine screws or welding (when specifically permitted or directed), or bolts, and to concrete by suitable inserts anchored to reinforcing steel, and poured in place unless other means are indicated on the plans. Power-actuated fasteners (shooting) will not be acceptable under any circumstances. If it is necessary to install a method of fastening a hanger after the structure has been installed, then only clamps or drilled anchors with torque nuts and washes shall be used.

- a. **Note: Under no circumstances will the use of plastic anchors or plastic expansion shields be permitted for any purpose whatsoever.**

- F. Rat proofing: The open space around all ductwork, piping, etc., passing through the ground floor and/or exterior walls shall be rat proofed in a manner acceptable to the Architect/Engineer.
- G. Weatherproofing: The annular space between a pipe and its sleeve in exterior walls or through floor to below grade shall be filled with polyurethane foam rods 50 percent greater in diameter than the space as backing and fill material and made watertight with a permanent elastic polysulfide compound. Seal both surfaces of wall or floor.
- H. Air Plenums: The space around piping, ductwork, etc., passing through air plenums shall be made airtight in a manner acceptable to the Architect/Engineer.
- I. Fireproofing: Each contractor shall seal duct, etc., penetrations through roof, fire rated walls and floors with a foam or sealant as described below or in Division 7 that will form a watertight, vermin tight barrier that is capable of containing smoke and fire up to 2000 degrees F for two hours. For wet locations, the foam material shall be silicone RTV foam or an approved equal. For dry locations, premixed putty equal to Nelson Flameseal Firestop putty may be used.

1.18 PROJECT/SITE CONDITIONS

- A. Install Work in locations shown on Drawings, unless prevented by Project conditions.
- B. Prepare drawings showing proposed rearrangement of Work to meet Project conditions, including changes to Work specified in other Sections. Obtain permission of Architect/Engineer before proceeding.

1.19 MANUFACTURER'S RECOMMENDATIONS

- A. The manufacturer's published directions shall be followed in the delivery, storage, protection, installation, piping, and wiring of all equipment and material. The Contractor shall promptly notify the Architect/Engineer, in writing, of any conflict between the requirements of the Contract Documents and the manufacturers' directions, and shall obtain the Architect/Engineer's instructions before proceeding with the work. Should the Contractor perform any such work that does not comply with the manufacturers' directions or such instructions from the Architect/Engineer, he shall bear all costs arising in connection with the deficiencies.

1.20 SPACE AND EQUIPMENT ARRANGEMENT

- A. The size of mechanical and electrical equipment indicated on the Drawings is based on the dimensions of a particular manufacturer. While other manufacturers may be acceptable, it is the responsibility of the Contractor to determine if the equipment he proposes to furnish will fit in the space. Fabrication Drawings shall be prepared when required by the Architect/Engineer or Owner to indicate a suitable arrangement.

- B. All equipment shall be installed in a manner to permit access to all surfaces. All valves, motors, drives, filters, and other accessory items shall be installed in a position to allow removal for service without disassembly of another part.

1.21 LARGE APPARATUS

- A. Any large piece of apparatus which is to be installed in any space in the building, and which is too large to permit access through stairways, doorways, or shafts shall be brought to the job and placed in the space before the enclosing structure is completed. Following placement in the space, such apparatus shall be thoroughly, completely protected from damage as hereinafter specified.

1.22 PROTECTION

- A. The Contractor shall at all times take such precautions as may be necessary to properly protect all materials and equipment from damage from the time of delivery until the completion of the work. This shall include the erection of all required temporary shelters and supports to adequately protect any items stored in the open on the site from the weather, the ground and surrounding work; the cribbing of any items above the floor of the construction; and the covering of items in the incomplete building with tarpaulins or other protective covering; the installation of electric heaters in electrical switchgear and similar equipment to prevent moisture damage. Failure on the part of the Contractor to comply with the above will be sufficient cause for the rejection of the items in question.
- B. Take particular care not to damage the building structure in performing work. All finished floors, step treads, and finished surfaces shall be covered to prevent any damage by workmen or their tools and equipment during the construction of the building.
- C. Equipment and materials shall be protected from rust both before and after installation. Any equipment or materials found in a rusty condition at the time of final inspection must be cleaned of rust and repainted as specified elsewhere in these Specifications.
- D. Storage of all equipment shall be per manufacturer's recommendations. All pumps, fans and motors shall be rotated by hand when received and when stored to maintain bearing lubrication

1.23 COOPERATION BETWEEN TRADES AND WITH OTHER CONTRACTORS

- A. Each trade, subcontractor, and/or Contractor must work in harmony with the various other trades, subcontractors and/or Contractors on the job as may be required to facilitate the progress to the best advantage of the job as a whole. Each trade, subcontractor, and/or Contractor must pursue its work promptly and carefully so as not to delay the general progress of the job. This Contractor shall work in harmony with Contractors working under other contracts on the premises.

1.24 ELECTRICAL WIRING OF MOTORS AND EQUIPMENT

- A. The Contractor shall note that the electrical design and Drawings are based on the equipment scheduled and indicated on the Drawings, and should any mechanical equipment be provided requiring changes to the electrical design, the required electrical changes shall be made at no cost to the Owner.
- B. The Electrical Trades shall provide all interconnecting wiring for the installation of all power. The Electrical Trades shall provide all disconnect switches as required for proper operation, as indicated on the Drawings or required by applicable code. All combination starters, individual starters, and other motor starting apparatus not specifically scheduled or specified as provided by the equipment manufacturer under the scope of Division 23, shall be provided under the scope of Division 26 or as directed by the General Contractor.

- C. The Mechanical Trades shall provide complete wiring diagrams indicating power wiring and interlock wiring. Diagrams shall be submitted for review within thirty (30) days after the submittals for equipment have been reviewed. Diagrams shall be based on accepted equipment and shall be complete full phase and interlock control Drawings, not a series of manufacturer's individual diagrams. After these diagrams have been reviewed, copies shall be transmitted to the Electrical Trades by the Contractor. They shall be followed in detail. See Section 230923, ENERGY MANAGEMENT SYSTEM (EMS), for additional clarification.

1.25 SUPERVISION

- A. Each Contractor and subcontractor shall keep a competent superintendent or foreman on the job at all times. (Refer to the Uniform General Conditions for additional information concerning supervision.)
- B. It shall be the responsibility of each superintendent to study all Drawings and familiarize himself with the work to be done by other trades. He shall coordinate his work with other trades and before material is fabricated or installed, make sure that his work will not cause an interference with another trade. Where interferences are encountered, they shall be resolved at the job site by the superintendents involved. Where interferences cannot be resolved without major changes to the design, the matter shall be referred to the A/E for ruling.

1.26 SITE OBSERVATION

- A. Site observation by the Architect/Engineer is for the express purpose of verifying compliance by the Contractor with the Contract Documents, and shall not be construed as construction supervision nor indication of approval of the manner or location in which the work is being performed as being a safe practice or place.

1.27 PRECEDENCE OF MATERIALS

- A. The specifications determine the nature and setting of materials and equipment. The drawings establish quantities, dimensions and details.
- B. The installation precedence of materials shall generally be as follows. Note that if interference is encountered, this shall guide the contractor in the determination of which trade shall be given the "Right-of-Way". This does not require elements with a lower preference to be relocated if such relocation is required to resolve interference or to provide better access.
- C. Building lines
- D. Structural Members
- E. Soil and Drain Piping
- F. Vent Piping
- G. Supply, Return, and Outside Air Ductwork
- H. Exhaust Ductwork
- I. HVAC Water and Steam Piping
- J. Condensate Piping

- K. Fire Protection Piping
- L. Natural Gas Piping
- M. Domestic Water (Cold and Hot)
- N. Refrigerant Piping
- O. Electrical Conduit

1.28 CONNECTIONS FOR OTHERS

- A. The Contractor shall rough in for and make all gas, water, steam, sewer, etc. connections to all fixtures, equipment, machinery, etc., provided by others in accordance with detailed roughing-in Drawings provided by the equipment suppliers, by actual measurements of the equipment connections, or as detailed.
- B. After the equipment is set in place, the Contractor shall make all final connections and shall provide all required pipe, fittings, valves, traps, etc.
- C. Provide all air gap fittings required, using materials hereinbefore specified. In each service line connected to an item of equipment or piece of machinery, provide a shutoff valve. On each drain not provided with a trap, provide a suitable trap.
- D. All pipe fittings, valves, traps, etc., exposed in finished areas and connected to chrome plated lines provided by others shall be chrome plated to match.
- E. Provide all sheet metal ductwork, transition pieces, etc., required for a complete installation of vent hoods, fume hoods, etc., provided by others.

1.29 INSTALLATION METHODS

- A. Where to Conceal: All pipes, conduits, etc., shall be concealed in pipe chases, walls, furred spaces, or above the ceilings of the building unless otherwise indicated.
- B. Where to Expose: In mechanical rooms, janitor's closets tight against pan soffits in exposed "Tee" structures, or storage spaces, but only where necessary, piping may be run exposed. All exposed piping shall be run in the most aesthetic, inconspicuous manner, and parallel or perpendicular to the building lines.
- C. Support: All piping, ducts and conduits shall be adequately and properly supported from the building structure by means of hanger rods or clamps to walls as herein specified.
- D. Maintaining Clearance: Where limited space is available above the ceilings below concrete beams or other deep projections, pipe and conduit shall be sleeved through the projection where it crosses, rather than hung below them in a manner to provide maximum above-floor clearance. Sleeves shall be as herein specified. Piping, ductwork and other installed materials shall be located so as to not obstruct maintenance clearance for mechanical components such as controls, filters and the like.
- E. All pipe, conduits, etc., shall be cut accurately to measurements established at the building and shall be worked into place without springing or forcing. All ducts, pipes and conduits run exposed in machinery and equipment rooms shall be installed parallel to the building lines, except that piping shall be sloped to obtain the proper pitch. Piping, ducts and conduits run in furred ceilings, etc., shall be similarly installed, except as otherwise shown. All pipe openings shall be kept closed until the systems are closed with final connections.

1. All piping not directly buried in the ground shall be considered as "interior piping".
2. Prior to the installation of any ceiling material, gypsum, plaster, or acoustical board, the Contractor shall notify the construction inspector so that arrangement can be made for an inspection of the above ceiling area about to be "sealed" off. The Contractor shall give as much advance notice as possible but no less than 10 working days.
3. All above ceiling areas will be subject to a formal inspection before ceiling panels are installed, or installation is otherwise concealed from view. All mechanical work at and above the ceiling, including items supported by the ceiling grid, such as air inlets or outlets, shall be complete and installed in accordance with contract requirements, including power to fans and other powered items. The inspection will include representatives of the following: General Contractor and each Subcontractor having work above the ceiling, Physical Plant, and the Resident Construction Manager's Construction Inspector Areas to be included and time of inspection shall be coordinated with the Construction Inspector.
4. The purpose of this inspection is to verify the completeness and quality of the installation of the air conditioning systems, the electrical systems, the plumbing systems, and any other special above ceiling systems such as pneumatic tube, vacuum systems, fire sprinkler piping and cable tray systems. The ceiling supports (tee bar or lath) shall be in place so that access panel and light fixture locations are identifiable and so that clearances and access provisions may be evaluated.
5. No ceiling materials may be installed until the resulting deficiency list from this inspection is worked off and the Construction Inspector has given approval.
6. Proper accessibility to equipment may be required to be demonstrated by the commissioning agent or inspector.

1.30 RECORDS FOR OWNER

- A. The Contractor shall maintain a set of prints in the Field Office for the sole purpose of recording "installed" conditions. Daily note all changes made in these Drawings in connection with the final installation including exact dimensioned locations of all new underground utilities, services and systems and all uncovered existing active and inactive piping outside the building.
- B. At Contract completion the Contractor shall provide a set of reproducible drawings in electronic format (PDF). The contractor shall transfer the information from the prints maintained as described above, and turn over this neatly marked set of reproducible Drawings representing the "as installed" work to the Architect/Engineers for verification and subsequent transmittal to the Owner. The Contractor shall refer to Division 1 of these Specifications, and to the Uniform General Conditions, for additional information. These Drawings shall include as a minimum:
 1. Addendum written drawing changes.
 2. Addendum supplementary drawings.
 3. Accurate, dimensioned locations of all underground utilities, services and systems.
 4. Identification of equipment work shown on Alternates as to whether alternates were accepted and work actually installed.
 5. Change Order written drawing changes.
- C. "As installed" PDF's shall bear a stamp or hand lettered title block generally located in lower right hand corner of Drawing entitled "AS INSTALLED DRAWING" with Company name of the installing trade Subcontractor and with a place for the date and the name of the responsible company representative.

- D. In addition to the above, the Contractor shall accumulate during the progress of the job the following data in electronic format (PDF) and turn over to the Architect/Engineer for review, and subsequent delivery to the Owner.
1. All warranties and guarantees and manufacturers' directions on equipment and material covered by the Contract.
 2. Operating instructions and preventative maintenance procedures for heating and cooling and other mechanical and electrical systems.
 3. Valve tag charts and diagrams specified herein.
 4. Approved wiring diagrams and control diagrams representing "as installed" conditions.
 5. Copies of approved Shop Drawings.
 6. Any and all other data and/or drawings required as submittals during construction.
 7. Repair parts list of all major items and equipment including name, address and telephone number of local supplier or agent.
- E. All of the above data shall be submitted to the Architect/Engineer for approval, and shall be corrected as instructed by the Architect/Engineer prior to submission of the final request for payment.
- F. Refer to additional requirements in the commissioning section of Division 1.
- G. Contractor shall reference approved BIMxP and provide the most stringent requirements noted with this section or in the BIMxP.

1.31 CUTTING AND PATCHING

- A. General: Cut and patch walls, floors, etc., resulting from work in existing construction or by failure to provide proper openings or recesses in new construction.
- B. Methods of cutting: Openings cut through concrete and masonry shall be made with masonry saws and/or core drills and at such locations acceptable to the Architect/Engineer. Impact-type equipment shall not be used except where specifically acceptable to the Architect/Engineer. Openings in precast concrete slabs for pipes, conduits, outlet boxes, etc., shall be core drilled to exact size.
- C. Restoration: All openings shall be restored to "as-new" condition under the appropriate Specification Section for the materials involved, and shall match remaining surrounding materials and/or finishes.
- D. Masonry: Where openings are cut through masonry walls, provide and install lintels or other structural supports to protect the remaining masonry. Adequate supports shall be provided during the cutting operation to prevent any damage to the masonry occasioned by the operation. All structural members, supports, etc., shall be of the proper size and shape, and shall be installed in a manner acceptable to the Architect/Engineer.
- E. Plaster: All mechanical work in areas containing plaster shall be completed prior to the application of the finish plaster coat. Cutting of finish plaster coat will not be permitted.
- F. Special Note: No cutting, boring, or excavating which will weaken the structure shall be undertaken. Rebar placement shall be determined prior to floor coring operations. Any rebar, which has been cut, shall be submitted in writing to the Architect/Engineer for evaluation.

1.32 ROOF PENETRATIONS AND FLASHING

- A. Pipe, conduit and duct sleeves, pitch pockets, and flashings compatible with the roofing installation shall be provided and installed by a qualified contractor for all roof penetrations. This shall be the responsibility of the General Contractor.

1.33 EXCAVATION, TRENCHING AND BACKFILL

- A. Excavation (See Divisions 0 and 1 for special requirements related to excavation and trenching.):
 - 1. The contractor shall perform all excavations of every description, for their particular installations and of whatever substances encountered, to the depths indicated on the Drawings and/or required for the installation of piping, conduit, utility systems, etc. All exterior lines shall be installed with a minimum cover of 24", unless otherwise indicated. Generally, more cover shall be provided if grade will permit. All excavation materials not required for backfill or fill shall be removed and wasted as acceptable to the Construction Inspector. All excavations shall be made only by open cut. The banks of trenches shall be kept as nearly vertical as possible and where required, shall be properly sheeted and braced. Trenches shall be not less than 12" wider nor more than 16" wider than the outside edges of the pipe to be laid therein, and shall be excavated true to line so that a clear space not less than 6" nor more than 8" in width is provided on each side of the pipe. For sewers, the maximum width of trench specified applies to the width at and below the level may be made as wide as necessary for sheeting and bracing, and the proper installation of the work.
 - 2. The bottom of trenches shall be accurately graded to provide proper fall and uniform bearing and support for each section of the pipe on undisturbed soil or 2" of sand fill at every point along its entire length, except for portions of the pipe sections where it is necessary to excavate for bell holes and for the proper sealing of pipe joints. Bell holes shall be dug after the trench bottom has been graded. Where inverts are not shown, grading shall be determined by the Plumbing Code for the service intended and the size used. Bell holes for lead pipe joints shall be 12" in depth below the trench bottom and shall extend from a point 6" back of the face of the bell. Such bell holes shall be of sufficient width to provide ample room for caulking. Bell holes for sewer tile and water pipe shall be excavated only to an extent sufficient to permit accurate work in the making of the joints and to insure that the pipe, for a maximum of its length, will rest upon the prepared bottom of the trench. Depressions for joints other than bell and spigot shall be made in accordance with the recommendations of the joint manufacturer for the particular type of joint used. In general, grading for electrical duct banks and conduits shall be from building to manhole, and from a high point between manholes to each manhole. Special pipe beds shall be provided as specified hereinafter.
 - 3. The lower 4" of the pipe trenches measuring from an overhead line set parallel to the grade line of the sewer shall be excavated only a few feet in advance to the pipe laying, by men especially skilled in this type of work. Where damage is likely to result from withdrawing sheeting, the sheeting shall be left in place. Except at locations where excavation of rock from the bottom of trenches is required, care shall be taken not to excavate below the depths required. Where rock excavation is required, the rock shall be excavated to a minimum over depth of 6" below the trench depths specified. The over depth rock excavation and all excess trench excavation shall be backfilled with sand. Whenever wet or otherwise unstable soil is incapable of properly supporting the pipe is encountered in the trench bottom, such soil shall be removed to a depth and for the trench lengths required, and then backfilled to trench bottom grade, as hereinafter specified, with sand.
 - 4. All grading in the vicinity of excavation shall be controlled to prevent surface ground water from flowing into the excavations. Any water accumulated in the excavations shall be removed by pumping or other acceptable method. During excavation, material suitable for backfilling shall be stacked in an orderly manner a sufficient distance back from edges of trenches to avoid overloading and prevent slides or cave-ins. Material unsuitable for backfilling shall be wasted and removed from the job site as directed by the Construction Inspector.

5. All shoring and sheeting required to perform and protect the excavations and to safeguard employees and/or adjacent structures shall be provided.
6. Excavate as required under the building in order that all piping, ductwork, etc., shall clear the ground a minimum of 12" for a distance of 24" on either side. Edges of such excavations shall slope at an angle of not over 45 degrees with the horizontal unless otherwise approved by the Construction Inspector. The bottom of such excavation shall be graded to drain in a manner acceptable to the Construction Inspector.
7. Trenches for cast iron drain, storm water and sewer lines inside the building shall be properly excavated, following, in general, the procedures set out for exterior lines. Where floors are to be poured over these lines, they shall be backfilled, tamped and settled with water. Where no flooring is to cover the lines, they shall be backfilled to form a level grade.
8. All surplus materials removed in these trenching operations becomes the property of the contractor, and shall be disposed of at the expense of the contractor, at a legal disposal site, off of the campus.

B. Backfilling:

1. Trenches shall not be backfilled until all required tests are performed and until the piping, utilities systems, etc., as installed are certified by the Owner's inspector to conform to the requirements specified hereinafter. The trenches shall be carefully backfilled with sand to a depth of 12 inches above the top of the pipe. The next layer and subsequent layers of backfill may be excavated materials approved for backfilling, consisting of earth, loam, sandy clay, sand and gravel, soft shale, or other approved materials free from large clods of earth or stones larger than 1-1/2" in diameter, flooded until the pipe has cover of not less than one foot. The remainder of the backfill material shall then be thrown into the trenches, moistened, and tamped or flooded in one foot layers. Blasted rock, broken concrete or pavement, and large boulders shall not be used as backfill material. Any trenches improperly backfilled, or where settlement occurs, shall be reopened to the depth required for proper compaction, then refilled and mounded over, and smoothed off.
2. Backfill under concrete slabs on fill shall be as specified above, shall be gravel, or shall be other such materials more suitable for the application. Installation and compaction shall be as required for compatibility with adjacent materials.

C.

D.

1.34 ACCESS DOORS

- A. General: Contractor shall provide wall or ceiling access doors for unrestricted access to all concealed items of mechanical equipment or devices.
- B. Doors: Access doors mounted in painted surfaces shall be of Milcor (Inland-Ryerson Construction Products Company) manufacture, Style K for plastered surfaces and Style M or DW for non-plastered surfaces. The Style K doors shall be set so that the finished surface of the door is even with the finished surface of the adjacent finishes. Access doors mounted on tile surfaces shall be of similar construction as noted above, except they shall be of stainless steel materials. Access doors shall be a minimum of 12" x 12" in size.

1.35 OPERATION PRIOR TO COMPLETION

- A. When any piece of mechanical equipment is operable and it is to the advantage of the Contractor to operate the equipment, he may do so, providing that he properly supervises the operation, and has the Construction Inspector's written permission to do so. The warranty period shall, however, not commence

until such time as the equipment is operated for the beneficial use of the Owner, or date of substantial completion, whichever occurs first.

- B. Regardless of whether or not the equipment has or has not been operated, the Contractor shall properly clean the equipment, install clean filter media, properly adjust, and complete all deficiency list items before final acceptance by the Owner. The date of acceptance and performance certification will be the same date.
- C. Air handling equipment shall only be operational with all specified filter media in place and additional filter media in place to prevent dust from entering the return and exhaust air systems.
- D. Additional requirements for operation of equipment prior to completion found in the commissioning sections of Division 1 and Division 23 shall be followed.

1.36 CHECKING AND TESTING MATERIALS AND/OR EQUIPMENT

- A. Before the work is accepted, an authorized representative of the manufacturer of the installed materials and/or equipment shall personally inspect the installation and operation of his materials and/or equipment to determine that it is properly installed and in proper operating order. The qualifications of the representative shall be appropriate to the technical requirements of the installation. The qualifications of the representative shall be submitted to the owner for approval. The decision of the owner concerning the appropriateness of the representative shall be final. Testing and checking shall be accomplished during the course of the work where required by work being concealed, and at the completion of the work otherwise. In addition, the Contractor shall submit to the Architect/Engineer a signed statement from each representative certifying as follows: **"I certify that the materials and/or equipment listed below have been personally inspected by the undersigned authorized manufacturer's representative and is properly installed and operating in accordance with the manufacturer's recommendations"**.
- B. Check inspections shall include plumbing equipment, heating, air conditioning, insulation, ventilating equipment, controls, mechanical equipment and such other items hereinafter specified or specifically designated by the Owner or the Architect/Engineer.
- C. Refer to the commissioning sections of Division 1 and Division 23 for additional start-up, testing, and acceptance requirements.

1.37 TESTS

- A. The Contractor shall make, at no additional cost to the Owner, any tests deemed necessary by the inspection departments having jurisdiction, and in the National Fire Protection Association, ASTM, etc. Standards listed. The Contractor shall provide all equipment, materials, and labor for making such tests. Reasonable amounts of fuel and electrical energy costs for system tests will be paid by the Owner. Fuel and electrical energy costs for system adjustment and tests which follow beneficial occupancy by the Owner will be borne by the Owner.
- B. Additional tests specified hereinafter under the various Specification Sections shall be made.
- C. The Construction Inspector shall be notified in writing at least 10 working days prior to each test and other Specification requirements requiring action on the part of the Construction Inspector. All equipment shall be placed in operation and tested for proper automatic control requirements before the balancing agency starts their work.
- D. Maintain Log of Tests as hereinafter specified.
- E. See Specifications hereinafter for additional tests and requirements.

1.38 LOG OF TESTS

- A. All tests shall have pertinent data logged by the Contractor at the time of testing. Data shall include date, time, personnel, description, and extent of system tested, test conditions, test results, specified results, and other pertinent data. Data shall be delivered to the Architect/Engineer as specified under "Requirements for Final Acceptance". All Test Log entries shall be legibly signed by the Project Contractor or his authorized job superintendent.

1.39 COOPERATION AND CLEANUP

- A. It shall be the responsibility of each trade to cooperate fully with the other trades on the job to help keep the job site in a clean and safe condition. At the end of each day's work, each trade shall properly store all of his tools, equipment and materials and shall clean his debris from the job. Upon the completion of the job, each trade shall immediately remove all of his tools, equipment, any surplus materials and all debris caused by that portion of the work.

1.40 CLEANING AND PAINTING

- A. All equipment, piping, conduit, ductwork, grilles, insulation, etc., furnished and installed in the tunnel and mechanical rooms under Division 23 of these Specifications and as hereinafter specified shall be cleaned, prepared, and painted according to the following specification. Natural gas piping shall be painted in its entirety. In the event of a conflict between the specifications referenced, the provisions of this specification shall prevail only for Division 23 work.
- B. All equipment furnished by the mechanical and electrical subcontractors shall be delivered to the job with a suitable factory protective finish and shall be painted, after installation, with the color hereinafter specified. The following materials shall not be painted: copper, galvanized metal, stainless steel, fiberglass, PVC, and PVDF.
- C. Before painting, materials and equipment surfaces shall be thoroughly cleaned of cement, plaster, and other foreign materials, and all oil and grease spots shall be removed. Such surfaces shall be carefully wiped and all cracks and corners scraped out. Exposed metal work shall be carefully brushed down with the steel brushes to remove rust and other spots and left smooth and clean.
- D. Color of finish painting in Mechanical Rooms shall be painted in accordance with The University of Texas Standard Color Schedule for machinery spaces using Kelly Moore or Pratt and Lambert enamel, or approved equal. For painting purposes, the equipment and piping inside of builtup air handling units shall be painted the same as if they were within the walls of a Mechanical Room. Two coats shall be applied with a light tint first coat and deep color for final coat. Refer to section 23 05 53 Mechanical Identification for additional requirements above the colors noted below:

ITEM	COLOR	PITTSBURGH PAINT #	PAINT PRODUCT/LAGGING
Fire Sprinkler Line	Red	134-7	SIREN RED 1700-62
Steam, LP	White		WHITE 5392
Steam, MP	White		White 5301 Activator
Condensate Return	White		White 1700-61
Softwater	Medium Blue		IRIS QUEEN 1700-222-V-23-2
Domestic Cold Water	Light Blue	152-4	ASJ Jacket & Casanova 1700-111-W-2-1
Domestic Hot Water	Green	205-7	
Domestic Hot Water Return	Light Green	105-5	
Natural Gas	Citrus Punch	112-5	DAISY YELLOW Y361M

Fuel Oil	Golden Glow	213-4	SUNBURST YELLOW 1700-63
Chilled Water Supply	Dark Blue	251-7	INDIGO BLUE 1700-64
Chilled Water Return	Light Blue	152-4	INDIGO BLUE 1700-64
Condenser Water Supply	Dark Green	205-7	FOREST GREEN 1700-51
Condenser Water Return	Light Green	105-5	FOREST GREEN 1700-51
Compressed Air	Light Green		SHAMROCK 1700-333-M-25-3
Heating Hot Water Supply	Orange	132-6	BMP294B01
Heating Hot Water Return	Light Orange	127-6	BMP294B01
Hanger Rods	Same as "PIPING" above		
Steam Traps & Metal Exposed to High Temperatures	Same as "PIPING" above High Temp Rated		
Atmospheric Relief Vents	Same as "PIPING" above		

- E. Aluminum jacketing on insulation shall not be painted.
- F. No nameplates on equipment shall be painted, and suitable protection shall be afforded to the plates to prevent their being rendered illegible due to the painting operation.
- G. Scope of painting for Division 23 work in areas other than those defined as "exposed" is as follows:
 - 1. All uncovered steel pipe, supports, exposed pipe and hanger rod threads, and hangers in under floor spaces shall be cleaned and painted with two coats of Tropical Paint Co. No. 77-black asphaltic emulsion. Galvanized steel and copper lines in these spaces shall not be painted.
 - 2. All canvas finishes including those under floor and in concealed spaces shall be painted with one sizing coat if not already sized, containing mildew resistant additive and Arabol adhesive prior to any other specified finish paint.
 - 3. All fuel piping shall be painted whether concealed or exposed, in all areas of the project without exception. Fuel piping shall be painted safety yellow, and fire protection piping shall be painted safety red. These "safety" colors shall be as defined by OSHA.
 - 4. If insulated, the piping shall be primed, only, prior to insulation, and the insulation jacketing shall be painted as specified for piping. The requirements of this paragraph are "primary" and have priority over any conflicting specification or instruction, should a conflict in the Construction Documents exist.
- H. In addition to painting in mechanical rooms, materials, piping, ductwork, conduit, gear, supports, foundations, equipment and appurtenances installed by the mechanical and electrical subcontractors in exposed areas shall be finish painted with two coats of Kelly Moore or Pratt and Lambert, Inc.'s "Effector" enamel of color selected by the Architect/Engineer.
- I. Additional areas to be defined as "exposed" for purposes of painting, are defined on the Drawings.
- J. The surfaces to be finish painted shall first be prepared as follows:
 - 1. On canvas finishes pretreated as specified above. Insulated surfaces having vapor barrier jacket exposed to view shall first be painted with one (1) coat of sealer.
 - 2. Galvanized and black steel surfaces shall first be painted with one (1) coat of P&L galvanized metal primer. Primer may be eliminated on concealed fire and gas piping.
 - 3. Aluminum surfaces shall first be painted with one (1) coat of P&L zinc chromate primer.

4. Cast iron pipe shall first be primed with a "non-bleed" primer.
 5. The underside of all cast iron sinks not recessed in a cabinet are included as items to be painted in exposed areas.
- K. All ferrous metal surfaces without a protective finish and not galvanized in exposed and concealed areas including chases, under floor and above ceilings shall be painted with two (2) coats of P&L zinc chromate primer as the construction progresses to protect against deterioration.

1.41 WARRANTY

- A. Refer to Division 1 and to individual specification sections for warranty requirements. Unless otherwise specified, a 5-year parts and labor warranty shall be provided on all systems and equipment from substantial completion.

1.42 CLEANING HVAC SYSTEMS

A. General Cleanup:

1. Upon completion of contract and progressively as work proceeds, clean up dirt, debris, old materials, etc., and remove from site, keeping premises in neat and clean condition to satisfaction of the Architect. See Division 01 of specifications for further requirements.
2. Seepage, discoloration or other damage to parts of the building, its finish, or furnishings due to Contractor's failure to properly clean piping systems or duct systems shall be repaired without cost to the Owner.

B. Factory Finishes:

1. Clean items with factory finishes. Touch up bare places, scratches and other minor damage to finishes. Use only factory supplied paint of matching color and formula. If finishes are badly damaged or if there are many damaged, scratched or bare places, refinish the entire item.

C. Ducts and Apparatus:

1. Thoroughly clean ducts and apparatus casings before fans and filters are operated.

D. HVAC Closed and Open Water Systems:

1. Initial flushing:
 - a. Remove loose dirt, mill scale, metal chips, weld beads, rust, and like deleterious substances without damage to any system components.
 - b. Bypass factory equipment unless acceptable means of protection are provided, or by subsequent inspection of water boxes and other "hide-out" areas takes place.
 - c. Isolate or protect "clean" system components including pumps and pressure vessels and any component that may be damaged.
 - d. Open all valves, drains, vents, strainers, and the like at all system levels. Close all valves isolating piping from the existing central piping systems.
 - e. Remove plugs, caps, spool pieces, and components to facilitate early discharge from the system.
 - f. Sectionalized system to obtain debris carrying velocity of six feet per second.

- g. Connect dead end supply and return headers and the like as necessary, or provide drains in dead end eccentric caps.
 - h. Install temporary strainers where necessary to protect downstream equipment.
 - i. Supply and drain-off "flushing" water by fire hoses, garden hoses, temporary or permanent piping, Contractor's booster pumps, and the like.
 - j. Flush for not less than four hours. Drain all dirt legs. If water drained is not visibly clean, repeat the above procedure until dirt legs are visibly clean.
 - k. Before starting the cleaning and flushing process, confirm a thermometer has been installed on the system. During flushing and cleaning process, monitor the water temperature to ensure the water temperature does not become too hot.
2. Cleaning (Closed Systems Only):
- a. Utilize defoamers to preclude damage to existing work, and specifically adjacent electrical equipment.
 - b. Utilize heat to maximize effectiveness of compounds or use live steam injection where practical and safe. Do not raise cleaning water temperature in excess of 150 degrees F. Install a thermometer in the piping system and constantly monitor the water during cleaning to prevent over-heating.
 - c. Install temporary strainers, reinforced against blowout, sized to not impair equipment performance, to preclude passing of particles larger than 60% of smallest radial and at a minimum to retain all particles larger than 1000 microns.
 - d. Permanent facility pumps shall not be used for circulating cleaning water. Contractor shall supply temporary pumps for this process.
 - 1) If the system construction, flow rates, and pressures are such that it is impractical for the Contractor to provide temporary pumps, the permanent facility pumps may be used with the specific express permission of the Owner, provided the guarantee on the entire pump assembly is unconditionally extended for two years after date of Substantial Completion. Leakage from pump seals or other damage resulting from circulating the uncleaned water shall require immediate rectification at no additional cost to the Owner.
 - e. Add 20 pounds of Garratt Callahan Formula 248, or equal, alkaline cleaner for each 1000 gallons of system water for chemical cleaning (approximate .2% solution). Formula 248 is a dry blend of buffered phosphates, a corrosion inhibitor, a surfactant, and an iron oxide sequestrant.
 - f. Circulate for a period of at least 72 hours.
 - g. Every eight hours, blow-down the condensers and system low points for three minutes.
 - h. Drain and flush the system. Rapid flushing from the lowest point in the system is needed to remove debris.
 - i. Inspect the system and repeat first four steps.
 - j. Begin the corrosion control program immediately at double the normal inhibitor dosage for one week. After one week, drop to the normal dosage.
 - k. Chemical treatment compound: Use Garratt Callahan Company Formula 12-L closed system inhibitor strictly following manufacturer's directions.
 - l. Before starting the cleaning and flushing process, confirm a thermometer has been installed on the system. During flushing and cleaning process, monitor the water temperature to ensure the water temperature does not become too hot.

E. Steam and Steam Condensate System:

1. Thoroughly blow out system and clean it by using steam under pressure of 5 psig. Allow condensate to be wasted to drains for a period of at least 8 hours. Cool condensate to 140 degree F maximum prior to discharge.

1.43 OPERATION OF HVAC SYSTEMS DURING CONSTRUCTION

- A. Install all specified filters prior to system operation. In addition to specified filters, install a roughing filter upstream of mixed air filter. Roughing filter shall consist of two layers of roll filter media clipped and sealed to entering side of filter frame. Change roughing filter as necessary to minimize dust collection on specified filters.
- B. Cover return and exhaust air grilles with temporary filter media. Attach media to avoid damage to grille or ceiling. Change temporary media as required to protect against dust buildup on ductwork. Remove temporary media from grilles after flooring is installed, walls are sanded and painted and other dust generating construction has been completed.
- C. During periods of excessive dust generation such as drywall sanding, seal off return and exhaust openings and grilles to prevent dust from accumulating in ductwork.
- D. If outside air source contains less dust than building air, adjust A/C unit dampers to operate with as much outside air as possible without causing a freezing condition for coil or exceeding capacity of coil to adequately condition supply air.
- E. Furnish and install a new set of specified filter media prior to start of system test and balance. Furnish a new, clean set of the specified media and turn over to Owner's Representative.

1.44 TESTING MECHANICAL SYSTEMS

- A. Test all systems and equipment installed to demonstrate proper operation.
- B. Advise Architect of scheduled systems testing and completed system demonstration/operation schedules so that he may witness, if desired.
- C. Correct and retest work found defective when tested.
- D. Make repairs to piping systems with new materials. Peening, doping, or caulking of joints or holes will not be acceptable.
- E. Domestic Water Piping: Test hot and cold water piping systems upon completion of rough-in, before fixtures are connected, at a hydrostatic pressure of 125 psig or 150% of working pressure whichever is greater for a period of two hours.
- F. Natural Gas Piping: Test piping with air, CO₂, or nitrogen at 100 psi pressure for two hours without leaking.
- G. HVAC Circulating Water Piping: Hydrostatically test piping at 150 psig pressure or at 1-1/2 times design pressure as indicated on drawings, whichever is greater, for a period of six hours without evidence of leaking.
- H. Food Service Heat Rejection Piping: Hydrostatically test piping at 125 psig pressure or at design pressure as indicated on drawings, whichever is greater, for a period of six hours without evidence of leaking.

- I. Steam and Steam Condensate Piping: Hydrostatically test piping at 150 psig pressure or 1-1/2 times design working pressure for a period of six hours without evidence of leaking.
- J. Fuel Oil Piping: Test piping with nitrogen at 100 psi pressure for two hours without evidence of leaking.
- K. Flush Valves: Test flush valves for proper operation.
- L. Ductwork Pressure Testing: Refer to Section 23 31 13 for required pressure testing for ductwork.
- M. Chemical Water Treatment Systems: Have equipment manufacturer field check installations of boiler water treatment systems, and cooling tower water treatment systems, including field calibration of pumps, controllers and other operable parts. Chemical water treatment for HVAC including cleaning, testing, and treatment shall be performed in accordance with Section 23 25 00.
- N. System Balance and Testing: Prepare to assist test and balance firm by assuring systems are complete and operational.
- O. Test all fire dampers by manually disconnecting linkage and observing blades fall into position.
- P. Test all smoke and combination fire/smoke, dampers by observing damper operation during fire alarm system commissioning.
- Q. Records of Testing: Maintain records of system testing and results thereof. Deliver results as part of project closing file and on an intermediate basis as requested by Architect.

1.45 INFECTION CONTROL REQUIREMENTS

- A. Coordinate with the Owner the exact requirements for the infection control measures to be executed and performed during the course of this Project.
- B. Prior to execution, present to the Owner for approval a written execution plan for each infection control measure.
- C. Coordinate infection control measures as needed with all other trades and disciplines.
- D. Provide documentation of infection control measures to the Owner, as required and specified in the ICRA.

PART 2 – PRODUCTS

2.1 ATTIC STOCK

- A. Provide one set of filters and two sets of belts at conclusion of the project to the Owner for attic stock

PART 3 – EXECUTION

NOT USED

END OF SECTION

SECTION 23 05 13

MOTORS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. This Specification applies to high-efficiency TEFC, horizontal and vertical, single-speed, squirrel cage polyphase induction motors, up to and including 500 hp, in NEMA frame sizes 143T and larger for severe duty applications (commonly referred to as severe duty motors).

1.2 REFERENCES

- A. AFBMA 9 - Load Ratings and Fatigue Life for Ball Bearings.
- B. AFBMA 11 - Load Ratings and Fatigue Life for Roller Bearings.
- C. ANSI/IEEE 112 - Test Procedure for Polyphase Induction Motors and Generators.
- D. ANSI/NEMA MG 1 - Motors and Generators.
- E. ANSI/NFPA 70 - National Electrical Code.
- F. Energy Policy Act of 1992

1.3 SUBMITTALS

- A. Submit product data under provisions of Section 23 00 00.
- B. Submit test results verifying efficiency and power factor for motors 1 horsepower and larger.
- C. Submit manufacturer's installation instructions under provisions of Section 23 00 00.

1.4 OPERATION AND MAINTENANCE DATA

- A. Submit operation and maintenance data under provisions of Section 23 00 00.
- B. Include assembly drawings, bearing data including replacement sizes, and lubrication instructions.

1.5 QUALIFICATIONS

- A. Manufacturer:
 - 1. Toshiba
 - 2. Baldor
 - 3. U.S. Motors
 - 4. General Electric

1.6 REQUIREMENTS

- A. IEEE Standard for Petroleum and Chemical Industry Severe Duty Totally Enclosed Fan-Cooled (TEFC) Squirrel Cage Induction Motors up to and Including 500 HP.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site under provisions of Section 23 00 00.
- B. Store and protect products under provisions of Section 23 00 00.
- C. Protect motors stored on site from weather and moisture by maintaining factory covers and suitable weather-proof covering. For extended outdoor storage, remove motors from equipment and store separately.

1.8 WARRANTY

- A. Provide five-year manufacturer's warranty under provisions of Section 23 00 00.
- B. Warranty: Include coverage for motors 1 horsepower and larger.

PART 2 - PRODUCTS

2.1 GENERAL CONSTRUCTION AND REQUIREMENTS

- A. Electrical Service: Refer to Drawing Schedules for required electrical characteristics.
- B. All Motors: Design for continuous operation in 40 degrees C environment, a service factor of 1.15.
- C. Explosion-Proof Motors: UL approved and labeled for hazard classification, with over temperature protection.
- D. Visible Nameplate: Indicating motor horsepower, voltage, phase, cycles, RPM, full load amps, locked rotor amps, frame size, manufacturer's name and model number, Service Factor, Power Factor, efficiency.
- E. Electrical Connection: Conduit connection boxes, threaded for conduit. For fractional horsepower motors where connection is made directly, provide threaded conduit connection in end frame.
- F. In addition, all motors shall be provided with adequately sized electrical connection box with threaded hub for attachment of flexible conduit, unless bus duct connection is indicated. Where motors are connected to driven equipment by the use of a V-belt drive, they shall be furnished with adjustable rails.
- G. Single phase motors, in general, shall be less than 3/4 horsepower and shall be 120 or 208 volt, 60-hertz motors. These motors shall have built-in thermal overload protection with automatic reset, and shall be rated for temperature rise as hereinbefore specified for 3-phase motors.
- H. All motors 3/4 horsepower and larger, unless smaller motors are indicated shall be supplied as 3-phase, and shall be squirrel cage premium efficiency induction type with standard NEMA frame sizes, 480 volt.
- I. Motors 1 HP and larger shall have integral frames.
- J. Efficiency: Premium
 - 1. For motors 1 horsepower and above. Premium efficiency motors shall have efficiency and losses determined in accordance with the latest revisions of IEEE Standard 112. Polyphase squirrel cage

motors rated 1 through 150 horsepower shall be tested by dynamometer method B. The efficiency shall be determined using segregated losses in which stray load loss is obtained from a linear regression analysis to reduce the effect of random errors in the test measurements.

- K. Motors 1 HP and larger shall be provided with a copper frame-grounding lug of hydraulic compression design, for installation by the electrical subcontractor.
- L. Motors 3/4 HP and larger shall have 120V space heater that is energized only when motor is idle.

2.2 MOTORS FOR VAIRALBE FREQUENCY APPLICATIONS

- A. Motors specified with variable frequency motor controllers shall comply with NEMA MG1, Part 31 for Definite Purpose, Inverter Fed motors including insulation meeting the requirement for 1600 Vpk at 0.1 μ S rise time. In addition to compliance with MG1, Part 31, motors also shall be designed for starting across the line and specifically designed to reduce inrush current
 - 1. To protect motor bearings and shafts from damage due to induced electrical currents along the motor shaft, provide Aegis shaft grounding ring (SGR), conductive microfiber motor shaft grounding ring on the driven end of all inverter fed motors. For inverter fed motors 100 HP and larger, also provide either an insulated motor bearing or a ceramic bearing on non-driven end of motor. Comply with manufacturer's installation instructions and with NEMA MG1, Part 31 for inverter fed motor bearings.

2.3 STARTING EQUIPMENT

- A. Each motor shall be provided with proper starting equipment. The trade furnishing the motor, unless hereinafter specified or scheduled to the contrary, shall provide this equipment. All motor starting equipment provided by any one trade shall be of the same manufacture unless such starting equipment is an integral part of the equipment on which the motor is mounted. The Mechanical Subcontractor shall furnish all starters for Division 23 work, except those starters scheduled to be provided in Division 26 Motor Control Centers.
- B. Motor starters shall conform to NEMA Standards for Industrial Control, #IC-1, latest issue, and shall be housed in NEMA Standard enclosures. Control voltage in each starter shall be either 24v or 120 volts to ground (as required), with an individual control transformer provided in each starter as required. Manual starters for fractional horsepower single phase motors shall be on-off or snap switch type combined with thermal overload device. The switch shall be so constructed so that it cannot be held closed under a sustained motor overload.
- C. Magnetic starters shall have thermal overload protection in each of the ungrounded legs and shall be solenoid operated. Provide the correct size heater element to protect the motor and allow it to operate based on motor nameplate amperes and ambient temperatures anticipated for each individual motor. Each starter shall be provided with a control power transformer or 120v control power circuit.
- D. Pushbuttons with or without pilot lights, hand-off-automatic switches and other scheduled apparatus shall be standard duty type mounted in NEMA enclosures or in cover of starter as specified or scheduled, and shall be furnished by the trade furnishing the starter except as specifically indicated elsewhere.
- E. Hand-Off-Automatic switches for equipment that could damage itself if left in the "hand" position (such as sump pumps), shall be spring return to "off" from the "hand" position.
- F. Motors 75 HP and above shall be equipped with A Cutler Hammer (IT) Soft Starter if not equipped with a VFD.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Install base mounted motors securely on firm foundation.
- C. Align motors on direct drive equipment using dial type gauges.
- D. Check line voltage and phase and ensure agreement with nameplate. Test motor for proper rotation under Division 26.

3.2 ADJUSTMENTS

- A. Motors, together with driven equipment, shall be dynamically and statically balanced. Imbalance shall be reduced to minimum specified by equipment manufacturers.

3.3 APPLICATION

- A. Motors drawing less than 250 Watts and intended for intermittent service may be germane to equipment manufacturer and need not conform to these specifications.

END OF SECTION

SECTION 23 05 15

MECHANICAL PIPING, VALVES, AND FITTINGS

PART 1 - GENERAL

1.1 TOPICS COVERED

- A. General requirements for all Mechanical piping systems. This section is supplemented by other sections with additional requirements and more detail on specific systems. In the case of conflict, the most stringent requirement will apply.

1.2 RELATED WORK

- A. Section 22 0050 - Common Work Results for HVAC
- B. Section 23 0516 - Piping Expansion Compensation.
- C. Section 23 0529 - Supports and Anchors.
- D. Section 23 0553 - Mechanical Identification.
- E. Section 22 0548 - Vibration Isolation.
- F. Section 23 0701 - Piping Insulation.

1.3 REFERENCES

- A. AGA - American Gas Association
- B. ANSI B31.1 - Power Piping
- C. ANSI B31.2 - Fuel Gas Piping
- D. ANSI B31.4 - Liquid Petroleum Transportation Systems
- E. ANSI B31.9 - Building service Piping
- F. ANSI/ASME B16.3 - Malleable Iron Threaded Fittings Class 150 and 300.
- G. ANSI/ASME B16.23 - Cast Copper Alloy Solder Joint Drainage Fittings - DWV.
- H. ANSI/ASME B16.29 - Wrought Copper and Wrought Copper Alloy Solder Joint Drainage Fittings - DWV.
- I. ANSI/AWS A5.8 - Brazing Filler Metal.
- J. ANSI/AWS D1.1 - Structural Welding Code.
- K. ANSI/AWWA C105 - Polyethylene Encasement for Ductile Iron Piping for Water and Other Liquids.
- L. ANSI/AWWA C111 - Rubber-Gasket Joints for Ductile Iron and Gray-Iron Pressure Pipe and Fittings.

- M. ASTM A53 - Pipe, Steel, Black and Hot-Dipped Zinc Coated, Welded and Seamless.
- N. ASTM A120 - Pipe, Steel, Black and Hot-Dipped Zinc Coated (Galvanized), Welded and Seamless, for Ordinary Uses.
- O. ASTM A234 - Pipe Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures.
- P. ASTM B32 - Solder Metal.
- Q. ASTM B88 - Seamless Copper Water Tube.
- R. ASTM D1785 - Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
- S. ASTM D2235 - Solvent Cement for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe and Fittings.
- T. ASTM D2310 - Machine-Made Reinforced Thermosetting Resin Pipe.
- U. ASTM D2680 - Acrylonitrile-Butadiene-Styrene (ABS) Composite-Sewer Piping.
- V. ASTM D2683 - Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing.
- W. ASTM D2751 - Acrylonitrile-Butadiene-Styrene (ABS) Sewer Pipe and Fittings.
- X. ASTM F477 - Elastomeric Seals (Gaskets) for Joining Plastic Pipe.

1.4 SUBMITTALS

- A. Submit product data under provisions of Division 1 and Section 22 0010.
- B. Include data on pipe fittings, valves, and accessories.
- C. Provide manufacturer's catalog information. Indicate valve rating and data.

1.5 QUALITY ASSURANCE

- A. Valves: Manufacturer's name and pressure rating marked on valve body.
- B. Welding Certificates, Materials and Procedures: Conform to ANSI/ASME SEC. 9 and applicable state labor regulations.
- C. Welders Certification: In accordance with ASME Section 9.
- D. Maintain one copy of the above listed documents on site for inspection.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site under provisions of Division 1 and Section 22 0050.
- B. Store and protect products under provisions of Division 1 and Section 22 0050.
- C. Deliver and store valves in shipping containers with labeling in place.

PART 2 - PRODUCTS

2.1 VALVES

- A. All valves shall be located such that the removal of their bonnets is possible.
- B. All flanged valves shown in horizontal lines with the valve stem in a horizontal position shall be positioned so that the valve stem is inclined one bolt hole above the horizontal position. Screw pattern valves placed in horizontal lines shall be installed with their valve stems inclined at an angle of a minimum of 30 degrees above the horizontal position. When installation configuration does not allow any other orientation except horizontal stem, valves with stem in horizontal position shall be drilled and tapped in accordance with MSS-SP-45 at Boss G to accommodate a drain valve.
- C. All valves must be true and straight at the time the system is tested and inspected for final acceptance. Valves shall be installed as nearly as possible to the locations indicated in the Construction Drawings. Any change in valve location must be so indicated on the Record Drawings.
- D. All valves must be of threaded or flanged type. No solder connected or grooved fitting valves shall be used on this project.
- E. All bronze and iron body gate and globe valves shall be the product of one manufacture for each project. Manufacturers of other types may not be mixed on the same project; i.e., all butterfly valves shall be of the same manufacture, all ball valves shall be of the same manufacture, etc.
- F. All valves used in circulating systems, plumbing and steam systems (low and medium pressure) shall be Class 150 SWP.
- G. Class 300 valves shall be constructed of all ASTM B-61 composition.
- H. All gate, globe and angle valves shall be union or screw-over-bonnet design. Metal used in the stems of all bronze gate, globe and angle valves shall conform to ASTM B371 Alloy 694, ASTM B99 Alloy 651, or other corrosion resistant equivalents. Written approvals must be secured for the use of alternative materials.
- I. All iron body valves shall have the pressure containing parts constructed of ASTM designated of 126 class B iron. Stem material shall meet ASTM B16 Alloy 360 or ASTM 371 Alloy 876 silicon bronze or its equivalent.
- J. Gates and globes shall be bolted bonnet with OS&Y (outside screw and yoke) and rising stem design. A lubrication fitting is preferred on yoke cap for maintenance lubrication of the yoke bushing. Gate and globe valves shall be bolted bonnet outside and screw and yoke design with pressure-temperature rating conforming to ANSI B16-34-1977. Stems shall meet ASTM designation A-186-F6 chromium stainless steel.
- K. Wedge (gate valves) may be solid or flexible type and shall meet ASTM A-182-F6 chromium stainless steel on valves from 2" to 6". Sizes 8" and larger may be A-216-WCB with forged rings or overlay equal to 182-F6. Seat ring shall be hard faced carbon steel or 13% chromium A-182-F6 stainless. Handwheels shall be A47 Grade 35018 malleable iron or Ductile Iron ASTM A536.
- L. All cast steel body valves shall have the pressure containing parts constructed of ASTM designation A-216-GR-WCB carbon steel.

- M. All forged steel body valves shall have the pressure containing parts constructed of ASTM 105, Grade 2 forged carbon steel. Seat and wedges shall meet ASTM A-182-F6 chromium stainless steel. Seat rings shall be hard faced. Valves shall conform to ANSI B16-34 pressure-temperature rating.
- N. All valves shall be repackable, under pressure, with the valve in the full open position. Packing for all valves shall be free of asbestos fibers and selected for the pressure-temperature service of the valve. It is incumbent upon the manufacturer to select the best quality, standard packing for the intended valve service. At the end of one year, period spot checks will be made, and should the packing show signs of hardening or causing stem corrosion then all valves supplied by the manufacturer shall be repacked by the Contractor, at no expense to the Owner, with a packing material selected by the Owner.
- O. All gate valves, globe valves, angle valves and shutoff valves of every character shall have malleable iron hand wheels, except iron body valves 2-1/2" and larger which may have either malleable iron or ASTM A-126 Class B, gray iron hand wheels.
- P. Balancing and/or Shutoff Valves for Closed Water Circulating Systems:
- Two inches and smaller, three piece bronze body, bronze or stainless steel ball and stem, Teflon seats, packing and gasket, bronze gland follower, adjustable stuffing box, steel lever type handle, with plastic operating handle, quarter turn stops, and shall be class 150. Manufacturer shall certify ball valves for use in throttling service. Stem extensions shall be furnished for use in insulated lines.
 - Valves 2-1/2" and larger shall be tapped full lug butterfly valves with aluminum bronze discs of ASTM B148 Alloy C955 and 316, 416, or 420 stainless steel shafts. Design must incorporate bushing between shafts and body of material suitable to provide a bearing surface to eliminate seizing or galling. Valve must be capable of providing a bubble tight seal at 200 psi for valves up to 12" (150 psi for larger valves) when used for end of line service without requiring the installation of a blind flange on the downstream side. Liners shall be resilient material suitable for 225°F temperature and bodies of ductile iron.
 - Butterfly valves 8" and larger and butterfly valves used for balancing service or control, regardless of size, shall have heavy duty weather proof encased gear operators, with malleable iron handwheel. Valves 2-1/2" through 6" shall have lever handles which can be set in interim positions between full open and full closed. All butterfly valves shall be absolutely tight against a pressure differential of 150 psi.
- Q. Check Valves: Bronze body, 2" and smaller, bronze body regrinding disc and seat with screw-in cap. Iron body, 2-1/2" and larger, bronze disc and seat or non-slam wafer type with stainless pins and springs, and bronze plate. Forged steel lift check valves, 2" and smaller shall be bolted cap and body, screwed end connections and conform to ANSI B16.34 and pressure temperature rating.

Standard of Quality for Valves:

<u>Size</u>	<u>Type</u>	<u>Service</u>	<u>Milwaukee</u>	<u>Nibco</u>	<u>Stockham or as Noted</u>
2" and smaller	Gate Valve	Chilled Water, LP Steam	1151	T-134	B105
2-1/2" & larger	Gate Valve	Plumbing, Chilled Water, LP Steam	F-2885-M	F-617-0	G-623
*2" & smaller	Ball Valve for shut-off	Domestic HW & CW Plbg. systems, Recirculating chilled water	BA100A	T-585-70	Appollo 77-100
2" &	Globe, Angle &	Chilled Water, LP Steam	597T	T-235	B-22

smaller	Balancing Valve				
2-1/2" & larger	Globe, Angle & Balancing Valve	Plumbing, Chilled Water, LP Steam	F-2981M	F-718-B	G-514-T
2-1/2" & larger	Butterfly Valve for shut-off	Domestic HW & CW Plbg. systems, Recirculating chilled water	ML-123B 642BG	LD2000	**DeZurik 632,L,D RS66-6 Demco NE-C, NF
2" & smaller	Gate Valve	Fire Protection	--	T-104-0	B-133
2-1/2" & larger	Gate Valve	Fire Protection	--	F-607-0	--
2-1/2" & larger	Gate Valve	***Fire Protection	--	F-908-W	G-940
2" & smaller	Check Valve	All Water Systems	510	T-433	B-345
2" & smaller	Globe Valve	Heating Hot Water	--	276AP	--
2-1/2" & larger	Globe Valve	Heating Hot Water	F-2983-M	F-768-B	--
2-1/2" & larger	Check Valve	All Water Systems	1400 Series	W-920-W	"Duo-Check"
1-1/2" & smaller	Gate Valve	MP & HP Steam	--	--	Vogt
2" & smaller	Gate Valve	MP & HP Steam			30-OF
2" to 12"	Hub End	(AWWA)	--	--	Mueller 2380-5
1-1/2" & smaller	Lubricated Gas Cock	Medical & Lab Gases	BB2-100	--	Rockwell 142 & 1796 (w/ wrench)
2" & Larger	Lubricated Gas Cock	Medical & Lab Gases	--	--	Rockwell 143 & 1797 (w/ wrench)
2" & smaller	Isolation Ball Valve	Medical & Lab Gases	--	--	Spirax Sarco Model 60

*= Requires extended stem in insulated lines

**= Valves 8" and Larger, and valves used for balancing services regardless of size, shall have heavy duty weather proof encased gear operators.

***= Requires ball drip assembly.

Contractor may submit substitution request for equal valves. Must show torque ratings for substitute and all other material equal to base spec.

Valve operators shall be rated for 2.5x the torque for full shut-off.

2.2 STRAINERS

- A. Strainers, 2" and smaller, bronze body, screwed ends, No. 10 mesh strainer, screwed cap with bronze blow-off valve (size to be determined by standard tap size in cap). Cast iron body, 2-1/2" and larger,

isolating type flanged ends where installed in copper lines, No. 7 perforated monel strainer, flanged cap with bronze ball blow-off valve (size of blow-off valve shall be determined by standard tap size in cap). Special Note: All strainers 6" and larger shall have studs mounted in the body flange in lieu of bolts for removal of cap. Baskets for strainers 6" and larger shall have stainless steel reinforcing bands at ends to prevent collapsing.

- B. Suction diffusers shall be Paco or approved equal, cast iron body and cover, steel diffuser, and stainless steel strainer, 125 pound ASA (flat face) flange for a working pressure of 175 psi and temperature of 300°F.

2.3 UNIONS

- A. Provide and install unions at proper points to permit removal of pipe and various equipment and machinery items without injury to other parts of the system. No unions will be required in welded lines or lines assembled with solder joint fittings except at equipment items, machinery items and other special pieces of apparatus. Unions in 2" and smaller in ferrous lines shall be Class 300 AAR malleable iron unions with iron to brass seats, and 2-1/2" and larger shall be ground flange unions. Unions in copper lines shall be Class 125 ground joint brass unions or Class 150 brass flanges if required by the mating item of equipment. Companion flanges on lines at various items of equipment, machines and pieces of apparatus shall serve as unions to permit removal of the particular items. See particular Specifications for special fittings and pressure.
- B. Unions connecting ferrous pipe to copper or brass pipe shall be dielectric type equal to Epco.
- C. In all domestic water lines where the material of the pipe is changed from ferrous to copper or brass, a dielectric coupling shall be used at the transition.

2.4 WELDED PIPING

- A. SCOPE: This section applies to all piping systems providing for welded piping, fittings, and other appurtenances. Specific systems requiring welded piping include, but are not limited to: chilled water, steam, steam condensate, and fire protection systems.
- B. MATERIALS: All weld fittings shall be USA factory made wrought carbon steel butt-welding fittings conforming to ASTM A234 and ASME/ANSI B16.9, latest edition, as made by Weld Bend, Tube Turn, Hackney, or Ladish Company. Each fitting shall be stamped as specified by ASME/ANSI B16.9 and, in addition, shall have the laboratory control number metal stenciled on each fitting for ready reference as to physical properties required for any fittings selected at random. Only one manufacturer of weld fittings will be approved for the project. Fittings which have been machined, remarked, printed or otherwise produced domestically from nondomestic forgings or materials will not be acceptable. Each in accordance with MSS SP-25. Markings shall be placed on the fittings at the farthest point from the edge to be welded to prevent disfiguring from the welding process. Submittal data for these fittings shall include a letter signed by an official of the manufacturing firm certifying compliance with these specifications.
- C. EXECUTION
 - 1. Piping and fittings shall be welded and fabricated in accordance with ASME/ANSI the latest editions of Standards B31.1 for all systems, and B31.3 for Steam and Condensate systems, from the Code for Pressure Piping. Machine beveling in shop is preferred. Field beveling may be done by flame cutting to recognized standards.
 - 2. Contractor shall ensure complete penetration of deposited metal with base metal. Contractor shall provide filler metal suitable for use with base metal. Contractor shall keep inside of fittings free from globules of weld metal. All welded pipe joints shall be made by the fusion welding process, employing a metallic arc or gas welding process. All pipe shall have the ends beveled 37-1/2

degrees and all joints shall be aligned true before welding. Except as specified otherwise, all changes in direction, intersection of lines, reduction in pipe size and the like shall be made with factory-fabricated welding fittings. Mitering of pipe to form elbows, notching of straight runs to form tees, or any similar construction will not be permitted.

3. Contractor shall align piping and equipment so that no part is offset more than 1/16 inch. Set all fittings and joints square and true, and preserve alignment during welding operation. Use of alignment rods inside pipe is prohibited.
4. Contractor shall not permit any weld to project within the pipe so as to restrict it. Tack welds, if used, must be of the same material and made by the same procedure as the completed weld. Otherwise, remove tack welds during welding operation.
5. Contractor shall not split, bend, flatten or otherwise damage piping before, during or after installation.
6. Contractor shall remove dirt, scale and other foreign matter from the inside of piping, by swabbing or flushing, prior to the connection of other piping sections, fittings, valves or equipment.
7. In no cases shall Schedule 40 pipe be welded with less than three passes including one stringer/root, one filler and one lacer. Schedule 80 pipe shall be welded with not less than four passes including one stringer/root, two filler and one lacer. In all cases, however, the weld must be filled before the cap weld is added.
8. TESTING
 - a. All welds are subject to inspection, visual and/or Xray, for compliance with specifications. The owner will, at the Owner's option, provide employees or employ a testing laboratory for the purposes of performing said inspections and/or Xray testing. Initial visual and Xray inspections will be provided by the owner. The contractor shall be responsible for all labor, material and travel expenses involved in the reinspection and retesting of any welds found to be unacceptable. In addition, the contractor shall be responsible for the costs involved in any and all additional testing required or recommended by ASME/ANSI Standards B31.1 and B31.3 due to the discovery of poor, unacceptable or rejected welds.
 - b. Welds lacking penetration, containing excessive porosity or cracks, or are found to be unacceptable for any reason, must be removed and replaced with an original quality weld as specified herein. All qualifying tests, welding and stress relieving procedures shall, moreover, be in accord with Standard Qualification for Welding Procedures, Welders and Welding Operators, Appendix A, Section 6 of the Code, current edition.
9. All weld fittings shall be USA factory made wrought carbon steel butt welding fittings conforming to ASTM Spec. A234 and ANSI standard B16.9-1964 as made by Grinnell, Tube Turn, Hackney, Taylor Forge, or Ladish Company. Each fitting shall be stamped as specified by ANSI B16.9 and, in addition, shall have the laboratory control number metal stenciled on each fitting for ready reference as to physical properties and chemical composition of the material. Complete test reports may be required for any fittings selected at random. Only one manufacturer of weld fittings will be approved for each project. Fittings which have been machined, remarked, printed or otherwise produced domestically from imported forgings or materials will not be acceptable. Each fitting shall have the manufacturer's trademark permanently identified in accordance with MSS SP-25. Markings shall be placed on the fittings at the farthest point from the edge to be welded to prevent disfiguring from the welding process. Submittal data for these fittings shall include a letter signed by an official of the manufacturing firm certifying compliance with these Specifications.

2.5 FLANGES

- A. All 150 lb. and 300 lb. ANSI flanges shall be weld neck and shall be domestically manufactured, forged carbon steel, conforming to ANSI B16.5 and ASTM A-181 Grade I or II or A-105-71 as made by Tube Turn, Hackney or Ladish Company. Slip on flanges shall not be used. Each fitting shall be stamped as specified by ANSI B16.9 and, in addition, shall have the laboratory control number stenciled on each fitting for ready reference as to physical properties and chemical composition of the material. Complete test reports may be required for any fitting selected at random. Flanges which have been machined, remarked, painted or otherwise produced domestically from imported forges will not be acceptable. Flanges shall have the manufacturer's trademark permanently identified in accordance with MSS SP-25. Contractor shall submit data for firm certifying compliance with these Specifications. Bolts used shall be carbon steel bolts with semi-finished hexagon nuts of American Standard Heavy dimensions. Allthread rods will not be an acceptable for flange bolts. Bolts shall have a tensile strength of 60,000 psi and an elastic limit of 30,000 psi. Flat faced flanges shall be furnished where required to match flanges on pumps, check valves, strainers, etc. Only one manufacturer of weld flanges will be approved for each project.
- B. All flanges shall be gasketed. Contractor shall place gasket between flanges of flanged joints. Gaskets shall fit within the bolt circle on raised face flanges and shall be full face on flat face flanges. Gaskets shall be cut from 1/16" thick, non-metallic, non-asbestos gasket material suitable for operating temperatures from -150°F to +750°F, Klingerseal C-4400, Manville Style 60 service sheet packing, or equal.

2.6 BACKFLOW PREVENTERS

- A. Backflow preventers (BFP) shall be reduced pressure type, Febco 825, or approved equal. A BFP shall be installed to isolate all non-potable water requirements from the building domestic water system. (All BFP's shall be installed within the building.)

PART 3 - EXECUTION

3.1 PREPARATION

- A. Ream pipe and tube ends. Remove burrs. Bevel plain end ferrous pipe.
- B. Remove scale and dirt on inside and outside before assembly.
- C. Prepare piping connections to equipment with flanges or unions.
- D. After completion, fill, clean, and treat systems.

3.2 INSTALLATION

- A. Route piping in orderly manner, plumb and parallel to building structure, and maintain gradient.
- B. Install piping to conserve building space, and not interfere with use of space and other work.
- C. Group piping whenever practical at common elevations.
- D. Install piping to allow for expansion and contraction without stressing pipe, joints, or connected equipment.
- E. Provide clearance for installation of insulation, and access to valves and fittings.

- F. Provide access where valves and fittings are not exposed.
- G. Slope piping and arrange systems to drain at low points. Use eccentric reducers to maintain top of pipe level.
- H. Where pipe support members are welded to structural building framing, scrape, brush clean, and apply one coat of zinc rich primer to welding.
- I. Prepare pipe, fittings, supports, and accessories for finish painting. Refer to Section 09900.
- J. Install valves with stems upright or horizontal, not inverted.
- K. Locate valves in readily accessible areas and where valve operation will not damage other components, materials, or devices.

3.3 FABRICATION OF PIPE

- A. All the various piping systems shall be made up straight and true and run at proper grades to permit proper flow of the contained material. Lines shall also be graded for proper drainage.
- B. Piping shall follow as closely as possible the routes shown on Drawings which take into consideration conditions to be met at the site.
- C. Should any unforeseen conditions arise, lines shall be changed or rerouted as required after proper approval has been obtained.
- D. All piping shall be installed with due regard to expansion and contraction and so as to prevent excessive strain and stress in the piping, in connections, and in equipment to which the lines are connected.
- E. All piping shall be clean when it is installed. Before installation it shall be checked, upended, swabbed, if necessary, and all rust or dirt from storage or from lying on the ground shall be removed.
- F. Procedure of Assembling Screw Pipe Fittings: All screw joints shall be made with taper threads, properly cut. Joints shall be made tight with Teflon applied to the pipe threads only and not to fittings. When threads are cut on pipes, the ends shall be carefully reamed to remove any burrs. Before installing pipe that has been cut and threaded, the lengths of pipe shall be upended and hammered to remove all shavings and foreign material.
- G. Procedure for Assembling Other Joints: Procedures for assembling joints in cast iron and copper lines have been set forth elsewhere in these Specifications. For any special materials, consult the manufacturers for the recommended procedures in assembling the joints.

3.4 APPLICATION

- A. Use grooved mechanical couplings and fasteners only in accessible locations.
- B. Install unions downstream of valves and at equipment or apparatus connections.
- C. Install brass male adapters each side of valves in copper piped system. Sweat solder adapters to pipe.
- D. Install gate valves for shut-off and to isolate equipment, part of systems, or vertical risers.
- E. Install ball valves for throttling, bypass, or manual flow control services.

- F. Provide spring loaded check valves on discharge of condenser water pumps. Provide non-slam or other system-appropriate check valves on all other pumps.
- G. Use plug cocks for throttling service. Use non-lubricated plug cocks only when shut-off or isolating valves are also provided.
- H. Use butterfly valves interchangeably with gate and globe valves.
- I. Use only butterfly valves in chilled and condenser water systems for throttling and isolation service.
- J. Use lug end butterfly valves to isolate equipment.
- K. Provide 3/4 inch (20 mm) ball drain valves at main shut-off valves, low points of piping, bases of vertical risers, and at equipment.

<u>Line</u>	<u>Testing Medium</u>	<u>Testing Pressure (PSIG)</u>	<u>Time in Hours</u>
Chilled Water	Water	1.5x working pressure	24
Steam MP & LP	Water	1.5x working pressure	24
Steam Condensate MP& LP	Water	1.5x working pressure	24
Pumped Condensate Return	Water	1.5x working pressure	24

3.5 PIPE PRESSURE TESTS

- A. Where leaks occur, the pipe shall be repaired and the tests repeated. No leaks shall be corrected by peening. Defective piping and joints shall be removed and replaced.

END OF SECTION

SECTION 23 05 16

PIPING EXPANSION COMPENSATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Flexible pipe connectors.
- B. Expansion joints and compensators.
- C. Pipe loops, offsets, and swing joints.

1.2 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Requirements and Division 01 Specification Sections, apply to this Section.

1.3 REFERENCES

- A. MIL-E-17814E - Expansion Joints, Pipe, Slip-Type, Packed.

1.4 PERFORMANCE REQUIREMENTS

- A. Provide structural work and equipment required to control expansion and contraction of piping. Verify that anchors, guides, and expansion joints provided, adequately protect system.
- B. Expansion Calculations:
 - 1. Installation Temperature: 50 degrees F (Hot Systems), 100 degrees F (Cold Systems)
 - 2. Chilled Water: 120 degrees F.
 - 3. Steam: 350 degrees F.
 - 4. Steam Condensate: 250 degrees F.
 - 5. Domestic Hot Water: 140 degrees F.
 - 6. Heating Hot Water: 180 degrees F.
 - 7. Safety Factor: 30 percent.
- C. Reactions from all expansion compensation devices anchors and guides to support structures shall be submitted to the structural engineer for review.

1.5 SUBMITTALS

- A. Submit under provisions of Section 230500.
- B. Product Data:

1. Flexible Pipe Connectors: Indicate maximum temperature and pressure rating, face-to-face length, live length, hose wall thickness, hose convolutions per foot and per assembly, fundamental frequency of assembly, braid structure, and total number of wires in braid.
 2. Expansion Joints: Indicate maximum temperature and pressure rating, and maximum expansion compensation.
 - C. Design Data: Submit calculations for the sizing of expansion loops and selection of expansion joints.
 - D. Shop Drawings: Submit shop drawings indicating the exact location and calculated axial and lateral loads for all anchors, guides, expansion joints and loops. Provide drawings indicating the proposed method of attachment to the building structure.
 - E. Manufacturer's Installation Instructions: Indicate special procedures, and external controls.
- 1.6 PROJECT RECORD DOCUMENTS
- A. Submit under provisions of Section 230500.
 - B. Record actual locations of flexible pipe connectors, expansion joints, anchors, and guides.
- 1.7 OPERATION AND MAINTENANCE DATA
- A. Submit under provisions of Section 230500.
 - B. Maintenance Data: Include adjustment instructions.
- 1.8 QUALIFICATIONS
- A. Manufacturer: Company specializing in manufacturing the products specified in this section with minimum three years documented experience.
 - B. Design expansion compensating system under direct supervision of a Professional Engineer experienced in design of this work and licensed in the State of Texas.
- 1.9 DELIVERY, STORAGE, AND HANDLING
- A. Deliver, store, protect and handle products to site under provisions of Section 230500.
 - B. Accept expansion joints on site in factory packing with shipping bars and positioning devices intact. Inspect for damage.
 - C. Protect equipment from exposure by leaving factory coverings, pipe end protection, and packaging in place until installation.
- 1.10 WARRANTY
- A. Provide five year warranty.
 - B. Warranty: Include coverage for leak free performance of packed expansion joints.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Microflex, Inc.
- B. Amber Booth Company
- C. The Metroflex Company
- D. ADSCO Manufacturing Corporation
- E. Substitutions: Under provisions of Div 1.

2.2 FLEXIBLE PIPE CONNECTORS

- A. Steel Piping :
 - 1. Inner Hose: Stainless Steel.
 - 2. Exterior Sleeve: Double braided stainless steel.
 - 3. Pressure Rating: 200 psig WOG and 250 degrees F.
 - 4. Joint: As specified for pipe joints.
 - 5. Size: Use pipe sized units.
 - 6. Maximum offset: 3/4 inch on each side of installed center line.

2.3 EXPANSION JOINTS

- A. Stainless Steel Bellows Type:
 - 1. Pressure Rating: 200 psig WOG and 250 degrees F.
 - 2. Joint: As specified for pipe joints.
 - 3. Size: Use pipe sized units.
 - 4. Application: Steel piping 3 inch and under.
- B. Externally-Internally Guided (Slip Joint) Piston-Ring Type:
 - 1. Pressure Rating: 300 psig steam and 800 degrees F.
 - 2. Joint: Flanged.
 - 3. Size: Use pipe sized units.
 - 4. Accessories: Internal flow liner.
 - 5. Application: Steel piping over 3 inch.

2.4 ACCESSORIES

- A. Pipe Alignment Guides: Two piece welded steel with enamel paint, bolted, with spider to fit standard pipe, frame with four mounting holes, clearance for minimum 1 inch thick insulation, minimum 3 inch travel.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Construct spool pieces to exact size of flexible connection for future insertion.
- C. Install flexible pipe connectors on pipes connected to equipment supported by vibration isolation. Provide line size flexible connectors.
- D. Install flexible connectors at right angles to displacement. Install one end immediately adjacent to isolated equipment and anchor other end. Install in horizontal plane unless indicated otherwise.
- E. Rigidly anchor pipe to building structure where necessary. Provide pipe guides so movement is directed along axis of pipe only. Erect piping such that strain and weight is not on cast connections or apparatus.
- F. Provide support and equipment required to control expansion and contraction of piping. Provide loops, pipe offsets, and swing joints, or expansion joints where indicated.

3.2 MANUFACTURER'S FIELD SERVICES

- A. Prepare and start systems under provisions of Section 230500.
- B. Provide inspection services by flexible pipe manufacturer's representative for final installing and certify installation is in accordance with manufacturer's recommendations and connectors are performing satisfactorily.

END OF SECTION

SECTION 23 05 19

METERS AND GAUGES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Pressure gages and pressure gage cocks. Thermometers and thermometer wells. Pressure/temperature test plugs.
- B. Ultrasonic flow meters used for balancing

1.2 REFERENCES

- A. ASME B40.1 - Gages - Pressure Indicating Dial Type - Elastic Element.
- B. ASTM E1 - Specification for ASTM Thermometers.
- C. ASTM E77 - Verification and Calibration of Liquid-in-Glass Thermometers.
- D. FS-GG-G-76 - Gages, Pressure and Vacuum, Dial Indicating (for Air, Steam, Oil, Water, Ammonia, Chloro- Flourhydrocarbon Gases, and Compressed Gases).
- E. UL 404 - Gages, Indicating Pressure, for Compressed Gas Service.

1.3 SUBMITTALS

- A. Submit under provisions of Division 1 and Section 230500.
- B. Product Data: Include list for every meter and gauge to be provided which indicates use, operating range, total range and location.
- C. Samples: Submit one of each type of instrument specified upon request by Engineer.
- D. Submit manufacturer's installation instructions.

1.4 PROJECT RECORD DOCUMENTS

- A. Submit record documents under provisions of Division 1 and Section 230500.
- B. Accurately record actual locations of meters and gauges.

1.5 ENVIRONMENTAL REQUIREMENTS

- A. Do not install instrumentation when areas are under construction, except for required rough-in, taps, supports and test plugs.

PART 2 - PRODUCTS

2.1 GAUGES AND GAUGE CONNECTIONS

- A. Pressure Gauges and Accessories: Provide Ashcroft gauges or approved equal, complete with lever handle shut-off cocks, and pulsation piston type dampeners, or approved equal. Porous type dampeners will not be accepted. Gauges shall have stainless steel movement and 1/2 of 1% accuracy. Gauges shall have back connection when used on a panel; otherwise they shall have bottom connections.
- B. Pressure Gauge Connections: Provide where noted or indicated on the Drawings or called for elsewhere in the Specifications, gauge connections complete with lever handle union shutoff cocks, or approved equal. All gauge connections shall be made up with brass pipe, nipples and brass screw fittings.
- C. Siphon: Brass, 1/4 inch angle or straight pattern. Model SY14b manufactured by Weiss.
- D. Pressure Test Tap: 1/4 inch full port brass ball valve, 150 psig.
- E. Minimum gauge face size shall be 3 inches in diameter.

2.2 THERMOMETER AND THERMOMETER WELLS

- A. Provide thermometers of not less than 9" length scale complete with brass separable sockets (wells) with extension neck to allow for insulation of piping. These thermometers shall be mercury red reading type in one piece glass tubes extending from top of scale to sensor, and shall be located so that they may be easily read. Field adjustable angle thermometers are acceptable. Thermometers shall be Weksler Industrial Thermometers, or approved equal.
- B. Thermometer test wells shall be 3/4" Weksler Thermal Wells, brass with stem of minimum length to extend beyond the mid-diameter of the pipe, 2-1/2" extension neck, and brass screw plug. Wells shall be suitable for use of industrial type thermometers.
- C. All thermometers shall be easily read from floor and readily visible from floor.

2.3 PRESSURE TEMPERATURE TEST PLUGS

- A. Test Plug: 1/4 inch brass fitting and cap for receiving 1/8 inch outside diameter pressure or temperature probe. Provide test plugs rated for 400 psig with neoprene core for temperatures up to 200 degrees F: and Nordel core for temperatures up to 350 degrees F and manufactured by Fairfax or approved equal.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Pressure gauges shall be provided on the suction and discharge of base mounted pumps as indicated on the Drawings.
- C. Single pressure gauges shall be installed on in-line pumps as indicated on the Drawings.
- D. Install pressure gages with pulsation dampers. Provide gage cock to isolate each gage. Provide siphon on gages in steam systems. Install thread-o-lets on piping for gage installation, do not weld nipple directly to pipe.

- E. Install thermometers in piping systems in thermometer wells. Enlarge pipes smaller than 2-1/2 inch for installation of -thermometer sockets.
- F. Thermometer wells and thermometers shall be located where noted on the Drawings and where called for in other sections of the Specifications. Thermometer test wells only shall be installed in a vertical position in horizontal lines and at 45 degrees, in vertical lines to hold a fluid in the well.
- G. Locate pressure/temperature test plugs on the inlet and outlet of each heating coil in air terminal units, unit heaters, duct heating coils, etc. Mount short nipple with 1/4 inch ball valve. Place in accessible locations.

END OF SECTION

SECTION 23 05 29

SUPPORTS AND ANCHORS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Pipe and equipment hangers and supports.
- B. Equipment bases and supports.
- C. Sleeves and seals.
- D. Flashing and sealing equipment and pipe stacks.

1.2 RELATED SECTIONS

- A. Section 21 3133 - Fire Protection Piping.
- B. Section 22 0548 - Vibration Isolation.
- C. Section 23 0701 - Piping Insulation.
- D. Section 23 0702 - Equipment Insulation.
- E. Section 23 0703 - Ductwork Insulation.
- F. Section 22 1116 – Plumbing Piping.
- G. Section 23 2113 - Hydronic Piping.
- H. Section 23 2213 - Steam and Steam Condensate Piping.

1.3 REFERENCES

- A. ASME B31.1 - Power Piping
- B. ASME B31.2 - Fuel Gas Piping
- C. ASME B31.5 - Refrigeration Piping
- D. ASME B31.9 - Building Services Piping
- E. ASTM F708 - Design and Installation of Rigid Pipe Hangers.
- F. MSS SP58 - Pipe Hangers and Supports - Materials, Design and Manufacturer.
- G. MSS SP69 - Pipe Hangers and Supports - Selection and Application.
- H. MSS SP89 - Pipe Hangers and Supports - Fabrication and Installation Practices.

1.4 SUBMITTALS

- A. Submit under provisions of Section 230500.
- B. Shop Drawings: Indicate system layout with location and detail of trapeze hangers.
- C. Product Data: Provide manufacturers catalog data including load capacity.
- D. Design Data: Indicate load carrying capacity of trapeze, multiple pipe, and riser support hangers.
- E. Manufacturer's Installation Instructions: Indicate special procedures and assembly of components.

1.5 REGULATORY REQUIREMENTS

- A. Conform to applicable code for support of plumbing, hydronic, steam and steam condensate piping.

PART 2 – PRODUCTS

2.1 PIPE HANGERS AND SUPPORTS

- A. Manufacturers:
 - 1. Anvil
 - 2. Other acceptable manufacturers offering equivalent products.
- B. Supports, hangers, anchors and guides shall be provided for all horizontal and vertical piping. Shop Drawings shall be provided, indicating locations and details of anchors, guides, expansion loops and joints, hangers, etc. The hanger design shall conform to the ASME Code for Pressure Piping.
- C. All auxiliary steel required for pipe supports, anchors, guides, etc. shall be provided by the Mechanical Trades unless specifically indicated to be provided by others.
- D. The supports, hangers, anchors, and guides for the chilled water supply and return piping, steam piping, condensate return piping, etc. of the Campus Loop System routed through utility tunnels and below buildings shall be provided as indicated on the Drawings.
- E. Contractor shall review all Drawings, including Structural Drawings, for details regarding pipe supports, anchors, hangers, and guides.
- F. All pipe supports shall be of type and arrangement to prevent excessive deflection, to avoid excessive bending stresses between supports, and to eliminate transmission of vibration.
- G. All rod sizes indicated in this Specification are minimum sizes only. This trade shall be responsible for structural integrity of all supports, anchors, guides, etc. All structural hanging materials shall have a minimum safety factor of 5 built in.
- H. Anchor points as indicated on Drawings or as required shall be located and constructed to permit the piping system to take up its expansion and contraction freely in opposite directions away from the anchored points.
- I. Guide points shall be located and constructed wherever required or indicated on Drawings and at each side of an expansion joint or loop, to permit free axial movement only.
- J. Pipe supports, hangers, anchors, and guides shall be fastened to the structure only at such points where the structure is capable of restraining the forces in the piping system.

- K. Hangers supporting and contacting brass or copper lines 3" in size and smaller shall be Grinnell Fig. CT-99c, adjustable, copper plated, tubing ring. Hangers supporting and contacting brass or copper lines 4" and larger shall be Grinnell Fig. 260, adjustable clevis, with a nut above and below the hanger, and approved neoprene isolating material between pipe (or tubing) and hanger on the support rod. For insulated copper or brass domestic water lines, hangers for all sizes of pipe shall be Grinnell Fig. 300, adjustable clevis, with a nut above and below the hanger, and approved neoprene isolating material between pipe (or tubing) and hanger on the support rod. Isolate all copper or brass lines from all ferrous materials with approved dielectric materials. Hangers supporting and contacting plastic or glass piping shall be of equal design, but shall be padded with neoprene material or equal. The padding material and the configuration of its installation shall be submitted for approval.
- L. Hangers supporting insulated lines where the outside diameter of the insulation is the equivalent of 8" diameter pipe or smaller in size and supporting all ferrous lines 6" and smaller in size shall be Grinnell Fig. 260, adjustable clevis, with a nut above and below the hanger on the support rod.
- M. Hangers supporting and contacting ferrous lines larger than 6" in size and outside of insulation on lines with the outside diameter equivalent to 10" diameter pipe shall be Grinnell Fig. 260, adjustable clevis, with a nut above and below the hanger on the support rod.
- N. Glass riser clamps shall be neoprene coated.
- O. Other special type of hangers may be employed where so specified or indicated on the Drawings, or where required by the particular conditions. In any case, all hangers must be acceptable to the owner.
- P. Each hanger shall be properly sized to fit the supported pipe or fit the outside of the insulation on lines where specified. Hangers for dual or low temperature insulation pipes shall bear on the outside of the insulation, which shall be protected by support shields as specified in Section 23 0701 - PIPING INSULATION. Protect insulation from crushing by means of a section of rigid insulation to be installed at hanger points. Hangers for high temperature insulated pipes and all insulated hot and cold domestic water pipes shall be encased in the insulation unless supported by trapezes in which case shield and rigid insulation shall be provided as specified above for low temperature insulated pipes.
- Q. Hangers for large diameter steam and chilled water piping shall be roller style where thermal expansion is expected.
- R. Supports for vertical piping in concealed areas shall be double bolt riser clamps, Grinnell Fig. 261, or other approved equal, with each end having equal bearing on the building structure, and located at each floor. Two-hole rigid pipe clamps at 4 ft. o.c. or Kindorf channels and Grinnell Fig. 261 riser clamps may be used to support pipe from vertical surfaces or members where lines are not subject to expansion and contraction. Where brass or copper lines are supported on trapeze hangers or Kindorf channels the pipes shall be isolated from these supports with plastic tape with insulating qualities, or strut clamps as manufactured by Specialty Products Company, Stanton, California.
- S. Supports for vertical piping in exposed areas (such as fire protection standpipe in stairwells) shall be attached to the underside of the building structure above the top of the riser, and the underside of the penetrated structure. The contractor shall use a drilled anchor as specified above, and use a Grinnell No. 595 Socket Clamp with Grinnell No. 594 Socket Clamp Washers, as a riser clamp. The top riser hanger shall consist of two (2) hanger rods (sized as specified) anchored to the underside of the building structure, supporting the pipe by means of the material specified. Risers penetrating floors shall be supported from the underside of the penetrated floor as specified for the top of the riser, but the hanger rods shall also include spring isolators (see Specification on Vibration Isolation for isolator specification). Floor penetrations in exposed areas shall be finished using Grinnell Fig. 395 "Ceiling Plates" painted to match the pipe if uninsulated; or if insulated, the penetration shall be covered using a chrome plated escutcheon.
- T. Provide hangers at all changes in direction of pipe.

U. Vibration Isolation: Resilient hangers shall be provided on all piping connected to rotating equipment (pumps, etc.). Piping that may vibrate and create an audible noise shall also be isolated. Spring hangers or supports shall be provided where indicated on the Drawings and/or specified under Section 230548.

V. Attachment:

1. The load and spacing on each hanger and/or insert shall not exceed the safe allowable load for any component of the support system, including the concrete, which holds the inserts. Reinforcement at inserts shall be provided as required to develop the strength required.
2. Inserts for piping shall be of a type which will not interfere with reinforcing as shown on the structural Drawings and which will not displace excessive amounts of structural concrete.
3. All pipe supports shall be designed and installed to avoid interference with other piping, hangers, ducts, electrical conduit, supports, building structures, equipment, etc. All piping shall be installed with due regard to expansion and contraction and the type of hanger method of support, location of support, etc. shall be governed in part by this Specification.
4. Pipe hangers shall be attached to the structure as follows:
 - a. Poured In Place Concrete: Where pipes and equipment are supported under poured in place concrete construction and the schedule allows for inserts, each hanger rod shall be fitted with a nut at its upper end, which nut shall be set into an Underwriters' Laboratories, Inc. listed universal concrete insert placed in the form work before concrete is poured. Where inserts are placed in the bottom faces of concrete joists which are too narrow to provide adequate strength of concrete to hold the insert properly or where a larger insert would require displacement of the bottom joist steel, the hanger rod shall be suspended from the center of a horizontal angle iron, channel iron, I-beam, etc. spanning across two adjacent joists. The horizontal support shall be bolted to nonadjustable concrete inserts of the "spot" type, of physical size small enough to avoid the bottom joist steel.
 - b. Steel Bar Joists: Where pipes and loads are supported under bar joists, hanger rods may be run through the space between the bottom angles and secured with a washer and two nuts. Where larger lines are supported beneath bar joists, hanger rods shall be secured to angle irons of adequate size; each angle shall span across two or more joists as required to distribute the weight properly and shall be welded to the joists or otherwise permanently fixed thereto.
 - c. Steel Beams: Where pipes and loads are supported under steel beams, approved type beam clamps shall be used.
 - d. Wood Framing: Where pipes and loads are supported from wood framing, hanger rods shall be attached to framing with side beam brackets or angle clips.
 - e. Precast Tee Structural Concrete: Hanger supports, anchors, etc. required for mechanical systems attached to the precast, double tee, structural concrete system are to be installed in accord with approved shop Drawings only. Holes required for hanger rods shall be core drilled in the "flange" of the double tee only; impact type tools are not allowed under any circumstances. Core drilling in the "stem" portions of the double tee is not allowed. Holes core drilled through the "flange" for hanger rods shall be no greater than 1/4" larger than the diameter of the hanger rod. Hanger rods shall be supported by means of bearing plates of size and shape acceptable to the Architect/Engineer, with welded double nuts on the hanger rod above the bearing plate. Cinch anchors, lead shields, expansion bolts, and studs driven by explosion charges are not allowed under any circumstances in the lower 15" of each stem and in the "shadow" of the stem on the top side of the "double tees".

W. Trapezes: Where multiple lines are run horizontally at the same elevation and grade, they may be supported on trapezes of Kindorf, Elcen, or approved equal, channel suspended on rods or pipes. Trapeze members including suspension rods shall each be properly sized for the number, size, and loaded weight of the lines they are to support.

- X. Finishes: All hangers on piping including clevis hangers, rods, inserts, clamps, stanchions, brackets, shall be dipped in Zinc Chromate Primer before installation. Rods may be galvanized or cadmium plated after threading, in lieu of dipping zinc chromate. Universal concrete inserts shall be cadmium plated.

2.2 PIPE SUPPORTS IN CHASES AND PARTITIONS

- A. Horizontal and vertical piping chases and partitions shall be supported by hangers or other suitable support. Pipes serving plumbing fixtures and equipment shall be securely supported

near the point where pipes penetrate the finish wall. Supports shall be steel plate, angles or special channels such as Unistrut mounted in vertical or horizontal position. Pipe clamps such as Unistrut P2426, P2008, P1109 or other approved clamps shall be attached to supports. Supports shall be attached to wall or floor construction with clip angles, brackets, or other approved method. Supports may be attached to cast iron pipe with pipe clamp, or other approved method. All copper or brass lines shall be isolated from ferrous metals with dielectric materials to prevent electrolytic action.

2.3 DUCT HANGERS AND SUPPORTS

- A. Ductwork: All ductwork shall be supported in accordance with the SMACNA recommendation for the service involved; however, all horizontal ductwork shall be supported at intervals not to exceed 8'- 0". Horizontal ducts shall be supported using galvanized steel bands extending up both sides and onto the construction above, where they shall turn over and be secured with bolts into nuts fitted in inserts set in the concrete bolted to angles secured to the construction above, or secured in another approved manner. For attaching methods for precast double tee structural concrete, refer to details on the Drawings and as specified herein. All ductwork over 36" wide shall be supported with vibration isolation integral with the hangers or supports.
- B. Terminal units weighing less than 150 lbs. shall be supported by four 16 gauge, 1" wide sheet metal straps with ends turned under bottom of box at corners and secured by one not over 3/4" in length, 1/4" diameter sheet metal screw per strap. The other strap end shall be attached to the structure by 1/4" diameter threaded bolt into the concrete insert or into drilled-hole threaded concrete expansion anchor. Boxes over 150 lbs. in weight shall be supported the same as described above except two 1/4" diameter sheet metal screws through turned end of strap box shall be provided. Where interferences occur, overhead of the box, not allowing direct vertical support by straps, provide trapezes of Kindorf, Unistrut, or Elcen channel suspended by 1/4" diameter galvanized threaded rods providing such channels do not block access panels of boxes. Threaded rods shall be supported from structure by concrete insert or by drilled-hole threaded concrete expansion anchor.
- C. Miscellaneous: Provide any other special foundations, hangers and supports indicated on the Drawings, specified elsewhere herein; or required by conditions at the site. Hangers and supporting structures for suspended equipment shall be provided as required to support the load from the building structure in a manner acceptable to the Architect/Engineer.

2.4 ACCESSORIES

- A. Hanger Rods: Galvanized mild steel threaded both ends, galvanized threaded one end, or galvanized continuous threaded.

2.5 INSERTS

- A. Inserts: Malleable iron case of galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms; size inserts to suit threaded hanger rods.

2.6 FLASHING

- A. Metal Flashing: 26 gage galvanized or stainless steel.
- B. Metal Counterflashing: 22 gage galvanized or stainless steel.
- C. Roofing Flashing: See specifications for Roofing, elsewhere in these Specifications.
- D. Caps: Steel, 22 gage minimum; 16 gage at fire resistant elements.

2.7 EQUIPMENT CURBS

- A. Fabrication: Welded 18 gage galvanized steel shell and base, mitered 3 inch cant, variable step to match roof insulation, factory installed wood nailer.

2.8 CONCRETE FOUNDATIONS ("HOUSEKEEPING PADS")

- A. Concrete foundations for the support of equipment such as floor mounted panels, pumps, fans, air handling units, etc., shall not be less than 3.5" high (formed by 2x4 lumber) and extend 4" on all sides beyond the limits of the mounted equipment unless otherwise noted and shall be poured in forms built of new dressed lumber. All corners of the foundations shall be neatly chamfered by means of sheet metal or triangular wood strips nailed to the form. Foundation bolts shall be placed in the forms when the concrete is poured, the bolts being correctly located by means of templates. Each bolt shall be set in a sleeve of size to provide 1/2" clearance around bolt. Allow 1" below the equipment bases for alignment and grouting. After grouting, the forms shall be removed and the surface of the foundations shall be hand rubbed with Carborundum. Foundations for equipment located on the exterior of the building shall be provided as indicated. Foundations shall be constructed in accordance with Shop Drawings submitted by the Contractor for review by the Architect/Engineer. For all floor mounted air handling units and outside air handling units, provide isolation under the housekeeping pad, with housekeeping pad poured on top of resilient isolation pads or resilient underlayment sheet material to decouple from floor structure. 0.2" minimum static deflection is recommended (use pads that have nominal 0.1" deflection stacked with steel plate or sheet shims between to distribute load).

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.

3.2 INSERTS

- A. Provide inserts for placement in concrete formwork.
- B. Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams.
- C. Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 4 inches.
- D. Where concrete slabs form finished ceiling, locate inserts flush with slab surface.
- E. Where inserts are omitted, drill through concrete slab from below and provide through-bolt with recessed square steel plate and nut recessed into and grouted flush with slab.

3.3 PIPE HANGERS AND SUPPORTS

- A. Support horizontal piping in accordance with recognized standard practices, minimum spacing as required by piping materials and size. In addition to minimum requirements, hangers and supports must be provided at the following locations:
 - 1. Within 12 inches of any change in direction
 - 2. Each side of 6" and larger valves
 - 3. Each side of in-line pumps.
- B. Install hangers to provide minimum 1/2 inch space between finished covering and adjacent work.
- C. Place hangers within 12 inches of each horizontal elbow.
- D. Use hangers with 1-1/2 inch minimum vertical adjustment.
- E. Support horizontal cast iron pipe adjacent to each hub, with 5 feet maximum spacing between hangers.
- F. Support vertical piping at every floor. Support vertical cast iron pipe at each floor at hub.
- G. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- H. Support riser piping independently of connected horizontal piping.
- I. Provide copper plated hangers and supports for copper piping.
- J. Design hangers for pipe movement without disengagement of supported pipe.
- K. Prime coat exposed steel hangers and supports. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed, but shall be corrosion protected with galvanized plating.
- L. Hanger Rods: (NOTE: All hanger rods shall be trimmed neatly so that no more than 1 inch of excess hanger rod protrudes beyond the hanger nut. In the event a rod is intentionally but temporarily left excessively long (for sloped or insulated lines for example), the contractor shall take appropriate measures to protect the pipe or other materials from damage.)

3.4 FLASHING

- A. Provide flexible flashing and metal counterflashing where piping and ductwork penetrate weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes projecting 3 inches minimum above finished roof surface with lead worked one inch minimum into hub, 8 inches minimum clear on sides with 24 x 24 inches sheet size. For pipes through outside walls, turn flanges back into wall and caulk, metal counterflash, and seal.
- C. Flash floor drains in floors with topping over finished areas with lead, 10 inches clear on sides with minimum 36 x 36 inch sheet size. Fasten flashing to drain clamp device.
- D. Seal floor, shower, mop sink, and drains watertight to adjacent materials.
- E. Provide acoustical lead flashing around ducts and pipes penetrating equipment rooms, installed in accordance with manufacturer's instructions for sound control.

- F. Provide curbs for mechanical roof installations 14 inches minimum high above roofing surface. Flash and counterflash with sheet metal; seal watertight. Attach counterflashing mechanical equipment and lap base flashing on roof curbs. Flatten and solder joints.
- G. Adjust storm collars tight to pipe with bolts; caulk around top edge. Use storm collars above roof jacks. Screw vertical flange section to face of curb.

3.5 SLEEVES

- A. Set sleeves in position in formwork. Provide reinforcing around sleeves.
- B. Size sleeves large enough to allow for movement due to expansion and contraction. Provide for continuous insulation wrapping.
- C. Extend sleeves through floors four inches above finished floor level. Caulk sleeves.
- D. Where piping, ductwork or conduit penetrates floor, ceiling, or wall, close space between pipe or duct and adjacent work with fire stopping insulation and caulk air tight. Provide close fitting metal collar or escutcheon covers, as appropriate, at both sides of penetration.
- E. Install chrome plated steel or stainless steel escutcheons at finished surfaces.

3.6 SCHEDULES

Pipe Size <u>INCHES</u>	Max. Hanger Spacing <u>FEET</u>	Hanger Rod Diameter <u>INCHES</u>
1/2 to 1-1/4	6.5	3/8
1-1/2 to 2	10	3/8
2-1/2 to 3	10	1/2
4 to 6	10	5/8
8 to 12	14	7/8
14 and over	20	1

Note: Insulated piping support spacing shall be reduced as necessary to meet the requirements of the “blocking” or insulation at the support.

- F. and at all Joints
- G. Note: Insulated piping support spacing shall be reduced as necessary to meet the requirements of the “blocking” or insulation at the support.

END OF SECTION

SECTION 23 05 48

VIBRATION ISOLATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Vibration isolators, pipe supports, and equipment anchors, of appropriate sizes and weight loading to meet the specified deflection requirements, in accordance with instructions of isolator manufacturer.
- B. Vibration isolation for all Division 22 and 23 systems as noted below. Provide all miscellaneous items (angle iron, bolts, rods, etc.) required for a complete system. Contractor and vendors shall thoroughly coordinate all vibration isolation systems.
- C. Coordination of installation with other trades (placement of anchor bolts in concrete slabs, etc.)

1.2 RELATED REQUIREMENTS

- A. Section 01 45 33 - Code-Required Special Inspections
- B. Section 03 30 00 - Cast-in-Place Concrete
- C. Division 22: Plumbing

1.3 MANUFACTURER RESPONSIBILITIES

- A. Manufacturer of vibration isolation and seismic control products shall have the following responsibilities:
 - 1. Manufacturer of vibration isolation shall have the following responsibilities:
 - a. Determine vibration isolation and restraint sizes and locations for mechanical and plumbing equipment.
 - b. Determine vibration isolation sizes and locations for mechanical and plumbing equipment.
 - c. Provide isolation systems for all plumbing and mechanical of equipment (vibration isolated and non-isolated) and systems (piping and ductwork).
 - d. Provide installation instructions and drawings.
- B. Vibration isolation specialist shall coordinate his work with that of other trades to verify that equipment speeds, in revolution per minute (rpm), are based upon actual equipment installed at the project site.
- C. Verify that equipment rpm and spring deflection selected are arranged so that resonance is avoided.
- D. Exact mounting sizes, dimensions and quantity of isolators and static deflection required shall be determined by the isolator manufacturer based upon equipment that will be furnished and installed by the contractor under this Contract.

1.4 SUBMITTALS

- A. Contractor's Certification: Vibration isolator submittals shall include a certification, signed by an officer representing the Contractor and stipulating that the submittal prepared by the manufacturer has been reviewed, and checked on an item by item basis against each piece of mechanical equipment, piping,

ductwork and panel shown or specified in the Contract Documents, which requires vibration isolation and/or support.

- B. Manufacturer's Certification: The manufacturer or manufacturers (if there are more than one) shall each certify that the selections of vibration isolation equipment are based upon the drawings and specifications, and that each piece of mechanical equipment has been examined for rotational speed, equipment type, mounting location, and supporting span between column centers, and that an appropriate isolator has been selected.
- C. Product Data: Furnish manufacturer's product data covering each isolator type for style, characteristic, and finish. Isolator quantities, dimensions, deflections, capacities and types shall remain the responsibility of the manufacturer and the contractor.
- D. Shop Drawings: Provide layout drawings, drawn to a scale of not less than 1/8-inch to 1-foot, showing the proposed layout of equipment and piping systems and the location and type of each vibration isolation and restraint device. Carefully examine other sections requiring coordinated shop drawings, including but not limited to Section 23 31 13, "Sheetmetal Ductwork", Section 23 31 14, "Sheetmetal - Special Ductwork", and prepare restraint/isolation shop drawings to the same scale showing the location of each vibration isolation equipment base, pipe hanger, flexible connection, and isolator restraint device.

1.5 QUALITY ASSURANCE

- A. Responsibility for Products: Select deflection for spring isolators in accordance with recommendations in the current issue of ASHRAE Handbook of Fundamentals, unless noted otherwise on drawings.
- B. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with not less than 10 years of documented experience.
 - 1. Member of Vibration Isolation and Seismic Control Manufacturers Association (VISCMA).

1.6 STORAGE AND PROTECTION

- A. Storage: Store vibration isolation equipment indoors in the manufacturer's original shipping containers. Preclude the entrance of construction dirt and debris. Vibration isolation equipment and bases, which show signs of rust, cement or concrete fouling, dirt and construction debris shall be disassembled and cleaned, approved or removed from the project site and replaced with new.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Amber Booth, Kinetics Noise Control, Korfund Company, Mason Industries, Vibration Eliminator Co., or Vibration Mountings & Controls.
- B. Furnish vibration isolators by single manufacturer.
- C. Substitutions: Not permitted.

2.2 PRODUCTS

- A. Type 1: Mason Super "W", 2 layers of 3/4" neoprene pad with 16 ga. galvanized shim.
- B. Type 2: Mason BR, two neoprene elements housed in a ductile iron casting.

- C. Type 3: Mason SLF, free standing spring isolator, 1/4" neoprene non-skid pad, leveling bolt, spring diameter no less than 0.8 of compressed height at rated load, minimal additional travel to solid equal to 50% or rated deflection.
- D. Type 4: Mason SLR, restrained spring isolator, vertical limit stops, internal isolation pad.
- E. Type 6: Mason 30N, spring and double neoprene hanger, 1-1/4" neoprene element at top of housing, spring seated in neoprene cup at bottom of housing, designed to allow 30 degrees arc from side to side of hanger rod.
- F. Type 10: Mason BMK rectangular steel frame form for concrete inertia base, 1/2" reinforcing bars on 6" centers, both ways.
- G. Type 11: Mason RSC, spring isolation curb for roof mounted equipment, heavy gauge Z section sheet metal base that supports adjustable and removable restrained spring mounts, top section to be continuous rail support for equipment; springs to rest on 1/4" neoprene pads; hardware shall be plated and springs furnished with rust resistant finish; curb to be waterproofed using continuous galvanized flexible counter flashing, joined at corners with EDAM bellows; spring locations to have removable, waterproof access ports.
- H. Type 12: Mason BBS flexible stainless steel braided hose; minimum lengths as follows:
 - 1. 1/2" - 1-1/2" : 12"
 - 2. 2" 4" : 18"
 - 3. 6" - 10" : 24"
 - 4. 12" - 16" : 32"
- I. Type 13: Mason HS spring hanger, spring seated in neoprene cup.
- J. Type 14: Mason WF steel frame base, with motor slide rail.
- K. Type 15: Mason SafeFlex flexible rubber pipe connection, peroxide cured EPDM with Kevlar tire cord reinforcement, raised face rubber flanges with encased solid steel rings.
 - 1. 14" diameter and below: Mason SFDEJ twin sphere with reinforcing ring; minimum pressure rating of 250 psi at 170 degrees F. and 215 psi at 250 degrees F.
 - 2. 16" diameter and above: Mason SFEJ single sphere; minimum pressure rating of 180 psi at 170 degrees F. and 150 psi at 250 degrees F.
 - 3. Control rods; Mason CR with 1/2" thick Neoprene washer bushings.

2.3 MATERIALS AND EQUIPMENT

- A. Materials and equipment shall conform to the respective specifications and other requirements specified below:
 - 1. Squarehead bolts and heavy hexagon nuts, ANSI B18.2.1 and ANSI B18.2.2, and ASTM A 307 or ASTM A 576.
 - 2. Sway Brace Material used for members shown on mechanical drawings, except for pipes, shall be structural steel conforming with ASTM A 36. Steel pipes shall conform to ASTM A 501.

PART 3 - EXECUTION

3.1 VIBRATION CONTROL

- A. Size vibration control equipment in accordance with weight distribution, pull or the imposed torque as shown on equipment shop drawings. Minimum static deflections may be revised subject to prior approval.
- B. Provide revised vibration control equipment to match revised or substituted equipment.
- C. Install vibration control equipment in accordance with the manufacturer's installation instructions and as specified.
- D. Install equipment on vibration isolation curbs to provide watertight seal.

3.2 APPLICATIONS

- A. Equipment: Use the vibration and restraint types listed above on the following applications:
 - 1. A/C units, indoor, not internally isolated
 - a. Type 3
 - b. Type 5
 - 2. A/C units, packaged rooftop: Type 11
 - 3.
 - 3. Fans, Floor mounted
 - a. Upper floors: Type 3, 10
 - 4. Fans, suspended
 - a. Type 6, 14
 - 5. Pumps
 - a. Slab on grade: Type 15; bolt frame directly to pad.
 - b. Upper floors: Type 3, 10, 15
 - 6. Unit heaters
 - a. Type 13
- B. Piping
 - 1. Provide Type 6 vibration isolation on following piping:
 - a. Closest two hangers on piping at air handling units and blower coil units.
 - b. HVAC water piping within 20 pipe diameters each side of HVAC pumps.
 - c. Food service heat rejection piping within 20 pipe diameters of heat rejection equipment.

- d. Condenser water piping within 20 pipe diameters of tower. Alternate: Use Mason BBF braided stainless steel hose at connections to tower.
 - C. Provide Type 12 flexible connections at following locations:
 - 1. Medical air lines within 50 pipe diameters each side of compressors
 - 2. Vacuum lines within 50 pipe diameters each side of vacuum pump
 - D. Use hold down clamps to attach multiple pipes to trapeze hangers.
- 3.3 ANCHORING
- A. Installation: Installation shall comply with manufacturer's published recommendations and shall be installed so that isolators are plumb and are operating at a manner for which they were designed.
 - B. Unless otherwise specified, all equipment shall be securely bolted to isolators, steel bases or concrete inertia bases.
- 3.4 ANCHOR BOLTS
- A. If the size and number of the anchor bolts are not shown on the drawings then anchor bolts shall conform to the schedule for the various equipment weights or the manufacturer's installation recommendations, whichever is the most stringent.
- 3.5 MISCELLANEOUS EQUIPMENT
- A. The following specific items of equipment to be furnished under this contract shall be manufactured and assembled, and constructed so as to be capable of withstanding the horizontal equivalent static force of 0.11 times the operating weight of the equipment, at vertical center of gravity of the equipment without causing permanent deformation, dislocations, separation of components, or other damage, which would render the equipment inoperative for significant periods of time following an earthquake.
 - 1. Air-Handling Units
- 3.6 INSTALLATION
- A. Set anchor bolts when concrete is placed.
 - B. Install isolators in accordance with recommendations of isolator manufacturer and equipment manufacturer.
 - C. Isolate mechanical equipment as indicated.
 - D. Remove all debris from under equipment, and thoroughly clean steel bases, inertia bases and check for free movement.

END OF SECTION

SECTION 23 05 53

MECHANICAL IDENTIFICATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Nameplates.
- B. Stencils.
- C. Pipe Markers.

1.2 RELATED SECTIONS

- A. Division 9 - Painting: Identification painting.
- B. Section 230500 - Common Work Results for HVAC.

1.3 REFERENCES

- A. ASME A13.1 - Scheme for the Identification of Piping Systems.

1.4 SUBMITTALS

- A. Submit under provisions of Division 1 and Section 230500.
- B. Submit list of wording, symbols, letter size, and color coding for mechanical identification.
- C. Obtain new equipment identification numbers from the Drawings or the Superintendent of Utilities.
- D. Product Data: Provide manufacturers catalog literature for each product required.
- E. Samples: Submit two of each type of label, tag, etc., of the approximate size specified or implied in the specification.
 - 1. Manufacturer's Installation Instructions: Indicate special procedures, and installation.

1.5 PROJECT RECORD DOCUMENTS

- A. Submit under provisions of Division 1 and Section 230500.
- B. Not used

PART 2 - PRODUCTS

2.1 NAMEPLATES

- A. Description: Laminated three-layer plastic with engraved black letters on light contrasting background color.

2.2 TAGS

- A. Metal Tags: Brass with stamped letters; tag size minimum 1-1/2 inch (40 mm) diameter with smooth edges.
- B. Chart: Typewritten letter size list in anodized aluminum frame.

2.3 PIPE MARKERS

- A. Color: Conform to ASME A13.1.
- B. Plastic Pipe Markers: Factory fabricated, flexible, semi- rigid plastic, preformed to fit around pipe or pipe covering; minimum information indicating flow direction arrow and identification of fluid being conveyed.
- C. Plastic Tape Pipe Markers: Flexible, vinyl film tape with pressure sensitive adhesive backing and printed markings.
- D. Underground Plastic Pipe Markers: Bright colored continuously printed plastic ribbon tape, minimum 6 inches (150 mm) wide by 4 mil (0.10 mm) thick, manufactured for direct burial service.

2.4 CEILING TACKS

- A. Description: Steel with 3/4 inch (20 mm) diameter color coded head.
- B. Color code as follows:
 - 1. Yellow - HVAC equipment
 - 2. Red - Fire dampers/smoke dampers
 - 3. Blue - Heating/cooling valves

2.5 GENERAL

- A. The Contractor shall make it possible for the personnel operating and maintaining the new equipment and systems in this project to readily identify the various pieces of equipment, major valves, piping, etc., by marking them. All items of equipment such as fans, pumps, etc., shall be clearly marked using engraved nameplates as hereinafter specified. The item of equipment shall indicate the same number as shown on the Drawings. The equipment identification will be shown on the Drawings or provided by the Superintendent of Utilities.

2.6 MECHANICAL

- A. All items of new mechanical equipment shall be identified by the attachment of engraved nameplates constructed from laminated phenolic plastic, at least 1/16" thick, 3-ply, with black surfaces and white core. Engraving shall be condensed Gothic, at least 1/2" high, appropriately spaced. Nomenclature on the label shall be in accordance with UTSW standards. Nomenclature shall be as described in this section. Equipment to be labeled shall include but not be limited to the following:
 - 1. Air Handling Units
 - 2. Ventilation Fans
 - 3. Exhaust Fans
 - 4. Fan and Coil Units
 - 5. Condensing Units

Hot Water Generators

Storage Tanks

- | | | |
|----|--------------------------|-------------------------|
| 6. | Converters | Compressors |
| 7. | Air Conditioning Control | Chillers |
| 8. | Panels and Switches | Miscellaneous - similar |
| 9. | and/or related items | |

2.7 PIPING: PIPE MARKERS AND ARROW MARKERS ALSO SHALL BE PROVIDED ON BUT NOT LIMITED TO THE PIPING OF THE FOLLOWING SYSTEMS AFFECTED BY THE PROJECT:

- | | | |
|----|------------------------------|-------------------|
| A. | Primary Chilled Water Supply | Steam Condensate |
| B. | Primary Chilled Water Return | Secondary Chilled |
| C. | Water Supply | |
| D. | Atmospheric Relief | Secondary Chilled |
| E. | Water Return | |
| F. | Low Pressure Steam | Compressed Air |
| G. | Medium Pressure | |
| H. | Steam | |

2.8 MARKERS: IN ADDITION, PIPE RUNS THROUGHOUT THE BUILDING INCLUDING THOSE ABOVE LIFT OUT CEILINGS, UNDER FLOOR, AND THOSE EXPOSED TO VIEW WHEN ACCESS DOORS OR ACCESS PANELS ARE OPENED SHALL BE IDENTIFIED BY MEANS OF SETON SETMARK OR BRADY MECHANICAL PIPE MARKERS. CONCEALED AREAS, FOR PURPOSES OF THIS IDENTIFICATION SECTION, ARE THOSE AREAS WHICH CANNOT BE SEEN EXCEPT BY DEMOLITION OF THE BUILDING ELEMENTS. IN ADDITION TO THE PIPE MARKERS, ARROW MARKERS SHALL BE USED TO INDICATE DIRECTION OF FLOW. THE FOLLOWING SPECIFIC INSTRUCTIONS SHALL APPLY TO THE APPLICATION OF THESE MARKERS:

- A. Provide a pipe marker at each valve to indicate proper identification of pipe contents. Where several valves exist on one header, it is necessary to mark only the header.
- B. Provide an arrow marker with each pipe marker pointing away from the pipe marker to indicate direction of flow.
- C. Provide a double ended arrow marker when flow can be in either or both directions.
- D. Provide a pipe marker and an arrow marker at every point of pipe entry or exit where line goes through a wall or service column.
- E. Provide pipe markers and arrow markers at intervals not exceeding 50 feet.
- F. Markers shall be located on the two lower quarters of the pipe where view is unobstructed.
- G. Use Seton Setmark Type SNA or Brady snap-on type identification for all piping systems, 3/4" thru 6". For piping systems larger than 6", use Seton or Brady strap on markers.
- H. Pipe Markers shall conform to ANSI A 13.1-1981 "Scheme for the Identification of Piping Systems". Arrow markers must have same ANSI background colors as their companion pipe markers, or be incorporated into the pipe identification marker.

2.9 NOMENCLATURE

- A. The building designators for the new hospital and ancillary buildings will be:
1. UH – University Hospital
 2. US – University Hospital Support Building
 3. UW – University Hospital Power Plant
- B. The new University Hospital floors will be numbered as follows:
1. UH0 (crawl space)
 2. UH1
 3. UH2
 4. UH3
 5. UH4
 6. UHM (mechanical floor)
 7. UH5
 8. UH6
 9. UH7
 10. UH8
 11. UH9
 12. UH10
 13. UH11
 14. UH12
 15. UHP (penthouse)
- C. UTSWMC Energy Management system Naming and Numbering Specification: The purpose of this document is to define the standard for naming items that become part of the EMS system as well as numbering the BACnet networks and devices.
1. Network Numbering: The UTSWMC BACnet configuration utilizes two of the defined Local Area Network (LAN) technologies for its interconnection of devices. Ethernet is used for Management level and Integration level device communication. That is to say that front-end servers and workstations (Management level devices) talk to floor and building controllers, routers, and BACnet Broadcast Management Devices (Integration level devices). The Field level controllers such as Room Controllers, Air Handler Controllers, and other building system controllers communicate to the Integration level devices via the Master Slave/Token-Passing [MS/TP] LAN. Including both of these LAN types, the UTSWMC BACnet control system can contain up to 65,535 (64K) interconnected networks each of which are required to have a unique network number. At UTSWMC, network numbers shall be assigned as follows:
 - a. NBBSS
 - b. Where: N = Network Type
 - c. 0 = Ethernet
 - d. 1 = MS/TP

- e. 2 = P1
- f. 3 = >6 not used
- g. SS = 00 to 35 for individual I/P segments in a building
- h. BB = the building number
- i. 00 = not used
- j. 01 = DC
- k. 02 = DE, CB2
- l. 03 = DF, CB1
- m. 04 = DH
- n. 05 = BL
- o. 06 = BP
- p. 07 = NTEP
- q. 08 = V
- r. 09 = BTEP
- s. 10 = NA
- t. 11 = NB
- u. 12 = NC
- v. 13 = ND
- w. 14 = NE
- x. 15 = NF
- y. 16 = NG
- z. 17 = NL
- aa. 18 = not used
- ab. 19 = ZL
- ac. 20 = HA, SP, St. Paul
- ad. 21 = HP, POB1
- ae. 22 = HQ, POB2
- af. 23 = WA, ASC, Ambulatory Surgical Center
- ag. 24 = not used
- ah. 25 = not used
- ai. 26 = EC
- aj. 27 = RC
- ak. 28 = not used
- al. 29 = not used

- am. 30 = A
- an. 31 = B
- ao. 32 = C
- ap. 33 = D
- aq. 34 = E
- ar. 35 = F
- as. 36 = G
- at. 37 = H
- au. 38 = J
- av. 39 = K
- aw. 40 = L
- ax. 41 = M
- ay. 42 = MA
- b`. 43 = P
- ba. 44 = R
- bb. 45 = S
- bc. 46 = U
- bd. 47 = X
- be. 48 = Y
- bf. 49 = CS
- bg. 50 = PE, STEP
- bh. 51 = unused
- bi. 52 = EB
- bj. 53 = unused
- bk. 54 = unused
- bl. 55 = unused
- bm. 56 = JA
- bn. 57 = PW
- bo. For example, a BACnet Ethernet network in P building on the primary segment would be:
 - 1) NBBSS
 - 2) 04300 or simply network number 4300

2. Device Numbering: A BACnet control system can contain up to 4,194,303 devices each of which are required to have a unique value for the Object Identifier property of the Device object. At UTSWMC, device numbers shall be assigned as follows:

- a. VBBYYY

- b. Where: V = Vendor
 - c. 0 = not used
 - d. 1 = Alerton controls
 - e. 2 = Siemens controls
 - f. 3 = Trane
 - g. 4 = not used
 - h. BB = Building number as defined in Section 1
 - i. YYYY = 0000 to 9999 for devices
 - j. For example, for the 112th Siemens device in NB building the device number would be:
 - k. 2110112
 - l. Or for the 86th Alerton device in CS building the device number would be:
 - m. 1490086
3. System Naming Convention: To create a consistent naming structure across legacy, proprietary, new/remodel construction, and open protocol implementations at UTSWMC, the following sections specify conventions for naming buildings and equipment as it relates to the EMS system. Deviation from this scheme requires prior approval in writing.
- a. All buildings shall be designated by a one-letter or two-letter name. Reference construction documents for the most current designation. The current exception to this rule is Thermal Energy Plants, whose names can contain TEP.
 - b. All items not specifically identified in this document shall be named as BB.NNN...N. Where BB is the one or two letter building designation and NNN...N is the industry standard designation for that equipment type. For example S.BBMD-01 is the name for BACnet Broadcast Management Device [BBMD] number 1 in S building, and described as S.BBMD number one.
 - c. Rooms shall be designated BBF.NNN. Where BB is the one or two letter designation, F is the floor number and is always a decimal value; NNN is the actual room number as designated on the official drawings and signage. For example, NB10.403 is the name for room 403 on the 10th floor of NB building. In the event of multiple room controllers in a large area, they shall be designated as -1, -2... and their relative positions in the room shall be described in the description. For example, NB10.403-2 described as NB 10 Room 403, TEC 2 in NW corner.
 - 1) Environmental rooms, Refrigerators, and Freezers shall have that designation as part of the name. For example, CY3.333 Refrigerator 33 is described as CY 3 333 Refrigerator #33. Or, F5.222 Freezer #2 is described as F 5 222 UL Freezer #2, NB8.408 Warm Room is described as NB 8 408 Warm Room.
 - 2) In a master/slave configuration of room controllers, the master and slave designations shall be added to the description, i.e., NB10.403-02 described as NB 10 Room 403, TEC 02 slaved to TEC 01 in NW corner.
 - 3) In the event that corridors or hallways do not have a room number designated, they shall be named as follows: BBF.N Corridor. Where BB is the one or two letter building designation, F is the floor number and N is an incrementing decimal value for controllers throughout that corridor or hallway. The relative location for the controller shall be described in the description. For example, X2.3 Corridor described as X 2, Corridor TEC 3 outside room X2.100.
 - 4) ARC rooms shall be designated as BBF.NNN ARC. For example J1.116 ARC described as J 1 116 ARC Room.
4. Air Handlers shall be designated BBF. (includes floor #) AHU-NN. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, and NN is an incrementing decimal value that designates the air handler number in that particular building.

- a. Air Handler Fans shall be designated BBF.AHUNN Type Fan. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, and NN is an incrementing decimal value that designates the air handler number in that particular building, and Type is either Supply Air or Return Air. For example, NC.AHU1 Supply Air Fan shall be described as NC AHU 1 Supply Air Fan.
 - b. Air Handler Valves shall be designated BBF.AHUNN Type VLV. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, NN is an incrementing decimal value that designates the air handler number in that particular building, and Type is CD for Cold Deck, HD for Hot Deck, PC for Precool, PH for Preheat or HUM for Humidity. For example, NC.AHU1 CD VLV shall be described as NC AHU 01 Cold Deck VLV.
 - c. Air Handler Dampers shall be designated BBF.AHUNN Type Damper. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, NN is an incrementing decimal value that designates the air handler number in that particular building, and Type is OA for Outside Air, MA for Mixed Air, RA for Relief Air or EA for Exhaust Air. For example, L.AHUB2 OA Damper shall be described as L AHU B2 Outside Air Damper.
 - d. Air Handler Safeties shall be designated BBF.AHUNN Type. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, NN is an incrementing decimal value that designates the air handler number in that particular building, and Type is Freeze Stat, High Static, Low Static, Smoke, and High Duct Temp. For example, NA.AHU4 Low Static shall be described as NA AHU 4 Low Static Pressure.
 - e. Air Handler Temperatures and set points shall be designated BB.AHUNN Type. Where BB is the one or two letter building designation, F is the floor number, AHU is a designation for the air handler, NN is an incrementing decimal value that designates the air handler number in that particular building, and Type is SAT for Supply Air, CD for Cold Deck, HD for Hot Deck, PH for Preheat, MAT for Mixed Air, OAT for Outside Air Temp and RAT for Return Air. For example, NA.AHU4 SAT shall be described as NA AHU 4 Supply Air Temperature.
5. Valves shall be designated BBF.T.Function VLV. Where BB is the one or two letter building designation, F is the floor number, T is Valve type such as:
- a. RHW = Reheat Water
 - b. DHW = Domestic Hot Water
 - c. CHW = Chilled Water
 - d. HUM = Humidity
 - e. SCHW = Secondary Chilled Water
 - f. PCHW = Process Chilled Water
 - g. CW = Condenser Water
 - h. GLY = Glycol
 - i. Function indicates any special purpose such as Sup for Supply, Ret for Return, BP for Bypass, ISO for Isolation. VLV is the designation for Valve. For example, NE.CHW.BPVLV shall be described as NE CHW Bypass VLV. Or, NL.HX1.1/3VLV shall be described as NL HX1 1/3 VLV.
6. Exhaust Fans shall be designated BBF.Type EFNN. Where BB is the one or two letter building designation, F is the floor number, Type is EF will be used for general purpose building exhaust, L for Lab, A for Animal, MRI for MRI. EF is a designation for the exhaust fan, NN is an incrementing decimal value that designates the exhaust fan number in that particular building. For example, NA.AEF13 shall be described as NA Animal EF 13. NE.MRIEF17 shall be described as NE MRI EF 17.

7. Fire Status shall be designated BB.Fire Type. Where BB is the one or two letter building designation, Type is either Fire Trouble or Fire Alarm or Fire Supervisor.
8. Fire/Smoke Dampers shall be designated BBF.FSD NN. Where BB is the one or two letter building designation, F is the floor number and is always a decimal value, FSD is a designation for the fire smoke damper, NN is an incrementing decimal value that designates the smoke damper number in that particular building.
9. Chillers shall be designated BB.CHLRNN. Where BB is the one or two letter building designation, CHLR is a designation for the chiller, NN is an incrementing decimal value that designates the chiller number in that particular building. For example, NTEP.CHLR5 shall be described as NTEP Chiller #5.
 - a. Associated chiller information points will be prefaced with the chiller designation in the name and followed by the information type. For example, PE.CHLR2 TONS shall be described as PE Chiller #2 Tons, or PE-CHLR-4 GPM shall be described as PE Chiller #4 Gallons per Minute.
10. Cooling Towers shall be designated BB.CTNN. Where BB is the one or two letter building designation, CT is a designation for the cooling tower, NN is an incrementing decimal value that designates the cooling tower number in that particular building. For example, NTEP.CT2 shall be described as NTEP Cooling Tower #2.
 - a. Associated cooling tower information points will be prefaced with the cooling tower designation in the name and followed by the information type. For example, PE.CT2 CW Flow shall be described as PE Cooling Tower #2 Condenser Water Flow, or PE.CT4 SS shall be described as PE Cooling Tower #4 Start/Stop.
11. Boilers shall be designated BB.BLRNN. Where BB is the one or two letter building designation, BLR is a designation for the boiler, NN is an incrementing decimal value that designates the boiler number in that particular building. For example, NTEP.BLR1 shall be described as NTEP Boiler #1.
 - a. Associated boiler information points will be prefaced with the boiler designation in the name and followed by the information type. For example, PE.BLR2 ECON OUT WATER TEMP shall be described as PE Boiler #2 Econ Out Water Temp, or NTEP.BLR4 GAS TEMP shall be described as PE Boiler #4 Gas Temperature.
12. All other items shall be described as BB. (ITEM DESCRIPTION). Where BB is the one or two letter building designation, (ITEM DESCRIPTION) is the complete name/description of the item. For example J.Control Air HI Press shall be described as J Control Air High Pressure, or NB.LAB VAC PMP STATUS shall be described as NB Lab Vacuum Pump Status.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Degrease and clean surfaces to receive adhesive for identification materials.
- B. Prepare surfaces in accordance with Division 9 for stencil painting.

3.2 INSTALLATION

- A. Install plastic nameplates with corrosive-resistant mechanical fasteners, or adhesive. Apply with sufficient adhesive to ensure permanent adhesion and seal with clear lacquer.
- B. Apply stencil painting in accordance with Division 9.
- C. Install plastic pipe markers in accordance with manufacturer's instructions.
- D. Install plastic tape pipe markers complete around pipe in accordance with manufacturer's instructions.

- E. Install underground plastic pipe markers 6 to 8 inches (150 to 200 mm) below finished grade, directly above buried pipe.
- F. Identify air handling units, pumps, heat transfer equipment, tanks, and water treatment devices with plastic nameplates. Small devices, such as in-line pumps, may be identified with tags.
- G. Identify control panels and major control components outside panels with plastic nameplates.

END OF SECTION

SECTION 23 05 93

TESTING, ADJUSTING AND BALANCING

PART 1 - GENERAL

1.1 SUMMARY

- A. Perform all work required to prepare the building HVAC systems for testing, adjusting and balancing indicated by the Contract Documents as follows:
 - 1. Responsibilities of project contractor.
 - 2. Preparation for balancing of air systems.
 - 3. Preparation for balancing of hydronic and steam systems.
- B. The scope of the TAB work as defined in Section 230593 is indicated in order that the Contractor will be advised of the coordination, adjustment, and system modification which will be required under the project work in order to complete the Owner's requirements for final TAB. The TAB firm will not have a contractual relationship with any Contractor referred to herein, but will be responsible to the Construction Inspector and the Owner for the satisfactory execution of the TAB work. The Contractor in his original bid shall allow for the costs required to cover all work which may be required in the TAB phases as defined herein and as may be necessary for the completion of the TAB work as defined by the TAB firm.

1.2 RELATED SECTIONS

- A. Section 22 0050 - Common Work Results for HVAC.
- B. Section 22 0548 - Vibration Isolation.
- C. Section 23 2115 - Hydronic Specialties.
- D. Section 23 3413 - Axial Fans.
- E. Section 23 8223 - Power Ventilators.
- F. Section 23 3113 - Ductwork.
- G. Section 23 3300 - Ductwork Accessories.
- H. Section 23 3600 - Air Terminal Units.
- I. Section 23 3315 - Air Outlets and Inlets.
- J. Section 23 0925 - Building Automation and DDC Control System.
- K. Section 23 0594 - System Testing, Adjusting and Balancing.

1.3 SCOPE OF WORK

- A. Testing, adjusting, and balancing (TAB) of the air conditioning systems and related ancillary equipment will be performed by an impartial technically qualified TAB firm selected and employed directly by the Owner, separate and apart from the Construction Contract. However, the preparation for and corrections

necessary for the Testing, Adjusting and Balancing of these systems, as described herein, are the responsibility of the Contractor.

- B. As a part of this project Construction Contract, the Contractor shall make any changes or replacements to the sheaves, belts, dampers, valves, etc. required for correct balance as advised by the TAB firm, at no additional cost to the Owner.
- C. The Contractor shall provide and coordinate the services of qualified, responsible Subcontractors, suppliers and personnel as required to correct, repair, and/or replace any and all deficient items or conditions found during the course of this project, including the testing, adjusting and balancing period.
- D. In order that all systems may be properly tested, balanced, and adjusted as required herein by these Specifications, the Contractor shall operate said systems at his expense for the length of time necessary to properly verify their completion and readiness for TAB. This length of time shall be subject to the approval of the Owner's Representative.
- E. Project Contract completion schedules shall allow for sufficient time to permit the completion of TAB services prior to Owner occupancy. The contractor shall allow adequate time for the testing and balancing activities of the owner provided services, during the construction period, and prior to Substantial Completion as defined in the Uniform General Conditions of this Construction Document.
- F. The Drawings and Specifications indicate valves, dampers and miscellaneous adjustment devices for the purpose of adjustment to obtain optimum operating conditions, and it will be the responsibility of the Contractor to install these devices in a manner that will leave them accessible and readily adjustable. Should any such device not be readily accessible, the Contractor shall provide access as requested by the TAB firm. Also, any malfunction encountered by TAB personnel and reported to the Contractor or the Owner's Representative shall be corrected by the Contractor immediately so that the balancing work can proceed with the minimum of delays.

1.4 RESPONSIBILITIES OF THE PROJECT CONTRACTOR

- A. The Contractor shall:
 - 1. Have the building and air conditioning systems in complete operational readiness for TAB work to begin.
 - 2. The contractor shall allow sufficient time for the TAB firm to perform his contracted work within the construction schedule. The contractor shall complete his work by systems or floors whichever is the most efficient for scheduling. After awarding of the contract and the contractor has developed a construction schedule, a TAB coordination meeting shall be held with the TAB firm, the general contractor and his primary subcontractors (i.e. mechanical, electrical, building automation etc.) to develop a testing schedule for the project. The contractor shall submit copies of the proposed schedule two (2) weeks prior to this meeting to the Owner and TAB firm.
 - 3. Promptly correct deficiencies of materials and workmanship identified as delaying completion of TAB work.
 - 4. The Contractor shall be responsible for any added costs to the Owner resulting from failure to have the building and air conditioning systems ready for TAB when scheduled, or from failure to correct deficiencies promptly.
- B. Complete operational readiness of the building requires that construction status of the building shall permit the closing of doors, windows, ceilings installed, etc., to obtain simulated or projected operating conditions.
- C. Complete operational readiness of the air conditioning systems also requires that the following be accomplished:

1. Air Distribution Systems:
 - a. Verify installation for conformity to design. All supply, return and exhaust ducts terminated and pressure tested for leakage as required by the Specification.
 - b. All volume, smoke and fire/smoke dampers are properly located and functional. Dampers serving requirements of minimum and maximum outside, return and relief air shall provide tight closure and full opening, smooth and free operation.
 - c. All supply, return, exhaust and transfer grilles, registers, diffusers and terminal devices installed.
 - d. Air handling systems, units and associated apparatus, such as heating and cooling coils, filter sections, access doors, etc., shall be blanked and/or sealed to eliminate excessive bypass or leakage of air.
 - e. All fans (supply, return and exhaust) operating and verified for freedom from vibration, proper fan rotation and belt tension; heater elements in motor starters to be of proper size and rating; record motor amperage and voltage on each phase at start-up and running, and verify they do not exceed nameplate ratings.
 - f. All single and/or double duct variable and constant volume terminal units ("mixing boxes") shall be installed and functional (i.e. controls functioning).
2. Water Circulating Systems:
 - a. Check and verify pump alignment and rotation.
 - b. Open all valves to their full open position, close bypass stop valves. Set mixing valves to full-flow through systems components. After the system is flushed and checked for proper operation, remove and clean all strainers. The Contractor shall repeat the operation until circulating water is clean.
 - c. Record each pump motor amperage on each phase and voltage after reaching rated speed. Readings shall not exceed nameplate rating.
 - d. Verify that the electrical heater elements are of the proper size and rating.
 - e. In preparation of TAB all water circulating systems shall be full and free of air, expansion tanks shall be set for proper water level, and all air vents shall be installed at high points of systems and operating freely. Systems shall be cleaned and flushed. Chemicals shall be added to closed systems to treat piping and inhibit corrosion.
 - f. Check and set operating parameters of the heat exchangers and control devices to the design requirements.
3. Automatic Controls:
 - a. The Contractor shall schedule a meeting with the Engineer, Control Contractor, TAB firm and Owner's representative for a pre-submittal review to establish that his interpretation of the sequences of operation are correct.
 - b. Verify that all control components are installed in accordance with project requirements and are functional, including all electrical interlocks, dampers sequences, air and water resets, fire and freeze stats, high and low temperature thermostats, safeties, etc.
 - c. Verify that all controlling instruments are calibrated and set for design operating conditions with the exception of room thermostats or sensors, which shall be calibrated at the completion of TAB services with cooperation between the TAB firm and Control Contractor.
 - d. The Automatic Temperature Control Contractor and/or Energy Management System Contractor shall thoroughly check all controls, sensors, operators, sequences, etc. before

notifying the TAB agency that the Automatic Temperature Controls and Energy Management System are operational. The Automatic Temperature Contractor and/or Energy Management System Contractor shall provide technical support (technicians and necessary computers) to the TAB firm for a complete check of these systems.

4. Tabulated Data: The motor amperages, voltages shall be recorded showing "actual" and "nameplate" voltage and amperage and submitted and actual RPM. This applies to each piece of electrically driven air conditioning equipment in the system including supply and exhaust fans, fans of fractional horsepower, pumps, etc. Include any additional relevant start-up information or documentation.

D. Notification of System Readiness:

1. After completion of the work in Paragraph above, the Contractor shall notify the Owner in writing, certifying that the work has been accomplished and that the building and the air conditioning systems are in operational readiness for testing, adjusting, and balancing. He shall include a copy of the tabulated data described above.
2. The Owner will, in turn, notify the TAB firm of the readiness for balancing and forward copies of the Contractor's certification.
3. Should the TAB firm be notified as described above, and the TAB work commenced and the systems are found NOT to be in readiness or a dispute occurs as to the readiness of the systems, the Contractor shall request an inspection be made by duly appointed representative of the Owner, Architect, TAB firm and the Contractor. This inspection will establish to the satisfaction of the represented parties whether or not the systems meet the basic requirements for TAB services. Should the inspection reveal the TAB services notification to have been premature, all cost of the inspection and wasted work accomplished by the TAB firm shall be reimbursed to the appropriated parties by the Project Contractor.

1.5 RESPONSIBILITIES OF THE TAB FIRM

- A. Refer to Section 23 0594 entitled "System Testing, Adjusting and Balancing".

PART 2 - PRODUCTS

2.1 NOT USED

PART 3 - EXECUTION

3.1 NOT USED

END OF SECTION

SECTION 23 05 94

SYSTEM TESTING, ADJUSTING AND BALANCING

PART 1 - GENERAL

1.1 SUMMARY

- A. Testing, adjusting and balancing (TAB) of the air conditioning systems and related ancillary equipment will be performed by an impartial technically qualified TAB firm selected and employed by the Owner, separate and apart from the construction contract.
- B. The firm shall be capable of performing the services specified at the location of the facility described within the time specified, of preparing and submitting the detailed report of the actual field work performed, and following up the basic work as may be required.

1.2 QUALIFICATIONS

- A. The Firm shall be one which is organized to provide professional services of this specified type in the State of Texas and as a minimum shall have one (1) professional engineer licensed in the State of Texas, with current registration, to perform such professional services. This engineer shall be personally responsible for developing the job site data as required in the test procedures outlined in these Specifications.
- B. The Firm shall have operated a minimum of five (5) years under its current Firm name, and shall be in good standing with the State of Texas, Franchise Tax Board. The firm shall submit their full incorporated name, Charter Number and Taxpayer's I.D. Number for proper verification of the firm's status.
- C. The Firm shall be capable of providing a performance bond, by a bonding company licensed to do business in the State of Texas, if determined by the Owner that such a bond is required. The amount of the bond which may be required shall be equal to the cost of the proposal submitted, or in the case of more than one proposal, the sum of all such proposals and any awarded work in progress.
- D. The Firm shall maintain current insurance coverages in the minimum amounts indicated below. If the Firm normally carries such insurance coverages (minimum or higher) incident to its operation, additional insurance for the specific proposal or proposals is not required. The minimum insurance coverages required are:
 - 1. Worker's Compensation as required by law.
 - 2. General Liability for not less than \$2,000,000 aggregate.
 - 3. Fire Damage, and Extended Coverage, Vandalism and Malicious Mischief, in the full amount of Contract. The above policies shall be carried with companies satisfactory to the Owner. Certificates of each of the above policies, together with a written statement by the issuing company, stating that said policy will not be canceled without ten (10) days prior written notice to the Board of Regents of the University of Texas system, shall be delivered to the Owner before any work is started.
- E. All personnel used on the job site shall be either professional engineers or engineering technicians, who shall have been permanent, full time employees of the firm for a minimum of six (6) months prior to the start of work for this specific project.
- F. The TAB firm shall submit biographical data on the individual proposed to directly supervise the TAB work, as well as other personnel scheduled to perform the technical work under the contract. It shall also submit a background record of at least five years of specialized experience in the field of air hydronic system

balancing, and shall possess properly calibrated instrumentation. The supervisory personnel for the TAB firm shall be registered engineers in the mechanical field and all of the employees used in the TAB firm shall be permanent, full-time employees of the firm.

1.3 REFERENCES

- A. AABC - National Standards for Total System Balance, sixth edition, 2002.
- B. ASHRAE - HVAC Applications Chapter 37: Testing, Adjusting and Balancing.
- C. ANSI/ASHRAE Standard 111-1988 - Practices for Measurement, Testing, Adjusting and Balancing of Buildings, Heating, Ventilation, Air Conditioning and Refrigeration Systems.

1.4 DOCUMENTS

- A. The Owner or Owner's Representative shall arrange with the Architect to provide one set of mechanical specifications, all pertinent change orders, and the following:
 - 1. One complete set of Drawings less the structural sheets.
 - 2. One set of mechanical floor plans of the conditioned spaces. These Drawings shall be ozalid type (blue or black on light background) reproductions to facilitate marking.
- B. Approved submittal data on equipment installed, and related changes as required to accomplish the test procedures outlined in Paragraphs 1.06 through 1.10 of this Specification will be available through the Owner's Representative.

1.5 RESPONSIBILITIES OF THE TAB FIRM

- A. The TAB personnel shall check, adjust, and balance the components of the air conditioning system which will result in optimal noise, temperature, and airflow conditions in the conditioned spaces of the building while the equipment of the system is operating economically. This is intended to be accomplished after the system components are installed and operating as provided for in the contract documents. It is the responsibility of the Mechanical Contractor to place the equipment into service. Variable air volume systems shall be balanced in accordance with AABC Standard 2002, Sixth Edition.
- B. Liaison and Early Inspection:
 - 1. The TAB firm personnel on the job shall act as liaison between the Owner, Architect and Contractor. The following reviews (observations) and tests shall be performed by the TAB Firm:
 - a. During the early design stages of the project, review the mechanical drawings and specifications for balanceability and provide commentary.
 - b. During construction, review all HVAC submittals such as control diagrams, air handling devices, etc., that pertain to commissioning work and balanceability.
 - c. Allow for a fixed number of trips to the project site, over and above those required for testing and balancing for inspection of installation of the mechanical piping systems, sheet metal work, temperature controls and other component parts of the heating, air conditioning and ventilating systems during the construction stage. These inspections shall be made prior to and/or at the above ceiling inspection. Commentary will be provided to the Owner's Representative of each observation.
 - d. Test one (1) 8" single duct terminal box for performance capability and leakage as described in Section 15930. The shipment of the box to the TAB Firm's lab will be at the manufacturer's cost and the test period will be for three (3) weeks from receipt of the box. Submittal data will not be approved until box testing passes. If the sample box is rejected for any reason the

second test will be at the Contractor's cost and the time allowed will restart when the box is received at the TAB Firm.

- e. Test one (1) 8" dual duct box for performance capability and leakage as described in Section 15930. The shipment of the box to the TAB Firm's lab will be at the manufacturer's cost and the test period will be for three (3) weeks from receipt of the box. Submittal data will not be approved until box testing passes. If the sample box is rejected for any reason the second test will be at the Contractor's cost and the time allowed will restart when the box is received by the TAB firm.
 - f. Test 10% of the single and dual duct boxes for casing and damper leakage when the shipment arrives at the project site. All testing (except for the initial boxes) shall be performed on site. Boxes requiring re-testing will be charged to the Contractor at the unit price provided to the Owner.
 - g. Test one (1) lab configuration including fume hood with air valve, general exhaust air with air valve and supply air with air valve for performance capability through a full range of inlet pressures. The tracking capability of the exhaust air versus the supply air will be with the submitted hood sash fully open and as the sash is closed in 2" increments until fully closed. Track the three (3) valve's response time in relation to sash movement and the lab differential.
 - h. Witness Air Handling Unit leakage and deflection testing as specified in other sections. Typical for Air Handling Units 10,000 cfm and larger. Provide test results to UTSW representative.
 - i. Witness Fan vibration testing as specified in other sections. Provide test results to UTSW representative.
 - j. Attend Commissioning meetings, as required, to support OFPC during all HVAC Commissioning phases.
2. During the balancing process, as abnormalities and malfunctions of equipment or components are discovered by the TAB personnel, the Owner's Representative shall be advised in writing so that the condition can be corrected by the Contractor. The written document need not be formal, but must be understandable and legible. Data from malfunctioning equipment shall not be recorded in the final TAB report. The TAB firm shall not instruct or direct the Contractor in any of the work, but will make such reports as are necessary to the Owner.

1.6 FINAL AIR BALANCE

- A. General: When systems are complete and ready for operation, the TAB firm will perform a final air balance for all air systems and record the results. The outside, supply, exhaust and return air volume for each air handling unit, supply fan and exhaust fan and the supply, exhaust or return air volume for each distribution device shall be adjusted to within +5% of the value shown on the drawings. Air handling unit and fan volumes shall be adjusted by changing fan speed and adjusting volume dampers associated with the unit. Air distribution device volume shall be adjusted using the spin-in tap damper for flexible duct connected devices and the device OBD for duct connected devices. Air distribution devices shall be balanced with air patterns as specified. Duct volume dampers shall be adjusted to provide air volume to branch ducts where such dampers are shown. The general scope of balancing by the TAB Consultant will include, but is not limited to, the following:
- 1. Filters: Check air filters and filter media and balance only system with essentially clean filters and filter media. The Division 15 Contractor shall install new filters and filter media prior to the final air balance.
 - 2. Blower Speed: Measure RPM at each fan or blower to design requirements. Where a speed adjustment is required, the Division 15 Contractor shall make any required changes.
 - 3. Ampere Readings: Measure and record full load amperes for motors.

4. Static Pressure: Static pressure gains or losses shall be measured across each supply fan, cooling coil, heating coil, return air fan, air handling unit filter and exhaust fan. These readings shall be measured and recorded for this report at the furthest air device or terminal unit from the air handler supplying that device. Static pressure readings shall also be provided for systems which do not perform as designed.
5. Equipment Air Flow: Adjust and record exhaust, return, outside and supply air CFM (s) and temperatures, as applicable, at each fan, blower and coil.
6. Coil Temperatures: Set controls for full cooling and for full heating loads. Read and record entering and leaving dry bulb and wet bulb temperatures (cooling only) at each cooling coil, heating coil and HVAC terminal unit. At the time of reading record water flow and entering and leaving water temperatures (In variable flow systems adjust the water flow to design for all the above readings).
7. Zone Air Flow: Adjust each zone of multizone units, each HVAC terminal unit and air handling unit for design CFM.
8. Outlet Air Flow: Adjust each exhaust inlet and supply diffuser, register and grille to within +5% of design air CFM. Include all terminal points of air supply and all points of exhaust. Note: For Labs and Rooms that are negative exhaust air flow shall be set to design +10% and supply to design -5%. Positive areas will have opposite tolerances.
9. Pitot Tube Traverses: For use in future troubleshooting by maintenance personnel, all exhaust ducts, main supply ducts and return ducts shall have air velocity and volume measured and recorded by the traverse method. Locations of these traverse test stations shall be described on the sheet containing the data.
10. Maximum and minimum air flow on terminal boxes.

1.7 FINAL CHILLED WATER BALANCE

- A. General: When systems are completed and ready for operation, the TAB Consultant will perform a final water balance for each chilled and hot water system. The general scope of balancing by the TAB Consultant will include, but not be limited to, the following:
 1. Adjusted System Tests: Adjust balancing valves at each coil for design flow, +5%. Adjust balancing valves at pumps to obtain design water flow. Record pressure rise across pumps and GPM flow from pump curve. Permanently mark the balanced position for each valve (Note: If discharge valves on the pumps are used for balancing record the head being restricted by the valves).
 2. Temperature Readings: Read and record entering and leaving water temperature at each water coil,. Adjust as necessary to secure design and conditions. Provide final readings at all thermometer well locations.
 3. Pressure Readings: Water pressure shall be recorded at all gauge connections. Pressure readings at coils and pumps shall be related to coil and pump curves in terms of GPM flow through flow measuring status, if provided and installed, at each air handler. The flow of water through all water coils shall be adjusted by manipulating valves until the rated pressure drops across each coil is obtained and total water flow is verified by flow measuring status. For coils equipped with 3 way valves, the rated pressure drop shall first be adjusted through the coils. The bypass valve shall then be adjusted on each coil until an equal pressure drop between supply and return connections is the same as with the flow through the coil.
 4. Ampere Readings: Reading and record full load amperes for each pump motor.

1.8 SOUND VIBRATION AND ALIGNMENT

- A. Sound: Read and record sound levels at up to 15 locations in the building designated by the Engineer. All measurements shall be made using an Octave Band Analyzer. All tests shall be conducted when the building is quiet in the presence of the Engineer, if he so desires.
- B. Vibration: Read and record vibration for all water circulating pumps which have motors larger than 10 HP. Include equipment vibration, bearing housing vibration, foundation vibration, building structure vibration, and other tests as directed by the Engineer. Readings will be made using portable IRD (or approved equal) equipment capable of filtering out various unwanted frequencies and standard reporting forms. Maximum vibration at any point listed above, or specified, shall not exceed 1 mil on fans and 1 mil on pumps unless otherwise specified. Equipment manufacturers shall rectify all systems exceeding vibration tolerances.

1.9 TESTING OF TEMPERATURE CONTROL SYSTEMS

- A. In the process of performing the TAB work, the TAB Firm shall:
 - 1. Work with the temperature control contractor to ensure the most effective total system operation within the design limitations, and to obtain mutual understanding of intended control performance.
 - 2. Verify that all control devices are properly connected.
 - 3. Verify that all dampers, valves and other controlled devices are operated by the intended controller.
 - 4. Verify that all dampers and valves are in the position indicated by the controller (open, closed or modulating).
 - 5. Verify the integrity of valves and dampers in terms of tightness of close-off and full-open positions. This includes dampers in multizone units, terminal boxes and fire/smoke dampers.
 - 6. Observe that all valves are properly installed in the piping system in relation to direction of flow and location.
 - 7. Observe the calibration of all controllers.
 - 8. Verify the proper application of all normally opened and normally closed valves.
 - 9. Observe the locations of all space thermostats and humidistats for potential erratic operation from outside influences such as sunlight, drafts or cold walls.
 - 10. Observe the locations of all sensors to determine whether their position will allow them to sense only the intended temperatures or pressures of the media. Control Contractor will relocate as deemed necessary by the TAB Firm.
 - 11. Verify that the sequence of operation for any control mode is in accordance with approved shop drawings and specifications. Verify that no simultaneous heating and cooling occurs.
 - 12. Verify that all controller setpoints meet the design intent.
 - 13. Check all dampers for free travel.
 - 14. Verify the operation of all interlock systems.
 - 15. Perform variable volume system verification to assure the system and its components track with changes from full flow to minimum flow.
- B. A systematic listing of the above testing and verification shall be included in the final TAB report. Each system will be tested in accordance with written control sequence verification procedures. The written control sequence verification will document the performance of the specified control sequence and the control manufacturers as built drawings. The written control sequence verification will identify each

components sequence, safety devices and alarms. The written control sequence verification will test multiple components in a manifold system. The written control sequence verification will test the systems in normal power and emergency power.

- C. List all the control points of each system. Verify back to the front end graphics that the point is calibrated and the graphics indicate a change in value with the correct point name. Record all the information on a point verification log.

1.10 REPORTS

- A. The activities described in this section shall culminate in a report to be provided in triplicate (3) individually bound and an electronic report formatted in the same manner as the bound report on a CD to the OFPC representative. Neatly type and arrange data. Include with the data the date tested, personnel present, weather conditions, nameplate record of test instrument and list all measurements taken after all corrections are made to the system. Record all failures and corrective action taken to remedy incorrect situation. The intent of the final report is to provide a reference of actual operating conditions for the Owner's operations personnel. The report will be organized in the following manner:
 - 1. Title Sheet
 - 2. Table of Contents (Per system)
 - 3. AHU Data (Numerical sequence)
 - a. Air Distribution Data, Traverse Data, All Supporting Data, etc.
 - 4. Fan Data with supporting data
 - 5. Control Verification (Sequences, Sensor Calibration, Point to Point, Graphics, etc.)
 - 6. Appendix (Miscellaneous supporting documentation relative to systems performance Fan and Pump Curves, Control Schematics, Correspondence, etc.)
- B. All measurements and recorded readings (of air, water, electricity, etc.) that appear in the reports must have been made onsite by the permanently employed technicians or engineers of the firm.
- C. At the option of the Owner's Representative, all data sheets tabulated each day by TAB personnel shall be submitted for initial by the Owner's Representative. Those work sheets so initialed, or copies thereof, shall be presented as a supplement to the final TAB report.
- D. Submit reports on hard copy and electronic forms approved by the Owner & Engineer which will include the following information as a minimum:
 - 1. Title Page
 - a. Company Name
 - b. Company Address
 - c. Company telephone number
 - d. Project name
 - e. Project location
 - f. Project Manager
 - g. Project Engineer
 - h. Project Contractor

- i. Project Identification Number
- 2. Instrument List
 - a. Instrument
 - b. Manufacturer
 - c. Model
 - d. Serial Number
 - e. Range
 - f. Calibration date
 - g. What test instrument was used for
- 3. Fan Data (Supply and Exhaust)
 - a. Location
 - b. Manufacturer
 - c. Model
 - d. Air flow, specified and actual
 - e. Total static pressure (total external), specified and actual
 - f. Inlet pressure
 - g. Discharge pressure
 - h. Fan RPM
- 4. Return Air/Outside Air Data (If fans are used, same data as for 3 above)
 - a. Identification/location
 - b. Design return air flow
 - c. Actual return air flow
 - d. Design outside air flow
 - e. Return air temperature
 - f. Outside air temperature
 - g. Required mixed air temperature
 - h. Actual mixed air temperature
- 5. Electric Motors
 - a. Manufacturer
 - b. HP/BHP
 - c. Phase, voltage, amperage, nameplate, actual
 - d. RPM
 - e. Service factor

- f. Starter size, heater elements, rating
- 6. V-Belt Drive
 - a. Identification/location
 - b. Required driven RPM
 - c. Driven sheave, diameter and RPM
 - d. Belt, size and quantity
 - e. Motor sheave, diameter and RPM
 - f. Center-to-center distance, maximum, minimum and actual
- 7. Duct Traverse
 - a. System zone/branch
 - b. Duct size
 - c. Area
 - d. Design velocity
 - e. Design air flow
 - f. Test velocity
 - g. Test air flow
 - h. Duct static pressure
 - i. Air temperature
 - j. Air correction factor
- 8. Air Monitoring Station Data
 - a. Identification/location
 - b. System
 - c. Size
 - d. Area
 - e. Design velocity
 - f. Design air flow
 - g. Test velocity
 - h. Test air flow
- 9. Air Distribution Test Sheet
 - a. Air terminal number
 - b. Room number/location
 - c. Terminal type
 - d. Terminal size

- e. Area factor
 - f. Design velocity
 - g. Design air flow
 - h. Test (final) velocity
 - i. Test (final) air flow
10. Pump Data
- a. Identification/number
 - b. Manufacturer
 - c. Size/model
 - d. Impeller
 - e. Service
 - f. Design flow rate, pressure drop, BHP
 - g. Actual flow rate, pressure drop, BHP
 - h. Discharge pressure
 - i. Suction pressure
 - j. Total operating head pressure
 - k. Shut off, discharge and suction pressure
 - l. Shut off, total head pressure
 - m. Pressure differential settings
11. Cooling Coil Data
- a. Identification/number
 - b. Location
 - c. Service
 - d. Manufacturer
 - e. Entering air DB temperature, design and actual
 - f. Entering air WB temperature, design and actual
 - g. Leaving air DB temperature, design and actual
 - h. Leaving air WB temperature, design and actual
 - i. Water pressure flow, design and actual
 - j. Water pressure drop, design and actual
 - k. Entering water temperature, design and actual
 - l. Leaving water temperature, design and actual
 - m. Air pressure drop, design and actual

12. Heating Coil Data
 - a. Identification/number
 - b. Location
 - c. Service
 - d. Manufacturer
 - e. Air flow, design and actual
 - f. Entering water or steam temperature, design and actual
 - g. Entering air temperature, design and actual
 - h. Leaving air temperature, design and actual
 - i. Air pressure drop, design and actual
13. Sound Level Report
 - a. Location (Location established by the design engineer)
 - b. NC curve for eight (8) bands - equipment off
 - c. NC curve for eight (8) bands - equipment on
14. Vibration Test on equipment having 10 HP motors or above
 - a. Location of points:
 - 1) Pump bearing, drive end
 - 2) Pump bearing, opposite end
 - 3) Motor bearing, center (if applicable)
 - 4) Motor bearing, drive end
 - 5) Motor bearing, opposite end
 - 6) Casing (bottom or top)
 - 7) Casing (side)
 - 8) Pipe after flexible connection (discharge), if applicable
 - 9) Pipe after flexible connection (suction), if applicable
 - b. Test readings:
 - 1) Horizontal, velocity and displacement
 - 2) Vertical, velocity and displacement
 - 3) Axial, velocity and displacement
 - c. Normally acceptable readings, velocity and acceleration
 - d. Unusual conditions at time of test
 - e. Vibration source (if non-complying)
15. Control verification indicating date performed and any abnormalities identified.
 - a. Point Location/Description
 - b. EMS Readout (Setpoint and Actual)
 - c. Actual Readout
 - d. Interlocks

- e. Safeties
 - 1) VSD Normal Operation
 - 2) VSD Bypass Operation
- f. Alarms
- g. Sequences of Operation

PART 2 – PRODUCTS

NOT USED

PART 3 – EXECUTION

NOT USED

END OF SECTION

SECTION 23 07 01

PIPING INSULATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Piping insulation.
- B. Jackets and accessories.

1.2 PRODUCTS FURNISHED BUT NOT INSTALLED UNDER THIS SECTION

- A. Section 21 3113 - Fire Protection Systems: Placement of hangers and hanger inserts.
- B. Section 22 1116 – Plumbing Piping: Placement of hangers and hanger inserts.
- C. Section 23 2113 - Hydronic Piping: Placement of hangers and hanger inserts.
- D. Section 23 2213 - Steam and Steam Condensate Piping: Placement of hangers and hanger inserts.

1.3 RELATED SECTIONS

- A. Section 22 0010 – Basic Mechanical Requirements.
- B. Section 23 0553 - Mechanical Identification.

1.4 REFERENCES

- A. ASTM B209 - Aluminum and Aluminum-Alloy Sheet and Plate.
- B. ASTM C177 - Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded- Hot-Plate Apparatus.
- C. ASTM C195 - Mineral Fiber Thermal Insulation Cement.
- D. ASTM C335 - Steady-State Heat Transfer Properties of Horizontal Pipe Insulation.
- E. ASTM C449 - Mineral Fiber Hydraulic-setting Thermal
- F. ASTM C518 - Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
- G. ASTM C533 - Calcium Silicate Block and Pipe Thermal Insulation.
- H. ASTM C534 - Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form.
- I. ASTM C547 - Mineral Fiber Preformed Pipe Insulation.
- J. ASTM C552 - Cellular Glass Block and Pipe Thermal Insulation.

- K. ASTM C578 - Preformed, Block Type Cellular Polystyrene Thermal Insulation.
- L. ASTM C585 - Inner and Outer Diameters of Rigid Thermal Insulation for Nominal Sizes of Pipe and Tubing (NPS System).
- M. ASTM C591 - Rigid Preformed Cellular Urethane Thermal Insulation.
- N. ASTM C610 - Expanded Perlite Block and Pipe Thermal Insulation.
- O. ASTM C640 - Corkboard and Cork Pipe Thermal Insulation.
- P. ASTM C921 - Properties of Jacketing Materials for Thermal Insulation.
- Q. ASTM D1056 - Flexible Cellular Materials - Sponge or Expanded Rubber.
- R. ASTM D1667 - Flexible Cellular Materials - Vinyl Chloride Polymers and Copolymers (Closed Cell Foam).
- S. ASTM D2842 - Water Absorption of Rigid Cellular Plastics.
- T. ASTM E84 - Surface Burning Characteristics of Building Materials.
- U. ASTM E96 - Water Vapor Transmission of Materials.
- V. NFPA 255 - Surface Burning Characteristics of Building Materials.
- W. UL 723 - Surface Burning Characteristics of Building Materials.

1.5 SUBMITTALS

- A. Submit under provisions of Section 22 0010.
- B. Product Data: Provide product description, list of materials and thickness for each service, and locations.
- C. Samples: Submit two samples of any representative size illustrating each insulation type.
- D. Manufacturer's Installation Instructions: Indicate procedures, which ensure acceptable workmanship and installation standards will be achieved.

1.6 QUALITY ASSURANCE

- A. Materials: Flame spread/smoke developed rating of 25/50 or less in accordance with NFPA 255.
- B. Manufacturer of product may be requested by Owner to inspect the installation.

1.7 QUALIFICATIONS

- A. Applicator: Company specializing in performing the work of this section with minimum three years' experience.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect, and handle products to site under provisions of Section 22 0010.

- B. Deliver materials to site in original factory packaging, labeled with manufacturer's identification, including product density and thickness.
- C. Store insulation in original wrapping and protect from weather and construction traffic.
- D. Protect insulation against dirt, water, chemical, and mechanical damage.

1.9 ENVIRONMENTAL REQUIREMENTS

- A. Maintain ambient temperatures and conditions required by manufacturers of adhesives, mastics, and insulation cements.
- B. Maintain temperature during and after installation for minimum period of 24 hours.
- C. All insulation materials to be asbestos free.

PART 2 - PRODUCTS

2.1 STEAM AND CONDENSATE PIPING - BUILDINGS

- A. Contractor shall install dimensional, Pittsburgh Corning high temperature foamglass, on steam and condensate piping. Prior to application of any insulation, all metal surfaces shall be thoroughly cleaned. Insulation shall be applied to the piping with butt joints staggered and all joints tightly butted. The insulation shall be held in place using stainless steel bands. The bands shall be spaced at 12 inches on center. A jacket of aluminum or stainless steel shall be applied to the completed installation, 8'-0" above finished floor. 15 mil PVC jacket shall be applied to pipes 8'-0" above finished floor. The Longitudinal joint of the jacketing shall be placed with overlap directed to bottom of pipe. The jacketing shall be overlapped a minimum of 3 inches, and it shall be held in place using 3/4 inch bands applied at 12 inches on center. Note that the use of asphaltic materials on higher-than-ambient service is specifically prohibited.
- B. Fiberglass (rigid) insulation on flanges, valves and other fitting shall consist of prefabricated fitting covers of the same thickness as specified for adjoining pipe insulation. Use Insultech blanket on all team valves and flanges 4" and larger.
- C. Fitting insulation shall be applied in same manner as pipe application. Fitting cover jacketing shall be equal to Gasco, Pabco or RPR Metals prefabricated fitting covers of 0.016" paper coated aluminum, secured as recommended by the manufacturer.
- D. The insulation thickness shall be as follows:

<u>Nominal IPS</u>	<u>Thickness</u>
2" & Less	3"
2-1/2" thru 4"	4"
5" thru 10"	4-1/2"

- E. Hangers shall bear on the outside of the insulation and shall not be in contact with the pipe. Curved metal shields shall be used between the hangers or support points and the bottom of the insulated pipe. Curved metal shields shall be designed to limit the bearing stress on the insulation to 35 psi and shall be curved to fit up to mid-perimeter of the insulated pipe. Required metal shield sizes are as follows:

		<u>Lengths of</u>
--	--	-------------------

<u>Nominal IPS</u>	<u>Metal Thickness</u>	<u>Shield</u>
up to 2"	14 gauge	12"
2-1/2" thru 6"	12 gauge	16"
8" and above	10 gauge	20"

- F. Shields shall be installed at all locations of support. They shall be made of galvanized iron, or black iron painted on both sides with two coats of aluminum paint.
- G. Where expansion bends occur in the lines, a two-foot (2') double layer contraction joint shall be provided in the main line starting two feet from the end of the main line ells on both sides of the expansion loop. Expansion joints shall consist of two 1-1/2" thick x 24" long pipe covering cuts into 17-1/4" and 6" lengths to provide a 3/4" space by 10-1/4".
- H. A 3/4" space on inside layer of expansion joints shall be filled with needled fiberglass blanket loose fill and the 3/4" space on the outer layer shall also be filled with the same loose fill. On two inch (2") or smaller IPS piping, both inner and outer layer shall be 1" thick "FOAMGLAS".
- I. Securing of aluminum jacket shall be made by the use of 3/4" x 0.016" aluminum or stainless steel bands and seals. The shields at support points shall be secured with 1/2" x 0.020" stainless steel bands and seals. The pipe primer shall be a red lead oxide primer. The metal jacketing and fitting covers shall be fabricated of 0.016" aluminum or stainless steel with a smooth finish.
- J. Insulation installed under this paragraph inside the building downstream of the pressure reducing station shall be insulated with 3 1/2 pound density preformed fiberglass insulation with a factory applied All Service Jacket, vapor sealing all joints, and factory performed fittings with vapor seal. Valves shall be insulated with mitered pipe covering with voids filled with glass fiber blanket insulation. Valves and fittings shall be vapor sealed with a water base asphaltic emulsion. Fittings on concealed insulation shall be built up to the thickness of adjacent insulation with glass fiber fitting wrap and shall be finished with Glasfab tape embedded in vapor barrier emulsion. Exposed fitting insulation shall be built up to same thickness as adjoining pipe insulation with one coat cement and after drying shall be finished with a white vapor seal and canvas jacket secured with "Arabol" adhesive and be suitable for painting. Seams in jacket shall be placed in the least noticeable locations. Where seams, joint or fittings are rough they shall be covered with an application of insulating cement trowelled on smoothly before the canvas is applied with Arabol adhesive. The canvas must be free of wrinkles and have a smooth, neat appearance. Thickness shall be as follows (as required by the IECC):

<u>MPS (30-90 PSI)</u>	
<u>Nominal IPS</u>	<u>Thickness</u>
2" & Less	3"
2-1/2" thru 4"	4"
5" thru 10"	4-1/2"

<u>MPS (0-30 PSI)</u>	
<u>Nominal IPS</u>	<u>Thickness</u>
2-1/2" & Less	2-1/2"
3" and over	3"

- K. No pipe supporting structures shall pierce the insulation except as anchor points as shown on the Drawings. At these points, the anchor member shall occur on the bottom of the piping to allow any moisture leakage to drain.
- L. The application of the protective shields at rack and guide points in tunnels and in central chilling stations shall be as detailed on the accompanying Drawings.

- M. Use prefabricated removable insulated, valve jackets on all valves in steam and steam condensate piping. Insulated jackets shall cover the entire valve including flanges and bonnet.
1. Acceptable manufacturers: Insultech or ThermaXX.
 2. Inner lining: heat treated fiberglass cloth (<450F).
 3. Insulating material: needled fiberglass mat (<1000F).
 4. Outer cover: aluminum foil laminated fiberglass cloth.
 5. Fastener: securing snaps or high temperature rated velco.

2.2 CHILLED WATER PIPING

- A. Chilled Water Piping – Insulation shall be installed on chilled water lines to the blending station at the building circulating pump shall be dimensional standard cellular glass insulation. The insulation will be installed on larger pipe sizes as appropriate. Prior to application of any insulation, all metal surfaces shall be thoroughly cleaned. The metal shall then be primed with an asphaltic primer consisting of one (1) coat of Foster No. 60-26 Primer or Pittcote 300 Primer. Cleaning and priming specified in this paragraph is not included in requirements for "Cleaning and Painting" specified in other sections of the Specifications. Regular insulation shall be applied to the piping with butt joints staggered and all joints tightly butted and sealed with a 1/4" bead of joint sealer 1/2" from outside edge. Hold in place with 14 ga. copper clad wire 9" o.c. After insulation has been wired in place, a 1/16" minimum thick, 3" wide band of asphaltic vapor seal mastic shall be brushed or trowelled on the outside of the insulation at the approximate location of the aluminum bands. (Note that the asphaltic material specified in this paragraph is intended to be an exception to the flame spread and smoke generation limitations found elsewhere in this specification.) . 15 mil PVC jacket shall be applied to pipes and shall be color coded to match existing systems.
- B. Provide insulation on flanges, valves and other fitting shall consist of prefabricated fitting covers of the same thickness as specified for adjoining pipe insulation.
- C. Fitting covers shall be applied in same manner as pipe application except that 16 ga. aluminum wire may be used to secure screwed fitting covers. Protruding metal parts (such as valve stems) shall be completely sealed off. Fitting cover jacketing shall be equal to Gasco, Papco or RPR Metals prefabricated fitting covers of 0.016" paper coated aluminum, secured as recommended by the manufacturer.
- D. The insulation thickness shall be as follows:

<u>Nominal IPS</u>	<u>Thickness</u>
2-1/2" & Less	2"
3" thru 14"	2-1/2"
16" and above	3"

- E. Hangers shall bear on the outside of the insulation and shall not be in contact with the pipe. Curved metal shields shall be used between the hangers or support points and the bottom of the insulated pipe. Curved metal shields shall be designed to limit the bearing stress on the insulation to 35 psi and shall be curved to fit up to mid-perimeter of the insulated pipe. Required metal shield sizes are as follows:

<u>Nominal IPS</u>	<u>Metal Thickness</u>	<u>Lengths of Shield</u>
up to 2"	14 gauge	12"
2-1/2" thru 6"	12 gauge	16"
8" and above	10 gauge	20"

- F. Shields shall be made of galvanized iron, or black iron painted on both sides with two coats of aluminum paint.
- G. Where expansion bends occur in the lines, a two foot (2') double layer contraction joint shall be provided in the main line starting two feet from the end of the main line ells on both sides of the expansion loop. Contraction joints shall consist of two 1-1/2" thick x 24" long pipe covering cuts into 17-1/4" and 6" lengths to provide a 3/4" space by 10-1/4". A slip joint mastic (Pittseal III) shall be placed between layers from the 3/4" space provided on the inside layer to the 3/4" space on the outside layer.
- H. The 3/4" space on inside layer shall be filled with mineral wool loose fill and the 3/4" space on the outer layer shall also be filled with same loose fill and joint sealer pressed 1/2" deep into space for sealing (Pittseal III). Around the outside layer at the 3/4" space, there shall be wrapped a 4" wide piece of glass fabric and sealed down with vapor seal mastic. On pipe sizes smaller than aforementioned, the same contraction joint shall be provided using one inch (1") thick insulation pipe covering for the inner layer. On two inch (2") IPS, both inner and outer layer shall be 1" thick insulation.
- I. The joint and vapor seal mastic shall be Pittsburgh Corning Corporation Pittcote 300. (Note that the asphaltic material specified in this paragraph is intended to be an exception to the flame spread and smoke generation limitations found elsewhere in this specification.) The mastic reinforcement shall be 10 x 10 mesh asphalt impregnated glass cloth. Securing of aluminum jacket shall be made by the use of 3/4" x 0.016" aluminum bands and seals. The shields at support points shall be secured with 1/2" x 0.020" stainless steel bands and seals. The pipe primer shall be Pittsburgh Corning Corporation Pittcote 300. The metal jacketing and fitting covers shall be fabricated of 0.016" aluminum with a smooth finish. The slip joint sealer shall be Pittsburgh Corning Corporation's Pittseal III.
- J. Insulation installed inside the building downstream of the blending station (or in the crawlspace, buildings, or mechanical rooms) shall be insulation with factory applied Class 1, Fire Rated, Kraft-Aluminum, vapor barrier jacketing as manufactured by Pittsburgh Corning. It shall be applied in strict accordance with the manufacturer's recommendations after the pipe has been primed with one (1) coat of Foster No. 60-26 Primer or Pittcoat 300 Primer. (Note that the asphaltic material specified in this paragraph is intended to be an exception to the flame spread and smoke generation limitations found elsewhere in this specification.) It will be necessary to include expansion joints at regular intervals on dual temperature service. Thickness shall be as follows:
1. Pipe sizes up to 2-1/2" - 2" thick
 2. Pipe sizes 3" and over - 2-1/2" thick
- K. Fitting covers shall be built up of shaped segments of foamglass insulation. These fitting covers shall be adhered in place using "Foster No. 30-35" vapor seals, then smoothly covered by a one-quarter inch (1/4") thick application of one coat white insulating cement. L. Valves, fittings, etc., in congested areas around coil and heat exchanger equipment, etc., shall be insulated by building up fitting segments and premolded sections, plus white vapor seal mastic, plus Manville No. 301 finishing cement to smooth surfaces, plus canvas applied and sized for painting with fire resistant adhesive. In addition, all manufactured vapor barrier jacketing in mechanical rooms and finished spaces shall be finished with canvas applied and sized for painting with fire resistant adhesive.
- M. No chilled water pipe supporting structures shall pierce the insulation except as anchor points as shown on the Drawings. At these points, the anchor member shall occur on the bottom of the piping to allow condensation to drain.
- N. The application of the protective shields at rack and guide points in tunnels and in central chilling stations shall be as detailed on the accompanying Drawings.

2.3 CONDENSATE DRAIN PIPING

- A. Condensate drain piping from fan and coil units, coil banks, drinking fountain refrigeration units, and other items of piping or equipment subject to condensation forming on the surface shall be insulated with a minimum of 3/8" thick "25-50" rated, elastomeric thermal insulation. Elastomeric products shall be supplied in a preslit tubular form with a pressure sensitive adhesive system for closure and vapor sealing of the longitudinal joint. All elastomeric insulating products shall be guaranteed not to react with copper piping.

2.4 HEATING HOT WATER PIPING –

- A. Contractor shall install dimensional, Pittsburgh Corning high temperature foamglass, on heating hot water piping, in the TEP.. Prior to application of any insulation, all metal surfaces shall be thoroughly cleaned. Insulation shall be applied to the piping with butt joints staggered and all joints tightly butted. The insulation shall be held in place using stainless steel bands. The bands shall be spaced at 12 inches on center. . 15 mil PVC jacket shall be applied to pipes 8'-0" above finished floor and shall be color coded to match existing systems. The Longitudinal joint of the jacketing shall be placed with overlap directed to bottom of pipe. The jacketing shall be overlapped a minimum of 3 inches, and it shall be held in place using 3/4 inch bands applied at 12 inches on center. Note that the use of asphaltic materials on higher-than-ambient service is specifically prohibited.
- B. Fiberglass (rigid) insulation on flanges, valves and other fitting shall consist of prefabricated fitting covers of the same thickness as specified for adjoining pipe insulation. Use Insultech blanket on all team valves and flanges 4" and larger.
- C. Fitting insulation shall be applied in same manner as pipe application. Fitting cover jacketing shall be equal to Gasco, Pabco or RPR Metals prefabricated fitting covers of 0.016" paper coated aluminum, secured as recommended by the manufacturer.
- D. The insulation thickness shall be as follows:

<u>Nominal IPS</u>	<u>Thickness</u>
2-1/2" & Less	2"
3" thru 14"	2-1/2"
16" and above	3"

- E. Hangers shall bear on the outside of the insulation and shall not be in contact with the pipe. Curved metal shields shall be used between the hangers or support points and the bottom of the insulated pipe. Curved metal shields shall be designed to limit the bearing stress on the insulation to 35 psi and shall be curved to fit up to mid-perimeter of the insulated pipe. Required metal shield sizes are as follows:

<u>Nominal IPS</u>	<u>Metal Thickness</u>	<u>Lengths of Shield</u>
up to 2"	14 gauge	12"
2-1/2" thru 6"	12 gauge	16"
8" and above	10 gauge	20"

- F. Shields shall be installed at all locations of support. They shall be made of galvanized iron, or black iron painted on both sides with two coats of aluminum paint.
- G. Where expansion bends occur in the lines, a two-foot (2') double layer contraction joint shall be provided in the main line starting two feet from the end of the main line ells on both sides of the expansion loop. Expansion joints shall consist of two 1-1/2" thick x 24" long pipe covering cuts into 17-1/4" and 6" lengths to provide a 3/4" space by 10-1/4".

- H. A 3/4" space on inside layer of expansion joints shall be filled with needled fiberglass blanket loose fill and the 3/4" space on the outer layer shall also be filled with the same loose fill. On two inch (2") or smaller IPS piping, both inner and outer layer shall be 1" thick "FOAMGLAS".
- I. Securing of aluminum jacket shall be made by the use of 3/4" x 0.016" aluminum or stainless steel bands and seals. The shields at support points shall be secured with 1/2" x 0.020" stainless steel bands and seals. The pipe primer shall be a red lead oxide primer. The metal jacketing and fitting covers shall be fabricated of 0.016" aluminum or stainless steel with a smooth finish.
- J. Insulation installed under this paragraph inside the buildings, crawlspaces, and mechanical rooms shall be insulated with 3 1/2 pound density preformed fiberglass insulation with a factory applied All Service Jacket, vapor sealing all joints, and factory performed fittings with vapor seal. Valves shall be insulated with mitered pipe covering with voids filled with glass fiber blanket insulation. Valves and fittings shall be vapor sealed with a water base asphaltic emulsion. Fittings on concealed insulation shall be built up to the thickness of adjacent insulation with glass fiber fitting wrap and shall be finished with Glasfab tape embedded in vapor barrier emulsion. Exposed fitting insulation shall be built up to same thickness as adjoining pipe insulation. Thickness shall be as follows:

<u>Nominal IPS</u>	<u>Thickness</u>
2" & Less	1-1/2"
2-1/2" thru 4"	2"
5" thru 10"	2-1/2"
12" and over	3"

- K. No pipe supporting structures shall pierce the insulation except as anchor points as shown on the Drawings. At these points, the anchor member shall occur on the bottom of the piping to allow any moisture leakage to drain.

2.5 CHILLED AND HEATING HOT WATER JACKETING

- A. Applications in the mechanical rooms up to 8 ft. above finished floor, an aluminum jacketing 0.016" thickness equal to Premetco precast, pre rolled Z lock Kraft paper lined pipe covering with zee type closure and 3/4" wide snap straps with permanent sealant shall then be fitted to O.D. of insulation and applied over the insulated pipe with 3" end and side caps secured with aluminum bands on 12" centers

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that piping has been tested before applying insulation materials.
- B. Verify that surfaces are clean, foreign material removed, and dry.

3.2 INSTALLATION

- A. Install materials in accordance with manufacturer's instructions.
- B. On exposed piping, locate insulation and cover seams in least visible locations, but not higher than at the side of the pipe at the "90°" position, with the seam lapped such that the lap is directed down.
- C. Insulated dual temperature pipes or cold pipes conveying fluids below ambient temperature:

1. Provide vapor barrier jackets, factory applied or field applied.
 2. Insulate fittings, joints, and valves with molded insulation of like material and thickness as adjacent pipe.
 3. Finish with glass cloth and vapor barrier adhesive.
 4. PVC fitting covers shall not be used.
 5. Continue insulation through walls, sleeves, pipe hangers, and other pipe penetrations.
 6. Insulate entire system including fittings, valves, unions, flanges, strainers, flexible connections, pump bodies, and expansion joints.
- D. For insulated pipes conveying fluids above ambient temperature:
1. Provide standard jackets, with or without vapor barrier, factory applied or field applied.
 2. Insulate fittings, joints, and valves with insulation of like material and thickness as adjoining pipe.
 3. PVC fitting covers shall not be used.
 4. For hot piping conveying fluids 140 degrees F or less, do not insulate flanges and unions at equipment, but bevel and seal ends of insulation.
 5. For hot piping conveying fluids over 140 degrees F, insulate flanges and unions, including those at equipment, but label the insulation to indicate a concealed flange or union.
- E. Inserts and Shields:
1. Application: Piping 2 inches diameter or larger.
 2. Shields: Galvanized steel between pipe hangers or pipe hanger rolls and inserts.
 3. Insert Location: Between support shield and piping and under the finish jacket.
 4. Insert Configuration: Minimum 6 inches long, of same thickness and contour as adjoining insulation; may be factory fabricated.
 5. Insert Material: Heavy density insulating material suitable for the planned temperature range, and the weight of the pipe.
- F. Finish insulation at supports, protrusions, and interruptions.
- G. For heat traced piping, insulate fittings, joints, and valves with insulation of like material, thickness, and finish as adjoining pipe. Size large enough to enclose pipe and heat tracer. Cover with aluminum jacket with seams located on bottom side of horizontal piping.
- H. All piping, equipment, ductwork, all plenums including metal and masonry construction, fans, etc., shall be insulated as indicated on the Drawings, as specified herein, and as required for a complete system. In each case, the insulation shall be equal to that specified and materials applied and finished as described in these Specifications.
- I. All insulation shall be applied by mechanics skilled in this particular work and regularly engaged in such occupation. All insulation shall be applied in strict accordance with these Specifications and with factory printed recommendations on items not herein mentioned. Unsightly, inadequate, or sloppy work will not be acceptable, and all such work shall be removed and replaced as many times as necessary to achieve an acceptable installation.

- J. All insulation, jacket, adhesives, mastics, sealers, etc., utilized in the fabrication of these systems shall meet NFPA for fire resistant ratings (maximum of 25 flame spread and 50 smoke developed ratings) and shall be approved by the insulation manufacturer for guaranteed performances when incorporated into their insulation system, unless a specific product is specified for a specific application, and is stated as an exception to this requirement. Certificates to this effect shall be submitted along with Contractor's submittal data for this section of the Specifications. No material may be used that, when tested by the ASTM E84-89 test method, is found to melt, drip or delaminate to such a degree that the continuity of the flame front is destroyed, thereby resulting in an artificially low flame spread rating.
- K. All surfaces to be insulated shall be clean and dry before applying the insulation. All sections of molded pipe covering shall be firmly butted together. Where an insulation covering is applied, it shall lap the adjoining section of insulation by at least three inches (3"). Where insulation terminates, it shall be neatly beveled and finished. No insulation shall be applied until the pipe, duct, etc., have been pressure tested and found tight. Piping, flexible connections, flanges, valves, strainers, and unions shall be covered unless specifically noted otherwise. Flexible connections on duct shall not be covered. All materials used shall be fire retardant or nonflammable. Refer to Section 15010.
- L. Where vapor barriers are required, the vapor barrier shall be on the outside. Extreme care shall be taken that the vapor barrier is unbroken. Joints, etc., shall be sealed. Where insulation with a vapor barrier terminates, it shall be sealed off with the vapor barrier being continuous to the surface being insulated. Ends shall not be left raw.
- M. Extreme care shall be taken in covering high and medium pressure (high and medium pressure ductwork shall be all ductwork between the fan discharge and all mixing boxes) ductwork to insure the duct is not pierced with sheet metal screws or other fasteners. All high and medium pressure ducts in these specifications are classified as high velocity ductwork.
- N. Where specified, aluminum bands shall be used on piping insulation. The bands shall be applied three (3) to a section of pipe. Fittings, valves, etc., shall have bands on each side.
- O. For purpose of definition in this Specification: "concealed" areas are those areas which cannot be seen by the building occupants, and "exposed" areas are all areas which are exposed to view by the building occupants, including under counter and inside cabinet areas, plus all mechanical rooms.
- P. The handling and installation of all insulation materials shall be performed in strict accordance with the manufacturer's recommendations.

END OF SECTION

SECTION 23 07 03

DUCTWORK INSULATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Ductwork insulation.
- B. Insulation jackets.

1.2 RELATED SECTIONS

- A. Section 23 0050 – Common Work Results for HVAC.
- B. Section 23 0553 - Mechanical Identification.

1.3 REFERENCES

- A. ASTM B209 - Aluminum and Aluminum-Alloy Sheet and Plate.
- B. ASTM C518 - Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
- C. ASTM C553 - Mineral Fiber Blanket and Felt Insulation.
- D. ASTM C612 - Mineral Fiber Block and Board Thermal Insulation.
- E. ASTM E84 - Surface Burning Characteristics of Building Materials.
- F. ASTM E96 - Water Vapor Transmission of Materials.
- G. NFPA 255 - Surface Burning Characteristics of Building Materials.
- H. SMACNA - HVAC Duct Construction Standards - Metal and Flexible.
- I. UL 723 - Surface Burning Characteristics of Building Materials.

1.4 SUBMITTALS

- A. Submit under provisions of Section 23 0050.
- B. Product Data: Provide product description, list of materials and thickness for each service, and locations.
- C. Samples: Submit two samples of any representative size illustrating each insulation type.
- D. Manufacturer's Installation Instructions: Indicate procedures which ensure acceptable workmanship and installation standards will be achieved.

1.5 QUALITY ASSURANCE

- A. Materials: Flame spread/smoke developed rating of 25/50 in accordance with NFPA 255.

1.6 QUALIFICATIONS

- A. Applicator: Company specializing in performing the work of this section with minimum three years experience.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect and handle products to site under provisions of Section 22 0050.
- B. Deliver materials to site in original factory packaging, labeled with manufacturer's density and thickness.
- C. Store insulation in original wrapping and protect from weather and construction traffic.
- D. Protect insulation against dirt, water, chemical, and mechanical damage.

1.8 ENVIRONMENTAL REQUIREMENTS

- A. Maintain ambient temperatures and conditions required by manufacturers of adhesives, mastics, and insulation cements.
- B. Maintain temperature during and after installation for minimum period of 24 hours.

PART 2 - PRODUCTS

2.1 INSULATION A

- A. Three pound per cubic foot minimum density glass fiber semi-rigid board insulation with fiber perpendicular to the surface and with factory applied white foil reinforced vapor barrier jacket (ASJ). Insulation shall be equal to E.O. Woods Company "Rigid-Wrap".
- B. The insulation shall be secured to the ducts with mechanical fasteners; "Stick-clips", Graham Pins or Speed Clips, and shall be spaced approximately 12" on center on bottom of duct and where required elsewhere to hold insulation securely against the duct as noted in the published recommendations of the Insulation Manufacturer.
- C. After insulation is in place, all joints and seams shall be sealed with Foster 30-35 white vapor barrier emulsion applied over a 3" wide strip of Duramesh Glass Fabric. All protrusions through the vapor barrier shall be thoroughly sealed.
- D. On ducts that are reinforced with standing seams or angle iron stiffeners 1" and over in height, the Contractor shall apply a strip of fiberglass board 1" thick by 6" wide, sealing same to the other insulation with mastic.
- E. Vapor sealing of joints and seams is not required on hot duct application where concealed.
- F. Exposed ductwork shall be provided with external Alumaguard jacketing.

2.2 INSULATION B

- A. Three pound per cubic foot minimum density glass fiber rigid board insulation with factory applied white foil reinforced All Service Jacket (ASJ).
- B. Insulation B shall be applied as specified for Insulation A.
- C. Contractor at his option may substitute Insulation A where Insulation B is called for.

2.3 INSULATION C

- A. Blanket insulation with a thermal conductivity (K) of 0.27 or less similar in construction to Owens-Corning Fiberglass Series one pound per cubic foot minimum density with foil reinforced Kraft (FRK) vapor barrier facing. Insulation shall be wrapped tightly on the ductwork with all circumferential joints butted and longitudinal joints overlapped a minimum of 2". Adhere insulation to metal with 4" strips of insulation bonding adhesive at 8" on center. On circumferential and longitudinal joints, the 2" flange of the facing shall be secured using 9/16" flare door staples applied 6" on center and taped with 4" wide fiberglass tape. All pin penetrations or punctures in facing shall also be taped. Vapor sealing of joints is not required on hot duct application where concealed.

2.4 ALL DUCTWORK IN THE BUILDING AND IN THE CRAWL SPACES EXCEPT TOILET EXHAUST AND FUME HOOD EXHAUST DUCTS SHALL BE INSULATED EXTERNALLY UNLESS SPECIFICALLY EXCLUDED. ONLY SOUND ATTENUATED RETURN DUCTING MAY BE INSULATED INTERNALLY, IF SPECIFICALLY DESIGNATED AS SUCH.

2.5 LOW PRESSURE SUPPLY DUCT TAPS TO CEILING DIFFUSERS SHALL BE EXTERNALLY INSULATED INCLUDING TOP OF CEILING DIFFUSER WITH 2" INSULATION C.

2.6 FLEXIBLE ROUND DUCTS ARE SPECIFIED IN SECTION 233300 AS FACTORY INSULATED.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that ductwork has been tested before applying insulation materials.
- B. Verify that surfaces are clean, foreign material removed, and dry.
- C. No interior walls of ductwork shall be insulated.

3.2 INSTALLATION

- A. Install materials in accordance with manufacturer's instructions.
- B. Insulated ductwork conveying air below ambient temperature:
 - 1. Provide insulation with vapor barrier jackets.
 - 2. Finish with tape and vapor barrier jacket.
 - 3. Continue insulation through walls, sleeves, hangers, and other duct penetrations.
 - 4. Insulate entire system including fittings, joints, flanges, fire dampers, flexible connections, and expansion joints.

- C. Insulated ductwork conveying air above ambient temperature:
 - 1. Provide with or without standard vapor barrier jacket.
 - 2. Insulate fittings and joints. Where service access is required, bevel and seal ends of insulation.
- D. For ductwork exposed in mechanical equipment rooms or in finished spaces, finish with Alumaguard jacketing.
- E. For exterior applications, provide insulation with vapor barrier jacket. Cover with caulked aluminum jacket with seams located on bottom side of horizontal duct section.
- F. External Duct Insulation Application:
 - 1. Secure insulation with vapor barrier with wires and seal jacket joints with vapor barrier adhesive or tape to match jacket.
 - 2. Secure insulation without vapor barrier with staples, tape, or wires.
 - 3. Install without sag on underside of ductwork. Use adhesive or mechanical fasteners where necessary to prevent sagging. Lift ductwork off trapeze hangers and insert spacers.
 - 4. Seal vapor barrier penetrations by mechanical fasteners with vapor barrier adhesive.
 - 5. Stop and point insulation around access doors and damper operators to allow operation without disturbing wrapping.
- G. All piping, equipment, ductwork, all plenums including metal and masonry construction, fans, etc., shall be insulated as indicated on the Drawings, as specified herein, and as required for a complete system. In each case, the insulation shall be equal to that specified and materials applied and finished as described in these Specifications.
- H. All insulation shall be applied by mechanics skilled in this particular work and regularly engaged in such occupation. All insulation shall be applied in strict accordance with these Specifications and with factory printed recommendations on items not herein mentioned. Unsightly, inadequate, or sloppy work will not be acceptable, and all such work shall be removed and replaced as many times as necessary to achieve an acceptable installation.
- I. All insulation, jacket, adhesives, mastics, sealers, etc., utilized in the fabrication of these systems shall meet NFPA for fire resistant ratings (maximum of 25 flame spread and 50 smoke developed ratings) and shall be approved by the insulation manufacturer for guaranteed performances when incorporated into their insulation system, unless a specific product is specified for a specific application, and is stated as an exception to this requirement. Certificates to this effect shall be submitted along with Contractor's submittal data for this section of the Specifications. No material may be used that, when tested by the ASTM E84-89 test method, is found to melt, drip or delaminate to such a degree that the continuity of the flame front is destroyed, thereby resulting in an artificially low flame spread rating.
- J. All surfaces to be insulated shall be clean and dry before applying the insulation. All sections of molded pipe covering shall be firmly butted together. Where an insulation covering is applied, it shall lap the adjoining section of insulation by at least three inches (3"). Where insulation terminates, it shall be neatly beveled and finished. No insulation shall be applied until the pipe, duct, etc., have been pressure tested and found tight. Piping, flexible connections, flanges, valves, strainers, and unions shall be covered unless specifically noted otherwise. Flexible connections on duct shall not be covered. All materials used shall be fire retardant or nonflammable. Refer to Section 22 0010.
- K. Where vapor barriers are required, the vapor barrier shall be on the outside. Extreme care shall be taken that the vapor barrier is unbroken. Joints, etc., shall all be sealed. Where insulation with a vapor barrier terminates, it shall be sealed off with the vapor barrier being continuous to the surface being insulated. Ends shall not be left raw.

- L. Extreme care shall be taken in covering high and medium pressure (high and medium pressure ductwork shall be all ductwork between the fan discharge and all mixing boxes) ductwork to insure the duct is not pierced with sheet metal screws or other fasteners. All high and medium pressure ducts in these specifications are classified as high velocity ductwork.
- M. Where specified, aluminum bands shall be used on piping insulation. The bands shall be applied three (3) to a section of pipe. Fittings, valves, etc., shall have bands on each side.
- N. For purpose of definition in this Specification: "concealed" areas are those areas which cannot be seen by the building occupants, and "exposed" areas are all areas which are exposed to view by the building occupants, including under counter and inside cabinet areas, plus all mechanical rooms.
- O. The handling and installation of all insulation materials shall be performed in strict accordance with the manufacturer's recommendations.

3.3 TOLERANCE

- A. Substituted insulation materials shall provide thermal resistance within 10 percent at normal conditions, as materials indicated.

3.4 DUCT INSULATION SCHEDULE

<u>Duct Type</u>	<u>Insulation Type</u>
Medium pressure supply ductwork (concealed)	3" Insulation C
Low pressure supply ductwork (concealed)	3" Insulation C
Return air ductwork	3" Insulation C (Note 1)
Supply, return and relief ductwork within equipment room	2" Insulation A (Note 1)
Ductwork exposed to weather/exterior	2" Insulation A (Note 1)
Outside air and relief air plenums	2" Insulation B
Supply diffuser housing/plenum	3" Insulation C
Exhaust air duct work handling moisture laden air	3" Insulation C

Note 1: Where approved by the Owner, insulation may be omitted from return and relief air ductwork except at the following locations:

1. Where return or relief air ductwork is installed in an unconditioned space below a roof.
2. Where return or relief air ductwork is installed in the crawlspace.
3. Relief air ductwork between the relief air plenum and the first motorized or backdraft isolation damper.

END OF SECTION

SECTION 23 09 13

INSTRUMENTATION AND CONTROL DEVICES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Provide input and output control devices to integrate with direct digital control and building automation system.
- B. Furnish instrumentation control devices as an integral part of the Building Automation Section specified in Section 23 09 23.

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 09 23 - Building Automation and Direct Digital Controls
- C. Section 23 20 00 - HVAC Piping
- D. Section 23 31 13 - Sheetmetal Ductwork
- E. Section 23 36 00 - Air Terminal Units
- F. Division 26: Electrical

1.3 SUBMITTALS

- A. Submit product data and schedules for all input/output devices.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Provide products and components by manufacturers listed. Where manufacturers are not listed, provide component that complies with specifications.
- B. Manufacturers listed must meet performance and material specifications of product or component. Listing of a manufacturer as an acceptable manufacturer does not grant permission to deviate from the specification requirements.

2.2 INPUT DEVICES

- A. General Requirements
 - 1. Installation, testing, and calibration of all sensors, transmitters, and other input devices shall be provided to meet the system requirements.
- B. Temperature Sensors
 - 1. Acceptable Manufacturers: Siemens or JCI or Honeywell or Intempco.

- a. Substitutions: Not permitted.
- 2. General Requirements:
 - a. Sensors and transmitters shall be provided, as outlined in the input/output summary and sequence of operations.
 - b. The temperature sensor shall be of the resistance type, and shall be either two-wire 1000 ohm nickel RTD, or two-wire 1000 ohm platinum RTD.
 - c. Accuracy values indicated include errors associated with the sensor, lead wire, and analog to digital conversion. Plus or minus 0.5 deg at calibration point.
- 3. Room Temperature Sensors
 - a. Refer to temperature sensor legend, schedules, floor plans, and control sequences for specific room temperature sensor requirements in each zone.
 - b. Room sensors shall be constructed for either surface or wall box mounting.
 - c. Room sensors shall have the following options when specified:
 - 1) Local setpoint adjustment providing a +/- 3 degree (adjustable) range.
 - 2) Timed override request push button with LED status for activation of after-hours operation.
 - 3) Flush mounting (sensor only, no local adjustment)
 - 4) Integral LCD display and keypad with the following capabilities:
 - (a) Display room temperature.
 - (b) Display room setpoint.
- 4. Stand Alone Thermostats
 - a. Stand alone, heavy-duty electric thermostats shall be provided for unit heaters, cabinet unit heaters, and ventilation fans, when equipment is not indicated to be connected to the BAS. Thermostats shall be provided with concealed adjustment and function to cycle the equipment fan, electric coil, and/or control valves as applicable to maintain the space temperature setpoint. Finish of covers for all room-type instruments shall match and, unless otherwise indicated or specified, covers shall be manufacturer's standard finish.
- 5. Thermo Wells
 - a. When thermo wells are required, the sensor and well shall be supplied as a complete assembly, including wellhead and Greenfield fitting.
 - b. Thermo wells shall be pressure rated and constructed in accordance with the system working pressure.
 - c. Thermo wells and sensors shall be mounted in a threadolet or 1/2" NPT saddle and allow easy access to the sensor for repair or replacement.
 - d. Thermo wells shall be constructed of 316 stainless steel.
- 6. Outside Air Sensors
 - a. Outside air sensors shall be designed to withstand the environmental conditions to which they will be exposed. They shall also be provided with a solar shield.
 - b. Sensors exposed to wind velocity pressures shall be shielded by a perforated plate that surrounds the sensor element.
 - c. Temperature transmitters shall be of NEMA 3R construction and rated for ambient temperatures.

7. Duct Mount Sensors

- a. Duct mount sensors shall mount in an electrical box through a hole in the duct, and be positioned so as to be easily accessible for repair or replacement.
 - b. Duct sensors shall be insertion type and constructed as a complete assembly, including lock nut and mounting plate.
 - c. For outdoor air duct applications, a weatherproof mounting box with weatherproof cover and gasket shall be used.
8. Averaging Sensors
- a. Provide at the following locations:
 - 1) Heating coils and cooling coils at air handling units and fan coil units.
 - 2) Ductwork greater in any dimension that 48 inches and/or where air temperature stratification exists.
 - b. For plenum applications, such as mixed air temperature measurements, a string of sensors mounted across the plenum shall be used to account for stratification and/or air turbulence. The averaging string shall have a minimum of 4 sensing points per 12-foot long segment and twice the diagonal length of the duct.
 - c. Capillary supports at the sides of the duct shall be provided to support the sensing string.
9. Low Limit Temperature Sensors
- a. Provide vapor charged sensing element that reacts to coldest 14" of sensor length.
 - b. Sensor shall have field adjustable setpoint.
10. Probe temperature sensors
- a. Matched Chilled water immersion sensors with LED display 2 – wire loop powered (One set per plant) Ordered as a matched pair. Equal to Intempco DTG5A-05(F)-CD-D-H101-___-N26-___-H-MC-B2-SF-___:
 - 1) Sensing element: RTD, Pt1000 DIN EN 60751, Class A
 - 2) Accuracy: Plus or minus 0.25 percent span.
 - 3) Display: 4 - Digit LCD, ½" high (12.7mm), decimal point selectable by software
 - 4) Ambient range: (32-122 Deg F)
 - 5) Wire: Twisted, shielded-pair cable.
 - 6) Insertion Elements for Liquids: Stainless steel 316, spring Loaded
 - 7) Housing Material Stainless steel 316
 - 8) Environmental protection: NEMA 4 / IP65
 - 9) Insertion Length: Length shall be sufficient to reach the center of the pipe
 - b. Chilled Water immersion sensors (entering and leaving for each chiller condenser and evaporator). Equal to Intempco DTG5A-05(F)-CD-D-H101-___-N26-___-H-MC-B2-SF-___:
 - 1) Sensing element: RTD, Pt1000 DIN EN 60751, Class A
 - 2) Accuracy: Plus or minus 0.25 percent span.
 - 3) Display: 4 - Digit LCD, ½" high (12.7mm), decimal point selectable by software
 - 4) Ambient range: (32-122 Deg F)
 - 5) Wire: Twisted, shielded-pair cable.
 - 6) Insertion Elements for Liquids: Stainless steel 316, spring Loaded
 - 7) Housing Material Stainless steel 316
 - 8) Environmental protection: NEMA 4 / IP65
 - 9) Insertion Length: Length shall be sufficient to reach the center of the pipe
 - c. Chilled Water immersion sensors (non-display type). Equal to Intempco R55B-L-T-2-S2-TB-D-S-___-EN-H10-AC2-___:

- 1) Sensing element: 100 ? @ 0 °C, ? =0.00385 DIN EN 60751
- 2) Element tolerance: 0.06% Class A
- 3) Ambient range: (32-122 Deg F)
- 4) Wire: Twisted, shielded-pair cable.
- 5) Insertion Elements for Liquids: Stainless steel 316, spring Loaded
- 6) Housing Material Stainless steel 316
- 7) Environmental protection: NEMA 4 / IP65
- 8) Insertion Length: Length shall be sufficient to reach the center of the pipe

C. Humidity Sensors

1. Acceptable Manufacturers: Johnson Controls or Veris Industries.
 - a. Substitutions: Not permitted.
2. The sensor shall be a solid-state type, relative humidity sensor of the Bulk Polymer Design. The sensor element shall resist service contamination.
3. The humidity transmitter shall be equipped with non-interactive span and zero adjustments, a 2-wire isolated loop powered, 4-20 mA, 0-100% linear proportional output.
4. The humidity transmitter shall meet the following overall accuracy, including lead loss and Analog to Digital conversion. 2% between 20% and 80% RH @ 77 Deg F unless specified elsewhere.
5. Outside air relative humidity sensors shall be installed with a rain proof, perforated cover. The transmitter shall be installed in a NEMA 3R enclosure with sealtite fittings and stainless steel bushings.
6. A single point humidity calibrator shall be provided, if required, for field calibration. Transmitters shall be shipped factory pre-calibrated.
7. Duct type sensing probes shall be constructed of 304 stainless steel, and shall be equipped with a neoprene grommet, bushings, and a mounting bracket.

D. Differential Pressure Transmitters

1. Acceptable Manufacturers: Setra or Siemens.
 - a. Substitutions: Not permitted.
2. General Air and Water Pressure Transmitter Requirements:
 - a. Pressure transmitters shall be constructed to withstand 100% pressure over-range without damage, and to hold calibrated accuracy when subject to a momentary 40% over-range input.
 - b. Pressure transmitters shall transmit a 0 to 5 VDC, 0 to 10 VDC, or 4 to 20 mA output signal.
 - c. Differential pressure transmitters used for flow measurement shall be sized to the flow sensing device, and shall be supplied with tee fittings and shut-off valves in the high and low sensing pick-up lines to allow the balancing Contractor and Owner permanent, easy-to-use connection.
 - d. A minimum of a NEMA 1 housing shall be provided for the transmitter. Transmitters shall be located in accessible local control panels wherever possible.
3. Differential Pressure Transmitters: (chilled water and hot water systems)
 - a. Provide self-contained, variable capacitance type differential pressure transmitters at chillers, boilers and air handling units to measure system and equipment differential pressures.
 - b. Installed by the contractor where indicated on the drawings including system DPs at AHU coils and across chillers. Wiring terminals and electronics shall be in separate compartments, so the electronics remain sealed during installation. Reverse polarity protection shall be included to

keep wiring mishaps from damaging the transmitter. Wiring installed by the contractor between the control system and the transmitter(s) shall be Belden 9320, two wire, shielded twisted cable, and shall not be included in conduit containing AC circuit wiring.

- c. Design range shall be as required by system. External zero and span adjustments, over-pressure to 2,000 PSI, and no humidity effects.
 - d. Minimum accuracy shall be 0.25% of calibrated span. Includes combined effects of linearity, hysteresis and repeatability. Stability shall be 0.25% of upper range limit for six months. No internal mechanical linkages shall be used in the transmitter(s).
 - e. Low Differential Water Pressure Applications (0" - 20" w.c.)
 - 1) The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of flow meter differential pressure or water pressure sensing points.
 - 2) The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - (a) .01-20" w.c. input differential pressure range.
 - (b) 4-20 mA output.
 - (c) Maintain accuracy up to 20 to 1 ratio turndown.
 - (d) Reference Accuracy: +0.2% of full span.
 - f. Medium to High Differential Water Pressure Applications (Over 21" w.c.)
 - 1) The differential pressure transmitter shall meet the low pressure transmitter specifications with the following exceptions:
 - (a) Differential pressure range 10" w.c. to 300 PSI.
 - (b) Reference Accuracy: +1% of full span (includes non-linearity, hysteresis, and repeatability).
 - 2) Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.
4. Building Differential Air Pressure Applications (-1" to +1" w.c.)
- a. The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of differential pressure or air pressure sensing points.
 - b. The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - 1) -1.00 to +1.00 w.c. input differential pressure ranges. (Select range appropriate for system application)
 - 2) 4-20 mA output.
 - 3) Maintain accuracy up to 20 to 1 ratio turndown.
 - 4) Reference Accuracy: +0.2% of full span.
5. Building Low Range Differential Air Pressure Applications (0" to 5" w.c.)
- a. The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of differential pressure or air pressure sensing points.
 - b. The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - 1) (0.00 - 1.00" to 5.00") w.c. input differential pressure ranges. (Select range appropriate for system application.)
 - 2) 4-20 mA output.
 - 3) Maintain accuracy up to 20 to 1 ratio turndown.
 - 4) Reference Accuracy: +0.2% of full span.

6. Building Medium Range Differential Air Pressure Applications (5" to 21" w.c.)

- a. The pressure transmitter shall be similar to the Low Air Pressure Transmitter, except that the performance specifications are not as severe. Differential pressure transmitters shall be provided that meet the following performance requirements:
 - 1) Zero & span: (c/o F.S./Deg. F): .04% including linearity, hysteresis and repeatability.
 - 2) Accuracy: 1% F.S. (best straight line) Static Pressure Effect: 0.5% F.S. (to 100 PSIG.
 - 3) Thermal Effects: <+.033 F.S./Deg. F. over 40 Deg. F. to 100 Deg. F. (calibrated at 70 Deg. F.).
- b. Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.

E. Air Flow Measuring Stations

1. Thermal Dispersion Air Flow/Temperature Measurement Devices

- a. Acceptable manufacturers: Air Monitor Corporation, Ebtron, or Johnson Controls.
 - 1) Substitutions: Not permitted.
 - 2) Vortex shedding, pitot tubes, pitot arrays, or piezo ring monitors are not acceptable.
- b. Basis of Design:
 - 1) Fan walls: Ebtron Model GTx108-F
 - 2) Duct/plenum mounted: Ebtron Model GTx116-P+
- c. Provide air flow/temperature measurement devices (AFMS) where indicated on the plans.
- d. Each AFMS shall consist of one or more sensor probes and a single, remotely mounted, microprocessor-based transmitter capable of independently processing up to 16 independently wired sensor nodes contained in one or more probe assemblies per measurement location.
 - 1) Each sensor node shall contain two individually wired, hermetically sealed bead-in-glass thermistors.
 - 2) Thermistors shall be mounted in the sensor node using a marine-grade, waterproof epoxy. Thermistor leads shall be protected and not exposed to the environment. Thermistor leads shall not be fastened to the thermistor semiconductor substrate by weld or solder connections.
 - 3) The air flow rate at each sensor node shall be equally weighted and arithmetically averaged by the transmitter prior to output. All integrated circuitry shall be temperature rated as 'industrial-grade'. Submissions containing 'commercial-grade' integrated circuitry are not acceptable.
 - 4) The temperature at each sensor node shall be arithmetically averaged or velocity weighted and averaged by the transmitter prior to output, either as field-selected by the contractor or user.
 - 5) Each transmitter shall have a 16-character alpha-numeric display capable of displaying air flow, temperature, system status, configuration settings and diagnostics.
 - 6) Other than the thermistor sensors, no other electronic components shall be located at the sensing node.
 - 7) Devices using chip-in-glass, epoxy-coated or diode-case chip thermistors are not acceptable.
 - 8) Devices with RJ-45 connections exposed to the environment or having electronic circuitry mounted in or at the sensor node are not acceptable.
- e. All Sensor Probes
 - 1) Each sensor node, consisting of two thermistor-sensors and their structural housing, shall independently determine the air flow rate and temperature at each measurement point.
 - 2) Each sensor node shall be factory calibrated at a minimum of 16 air flow rates and 3 temperatures to standards that are traceable to the national Institute of Standards and

Technology (NIST). Thermistor sensor calibrations traceable only to temperature standards are not acceptable.

- 3) Temperature accuracy shall be ± 0.14 degrees F (0.08 degrees C) over the entire operating temperature range of -20 degrees F to 160 degrees F (-28.9 degrees C to 71 degrees C).
- 4) The operating humidity range for each sensor probe shall be 0-99% RH (non-condensing). Product design shall consider direct exposure to or immersion in liquid water and temporary exposure shall not damage the sensing elements.
- 5) Each sensor or probe assembly shall not require matching to the transmitter in the field.
- 6) A single manufacturer shall provide both the air flow/temperature measuring probe(s) and transmitter for each measurement location.

f. Duct and Plenum Probes

- 1) Probes shall be constructed of extruded, gold anodized, 6063 aluminum tubes. All internal wires within the tube shall be Kynar coated. PVC insulated conductors are not acceptable.
- 2) Sensor probe design shall be capable of providing up to 4 sensor nodes per probe.
- 3) Air flow accuracy shall be $\pm 2\%$ of reading over the entire operating air flow range of not less than zero to 5,000 feet per minute.
 - (a) Devices whose overall performance at the host controller input terminals is the combined accuracy of the transmitter and sensor probes shall demonstrate that the total accuracy meets the performance requirements of this specification throughout the measurement range.
- 4) Each ducted sensor probe shall have an integral, U.L. listed, plenum rated cable. Cable jackets and conductor insulation shall be FEP, Teflon-FEP or Neoflon-FEP. Cables shall include a terminal plug for connection to the remotely mounted transmitter. All terminal plug interconnecting pins shall be gold plated. PVC jacketed cables of PVC insulated conductors are not acceptable with ducted sensor probes.

g. Fan Inlet Sensor Probes

- 1) Each sensor node, consisting of at least two thermistor-sensors and their structural housing, shall independently determine the air flow rate and temperature at each measurement point.
- 2) Air flow accuracy shall be $\pm 2\%$ of reading over the entire operating air flow range of not less than zero to 10,000 feet per minute.
 - (a) Devices whose overall performance at the host controller input terminals is the combined accuracy of the transmitter and sensor probes shall demonstrate that the total accuracy meets the performance requirements of this specification throughout the measurement range.
- 3) Sensor node assemblies shall be mounted on Type 304 stainless steel housings.
- 4) Mounting rods shall be field adjustable to fit the fan inlet and constructed of zinc plated steel.
- 5) Mounting feet shall be constructed of Type 304 stainless steel.
- 6) Mounting style shall be "face mount" or "throat mount" unless noted otherwise.
 - (a) Face mounting style shall provide no mechanical fastening in the throat or on the surface of the inlet cone and shall be used on all inlet condition-sensitive plenum-type or plug fans.
- 7) Fan inlet probe cables shall be UL plenum-rated and may be FEP jacketed. Provide length as required based on transmitter location.

2. Thermal Dispersion Sensor Transmitters

- a. The transmitter shall have an integral LCD display capable of simultaneously displaying air flow and temperature. The LCD display shall be capable of displaying individual air flow and temperature readings of each independent sensor node.
- b. The transmitter shall be capable of field configuration and diagnostics using an on-board pushbutton interface and LCD display.

- c. The transmitter shall have an on-off power switch and operate on 24 VAC. Isolation transformers shall not be required. Power shall be extended from local BAS panel.
 - 1) The transmitter shall use a switching power supply, fused and protected from transients and power surges.
 - 2) The transmitter shall use "watch-dog" circuitry to assure automatic reset after power disruption, transients and brown-outs.
 - d. All interconnecting pins, headers and connections on the main circuit board, option cards and cable receptacles shall be gold plated.
 - e. The operating temperature range for the transmitter shall be -20 degrees F to 120 degrees F (-28.9 degrees C to 48.9 degrees C). The transmitter shall be installed at a location that is protected from weather and water.
 - f. The transmitter shall be capable of communicating with other devices using one of the following interface options:
 - 1) Linear analog output signals for air flow and temperature: Field selectable, fuse protected and electrically isolated from all other circuitry, 0-5VDS / 0-10VDC / 4-20mA (4-wire).
 - 2) RS-485: Field selectable BACnet/MS/TP, BACnet-ARCNET, Modbus-RTU or Johnson Controls N2-Bus.
 - (a) BACnet devices shall provide analog variables for air flow and temperature containing individual sensor air flow rate and temperature data.
 - 3) 10 Base-T Ethernet: Field selectable BACnet Ethernet, BACnet-IP, Modbus-TCP and TCP/IP.
 - g. Provide dynamic link libraries and VBA functions to interface Ethernet devices to Microsoft Excel for remote monitoring of air flow and temperature using a MS Windows-based PC.
3. Static Pressure Traverse Probe
- a. Acceptable manufacturers: Air Monitor Corp. or Cleveland Controls.
 - 1) Substitutions: Refer to Division 01.
 - b. Duct static traverse probes shall be provided where required to monitor duct static pressure. The probe shall contain multiple static pressure sensors located along exterior surface of the cylindrical probe.
 - c. Shielded Static Air Probe
 - 1) A shielded static pressure probe shall be provided at each end of the building. The probe shall have multiple sensing ports, an impulse suppression chamber, and air flow shielding. A suitable probe for indoor and outdoor locations shall be provided.
4. Water Flow Monitoring
- a. Acceptable manufacturers: Omega, Onicon, or Siemens.
 - 1) Substitutions: Refer to Division 01.
 - b. Water flow meters shall be electromagnetic type with integral microprocessor-Based electronics. The meter shall have an accuracy of 0.25%.
 - c. Ultrasonic flow meters on each chilled water supply leaving each chillers: Equal to Siemens FUS1010:
 - 1) Flow Range: <12 m/s (< 40 ft/s), bidirectional
 - 2) Flow sensitivity: 0.0003 m/s (0.001 ft/s), flow rate independent
 - 3) Pipe size: 6.4 mm ... 9.14 m (0.25" ... 360")
 - 4) Output: BACnet MSTP/ BACbet IP
 - 5) Accuracy: < 0.5 % ... 1.0 % of flow,
 - 6) Batch repeatability: < 0.15 % of flow
 - 7) Display 128 x 240 pixel LCD with backlight

- 8) Environmental protection: NEMA 4x / IP65
- 9) Ambient range: 0-140 Deg F

F. Power Monitoring Devices

- 1. Acceptable manufacturers: Veris Industries.
 - a. Substitutions: Refer to Division 01.
- 2. Current Measurement (Amps)
 - a. Current measurement shall be by a combination current transformer and a current transducer. The current transformer shall be sized to reduce the full amperage of the monitored circuit to a maximum 5 Amp signal, which will be converted to a 4-20 mA DDC compatible signal for use by the Facility Management System.
 - b. Current Transformer - A split core current transformer shall be provided to monitor motor amps.
 - 1) Operating frequency - 50 - 400 Hz.
 - 2) Insulation - 0.6 Kv class 10Kv BIL.
 - 3) UL recognized.
 - 4) Five amp secondary.
 - 5) Select current ration as appropriate for application.
 - 6) Acceptable manufacturers: Veris Industries
 - c. Current Transducer - A current to voltage or current to mA transducer shall be provided. The current transducer shall include:
 - 1) 6X input over amp rating for AC inrushes of up to 120 amps.
 - 2) Manufactured to UL 1244.
 - 3) Accuracy: +.5%, Ripple +1%.
 - 4) Minimum load resistance 30kOhm.
 - 5) Input 0-20 Amps.
 - 6) Output 4-20 mA.
 - 7) Transducer shall be powered by a 24VDC regulated power supply (24 VDC +5%).

G. Smoke Detectors

- 1. Ionization type air duct detectors shall be furnished as specified elsewhere in Division 28. for installation under Division 23. All wiring for air duct detectors shall be provided under Division 28, Fire Alarm System. Coordinate interface with BAS and Fire Alarm System.

H. Status and Safety Switches

- 1. General Requirements
 - a. Switches shall be provided to monitor equipment status, safety conditions, and generate alarms at the BAS when a failure or abnormal condition occurs. Safety switches shall be provided with two sets of contacts and shall be interlock wired to shut down respective equipment.
- 2. Current Sensing Switches
 - a. Acceptable manufacturers: Veris Industries.
 - 1) Substitutions: Refer to Division 01.
 - b. The current sensing switch shall be self-powered with solid-state circuitry and a dry contact output. It shall consist of a current transformer, a solid state current sensing circuit, adjustable trip point, solid state switch, SPDT relay, and an LED indicating the on or off status. A conductor of the load shall be passed through the window of the device. It shall accept over-current up to twice its trip point range.

- c. Current sensing switches shall be used for run status for fans, pumps, and other miscellaneous motor loads.
 - d. Current sensing switches shall be calibrated to show a positive run status only when the motor is operating under load. A motor running with a broken belt or coupling shall indicate a negative run status.
3. Air Filter Status Switches
- a. Acceptable manufacturers: Automated Logic, Cleveland Controls, Johnson Controls, or Siemens.
 - 1) Substitutions: Refer to Division 01.
 - b. Differential pressure switches used to monitor air filter status shall be of the automatic reset type with SPDT contacts rated for 2 amps at 120VAC.
 - c. A complete installation kit shall be provided, including: static pressure tops, tubing, fittings, and air filters.
 - d. Provide appropriate scale range and differential adjustment for intended service.
4. Air Flow Switches
- a. Acceptable manufacturers: Automated Logic, Cleveland Controls, Johnson Controls, or Siemens.
 - 1) Substitutions: Refer to Division 01.
 - b. Differential pressure flow switches shall be bellows actuated mercury switches or snap acting micro-switches with appropriate scale range and differential adjustment for intended service.
 - c. Acceptable manufacturers: Johnson Controls, Cleveland Controls
5. Air Pressure Safety Switches
- a. Acceptable manufacturers: Automated Logic, Cleveland Controls, Johnson Controls, or Siemens.
 - 1) Substitutions: Refer to Division 01.
 - b. Air pressure safety switches shall be of the manual reset type with SPDT contacts rated for 2 amps at 120VAC.
 - c. Pressure range shall be adjustable with appropriate scale range and differential adjustment for intended service.
6. Differential Pressure / Water Flow Switches
- a. Acceptable manufacturers: Johnson Controls (model P74).
 - 1) Substitutions: Refer to Division 01.
 - b. Switch shall contain heavy duty pressure elements and be used to measure and relay the pressure difference between two sources or across the inlet/outlet of equipment.
 - c. Device shall be an automatic reset device and shall provide a control output at the differential pressure setpoint.
7. Low Temperature Limit Switches
- a. Acceptable manufacturers: Johnson Controls (model A70).
 - 1) Substitutions: Refer to Division 01.
 - b. The low temperature limit switch shall be of the manual reset type with Double Pole/Single Throw snap acting contacts rated for 16 amps at 120VAC.

- c. The sensing element shall be a minimum of 15 feet in length and shall react to the coldest 18-inch section. Element shall be mounted horizontally across duct in accordance with manufacturers recommended installation procedures.
- d. For large duct areas where the sensing element does not provide full coverage of the air stream, additional switches shall be provided as required to provide full protection of the air stream.

2.3 OUTPUT DEVICES

A. Actuators

1. Acceptable manufacturers: Johnson Controls, Mamac, or Siemens.
 - a. Substitutions: Refer to Division 01.
2. General Requirements
 - a. Damper and valve actuators shall be electronic and/or pneumatic, as specified in the System Description section.
3. Electronic Damper Actuators
 - a. Electronic damper actuators shall be direct shaft mount.
 - b. Modulating and two-position actuators shall be provided as required by the sequence of operations. Damper sections shall be sized Based on actuator manufacturer's recommendations for face velocity, differential pressure and damper type. The actuator mounting arrangement and spring return feature shall permit normally open or normally closed positions of the dampers, as required. All actuators (except terminal units) shall be furnished with mechanical spring return unless otherwise specified in the sequences of operations. All actuators shall have external adjustable stops to limit the travel in either direction, and a gear release to allow manual positioning.
 - c. Modulating actuators shall accept 24 VAC or VDC power supply, consume no more than 15 VA, and be UL listed. The control signal shall be 2-10 VDC or 4-20 mA, and the actuator shall provide a clamp position feedback signal of 2-10 VDC. The feedback signal shall be independent of the input signal and may be used to parallel other actuators and provide true position indication. The feedback signal of one damper actuator for each separately controlled damper shall be wired back to a terminal strip in the control panel for trouble-shooting purposes.
 - d. Two-position or open/closed actuators shall accept 24 or 120 VAC power supply and be UL listed. Isolation, smoke, exhaust fan, and other dampers, as specified in the sequence of operations, shall be furnished with adjustable end switches to indicate open/closed position or be hard wired to start/stop associated fan. Two-position actuators, as specified in sequences of operations as "quick acting," shall move full stroke within 20 seconds. All smoke damper actuators shall be quick acting.
4. Electronic Valve Actuators
 - a. Electronic valve actuators shall be manufactured by the valve manufacturer.
 - b. Each actuator shall have current limiting circuitry incorporated in its design to prevent damage to the actuator.
 - c. Modulating and two-position actuators shall be provided as required by the sequence of operations. Actuators shall provide the minimum torque required for proper valve close-off against the system pressure for the required application. The valve actuator shall be sized Based on valve manufacturer's recommendations for flow and pressure differential. All actuators shall fail in the last position unless specified with mechanical spring return in the

sequence of operations. The spring return feature shall permit normally open or normally closed positions of the valves, as required. All direct shaft mount rotational actuators shall have external adjustable stops to limit the travel in either direction.

- d. Modulating Actuators shall accept 24 VAC or VDC and 120 VAC power supply and be UL listed. The control signal shall be 2-10 VDC or 4-20 mA and the actuator shall provide a clamp position feedback signal of 2-10 VDC. The feedback signal shall be independent of the input signal, and may be used to parallel other actuators and provide true position indication. The feedback signal of each valve actuator (except terminal valves) shall be wired back to a terminal strip in the control panel for trouble-shooting purposes.
- e. Two-position or open/closed actuators shall accept 24 or 120 VAC power supply and be UL listed. Butterfly isolation and other valves, as specified in the sequence of operations, shall be furnished with adjustable end switches to indicate open/closed position or be hard wired to start/stop the associated pump or chiller.

B. Control Dampers

- 1. The BAS Contractor shall furnish all automatic dampers. All automatic dampers shall be sized for the application by the BAS Contractor or as specifically indicated on the Drawings.
- 2. All dampers used for throttling air flow shall be of the opposed blade type arranged for normally open or normally closed operation, as required. The damper is to be sized so that, when wide open, the pressure drop is a sufficient amount of its close-off pressure drop to shift the characteristic curve to near linear.
- 3. All dampers used for two-position, open/close control shall be parallel blade type arranged for normally open or closed operation, as required.
- 4. Damper frames and blades shall be constructed of either galvanized steel or aluminum. Maximum blade length in any section shall be 60". Damper blades shall be 16-gauge minimum and shall not exceed eight (8) inches in width. Damper frames shall be 16-gauge minimum hat channel type with corner bracing. All damper bearings shall be made of reinforced nylon, stainless steel or oil-impregnated bronze. Dampers shall be tight closing, low leakage type, with synthetic elastomer seals on the blade edges and flexible stainless steel side seals. Dampers of 48"x48" size shall not leak in excess of 8.0 cfm per square foot when closed against 4" w.g. static pressure when tested in accordance with AMCA Std. 500.
- 5. Airfoil blade dampers of double skin construction with linkage out of the air stream shall be used whenever the damper face velocity exceeds 1500 FPM or system pressure exceeds 2.5" w.g., but no more than 4000 FPM or 6" w.g. Acceptable manufacturers are Johnson Controls D-7250 D-1250 or D-1300, Ruskin CD50, and Vent Products 5650.
- 6. One piece rolled blade dampers with exposed or concealed linkage may be used with face velocities of 1500 FPM or below. Acceptable manufacturers are: Johnson Controls D-1600, Ruskin CD36, and Vent Products 5800.
- 7. Multiple section dampers may be jack-shafted to allow mounting of piston pneumatic actuators and direct connect electronic actuators. Each end of the jackshaft shall receive at least one actuator to reduce jackshaft twist.

C. Control Relays

- 1. Control Pilot Relays
 - a. Acceptable manufacturers: Johnson Controls, Lectro, or Siemens.
 - 1) Substitutions: Refer to Division 01.
 - b. Control pilot relays shall be of a modular plug-in design with retaining springs or clips.
 - c. Mounting Bases shall be snap-mount.

- d. DPDT, 3PDT, or 4PDT relays shall be provided, as appropriate for application.
 - e. Contacts shall be rated for 10 amps at 120VAC.
 - f. Relays shall have an integral indicator light and check button.
2. Lighting Control Relays
- a. Lighting control relays shall be latching with integral status contacts.
 - b. Contacts shall be rated for 20 amps at 277 VAC.
 - c. The coil shall be a split low-voltage coil that moves the line voltage contact armature to the ON or OFF latched position.
 - d. Lighting control relays shall be controlled by:
 - 1) Pulsed Tri-state Output - Preferred method.
 - 2) Pulsed Paired Binary Outputs.
 - 3) A Binary Input to the Facility Management System shall monitor integral status contacts on the lighting control relay. Relay status contacts shall be of the "dry-contact" type.
 - e. The relay shall be designed so that power outages do not result in a change-of-state, and so that multiple same state commands will simply maintain the commanded state. Example: Multiple OFF command pulses shall simply keep the contacts in the OFF position.

D. Control Valves

- 1. Acceptable manufacturers: Johnson Controls or Siemens or Bray.
 - a. Substitutions: Refer to Division 01.
- 2. All automatic control valves shall be fully proportioning and provide near linear heat transfer control. The valves shall be quiet in operation and fail-safe open, closed, or in their last position. All valves shall operate in sequence with another valve when required by the sequence of operations. All control valves shall be sized by the control manufacturer, and shall be guaranteed to meet the heating and cooling loads, as specified. All control valves shall be suitable for the system flow conditions and close against the differential pressures involved. Body pressure rating and connection type (sweat, screwed, or flanged) shall conform to the pipe schedule elsewhere in this Specification.
- 3. Chilled water control valves shall be modulating plug, ball, and/or butterfly, as required by the specific application. Modulating water valves shall be sized per manufacturer's recommendations for the given application. In general, valves (2 or 3-way) serving variable flow air handling unit coils shall be sized for a pressure drop equal to the actual coil pressure drop, but no less than 5 PSI. Valves (3-way) serving constant flow air handling unit coils with secondary circuit pumps shall be sized for a pressure drop equal to 25% the actual coil pressure drop, but no less than 2 PSI. Mixing valves (3-way) serving secondary water circuits shall be sized for a pressure drop of no less than 5 PSI. Valves for terminal reheat coils shall be sized for a 2 PSIG pressure drop, but no more than a 5 PSI drop.
- 4. Ball valves shall be used for hot and chilled water applications, water terminal reheat coils, radiant panels, unit heaters, package air conditioning units, and fan coil units except those described hereinafter.
- 5. Modulating plug water valves of the single-seat type with equal percentage flow characteristics shall be used for all special applications as indicated on the valve schedule. Valve discs shall be composition type. Valve stems shall be stainless steel.
- 6. Butterfly valves shall be acceptable for modulating large flow applications greater than modulating plug valves, and for all two-position, open/close applications. In-line and/or three-way butterfly valves shall be heavy-duty pattern with a body rating comparable to the pipe rating, replaceable

lining suitable for temperature of system, and a stainless steel vane. Valves for modulating service shall be sized and travel limited to 50 degrees of full open. Valves for isolation service shall be the same as the pipe. Valves in the closed position shall be bubble-tight.

E. Control valves, Bray Butterfly Valves

1. Provide Chiller and Condenser water control valves equal to as follows:
 - a. Bray NYL2-Cxxx with Actuator 70-1316-113DO-536
 - b. Service: Chilled Water, Condenser Water up to 50% Glycol
 - c. Size Range: 2" through 30" (DN 50 to 750)
 - d. Body Style: Lug, 2-way and 3-way, for ANSI 125 and ANSI 150 flanges
 - e. Flow Characteristics: Modified Equal Percentage
 - f. Fluid Temperature Limits: -40 to 250F (-40 to 121C)
 - g. Leakage: Bubble tight at rated maximum differential pressure
 - h. Maximum Fluid Velocity: 30 ft/second (9 m/second)
 - i. Materials: Body: Cast Iron
 - j. Materials: Disk: NY Series - Ductile Iron, Nylon 11 coated
 - k. Materials: Seat: EPDM (Ethylene Propylene Diene Monomer) - Peroxide Cured
 - l. Materials: Stem: 416 Stainless Steel

F. Control Valves, Segmented Ball Valves

1. Acceptable manufacturers: Fisher, Kele, Nelis-Jamesbury, or Valve Solutions, Inc.
 - a. Substitutions: Refer to Division 01.
2. High performance segmented V-Ball control valve: Carbon steel body, stainless steel V-notch ball and shaft, low friction bearings and a low friction graphite ball set.; ANSI Class 150 or 300 flanges as required by application.
3. Rated ANSI Class VI leakage rate, -20 degrees F. to 450 degree F. temperature range and minimum 285 PSI allowable shutoff pressure drop at -20 to 100 F.
4. Rotation: 90 degrees; rangeability: 300 to 1 with equal percentage control characteristic; valve shall accommodate standard electric actuators.
5. Warranty: Valve and linkage, 3 year warranty from date of installation.

G. Electronic Signal Isolation Transducers

1. Acceptable manufacturers: Advanced Control Technologies.
 - a. Substitutions: Refer to Division 01.
2. A signal isolation transducer shall be provided whenever an analog output signal from the BAS is to be connected to an external control system as an input (such as a chiller control panel), or is to receive as an input signal from a remote system.
3. The signal isolation transducer shall provide ground plane isolation between systems.
4. Signals shall provide optical isolation between systems.

H. External Manual Override Stations

1. External manual override stations shall provide the following:
 - a. An integral HAND/OFF/AUTO switch shall override the controlled device pilot relay.
 - b. A status input to the Facility Management System shall indicate whenever the switch is not in the automatic position.
 - c. A Status LED shall illuminate whenever the output is ON.
 - d. An Override LED shall illuminate whenever the HOA switch is in either the HAND or OFF position.
 - e. Contacts shall be rated for a minimum of 1 amp at 24 VAC.

2.4 MISCELLANEOUS DEVICES

A. Local Control Panels

1. All control panels shall be factory constructed, incorporating the BAS manufacturer's standard designs and layouts. All control panels shall be UL inspected and listed as an assembly and carry a UL 508 label listing compliance. Control panels shall be fully enclosed, with perforated sub-panel, hinged door, and slotted flush latch. Provide common keying for all new panels and match keying when existing panels are present.
2. Control panels shall consist of the DDC controller(s), display module as specified and indicated on the plans, and I/O devices-such as relays, transducers, and so forth-that are not required to be located external to the control panel due to function. Where specified the display module shall be flush mounted in the panel face unless otherwise noted.
3. All I/O connections on the DDC controller shall be provide via removable or fixed screw terminals.
4. Low and line voltage wiring shall be segregated. All provided terminal strips and wiring shall be UL listed, 300-volt service and provide adequate clearance for field wiring.
5. All wiring shall be neatly installed in plastic trays or tie-wrapped.
6. A 120 VAC duplex convenience receptacle and required transformers shall be provided in each enclosure.

B. Power Supplies

1. DC power supplies shall be sized for the connected device load. Total rated load shall not exceed 75% of the rated capacity of the power supply.
2. Input: 120 VAC +10%, 60Hz.
3. Output: 24 VDC.
4. Line Regulation: +0.05% for 10% line change.
5. Load Regulation: +0.05% for 50% load change.
6. Ripple and Noise: 1 mV rms, 5 mV peak to peak.
7. An appropriately sized fuse and fuse block shall be provided and located next to the power supply.
8. A power disconnect switch shall be provided next to each power supply.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Actuation / Control Type

1. Primary Equipment

- a. Controls shall be provided by equipment manufacturer as specified herein.
- b. All damper and valve actuation shall be electric.

2. Air Handling Equipment

- a. All air handlers shall be controlled with a HVAC-DDC Controller
- b. All damper and valve actuation shall be electric.

3. Terminal Equipment:

- a. Terminal Units (ATU, UV, etc.) shall have electric damper and valve actuation.
- b. All Terminal Units shall be controlled with HVAC-DDC Controller.

B. HVAC Input Devices - General

1. All Input devices shall be installed per the manufacturer recommendation.

2. Locate components of the BAS in accessible local control panels wherever possible.

- a. The mechanical contractor shall install all in-line devices such as temperature wells, pressure taps, air flow stations, etc.

3. Flow Measuring Devices shall be installed in strict compliance with ASHRAE and ASME guidelines affecting non-standard approach conditions.

4. Outside Air Sensors

- a. Sensors shall be mounted on the North wall to minimize solar radiant heat impact or located in a continuous intake flow adequate to monitor outside air conditions accurately.
- b. Sensors shall be installed with a rain proof, perforated cover.

5. Water Differential Pressure Sensors

- a. Differential pressure transmitters used for flow measurement shall be sized to the flow-sensing device.
 - 1) Differential pressure transmitters shall be supplied with tee fittings and shut-off valves in the high and low sensing pick-up lines.
- b. The transmitters shall be installed in an accessible location wherever possible.

6. Medium to High Differential Water Pressure Applications (Over 21" w.c.)

- a. Air bleed units, bypass valves and compression fittings shall be provided.

7. Building Differential Air Pressure Applications (-1" to +1" w.c.)

- a. Transmitters exterior sensing tip shall be installed with a shielded static air probe to reduce pressure fluctuations caused by wind.
- b. The interior tip shall be inconspicuous and located as shown on the drawings.

8. Air Flow Measuring Stations

- a. Install air flow measuring stations in accordance with manufacturer's instructions at the locations indicated on the plans including clear distances to adjacent fittings, elbows, inlets, or other interference. A written report shall be submitted to the Engineer if any discrepancies exist or if installation cannot be completed per the manufacturer's recommendations.
 - b. Station flanges shall be two inch to three inch to facilitate matching connecting ductwork.
- 9. Duct Temperature Sensors
 - a. Duct mount sensors shall mount in an electrical box through a hole in the duct and be positioned so as to be easily accessible for repair or replacement.
 - b. The sensors shall be insertion type and constructed as a complete assembly including lock nut and mounting plate.
 - c. For ductwork greater in any dimension than 48 inches or where air temperature stratification exists such as a mixed air plenum, utilize an averaging sensor.
 - d. The sensor shall be mounted to suitable supports using factory approved element holders.
- 10. Space Sensors
 - a. Mounted per ADA requirements.
- 11. Averaging and Low Temperature Limit Switches
 - a. Install as indicated in the control diagram.
 - b. Mount element horizontally across duct in a serpentine pattern insuring each square foot of coil is protected by at least 1 foot of sensor.
 - c. For large duct areas where the sensing element does not provide full coverage of the air stream, provide additional switches as required to obtain full coverage.
- 12. Air Differential Pressure Status Switches
 - a. Install with static pressure tips, tubing, fittings, and air filter.
- 13. Water Differential Pressure Status Switches
 - a. Install with shut off valves for isolation.

C. HVAC Output Devices

- 1. All output devices shall be installed per the manufacturer's recommendation. The mechanical contractor shall install all in-line devices such as control valves, dampers, air flow stations, pressure wells, etc.
- 2. Actuators: All control actuators shall be sized capable of closing against the maximum system shut-off pressure. The actuator shall modulate in a smooth fashion through the entire stroke.
- 3. Control Dampers: Shall be opposed blade for modulating control of air flow. Parallel blade dampers shall be installed for two position applications.
- 4. Control Valves: Shall be sized for proper flow control with equal percentage valve plugs. The maximum pressure drop for water applications shall be 5 PSI. The maximum pressure drop for steam applications shall be 7 PSI.
- 5. Electronic Signal Isolation Transducers: Whenever an analog output signal from the BAS is to be connected to an external control system as an input (such as a chiller control panel), or is to receive as an input a signal from a remote system, provide a signal isolation transducer. Signal isolation transducer shall provide ground plane isolation between systems. Signals shall provide optical isolation between systems

3.2 TRAINING

- A. The BAS contractor shall provide the following training services:
 - 1. One day of on-site orientation by a system technician who is fully knowledgeable of the specific installation details of the project. This orientation shall, at a minimum, consist of a review of the project as-built drawings, the BAS software layout and naming conventions, and a walk through of the facility to identify panel and device locations.

3.3 COMMISSIONING

- A. Fully commission all aspects of the Building Management System work.
- B. Acceptance Check Sheet
 - 1. Prepare a check sheet that includes all points for all functions of the BAS as indicated on the point list included in this specification.
 - 2. Submit the check sheet to the Engineer for approval
 - 3. The Engineer will use the check sheet as the basis for acceptance with the BAS Contractor.
- C. ATU performance verification and documentation:
 - 1. The BAS Contractor shall test each air terminal unit for operation and correct flow. At each step, after a settling time, box air flows and damper positions will be sampled. Following the tests, a pass/fail report indicating results shall be produced. Possible results are Pass, No change in flow between full open and full close, Reverse operation or Maximum flow not achieved. The report shall be submitted as documentation of the installation.
 - 2. The BAS Contractor shall issue a report based on a sampling of the ATU calculated loop performance metrics. The report shall indicate performance criteria, include the count of conforming and non-conforming boxes, list the non-conforming boxes along with their performance data, and shall also include graphical representations of performance.
- D. Promptly rectify all listed deficiencies and submit to the Engineer that this has been done.

END OF SECTION

SECTION 23 09 23

BUILDING AUTOMATION AND DIRECT DIGITAL CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to Work covered by this Section.
- B. Comply with other Division 23 Sections, as applicable. Refer to other Divisions for coordination of the Work.

1.2 SUMMARY

- A. This Section includes control equipment for HVAC systems, components and other systems shown to be controlled by the Building Automation System (BAS), including, but not limited to, all computer software and hardware, controllers, sensors, transmission equipment, local panels, installation, engineering, supervision, commissioning, acceptance testing, training and warranty service necessary for a complete and working system.

1.3 SCOPE OF WORK

- A. The existing building automation systems shall be expanded to accommodate the new project scope.
- B. All components of the system - workstations, network controllers, local controllers, etc. shall communicate using a standard protocol, as defined by ASHRAE Standard 135-2001 and as specified herein.
 - 1. Level 1 communication protocol shall be BACnet IP.
 - 2. Level 2 communication protocol shall be BACnet IP, BACnet MS/TP Modbus IP or Modbus RTU, Ethernet IP.
 - 3. Proprietary communications is allowed as an alternate based on the project with owner approval.
 - 4. LON communication protocol is not acceptable at any level in the BAS system.
- C. The BAS contractor shall review and study all HVAC drawings and the entire specification to become familiar with the equipment and system operation and to verify the quantities and types of dampers, operators, alarms, etc. to be provided.
- D. All interlocking, wiring and installation of control devices associated with the equipment described in the sequence of operations, points list and control diagrams shall be provided under this Contract.
- E. Provide services and manpower necessary for commissioning of system in coordination with the Commissioning Authority, HVAC Contractor, Testing and Balancing Contractor, Electrical Contractor and Owner's Representative. Refer Section 23 08 00.
- F. All work performed under this section of the specifications will comply with all codes, laws and governing bodies. If the drawings and/or specifications are in conflict with governing codes, the Contractor shall submit a proposal with appropriate modifications to the project to meet code restrictions. If this specification and associated drawings exceed governing code requirements, the specification will govern. The Controls Contractor shall obtain and pay for all necessary construction permits and licenses associated with this scope of work.

1.4 REFERENCE STANDARD

- A. The control system components shall be new and in conformance with the following applicable standards for products specified:
 - 1. American Society for Testing and Materials, ASTM
 - 2. Institute of Electrical and Electronic Engineers, IEEE
 - 3. National Electrical Manufacturers Association, NEMA
 - 4. Underwriters Laboratory, UL (UL 916 & 864)
 - 5. FCC Regulation, Part 15, Section 156
 - 6. National Fire Protection Association, NFPA
 - 7. Local Building Codes

1.5 PROTECTION OF SOFTWARE RIGHTS

- A. Prior to delivery of software, the Owner and the party providing the software will enter into a software license agreement with provisions for the following:
 - 1. Limiting use of software to equipment provided under these Specifications
 - 2. Limiting copying.
 - 3. Preserving confidentiality.
 - 4. Prohibiting transfer to a third party.

1.6 SUBMITTALS

- A. Product Data: Include manufacturer's technical literature for each control device. Indicate dimensions, capacities, performance characteristics, electrical characteristics, finishes for materials, and installation and startup instructions for each type of product indicated.
 - 1. DDC System Hardware: Bill of materials of equipment indicating quantity, manufacturer, and model number. Include technical data for system architecture, operator workstation equipment, interface equipment, control units, transducers/transmitters, sensors, actuators, valves, relays/switches, control panels, communication methods and operator interface equipment.
 - 2. Control System Software: Include technical data for operating system software, operator interface, color graphics, and other third-party applications including all software licensing agreements.
 - 3. Controlled Systems: Instrumentation list with element name, type of device, manufacturer, model number, and product data. Include written description of sequence of operation including schematic control diagram.
- B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 1. Riser diagram of main network architecture depicting all controllers, workstations and associated network wiring.
 - 2. Bill of materials of equipment indicating quantity, manufacturer, and model number.
 - 3. Schematic flow/control diagrams showing fans, pumps, coils, dampers, valves, and control devices.
 - 4. Wiring Diagrams: Power, signal, and control wiring.

5. Details of control panel faces, including controls, instruments, and labeling.
 6. Floor plans indicating control panel locations.
 7. Written description of sequence of operation.
 8. Schedule of dampers including size, leakage, and airflow characteristics.
 9. Schedule of valves including flow characteristics.
 10. Schedule of airflow monitoring stations including airflow characteristics.
 11. DDC System Hardware:
 - a. Wiring diagrams for control units with termination numbers.
 - b. Schematic diagrams and floor plans for field sensors and control hardware.
 - c. Schematic diagrams for control, communication, and power wiring, showing trunk data conductors and wiring between operator workstation and control unit locations.
 12. Control System Software: Graphics outline and "Print Page" examples of final product indicating monitored systems, data (connected and calculated) point addresses, output schedule, and operator notations.
 13. Controlled Systems:
 - a. Schematic diagrams of each controlled system with control points labeled and control elements graphically shown, with wiring.
 - b. Scaled drawings showing mounting, routing, and wiring of elements including bases and special construction.
 - c. Written description of sequence of operation including schematic diagram.
 - d. Points list.
- C. Data Communications Protocol Certificates: Certify that each proposed DDC system component complies with ASHRAE Standard 135-2001 for each protocol.
- D. Samples for Initial Selection: For each type of sensor cover with factory-applied color finishes.
- E. Software and Firmware Operational Documentation: Include the following:
1. Software operating and upgrade manuals.
 2. Program Software Backup: On CD, complete with data files.
 3. Device address list.
 4. Printout of software application and graphic screens.
 5. Software license required by and installed for operator workstations and control systems.
- F. Software Upgrade Kit: For Owner to use in modifying software to suit future systems revisions or monitoring and control revisions.
- G. Field quality-control test report forms.
- H. Operation and Maintenance Data: Include emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:
1. Maintenance instructions and lists of spare parts for each type of control device.

2. Interconnection wiring diagrams with identified and numbered system components and devices.
3. Keyboard illustrations and step-by-step procedures indexed for each operator function.
4. Inspection period, cleaning methods, cleaning materials recommended, and calibration tolerances.
5. Calibration records and list of set points.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: Automatic control system manufacturer's authorized representative who is trained and approved for installation of system components required for this Project.
- B. Warranty: Controls Contractor shall guarantee all system components and installations to be free from defects for one (1) year from the date of acceptance as determined by the Owner. Any defects found during this period shall be repaired and/or replaced at no cost to the Owner. The Controls Contractor shall provide maximum of 24-hour response time for trouble calls or maintenance.
- C. Upon completion of the installation, the Contractor shall thoroughly inspect, check, adjust, calibrate, and make ready for use all devices/sensors comprising the control system and certify that they are installed in accordance with "Record" Drawings.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- E. Data Communications Protocol: Certify that each proposed DDC system component complies with ASHRAE Standard 135-2001 for each protocol.
- F. DDC system component testing: Comply with ASHRAE 135.1-2001 for all DDC controllers.
- G. All controllers used to control or monitor equipment and/or field devices shall be tested, compliant with and carry a testing seal:
 1. Building Controllers.
 2. Advanced Application Controllers.
 3. Application Specific Controllers.
- H. System Software: Provide latest version of software at Project completion. Provide all software updates for one (1) year after date of acceptance as determined by the Owner.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Factory-Mounted Components: Where control devices specified in this Section are indicated to be factory mounted on equipment in other Sections, arrange for shipping of control devices to equipment manufacturer. Upon delivery the equipment manufacturer shall inspect shipment for visual damages. The Controls Contractor shall replace any damaged control equipment at no cost to the Owner.
- B. Provide factory shipping containers for each piece of equipment. Provide factory applied plastic end caps on each length of pipe and tube. Maintain cartons and end caps through shipping, storage and handling as required to prevent equipment and pipe-end damage, and to eliminate dirt and moisture from equipment and inside of pipe and tube. Where possible store equipment and materials inside and protected from weather. When necessary, to store outside, elevate well above grade and enclose with durable water-proof wrapping.

1.9 WORK BY OTHERS

- A. The installation of motor starters that are not factory installed, thermal overload switches, and power wiring to motors, starters, and thermal overload switches, is specified in another section. This section includes the furnishing and installing of all controls, devices, interlocks, and wiring to provide a complete operating system as outlined in the sequence of operation.
- B. The following general work scope of Contractors requiring coordination by the Controls Contractor includes, but is not limited to:
 - 1. The Piping Contractor shall:
 - a. Install automatic valves, flow meters and separable wells that are specified to be supplied by the Controls Contractor.
 - b. Furnish and install all necessary pressure taps, wells, as directed by the Controls Contractor.
 - c. Furnish and install all necessary drain and overflow connections and piping.
 - d. Furnish and install all necessary fittings and piping connections required for flow devices.
 - 2. The Sheet Metal Contractor shall:
 - a. Install all automatic dampers and provide necessary blank-off plates or transitions required to install dampers that are smaller than duct size.
 - b. Assemble multiple section dampers with required interconnecting linkages and extend required number of shafts through duct for external mounting of damper motors.
 - c. Furnish and install necessary sheet metal baffle plates to eliminate stratification and provide air volumes specified. Locate baffles by experimentation. Affix and seal permanently in place once stratification problems have been eliminated.
 - d. Furnish and install access doors or other approved means of access through ducts for service to control equipment.
 - e. Install duct mounted airflow monitoring stations.
 - f. Install AHU and duct mounted heat and smoke detectors.
 - 3. The General Contractor shall:
 - a. Provide access doors or other means of access through ceilings and walls for services to control equipment.
 - b. Provide necessary housekeeping pads and, where required, concrete inertia bases.
 - 4. Air Terminal Box Manufacturers shall:
 - a. Factory mount, wire and configure the terminal box DDC controller and actuator.
 - b. Furnish static pressure probes.
 - c. Furnish 24 volt transformers for terminal box controller power.
 - 5. Electrical Contractor shall:
 - a. Furnish fire alarm system compatible duct mounted heat and smoke detectors and wire to Fire Alarm System. Coordinate with the mechanical contractor the exact placement of duct mounted detectors.
 - b. Provide relay cabinets, required for lighting control and wiring/conduits to the EMS panels.

- c. Provide dedicated 120VAC circuits in j-boxes throughout all building areas for control panel and terminal box control power.

PART 2 - PRODUCTS

2.1 ACCEPTABLE BIDDERS

- A. The specifications are intended to describe the microprocessor based Energy Management System - System 600 APOGEE[®] and Siemens Building Technologies is the acceptable manufacturer/installer.
- B. Alternate bidders are acceptable with ownership approval. Any alternate bidder must be able to demonstrate to UTSWMC the ability to integrate with the existing Siemens Building Technologies Energy Management System prior to bidding.

2.2 NETWORKING

- A. The design of the EMS shall network operator workstations and stand-alone DDC Controllers. The network architecture shall consist of three levels, a campus-wide (Management Level Network - MLN) Ethernet network based on TCP/IP protocol, high performance peer-to-peer Building Level Network (BLN) and Application Specific Controller Floor Level Networks (FLN) with access being totally transparent to the user when accessing data or developing control programs.
- B. The design of EMS shall allow the co-existence of new DDC Controllers with existing DDC Controllers in the same network without the use of gateways or protocol converters.
- C. All operator devices either network resident or connected via remote connection shall have the ability to access all point status and application report data or execute control functions for any and all other devices via the peer-to-peer network. No hardware or software limits shall be imposed on the number of devices with global access to the network data at any time.
- D. All Networks shall be dynamically connected to allow access to points on different BLN's simultaneously.

2.3 SYSTEM PERFORMANCE

- A. Comply with the following performance requirements:
 - 1. Object Command: Reaction time of less than two seconds between operator command of a binary object and device reaction.
 - 2. Object Scan: Transmit change of state and change of analog values to control units or workstation within six seconds.
 - 3. Alarm Response Time: Annunciate alarm at workstation within 45 seconds. Multiple workstations must receive alarms within five seconds of each other.
 - 4. Program Execution Frequency: Run capability of applications as often as five seconds, but selected consistent with mechanical process under control.
 - 5. Performance: Programmable controllers shall execute DDC PID control loops, and scan and update process values and outputs at least once per second.
 - 6. Reporting Accuracy and Stability of Control: Report values and maintain measured variables within tolerances as follows:
 - a. Water Temperature: Plus or minus 1 deg F.
 - b. Water Flow: Plus or minus 5 percent of full scale.

- c. Water Pressure: Plus or minus 2 percent of full scale.
- d. Space Temperature: Plus or minus 1 deg F.
- e. Ducted Air Temperature: Plus or minus 1 deg F.
- f. Outside Air Temperature: Plus or minus 2 deg F.
- g. Dew Point Temperature: Plus or minus 3 deg F.
- h. Temperature Differential: Plus or minus 0.25 deg F.
- i. Relative Humidity: Plus or minus 5 percent.
- j. Airflow (Pressurized Spaces): Plus or minus 3 percent of full scale.
- k. Airflow (Measuring Stations): Plus or minus 5 percent of full scale.
- l. Airflow (Terminal): Plus or minus 10 percent of full scale.
- m. Air Pressure (Space): Plus or minus 0.01-inch wg.
- n. Air Pressure (Ducts): Plus or minus 0.1-inch wg.
- o. Carbon Monoxide: Plus or minus 5 percent of reading.
- p. Carbon Dioxide: Plus or minus 50 ppm.
- q. Electrical: Plus or minus 5 percent of reading.

B. Graphics User Interface:

- 1. Menu: Selectable for command entry, information management, network alarm management and database management functions including, but not limited to:
 - a. Configuration.
 - b. Data Archiving.
 - c. Commanding.
 - d. System Diagnostics.
 - e. Graphic Display Tree.
 - f. Alarm Logs.
 - g. Reports.
 - h. Schedules.
- 2. Graphic Display: Display graphic with current state of the art dynamic points and refresh speed. As a minimum provide the following graphics pages:
 - a. Overall Building Level: Display building, building name and all exterior equipment.
 - b. Floor Level: Display floor level derived from CAD drawing, active links to monitored and/or controlled equipment on that floor.
 - c. AHU System Level: Similar to floor level but displaying the area of a single AHU system and embedded links for all associated equipment including, but not limited to:
 - 1) AHU.
 - 2) Terminal boxes.
 - 3) Exhaust fans.

- 4) Dampers.
- 5) Thermostats.
- 6) Field sensors.
- d. Equipment Level: Display associated equipment setpoints and real-time sensor readings as defined by the Owner.
- e. Equipment Support Data: Link the following information directly from the Equipment Level graphic display or menu graphic display tree:
 - 1) Operation & maintenance manual.
 - 2) Equipment schedules.
 - 3) Sequence of operations.
- f. Other graphic displays
 - 1) Alarm log.
 - 2) Energy overview: costs, consumption, production.
 - 3) Equipment runtimes.

2.4 DDC CONTROLLERS

- A. Building Controllers shall be 32-bit, multi-tasking, multi-user, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers, power supplies and input/output point modules. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
- B. Each Building Controller shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, operator I/O, and dial-up communications.
- C. Provide Universal I/O capability, including software configurable universal inputs and universal outputs.
- D. Shall support a minimum of one directly connected Secondary Network.
- E. Building Controller shall have an integral real-time clock.
- F. Each Building Controller shall support firmware upgrades without the need to change hardware.
- G. Each Building Controller shall support:
 - 1. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the Building Controller cabinet.
 - 2. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the Building Controller cabinet.
- H. Spare Point Capacity. Each Building Controller shall have a minimum of 10 percent spare point capacity.
 - 1. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
 - 2. Provide all processors, power supplies, and communication controllers so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - a. Expansion modules

- b. Sensor/actuator
 - c. Field wiring/tubing
- I. Serial Communication. Building Controllers shall provide at least one EIA-232C serial data communication ports for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. Building Controllers shall allow temporary use of portable devices without interrupting the normal operation of permanently connected printers or terminals. A USB port shall alternatively be available to support local HMI tools connection.
- J. Printer support. A USB port shall be available to support printers for printed alarm record keeping.
- K. Manual Override. The operator shall have the ability to manually override automatic or centrally executed commands at the Building Controller via local, point discrete, integral hand/off/auto operator override switches for all digital control type points and gradual switches for all analog control type points. These override switches shall be operable whether the panel processor is operational or not. Each Building Controller shall monitor and alarm the hand, off and auto positions of integral HOA switches.
- L. I/O Status and Indication. Building Controllers shall provide local LED status indication for each digital input and output for constant, up-to-date verification of all point conditions without the need for an operator I/O device. Graduated intensity LEDs or analog indication of value shall also be provided for each analog output. All wiring connections shall be made to field-removable terminals.
- M. Shall provide I/O modules with LCD's capable of displaying information faults including but not limited to open circuit, short circuit, unreliable input signal, signal under range, and signal over range via informative symbols.
- N. Self Diagnostics. Each Building Controller shall continuously perform self diagnostics, communication diagnosis, and diagnosis of all panel components. The Building Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
- O. Power loss. In the event of the loss of power, there shall be an orderly shutdown of all Building Controllers to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 30 days.
- P. Environment.
 - 1. Controller hardware shall be suitable for the anticipated ambient conditions.
 - 2. Controllers used outdoors and/or in wet ambient conditions shall be mounted within waterproof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
 - 3. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
- Q. Immunity to power and noise.
 - 1. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 - 2. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).
 - 3. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - a. RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3V.

- b. Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact.
 - c. Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500V signal, 1 kV power.
 - d. Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max).
- 4. Isolation shall be provided at all Building Controller's AC input terminals to suppress induced voltage transients consistent with:
 - a. IEEE Standard 587 1980
 - b. UL 864 Supply Line Transients
 - c. Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000-4-11)
- R. Local Controller Interface. A local user interface to the controller shall be provided. The interface may be mounted on any building controller and automatically read and initiate commands of local database points without further set-up or configuration. The Controller Interface shall be provided for interrogating and editing data, commanding point values at user defined priorities, viewing and acknowledging alarms, and viewing point monitoring reports. An optional system security password shall be available to prevent unauthorized use of the local controller interface and display.

2.5 BUILDING CONTROLLER SOFTWARE

A. General:

1. Furnish the following applications software to form a complete operating system for building and energy management as described in this specification.
2. The software programs specified in this section shall be provided as an integral part of Building Controllers and shall not be dependent upon any higher level computer or another controller for execution.
3. All points, panels and programs shall be identified by a 30-character name. All points shall also be identified by a 16-character point descriptor. The same names shall be displayed at both Building Controller and the Operator Interface.
4. All digital points shall have a user defined two-state status indication with 8 characters minimum (e.g., Summer, Enabled, Disabled, Abnormal).
5. The Building Controller Software shall be capable of BACnet communications. The BACnet Building Controller (B-BC) shall have demonstrated interoperability during at least one BTL Interoperability Workshop, have demonstrated compliance to BTL through BTL listing and shall substantially conform to BACnet Building Controller (B-BC) device profile as specified in ANSI/ASHRAE 135-2004 or ANSI/ASHRAE 135-2008.
6. Control product shall support the following BACnet Interoperability Building Blocks to facilitate an open and interoperable system:

BIBB	Name	Initiat	Execute
Data Sharing			
DS-RP-A	Data Sharing-ReadProperty-A	☐	
DS-RP-B	Data Sharing-ReadProperty-B		☐

BIBB	Name	Initiat	Execute
DS-RPM-A	Data Sharing-ReadPropertyMultiple-A	☐	
DS-RPM-B	Data Sharing-ReadPropertyMultiple-B		☐
DS-WP-A	Data Sharing-WriteProperty-A	☐	
DS-WP-B	Data Sharing-WriteProperty-B		☐
DS-WPM-B	Data Sharing-WritePropertyMultiple-B		☐
DS-COV-A	Data Sharing-COV-A	☐	
DS-COV-B	Data Sharing-COV-B		☐
DS-COVU-A	Data Sharing-COV-Unsolicited-A	☐	
DS-COVU-B	Data Sharing-COV-Unsolicited-B		☐
Scheduling			
SCHED-I-B	Scheduling-Internal-B		☐
SCHED-E-B	Scheduling-External-B		☐
Alarm and Event Management			
AE-N-A	Alarm and Event-Notification-A	☐	
AE-N-I-B	Alarm and Event-Notification Internal-B		☐
AE-N-E-B	Alarm and Event-Notification External-B		☐
AE-ACK-A	Alarm and Event-ACK-A	☐	
AE-ACK-B	Alarm and Event- ACK-B		☐
AE-ASUM-B	Alarm and Event-Alarm Summary-B		☐
AE-ESUM-A	Alarm and Event-Enrollment Summary-A	☐	
AE-ESUM-B	Alarm and Event-Enrollment Summary-B		☐
AE-INFO-A	Alarm and Event-Information-A	☐	
AE-INFO-B	Alarm and Event-Information-B		☐

BIBB	Name	Initiat	Execute
Trending			
T-VMT-A	Trending-Viewing and Modifying Trends-A	☐	
T-VMT-I-B	Trending-Viewing and Modifying Trends-Internal-B		☐
T-VMT-E-B	Trending-Viewing and Modifying Trends-External-B		☐
T-ATR-B	Trending-Automated Trend Retrieval-B		☐
Network Management			
NM-CE-A	Network Management-Connection Establishment-A	☐	
Device Management			
DM-DDB-A	Device Management-Dynamic Device Binding-A	☐	
DM-DDB-B	Device Management-Dynamic Device Binding-B		☐
DM-DOB-A	Device Management-Dynamic Object Binding-A	☐	
DM-DOB-B	Device Management-Dynamic Object Binding-B		☐
DM-DCC-B	Device Management-DeviceCommunicationControl-B		☐
DM-PT-A	Device Management-Private Transfer-A	☐	
DM-PT-B	Device Management-Private Transfer-B		☐
DM-TM-A	Device Management-Text Message-A	☐	
DM-TM-B	Device Management-Text Message-B		☐
DM-TS-B	Device Management-TimeSynchronization-B		☐
DM-RD-B	Device Management-ReinitializeDevice-B		☐
DM-BR-B	Device Management-Backup and Restore-B		☐
DM-LM-B	Device Management-List Manipulation-B		☐
DM-OCD-B	Device Management-Object Creation and Deletion-B		☐

7. Building Controllers shall have the ability to perform energy management routines including but not limited to time of day scheduling, calendar-based scheduling, holiday scheduling, temporary schedule overrides, start stop time optimization, automatic daylight savings time switch over, night

setback control, enthalpy switch over, peak demand limiting, temperature-compensated duty cycling, heating/cooling interlock, supply temperature reset, priority load shedding, and power failure restart.

8. The Building Controllers shall have the ability to perform the following pre tested control algorithms:
 - a. Two position control
 - b. Proportional control
 - c. Proportional plus integral control
 - d. Proportional, integral, plus derivative control
 - e. Automatic tuning of control loops
 - f. Model-free adaptive control
9. Each controller shall be provided with an interactive HELP function to assist operators using POTs and remote connected operators.
10. Building Controllers shall not be susceptible to Microsoft Windows operating systems based viruses.

B. System Security

1. User access shall be secured using individual security passwords and user names.
2. Passwords shall restrict the user to the objects, applications, and system functions as assigned by the system manager.
3. User Log On/Log Off attempts shall be recorded.
4. The system shall protect itself from unauthorized use by automatically logging off following the last keystroke. The delay time shall be user-definable.
5. Use of workstation resident security as the only means of access control is not an acceptable alternative to resident system security in the field panel.

C. User Defined Control Applications:

1. Controllers shall be able to execute custom, job-specific processes defined by the user, to automatically perform calculations and special control routines.
2. It shall be possible to use any system measured point data or status, any system calculated data, a result from any process, or any user-defined constant in any controller in the system.
3. Any process shall be able to issue commands to points in any and all other controllers in the system.
4. Processes shall be able to generate operator messages and advisories to other operator I/O devices. A process shall be able to directly send a message to a specified device or cause the execution of a dial-up connection to a remote device such as a printer or pager.
5. Each controller shall support plain language text comment lines in the operating program to allow for quick troubleshooting, documentation, and historical summaries of program development.
6. Controller shall provide a HELP function key, providing enhanced context sensitive on-line help with task oriented information from the user manual.

D. Alarm Management:

1. Alarm management shall be provided to monitor and direct alarm information to operator devices. Each Building Controller shall perform distributed, independent alarm analysis, minimize network traffic and prevent alarms from being lost. At no time shall the Building Controllers ability to report

alarms be affected by either operator or activity at a PC workstation, local I/O device or communications with other panels on the network.

2. Conditional alarming shall allow generation of alarms based upon user defined multiple criteria.
3. An Alarm "shelving" feature shall be provided to disable alarms during testing. (Pull the Plug, etc.).
4. Binary Alarms. Each binary object shall be set to alarm based on the operator-specified state. Provide the capability to automatically and manually disable alarming.
5. Analog Alarms. Each analog object shall have both high and low alarm limits. Alarming must be able to be automatically and manually disabled.
6. All alarm or point change reports shall include the point's user-defined language description and the time and date of occurrence.
7. Alarm reports and messages shall be routed to user-defined list of operator workstations, or other devices based on time and other conditions. An alarm shall be able to start programs, print, be logged in the event log, generate custom messages, and display graphics.
8. In addition to the point's descriptor and the time and date, the user shall be able to print, display or store a 200-character alarm message to more fully describe the alarm condition or direct operator response.
9. Each Building Controller shall be capable of storing a library of at least 50 alarm messages. Each message may be assigned to any number of points in the Controller.
10. Operator-selected alarms shall be capable of initiating a call to a remote operator device.

E. Scheduling:

1. Provide a comprehensive menu driven program to automatically start and stop designated multiple objects or events in the system according to a stored time.
2. Schedules shall reside in the building controller and shall not rely on external processing or network.
3. For points assigned to one common load group, it shall be possible to assign variable time delays between each successive start and/or stop within that group.
4. The operator shall be able to define the following information:
 - a. Time, day
 - b. Commands such as on, off, auto, etc.
 - c. Time delays between successive commands.
 - d. There shall be provisions for manual overriding of each schedule by an authorized operator.
5. It shall be possible to schedule calendar-based events up to one year in advance based on the following:
 - a. Weekly Schedule. Provide separate schedules for each day of the week. Each of these schedules should include the capability for start, and stop.
 - b. Exception Schedules. Provide the ability for the operator to designate any day of the year as an exception schedule. Exception schedules may be defined up to a year in advance. Once an exception schedule is executed, it will be discarded and replaced by the standard schedule for that day of the week.

F. Peak Demand Limiting (PDL):

1. The Peak Demand Limiting (PDL) program shall limit the consumption of electricity to prevent electrical peak demand charges.
 2. PDL shall continuously track the amount of electricity being consumed, by monitoring one or more electrical kilowatt-hour/demand meters. These meters may measure the electrical consumption (kWh), electrical demand (kW), or both.
 3. PDL shall sample the meter data to continuously forecast the demand likely to be used during successive time intervals.
 4. If the PDL forecasted demand indicates that electricity usage is likely to exceed a user preset maximum allowable level, then PDL shall automatically shed electrical loads.
 5. Once the demand peak has passed, loads that have been shed shall be restored and returned to normal control.
- G. Temperature-compensated duty cycling.
- H. The DCCP (Duty Cycle Control Program) shall periodically stop and start loads according to various patterns.
- I. The loads shall be cycled such that there is a net reduction in both the electrical demands and the energy consumed.
- J. Automatic Daylight Savings Time Switchover. The system shall provide automatic time adjustment for switching to/from Daylight Savings Time.
- K. Night setback control. The system shall provide the ability to automatically adjust setpoints for night control.
- L. Enthalpy switchover (economizer). The Building Controller Software (BCS) shall control the position of the air handler relief, return, and outside air dampers. If the outside air dry bulb temperature falls below changeover setpoint the BCS will modulate the dampers to provide 100 percent outside air. The user will be able to quickly change over to an economizer system based on dry bulb temperature and will be able to override the economizer cycle and return to minimum outside air operation at any time.
- M. Loop Control. A Model-Free Adaptive Control algorithm or alternatively a PID (proportional-integral-derivative) closed-loop control algorithm with direct or reverse action and anti-windup shall be supplied. The algorithm shall calculate a time-varying analog value that is used to position an output or stage a series of outputs. The controlled variable, setpoint, and weighting parameters shall be user-selectable.
- N. Sequencing. Provide application software based upon the sequences of operation specified to properly sequence equipment.
- O. Staggered Start:
1. This application shall prevent all controlled equipment from simultaneously restarting after a power outage. The order in which equipment (or groups of equipment) is started, along with the time delay between starts, shall be user definable.
 2. Upon the resumption of power, each Building Controller shall analyze the status of all controlled equipment, compare it with normal occupancy scheduling and turn equipment on or off as necessary to resume normal operations.
- P. Totalization:
1. Run-Time Totalization. Building Controllers shall automatically accumulate and store run-time hours for all digital input and output points. A high runtime alarm shall be assigned, if required, by the operator.

2. Consumption totalization. Building Controllers shall automatically sample, calculate and store consumption totals on a daily, weekly or monthly basis for all analog and digital pulse input type points.
3. Event totalization. Building Controllers shall have the ability to count events such as the number of times a pump or fan system is cycled on and off. Event totalization shall be performed on a daily, weekly or monthly basis for all points. The event totalization feature shall be able to store the records associated with events before reset.

Q. Data Collection:

1. A variety of historical data collection utilities shall be provided to manually or automatically sample, store, and display system data for all points.
2. Building Controllers shall store point history data for selected analog and digital inputs and outputs:
 - a. Any point, physical or calculated may be designated for trending. Any point, regardless of physical location in the network, may be collected and stored in each Building Controllers point group.
3. Trend data shall be stored at the Building Controllers and uploaded to the workstation when retrieval is desired. Uploads shall occur based upon either user-defined interval, manual command or when the trend buffers are full. All trend data shall be available for use in third-party personal computer applications
4. Loop Tuning. Building Controllers shall also provide high resolution sampling capability
 - a. For Model-Free Adaptive Control loops, evidence of tuned control loop performance shall be provided via graphical plots or trended data logs. Graphical plots shall minimally include depictions of setpoint, process variable (output), and control variable (e.g., temperature). Other parameters that may influence loop control shall also be included in the plot (e.g., fan on/off, mixed-air temp).
 - b. For PID control loops, operator-initiated automatic and manual loop tuning algorithms shall be provided for all operator-selected PID control loops. Evidence of tuned control loop performance shall be provided via graphical plots or trended data logs for all loops.
 - 1) In automatic mode, the controller shall perform a step response test with a minimum one-second resolution, evaluate the trend data, calculate the new PID gains and input these values into the selected LOOP statement.
 - 2) Loop tuning shall be capable of being initiated either locally at the Building Controller, from a network workstation or remotely using dial-in modems. For all loop tuning functions, access shall be limited to authorized personnel through password protection.

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2.6 ADVANCED APPLICATION CONTROLLERS (AAC)

- A. Advanced Application Controllers shall be 32-bit, multi-tasking, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers, power supplies and input/output point modules. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
- B. Each Advanced Application Controller shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, and operator I/O.
- C. Provide Universal I/O capability, including software configurable universal inputs and universal outputs
- D. Advanced Application Controller shall have an integral real-time clock.

- E. Each Advanced Application Controller shall support firmware upgrades without the need to change hardware.
- F. Each Advanced Application Controller shall support:
 - 1. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the controller cabinet.
 - 2. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the controller cabinet.
- G. Spare Point Capacity. Each Advanced Application Controller shall have a minimum of 10 percent spare point capacity.
 - 1. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
 - 2. Provide all processors, power supplies, so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - a. Sensor/actuator
 - b. Field wiring/tubing
- H. Serial Communication. Advanced Application Controllers shall provide at least one EIA-232C serial data communication ports for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. Advanced Application Controllers shall allow temporary use of portable devices without interrupting the normal operation of permanently connected printers or terminals. A USB port shall alternatively be available to support local HMI tools connection.
- I. Printer support. A USB port shall be available to support printers for printed alarm record keeping.
- J. Manual Override. The operator shall have the ability to manually override automatic or centrally executed commands at the controller via local, point discrete, integral hand/off/auto operator override switches for all digital control type points and gradual switches for all analog control type points. These override switches shall be operable whether the panel processor is operational or not. Each controller shall monitor and alarm the hand, off and auto positions of integral HOA switches.
- K. I/O Status and Indication. Advanced Application Controllers shall provide local LED status indication for each digital input and output for constant, up-to-date verification of all point conditions without the need for an operator I/O device. Graduated intensity LEDs or analog indication of value shall also be provided for each analog output.
- L. Self Diagnostics. Each Advanced Application Controller shall continuously perform self diagnostics, communication diagnosis, and diagnosis of all panel components. The controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
- M. Power loss. In the event of the loss of power, there shall be an orderly shutdown of all controllers to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 30 days.
- N. Environment.
 - 1. Controller hardware shall be suitable for the anticipated ambient conditions.
 - 2. Controllers used outdoors and/or in wet ambient conditions shall be mounted within waterproof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).

3. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
- O. Immunity to power and noise.
1. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 2. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).
 3. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - a. RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3V.
 - b. Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact.
 - c. Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500V signal, 1 kV power.
 - d. Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max).
 4. Isolation shall be provided at all Advanced Application Controller's AC input terminals to suppress induced voltage transients consistent with:
 - a. IEEE Standard 587 1980
 - b. Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000-4-11)

2.7 ADVANCED APPLICATION CONTROLLER SOFTWARE

A. General:

1. Furnish the following applications software to form a complete operating system for building and energy management as described in this specification.
2. The software programs specified in this section shall be provided as an integral part of Advanced Application Controllers and shall not be dependent upon any higher level computer or another controller for execution.
3. The Advanced Application Controller Software shall be capable of BACnet communications. The BACnet Advanced Application Controller (B-AAC) shall have demonstrated compliance to BTL through BTL listing and shall substantially conform to BACnet Advanced Application Controller (B-AAC) device profile as specified in ANSI/ASHRAE 135-2004 or ANSI/ASHRAE 135-2008.
4. Control product shall support the following BACnet Interoperability Building Blocks to facilitate an open and interoperable system:
5. Advanced Application Controllers shall have the ability to perform energy management routines including but not limited to scheduling, calendar-based scheduling, holiday scheduling, temporary schedule overrides, automatic daylight savings time switch over, night setback control, enthalpy switch over, peak demand limiting, temperature-compensated duty cycling, heating/cooling interlock, supply temperature reset, priority load shedding, and power failure restart.
6. The Advanced Application Controllers shall have the ability to perform the following pre tested control algorithms:
 - a. Two position control
 - b. Proportional control

- c. Proportional plus integral control
- d. Proportional, integral, plus derivative control
- e. Automatic tuning of control loops
- f. Model-free adaptive control

B. System Security

- 1. User access shall be secured using individual security passwords and user names.
- 2. Passwords shall restrict the user to the objects, applications, and system functions as assigned by the system manager.
- 3. The system shall protect itself from unauthorized use by automatically logging off following the last keystroke. The delay time shall be user-definable.
- 4. Use of workstation resident security as the only means of access control is not an acceptable alternative to resident system security in the field panel.

C. User Defined Control Applications:

- 1. Controllers shall be able to execute custom, job-specific processes defined by the user, to automatically perform calculations and special control routines.
- 2. Each controller shall support plain language text comment lines in the operating program to allow for quick troubleshooting, documentation, and historical summaries of program development.
- 3. Controller shall provide a HELP function key, providing enhanced context sensitive on-line help with task oriented information from the user manual.

2.8 APPLICATION SPECIFIC CONTROLLERS (ASC)

A. General:

- 1. Provide for control of each piece of equipment , including, but not limited to the following:
 - a. Variable Air Volume (VAV) boxes
 - b. Constant Air Volume (CAV) boxes
 - c. Room Pressurization
- 2. Each Building Controller shall be able to communicate with application specific controllers (ASCs) over the Secondary Network to control terminal equipment only.
- 3. The use of Secondary Network controllers with custom program applications to control AHU's, water systems, etc. is not acceptable.
- 4. Each ASC shall operate as a stand-alone controller capable of performing its specified control responsibilities independently of other controllers in the network. Each ASC shall provide standard applications and programmability to provide both reliability and flexibility. Each ASC shall be a microprocessor-based, multi-tasking, real-time digital control processor.
- 5. The Application Specific Controller Software shall be capable of BACnet communications. The BACnet Application Specific Controller (B-ASC) shall have demonstrated interoperability during at least one BTL Interoperability Workshop, have demonstrated compliance to BTL through BTL listing and shall substantially conform to BACnet Application Specific Controller (B-ASC) device profile as specified in ANSI/ASHRAE 135-2004.

6. Control product shall support the following BACnet Interoperability Building Blocks to facilitate an open and interoperable system:

BIBB	Name	Tested
DS-RP-B	Data Sharing-ReadProperty-B	●
DS-RPM-B	Data Sharing-ReadPropertyMultiple-B	●
DS-WP-B	Data Sharing-WriteProperty-B	●
DM-DDB-B	Device Management-Dynamic Device Binding-B	●
DM-DOB-B	Device Management-Dynamic Object Binding-B	●
DM-DDC-B	Device Management-DeviceCommunicationControl-B	●

7. Each ASC shall include all point inputs and outputs necessary to perform the specified control sequences. The ASC shall accept input and provide output signals that comply with industry standards. Controllers utilizing proprietary control output signals shall not be acceptable. Outputs utilized either for two-state, modulating floating, or proportional control, allowing for additional system flexibility.
8. Space Temperature Sensors. Each controller performing space temperature control shall be provided with a matching room temperature sensor.
- Wired temperature sensor specifications. The sensing element for the space temperature sensor must be IC-based and provide the following.
 - Digitally communicating with the Application Specific Controller.
 - Mountable to and fully covering a 2 x 4 electrical junction box without the need for an adapter wall plate.
 - IC Element Accuracy: +/- 0.9°F
 - Operating Range: 55 to 95°F
 - Setpoint Adjustment Range: User limiting, selectable range
 - Display of temperature setpoint with numerical temperature values.
 - Calibration: Single point, field adjustable at the space
 - Installation: Up to 100 ft. from controller
 - Auxiliary Communications Port: included
 - Local OLED Temperature Display: included
 - Display of Temperature to one decimal place
 - Temperature Setpoint Adjustment included
 - Occupancy Override Function included
 - Setpoint Modes:
 - Independent Heating, Cooling
 - Night Setback-Heating
 - Night Setback-Cooling
 - Auxiliary Communication Port. Each room temperature sensor shall include a terminal jack integral to the sensor assembly. The terminal jack shall be used to connect a portable operator's terminal to control and monitor all hardware and software points associated with the controller. RS-232 communications port shall allow the operator to query and modify operating parameters of the local room terminal unit from the portable operator's terminal.
 - LCD Display. Interactive, two-line liquid crystal display shall allow the operator to query and modify operating parameters of the local room terminal unit from the room sensor. The display shall indicate the space temperature and associated ASC point when not being used to query or modify operating parameters.
 - Setpoint Adjustment Dial. The setpoint adjustment function shall allow for modification of the temperature by the building operators. Setpoint adjustment may be locked out, overridden, or limited as to time or temperature through software by an authorized operator at any central

workstation, Building Controller, room sensor two-line display, or via the portable operator's terminal.

- f. **Override Switch.** An override button shall initiate override of the night setback mode to normal (day) operation when activated by the occupant and enabled by building operators. The override shall be limited to two (2) hours (adjustable.) The override function may be locked out, overridden, or limited through software by an authorized operator at the operator interface, Building Controller, room sensor two-line display or via the portable operator's terminal.
9. **Space Combination Carbon Dioxide (CO₂)/Temperature or Carbon Dioxide/Temperature/Relative Humidity Sensors.** Each controller performing space temperature, relative humidity or CO₂ control shall be provided with a single room sensor, which includes the ability to measure CO₂ and temperature or CO₂, temperature and relative humidity for either monitoring or control purposes. The combination sensors shall have the same appearance as the space temperature sensors. Room sensing devices shall be digitally communicating with the zone controller, and multiple sensed values (CO₂, RH) should not utilize additional analog inputs on the controller. CO₂ elements shall be of the non-dispersive infrared (NDIR) technology type, and shall measure over the range of 0 to 2000 parts per million (PPM), with an accuracy of +/- 50PPM + 2% of reading. Where display is needed, display shall be of the organic light emitting diode (OLED) type. Room sensor shall be capable of showing or hiding any displayed value. Measured values (CO₂, humidity and temperature) shall be capable of being field calibrated at a single point, as a one-point bias/offset calibration, without the use of additional calibration software. Humidity elements shall measure relative humidity with a +/- 2% accuracy over the range of 10 to 90% relative humidity. Humidity element shall be an IC (integrated circuit) sensing element. Humidity sensing elements shall be removable and field replaceable if needed.
10. **Communication.** Each controller shall perform its primary control function independent of other Secondary Network communication, or if Secondary Network communication is interrupted. Reversion to a fail-safe mode of operation during Secondary Network interruption is not acceptable.
11. **Control Algorithms.** The controller shall receive its real-time data from the Building Controller time clock to ensure Secondary Network continuity. Each controller shall include algorithms incorporating proportional, integral and derivative (PID) gains for all applications. All PID gains and biases shall be field-adjustable by the user via room sensor LCD or the portable operator's terminal as specified herein. Controllers that incorporate proportional and integral (PI) control algorithms only shall not be acceptable.
12. **Control Applications.** Operating programs shall be field-selectable for specific applications. In addition, specific applications may be modified to meet the user's exact control strategy requirements, allowing for additional system flexibility. Controllers that require factory changes of all applications are not acceptable.
13. **Programmability.** Application Specific Controllers shall be programmable, using software provided by the BMS manufacturer. Software shall be field-installable on any standard laptop or Portable Operator's Terminal. Program language shall be text-based and allow up to 200 lines of code for programming. Programming shall allow for changing sequence of operation, commanding and releasing points, additional monitoring, and command priority management within the Application Specific Controller.
14. **Calibration.** Each controller shall include provisions for manual and automatic calibration of the differential pressure transducer in order to maintain stable control and ensuring against drift over time.
 - a. Manual calibration may be accomplished by either commanding the actuator to 0% via the POT or by depressing the room sensor override switch. Calibration of the transducer at the controller location shall not be necessary.
 - b. Calibration shall be accomplished by stroking the terminal unit damper actuator to a 0% position so that a 0 cfm air volume reading is sensed. The controller shall automatically

accomplish this whenever the system mode switches from occupied to unoccupied or vice versa.

- c. Calibration shall be accomplished by zeroing out the pressure sensor and holding damper at last known position until calibration is complete. The controller shall automatically accomplish this whenever the system mode switches from occupied to unoccupied or vice versa.

15. Memory.

- a. Provide each ASC with sufficient memory to accommodate point databases, operating programs, programming capability, local alarming and local trending. All databases and programs shall be stored in non-volatile EEPROM, EPROM and PROM, or minimum of 72-hour battery backup shall be provided. The controllers shall be able to return to full normal operation without user intervention after a power failure of unlimited duration.
- b. Upon replacement, new ASCs shall recover control function and site specific defaults automatically and resume normal operation.

16. Power Supply. The ASCs shall be powered from a 24 Vac source and shall function normally under an operating range of 18 to 28 Vac, allowing for power source fluctuations and voltage drops. Power supply for the ASC must be rated at a minimum of 125% of ASC power consumption and shall be of the fused or current limiting type. The BMS contractor shall provide 24 Vac power to the terminal units by utilizing:

- a. The existing line voltage power trunk and installing separate isolation transformers for each controller.
- b. Dedicated line voltage power source and isolation transformers at a central location and installing 24 Vac power trunk to supply multiple ASCs in the area.

17. Environment. The controllers shall function normally under ambient conditions of 32 to 122°F (0 to 50°C) and 10% to 95% rh (non-condensing). Provide each controller with a suitable cover or enclosure to protect the circuit board assembly.

18. Immunity to noise. Operation shall be protected against electrical noise of 5-120 Hz and from keyed radios up to 5 W at 1 m (3 ft).

19. Manufacturer Installed Controls.

- a. BMS manufacturer shall furnish ASC and actuator for factory mounting to equipment manufacturer.
- b. Cost of factory mounting shall be borne by equipment manufacturer.
- c. For VAV terminals, equipment manufacturer shall provide and install flow-cross sensor, 24 Vac transformer, controls enclosure, fan relay, SCR and factory install, wire and tube ASC controller and actuator.
- d. Fan powered VAV terminals shall be equipped with a fan speed controller and relay to change summer and winter speed setpoint.

B. Controllers for VAV terminals.

- 1. All VAV terminal control applications shall be field selectable such that a single controller may be used in conjunction with any of the above types of terminal units to perform the specified sequences of control. . In addition to the field selectable applications, additional programming flexibility to meet sequences of control is required in all ASCs. ASCs that require factory application changes are not acceptable. The VAV terminal ASC shall support the following types of pressure independent terminal boxes as a minimum:

- a. VAV cooling only

- b. VAV with hot water or electric reheat
 - c. Fan-powered VAV
 - d. Fan-powered VAV with hot water or electric reheat
 - 2. VAV applications shall be able to monitor relative space humidity and CO2.
 - 3. The controller shall include a differential pressure transducer that shall connect to the terminal unit manufacturer's standard averaging air velocity sensor to measure the average differential pressure in the duct. The controller shall convert this value to actual airflow. Single point air velocity sensing is not acceptable. The differential pressure transducer shall have a measurement range of 0 to 4000 fpm (0 to 20.4 m/s) and measurement accuracy of +5% at 400 to 4000 fpm (2 to 20 m/s), insuring primary airflow conditions shall be controlled and maintained to within +5% of setpoint at the specified parameters. The BMS contractor shall provide the velocity sensor if required to meet the specified functionality.
- C. Controllers for CV terminals.
- 1. Constant volume ASCs shall meet all requirements of paragraphs as previously specified for VAV terminals. The controllers shall have a minimum and maximum flow setpoint, which shall be selected based on interior ventilation requirements. Under normal conditions, the setpoint shall be set to minimum setpoint. When the floor area requires additional ventilation (high CO2 level, manual command, etc.) the setpoint shall be set to maximum setpoint.
 - 2. Constant volume applications shall be able to monitor relative space humidity and CO2.
- D. LABORATORY CONTROLS
- 1. FUME HOOD CONTROLLER (FHC)
 - a. The DDC controller shall control Variable Air Volume laboratory fume hoods.
 - b. The controller will maintain constant face velocity as the sash is raised and lowered.
 - c. An operator display panel will provide the user the operating status of the hood, alarm horn, and emergency purge function.
 - d. The hood controller will interface with the energy management system.
 - 2. BIO-SAFETY CABINET CONTROLLER (BSCC)
 - a. Ducted bio-safety cabinets will be controlled to a constant CFM specified by the cabinet's manufacturer.
 - b. Alarms and flows will be reported to the EMS.
 - 3. ROOM PRESSURE CONTROLLER (RPC)
 - a. The DDC controller will provide room pressurization and temperature control to rooms or labs without fume hoods where positive or negative room pressure is critical.
 - b. The RPC will maintain differential flow between the supply and exhaust terminal boxes.
 - c. The Room Pressurization Controller shall operate as a networked component of the EMS.
 - 4. LABORATORY ROOM CONTROLLER (LRC)
 - a. The DDC controller will provide control for laboratory space pressurization and temperature control with control sequences for both single duct and dual duct supply systems.

- b. The controller will maintain user defined differential airflow between lab supply air and fume hood controller exhaust and general exhaust terminals by measuring the airflow and controlling the damper position of the supply and general exhaust terminals.
 - c. Lab temperature control will be maintained by measuring the room temperature and controlling the reheat valve and adjusting the air flow.
 - d. Closed loop PID control will be used to maintain tighter air volume and temperature control.
 - e. All air flows will be reported in CFM's from physical air flow stations or flow sensors and not calculated base on valve or damper position.
 - f. The Laboratory Room Controller shall operate as a networked component of the EMS. The LRC shall communicate differential pressure values, air flow values, temperature values and alarm status.
5. DIFFERENTIAL PRESSURE MONITOR (DPM)
- a. The monitor measures and displays the differential pressure between a room and its adjoining space.
 - b. The Differential Pressure Monitor (DPM) shall operate as a networked component of the LCS. The DPM shall communicate alarm status, differential pressure values, and door status. The alarm setpoint shall be adjustable from the LCS Workstation.
 - c. The Differential Pressure Transmitter shall have an accuracy of ± 0.001 inches of water over a range of - 0.2 to + 0.2 inches of water.
6. CAV FUME HOOD MONITOR
- a. The DDC Fume Hood Monitor shall continuously monitor the fume hood face velocity
 - b. Monitor shall be mounted to accurately monitor the face velocity.C
 - c. Monitor will be visible to the fume hood user and display the actual face velocity.
 - d. Monitor will issue an audible alarm when the face velocity is below operating parameters.
 - e. The monitor will connect to the EMS system so fume hood face velocity and alarms can be monitored remotely

2.9 CONTROL CABLE

- A. Electronic and fiber-optic cables for control wiring are specified in Division 27 Section "Communications Horizontal Cabling."

2.10 LOCAL CONTROL PANELS

- A. Provide control panels with suitable brackets for wall mounting, for each miscellaneous control system. Locate panel adjacent to systems served.
- B. Fabricate panels of 14-gauge furniture-grade steel, or 6063-T5 extruded aluminum alloy, totally enclosed, with hinged doors and keyed lock, with manufacturer's standard shop-painted finish and color. Provide UL listed cabinets for use with line voltage devices.
- C. Panel Mounted Equipment: Include temperature controllers, relays, and other devices excluded in the sequence of operation. Mount devices with adjustments accessible through the fronts of panels.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that power supply is available to control units and OWS.

3.2 COORDINATION

- A. Coordinate location of temperature sensors, humidistats, and other exposed control sensors with plans and room finish details before installation.
- B. Coordinate equipment with Division 28 Section "Intrusion Detection" to achieve compatibility with equipment that interfaces with that system and with building master clock.
- C. Coordinate equipment with Division 28 Section "Access Control" to achieve compatibility with equipment that interfaces with that system.
- D. Coordinate equipment with Division 28 Section "PLC Electronic Detention Monitoring and Control Systems" to achieve compatibility with equipment that interfaces with that system.
- E. Coordinate equipment with Division 26 Section "Network Lighting Controls" to achieve compatibility with equipment that interfaces with that system.
- F. Coordinate equipment with Division 28 Section "Fire Detection and Alarm" to achieve compatibility with equipment that interfaces with that system.
- G. Coordinate supply of conditioned electrical branch circuits for control units and OWS.
- H. Coordinate equipment with Division 26 Section "Electrical Power Monitoring and Control" to achieve compatibility of communication interfaces.
- I. Coordinate equipment with Division 26 Section "Panelboards" to achieve compatibility with starter coils and annunciation devices.
- J. Coordinate equipment with Division 26 Section "Motor-Control Centers" to achieve compatibility with motor starters and annunciation devices.
- K. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03 Section "Cast-in-Place Concrete."

3.3 INSTALLATION

- A. Install software in control units and OWS. Implement all features of programs to specified requirements and as appropriate to achieve sequence of operations.
- B. Connect and configure equipment and software to achieve sequence of operations specified.
- C. Mount all wall thermostats, humidistats, and other exposed control sensors on dedicated electrical backboxes.
- D. Install averaging elements in ducts and plenums in crossing or zigzag pattern.
- E. Install automatic dampers according to Division 23 Section "Air Duct Accessories."
- F. Install damper motors on outside of duct in warm areas, not in locations exposed to outdoor temperatures.

- G. Install labels and nameplates to identify control components according to Division 23 Section "Identification for HVAC Piping and Equipment."
- H. Install hydronic instrument wells, valves, and other accessories according to Division 23 Section "Hydronic Piping."
- I. Install refrigerant instrument wells, valves, and other accessories according to Division 23 Section "Refrigerant Piping."
- J. Install duct volume-control dampers according to Division 23 Sections specifying air ducts.
- K. Install electronic and fiber-optic cables according to Division 27 Section "Communications Horizontal Cabling."

3.4 ELECTRICAL WIRING AND CONNECTION INSTALLATION

- A. Install systems and materials in accordance with manufacturer's instructions, rough-in drawings and equipment details. Install electrical components and use electrical products complying with requirements of applicable Division 26, 27 & 28 Sections of these Specifications except where specifically stated in this Section.
- B. The term "control wiring" is defined to include providing of wire, conduit, and miscellaneous material as required for mounting and connecting electric or electronic control devices.
- C. Install all control wiring in conduit for electric/electronic control systems. Conceal wiring, except in mechanical rooms and areas where other conduit and piping are exposed. UL plenum rated cable shall be allowed above accessible lift out ceiling, in air plenums, and in other areas as approved by Architect and local and NEC codes.
- D. Stub conduit to above lift out ceilings. Plastic bushing shall be installed where the sensor wire exits the conduit to prevent damage.
- E. Number-code or color-code conductors, excluding those used for individual zone controls, appropriately for future identification and servicing of control system.
- F. This section shall provide all line voltage power wiring required because of substitution of equipment specified in this section.
- G. Install raceways, boxes, and cabinets according to Division 26 Section "Raceway and Boxes for Electrical Systems."
- H. Install building wire and cable according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."
- I. Install signal and communication cable according to Division 27 Section "Communications Horizontal Cabling."
 - 1. Bundle and harness multi-conductor instrument cable in place of single cables where several cables follow a common path.
 - 2. Fasten flexible conductors, bridging cabinets and doors, along hinge side; protect against abrasion. Tie and support conductors.
 - 3. Number-code or color-code conductors for future identification and service of control system, except local individual room control cables.

- 4. Install wire and cable with sufficient slack and flexible connections to allow for vibration of piping and equipment.
- J. Connect manual-reset limit controls independent of manual-control switch positions. Automatic duct heater resets may be connected in interlock circuit of power controllers.
- K. Connect hand-off-auto selector switches to override automatic interlock controls when switch is in hand position.
- L. Serve only one DDC controller from any 24V control power transformer.

3.5 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including connections. Report results in writing.
- B. Perform the following field tests and inspections and prepare test reports:
 - 1. Operational Test: After electrical circuitry has been energized, start units to confirm proper unit operation. Remove and replace malfunctioning units and retest
 - 2. Test and adjust controls and safeties.
 - 3. Test each point through its full operating range to verify that safety and operating control set points are as required.
 - 4. Test each control loop to verify stable mode of operation and compliance with sequence of operation. Adjust PID actions.
 - 5. Test each system for compliance with sequence of operation.
 - 6. Test software and hardware interlocks.
- C. DDC Verification:
 - 1. Verify that instruments are installed before calibration, testing, and loop checks.
 - 2. Check instruments for proper location and accessibility.
 - 3. Check instrument installation for direction of flow, elevation, orientation, insertion depth, and other applicable considerations.
 - 4. Check flow instruments. Inspect tag number and line and bore size, and verify that inlet side is identified and that meters are installed correctly.
 - 5. Check pressure instruments, piping slope, installation of valve manifold, and self-contained pressure regulators.
 - 6. Check temperature instruments and material and length of sensing elements.
 - 7. Check control valves. Verify that they are operating in the correct direction.
 - 8. Check dampers. Verify that proper blade alignment, either parallel or opposed, has been provided.
 - 9. Check DDC system as follows:
 - a. Verify that DDC controller power supply is from emergency power supply, if applicable.
 - b. Verify that wires at control panels are tagged with their service designation and approved tagging system.

- c. Verify that spare I/O capacity has been provided.
- d. Verify that DDC controllers are protected from power supply surges.
- 10. Replace damaged or malfunctioning controls and equipment and repeat testing procedures.

3.6 ADJUSTING

A. Calibrating and Adjusting:

- 1. Calibrate instruments.
- 2. Make three-point calibration test for both linearity and accuracy for each analog instrument.
- 3. Calibrate equipment and procedures using manufacturer's written recommendations and instruction manuals. Use test equipment with accuracy at least double that of instrument being calibrated.
- 4. Control System Inputs and Outputs:
 - a. Check digital inputs using jumper wire.
 - b. Check digital outputs using ohmmeter to test for contact making or breaking.
 - c. Check resistance temperature inputs at 0, 50, and 100 percent of span using a precision-resistant source.
- 5. Flow:
 - a. Set differential pressure flow transmitters for 0 and 100 percent values with 3-point calibration accomplished at 50, 90, and 100 percent of span.
 - b. Manually operate flow switches to verify that they make or break contact.
- 6. Pressure:
 - a. Calibrate pressure transmitters at 0, 50, and 100 percent of span.
 - b. Calibrate pressure switches to make or break contacts, with adjustable differential set at minimum.
- 7. Temperature:
 - a. Calibrate resistance temperature transmitters at 0, 50, and 100 percent of span using a precision-resistance source.
 - b. Calibrate temperature switches to make or break contacts.
- 8. Stroke and adjust control valves and dampers without positioners, following the manufacturer's recommended procedure, so that valve or damper is 100 percent open and closed.
- 9. Stroke and adjust control valves and dampers with positioners, following manufacturer's recommended procedure, so that valve and damper is 0, 50, and 100 percent closed.
- 10. Provide diagnostic and test instruments for calibration and adjustment of system.
- 11. Provide written description of procedures and equipment for calibrating each type of instrument. Submit procedures review and approval before initiating startup procedures.

B. Adjust initial temperature and humidity set points.

3.7 SYSTEM EXCEPTANCE

- A. General: The system installation shall be complete and tested for proper operation prior to acceptance testing for the Owner's authorized representative. A letter shall be submitted to the Architect requesting system acceptance. This letter shall certify all controls are installed and the software programs have been completely exercised for proper equipment operation. Acceptance testing will commence at a mutually agreeable time within ten (10) calendar days of request. When the field test procedures have been demonstrated to the Owner's representative, the system will be accepted. The warranty period will start at this time.
- B. Field Equipment Test Procedures: DDC control panels shall be demonstrated via a functional end-to-end test. Such that:
 - 1. All output channels shall be commanded (on/off, stop/start, adjust, etc.) and their operation verified.
 - 2. All analog input channels shall be verified for proper operation.
 - 3. Changing the state of the field device and observing the appropriate change of displayed value shall verify all digital input channels.
 - 4. If a point should fail testing, perform necessary repair action and retest failed point and all interlocked points.
 - 5. Introducing an error into the system and observing the proper corrective system response shall verify automatic control operation.
 - 6. Changing the schedule and observing the correct response on the controlled outputs shall verify selected time and setpoint schedules.
- C. Workstation Test Procedures: The system workstation test procedures shall be as follows:
 - 1. Communication with each DDC control panel shall be demonstrated.
 - 2. Operator commands will be explained and demonstrated.
 - 3. Control sequences shall be demonstrated for proper operation.
 - 4. All available system reports and logs shall be demonstrated at the system workstation.
 - 5. Correct system start-up and shutdown procedures shall be demonstrated.
 - 6. All controllers shall be demonstrated to operate in a standalone mode.
- D. Record Documentation: After a successful acceptance demonstration, the Contractor shall submit as-built drawings of the completed project for final approval. After receiving final approval, supply "3" complete 11 x 17 as-built drawings sets.
- E. Operation and Maintenance Manuals: Submit three copies of operation and maintenance manuals. Include the following:
 - 1. Manufacturer's catalog data and specifications on sensors, transmitters, controllers, control valves, damper actuators, gauges, indicators, terminals and any miscellaneous components used in the system.
 - 2. An operator's manual that will include detailed instructions for all operations of the system.
 - 3. An operator's reference table listing the addresses of all connected input points and output points. Settings shall be shown where applicable.
 - 4. A programmer's manual that will include all information necessary to perform programming functions.

5. A language manual that will include a detailed description of the language used and all routines used by the system.
6. Complete program listing file and parameter listing file for all programs.
7. A copy of the warranty.
8. Operating and maintenance cautions and instructions.
9. Recommended spare parts list.

3.8 TRAINING

- A. Contractor shall provide to the engineer a training class outline prior to any scheduled training.
- B. Factory trained control engineers and technicians shall provide training sessions for the Owner's personnel.
- C. The control contractor shall conduct five six-hour training sessions on the DDC System for the designated Owner's personnel in the maintenance and operation of the Systems. The class shall be given upon system acceptance.
- D. The course shall include instruction on specific systems and instructions for operating the installed system to include as a minimum:
 1. HVAC system overview
 2. Operation DDC Systems
 3. Function of each Component
 4. System Operating Procedures
 5. Programming Procedures
 6. Maintenance Procedures

3.9 SERVICE AND GUARANTEE

- A. This system specified under this Section of the Specifications shall be guaranteed from defects in workmanship and material under normal use and service for a period of twelve (12) months from the date of acceptance. If, during the one year period, any of the factory equipment or materials provided in the system is found to be defective in materials or workmanship, it shall be replaced or repaired by the DDC Manufacturer at no additional cost to the Owner.
- B. Upon completion of the installation, the Contractor shall thoroughly inspect, check, adjust, calibrate, and make ready for use all devices/sensors comprising the control system and certify that they are installed in accordance with "Record" Drawings.

END OF SECTION

SECTION 23 21 13

HVAC PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Piping and pipe fittings for:
 - 1. Chilled Water Piping
 - 2. Heating Water Piping
 - 3. Steam Piping
 - 4. Steam Condensate Piping

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 21 16 - Hydronic Piping Specialties
- C. Section 23 05 15 - Mechanical Piping Valves and Fittings
- D. Section 23 07 01 - Piping Insulation
- E. Section 23 22 14 - Steam and Condensate Heating Specialties

1.3 REFERENCE STANDARDS

- A. ASME B1.1 - Unified Inch Screw Threads; 2008.
- B. ASME B16.3 - Malleable Iron Threaded Fittings: Classes 150 and 300; 2011.
- C. ASME B16.5 - Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard; 2013.
- D. ASME B16.9 - Factory Made Wrought Butt welding Fittings; 2012.
- E. ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings; 2012.
- F. ASME B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings; 2013.
- G. ASME B18.2 - Square, Hex, Heavy Hex, and Askew Head bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws; 2013.
- H. ASME B31.1 - Power Piping; 2014.
- I. ASME B31.3 - Process Piping; 2012.
- J. ASME B31.9 - Building Services Piping; 2014 (ANSI/ASME B31.9).

- K. ASTM A21.11 - Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings; 2010.
- L. ASTM A21.51 - American National Standard for Ductile-Iron Pipe, Centrifugally Cast, for Water; 2002.
- M. ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless; 2012.
- N. ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service; 2013.
- O. ASTM A193 - Standard Specification for alloy Steel and Stainless Steel Bolting Materials for High Temperature Service; 2014.
- P. ASTM A194 - Standard Specification for Carbon and Alloy Steel Nuts for bolts for High Pressure or High Temperature Service, or Both; 2014.
- Q. ASTM A307 - Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod; 2014.
- R. ASTM A563 - Standard Specification for Carbon and Alloy Steel Nuts; 2014.
- S. ASTM A449 - Standard Specification for Quenched and Tempered Steel bolts and Studs; 2004.
- T. ASTM B88 - Standard Specification for Seamless Copper Water Tube; 2009.
- U. ASTM F1476 - Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications; 2013.
- V. AWWA C110/A21.10 - American National Standard for Ductile-Iron and Gray-Iron Fittings, 3 In. Through 48 In. (75 mm Through 1200 mm), for Water and Other Liquids; American Water Works Association; 2012.
- W. AWWA C151/A21.51 - Ductile-Iron Pipe, Centrifugally Cast, for Water; American Water Works Association; 2009 (ANSI/AWWA C151/A21.51).
- X. MSS SP-25 - Standard Marking System for Valves, Fittings, Flanges, and Unions; 2013.

1.4 SUBMITTALS

- A. Submit product data for review on piping and fittings. Submittal data shall include:
 - 1. Manufacturer of pipe.
 - 2. Tests or listings by recognized testing laboratory that certifies material composition is in accordance with ANSI/ASTM requirements.
 - 3. Product data for pipe and fittings to be used on each piping system.
 - 4. Welding procedures for steel pipe.
 - 5. Solder and brazing product data and installation procedures for copper pipe.

PART 2 - PRODUCTS

2.1 STEEL PIPE

- A. Black steel: Electric resistance welded or seamless, ASTM A53 or ASTM A106 Grade B. Mill wrap uninsulated underground steel pipe with Republic X-Tru-Coat or equal.

1. Through 10" standard weight Schedule 40
 2. 12" pipe and larger: standard weight with 0.375" wall thickness
- B. Provide for the following services:
1. Chilled water supply and return piping, 1-1/4" diameter and larger. (Contractor option: copper up to 2.5" diameter; see "COPPER PIPE" below.)
 2. Heating water supply and return piping, 1-1/4" diameter and larger. (Contractor option: copper up to 2.5" diameter. See "COPPER PIPE" below.)
- C. Schedule 40, seamless: ASTM A53 or ASTM A106, Grade B. Provide for the following services:
1. Steam piping
- D. Schedule 80, seamless: ASTM A53 or ASTM A106, Grade B. Provide for the following services:
1. Steam Condensate piping
- E. Schedule 40, A53 or A106 galvanized pipe for:
1. Cooling Coil condensate drain piping. (Contractor option: Copper; see "COPPER PIPE" below.)
 2. Drain piping from equipment 1-1/2" diameter and smaller.

2.2 STEEL PIPE FITTINGS

- A. Flanges, Fittings, and Unions: Mark in accordance with MSS-SP-25.
- B. Fittings:
1. 2-1/2" and larger: Class 150, wrought steel, butt welded fittings, ASME B16.9
 2. 2" and Smaller
 - a. Water Service: Class 150, malleable iron, screwed.
 - b. Steam, 125 psi and less: Class 150, A53 malleable iron, screwed, ASME B16.3
 - c. Condensate: Class 150 malleable iron, A53 screwed, ASME B16.3
- C. Flanges, 2-1/2" and larger: Class 150, A53 wrought forged steel, slip-on or weld neck, ASME/ANSI B16.5. Flange faces shall match equipment or mating flanges (i.e. flat-faced flanges shall be used adjacent to equipment with integral flat-faced flanges, such as pumps, control valves, etc.)
- D. Gaskets:
1. Inorganic fibers, 1/16 or 1/8 inch thick, reinforced EPDM binder, 550 deg F (continuous) and 700 psig operation, Garlock 5507 or equal:
 - a. Chilled water
 - b. Heating water
 2. Spiral wound "chevron" metallic gaskets, flexible graphic filler, class 150 and 300 service, Flexitallic LS, CG or equal
 - a. Steam
- E. Unions, 2" and smaller: Material as specified under fittings, screwed with brass seat.

- F. Branch connections from mains or headers, 2-1/2" and larger: Welded tees or welding outlets, Bonney Forge Weldolets or Thredolets. Use forged outlets only if branch line is at least one pipe size smaller than main or header.
- G. Galvanized steel pipe fittings: Same as above, except galvanized coated.
 - 1. Provide drainage pattern type fittings for drain piping.
- H. Bolting Materials: Torque all bolts to 50% of yield strength or per equipment manufacturer's recommendation, whichever is lower. Use anti-seize lubricant on all bolt threads. Same finished carbon steel bolts and hex nuts, ASTM A307. Threads and Dimensions: ASME/ANSI B1.1 and B18.2.
 - 1. Systems 210 degrees F and less: Use ASTM A449 studs or bolts and ASTM-A563 Grade B hex nuts
 - 2. Systems above 210 degrees F: Use ASTM-A193 Grade B7 studs or bolts and ASTM-A194 Grade 2 Heavy Hex Nuts
 - 3. Use galvanized bolts and nuts on piping outside the building, inside tunnels, and inside manholes.
- I. Thread Lubricant: Similar to Crane "Formula 425".

2.3 IPS GROOVED PIPING SYSTEM

- A. Grooved piping shall only be allowed in EXPOSED locations and for straight run piping only. Example: Mechanical rooms where piping is visible above/below ductwork.
 - 1. Applicable Systems:
 - a. Chilled Water Supply/Return
 - b. Heating Hot Water Supply/Return
- B. Grooved mechanical pipe couplings, fittings, valves and other grooved components may be used as an option to welding, threading or flanged methods as specified herein. All grooved components shall be of one manufacturer and conform to local code approval and as listed by ASME/ANSI B-31.1, B-31.3, B-31.9, ASME, UL/ULC, FM, IAPMO or applicable Building Code. Grooved end product manufacturer to be ISO-9001 certified. Grooved couplings shall meet the requirements of ASTM F-1476.
- C. Grooved Pipe: Carbon Steel, as specified with roll or cut grooved-ends as appropriate to pipe material, wall thickness, pressures, size and method of joining. Pipe ends shall be grooved in accordance with manufacturer's current listed standards conforming to ANSI/AWWA C-606.
- D. Mechanical Couplings for Steel Pipe
 - 1. Acceptable manufacturers: Victaulic. Victaulic model numbers are used to establish product type, quality and performance.
 - 2. Mechanical couplings shall be manufactured in two segments of cast ductile iron, conforming to ASTM A-536, Grade 65-45-12. Gaskets shall be pressure-responsive synthetic rubber, grade to suit the intended service, conforming to ASTM D-2000. Mechanical coupling bolts shall be zinc plated (ASTM B-633) heat treated carbon steel track head conforming to ASTM A-449 and ASTM A-183. Couplings shall comply with ASTM F1476 - Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications.
 - a. Rigid Type: coupling housings with offsetting, angle-pattern bolt pads shall be used to provide system rigidity, support, and hanging in accordance with ANSI B31.1, B31.9, with Victaulic Style 107H/107N (Quick-Vic) installation ready rigid coupling for direct stab installation without

field disassembly. Gasket shall be Grade "EHP" EPDM designed for operating temperature from -30 deg. F (-34 deg. C) to 250 deg. F (120 deg. C).

- b. Victaulic AGS mechanical couplings, 14 inch (DN350) through 60 inch (DN1500): Couplings shall consist of two ASTM A-536 ductile iron housing segments with lead-in chamfer on housing key and a wide-width elastomer pressure responsive gasket. Victaulic Style W07 AGS Rigid and Style W77 AGS Flexible Coupling.

2.4 COPPER PIPE

- A. Conform to ASTM B-88 specification for wrought seamless copper.
- B. Type L, hard for:
 - 1. HVAC reheat water pipe, 1" and smaller; Contractor option for copper or steel pipe 1-1/4" to 2.5".
 - 2. HVAC chilled water pipe, 1" and smaller. Contractor option for copper or steel pipe 1-1/4" to 2.5".
- C. Type M, hard for:
 - 1. For non-pressurized drain piping.
 - 2. Cooling coil condensate piping.

2.5 COPPER PIPE FITTINGS

- A. Sweat type, wrought copper, ASTM B62, with dimensions conforming to ASTM/ANSI B16.22 and sweep patterns for copper tubing.
- B. Dielectric Connections:
 - 1. Provide at junction of copper pipe and equipment with steel piping systems.
 - 2. Central, Dielectric insulating unions, and insulating flange unions, as manufactured by Central Plastic Company or CTS Fabrication USA (1-1/2" thru 8").
 - 3. Provide copper solder joint to plated female iron pipe for sizes 1/2" through 2".
 - 4. Provide insulating flange unions, malleable female iron pipe thread to copper solder joint flange unions for sizes 2-1/2" through 4".
 - 5. Brass fittings and valves may not be used for dielectric union locations.
- C. Unions: Brass ground joint, 250 lb. working pressure.
- D. Nipples: Brass.

2.6 DUCTILE IRON WATER PIPE AND FITTINGS

- A. Manufactured in accordance with latest revision of ANSI A21.51/AWWA C151.
- B. Use Class 52 thickness or heavier for 12" diameter and less and Class 51 thickness or heavier for 14" diameter and larger.
- C. Furnish pipe with standard thickness cement lining on inside with a bituminous seal coat and a bituminous coating on the outside.

- D. Clearly mark pipe exterior to indicate the manufacturer, date of manufacture, pipe class and weight.
- E. Provide for the following services:
 - 1. Buried condenser water piping
 - 2. Buried chilled water piping
- F. Fittings:
 - 1. Acceptable Manufacturers: Fastite by Acipco, Tyton by U.S. Pipe Co., or Bell-Tite by James B. Clow and Sons.
 - 2. Ductile iron, cement lined, bituminous coated, manufactured in accordance with the latest revision of ANSI/AWWA C110.
 - 3. Furnish with mechanical joints conforming to ANSI A21.11 with 250 psi pressure rating.

2.7 MISCELLANEOUS PIPE ACCESSORIES

- A. Escutcheons: Chrome pipe escutcheons, slip-on or split type where pipe passing through finished walls or ceiling may be visible.
- B. Exposed Metal Pipe and Trim: Chrome plated.
- C. Control System Connectors: Crane No. 386, 1" steel half couplings, or 1" female pipe thread connectors.
- D. Install 18 gauge sheetmetal or galvanized steel pipe saddles to protect insulation.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Clean inside of pipe before installation. Keep installed piping clean, and protect ends from foreign matter by capping or plugging them.
- B. Install pipe so that it does not interfere with opening of doors or apparatus, access to equipment, or to electrical equipment.
- C. Do not install pipes in such a way that they will apply torque to pumps. After pumps have been installed and pumps have been operated, recheck and realign pumps if necessary.
- D. Run pipes in straight lines and square with building. Install risers plumb. Make offsets only where indicated and where necessary.
- E. Install branch connections using separate tee or lateral fittings for each branch. Do not combine branches into "bullhead tee" arrangement.
- F. Do not install water pipes in electric rooms, tele/data rooms, transformer rooms, audio/visual rooms or elevator equipment rooms. Fire protection piping runouts serving only these rooms shall be installed in these rooms.
- G. Do not install piping above electrical equipment such as starters, variable frequency motor controllers, motor control centers, or disconnects. Maintain code required clearance above, below and to sides of electrical equipment.

- H. Provide flanges or unions throughout the pipe systems at all equipment. Make provisions for servicing and removal of equipment without dismantling piping.
- I. In so far as possible, drainage piping shall not be installed overhead, whether exposed or above ceiling, in operating rooms, delivery rooms, nurseries, food preparation or serving areas, or in rooms listed above. Where unavoidable, provide drain troughs or other means to carry away leakage.
 - 1. Grading Pipes for Drainage:
 - 2. Slope cooling coil condensate drains at 1/8" per foot.
- J. Slope steam pipes and steam condensate drain pipes at 1/4" per 10 feet. Do not raise condensate pipe except at ends of main drips.
- K. Branch Lines:
 - 1. Where possible branch lines shall come off top of mains to prevent sediment, welding slag, or pipe burrs from entering the branch lines and causing valve leakage or failure.

3.2 PIPE JOINTING

A. Preparing Pipe Ends:

- 1. Machine cut pipe ends square.
- 2. Ream pipe ends, after cutting, to full diameter.
- 3. Where pipe is to be threaded, secure pipe in pipe stand, die cut, full depth, right hand threads. Threads to be taper type.
- 4. All threaded pipe joints to have suitable pipe sealant applied to threads prior to assembly of joint. Joints shall be leak proof.
- 5. Where pipe is to be welded, die-cut end of butt joints at 30 degree taper. Weld should have a full penetration with no bubbles or holes. Remove all slag.

B. Welded Steel Piping:

- 1. Where welded piping is specified, make welds by oxy-acetylene process or electric process in accordance with ASME/ANSI B31.1.
 - a. Welding Rods: Grade recommended for purpose by manufacturer's and identification.
- 2. Line welds, single V-butt type:
 - a. Mill or machine bevel pipe at 37 1/2 degrees to within 1/16" of inside wall, except that in field limited amount of pipe may be flame beveled.
 - b. Pipe with a wall thickness of 3/16" or less need not be beveled but may be welded by melting down into building up over abutting ends.
 - c. Separate abutting ends of joints before welding to permit complete fusion to bottom without overlapping.
 - d. Tack in two or more points to maintain alignment, and fusion weld.
- 3. Make all welds of sound weld metal, thoroughly fused into ends of pipe, and to bottom of vee.
 - a. Build in excess of pipe wall to give reinforcement to one fourth pipe wall thickness.
 - b. Weld metal shall present a gradual increase in thickness from surface of pipe to center of weld.

- c. Minimum weld width: Two and one half times thickness of pipe wall.
 - 4. Use welding ells at turns in welded lines
 - 5. Do not weld pipe couplings in place of welding fittings for any branch connections.
 - 6. Weld-o-lets and thread-o-lets:
 - a. Scribe and cut openings in main pipes for welded branches accurately taking care to remove all of plugs and cuttings from main pipe.
 - b. Full weld fillet welds for full depth of fillet, with additional beads to form well rounded connection as recommended by weld-o-let manufacturer.
 - 7. Cut openings into pipe for welded connections accurately to give matched intersections.
 - 8. Make welded fittings of same material with same pressure and temperature rating as pipe with which they are used.
 - 9. Make flanged connections to control valves, pump suction and specialties with ANSI standard welding neck flanges. All other flange connections may be made with slip-on flanges provided they are seal welded on inside.
 - 10. Fuse all fillet welds for flanges or fittings into pipe and plate for minimum distance of 1-1/2 times pipe wall thickness and depth weld on 1-1/4 times pipe wall thickness.
- C. Grooved Coupling Installation:
- 1. Pipe ends shall be clean and free from indentations, projections and roll marks in the area from pipe end to groove for proper gasket sealing.
 - 2. The gasket style and elastomeric material (grade) shall be verified as suitable for the intended service as specified.
 - 3. All grooved components (couplings, fittings, valves, gaskets, bolts and nuts) shall be of one manufacturer.
 - 4. Grooved joints shall be installed in accordance with manufacturer's latest published installation instructions.
 - 5. Grooved ends shall be clean and free from indentations, projections, and roll marks.
 - 6. A factory trained field representative (direct employee) of the mechanical joint manufacturer shall provide on-site training for contractor's field personnel in the proper use of grooving tools and installation of grooved piping products. The factory trained representative shall periodically review the product installation and ensure best practices are being followed. Contractor shall remove and replace any improperly installed products at their cost. A distributor's representative is not considered qualified to conduct the training.
- D. Soldered and Brazed Joints:
- 1. Make Type L and M copper pipe joints with suitable flux and 95/5, lead free solder.
 - 2. Make Type K copper pipe joints with silver (BAG Series) brazing filler material with flux or copper-phos (BCup Series) brazing filler material without flux per the recommendations of the Copper Development Association.
- E. "T" Drill Branch Connections are not allowed.
- F. Bracing Joints:
- 1. Provide braces and bridle rods as required to reinforce joints.

2. If mechanical lock type couplings are used, then prepare pipe ends and make joints in accordance with pipe coupling manufacturer's printed instructions.
3. Where large pipes underground are subject to shock because of sudden changes in liquid flow rate, provide concrete "kicker" blocks at joints, fittings, and changes of pipe direction. Provide "kicker" blocks in accordance with applicable pipe industry trade or research organization recommendations.
 - a. For example, for ductile iron pipe follow recommendations of Ductile Iron Pipe Research Association.

3.3 ESCUTCHEONS

- A. Provide chrome plated escutcheons where uninsulated pipes penetrate walls or ceilings of finished spaces.

3.4 STRAINERS

- A. Install strainers so the strainer basket can be removed without spilling water on motors and electrical equipment.

3.5 AIR VENTING

- A. Provide manually operated air vents at high points in vertical risers and at water coils to eliminate air from systems. Air vents are not required at reheat coils.
- B. Use ball valves for manual air vents.

3.6 VALVE ACCESS

- A. Locate ceiling/wall access panels at shut-off and control valves for proper access and operation. Furnish and install access doors in accordance with Section 23 05 00 and other Divisions as applicable.

3.7 CONTROL SYSTEM CONNECTORS

- A. Weld connectors at points indicated, and at other points where necessary for installation of thermometers, sensors, and automatic controls.

3.8 TESTING

- A. Before piping is concealed or insulated, recheck it for leaks.
- B. Rework or replace defective and leaking joints, and joints which are otherwise unsatisfactory. Peening, caulking, and doping are not permitted.

END OF SECTION

SECTION 23 21 16

HYDRONIC PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. This section includes requirements for:
 - 1. P.T. test plugs
 - 2. Pressure test kit
 - 3. Relief valves
 - 4. Strainers
 - 5. Auto-Flow Balance Valve

1.2 RELATED REQUIREMENTS

- A. Section 23 21 13 - HVAC Piping
- B. Section 23 21 23 - HVAC Pumps

1.3 SUBMITTALS

- A. Submit product data for review in accordance with the requirements of Division 01 including:
- B. Operation and Maintenance Data

1.4 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in the manufacture of the products specified herein for a minimum of five years of documented experience.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. As noted in the individual product paragraphs.

2.2 P.T. TEST PLUGS

- A. Acceptable Manufacturers: Peterson Engineering or approved equal.
- B. Provide 1/4" solid brass pressure/temperature test plugs at locations shown on drawings.
- C. Provide N2 Nordel self-closing valve rated for 275 degrees F service.
- D. P.T. test plugs shall be installed at the following locations and elsewhere as shown on the drawings.

1. Entrance and exit of building when connected to the chilled water central plant.
2. At suction and discharge of pumps.
3. At water inlet and outlet of heat exchanger.
4. At water inlet and outlet of air handling unit coils.
5. Adjacent to each pressure gauge and thermometer.

2.3 PRESSURE TEST KIT

- A. Acceptable Manufacturers: Bell & Gossett or Taco.
- B. Provide two complete portable pressure and temperature test kits to include at least the following:
 1. Prior to commissioning, provide the Owner one complete portable pressure and temperature test kit to include at least the following: pressure test gauge, necessary connector hoses, temperature test thermometer with adapter, shutoff and vent valves and carry case. Select thermometers and gauges to read mid-range of design operating temperatures and pressures where PT plugs are installed.
 2. Pressure test gauge, necessary connector hoses, temperature test thermometer with adapter, shutoff and vent valves and carrying case. Select thermometers and gauges to read mid-range at design operating temperatures and pressures where PT plugs are installed.

2.4 RELIEF VALVES

- A. Acceptable Manufacturers: Bell & Gossett or approved equal.
- B. Provide relief valves for each hydronic system as shown on drawings.
- C. Construct valve to ASME requirements, tested by National Board of pressure vessels, and labeled with ASME symbol.
- D. Construct valve body of non-ferrous material.
- E. Provide diaphragm type valve operating with slow opening and closing feature.
- F. Valve to seat against face of EPDM rubber.
- G. Set differential between opening and closing pressure to prevent water flash and water hammer.
- H. Include manual lever for testing valve.

2.5 STRAINERS

- A. Acceptable Manufacturers: Elliott, Mueller, Keckley, or Wheatley.
- B. Cast semi-steel body or cast iron construction for steel piping and bronze body construction for copper piping; equipped with removable, Monel or stainless steel 20 mesh, water screen; maximum pressure drop 2 psi with free area at least four times area of pipe. Provide with blow-off outlet piped to nearest floor drain.

2.6 AUTO-FLOW BALANCE VALVE

- A. Acceptable Manufacturers: Flow Design model AC, Griswold Isolator Series, or Victaulic Series 76.

- B. Valve shall contain a stainless steel spring cartridge.
- C. Spring cartridge valves shall have a 2-32 psi spring range.
- D. Factory set to automatically limit flow to within 5% of flow range. Pressure drop through valve shall not exceed 5 ft. for 2 inch size. Valves flow range shall be independently tested and certified by professional engineer.
- E. Provide with removable cartridge without the use of special tools or cutting piping. Provide two extended pete ports on each valve to clear insulation.
- F. Valve shall be constructed for 400 psig design at 250 degrees F. The flow cartridge shall have stainless steel internal parts. Machined threads shall be provided to adjust cartridge height without the use of shims or crimped sheet metal. Valve body shall be brass with brass ball valve and 'O' ring type union. All valves shall be factory leak tested.
- G. Provide integral union on the downstream side of the assembly.
- H. Provide valve with engraved tag attached indicating design flow, pressure, and flow characteristic of station.
- J. Provide electronic pressure (0-75 psi) and temperature (-10 - 230 deg F) test kit.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install hydronic specialty items in locations shown on drawings.
- B. Insulate any components subject to sweating or any hot or cold service in accordance with Section 23 07 00.
- C. Utilize manufacturer's instructions to install specialty items. Manufacturer's instructions must be adhered to for proper operation of air removal devices and flow control valves.
- D. Provide manual air vents at top of pipe risers and other locations where air can be trapped or collected.
- E. Pipe relief valve outlets and drain connections from hydronic systems to nearest floor drain.
- F. Support pump inlet and strainer fittings with floor mounted pipe and flange supports.
- G. Locate thermometers and pressure gauges no higher than 7 feet above finished floor elevation, positioned to be read from the floor.
- H. After systems are started up, placed in service and tested/adjusted to perform as designed, contractor shall provide competent representative to demonstrate proper operation and provide instruction to maintenance personnel to include performance on both cooling and heating cycles.
- I. At a minimum, pressure gauges and thermometers shall be installed at the following locations:
 - 1. Suction and discharge of pumps.
 - 2. Water side, Inlet and outlet of heat exchangers.

3. Water side, inlet and outlet of air handling units.

END OF SECTION

SECTION 23 21 23

HVAC PUMPS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. In-line centrifugal pumps.

1.2 RELATED WORK

- A. Section 22 0010 - Basic Mechanical Requirements.
- B. Section 23 0513 - Motors.
- C. Section 22 0548 - Vibration Isolation.
- D. Section 23 0701 - Piping Insulation.
- E. Section 23 0702 - Equipment Insulation.
- F. Section 23 2113 - Hydronic Piping.
- G. Section 23 2115 - Hydronic Specialties.

1.3 REFERENCES

- A. ANSI/UL 778 - Motor Operated Water Pumps.

1.4 QUALITY ASSURANCE

- A. Manufacturer: Company specializing in manufacture, assembly, and field performance of pumps with minimum three years experience.
- B. Alignment: Base mounted pumps shall be aligned by qualified millwright and alignment certified.

1.5 SUBMITTALS

- A. Submit shop drawings and product data under provisions of Division 1 and Section 22 0010.
- B. Submit certified pump curves showing performance characteristics with pump and system operating point plotted. Include NPSH curve when applicable.
- C. Submit manufacturer's installation instructions under provisions of Division 1 and Section 22 0010.

1.6 OPERATION AND MAINTENANCE DATA

- A. Submit operation and maintenance data under provisions of Division 1 and Section 22 0010.
- B. Include installation instructions, assembly views, lubrication instructions, and replacement parts list.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site under provisions of Division 1 and Section 22 0010.
- B. Store and protect products under provisions of Division 1 and Section 22 0010.

1.8 EXTRA PARTS

- A. Provide one extra set of mechanical seals for pumps under provisions of Section 22 0010.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. ITT Bell & Gossett.
- B. Aurora.
- C. Taco.
- D. Grundfos
- E. Armstrong
- F. Substitutions: Under provisions of Division 1 and Section 22 0010.

2.2 IN-LINE CENTRIFUGAL PUMPS

- A. Type: In-line, close-coupled, for installation in vertical or horizontal position, for 175 psig maximum working pressure. The motor shall be non-overloading at any point on the pump curve.
- B. Casing: Class 30 cast iron with gauge ports at nozzles, and vent and drain ports in casing.
- C. Impeller: Brass or bronze, dynamically balanced and keyed to shaft.
- D. Bearings: Heavy-duty grease lubricated ball bearings.
- E. Shaft: Stainless steel with stainless steel sleeve, integral thrust collar.
- F. Seal: Carbon rotating against a stationary ceramic seat, 225 degrees F maximum continuous operating temperature.
- G. Motor: Energy efficient.
- H. Performance: Refer to schedule on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install pumps in accordance with manufacturer's instructions.
- B. Provide access space around pumps for service. Provide no less than a minimum of three feet, not including piping and piping appurtenances.

- C. Ensure pumps operate at specified system fluid temperatures without vapor binding and cavitation, are non-overloading in parallel or individual operation, and operate within 15 percent of midpoint of published maximum efficiency curve.
- D. Decrease from line size with long radius reducing elbows or reducers. Support piping adjacent to pump such that no weight is carried on pump casings. For close coupled or base mounted pumps, provide supports under elbows on pump suction and discharge line sizes 4 inches and over.
- E. Provide line sized shut-off (ball or butterfly) valve and strainer on pump suction, and line sized soft seat check valve and balancing valve (ball valve memory stop, or butterfly valve with memory stop) on pump discharge.
- F. Provide air cock and drain connection on horizontal pump casings.
- G. Provide drains for bases and seals, piped to and discharging into floor drains.
- H. Lubricate pumps before start-up.
- I. Alignment: A qualified millwright shall check, align, and certify base mounted pumps. A reverse alignment procedure utilizing laser instruments shall be used. Alignment shall be performed in both the hot and cold operating temperature extremes. The maximum parallel and angular misalignment shall not exceed 0.002 inch. Record and deliver copies of the alignment report to the Owner and include report in the operations and maintenance manuals.
- J. Vibration Testing: Vibration velocity reading shall be taken at all bearing locations of base mounted pumps. Pumps driven by variable speed drives shall be tested throughout their range of speeds. Vibration velocity shall not exceed 0.15 inch/second (peak). Record and deliver copies of the vibration test report to the Owner and include report in the operations and maintenance manuals.

END OF SECTION

SECTION 23 22 23

STEAM AND STEAM CONDENSATE SPECIALTIES

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Steam traps.
- B. Flash tanks.
- C. Steam pressure reducing valves.
- D. Steam relief valves.
- E. Steam safety valve discharge elbows.
- F. Steam pipe anchors.
- G. Steam pipe guides.
- H. Drip traps.
- I. Sediment strainers.

1.2 RELATED WORK

- A. Section 23 0515 - Hydronic Piping Expansion Compensation.
- B. Section 23 0513 - Motors.
- C. Section 23 0701 - Piping Insulation.
- D. Section 23 0702 - Equipment Insulation.
- E. Section 23 2115 - Hydronic Specialties.
- F. Section 23 2213 - Steam and Steam Condensate Piping.

1.3 REFERENCES

- A. ANSI/ASTM - Boilers and Pressure Vessels Code.
- B. ASTM A105 - Forgings, Carbon Steel, for Piping Components.
- C. ASTM A126 - Gray Iron Casings for Valves, Flanges, and Pipe Fittings.
- D. ASTM A216 - Steel Casings, Carbon, Suitable for Fusion Welding, for High Temperature Service.
- E. ASTM A395 - Ferric Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures.

F. ASME B31.9 - Building Services Piping.

1.4 REGULATORY REQUIREMENTS

A. Conform to ASME B31.9 - Building Services Piping.

1.5 QUALITY ASSURANCE

A. Manufacturer For each product specified, provide components by same manufacturer throughout.

1.6 SUBMITTALS

- A. Submit shop drawings and product data under provisions of Division 1 and Section 22 0010.
- B. Submit shop drawings and product data for manufactured products and assemblies required for this project
- C. Include product description, model, dimensions, component sizes, rough-in requirements, service sizes, and finishes.
- D. Submit schedule indicating manufacturer, model number, size, location, rated capacity, and features for each specialty.
- E. Submit manufacturer's installation instructions under provisions of Division 1 and Section 22 0010.

1.7 OPERATION AND MAINTENANCE DATA

A. Submit operation and maintenance data under provisions of Section Division 1 and Section 22 0010.

1.8 EXTRA STOCK

A. Provide two service kits for each size and type of steam trap under provisions of Division 1 and Section 22 0010.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS - STEAM TRAPS

- A. Armstrong.
- B. Spirax/Sarco.
- C. ITT Hoffman.
- D. Substitutions: Under provisions of Division 1 and Section 22 0010.

2.2 INVERTED BUCKET TRAPS

- A. Cast iron or semi-steel body and bolted cover for 250 psig WSP; provide access to internal parts without disturbing piping; with top test plug and bottom drain plugs, brass or stainless steel bucket, stainless steel seats and plungers, and stainless steel lever mechanism with knife edge operating surfaces, integral inlet strainer of monel or stainless steel.

1. Basis of Design is Armstrong 800-813 Series.

2.3 FLOAT AND THERMOSTATIC TRAPS

- A. ASTM A126, cast iron or semi-steel body and bolted cover for 250 psig WSP; provide access to internal parts without disturbing piping; with bottom drain plug, stainless steel or bronze bellows type air vent, stainless steel or copper float stainless steel lever and valve assembly.

2.4 THERMOSTATIC TRAPS

- A. Pressure balanced type with ASTM A216 WCB cast steel body and bolted or screwed cover, and integral ball joint union, for 300 psig WSP; monel or stainless steel bellows, stainless steel valve and seat, integral stainless steel strainer.
- B. Freeze proof type with cast iron body for 300 psig WSP, bronze bellows, stainless steel valve and seat, external adjustment.
- C. Bi-metallic type with ASTM A105 forged steel body and cover, for 300 psig WSP, bi-metal element with stainless steel components, integral Type 304 stainless steel strainer screen, 1/4 inch blow down valve.

2.5 FLASH TANKS

- A. Closed type, welded steel construction, tested and stamped in accordance with Section 8D of ANSI/ASME Boilers and Pressure Vessels Code for 125 psig working pressure; cleaned, prime coated, and supplied with steel support legs. Construct with nozzles and tapings for installation of accessories and piping connections.

2.6 STEAM PRESSURE REDUCING VALVES

- A. All pressure reducing valves shall be capable of maintaining the set pressure from zero to the maximum steam flow within reasonable limits when subjected to usual steam pressure fluctuations. They shall be single seated valves with stainless steel trim, with renewable valve, lugs and seats. Valve bodies shall be cast steel for high pressure service and cast iron for medium and low pressure service. These valves shall be self contained type with upstream and downstream pressure gauges and shall be installed as per manufacturer's recommendations. Valve capacities are scheduled on the drawings. Each stage of pressure reduction shall consist of two PRVs sized 1 1/3 and 2/3 capacity. Discharge pressure shall be adjustable to any value between 10 psig and 75% of the supply pressure.
- B. All pressure regulators 2-1/2" and larger shall have flanged connections and those 2" and smaller may have screwed connections. Unions shall be installed on each side of any screwed pattern regulators installed.
- C. Each reducing valve shall be preceded by a sediment strainer complete with a full-sized blow off valve with threaded end for hose connection.
- D. These valves shall be Leslie, Spence, Spirax Sarco, Fisher, Mason Neilan, or approved equal, with suitable automatic controllers.

2.7 STEAM RELIEF VALVES

- A. Relief valves 2" and smaller shall have brass bodies and arranged for screwed connections. Such relief valves shall be Crane No. 2501 or Spirax Sarec 6010 Brass Safety Valves for steam or approved equal. Bushings shall not be used.

- B. Relief valves 2-1/2" and larger shall in the case of all medium and low pressure steam piping systems be arranged for flanged inlet and screwed outlet connections. Such relief valves shall be Consolidated Type 1511 or Spirax Sarco 252, ASME Standard Cast Iron Safety Valves, or approved equal.
- C. The pressure at which each relief valve shall open is designated on the Drawings. When such valves are ordered by the Contractor, he shall definitely specify the pressure at which each relief valve is to be set. Each valve shall have a metal tag attached stamped with the valve identification plus the pressure setting.

2.8 STEAM SAFETY VALVE DISCHARGE ELBOWS

- A. All vent lines from safety valves shall be provided with safety valve discharge elbows at the point at which such lines rise to an elevation higher than that of the safety valve. The nature and design of the piping systems involved shall be such as to drain effectively all condensate from the discharge side of all relief valves. These safety valve discharge elbows shall be Grinnell Company's Safety Valve Drip Pan Elbows Figure No. 1538F, Spirax Sarco No. 299, or approved equal. No force shall be exerted on the safety valve by the discharge piping.

2.9 STEAM PIPE ANCHORS

- A. All steam lines shall be securely anchored at points designated on the Drawings and/or at such points as required to assure proper control of the expansion and contraction of such systems.
- B. See Section 23 0529 for additional requirements.

2.10 STEAM PIPE GUIDES

- A. All steam piping systems shall be properly guided as shown on the Drawings and/or as required.

2.11 DRIP TRAPS

- A. High pressure drip trap assemblies shall be provided wherever called for on the Drawings and where required to keep such piping systems completely drained of condensate. Traps used in assemblies shall be 3/4" traps unless specifically shown to the contrary, i.e., they shall have 3/4" inlet and outlet connections. They shall have semi-steel bodies and the internal operating mechanisms shall be made of heat treated chrome steel. The caps shall be bolted to the bodies by the use of alloy steel heat treated machine bolts. These No. 213 Armstrong Traps, manufactured by Armstrong Machine Works, or approved equal, shall have a capacity for discharging at least 3,500 pounds of condensate per hour when operating at a pressure of 250 pounds per square inch. Where drip traps are installed in conjunction with 3" and larger steam lines, a drip pocket of the nature detailed on the Drawings shall be provided where a natural pocket does not exist. The piping and valves in trap assemblies shall be arranged as detailed on the Drawings; extra strong pipes shall be used on both sides of the trap.
- B. All drip traps used in medium pressure steam piping systems where automatic steam control valves are not employed shall be arranged as shown on the Drawings. They shall be 3/4" Armstrong No. 811 Inverted Bucket Traps, or approved equal, with cast iron bodies, vacuum breakers and stainless steel trim. Each trap shall be provided with a valved test line and shall be preceded by a sediment strainer.
- C. Condensate from coils, converters, hot water generators, low pressure drips and from all other devices where modulating steam valves are employed shall be of the float and thermostatic type. These traps shall be sized to handle 200% of the load with an inlet pressure drop of 0.5 psig and shall be equal to Armstrong "A" or "B" series, with vacuum breaker suitable for the system pressures. Installed traps with less than 12" of height between equipment outlet and trap inlet shall be sized for not less than 300 percent of the load. Each trap shall be provided with a 1/2" valve test line and shall be preceded by a sediment strainer. Under no circumstances shall a float and thermostatic trap be installed in a manner to lift condensate up in a return line.

- D. Shop Drawing submittal of traps shall contain an itemized list with a tabulation of the load, trap type, and trap size.

2.12 SEDIMENT STRAINERS

- A. Each drip trap assembly, each control valve, for steam and each pressure reducing valve assembly regardless of its size shall be preceded by a sediment strainer. The arrangement of these sediment strainers shall be such that the screens may be removed for cleaning with ease through a gasketed plug.
- B. Sediment strainers shall be placed in steam piping systems wherever shown on the Drawings and at such other points as may be required for the removal of foreign material from the piping systems.
- C. Strainers in high pressure steam piping shall be cast steel sediment strainers and shall be suitable for working steam pressures as high as 300 pounds per square inch and temperatures not in excess of 750 degrees F. These strainers shall be the size designated on the Drawings. In the case of pipe sizes 2-1/2" and larger, Ranged pattern sediment strainers shall be used. In the case of pipe smaller than 2-1/2", screwed pattern shall be used. Such strainers shall be Yarway No. 821 or 822 strainers manufactured by Yarnall Waring Company, or approved equal. The flanges of Ranged strainers shall be dimensioned, faced, dolled, and spot faced to conform to the 300 pound American Standard for Steel Pipe Flanges and Flanged Fittings (B16e-1939).
- D. Strainers in low and medium pressure steam piping systems 2-1/2" and larger shall be flanged iron body strainers having bolted covers. These strainers shall be suitable for operating pressures as high as 125 psig. They shall be Crane Company No. 989-1/2 Sediment Separators, or approved equal.
- E. Sediment strainers in low and medium pressure steam piping systems 2" and smaller shall be arranged for screwed pipe connections. They shall be Crane No. 988-1/2 Sediment Separators, or approved equal.
- F. Full sized blow off valves shall be installed on all strainers in steam, condensate, chilled and hot water lines.

2.13 GAUGES AND GAUGE CONNECTIONS

- A. See Section 23 0519.

2.14 THERMOMETER AND THERMOMETER WELLS

- A. See Section 23 0519.

PART 3 - EXECUTION

3.1 INSTALLATION AND APPLICATION

- A. Install specialties in accordance with manufacturer's instructions.
- B. Install thermostatic steam traps to drain condensate from Steam radiation units, convectors, and other terminal heating units.
- C. Install float and thermostatic traps to drain condensate from unit heaters, convertors, heating coils, steam separators, flash tanks, steam jacketed equipment and direct steam injected equipment.
- D. Install inverted bucket steam traps to drain condensate from steam main headers and branch lines.
- E. Size steam traps to handle minimum of two times maximum condensate load of apparatus served.

- F. Traps used on steam mains and branches shall be minimum 3/4 inch size.
- G. Install steam traps with union or flanged connections at both ends.
- H. Provide gate valve and strainer at inlet, and gate valve at discharge of steam traps.
- I. Provide minimum 10 inch long dirt pocket of same pipe sizes as apparatus return connection between apparatus and steam trap.
- J. Remove thermostatic elements from steam traps during temporary and trial usage, and until system has been operated and dirt pockets cleaned of sediment and scale.
- K. Provide pressure reducing stations with pressure reducing valve, valved bypass, strainer and pressure gage on upstream side, relief valve and pressure gage on downstream side of pressure reducing valve.
- L. Pressure reducing station shall be one or two stages as indicated, to produce flat reduced pressure curve over range of capacity.
- M. Rate relief valves for pressure upstream of pressure reducing station, for full operating capacity. Refer to schedules on Drawings for relief valve settings.
- N. Terminate relief valves to outdoors with muffler attachment. Provide drip pan elbow with drain connection to nearest floor drain.
- O. When several relief valve vents are connected to a common header, header cross section area shall equal sum of individual vent outlet areas.

END OF SECTION

SECTION 23 31 13

DUCTWORK

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Low pressure ductwork.
- B. Medium pressure ductwork.
- C. Duct cleaning.

1.2 RELATED WORK

- A. Section 22 0500 - Common Work Results for HVAC.
- B. Section 23 0529 - Supports and Anchors: Sleeves.
- C. Section 22 0548 - Vibration Isolation.
- D. Section 23 0703 - Duct Insulation.
- E. Section 23 3300 - Ductwork Accessories.
- F. Section 23 3600 - Air Terminal Units.
- G. Section 23 0593 - Testing, Adjusting, and Balancing.

1.3 REFERENCES

- A. ASHRAE - Handbook of Fundamentals; Duct Design.
- B. ASHRAE - Handbook of Equipment; Duct Construction.
- C. ASTM A 90 - Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles.
- D. ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
- E. ASTM A 525 - General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process.
- F. ASTM A 527 - Steel Sheet, Zinc-Coated (Galvanized) by Hot-Dip Process, Lock Forming Quality.
- G. ASTM B209 - Aluminum and Aluminum Alloy Sheet and Plate.
- H. ASTM C 14 - Concrete Sewer, Storm Drain, and Culvert Pipe.
- I. ASTM C 443 - Joints for Circular Concrete Sewer and Culvert Pipe, Using Rubber Gaskets.
- J. NFPA 90A - Installation of Air Conditioning and Ventilating Systems.

- K. NFPA 90B - Installation of Warm Air Heating and Air Conditioning Systems.
- L. SMACNA - Low Pressure Duct Construction Standards.
- M. SMACNA - High Pressure Duct Construction Standards.
- N. UL 181 - Factory-Made Air Ducts and Connectors.

1.4 REFERENCES

- A. Fundamentals Handbook, American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE).
- B. Equipment Handbook, ASHRAE.
- C. HVAC Duct Construction Standards, Sheet Metal and Air Conditioning Contractors National Association (SMACNA).
- D. HVAC Duct System Design, SMACNA.
- E. Round Industrial Duct Construction Standards, SMACNA.
- F. Engineering Design Manual for Air Handling Systems, United McGill Corporation (UMC).
- G. Assembly and Installation of Spiral Duct and Fittings, UMC.
- H. Engineering Report No. 132 (Spacing of Duct Hangers), UMC.

1.5 DEFINITIONS

- A. Duct Sizes: Inside clear dimensions. For lined ducts, maintain sizes inside lining.
- B. Low Pressure: 2 inch WG positive or negative static pressure and velocities less than 1,500 fpm.
- C. Medium Pressure: 6 inch WG positive static pressure and velocities greater than 1,500 fpm.
- D. High Pressure: 10 inch WG positive static pressure and velocities greater than 2,500 fpm.

1.6 REGULATORY REQUIREMENTS

- A. Construct ductwork to NFPA 90A, NFPA 90B and NFPA 96 standards.

1.7 SUBMITTALS

- A. Shop Drawings shall be submitted on all items of sheet metal work specified herein. Shop Drawings of ductwork at air units shall be submitted at a minimum scale of 3/8" equal to one foot.
- B. Shop Drawings shall be submitted on all other ductwork at a minimum scale of 1/4" = 1 foot. Shop Drawings shall indicate location of all supply, return, exhaust and light fixtures from the approved reflected ceiling plans.
- C. Submit shop drawings and product data under provisions of Section 22 0010.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site under provisions of Section 22 0010.
- B. Store and protect products under provisions of Section 22 0010.

PART 2 - PRODUCTS

2.1 DUCTWORK GENERAL

- A. All ductwork indicated on the Drawings, specified or required for the air conditioning and ventilating systems shall be of materials as hereinafter specified unless indicated otherwise. All air distribution ductwork shall be fabricated, erected, supported, etc., in accordance with all applicable standards of SMACNA Duct Manuals where such standards do not conflict with NFPA 90A and where class of construction equals or exceeds that noted herein. All exhaust ductwork including toilet room exhausts shall be constructed and leak tested as specified for medium pressure supply ducts.
- B. All ductwork shown on the Drawings, specified or required for the heating, ventilating and air conditioning systems shall be constructed and erected in a first class workmanlike manner. The work shall be guaranteed for a period of one (1) year from and after the date of acceptance of the job against noise, chatter, whistling, vibration, and free from pulsation under all conditions of operation. After the system is in operation, should these defects occur, they shall be corrected as directed by the Architect.
- C. All duct sizes shown on the Drawings are metal sizes.
- D. All holes in ducts for damper rods and other necessary devices shall be either drilled or machine punched (not pin punched), and shall not be any larger than necessary. All duct openings shall be provided with sheet metal caps if the openings are to be left unconnected for any length of time.
- E. Except for special ducts specified elsewhere herein, all sheet metal used on the project shall be constructed from prime galvanized steel sheets and/or coils up to 60" in width. Each sheet shall be stenciled with manufacturer's name and gauge. Coils of sheet steel shall be stenciled throughout on ten foot (10') centers with manufacturer's name and must be visible after duct is installed. Sheet metal must conform to SMACNA sheet metal tolerances as outlined in SMACNA's "HVAC Duct Construction Standards."
- F. Where ducts, exposed to view (including equipment rooms), pass through walls, floors or ceilings, furnish and install sheet metal collars around the duct.
- G. Branch take offs shall not be made directly across from one another but shall be offset at least one duct diameter.

2.2 PRESSURE CLASSIFICATION

- A. Ductwork where maximum dimension is less than 97" shall be constructed based on applicable pressure classification in accordance with SMACNA Manual including sheetmetal gauge, reinforcement gauge and spacing.
- B. Construct the following for 1" pressure classification, Table 1- 4:
 - 1. Supply ductwork downstream of air terminal units
 - 2. Low pressure supply ductwork to reheat coils
 - 3. Low pressure supply and return air ductwork at fan coil units
- C. Construct the following for 2" pressure classification, Table 1- 5:

1. Return ductwork between grilles and fire/smoke damper at shaft riser.
 2. Exhaust ductwork between grilles and fire/smoke damper at shaft riser.
 3. Make-up air ductwork between grilles/appliances and fire/smoke damper at shaft riser.
- D. Construct the following for 4" w.g. pressure classification, Table 1-7:
1. Return in vertical shafts.
 2. Exhaust ductwork between laboratory hoods and exhaust fans.
 3. Make-up air ductwork between AHU and fire/smoke dampers located at shaft openings and make-up air ductwork in mechanical rooms.
 4. Outside air ductwork between OAU's and AHU's on in mechanical rooms
- E. Construct the following for 6" w.g. pressure classification Table 1-8:
1. Supply ductwork and plenums between shaft riser fire/smoke damper and terminal boxes.
 2. Return ductwork between AHU first fire/smoke dampers located in the shaft including ductwork in mechanical rooms

2.3 DUCTWORK LOW PRESSURE (INCLUDES ALL NON-LAB "GENERAL" AND TOILET EXHAUST DUCTWORK)

- A. The scope of low pressure ductwork is defined as all ductwork downstream of terminal units, and all non-lab exhaust ductwork. Construction of all low pressure duct shall be in accordance with Low Velocity Duct Construction Standards as published by Sheet Metal and Air Conditioning Contractors National Association (SMACNA) and shall be sealed and tested at 2" static with the same test procedures as medium pressure ductwork.
- B. Spiral wound round duct shall be as manufactured by United McGill Sheet Metal Company or approved equal.
- C. The metal gauges listed in the 1985 SMACNA HVAC Duct Construction Standards for Metal and Flexible Duct are the minimum which shall be used for this project. It shall be noted that the Contractor is responsible that the metal gauge selected is heavy enough to withstand the physical abuse of the installation.
- D. Elbows shall be radius type and have a centerline radius of 1-1/2 times the duct diameter or width. Elbows in round ducts may be smooth radius as described above or 5-piece 90 degree elbows and 3-piece 45 degree elbows. Joints in round ducts shall be slip type with a minimum of three sheet metal screws. Joints in sectional elbows shall be sealed as specified for duct sealing.
- E. Sealant: All ductwork (except welded exhaust duct) shall be sealed with either "MP" (Multi Purpose), Hardcast "Iron grip 601", Polymer Adhesive "Airseal/AirSeal #11", or "United Duct Seal" (United McGill Corp.), or Ductmate 5511M pumpable duct sealant in the longitudinal joints on the coil line and DP1010 duct sealer for field applied joints, water base, latex or acrylic type sealant to provide a seal class "A". Note that, except as noted, oil or solvent based sealants are specifically prohibited for use on this project. For exterior applications, "Uni Weather" (United McGill Corp.) neoprene based sealant shall be used. No other sealants may be used. All sealants shall be UL rated at no more than flame spread of 5 and smoke developed of 0.

2.4 DUCTWORK MEDIUM PRESSURE (INCLUDES ALL LAB EXHAUST DUCTWORK)

- A. The scope of medium pressure ductwork is defined as all ductwork downstream of all air handlers, up to and including terminal units, all return air ductwork, and all lab exhaust (LE) ductwork. Construction of all ducts shall be in accordance with High Velocity Construction Standards as published by SMACNA. All round and rectangular duct construction, duct fittings, dampers, etc., are covered in this manual and it is to be adhered to.
1. Spiral wound round duct shall be as manufactured by United McGill Sheet Metal Company or approved equal.
 2. The metal gauges are listed herein for round duct and for rectangular duct.
- B. All ductwork (except welded exhaust duct) shall be sealed with either "MP" (Multi-Purpose), Hardcast "Iron-grip 601", "United Duct Seal" (United McGill Corp.), or Ductmate 5511M pumpable duct sealant in the longitudinal joints on the coil line and DP1010 duct sealer for field applied joints, water base, latex or acrylic type sealant. Note that, except as noted, oil or solvent based sealants are specifically prohibited for use on this project. For exterior applications, "Uni Weather" (United McGill Corp.) solvent based sealant shall be used. No other sealants may be used. All sealants shall be UL rated at no more than flame spread of 5 and smoke developed of 0.
- C. Oval ducts shall be spiral flat oval or welded flat oval equal to those of United McGill Sheet Metal Company with gauges and reinforcing as recommended by the manufacturer for medium pressure or the ducts may be Shop fabricated of completely welded construction of the following gauge:
- | | |
|----------------------|--------------|
| Major Axis 12 to 20 | No. 24 gauge |
| Major Axis 20 to 30 | No. 22 gauge |
| Major Axis 30 to 46 | No. 20 gauge |
| Major Axis 46 to 50 | No. 18 gauge |
| Major Axis 50 and Up | No. 16 gauge |
- D. Oval fittings shall be equal to those of United McGill Sheet Metal Company with requirements, sealing, etc., similar to that specified for round medium pressure work.
- E. Oval duct reinforcing methods shall be submitted as Shop Drawings for approval. Reinforcing galvanized angles shall be of sizes specified for same size rectangular ducts. Galvanized angles shall be used where standing seams are specified for rectangular ducts. Attaching methods shall be shown on Shop Drawings and submitted for approval.
- F. Low, medium and high pressure ducts, round and rectangular, up to 48" wide shall be hung with 22 gauge strap per SMACNA standards. Ducts 48" wide or larger shall be supported with trapeze hangers of angle iron or unistrut and rods, also per SMACNA.
- G. All lab exhaust ductwork shall be constructed as for medium pressure ducts and shall be tested for leaks in the same manner as for medium pressure supply ducts.
- H. Contractor may at his option use DUCTMATE coupling system on rectangular ductwork. Contractor may at his option where space permits use rectangular ductwork with DUCTMATE system in lieu of oval ductwork. No other flange-type duct joining system may be used.
- I. Rectangular 90 degree elbows shall be constructed with single thickness turning vanes. Radius type rectangular elbows shall have a centerline radius of 1-1/2 times the duct diameter or width. Contractor shall have the option to substitute short radius vaned elbows, but shall request the substitution at the time

of submittal of Shop Drawings, and shall request the substitution as required in Section 15010. Elbows in round or oval ducts may be smooth long radius as described above or 5-piece 90 degree elbows and 3-piece 45 degree elbows. Joints in round ducts shall be slip type with a minimum of three sheet metal screws. Joints in sectional elbows shall be sealed as specified for duct sealing.

2.5 ELBOWS

- A. Where rectangular elbows are shown, or are required for good air flow, contractor shall provide and install turning vanes. Job fabricated turning vanes, if used, shall be fabricated of the same gauge and type of material as the duct in which they are installed. Vanes must be fabricated for same angle as duct offset. Radius elbows shall have a centerline radius of not less than one and one-half (1-1/2) times the duct width. Submit Shop Drawings on factory fabricated and job fabricated turning vanes. Provide turning vanes in all rectangular radius elbows and offsets.
- B. All turning vanes shall be anchored to the cheeks of the elbow in such a way that the cheeks will not breathe at the surfaces where the vanes touch the cheeks. In most cases, this will necessitate the installation of an angle iron support on the outside of the cheek parallel to the line of the turning vanes.
- C. Turning vanes:
 - 1. Acceptable manufacturers: Aero Dyne
 - 2. Substitutions: Not permitted.
 - 3. True airfoil design; smoothly-rounded entry nose with extended trailing edge. Generated sound power level shall not exceed 54 decibels in band 4 at 2000 FPM in a 24"x24" duct.
 - 4. Fabricate assemblies with Aero Dyne Co. side rails; install vanes on design centers of 2.4 inches across the full diagonal dimension of the elbow.
 - 5. Submit Aero Dyne product and performance data for review.

2.6 FLEXIBLE DUCTS

- A. Low Pressure Insulated Flexible Duct may be used where shown on the drawings. Duct shall be made with factory preinsulated duct composed of dead soft, spiral wound, triple locked corrugated aluminum core covered with a minimum of 1-1/2" thick, 3/4 lb. density fiberglass blanket sheathed in a vapor barrier of fiberglass reinforced aluminum foil and mylar laminate. The insulation shall have a minimum "K" factor of 0.29 at 60 degrees F. mean and a vapor barrier permeability rating of 0.05 per ASTM method E96-66, Procedure A. The C factor shall be 0.24 to meet HUD requirements. The duct shall be rated for a positive working pressure of 10" w.g. and a temperature of up to 250 degrees F. The duct must comply with the latest NFPA Bulletin 90A and be listed and labeled by Underwriter's Laboratories, Inc., as Class I Air Duct, Standard 181, and meet GSA, FHA and other U. S. Government standards; flame spread, not over 15; smoke developed, not over 10. Flexible ducts shall be not more than 51" in length, shall be installed in straight runs only, and shall be Flexmaster Type TL-M or approved equal. Flexmaster type 3M with tri-laminate aluminum foil, fiberglass and aluminized polyester core and jacket as specified above may be used for connections to supply diffusers and return grilles. For the first six (6) patient room toilet grilles tapped from the main branch line on each floor (each side) shall be provided with Flexmaster type 6M with metalized vapor barrier, fiberglass insulation and mechanical lock helix. Provide a minimum of one (1) elbow in flex duct, this is for sound attenuation.
 - 1. The terminal ends of the duct core shall be secured by compression coupling or stainless steel worm gear type clamp equal to Ideal Series 56 Snaplock. The fittings on air mixing devices and on sheet metal duct shall be coated with the sealant specified for low pressure ductwork, then flexible duct core slipped over duct and coupling or clamp tightened, then connection sealed with more sealant. Insulation of flexible duct shall be slipped over connection to point where insulation abuts mixing box or insulation on duct. These insulation connections shall be sealed by imbedding fiberglass tape in the sealant specified for medium pressure ductwork and coating with more sealant

to provide a vapor barrier. (This applies to all flex connections to diffusers, grilles, etc. when allowed on the drawings.)

2. Low Pressure Taps (Conical Bell Mouth Fittings)

- a. Conical fittings may be used for duct taps and shall include quadrant dampers on all lines to air devices (diffusers and grilles) even though a volume damper is specified for the air device. (This does not apply to medium pressure duct.) Spin-in fittings shall be sealed at the duct tap with a gasket, or compression fit, or sealed with sealant specified for medium pressure ductwork. The location of spin-in fittings in the ducts shall be determined after dual or single duct terminal units are hung or the location of the light fixtures is known so as to minimize flexible duct lengths and sharp bends.
 - b. The conical fitting shall be made of at least 26 gage sheet metal. The construction to be a two-piece fitting with a minimum overall length of 6 inches and shall be factory sealed for high pressure requirements. Average loss coefficient for sizes 6, 8, and 10 shall be less than 0.055.
 - c. Each to be provided with minimum 24 gage damper plate with locking quadrant operator and sealed end bearings. Damper blade shall be securely attached to shaft to prevent damper from rotating around shaft.
 - d. Provide flange and gasket with adhesive peel-back paper for ease of application. The fitting shall be further secured by sheet metal screws spaced evenly at no more than 4 inches on-center with a minimum of four screws per fitting.
 - e. The conical bellmouth fitting shall be Series 3000G as manufactured by Flexmaster U.S.A., Inc., or Buckley Air Products, Inc., 'AIR-TITE'.
- B. Medium and High Pressure Insulated Flexible Duct shall be the same construction as the Low Pressure Duct, factory applied insulation of 1" minimum thickness, 3/4 lb. density with a permeability rating of 0.30. The duct shall be supported by a corrosion resistant metal spiral, or a coated spring steel helix and solid innerliner mechanically interlocked or permanently bonded to the helix wire. Ratings shall be as described for Low Pressure Duct above. Flexible ducts shall be not more than 2'-0" in length, used for alignment or sound/vibration purposes only, and may only be installed in straight runs. Flexible duct shall NOT be used for changes of direction of air flow. Installation, clamps and sealing shall be the same as specified for rigid duct.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Refer also to requirements included in Part 2 of this specification.
- B. Provide openings in ductwork where required to accommodate thermometers and controllers. Provide pilot tube openings where required for testing of systems, complete with metal can with spring device or screw to ensure against air leakage. Where openings are provided in insulated ductwork, install insulation material inside a metal ring.
- C. Locate ducts with sufficient space around equipment to allow normal operating and maintenance activities.
- D. Slope underground ducts to plenums or low pumpout points at 1:500. Provide access doors for inspection.
- E. Coat buried, metal ductwork without factory jacket with one coat and seams and joints with additional coat of asphalt base protective coating.

- F. Connect terminal units to medium or high pressure ducts directly or with two feet maximum length of flexible duct. Do not use flexible duct to change direction. Allow for a minimum of 4 diameters of straight duct to the entrance of all terminal units.
- G. Connect diffusers or troffer boots to low pressure ducts with 2 feet maximum length of flexible duct. Hold in place with strap or clamp, and seal as specified.
- H. During construction provide temporary closures of metal or taped polyethylene on open ductwork to prevent construction dust from entering ductwork system.

3.2 DUCT SUPPORTS

- A. Hanger straps on duct width of 60 inches and under shall lap under the duct a minimum of 1 inch and have minimum of one fastening screw on the bottom and two on the side.
- B. Hanger straps on duct widths over 60 inches shall be bolted to duct reinforcing with 3/8" bolts minimum.
- C. Use 3/8" minimum bolts for securing round duct hanger straps to band straps.

3.3 DUCTWORK APPLICATION SCHEDULE

<u>Air Steam</u>	<u>Material</u>
Low Pressure Supply	Steel, Aluminum
Medium and High Pressure Supply	Steel
Return and Relief	Steel, Aluminum
General and Toilet Exhaust	Steel, Aluminum (branch duct in toilets and showers shall be aluminum)
General Lab Exhaust	Stainless Steel, Aluminum
Fume Hood Exhaust	Stainless Steel, Aluminum
Outside Air Intake	Steel
Biosafety Hood Exhaust	Stainless Steel

3.4 MAINTAINING DUCT SYSTEMS CLEANLINESS

- A. Submit and execute a plan for temporary operation of HVAC systems, including the following as minimum.
 - 1. Protect all air handling and distribution equipment, and air supply and return ducting during construction. Inspect work and monitor as required for compliance.
 - 2. All ductwork arriving on site shall be cleaned and sealed in the shop with plastic sheeting and stored on pallets or dunnage until installed.
 - 3. Cover and protect all exposed air inlets and outlets, openings, grilles, ducts, plenums etc, to prevent water, moisture, dust and other contaminate intrusion.
 - 4. Apply protection immediately after installation of equipment and ducting.
 - 5. Ducting runs will be protected at the end of each day's work.
 - 6. If air handling units will be used during construction, install and maintain specified filters including 90% and 99.97% efficiency filters where required. Use filtered outside air only for ventilation until construction in area served is complete and area is thoroughly cleaned. In addition, install and weekly maintain temporary filtration media with at least a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2 - 1999 on all outside air and return air intakes to protect the return air ductwork and plenums within the construction zone.
 - 7. Maintain minimum MERV 13 filtration on all air handling units used during construction.

8. Inspect temporary filtration weekly and replace as required to maintain the proper ventilation rates in the building.
9. Coordinate use of air handling equipment with ICRA plan, if applicable. Maintain required pressure relationships in construction areas adjacent to occupied areas.

3.5 LEAKAGE TESTING OF INSTALLED SYSTEMS

- A. Ductwork shall be pressure tested by HVAC Test and Balance Contractor. Refer to Section 23 05 93.
- B. Test duct for leakage in accordance with SMACNA HVAC Air Duct Leakage Test Manual. Use prescribed test kit containing test blower, two U-tube manometers and calibrated curve attached to the orifice tube assembly.
- C. Pressure testing shall include taps/take-offs to air terminal units in medium pressure ductwork and taps/take-offs to air devices in supply, return, and exhaust ductwork.
- D. Pressurize all installed duct systems for each pressure class to maximum pressure for fabrication classification. The leakage amount shall not exceed the allotted amount for the pressure class or the allotted amount for that portion of the system as follows:
 1. 1" Pressure Class - Leakage Class 6; Max. Leakage Factor - 6.0 CFM/100 SF
 2. 2" Pressure Class - Leakage Class 6; Max. Leakage Factor - 9.4 CFM/100 SF
 3. 1" and 2" Pressure Class exhaust ductwork connected to or serving fume hoods, bio-safety cabinets, chemical or hazardous storage rooms, smoke removal/purge systems, laboratory spaces, isolation rooms, bronchoscopy rooms, and nuclear medicine rooms shall be construction and tested as follows:
 - a. 1" Pressure Class - Leakage Class 3; Max. Leakage Factor - 3.0 CFM/100 SF
 - b. 2" Pressure Class - Leakage Class 3; Max. Leakage Factor - 4.7 CFM/100 SF
 4. 4" Pressure Class - Leakage Class 3; Max. Leakage Factor - 7.4 CFM/100 SF
 5. 6" Pressure Class - Leakage Class 3; Max. Leakage Factor - 9.6 CFM/100 SF
- E. All ductwork shall be leak tested first before being enclosed in a shaft or above other inaccessible areas.
- F. Correct leaks found in excess of allowable limits. Retest until acceptable leakage is witnessed.
- G. Have test results available for review on a progressive and final basis. Include all test results in project closing file along with name, signature, and date of independent witness to testing. Test results shall show preliminary and final test results and include all calculations used to determine system compliance with the maximum specified leakage rate.

3.6 AIR TEST AND BALANCE

- A. Prepare the system for tests as specified in Section 23 05 93 and 23 05 94 and correct deficiencies found by the Test and Balance firm.

END OF SECTION

SECTION 23 31 14

SHEETMETAL - SPECIAL DUCTWORK

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Wet Air Exhaust Duct
- B. Stainless Steel Ductwork

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
- C. Section 23 31 13 - Sheetmetal Ductwork
- D. Section 23 33 00 - Air Duct Accessories

1.3 REFERENCE STANDARDS

- A. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials.
- B. ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
- C. ASTM B209 - Aluminum and Aluminum Alloy Sheet and Plate
- D. NFPA 90A - Standard for the Installation of Air-Conditioning and Ventilating Systems; National Fire Protection Association ; 2012.
- E. NFPA 90B - Standard for the Installation of Warm Air Heating and Air Conditioning Systems; National Fire Protection Association ; 2012.
- F. NFPA 96 - Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations; National Fire Protection Association.
- G. SMACNA 1972 - HVAC Air Duct Leakage Test Manual; Sheet Metal and Air Conditioning Contractors' National Association ; 2012, 2nd Edition.
- H. SMACNA 1966 - HVAC Duct Construction Standards; Sheet Metal and Air Conditioning Contractors' National Association ; 2005.
- I. SMACNA 1767 - Kitchen Ventilation Systems and Food Service Equipment Fabrication & Installation Guidelines ; 2001.
- J. UL 1978 - Grease Ducts ; Current Edition, Including All Revisions.
- K. UL 2221 - Tests of Fire Resistive Grease Duct Enclosure Assemblies ; Current Edition, Including All Revisions.

1.4 SUBMITTALS

- A. Submit manufacturer's product data for review in accordance with the provisions of Division 01.
- B. Ductwork shop drawings for systems listed herein shall be coordinated and submitted along with ductwork shop drawings in Section 23 31 13 - Sheetmetal Ductwork.
- C. Coordination drawings required in Section 23 31 13 - Sheetmetal Ductwork shall also include all Special Ductwork systems.

PART 2 - PRODUCTS

2.1 MATERIAL AND FABRICATION

- A. Wet Exhaust Ductwork:
 - 1. Ducts fume hoods; sterilizer exhaust; shall be minimum 18 gauge 304 stainless steel made liquid tight with continuous external weld for all seams and joints. Provide neoprene gaskets at flanged connections. Where ducts are not self draining back to the equipment, provide low point drain pocket with 1" copper or stainless steel drain connection. Drain connection shall be capped or routed as shown on the drawings. Provide access door of the same material at each drain pocket for inspection and cleaning.

2. All exposed ductwork and fittings below the finished ceiling shall be 316 stainless steel.
 3. Where volume dampers are installed in this duct system, the damper and blade material shall be of the same material as the duct in which it is installed.
- B. Stainless Steel Ductwork:
1. Stainless steel ductwork shall be Type 304 or 316L with welded longitudinal seams and welded transverse joints. Welds on exposed ductwork shall be positioned for minimum view and shall be ground and polished. Duct sealant shall not be used to seal this ductwork.
 2. All duct risers shall be installed as vertical as possible within the constraints of the design indicated on the drawings.
 3. Metal gauges shall be not less than the following: 30 inch diameter and less--18 gauge; 31 to 60 inch diameter--16 gauge; 61 inch diameter and greater--14 gauge.
 4. In all cases, ductwork shall be installed so that any washdown water, condensation or moisture shall drain back to the hood or drain low point.
 5. Where volume dampers are installed in this duct system, the damper and blade material shall be of the same material as the duct in which it is installed.
 6. Where fire dampers or fire smoke dampers are installed in this duct system, the dampers shall be of stainless steel construction.
- C. Refer to Section 23 31 13 - Sheetmetal Ductwork for pressure classifications.

PART 3 - EXECUTION

3.1 INSTALLATION, APPLICATION, ERECTION

- A. All special ductwork shall be stored in a dry environment and protected from construction dust, debris, and damage. Open ends of ductwork shall remain covered and sealed.

3.2 CLEANING AND TESTING

- A. Clean and inspect work as specified in Section 23 05 00 - Common Work Results for HVAC.
- B. Pressure test ductwork as specified in Section 23 31 13 - Sheetmetal Ductwork.

END OF SECTION

SECTION 23 33 00

DUCTWORK ACCESSORIES

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Volume control dampers.
- B. Fire dampers.
- C. Combination fire and smoke dampers.
- D. Backdraft dampers.
- E. Air turning devices.
- F. Flexible duct connections.
- G. Duct access doors.
- H. Duct test holes.
- I. Bubble Tight Dampers

1.2 RELATED WORK

- A. Section 22 0548 - Vibration Isolation.
- B. Section 23 3113 - Ductwork.
- C. Section 23 3600 - Air Terminal Units: Medium and high pressure damper assemblies.

1.3 REFERENCES

- A. NFPA 90A - Installation of Air Conditioning and Ventilating Systems.
- B. SMACNA - Low Pressure Duct Construction Standards.
- C. UL 33 - Heat Responsive Links for Fire Protection Service.
- D. UL 555 - Fire Dampers and Ceiling Dampers.

1.4 SUBMITTALS

- A. Submit shop drawings and product data under provisions of Section 22 0010.
- B. Provide shop drawings for shop fabricated assemblies indicated, including volume control dampers duct access doors duct test holes. Provide product data for hardware used.

- C. Submit manufacturer's installation instructions under provisions of Section 22 0010 for fire dampers and combination fire and smoke dampers.
 - 1. Submittal(s) for fire and combination fire/smoke dampers shall include the following:
- D. Provide a schedule for all fire and combination fire and combination fire/smoke dampers Assign identification numbers for each damper. Provide a set of 11" x 17" drawings bound with the schedule identifying each damper in plan with the unique identification number.
- E. Provide air quantity, size, free area of damper, pressure drop and proposed velocity through each damper.
- F. Provide manufacturer's data of damper and its accessories or options.

PART 2 -- PRODUCTS

2.1 DAMPERS

- A. Furnish and install dampers where shown on the Drawings and wherever necessary for complete control of the air flow, including all supply, return and exhaust branches, "division" in main supply, return and exhaust ducts, each individual air supply outlet and fresh air ducts. Where access to dampers through a fixed suspended ceiling is necessary, the Contractor shall be responsible for the proper location of the access doors.
- B. Splitter dampers shall be fabricated of steel not lighter than 16 gauge. The leading edge of the damper shall be hemmed. Each splitter shall be a minimum of 12" long or 1-1/2 times the width of the smaller of the two branches it controls, whichever is greater. Dampers shall be carefully fitted, and shall be controlled by locking quadrants equal to Ventlok No. 555 on exposed uninsulated ductwork, No. 644 on exposed externally insulated ductwork and No. 677 (2-5/8" diameter) chromium plated cover plate for concealed ductwork not above lay-in accessible ceilings. Furnish and install end bearings for the damper rods on the end opposite the quadrant when No. 555 or No. 644 regulators are used, and on both ends when No. 677 regulators are used.
- C. On concealed ductwork above lay-in accessible ceilings use Ventlok No. 555 or No.644 locking quadrant for splitter dampers.
- D. Dampers larger than three (3) square feet in area shall be controlled by means of rods hinged near the leading edge of the damper with provisions for firmly anchoring the rod and with end bearings supporting the axle.
- E. Volume dampers shall be equal to those of American Foundry. Blades shall not exceed 48 inches (48") in length or twelve inches (12") in width and shall be of the opposed interlocking type. The blades shall be of not less than No. 16 gauge galvanized steel supported on one-half inch (1/2") diameter rust-proofed axles. Axle bearings shall be the self-lubricating ferrule type.
- F. Install all dampers furnished by the Temperature Control manufacturer's in strict accordance with the manufacturer's recommendations and requirements of these Specifications. All adjustable dampers installed in externally insulated ductwork shall be installed with Ventlok No. 639, or equal, elevated dial operators. Insulation shall extend under the elevated dial. All adjustable dampers installed in internally insulated ductwork shall be installed with Ventlok No. 635, or equal, dial operators. All damper shaft penetrations in the ductwork shall be installed with Ventlok No. 609 end bearings.

2.2 FIRE AND FIRE/SMOKE DAMPERS

A. SLEEVES FOR RATED DAMPERS

1. Unless otherwise required by the authority having jurisdiction, sleeves for fire dampers, smoke dampers and combination fire and smoke dampers shall be provided by the damper manufacturer and be of rigid type construction recommended in Schedule 2 of SMACNA Publication for "Fire Damper and Heat Stop Guide for Air Handling Systems". Use 16 gauge for ducts 24" or less in diameter or either rectangular dimension and 14 gauge for ducts over 24". Provide minimum 18" long sleeves. Coordinate required length with wall thicknesses.
2. Install 1-1/2"x1-1/2"x1/8" angle bar on four sides of sleeves and both sides of wall. Fasten angles to sleeve only. Do not fasten to the wall.

B. Fire Dampers:

1. Furnish and install where shown on the drawings or required by the Specifications, fire dampers meeting the following requirements.
2. Each fire damper shall be constructed and tested in accordance with Underwriters Laboratories Safety Standard 555, 4th Edition. All dampers shall possess a 1-1/2 hour or 3 hour (as appropriate for the construction shown in the architectural) drawing protection rating, 160 or 165 degree F fusible link, and shall bear a U.L. label in accordance with Underwriters' Laboratories labeling, procedures. Fire dampers shall be constructed such that the damper frame material and the curtain material shall be galvanized.
3. Fire dampers shall be curtain blade type and the damper shall be so constructed that the blades are out of the air stream to provide a 100 percent free area of the duct in which the damper is housed.
4. The damper manufacturer's literature submitted for approval prior to the installation shall include performance data developed from testing, in accordance with AMCA 500 Standards and shall show the pressure drops for all sizes of dampers required at anticipated air flow rates. Maximum pressure drop through the damper shall not exceed 0.05 inch water gauge.
5. Fire dampers shall be equipped for vertical or horizontal installation as required by the locations shown in the drawings. Fire dampers shall be installed in wall and floor openings utilizing steel sleeves, angles and other material and practices required to provide an installation equivalent to that utilized by the manufacturer when the respective dampers were tested by Underwriters Laboratories. (Note: Prefco Model 5500 E6-M fire dampers do not require sleeves.) Mounting angles shall be a minimum of 1-1/2 inch by 1-1/2 inch by 14 gauge and bolted, tack welded or screwed to the sleeve at maximum spacings of 12 inches and with a minimum of two connections at all sides. Mounting angles shall overlap at least equal to the gauge of the duct defined by the appropriate SMACNA Duct Construction Standard, latest edition, and as described in NFPA 90A. The entire assembly, following installation, shall be capable of withstanding 6" water gauge static pressure.
6. The damper installation shall be in accordance with the damper manufacturer's instructions.
7. All fire dampers shall comply with the specification as written above and shall be Ruskin Model IBD2 (Style C, CR or CO), Greenheck Model FD-150 (Type C, CR or CO), or Prefco Model 5500 (E6-M, CR, or CO).
8. The Contractor shall completely caulk the damper into the sleeve if recommended by the damper manufacturer using manufacturer recommended material(s).

C. Combination Fire/Smoke Dampers:

1. Furnish and install where shown on the Drawings, or as required by the Specifications, combination fire/smoke dampers meeting the following requirements.
2. Each combination fire/smoke damper shall be 1-1/2 hour fire rated under UL Standard 555, 4th Edition, and shall be further classified by Underwriters Laboratories as a Leakage Rated Damper for use in control systems under smoke the latest version of UL 555S, and bear a UL label attesting to same. The damper manufacturer shall have tested, and qualified with UL, a complete range of

damper sizes covering all dampers required by this Specification. Testing and UL qualifying a single damper size is not acceptable. The leakage rating under UL555S shall be no higher than Leakage Class I (4 cfm per square foot at one inch water gage pressure and 8 cfm per square foot at 4 inches water gage pressure). The maximum air pressure drop through each combination fire/smoke damper shall not exceed 0.10 inch water gage at the design air quantity. (Note that this may require a larger damper than the connected duct size.)

3. The damper frame shall be a minimum of 20 gauge galvanized steel formed into a structural hat channel shape with tabbed corners for reinforcement, as approved in testing by Underwriters Laboratories. Bearings shall be integral high surface area non electrolytic materials construction to incorporate a friction free frame blade lap seal, or molybdenum disulfide impregnated stainless steel or bronze oilite sleeve type turning in an extruded hole in the frame or an extruded frame raceway. The dampers may be either parallel or opposed blade type. The blades shall be constructed with a minimum of 14 gauge equivalent thickness. The blade edge seal material shall be able to withstand 450 degrees F. The jamb seals shall be flexible stainless steel compression type or lap seal type.
4. In addition to the leakage ratings specified herein, the combination fire/smoke dampers and their operators shall be qualified under UL555S to an elevated temperature of 250 degrees F. Electric operators shall be installed by the damper manufacturer at the time of damper fabrication. The damper and operator shall be supplied as a single entity which meets all applicable UL555 and UL555S qualifications for both dampers and operators. The manufacturer shall provide a factory assembled sleeve. The sleeve shall be a minimum of either 20 gauge for dampers where neither width nor height exceeds 48 inches or 16 gauge where either dimension equals or exceeds 48 inches.
5. As part of the UL qualification, dampers shall have demonstrated a capacity to operate (open and close) under HVAC system operation conditions, with pressures of at least 4 inches water gage in the closed position, and 2500 fpm air velocity in the open position.
6. Each combination fire/smoke damper shall be equipped with a UL Classified Firestat/releasing device. The firestat/releasing device shall electrically and mechanically lock the damper in a closed position when the duct temperatures exceed 165 degrees F and still allow the appropriate authority to operate the damper as may be required for smoke control functions. The damper must be operable while the temperature is above 250 degrees F. The actuator/operator package shall include two damper position indicator switches linked directly to damper blade to provide capability of remotely indicating damper position. One switch shall close when the damper is fully open, and the other switch shall close when the damper is fully closed. The firestat/releasing device and position indicator switches shall be capable of interfacing electrically with the smoke detectors, building fire alarm systems, and remote indicating/control stations.
7. The damper releasing device shall be mounted within the airstream. The device shall be activated and the damper shall close and lock when subjected to duct temperatures in excess of approximately 285 degrees F.
8. Motors for operation of smoke dampers shall be smoke system fail safe, spring return normally open supplies and normally closed returns, or as indicated in the plans, and shall be furnished and installed by the damper manufacturer as required by the U.L. rating, mentioned above. Motors shall be electric or pneumatic to match the type of temperature control system specified elsewhere in this specification. All required relays, EP switches, wiring piping and other labor and material necessary to completely interconnect the smoke detector system shall be furnished by the Contractor.
9. Each damper shall be furnished in a square or rectangular configuration. The Contractor shall furnish and install sleeves manufactured by the approved damper manufacturer for each damper. The sleeves shall be constructed with square or rectangular to square, rectangular, round, or oval adapters as required. Dampers shall be installed in the sleeves in accordance with manufacturers U.L. installation instructions. The entire assembly, following installation, shall be capable of withstanding 6" w.g. static pressure.

10. All combination fire/smoke dampers shall comply with the specification as written above and shall be Ruskin Model FSD-60, Greenheck Model FSD-33, or Prefco Model 5020-1 (or Model 5010 for manual reset dampers).
11. The Contractor shall completely caulk the damper into the sleeve as recommended by the damper manufacturer using manufacturer recommended material(s).

D. Submittal and Installation:

1. The air quantity through each fire and combination fire and smoke damper will be identified in Part 2 of the construction. Dampers to be full size of the duct size or wall opening as shown on the drawings.
2. Access doors as specified elsewhere shall be provided to make all parts of the damper accessible. Doors shall open not less than 90 degrees following installation and shall be insulated type where installed in insulated ducts.
3. Contractor shall install each damper square and true to the building. The installation shall not place pressure on the damper frame, but shall enclose the damper as required by UL555.
4. After each fire damper and combination fire and smoke damper has been installed and sealed in their prescribed opening, and prior to the installation of the ceilings, the Contractor shall, as directed by the Construction Inspector, activate part or all the dampers as required to verify "first-time" closure. Activation of the damper shall be accomplished by manually operating the resettable link, disconnecting the linkage at the fusible link of the fire damper, and manually operating the fire/smoke damper through the pneumatic or electronic controls as appropriate. Failure of the damper to close properly and smoothly on the first attempt will be cause to replace the entire damper assembly.

2.3 FLEXIBLE CONNECTIONS

- A. Where ducts connect to fans, including roof exhausters, flexible connections shall be made using "Ventglas" fabric that is fire resistant, waterproof, mildew resistant and practically air tight, and shall weigh approximately thirty ounces (30 oz.) per square yard. There shall be a minimum of one-half inch (1/2") slack in the connections, and a minimum of two and one-half inches (2-1/2") distance between the edges of the ducts except that there shall also be a minimum of one inch (1") of slack for each inch of static pressure on the fan system. This does not apply to Air Handling Units with internal isolation.

2.4 ACCESS DOORS

- A. Furnish and install in the ductwork, hinged rectangular or round "spin-in" access doors to provide access to all fire dampers mixed air plenums, upstream of steam reheat coils, automatic dampers, etc. Where the ducts are insulated, the access doors shall be double skin doors with one inch (1") of insulation in the door. Where the size of the duct permits, the doors shall be eighteen inches (18") by sixteen inches (16"), or eighteen inches in diameter, and shall be provided with Ventlok No. 260 latches (latches are not required in round doors). Latches for rectangular doors smaller than 18" x 16" shall be Ventlok No. 100 or 140. Doors for zone heating coils shall be Ventlok, stamped, insulated access doors, minimum 10" x 12", complete with latch and two (2) hinges, or twelve inches (12") in diameter. Round access doors shall be "Inspector Series" spin-in type door as manufactured by Flexmaster USA, or approved equal. Doors for personnel access to ductwork shall be nominal twenty-four inches (24") in diameter.
- B. Where these access doors are above a suspended ceiling, this Contractor shall be responsible for the proper location of the ceiling access doors.

2.5 SCREENS

- A. Furnish and install screens on all duct, fan, etc., openings furnished by this Contractor which lead to, or are, outdoors. Screens shall be No. 16 gauge, one-half inch (1/2") mesh in removable galvanized steel frame. Provide safety screens meeting OSHA requirements for protection of maintenance personnel on all fan inlets and fan outlets to which no ductwork is connected.

2.6 TEST OPENINGS

- A. Furnish and install in the return air duct and in the discharge duct of each fan unit Ventlok No. 699 instrument test holes. The test holes shall be installed in locations as required to measure pressure drops across each item in the system, e.g., O.A. louvers, filters, fans, coils, intermediate points in duct runs, etc.

2.7 BUBBLE TIGHT DAMPERS

- A. Acceptable manufacturers: EB Air Control Inc - ICPSP series or Greenheck - HBTR series.
 - 1. Substitutions: Not permitted.
- B. Fabricate in accordance with NFPA 90A, UL 555, and as indicated.
- C. Material: 304 stainless steel
- D. Damper casing shall be round or square to match adjacent ductwork. Seals around actuator shaft shall be leakproof and rated at no less than 250,000 open/close cycles. Actuator shall be located outside the airstream.
- E. Bearings shall be nylon or ball bearing with grease zerkl type accessible from outside the damper.
- F. Dampers shall be rated for zero bubble leakage at 10 in wg. Damper shall be leak tested in accordance with AMCA 500-D figure 5.8 before it leaves the factory to ensure bubble-tight performance.
- G. Dampers shall be rated for 3900 feet/minute velocity and operation at up to 250 deg. F.
- H. Pressure loss thru open damper shall not exceed 0.06" w.g. at 2000 FPM and 0.15" w.g. at 2500 FPM.
- I. Provide two-position 120V actuator. Damper shall fail normally closed unless noted otherwise.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install accessories in accordance with manufacturer's instructions.
- B. Provide balancing dampers at points on low pressure supply, return, and exhaust systems where branches are taken from larger ducts as required for air balancing. Use splitter dampers only where indicated.
- C. Provide balancing dampers on medium and high pressure systems where indicated. Refer to Section 15930 - Air Terminal Units.
- D. Provide fire dampers, and combination fire and smoke dampers at locations indicated, where ducts and outlets pass through fire-rated components. Install with required perimeter mounting, angles, sleeves, breakaway duct connections, corrosion resistant springs, bearings, bushings and hinges.

- E. Demonstrate re-setting of fire dampers to Owner's representative.
- F. Provide backdraft dampers on exhaust fans or exhaust ducts where indicated.
- G. Provide flexible connections immediately adjacent to equipment in ducts associated with fans and motorized equipment. Cover connections to medium and high pressure fans with leaded vinyl sheet, held in place with metal straps.
- H. Provide duct access doors for inspection and cleaning before and after duct mounted filters, coils, fans, automatic dampers, at fire dampers, and elsewhere as indicated. Provide minimum 8 x 8 inch (200 x 200 mm) size for hand access, 18 x 18 inch (450 x 450 mm) size for shoulder access, and as indicated.
- I. Provide duct test holes where indicated and where required for testing and balancing purposes. Refer also to Section 23 0593.
- J. On any individual floor or suite of floors, all air distribution devices visible from occupied spaces must be by the same manufacturer, mixing of similar devices by different manufacturers is not allowed.

END OF SECTION

SECTION 23 34 17

BELTED VENT SETS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Belted Vent Sets

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 05 13 - Common Motor Requirements for HVAC Equipment
- C. Section 23 05 48 - Vibration Isolation
- D. Division 26: Electrical

1.3 REFERENCE STANDARDS

- A. AMCA 99 - Standards Handbook
- B. AMCA 211 - Laboratory Methods of Testing Fans for Rating Purposes
- C. AMCA 300 - Test Code for Sound Rating Air Moving Devices
- D. AMCA 301 - Method of Publishing Sound Ratings for Air Moving Devices
- E. ANSI/AFBMA 9 - Load Ratings and Fatigue Life for Ball Bearings

1.4 SUBMITTALS

- A. Submit manufacturer's product data for review.
- B. Fan curves shall include entire range of RPM curves, scheduled operating point, brake horsepower, motor horsepower, and sound performance data.

1.5 QUALITY ASSURANCE

- A. Performance Ratings: Conform to AMCA 211 and bear the AMCA Certified Rating Seal.
- B. Sound power ratings: Provide in 8 octave bands, based on AMCA Standard 300-67, test set up #1.

1.6 OPERATION AND MAINTENANCE DATA

- A. Submit operation and maintenance data as specified in Division 01.
- B. Include instructions for lubrication, motor, spare parts list, and wiring diagrams.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to site under provisions of Division 01.
- B. Store and protect products under provisions of Division 01.
- C. Protect motors, shafts, and bearings from weather and construction dust.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Greenheck or Loren Cook.
- B. Substitutions: Refer to Division 01.

2.2 EQUIPMENT REQUIREMENTS

- A. Assembly: Packaged unit including fan, motor, adjustable belt drive or direct drive, and motor base.
- B. Material: Low carbon steel.
- C. Finish: Factory applied, high quality corrosion resistant enamel paint.
- D. Wheel diameters and discharge areas: Sized in accordance with AMCA standards.
- E. Housings:
 - 1. Air tight lock seam construction using heavy gauge steel, reinforced to prevent vibration and pulsation.
 - 2. Field rotatable to eight standard discharge positions.
 - 3. Suitable for outdoor mounting with weatherproof motor and drive cover.
- F. Fan wheels: Non-overloading backward inclined type.
- G. Shafts: ASTM A-108 steel, grade 1018 or 1045.
- H. Bearings:
 - 1. Design for heavy duty service with a nominal L-10 life of 40,000 hours based on maximum catalogued operating speed.
 - 2. Lubricatable anti-friction, self-aligning pillow block ball bearings.
 - 3. Support to directly oppose drive tension and transmit loads to the fan base.
- I. Special construction: Spark-proof constructed fans with explosion proof motors as noted on drawings.
- J. Accessories: Punched inlet and outlet flanges, quick opening access door, and drain.
- K. Discharge dampers or shutters: As noted on drawings.
- L. Balancing

1. Dynamically balance at factory after assembly:
 - a. Use IRD or PMC analyzer to measure velocity. Final reading not to exceed 0.1 inches per second.
 - b. Record exact level of vibration as proof of the final dynamic balance.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install fans in locations shown on drawings and in accordance with manufacturer's instructions.
- B. Provide flexible connection between ductwork and fan inlet.
- C. Support associated suction and discharge ductwork from roof to prevent bearing on fan.

3.2 TEST AND ACCEPTANCE

- A. Start-up and checkout for proper motor phasing, alignment, and vibration free operation. Correct improperly aligned fans. Change unmatched belts.
- B. Verify control interlocks.
- C. Demonstrate and instruct system operation to maintenance personnel.

END OF SECTION

SECTION 23 36 00

AIR TERMINAL UNITS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Single duct terminal units.

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC.
- B. Section 23 05 13 - Common Motor Requirements for HVAC Equipment
- C. Section 23 05 23 - General Duty Valves for HVAC
- D. Section 23 05 29 - Hangers for HVAC Piping
- E. Section 23 05 53 - Identification for HVAC Piping and Equipment
- F. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
- G. Section 23 07 00 - HVAC Insulation
- H. Section 23 09 13 - Instrumentation and Control Devices for HVAC
- I. Section 23 21 13 - Hydronic Piping
- J. Section 23 21 14 - Hydronic Specialties
- K. Section 23 31 13 - Sheetmetal Ductwork
- L. Section 23 33 00 - Air Duct Accessories
- M. Section 26 27 17 - Equipment Wiring: Electrical characteristics and wiring connections.

1.3 REFERENCE STANDARDS

- A. NFPA 90A - Standard for the Installation of Air-Conditioning and Ventilation Systems; National Fire Protection Association; 2012.
- B. UL 181 - Standard for Factory-Made Air Ducts and Air Connectors; Underwriters Laboratories Inc.; Current Edition, Including All Revisions.
- C. Acoustical Liner: Meet requirements of NFPA 90A, UL 181, and ASTM C665 as specified.
- D. Air Diffusion Council, ADC Standard 1062R2, Air Diffusing Equipment Test Code.
- E. Air Moving and Conditioning Association, AMCA Standard 210, Test Code for Air Moving Devices.
- F. SMACNA HVAC Duct Construction Standards; Current Edition.

1.4 SUBMITTALS

- A. See Division 01 requirements.
- B. Product Data: Provide data indicating configuration, general assembly, materials used in fabrication, access door location and size, insulation thickness, density, and R-value. Include specific performance ratings that indicate unit ID, airflow setpoints, coil performance, air pressure drop, NC rating, and electrical characteristics and connection requirements.
- C. Manufacturer's Installation Instructions: Indicate support and hanging details, and service clearances required.
- D. Project Record Documents: Record actual locations of units.
- E. Operation and Maintenance Data: Include manufacturer's descriptive literature, operating instructions, maintenance and repair data, and parts lists.
- F. Substitutions: Refer to Division 1.
- G. Warranty: Submit manufacturer warranty and ensure forms have been completed in Owner's name and registered with manufacturer.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing the type of products specified in this section, with minimum three years of documented experience.
- B. Products Requiring Electrical Connection: Listed and classified by Underwriters Laboratories Inc. as suitable for the purpose specified and indicated.

1.6 WARRANTY

- A. See Division 1 for additional warranty requirements.
- B. Provide five year manufacturer warranty for air terminal units.

1.7 GUARANTEE

- A. Manufacturer guarantees resultant noise levels to be within NC rating published by manufacturer.

PART 2 PRODUCTS

2.1 SINGLE DUCT AIR TERMINAL UNITS

- A. Acceptable Manufacturers:
 - 1. Krueger, Metalaire, Nailor, Price, Titus, and Trane.
- B. Basic Assembly:
 - 1. Casings: Minimum 22 gauge galvanized steel.
 - 2. Lining:
 - a. Dual Wall: Interior liner of minimum 22 gauge phosphatized steel covering the insulation. All cut edges of insulation shall be covered with metal flange. All wire penetrations shall be covered by grommets. High density, glass fiber insulation, 1" thick, 1.9 lb/cu.ft., R-Value of 4.2.

- b. Insulation shall comply with the requirements of UL 181, NFPA 90A, and ASTM C665.
 - 3. Provide insulated gasketed access panel on bottom of terminal unit for access to internal air valve and heating coil inspection. Adjacent duct insulation shall not block access door.
 - 4. Leakage: Maximum 1% of maximum rated airflow at 1" wg. inlet static pressure.
 - 5. Multi-point, multi-axis flow ring or cross sensor at box inlet.
 - 6. Provide integral flow taps and calibration chart on each unit.
 - 7. Factory calibrate sensor and controller for maximum, minimum, heating, and unoccupied design airflow according to the air terminal unit schedule. Terminal units scheduled for constant volume operation shall be provided with controls capable of variable volume operation.
 - 8. Factory mount, wire, connect, calibrate, setup and test DDC controller, pressure transducer, and electronic damper actuator furnished to box manufacturer under Section 23 09 13. Damper actuators integral with terminal unit may be furnished by terminal unit manufacturer and operation coordinated with DDC controller.
- C. Actuator / Controls: Electronic
- 1. Configuration: Air volume damper assembly inside unit casing. Locate control components inside protective metal shroud with removable cover.
 - 2. Volume Damper: Construct of galvanized steel with peripheral gasket and self lubricating bearings; maximum damper leakage: 1 percent of design air flow at 3 inches rated inlet static pressure. Damper position shall be indicated on the end of the shaft on the outside of the casing
 - 3. Controller to provide consistent air delivery within 5% of nominal airflow down to 25% of unit rated CFM, independent of changes in system static pressure.
 - 4. The actuator shall be directly coupled to the damper shaft.
- D. Attenuator:
- 1. Where indicated, attenuators shall be provided as required to reduce the discharge and radiated NC levels to 35 or less based on selections made at 3.0" w.c. inlet static pressure.
 - 2. The attenuator lining shall match the specified air terminal unit lining.
 - 3. Provide attenuator on all air terminal units in the following spaces: see schedule.
- E. Heating Water Coil:
- 1. Selection criteria:
 - a. Unless noted otherwise, coils shall be selected at the scheduled heating water supply temperature to meet the scheduled performance criteria in the following priority: total capacity (BTUH), leaving air temperature, heating water return temperature.
 - b. Coils rows shall be selected to obtain the performance closest to the scheduled values while meeting the scheduled capacity and without exceeding the leaving air temperature.
 - 2. Construction:
 - a. Fins: Aluminum, maximum 10 fins per inch.
 - b. Tubes: 1/2 inch, 5/8 inch, or 7/8 inch seamless, copper tubes mechanically expanded into the fin collars; arranged for counter-flow of heating water.

- c. Water velocity maximum 8 feet per second with water pressure loss not greater than indicated on the drawings.
- d. Rows: Provide coil rows scheduled or as required to achieve scheduled performance.
- 3. Coils shall be ARI certified and leak tested at 300 PSIG under water.
- 4. Casing shall be minimum 22 gauge galvanized steel.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Provide ceiling access doors or locate units above easily removable ceiling components.
- C. Support units independently from structure. Do not support from adjacent ductwork.
- D. Connect to ductwork in accordance with Section 23 31 13.
- E. Provide insulation and engraved equipment nameplate as specified.
- F. Provide insulation in accordance with Section 23 07 13 and Section 23 07 19.
- G. Verify that electric power is available and of the correct characteristics.
- H. Coordinate control installations with temperature controls vendor.

3.2 CLEANING, TESTING, STARTUP, AND DEMONSTRATION

- A. Clean and test units in accordance with Section 23 05 00.
 - 1. Include flushing of connected piping and cleaning of water control valves.
- B. Start-up units, check for proper operation as a system with air handling unit, fan, and connected ductwork.
- C. Check for clear access to control panel, isolation valves, control valves, balancing valves, and access panels. Verify required working clearance for control panels.
- D. Prepare units for Test and Balance as required by Section 23 05 93, correct any deficiencies found and retest.
- E. Demonstrate operation of units as a complete system to maintenance personnel and instruct them in the operation, adjustment and repair of the system.
- F. Check connections to insure they are tight and without noticeable leakage. Correct any deficiencies found.

END OF SECTION

SECTION 23 36 10

AIR VALVES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Air Valves

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 21 13 - Hydronic Piping
- C. Section 23 05 23 - General Duty Valves for HVAC
- D. Section 23 05 53 - Identification for HVAC Piping and Equipment
- E. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
- F. Section 23 07 00 - HVAC Insulation
- G. Section 23 31 13 - Sheetmetal Ductwork
- H. Section 26 27 17 - Equipment Wiring: Electrical characteristics and wiring connections.

1.3 REFERENCE STANDARDS

- A. UL 181 - Standard for Factory-Made Air Ducts and Air Connectors; Underwriters Laboratories Inc.; Current Edition, Including All Revisions.
- B. Air Diffusion Council, ADC Standard 1062R2, Air Diffusing Equipment Test Code.
- C. Air Moving and Conditioning Association, AMCA Standard 210, Test Code for Air Moving Devices.
- D. ASHRAE Standard 70-72, Method of Testing for Rating the Air Flow Performance of Outlets and Inlets.
- E. NFPA 90A - Standard for the Installation of Air-Conditioning and Ventilation Systems; National Fire Protection Association; 2012.
- F. SMACNA HVAC Duct Construction Standards; Current Edition.

1.4 SUBMITTALS

- A. See Division 01 requirements.
- B. Submit product, performance data, and control diagrams for review.
- C. Submit test reports as specified herein.
- D. Submit certificates as specified herein.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing the type of products specified in this section, with minimum eight years of documented experience.
- B. Products Requiring Electrical Connection: Listed and classified by Underwriters Laboratories Inc. as suitable for the purpose specified and indicated.

1.6 WARRANTY

- A. See Division 01 for additional warranty requirements.
- B. Provide five year manufacturer warranty for air valves and integral controls.

1.7 GUARANTEE

- A. Manufacturer guarantees resultant noise levels to be within NC rating published by manufacturer.

PART 2 PRODUCTS

2.1 AIR VALVES (SUPPLY, RETURN, EXHAUST, FUME HOOD EXHAUST)

- A. Acceptable Manufacturer: Phoenix or Siemens.
- B. All materials and equipment used shall be standard components, regularly manufactured for this and/or other systems and shall not be custom designed especially for this project. All components shall have been thoroughly tested and proven in actual use.
- C. The system provided shall be by one manufacturer, complete in all respects, including required controller, accessories, hardware, software and peripheral devices necessary to execute the sequence of operations.
- D. Air valves shall be provided for hood exhaust as scheduled and indicated on the drawings. Supply valves shall be equipped with heating water coils as scheduled. Heating water coils shall be as specified in Section 23 36 00.
- E. The airflow control device shall be pressure independent over its specified differential static pressure operating range. An integral pressure independent assembly shall respond and maintain specific airflow within one second of a change in duct static pressure irrespective of the magnitude of pressure and/or flow change or quantity of airflow controllers.
- F. The airflow control device shall include factory-mounted digital vortex-shedding airflow measuring device and factory-mounted high speed actuator.
- G. The airflow control device and factory controls shall respond within one second of a change in duct static pressure.
- H. The airflow control device shall be linear type and maintain accuracy within $\pm 5\%$ of signal over an airflow turndown range of no less than 8 to 1.
- I. Two duct diameters for entrance shall be required to ensure accuracy and/or pressure independence.
- J. Pressure drop is not to exceed 0.3" wg on any single valve.

- K. Valve body material for fume hood air valves or other corrosive service shall be 304SS; 20 gauge for body and 16 gauge for blades.
- L. Valve body material for non-corrosive service such as for supply, return, and general exhaust shall be aluminum.
- M. Furnish labor, materials, tools, equipment, controls, and services for air valve system. The control system shall vary the amount of supply and return/exhaust air into the room to operate the space at the lowest possible airflow rates necessary to maintain temperature control, achieve minimum ventilation rates, and maintain room pressurization in relation to adjacent spaces (positive or negative as indicated on the drawings).
- N. For lab areas, the system provided shall be based on variable fume hood exhaust volume with sash position reset control of fume hoods (as applicable), on / off status of fume hoods, synchronization of supply and exhaust airflow, and control of lab room temperature via reheat coil(s) and airflow reset.
- O. Any changes required in the ductwork, air handling equipment or other mechanical systems, which would be required for the application of proposed equipment other than the specified basis of design shall be the responsibility of the laboratory airflow control system supplier. All changes shall be subject to prior approval by the Architect/Engineer.
- P. The laboratory airflow control systems shall digitally interface to the building automation system (BAS). The laboratory airflow control contractor shall be responsible to provide an interface device between the air valve airflow controls and the BAS via BACnet protocol.
- Q. Coordinate work of this section with control drawings and sequences as indicated on the drawings.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Install all units as indicated and in accordance with manufacturer's recommendations and instructions.
- B. Check connections to insure they are tight with all leakage of 1% or less.
- C. Provide insulation in accordance with Section 23 07 00.

3.2 CLEANING, TESTING, START-UP, DEMONSTRATION

- A. Clean units in accordance with Section 23 05 00, including flushing of connected piping and cleaning of water control valves.
- B. Start-up units, check for proper operation as a system with air handling unit, fans, and connected ductwork.
- C. Test units in accordance with Section 23 05 00.
- D. Prepare units for Test and Balance as required by Section 23 05 93, correct any deficiencies found and retest.
- E. Demonstrate operation of units as a complete system to maintenance personnel and instruct them in the operation, adjustment and repair of the system.
- F. Air valve airflow control system shall be commissioned by provider and tested as required by commissioning agent under Division 01.

- G. The air valve airflow control contractor shall demonstrate to the Engineer the ability of the controls to maintain setpoints in all modes of operation.

END OF SECTION

SECTION 23 37 00

AIR OUTLETS AND INLETS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Diffusers.
- B. Registers/grilles.

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC
- B. Section 23 33 00 - Air Duct Accessories
- C. Section 23 31 13 - Sheetmetal Ductwork
- D. Section 23 31 14 - Sheetmetal - Special Ductwork

1.3 REFERENCE STANDARDS

- A. AMCA 500-L - Laboratory Methods of Testing Louvers for Rating; Air Movement and Control Association International, Inc.; 2012.
- B. ASHRAE Std 70 - Method of Testing the Performance of Air Outlets and Inlets; American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc.; 2006 (R2011).
- C. SMACNA 1966 - HVAC Duct Construction Standards; 2005.

1.4 SUBMITTALS

- A. Product Data: Submit product data for review. Review outlets and inlets as to size, finish, and type of mounting prior to submission. Submit schedule of outlets and inlets showing type, size, location, application, and noise level.
- B. Samples: Submit two of each required air outlet and inlet type upon request.
- C. Project Record Documents: Record actual locations of air outlets and inlets.

1.5 QUALITY ASSURANCE

- A. Test and rate air outlet and inlet performance in accordance with ASHRAE Std 70.
- B. Test and rate louver performance in accordance with AMCA 500-L.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Krueger, Metalaire, Nailor, Price, or Titus unless noted otherwise.
- B. Substitutions: Refer to Division 01.
- C. Air devices shall meet these specifications and the requirements noted in the Air Distribution Device Schedule. Titus model numbers are not inclusive of all listed requirements.

2.2 SUPPLY DIFFUSERS

- A. Type S1 (based on Titus TMS)
 - 1. Square, 3-cone louvered face ceiling diffuser, four way directional blow. For diffusers noted on drawings to be 2 or 3 way blow, provide blank off plates in diffuser. Provide panel, face and neck size scheduled.
 - 2. Material: Steel with baked acrylic finish.
 - 3. Color: White.
 - 4. Borders and mounting: Coordinate ceiling device frame type with architectural ceiling type.
 - 5. Dampers: Provide ceiling diffusers complete with opposed blade volume dampers where diffuser is installed in inaccessible ceiling and spin-in fitting manual volume damper at branch ductwork tap is not accessible.
 - 6. Accessories: None.
- B. Type S8 (based on Titus TriTec-ss)
 - 1. Acceptable manufacturers:
 - a. Nailor, Precision Air Products, Price, or Titus.
 - b. Substitutions: Not permitted.
 - 2. Square or rectangular radial flow diffuser with 6" deep backpan and integral internal baffle for evenly distributing air over the entire perforated face of the diffuser. The face shall be secured in place by quarter-turn fasteners for quick removal and sanitizing. Throw for terminal velocities of 75, 50, and 30 feet/minute along with corresponding pressure drop shall be provided with submittal data. Provide panel, face and neck size scheduled.
 - 3. Material: Stainless with continuously welded seams and baked acrylic finish.
 - 4. Color: White.
 - 5. Borders and mounting: Coordinate ceiling device frame type with architectural ceiling type.
 - 6. Accessories: Safety chain.

2.3 RETURN AND EXHAUST GRILLES

- A. Type R1 / E1 (based on Titus 50F)
 - 1. Square or rectangular frame with perforated 1/2" x 1/2" x 1/2" grid. Provide frame and face size scheduled with minimum 3" deep fabricated steel backpan with centered ductwork connection equal

to neck size scheduled. The visible surface of the backpan shall be painted flat black, unless noted otherwise. Provide panel, face and neck size scheduled.

2. Material: Aluminum frame and grid with baked acrylic finish.
3. Color: White.
4. Borders and mounting: Coordinate ceiling device frame type with architectural ceiling type.
5. Dampers: Provide ceiling grille complete with opposed blade volume dampers where diffuser is installed in inaccessible ceiling and spin-in fitting manual volume damper at branch ductwork tap is not accessible.
6. Accessories: None.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Install accessories in accordance with manufacturer's published recommendations as well as applicable sections of SMACNA manual and other standards set forth in Part 1.
- C. Provide all screws, bolts, nuts, inserts, and material required for attaching sheet metal to duct, walls, floors, and ceilings.
- D. Check location of outlets and inlets and make necessary adjustments in position to conform with architectural features, symmetry, and lighting arrangement.
- E. Where diffusers or grilles and registers are not provided with volume dampers, install spin-in fitting with balancing damper in duct runout.

3.2 TESTING

- A. Check work for satisfactory installation and performance.
- B. Check duct connections at air inlets and outlets air leakage or condensation. Correct conditions found.

3.3 INSPECTION

- A. Air inlets and outlets shall be clean and free from scratches and dents. Repair or replace damaged devices as required.

END OF SECTION

SECTION 23 41 00

PARTICULATE AIR FILTRATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Sheet Filters
- B. Panel Filters
- C. Pleated Filters
- D. Cartridge Filters
- E. Roll Media Filters
- F. Housing and Frames
- G. Filter Pressure Gauges

1.2 RELATED REQUIREMENTS

- A. Section 23 05 00 - Common Work Results for HVAC.
- B. Section 23 73 13 - Modular Indoor Central-Station Air Handling Units.
- C. Section 23 73 23 - Custom Indoor Central-Station Air Handling Units.

1.3 REFERENCE STANDARDS

- A. ASHRAE 52.2; 2012: Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size.
- B. ASHRAE 52.1; 1992: Gravimetric and Dust-Spot Procedures for Testing Air-Cleaning Devices Used in General Ventilation for Removing Particulate Matter.
- C. NFPA 90A: Standard for the Installation of Air-Conditioning and Ventilating Systems.
- D. UL 900: Test Performance of Air Filter Units.
- E. ISO 9001-2000: Certified manufacturing facility

1.4 SUBMITTALS

- A. Submit manufacturer's product data for review.
- B. Submit evidence of manufacturing facility certification with ISO 9001-2000.

1.5 QUALITY ASSURANCE

- A. MERV Ratings: Minimum Efficiency Reporting Value of MERV when evaluated under the guidelines of ASHRAE Standard 52.2; 2012.
- B. Average atmospheric dust spot and arrestance: Average dust spot efficiency of and a minimum arrestance based evaluation ASHRAE Standard 52.1; 1992.
- C. Performance: Media to maintain or increase in efficiency over the life of the filter.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Airguard, American Air Filter (AAF), Camfil, Flanders/Precisionaire, or Columbus Industries.

2.2 PLEATED PANEL FILTERS

- A. Scheduled as AAF Perfect Pleat HC M8 - MERV 8.
- B. Construction
 - 1. Media: Cotton and synthetic blend, lofted to a uniform depth of 0.18", and formed into a uniform radial pleats. There shall be at least 15 pleats per linear foot for 2" deep filters.
 - 2. Support: Welded wire grid, spot-welded on one-inch centers, treated for corrosion resistance, bonded to the downstream side of the media to maintain the radial pleat and prevent media oscillation.
 - 3. Frame: Minimum 28-point high wet-strength beverage board. Bond frame to media to prevent air bypass. Include integral diagonal support members on the air entering and air existing side to maintain uniform pleat spacing in varying airflow.
 - 4. Filter shall be rated by Underwriters Laboratories as UL Class 900.
- C. Performance
 - 1. The filter shall have a Minimum Efficiency Reporting Value of MERV 8 and MERV-A of 8 when evaluated under the guidelines of ASHRAE Standard 52.2-2012 must include -B with appendix J. Minimum arrestance of 92% The media shall maintain or increase in efficiency over the life of the filter.
 - 2. Initial resistance to airflow: Not to exceed 0.25" w.g. at airflow velocity of 500 feet/minute on 2" deep model.

2.3 CARTRIDGE TYPE FILTERS

- A. Scheduled as AAF Varicel RF - MERV 14.
 - 1. Construction
 - a. Media: Microfine glass laminated to a reinforcing backing to form a uniform lofted media blanket.
 - b. Blanket: Form into uniform tapered radial pleats and bonded to a stiffened backing that is bonded to the downstream side of the media to preclude media oscillation.

- c. Bonding: Mechanically and chemically bond media within the frame to prevent air bypass.
 - d. Frame: Constructed of corrosion resistant galvanized steel. Media support contour stabilizers shall be mechanically fastened to diagonal support members of the same construction shall create a rigid and durable filter enclosure. There shall be a minimum of four contour stabilizers on the air entering side and four on the air exiting side.
 - e. Filter shall be rated by Underwriters Laboratories as UL Class 900.
2. Performance
- a. Performance: The filter media shall have an average efficiency of MERV 14 by ASHRAE 52.2-2012-B with Appendix J. It shall have an average arrestance of not less than 98%, 99 or 100 in accordance with that standard. The dust holding capacity shall not be less than 142 grams on the same standard.
 - b. Initial resistance to airflow: Not to exceed 0.53" wg at airflow velocity of 500 feet/minute.
 - c. Maximum pressure withstand rating: 10" w.g. without failure of the media pack.

2.4 CARTRIDGE TYPE, HEPA FILTER

A. Scheduled as AAF Astrocel I - HCX.

B. Construction

- 1. Media: Continuous pleating of micro fine glass fiber media. Pleats shall be uniformly separated by corrugated aluminum separators incorporating a hemmed edge to prevent damage to the media.
- 2. Media pack: Potted into the enclosing frame with a fire-retardant urethane sealant.
- 3. Frame: 16-gauge steel, with a zinc aluminum alloy finish, and shall be bonded to the media pack to form a rugged and durable enclosure. The filter shall be assembled without the use of fasteners to ensure no frame penetrations. Overall dimensional tolerance shall be correct within -1/8", +0", and square within 1/8"
- 4. Gasket: Include a poured-in-place seamless sealing gasket on the downstream side of the enclosing frame to form a positive seal upon installation.
- 5. Filter shall be rated by Underwriters Laboratories as UL Class 900.

C. Performance

- 1. Efficiency: Minimum tested efficiency of 99.97% when evaluated according to IEST Recommended Practice RP-CC001.
- 2. Initial resistance to airflow: Not exceed 1.4" w.g. at airflow velocity of 500 feet/minute.
- 3. Maximum pressure withstand rating: 10" w.g. without failure of the media pack.
- 4. Supporting Data: Label filters as to tested efficiency, rated/tested cfm, pressure drop and shall be serialized for identification. Each filter must be individually tested, no batch testing will be allowed.

2.5 ROLL MEDIA FILTERS

A. Schedules as Columbus P-100

B. Construction

1. Media: air laid & multi-denier, single layer polyester with heavy tackifier provides two densities and three zones of filtration

C. Performance

1. Efficiency: The filter media shall have an average efficiency of MERV 10 by ASHRAE 52.2-2012-B with Appendix J. It shall have an average arrestance of not less than 98%, 99 or 100 in accordance with that standard. The dust holding capacity shall not be less than 432 grams on the same standard.
2. Initial resistance to airflow: Not exceed 0.37" w.g. at airflow velocity of 500 feet/minute.

2.6 HOUSINGS AND FRAMES FOR CARTRIDGE FILTERS

A. General

1. Filter housing: Two-stage filter system consisting of 16-gauge galvanized steel enclosure, aluminum filter mounting track, universal filter holding frame, dual-access doors, static pressure tap, filter gaskets and seals. In-line housing depth shall not exceed 21".

B. Construction

1. 16-gauge galvanized steel with pre-drilled standing flanges to facilitate attachment to other system components. Corner posts of Z-channel construction shall ensure dimensional adherence. The housing shall be weatherproof and suitable for rooftop/outdoor installation.
2. The housing shall incorporate the capability of two stages of filtration without modification to the housing. A filter track, of aluminum construction shall be an integral component of housing construction. The track shall accommodate a 2" deep prefilter, a 12" deep rigid final filter, or a pocket filter with header.
3. Dual access doors, swing-open type: High-memory sponge neoprene gasket to facilitate a door-to-filter seal. Each door shall be equipped with adjustable and replaceable positive sealing UV-resistant star-style knobs and replaceable door hinges.
4. Universal holding frame: Constructed of 18-gauge galvanized steel, equipped with centering dimples, multiple fastener lances, and polyurethane filter sealing gasket, shall be included to facilitate installation of high-efficiency filters.
5. Fittings: Housing shall incorporate a pneumatic fitting to allow the installation of a static pressure gauge to evaluate pressure drop across a single filter or any combination of installed filters.

C. Performance

1. Leakage at rated airflow, upstream to downstream of filter, holding frame, and slide mechanism shall be less than 1% at 3.0" w.g. Leakage in to or out of the housing shall be less than one half of 1% at 3.0" w.g.
2. Accuracy of pneumatic pressure fitting, when to evaluate a single-stage, or multiple filter stages, shall be accurate within $\pm 3\%$ at 0.6" w.g.

2.7 HOUSING AND FRAMES FOR HEPA FILTERS

A. General

1. Side-access absolute filter housing shall be constructed of 14-gauge 304 stainless steel and shall include removeable dual access doors, challenge injection port, door gasketing and swing bolt filter retainers. An appropriate number of swing bolts to match air filters shall also be included.

B. Construction

1. Absolute filter housing: All-welded construction of 14-gauge galvanized steel (304 SST); reinforced with channel bracing to withstand 8.0" w.g. positive or negative pressure.
2. Standing flanges: Provide to mate housing to other housings or ductwork.
3. Housing shall be weatherproof without modification.
4. Doors: Pin-hinged removable access doors for service from either side of the housing. Doors shall include high-memory door gasketing to prevent leaks to ± 8.0 " w.g.
5. Filter securing swing bolt assemblies: Same construction as the housing, and with equip-bearing filter clamps, to secure filters into the housing. There shall be a minimum of four assemblies per filter unit.
6. Housing shall include integral challenge injection port for introduction of filter evaluation challenge.
7. Housings shall include a filter track for application of a nominal 2" deep prefilter.
8. Housing shall include knife-edge gel seal opening for sealing to filter frame. Filters edge seal method shall be coordinated with housing.

C. Performance

1. The sealing assembly shall be capable of sealing each element with 30 inch/lbs. of torque to 50% filter gasket compression. The sealing assembly shall create a scan capable filter to housing seal.

2.8 FILTER PRESSURE GAUGES

- A. Acceptable manufacturers: Dwyer Instruments, H.O. Trerice, or Weiss Instruments.
- B. Magnehelic: Direct reading 3-1/2 inch diameter diaphragm actuated dial in metal case, vent valves, black figures on white background, front recalibration adjustment, $\pm 3\%$ full-scale accuracy. Range shall start at zero and have a maximum of 0.25" to 1.0" w.c. above the scheduled final resistance. Mark scheduled clean and dirty resistance on face of dial. If two filter beds are installed in tandem, provide magnehelic across each filter bed.
- C. Provide filter pressure gauges as specified in addition to BAS differential pressure sensor across each filter bank.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide filters in locations as shown on drawings. Provide quantity and sizes to comply with scheduled performance.
- B. Upon completion of ductwork and fan system, clean systems as required in Section 23 05 00 and install specified filter media prior to placing system in operation.
- C. All filters shall be installed prior to operating the HVAC system. Provide a complete change in filter media as required during construction and prior to the HVAC test and balance process. If equipment and/or ductwork is found to be contaminated at any point during construction, an independent NADCA certified contractor shall be retained to clean the ductwork and/or equipment at the contractors expense.

- D. Install filters in accordance with manufacturer's published installation instructions. Provide manufacturer's recommended media change data to maintenance personnel.
- E. Install filters in frames or apparatus casing so as to be leak free. Verify with light test from both sides.
- F. Install and level filter gauges outside air stream for each bank of filters.
- G. Protect cooling and/or heating coils with temporary media during construction.
- H. Deliver one complete change of media to the maintenance personnel at Substantial Completion. Store spare media in a clean and dry place adjacent to equipment served or as coordinated with the Owner.
- I. Provide insulation as required on filter housing to prevent condensation.
- J. Insulate and make leak-proof filter access doors.

END OF SECTION

SECTION 23 44 13

ULTRAVIOLET FIXTURES

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Ultraviolet Fixtures

1.2 RELATED REQUIREMENTS

- A. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
- B. Section 23 09 23 - Building Automation and Direct Digital Controls
- C. Section 23 73 13 - Modular Central-Station Air-Handling Units
- D. Section 23 73 23 - Custom Indoor Central-Station Air Handling Units
- E. Division 26: Electrical

1.3 REFERENCE STANDARDS

- A. NFPA 90A: Standard for the Installation of Air-Conditioning and Ventilating Systems.
- B. UL Standard 1598- Luminaires
- C. UL Standard 1995 - Heating and Cooling Equipment
- D. NFPA 70 - National Electric Code

1.4 SUBMITTALS

- A. Submit product data for review including installation lamp performance, certification of UL listing, wiring diagrams, and installation and operating instructions.

1.5 QUALITY ASSURANCE

- A. UL Compliance: Comply with UL Standard 1995 as applicable to usage of UVC Emitters in HVAC Equipment.

1.6 WARRANTY

- A. Warrant UVC fixtures to be free from factory defects for no less than one year from date of installation. "Green Lamps" shall be warranted for no less than two years (17,000 hours) from date of start-up and ballast shall be warranted for no less than non-pro-rated 5 years.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. American Ultraviolet Company, Sanuvox, Steril-Aire, or UVDI.
- B. Substitutions: Refer to Division 01.

2.2 FIXTURES

- A. High-Output utility fixtures: Designed specifically for HVAC applications and mounted in arrangements for cooling coil and drain pan irradiation as well as for pass-by air decontamination.
- B. UVC output: As specified herein and guaranteed for continuous destruction of all known molds, bacteria and viruses and shall produce the specified UVC energy to prevent coil biological fouling and eliminate the need for coil cleaning and drain pan biocides.
- C. Moisture-Proof design: Use UVC "Cell" lamp design to incorporate an outer quartz sleeve with protective boot on power end for water-tight sealing.
- D. Configuration: Single lamp configuration, sized to provide adequate coverage for air handling unit chilled water coils.
- E. Fixture construction:
 - 1. Factory assembled and tested prior to shipment.
 - 2. Each assembly consists of housing, reflector, electronic ballast, lamp brackets, water-tight, plug-in power connectors for fixture to fixture connection as well as lamp to fixture connection and high output lamp.
 - 3. Housings shall be constructed of heavy gauge hospital grade stainless steel.
 - 4. Reflectors shall be fabricated from the highest grade polished aluminum lighting sheet material, which has a reflectivity rate of 88% when exposed to short-wave UVC in the range of 254 nm.

2.3 BALLASTS

- A. Solid-state electronic design, Class P rapid start with a 0.95 power factor minimum.
- B. Voltage: 277V AC.
- C. Designed for maximize photon production in air temperatures of 35 to 175 degrees F.
- D. Minimum ballast start temperature of -20 degrees F.
- E. RFI/EMI rating: As defined by FCC part 18A for industrial / commercial applications in regards to suppression.
- F. Ballasts shall be UL listed.

2.4 UVC LAMPS

- A. Rated for a two years (17,000 hours) continuous operational life with no greater than a 20% drop in UVC output at end of lamp life.

- B. Provide UVC "Green Lamps" with mercury content not greater than 8 mg.
- C. Guaranteed to produce no ozone.
- D. Minimum of 11.6 microwatts/cm² per linear inch of lamp arc length at a distance of one meter as per IES Test Standards. Output shall be independently tested in airstreams of 400 feet per minute velocities at temperatures of 45 degrees F. Successful supplier shall submit independent documentation of specified performance with their submittals.
- E. High-output variety, T5 lamp diameter, and shall be constructed from hard quartz tubing for superior UVC transmittance. Electrodes shall be designed to maximize plasma convection and stability for high output lamp performance.
- F. UL listed.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide factory technician for startup assistance and owner training at the project site to insure provide instructions on correct installation and operation of the UVC System.
- B. Provide factory startup report that UVC system is installed and functioning in accordance with manufacturer's operating parameters.
- C. Comply with manufacturer's installation instructions placement, wiring and testing.
- D. Provide an interlock switch on the access to the UVC Emitters to turn the lights off when the access is opened. Provide disconnect switch on unit exterior to turn off UVC lamps prior to entering UVC compartment. Install Caution Labels on all accesses to the fixtures.

3.2 TRAINING

- A. Factory technician shall train Owner's Representative's operating personnel in UVC system operation, maintenance and safety procedures.

END OF SECTION

SECTION 23 73 23

CUSTOM INDOOR CENTRAL-STATION AIR HANDLING UNITS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Factory fabricated, custom construction, indoor air handling units with internal components including fans with belt or direct drives, heating and cooling coils, filter racks, access doors, and lights.
- B. Refer to drawings for additional required components and configuration of air handling unit(s).

1.2 REFERENCES

- A. AFBMA 9 Load Ratings and Fatigue Life for Ball Bearings.
- B. AFBMA 11 Load Ratings and Fatigue Life for Roller Bearings.
- C. AMCA 99 Standards Handbook.
- D. AMCA 210 Laboratory Methods of Testing Fans for Rating Purposes.
- E. AMCA 300 Test Code for Sound Rating Air Moving Devices.
- F. AMCA 301 Method of Publishing Sound Ratings for Air Moving Devices.
- G. AMCA 500 Test Methods for Louver, Dampers, and Shutters.
- H. ARI 410 Forced Circulation Air Cooling and Air Heating Coils.
- I. ARI 430 Central Station Air Handling Units.
- J. ARI 435 Application of Central Station Air Handling Units.
- K. ARI 610 Central System Humidifiers.
- L. NEMA MG1 Motors and Generators.
- M. NFPA 70 National Electrical Code.
- N. SMACNA HVAC Duct Construction Standards Metal and Flexible.
- O. UL 900 Test Performance of Air Filter Units.

1.3 SUBMITTALS

- A. Submit to Owner's designated representative minimum 1 electronic soft copy or 5 bound copies for approval prior to release to fabrication. Refer to Section 019100 for requirements for submittals to the Commissioning Agent.

- B. Shop Drawings: Indicate assembly, unit dimensions, weight loading, required clearances, construction details, field connection details, electrical characteristics, connection requirements, and connection requirements.
- C. Product Data:
- D. Provide literature that indicates dimensions, weights, capacities, ratings, fan performance, gages and finishes of materials, electrical characteristics and connection requirements.
- E. Provide data of filter media, filter performance data, filter assembly, and filter frames as tested and certified per ASHRAE standards.
- F. Provide fan curves with specified operating point clearly plotted, as tested and certified per AMCA standards. Ratings shall include system effects. Bare fan ratings will not satisfy this requirement, but shall be submitted for comparison purposes.
- G. Submit sound power level data for both fan outlet and casing radiation at rated capacity, as certified per ARI 260.
- H. Provide data on all coils as tested and certified per ARI standards.
- I. Submit electrical requirements for power supply wiring including wiring diagrams for interlock and control wiring, clearly indicating factory installed and field installed wiring.
- J. All materials shall have NFPA-90 rating of 25/50 or better.
- K. Submit condensation calculations or certification as specified hereinafter.
 - 1. Manufacturer's Installation Instructions.

1.4 OPERATION AND MAINTENANCE DATA

- A. Submit to Owner's designated representative minimum 1 electronic soft copy and 2 bound copies for prior to request for final payment. Refer to Section 019100 for requirements for submittals to the Commissioning Agent.
- B. Maintenance Data: Include instructions for lubrication, filter replacement, motor and drive replacement, spare parts lists, and wiring diagrams.

1.5 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing the Products specified in this section with minimum 10 years documented experience, who issues complete catalog data on total product.
- B. Each units shall bear and ETL label, conforming to UL standards 1995.
- C. Units shall comply with the requirements of UL 1995 and NFPA 90.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, and handle products to site location as directed by owner. Owner shall be given minimum 5 days notice of delivery. Coordinate delivery times with owner, make any necessary provisions for protected local storage not to exceed one month as required to meet owner's required receipt of delivery schedule.

- B. Deliver products to site wrapped in heavy duty protective plastic on factory installed shipping skids. Contractor shall inspect for damage upon receipt of delivery and provide written notice of concerns or exceptions within 10 business days of receipt of equipment. Owner's representative shall be given notice of delivery and an opportunity to inspect the equipment.
- C. Equipment will be stored in clean dry place and protected from weather and construction traffic. Handle carefully to avoid damage to components, enclosures, and finish.
- D. Refer to Specification Section 01 91 00, Paragraph 1.23 for procedures to be followed when using permanent equipment for operation by the Contractor.

1.7 ENVIRONMENTAL REQUIREMENTS

- A. The units will not be operated on site for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings lubricated, and fan has been test run under observation of the Owner's Representative.

1.8 EXTRA MATERIALS

- A. Provide two sets for each unit of fan belts and one set for each unit of filters. Tag to identify associated unit.

1.9 SCHEDULES

- A. In general, all capacities of equipment, and motor and starter characteristics are shown in schedules included in this specification. The capacities shown are minimum capacities. Capacity variations lower than scheduled for equipment supplied under this contract will be permitted only with the written direction of the owner. Variations in the capacities up to 25% above scheduled are allowed, variation in capacity above 25% higher require written permission from owner.

1.10 OTHER REQUIREMENTS

- A. Where equipment is to be installed outdoors, or is required to be painted on the drawings, it shall be shipped to the job with not less than a prime coat of paint or as specified hereinafter.
- B. Insofar as is possible, all items of the same type (i.e., pumps, fans, etc.) shall be by the same manufacturer.
- C. Where installation instructions are not included in these Specifications or on the Drawings, the manufacturer's instructions shall be followed.
- D. All equipment affected by altitude shall be rated to operate at the altitude where it is installed.
- E. Provide warrantee minimum 5 year parts and labor on all components.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturer's listed make acceptable products:
- B. Temtrol
- C. Climate Craft

D. Energy Labs

2.2 GENERAL DESCRIPTION

A. Fabricate with fan and coil sections as scheduled plus accessories, including where applicable:

B. Outdoor Air Intake Section

C. Mixing Box Section

D. Filter Section

E. Return Fan Section

F. Economizer/Relief Section

G. Preheating coil section.

H. Cooling Coil Section

I. Fan and Motor Section

J. Reheat Coil Section

K. Hot Deck Coil Section

L. Humidifier Section

1. Performance Base: Sea level conditions.
2. Fabrication: Conform to AMCA 99 and ARI 430 in the absence of direction in this specification.
3. Performance: Refer to schedule.
4. Typical Configuration (Refer to Drawings for Specific Applications):

Component Position:	1	2	3	4	5	6	7	8	9	10	11
Component											
Inlet Plenum	X										
Mixing Box/w access door											
Preheat Coil			X								
Filter		X									
Access Section with Door											
Return Fan											
Economizer/Relief											
Cooling Coil					X						
Fan and Motor						X					
Reheat Coil											
Hot Deck Coil											
Final Filter							X				
Humidifier				X							
Access section											
Outlet Plenum								X			

2.3 NOTE: COMPONENT POSITIONS ARE GIVEN IN DIRECTION OF AIR FLOW. COMPONENT TYPE MAY BE USED MORE THAN ONCE.

2.4 AIR HANDLER CASING AND GENERAL CONSTRUCTION

- A. Exterior wall shall be minimum 16 gauge galvanized steel. Interior wall shall be minimum 20 gauge solid galvanized steel. Construction shall contain thermal breaks between all interior and exterior steel surfaces.
- B. Casing Insulation: Insulation shall comply with NFPA-90A requirements for flame spread and smoke generation. Casing panels shall be double wall with throughout. Internal insulation shall have an R-value of at least 13 and shall be minimum 2" thick injected foam.
- C. Casing Panel Liner: Solid Galvanized steel.
- D. All sheet metal joints throughout the air handler, and between panelized sections, shall be gasketed with closed cell, soft rubber gaskets, fabricated from neoprene, EPDM, or other approved material.
- E. Drain pan liners shall be constructed of No. 20 gauge 304 stainless steel or heavier as standard with the manufacturer. Drain pans shall be sloped in two planes to the outside edge of the unit and shall extend sufficiently beyond the face of the coil to capture all condensate carryover. The minimum R-13 insulation shall be installed and sealed as is appropriate for the equipment construction.
- F. The floor of the fan/access sections of draw through units immediately downstream of cooling coils shall be constructed to manage condensate carryover and shall be surfaced with non-skid aluminum tread plate.
- G. Channel Base: Units shall be constructed and delivered in as few sections as necessary to deliver and install. Single sections units shall have a complete perimeter channel base of at least 6" galvanized steel or 6" carbon steel with marine quality primer. If installation path requires multiple sections, continuous rails as specified for single sections shall be provided on each section. Base channel shall be greater than 6" as required to accommodate the p-trap drain requirements per the construction drawings.
- H. All floors shall be insulated with minimum R-13 insulation accessible sections and shall be surfaced with 1/8" non-skid aluminum tread plate. A galvanized sheet shall enclose and form a vapor barrier for the insulation on the bottom of the unit. All points of contact between the floor, vapor barrier and structure shall be thermally isolated with gasketing of closed cell soft rubber or EPDM. Floor structure of accessible sections shall limit deflection to L/240 due to the weight of a 300 pound worker.
- I. Hinged, gasketed, quick-access doors of adequate size for a man to enter shall be provided for each space between coils, filters and other components. Access doors shall be provided to allow access to both sides (upstream and downstream) of the filter racks, into the fan section, and both sides of all coils. Access doors shall be double wall, insulated, and the opening framed, with full perimeter gasket. Door size shall be at least 17" wide and full height of the panel up to 5'-0" tall. The construction of the access doors shall equal or exceed the quality and quantity of the air handler casing materials as specified herein. Each door shall have a minimum of 8 inch by 6 inch double glazed view window, capable of withstanding the total developed pressure of the unit. The doors shall be hinged using either heavy duty steel butt hinges or stainless steel piano hinges, extending along the entire edge of the door, except for a maximum of two inches at each end. If butt hinges are used, provide two per door for up to 36" high doors and three per door for longer doors. There shall be a minimum of two latches on doors longer than 18", and three latches in doors over 36" long. Latches shall be Ventlok 310, heavy duty latch, or approved equal. All access doors shall open against air pressure, unless approved by the owner in writing. All doors with multiple points of latching shall be provided with a single break handle that is located at an accessible height with the units on an equipment pad.

1. For doors over 6'-0" tall a single point access hinged handle shall be provided to allow door access without the use of a ladder for upper hinge contact points.
- J. Panels shall be double wall, reinforced construction with sufficient internal bracing to prevent excessive deflection of the panels. Maximum deflection at any point on the unit casing shall be limited to L/240 of the overall panel width or height at +/- 8" pressure or 125% of the listed TSP on the schedule, whichever is greater. Floors and ceilings shall be supported as required to limit deflection to L/240 while supporting the weight of a 300 pound worker.
- K. Unit manufacturer shall submit full sound performance data to the Project sound consultant for evaluation. Units shall not exceed the following:
- L. 50,000 CFM Fan Array Unit at 7" TSP - Outside Air:

	63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz
Inlet	88	85	95	82	72	74	69	66
Discharge	86	96	83	96	97	77	59	60
Casing	91	91	84	97	101	78	63	61

1. 50,000 CFM Fan Array Unit at 9" TSP with Return and Supply Fans:

	63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz
Inlet	80	85	99	85	88	87	84	77
Discharge	89	87	101	86	83	81	77	75
Casing	93	91	84	97	101	78	63	61

1. 27,000 CFM Fan Array Unit at 9" TSP with Return and Supply Fans:

	63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz
Inlet	82	77	93	71	73	85	75	70
Discharge	77	73	90	65	72	75	67	66
Casing	77	73	90	65	72	75	67	66

1. Surface Condensation per ASTM D4230 "Measuring Humidity with Cooled Surface Condensation". Provide unit housing including joints and seams with thermal break construction. Thermal breaks shall be provided in framing/panel construction and at doors and access panels with no through metal that will not condense moisture when:
 - a. $(T_h - T_l) = \text{Less than } 3.3$
 - b. $(T_h - T_{dp})$
2. Where:
3. T_h = Higher temperature internal or external to casing
4. T_l = Lower temperature internal or external to casing
5. T_{dp} = Dew point at location corresponding to T_h
6. The unit shall be supplied with factory installed and sealed flanges for connection to ductwork.

2.5 FANS

- A. The fan type shall be provided as required for stable operation over all utilized frequencies and to optimize energy efficiency.
- B. Fan arrays: Where fan arrays are scheduled, provide a single (duplex redundant) variable frequency drive for the array. A unit mounted NEMA 4 UL508A listed control panel shall be provided with disconnects and all required motor protection prewired to each fan motor located

downstream of the VFD(s). Adequate space shall be provided in the panel for current sensing relays on each fan circuit. A single power distribution block shall be provided for connection of the field mounted VFDs. Drives shall be provided with the necessary contactors and shall be arranged so that each drive serves all fans and automatically transfers from the primary to the back-up drive upon failure of the primary drive. Class J or Class CC fusing and an electronic motor overload protector with manual isolation switch shall be provided for each motor in the array. Each motor in the array shall be independently grounded with a dedicated green conductor. A minimum of one open ground lug shall be provided for field use.

- C. Where fan arrays are provided, each fan/motor assembly shall be removable through an adequately sized access door located on either the inlet or the discharge side of the array. Individual fan/motor assemblies shall be removable on a slide-out base rail. A structural beam, capable of handling weight of fan/motor/base sections, shall be mounted at ceiling of fan section, to facilitate removal of assembly with Owner furnished chain fall.
- D. Motors shall be high efficiency or premium efficiency type, suitable for continuous duty and variable frequency drive operation. The motor mounting for each unit shall be an integral part of the fan support frame. Fans shall have an appropriately designed fabric duct vibration isolator installed within the air handler casing.
- E. Fans shall be both dynamically and statically balanced at the factory as a complete assembly. The fan/motor unit shall be mounted on spring isolators within the air handler casing.
- F. Fans shall have an appropriately designed fabric duct vibration isolator installed within the air handler casing.
- G. The fan unit bearings shall be of the antifriction type, either ball or roller, lubricated at the factory, and shall be equipped with means for lubrication with a grease fitting on the outside of the bearing housing. Both inboard and outboard bearings shall be the same, identical size. The bearings shall be a catalogued type as manufactured by Fafnir, SKF, Sealmaster, or approved equal, and stocked locally. Bearings shall have an L 10 minimum life of 250,000 hours. Intermediate bearings will not be acceptable. Grease fittings for bearings shall be remotely mounted within line of sight of the bearing, where possible. Where it is not possible, then the fitting shall be mounted where it is most easily accessible for service. The tubing used for removing the fitting shall be stainless steel, brass, or clear nylon installed with a strain relief pigtail approximately 4 inches in diameter.
- H. After final approved TAB, fan and motor shall be given an electronic vibration analysis while operating over the entire speed range. Vibration amplitude and frequency shall be recorded in the horizontal, vertical and axial planes. Maximum allowable vibration velocity shall be 0.2" per second. Full range of frequencies (500 to 50,000 cpm) shall be scanned to detect misalignment, bearing defects, mechanical looseness or foundation weakness. Each bearing shall be tested. Fans with VFD's shall be checked from 33% to 100% of the rated speed. "Lock-out" ranges may be used to correct up to two ranges of excess vibration. The span of each "lock-out" range shall be limited to an effective fan speed of 50 RPM. Any "lock-out" ranges used shall be clearly identified in the test report and shall be prominently displayed on a typed, laminated legend mounted inside the VFD controller cabinet.

- I. The system shall provide a positive shutoff for each fan in case of a fan failure. Each fan shall be provided with an isolation or backdraft damper to prevent bypass in the event of a motor failure. Blank off plates requiring manual installation are not acceptable
- J. Fans shall be provided with unloading technology to allow fan modulation without surge from 100% to 10%; while maintaining the part load static pressure requirements of the system. There shall be no static pressure or intake plenum losses or any horsepower penalty associated with the system.
- K. Provide one complete spare fan/motor assembly for each size of assembly provided on the project for emergency replacement.

2.6 DRIVES

- A. Variable speed drives: Drives shall be furnished for mounting separately from the units. Refer to Section 26 29 23.

2.7 COILS

- A. Coils in the air handling units shall be individually removable and shall not be used to provide structural stability for the casing. All coil frames shall be fabricated of stainless steel. The coils shall be completely enclosed within the coil housing of the air unit casing. All penetrations of the air handler casing shall be neatly sealed using a resilient sealant.
- B. All coils shall be arranged for and piped to provide counterflow operation.
- C. Coils shall be selected, manufactured and tested as appropriate for duty. Coils shall have aluminum fins and seamless copper tubes. Fins shall have collars drawn, belled and firmly bonded to tubes by mechanical expansion of the tubes. The coils shall be proof tested and designed for the working pressure of 250 psig. Submit certification that coils are suitable for testing at pressures of 450 psig at 100°F.
- D. Coils shall either be self venting or shall be provided with a coil vent extended to the exterior of the casing. Coil drains shall be provided and piped to the exterior of the casings. Ball valves shall be provided on the coil vents and drains.
- E. Coils shall be designed so that the average leaving temperature over each square foot of face area is not be more that 2° F from the average leaving air temperature over the entire coil face area.
- F. Chilled water coils
 - 1. Water coil capacities, pressure drops and selection procedures shall be certified for the capacity scheduled in accordance with ARI Standard 410-87. Non certified coils will not be accepted.
 - 2. Chilled water coils shall be of the extended surface type meeting all conditions and having the minimum face area and pressure drops scheduled on the Drawings, and shall have same end supply and return connections unless otherwise indicated. Coils shall be constructed of copper tubes 5/8" O.D. with .035" thick minimum wall thickness and aluminum fins permanently bonded to the tubes by mechanical expansion. Coils shall have a maximum of 11 fins per inch, and a maximum of 8 rows. If additional capacity is necessary, the additional capacity shall be provided by an additional coil, with an additional access section. If coils are not self venting, a coil vent shall be extended to the exterior of the casing. Coil drains shall be extended to the exterior of the casing. Ball valves shall be installed by the Contractor on the drain and vent piping.
 - 3. Coil headers and connections shall be of I.P.S. brass or heavy gauge seamless hard drawn copper tubing. with penetrations for connection of core tubing by die formed intrusion process with resulting contact depth between the header wall and core tubing of not less than .090". Joints between core tubing and header shall be of recess swage design to allow a large mating area for build up of

brazing materials to give increased strength to the joint. Supply and return connection of brass or copper shall be terminated with National Pipe Threads with wrench flats.

4. Coils shall be designed and certified by the manufacturer to operate to scheduled face velocity plus 10% without moisture carry over. Each coil section shall be provided with a 316 stainless steel frame/casing, including tube sheets, no lighter than 18 gauge. Frame members shall extend over the ends and edges of the coils and shall be constructed with formed holes for tubes, permitting free expansion and contraction of coil sections while supported by an extended surface of the frame. Intermediate tube support sheets of 316 stainless steel shall be provided in all coils having tube lengths in excess of 48": on long coil sections the spacing of coil supports shall not exceed 48". All intermediate supports shall be welded to coil frame members and fabricated with formed tube holes to support the penetrating tubes.

2.8 FILTERS HOUSINGS

- A. General: Provide filter sections appropriate for the specified filters with the specified access.
- B. Provide housings that do not require blank-off plates to prevent air from by-passing the filters.
- C. Provide front or side loading pre-filter housings and front loading final filter housings.
- D. Refer to Specification Section 23 41 00 for additional requirements.

2.9 ACCESSORIES

- A. Ultraviolet Lights
 1. Refer to section 23 44 13 for UV light specifications. Provide UV light(s) to shine on the wetted side of the chilled water coil.
 - a. Provide germicidal UV power density for units listed in schedule.
 2. Factory assemble and test all UV-C fixtures in the air handling unit.
 3. Treat view port and other windows to assure the UV-C energy emitted is below the threshold limit specified by American Conference of Governmental Industrial Hygienists (ACGIH).
 - a. Provide a mechanical interlock switch to disconnect power to the UV-C fixtures when the opening of an access door may pose an exposure risk to UV-C light.
 - b. Provide externally mounted disconnect switch with lockout/tagout that disconnects UV-C power and prevents unwanted operation of UV-C lights.
- B. Marine Lights: Provide factory mounted, water and dust resistant LED marine observation lights in all access, intake, discharge, and fan sections. Marine lights shall be prewired in conduit to a single timer switch located on the exterior of the module adjacent to the access door. Switches shall be factory mounted complete with outlet box for wiring connection by Division 26 and switch face plate. One switch shall control all lights within the unit.

2.10 ELECTRICAL

- A. Wire and test unit at factory before shipment. Comply with NEC requirements and conform with applicable U.L. Standards. Number code wiring per electrical wiring diagrams. Label electrical components according to electrical diagram.
- B. Fan motors shall be factory mounted and wired to an external disconnect switch adjacent to the motor access door.

- C. Vapor proof lights and GFI convenience outlets shall be provided in each accessible compartment. Lights shall have a single switch on the unit exterior. Wire lights and outlets to two external 120v, 20a power connections.
- D. All wiring shall be 600v rated type MTW/THWN stranded copper in EMT or LiquidTite conduit (max 3 feet). All junction boxes shall be UL approved and gasketed. Connections to rotating equipment shall be flexible liquid tight.
- E. Provide main control panel in an enclosed housing with a dead front cover over the main power circuit controls. Provide terminal blocks for main power connection and low voltage control wiring as required.
 - 1. Provide multiple terminal blocks are required to provide multiple VFD arrangement per fan array. Refer to schedule for number of VFDs. Each VFD shall serve an equal number of fans. Terminal blocks shall be provided and sized as such that the NEC required wire sizing can be terminated at the terminal block
- F. Provide disconnect switch to disable power before control panel door can be opened.

2.11 FINISH

- A. All external parts of the unit shall be thoroughly cleaned and painted, or provided with manufacturer's standard, galvanized, enamel or epoxy finish.

2.12 ELECTRONIC AIR PURIFICATION SYSTEM

- A. Where scheduled a third party electronic air purification system will be required. Unit manufacturer shall coordinate with the mechanical contractor to ensure that the required units section is provided directly down stream of the pre-filter section.
 - 1. Basis of Design is to provide a CosaTron filtration section.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Units shipped in multiple sections shall have manufacturer's representative on site for assembly. Assemble high pressure units by bolting sections together. Isolate fan section with flexible duct connections.
- B. Penetrations of the unit casing for pressure or temperature sensors and test ports shall be factory or field installed Ventlock 699 penetration devices.
- C. Unit shall be set on a Frame and then on the housekeeping pad to allow for an adequate trap size to provide for free flow of condensate to drain. Frame and pad heights shall be provided that allow for proper drainage. Refer to details on the drawings for condensate trap height requirements.
- D. Air handling unit equipment interiors shall be completely weather protected by the unit manufacturer from the elements, dirt and other damage prior to shipment from the manufacturer's factory. The Division 23 Contractor shall protect the air handling unit from the point of receipt or storage until final system acceptance by Owner.
- E. All units shall be cleaned prior to turnover. Cleaning shall include, but not be limited to, Fan Casings, Fans, Coils, Pans, etc. and the system be turned over in like new condition.

- F. Storage of all units shall be per manufacturer's recommendations during site storage and during installation. All fans and motors shall be rotated by hand once units are received and are stored on or off site from the factory to ensure lubrication of all bearings is maintained.

3.2 AIR MOVING EQUIPMENT OPERATION DURING CONSTRUCTION

- A. The use of air handling units, fans, or other permanent air moving equipment during construction is prohibited unless approved by the owner. If approved for use during construction, the following procedures shall be followed:
1. The contractor shall protect the interior of all ductwork, air handling units, and other equipment from the accumulation of dirt and dust and other contaminants. If the permanent equipment cannot be adequately protected, temporary air moving/ conditioning equipment and distribution systems shall be utilized as required for finishing trades.
 2. Provide temporary filters on all return and exhaust air grilles, open ductwork, and transfer openings in the work area.
 3. The contractor shall remove all filters used during construction and replace them with new filters prior to test and balance work and prior to substantial completion.
 4. If the ductwork and/or equipment is found to be contaminated at any point during construction, an independent NADCA certified contractor shall be retained to clean the ductwork and/or equipment at the contractors expense. Refer to Section 23 01 30.51.
 5. System operating temperatures shall be maintained to avoid condensation on ductwork and equipment surfaces. New or existing insulation found damaged shall be replaced.
- B. Coordinate use of air handling equipment with ICRA plan, if applicable. Maintain required pressure relationships in construction areas adjacent to occupied areas.

3.3 TESTING

- A. Units with cabinet mounted fans shall be certified at rated conditions per AMCA standards. Bare fan data will not be accepted.
- B. Provide manufacturer representative on site for start-up and testing.
- C. With unit set in place, leveled and ready to receive duct work connections, unit shall be tested for casing leakage by sealing all openings and pressurizing to 125% of the listed TSP on the schedule or +/- 8" pressure, which ever is greater. Maximum allowable leakage rate is 1.0% of rated unit flow. Test is to be performed by the manufacturer using certified flow measurement devices and shall be witnessed by a representative of the Owner's Test & Balance firm. Deflection limit of L/240 will also be demonstrated at this time.
- D. Drain Pan Test: Each cooling coil drain pan shall be tested for leaks and proper pitch. The coil drain pan test will be as follows: Before each test, the unit floor will be verified to be level. Each drain pan will be continuously filled with water and the drainage is to be observed. Upon the satisfaction of the Owner's representative, the filling of water will be discontinued at which time the drain pan will be observed to whether it holds any water after a ½ minute time period. The test will be complete when the Owner's representative is satisfied with the result. Any defects or failure to satisfy the drain pan test will require immediate correction to the satisfaction of the owner's representative. If supplemental drain pans are specified in any section, a similar test for water tightness (excluding drainage) will be performed. This test shall be performed on all units.

END OF SECTION

SECTION 23 84 13

HUMIDIFIERS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Direct Injected Steam Humidifiers
- B. Electric Resistive Element Steam Humidifiers
- C. Condensate Drain Cooler

1.2 RELATED REQUIREMENTS

- A. Section 23 07 00 - HVAC Insulation
- B. Section 23 21 13 - HVAC Piping
- C. Section 23 22 16 - Steam and Condensate Specialties
- D. Section 23 31 13 - Sheetmetal Ductwork
- E. Division 26: Electrical

1.3 SUBMITTALS

- A. Submit product data and information for review in accordance with the provisions of Division 01.
- B. Product Data: Provide catalog sheets indicating general assembly, dimensions, weights, materials, and certified performance ratings.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Armstrong; DriSteem; Condair
 - 1. Humidifiers and drain cooler shall be provided from the same manufacturer.

2.2 ELECTRIC RESISTIVE ELEMENT STEAM HUMIDIFIER

- A. Furnish a self-contained electric element steam humidifier to meet scheduled capacity. The humidifier shall be capable of operating on normal or demineralized water without modification.
- B. Unit(s) shall have a serviceable stainless steel steam canister with corrosion resistant heating elements, water level sensor, and a high-temperature limit switch.
 - 1. The steam tank shall be constructed of stainless steel and include a drain strainer. The tank shall also have an integral surface skimming port to remove floating debris and foam during fill cycles when non-treated water is used. This feature shall have user adjustable interval & duration. The cover shall include a silicone gasket and shall be removable by loosening threaded fasteners. The steam tank shall have a manual reset high temperature limit switch and act to de-energize the power contactor in the event of an over-temperature condition.
 - 2. The electric heating elements shall be constructed with corrosion resistant, alloy sheaths and have screw down type connections.
 - 3. The steam generator tank shall include water level controls in the form of two float switches and be located in a portion of the tank segregated from the main steam generating portion of the tank. Water level electrodes, float switches, and/or valves located in the main portion of the tank are subject to excessive mineral build-up are not acceptable.
 - 4. Water make-up system shall include inlet strainer, solenoid valve with flow regulator, and fill tee with built-in air gap to prevent back siphoning.
 - 5. Drain system shall consist of a solenoid valve, with shielded core dry operator.
- C. Humidifier cabinet shall be minimum 18 gauge steel and power-coat painted to protect against corrosion. The electrical and steam generating compartments shall be separate, but contained within the same cabinet to minimize field wiring. The access doors shall be lockable with a key.
 - 1. Power to the resistive heating elements shall be controlled through the actuation of contactors and solid state relays.

2. Terminal connection points for all line and low voltage (control) connections. A system on/off switch shall be externally available. A door interlock switch shall prevent operation when the access door is open.
3. The unit(s) will utilize one of the following humidity control systems (shown with corresponding input method):
 - a. On/Off (Humidistat)
 - b. Proportional (Transmitter or signal-ohms, VDC, mADC)
 - c. Proportional + Integral (Sensor or signal VDC , mADC)
4. Unit controls shall also include:
 - a. A proportional high limit device (or signal) to enable modulating control systems to adapt the capacity setpoint to variable airflow.
 - b. Duct mounted high level limit humidistat.
 - c. Air proving switch
5. A Native BACnet microprocessor control, monitor and diagnostic system. The microprocessor control shall:
 - a. Sense humidifier demand and adjust water level and output capacity.
 - b. Include a backlit LCD display -- (2) lines, 16 alphanumeric characters per line -- to monitor system operation/performance.
 - 1) Note: Only parameters pertinent to the specific humidity control system shall be displayed.
 - 2) Provide programmable set points with built-in default values for simplified startup. Only set-points pertinent to the control mode are accessible.
 - (a) Note: Certain set-points shall be password secured to prevent tampering; such as PID loop settings.
 - 3) Provide adjustable drain tempering capability.
 - 4) Provide "fault" sensing, record fault history, and display the type of fault. Certain faults shall result in system shutdown. Other faults reflect a potential maintenance problem and allow the unit to continue operation.
 - 5) Network ability to common FMS systems such as BACnet, Modbus, N2, or Lonworks.
- D. Humidistat: As shown on drawings
- E. Provide multiple tube dispersion assembly as follows:
 1. Assembly shall be sized as required to comply with maximum absorption distance scheduled.
 2. Provide complete assembly with the following:
 - a. Fabricated stainless steel horizontal header/separator at bottom of assembly, sloped for sufficient drainage.
 - b. Vertical stainless steel dispersion tubes with top support connections.
 - c. The distribution manifold shall provide uniform distribution over its entire length and be jacketed by steam to assure that vapor discharged is free of water droplets.

2.3 CONDENSATE DRAIN COOLER

- A. Provide external stainless steel condensate drain cooler for humidifiers that do not contain internal drain coolers to temper the steam condensate to less than 130 deg. F using domestic cold water, hot water, or tempered water.
- B. Drain cooler shall have the following options:
 1. Floor mount bracket; or Suspension mount kit
 2. 1" air gap fitting for discharge piping
 3. Integral temperature actuated cold water valve.
 4. Vacuum breaker to prevent backflow into potable water system.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Insulate humidifier, dispersion tube(s), and condensate piping in accordance with manufacturer's written recommendations and as specified in Section 23 07 00 - HVAC Insulation. Provide working NEC required working clearance.
- B. Install humidifier dispersion tube(s) in ductwork with recommended distance from elbows, transitions, and offsets to insure absorption of moisture into air stream.

- C. Install steam and condensate piping to insure dry steam is supplied to humidifier. Install steam supply end of main traps as required for proper operation.
- D. Install humidifiers and limit humidistat in strict accordance with manufacturer's written instructions.
- E. Install drain cooler in strict accordance with manufacturer's written instructions. Size make-up water piping based on anticipated temperature of make-up water. Support unit body separately from connected piping.

3.2 START-UP AND TESTING

- A. Start-up and test humidifiers, humidistats, and condensate cooler for satisfactory operation as part of steam/condensate system and control system. Check to ensure all valves are open and that humidifier responds to humidistat over selected ranges. Check high limit humidistat for proper operation. Check condensate piping for proper slope back to generating unit or condensate cooler. Set drain/fill and skimming controls as appropriate and as recommended by manufacturer for the specific application and water quality.
- B. Demonstrate to and instruct facility engineering personnel in system operation.
- C. Provide manufacturer's operation and maintenance manuals and parts list as required by Section 23 05 00 - Common Work Results for HVAC.

END OF SECTION

SECTION 26 00 00

BASIC ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Provisions established within the Conditions of the Contract and Division 1, the remaining Sections of the Specifications, and the Contract Drawings are collectively applicable to this Section.

1.2 SECTION INCLUDES

- A. General Requirements specifically applicable to Division 26 in addition to Division 1 requirements.

1.3 REFERENCES

- A. ADA - Americans with Disabilities Act.
- B. EIA - Electronics Institute of America.
- C. FM - Factory Mutual.
- D. NEC - National Electrical Code.
- E. NECA's Standard of Installation.
- F. NFPA - National Fire Protection Association.
- G. OSHA - Occupational Safety and Health Act.
- H. TDH - Texas Department of Health (Hospital Licensing Standards).
- I. UL - Underwriters Laboratories, Inc.
- J. NEMA – National Electrical Manufacturers Association

1.4 SCOPE OF WORK

- A. The work included consists of the furnishing of all materials, labor, tools, transportation, services, permits, fees, etc. which are required and/or necessary for a complete and proper installation of the electrical system(s), which shall include:
- B. Installation of branch circuit panelboards, breakers, and feeders. Including engraved nameplates with name, voltage, and source.
- C. Connection of electric power to heating, ventilating, and air conditioning equipment.
- D. Interior lighting.
- E. Connection of electric power to owner-furnished equipment and other equipment indicated on the Contract Documents.

- F. Raceways with pull string and conduit stubs for use in future design release.
- G. Temporary power service for construction work.
- H. Wall receptacles with stainless steel wall plates, engraved with the source panelboard name and circuit number.
- I. Metal Clad type MC cable not allowed.

1.5 WORK SEQUENCE

- A. Install work in phases to accommodate Owner's occupancy requirements during the construction period and as outlined in Division 1.

1.6 ALLOWANCES

- A. Refer to Division 1.

1.7 ALTERNATES

- A. Refer to Division 1.

1.8 REGULATORY REQUIREMENTS

- A. Conform to applicable codes and ordinances. Refer to paragraph 1.3 above.
- B. Obtain permits and request inspections from authorities having jurisdiction.
- C. Perform all work to comply with applicable codes, ordinances, and regulations of authorities having jurisdiction.
- D. In case of differences between the Contract Documents and applicable codes and ordinances, the more stringent shall apply.

1.9 PROJECT/SITE CONDITIONS

- A. Install Work in locations shown on Drawings unless prevented by Project conditions.
- B. Prepare drawings showing proposed rearrangement of Work to meet Project conditions, including changes to Work specified in other Sections. Obtain permission of Owner before proceeding.
- C. In some cases the existing system(s) will be expanded or replaced. Contractor shall thoroughly familiarize himself with the existing system(s) and bring to the attention of the Architect/Engineer any situations, which deviate from those, indicated in the Contract Documents.

1.10 ELECTRICAL COORDINATION DRAWINGS

- A. Prepare and submit a set of coordination drawings showing major elements, components, and systems of electrical equipment and materials in relationship with other building components.
- B. Prepare drawings to an accurate scale of 1/4 inch = 1 foot, 0 inches or larger. Indicate the locations of all equipment and materials, including clearances for servicing and maintaining equipment. Indicate movement and positioning of large equipment into the building during construction.

- C. Prepare floor plans, reflected ceiling plans, elevations, sections, and details to coordinate and integrate all installations. Indicate locations where space is limited and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
- D. Variable Frequency Drives and Motor control centers.
- E. Panelboards.
- F. Equipment connections.
- G. Control panels.
- H. Wiring diagrams (indicating field installed electrical power and control wiring and cabling layouts, overcurrent protective devices, equipment, and equipment connections).
- I. Work in crawl spaces.
- J. Exterior wall penetrations.
- K. Ceiling plenums, which contain piping, ductwork, or equipment in congested arrangement.
- L. Work in existing electrical rooms / closets
- M. Building ground system.

1.11 SEQUENCING AND SCHEDULING

- A. Construct Work in sequence under provisions of Division 1.
- B. Any Work requiring a shutdown of an existing system shall be scheduled and approved by the Owner.
- C. Early in the construction process, and before beginning any electrical related work, Contractor shall submit both a written and oral discussion to the Engineer, Architect, and Owner representative a Sequencing and Scheduling Plan for all medium voltage circuit and switchgear work which will involve outages. It is critical that electric supply be maintained on all Campuses. Any special provisions or coordination with other disciplines as may be needed to meet these supply needs shall be provided by the Contractor and included in the contract.

1.12 TEMPORARY POWER REQUIREMENTS

- A. Provide power distribution system sufficient to accommodate construction operations requiring power, use of power tools, electrical heating, lighting, and start-up/testing of permanent electric-powered equipment prior to its permanent connection to electrical system. Provide overload protection.
- B. Provide sufficient temporary lighting to ensure proper workmanship by combined use of daylight, general lighting, and portable plug-in task lighting.
- C. Provide temporary power, including emergency generators, as may be required to accommodate power supply requirements during periods of construction.

1.13 TESTING

- A. Upon completion of the work test the individual components for proper voltage and operation.

- B. Balance loads connected to panel to achieve a maximum imbalance of 10 percent or less.

1.14 WARRANTIES

- A. Refer to Division 1 for procedures and submittal requirements for warranties. Refer to individual equipment specifications for warranty requirements.
- B. Compile and assemble the warranties specified in Division 26 into a separated set of vinyl covered three ring binders, tabulated and indexed for easy reference.
- C. Provide complete warranty information for each item (product or equipment) to include:
 - D. Date of beginning of warranty or bond.
 - E. Duration of warranty or bond.
 - F. Name, address, and telephone number of manufacturer.
 - G. Procedures for filing a claim and obtaining warranty services.

1.15 RECORD DRAWINGS AND SPECIFICATIONS

- A. Keep a complete set of Drawings and Specifications in job site office for showing actual installation of electrical systems and equipment. Use this set of Documents for no other purpose.
- B. Record Drawings: Where material, equipment, or system components are installed differently from that shown, indicate differences clearly and neatly on Record Drawings using ink or indelible pencil.
- C. Record Specifications: Mark specifications to indicate approved substitutions, change orders, actual equipment and materials used.
- D. At project completion submit Record Drawings and Specifications to Engineer. All electrical drawings shall be submitted in mylar reproducible.
- E. Electronic Media Transfer: AutoCAD files specific to each floor shall be turned over (under formal transmittal letter) to the owner within 1 week of each floor acceptance. The electronic files shall have the latest changes incorporated, and represent the most accurate design issued for construction. The files shall comply with Specification 220011. The files shall be in AutoCAD 2016.

1.16 TRAINING

- A. Refer to other sections for specific training requirements.
- B. Where training is called for in other sections provide a minimum of 8 hours on site training for Owner's representatives.
- C. Training shall be presented by a qualified instructor with training experience and technical knowledge of the product.
- D. Submit a training agenda, proposed date, and instructor qualifications to the Owner for approval.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Materials and equipment shall be acceptable to the authority having jurisdiction as suitable for the use intended.
- B. Furnish products listed and classified by Underwriter's Laboratory, Inc. as suitable for purpose specified and shown.
- C. Unless otherwise specified materials shall be new and free from any defects.

PART 3 - EXECUTION

3.1 WORKMANSHIP

- A. Arrange electrical work in a neat, well-organized manner:
- B. Do not block future connection points of electrical service.
- C. Locate equipment to provide easy access.
- D. Arrange electrical work to provide adequate access for operation and maintenance.
- E. Install work parallel or perpendicular to building lines, unless noted otherwise.
- F. Locate conduit and piping as high as possible or per the hierarchy drawings.
- G. Support work with appropriate fasteners and/or support systems.
- H. Conceal conduit and piping in finished spaces.
- I. Apply, install, connect, erect, use, clean, adjust and condition materials and equipment as recommended by the manufacturers in their published literature.

3.2 CUTTING AND PATCHING

- A. Cut and patch walls, floors, etc., resulting from work in existing construction.
- B. Make opening through masonry and concrete by core drilling in locations acceptable to the Owner's Agent.
- C. Restore openings to "as new" condition to match existing surrounding materials.
- D. Penetrations through fire-rated elements shall conform to applicable UL classifications.

3.3 SLEEVES

- A. Generally, where pipes and conduits pass through non-fire rated walls or floors, use 18 ga. galvanized sheet steel sleeves for mechanical items and E.M.T. conduit for electrical items. Size sleeves to permit the subsequent insertion of pipe and to fill the annular space between the pipe and its sleeve with fire rated packing. Sleeves through floor shall extend 2 inches above finished floor.

END OF SECTION

SECTION 26 00 10

UTSW ELECTRICAL DESIGN REQUIREMENTS

PART 1 - GENERAL

- 1.1 PURPOSE
- 1.2 THIS DOCUMENT ESTABLISHES DESIGN REQUIREMENTS FOR ELECTRICAL PROJECTS ON UTSW CAMPUS.
- 1.3 THIS DOCUMENT ADDRESSES DESIGN CRITERION NOT SPECIFICALLY COVERED BY THE NATIONAL ELECTRIC CODE (NEC), OR SET REQUIREMENTS THAT MAY MEET THE MINIMUM REQUIREMENTS OF THE NEC.
- 1.4 APPLICABILITY: THIS DOCUMENT APPLIES TO ALL ELECTRICAL PROJECTS DESIGNED AND CONSTRUCTED BY PHYSICAL PLANT PERSONNEL, OUTSIDE ARCHITECT/ENGINEERING (A/E) FIRMS, AND ALL CONSTRUCTION CONTRACTORS. NO SUBSTITUTIONS TO THE STANDARD ARE ACCEPTABLE WITHOUT THE WRITTEN AUTHORIZATION OF THE ENGINEER.
- 1.5 TERMS AND DEFINITIONS
 - A. UTSW - University of Texas Southwestern Medical Center at Dallas.
 - B. Wet Lab - Any laboratory equipped with sinks, fume hoods, biological safety cabinets, or other equipment, which requires multiple utilities (such as gas, air or vacuum), and a greater degree of HVAC than a dry lab.
 - C. Dry Lab - Laboratories that are not equipped with multiple utilities but require a greater degree of electrical power and HVAC than an office area.
 - D. CDAS - Central Data Acquisition System. CDAS is the campus-wide central monitoring station located in the Physical Plant Office (P Building).
 - E. ENGINEER - UTSW Engineer.
 - F. INSPECTOR - UTSW Inspector
- 1.6 ELECTRICAL CONDUIT HOME RUNS
 - A. All wiring on UTSW campus shall be rated for 90 degrees C service and be copper conductor.
 - B. Electrical conduit home run diameter shall not be less than ¾ inch diameter for any circuit.
 - C. Metal Clad (MC) cable is prohibited for use on UTSWMC campus, except in special cases. Written approval of the Engineer and Inspector is required before MC can be installed. Unlike rigid metallic conduit, a new set of wires cannot be pulled through MC cable making it unsuitable for renovation purposes.

1.7 ELECTRICAL BRANCH PANELS

- A. Electrical branch panels rated less than 225 amps and less than 42 circuits shall not be used, unless special circumstances require a lower rated panel. Approval is required from the ENGINEER. Manufacturer to match existing (SquareD).
- B. Branch panels shall be equipped with main circuit breaker where indicated, and designed for 4-wire, three-phase use. A lockable door-in-door hinged factory cover is required. See Specification 262417.
- C. MLO panels shall only be used in multi-tub configurations as specified on the drawings.

1.8 VARIABLE FREQUENCY DRIVES (VFD)

- A. UTSW uses VFD's, which readily interface with Siemens Control Systems. UTSW Specification 262923 shall be used for all VFD's. Manufacturer to match existing (Yaskawa) and be 6 pulse.

1.9 ELECTRICAL MOTORS

- A. The supplier of any component driven by an electrical motor shall provide the motor and motor starter correctly sized for the component.
- B. Refer to Division 23 for electric motor requirements. Otherwise, all components and motors shall be sized for 120% of design. With the introduction of the premium efficiency motors, a new problem has arisen which is generally unknown by many end users. Designs of premium efficiency motors vary by manufacturers. In short, designing a motor with improved efficiency has resulted in a lower X/R ratio, causing higher inrush currents. These higher currents can cause circuit breaker trip when using a full voltage starter.
- C. Therefore, all motors greater than 5 HP shall be premium efficiency TEFC and IEEE 841 (severe duty) rated. Motors greater than 75 HP shall additionally use a soft starter if a VFD is not specified.
- D. All soft starters (if specified) to be SquareD. All VFD(s) to be 6-pulse Yaskawa.

1.10 ELECTRICAL DISCONNECTS

- A. Disconnects shall be located in plain view of the electrical load (within six feet). Where the electrical load is less than 1/8 Horsepower or 300 volt amperes, a toggle switch rated for the load may be used as the disconnect if it is fully visible from the load (within six feet) or is capable of being locked.
- B. Electrical disconnects for above the ceiling equipment shall be located within 6 feet of the equipment.
- C. All disconnects to be individually enclosed, Heavy Duty, SquareD.

END OF SECTION

SECTION 26 05 00

BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Provisions established within the Conditions of the Contract and Division 1, the remaining Sections of the Specifications, and the Contract Drawings are collectively applicable to this Section.

1.2 WORK INCLUDED

- A. Hinged cover enclosures and cabinets
- B. Penetration sealing systems (fire stops)
- C. Electrical/control portion of HVAC work covered by Division 23 pertaining to basic electrical materials and methods shall follow the requirement set forth by this specification.

1.3 APPLICABLE CODES AND STANDARDS

- A. NFPA 70, National Electrical Code (latest edition)
- B. American National Standard, National Electrical Safety Code, (latest edition)
- C. Applicable publications of NEMA, ANSI, IEEE, and ICEA
- D. Underwriters Laboratories, Inc. Standards (UL)
- E. Federal, city, state, and local codes and regulations having jurisdiction
- F. OSHA requirements
- G. NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)
- H. NEMA WD 1 - General-Purpose Wiring Devices
- I. UL 98 - Enclosed Switches

1.4 INTENT

- A. This Section is not, and shall not be interpreted to be, a complete listing of all materials or equipment that is Contractor furnished and erected. It is intended to clarify and further define the Contractor scope of work, procurement, and responsibilities for those incidental materials that are not specified by other specifications, but important to a complete and operational system.
- B. The Contractor shall furnish all equipment and materials, whether or not specified in other Sections of specification and on drawings, for installation and connection required to place equipment into satisfactory operating service. The Contractor shall review the Drawings and specifications for clarification of his responsibility in the handling and installation of equipment and material. Where applicable, and not in

contradiction with the Drawings and Specifications, the Contractor shall install and connect the equipment in accordance with the manufacturer's recommendations and instructions.

- C. All materials and equipment shall be of types and manufacturer specified wherever practical. Should materials or equipment so specified be unattainable, the Contractor shall submit the description and manufacturer's literature, reason for substitution request and shall secure the approval of the Engineer before substitution of other material or equipment is purchased. This Section establishes performance requirements and the quality of equipment acceptable for use and shall in no way be construed to limit procurement from other manufacturer.

1.5 SUBMITTALS

- A. Provide submittals in addition and in accordance with Section 260000, Basic Electrical Requirements, and Division 01 for submittal requirement.
- B. Submit manufacturer's literature and specification data sheets for each type of basic material, which is applicable to the project.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Provide factory-wrapped waterproof flexible barrier material for covering materials, where applicable, to protect against physical damage in transit. Damaged materials shall be removed from project site.
- B. In their factory-furnished coverings, store materials in a clean, dry indoor space, which provides protection against the weather.

PART 2 - PRODUCTS

2.1 ENCLOSURES AND CABINETS

- A. Enclosures and cabinets for all Contractor furnished electrical equipment and devices shall be suitable for the location and environmental conditions and shall be of the NEMA type as shown in Table 260500-1. Exceptions shall be specifically designated on the Drawings.

Table 260500-1 Enclosures		
Location	Environment	Enclosure Type
Indoor Utility	Dry, subject to dust, falling dirt and dripping non-corrosive liquids	NEMA 12
Indoor	Clean, Dry	NEMA 1
Outdoor	Subject to windblown dust and rain, splashing water, and hose-directed water	NEMA 4
Indoor	Wet, subject to hose-directed water	NEMA 4
Outdoor	Subject to falling rain, sleet, and external ice formation	NEMA 3R
Indoor or Outdoor	Subject to corrosion, windblown dust and rain, splashing water and hose-directed water	NEMA 4X

1. Enclosures shall have the following properties:

- B. Hinged Cover Enclosures: NEMA 250.

1. Type 1: Steel.
2. Type 3R: Steel.
3. Type 4: Steel with gasket door, rain tight.
4. Type 4X: Stainless steel, (polycarbonate or fiberglass reinforced polyester (FRP) in corrosive areas).
5. Type 12: Steel with gasketed door, dust-tight.
6. Finish: Exterior, manufacturer's standard gray enamel finish; interior, white enamel finish.
7. Covers: Continuous hinge, held closed by flush latch operable by hasp and staple for padlock. Where required for NEMA ratings, gaskets shall be neoprene rubber.
8. Interior Panel for Mounting Terminal Blocks or Electrical Components: 14 gauge steel, white enamel finish.
9. Provide protective pocket inside front cover with schematic diagram, connection diagram, and layout drawing of control wiring and components within enclosure.

2.2 PENETRATION SEALING SYSTEMS (FIRE STOPS)

- A. Provide penetration sealing where conduit, cable tray, etc. pass through rated walls, ceilings, and floors. See Section 078413, Fire Stopping, and Section 078446, Joint Sealants, for sealing requirements and systems.

2.3 UL LISTING

- A. All equipment and materials shall be new and conform to the requirements of this Section. All equipment and materials shall be UL listed, and shall bear their label whenever standards have been established and level service is regularly furnished. All equipment and materials shall be of the best grade of their respective kind for the purpose.

PART 3 - EXECUTION

3.1 FABRICATION - CONTROL ENCLOSURES AND CABINETS

- A. Shop assemble enclosures and cabinets housing terminal blocks or electrical components in accordance with NEMA ICS 6.

3.2 INSTALLATION - ENCLOSURES AND CABINETS

- A. Install cabinets and enclosures plumb; anchor securely to wall and structural supports at each corner, minimum. Direct attachment to dry wall is not permitted.
- B. Provide accessory feet for freestanding equipment enclosures.
- C. Install trim plumb.

3.3 ERECTION OF EQUIPMENT

- A. Manufacturer's Installation Instructions: Where furnished or called for by the manufacturer equipment manufacturer's installation instructions shall be considered a part of this specification and fully complied

with. Where the Contractor damages the finishing coat of paint in existing or completed areas, he shall refinish with matching paint.

- B. Mounting Heights: Individual safety switches and buttons and devices shall normally be installed at the following mounting heights, when not specified on the Drawings.
- C. Safety Switches: 6 feet 0 inches (to top).
- D. Pushbuttons: 4 feet 0 inches (to center).
- E. Control Panels: 6 feet 0 inches (to top).
 - 1. Mounting: Equipment and control devices shall be supported independent of conduit connections. Panels or cabinets shall be mounted on metal frame supports independently of equipment. Control devices and metal enclosures shall be bolted or welded to steel channel or steel plate. All electrical equipment and devices not covered by the above, such as miscellaneous switches, thermostats, duct switches, temperature switches, floats, photoelectrical devices, and similar electrical devices shall be located and set as suitable for the application. Where control panels are provided as part of the equipment racks mounted on the floor, they shall be provided to support conduits and flexible connections to control panels.

3.4 COORDINATION

- A. Exact location of all electrical equipment, devices and fixtures shall be determined in field by contractor and verified by Engineer's field representative prior to installation.

END OF SECTION

SECTION 26 05 01

ELECTRICAL DEMOLITION

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Electrical demolition for remodeling.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for minor electrical demolition for remodeling.
- B. Section 260000 - Basic Electrical Requirements.
- C. In the event of conflict regarding minor electrical demolition requirements between this Section and any other section, the provisions of this Section shall govern.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Materials and equipment for patching and extending work: as specified in individual sections.
- B. Provide all materials necessary for work.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify field measurements and circuiting arrangements.
- B. Verify that abandoned wiring and equipment serve only abandoned facilities.
- C. Demolition drawings are based on casual field observation. Report discrepancies to Engineer before disturbing existing installation.
- D. Beginning of demolition means Contractor accepts existing conditions.
- E. All abandoned wiring must be removed.

3.2 PREPARATION

- A. Disconnect electrical systems in walls, floors, and ceilings scheduled for removal. Provide temporary wiring and connections to maintain existing systems in service during construction.
- B. Existing Electrical Service: Maintain existing system in service until new system is complete and ready for service. Disable system only to make switchovers and connections. Obtain written permission from Owner at least (2) weeks before partially or completely disabling system. Minimize outage duration. Make temporary connections to maintain service in areas adjacent to work area.

- C. Existing Fire Alarm System: Maintain existing system in service until new system is accepted. Disable system only to make switchovers and connections. Obtain written permission from Owner at least (2) weeks before partially or completely disabling system. Minimize outage duration. Make temporary connections to maintain service in areas adjacent to work area.
- D. Existing Telephone System: Maintain existing system in service until new system is complete and ready for service. Disable system only to make switchovers and connections. Obtain written permission from Owner at least (2) weeks before partially or completely disabling system. Minimize outage duration. Make temporary connections to maintain service in areas adjacent to work area.

3.3 DEMOLITION AND EXTENSION OF EXISTING ELECTRICAL WORK

- A. Remove existing installations to accommodate new construction.
- B. Remove abandoned conductors to source of supply.
- C. Remove exposed abandoned conduit, including abandoned conduit above accessible ceiling finishes. Cut embedded or concealed conduit flush with walls and floors, and patch surfaces.
- D. Disconnect abandoned outlets and remove devices. Remove abandoned outlets if conduit servicing them is abandoned and removed. Provide blank cover for abandoned outlets that are not removed.
- E. Disconnect and remove abandoned panelboards and distribution equipment.
- F. Disconnect and remove electrical devices and equipment serving utilization equipment that has been removed.
- G. Disconnect and remove abandoned luminaires. Remove brackets, stems, hangers, and other accessories.
- H. Repair adjacent construction and finishes damaged during demolition and extension work.
- I. Maintain access to existing electrical installations that remain active. Modify installation or provide access panel as appropriate.
- J. Existing installations shall not be extended for this project except as noted.

3.4 CLEANING AND REPAIR

- A. Clean and repair existing materials and equipment that remain or are to be reused.
- B. Panelboards: Clean exposed surfaces and check tightness of electrical connections. Replace damaged circuit breakers and provide closure plates for vacant positions. Provide typed circuit directory showing revised circuiting arrangement.
- C. Luminaires Indicated to be Reused: Remove existing luminaires for cleaning. Use mild detergent to clean all exterior and interior surfaces; rinse with clean water and wipe dry. Replace lamps, ballasts, and broken electrical parts.

3.5 DISPOSITION OF MATERIAL AND EQUIPMENT

- A. Review with the Owner materials that have been removed and are no longer required, to determine any which the Owner may desire to keep. Deliver those materials that the Owner desires to the Owner's specified location.

- B. For those materials not required by the Owner, dispose of them in accordance with applicable regulations.

END OF SECTION

SECTION 26 05 19

BUILDING WIRE, CABLE AND CONNECTORS (600 VOLTS AND BELOW)

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Provisions established within the Conditions of the Contract and Division 01, the remaining Sections of the Specifications, and the Contract Drawings are collectively applicable to this Section.

1.2 REFERENCES

- A. ANSI - American National Standards Institute.
- B. NEC - National Electrical Code.
- C. NFPA - National Fire Protection Association.
- D. UL - Underwriters Laboratories, Inc.
- E. NEMA WC 3 - Rubber-Insulated wire and Cable for the Transmission and Distribution of Electrical Energy.
- F. NEMA WC 5 - Thermoplastic-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

1.3 SUBMITTALS

- A. Submit under provisions of Division 01.
- B. Product Data: Provide for each cable assembly type.
- C. Test Reports: Indicate procedures and values obtained.
- D. Manufacturer's Installation Instructions: Indicate application conditions and limitations of use stipulated by product testing agency specified under Regulatory Requirements.

1.4 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing products specified in this Section with minimum ten years' experience.

1.5 REGULATORY REQUIREMENTS

- A. Conform to requirements of ANSI/NFPA 70.
- B. Furnish products listed by UL as suitable for purpose specified and shown.

1.6 PROJECT CONDITIONS

- A. Verify that field measurements are as shown on Drawings.
- B. Conductor sizes are based on copper.
- C. Wire and cable routing shown on Drawings is approximate unless dimensioned. Route wire and cable as required to meet Project Conditions.
- D. Where wire and cable routing is not shown, and destination only is indicated, determine exact routing and lengths required.

1.7 COORDINATION

- A. Coordinate Work under provisions of Division 01.
- B. Determine required separation between cable and other work.
- C. Determine cable routing to avoid interference with other work.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

2.2 BUILDING WIRE:

- A. AFC Co.
- B. American Insulated Wire Corp.
- C. Belden Wire & Cable Co.
- D. Houston Wire & Cable Co.
- E. Substitutions: Under provisions of Division 01.

2.3 BUILDING WIRE

- A. Description: Single conductor, soft drawn stranded copper insulated wire.
- B. Conductor: Copper (Aluminum wire is not acceptable).
- C. Insulation Voltage Rating: 600 Volts.
- D. Insulation: ANSI/NFPA 70; Type THWN, THHN, or XHHW insulation for feeders and branch circuits larger than 8 AWG; Type THWN, or THHN insulation for feeders and branch circuits 10 AWG and smaller.
- E. No conductor smaller than No. 12 wire shall be used for lighting or power purposes. In the case of "home runs" over 70' in length (150' for 277 volt) no conductor smaller than a No. 10 wire shall be used. The sizing of all wire except remote control wire shall be accomplished in the case of both feeder and branch circuits by conforming to the following provisions.

1. 480 Volt Branch Circuits: The voltage drop in the case of 480/277 volt circuits shall not exceed 3.0% at maximum load and 70.0% power factor.
 2. 208/120 Volt Branch Circuits: The voltage drop in the case of 208/120 volt circuits shall not exceed 3.0% at maximum load and 70.0% power factor.
 3. Feeder circuits shall have a maximum voltage drop of 2%.
- F. Remote control wires shall be no smaller than No. 14 conductors. Control wires shall be run in separate conduits. The National Electrical Code may require the use of larger conductors.

2.4 REMOTE CONTROL AND SIGNAL CABLE

- A. Control Cable for Class 1 Remote Control and Signal Circuits: 98 percent conductivity copper conductor, 600 Volt insulation, rated 60 degrees C, individual conductors twisted together, shielded, and covered with a PVC jacket; UL listed.
- B. Control Cable for Class 2 or Class 3 Remote Control and Signal Circuits: 98 percent conductivity copper conductor, 300 Volt insulation, rated 60 degrees C, individual conductors twisted together, shielded, and covered with a PVC jacket; UL listed.
- C. Plenum Cable for Class 3 Remote Control and Signal Circuits: 98 percent conductivity copper conductor, 300 Volt insulation, rated 60 degrees C, individual conductors twisted together, shielded, and covered with a nonmetallic jacket; UL listed for use in air handling ducts, hollow spaces used as ducts, and plenums.

2.5 WIRING CONNECTORS

- A. Solderless Pressure Connectors:
 1. 3M Scotchlok Spring Connectors.
 2. Buchanan model B-1 through B-4.
 3. Ideal Wire-Nut model 71-B through 76-B.
 4. Substitutions: Under provisions of Division 01.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that interior of building has been protected from weather.
- B. Verify that mechanical work likely to damage wire and cable has been completed.

3.2 PREPARATION

- A. Completely and thoroughly swab raceway before installing wire.

3.3 WIRING METHODS

- A. Use wiring methods indicated on Drawings.

3.4 INSTALLATION

- A. Install products in accordance with manufacturer's instructions.
- B. Install all line voltage cable in conduit.
- C. Use only stranded conductors for feeders and branch circuits.
- D. Use stranded conductors for control circuits and all circuits.
- E. Use conductor not smaller than 12 AWG for power and lighting circuits.
- F. Use conductor not smaller than 14 AWG for control circuits.
- G. Use 10 AWG conductors for 20 ampere, 120 Volt branch circuits longer than 70 feet. The voltage drop of any 120/208 Volt branch circuit shall not exceed 2 percent at maximum load and 70 percent power factor.
- H. Use 10 AWG conductors for 20 ampere, 277 Volt branch circuits longer than 150 feet. The voltage drop of any 277/480 Volt branch circuit shall not exceed 3 percent at maximum load and 70 percent power factor.
- I. Pull all conductors into raceway at same time.
- J. Use suitable wire pulling lubricant for building wire 4 AWG and larger. Do not use compound containing petroleum or other products that may deteriorate insulation.
- K. Install exposed wire and cable parallel and perpendicular to surface or exposed structural members and follow the surface contours, where possible. Protect exposed cable from damage.
- L. Support cables in conduit above accessible ceiling using spring metal clips or metal cable ties.
- M. Do not rest cable / conduit on ceiling panels.
- N. Use suitable cable fittings and connectors.
- O. Neatly train and lace wiring inside boxes, equipment, and panelboards.
- P. Clean conductor surfaces before installing lugs and connectors.
- Q. Make splices, taps, and terminations to carry full ampacity of conductors with no perceptible temperature rise.
- R. Use NSI Polaris connectors for copper conductor splices and taps 6 AWG and larger. Tape uninsulated conductors and connector with electrical tape to 150 percent of insulation rating of conductor.
- S. Use solderless pressure connectors with insulating covers for copper conductor splices and taps 8 AWG and smaller.

3.5 INTERFACE WITH OTHER PRODUCTS

- A. Identify wire and cable under provisions of all applicable Division 26 sections.
- B. Identify each conductor with its circuit number or other designation indicated on Drawings.

3.6 FIELD QUALITY CONTROL

- A. Perform field inspection and testing under provisions of Division 01.
- B. Inspect wire and cable for physical damage and proper connection.
- C. Measure tightness of bolted connections and compare torque measurements with manufacturer's recommended values.
- D. Verify continuity of each branch circuit conductor. Verify proper phasing connections.
- E. Conductors in vertical conduits or raceways shall be supported in a manner set forth in the appropriate section of the latest revision of the National Electrical Code.
- F. Conductors may be run in parallel on sizes 1/0 to 500 KCMIL inclusive provided all paralleled conductors are the same size, length, and type of insulation.
- G. Test feeder conductors clear of faults, high resistance connections, and megger test same at 600 Volts DC. Test results below 30 megohms shall be cause for rejection of the wiring installation. Replace and retest all such rejected conductors.
- H. At the completion of this project, the Contractor shall provide for the Owner three (3) complete and finally corrected sets of working drawings. These sets of working drawings shall be new, unused, and in good condition, and shall include the nature, destination, path, size, and type of wire and all other characteristics for complete identification of each and every conduit and circuit.

3.7 COLOR CODE

- A. Color code all conductors in accordance with NEC.
- B. Refer to the following table for color-coding of lighting and power wiring.

		<u>208/120V</u>	<u>480/277V</u>
Phase A		Black	Brown
Phase B		Red	Orange
Phase C		Blue	Yellow
Neutral		White	Gray
Ground		Green	Green
Isolated ground		Green/yellow	Green/yellow

- C. For other systems, match the color coding convention presently used by the hospital.
- D. Color coding of ends only will be acceptable for feeder phase conductors. Use 1 inch wide band of colored tape.
- E. Color coding of ends only will be acceptable for neutral and grounding conductors No. 4 and larger.

END OF SECTION

SECTION 26 05 26

GROUNDING

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Power system grounding
- B. Communication system grounding
- C. Electrical equipment and raceway grounding and bonding

1.2 SYSTEM DESCRIPTION

- A. Ground each separately-derived system neutral to nearest grounding electrode bus bar.
- B. Bond together system neutrals, service equipment enclosures, exposed non-current carrying metal parts of electrical equipment, metal raceway systems, grounding conductor in raceways and cables, receptacle ground connectors, and plumbing systems.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Grounding system components shall be as required to comply with the design and construction of the system indicated. Components shall be as indicated in manufacturer's submittal data.
- B. Ground conductors shall be stranded tinned, annealed copper cable in RMC or IMC conduit for #6 AWG or larger. Bond both ends of metallic conduit.
- C. Ground Rods shall be copper-encased steel, 3/4" diameter, minimum length 10 feet.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Extend existing ground system as indicated, in accordance with the applicable requirements of the National Electrical Code and the National Electrical Contractors Association's "Standard of Installation".
- B. Install grounding conductors continuous, without splice or connection, between equipment and grounding electrodes. Where splices cannot be avoided, splices using exothermic welding may be used with permission of the Engineer.
- C. In all feeder and branch circuits, provide a separate, insulated equipment grounding conductor. Terminate each end on a grounding lug, bus, or bushing.
- D. Connect grounding electrode conductors to metal water pipe using suitable ground clamp. Make connections to flanged piping at street side of flange. Provide bonding jumper around water meter.
- E. Install Fusion Welded ground connectors where they are concealed or inaccessible.

- F. Ground each outlet by the use of an approved grounding pig tail attached to the junction box.
- G. No strap grounding clamps shall be used; connections requiring bolting shall be made up with Monel metal bolts, washers and nuts. Connections shall be made only after surfaces have been cleaned, or ground to expose virgin metal.
- H. Install external ground wire on liquid tight flexible metal conduit with grounding bushings in addition to the internal equipment grounding conductor. Connectors shall have a factory external lug and be listed.
- I. Conductor connections shall be made by means of solderless connectors such as serrated bolted clamps or split bolt and nut type connectors.
- J. The neutral of each transformer shall be bonded to system ground at one point only. This point shall be ahead of the first secondary protective device.
- K. Use minimum 6 AWG copper conductor for connection busbar to communications devices, racks, etc. Allow 10 feet slack conductor at cabinet.
- L. Terminate equipment ground conductor at disconnects and other electrical equipment using a factory equipment ground bar. Field installed lugs are not acceptable.
- M. Provide a UL listed ground bar in all enclosures containing overcurrent protective devices. These ground bars shall be provided by the manufacturer of the equipment and may be field installed. The individual equipment ground bar shall accept at least two conductors. Individual conductor lugs are not acceptable.

3.2 FIELD QUALITY CONTROL

- A. Inspect grounding and bonding system conductors and connections for tightness and proper installation.
- B. Measure ground resistance from system neutral connection at service entrance to convenient ground reference point using suitable ground testing equipment. Resistance shall not exceed 10 ohms.

END OF SECTION

SECTION 26 05 29

SUPPORTING DEVICES

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish and install all necessary. hangers, supports, bases, and connections for properly installing all electrical equipment and materials.

1.2 REFERENCE DOCUMENTS

- A. The Basic Electrical Requirements are hereby made a part of this section of the work. Refer to Section 260000.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Supports for single runs of conduit shall be one-hole pipe straps, beam clamps, steel rod hangers, or trapeze hangers.
- B. Pipe straps for rigid conduit shall be malleable iron and plated. Pipe straps for EMT may be plated stamped steel.
- C. Rod hangers shall be rust inhibited and selected for weight supported but shall not be smaller than No. 8. Rod hangers for conduits shall be with bolted fastening equal to Minerallac conduit hangers.
- D. "Caddy Clips" are not allowed.
- E. Supports for multiple runs of conduit and other raceways shall be continuous channel inserts or trapeze hangers of steel framing channel and attached with single bolt channel pipe straps.
- F. Supports of vertical runs of conduit of more than 12'-0" shall be galvanized U-bolts or Kindorf C-210 riser pipe clamps on channel iron bearing plates.
- G. Supports for panelboards, cabinets, terminal boards, junction and pull boxes shall be angle iron or framing channel equal to B-Line B22 galvanized.
- H. Attachments to concrete for supports shall be equal to B-Line B22-I galvanized continuous inserts, B-Line B2500 spot inserts, Phillips carbon steel Type WS wedge anchors, or Phillips concrete fasteners selected for support required.
- I. Attachment to structural steel for supports shall be equal to Grinnell 131 I-beam clamps, 226 channel clamps and 93 C-clamps selected for support required.
- J. Supports on roofs shall be Miro Industries Model 02 for conduits 2" and smaller and Model 24-R for larger conduits, each fastened to the roof.
- K. Concrete pads shall provide support for floor mounted electrical equipment.

- L. Non-ferrous magnetic material shall be used in magnet areas such as: fiberglass, aluminum, copper, or stainless steel.
- M. Vertical conductor supports per NEC 300.19 in accordance with NEC Table 300.19(A) shall be OZi grip wedge type supports or approved equal.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Securely fasten and support conduits, raceways of all types and all electrical boxes, devices, and equipment from the main building structure except as specifically indicated otherwise. Use materials specified hereinbefore. Cable, strap, or wire hangers or fasteners shall not be used.
- B. Maintain horizontal and vertical alignment of raceways and to not adversely affect the building structure in strength or appearance.

3.2 CONDUIT SUPPORTS

- A. Support conduits within 3' of each end of each bend, of each termination and at intervals along the run that will maintain true raceway alignment without sag or deformation during pull-in of conductors and after conductors are in place. Support vertical run conduits at not more than 10'-0" on center in addition to the above. Place conduits running exposed on and adjacent to walls after wall surface is installed and on spacers to allow wall to be painted after conduit is installed.
- B. Support runs of flexible conduits that are more than 30" long at 36" on center and such that flexible conduit does not rest on any equipment, pipes, ductwork, walls, floors, or ceilings.

3.3 EQUIPMENT

- A. Support cabinets and boxes to the floor and to the structure above independent of all raceways entering the boxes. Structural walls or columns may be used to support cabinets or boxes only after specific permission is given by the Architect.
- B. Mount devices such as cabinets, boxes, panelboards, disconnects, and motor controls indicated other than at walls on channel iron racks fastened to the floor and the structure above. Plywood backboards of 3/4" thickness and BC grade painted to match the equipment finish may be used as a part of the rack. Plywood shall be fire retardant/rated with metal backing.
- C. Support outlet boxes and junction boxes 100 cubic inches and smaller independently from but equal to supports specified for raceways. Locate outlet and junction boxes above accessible ceilings so they will not interfere with the installation of a lay-in type lighting fixture in any space in the ceiling. Do not use ceiling wire for support of any device or conduit.
- D. Arrange that all supports for boxes, devices, cabinets or equipment that are flush mounted to be concealed behind the walls or ceiling where mounted.
- E. Provide concrete bases for all floor mounted equipment such as switchboards and floor mounted transformers to aid in housekeeping and to provide that the equipment has a flat and level surface on which to be mounted and anchored. Concrete work shall conform to the requirements for finished concrete work indicated under other Divisions of Work. Unless indicated otherwise, bases shall be reinforced concrete with No. 4 bars 9" o.c. each way and 1-1/2" minimum cover; be 3-1/2" high above finished floors; be 6" high above other finished grades; protrude 3" beyond all sides of the equipment mounted thereon; have exposed comers chamfered; and have exposed portions rub finished.

3.4 ATTACHMENT TO STRUCTURE

- A. Hangers and other supports from structures shall be of type and design that will safely support the materials and equipment in its operating state without interfering with the physical requirements of the structure.
- B. Supports shall be from concrete from proper inserts, or after set fittings that are located such they will not interfere with the structural requirements or the oxidizing or other deterioration of the reinforcement and concrete.
- C. Hangers and other supports from steel structure shall be chosen to clamp to the steel and not to weaken or deface the steel with drilling, welding or other similar marks. Attach the support as required by the device chosen in the manner prescribed by the device manufacturer.
- D. The type and location of each support shall be subject to review by the structural engineer and any support that may, in the engineer's sole opinion, reduce the quality of the structure, shall be removed and the structure repaired.

3.5 HANGERS

- A. Rod hangers shall be of proper size to handle the weight of the materials being supported, shall be vertical aligned, maintained straight and shall be length required for the support height and trimmed off immediately below the support.
- B. Threaded rod hangers for trapeze supports shall have nuts and washers above and below the trapeze to secure the trapeze in position. Trapeze hangers shall be level and rods shall be trimmed of excess length immediately below the bottom nut.

3.6 PROTECTION

- A. Rust inhibit all supports by galvanizing or other acceptable means. Supports shall be job rust inhibited at all cuts, breaks, welds, or other points where rust inhibitor coating is broken.

END OF SECTION

SECTION 26 05 33

CONDUITS AND BOXES

PART I - GENERAL

1.1 WORK INCLUDED

- A. Conduit:
 - 1. Rigid metal conduit and fittings
 - 2. Intermediate metal conduit and fittings
 - 3. Electrical metallic tubing and fittings
 - 4. Flexible metal conduit and fittings
 - 5. Liquid-tight flexible metal conduit and fittings
- 6. Boxes:
 - a. Wall and ceiling outlet boxes
 - b. Pull and junction boxes

1.2 REFERENCES

- A. ANSI C80.1 - Rigid Steel Conduit, Zinc-Coated
- B. ANSI/NEMA FB 1 - Fittings and Supports for Conduit and Cable Assemblies
- C. NEMA TC 3 - PVC Fittings for Use with Rigid PVC Conduit and Tubing
- D. ANSI/NEMA OS 1 - Sheet-Steel Outlet Boxes, Device Boxes, Covers and Box Supports
- E. NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

1.3 SUBMITTALS

- A. Submit manufacturer's product data floor boxes, outlet boxes, surface metal raceways, multi-outlet assemblies and wireways

1.4 STORAGE AND HANDLING

- A. Handle raceways carefully to avoid damage, breaking, denting and scoring. Damaged equipment or materials shall not be installed.
- B. Store raceways in a clean dry space and protected from the weather.

PART 2 - PRODUCTS

2.1 CONDUIT AND FITTINGS

- A. Conduit and fittings for all electrical systems on this project shall include the following:
 - 1. Electrical power and lighting feeders
 - 2. Electrical power and lighting circuits
 - 3. Technology systems
 - 4. Control systems (other than HVAC)
 - 5. Fire alarm and signaling systems
 - 6. Other electrical systems
- B. For each electrical wireway system indicated, provide a complete assembly of conduit, tubing or duct with fittings including, but not necessarily limited to, connectors, nipples, couplings, locknuts, bushings, expansion fittings, other components and accessories as needed to form a complete system of the type indicated.
- C. Conduit fittings shall be designed and approved for the specific use intended. Conduit fittings, including flexible, shall have malleable zinc-coated steel insulated throats or bushings. Rigid conduits shall have insulated bushings, unless grounding bushings are required by NEC Article 250-92. Grounding bushings shall have insulated throats.
- D. Rigid and intermediate metal conduit shall be hot-dipped galvanized. Fittings shall be threaded type. Expansion fittings shall be OZ Type DX.
- E. Electrical metallic tubing shall be galvanized. Fittings shall be all steel compression type. Expansion fittings shall be OZ Type TX.
- F. Flexible metal conduit shall be heavy-duty zinc-coated steel. Fittings shall be heavy-duty zinc-coated malleable steel. Flexible metal conduit fittings shall have external grounding lugs for installations other than lighting whips. Provide an external bonding conductor connected to both flexible connectors.
- G. Liquid-tight flexible conduit and fittings shall consist of single strip, continuous, flexible interlocked, double-wrapped steel, galvanized inside and outside, forming smooth internal wiring channel with liquid-tight covering of flexible polyvinyl chloride (PVC).
- H. Liquid-tight flexible conduit fittings shall be heavy-duty zinc-coated steel with external factory ground lugs. Provide an external bonding conductor connected to both liquid-tight flexible connectors.
- I. Crimp or set-screw type fittings are not acceptable.
- J. Minimum conduit size shall be 3/4", except for:
 - 1. 1/2" flexible metallic conduit may be used as fixture whips
 - 2. 1/2" EMT conduit may be used for individual drops to lighting switches with four maximum #12 AWG current carrying conductors. Each box shall have a separate drop.
 - 3. 1/2" EMT conduit may be used for individual drops to low voltage thermostats and similar HVAC controls. Each box shall have a separate drop.
- K. No MC conduit is allowed.

2.2 WALL AND CEILING OUTLET BOXES

- A. Galvanized steel interior outlet wiring boxes of the type, shape and size, including depth of box, to suit each respective location and installation; constructed with stamped knockouts in back and sides, and with threaded holes with screws for securing box covers or wiring devices.
- B. Outlet box accessories as required for each installation, including mounting brackets, wallboard hangers, extension rings, fixture studs, cable clamps and metal straps for supporting outlet boxes, compatible with outlet boxes, compatible with outlet boxes being used and meeting requirements of individual situations.
- C. Corrosion-resistant cast metal weatherproof exterior outlet wiring boxes of the type, shape and size, including depth of box, with threaded conduit ends, cast metal face plate with spring-hinged waterproof cap suitably configured for each application, including face plate gasket and corrosion-proof fasteners.
- D. Outlet boxes in poured concrete shall be plenum type without any holes and with reset knockouts. Where extension rings are used to offset conduit between wall reinforcing steel, joint between extension ring and box shall be sealed to prevent concrete from entering box during pour.

2.3 PULL AND JUNCTION BOXES

- A. Boxes shall be galvanized sheet metal with screw-on cover and welded seams, stainless steel nuts, bolts, screws and washers.
- B. Boxes larger than 12 inches in any dimension shall be panelboard code gasketed galvanized steel with hinged cover.
- C. Boxes shall be sized in accordance with NEC.

PART 3 - EXECUTION

3.1 INSTALLATION - CONDUIT

- A. Install products as indicated, in accordance with the applicable requirements of NEC, NEMA and the National Electrical Contractors Association's "Standard of Installation".
- B. Install raceway and conduit system from point of origin in outlets shown, complete with offsets, pull boxes, junction boxes and fittings.
- C. Install rigid wall hot-dipped galvanized steel conduit or hot-dipped galvanized Intermediate metal conduit. The following exceptions permitted:
 - 1. EMT
 - a. In sizes up to and including 4", may be used inside dry locations where not subject to mechanical damage. EMT may be used in air conditioned spaces, such as accessible ceilings, dry wall partitions and exposed where 6' above the floor. EMT may not be used outside, in concrete, underground, in underfloor spaces, in masonry walls, in locations likely to be damp, or exposed within 6' of the floor. EMT shall not be used for medium voltage circuits.
 - b. Where used for feeder circuits, receptacle branch circuits, and motor branch circuits, EMT shall also contain a NEC equipment grounding conductor.
 - 2. Liquid-tight Flexible Metal Conduit

- a. Install liquid-tight flexible metal conduit for connections to rotating, vibrating, moving or movable equipment, including dry-type transformers. Install external equipment ground wire on flexible conduit with grounding bushings. Maximum length shall be 6'-0" minimum of 3'-0".
3. Flexible Metal Conduit
 - a. Install standard flexible metal conduit (not liquid-tight) with internal ground wire, in spaces above ceilings. Install flexible conduit connection such that vibration is not transmitted to adjoining conduit or building structure. Maximum length shall be 6'-0" minimum of 3'-0", minimum size shall be 3/4" except for lay-in light fixtures where it may be 1/2".
 - 1) Each wall box shall have a separate drop extending up to accessible ceiling. Conduit shall not be installed horizontally to serve more than one box unless noted on plans.
4. Install conduits parallel and supported on Unistrut, or equal, trapezes and anchored with split ring hangers, conduit straps or other devices specifically designed for the purpose. Wire ties are not permitted. Fasten conduit with the following material:
 - a. Wood screws on wood
 - b. Toggle bolts on hollow masonry
 - c. Bolts and expansion anchors in concrete or brick
 - d. Machine screws, threaded rods and clamps on steel
 - e. 4" x 4" penta-treated pine installed in pitch pans on roof, spaced at intervals not to exceed 5'-0"
5. Metal conduit fittings shall be approved for grounding purposes or shall be jumpered with copper grounding conductors of appropriate ampacity. Leave termination of such jumpers exposed.
6. Install expansion fittings in metal conduit as follows:
 - a. Conduit Crossing Building Expansion Joints:
 - 1) EMT all sizes
 - 2) IMC all sizes
 - 3) RMC all sizes
 - b. Conduit Straight Runs in Excess of 100 Feet:
 - 1) EMT all sizes
 - 2) IMC all sizes
 - 3) RMC all sizes
 - c. Conduits entering environmental rooms and other locations subject to thermal expansion and as required by NEC.
 - d. Unless expansion fitting has an integral bonding braid, as in Crouse-Hinds Type XC, a green insulated grounding conductor shall be pulled in the conduit. Both ends of this green grounding conductor shall be accessible for inspection.
 - e. Expansion fittings are not required where offsets, expansion loops, or flexible conduit are placed in conduit runs.
7. Install conduit concealed in walls, partitions and above ceilings. Install conduit exposed in ceiling area (at structure) of boiler rooms, mechanical rooms and in other similar rooms where ceilings are not called for. Install conduit concealed in slab when finished areas below do not have ceiling.
8. Avoid moisture traps where possible; where unavoidable, provide junction box with drain fitting at conduit low point.
9. Use suitable conduit caps to protect installed conduit against entrance of dirt and moisture.

10. Provide 200 lb. nylon cord in empty conduit.
11. Where conduit penetrates fire-rated walls and floors, provide mechanical fire-stop fittings with UL listed fire-rating.

3.2 INSTALLATION - SURFACE METAL RACEWAY AND MULTI-OUTLET

- A. Use suitable insulating bushings and inserts at connections to outlets and corner fittings on multi-outlet assembly.
- B. Maintain grounding continuity between raceway components to provide a continuous grounding path.

3.3 INSTALLATION - WIREWAYS

- A. Bolt wireways to steel channels fastened to the wall or in self-supporting structure. Install level.
- B. Gasket each joint in oil-tight wireway.
- C. Mount raintight wireway in horizontal position only.

3.4 INSTALLATION - BOXES

- A. Provide electrical boxes as shown in Drawings, and as required for splices, taps, wire pulling, equipment connections, and code compliance.
- B. Provide outlet box accessories as required for each installation, including mounting brackets, wallboard hangers, extension rings, fixture studs, cable clamps and metal straps for supporting outlet boxes, compatible with outlet boxes being used and meeting requirements of individual situations.
- C. Electrical box locations shown on Contract Drawings are approximate unless dimensioned. Verify location of outlets prior to rough-in.
- D. Locate and install boxes to allow access.
- E. Do not install boxes back-to-back in walls. Provide a minimum 6-inch separation. Boxes mounted within 24-inches of each other and on both sides of walls shall have a minimum 6-inch separation and each box shall have a listed acoustic rated "putty pack" installed.
- F. Boxes serving the same side of a wall shall have a minimum 6-inch separation without a requirement for listed acoustic putty packs when not installed within 24-inches of boxes on the other side of the wall.
- G. Secure boxes rigidly to the substrate upon which they are being mounted, or solidly imbed boxes in concrete or masonry. Boxes shall not be permitted to move laterally. Boxes shall be secured between two studs. Boxes connected to one stud are not permitted.
- H. Boxes mounted to millwork panels shall be secured rigidly to a mounting bar attached to the boxes and to the panel with a minimum of two wood screws on each side of the box(es).
- I. Use multiple-gang boxes where more than one device is mounted together; do not use sectional boxes. Provide barriers to separate wiring of different voltage systems.
- J. Sectional boxes shall not be used individually or in gangs without written permission of the Engineer or as allowed by these specifications.
- K. Install boxes in walls without damaging wall insulation.

- L. Where required, provide plaster rings, 1-piece, stamped steel construction.
- M. Provide knockout plugs for unused openings.
- N. Outlet boxes in plaster partitions shall be "shallow-type" set flush in wall so there is at least 5/8" plaster covering back of box.
- O. Switch boxes shall not be used as junction boxes.
- P. Coordinate mounting heights and locations of outlets mounted above counters, benches and backsplashes.
- Q. In inaccessible ceiling areas, position outlets and junction boxes within 6 inches of recessed luminaire, to be accessible through luminaire ceiling opening.
- R. Outlet boxes supporting fixtures shall be securely anchored in place in an approved manner. Support outlet boxes and fixtures in acoustic ceiling areas from building structures, not from acoustic ceilings and no more than 3 feet above lay-in ceilings. Lighting fixture outlets shall be coordinated with mechanical and architectural equipment and elements to eliminate conflicts and provide a workable neat installation.
- S. Set floor boxes level and flush with finish flooring material.
- T. Locate pull boxes and junction boxes above accessible ceilings or in unfinished areas.
- U. Support pull and junction boxes independent of conduit.
- V. For each outlet box containing a lighting switch or controller, provide a neutral for each circuit contained.

END OF SECTION

SECTION 26 05 50

SYSTEM COORDINATION AND ANALYSIS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

- A. Provide a complete analysis of the operation of the electrical power system under overcurrent (coordination), short circuit conditions (L-G, L-L and 3Ø bolted fault) and calculation of flash protection boundary and Arc Flash per NFPA 70E..

1.2 SUBMITTALS

- A. Analysis may be performed by the following, but the accuracy, interpretation, application, responsibility and liability resulting from the use of any program, lies with the registered Engineer performing the studies:
 - 1. The studies shall be performed using the latest revision of the SKM Systems Analysis Power*Tools for Windows (PTW) software program. Other commercially available products may be considered under certain circumstances.
- B. Submit a preliminary analysis of the system for approval prior to a release for fabrication of electrical equipment.
- C. Submit final analysis of the system prior to the installation or energization of equipment.

1.3 REFERENCED STANDARDS

- A. The analysis of overcurrent operation shall be based on IEEE "Overcurrent Protective Device Coordination by Computer".
- B. The analysis of short circuit current operation shall be based on IEEE "Procedure for Determining Maximum Short Circuit Value in Electrical Distribution Systems".
- C. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
 - 1. IEEE 141 - Recommended Practice for Electric Power Distribution and Coordination of Industrial and Commercial Power Systems
 - 2. IEEE 242 - Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems
 - 3. IEEE 399 - Recommended Practice for Industrial and Commercial Power System Analysis
 - 4. IEEE 241 - Recommended Practice for Electric Power Systems in Commercial Buildings
 - 5. IEEE 1015 - Recommended Practice for Applying Low-Voltage Circuit Breakers Used in Industrial and Commercial Power Systems
 - 6. IEEE 1584 - Guide for Performing Arc-Flash Hazard Calculations
- D. American National Standards Institute (ANSI):

1. ANSI C57.12.00 - Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
 2. ANSI C37.13 - Standard for Low Voltage AC Power Circuit Breakers Used in Enclosures
 3. ANSI C37.010 - Standard Application Guide for AC High Voltage Circuit Breakers Rated on a Symmetrical Current Basis
 4. ANSI C 37.41 - Standard Design Tests for High Voltage Fuses, Distribution Enclosed Single-Pole Air Switches, Fuse Disconnecting Switches and Accessories
- E. The National Fire Protection Association (NFPA)
1. NFPA 70 - National Electrical Code
 2. NFPA 70E - Standard for Electrical Safety in the Workplace

PART 2 - PRODUCTS -- NOT USED

PART 3 - EXECUTION

3.1 SCOPE OF ANALYSIS

- A. Furnish short-circuit and protective device coordination studies as prepared by the electrical equipment manufacturer or a professional electrical engineering firm.
- B. Furnish an Arc Flash Hazard Analysis Study per NFPA 70E - Standard for Electrical Safety in the Workplace.
- C. The analysis shall be comprehensive from the main 13.8 KV substation through the distribution system to the last overcurrent device serving equipment or outlets.
- D. The contribution of all motors one horsepower and larger shall be included.
- E. Where operation from one or more sources is possible, all operating configurations shall be analyzed.
- F. Also provide comprehensive THD voltage and TDD current harmonic analysis of the 480 volt systems through the 19th harmonic.

3.2 BASIS OF COMPUTATION

- A. Computation shall be based on infinite bus method.
- B. Device characteristics and equipment impedances shall be obtained in writing from the equipment supplier.
- C. The preliminary analysis shall be based on the Contractor's estimation of feeder types and lengths and the proposed equipment characteristics.
- D. The final analysis shall be based on the equipment and materials actually installed at the project. Conductor and raceway type, lengths, and characteristics shall be supplied by the Contractor on the actual materials and routings to be installed.
- E. Coordination TCC's (Time Current Curves) shall include all pertinent data including MAG-I, FLA, cable damage limit, fault current, partial one-line of devices plotted, motor stall, etc.

3.3 RESULTS

- A. The overcurrent device coordination analysis shall present a graphic representation of the required time-current settings for every protective device in the system and a tabular listing for equipment calibration. All devices that are not able to be fully selectively coordinated shall be noted along with recommended action.
- B. The short circuit analysis shall list the phase and ground fault current available at each switchgear, switchboard and panelboard bus in the system, and define whether each device in the system is adequately rated for the duty imposed. Contractor shall furnish equipment with AIC rating that exceeds maximum available fault current regardless of rating specified on drawings. Equipment ratings on drawings are minimum AIC duty and shall not be reduced. Contractor shall apply label with short circuit rating and available fault current.
- C. Series rating is not permitted.

END OF SECTION

SECTION 26 05 53

ELECTRICAL IDENTIFICATION

PART 1 - GENERAL

1.1 DESCRIPTION OF WORK

- A. The extent of the electrical systems and equipment requiring identification is shown on the drawings, and the extent of identification required is specified herein and in individual sections of work requiring identification.
- B. The types of electrical identification specified in this section include the following:
 - 1. Conductor color coding.
 - 2. Operational instructions and warnings.
 - 3. Danger signs.
 - 4. Equipment/system identification signs.

PART 2 - PRODUCTS

2.1 IDENTIFICATION MATERIALS

- A. Provide engraving stock melamine plastic laminate, complying with FS L-P-387, engraved with engraver's standard letter style unless otherwise indicated. Plastic laminate shall be 1/16" thick up to 20 sq. in. and 1/8" for larger size. Letters shall be 1/2" high, black on white background.
 - 1. Nameplates on emergency panels shall be red with white letters.

2.2 LETTERING AND GRAPHICS

- A. Provide numbers, lettering and wording as indicated or, if not otherwise indicated, as required for proper identification and operation/maintenance of the electrical systems and equipment.

PART 3 - EXECUTION

3.1 APPLICATION AND INSTALLATION

- A. Where identification is to be applied to surfaces which require finish, install identification after completion of finishing.
- B. Comply with governing regulations and the requests of governing authorities for the identification of electrical work.

3.2 CONDUCTOR COLOR CODING

- A. Conductors #10 and Smaller: Insulation continuously colored throughout.
- B. Conductors Larger than #10: Three continuous wraps of 1/2" wide 3M #191 tape at each end and at pull and junction boxes.

- C. 208/120 volt, 3 phase, 4 wire system:
 - 1. Phase A - black
 - 2. Phase B - red
 - 3. Phase C - blue
 - 4. Neutral - white
 - 5. Ground - green
- D. 480/277 volt, 3 phase, 4 wire system:
 - 1. Phase A - brown
 - 2. Phase B - orange
 - 3. Phase C - yellow
 - 4. Neutral - gray
 - 5. Ground - green
- E. Conductors installed with color different than that indicated above are not acceptable.
- F. Control and special systems, 600 volt and less, #14 AWG and larger:
 - 1. Colors other than green and white.
 - 2. Include tracer color for identification.
 - 3. Keep colors selected continuous throughout project.
 - 4. Isolated systems, as required by NEC Section 517.
- G. Note colors on record drawings, including switch leg and traveler colors.

3.3 DANGER SIGNS

- A. Install danger signs required by governing regulations and authorities.
- B. Install appropriate danger signs at locations of electrical work constituting similar dangers for persons in or about the project.
- C. Install danger signs on switches and similar controls, regardless of whether concealed or locked up, where untimely or inadvertent operation (by anyone) could result in significant danger to persons, or damage to or loss of property.

3.4 EQUIPMENT/SYSTEM IDENTIFICATION SIGNS

- A. Identify with engraved laminated nameplates, designating load served, on each electrical item on the project. Items to be identified and location of nameplates are as follows:
 - 1. Each main switch unit or circuit breaker adjacent to switch or circuit breaker.
 - 2. Each switch unit or circuit breaker in each distribution panel-- adjacent to switch unit or circuit breaker.
 - 3. Spares shall be labeled "Spare" and shall be left in the "off" position, locked out.

4. Each lighting and appliance panel -- panel name on panel trim cover immediately above panel door. Circuit numbers shall be permanently labeled at factory. Stick-on decals for field installation are not acceptable.
 5. Each safety switch -- on outside of cover.
 6. Each exhaust fan switch -- custom engraved on outside of switch cover plate (high and low if required).
 7. Each motor starter -- on outside of cover.
 8. Outside light switches - custom engraved on outside of switch cover plate.
 9. Switch for load that cannot be seen from the control point -- custom engraved on outside of switch cover plate.
- B. Nameplates shall be securely attached with an approved mechanical fastener. Adhesive attachment shall not be permitted.

3.6 PAINTING

- A. Junction boxes shall be painted as follows:
1. Emergency system - yellow.
 2. Fire alarm system - red.
 3. Critical System -- orange.
 4. Equipment System -- green.

3.7 JUNCTION BOX AND DEVICE PLATES

- A. The appropriate panel and circuit number shall be written in indelible ink on the back of all switch and receptacle faceplates. Junction boxes for circuiting distribution shall also indicate panel and circuit numbers on front of cover plate where junction boxes are above ceiling and on back for exposed junction boxes.
- B. Device cover plates for receptacles and lighting switches shall also have the panel and circuit number engraved on the outside of the cover plate with 3/8" high, black lettering unless noted otherwise on the plans.
- C. Provide engraved legend on coverplates for light switches in patient rooms indicating which lights are controlled within the rooms.

END OF SECTION

SECTION 26 24 17

BRANCH CIRCUIT PANELBOARDS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Lighting and appliance branch circuit panelboards.

1.2 REFERENCES

- A. NEMA AB 1 - Molded Case Circuit Breakers.
- B. NEMA PB 1 - Panelboards.
- C. NEMA PB 1.1 - Instructions for Safe Installation, Operation and Maintenance of Panelboards Rated 600 Volts or Less.

1.3 SUBMITTALS

- A. Submit manufacturer's product data.
- B. Submit dimensioned drawings showing size, circuit breaker and equipment arrangement and ratings, including but not limited to, voltage, main bus ampacity, integrated short circuit ampere rating.

1.4 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Deliver panelboards in factory-fabricated water-resistant wrapping.
- B. Handle panelboards carefully to avoid damage to material components, enclosure and finish.
- C. Store in a clean, dry space and protected from the weather.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Schnieder Electric / Square-D (Basis of Design)
- B. General Electric Company
- C. Eaton / Cutler Hammer
- D. Other manufacturers equal in design and function will be considered upon submittal of manufacturer's data.

2.2 PANELBOARD CONSTRUCTION

- A. Provide circuit breaker type panelboards as scheduled.

- B. Enclosure shall be NEMA Type 1 unless shown otherwise on the drawings.
- C. Provide cabinet front with door in door construction. One door over interior and additional door over wiring gutters. Front cover hinged to box with concealed trim clamps. Provide flush locks.
- D. Bus shall be copper and braced for the maximum available fault current. Minimum bus capacity shall be 225 amp.
- E. Provide copper ground bus in all panelboards.
- F. Neutral bars shall be full capacity.
- G. Exterior and interior steel surfaces shall be cleaned and finished with gray enamel over a rust inhibiting phosphatized coating. Color shall be ANSI 61 gray.
- H. Panelboard electrical ratings and configurations as scheduled on the drawings.
- I. Panelboards shall be provided with an additional 25% of actual circuit requirements as single pole, 20 amp spares.
- J. Circuit directory shall be typewritten and mounted behind clear plastic, in metal frame on inside of each panel door.
- K. Panelboards shall be fully bussed, entire length of panel, 42 spaces, or number of indicated spaces on plans.

2.3 SWITCHING AND OVERCURRENT PROTECTIVE DEVICES

- A. Provide molded case circuit breakers of manufacturer's standard industrial construction, bolt-on type, integral inverse time delay thermal and instantaneous trip.
- B. Multi-pole breakers shall be two or three pole as specified. Handle ties are not permitted.
- C. Interrupting capacity shall be as noted on the drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install panelboards in accordance with manufacturer's written instructions and the applicable requirements of the NEC, NEMA, ANSI and the National Electrical Contractors Association's "Standard of Installation".
- B. Anchor enclosures firmly to walls and structural surfaces, ensuring that they are permanently and mechanically secured.
- C. At the completion of the electrical system this Contractor shall check each phase of all panels under full load and arrange so that all phases shall carry the same load as near as possible.
- D. Contractor shall provide one typewritten and one photocopy of the as-built panel schedule to the Construction Coordinator.

END OF SECTION

SECTION 26 27 26

WIRING DEVICES AND FLOOR BOXES

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Wiring Devices:
 - 1. Wall switches
 - 2. Receptacles
 - 3. Device plates and box covers
 - 4. Occupancy sensors

1.2 REFERENCES

- A. NEMA WE1 - General-Purpose Wiring Devices
- B. NEMA WD5 - Specific-Purpose Wiring Devices

1.3 SUBMITTALS

- A. Submit manufacturer's product data for wiring devices and floor boxes

1.4 DELIVERY, STORAGE AND HANDLING

- A. Deliver wiring devices individually wrapped in factory-fabricated containers
- B. Handle wiring devices carefully to avoid damage, breaking and scoring.
- C. Store in a clean dry space and protected from the weather

PART 2 - PRODUCTS

2.1 GENERAL

- A. Provide factory fabricated wiring devices in the type and electrical rating for the service indicated. Where type and grade are not indicated, provide proper selection to correspond with branch circuit wiring and overcurrent protection. Attachment of wires to devices shall be by screw pressure under the head of binding screws. Arrangements depending on spring pressure or tension are not acceptable. All binding screws shall be brass or bronze.

2.2 ACCEPTABLE MANUFACTURERS - WALL SWITCHES

- A. Bryan
- B. Hubbell
- C. Leviton

- D. Arrow-Hart
- E. Other manufacturers equal in design and function will be considered upon submittal of manufacturer's data.

2.3 WALL SWITCHES

- A. Wall switches for lighting circuits and motor loads under 1/3 hp shall be AC general use snap switch with toggle handle, Ivory, 20 amperes and 120/277 volt AC with number of poles as required, equal to Arrow-Hart #1991.
- B. Horsepower rated switch shall be 30 ampere, equal to Arrow-Hart #3999I (number of poles as required).
- C. Use horsepower rated switches approved for motor control or disconnect service when controlling or disconnecting motor loads in excess of 1/3 hp.
- D. Switch terminal screws or connectors shall be designed to accommodate No. 10 solid conductor.

2.4 ACCEPTABLE MANUFACTURERS - RECEPTACLES

- A. Arrow-Hart
- B. Hubbell
- C. Pass & Seymour
- D. Leviton
- E. Other manufacturers equal in design and function will be considered upon submittal of manufacturer's data.

2.5 RECEPTACLES

- A. Duplex Hospital Grade receptacles shall be white, 20 amperes and 125 volt AC, equal to Arrow-Hart #8300V.
- B. Duplex receptacles in Waiting Areas, Play Areas, Pediatric Treatment Areas, and other designated areas shall be Hospital Grade Tamper Resistant, equal to Hubbell #8300IVTRA.
- C. GFCI receptacles shall be ivory, 20 amperes, 120 volt with integral ground fault current interrupter equal to Arrow-Hart #VGFH20V or TRVGFH20V. Do not use feed through feature.
- D. Heat tape receptacles shall be Arrow-Hart #5262CRGRY with Crouse Hinds #WLRD-1 cover. Install round plug on heat tape cord to match weatherproof bushing on receptacle cover for watertight installation.
- E. Specific-use receptacles shall have volts, amps, poles and NEMA configuration as noted on drawings.
- F. Corridor receptacles shall be Hospital Grade; Arrow-Hart #8300I.
- G. Receptacles with integral USB charging port; Hubbell # USB8300I.

2.6 ACCEPTABLE MANUFACTURERS - WALL PLATES

- A. Arrow-Hart

- B. Hubbell
- C. Pass & Seymour
- D. Leviton
- E. Other manufacturers equal in design and function will be considered upon submittal of manufacturer's data.

2.7 WALL PLATES

- A. Decorative wall plates shall be stainless steel with cutouts required for devices indicated on drawings. Where switches or outlets are shown adjacent to each other, they shall be ganged with partitions between different type services and covered by a single custom wall plate.
- B. Jumbo plates are not permitted.
- C. Weatherproof cover plate shall be gasketed cast aluminum or feraloy (by Crouse-Hinds) with hinged gasketed device covers.

2.8 LIGHTING OCCUPANCY SENSORS

- A. Lighting occupancy sensors shall be installed as a functioning system per the Contract Documents and manufacturer's installation instructions.
- B. Dual technology type that operate using Passive Infrared (PIR) and Ultrasonic (US) shall be provided unless otherwise noted.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Installer must examine the areas and conditions under which wiring devices and floor boxes are to be installed and notify the Contractor in writing of conditions detrimental to the proper and timely completion of the work. Inspect devices for physical damage. Do not proceed with the work until unsatisfactory conditions have been corrected.
- B. Contractor to test and calibrate each occupancy sensor to operate as intended for the area or room installed.

3.2 INSTALLATION

- A. Install wiring devices as indicated, in accordance with the applicable requirements of the NEC, NEMA, ANSI and the National Electrical Contractors' Association's "Standard of Installation."
- B. The approximate location of switches, power outlets, etc., is indicated on the drawings. These drawings, however, may not give complete and accurate information in regard to locations of such items. Determine exact locations by reference to the general building drawings and by actual measurements during construction of the building before rough-in, subject to the approval of the Constructor Inspector.
- C. Install wall switches 48 inches above floor, OFF position down.
- D. Install wall dimmers 48 inches above floor; derate ganged dimmers as instructed by manufacturer; do not use common neutral.

- E. Where shown near doors, install switches and dimmers not less than 2" and not more than 12" from door trim.
- F. Verify all door swings before rough-in and locate switches and dimmers on strike side of door.
- G. Install convenience receptacles, telephone, computer, and CCTV outlets 18 inches above floor, 6 inches above counters, or at the backsplash level.
- H. Install specific-use receptacles at heights shown on Drawings.
- I. Install devices and wall plates flush and level.
- J. Proper commissioning of occupancy sensors shall be completed prior to Substantial Completion.
- K. Make sensitivity and delay-off adjustments to occupancy sensors after installation. Test each sensor after adjustments are made to ensure proper operation.

END OF SECTION

SECTION 26 28 13

FUSES, 600 VOLT

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Dual-element, current limiting Class R fuses for loads up to 600 volts, 0-600 Amps.
- B. Time delay, current limiting Class L fuses for loads up to 600 volts, 601-6000 Amps.

1.2 RELATED WORK

- A. Section 262816 - Disconnects Switches.

1.3 REFERENCES

- A. ANSI/UL 198E - Class R Fuses.
- B. ANSI/UL 198C - High-Interrupting-Capacity Fuses, Current Limiting Types, Class L.
- C. Where application of local codes, trade association standard or publications appears to be in conflict with the requirements of this section, the Architect/Engineer shall be asked for an interpretation.

1.4 SUBMITTALS

- A. Submit manufacturer's data on fuses.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Store fuses in a clean and dry space and protected from weather. When necessary to store outdoors, elevate materials well above grade and enclose with durable, waterproof wrapping.

PART 2 - PRODUCTS

2.1 MATERIAL AND EQUIPMENT

- A. Furnish fuses manufactured by Buss, or equal, in accordance with the following:
 - 1. Motors and Transformers, 0 to 600 Amp:
- B. 250 volt - Buss LPN-R, UI Class RK1
- C. 250 volt - Buss KTN-R, UL Class RK1
- D. 600 volt - Buss KTS-R, UL Class RK1
 - 1. All Applications, 601 to 6000 Amp:
- E. 600 volt - Buss KRP-C, UL Class L
- F. Interrupting Rating: 200,000 RMS Amps.

G. Maintenance Stock, Fuses:

1. Furnish the following:

H. Three spare fuses of each size and type for a spare set.

I. Furnish spare fuses of each size and type for a spare set.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install fuses where indicated, in accordance with the manufacturer's written instructions, the applicable requirements of NEC and the National Electrical Contractors Association's "Standard of Installation".

END OF SECTION

SECTION 26 28 16

DISCONNECT SWITCHES - HEAVY DUTY

PART 1 GENERAL

1.1 WORK INCLUDED

- A. Disconnect switches, fusible and non-fusible.
- B. Enclosures.
- C. Switches shall be furnished and installed at locations as shown on the drawings. Switches shall be of the type approved, indicated, and specified herein.

1.2 REFERENCES

- A. NEMA 250 - Enclosures for Electrical Equipment
- B. NEMA KS 1 - Enclosed Switches.
- C. NFPA 70 - National Electrical Code
- D. NFPA 70E - Electrical Safety Requirement for Employee Workplaces
- E. UL 98 - Enclosed and Dead Front Switches.

1.3 SUBMITTALS

- A. Provide submittals in accordance with and in addition to Section 260000, Basic Electrical Requirements, and Division 01.
- B. Submit manufacturer's product data. Submit dimensioned drawings and equipment ratings for voltage, capacity, horsepower, and short circuit.

1.4 DELIVERY, STORAGE AND HANDLING

- A. Deliver switches individually wrapped in factory-fabricated water-resistant type containers.
- B. Handle switches carefully to avoid damage to material components, enclosure and finish. Damaged switches shall not be installed on project.
- C. Store switches in a clean and dry space and protected from weather.

PART 2 PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Schnieder Electric / Square-D
- B. General Electric Company
- C. Eaton / Cutler Hammer

- D. Other manufacturers equal in design and function will be considered upon submittal of manufacturer's data.

2.2 SWITCH INTERIOR

- A. All switches shall have switch blades which are visible when the switch is OFF and the cover is open.
- B. Lugs shall be front removable and UL Listed for 75 degrees C conductors.
- C. All current carrying parts shall be plated to resist corrosion.
- D. Switches shall have removable arc suppressors to facilitate easy access to line side lugs.
- E. Switches shall have provisions for a field installable electrical interlock.
- F. Voltage rating: 240VAC or 600VAC as per drawings.
- G. Use fuse clips on fusible switches that are rejecting type to accept Class RK or L fuses only.
- H. Use switches that have number of poles required as per drawings.

2.3 SWITCH MECHANISM

- A. Switch operating mechanism shall be quick-make, quick-break such that, during normal operation of the switch, the operation of the contacts shall not be capable of being restrained by the operating handle after the closing or opening action of the contacts has started. Switch mechanism shall conform to NEMA KS1.
- B. The operating handle shall be an integral part of the box, not the cover.
- C. Provisions for padlocking the switch in the OFF position with at least three padlocks shall be provided.
- D. All switches shall have a dual cover interlock mechanism to prevent unintentional opening of the switch cover when the switch is ON and prevent turning the switch ON when the cover is open. The cover interlock mechanism shall have an externally operated override but the override shall not permanently disable the interlock mechanism. The tool used to override the cover interlock mechanism shall not be required to enter the enclosure in order to override the interlock.

2.4 SWITCH ENCLOSURES

- A. Switch covers shall be attached with welded pin-type hinges for: Type 1, 12, 12K, 4-4X-5 stainless steel; top hinged, attached with removable screws and securable in the open position for Type 3R; attached by molded hinges and type 316 stainless steel hinge pins for Type 4X polyester; or attached by type 316 stainless steel bolts for Type 7/9.
- B. The enclosure shall be finished with gray baked enamel paint which is electrodeposited on cleaned, phosphate pre-treated steel for Type 1, gray baked enamel paint which is electrodeposited on cleaned, phosphate pre-treated galvanized steel for Type 3R, 12, 12K, brush finish on type 304 stainless steel for Type 4-4X-5 stainless steel, or gray baked enamel on copper free cast aluminum alloy for Type 7/9
- C. The enclosure shall have permanent ON and OFF markings.
- D. Identify switches, as to equipment served, with engraved laminated plastic plates. Refer to 260553 Electrical Identification Section of this specification.

- E. NEMA 3R switches shall be provided where exposed to weather. NEMA 3R switches shall have weatherproof threaded hubs for all conduit entries into the switch.
- F. Identify switches, as to equipment served, with engraved laminated plastic plates. Refer to 260553 Electrical Identification Section of this specification.

2.5 SWITCH RATINGS

- A. Fusible and non-fusible switches shall be horsepower rated for ac and/or dc as indicated on the plans.
- B. The UL Listed short circuit current rating for fusible switches shall be 200,000 rms symmetrical amperes minimum.
- C. All current carrying parts shall be plated to resist corrosion.
- D. Switches shall be Underwriters' approved for duty shown and enclosure type per drawings.

PART 3 EXECUTION

3.1 INSPECTION

- A. Installer shall examine the areas and conditions under which safety and disconnect switches are to be installed and notify the Contractor in writing of conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF SAFETY AND DISCONNECT SWITCHES

- A. Provide switches: where required by NEC; where indicated on drawings; and where required by equipment manufacturer; in a location convenient for maintenance on switch; and adjacent equipment being protected.
- B. Install switches in accordance with manufacturer's written instructions and the applicable requirements of the NEC, NEMA, ANSI, and the National Electrical Contractors Association's "Standard of Installation."
- C. Install fuses in fusible disconnect switches. Provide permanent marking inside switch enclosure for fuse type.
- D. Provide fused disconnect switches whether or not indicated on drawings and when required to maintain equipment manufacturer's warranty. Coordinate with Division 23 for warranty requirements of equipment approved by submittal.
- E. Install fuses in fusible disconnect switches. Provide permanent marking inside switch enclosure for fuse type.
- F. Wall mount switches, where possible, or mount on unistrut supports. Switches shall not be mounted on equipment without written permission of the Engineer of Record.
- G. For equipment with motors larger than 1/8 HP, install disconnect switches within sight of the motor.
- H. Install a UL approved equipment ground bar in each switch. The individual equipment ground bar shall accept at least two conductors. Individual conductor lugs are not acceptable.
- I. Install a UL approved neutral bar in each switch where a neutral is present.

END OF SECTION

SECTION 26 29 20

MOTORS, CONTROLLERS AND ELECTRIC POWERED EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish and install all wiring necessary to completely connect all motors, electric powered equipment and electric control equipment that is furnished by the Owner, other contractors, or other Divisions of Work. This includes HVAC equipment, plumbing equipment, elevators, hoists, signs, door operators, and similar items that are furnished and installed by others.
- B. The Owner, other contractors, or other Divisions of the Work will furnish locations of equipment and instructions and wiring diagrams necessary to select the materials required to install this equipment properly. Furnish and install all conduit, wire, boxes and common wiring materials to complete the installation and place the equipment in operation.

1.2 REFERENCE DOCUMENTS

- A. The Basic Electrical Requirements are hereby made a part of this section of the work. Refer to Section 26 00 00, and for Disconnects, refer to Section 26 28 16.
- B. Refer to other Divisions of the Drawings and Specifications for information as to the scope of this work. All notations for electrical work to be "By Electrical" or "See Electrical" shall be deemed instructions for work in Division 26.

PART 2 - PRODUCTS

2.1 FURNISHED EQUIPMENT

- A. The Owner or other contractors will furnish and deliver to the job site the motors, controllers, switches, and other controls for the equipment they furnish except as indicated otherwise.

2.2 START-STOP STATIONS

- A. Start-stop stations indicated remote from motor controllers shall, unless otherwise indicated, be flush mounted momentary contact push button and pilot units with stainless steel plate.
- B. Units shall be equal to Square D 9001-BF308.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Set in place all controllers, switches, and control devices. Furnish and install all supports, conduit, wire, boxes and common wiring materials, etc., as required. Furnish and install all interlocks and interconnecting wiring for equipment controls and safeties and make all other electrical connections for proper operation.

- B. Furnish and install a suitable disconnect switch for each motor and electric powered equipment which does not have such a disconnect as an integral part of the equipment or which is not within sight of a feeding branch circuit protective device which meets the requirements of a disconnect.
- C. Furnish and install remote start-stop stations. Verify the control voltage and specified hook-up and provide suitable unit to interface with starter and other controls.

3.2 EQUIPMENT CONNECTIONS

- A. The Contract Documents indicate electrical connection to equipment. Refer to the other trade construction requirements and make connections to and hook-up of the equipment and devices by furnishing all common wiring materials necessary to complete the wiring of and make ready for the electrical operation of the equipment.
- B. Locate outlet boxes, conduit stubs, and connections for the equipment and its related devices to suit the power, control and signal wiring of the equipment and devices.

3.3 LABELING

- A. Provide an engraved plastic identification nameplate on each motor controller and disconnect.

END OF SECTION

SECTION 26 29 23

VARIABLE FREQUENCY MOTOR CONTROLLERS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. For motors as scheduled on the drawings with variable frequency drives, provide variable frequency motor controllers as specified herein.
- B. Provide Variable Frequency motor Drive (VFD) consisting of a pulse width modulated (PWM) inverter designed for use on a standard NEMA Design B induction motor. Drive manufacturer shall have the following support organizational structure:
 - 1. Sales representative exclusively for HVAC products, with expertise in HVAC systems and controls.
 - 2. An independent service organization that is factory trained and certified to start, test and diagnose the VFD provided
- C. The drive manufacturer shall supply the drive and all necessary controls as herein specified. The manufacturer shall have been engaged in the production of this type of equipment for a minimum of twenty years.

1.2 RELATED REQUIREMENTS

- A. Section 23 09 23 Building Automation and Direct Digital Control

1.3 REFERENCE STANDARDS

- A. IEEE Standard 519-2014 - IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems
- B. UL 508C or UL 61800-5-1 - Standard for Power Conversion
- C. NEMA ICS 7.0-2006 - AC Adjustable Speed Drives
- D. NEMA ICS 7.1-2006 - Safety Standards for Construction and Guide for Selection, Installation and Operation of Adjustable Speed Drive Systems
- E. IEC 16800-5-1 Parts 1 and 2: Adjustable Speed Electrical Power Drive Systems – Part 5-1: Safety Requirements – Electrical, Thermal, and Energy.

1.4 SUBMITTALS

- A. Submittals shall include the following information:
 - 1. Outline dimensions, conduit entry locations and weight.
 - 2. Customer connection and power wiring diagrams.
 - 3. Complete technical product description - include a complete list of options provided.

4. Compliance to IEEE 519 - harmonic analysis for particular jobsite including total harmonic voltage distortion and total harmonic current distortion (TDD). For this harmonic analysis the Point of Common Coupling (PCC) is defined as the unit substation busses.
 - a. VFD manufacturer shall analyze and provide calculations, specific to this installation with his VFDs, indicating total harmonic voltage distortion is less than 3% with total current distortion less than 8%.
 - b. Electrical design engineer shall provide specific electrical distribution system data to VFD manufacturer necessary to perform harmonic calculations.
 - c. Input line filters shall be sized and provided as required by the VFD manufacturer to ensure compliance with IEEE Standard 519-2014.
 - d. Demonstrate compliance with IEEE Standard 519-2014 when supplied by the normal utility source, as well as when supplied by the alternate emergency source.

1.5 QUALIFICATIONS

- A. VFDs and options shall be UL listed as a complete assembly. VFD's that require the customer to supply external fuses for the VFD to be UL listed are not acceptable. The base VFD shall be UL listed for 100 KAIC without the need for input fuses.
- B. VDFs shall provide for harmonic mitigation associated with each individual drive in compliance to IEEE 519 as specified in section 1.4 Submittals of this specifications. VFDs submitted with any exceptions to provide for harmonic mitigation will not be accepted.

1.6 WARRANTY

- A. Warranty shall be 60 months from the date of substantial completion of the entire building construction. The warranty shall include all parts, labor, travel time and expenses. There shall be 365/24 support available via a toll free phone number.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Yaskawa (Basis of Design)
- B. ABB

2.2 ADJUSTABLE FREQUENCY DRIVES

- A. VFD package as specified herein shall be enclosed in a UL Listed Type 1 or 3R enclosure unless otherwise indicated, completely assembled and tested by the manufacturer in an ISO9001 facility.
- B. Provide VFDs with the following rectifier front ends:
 1. Total Motor HP of 0.5 HP to 10 HP: 6-Pulse drives.
 2. Total Motor HP of > 11 HP to 49 HP: 12-Pulse drives.
 3. Total Motor HP > 50 HP: 18-Pulse drives.
- C. The VFD tolerated voltage window shall allow the VFD to operate from a line of +30% nominal, -35% nominal voltage as a minimum.

- D. Environmental operating conditions: 0 to 40 degree C (32 to 104 degree F) continuous.
1. VFDs that can operate at 40C intermittently (during a 24 hour period) are not acceptable and must be oversized.
- E. Altitude: 0 to 3300 feet above sea level, less than 95% humidity, non-condensing.
- F. Provide VFDs with the following standard features:
1. The same customer interface, including digital display, and keypad, regardless of horsepower rating. The keypad shall be removable, capable of remote mounting and allow for uploading and downloading of parameter settings as an aid for start-up of multiple VFDs.
 2. Hand-Off-Auto selections and manual speed control. The drive shall incorporate "bumpless transfer" of speed reference when switching between "Hand" and "Auto" modes. There shall be fault reset and "Help" buttons on the keypad. The Help button shall include "on-line" assistance for programming and troubleshooting.
 3. Built-in time clock in the VFD keypad. The clock shall have a battery backup with 10 years minimum life span. The clock shall be used to date and time stamp faults and record operating parameters at the time of fault. If the battery fails, the VFD shall automatically revert to hours of operation since initial power up. The clock shall also be programmable to control start/stop functions, constant speeds, PID parameter sets and output relays. Include a digital input that allows an override to the time clock (when in the off mode) for a programmable time frame and provide four (4) separate, independent timer functions that have both weekday and weekend settings.
 4. VFD's shall utilize pre-programmed application macro's specifically designed to facilitate start-up. The Application Macros shall provide one command to reprogram all parameters and customer interfaces for a particular application to reduce programming time. The VFD shall have two user macros to allow the end-user to create and save custom settings.
 5. Cooling fans, designed for easy replacement. The fans shall be designed for replacement without requiring removing the VFD from the wall or removal of circuit boards. The VFD cooling fans shall operate only when required. To extend the fan and bearing operating life, operating temperature will be monitored and used to cycle the fans on and off as required.
 6. The VFD shall be capable of starting into a coasting load (forward or reverse) up to full speed and accelerate or decelerate to setpoint without safety tripping or component damage (flying start).
 7. Ability to automatically restart after an over-current, over-voltage, under-voltage, or loss of input signal protective trip. The number of restart attempts, trial time, and time between attempts shall be programmable.
 8. The overload rating of the drive shall be 110% of its normal duty current rating for 1 minute every 10 minutes, 130% overload for 2 seconds. The minimum FLA rating shall meet or exceed the values in the NEC/UL table 430-150 for 4-pole motors.
 9. Fixed impedance added in order to reduce the harmonics to the power line as well as add protection from AC line transients shall consist of either dual (positive and negative DC bus) DC bus reactors, or AC line reactors.
 - a. Reactors shall provide rated impedance from 100% full load down to 50% full load.
 - b. Reactors providing rated impedance only at 100% full load are not acceptable.
 - c. VFDs with only one DC bus reactor shall add an AC line reactor.
 10. Include a coordinated AC transient protection system consisting of 4-120 joule rated MOV's (phase to phase and phase to ground), a capacitor clamp, and 5% impedance reactors.

11. Provide capability of sensing a loss of load (broken belt / broken coupling) and signal the loss of load condition. The drive shall be programmable to signal this condition via a keypad warning, relay output and/or over the serial communications bus. Relay outputs shall include programmable time delays that will allow for drive acceleration from zero speed without signaling a false underload condition.
12. If the input reference (4-20mA or 2-10V) is lost, the VFD shall give the user the option of either (1) stopping and displaying a fault, (2) running at a programmable preset speed, (3) hold the VFD speed based on the last good reference received, or (4) cause a warning to be issued, as selected by the user. The drive shall be programmable to signal this condition via a keypad warning, relay output and/or over the serial communication bus.
13. Programmable "Sleep" and "Wake up" functions to allow the drive to be started and stopped from the level of a process feedback signal.
14. For motors indicated on drawings, provide an integral bypass contactor for operation of motor at constant speed, electrically independent of the inverter. Include motor overload protection when in bypass mode. Include necessary control relays and switches to allow automatic controls and safeties to operate when drive is in bypass mode.

G. VFDs shall have the following adjustments:

1. Three (3) programmable critical frequency lockout ranges to prevent the VFD from operating the load continuously at an unstable speed.
2. Two (2) PID Setpoint controllers shall be standard in the drive, allowing pressure or flow signals to be connected to the VFD, using the microprocessor in the VFD for the closed loop control. The VFD shall have 250 ma of 24 VDC auxiliary power and be capable of loop powering a transmitter supplied by others. The PID setpoint shall be variable from the VFD keypad, analog inputs, or over the communications bus. There shall be two parameter sets for the first PID that allow the sets to be switched via a digital input, serial communications or from the keypad for night setback, summer/winter setpoints, etc. There shall be an independent, second PID loop that can utilize the second analog input and modulate one of the analog outputs to maintain setpoint of an independent process (i.e. valves, dampers, etc.). All setpoints, process variables, etc. to be accessible from the serial communication network. The setpoints shall be set in Engineering units and not require a percentage of the transducer input.
3. Two (2) programmable analog inputs shall accept current or voltage signals.
4. Two (2) programmable analog outputs (0-20ma or 4-20 ma). The outputs may be programmed to output proportional to Frequency, Motor Speed, Output Voltage, Output Current, Motor Torque, Motor Power (kW), DC Bus voltage, Active Reference, and other data.
5. Six (6) programmable digital inputs for maximum flexibility in interfacing with external devices, typically programmed as follows:
 - a. There shall be a run permissive circuit for damper or valve control. Regardless of the source of a run command (keypad, input contact closure, time-clock control, or serial communications) the VFD shall provide a dry contact closure that will signal the damper to open (VFD motor does not operate). When the damper is fully open, a normally open dry contact (end-switch) shall close. The closed end-switch is wired to an VFD digital input and allows VFD motor operation. Two separate safety interlock inputs shall be provided. When either safety is opened, the motor shall be commanded to coast to stop, and the damper shall be commanded to close. The keypad shall display "start enable 1 (or 2) missing". The safety status shall also be transmitted over the serial communications bus. All digital inputs shall be programmable to initiate upon an application or removal of 24VDC.
6. Three (3) programmable digital Form-C relay outputs. The relays shall include programmable on and off delay times and adjustable hysteresis. Default settings shall be for run, not faulted (fail

safe), and run permissive. The relays shall be rated for maximum switching current 8 amps at 24 VDC and 0.4 A at 250 VAC; Maximum voltage 300 VDC and 250 VAC; continuous current rating 2 amps RMS. Outputs shall be true form C type contacts; open collector outputs are not acceptable.

7. Seven (7) programmable preset speeds.
 8. Two independently adjustable accel and decel ramps with 1 - 1800 seconds adjustable time ramps.
 9. Include a motor flux optimization circuit that will automatically reduce applied motor voltage to the motor to optimize energy consumption and audible motor noise.
 10. Include a carrier frequency control circuit that reduces the carrier frequency based on actual VFD temperature that allows the highest carrier frequency without derating the VFD or operating at high carrier frequency only at low speeds.
 11. Include password protection against parameter changes.
- H. Provide a backlit LCD display. The display shall be in complete English words for programming and fault diagnostics (alpha-numeric codes are not acceptable). The keypad shall utilize the following assistants:
1. Start-up assistants.
 2. Parameter assistants
 3. Maintenance assistant
 4. Troubleshooting assistant
- I. Applicable operating values shall be capable of being displayed in engineering (user) units. A minimum of three operating values from the list below shall be capable of being displayed at all times. The display shall be in complete English words (alpha-numeric codes are not acceptable):
1. Output Frequency
 2. Motor Speed (RPM, %, or Engineering units)
 3. Motor Current
 4. Calculated Motor Torque
 5. Calculated Motor Power (kW)
 6. DC Bus Voltage
 7. Output Voltage
- J. The VFD shall include a fireman's override input. Upon receipt of a contact closure from the fireman's control station, the VFD shall operate at an adjustable preset speed. The mode shall override all other inputs (analog/digital, serial communication, and all keypad commands) and force the motor to run at the adjustable, preset speed. "Override Mode" shall be displayed on the keypad. Upon removal of the override signal, the VFD shall resume normal operation.
- K. Serial Communications
1. The VFD shall have an RS-485 port as standard. The standard protocols shall be BACnet, Modbus, Johnson Controls N2 bus, and Siemens Building Technologies FLN. Optional protocols for LonWorks, Profibus, Ethernet, and DeviceNet shall be available. Each individual drive shall have the protocol in the base VFD. The use of third party gateways and multiplexers is not acceptable. All protocols shall be "certified" by the governing authority. Use of non-certified protocols is not allowed.

2. Serial communication capabilities shall include, but not be limited to; run-stop control, speed set adjustment, proportional/integral/derivative PID control adjustments, current limit, accel/decel time adjustments, and lock and unlock the keypad. The drive shall have the capability of allowing the DDC to monitor feedback such as process variable feedback, output speed / frequency, current (in amps), % torque, power (kW), kilowatt hours (resettable), operating hours (resettable), and drive temperature. The DDC shall also be capable of monitoring the VFD relay output status, digital input status, and all analog input and analog output values. All diagnostic warning and fault information shall be transmitted over the serial communications bus. Remote VFD fault reset shall be possible. The following additional status indications and settings shall be transmitted over the serial communications bus - keypad "Hand" or "Auto" selected, bypass selected, the ability to change the PID setpoint, and the ability to force the unit to bypass (if bypass is specified). The DDC system shall also be able to monitor if the motor is running in the VFD mode or bypass mode (if bypass is specified) over serial communications. A minimum of 15 field parameters shall be capable of being monitored.
 3. The VFD shall allow the DDC to control the drive's digital and analog outputs via the serial interface. This control shall be independent of any VFD function. For example, the analog outputs may be used for modulating chilled water valves or cooling tower bypass valves. The drive's digital (relay) outputs may be used to actuate a damper, open a valve or control any other device that requires a maintained contact for operation. In addition, all of the drive's digital and analog inputs shall be capable of being monitored by the DDC system.
 4. The VFD shall include an independent PID loop for customer use. The independent PID loop may be used for cooling tower bypass valve control, chilled water valve control, etc. Both the VFD control PID loop and the independent PID loop shall continue functioning even if the serial communications connection is lost. The VFD shall keep the last good set-point command and last good DO & AO commands in memory in the event the serial communications connection is lost.
- L. Provide integral EMI/RFI filters for all VFDs. The onboard filters shall allow the VFD assemble to be CE Marked and the VFD shall meet product standard EN 61800-3 for the First Environment restricted level.
- M. All VFD's through 50HP shall be protected from input and output power mis-wiring. The VFD shall sense this condition and display an alarm on the keypad.
- N. Features specified herein and noted on drawings shall be furnished and mounted by the drive manufacturer. All Features specified shall be UL Listed by the drive manufacturer as a complete assembly and carry a UL508 label.
- O. Provide factory wired and tested bypass system consisting of an output contactor and bypass contactor. Overload protection and shall be provided in both drive and bypass modes.
- P. Door inter-locked, padlock capable circuit breaker that will disconnect all input power from the drive and all internally mounted options.
- Q. Fused VFD only disconnect (service switch). Fast acting fuses exclusive to the VFD - fast acting fuses allow the VFD to disconnect from the line prior to clearing upstream branch circuit protection, maintaining bypass capability.
1. Bypass designs, which have no such fuses, or that incorporate fuses common to both the VFD and the bypass will not be accepted.
 2. Three contactor bypass schemes and designs will be not accepted.
- R. Provide single-phase motor protection.
1. Single phase motor protection shall protection drive in bypass mode.

- S. The following operators shall be provided:
1. Bypass Hand-Off-Auto
 2. Drive mode selector
 3. Bypass mode selector
 4. Bypass fault reset
- T. The following indicating lights (LED type) shall be provided. A test mode or push to test feature shall be provided.
1. Power-on (Ready)
 2. Run enable (safeties) open
 3. Drive mode select damper opening
 4. Bypass mode selected
 5. Drive running
 6. Bypass running
 7. Drive fault
 8. Bypass fault
 9. Bypass H-O-A mode
 10. Automatic transfer to bypass selected
 11. Safety open
 12. Damper opening
 13. Damper end-switch made
- U. The following relay (form C) outputs from the bypass shall be provided:
1. System started
 2. System running
 3. Bypass override enabled
 4. Drive fault
 5. Bypass fault (motor overload or underload (broken belt))
 6. Bypass H-O-A position
- V. The digital inputs for the system shall accept 24V or 115VAC (selectable). The bypass shall incorporate internally sourced power supply and not require an external control power source.
- W. Customer Interlock Terminal Strip - provide a separate terminal strip for connection of freeze, fire, smoke contacts, and external start command. All external safety interlocks shall remain fully functional whether the system is in Hand, Auto, or Bypass modes (not functional in Fireman's Override 2). The remote start/stop contact shall operate in VFD and bypass modes.

- X. Dedicated digital input that will transfer motor from VFD mode to bypass mode upon dry contact closure for fireman's override. Two modes of operation are required.
 - 1. One mode forces the motor to bypass operation and overrides both the VFD and bypass H-O-A switches and forces the motor to operate across the line (test mode). The system will only respond to the digital inputs and motor protections.
 - 2. The second fireman's override mode remains as above, but will also defeat the overload and single-phase protection for bypass and ignore all keypad and digital inputs to the system (run until destruction).
- Y. The VFD shall include a "run permissive circuit" that will provide a normally open contact whenever a run command is provided (local or remote start command in VFD or bypass mode). The VFD system (VFD or bypass) shall not operate the motor until it receives a dry contact closure from a damper or valve end-switch. When the VFD system safety interlock (fire detector, freezestat, high static pressure switch, etc.) opens, the motor shall coast to a stop and the run permissive contact shall open, closing the damper or valve.
- Z. Class 20 or 30 (selectable) electronic motor overload protection shall be included. There shall be an internal switch to select manual or automatic bypass.
- AA. There shall be an adjustable current sensing circuit for the bypass to provide loss of load indication (broken belt) when in the bypass mode.

2.3 SOFT STARTERS

- A. Provide "soft-starters" for mechanical equipment as indicated on the plans.
 - 1. ABB PSTX Series.
 - 2. Approved equal.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install drives under this Division in accordance with the recommendations of the VFD manufacturer as outlined in the installation manual.
- B. Provide power wiring under Division 26. Wire and terminate connections in accordance with the recommendations of the VFD manufacturer as outlined in the installation manual.
- C. Coordinate control interface and wiring with Building Automation supplier.

3.2 START-UP

- A. Certified factory start-up shall be provided for each drive by a factory authorized service center. A certified start-up form shall be filled out for each drive with a copy provided to the owner, and a copy kept on file at the manufacturer.

3.3 PRODUCT SUPPORT

- A. Factory trained application engineering and service personnel that are thoroughly familiar with the VFD products offered shall be locally available at both the specifying and installation locations. A 24/365 technical support line shall be available on a toll-free line.

3.4 **TRAINING**

- A. Training of the Owner's operation and maintenance personnel is required in cooperation with the Owner and the Construction Management Team. Provide competent, factory-authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. The instruction shall be scheduled with the Owner after submission and approval of formal training agendas.

END OF SECTION

SECTION 26 41 13

LIGHTNING PROTECTION SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

- A. Furnish and install a complete lightning protection system for each new structure of the facility and connect to comply with all requirements of the NFPA.
- B. The system shall be as described herein and shall be installed by or under the supervision of an accredited lightning protection subcontractor.
- C. Interconnect the system with the existing adjacent building system in such a manner that both structures can be labeled.

1.2 REFERENCE DOCUMENTS

- A. The Basic Electrical Requirements are hereby made a part of this Section of the Work. Refer to Section 26 0000.
- B. All materials and installation shall be in accord with the applicable requirements of latest NFPA-780 and the Lightning Protection Institute for a Class II installation.

1.3 REQUIREMENTS OF DESIGN

- A. Equipment and accessories shall be the standard products of a single manufacturer and shall be equal in all respects to those manufactured by Thompson Lightning Protection, Inc. of St. Paul, Minnesota. Catalog numbers and model designations indicate design, quality and type of material as well as required operating characteristics.
- B. All materials, equipment and installations shall meet the requirements of Underwriters' Laboratories, Inc.
- C. The completed installation shall be inspected by the Underwriters' Laboratories, Inc. and their Master Label issued and attached to each completed building. All arrangements and costs for inspection, issuance of label and installation of the label is a part of this section of the work.

1.4 SUBMITTALS

- A. Submit complete manufacturers' data on each component of the lightning protection system.
- B. Provide a layout of the points, conductance and earthing systems and interconnection indicating where each of the submitted components is to be installed. Show locations and techniques for connecting to existing CUH Building lightning protection system.
- C. Submit the design and method of connection of each metallic device to the system.
- D. Submit the qualifications of the installer to provide an installation that will have a UL Master Label. This includes re-labelling of the existing and new lightning protection systems as one system.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Main conductors shall be stranded copper cables weighing not less than 375 pounds per 1,000 feet with no conductor strand smaller than No. 15 AWG.
- B. Roof penetration units shall be brass threaded rods of greater diameter than the down leads.
- C. Air terminals shall be solid copper with nickel plated tapered point, not less than 12" height and not less than 1/2" in diameter.
- D. Bases on flat roofs shall be cast bronze adhesive type equal to Thompson No. 688. Bases on parapets shall be cast bronze bolt-on type equal to Thompson No. 678. Bases on sloping walls or decks shall be cast bronze hinged type bolt-on equal to Thompson No. 30. Each shall be complete with all necessary anchoring bolts and accessories.
- E. Ground shall be main size conductor cable.
- F. Rods for main conductors passing through structural concrete, piers, grade beams, from piers into columns and through waterproofing membranes shall be 1/2" diameter copper bars.
- G. Bonding wires shall be 14 strand No. 17 AWG copper wire.
- H. Cable connectors for connecting cables in accessible locations shall be bronze bolted mechanical clamps for "end-to-end", "tee", "parallel" or "Y" splices. Connectors for bonding metals such as roof mechanical equipment shall be bronze bolted lugs with plated bolted surfaces where the lugs connect to dissimilar metals. Connectors for conductors in inaccessible locations including underground and behind walls shall be thermionic type as furnished by Cadweld or Burndy Thermoweld.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Mount air terminals on the roof deck and parapets at a maximum 20'-0" on centers around the edge of the roof. Install air terminal on all areas of the flat roof such that the terminals will not be more than 50 feet on centers each way over the roof. Interconnect all air terminal with a roof conductor in a closed loop. Course roof conductors over the roof structure, the roof insulation and the roof waterproofing membrane and under the ballast. Install down conductors on the perimeter of the building at locations that are less than every one hundred feet (100') of building perimeter and connect to roof conductors. Route all down conductors concealed in piers. Terminate down conductors to counterpoise conductor embedded in concrete below foundation walls. Coordinate with Structural. From the wall outlet, make a straight run with a main size conductor cable to the perimeter grounding cable and mechanically connect. Maintain a horizontal course for all roof conductors and vertical course for all down conductors that is free from sharp bends and "U" pockets.
- B. Install grounding of a perimeter counterpoise and under building ground system. Run a main conductor size cable under the base of the perimeter wall and across underneath the building. This cable shall be run in the granular fill and encased in concrete such that the conductor is not in direct contact with the earth or granular fill.
- C. Provide 3/4" diameter copper rods at each location where the main down lead counterpoise or cross conductor must pass through a pier, pass from a pier to a column or ran through a waterproofing membrane. Connections between these rods and between the rods and cables shall be carefully joined by the exothermic reaction process using Cadweld or Burndy Thermoweld process.

- D. Make all cable joints mechanically strong and secure providing a contact area not less than twice the conducting cross sectional area of the conductor. Connections in inaccessible locations shall be Cadweld or Burndy Thermoweld. Others may be bolted pressure connectors.
- E. Connect new lightning protection system to the existing tower lightning protection system. The entire LPS (new and existing) shall be UL certified as a single lightning protection system.

3.2 AUXILIARY DEVICE CONNECTIONS

- A. Connect every metallic mass including exhaust fans, metal ventilators, roof drains, and gravel guards exposed on the roof to the main conductor system with bonding wire. Bond each plumbing vent pipe projecting through roof, all metal masts, and any metallic masses having a low resistance to the ground to the main conductor system.
- B. All interior metal bodies that at any point are within six feet of the main conductor shall be connected to that system.
- C. Use proper fittings on all connections to guard against the corrosive effect of dissimilar metals.

3.3 CERTIFICATION

- A. Upon completion of the installation, the lightning protection system shall be inspected by a representative of the Underwriters' Laboratories, Inc.
- B. Submit a copy of a letter of acceptance as a part of the Records for Owner as indication of a satisfactory installation
- C. Install the Underwriters' Laboratories Inc. approved Master Label on the building

END OF SECTION

SECTION 26 51 00

LIGHTING FIXTURES

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Interior lighting fixtures.
- B. Exterior lighting fixtures.

1.2 REFERENCES

- A. NEPA 101 - Code for Safety to Life from Fire in Buildings and Structures.
- B. NEMA WD1 - General-Purpose Wiring Devices.
- C. ANSI C78.377 LED Standards.

1.3 SUBMITTALS

- A. Submit manufacturer's data on interior and exterior lighting fixtures in booklet form, with separate sheet for each fixture, assembled by luminaire "type" in alphabetical order, with the proposed fixture and accessories clearly labeled. Ballast and lamp product data shall accompany fixture submittals.
- B. Submit dimensioned drawings and performance data including coefficients of utilization, candela distribution, spacing to mounting height ration, efficiency and visual comfort probability.

1.4 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Deliver lighting fixtures individually wrapped in factory-fabricated fiberboard type containers. Parabolic louvers shall be shipped in thermally sealed polyethylene wrapper.
- B. Handle lighting fixtures carefully to prevent breakage, denting and scoring the fixture finish. Do not install damaged lighting fixtures.
- C. Store lighting fixtures in a clean, dry space and protected from the weather.

1.5 WARRANTY

- A. Provide five year manufacturer warranty for all LED luminaires, including drivers.
- B. Provide five year manufacturer warranty for all fluorescent ballasts.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Lighting fixtures and accessories shall comply with the design and function requirements of the project. Design characteristics shall be as noted in manufacturer's submittal data.

- B. Provide lighting fixtures of the size, type and rating as scheduled.

2.2 INTERIOR LIGHTING FIXTURES

- A. LED Fixtures:
- B. Lenses shall be 0.125 inch thick virgin acrylic. Provide a minimum of 8 hold-down lens retaining clips for troffers utilizing framed diffuser lenses.
- C. Frames shall be flush, steel hinged and equipped with rotary-action cam latches.

2.3 EXTERIOR LIGHTING FIXTURES

- A. Enclosures shall be complete with gaskets to form weatherproof seal and UL approved for wet locations.
- B. Provide low temperature ballasts or driver with reliable starting to 0 degrees F.

2.4 LAMPS

- A. LED for all light fixtures in new building construction as indicated on the drawings.
 - 1. Include estimated useful life, calculated based on IES LM-80 test data.
 - 2. Include IES LM-79 test report.
- B. Maintenance Stock: Furnish a stock of replacement lamps in the original cartons or packing sleeves, amounting to 10% (but not less than two lamps in each case) of each type and size lamp used in each fixture type. Deliver replacement stock as directed to Owner's storage space.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Installer shall examine the areas and conditions which light fixtures are to be installed and notify the Contractor in writing of conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF LIGHTING FIXTURES

- A. Install light fixtures in accordance with the manufacturer's written instructions, the applicable requirements of NEC and the National Electrical Contractors Association's "Standard of Installation".
- B. If a fixture type designation is omitted, furnish fixture of the same type as shown for rooms of similar usage. Verify with Construction Inspector before purchase and installation.
- C. Check the building electrical system requirements and architectural finishes, and regardless of the catalog number prefixes and suffixes shown, furnish fixtures with the proper trim, frames, supports, hangers, ballasts, voltage rating, and other miscellaneous appurtenances to properly coordinate with said conditions. Verify with Construction Inspector prior to ordering.
- D. Check the type of ceilings to be installed in each room and verify that the recessed light fixtures are proper for the type of ceiling to be installed before ordering fixtures.

- E. Fasten fixtures securely to structural support members of the building; solid pendant fixtures shall be plumb.
- F. Immediately before final observation, clean all fixtures, inside and out, including plastics and glassware, and adjust all trim to properly fit adjacent surface, replace broken or damaged parts, and lamp and test all fixtures for electrical as well as mechanical operation.
- G. Protect installed fixtures from damage during the remainder of the construction period.
- H. Upon completion of installation of interior lighting fixtures, and after circuitry has been energized; apply electrical energy to demonstrate capability and compliance with requirements. When possible, correct malfunctioning units at the site, then retest to demonstrate compliance; otherwise, remove and replace with new units, and proceed with retesting.

3.3 LIGHT FIXTURE SCHEDULE

- A. Refer to Drawings for Light Fixture Schedule.

END OF SECTION

SECTION 27 05 00

COMMON WORK RESULTS FOR COMMUNICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes

1. Basic materials and methods, along with General Requirements of Division 1, applicable to Division 27 sections.
2. Drawings and general provisions of the contract, including General and Supplementary Conditions of Division 1, applicable to Division 27 sections.

B. Related Sections

1. Section 27 05 28 – Communications Pathways
2. Section 27 08 00 – Communications Testing
3. Section 27 11 00 – Communications Rooms
4. Section 27 13 13 – Communications Backbone Copper Cabling
5. Section 27 13 23 – Communications Backbone Fiber Cabling
6. Section 27 15 13 – Communications Horizontal Copper Cabling
7. Section 27 15 33 – Communications Horizontal Coaxial Cabling
8. Section 27 51 13 – Overhead Paging System
9. Section 27 51 23 – Intercommunications System

1.2 CONTRACTOR REQUIREMENTS AND QUALITY ASSURANCE

- A. Comply with applicable local, state and federal codes.
- B. Comply with applicable requirements of recognized industry associations which produce standards for the various trades.
- C. Warrant work under this specification against faulty material or Workmanship in accordance with Division 1. If the project is occupied or the systems placed in operation in several phases at the request of Owner's Representative, then the warranty of each system or piece of equipment used shall begin on the date of substantial completion for each phase. The use of building equipment for temporary service and testing does not constitute the beginning of the warranty.
- D. Equipment and material provided under this Division shall be periodically inspected and serviced by competent installers. This function becomes the responsibility of the Owner once the system is accepted by the Owner. The one year material and Workmanship warranty is not intended to supersede normal inspection or service and shall not be construed to mean the Contractor shprovide free service for normal maintenance items such as periodic cleaning and adjustment due to normal use, nor to correct without charge, breakage, maladjustment, and other trouble caused by improper maintenance.
- E. Upon completion of contract and progressively as work proceeds, clean-up and remove dirt, debris and scrap materials. Maintain the premises in a neat and clean condition at all times during construction. Protect and

preserve access to Head-end equipment at all times. Clean items with factory finishes. Touch-up minor damage to surfaces; refinish entire piece of equipment when sustained major damage. All electronics must be protected from dust and other airborne debris.

1.3 STANDARDS

- A. The Contractor's performance of the work shall comply with applicable federal, state and local laws, rules and regulations. The Contractor shall give required notices, shall procure necessary governmental licenses, permits, and inspections and shall pay without burden to Owner's Representative, all fees and charges in connection therewith unless specifically provided otherwise. In the event of violation, the Contractor shall pay all fines and penalties, including attorney's fees and other defense costs and expenses in connection therewith.
- B. Federal Communications Commission
 - 1. Equipment requiring FCC registration or approval shall have received such approval and shall be appropriately identified.
- C. Codes, Standards and Ordinances: Design, manufacture, test, and install telecommunications cabling networks per manufacturer's requirements and in accordance with NFPA-70 (National Electrical Code), state codes, local codes, requirements of authorities having jurisdiction, and particularly the following standards:
 - 1. ANSI/TIA Standards:
 - a. ANSI/TIA-526-7: Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant
 - b. ANSI/TIA-568-C.0: Generic Telecommunications Cabling for Customer Premises
 - c. ANSI/TIA-568-C.1: Commercial Building Telecommunications Cabling Standards
 - d. ANSI/TIA-568-C.2: Balanced Twisted-Pair Telecommunications Cabling and Components Standard
 - e. ANSI/TIA-568-C.3: Optical Fiber Cabling Components Standard
 - f. ANSI/TIA-569-D: Telecommunications Pathways and Spaces
 - g. ANSI/TIA-606-B: Administration Standard for Telecommunications Infrastructure
 - h. ANSI/TIA-607-C: Generic Telecommunications Bonding and Grounding
 - i. ANSI/TIA-758: Customer-Owned Outside Plant Telecommunications Infrastructure Standard
 - j. ANSI/TIA-1179: Healthcare Facilities Telecommunications Infrastructure Standard
 - k. ANSI/TIA-TSB67: Transmission Performance Specifications for Field Testing of Unshielded Twisted Pair Cabling Systems
 - 2. Install cabling in accordance with the most recent edition of BICSI publications:
 - a. BICSI – Telecommunications Distribution Methods Manual
 - b. BICSI – Cabling Installation Manual
 - c. BICSI – LAN Design Manual
 - d. BICSI – Customer-Owned Outside Plant Design Manual

3. UT Southwestern Medical Center

a. Information Resources Cable Installation Standard – August 2, 2017

4. Federal, state, and local codes, rules, regulations, and ordinances governing the work, are as fully part of the specifications as if herein repeated or hereto attached. If the contractor should note items in the drawings or the specifications, construction of which would be code violations, promptly call them to the attention of Owner's Representative in writing. Where the requirements of other sections of the specifications are more stringent than applicable codes, rules, regulations, and ordinances, the specifications shall apply.

a. American Society for Testing and Materials (ASTM): ASTM E. 814 - Fire Tests of Through - Penetration Firestops.

b. Underwriters Laboratories, Inc. (UL): U.L. 1479 - Fire Tests of Through - Penetrations Firestops.

c. National Fire Protection Association (NFPA): NFPA 70 - National Electrical Code.

d. Americans with Disabilities Accessibility Guidelines.

1.4 COMPLETENESS OF WORK

A. The Contract Documents depict low voltage systems which are intended to be complete and functioning systems. All products, materials, labor, and programming necessary to render a fully functional system to fulfill the design intent shown on the documents shall be provided by the contractor.

B. Catalog numbers referenced throughout this Division's drawings and specifications are intended to convey a general understanding of the type of quality of the product required. Where written descriptions differ from information conveyed by a catalog number, the written description shall govern. No extra charge shall be allowed because a catalog number is found to be incomplete or obsolete.

1.5 SUBMITTALS

A. Comply with provisions of Division 1.

B. Submit shop drawings as called for in the Specification Sections that follow.

C. Within thirty (30) days after contract has been awarded, Contractor to submit to Designer for review a complete list of materials, equipment, and accessories proposed for use, listing the item and manufacturer's name only.

D. AutoCAD or Revit files requested by Contractor for use in submittals, coordination drawings, as- builds, etc., will be subject to executing a release document from the Engineer prior to release of electronic media.

E. Based upon aforementioned approved listing, Contractor to submit four (4) copies of complete brochures and shop drawings of all materials, fixtures, and equipment that he proposes to use giving the names of manufacturers, trade name and specific catalog numbers.

F. Product Data Brochures

1. Brochures to be submitted in time to allow fifteen (15) days from date of receipt in the consultant's office before final approval or disapproval is required to meet construction schedule. Submittal to bear Contractor's stamp of approval evidencing they have been examined and checked same and information contained therein is in accordance with contract requirements. Deviations to be clearly marked. "No Exception taken" response to shop drawings is not to be construed as permitting departure from the contractual documents.

2. Above-mentioned brochures to be submitted and approved before any materials are ordered.

3. Proposed items to be clearly indicated when other items are shown on same sheet. When proposing items other than those specified, brochures to contain both specified item sheets and proposed item sheets for ease of comparison. On request from Designer, samples shall be submitted and/or set up, as directed, for inspection and approval. Samples will be returned to Contractor.

G. Shop Drawings

1. Point-to-point wiring diagrams for cables to be installed.
2. Detailed plan views of telecommunications rooms showing equipment racks, cable supports, and termination hardware.
3. Shop drawings shall be numbered in accordance with contract drawing numbers and appended to indicate the system covered. Provide a legend of drawings on the cover sheet.
4. Shop drawing sheets shall contain the project name, project location, project number, submittal number (and letter) and name and address of vendor contact responsible for answering questions related to the submittal.
5. Shop drawings shall be produced in AutoCAD or Revit on architectural backgrounds provided by the project design team.
6. Shop drawings shall include equipment and device layers as shown on the contract drawings for the system submitted and all related equipment and devices (example: television system shop drawings should show related nurse call stations and vice-versa). All unnecessary layers should be turned off.
7. Proposed system layout on project floor plan.
8. All applicable riser diagrams.
9. Rack layouts and wall elevations.
10. Legend indicating all symbols used in shop drawings correlating to product data booklet.
11. Shop drawing symbols shall be identical to contract drawings. If different symbols are necessary, provide table showing which shop drawing symbol equates to each contract drawing symbol.
12. Proposed cable routing indicated on floor plan.
13. Proposed cable numbering indicated on the floor plans.
14. Clearly indicate all power, conduit and sleeve requirements.
15. Interior elevations of proposed equipment located in all communications rooms.
16. Identify equipment dimensions and their distance from adjacent walls, floor, ceilings, door frames, counters, back-splash, etc. Clearly denote electrical requirements on interior elevations.
17. Provide equipment rack layouts and installation details.
18. Indicate cable management hardware and cable routing within all communications rooms.
19. Provide all back-box requirements specifically noting any deviations from the back boxes shown on the telecommunications drawings.

H. Assurance/Quality Control Submittals

1. Proposed test forms for fiber riser, copper riser and horizontal UTP cable.

2. Documentation of manufacturer's qualification of contractor as an approved Value-Added Reseller (VAR) or installer of the proposed cabling solution.
3. Certificate of insurance. Contractor cannot begin installation of the system or be paid for any material or labor until this document is provided.
4. Contractor shall submit faceplate sample to General Contractor for coordination with electrical device cover plates for color and texture. Approval must be obtained before purchasing and installing face plates.
5. Documentation to provide certification for a current BICSI RCDD to manage all installations and testing procedures. The BICSI representative must be an employee of the Contractor.
6. Record drawings shall be kept on site. Record drawings shall include marked up floor plans showing outlet locations, type of cable, and cable label identification.

1.6 DELIVERY AND STORAGE

- A. Insofar as possible, deliver items in manufacturers' original unopened packaging. Where this is not practical, cover items with protective materials, to keep them from being damaged. Use care in loading, transporting, unloading, and storage to keep items from being damaged.
- B. Store items in a clean dry place and protect from damage.

1.7 OPERATING AND MAINTENANCE MANUALS

- A. Prior to final acceptance of the project, furnish to Owner complete bound sets of operation and maintenance manuals of instructions for operation and maintenance of all pieces of equipment and systems provided under this division of specifications.
- B. Manuals to also include all submittal data on all materials and equipment. Clearly indicate items provided on this project. Included a list giving name and address of nearest supply house which carry the spare parts and name and address and phone number of Installation Contractor to be given to Owner.
- C. The supplier or manufacturer shall provide thorough training of all staff assigned to those areas receiving new systems and equipment. This training shall be developed and implemented to address two different types of staff. The end users will be trained on the features and functions of the systems, and the engineering or maintenance department will be trained on the routine maintenance and basic programming of the systems. All training will take place on-site using the fully installed and certified system. Training shall be videotaped. Two copies of each type of training class shall be given to the owner.
- D. Three sets of the following data are required:
 1. Operating and maintenance instructions.
 2. Spare parts lists.
 3. Copies of approved submittal data.
 4. Warranty information including any required test results.
- E. Arrange each set of data in an orderly way, and bind each set in a separate 3-ring, hard-cover binder.
- F. As soon as data accumulates, prepare one of the sets and deliver to the Owner's Representative, continuously updating this set as additional data is obtained.
- G. At completion of work, submit two complete sets of data to the Owner's Representative for distribution to the proper parties.

1.8 RECORD DRAWINGS

- A. Record locations and depths of buried and concealed conduits from fixed, easily identifiable objects, such as building walls. Where conduits are concealed in walls, indicate distances off of building corners or other building features not likely to be disturbed by future alterations.
- B. Mark deviations in colored pencils so that work of various systems can be easily identified.
- C. When work is completed, record all deviations in an electronic format using AutoCAD or Revit in a format usable to Owner.
- D. Submit four copies of completed "record drawings" on compact disc to Owner's Representative for distribution.

1.9 WARRANTY

- A. The Contractor shall adhere to the warranty requirement for all installations.
- B. Warrant work under this specification against faulty material or Workmanship in accordance with Division 1. If the project is occupied or the systems placed in operation in several phases at the request of the Owner, then the warranty of each system or piece of equipment used shall begin on the date of substantial completion for each phase. The use of building equipment for temporary service and testing does not constitute the beginning of the warranty.
- C. The Contractor shall install all copper components of the installed manufacturer's system-wide solution to the specifications and requirements needed to extend the longest and most extensive performance warranty available under the installed solution(s). Data cabling shall adhere to the warranty requirements of the manufacture. Universal cabling shall adhere to the above data cabling requirements.
- D. The Contractor shall install all fiber components of the installed manufacturer's system-wide to the specifications and requirements needed to extend the longest and most extensive performance warranty available under the installed solution(s).
- E. The Contractor shall submit in the bid documents any additional, contractor-specific warranties or guarantees to be offered on the project.
- F. The Contractor shall provide any and all necessary documentation needed to implement this warranty and to verify the solution installation.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. All materials and equipment used in carrying out these specifications are to be new and have UL listing or listing by other recognized testing laboratory when such listings are available. Specifications and drawings indicate name, type and catalog numbers of materials and equipment to be used as "standards" shall not be construed as limiting competition. Unless otherwise indicated, Contractor may, at his option, use materials and equipment when, in the judgment of the Owner's Representative, they are equivalent to that specified. Prior approval of substitutions must be given prior to submittals.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Insofar as it is possible to determine in advance, advise the general contractor to leave proper chases and openings. Place all outlets, anchors, sleeves, and supports prior to pouring concrete or installation of masonry work. Should contractor neglect doing this, any cutting and/or patching required is to be done at this contractor's expense. Visit site and be informed of conditions under which work must be performed. No subsequent allowance will be made because of error or failure to obtain necessary information to completely estimate and perform work involved.

- B. Carefully coordinate with other divisions to ensure proper power requirements, grounding, fireproofing and interlocking controls between the fire alarm system, nurse call system, and owner furnished systems.
- C. Notify other tradesmen of any deviations or special conditions necessary for the installation of work. Interferences between work of various contractors to be resolved prior to installation. Work installed not in compliance with specifications and drawings and without properly checking and coordinating as specified above shall, if necessary, be removed and properly reinstalled without additional cost to Owner.
- D. Intent:
 - 1. These sections of specifications and drawings form a complete set of documents for the medical communication systems for this project. Neither is complete without the other. Any item mentioned in one shall be as binding as though mentioned in both.
 - 2. The intent of these specifications and drawings is to form a guide for a complete systems installation. Where an item is reasonably necessary for a complete system but not specifically mentioned, such as pull boxes, fittings, expansion fittings, support hangers, etc. provide same without additional cost to Owner.
 - 3. Device layouts indicated on drawings are diagrammatical only. Exact location of outlets and equipment to be coordinated and governed by project conditions. The Designer reserves the right to make any reasonable changes (approximately 6 feet) in location of junction boxes, or equipment prior to roughing in of such without additional cost to Owner.
- E. Deviations:
 - 1. No deviations from specifications and drawings to be made without full knowledge and consent of Designer.
 - 2. Should Contractor find during progress of work that existing conditions make desirable a modification of the requirements of any particular item, report such item promptly to Designer for his decision and instructions.

3.2 PLYWOOD BACKBOARD AND WALL BACKING

- A. Under other Divisions, provide 4' W x 8' H x 3/4" D fire retardant plywood backboard as indicated in all Communication Rooms. Plywood is to be installed 6" above finished floor and painted with two coats of white fire retardant paint.
- B. Under other Divisions, provide appropriate backing in walls as required for mounting brackets and other wall mounted equipment per manufacturer requirements.

3.3 FIRESTOPPING

- A. Select appropriate type or types of through penetration firestop devices or systems appropriate for each type of communications penetration and base each selection on criteria specified herein.
- B. Selected systems shall not be less than the hourly time delay ratings indicated in the Contract Documents for each respective fire-rated floor, wall, or other partition of building construction. Firestop for each type of communications penetration shall conform to requirements of an independent testing laboratory design drawing or manufacturer's approved modification when used in conjunction with details shown on the Drawings.
- C. Perform all necessary coordination with trades constructing floors, walls, or other partitions of building construction with respect to size and shape of each opening to be constructed and device or system approved for use in each instance.
- D. Coordinate each firestop selection with adjacent Work for dimensional or other interference and for feasibility. In areas accessible to public and other "finished" areas, firestop systems Work shall be selected, installed, and finished to the quality of adjacent surfaces of building construction being penetrated.

- E. Use materials that have no irritating or objectionable odors when firestopping is required in existing buildings and areas that are occupied.
- F. Provide damming materials, plates, wires, restricting collars, and devices necessary for proper installation of firestopping. Remove combustible installation aids after firestopping material has cured.
- G. All firestops shall be installed in accordance with the manufacturer's instructions in order to maintain the specific rating assigned by the independent testing laboratory.
- H. If required by inspecting authorities:
 - 1. Expose and remove firestopping to the extent directed by inspecting authority to permit his or her inspection.
 - 2. Reinstall new firestopping and restore Work where removed for inspection.

3.4 TESTS

- A. On completion of work, installation to be entirely free of damaged conductors, software errors, incomplete jack termination including labeling and faceplates and dust. Perform a thorough operation test in the presence of the Owner's Representative. Provide documentation of all test results as outlined in each system's specifications. Include labor, materials and instruments for above tests.
- B. Furnish Owner's Representative, as a part of closing documents, a copy of such tests including identification of each cable, also the dedicated communication service ground test as required by each system's individual manufacturer indicating compliance with their requirements.
- C. Prior to final observation and acceptance, test and leave in satisfactory operating condition, all systems and equipment including but not limited to the following:
 - 1. Grounding.
 - 2. Firestopping of all sleeves and conduits.
 - 3. Telephone and LAN systems.
 - 4. CATV/MATV distribution systems.

3.5 INSPECTION FEES AND PERMITS

- A. Obtain and pay for all necessary permits and inspection fees required for communication systems installation.

END OF SECTION

SECTION 27 05 28

COMMUNICATIONS PATHWAYS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes

1. Hangers, supports, anchors, and fasteners to support the communications and low voltage cabling system.

B. Related Sections

1. Section 07 84 00 – General Firestopping
2. Section 26 05 29 – Supporting Devices
3. Section 26 05 33 – Conduits and Boxes

1.2 SUBMITTALS

A. Comply with the submittal procedures of Section 27 05 00 – Common Work Results for Communications.

B. Submittal requirements:

1. Product Data Brochures
2. Shop Drawings
 - a. When primary pathways are shown on project drawings, include any proposed changes to cable routing plans on shop drawings.
 - b. When primary pathways are not shown on project drawings or when project drawings are not developed, include all cable routing pathways on shop drawings.
3. Record Drawings

1.3 SYSTEM DESCRIPTION

- A. Provide a system of supporting devices, hangers, and pathways for support of communications and low voltage cabling. Include engineering, labor, and materials necessary for a complete installation.
- B. Provide fire-rated penetration assemblies for communications and low voltage cabling penetrating rated walls.
- C. Provide non-rated, acoustical penetration assemblies for communication and low voltage cabling penetrating non-rated walls.
- D. Provide label nameplates for each 4" conduit shown on project drawings.

PART 2 - PRODUCTS

2.1 ELECTRICAL METALLIC TUBING (EMT) CONDUITS

- A. Application: Support for horizontal communications and low voltage cabling concealed in walls and suspended ceilings. Support for interior intra-building backbone communications cabling installed above

suspended ceilings.

- B. Refer to Section 26 05 33 – Conduits and Boxes for product description and manufacturers.

2.2 POLYVINYL CHLORIDE (PVC) CONDUITS

- A. Application: Support for exterior horizontal communications and low voltage cabling installed underground. Support for exterior inter-building backbone communications cabling installed underground or within concrete encased ductbanks.
- B. Refer to Section 26 05 33 – Conduits and Boxes for product description and manufacturers.

2.3 CABLE TRAY

- A. Application: Support for large bundles of communications and low voltage cabling in an indoor installation within ceiling space.
- B. Description:
 - 1. Construction from carbon steel wire, ASTM A 510, Grade 1008 with a electroplated zinc finish, ASTM B633.
 - 2. Consists of a continuous, rigid, welded steel wire mesh cable management system, to allow continuous ventilation of cables and maximum dissipation of heat, with UL Classified splices where tray acts as Equipment Grounding Conductor. Wire mesh wire tray will have continuous welded top and side wires to protect cable insulation and installers.
 - 3. Supports a load of 50 pounds per linear foot when supported on 6 foot centers.
 - 4. Provide splices, supports, and other fittings necessary for a complete, continuously grounded system.
 - 5. Provide “side car” bracket to support low-voltage cabling.
 - 6. Size:
 - a. Depth: 4 inches (150 mm), unless otherwise noted on project drawings
 - b. Width: 12 inches (300 mm), 18 inches (450 mm), and 24” inches (600 mm)
- C. Manufacturer:
 - 1. WBT (no substitutions)
 - a. #WBT4X12S, shaped cable tray system, 12"W x 4"D
 - b. #WBT4x18S, shaped cable tray system, 18"W x 4"D
 - c. #WBT4x24S, shaped cable tray system, 24"W x 4"D
 - d. #SIDECAR4, additional cable support, 4"D

2.4 J-HOOK CABLE SUPPORT

- A. Application: Support for small bundles of communications and low voltage cabling in an indoor installation within ceiling space.
- B. Description:

1. Non-continuous cable supports shall provide a bearing surface of sufficient width to comply with required bend radii of high-performance cable.
2. Non-continuous cable supports shall have flared edges to prevent damage while installing cables.
3. Non-continuous cable supports sized 1 5/16" and larger shall have a cable retainer strap to provide containment of cables within the hanger. The cable retainer strap shall be removable and reusable and be suitable for use in air handling spaces.
4. Non-continuous cable supports shall have an electro-galvanized or G60 finish and shall be rated for indoor use in non-corrosive environments.
5. Stainless Steel non-continuous cable supports are intended for indoor and outdoor use in non-corrosive environments or where only mildly corrosive conditions apply.

C. Manufacturer:

1. Legrand
 - a. #CJ21, "Cablofil" series j-hook, 1 5/16"
 - b. #CJ32HP, "Cablofil" series j-hook with retainer flap, 2"
 - c. #CJ48HP, "Cablofil" series j-hook with retainer flap, 3"
2. Or equal

2.5 FIRE RATED PENETRATION ASSEMBLY

A. Application: Pathway device designed to allow communications and low voltage cabling to penetrate fire-rated walls and floors without the need for firestopping.

B. Description:

1. Consists of an enclosed heavy gauge galvanized steel pathway lined with intumescent material engineered for rapid expansion when exposed to fire or high temperatures, quickly sealing the pathway and preventing the passage of flames and smoke.
2. Pathway device contains a built-in fire sealing system sufficient to maintain the hourly fire rating of the barrier being penetrated.
3. Utilizes a self-contained sealing system to automatically adjust to the installed cable loading and allows for cables to be installed, removed, or retrofitted without the need to remove or reinstall firestop materials.
4. UL tested and classified in accordance with ASTM E814. Equal F and T ratings.
5. Physical Properties:
 - a. Size: 4"
 - b. Allowable Cable Fill: 0-100% visual fill
 - c. Fire Ratings: Up to 3 hours in concrete floor, up to 4 hours in gypsum walls
 - d. Expansion Begins: 350°F (177°C)
 - e. Volume Expansion: 800%

C. Manufacturer:

1. Specified Technologies Inc. (no substitutions)
 - a. #EZDP44S2, "EZ-Path Series 44+" single horizontal pathway kit
 - b. #EZP544W, "EZ-Path Series 44+" ganged bracket for multiple horizontal pathways
 - c. #EZRCM44S, "EZ-Path Series 44+" radius control module
 - d. #EZDG844S2, "EZ-Path Series 44+" two-slot (8 pathways) vertical pathway kit

2.6 NON-RATED PENETRATION ASSEMBLY

A. Application: Pathway device designed to allow communications and low voltage cabling to penetrate smoke and acoustical walls without the need for smoke sealing.

B. Description:

1. Utilizes a self-contained smoke sealing system to automatically adjust to the installed cable loading and allows for cables to be installed, removed, or retrofitted without the need to remove or reinstall caulking materials.
2. UL tested and classified in accordance with ASTM E90-09 for L rating.
3. Physical Properties:
 - a. Size: 4"
 - b. Allowable Cable Fill: 0-100% visual fill
 - c. Seal: Flame retardant neoprene rubber

C. Manufacturer:

1. Specified Technologies Inc. (no substitutions)
 - a. #NEZ33, single pathway
 - b. #NEXP433, four-gang wall plate

2.7 FIRE-RATED RETROFIT ASSEMBLY

A. Application: Two-piece fire-rated assembly installed around existing, overfilled conduit penetrations that restores fire and smoke ratings.

B. Description:

1. Consists of a heavy gauge galvanized steel housing containing intumescent materials that expand rapidly when exposed to fire or high temperatures to close off the void area.
2. Two-piece device that installs around cable bundles and existing sleeves. Contains a built-in fire sealing system sufficient to maintain the hourly fire rating of the barrier being penetrated.
3. Utilizes a self-contained sealing system to automatically adjust to the installed cable loading and allows for cables to be installed, removed, or retrofitted without the need to remove or reinstall firestop materials.

4. UL tested and classified in accordance with ASTM E814. Equal F and T ratings.
5. Physical Properties:
 - a. Size: Suitable for installation over a 4" conduit
 - b. Allowable Cable Fill: 50-100% visual fill
 - c. Fire Ratings: 1 or 2 Hours
 - d. Expansion Begins: 250°F (121°C)

C. Manufacturer:

1. Specified Technologies Inc. (no substitutions)
 - a. #EZDR400, "EZ-Path" retrofit assembly, 4" conduits

PART 3 - EXECUTION

3.1 INSTALLATION

A. Cable Tray

1. Install wire tray level and plumb according to manufacturer's written instructions, Coordination Drawings, original design, and referenced standards.
2. Cutting: Field-fabricate changes in direction & elevation by cutting & bending wire tray.
 - a. Cut wire tray wires in accordance with manufacturer's instructions.
 - b. Wire tray wires must be cut with side-action bolt cutters with offset head to ensure integrity of protective galvanic layer.
 - c. Remove burrs and sharp edges from wire trays.
3. Cable tray shall be supported to structure using threaded rods
4. Cable tray shall have a minimum of 12" clearance above top of tray. Any point where trays cross ductwork and piping in an area where 12" clearance is not practical, they should be routed to minimize total distance where clearance is compromised. Any section where the minimum clearance will not be met shall be brought to the attention of the Consulting Engineer and the Owner.
5. Cable tray shall have a minimum clearance of 6" from bottom of tray to finished ceiling.

B. J-Hook Cable Supports

1. J-hook must be installed every 4-5 feet at irregular intervals.
2. Installation of supports at a repeating interval (i.e. every 4 feet exactly) will establish a standing wave induction of interference on the cable. Attaching support to alternating sides of structural steel can accomplish this as can a non-linear installation.
3. The Contractor shall "close" J-hook supports with manufacturer provided bars and not with cable ties. The Contractor shall not use cable ties to strap cable to J-hook supports. The Contractor shall not install cables under such strain as to require tying to supports.

C. Fire Rated Penetration Assembly

1. Install fire rated pathway device when penetrating rating walls and floors as shown on project documents.
2. Utilize radius control module for cable support inside of communications rooms.
3. Comply with manufacturer's instructions for installation of products.

D. Non-Rated Penetration Assembly

1. Install pathway device when penetrating non-rated smoke and acoustical walls whether shown on project drawings or not.
2. Utilize gang brackets for installations utilizing two or four pathway devices.
3. Comply with manufacturer's instructions for installation of products.

E. Fire-Rated Retrofit Assembly

1. Install retrofit device on 4" conduits containing communications or low voltage cabling with a minimum fill ratio of 50%. For conduits with less than a 50% fill ratio, provide firestopping materials as required to meet applicable rating. Refer to Section 07 84 00 – General Firestopping for additional requirements.
2. For horizontal conduits, provide fasteners and plate kits as required to secure assembly to finished wall.
3. Comply with manufacturer's instructions for installation of products.

F. Nameplates

1. Provide engraved, plastic nameplates for each conduit at every pull-box and wall- penetration.
2. Labels to indicate designation and conduit number. Adhere to UT Southwestern Information Resources Cable Installation Standards to label conduits.

END OF SECTION

SECTION 27 08 00
COMMUNICATIONS TESTING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes
 - 1. Testing of backbone copper and fiber optic cabling.
- B. Related Sections
 - 1. Section 27 13 13 – Communications Backbone Copper Cabling
 - 2. Section 27 13 23 – Communications Backbone Fiber Optic Cabling
 - 3. Section 27 15 13 – Communications Horizontal Copper Cabling
 - 4. Section 27 15 33 – Communications Horizontal Coaxial Cabling

1.2 SUBMITTALS

- A. Comply with the submittal procedures of Section 27 05 00 – Common Work Results for Communications Systems.
- B. Submittal requirements:
 - 1. Record Documents
 - a. Include copies of test reports as described in this section.
 - b. Include warranty certificate(s) from cable manufacturer(s).

1.3 SYSTEM DESCRIPTION

- A. The Owner and the Engineer shall require validation testing on all voice and data communications cable installed as part of any project. This testing shall verify that the cabling has been installed properly and has the specific characteristics required by the project.

1.4 PROJECT CONDITIONS

- A. Testing of existing and active connections may be restricted until after normal working hours. Coordinate with the Owner's Representative to determine if testing can occur during normal business hours.
- B. Testing shall take place only when the channel is fully installed. If channel components may be moved or re-positioned after testing, the Contractor shall delay testing until cables, faceplates, and other components are in their final positions. This includes any repositioning by other trades prior to occupancy by the Owner.

PART 2 - PRODUCTS – NOT USED

PART 3 - EXECUTION

3.1 GENERAL

- A. The Contractor shall perform validation testing on all voice and data communications cable installed as part of any project. This testing shall verify that the cable has been installed properly and has the specific

characteristics required by the project.

- B. The Contractor shall provide all required test equipment and personnel necessary to support the certification and validation tests prescribed in this section.
- C. The Contractor shall provide a listing of the test equipment proposed for use for all certification testing.

3.2 CATEGORY 3 TESTING

- A. All Category 3 field testing shall be performed with an approved twisted-pair field test device with a multi-pair adapter.
- B. The Contractor shall test continuity and wire map for all pairs.
- C. Test results shall be unedited and as presented by the tester software. With the test results, the contractor shall provide software from the tester manufacturer to enable viewing of test results in native format.

3.3 CATEGORY 6A TESTING

- A. All Category 6A field testing shall be performed with an approved Level III balanced twisted-pair field test device.
- B. All installed Category 6A channels shall perform equal to or better than the minimum requirements as specified by the current ANSI/TIA/EIA standards for Category 6A. If the cable manufacturer has a separate, more stringent set of test standard required to certify the total solution being installed, the Contractor shall use the more stringent requirements.
- C. Category 6A balanced twisted-pair horizontal and backbone cables, whose length does not exceed 90 m (295 ft) for the basic link, and 100 m (328 ft) for the channel shall be 100 percent tested according to ANSI/TIA/EIA-568-C.2. Test parameters include wire map plus ScTP shield continuity (when present), length, NEXT loss (pair-to-pair), NEXT loss (power sum), ELFEXT (pair-to-pair), ELFEXT loss (power sum), return loss, insertion loss, propagation delay, and delay skew.
- D. Test results shall be unedited and as presented by the tester software. With the test results, the contractor shall provide software from the tester manufacturer to enable viewing of test results in native format.

3.4 COPPER TEST EQUIPMENT

- A. All balanced twisted-pair field testers shall be factory calibrated each calendar year by the field test equipment manufacturer as stipulated by the manuals provided with the field test unit. The calibration certificate shall be provided for review prior to the start of testing.
- B. The Contractor shall set the testers to the correct cable, by manufacturer and name, to ensure correct parameters are used during testing. Test settings selected from options provided in the field testers shall be compatible with the installed cable under test.

3.5 FIBER OPTIC TESTING

- A. The Contractor shall test all optical fiber strands for insertion loss, continuity, and length. The Contractor shall perform bi-directional OTDR tests on all OSP optical fiber strands.
- B. The Contractor shall test insertion loss at 850 nm and 1300 nm for 50/125m multimode cabling in at least one direction using the Method B (1-jumper) test procedure as specified in ANSI/TIA/EIA-526-14A.
- C. The Contractor shall test insertion loss at 1310 and 1550 for singlemode cabling using the Method A.1 (1-jumper) test procedure as specified in ANSI/TIA/EIA-526-7.
- D. The Contractor shall test continuity of each cable using a Visual Fault Locator tester.
- E. The Contractor shall determine and record length using an OTDR, optical length test measurement device

or sequential cable measurement markings.

- F. The Contractor shall assure that all cables installed do not exceed the allowable attenuated loss based on final installed length, attenuation coefficient, and connector loss. The Contractor shall remediate any strands testing above calculated limit as determined by the manufacturer.
- G. The Contractor may, at their expense, have a third party terminate the fiber in order to achieve this limit. Third party installers must be listed as subcontractors in the bid documents. Contractors may list this subcontractor as "tentative."
- H. Test results shall be unedited and as presented by the tester software. The Contractor may provide supplemental summaries generated by the Contractor. The Contractor shall provide Fiber performance calculation worksheets and fiber link attenuation records as illustrated in Section 21 (Figures 21.14 and 21.15) of the BICSI Telecommunications Cabling Installation Workbook, Technician, 2nd Edition.

3.6 FIBER TEST EQUIPMENT

- A. All optical fiber test equipment shall be factory calibrated as recommended by the field test equipment manufacturer. The calibration certificates shall be provided for review prior to the start of testing.

END OF SECTION

SECTION 27 15 13
COMMUNICATIONS HORIZONTAL COPPER CABLING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Minimum requirements for horizontal copper twisted pair cabling.
- B. Horizontal Cabling:
 - 1. Communication:
 - a. Communication horizontal cabling for project shall use conventional hierarchical star topology that home runs cables, from the communications room (FDs) to the WAO locations.
 - 2. Use point-to-point cable runs as indicated on drawings.
- C. Minimum composition requirements and/or installation methods for following materials and work are included in this section:
 - 1. Category 6A/6 UTP Cable
 - 2. #23 AWG Shielded Twisted Pair Cable.
 - 3. Data/Power Cable.
- D. Related Requirements: Comply with following:
 - 1. Section 270500 - Common Work Results for Communications.
 - 2. Section 274100 - Common Work Results for Audiovisual Systems and Equipment.
 - 3. Section 270526 - Grounding and Bonding for Communications.
 - 4. Section 270553 - Identification for Communications Systems.
 - 5. Section 270700 - Communications Testing
 - 6. Section 271119 - Communications Termination Blocks and Patch Panels.

1.2 REFERENCES

- A. Reference Standards: See Section 014200 – References and Section 270500 - Common Work Results for Communications.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 013300 – Submittal Procedures and Section 270500 - Common Work Results for Communications.
- B. Product Data: Submit product data for each type of cable to be installed including but not limited to physical dimensions, configurations, construction and performance specifications
- C. Product Samples: As required.
- D. Selection of the cable Category in accordance with section 270500 - Common Work Results for Communications
- E. Extra Materials: Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Cable: 1000 feet of each type used for project. Furnish on reels.

PART 2 - PRODUCTS

2.1 ALL CABLES

- A. All cables shall be plenum rated.

2.2 CATEGORY 6A UTP CABLE

- A. Physical Characteristics:
 - 1. Category 6A/6 copper cable.
 - 2. Pairs: 4.
 - 3. Conductor Size: 23 AWG
 - 4. Color Coding of Pairs:
 - a. Pair 1: W-BL; BL.
 - b. Pair 2: W-O; O.

- c. Pair 3: W-G; G.
 - d. Pair 4: W-BR; BR.
 - 5. Comply with ANSI/ICEA S-102-732-2009 for mechanical performance requirements, testing and testing methods.
 - 6. Ultimate Breaking Strength, ASTM D4565: 90 lbf minimum.
 - 7. Cold Bend Radius, ASTM D4565: 4 times cable diameter minimum at $-20^{\circ}\text{C} \pm 1^{\circ}\text{C}$ without jacked or insulation cracking.
 - 8. Jacket type: FEP. Color: yellow for Cat6A and Blue for Cat6
 - 9. Labeled and third party verified category 6A/6 cables.
 - B. Transmission Characteristics:
 - 1. DC Resistance of any Conductor, ASTM D4566: Shall not exceed 9.38 Ohms per 328 ft at 20°C .
 - 2. DC Resistance Unbalance Between any Two Conductors of any Pair, ASTM D4566: Not exceed 4 percent at 20°C .
 - 3. Mutual Capacitance at 1 kHz, ASTM D4566: Shall not exceed 5.6 nF per 328 ft at 20°C .
 - 4. Capacitance Unbalance, Pair-To-Ground, at 1 kHz, ASTM D4566: Shall not exceed 330 pF per 328 ft at 20°C .
 - C. Acceptable Manufacturers:
 - 1. CommScope Systimax 2091SD yellow in color plenum rated.
 - 2. CommScope Systimax 2071 Blue in color plenum rated
 - 3. No Substitutions.
- 2.3 #23 AWG SHIELDED TWISTED PAIR CABLE
- A. Physical Characteristics:
 - 1. #23 AWG conductors.
 - 2. Twisted pair, stranded, color-coded conductors.
 - 3. 100% overall shield with stranded copper or tinned-copper drain wire.
 - 4. Insulation Thickness:
 - a. 0.022 to 0.016 inches
 - B. Transmission Characteristics:
 - 1. Capacitance: 24 pF/ft between conductors, 79 pF/ft between one conductor and the other connected to the shield.
 - C. Acceptable Manufacturers:
 - 1. Commscope

PART 3 - EXECUTION

- 3.1 CATEGORY CABLE
- A. Cable distributions
 - 1. Copper Horizontal Cabling shall terminate in Voice/Data TRs, unless otherwise noted.
 - 2. Each Copper Horizontal Cable for every faceplate shall terminate in TRs.
 - 3. Copper Horizontal Cables for IP-based cameras shall terminate in TRs.
 - B. Comply with the manufacturer's installation instructions, BICSI Information Transport Systems Installation Manual and best industry practices.
 - C. Conceal in walls or soffits and install in metal conduits:
 - 1. Exposed Cabling: Installed in surface raceway.
 - 2. Cabling Below Raised Floors: Installed in cable tray and J-Hooks out to Work Area Outlet.
 - D. Schedule work in manner to complete above ceiling work/below raised floor work prior to tile/panel installation. In event installer is required to remove tiles/panels, coordinate with the contractor and do not break or disturb grid.
 - 1. Cable Above Accessible Ceilings: Supported 4 to 5 ft on center from cable support attached to building structure.
 - 2. Cable Below Raised Floor: Supported every 2 ft on center from the cable support attached to floor pedestals.
 - E. Cables shall not lay on the ceiling or the ceiling support structure. Anchor cables to not interfere with other services or space access.
 - F. Replace horizontal copper cables that do not pass Category 6A/6 requirements.

1. Maximum Length: Not exceed 295 ft.
 - G. No physical defects such as cuts, tears or bulges in outer jacket. Replace cables with defects.
 - H. Install cable in neat and workman-like manner. Neatly bundle and tie cable in closets. Leave sufficient cable for 90 degree sweeps at vertical drops.
 - I. Velcro ties and other cable management clamps shall be no more than hand tightened and shall fit snugly, but not compress, crimp, or otherwise change the physical characteristics of the cable jacket or distort the placement of twisted-pair components. Replace any cable exhibiting stresses due to over tightening of cable management devices.
 - J. Each Run of Horizontal Copper Cable between Terminating Patch Panel and WAO: Continuous without any joints or splices. No MUTO's allowed.
 - K. Do not untwist horizontal copper cable pairs more than 0.5 inch when terminating.
 - L. Make use of raceways built into furniture for open office furnished work areas.
 - M. Brushed aluminum raceways used in labs must have a divider between electrical and communication cabling.
 - N. Raceway adapters must be provided by GC for Commscope jacks/faceplates.
 - O. Do not install cable in common cable hangers with speaker cable.
 - P. Maintain following clearances from possible sources of electromagnetic interference (EMI) exceeding 5 kVA:
 1. Power lines enclosed in a grounded metal conduit (or equivalent shielding) in proximity to a grounded metal conduit pathway: 6 inch.
 2. Unshielded power lines or electrical equipment in proximity to a grounded metal conduit pathway: 12 inch.
 3. Unshielded power lines or electrical equipment in proximity to open or nonmetal pathways: 24 inch.
 4. Electrical motors and transformers: 47 inch.
 - Q. Do not install Category 6A/6 cable using more than 25 lbs pull force in accordance with TIA/EIA and BICSI TDMM practices. Utilize appropriate cable lubricant in sufficient quantity to reduce pulling friction to acceptable levels on long pulls inside conduit. Use tensile rated cords (i.e. fishing line) for difficult or questionable pulls - to judge to go/no-go condition of conduit and pulling setup.
 - R. Replace cables with jackets that are chaffed or burned, exposing internal conductor insulation or have bare copper (shiners).
 - S. If deviations from the drawings are required, they shall be approved by Owner's Representative prior to placement of affected cables.
- 3.2 NON-CATEGORY TWISTED PAIR CABLE
- A. Cable distributions
 1. Centralized Copper Horizontal Cabling for audiovisual systems shall terminate in Low Voltage TRs, unless otherwise noted.
 - B. Maintain separation between each of the following signal types throughout the system to avoid possible interference:
 1. Microphone signals.
 2. Line Level Audio signals.
 3. Loudspeaker signals.
 4. Video and RF signals.
 5. Control signals.
 - C. There shall be no splices in the system without the prior approval by the Owner's Representative. Splices shall never occur inside conduit, but shall only occur in accessible junction boxes or equipment racks. Cable splices shall not be wrapped with adhesive tape.
 - D. Install plenum rated cables where required by code.
 - E. Adhere to the manufacturers' guidelines for cable bend radii and maximum pulling tensions.
 - F. Maintain separation between varying cable types to avoid hum, crosstalk, and interference.
 - G. Maintain separation between cables and devices with a potential electromagnetic field to avoid interference.
 - H. Provide rubber or nylon grommets to protect cables passing through racks, panels, furniture, conduits, and junction boxes.
 - I. Arrange cables in a neat and orderly manner in junction boxes, cable trays, conduits, and accessible ceilings.

3.3 TESTING

- A. Test all cables in accordance with Section 270700 – Communications Testing.

3.4 IDENTIFICATION

- A. Label all cables in accordance with Section 270553 – Identification for Communications Systems.

3.5 FIRESTOPPING

- A. Firestop openings and penetrations through fire and smoke rated wall and floor assemblies in accordance with Section 078400 – Firestopping and Section 270500 - Common Work Results for Communications.

3.6 BONDING AND GROUNDING

- A. Bond and ground shielded cables in accordance with Section 270526 – Grounding and Bonding for Communications.

END OF SECTION

SECTION 27 15 33

COMMUNICATIONS HORIZONTAL COAXIAL CABLING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes
 - 1. Horizontal cabling system including coaxial cable, amplifiers, taps, and connectors.
- B. Related Sections
 - 1. Section 27 05 28 – Communications Pathways
 - 2. Section 27 08 00 – Communications Testing

1.2 SUBMITTALS

- A. Comply with the submittal procedures of Section 27 05 00 – Common Work Results for Communications.
- B. Submittal requirements:
 - 1. Product Data Brochures
 - 2. Record Drawings
 - a. Primary cable routes
 - b. Outlet or termination locations
 - c. Labeling
 - 3. Test Reports

1.3 SYSTEM DESCRIPTION

- A. Provide RG-6 horizontal cables for TV and audiovisual outlets. Include taps, connectors, adapters, and components necessary for a complete installation.
- B. The backbone coaxial cabling system is provided directly by Time Warner Cable.
- C. Design Criteria
 - 1. Provide CATV system components rated for a minimum of 50-1000MHz.
 - 2. Signal flow shall be on a bi-directional basis.
 - 3. CATV outlet locations to measure between 3-6 dBmV. Contractor to perform loss calculations to ensure tap values meet or exceed the design intent of the CATV system.
- D. Design Guidelines for Coaxial Cabling
 - 1. Horizontal cable shall be installed in a star topology in a continuous run directly from the Low Voltage rooms.
 - 2. Cables shall not traverse floors. Horizontal cabling shall be served from a room on the same floor as

the WAO, unless otherwise noted. Cables that must be routed through floors and utilize the ceiling space of the floor below shall not be served from the Low Voltage rooms on the lower floor.

3. Cabling shall be free of surface damage, kinks, excessive twists, and visible anomalies.
4. All wire and cable are to be plenum rated.

PART 2 - PRODUCTS

2.1 HORIZONTAL COAXIAL CABLE

- A. Application: Horizontal RG-6 coaxial cabling suitable for providing CATV service in an indoor environment.
- B. RG-6 Coaxial Cable:
 1. Consists of an 18 AWG center conductor with a foam FE dielectric material and quad shielding.
 2. Conforms to performance requirements for Series 6 coaxial cable.
 3. UL tested and listed as CMR.
- C. Manufacturer:
 1. CommScope
 - a. 741690074/10 | 5740 BKRB RG6 QD 1000, RG-6 riser rated coaxial cable, quad shield, black
 2. Or equal

2.2 CATV DIRECTIONAL TAP

- A. Application: Termination of horizontal RG-6 coaxial cabling onto a wall mounted apparatus.
- B. Directional Taps:
 1. Frequency: Performance to 1000 MHz
 2. Outputs: 8 x 8 F-type connectors
 3. Contains a built-in ground block and RFI shielding for up to 120 dB.
- C. Manufacturer:
 1. Blonder Tongue
 - a. #DGT-8, indoor directional tap, 8-output
 2. Or equal

2.3 CATV CONNECTORS

- A. Application: Termination of horizontal coaxial cabling using an F-type connector.
- B. F-Type connectors for WAOs:
 1. UL listed 75 Ohm F-type connector for RG-6 and RG-59 coaxial cabling.
 2. Performance rating up to 3 GHz.

3. Supports both push and screw-style termination.
- C. F-Type connectors for termination to directional tap:
 1. Hex crimp style
 2. Female to female coupler.
 3. UL listed 75 Ohm F-type connector for RG-6 coaxial cabling.
- D. Manufacturer:
 1. F-Type connectors for WAOs:
 - a. CommScope (no substitutions)
(1) #108009432 | M81C-COUPLER, F-type connector, white
 2. F-Type connectors for termination to directional tap:
 - a. Blonder Tongue
(1) #3587, "BTF-56" F-type connectors, hex crimp
 - b. Or equal

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General
 1. Install in accordance with manufacturer's instructions to ensure a manufacturer certified solution.
 2. The Contractor shall be responsible for verifying actual footages and distance identified on drawings.
 3. The Contractor shall be responsible for verifying that conduits and raceways are ready for occupancy before cable placement.
 4. The Contractor shall assume responsibility for difficulties or damage to cable during installation.
 5. The Contractor shall install all horizontal cable free of surface damage, kinks, excessive twists, and visible anomalies. Damaged cable shall be replaced at Contractor's expense. Horizontal cables damaged by other during construction shall be replaced by the Contractor under unit pricing specified in bid documents.
- B. Horizontal Coaxial Cabling
 1. Provide coaxial cabling to TV and audiovisual outlets as indicated on construction drawings.
 2. Provide a 10 foot service loop at the end of each cable. For the service loop within the Low Voltage rooms, route slack in the overhead cable runway.
 3. Neatly dress cables to their respective wall-mounted directional tap using Velcro cable ties and/or rack cable management loops.
 4. Bond components to communications grounding systems.
 5. Terminate horizontal RG-6 coaxial cables with F-type connection jacks onto shared WAO faceplates

with CAT6 cabling.

C. Testing and Inspection

1. Frequency response of the system shall pass 5 MHz to 750MHz.
2. Amplitude response for this spectrum shall be +/- 5 dB with respect to the line represented by normal cable tilt.
3. Taps, splitters and other passive equipment shall be of the total shielding type, using a sealed metal or aluminum case so as to minimize re-radiation and ingress.
4. Terminating resistors with 75 ohm impedance shall be installed at unused ports and feeder line ends.

3.2 LABELING

- A. In accordance with the UTSW IR Department standards for labeling, the Contractor shall label the communications faceplates and patch panels using a mechanical labeler.
- B. Coordinate with UTSW IR Department for existing labeling schemes and requirements. At a minimum, the following shall be reflected in the faceplate and patch panel labeling schemes:
 1. Alpha or numeric designator
 2. Text size shall be a minimum of 1/4" high
 3. Handwritten labels are not acceptable for final configurations
- C. Labeling shall be done in accordance with relevant standards such as ANSI/TIA-606-B. Any deviations shall be approved by the UTSW IR Department Representative.

END OF SECTION

SECTION 27 15 43

COMMUNICATIONS FACEPLATES AND CONNECTORS

PART 1 - GENERAL

1.1 SUMMARY

- A. Minimum composition requirements and/or installation methods for following materials and work are included in this section:
 - 1. Category 6A Insert Commscope X10D MGS600 6A Jacks.
 - 2. Category 6 Insert Commscope XL MGS400 6 Jacks.
 - 3. BNC-Type Connector.
 - 4. BNC-Type Insert.
 - 5. F-Type Connector.
 - 6. F-Type Insert.
 - 7. N-Type Connector.
 - 8. Wall/Modular Furniture Faceplate.
 - 9. Wall Phone Jack Assembly.
 - 10. Floorbox Assembly Mounting Plate.
 - 11. Recessed Table Box.
 - 12. Surface Mount Box (Biscuit Box).
- B. Related Requirements: Comply with following:
 - 1. Section 270500 – Common Work Results for Communications.
 - 2. Section 270526 – Grounding and Bonding for Communications.
 - 3. Section 270553 – Identification for Communications Systems.

1.2 REFERENCES

- A. Reference Standards: See Section 014200 – References and Section 270500 - Common Work Results for Communications.

1.3 SUBMITTALS

- A. General: Submit in accordance with Section 013300 – Submittal Procedures and Section 270500 – Common Work Results for Communications.
- B. Product Data and Installation Instructions: Submit for products included within this specification section.
- C. Colors and Finishes: Submitted and approved by Architect prior to order.
- D. Product Samples: As required.
- E. If providing pre-standards manufacturer system solution, submit installer/contractor certification documentation and channel certification information and requirements from manufacturer.

1.4 EXTRA MATERIALS

- A. Outlet assemblies: Provide one of each type for every twenty-five (25) installed, but not less than one.

PART 2 - PRODUCTS

2.1 CATEGORY 6/6A INSERT

- A. Functional from 14 to 140 degrees F.
- B. Tested: TIA/EIA-568-C.2.
- C. Connector Body: High-impact, fire-retardant plastic rated UL 94V-0.
- D. Unscreened per IEC 60603-7-41 8-position/8-conductor connector.
- E. Modular Connector: Individual snap-in-style.
 - 1. Comply with National Electrical Code; compliant with FCC Part 68; UL listed; and independently UL Certified.
 - 2. In addition to Category 6/6A Compliance, Connector: Ability to support high megabit and shared sheath applications.
 - 3. Connector Wiring Label: Provide installation color codes for both T568A and T568B wiring schemes.

- F. Modular Connector: Consult with owner on termination scheme.
 - G. Acceptable Manufacturers:
 - 1. Cat6A: CommScope Systimax 360 GigaSPEED X10D MGS600-262
 - 2. Cat6: CommScope Systimax 360 GigaSPEED XL MGS400-262
 - 3. No Substitutions.
- 2.2 BNC-TYPE CONNECTOR
- A. Crimp-type, male BNC-type connector.
 - 1. Terminate 75ohm coaxial cable.
 - 2. Meet SCTE and Bellcore specifications.
 - B. Acceptable Manufacturers:
 - 1. AMP Netconnect.
 - 2. Hubbell.
 - 3. Leviton.
 - 4. Ortronics.
 - 5. Panduit.
 - 6. Pass & Seymour.
 - 7. Siemon.
 - 8. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.3 BNC-TYPE INSERT
- A. Connector Body: High-impact, fire-retardant plastic rated UL 94V-0.
 - B. BNC-type Bulkhead: Designed for Series 6 applications.
 - C. Modules: Individual snap-in style, female-to-female bulkhead adapters.
 - D. Connectors: UL listed; and independently UL Certified.
 - E. Modular Connector: Fit other installed telecommunications wallplates, outlets and field-configurable patch panels and patch blocks.
 - F. Acceptable Manufacturers:
 - 1. AMP Netconnect.
 - 2. Hubbell.
 - 3. Leviton.
 - 4. Ortronics.
 - 5. Panduit.
 - 6. Pass & Seymour.
 - 7. Siemon.
 - 8. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.4 F-TYPE CONNECTOR
- A. Series 6, Series 11 - crimp-type, male F-type connector.
 - 1. Terminate 75ohm coaxial cable.
 - 2. Meet SCTE and Bellcore specifications.
 - B. Acceptable Manufacturers:
 - 1. Hubbell.
 - 2. Leviton.
 - 3. Ortronics.
 - 4. Panduit.
 - 5. Pass & Seymour.
 - 6. Siemon.
 - 7. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.5 F-TYPE INSERT
- A. Connector Body: High-impact, fire-retardant plastic rated UL 94V-0.
 - B. F-type Bulkhead: Designed for Series 6/Series 11 applications.
 - C. Modules: Individual snap-in style, female-to-female bulkhead adapters.

- D. Connectors: Comply with National Electrical Code; comply with FCC Part 68; UL listed; and independently UL Certified.
 - E. Modular Connector: Fit other installed telecommunications wallplates, outlets and field-configurable patch panels and patch blocks.
 - F. Acceptable Manufacturers:
 - 1. AMP Netconnect.
 - 2. Hubbell.
 - 3. Leviton.
 - 4. Ortronics.
 - 5. Panduit.
 - 6. Pass & Seymour.
 - 7. Siemon.
 - 8. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.6 N-TYPE CONNECTOR
- A. Impedance: 50 Ohms.
 - B. Style: straight female.
 - C. Connector shall be same manufacturer as 50 ohm coaxial cable.
 - D. Connector shall be for same size as 50 ohm coaxial cable.
 - E. Acceptable Manufacturers:
 - 1. Andrew (Commscope).
 - 2. RFS.
 - 3. Trilogy Communications.
 - 4. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.7 WALL/MODULAR FURNITURE FACEPLATE
- A. Wall-mounted Faceplates: High impact, thermoplastic or stainless, flush-mounted design.
 - 1. Capable of accepting mixed multimedia.
 - 2. Rear-loading inserts.
 - 3. Single gang – Systimax L Type Flush Mounted 4 Port Faceplate M14L-262
 - 4. Double gang – Systimax L Type Flush Mounted Double Gang Faceplate
 - 5. Readily accept Systimax outlets without modification to device.
 - B. Modular Furniture Faceplates: Snap-in type accepted by furniture manufacturer.
 - 1. Capable of accepting mixed multimedia.
 - 2. Rear-loading inserts.
 - 3. Contain at least one designation ID window for labeling.
 - 4. Readily accept Systimax outlets without modification to device.
 - C. Faceplates: UL listed.
 - D. Accessories:
 - 1. Blank inserts
 - E. Acceptable Manufacturers:
 - 1. CommScope Systimax
 - 2. Manufacturer of modular furniture
 - 3. Accepted Substitute in accordance with Section 012500 – Substitution Procedures.
- 2.8 WALL PHONE JACK ASSEMBLY
- A. Mount to single gang outlet box.
 - B. Mounting lugs designed to mate with corresponding telephone base plate or adapter.
 - C. 8-pin jack configuration.
 - D. UL listed.
 - E. Acceptable Manufacturers:
 - 1. CommScope Systimax
 - 2. No Substitutions.

2.9 FLOOR BOX ASSEMBLY MOUNTING PLATE

- A. Floor Box Type and Manufacturer:
 - 1. Accept snap-in type inserts.
 - 2. Same manufacturer as floorbox or designed to fit with floorbox manufacturer.
 - 3. Mount permanently into floor box while maintaining proper cable bend radius.
 - 4. UL listed.
 - 5. Readily accept Systimax outlets without modification to device.
- B. Acceptable Manufacturers:
 - 1. Legrand
 - 2. No Substitutions.

2.10 RECESSED TABLE BOX

- A. Flush-mount recessed table box to house SCS communications inserts and power receptacles.
- B. Colors and Finishes: Black, Brass, Bronze, Nickel. Confirm finish with architect.
- C. Flip up cover, Pop up box. Confirm type with architect.
- D. Readily accept Systimax outlets without modification to device.
- E. Acceptable Manufacturers:
 - 1. FSR
 - 2. Hubbell
 - 3. Mockett
 - 4. Wiremold
 - 5. No Substitutions.

2.11 SURFACE MOUNT BOX (BISCUIT BOX)

- A. Surface mount with snap-on cover.
- B. Accepts snap-in type SCS inserts.
- C. Side breakouts for cable or raceway entry.
- D. Rear cable access.
- E. Fastens to mounting surface utilizing screws or modular furniture mounting bracket.
- F. Cable tie anchor points for cable management and strain relief.
- G. Includes port identification labels.
- H. Acceptable Manufacturers:
 - 1. CommScope Systimax
 - 2. No Substitutions.

PART 3 - EXECUTION

3.1 8P8C INSERT

- A. UTP Cables: Terminated with high density modular jacks that snap into faceplate mounted on wall outlet box, modular furniture raceway or in floor box bracket.
- B. Fill extra openings with blank inserts.
- C. Coordinate with Owner's Representative for proper termination standard (T568A or T568B).

3.2 BNC/F-TYPE CONNECTOR

- A. Stray Strands of Shield: Clipped or moved away from center conductor.
- B. White Center Conductor of Cable: Even with shoulder of attachment nut.
- C. BNC/F-type Connector: Completely cover exposed braid.
- D. Use proper crimp tool to do termination.

3.3 BNC/F-TYPE INSERT

- A. Horizontal Cable: Screw onto backside of bulkhead.
- B. Bulkhead Module: Snap into adapter plate.

- C. Adapter Plate: Rear-loaded into faceplate.
 - D. Fill extra openings with blank inserts.
- 3.4 N-TYPE CONNECTOR
- A. Comply with manufacturer installation recommendations.
- 3.5 FACEPLATE/SURFACE MOUNT BOX (BISCUIT BOX)
- A. Secure outlet boxes to building with mechanical fasteners. Adhesive fasteners not allowed.
 - B. Coordinate modular furniture faceplates with furniture provider.
 - C. Fill extra openings with blank inserts.
- 3.6 RECESSED TABLE BOX
- A. Coordinate exact location with Architect.
 - B. Install per manufacturers' recommendations.
- 3.7 TESTING
- A. Test all connectors and inserts in accordance with Section 270700 – Communications Testing.
- 3.8 IDENTIFICATION
- A. Label all faceplates in accordance with Section 270553 – Identification for Communications Systems.

END OF SECTION

SECTION 28 05 00

COMMON WORK RESULTS FOR ELECTRONIC SAFETY & SECURITY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes

1. Basic materials and methods, along with General Requirements of Division 1, applicable to Division 28 sections.
2. Drawings and general provisions of the contract, including General and Supplementary Conditions of Division 1, applicable to Division 28 sections.

B. Related Sections

1. Section 28 13 00 – Access Control System

1.2 CONTRACTOR REQUIREMENTS AND QUALITY ASSURANCE

A. Comply with applicable local, state and federal codes.

B. Comply with applicable requirements of recognized industry associations which produce standards for the various trades.

C. Warrant work under this specification against faulty material or Workmanship in accordance with Division 1. If the project is occupied or the systems placed in operation in several phases at the request of Owner's Representative, then the warranty of each system or piece of equipment used shall begin on the date of substantial completion for each phase. The use of building equipment for temporary service and testing does not constitute the beginning of the warranty.

D. Equipment and material provided under this Division shall be periodically inspected and serviced by competent installers. This function becomes the responsibility of the Owner once the system is accepted by the Owner. The one year material and Workmanship warranty is not intended to supersede normal inspection or service and shall not be construed to mean the Contractor shall provide free service for normal maintenance items such as periodic cleaning and adjustment due to normal use, nor to correct without charge, breakage, maladjustment, and other trouble caused by improper maintenance.

E. Upon completion of contract and progressively as work proceeds, clean-up and remove dirt, debris and scrap materials. Maintain the premises in a neat and clean condition at all times during construction. Protect and preserve access to Head-end equipment at all times. Clean items with factory finishes. Touch-up minor damage to surfaces; refinish entire piece of equipment when sustained major damage. All electronics must be protected from dust and other airborne debris.

1.3 STANDARDS

A. The Contractor's performance of the work shall comply with applicable federal, state and local laws, rules and regulations. The Contractor shall give required notices, shall procure necessary governmental licenses, permits, and inspections and shall pay without burden to Owner's Representative, all fees and charges in connection therewith unless specifically provided otherwise. In the event of violation, the Contractor shall pay all fines and penalties, including attorney's fees and other defense costs and expenses in connection therewith.

B. Codes, Standards and Ordinances: Design, manufacture, test, and install electronic safety and security systems per manufacturer's requirements and in accordance with state codes, local codes, requirements of authorities having jurisdiction, and particularly the following standards:

1. National Fire Protection Association (NFPA):

- a. NFPA 70 – National Electrical Code
- b. NFPA 72 – National Fire Alarm and Signaling Code
- c. NFPA 101 – National Life Safety Code
- 2. Underwriters Laboratories, Inc. (UL):
 - a. UL 1479 – Fire Tests of Through - Penetrations Firestops
 - b. UL 294 – Access Control System Units
- 3. Americans with Disabilities Accessibility Guidelines.

1.4 COMPLETENESS OF WORK

- A. The Contract Documents depict low voltage systems which are intended to be complete and functioning systems. All products, materials, labor, and programming necessary to render a fully functional system to fulfill the design intent shown on the documents shall be provided by the contractor.
- B. Catalog numbers referenced throughout this Division's drawings and specifications are intended to convey a general understanding of the type of quality of the product required. Where written descriptions differ from information conveyed by a catalog number, the written description shall govern. No extra charge shall be allowed because a catalog number is found to be incomplete or obsolete.

1.5 SUBMITTALS

- A. Comply with provisions of Division 1.
- B. Submit shop drawings as called for in the Specification Sections that follow.
- C. Within thirty (30) days after contract has been awarded, Contractor to submit to Designer for review a complete list of materials, equipment, and accessories proposed for use, listing the item and manufacturer's name only.
- D. AutoCAD or Revit files requested by Contractor for use in submittals, coordination drawings, as- builds, etc., will be subject to executing a release document from the Engineer prior to release of electronic media.
- E. Based upon aforementioned approved listing, Contractor to submit four (4) copies of complete brochures and shop drawings of all materials, fixtures, and equipment that he proposes to use giving the names of manufacturers, trade name and specific catalog numbers.
- F. Product Data Brochures
 - 1. Furnish original color product datasheets, describing the specification and technical requirements of the system and each component. When multiple products are listed, denote by highlighting or other form which product they intent to use.
 - 2. Miscellaneous cut sheets for wire, cable; terminal block connector's etc. clearly identifies model numbers, manufacturer names and miscellaneous engineering data.
 - 3. Provide equipment quantities and model numbers for all equipment being proposed.
 - 4. Provide engineering calculations, such as voltage drop, electrical load calculations, etc. as it relates to devices.
- G. Shop Drawings

1. Provide detailed system drawings, including location and mounting requirements.
2. Complete and legible legends should be provided for each of the drawings.
3. Provide functional block diagrams.
4. Provide mounting details, indicating mounting, protective housings etc.
5. Provide wiring diagrams and point-to-point wiring schematics to panels, indicating utilized inputs and spare inputs.
6. Details of connections to power sources including power supplies.
7. Console/Rack installation, block diagrams, and wiring diagrams.
8. Denote main cable pathways or individual home runs.

H. Assurance/Quality Control Submittals

1. Certification letters stating the Contractor is an authorized reseller, installer, and extended warranty provider for the specified systems.
2. Evidence of technicians qualified for work in the form of specified manufacturer's training certification.

1.6 DELIVERY AND STORAGE

- A. Insofar as possible, deliver items in manufacturers' original unopened packaging. Where this is not practical, cover items with protective materials, to keep them from being damaged. Use care in loading, transporting, unloading, and storage to keep items from being damaged.
- B. Store items in a clean dry place and protect from damage.

1.7 OPERATING AND MAINTENANCE MANUALS

- A. Prior to final acceptance of the project, furnish to Owner complete bound sets of operation and maintenance manuals of instructions for operation and maintenance of all pieces of equipment and systems provided under this division of specifications.
- B. Manuals to also include all submittal data on all materials and equipment. Clearly indicate items provided on this project. Included a list giving name and address of nearest supply house which carry the spare parts and name and address and phone number of Installation Contractor to be given to Owner.
- C. The supplier or manufacturer shall provide thorough training of all staff assigned to those areas receiving new systems and equipment. This training shall be developed and implemented to address two different types of staff. The end users will be trained on the features and functions of the systems, and the engineering or maintenance department will be trained on the routine maintenance and basic programming of the systems. All training will take place on-site using the fully installed and certified system.
- D. Three sets of the following data are required:
 1. Operating and maintenance instructions.
 2. Spare parts lists.
 3. Copies of approved submittal data.
 4. Warranty information including any required test results.
- E. Arrange each set of data in an orderly way, and bind each set in a separate 3-ring, hard-cover binder.

- F. As soon as data accumulates, prepare one of the sets and deliver to the Owner's Representative, continuously updating this set as additional data is obtained.
- G. At completion of work, submit two complete sets of data to the Owner's Representative for distribution to the proper parties.

1.8 RECORD DRAWINGS

- A. Provide updated drawings, denoting all changes to the construction documents and the date at which the change took place.
- B. Provide updated drawings, which illustrate final field conditions; these drawings shall incorporate all changes to the construction documents.
- C. Panel power schedules, denoting circuits dedicated for use.
- D. Wire and cable run sheets including cable identification numbers and terminal strip designations.
- E. Legends which indicate device types, symbols, abbreviations and manufacturer's model numbers.
- F. Functional block diagrams for all subsystems. Schematic diagrams for all custom circuitry and interfaces to work not in contract.
- G. Elevations for equipment and/or riser closets, which show the exact configurations and physical installation of related equipment, interface panels, power supplies, junction boxes and equipment cabinetry.

1.9 WARRANTY

- A. The Contractor shall adhere to the warranty requirement for all installations.
- B. Warrant work under this specification against faulty material or Workmanship in accordance with Division 1 for a period of one (1) year that shall begin after system acceptance by Owner. If the project is occupied or the systems placed in operation in several phases at the request of the Owner, then the warranty of each system or piece of equipment used shall begin on the date of substantial completion for each phase. The use of building equipment for temporary service and testing does not constitute the beginning of the warranty.
- C. The Contractor shall submit in the bid documents any additional, contractor-specific warranties or guarantees to be offered on the project.
- D. The Contractor shall provide any and all necessary documentation needed to implement this warranty and to verify the solution installation.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. All materials and equipment used in carrying out these specifications are to be new and have UL listing, or listing by other recognized testing laboratory when such listings are available. Specifications and drawings indicate name, type and catalog numbers of materials and equipment to be used as "standards" shall not be construed as limiting competition. Unless otherwise indicated, Contractor may, at his option, use materials and equipment when, in the judgment of the Owner's Representative, they are equivalent to that specified. Prior approval of substitutions must be given prior to submittals.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Insofar as it is possible to determine in advance, advise the general contractor to leave proper chases and openings. Place all outlets, anchors, sleeves, and supports prior to pouring concrete or installation of

masonry work. Should contractor neglect doing this, any cutting and/or patching required is to be done at this contractor's expense. Visit site and be informed of conditions under which work must be performed. No subsequent allowance will be made because of error or failure to obtain necessary information to completely estimate and perform work involved.

- B. Carefully coordinate with other divisions to ensure proper power requirements, grounding, fireproofing and interlocking controls between the fire alarm system, nurse call system, and owner furnished systems.
- C. Notify other tradesmen of any deviations or special conditions necessary for the installation of work. Interferences between work of various contractors to be resolved prior to installation. Work installed not in compliance with specifications and drawings and without properly checking and coordinating as specified above shall, if necessary, be removed and properly reinstalled without additional cost to Owner.
- D. Intent:
 - 1. These sections of specifications and drawings form a complete set of documents for the electronic safety and security systems for this project. Neither is complete without the other. Any item mentioned in one shall be as binding as though mentioned in both.
 - 2. The intent of these specifications and drawings is to form a guide for a complete systems installation. Where an item is reasonably necessary for a complete system but not specifically mentioned, such as pull boxes, fittings, expansion fittings, support hangers, etc. provide same without additional cost to Owner.
 - 3. Device layouts indicated on drawings are diagrammatical only. Exact location of outlets and equipment to be coordinated and governed by project conditions. The Designer reserves the right to make any reasonable changes (approximately 6 feet) in location of junction boxes, or equipment prior to roughing in of such without additional cost to Owner.
- E. Deviations:
 - 1. No deviations from specifications and drawings to be made without full knowledge and consent of Designer.
 - 2. Should Contractor find during progress of work that existing conditions make desirable a modification of the requirements of any particular item, report such item promptly to Designer for his decision and instructions.

3.2 TRENCHING, EXCAVATION, BACKFILLING, AND REPAIRS

- A. Under other Divisions, provide trenching, excavation, and backfilling. Coordinate all requirements with trades. Failure to properly coordinate this effort resulting in additional trenching, excavation, backfilling, or repairs shall be performed without additional cost to Owner.

3.3 TESTS

- A. On completion of work, installation to be entirely free of damaged materials, software errors, incomplete cable installation including terminations, labeling, and dust. Perform a thorough operation test in the presence of the Owner's Representative. Provide documentation of all test results as outlined in each system's specifications. Include labor, materials and instruments for above tests.
- B. Prior to final observation and acceptance, test and leave in satisfactory operating condition, all systems and equipment.

3.4 INSPECTION FEES AND PERMITS

- A. Obtain and pay for all necessary permits and inspection fees required for electronic safety and security systems installation.

3.5 OBSERVATIONS

- A. When field observation services are a part of the project scope, the Architect office will provide periodic

observation of the progress of work specified herein. Purpose of the observation is to ensure compliance of Contractor's work with specifications and drawings. The Architect office may also observe tests required of this Contractor as called for in other sections of specifications.

- B. Specifications and drawings represent work to be done in view of total project requirements. Final location of card readers, cameras, duress buttons, etc., to eliminate possible conflict with other trades is responsibility of this Contractor. Contractor to provide all supervision required for his personnel to ensure that installation is made in accordance with specifications and drawings and all safety rules and regulations are observed. In event of conflicts of work on project with other trades, Contractor to make every reasonable effort to resolve conflict through meetings and discussions with other parties involved, by preparation of drawings or other appropriate action. Only after this has been done shall Architect assistance be requested through the RFI process.

END OF SECTION

SECTION 28 13 00
ACCESS CONTROL SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Access control panels, input/output modules, and card readers
2. Access control power supplies
3. Alarm initiating devices, including: door position switches, request-to-exit sensors, duress buttons, and general alarm points
4. Integration with the video surveillance system.
5. Integration with the security intercom system.
6. Integration with the emergency intercom system.
7. Interface to electrified door hardware and operators.
8. Interface to elevator controllers.

B. Related products provided under another Section:

1. 120V power
2. Conduit, junction boxes, and metallic raceways
3. Door hardware, operators, and pushbuttons
4. Fire alarm system interface relays
5. Electromagnetic door holders

C. Related Sections:

1. Section 28 15 23 – Security Intercom System
2. Section 28 15 33 – Emergency Intercom System
3. Section 28 23 00 – Video Surveillance System

1.2 SUBMITTALS

A. Comply with the submittal procedures of Section 28 05 00 – Common Work Results for Electronic Safety & Security.

B. Submittal requirements:

1. Product Data Brochures
2. Shop Drawings

3. Record Drawings

1.3 SYSTEM DESCRIPTION

A. General:

1. The intent of this document is to provide all pertinent information to allow the contractor to bid the labor, supervision, tooling, and miscellaneous consumables to provide a complete access control system. It is the responsibility of the contractor to include any and all items required for a complete system if not identified in these specifications or drawings.

B. Overview:

1. UTSW utilizes an existing Lenel OnGuard access control system that is monitored and controlled by the UTSW Police Department.
2. Card readers and other alarm initiating devices will route to wall-mounted access control panels located in the low voltage rooms and utilize the UTSW Security VLAN for communication with the existing Lenel OnGuard access control server in the Data Center.

C. Scope of Work:

1. Provide access control software with associated license fees to support the card readers and security devices shown on the project drawings.
2. Provide software integration licenses to integrate the access control software with the specified video management software (refer to Section 28 23 00 – Video Surveillance System for additional requirements) to enable automatic camera call-up upon event or alarm.
3. Provide access control panels with associated power supplies located in the telecommunications rooms. Panels to include controllers, reader modules, and input/output modules to support the devices shown on project drawings. Alarm monitored only doors that do not contain a card reader will terminate onto reader modules (not an input module) to match the UTSW Security standard.
4. Provide multi-technology card readers as indicated on project drawings. Card readers to accept 125 kHz and 13.56 MHz formats.
5. Provide interface to request-to-exit switch within electrified door hardware for access controlled doors. Provide request-to-exit motion sensor for access controlled doors that do not contain integrated switch.
6. Provide door position switches for access controlled and monitored doors shown on project drawings.
7. Provide under-counter duress buttons as shown on project drawings.
8. Provide riser rated composite cabling for access controlled and alarm monitored doors. This includes doors that are alarm monitored only and do contain a card reader.
9. Provide riser rated device cabling for duress buttons and other alarm initiating devices not associated with an access controlled or alarm monitored door.

D. Interface to Elevators:

1. Card readers within elevator cars to restrict access to individual floors. A valid card read enables selected floor buttons to operate based on cardholder security level.
2. Provide enclosure within elevator machine room for demarcation point between security cabling and elevator controller.
3. Provide coordination during installation of card readers and cable terminations. Furnish card readers

to elevator contractor for installation within elevator car. Elevator contractor responsible for traveler cable, connection from elevator controllers to demarcation point, and installation of card readers within elevator car.

E. Interface to Fire Alarm System:

1. Coordinate with fire alarm system contractor to automatically drop power from stairwell, elevator vestibule lobby, and other doors within the designated path of egress upon activation of the fire alarm system.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. General

1. It shall be the Contractor's burden to prove that any manufacturer other than the ones listed for each product meet or exceed that sections requirements, if submitted.
2. This specification is based on Lenel's "OnGuard" 7.3 access control software – the most current software provided by Lenel at the time of this specification's publication. Provide the latest release of electronic hardware and software at the time of installation. Using older versions of software and materials must be approved in writing by the Owner.

2.2 ACCESS CONTROL SOFTWARE

A. Application: Software that provides monitoring and management of access control panels, card readers, alarms and events, cardholder information, and allows integration with other security subsystems.

B. Access Control Software:

1. General:
 - a. Open architecture software platform that supports Microsoft Windows Server 2016 and 2016. Client component supports Microsoft Windows 10 operating systems for workstations.
 - b. Supports Active Directory, single sign-on, and capable of running on a Microsoft Windows Workgroup environment.
 - c. Capable of being installed in a virtual environment using industry leading standards, such as VMware.
 - d. Software is hardware agnostic and supports commercial off-the-shelf servers (COTS) and storage attachments.
2. Interfaces:
 - a. Supports an unlimited number of card readers, inputs, and output points.
 - b. Supports elevator control.
 - c. Supports desktop and browser clients.
 - d. Supports IP, serial, and modem communications.
 - e. Supports mustering, anti-passback, and guard tour functions.
 - f. Supports high-availability or five-nine compliant (99.999%) server solutions.

- g. Supports integration with Owner's HR database for automatic provisioning and revocation of cardholder privileges. Software allows for automated communication in either real time or user configurable timed intervals.
- h. Supports integration with the specified video management software (refer to Section 28 23 00 – Video Surveillance System for additional requirements).

3. Features:

- a. Supports fully customizable dynamic views for displaying system components, devices, and database information. Views display changes to the system in real time.
- b. Supports scalable and editable maps by importing vector and raster files to create complex floor plans that graphically illustrate the security system. Maps contain hyperlinks to other maps to create a hierarchy of interlinked files.
- c. Supports secure database partitioning to allow for multiple independent entities to share a single database.
- d. Supports intrusion zones and alarm/disarm commands through card readers with integrated keypads. Keypad commands can be configured to require valid card read and/or valid PIN to activate command.
- e. Includes badge design software and smart card enrollment functionality.

C. Manufacturers:

1. Lenel (no substitutions)

- a. #PRO-64RUP, "OnGuard PRO" add-on license for 64-readers
- b. #SWG-1340, interface license for Vingtor-Stetofon Alphacom servers

2.3 ACCESS CONTROL PANELS

A. Application: An intelligent controller with integrated battery backup, database, and communication ports that supports multiple card readers.

B. Features:

- 1. Supports multiple communication channels to which control panels, card readers, and alarm monitoring devices connect.
- 2. Supports hardware modules used for additional memory and/or future feature enhancements.
- 3. Supports flash upgrades for firmware updates.
- 4. On-board memory stores a minimum of 50,000 transactions.
- 5. Utilizes AES 128-bit data encryption for compliance with HSPD-12 and FIPS201 requirements.
- 6. Compliant with UL294 requirements.
- 7. Supports HID proximity, HID iCLASS SE, MIFARE, and DESFire card reader formats.
- 8. Supports up to 32 access levels per cardholder.
- 9. Supports redundant backup of configuration data.

10. Utilizes an onboard Ethernet NIC with DHCP support.

C. Manufacturer:

1. Lenel (no substitutions)

a. #LNL-3300, "Intelligent System Controller (ISC)", 32-interfaces (up to 64 readers)

b. #LNL-1320, interface module, 2-readers

c. #LNL-1100, interface module, 16-inputs

2.4 ACCESS CONTROL ENCLOSURES & POWER SUPPLIES

A. Application: Wall-mounted enclosures with integrated power supplies to support up to 8 security access control modules.

B. Enclosures:

1. Type: NEMA Type-1

2. Size: 30"W x 48"L x 6"D

3. Contains perforated back-panel for mounting control panels and modules.

4. Contains integrated tamper switch.

5. Contains surface mounted 120V 15A duplex receptacle.

6. Utilize finger duct wireways within enclosure to dress security cabling.

7. Customized for UTSW with screen printed logo on enclosure cover.

C. Device Power Supply Charger:

1. Power Input: 120VAC 60Hz, 2.6A

2. Power Output: 12VDC, 10A

3. AC fail supervision

4. Low battery and battery presence supervision

D. Device Power Distribution Module:

1. Input Power: 12/24VDC

2. Outputs: 8 individually fused outputs (outputs rated at 3.5 amps) with surge suppression

E. Lock Power Supply Charger:

1. Power Input: 120VAC 60Hz, 4.2A

2. Power Output: 24VDC, 10A

3. AC fail supervision

4. Low battery and battery presence supervision

F. Lock Power Controller:

1. Power: 12/24VDC
2. Inputs: 8 normally open inputs
3. Outputs: 8 auxiliary power outputs (outputs rated at 3.5 amps), thermal and short circuit protection with auto reset
4. UL listed fire alarm disconnect for any or all of the 8 outputs

G. Batteries:

1. Sealed, non-spillable lead acid battery with spaded terminals.
2. Output: 12VDC 7 ampere hour

H. Manufacturer:

1. Unity (no substitutions)
 - a. #UTPDAC16-LOGO-R1, access control enclosure and power supplies customized specifically for UTSW

2.5 CARD READERS

A. Application: Reader capable of supporting 125 kHz low-frequency and 13.56 MHz high- frequency formats.

B. Card Readers:

1. Rugged, weatherized polycarbonate enclosure, designed to withstand an operating temperatures of - 40° to 150° F (-40° to 65° C).
2. Utilizes a Wiegand protocol for communication to access control panels.
3. Utilizes custom site keys for increased security.
4. Utilizes a multi-color LED and an audible sounder to indicate door status.
5. Capable of reading the following formats:
 - a. 125 kHz – HID proximity
 - b. 13.56 MHz – iCLASS SE (secure identity object)

C. Manufacturer:

1. HID Global (no substitutions)
 - a. #920L, "multiCLASS SE RP40" series card reader, standard wall mount

2.6 POSITION SWITCHES

A. Application: Contact switches for use in access controlled and alarm monitored doors to provide door position functionality.

B. Door Position Switches:

1. Mounting: Recessed
2. Size: 3/4" diameter
3. Loop Type: Normally closed
4. Contacts: Single pole double throw
5. Gap Distance: Up to 0.5" in steel door applications

C. Manufacturer:

1. Honeywell
 - a. #947-75TBR, recessed 3/4" door contact switch with screw terminals, brown
2. Or equal

2.7 DURESS BUTTONS

A. Application: Latching hold up button for use in under counter applications.

B. Duress Buttons:

1. Double-pole double-throw contacts for multi-notifications
2. Switch action provides silent operation
3. Stainless steel housing

C. Manufacturer:

1. Honeywell (no substitutions)
 - a. #269R, duress button with stainless steel housing

2.8 WIRE & CABLE

A. Application: Horizontal security cabling to access controlled doors and alarm initiating devices.

B. Access Control System Cabling:

1. Composite Cabling:
 - a. Composite cabling for access controlled and alarm monitored doors consists of the following within a single PVC jacket:
 - (1) Lock Power: 4C 18 AWG, shielded
 - (2) Card Reader: 3TP 22 AWG, overall shielded
 - (3) Door Contact: 2C 22 AWG, shielded
 - (4) Request-to-Exit: 4C 22 AWG, shielded
 - b. UL tested and listed as CMP.
 - c. Cable installed indoors to contain a blue jacket.
2. Device Cabling:

- a. Cabling for duress buttons and other alarm initiating devices not associated with card reader controlled doors will utilize a minimum of 18 AWG stranded conductors. Increase conductor size as required to allow for voltage drop on long runs.
 - b. UL tested and listed as CMR.
 - c. Cable installed indoors to contain a blue jacket.
- C. Manufacturer:
 - 1. Superior Essex
 - a. #AC-A23-55, access control composite riser rated cable, custom blue jacket
 - b. #2F-C15-32, 2C 18 AWG, shielded, custom blue jacket
 - c. #2F-C35-32, 4C 18 AWG, shielded, custom blue jacket
 - 2. Or equal

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Install all system components including Owner furnished equipment, and appurtenances in accordance with the manufacturer's instructions, and as shown, and shall furnish all necessary connectors, terminators, interconnections, services, and adjustments required for a complete and operable system.
- B. Visit the site and verify that site conditions are in agreement with the design package. Report all changes to the site or conditions that will affect performance of the system to the Owner in a report. The Contractor shall not take any corrective action without written permission from the Owner and Owner.
- C. The Contractor shall perform a field survey, and furnish a report to the Owner as part of the site survey report. The Contractor shall be held responsible for repair costs due to Contractor negligence or abuse of Owner equipment.
- D. Verify that all surfaces and areas are ready to receive work.
- E. Verify field measurements as shown on drawings and as instructed by manufacturer.
- F. Verify that required utilities are available, in proper location, and ready for use.

3.2 INSTALLATION

- A. General:
 - 1. Install the access control system related equipment as specified by the manufacturer and as shown, provide mounting hardware sized appropriately to secure each device for conditions encountered at the site; connect signal lines and AC power to equipment interfaces.
 - 2. Install system in accordance with manufacturer's recommendations. This includes, but is not limited to, the following:
 - a. Wiring with racks shall be routed horizontally and vertically in neatly tied bundles. Point-to-point wiring shall not be used.
 - b. Cable bundles shall cross from one rack to the next at the top, center, or bottom only, leaving sufficient working room within the rack.

- c. Wiring for shall be tie-wrapped so that all connectors in a bundle can be removed and re-installed without the possibility of cross connecting.
 - d. Where wiring is routed through sheet metal or over frame members, the metal edges shall be covered with flexible grommeting or edge dressing (such as automobile door edge trim).
 - e. Double-sided foam tape shall not be used to secure any terminal boxes, relay bases or circuit boards, etc. All device mounting shall be of a permanent nature.
 - f. All excess length AC cords are to be tie-wrapped out of the way, and new plugs installed.
 - g. All cable shall be in conduit, present a professional appearance, and maintain the aesthetics of the installation area.
 - h. Care shall be exercised at all times to protect Owner's property. For example, ladders shall not be placed against wallpapered or finished surfaces, equipment or furnishings; desks or countertops shall not be used in lieu of ladders.
3. Provide all panel wiring required, including temporary wiring. Install wiring in accordance with NEC and NFPA regulations (as applicable), local building codes and ordinances, and all Owner wiring standards. Contractor shall be responsible for obtaining and adhering to applicable regulations, codes, ordinances, and standards. Label conductors at each end with legible, write-on type, self-laminating vinyl labels.
4. Ensure that:
- a. All applicable statutes, ordinances, regulations, license requirements and codes are fully complied with.
 - b. All required permits are obtained.
 - c. All required inspections are conducted.
 - d. All necessary certificates are issues, obtained, and delivered to the Owner.
 - e. All equipment installations and mounting are in strict accordance with requirements for applicable seismic classification.
- B. Access Control Software:
1. Coordinate with UTSW Police Department to set the following criteria:
- a. Access card levels and door groupings
 - b. Priority levels for alarm and events
 - c. Schedules and time zones
 - d. Holidays and holiday types
 - e. Actions and responses from individual alarm points
 - f. Routing of alarm points to operator's workstations
 - g. Custom reporting options
2. Integrate the access control software with the specified video management software to enable the

following features:

- a. Automatic camera call-up upon event (i.e. door forced alarm, door held open too long, etc.) within the access control software to provide a single-user interfaced for responding to alarms.
 - b. View live and recorded video within the access control software.
 - c. Control PTZ cameras.
 - d. Time synchronization of events within the access control and video management software.
3. Incorporate building floor plans from architectural documents to develop dynamic maps complete with device icons and applicable text within the access control software.
 - a. Edit CAD or BIM drawing files so that non-applicable layers are removed so that the floor plans can be viewed in a simple manner.
 - b. Convert drawing files to a graphic format that is compatible with access control system software.
 - c. Utilize device icons on each graphical floor plan to indicate card readers, cameras, duress buttons, and other alarm points. Each icon to provide real time status of associated device with command and control functionality.

C. Access Control Panels:

1. Install access control panels and power supplies in the telecommunications rooms as indicated on project drawings.
2. Install resistors at device end of line (EOL) per manufacturer's instructions to provide four- state EOL supervision of devices and cabling.

D. Access Controlled Doors:

1. Card Readers:
 - a. Connect horizontal security cabling to card reader terminal blocks to enable the following functions:
 - (1) Displays red LED when door is locked and is in secure position.
 - (2) Displays green LED when door is unlocked.
 - (3) Produces audible tone to indicate valid and invalid reads.
 - b. Utilize mounting plate to install card reader onto electrical back-box. Install plumb and square so that there are no obvious gaps between the wall surface and card reader.
2. Electrified Locks:
 - a. Terminate cabling from electrified door hardware onto designated output points of lock power supplies so that relays within the access control panels are thermally protected.
 - b. Utilize UL listed disconnect switch on lock power supplies so that access controlled doors within the path of egress can automatically drop power upon activation of fire alarm system.

3. Door Position Switches:

- a. Install recessed door contacts 6" from leading edge of door, unless otherwise noted.

4. Request-to-Exit Sensors:

- a. Connect to electrified door hardware's "RX" trigger wire for request-to-exit functionality. Install passive infrared request-to-exit motion sensors for doors that do not contain RX functionality within electrified door hardware.
- b. Install motion sensor on protected side of door. Ensure that detection beam is not obstructed by exit signs, fixtures, or other objects.
- c. Adjust relay times and sensitivity on each door so that it properly detects valid exits and eliminates false alarms.

E. Duress Buttons:

1. Install duress buttons under counter space. Ensure that duress button is not obstructed by shelving, PC holders, or other equipment.
2. Coordinate with furniture and/or millwork contractor to field determine exact location of device prior to installation.

F. Enclosure Penetrations:

1. All enclosure penetrations shall be from the bottom unless the system design requires penetrations from other directions. Penetrations of interior enclosures involving transitions of conduit from interior to exterior, and all penetrations on exterior enclosures shall be sealed with rubber silicone sealant to preclude the entry of water. The conduit riser shall terminate in a hot-dipped galvanized metal cable terminator. The terminator shall be filled with an approved sealant as recommended by the cable manufacturer and in such a manner that the cable is not damaged.

G. Control Line Surge Protection:

1. Cables and conductors, except fiber optic cables, which serve as communication, control, or signal lines shall be protected against surges and shall have surge protection installed at each end.
2. Protection shall be provided at the equipment and additional triple electrode gas surge protectors rated for the application on each wire line circuit shall be installed within three (3) feet of the building cable entrance.
3. Fuses shall not be used for surge suppression.

3.3 LABELING

A. General:

1. The Contractor shall generate all labeling (no labels will be furnished by the Owner) for access control system components. Components include, but are not limited to, the following:
 - a. Control panels, equipment enclosures, and power supplies
 - b. Conduits designated for security
 - c. Card readers and security devices
 - d. Wires and cables

- e. Terminal blocks and relays
- 2. Labels shall coincide with identifications used on contractor generated as-built drawings.
- 3. Labels utilized shall be of sufficient size and have an adhesive backing for permanent attachment.
- B. Device Labeling:
 - 1. Coordinate with UTSW Police Department for device label format.
 - 2. Labeling of security devices shall match labeling of device icons within access control system software.
- C. Cable Labeling:
 - 1. Utilize permanent machine-generated labels for security cabling with applicable font and text size so that it can be easily legible.
 - 2. Position labels so that they are visible without the need to remove obstructions such as wire management or other materials.
 - 3. Label security cabling at both ends (device and control panel).

3.4 TESTING

- A. General:
 - 1. The Contractor shall provide all personnel, equipment, instrumentation, and supplies necessary to perform all site testing. The Owner will witness all performance verification. Written permission shall be obtained from the Owner before proceeding with the next phase of testing. Original copies of all data produced during performance verification shall be turned over to the Owner at the conclusion of each phase of testing prior to Owner approval of the test.
 - 2. Coordinate testing requirements with the General Contractor and provide specific information on pre-acceptance and final acceptance testing activities so that they can be entered into the overall construction schedule.
- B. Cable Testing:
 - 1. All cables and termination hardware shall be 100% tested for defects in the installation and the materials used in order to verify performance under installed conditions. All conductors of each installed cable and system component shall be verified usable by the contractor.
 - 2. All cables shall be tested in accordance with this document, and the best industry standard practices. If any of these are in conflict, the Contractor shall be responsible to bring any discrepancies to the attention of the designer for clarification and/or resolution.
 - a. All cables shall be tested for acceptable signal strength and passage.
 - b. All test results shall be well documented and the documentation given to the designer for proper distribution.
 - 3. The access control cabling system shall be tested by using an approved using an approved test unit capable of performing the required tests.
- C. Pre-Acceptance Testing:
 - 1. The Contractor shall perform a 100% pre-acceptance test to verify operation of the access control system prior to the final acceptance test with the Owner.

2. Pre-Acceptance Testing activities shall not occur until Contractor has ensured that all punch list items have been remedied. If conditions exist that may cause degradation or interference with any security device, the Contractor shall inform the Owner.
3. The pre-acceptance testing shall, as a minimum, include:
 - a. Verification that any signal or control cabling have been installed, tested, and approved as specified.
 - b. Verification that electrified door hardware functions properly on access controlled doors. Coordinate testing with door hardware contractor.
 - c. Verification that card readers, position switches, and request-to-exit sensors have been properly installed to provide: valid card reads, invalid card reads, door forced open, and door propped alarms.
 - d. Verification that access control software has been properly programmed per the Owner's instructions and configured to integrate third-party databases.
 - e. Verify integration with specified video surveillance system for automatic camera call-up on alarm with proper alarm/event recording parameters.
4. Document the results of the pre-acceptance testing using approved test forms. Report shall indicate the system has been properly calibrated, tested, and is ready to begin final acceptance testing with the Owner.

D. Acceptance Testing

1. Upon successful completion of Pre-Acceptance Testing, the Contractor shall demonstrate to the Owner that the completed access control system complies with the contractor requirements. Acceptance Testing shall not commence until receipt of approved Pre- Acceptance Testing activities based on the Contractor's written report.
2. Using approved test procedures, all physical and functional requirements of the project shall be demonstrated and shown.
3. The Owner may terminate Acceptance Testing at any time when the system fails to perform as specified. Upon termination of testing by the Owner or by the Contractor, the Contractor shall submit a report outlining the required repairs to the Owner then commence system repairs upon direction by the Owner. Upon successful completion of the Acceptance Testing, the Contractor shall deliver test reports and other documentation as specified to the Owner.

END OF SECTION

SECTION 283100

ADDRESSABLE FIRE ALARM SYSTEM

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
- B. Related Sections:
 - 1. Division 01 General Requirements
 - 2. Division 07 Thermal and Moisture Protection, Section 078400 Firestopping
 - 3. Division 08 Openings, Section 087100a Door Hardware Sets
 - 4. Division 21 Automatic Fire Sprinkler
 - 5. Division 23 Heating, Ventilating and Air Conditioning
 - 6. Division 26 Electrical
 - 7. Division 27 Communications

1.2 SUMMARY

- A. Section Includes:
 - 1. This specification describes an addressable fire detection and alarm signaling system, current scope is modification of the existing fire alarm system to cover the reconfiguration of the existing space.
 - 2. The system modifications shall be in full compliance with National and Local Codes.
 - 3. The system modifications shall include all required hardware, raceways, interconnecting wiring and software to accomplish the requirements of this specification and the contract drawings, whether or not specifically itemized herein.
 - 4. All equipment furnished shall be new and the latest state of the art products of a single manufacturer (Edwards EST), engaged in the manufacturing and sale of analog fire detection devices for over ten years.
 - 5. The system as specified shall be supplied, installed, tested and turned over to the owner in an operational condition without system troubles, supervisory, or alarm signals.
 - 6. In the interest of job coordination and responsibilities the installing contractor shall contract with a single State of Texas licensed supplier for fire alarm equipment, engineering, programming, inspection and tests.
 - 7. The fire alarm system modifications shall be provided as turnkey to the project. All necessary conduit, wiring, boxes, 120 volt power, breakers, etc. shall be provided for a complete and operative system. Coordinate requirements with the general contractor for each building.
 - 8. Fire alarm contractor shall provide all necessary equipment for a complete system conforming to all applicable codes; layout of equipment shall be reflected in the shop drawings.

1.3 UNIT PRICES

- A. Provide unit pricing for devices required to complete the modifications to the existing system, not all devices listed below will be part of that system. Unit pricing shall include all materials and labor to completely install the device, connect it to the system, provide appropriate programming and completely test its operation.
 - 1. Smoke detector
 - 2. Heat Detector
 - 3. Duct Mounted Smoke Detector

4. Pull Station
5. Ceiling Mounted Horn
6. Ceiling Mounted Horn/Strobe (any candela rating)
7. Ceiling Mounted Strobe (any candela rating)
8. Ceiling Mounted Speaker
9. Ceiling Mounted Speaker/Strobe (any candela rating)
10. Addressable Monitor Module
11. Addressable Control Module
12. Remote Power Supply
13. Audio Amplifier (Minimum of 1 amplifier per floor depending on the size of the building)

1.4 DEFINITIONS

- A. ASME: American Society of Mechanical Engineers
- B. FACP: Fire alarm control panel.
- C. Furnish: To supply the stated equipment or materials.
- D. Install: To set in position and connect or adjust for use.
- E. LED: Light-emitting diode.
- F. NFPA: National Fire Protection Association. Definitions in NFPA 72 apply to fire alarm terms used in this Section.
- G. NICET: National Institute for Certification in Engineering Technologies.
- H. Provide: To furnish and install the stated equipment or materials.
- I. UL: Underwriters Laboratories

1.5 SYSTEM DESCRIPTION

- A. Basic System - The system shall be a complete, electrically supervised fire detection and notification system, with a microprocessor based operating system having the following capabilities, features, and capacities:
 1. Support of walk test system capable of providing point test reports in NFPA standard format without manual report entries.
 2. System shall provide an output port for monitoring purposes by external systems. Communications to an external system shall be RS-232 or RS-485 communications.
 3. A single node or system shall support at least 50 remote control panels
 4. At least 59 nodes shall be networkable.
 5. Communications between network nodes, each supporting an interactive, self-standing, intelligent local control panel, with system wide display. Any network node shall be capable of supporting a local system of 2500 points or more.
 6. The local system shall provide status indicators and control switches for all of the following functions:
 - a. Audible and visual notification alarm circuit zone control.
 - b. Status indicators for sprinkler system water-flow and valve supervisory devices.
 - c. Any additional status or control functions as indicated on the drawings, including but not limited to; emergency generator functions, fire pump functions, door unlocking and security with bypass capabilities, damper smoke duct detectors.
 - d. The system shall be listed by UL for configuration as an approved release system for deluge, pre-action or clean agent extinguishing agent release system.
 7. The system shall be UL 1076 listed for monitoring and reporting security System Zoning.

8. Each intelligent addressable device or conventional zone on the system shall be displayed at the Central Alarm Receiving Terminal and the local fire alarm control panel by a unique alphanumeric label identifying its location.
 9. Batteries shall be supplied to operate the system for 24 hours with the capability to announce a single alarm event at the end of this duration for 15 minutes.
- B. FACP with Digital Voice Evacuation - The system shall be complete, electrically supervised fire detection and evacuation system using one way communication and smoke control systems with microprocessor based operating system having the following capabilities, features and capacities:
1. Voice amplification shall be supervised.
 2. Amplifiers shall be rated for 25V or 70.7V RMS.
 3. Amplifiers shall be sized as minimum, to accommodate speakers in corridors at 2 watt and other locations 1 watt. All speakers shall be ceiling mounted to provide maximum coverage of the area.
 4. The system shall have the capability to support Peer-to-Peer network and voice configuration.
 5. Multiple nodes shall provide Peer-to-Peer voice capability in order to eliminate a single point of failure.
 6. Audio shall be synchronized between nodes in order to take into account common areas.
 7. The network and audio risers between nodes shall be copper and shall be Class A loop configuration to allow communication to continue in the event of a fault.
 8. Speakers shall have the ability to play coded audio tones.
 9. The local system shall provide status indicators and control switches for all of the following functions:
 - a. The system shall provide inputs to the firefighters smoke management panel furnished under Division 23.
 - b. Audible and visual notification alarm circuit zone control.
 - c. Speaker circuit zone control.
 - d. Status indicators for sprinkler system waterflow and valve supervisory devices.
 - e. Any additional status or control functions as indicated on the drawings, including but not limited to emergency generator functions, fire pump functions, door unlocking and security with bypass capabilities.
- C. FACP with Clean Agent Suppression Releasing.
1. The system shall be listed by Underwriters Laboratories for configuration as an approved Clean Agent (NFPA 2001) extinguishing agent release system.
 2. Any additional status or control functions as indicated on the drawings, including but not limited to: HVAC shut down, doors closing, Clean Agent discharge.
 3. The fire suppression system shall be listed by Underwriters Laboratories for use as a Clean Agent fire extinguishing agent system.
 4. The fire suppression system and the fire alarm releasing system shall be compatibility listed by UL for use with each other.
 5. Any clean agent requirements shall be provided by separate specification – Fire Suppression System

1.6 PERFORMANCE REQUIREMENTS

- A. General Performance: Comply with NFPA 72 and all contract documents and specification requirements.
- B. All intercommunications between this system and the monitoring system shall be arranged so that the entire system can be UL Certified.
- C. System shall be complete, supervised, non-coded, addressable multiplex fire alarm system conforming to NFPA 72.
- D. The system shall have Style 6 circuits for each floor. The system shall operate in the alarm mode upon

actuation of any alarm initiating device. The system shall remain in the alarm mode until all initiating device(s) are reset and the fire alarm control panel is manually reset and restored to normal. The ability to bypass devices in "alarm" shall be provided.

- E. The system shall provide the following functions and operating features:
1. The FACP and auxiliary power panels shall provide power, annunciation, supervision and control for the system.
 2. Provide Class B initiating device circuits.
 3. Provide Class A Signaling Line Circuits for initiating devices.
 4. Provide Style 7 signaling line circuits for the network.
 5. Provide Class B notification appliance circuits (NAC). Arrange circuits to allow individual, selective, and all-call voice and visual notification by zone. Notification Appliance circuits shall be zoned to correspond with the building fire barriers.
 6. Stair Towers: Each stair tower NAC shall be separately zoned.
 7. Strobes shall be synchronized throughout the entire building.
 8. The system amplifiers for the voice evacuation system shall be configured as distributed, bulk or a combination of distributed and bulk audio.
 9. Provide multiple channels for live and recorded voice messaging.
 10. Provide electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within the control panel.
 11. Bypass for audio, visual, AHU shutdown, door release, fire suppression, flow and tamper switches, fire pump running, devices in "alarm", audible and visual notification by floor, smoke control and elevator recall. System shall have the capacity to provide 40 bypass function buttons.
- F. The system shall provide a field test function where one person can test the complete system or a specific area while maintaining full operational function of other areas not being tested. Alarms, supervisory signals, and trouble signals shall be logged on the system printer and in system history during the walk test.
- G. Alarm functions shall override trouble or supervisory functions. Supervisory function shall override trouble functions.
- H. Fire alarm signal initiation shall be by one or more of the following devices:
1. Manual pull stations
 2. Heat detector
 3. Addressable area smoke detector
 4. Duct smoke detector
 5. Automatic sprinkler system water flow switch
 6. Fire protection system activation (Ansul, FM200/ Halon, or similar)
- I. Activation of any system fire, supervisory, trouble or status initiating devices shall cause the following actions and indications at all fire alarm control panels.
1. Fire alarm condition:
 - a. Sound and audible alarm display a custom screen/message defining the building in alarm and the specific alarm point initiating the alarm.
 - b. Log into the system history archives all activity pertaining to the alarm condition.
 - c. Print alarm condition on system printer.
 - d. Audible and visual signals shall be silenced from the fire alarm control panel by an alarm silence switch in accordance with NFPA 72.
 - e. A signal dedicated to sprinkler system water flow alarm shall not be silenced while the sprinkler system is flowing at a rate of flow equal to a single head.
 - f. Create a dry contact output to the Energy Management System (EMS).
 - g. Notification to the University Police, CDAS, and Safety Department via Fireworks System.

2. Additional system operation for fire alarm condition for voice:
 - a. Sound a pre-announce tone on a general alarm. Voice evacuation mode will be self initiated if an actual alarm condition exists. The visual signals shall operate in a similar pattern.
 - b. Selective voice paging to speakers shall be done by floor and installed in stairways to inform occupants of the imminent shutdown of elevator circuits and the expected high traffic load in the stairwells.
 - c. An automatic announcement (UTSW Safety Message) or tone evacuation signal (Hospitals only) shall be capable of interruption by the operation of the system microphone to give voice evacuation instructions overriding the pre-programmed instructions.
 - d. Status lights next to speaker selection switches on the control panel shall indicate speaker circuit selection.
3. Supervisory Condition:
 - a. Audible signals shall be used for supervisory signals. Once silenced, the devices shall remain silenced. Internal clocks shall not be programmed to reactivate audible signals.
 - b. Record within system history the initiating device and time of occurrence of the event.
 - c. Print supervisory condition to system printer.
 - d. Create a dry contact output to the Siemens Energy Management System (EMS).
 - e. Notification to the University Police, CDAS, and Safety Department via Fireworks System.
4. Trouble Condition:
 - a. Audible signals shall be used for trouble signals. Once silenced, the devices shall remain silenced. Internal clocks shall not be programmed to reactivate audible signals.
 - b. Trouble conditions that have been restored to normal shall be automatically removed from the trouble display queue and not require operator intervention. This feature shall be software selectable and shall not preclude the logging of trouble events to the historical file.
 - c. Trouble reports for primary system power failure to the master control shall be automatically delayed for a period of time equal to 25% of the system standby battery capacity to eliminate spurious reports as a result of power fluctuations.
 - d. Record within system history, the occurrence of the event, the time of occurrence and the device initiating the event.
 - e. Print trouble conditions to system printers.
 - f. Create a dry contact output to the Energy Management System (EMS).
 - g. Notification to the University Police, CDAS, and Safety Department via Fireworks System.

1.7 SUBMITTALS

- A. Product Data: For each type of product indicated. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories. Complete manufacturer's catalog data including supervisory power usage, alarm power usage, physical dimensions, and finish and mounting requirements. The submittal will be reviewed and approved by The Director of Fire and Occupational Safety (University Fire Marshal) or his designee.
- B. Power calculations. Battery capacity calculations. Battery size shall be a minimum of 125% of the calculated requirement. Provide the following supporting information:
 1. Power supply rating justification showing power requirements for each of the system power supplies. Power supplies shall be sized to furnish the total connected load in a worst-case condition plus 20% spare capacity by zone.
 2. NAC circuit design shall incorporate a 20% spare capacity per zone for future expansion.
- C. Submit manufacturer's requirements for testing signaling line circuits and device addresses prior to

connecting to control panel. At a minimum the following tests shall be required; device address, the usage (Alarm, Supervisory etc.), environmental compensation, temperature ratings for thermal detectors and smoke detector sensitivities. This requirement shall need approval before any wiring is connected to the control panel.

- D. Shop Drawings: Include plans, elevations, sections, details, reflective ceiling plans and attachments to other work.
 - 1. Wiring Diagrams: For power, signal, and control wiring.
 - 2. Detailed equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components and location and size of each field connection.
 - 3. Indicated initiating and indicating device labeling sequence to be implemented to label every fire alarm system device. Device labeling shall match labeling on submittal documents.
 - 4. Complete drawings covering the following shall be submitted by the contractor for the proposed system:
 - a. Floor plan in a CAD compatible format at a scale of 1/8"=1'-0" showing all equipment and major raceways, marked for size, conductor type with type and size.
 - b. Provide a fire alarm system function matrix as referenced by NFPA 72, Figure A-7-5.5.5(9). Matrix shall illustrate alarm input/output events in association with initiation devices. Matrix summary shall include system supervisory and trouble output functions. Include any and all departures, exceptions, variances or substitutions from the specifications and/or drawings at time of bid.
 - 5. Installation drawings shop drawings, and as-built drawings shall be prepared by an individual experienced with the work specified herein and has the appropriate certification level from the state.
 - 6. Incomplete submittals shall be returned without review.
 - 7. Manufactures specifications for each device, component, or part which is associated with installation of this system.
 - 8. Calculations prepared by the FAC (Fire Alarm Contractor) detailing power dispersion, voltage drop, battery load throughout the system
- E. Coordination Drawings: Reflected ceiling plan(s) and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from Installers of the items involved:
 - 1. Light fixtures.
 - 2. HVAC registers
 - 3. Fire protection equipment interfaces
 - 4. Special suppression system by fire alarm contractor
- F. Qualification Data: For qualified Texas licensed Installer, Applicator, manufacturer, fabricator, professional engineer, testing agency, and factory-authorized service representative.
- G. Source quality-control reports.
- H. Field quality-control reports.
- I. Operation and Maintenance Data: For all fire alarm equipment, to include in operation and maintenance manuals.
- J. Software and firmware operational documentation
 - 1. Software operating and upgrade manuals
 - 2. Program software backup – On magnetic media or compact disk, complete with data files.
 - 3. Device address list
 - 4. Printout of software application and graphic screens

- K. Warranty: Sample of special warranty.
- 1.8 QUALITY ASSURANCE
- A. Manufacturer Qualifications: The publications listed below form a part of this publication to the extent referenced. The publications are referenced in the text by the basic designation only. The State of Texas adopted edition shall be used as a guide.
1. National Fire Protection Association (NFPA)
 - a. NFPA 70 National Electrical Code
 - b. NFPA 72 National Fire Alarm Code
 - c. NFPA 90A Standard for The Installation of Air Conditioning and Ventilating Systems
 - d. NFPA 92A Recommended Practice for Smoke Control Systems
 - e. NFPA 101 Life Safety Code
 2. Underwriters' Laboratories, Inc. (UL) equipment standards, Latest Edition
 - a. UL Fire Protection Equipment Directory
 - b. UL Electrical Construction Materials Directory
 - c. UL 38 – Manually Actuated Signaling Boxes for Use With Fire Protection Signaling Systems
 - d. UL 228 – Door Holding Devices
 - e. UL 268 - Smoke Detectors for Fire Protective Signaling Systems
 - f. UL 268A - Smoke Detectors for Duct Application
 - g. UL 464 - Audible Signal Appliances
 - h. UL 497A – Secondary Protectors for Communications Circuits
 - i. UL 521 - Heat Detectors for Fire Protective Signaling Systems
 - j. UL 864 - Control Units for Fire Protective Signaling Systems
 - k. UL 1076 – Security
 - l. UL 1283 – Electromagnetic Interference Filters
 - m. UL 1449 - Transient Voltage Surge Suppressors
 - n. UL 1480 - Speakers for Fire Protective Signaling Systems
 - o. UL 1971 - Signaling Devices for the Hearing Impaired
 3. International Code Council
 - a. International Building Code
 4. The manufacturer shall have a minimum of 15 years production experience in the manufacture and design of high sensitivity aspiration-type smoke detection systems.
 5. ISO 9002
- B. Supplier Qualifications
1. Provide the services of a factory trained and certified representative or technician, experienced in the installation and operation of the type of system and possess a valid Alarm Certificate of Registration (ACR) within the State of Texas.
 2. The technician shall supervise installation, software documentation, adjustment, preliminary testing, final testing and certification of the system. The technician shall provide the required instruction to the owner's personnel in the system operation and maintenance. The technician shall be in possession of a valid Fire Alarm Technician (FAL) license.
 3. The suppliers shall furnish evidence they have an experienced service organization, which carries a stock of spare and repair parts for the system being furnished.
 4. The equipment supplier shall be authorized and trained by the manufacturer to calculate, design, install, test, and maintain the air sampling system and shall be able to produce a certificate stating such upon request.
- C. Installer Qualifications:

1. Before commencing work, submit data showing that the manufacturer has successfully installed fire alarm systems of the same scope, type and design as specified.
 2. The contractor shall submit copies of all required Licenses and Bonds as required by the State of Texas.
 3. The contractor shall employ on staff a minimum of one NICET level III technician or a professional engineer, registered in the State of Texas.
 4. The contractor shall be qualified by UL for certifying fire alarm systems. Upon completion of the installation the contractor shall certify the final system meets UL ongoing maintenance.
 5. Contractors unable to comply with the provisions of Qualification of Installers shall present proof of engaging the services of a subcontractor qualified to furnish the required services.
- D. Testing Agency Qualifications: Qualified for testing indicated.
- E. Source Limitations for fire alarm equipment: Obtain fire alarm equipment from single source.
- F. Surface-Burning Characteristics: As determined by testing identical products according to ASTM E 84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
1. Flame-Spread Index: 25 or less.
 2. Smoke-Developed Index: 50 or less.
 3. Combustion Characteristics: ASTM E 136.
- G. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- H. Pre-installation Conference: Conduct conference at Project site.
- 1.9 DELIVERY, STORAGE, AND HANDLING
- A. Deliver products to project site in original. Unopened packages with intact and legible manufacturers labels identifying product and manufacturer, date of manufacturer and shelf life if applicable.
- B. Store materials inside, under cover, above ground, and kept dry and protected from physical damage until ready for use. Remove from site and discard wet or damaged materials.
- 1.10 PROJECT CONDITIONS
- A. Installed products or materials shall be free from any damage including, but not limited to, physical insult, dirt and debris, moisture, and mold damage. At time of system acceptance, all products shall be in their factory delivery condition without exception. Cost to replace will be the responsibility of the contractor, not UTSW.
- B. Environmental Limitations: Do not deliver or install products or materials until spaces are enclosed and weathertight, wet work in spaces is complete and dry, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.
- 1.11 WARRANTY
- A. Warranty for parts and labor shall be for a period of one year from the date of substantial completion
- 1.12 EXTRA MATERIALS
- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. One (1) smoke detector head
2. One (1) heat detector head
3. One (1) pull station
4. One (1) speaker strobe
5. One (1) strobe
6. One (1) duct detector

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Edward Systems Technology (EST)

2.2 CONTROL PANEL

- A. The fire alarm control panel shall be microprocessor based using multiple microprocessors throughout the system providing rapid processing of smoke detector and other initiation device information to control system output functions.
- B. There shall be a watchdog circuit, which shall verify the system processors and the software program. Problems with either the processors or the system program the panel shall activate a trouble signal, and reset the panel.
- C. The system modules shall communicate with an RS 485 network communications protocol. All module wiring shall be to terminal blocks, which will plug into the system card cage. The control panel shall be capable of expansion via up to 100 SLCs. Maximum system capacity shall be a minimum of 2500 intelligent initiation devices.
- D. System Components:
 1. The Device Loop Card shall be capable of 252 intelligent devices distributed between two SLC circuits. Any trouble on one circuit shall not affect the other circuit. This module controls the signaling from the initiation devices reporting alarms and troubles to the control panel. This module shall also provide the signaling to the field devices for the controlling the output of specific initiation devices. The on board microprocessor provides the device loop card with the ability to function even if the main microprocessor fails. LED's on the board shall provide annunciation for the following; Power, Card Failure, Network Failure, Gnd. Fault, Alarm, Trouble, Short Zone 1, Short Zone 2, Style 6 Open Zone 1, Style 6 Open Zone 2. This card shall plug into the system card cage. Circuit shall be capable of either input or output devices on any address without limitations. You can split the SLC in any ration while maintaining short circuit isolation between the two legs of the circuit
 2. The signal line circuits shall be tested for opens, shorts and communications with all addressable devices installed before connection to the control panel.
 3. Alphanumeric Display and System Controls: Arranged for interface between human operator at fire-alarm control unit and addressable system components including annunciation and supervision. Display alarm, supervisory, monitor, trouble and component status messages and control menu.
 - a. The common control switches and with corresponding LEDs provided as minimum will be; Reset Alarm Silence, Panel Silence. It shall be able to add additional switches/LEDs as required.
 - b. The main control panel shall have a graphic LCD display that is backlit when active.
 - c. Each point shall have custom event message.
 - d. Provide a minimum of 6 simultaneous events to be displayed. The first five (5) highest priority events in addition to the most recent event. The events shall be automatically placed in event types (Alarm, Supervisory, Monitor & Trouble) for easy access and shall be possible to view the specific event type separately.
 - e. Provide an internal audible signal with different programmable patterns to distinguish

between alarm, supervisory, trouble and monitor conditions.

4. The LCD shall have a keyboard screen to allow the technician ability to enter test and numbers for passwords or text changes. The interface screen module shall also have a Context Sensitive Help button. The interface screen module shall have the ability to view events, acknowledge, silence and reset networked systems. The interface screen module shall also have the ability to arm and disarm input and output points on FACP's. The interface screen module shall have the ability to be configured for control of the entire network, control of the local FACP, or annunciation only. In a networked configuration, the Partial System Disable LED shall be indicative of all networked FACP's. An interface screen module in a networked configuration shall have the ability to store 6 maps for every panel. At least 10 globally configured interface screen modules shall be supported in a network.
5. The Network Interface Card (NIC-C) shall provide either intranode (HNET) communication or internode (XNET) communication between enclosures. HNET and XNET communication shall support Class A Style 7 wiring. NIC-C shall have the ability to be configured as an electrical repeater in order to increase communication distances. In addition, the NIC-C shall support CAN network communication. This card shall plug into the system card cage.
6. The network ring card shall provide the ability to network systems in a Class A Style 7 Ring configuration. A single card per system (node) shall provide peer-to-peer communication between systems allowing a total of 59 systems to be networked together. The card shall reside in the same enclosure as the interface screen module. The network ring card shall supervise the ring network to ensure proper operation. The card shall also isolate a short-circuit fault to each individual segment of the network and perform ground fault detection on its outgoing ring port. Any faults that are detected by the card shall be reported to the panel for annunciation. The card shall isolate faults only to the individual node in trouble allowing communication on the network ring to continue. The card shall act as an electrical repeater for each network pair.
7. The System Status Display shall provide a remote LED/LCD display that shows the local status of a system. An LED shall illuminate when alarm, supervisory, trouble and security events occur on the system. The display shall consist of a LCD display that has four lines of forty characters each that provides details of the event in alphanumeric form. The display shall have three additional control buttons for acknowledging events, silencing audible circuits, and resetting the system. The display shall have an integral 5 digit code that enables these control buttons to operate. The display shall be located within a locked cabinet. The display shall be mountable in a 2-gang electrical box or 4-inch square electrical box.
8. The Zone Indicating Card shall contain 4 NAC circuits rated at 4 amps each with power-limited outputs. The zone inputs for the card shall be isolated and independently supervised. There shall be at least 3 unique codes/signals for each circuit based on system logic. These signals shall be a notification tone prior to voice instructing notification, Steady (Such as "Recall"), and Alert (Such as "Tornado Alert"). The card shall be listed for notification appliances, horns, bells, strobes, and speakers. The card shall be listed for NFPA 13 pre-action release, clean agent. The card shall have the ability to wire the circuits Style Y with outputs synchronized. The card shall have the following LED's to provide trouble shooting and annunciation: Power, Card Failure, Network Failure, Gnd. Fault, Zone Activation or Trouble. This card shall plug into the system card cage.
9. The Control Relay Card shall contain 6 fully programmable relays each rated at 4A, 30VDC/120VAC resistive and 3.5A, 120VAC 0.6 PF inductive. The card shall have the following LED's to provide trouble shooting and annunciation; Power, Card Fail, Network Fail, Relay 1 Active, Relay 2 Active, Relay 3 Active, Relay 4 Active, Relay 5 Active, Relay 6 Active.
10. The system card cage shall provide the mounting of all system cards, field wiring, and panel's inter-card wiring. All power limited field wiring shall connect to the top of the card cage. All non-power limited internal wiring shall be connected to the bottom of the card cage. The card cage shall hold the systems cards and have capability of connecting multiple card cages to meet system demands. All terminal blocks are removable.
11. The Supervised Input Module shall provide sixteen input circuits for remote system monitoring. Each input shall have the ability to be individually programmed as supervised (dry contact only) or unsupervised (general purpose input). The input module shall provide two programmable Form C relays. The input module shall be mountable in an enclosure that is remotely located from the main control panel. The input module shall be capable of supervising inputs 500 feet away.

12. The Output Control Module shall provide sixteen open collector outputs to drive LEDs, incandescent lamps or external relays. There shall also be an additional output for a local audible and two inputs for momentary lamp test as well as local audible silence switches. The module shall be mountable in an enclosure that is remotely located from the main control panel.
 13. The switch control module shall be a supervised module with 8 switches and two LED's per switch for controlling such items as speaker/strobe circuits. The switches shall also be used as generic inputs into the system. The module shall be mounted in the door for easy access. These modules shall be connected to the control area network, and have a maximum distance of 1000 ft.
 14. The LED Control Module shall contain eight groups of 2 LED's that shall be programmable. Eight LED's shall be dual color capable that can be lighted either RED or GREEN flashing or steady. The remaining LED's shall be AMBER flashing or steady. A space shall be provided for labeling of LED functions. The label shall slide behind a clear protective membrane. The module shall be mounted in the door for easy access. These modules shall be connected to the control area network, and have a maximum distance of 1000 ft.
 15. The Live Voice Module shall have the supervised Microphone and pre-amplifier. The module shall contain 6 programmable switches. The module shall connect to the control area network, and also connect to the interface screen module. The module shall have remote capability using a Remote Network Interface. The module requires a Local Page Board to convert the audio to a digital signal. The module shall contain a Local Speaker w/control, Ready to Page LED, and a Pre-Announcement tone LED. The module shall be mounted in the door for easy access and viewing.
 16. The Digital Audio Card shall provide Single channel of digital audio and control of all voice functions including Background Music or Conventional Paging. This module communicates with the interface screen module and to all other audio modules via the control area network. The module also contains an async serial interface buss to distribute the audio signals between the zone amplifiers and the Audio Input Card. The Local Page Board shall plug into this card converting the audio to a digital signal. The audio card shall plug in to the system card cage.
 17. The zone amplifier module shall support Class A or B speaker zone wiring configurations. The module shall be able to reproduce signals from 200 hertz to 12,000 hertz for improved intelligibility of voice communication. Audio amplifiers shall be power limited.
 18. The Audio Input Card (AIC) allows up to 2 external audio sources for any source such as CD, and tape player. There are two relays on the module to control the inputs. The system shall support multiple AIC cards. This card shall plug into the system card cage. The card shall be model number AIC. Output amplifiers and audio zone cards shall perform dynamic supervision, meaning, they can supervise the speaker circuit during operation, meeting UL requirements for the use of fire speakers with background music/paging.
 19. An audio level conversion card shall provide the capability of conducting a global-emergency page (All Call) over copper across multiple, remote nodes with each audio riser holding a maximum of 63 nodes. The emergency page originates at a global-paging station, where it is broadcast at 70VRMS over an audio riser by an amplifier. A single, remote node containing one card, steps the incoming speaker-level audio down to line-level audio that is compatible with the audio input (0.775VRMS, independently adjustable to $\pm 6\text{dBu}$).
 20. The audio level conversion card shall provide the ability to output line-level audio messages originating from the fire alarm system to other compatible 3rd party audio systems for enhanced integration
- E. System response time from alarm to output shall not exceed three (3) seconds. To expedite troubleshooting, the system cards shall have ground fault and diagnostics LED's on the cards. All system cards and modules shall have flash memory for downloading the latest module firmware.
- F. Passwords:
1. Maintenance/Control Password - There shall be a 5 character password that a user must enter into the control panel in order to perform such maintenance- and control-related functions at the panel as:
 - a. Arming and disarming devices.
 - b. Activating, deactivating or modifying detector ASD and sensitivity settings.

- c. Activating and deactivating the History Log function, and deleting obsolete entries.
 - d. Changing the system time and date.
 - 2. Function Key Password - There shall be a 5 character password that a user must enter into the control panel in order to access the panel's Function Keys: touch screen buttons which perform custom-programmed system functions.
 - 3. Reports Password - There shall be a 5 character password that a user must enter into the control panel in order to access the panel's reporting functions.
 - 4. Walk test Password - There shall be a 5 character password that a user must enter into the control panel in order to access the panel's walk testing functions.
 - 5. Acknowledge Silence able Reset Password - There shall be a 5 character password that a system user must enter into the control panel in order to acknowledge events, turn silence able audible and visuals on and off, and perform panel resets.
- G. Networking:
- 1. Digital communication capabilities supporting Style 4 (Class B) or Style 7 (Class A) communications using either DC digital or fiber optics technologies or combinations of both as required for the control panel to communicate with at least 50 remote transponders.
 - 2. Digital communication capabilities supporting Style 4 (Class B) or Style 7 (Class A) communications using either DC digital or fiber optics technologies or combinations of both as required for the control panel to communicate with at least 50 network nodes.
 - 3. Capability shall exist within the system to extend the network at any node. The system shall support a maximum of two network extension circuits in series on any system branch, extending the inherent distance limitations for network communications.
 - 4. Communication protocol shall be of the CSMA/CD (carrier sense, multiple access, collision detect) type, eliminating delays incorporated into other protocols. Communication techniques using token passing and requiring sensing of delays and re-generation of the token to re-establish network communications in the event of a fault shall not be acceptable.
- H. Degrade Mode Alarm Activation:
- 1. Each data gathering panel shall support the ability to have its corresponding zone indicating card and output devices on an SLC loop activate when the SLC or conventional detector card is in Degrade Mode (has lost communication with the FACP). For example, if the device loop includes detectors with relay bases and lamps, the relays and lamps will activate upon any system alarm when the SLC is in Degrade Mode.
 - 2. Degrade Mode Alarm Activation with Voice: Each data gathering panel shall support the ability to have its corresponding DAC-NET turn on audio when the SLC or conventional detector card is in Degrade Mode (has lost communication with the FACP).
- I. Smoke Control: The fire alarm panels shall have the ability to be configured as a smoke control station that complies with UL/UUKL (UL 864) and NFPA 92A and ULC/ORD-C100 requirements. The system shall have the capability to monitor and override smoke control systems and equipment provided at designated locations within the same building. Where installed, a graphical smoke control panel shall be provided and shall incorporate all systems associated with smoke control.
- J. Digital Voice Command:
- 1. The Digital Voice Command Center located with the FACP, shall contain all equipment required for all audio control, signaling and supervisory functions. This shall include speaker zone indication and control, digital voice units and microphone.
 - 2. Up to 5 digital voice command centers shall be supported per system with the ability to limit control to one digital voice command center via a Request/Grant/Deny mechanism.
 - 3. Function: The Voice Command Center equipment shall perform the following functions:
 - a. Operate as a supervised multi-channel emergency voice communication system.
 - b. Audibly and visually annunciate the active or trouble condition of every speaker circuit.

- c. Audibly and visually annunciate any trouble condition for digital tone and voice units required for normal operation of the system.
 - d. Provide all-call Emergency Paging activities through activation of a single control switch.
 - e. As required, provide vectored paging control to specific audio zones via dedicated control switches.
 - f. Provide a factory recorded "library" of voice messages and tones in standard WAV. File format, which may be edited and saved on a PC.
 - g. Provide a software utility capable of off-line programming for the VCC operation and the audio message files. This utility shall support the creation of new programs as well as editing and saving existing program files. Uploading or downloading the VCC shall not inhibit the emergency operation of other nodes on the fire alarm network.
 - h. The Digital Voice Command shall be modular in construction, and shall be capable of being field programmable without requiring the return of any components to the manufacturer and without requiring use of any external computers or other programming equipment.
 - i. The Digital Voice Command and associated equipment shall be protected against unusually high voltage surges or line transients.
- K. Software Modifications: The system structure and software shall place no limit on the type or extent of software modifications on-site. Modification of software shall not require power-down of the system or loss of system fire protection while modifications are being made. Systems that require the use of external programmers or change of EPROMs are not acceptable.
- L. Computer Interface: The fire alarm control panel shall network to the computer.
- M. Logic: The fire alarm system shall support generic functions that deal with binary states (True/False, high/low), and produce desired outputs from one or more binary inputs (for example, alarm outputs from detector or manual station inputs). AND, OR, NOT, Any N, D Latch, RS Latch, Time Base Control, Start Timer, Restart Timer are generic functions. Generic functions can be used as inputs to other function. The system shall support 1500 logic functions.
- N. History – The system shall store 5000 events in history while in straight mode and 4500 in circular mode. In straight mode, trouble warnings will occur at 4000 and 4500 events. In circular mode, the control panels shall maintain a 2000 event alarm history, which consists of 2000 most recent alarm events from the 4500 event history file.
- O. Reports:
- 1. The system shall have the ability to provide configuration, status, queue and history reports.
 - 2. Configuration reports shall provide the following information:
 - a. Custom Messages
 - b. Entity Type
 - c. Device Usage
 - d. Device Category
 - e. Firmware revision
 - 3. Status reports shall provide the following information:
 - a. Disarmed cards and devices
 - b. ASD settings
 - c. Sensitivity in %/foot
 - d. Alarm threshold in %/foot
 - e. Temperature in degrees C
 - f. Walktest
 - 4. Queue reports shall provide the following information:

- a. Alarm events with custom message and event time
 - b. Supervisory events with custom message and event time
 - c. Security events with custom message and event time
 - d. Trouble events with custom message and event time
5. History reports shall provide Address, History Type, Description, Time & Date and Custom Message. The following event types shall be reported:
- a. Alarm events
 - b. Supervisory events
 - c. Security events
 - d. Status changes
 - e. Alarm verification
 - f. Output activation from logic
 - g. System Reset
 - h. Event Acknowledgements
 - i. Block Acknowledgements
 - j. Audible Silence System Flag Changes
 - k. Sensitivity Changes
 - l. Arm / Disarm Commands
 - m. Arm / Disarm By Logic
 - n. Manual Output Overrides
 - o. Output Overrides By Logic
 - p. Time Changes
 - q. Menu Logins
 - r. ASD Changes
 - s. Walktest
 - t. Device Input to Logic Activations/Deactivations

- P. A UTSW Fire Safety lock shall be applied to the fire alarm control panel at the cost of the contractor. This is a separate key from what will be utilized on the battery boxes, fire alarm pull stations, and similar devices.
- Q. A UTSW Information Resources wire drop shall be provided for communications with the Edwards Fireworks system. Additionally, a "traditional" phone line shall be provided to meet the compliance of NFPA 72 regarding primary and secondary communications from the fire alarm panel.

2.3 POWER SUPPLY

- A. The system Power Supply/Charger shall be a 12-amp supply with battery charger. The power supply shall be filtered and regulated. The power supply shall have a minimum of 1 power limited output rated at 4 amps, and a minimum of 1 output rated at 12 amps. The system power supply can be expanded up to 48 amps. The auxiliary power supply module shall share common batteries with the primary power supply. The system power supply shall have 4 relays, 1 for common alarm, one for common trouble and two programmable relays. The power supply shall be rated for 120/240 VAC 50/60 Hz.
- B. The battery charger shall be able to charge the system batteries up to 100 AH batteries. Battery charging shall be microprocessor controlled and programmed with a special software package to select charging rates and battery sizes. Thermistor for monitoring battery temperature to control charging rate shall be available.
- C. The power supply shall have a plug for an AC adapter cable, which allows a technician to plug in a laptop computer for up or down loading program information or test equipment.
- D. Transfer from AC to battery power shall be instantaneous when AC voltage drops to a point where it is not sufficient for normal operation.

2.4 SYSTEM ENCLOSURE

- A. Provide the enclosure needed to hold all the cards and modules as specified with at least spare capacity for two cards. The enclosure outer door shall be either black or red. Provide the color per UTSW requirements. The outer doors shall be capable of being a left hand open or a right hand open. The inner door shall have a left hand opening. System enclosure doors shall provide where required ventilation for the modules or cards in the enclosure.
- B. Provide system enclosure for all amplifiers. Where required by the manufacturer, provide means for venting heat from the enclosure either by having enclosure sides and top vented or the doors vented.

2.5 INTELLIGENT INITIATING DEVICES

A. Intelligent Initiation Devices – General

- 1. All initiation devices shall be insensitive to initiating loop polarity. Specifically, the devices shall be insensitive to plus/minus voltage connections on either Style 4 or Style 6 circuits.

B. Smoke Detectors – Addressable

- 1. The detector shall have an LED to identify normal operation, maintenance required and alarm.
- 2. Provide intelligent analog addressable multi-sensor smoke detectors at the locations shown on the drawings. The intelligent detector gathers analog information from each of its two fire sensing elements and converts it into digital signals. The detectors on-board microprocessor measures and analyzes these signals separately with respect to a third element – Time. It compares the information to historical readings, time patterns and known fire characteristics to make an alarm decision. Digital filters remove signal patterns that are not typical of fires.
- 3. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.

C. Heat Detectors – Addressable

- 1. Thermal detectors shall be rated at 135 degrees fixed temperature and 15 degrees per minute rate of rise. Detectors shall be constructed to compensate for the thermal lag inherent in conventional type detectors due to the thermal mass, and alarm at the set point of 135 degrees Fahrenheit. The choice of alarm reporting as a fixed temperature detector or a combination of fixed and rate of rise shall be made in system software and be changeable at any time without the necessity of hardware replacement.
- 2. The detectors furnished shall have a listed spacing for coverage up to 2,500 square feet and shall be installed according to the requirements of NFPA 72 for open area coverage.
- 3. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.

D. Duct Smoke Detectors – Addressable

- 1. The detector shall have an LED to identify normal operation, maintenance required and alarm.
- 2. Provide intelligent analog addressable multi-sensor smoke detectors at the locations shown on the drawings. The intelligent detector gathers analog information from each of its two fire sensing elements and converts it into digital signals. The detectors on-board microprocessor measures and analyzes these signals separately with respect to a third element – Time. It compares the information to historical readings, time patterns and known fire characteristics to make an alarm decision. Digital filters remove signal patterns that are not typical of fires.
- 3. Duct Detectors are to be supplied by the fire alarm contractor, installed by the mechanical contractor. AHU shutdown is to be wired by the Division 26 contractor and tested by both the mechanical and fire alarm contractor. Fire Smoke Damper Operation is the responsibility of the Mechanical Contractor.
- 4. If associated remote indicators are required by code, they will be ceiling mounted +/- 1" f center of tile with back box and supports.

Devices shall be accessible at all times as required by NFPA. A remote test switch shall be provided where the devices are not accessible from the floor level. However, for maintenance purposes, devices shall be accessible

without moving other contractor's equipment, wiring, etc., nor be located behind construction features which would render it in accessible.

E. Detector Bases – Addressable

1. Detector bases shall be low profile twist lock type with screw clamp terminals and self-wiping contacts. Bases shall be installed on an industry standard, 4" square or octagonal electrical outlet box.
2. Where selective localized control of electrical devices is required for system operation, furnish and install detector base with software programmed addressable relay integral to the base. The relay shall switch electrical loads within relay ratings, as indicated on the drawings. Operation of the addressable control circuit shall be independent of the number of detectors and relays on the circuit or the number in an alarm state. Relay bases shall be rated for resistive or inductive load (120VAC or 30VDC) 3 amps.

F. Manual Pull Stations – Addressable

1. Provide addressable manual stations where shown on the drawings, to be flush or surface mounted as required. Manual stations shall contain the intelligence for reporting address, identity, alarm and trouble to the fire alarm control panel. The manual station communications shall allow the station to provide alarm input to the system and alarm output from the system within less than four (4) seconds.
2. The manual station shall be equipped with terminal strip and pressure style screw terminals for the connection of field wiring. Surface mounted stations where indicated on the drawings shall be mounted using a manufacturer's prescribed matching red enamel outlet box.
3. Provide double action pull station (keyswitch operated).
4. All pull stations shall be mounted to a backbox mounted in the wall, and at the height compliant with the code.
5. Clear security covers with alarms shall be provided where indicated to prevent the activation of false alarms.
6. Provide lockable institutional pull stations for the Psychiatric Floor(s).
 - a. Simplex 2099 Series Double Action Operation.

G. Addressable Interface Devices

1. Addressable interface devices shall be provided to monitor contacts for such items as water-flow, tamper, and PIV switches connected to the fire alarm system. These interface devices shall be able to monitor a single or dual contacts. An address will be provided for each contact. Where remote supervised relay is required the interface shall be equipped with a SPDT relay rated for 4 amps resistive and 3.5 amps inductive.
2. Where needed a conventional zone module shall connect to the signal line circuit, which will allow the use of conventional initiation devices. This module shall have the ability to support up to 15 convention smoke detectors and an unlimited number of contact devices. This module shall also be capable of monitoring linear beam detectors and conventional Flame detectors. Where required, there shall be an intrinsically safe detection solution for NEMA defined intrinsically safe installations compatible with the conventional zone module.
3. Single device damper monitoring and control: When connected to the FACP, a single switch input shall be able to monitor all 3 states of a damper – open, closed, and in transit. When connected to a FACP, a single relay shall be able to fully control a damper (through the relay connected to the motor control) while also using its switch input for monitoring all 3 states of the damper.

2.6 DEVICE PROGRAMMING UNIT

- A. The programming tool shall program the intelligent devices with addresses. The unit shall test the device to respond to its address. Dipswitches and rotary switches shall not be acceptable. The programmer shall have carrying case.

2.7 NOTIFICATION APPLIANCES

A. Strobes

1. The strobes shall meet and be listed for UL Standard 1971 (Emergency Devices for the Hearing-Impaired) for Indoor Fire Protection Service
2. Strobe shall be listed for indoor use, and shall meet the requirements of FCC Part 15 Class B
3. Strobe appliances shall produce a flash rate of one (1) flash per second over the Regulated Voltage Range, and shall incorporate a Xenon flashtube enclosed in a rugged Lexan® lens
4. All inputs shall be compatible with standard, reverse polarity supervision of circuit wiring by a Fire-Alarm Control Panel (FACP)
5. The Strobe shall be of low-current design
6. The strobe intensity shall have field-selectable settings, and shall be rated per UL Standard 1971 for 15/30/75/95cd or 115/177cd for ceiling mount where Multi-Candela appliances are specified
7. The selector switch for selecting the candela shall be tamper resistant
8. The appliance shall be compatible with power supplies with built-in sync protocol when synchronization is required
9. The strobes shall not drift out of synchronization at any time during operation
10. If the sync module or Power Supply fails to operate, (i.e. - contacts remain closed), the strobe shall revert to a non-synchronized flash rate
11. The strobes shall be designed for indoor surface of flush mounting
12. The Strobe Appliances shall incorporate a Patented, Integral Strobe Mounting Plate that shall allow mounting to single-gang, double-gang, 4-inch square, 100mm European type back boxes, or the Surface Back box
13. The Strobe Plate shall mount to either a standard, 4-inch square back box for flush mounting, or shall mount to the back box for surface mounting
14. All notification appliances shall be backward compatible
15. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.
16. All ceiling devices shall have backboxes. All ceiling backboxes must be supported from the structure or the ceiling grid, not the ceiling tile.

B. Speaker and Speaker Strobes

1. Speakers shall be UL Listed under Standard 1480 for Fire Protective Service, and speakers equipped with strobes shall be listed under UL Standard 1971 for Emergency Devices for the Hearing-Impaired
2. Speaker with strobes shall be certified to meet the requirements of FCC Part 15, Class B
3. All speakers shall be designed for a field-selectable input of either 25 or 70 VRMS; with selectable power taps from 1/8 watt to 2 watts
4. All ceiling-mount models shall have listed sound output of up to 87 dB at 10 feet and a listed frequency response of 400 to 4000 Hz
5. Speaker shall incorporate a sealed-back construction
6. All inputs shall employ terminals that accept #12 to #18 AWG wire sizes
7. Strobe intensity, where Multi-Candela appliances are specified, shall have field-selectable settings, and shall be rated per UL Standard 1971 for:
 - a. 15/30/75/110cd (ceiling mounting)
 - b. 135/185cd (ceiling mounting)
8. Selector switch for selecting the candela shall be tamper resistant
9. The strobe portion, when synchronization is required, shall be compatible with power supply with built-in protocol.
10. The strobes shall not drift out of synchronization at any time during operation
11. The strobes shall revert to a non-synchronized flash-rate, if the sync module or Power Supply should fail to operate (i.e. – contacts remain closed)

12. Wall-mount speaker and speaker-strobe appliances shall be designed for indoor-flush mounting to 4" x 2-1/8" electrical boxes without need for an extension ring or surface mounting
13. Ceiling-mount, speaker-strobe appliances shall be designed for indoor-flush mounting
14. Speaker and speaker strobe shall incorporate a speaker-mounting plate with a snap-on grille cover
15. The finish of the speakers and speaker strobes shall be red.
16. All speaker and speaker-strobe appliances shall listed for Special Applications: Strobes are designed to flash at 1-flash-per-second minimum over their "Regulated Input Voltage Range"
17. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.
18. All ceiling devices shall have backboxes. All ceiling backboxes must be supported from the structure or the ceiling grid, not the ceiling tile.
19. Speakers in the stairwells and elevator cabs shall be set up for paging only from the fire alarm control panel.

C. Vandal-Resistant Speakers

1. Speakers shall be UL Listed under Standard 1480 for Fire Protective Service, and speakers equipped with strobes shall be listed under UL Standard 1971 for Emergency Devices for the Hearing-Impaired
2. Speaker with strobes shall be certified to meet the requirements of FCC Part 15, Class B
3. Models shall be listed for outdoor use under UL Standard 1480
4. All models shall have listed sound output of up to 87 dB at 10 feet and a listed frequency response of 400 to 4000 Hz
5. All speakers shall include both 25 and 70 volt VRMS inputs with field-selectable power taps from 1/8 to 8 watts, with listed sound output up to 94 dB for speakers or speaker strobes
6. All models shall have provisions for standard NAC supervision and IN / OUT field wiring using terminals that accept #12 to #18 AWG wiring
7. Combination speaker strobe appliances shall incorporate a Xenon flashtube enclosed in a rugged Lexan® lens or equivalent with solid-state circuitry
8. Strobes shall meet UL Standard 1971, and produce a flash rate of one (1) flash per second minimum over the listed input voltage (16VDC-33VDC) range
9. Multi-candela strobe intensity shall be rated per UL Standard 1971 at 15/30/75/110cd
10. All strobe versions shall incorporate circuitry for synchronized strobe flash, and shall be designed for compatibility with power supply.
11. All UL Standard 1971 Listed strobe appliances shall be verified to meet FCC Part 15, Class "B"
12. Strobe activation shall be via independent input from the speaker notification appliance circuit
13. Speaker and speaker-strobe appliances shall be designed for indoor surface or flush mounting
14. All speaker and speaker-strobe appliances shall be listed for Special Applications: Strobes are designed to flash at 1-flash-per-second minimum over their "Regulated Input Voltage Range"
15. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.
16. All ceiling devices shall have backboxes. All ceiling backboxes must be supported from the structure or the ceiling grid, not the ceiling tile.

D. Horn Strobes and Horns

1. Appliances shall meet and be listed for UL Standard 1971(Emergency Devices for the Hearing-Impaired for Indoor Fire Protection Service)
2. Horn shall be UL Listed under Standard 464 (Fire Protective Signaling)
3. Horn strobe shall be listed for indoor use, and shall meet the requirements of FCC Part 15 - Class B.
4. All inputs shall be compatible with standard reverse polarity supervision of circuit wiring by the Fire Alarm Control Panel (FACP)
5. Audible portion of the appliance shall have a minimum of two (2) field-selectable settings for dBA levels (90 and 95 dBA), and shall have a choice of continuous or temporal (Code 3) audible output
6. Strobe portion of the appliance shall produce a flash rate of one (1) flash per second over the Regulated Input Voltage Range, and shall incorporate a Xenon flashtube enclosed in a rugged Lexan® lens
7. Horn shall be of low-current design

8. Strobe intensity – where ceiling mount, Multi-Candela appliances are specified – shall have field-selectable settings, and shall be rated per UL Standard 1971 for:
 - a. 15/30/75/95cd
 - b. 115/117cd
9. The selector switch for selecting the candela setting shall be tamper resistant
10. The appliance, when synchronization is required, shall be compatible with Power Supplies.
11. The strobes shall not drift out of synchronization at any time during operation
12. The strobes shall revert to a non-synchronized flash-rate, if the sync module or Power Supply should fail to operate (i.e. – contacts remain closed)
13. The appliance shall also be designed so that the audible signal may be silenced while maintaining strobe activation
14. Horn Strobes and horn shall incorporate a Universal Mounting Plate that shall allow mounting to a single-gang, double-gang, 4-inch square, 100mm European type backboxes, or the Surface Backbox.
15. If required, an NATP (Notification Appliance Trim Plate) shall be provided
16. All notification appliances shall be listed for Special Applications:
 - a. Strobes are designed to flash at 1-flash-per-second minimum over their “Regulated Input Voltage Range”
17. All candela ratings represent minimum-effective Strobe intensity, based on UL Standard 1971.
18. Series NS Strobe products are listed under UL Standard 1971 for indoor use with a temperature range of 32°F to 120°F (0°C to 49°C) and maximum humidity of 93% (±2%)
19. Horns shall be listed under UL Standard 464 for audible signal appliances (Indoor use only)
20. Devices ceiling mounted shall be installed within 1" of the center of the ceiling tiles.
21. All ceiling devices shall have backboxes. All ceiling backboxes must be supported from the structure or the ceiling grid, not the ceiling tile.

2.8 ACCESSORIES

A. Door Holders

1. Door holders shall be semi recessed, 24 volt. The unit shall operate in conjunction with the fire alarm system to provide automatic release of the door as indicated in the sequence of operation. The unit shall be UL listed. The door holder shall be wired to the fire alarm system in accordance with the National Electrical Code wiring practice.

B. Remote Annunciator

1. Remote LCD Alphanumeric Annunciator: Shall have LCD display functions for alarm, supervisory, and trouble indications as well as event location information.

2.9 DIGITAL COMMUNICATOR

A. The Multi-Point Digital Alarm Communicator shall be UL864 listed to provide point identification of alarm, supervisory, security and trouble events to a Central or Remote Receiving Station. The Communicator shall support the following:

1. Ademco Contact ID or SIA protocol
2. Ademco Contact ID selection shall provide the ability to transmit events for up to 999 individual points
3. SIA selection shall provide the ability to transmit events for up to 2040 individual points
4. Programming of accounts and phone numbers
5. Dual phone line interface
6. Line fault monitoring.
7. Automatic 24-hour test

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Perform work in accordance with the requirements of NFPA 70, NFPA 72 and NECA 1-2006, Standard of Good Workmanship in Electrical Contracting. Contracting company shall maintain a valid Alarm Certificate of Registration (ACR) and the technician(s) shall maintain a valid Fire Alarm Technician License (FAL) from the State of Texas throughout the installation and through the acceptance process.
- B. Fasten equipment to structural members of building or metal supports attached to structure, or to concrete surfaces.
- C. In the event that limited energy cable installation is allowed, all cable runs shall be run at right angles to building walls, supported from structure at intervals not exceeding 5 feet and where installed in environmental air plenums, be rated for such use and tied/supported by components listed for environmental air plenums installation.
- D. Wiring Method: Install cables in raceways and cable trays except within consoles, cabinets, desks, and counters and except in accessible ceiling spaces and in gypsum board partitions where unenclosed wiring method may be used. Conceal raceway and cables except in unfinished spaces.
- E. Wiring Method: Conceal conductors and cables in accessible ceilings, walls, and floors where possible.
- F. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points with no excess and without exceeding manufacturer's limitations on bending radii. Provide and use lacing bars and distribution spools.
- G. Provide primary power for each panel from normal/ emergency panels as indicated on the Electrical Power Plans. Power shall be 120 VAC service, transformed through a two-winding, isolation type transformer and rectified to low voltage DC for operation of all circuits and devices.

3.3 BOXES, ENCLOSURES AND WIRING DEVICES

- A. Boxes shall be installed plumb and firmly in position.
- B. Extension rings with blank covers shall be installed on junction boxes where required.
- C. Junction boxes served by concealed conduit shall be flush mounted.
- D. Upon initial installation, all wiring outlets, junction, pull and outlet boxes shall have dust covers installed. Dust covers shall not be removed until wiring installation when permanent dust covers or devices are installed.
- E. Fire alarm box covers shall be red in color unless it is exposed as part of the final interior finish of the building, in which case it must match its background. "Fire alarm system" decal or silk-screened label shall be applied to all junction box covers. The back box and conduit can be standard grey in color.

3.4 CONDUCTORS

- A. Each conductor shall be identified as shown on the drawings at each with wire markers at terminal points. Attach permanent wire markers within 2 inches of the wire termination. Marker legends shall be visible.
- B. All wiring shall be supplied and installed in compliance with the requirements of the National Electric Code, NFPA 70, Article 760, and that of the manufacturer.
- C. Wiring for strobe and audible circuits shall be a minimum 14 AWG, signal line circuits; 18 AWG twisted shielded, speaker circuits; 16 AWG twisted.
- D. All splices shall be made using solderless connectors. All connectors shall be installed in conformance with the manufacturer recommendations.
- E. The installation contractor shall submit for approval prior to installation of wire, a proposed color code for system conductors to allow rapid identification of circuit types.
- F. Wiring within sub panels shall be arranged and routed to allow accessibility to equipment for adjustment and maintenance.
- G. Wiring for notification appliance circuits and associated circuits required to activate the notification appliance circuits shall be installed in a 2 hour enclosure or rated for 2 hour survivability until the circuit enters the associated evacuation zone (smoke zone).

3.5 DEVICES

- A. Relays and other devices to be mounted in auxiliary panels are to be securely fastened to avoid false indications and failures due to shock or vibration.
- B. Wiring within panels shall be arranged and routed to allow accessibility to equipment for adjustment and maintenance.
- C. All devices and appliances shall be mounted to or in an approved electrical box.

3.6 IDENTIFICATION

- A. Identify system components, wiring, cabling, and terminals. Comply with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."
- B. Permanently label or mark each conductor at both ends with permanent alphanumeric wire markers.
- C. A consistent color code for fire alarm system conductors throughout the installation.
- D. All initiating, indicating, monitor and relay modules will have self adhesive label attached to each device. Label will match labeling scheme on submittal drawings.

3.7 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- B. Testing General:
 - 1. All Alarm Initiating Devices shall be observed and logged for correct zone and sensitivity. These devices and their bases shall be tagged with adhesive tags located in an area not visible when installed, showing the initials of the installing technician and date.
 - 2. Wiring runs shall be tested for continuity, short circuits and grounds before system is energized.

Resistance, current and voltage readings shall be made as work progresses.

3. The acceptance inspector shall be notified before the start of the required tests. All items found at variance with the drawings or this specification during testing or inspection by the acceptance inspector shall be corrected.
4. Test reports shall be delivered to the acceptance inspector as completed.
5. All test equipment, instruments, tools and labor required to conduct the system tests shall be made available by the installing contractor. The following equipment shall be a minimum for conducting the tests:
 - a. Ladders and scaffolds as required to access all installed equipment.
 - b. Multi-meter for reading voltage, current and resistance.
 - c. Two way radios, and flashlights.
 - d. A manufacturer recommended device for measuring air flow through air duct smoke detector sampling assemblies.
 - e. Decibel meter.
 - f. In addition to the testing specified to be performed by the installing contractor, the installation shall be subject to test by the acceptance inspector.

3.8 ACCEPTANCE TESTING

- A. A written acceptance test procedure (ATP) for testing the fire alarm system components and installation will be prepared by the contractor and approved by the UTSW Director of Fire and Occupational Safety or his designee in accordance with NFPA 72 and this specification. The contractor shall be responsible for the performance of the ATP, demonstrating the function of the system and verifying the correct operation of all system components, circuits, and programming. Acceptance testing will be performed by the AHJ (UTSW Director of Fire and Occupational Safety or his designee) at UTSW.
- B. A program matrix shall be prepared by the installing contractor referencing each alarm input to every output function affected as a result of an alarm condition on that input.
- C. The installing contractor prior to the ATP shall prepare a complete listing of all device labels for alphanumeric annunciator displays.
- D. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests shall be witnessed and approved by the AHJ (UTSW Director of Fire and Occupational Safety or his designee and test results recorded for use at the final acceptance test.
- E. Preliminary Testing: Conduct preliminary tests to ensure that all devices and circuits are functioning properly. After preliminary testing is complete, provide a letter certifying that the installation is complete and fully operable. The letter shall state that each initiating and indicating device was tested in place and functioned properly. The letter shall also state that all panel functions were tested and operated properly. The Contractor and an authorized representative from each supplier of equipment shall be in attendance at the preliminary testing to make necessary adjustments.
- F. Final Acceptance Test: Final acceptance testing shall be scheduled through the UTSW project manager with a minimum of three business days notification provided to UTSW Fire and Safety.
 1. Verify that the control unit is in the normal condition as detailed in the manufacturer's operating and maintenance manual.
 2. Test each initiating and indicating device and circuit for proper operation and response. Disconnect the confirmation feature for smoke detectors during tests to minimize the amount of smoke or test gas needed to activate the detector.
 3. Test the system for all specified functions in accordance with the contract drawings and specifications and the manufacturer's operating and maintenance manual.
 4. Visually inspect all wiring.
 5. Verify that all software control and data files have been entered or programmed into the FACP.

6. Measure the current in circuits to assure that there is the calculated spare capacity for the circuits.
 7. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
 8. Measure the voltage drop at the most remote appliance on each notification appliance circuit.
 9. Verify that as built drawings reflect the actual installation.
- G. The UTSW Fire and Safety representative shall use the system record drawings in combination with the documents specified in this specification during the testing procedure to verify operation as programmed. In conducting the ATP, the UTSW Fire and Safety representative shall request demonstration of any or all input and output functions. The items tested shall include but not be limited to the following:
1. System wiring shall be 100% tested to demonstrate correct system response and correct subsequent system operation in the event of:
 - a. Open, shorted and grounded signal line circuits.
 - b. Open, shorted and grounded notification, releasing circuits.
 - c. Primary power or battery disconnected.
 2. System notification appliances shall be demonstrated as follows:
 - a. All alarm notification appliances actuate as programmed
 - b. Audibility and visibility at required levels.
 3. System indications shall be demonstrated as follows:
 - a. Correct message display for each alarm input at the control display.
 - b. Correct annunciator light for each alarm input at each annunciator and graphic display as shown on the drawings.
 - c. Correct history logging for all system activity.
 4. System off-site reporting functions shall be demonstrated as follows:
 - a. Correct zone transmitted for each alarm input
 - b. Trouble signals received for disconnect
 - c. Supervisory signals for all devices
 5. Secondary power capabilities shall be demonstrated as follows:
 - a. System primary power shall be disconnected for a period of time as specified herein. At the end of that period, an alarm condition shall be created and the system shall perform as specified for a period as specified.
 - b. System primary power shall be restored for forty-eight hours and system-charging current shall be normal trickle charge for a fully charged battery bank.
 - c. System battery voltages and charging currents shall be checked at the fire alarm control panel.

3.9 DOCUMENTATION

- A. System documentation shall be furnished to UTSW Fire and Safety and shall include but not be limited to the following:
1. System record drawings and wiring details including one set of reproducible drawings, and a CD ROM with copies of the record drawings in DXF format for use in a CAD drafting program.
 2. System operation, installation and maintenance manuals.
 3. System matrix showing interaction of all input signals with output commands.
 4. Documentation of system voltage, current and resistance readings taken during the installation, testing and ATP phases of the system installation.
 5. System program showing system functions, controls and labeling of equipment and devices.

3.10 PROTECTION

- A. Remove and replace devices and panel components that are wet, moisture damaged, or mold damaged.

3.11 DEMONSTRATION

- A. Instructor: Include in the project the services of an instructor, who shall have received specific training from the manufacturer for the training of other persons regarding the inspection, testing and maintenance of the system provided. The instructor shall train the employees designated by the owner, in the care, adjustment, maintenance, and operation of the fire alarm system.
- B. Training sessions shall cover all aspects of system performance, including system architecture, signaling line circuit configurations, sensor and other initiating device types, locations, and addresses, fire alarm control panel function key operation, and other functions as designated by the owner.
- C. Required Instruction Time: Provide 8 hours of instruction after final acceptance of the system. The instruction shall be given during regular working hours on such dates and times as are selected by the owner.
- D. Comprehensive system troubleshooting training shall be provided for a single individual designated by the owner. This session shall be separate and distinct from the above described sessions.
- E. All training sessions shall be conducted following final system certification and acceptance. Three additional training sessions shall be provided for all security personnel on all shifts six months after final system certification.
- F. All training sessions shall be conducted by an authorized fire alarm system distributor representative, who has received specific training from the manufacturer for the training of other persons regarding the inspection, testing, and maintenance of the system provided.

PART 4 - PERMITTING

4.1 PLAN REVIEW & PERMITTING

- A. Plans shall be submitted for review and approval a minimum of 3 business days through the UT Southwestern Project Manager.
- B. Plans shall include all appropriate information, calculations, and manufactures information as required in National Fire Protection Association National Fire Alarm and Signaling Code (NFPA 72). Failure to provide the necessary information will result in a delay in the review of the project until all of the information is provided.
- C. Submittal of plans does not constitute approval. Contractors working on the project will do so at their own risk and will replace, without cost to the owner, any components necessary as required by the approved plans.
- D. Three copies of submittal information shall be provided to the UT Southwestern Medical Center Office of Safety and Business Continuity – Attention Director of Fire and Occupational Safety. A copy of the approved, stamped, plans shall be maintained on the job at all times.
- E. A permit shall be provided with the approval of the submitted plans

4.2 INSPECTION

- A. All state paperwork for testing shall be copied at the time of the conclusion for the test and signed by the contractor and UTSW Fire Safety representative. A copy of the test shall be maintained by the contractor and provided immediately to the Fire Safety representative. The test will not be considered passed under the paperwork for the test is complete.

4.3 AS BUILTS

- A. As built drawings shall be provided on the same day of the final approval of the building.

END OF SECTION